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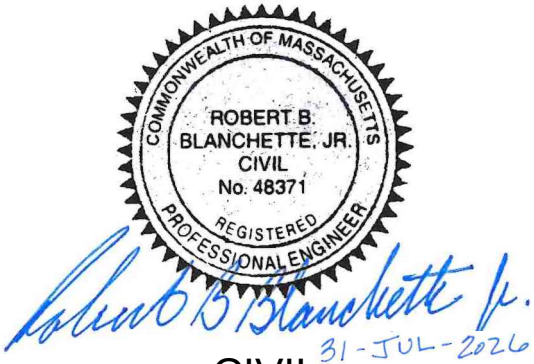
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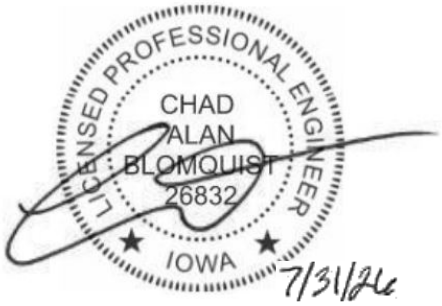
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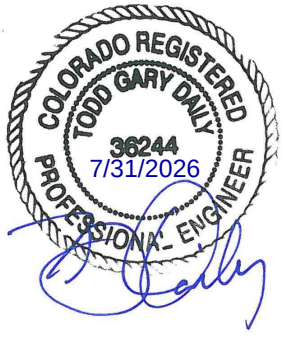
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FIRE PROTECTION

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SECTION 21 08 00.01
8/19

DVA/USACE PROJECTS COMMISSIONING OF FIRE SUPPRESSION SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

a. The requirements of this Section apply to all sections of Division 21.

b. This project will have selected building systems commissioned. The complete list of equipment and systems to be commissioned is specified in Section 01 91 00.01 DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS. The commissioning process, which the Contractor is responsible to execute, is defined in Section 01 91 00.01 DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIRMENTS. A Commissioning Agent (CxA) appointed by the Government and Commissioning Manager (CxM) as indicated in Section 01 91 00 GENERAL COMMISSIONING REQUIRMENTS will manage the commissioning process.

1.2 RELATED WORK

a. UFGS GENERAL REQUIREMENTS sections

b. Section 01 91 00 GENERAL COMMISSIONING REQUIREMENTS

c. UFGS Section 01 33 00 SUBMITTAL PROCEDURES

1.3 SUMMARY

a. This Section includes requirements for commissioning the Fire Suppression systems, subsystems and equipment. This Section supplements the general requirements specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

b. Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for more details regarding processes and procedures as well as roles and responsibilities for all Commissioning Team members.

1.4 DEFINITIONS

Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for definitions.

1.5 COMMISSIONED SYSTEMS

a. Commissioning of a system or systems specified in Division 21 is part of the construction process. Documentation and testing of these systems, as well as training of the VA’s Operation and Maintenance personnel in accordance with the requirements of Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and of Division 21, is required in cooperation with the VA and the Commissioning Agent.

b. The Fire Suppression systems commissioning will include the systems listed in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS:

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are forinformation only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Pre-Functional Checklists; G, DO

Contractors Tests; G, DO

Systems Functional Performance Testing; G, DO

SD-08 Manufacturer's Instructions

Training Agendas And Trainer Resumes; G, DO

Training Plans; G, DO

1.7 SUBMITTAL INSTRUCTIONS

a. The commissioning process requires review of selected Submittals. The CxA will provide a list of submittals that will be reviewed by the CxA and CxM. This list will be reviewed and approved by the DVA/USACE prior to forwarding to the Contractor. Refer to UFGS Section 01 33 00 SUBMITTAL PROCEDURES for further details.

b. The commissioning process requires Submittal review simultaneously with engineering review. Specific submittal requirements related to the commissioning process are specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 CONSTRUCTION INSPECTIONS

Commissioning of the building fire suppression systems will require inspection of individual elements of the fire suppression construction throughout the construction period. The Contractor shall coordinate with the CxA and CxM in accordance with Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS and the Commissioning plan to schedule inspections as required to support the Commissioning Process.

3.2 PRE-FUNCTIONAL CHECKLISTS

The Contractor shall complete Pre-Functional Checklists to verify systems, subsystems, and equipment installation is complete and systems are ready for Systems Functional Performance Testing. The CxM will prepare Pre-Functional Checklists to be used to document equipment installation. The Contractor shall complete the checklists. Completed checklists shall be submitted to the DVA/USACE, and CxA for review. The CxM and CxA may spot check a sample of completed checklists. If the CxM and CxA determine that the information provided on the checklist is not accurate, the CxM

will return the marked-up checklist to the Contractor for correction and resubmission. If the CxM and CxA determine that a significant number of completed checklists for similar equipment are not accurate, the CxM in coordination with CxA will select a broader sample of checklists for review. If the CxM and CxA determine that a significant number of the broader sample of checklists is also inaccurate, all the checklists for the type of equipment will be returned to the Contractor for correction and resubmission. Refer to SECTION 01 91 00.01 DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS for submittal requirements for Pre-Functional Checklists, Equipment Startup Reports, and other commissioning documents.

3.3 CONTRACTORS TESTS

Contractor tests as required by other sections of Division 21 shall be scheduled and documented in accordance with UFGS GENERAL REQUIREMENTS sections. All testing shall be incorporated into the project schedule. Contractor shall provide no less than 7 calendar days' notice of testing. The CxM will witness selected Contractor tests at the sole discretion of the CxM and CxA. The CxA and other Government representative(s) may choose to participate in some or all the testing procedures as observers. Contractor tests shall be completed prior to scheduling Systems Functional Performance Testing.

3.4 SYSTEMS FUNCTIONAL PERFORMANCE TESTING

The Commissioning Process includes Systems Functional Performance Testing that is intended to test systems functional performance under steady state conditions, to test system reaction to changes in operating conditions, and system performance under emergency conditions. The CxM will prepare final and detailed Systems Functional Performance Test procedures for review and approval by the Contracting Officer. The Contractor shall review and comment on the tests prior to approval. The Contractor shall provide the required labor, materials, and test equipment identified in the test procedure to perform the tests. The CxM will witness and document the testing. The Contractor shall sign the test reports to verify tests were performed. See Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS, for additional details.

3.5 TRAINING OF VA PERSONNEL

Training of the VA operation and maintenance personnel is required in cooperation with the Contracting Officer, VA Resident Engineer, and CxM. Provide competent, factory authorized personnel to provide instruction to operation and maintenance personnel concerning the location, operation, and troubleshooting of the installed systems. Contractor shall submit training agendas and trainer resumes in accordance with the requirements of Section 01 91 00.01 DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIRMENTS. The instruction shall be scheduled in coordination with the Contracting Officer and VA Resident Engineer after submission and approval of formal training plans. Refer to Section 01 91 00.01 DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS and Division 21 Sections for additional Contractor training requirements.

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DIVISION 21 - FIRE SUPPRESSION

SECTION 21 13 13

WET-PIPE SPRINKLER SYSTEMS

6/15

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3.2 INSPECTION AND TEST
3.3 INSTRUCTIONS

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SECTION 21 13 13
6/15

WET-PIPE SPRINKLER SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

a. Design, installation and testing shall be in accordance with NFPA 13.

b. The design and installation of a hydraulically calculated automatic wet-pipe system complete and ready for operation, , including the penthouse, attic space, elevator machine rooms, elevator pits, .

c. Modification of the existing sprinkler system as indicated on the drawings and as further required by these specifications.

1.2 RELATED WORK

a. Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

b. Section 33 10 00, WATER UTILITIES.

c. Section 07 84 00, FIRESTOPPING.

d. Section 09 91 00, PAINTING.

e. Section 22 05 23, GENERAL-DUTY VALVES FOR PLUMBING PIPING.

f. Section 28 31 00, FIRE DETECTION AND ALARM.

g. Section 21 30 13, ELECTRIC-DRIVEN FIRE PUMPS.

1.3 DESIGN CRITERIA

a. Design Basis Information: Provide design, materials, equipment, installation, inspection, and testing of the automatic sprinkler system in accordance with the requirements of NFPA 13.

(1) Perform hydraulic calculations in accordance with NFPA 13 utilizing the Area/Density method. Do not restrict design area reductions permitted for using quick response sprinklers throughout by the required use of standard response sprinklers in the areas identified in this section.

(2) Sprinkler Protection: Sprinkler hazard classifications shall be in accordance with NFPA 13. The hazard classification examples of uses and conditions identified in the Annex of NFPA 13 shall be mandatory for areas not listed below. Request clarification from the Government for any hazard classification not identified. To determining spacing and sizing, apply the following coverage classifications:

(a) Light Hazard Occupancies: Patient care, treatment, and

customary access areas.

(b) Ordinary Hazard Group 1 Occupancies: Laboratories, Mechanical Equipment Rooms, Transformer Rooms, Electrical Switchgear Rooms, Electric Closets, and Repair Shops.

(c) Ordinary Hazard Group 2 Occupancies: Storage rooms, trash rooms, clean and soiled linen rooms, pharmacy and associated storage, laundry, kitchens, kitchen storage areas, retail stores, retail store storage rooms, storage areas, building management storage, boiler plants, energy centers, warehouse spaces, file storage areas for the entire area of the space up to (1500 square feet) and Supply Processing and Distribution (SPD).

(3) Hydraulic Calculations: Calculated demand including hose stream requirements shall fall no less than 10 percent below the available water supply curve.

(4) Water Supply: Base water supply on hydraulic analysis of new water tanks and fire pump.

(5) Zoning:

(a) For each sprinkler zone provide a control valve, flow switch, and a test and drain assembly with pressure gauge. For buildings greater than two stories, provide a check valve at each control valve.

(b) Sprinkler zones shall conform to the smoke barrier zones shown on the drawings.

(6) Provide seismic protection in accordance with NFPA 13. Contractor shall submit load calculations for sizing of sway bracing for systems that are required to be protected against damage from earthquakes.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications;

SD-02 Shop Drawings

Drawings;

SD-03 Product Data

Manufacturer’s Data Sheets;

SD-05 Design Data

Calculation Sheets;

Valve Charts;

SD-11 Closeout Submittals

Final Document Submittals;

1.5 SUBMITTAL INSTRUCTIONS

a. Submit as one package in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES. Prepare detailed working drawings that are signed by a NICET Level III or Level IV Sprinkler Technician or stamped by a Registered Professional Engineer licensed in the field of Fire Protection Engineering. As the Government review is for technical adequacy only, the installer remains responsible for correcting any conflicts with other trades and building construction that arise during installation. Partial submittals will not be accepted. Material submittals shall be approved prior to the purchase or delivery to the job site. Suitably bind submittals in notebooks or binders and provide an index referencing the appropriate specification section. In addition to the hard copies, provide submittal items in Paragraphs 1.5(A) 1 through 1.5(A) 5 electronically in pdf format on a compact disc or as directed by the Contracting Officer. Submittals shall include, but not be limited to, the following:

(1) Qualifications:

(a) Provide a copy of the installing contractors fire sprinkler and state contractor’s license.

(b) Provide a copy of the NICET certification for the NICET Level III or Level IV Sprinkler Technician who prepared and signed the detailed working drawings unless the drawings are stamped by a Registered Professional Engineer licensed in the field of Fire Protection Engineering.

(c) Provide documentation showing that the installer has been actively and successfully engaged in the installation of commercial automatic sprinkler systems for the past ten years.

(2) Drawings: Submit detailed 1:100 1/8 inch scale (minimum) working drawings conforming to the Plans and Calculations chapter of NFPA 13. Drawings shall include graphical scales that allow the user to determine lengths when the drawings are reduced in size. Include a plan showing the piping to the water supply test location.

(3) Manufacturer’s Data Sheets: Provide data sheets for all materials and equipment proposed for use on the system. Include listing information and installation instructions in data sheets. Where data sheets describe items in addition to those proposed to be used for the system, clearly identify the proposed items on the sheet.

(4) Calculation Sheets:

(a) Submit hydraulic calculation sheets in tabular form conforming to the requirements and recommendations of the Plans and Calculations chapter of NFPA 13.

(b) Submit calculations of loads for sizing of sway bracing in accordance with NFPA 13.

(5) Valve Charts: Provide a valve chart that identifies the location of each control valve. Coordinate nomenclature and identification of control valves with Contracting Officer. Where existing nomenclature does not exist, the chart shall include no less than the following: Tag ID No., Valve Size, Service (control valve, main drain, aux. drain, inspectors test valve, etc.), and Location.

(6) Final Document Submittals: Provide as-built drawings, testing and maintenance instructions in accordance with the requirements in Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES. In addition, submittals shall include, but not be limited to, the following:

(a) A complete set of as-built drawings showing the installed system with the specific interconnections between the system switches and the fire alarm equipment. Provide a complete set in the formats as follows. Submit items 2 and 3 below on a compact disc or as directed by the Contracting Officer.

1. One full size (or size as directed by the Contracting Officer) printed copy.

2. One complete set in electronic pdf format.

3. One complete set in AutoCAD format or a format as directed by the Contracting Officer.

(b) Material and Testing Certificate: Upon completion of the sprinkler system installation or any partial section of the system, including testing and flushing, provide a copy of a completed Material and Testing Certificate as indicated in NFPA 13. Certificates shall be provided to document all parts of the installation.

(c) Operations and Maintenance Manuals that include step-by-step procedures required for system startup, operation, shutdown, and routine maintenance and testing. The manuals shall include the manufacturer's name, model number, parts list, and tools that should be kept in stock by the owner for routine maintenance, including the name of a local supplier, simplified wiring and controls diagrams, troubleshooting guide, and recommended service organization, including address and telephone number, for each item of equipment.

(d) One paper copy of the Material and Testing Certificates and the Operations and Maintenance Manuals above shall be provided in a binder. In addition, these materials shall be provided in pdf format on a compact disc or as directed by the Contracting Officer.

(e) Provide one additional copy of the Operations and Maintenance Manual covering the system in a flexible protective cover and mount in an accessible location adjacent to the riser or as directed by the Contracting Officer.

1.6 QUALITY ASSURANCE

a. Installer Reliability: The installer shall possess a valid State of Connecticut contractor's license. The installer shall have been actively and successfully engaged in the installation of commercial automatic sprinkler systems for the past ten years.

b. Materials and Equipment: All equipment and devices shall be of a make and type listed by UL or approved by FM, or other nationally recognized testing laboratory for the specific purpose for which it is used. All materials, devices, and equipment shall be approved by the VA. All materials and equipment shall be free from defect. All materials and equipment shall be new unless specifically indicated otherwise on the contract drawings.

1.7 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 13	(2025; TIA 24-1; TIA 24-2; TIA 24-3) Standard for the Installation of Sprinkler Systems
NFPA 25	(2020; TIA 21-1) Standard for the Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems
NFPA 101	(2024) Life Safety Code
NFPA 170	(2024; ERTA 1 2023) Standard for Fire Safety and Emergency Symbols

UL SOLUTIONS (UL)

UL Fire Prot Dir	UL Product IQ (updated online) at https://productiq.ulprospector.com/en

FM GLOBAL (FM)

FM APP GUIDE	(updated on-line) Approval Guide https://www.approvalguide.com/

PART 2 PRODUCTS

2.1 PIPING & FITTINGS

a. Piping and fittings for private underground water mains shall be in accordance with NFPA 13.

(1) Pipe and fittings from inside face of building (12 in.) above finished floor to a distance of approximately (5 ft.) outside building: Ductile Iron, flanged fittings and 316 stainless steel bolting.

b. Piping and fittings for sprinkler systems shall be in accordance with NFPA 13.

(1) Plain-end pipe fittings with locking lugs or shear bolts are not permitted.

(2) Piping sizes 2 inches and smaller shall be black steel Schedule 40 with threaded end connections.

(3) Piping sizes (2 ½ inches) and larger shall be black steel Schedule 10 with grooved connections. Grooves in Schedule 10 piping shall be rolled grooved only.

(4) Use nonferrous piping in MRI Scanning Rooms.

(5) Plastic piping shall not be permitted except for drain piping.

(6) Flexible sprinkler hose shall be FM Approved and limited to hose with threaded end fittings with a minimum inside diameter or 1-inch and a maximum length of 6-feet.

2.2 VALVES

a. General:

(1) Valves shall be in accordance with NFPA 13.

(2) Do not use quarter turn ball valves for 2 inch or larger drain valves.

b. Control Valve: The control valves shall be a listed indicating type. Control valves shall be UL Fire Prot Dir or FM APP GUIDE for fire protection installations. System control valve shall be rated for normal system pressure but in no case less than 175 PSI.

c. Check Valve: Shall be of the swing type with a flanged cast iron body and flanged inspection plate.

d. Automatic Ball Drips: Cast brass 3/4 inch in-line automatic ball drip with both ends threaded with iron pipe threads.

e.

f.

2.3 FIRE DEPARTMENT SIAMESE CONNECTION

Brass, exterior fire department connection with brass escutcheon plate, and a minimum of two (2-1/2 inch) connections threaded to match those on the local fire protection service, with polished brass caps and chains. Provide escutcheon with integral raised letters

2.4 SPRINKLERS

a. All sprinklers shall be FM approved quick response except “institutional” type sprinklers shall be permitted to be UL Listed quick response. Provide FM approved quick response sprinklers in all areas, except that standard response sprinklers shall be provided in freezers, refrigerators, elevator hoistways, elevator machine rooms, and generator rooms.

b. Temperature Ratings: In accordance with NFPA 13 except that sprinklers in elevator shafts and elevator machine rooms shall be no less than intermediate temperature rated and sprinklers in generator rooms shall be no less than high temperature rated.

c. Provide sprinkler guards in accordance with NFPA 13 and when the elevation of the sprinkler head is less than 7 feet 6 inches above finished floor. The sprinkler guard shall be UL listed or FM approved for use with the corresponding sprinkler.

2.5 SPRINKLER CABINET

a. Provide sprinkler cabinet with the required number of sprinkler heads of all ratings and types installed, and a sprinkler wrench for each type of sprinkler in accordance with NFPA 13. Locate adjacent to the riser.

b. Provide a list of sprinklers installed in the property in the cabinet. The list shall include the following:

(1) Manufacturer, model, orifice, deflector type, thermal sensitivity, and pressure for each type of sprinkler in the cabinet.

(2) General description of where each sprinkler is used.

(3) Quantity of each type present in the cabinet.

(4) Issue or revision date of list.

2.6 SPRINKLER SYSTEM SIGNAGE

Rigid plastic, steel or aluminum signs with white lettering on a red background with holes for easy attachment. Sprinkler system signage shall be attached to the valve or piping with chain.

2.7 SWITCHES

a. OS&Y Valve Supervisory Switches shall be in a weatherproof die cast/red baked enamel, oil resistant, aluminum housing with tamper resistant screws, 1/2 inch conduit entrance and necessary facilities for attachment to the valves. Provide two SPDT switches rated at 2.5 amps at 24 VDC.

b. Water flow Alarm Switches: Mechanical, non-coded, non-accumulative retard and adjustable from 0 to 60 seconds minimum. Set flow switches at an initial setting between 20 and 30 seconds.

c.

d. Valve Supervisory Switches for Ball and Butterfly Valves: May be integral with the valve.

2.8 GAUGES

Provide gauges as required by NFPA 13. Provide gauges where the normal pressure of the system is at the midrange of the gauge.

2.9 PIPE HANGERS, SUPPORTS AND RESTRAINT OF SYSTEM PIPING

Pipe hangers, supports, and restraint of system piping shall be in accordance with NFPA 13.

2.10 WALL, FLOOR AND CEILING PLATES

Provide chrome plated steel escutcheon plates.

2.11 VALVE TAGS

Engraved black filled numbers and letters not less than 1/2 inch high for number designation, and not less than 1/4 inch for service designation on 19 gage, (1-1/2 inches) round brass disc, attached with brass "S" hook, brass chain, or nylon twist tie.

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation shall be accomplished by the licensed contractor. Provide a qualified technician, experienced in the installation and operation of the type of system being installed, to supervise the installation and testing of the system.

b. Installation of Piping: Accurately cut pipe to measurements established by the installer and work into place without springing or forcing. In any situation where bending of the pipe is required, use a standard pipe-bending template. Concealed piping in spaces that have finished ceilings. Where ceiling mounted equipment exists, such as in operating and radiology rooms, install sprinklers so as not to obstruct the movement or operation of the equipment. Sidewall heads may need to be utilized. In stairways, locate piping as near to the ceiling as possible to prevent tampering by unauthorized personnel and to provide a minimum headroom clearance of (seven feet six inches). Piping shall not obstruct the minimum means of egress clearances required by NFPA 101. Pipe hangers, supports, and restraint of system piping, and seismic bracing shall be installed accordance with NFPA 13. Sprinkler piping shall not be used to support nonsystem components

c. Welding: Conform to the requirements and recommendations of NFPA 13.

d. Drains: Provide drips and drains, including low point drains, in accordance with NFPA 13. Pipe drains to discharge at safe points outside of the building or to sight cones attached to drains of adequate size to readily carry the full flow from each drain under maximum pressure. Do not provide a direct drain connection to sewer system or discharge into sinks. Install drips and drains where necessary and required by NFPA 13. The drain piping shall not be restricted or reducedand shall be of the same diameter as the drain collector.

e. Supervisory Switches: Provide supervisory switches for sprinkler control valves.

f. Waterflow Alarm Switches: Install waterflow alarm switches and valves in stairwells or othereasily accessible locations.

g. Inspector's Test Connection: Install and supply in accordance with

NFPA 13, locate in a secured area, and discharge to the exterior of the building.

h. Affix cutout disks, which are created by cutting holes in the walls of pipe for flow switches and non-threaded pipe connections to the respective waterflow switch or pipe connection near to the pipe from where they were cut.

i. Provide escutcheon plates for exposed piping passing through walls, floors or ceilings.

j. Clearances: For systems requiring seismic protection, piping that passes through floors or walls shall have penetrations sized 2 inches nominally larger than the penetrating pipe for pipe sizes 1 inch to (3 ½ inches) and 4 inches nominally larger for penetrating pipe sizes 4 inches and larger.

k. Sleeves: Provide for pipes passing through masonry or concrete. Provide space between the pipe and the sleeve in accordance with NFPA 13. Seal this space with a UL Listed through penetration fire stop material in accordance with Section 07 84 00, FIRESTOPPING. Where core drilling is used in lieu of sleeves, also seal space. Seal penetrations of walls, floors and ceilings of other types of construction, in accordance with Section 07 84 00, FIRESTOPPING.

l. Where dry pendent sprinklers are used for freezers or similar spaces and they are connected to the wet pipe system, provide an EPDM boot around the dry pendent sprinkler on the heated side and securely seal to the pipe and freezer to prevent condensation from entering the freezer.

m. Provide pressure gauges at each water flow alarm switch location and at each main drain connection.

n. For each fire department connection, provide the symbolic sign given in NFPA 170 and locate (8 to 10 feet) above each connection location. Size the sign to (18 by 18 inches with the symbol being at least (14 by 14 inches).

o. Firestopping shall be provided for all penetrations of fire resistance rated construction. Firestopping shall comply with Section 07 84 00, FIRESTOPPING.

p. MRI Suite: Provide no more than one penetration of the MRI shield enclosure.

q. Painting of Pipe: In finished areas where walls and ceilings have been painted, paint primed surfaces with two coats of paint to match adjacent surfaces, except paint valves and operating accessories with two coats of gloss red enamel. Exercise care to avoid painting sprinklers. Painting of sprinkler systems above suspended ceilings and in crawl spaces is not required. Painting shall comply with Section 09 91 00, PAINTING. Any painted sprinkler shall be replaced with a new sprinkler.

r. Sprinkler System Signage: Provide rigid sprinkler system signage in accordance with NFPA 13 and NFPA 25. Sprinkler system signage shall include, but not limited to, the following:

(1) Identification Signs:

(a) Provide signage for each control valve, drain valve, sprinkler cabinet, and inspector’s test.

(b) Provide valve tags for each operable valve. Coordinate nomenclature and identification of operable valves with C ontracting Officer. Where existing nomenclature does not exist, the Tag Identification shall include no less than the following: (FP-B-F/SZ-#) Fire Protection, Building Number, Floor Number/Smoke Zone (if applicable), and Valve Number. (E.g., FP-500-1E-001) Fire Protection, Building 500, First Floor East, Number 001.)

(2) Instruction/Information Signs:

(a) Provide signage for each control valve to indicate valve function and to indicate what system is being controlled.

(b) Provide signage indicating the number and location of low point drains.

(3) Hydraulic Placards:

(a) Provide signage indicating hydraulic design information. The placard shall include location of the design area, discharge densities, required flow and residual pressure at the base of riser, occupancy classification, hose stream allowance, flow test information, and installing contractor. Locate hydraulic placard information signs at each alarm check valve.

s. Repairs: Repair damage to the building or equipment resulting from the installation of the sprinkler system by the installer at no additional expense to the Government.

t. Interruption of Service: There shall be no interruption of the existing sprinkler protection, water, electric, or fire alarm services without prior permission of the Contracting Officer. Contractor shall develop an interim fire protection program where interruptions involve occupied spaces. Request in writing at least one week prior to the planned interruption.

3.2 INSPECTION AND TEST

a. Preliminary Testing: Flush newly installed systems prior to performing hydrostatic tests in order to remove any debris which may have been left as well as ensuring piping is unobstructed. Hydrostatically test system, including the fire department connections, as specified in NFPA 13, in the presence of the Contracting Officers Representative (Contracting Officer) or his designated representative. Test and flush underground water line prior to performing these hydrostatic tests.

b. Final Inspection and Testing: Subject system to tests in accordance with NFPA 13, and when all necessary corrections have been accomplished, advise Contracting Officer to schedule a final inspection and test. Connection to the fire alarm system shall have been in service for at least ten days prior to the final inspection, with adjustments made to prevent false alarms. Furnish all

instruments, labor and materials required for the tests and provide the services of the installation foreman or other competent representative of the installer to perform the tests. Correct deficiencies and retest system as necessary, prior to the final acceptance. Include the operation of all features of the systems under normal operations in test

3.3 INSTRUCTIONS

Furnish the services of a competent instructor for not less than two hours for instructing personnel in the operation and maintenance of the system, on the dates requested by the Contracting Officer.

-- End of Section --

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-- End of Section Table of Contents --

SECTION 21 30 13
5/15

ELECTRIC-DRIVEN, FIRE PUMPS

PART 1 GENERAL

1.1 DESCRIPTION

Installation of a new fire pump system with bypass and with an automatic transfer switch in accordance with NFPA 20 and NFPA 70.

1.2 RELATED WORK

a. Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

b. Section 07 84 00, FIRESTOPPING.

c. Section 09 90 00, PAINTING.

d. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES.

e. Section 28 31 00, FIRE DETECTION AND ALARM.

1.3 DESIGN CRITERIA

a. The design, materials, equipment, installation, inspection, and testing of the fire pump shall be in accordance with the required provisions of NFPA 20 and NFPA 70.

b. Seismic Protection: Seismically brace all fire pump piping systems in accordance with NFPA 13 and NFPA 20.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications;

SD-02 Shop Drawings

Drawings;

SD-03 Product Data

Manufacturers Data Sheets;

SD-05 Design Data

Calculation Sheets;

Valve Charts;

SD-07 Certificates

Factory Authorization Of Fire Pump Test Representative;

SD-11 Closeout Submittals

Final Document Submittals;

1.5 SUBMITTAL INSTRUCTIONS

a. Submit as one package in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES. Prepare detailed working drawings that are signed by a NICET Level III or Level IV Sprinkler Technician or stamped by a Registered Professional Engineer licensed in the field of Fire Protection Engineering. As Government review is for technical adequacy only, the installer remains responsible for correcting any conflicts with other trades and building construction that arise during installation. Partial submittals will not be accepted. Material submittals shall be approved prior to the purchase or delivery to the job site. Suitably bind submittals in notebooks or binders and provide an index referencing the appropriate specification section. In addition to the hard copies, provide submittal items in Paragraphs 1.5 (A) 1 through 1.5 (A) 5 electronically in pdf format on a compact disc or as directed by the Contracting Officer's Representative (Contracting Officer). Submittals shall include, but not be limited to, the following:

(1) Qualifications:

(a) Provide a copy of the installing contractors fire sprinkler and state contractors license.

(b) Provide a copy of the NICET certification for the NICET Level III or Level IV Sprinkler Technician who prepared and signed the detailed working drawings unless the drawings are stamped by a Registered Professional Engineer licensed in the field of Fire Protection Engineering.

(c) Provide documentation showing that the installer has been actively and successfully engaged in the installation of commercial automatic fire pump systems for the past 10 years.

(2) Drawings: Submit detailed 1/8 inch (1:100) scale (minimum) working drawings conforming to NFPA 20. Provide drawings showing pump room configuration. Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection. Provide wiring diagrams for connections to power, signal, and control wiring. Drawings shall include graphical scales that allow the user to determine lengths when the drawings are reduced in area.

(3) Manufacturers Data Sheets: Provide data sheets for all materials and equipment proposed for use on the system. Include listing information and installation instructions in data sheets. Where data sheets describe items in addition to those proposed to be used for the system, clearly identify the proposed items on the sheet. Data sheets shall be provided for, but not limited to, the

following:

(a) Pipe and fittings

(b) Valves

(c) Pressure Gauges

(d) Pipe Hangers and Supports

(e) Switches

(f) Fire Pump

(g) Jockey Pump

(h) Jockey Pump Controller

(i) Test Header

(j) Fire Pump Controller and Automatic Transfer Switch

(k) Certified Pump Curve

(l) Electric Motor

(m) Flow Meter

(4) Calculation Sheets: Submit seismic load calculations for sizing of sway bracing.

(5) Valve Charts: Provide a valve chart that identifies the location of each control valve. Coordinate nomenclature and identification of control valves with Contracting Officer. Where existing nomenclature does not exist, the chart shall include no less than the following: Tag ID No., Valve Size, Service (control valve, test header valve, etc.), and Location.

(6) Factory Authorization of fire pump test representative: Two weeks prior to final inspection and testing, provide a copy of the necessary factory authorizations of the representative to be present at the acceptance testing. Authorizations shall include manufacturer’s representative for the fire pump, the fire pump controller and transfer switch in accordance with NFPA 20 requirements.

(7) Final Document Submittals: Provide as-built drawings, testing and maintenance instructions in accordance with the requirements in Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES. Submittals shall include, but not be limited to, the following:

(a) A complete set of as-built drawings showing the installed system with the specific interconnections between the system switches and the fire alarm equipment. Provide a complete set in the formats as follows. Submit items 2 and 3 below on a compact disc or as directed by the Contracting Officer.

1. One full size (or size as directed by the Contracting Officer printed copy).

2. One complete set in electronic pdf format.

3. One complete set in AutoCAD format or a format as directed by the Contracting Officer.

(b) Centrifugal Fire Pump Acceptance Test Form and Certificates: Upon completion of the fire pump system, including testing and flushing, provide a copy of a completed Centrifugal Fire Pump Acceptance Test Form and all Contractor’s Material and Test Certificates as indicated in NFPA 20. The graph included on the Centrifugal Fire Pump Acceptance Test Form shall be annotated to show 1) the water supply available, 2) the manufacturer’s certified pump curve, 3) the acceptance test curve, and 4) a curve showing the water supply and acceptance test curve combined.

(c) Operating and Maintenance Manuals that include step-by-step procedures required for system startup, operation, shutdown, and routine maintenance and testing. The manuals shall include the manufacturer's name, model number, parts list, and tools that should be kept in stock by the owner for routine maintenance, including the name of a local supplier, simplified wiring and controls diagrams, troubleshooting guide, and recommended service organization, including address and telephone number, for each item of equipment.

(d) One paper copy of the Centrifugal Fire Pump Acceptance Test Form, including the graph identified in 7(b) above, the Contractor’s Material and Test Certificates and the Operating and Maintenance Manuals above shall be provided in a binder. In addition, these materials shall be provided in pdf format on a compact disc or as directed by the Contracting Officer.

(e) Provide one additional copy of the Operating and Maintenance Manual covering the system in a flexible protective cover and mount in an accessible location adjacent to the fire pump controller.

1.6 QUALITY ASSURANCE

a. Installer Reliability: The installer shall possess a valid State of Connecticut contractor's license. The installer shall have been actively and successfully engaged in the installation of commercial automatic fire pumps for the past ten years.

b. Materials and Equipment: All equipment and devices shall be of a make and type listed by UL or approved by FM, or other nationally recognized testing laboratory for the specific purpose for which it is used. All materials, devices, and equipment shall be approved by the VA. All materials and equipment shall be free from defect. All materials and equipment shall be new unless specifically indicated otherwise on the contract drawings.

1.7 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 13	(2025; TIA 24-1; TIA 24-2; TIA 24-3) Standard for the Installation of Sprinkler Systems
NFPA 20	(2025) Standard for the Installation of Stationary Pumps for Fire Protection
NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 101	(2024) Life Safety Code
NFPA 170	(2024; ERTA 1 2023) Standard for Fire Safety and Emergency Symbols

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 2	Enclosures constructed for indoor use to provide a degree of protection to the equipment inside the enclosure against falling dirt and limited amounts of falling water

PART 2 PRODUCTS

2.1 GENERAL

Fire pump systems shall be in accordance with NFPA 20.

2.2 PIPING & FITTINGS

a. General:

(1) Piping and fittings for the fire pump systems shall be in accordance with NFPA 20.

b. Piping:

(1) Piping Sizes 2 ½ inches through 6 inches shall be black steel Schedule 40 piping. Steel pipe shall be joined by means of flanges welded to the pipe or mechanical grooved joints only.

(2) Piping Sizes 8 inches and larger shall be black steel Schedule 30. Steel pipe shall be joined by means of flanges welded to the pipe or mechanical grooved joints only.

c. Fittings:

(1) Fittings, mechanical couplings, and rubber gaskets shall be of the same manufacturer.

2.3 VALVES

a. General:

(1) Valves shall be in accordance with NFPA 20.

b. Control Valves:

(1) Outside Screw & Yoke Valves (OS&Y)

(a) The OS&Y valve shall be of cast iron construction.

(2) Butterfly Valves

(a) Butterfly valves shall not be installed in any location on the suction side of the fire pump.

(b) The butterfly valve shall be of cast iron construction.

(c) The valve tamper switch is permitted to be integral to the butterfly valve.

c. Automatic Air-Relief Valve

(1) The automatic air-relief valve shall be discharged to the atmosphere.

d. Circulation Relief Valve

(1) The circulation relief valve shall be of brass construction and be spring loaded.

(2) The circulation relief valve shall be adjustable.

(3) The circulation relief valve shall discharge to the atmosphere and not be recirculated to the suction side of the fire pump.

e. Check Valve:

(1) Shall be of the swing type with a flanged cast iron body and flanged inspection plate.

f. Automatic Ball Drips: Cast brass 3/4 inch in-line automatic ball drip with both ends threaded with iron pipe threads.

2.4 FIRE PUMP

a. General:

(1) The fire pump shall be electric motor driven. The pump shall have a capacity of 1500 gpm with a rated net pressure of 100 psi . The fire pump shall furnish not less than 150% of the rated flow capacity at not less than 65% of rated net pressure.

(2) The fire pump shall be centrifugal horizontal split case fire pump.

(3) The fire pump shall be automatic start and manual stop. The fire pump shall start automatically at 10 psi below jockey pump start pressure.

(4) The fire pump shall be 480 Volts, 3 phase at 60 Hertz.

b. Electric Motor Driver

(1) The electric motor driver and fire pump controller shall be fully

compatible.

(2) The electric motor driver shall be rated 133 horsepower and 1750 rpm.

2.5 FIRE PUMP CONTROLLER

a. The fire pump controller shall be an automatic solid state soft starter starting type. The controller shall be completely wired, ready for field connections, and be mounted in a NEMA 2 enclosure.

b. Limited-service controllers are not permitted.

c. Provide a minimum run timer to prevent short cycling.

d. The fire pump controller shall be provided with digital readouts of the voltage of each phase, amperage of each phase, and frequency.

e. The fire pump controller shall monitor isolation switch open on secondary source, secondary source operation, fire pump running, loss of phase or line power, and phase reversal. Alarms shall be individually displayed on the front of the fire pump controller by lighting of visual lamps. The fire pump controller shall be equipped with terminals for remote monitoring of secondary power operation, pump running, loss of power, and phase reversal.

f. The fire pump controller shall be provided with voltage surge arrestors installed in accordance with NFPA 20.

g. The fire pump controller shall be equipped with an USB port for information download. The controller shall be provided with a minimum 3,000 events recorder.

2.6 AUTOMATIC TRANSFER SWITCH

Automatic transfer switch shall be factory assembled and packaged as a unit with the fire pump controller.

2.7 PRESSURE SENSING LINE

The fire pump controller and jockey pump controller shall be provided with completely separate pressure sensing lines in accordance with NFPA 20.

2.8 JOCKEY PUMP

a. The jockey pump shall be an electric motor driven shaft type.

b. The jockey pump flow shall be rated a minimum of 60 gpm .

c. Pressure provided by the jockey pump shall be in accordance with NFPA 20.

d. OS&Y valves shall be provided on the supply and discharge side of the jockey pump.

2.9 JOCKEY PUMP CONTROLLER

a. Jockey pump controller shall be arranged for automatic and manual starting and stopping. The jockey pump shall be equipped with a

bourbon tube pressure switch or solid state pressure switch with independent high and low adjustments for automatic stopping and starting.

b. The controller shall be equipped with a “manual-off-automatic” switch. The controller shall be factory assembled and pre-wired, and ready for field connections, and be mounted in a NEMA 2 enclosure.

c. No minimum run timer is allowed.

2.10 TEST HEADER

a. The body of the test header shall be a surface type and constructed of brass or ductile iron.

b. The finish of the test header finish plate shall be polished brass.

c. The number of valves shall be in accordance with NFPA 20.

d. Provide a cap and chain for each valve.

e. The test header shall be piped directly to the exterior through a straight-type header.

f. An automatic ball drip valve shall be provided at the low point of the test header between the test header and check valve.

2.11 FLOW METER

a. The flow meter shall be a venturi-type.

b. Provide a meter throttle valve and meter control valves. The throttle valve and control valves shall be OS&Y valves.

c. The flow meter shall be suitable for flow between 50% and 175% of the rated pump capacity.

d. Arrange piping to permit flow meter to discharge to pump suction and to discharge through test header.

e. Provide a circulation relief valve between the flow meter and the reconnection to the suction piping.

f. Provide calibrated gauges on the inlet and outlet of the flow meter.

g. Provide a laminated flow chart to convert the flow meter’s velocity pressure reading into gpm flow rate. The flow chart shall be affixed by metal chain to the pipe adjacent to the flow meter

2.12 IDENTIFICATION SIGNS

a. Valves: Rigid, plastic, steel or aluminum signs with white lettering on a red background with holes for easy attachment. Identification signs shall be attached to the valve or piping with chain.

b. Pipe: Pretensioned pipe labels consisting of pre-coiled, semi-rigid plastic formed to cover the full circumference of pipe and attach to pipe without fasteners or adhesive. Labels shall be red background with white letters, with the words “Fire Protection”. Labels shall be

provided at a maximum interval of 20 feet of pipe length.

2.13 SWITCHES

Contain in a weatherproof die cast/red baked enamel, oil resistant, aluminum housing with tamper resistant screws, 1/2 inch conduit entrance and necessary facilities for attachment to the valves. Provide two SPDT switches rated at 2.5 amps at 24 VDC.

2.14 GAUGES

a. Provide gauges as required by NFPA 20.

b. The gauges shall be liquid filled. Provide gauges where the normal pressure of the system is at the midrange of the gauge.

2.15 PIPE HANGERS, SUPPORTS, AND RESTRAINT OF SYSTEM PIPING

Pipe hangers, supports, and restraint of system piping shall be in accordance with NFPA 13 and NFPA 20.

2.16 WALL, FLOOR AND CEILING PLATES

Provide chrome plated steel escutcheon plates.

2.17 PUMP BASE PLATE AND PAD

a. The pump pad shall be provided with a common base plate for the pump and motor.

b. Construct the base plate of cast iron with a raised lip tapped for drainage or welded steel shapes with suitable drainage.

c. Each base plate shall have a 1 inch steel drain line piped to the nearest floor drain.

d. Mount pump units and bases on a raised reinforced concrete pad that is an integral part of the reinforced concrete floor.

2.18 VALVE TAGS

Engraved black filled numbers and letters not less than 1/2 inch high for number designation, and not less than 1/4 inch for service designation on 19 gage, 1-1/2 inches round brass disc, attached with brass “S” hook, brass chain, or nylon twist tie.

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation shall be accomplished by the licensed contractor. Provide a qualified technician, experienced in the installation and operation of the type of system being installed, to supervise the installation and testing of the system.

b. Installation of Piping: Accurately cut pipe to measurements established by the installer and work into place without springing or forcing. Piping shall not obstruct the minimum means of egress clearances required by NFPA 101. Pipe hangers, supports, and

restraint of system piping, and seismic bracing shall be installed in accordance with NFPA 13 and NFPA 20.

c. Welding: Conform to the requirements and recommendations of NFPA 13.

d. Drains: Install drains where necessary and required by NFPA 20. Drain piping shall be routed to properly discharge over floor drains or to site cones attached to floor drains. Such floor drains shall be of adequate size to readily accept the discharge from each drain under full flow and maximum pressure conditions. Drain piping shall also be allowed to be routed to the outside of the building, provided its discharge will not negatively impact the exterior conditions. Do not provide a direct drain connection to sewer system.

e. Supervisory Switches: Provide supervisory switches for control valves, including the test header control valve.

f. Provide escutcheon plates for exposed piping passing through walls, floors, or ceilings.

g. Sleeves: Provide for pipes passing through masonry or concrete. Provide space between the pipe and the sleeve in accordance with NFPA 13. Seal this space with a UL Listed through penetration fire stop material in accordance with Section 07 84 00, FIRESTOPPING. Where core drilling is used in lieu of sleeves, also seal space around penetrations. Seal penetrations of walls, floors and ceilings of other types of construction, in accordance with Section 07 84 00, FIRESTOPPING.

h. For the fire pump test header, provide the symbolic sign given in NFPA 170 and locate 8 to 10 feet above the header location. Size the sign to 18 by 18 inches with the symbol being at least 14 by 14 inches .

i. Firestopping shall be provided for all penetrations of fire resistance rated construction. Firestopping shall comply with Section 07 84 00, FIRESTOPPING.

j. Securely attach identification signs to control valves, test header, pump suction, pump discharge, and bypass valves.

k. Securely attach valve tags to each control valve.

l. Repairs: Repair damage to the building or equipment resulting from the installation of the fire pump system by the installer at no additional expense to the Government.

m. Interruption of Service: There shall be no interruption of the existing sprinkler protection, water, electric, or fire alarm services without prior permission of the Contracting Officer. Contractor shall develop an interim fire protection program where interruptions involve occupied spaces. Request in writing at least one week prior to the planned interruption.

n. The fire pump controller shall be located as close as practical and within site of the fire pump motor.

o. Painting of Pipe: Pipe shall be painted only where located in areas with corrosive conditions or in finished areas where walls and

ceilings have been painted. Paint primed surfaces with two coats of gloss red enamel. Paint valves and operating accessories with two coats of gloss red enamel. Exercise care to avoid painting sprinklers. Painting of sprinkler systems above suspended ceilings and in crawl spaces is not required. Painting shall comply with Section 09 90 00, PAINTING.

p. All fire pump alarm and signals required by NFPA 20 for remote monitoring shall be supervised by the fire alarm system, including isolation switch open on secondary source, secondary source operation, fire pump running, loss of phase or line power, and phase reversal in accordance with Section 28 31 00, FIRE DETECTION AND ALARM. Isolation switch open on secondary source, secondary source operation, fire pump running, loss of phase or line power, and phase reversal shall be supervisory fire alarm signals.

q. The electric drive for pumps shall comply with the requirements of NFPA 20 and NFPA 70 and be in accordance with Section 26 05 19, LOW VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES. The fire pump supply conductors shall be protected in accordance with NFPA 70.

3.2 INSPECTION AND TEST

a. Preliminary Inspection and Testing: Subject system to all inspections and tests in accordance with NFPA 20. When all necessary corrections have been accomplished, advise Contracting Officer to schedule a final acceptance inspection and test. Flushing and hydrostatic testing and fire alarm monitoring of fire pump controller alarms and signals shall be witnessed by the Contracting Officer or his designated representative.

b. Final Acceptance Inspection and Testing: Perform in accordance with NFPA 20 in the presence of the Contracting Officer or his designated representative. Furnish all labor and materials as required by NFPA 20 for the final acceptance test, including verification of fire alarm system monitoring of fire pump controller alarm and signals. The final acceptance test will not be conducted unless the required test equipment and equipment manufacturers or the equipment authorized representatives are present

c. Gauges used for preliminary testing and final inspection and testing shall have been calibrated by an accredited laboratory within one year of the testing. Date of latest calibration shall be listed on the pressure gauge.

3.3 INSTRUCTIONS

Furnish the services of a factory trained instructor for not less than two hours for instructing personnel in the operation and maintenance of the system, on the dates requested by the Contracting Officer.

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-- End of Section --

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SECTION 22 05 11
1/23

COMMON WORK RESULTS FOR PLUMBING

PART 1 GENERAL

1.1 DESCRIPTION

a. The requirements of this Section shall apply to all Sections of Division 22.

b. Definitions:

(1) Exposed: Piping and equipment exposed to view in finished rooms.

(2) Exterior: Piping and equipment exposed to weather be it temperature, humidity, precipitation, wind or solar radiation.

c. Abbreviations/Acronyms:

(1) ABS: Acrylonitrile Butadiene Styrene

(2) AC: Alternating Current

(3) ACR: Air Conditioning and Refrigeration

(4) A/E: Architect/Engineer

(5) AFF: Above Finish Floor

(6) AFG: Above Finish Grade

(7) AI: Analog Input

(8) AISI: American Iron and Steel Institute

(9) AO: Analog Output

(10) ASHRAE: American Society of Heating Refrigeration, Air Conditioning Engineers

(11) ASJ: All Service Jacket

(12) ASME: American Society of Mechanical Engineers

(13) ASPE: American Society of Plumbing Engineers

(14) AWG: American Wire Gauge

(15) BACnet: Building Automation and Control Network

(16) BAg: Silver-Copper-Zinc Brazing Alloy

(17) BAS: Building Automation System

(18) BCuP: Silver-Copper-Phosphorus Brazing Alloy

(19) bhp: Brake Horsepower

(20) Btu: British Thermal Unit

(21) Btu/h: British Thermal Unit per Hour

(22) BSG: Borosilicate Glass Pipe

(23) C: Celsius

(24) CA: Compressed Air

(25) CD: Compact Disk

(26) CDA: Copper Development Association

(27) CGA: Compressed Gas Association

(28) CFM: Cubic Feet per Minute

(29) CI: Cast Iron

(30) CLR: Color

(31) CO: Contracting Officer

(32) COND: Condensate

(33) CPVC: Chlorinated Polyvinyl Chloride

(34) CR: Chloroprene

(35) CRS: Corrosion Resistant Steel

(36) CWP: Cold Working Pressure

(37) CxA: Commissioning Agent

(38) dB: Decibels

(39) db(A): Decibels (A weighted)

(40) DCW: Domestic Cold Water

(41) DDC: Direct Digital Control

(42) DFU: Drainage Fixture Units

(43) DHW: Domestic Hot Water

(44) DHWR: Domestic Hot Water Return

(45) DHWS: Domestic How Water Supply

(46) DI: Digital Input

(47) DI: Deionized Water

(48) DISS: Diameter Index Safety System

(49) DN: Diameter Nominal

(50) DO: Digital Output

(51) DOE: Department of Energy

(52) DVD: Digital Video Disc

(53) DWG: Drawing

(54) DWH: Domestic Water Heater

(55) DWS: Domestic Water Supply

(56) DWV: Drainage, Waste and Vent

(57) ECC: Engineering Control Center

(58) EL: Elevation

(59) EMCS: Energy Monitoring and Control System

(60) EPA: Environmental Protection Agency

(61) EPACT: Energy Policy Act

(62) EPDM: Ethylene Propylene Diene Monomer

(63) EPT: Ethylene Propylene Terpolymer

(64) ETO: Ethylene Oxide

(65) F: Fahrenheit

(66) FAR: Federal Acquisition Regulations

(67) FD: Floor Drain

(68) FDC: Fire Department (Hose) Connection

(69) FED: Federal

(70) FG: Fiberglass

(71) FNPT: Female National Pipe Thread

(72) FOR: Fuel Oil Return

(73) FOS: Fuel Oil Supply

(74) FOV: Fuel Oil Vent

(75) FPM: Fluoroelastomer Polymer

(76) FSK: Foil-Scrim-Kraft Facing

(77) FSS: VA Construction and Facilities Management, Facility

Standards Service

(78) FU: Fixture Units

(79) GAL: Gallon

(80) GCO: Grade Cleanouts

(81) GPD: Gallons per Day

(82) GPH: Gallons per Hour

(83) GPM: Gallons per Minute

(84) HDPE: High Density Polyethylene

(85) HEFP: Healthcare Environment and Facilities Program (replacement for OCAMES)

(86) HEX: Heat Exchanger

(87) Hg: Mercury

(88) HOA: Hands-Off-Automatic

(89) HP: Horsepower

(90) HVE: High Volume Evacuation

(91) Hz: Hertz

(92) ID: Inside Diameter

(93) IE: Invert Elevation

(94) INV: Invert

(95) IPC: International Plumbing Code

(96) IPS: Iron Pipe Size

(97) IW: Indirect Waste

(98) IWH: Instantaneous Water Heater

(99) Kg: Kilogram

(100) kPa: Kilopascal

(101) KW: Kilowatt

(102) KWH: Kilowatt Hour

(103) lb: Pound

(104) lbs/hr: Pounds per Hour

(105) LNG: Liquid Natural Gas

(106) L/min: Liters per Minute

(107) LOX: Liquid Oxygen

(108) L/s: Liters per Second

(109) m: Meter

(110) MA: Medical Air

(111) MAWP: Maximum Allowable Working Pressure

(112) MAX: Maximum

(113) MBH: 1000 Btu per Hour

(114) MED: Medical

(115) MER: Mechanical Equipment Room

(116) MFG: Manufacturer

(117) mg: Milligram

(118) mg/L: Milligrams per Liter

(119) ml: Milliliter

(120) mm: Millimeter

(121) MIN: Minimum

(122) MV: Medical Vacuum

(123) N2: Nitrogen

(124) N20: Nitrogen Oxide

(125) NC: Normally Closed

(126) NF: Oil Free Dry (Nitrogen)

(127) NG: Natural Gas

(128) NIC: Not in Contract

(129) NO: Normally Open

(130) NOM: Nominal

(131) NPTF: National Pipe Thread Female

(132) NPS: Nominal Pipe Size

(133) NPT: Nominal Pipe Thread

(134) NTS: Not to Scale

(135) O2: Oxygen

(136) OC: On Center

(137) OD: Outside Diameter

(138) OSD: Open Sight Drain

(139) OS&Y: Outside Stem and Yoke

(140) PA: Pascal

(141) PBPU: Prefabricated Bedside Patient Units

(142) PD: Pressure Drop or Difference

(143) PDI: Plumbing and Drainage Institute

(144) PH: Power of Hydrogen

(145) PID: Proportional-Integral-Differential

(146) PLC: Programmable Logic Controllers

(147) PP: Polypropylene

(148) ppb: Parts per Billion

(149) ppm: Parts per Million

(150) PSI: Pounds per Square Inch

(151) PSIA: Pounds per Square Inch Atmosphere

(152) PSIG: Pounds per Square Inch Gauge

(153) PTFE: Polytetrafluoroethylene

(154) PVC: Polyvinyl Chloride

(155) PVDF: Polyvinylidene Fluoride

(156) RAD: Radians

(157) RO: Reverse Osmosis

(158) RPM: Revolutions Per Minute

(159) RTD: Resistance Temperature Detectors

(160) RTRP: Reinforced Thermosetting Resin Pipe

(161) SAN: Sanitary Sewer

(162) SCFM: Standard Cubic Feet per Minute

(163) SDI: Silt Density Index

(164) SMACNA: Sheet Metal and Air Conditioning Contractors National Association

(165) SPEC: Specification

(166) SPS: Sterile Processing Services

(167) SQFT/SF: Square Feet

(168) SS: Stainless Steel

(169) STD: Standard

(170) SUS: Saybolt Universal Second

(171) SWP: Steam Working Pressure

(172) TD: Temperature Difference

(173) TDH: Total Dynamic Head

(174) TEFC: Totally Enclosed Fan-Cooled

(175) TEMP: Temperature

(176) TFE: Tetrafluoroethylene

(177) THERM: 100,000 Btu

(178) THHN: Thermoplastic High-Heat Resistant Nylon Coated Wire

(179) THWN: Thermoplastic Heat and Water Resistant Nylon Coated Wire

(180) TIL: Technical Information Library
https://www.cfm.va.gov/til/catalog.asp

(181) T/P: Temperature and Pressure

(182) TYP: Typical

(183) USDA: U.S. Department of Agriculture

(184) V: Vent

(185) V: Volt

(186) VA: Veterans Administration

(187) VA CFM: VA Construction and Facilities Management

(188) VA CFM CSS: VA Construction and Facilities Management, Consulting Support Service

(189) VAC: Vacuum

(190) VAC: Voltage in Alternating Current

(191) VAMC: Veterans Administration Medical Center

(192) VHA OCAMES: This has been replaced by HEFP.

(193) VSD: Variable Speed Drive

(194) VTR: Vent through Roof

(195) W: Waste

(196) WAGD: Waste Anesthesia Gas Disposal

(197) WC: Water Closet

(198) WG: Water Gauge

(199) WOG: Water, Oil, Gas

(200) WPD: Water Pressure Drop

(201) WSFU: Water Supply Fixture Units

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 74 19, CONSTRUCTION WASTE MANAGEMENT.

d. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

e. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

f.

g. Section 03 30 00, CAST-IN-PLACE CONCRETE: Concrete and Grout.

h. Section 05 31 00, STEEL DECKING: Building Components for Attachment of Hangers.

i. Section 05 36 00, COMPOSITE METAL DECKING: Building Components for Attachment of Hangers.

j. Section 05 50 00, METAL FABRICATIONS.

k. Section 07 60 00, FLASHING AND SHEET METAL: Flashing for Wall and Roof Penetrations.

l. Section 07 84 00, FIRESTOPPING.

m. Section 07 92 00, JOINT SEALANTS.

n. Section 09 91 00, PAINTING.

o.

p. Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT.

q. Section 22 07 11, PLUMBING INSULATION.

r. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING

SYSTEMS.

s. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

t.

u. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

v. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES.

w. Section 26 29 11, MOTOR CONTROLLERS.

x. Section 31 20 00, EARTH MOVING: Excavation and Backfill.

1.3 REFERENCES

The publications listed below shall form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B31.1	(2024) Power Piping
ASME BPVC SEC IX	(2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A575	(2020) Standard Specification for Steel Bars, Carbon, Merchant Quality, M-Grades

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-58	(2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1	(2021) Motors and Generators

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 51B	(2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work
NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NSF INTERNATIONAL (NSF)

NSF/ANSI 14	(2023) Plastics Piping System Components

and Related Materials

NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects
NSF 372	(2022) Drinking Water System Components - Lead Content

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-P-21035	(1991; Rev B; Notice 2 2003; Notice 3 2021) Paint, High Zinc Dust Content, Galvanizing Repair (Metric)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

PL 109-58	Energy Policy Act of 2005 (EPAct05)

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Phasing Plan/Schedule With Manpower Levels; G, AE

Rigging Plan; G, AE

Welder Certificate; G, AE

Installer Qualifications; G, AE

SD-02 Shop Drawings

Submittals And Shop Drawings For Interdependent Items; G, AE

Coordination/Shop Drawings; G, AE

Complete Layout Drawings; G, AE

SD-03 Product Data

Manufacturer’s Literature And Data Updated; G, AE

SD-05 Design Data

The Support System Of Suspended Equipment; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

Operating And Performance Tests; G, AE

SD-08 Manufacturer's Instructions

Manufacturer's Literature And Data; G, AE

SD-10 Operation and Maintenance Data

Plumbing Maintenance Data And Operating Instructions; G, AE

Names, Mail And E-Mail Addresses And Phone Numbers Of Service Organizations Providing Service; G, AE

Operation And Maintenance Data Updated; G, AE

SD-11 Closeout Submittals

Training Plans; G, AE

Trainer Qualifications; G, AE

Instructor Qualifications; G, AE

As-Built Documentation; G, AE

Certification Documentation; G, AE

Operation And Maintenance Manuals; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this Section shall be marked "SUBMITTED UNDER SECTION 22 05 11, COMMON WORK RESULTS FOR PLUMBING", with applicable paragraph identification.

c. If the project is phased, Contractors shall submit complete phasing plan/schedule with manpower levels prior to commencing work. The phasing plan shall be detailed enough to provide milestones in the process that can be verified.

d. Contractor shall make all necessary field measurements and investigations to assure that the equipment and assemblies will meet Contract requirements, and all equipment that requires regular maintenance, calibration, etc. are accessible from the floor or permanent work platform. It is the Contractor’s responsibility to ensure all submittals meet the VA specifications and requirements and it is assumed by the VA that all submittals do meet the VA specifications unless the Contractor has requested a variance in writing and approved by Contracting Officer prior to the submittal. If at any time during the project it is found that any item does not meet the VA specifications and there was no variance approval the Contractor shall correct at no additional cost or time to the Government even if a submittal was approved.

e. If equipment is submitted which differs in arrangement from that shown, provide documentation proving equivalent performance, design standards and drawings that show the rearrangement of all associated systems. Additionally, any impacts on ancillary equipment or services such as foundations, piping, and electrical shall be the Contractor’s responsibility to design, supply, and install at no additional cost or

time to the Government. VA approval will be given only if all features of the equipment and associated systems, including accessibility, are equivalent to that required by the Contract.

f. Prior to submitting shop drawings for approval, Contractor shall certify in writing that manufacturers of all major items of equipment have each reviewed drawings and specifications, and have jointly coordinated and properly integrated their equipment and controls to provide a complete and efficient installation.

g. Submittals and shop drawings for interdependent items, containing applicable descriptive information, shall be furnished together and complete in a group. Coordinate and properly integrate materials and equipment in each group to provide a completely compatible and efficient installation. Final review and approvals will be made only by groups.

h. Manufacturer's Literature and Data including: Manufacturer’s literature shall be submitted under the pertinent Section rather than under this Section.

(1) Electric motor data and variable speed drive data shall be submitted with the driven equipment.

(2) Equipment and materials identification.

(3) Firestopping materials.

(4) Hangers, inserts, supports and bracing. Provide load calculations for variable spring and constant support hangers.

(5) Wall, floor, and ceiling plates.

i. Coordination/Shop Drawings:

(1) Submit complete consolidated and coordinated shop drawings for all new systems, and for existing systems that are in the same areas.

(2) The coordination/shop drawings shall include plan views, elevations and Sections of all systems and shall be on a scale of not less than 1:32 (3/8-inch equal to 1 foot). Clearly identify and dimension the proposed locations of the principal items of equipment. The drawings shall clearly show locations and adequate clearance for all equipment, piping, valves, control panels and other items. Show the access means for all items requiring access for operations and maintenance. Provide detailed coordination/shop drawings of all piping and duct systems. The drawings should include all lockout/tagout points for all energy/hazard sources for each piece of equipment. Coordinate lockout/tagout procedures and practices with local VA requirements.

(3) Do not install equipment foundations, equipment or piping until coordination/shop drawings have been approved.

(4) In addition, for plumbing systems, provide details of the following:

(a) Mechanical equipment rooms.

(b)

(c) Hangers, inserts, supports, and bracing.

(d) Pipe sleeves.

(e) Duct or equipment penetrations of floors, walls, ceilings, or roofs.

j. Rigging Plan: Provide documentation of the capacity and weight of the rigging and equipment intended to be used. The plan shall include the path of travel of the load, the staging area and intended access, and qualifications of the operator and signal person.

k. Plumbing Maintenance Data and Operating Instructions:

(1) Maintenance and operating manuals in accordance with Section 01 00 00, GENERAL REQUIREMENTS, Article, INSTRUCTIONS, for systems and equipment.

(2) Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(a) Include complete list indicating all components of the systems.

(b) Include complete diagrams of the internal wiring for each item of equipment.

(c) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

(3) Provide a listing of recommended replacement parts for keeping in stock supply, including sources of supply, for equipment. Include in the listing belts for equipment: Belt manufacturer, model number, size and style, and distinguished whether of multiple belt sets.

l. Provide copies of approved plumbing equipment submittals to the TAB and Commissioning Subcontractor.

m. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

n. Submit training plans, trainer qualifications and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

a. Mechanical, electrical, and associated systems shall be safe, reliable, efficient, durable, easily and safely operable and maintainable, easily and safely accessible, and in compliance with applicable codes as specified. The systems shall be comprised of high quality institutional-class and industrial-class products of manufacturers that are experienced specialists in the required product lines. All construction firms and personnel shall be experienced and

qualified specialists in industrial and institutional plumbing.

b. Products Criteria:

(1) Standard Products: Material and equipment shall be the standard products of a manufacturer regularly engaged in the manufacture, supply and servicing of the specified products for at least five years. However, digital electronics devices, software and systems such as controls, instruments, computer work station, shall be the current generation of technology and basic design that has a proven satisfactory service record of at least five years.

(2) Equipment Service: There shall be permanent service organizations, authorized and trained by manufacturers of the equipment supplied, located within 100 miles of the project. These organizations shall come to the site and provide acceptable service to restore operations within four hours of receipt of notification by phone, e-mail or fax in event of an emergency, such as the shut-down of equipment; or within 24 hours in a non-emergency. Names, mail and e-mail addresses and phone numbers of service organizations providing service under these conditions for (as applicable to the project): pumps, compressors, water heaters, critical instrumentation, computer workstation and programming shall be submitted for project record and inserted into the operations and maintenance manual.

(3) All items furnished shall be free from defects that would adversely affect the performance, maintainability and appearance of individual components and overall assembly.

(4) The products and execution of work specified in Division 22 shall conform to the referenced codes and standards as required by the specifications. Local codes and amendments enforced by the local code official shall be enforced, if required by local authorities such as the natural gas supplier. If the local codes are more stringent, then the local code shall apply. Any conflicts shall be brought to the attention of the Contracting Officer.

(5) Multiple Units: When two or more units of materials or equipment of the same type or class are required, these units shall be of the same manufacturer and model number, or if different models are required they shall be of the same manufacturer and identical to the greatest extent possible (i.e., same model series).

(6) Assembled Units: Performance and warranty of all components that make up an assembled unit shall be the responsibility of the manufacturer of the completed assembly.

(7) Nameplates: Nameplate bearing manufacturer's name or identifiable trademark shall be securely affixed in a conspicuous place on equipment, or name or trademark cast integrally with equipment, stamped or otherwise permanently marked on each item of equipment.

(8) Asbestos products or equipment or materials containing asbestos is prohibited.

(9) Bio-Based Materials: For products designated by the USDA’s bio-based Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as

products meet all performance requirements in this specifications Section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

c. Welding: Before any welding is performed, Contractor shall submit a welder certificate certifying that welders comply with the following requirements:

(1) Qualify welding processes and operators for piping according to ASME BPVC SEC IX, "Welding and Brazing Qualifications". Provide proof of current certification to CO.

(2) Comply with provisions of ASME B31.1 series "Code for Pressure Piping".

(3) Certify that each welder and welding operator has passed American Welding Society (AWS) qualification tests for the welding processes involved, and that certification is current.

(4) All welds shall be stamped according to the provisions of the AWS or ASME as required herein and by the association code.

d. Manufacturer's Recommendations: Where installation procedures or any part thereof are required to be in accordance with the recommendations of the manufacturer of the material being installed, printed copies of these recommendations shall be furnished to the Contracting Officer prior to installation. Installation of the item will not be allowed to proceed until the recommendations are received. Failure to furnish these recommendations can be cause for rejection of the material.

e. Execution (Installation, Construction) Quality:

(1) All items shall be applied and installed in accordance with manufacturer's written instructions. Conflicts between the manufacturer's instructions and the Contract documents shall be referred to the Contracting Officer for resolution. Printed copies or electronic files of manufacturer’s installation instructions shall be provided to the Contracting Officer at least 10 working days prior to commencing installation of any item.

(2) All items that require access, such as for operating, cleaning, servicing, maintenance, and calibration, shall be easily and safely accessible by persons standing at floor level, or standing on permanent platforms, without the use of portable ladders. Examples of these items include but are not limited to: all types of valves, filters and strainers, transmitters, and control devices. Prior to commencing installation work, refer conflicts between this requirement and Contract documents to Contracting Officer for resolution. Failure of the Contractor to resolve or call attention to any discrepancies or deficiencies to the Contracting Officer will result in the Contractor correcting at no additional cost or time to the Government.

(3) Complete layout drawings shall be required by Paragraph, SUBMITTALS. Construction work shall not start on any system until the layout drawings have been approved by VA.

(4) Installer Qualifications: Installer shall be licensed and shall

provide evidence of the successful completion of at least five projects of equal or greater size and complexity. Provide tradesmen skilled in the appropriate trade.

(5) Workmanship/craftsmanship will be of the highest quality and standards. The VA reserves the right to reject any work based on poor quality of workmanship this work shall be removed and done again at no additional cost or time to the Government.

f. Upon request by Government, provide lists of previous installations for selected items of equipment. Include contact persons who will serve as references, with current telephone numbers and e-mail addresses.

g. Guaranty: Warranty of Construction, FAR clause 52.246-21.

h. Plumbing Systems: International Plumbing Code (IPC). Unless otherwise required herein, perform plumbing work in accordance with the latest version of the IPC. For IPC codes referenced in the Contract documents, advisory provisions shall be considered mandatory, the word “should” shall be interpreted as “shall”. Reference to the “code official” or “owner” shall be interpreted to mean the Contracting Officer.

i. Cleanliness of Piping and Equipment Systems:

(1) Care shall be exercised in the storage and handling of equipment and piping material to be incorporated in the work. Debris arising from cutting, threading and welding of piping shall be removed.

(2) Piping systems shall be flushed, blown or pigged as necessary to deliver clean systems.

(3) The interior of all tanks shall be cleaned prior to delivery and beneficial use by the Government. All piping shall be tested in accordance with the specifications and the IPC. All filters, strainers, fixture faucets shall be flushed of debris prior to final acceptance.

(4) Contractor shall be fully responsible for all costs, damage, and delay arising from failure to provide clean systems.

1.7 DELIVERY, STORAGE AND HANDLING

Protection of Equipment:

a. Equipment and material placed on the job site shall remain in the custody of the Contractor until phased acceptance, whether or not the Government has reimbursed the Contractor for the equipment and material. The Contractor is solely responsible for the protection of such equipment and material against any damage or theft.

b. Damaged equipment shall be replaced with an identical unit as determined and directed by the Contracting Officer. Such replacement shall be at no additional cost or additional time to the Government.

c. Interiors of new equipment and piping systems shall be protected against entry of foreign matter. Both inside and outside shall be

cleaned before painting or placing equipment in operation.

d. Existing equipment and piping being worked on by the Contractor shall be under the custody and responsibility of the Contractor and shall be protected as required for new work.

e. Protect plastic piping and tanks from ultraviolet light (sunlight) while in pre-construction. Plastic piping and tanks shall not be installed exposed to sunlight without metal jacketing to block ultraviolet rays.

1.8 AS-BUILT DOCUMENTATION

a. Submit manufacturer’s literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, VA approved substitutions and construction revisions shall be in electronic version on CD or DVD inserted into a three-ring binder. All aspects of system operation and maintenance procedures, including applicable piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing Contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. Should the installing Contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the ‘third party testing company’ requirement. Provide record drawings as follows:

(1) Red-lined, hand-marked drawings are to be provided, with one paper copy and a scanned PDF version of the hand-marked drawings provided on CD or DVD.

(2)

(3)

d. The as-built drawings shall indicate the location and type of all lockout/tagout points for all energy sources for all equipment and pumps to include breaker location and numbers, valve tag numbers, etc. Coordinate lockout/tagout procedures and practices with local VA requirements.

e. Certification documentation shall be provided to Contracting Officer 21 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and provide documentation/certification that all results of tests were within

limits specified. Test results shall contain written sequence of test procedure with written test results annotated at each step along with the expected outcome or setpoint. The results shall include all readings, including but not limited to data on device (make, model and performance characteristics), normal pressures, switch ranges, trip points, amp readings, and calibration data to include equipment serial numbers or individual identifications, etc.

1.9 JOB CONDITIONS – WORK IN EXISTING BUILDING

a. Building Operation: Government employees will be continuously operating and managing all facilities, including temporary facilities that serve the VAMC.

b. Maintenance of Service: Schedule all work to permit continuous service as required by the VAMC.

c. Steam and Condensate Service Interruptions: Limited steam and condensate service interruptions, as required for interconnections of new and existing systems, will be permitted by the Contracting Officer during periods when the demands are not critical to the operation of the VAMC. These non-critical periods are limited to between 8 pm and 5 am in the appropriate off-season (if applicable). Provide at least 10 working days advance notice to the Contracting Officer. The request shall include a detailed plan on the proposed shutdown and the intended work to be done along with manpower levels. All equipment and materials shall be onsite and verified with planfive work days prior to the shutdown or it will need to be rescheduled.

d. Phasing of Work: Comply with all requirements shown on Contract documents. Contractor shall submit a complete detailed phasing plan/schedule with manpower levels prior to commencing work. The phasing plan shall be detailed enough to provide milestones in the process that can be verified.

e. Building Working Environment: Maintain the architectural and structural integrity of the building and the working environment at all times. Maintain the interior of building at 65 degrees F minimum. Limit the opening of doors, windows or other access openings to brief periods as necessary for rigging purposes. Storm water or ground water leakage is prohibited. Provide daily clean-up of construction and demolition debris on all floor surfaces and on all equipment being operated by VA. Maintain all egress routes and safety systems/devices.

f. Acceptance of Work for Government Operation: As new equipment, systems and facilities are made available for operation and these items are deemed of beneficial use to the Government, inspections and tests will be performed. Based on the inspections, a list of Contract deficiencies will be issued to the Contractor. After correction of deficiencies as necessary for beneficial use, the CO will process necessary acceptance and the equipment will then be under the control and operation of Government personnel.

g. Temporary Facilities: Refer to Paragraph, TEMPORARY PIPING AND EQUIPMENT in this Section.

PART 2 PRODUCTS

2.1 MATERIALS FOR VARIOUS SERVICES

a. Non-pressure PVC pipe shall contain a minimum of 25 percent recycled content. Steel pipe shall contain a minimum of 25 percent recycled content.

b. Plastic pipe, fittings and solvent cement shall meet NSF 14 and shall bear the NSF/ANSI 14 seal “NSF-PW”. Polypropylene pipe and fittings shall comply with NSF/ANSI 14 and NSF/ANSI 61. Solder or flux containing lead shall not be used with copper pipe.

c. Material or equipment containing a weighted average of greater than 0.25 percent lead shall not be used in any potable water system intended for human consumption and shall be certified in accordance with NSF/ANSI 61 or NSF 372.

d. In-line devices such as water meters, building valves, check valves, stops, valves, fittings, tanks and backflow preventers shall comply with NSF/ANSI 61 and NSF 372.

e. End point devices such as drinking fountains, lavatory faucets, kitchen and bar faucets, ice makers supply stops, and end-point control valves used to dispense drinking water shall meet requirements of NSF/ANSI 61 and NSF 372.

2.2 FACTORY-ASSEMBLED PRODUCTS

a. Standardization of components shall be maximized to reduce spare part requirements.

b. Manufacturers of equipment assemblies that include components made by others shall assume complete responsibility for final assembled unit.

(1) All components of an assembled unit need not be products of same manufacturer.

(2) Constituent parts that are alike shall be products of a single manufacturer.

(3) Components shall be compatible with each other and with the total assembly for intended service.

(4) Contractor shall guarantee performance of assemblies of components and shall repair or replace elements of the assemblies as required to deliver specified performance of the complete assembly at no additional cost or time to the Government.

c. Components of equipment shall bear manufacturer's name and trademark, model number, serial number and performance data on a name plate securely affixed in a conspicuous place, or cast integral with, stamped or otherwise permanently marked upon the components of the equipment.

d. Major items of equipment, which serve the same function, shall be the same make and model.

2.3 COMPATIBILITY OF RELATED EQUIPMENT

Equipment and materials installed shall be compatible in all respects with other items being furnished and with existing items so that the result will be a complete and fully operational system that conforms to Contract requirements.

2.4 SAFETY GUARDS

a. Pump shafts and couplings shall be fully guarded by a sheet steel guard, covering coupling and shaft but not bearings. Material shall be minimum 16-gauge sheet steel; ends shall be braked and drilled and attached to pump base with minimum of four 1/4 inch bolts. Reinforce guard as necessary to prevent side play forcing guard onto couplings.

b. All Equipment shall have moving parts protected from personal injury.

2.5 LIFTING ATTACHMENTS

Equipment shall be provided with suitable lifting attachments to enable equipment to be lifted in its normal position. Lifting attachments shall withstand any handling conditions that might be encountered, without bending or distortion of shape, such as rapid lowering and braking of load.

2.6 ELECTRIC MOTORS, MOTOR CONTROL, CONTROL WIRING

a. All material and equipment furnished and installation methods used shall conform to the requirements of Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT; Section 26 29 11, MOTOR CONTROLLERS; and, Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES. All electrical wiring, conduit, and devices necessary for the proper connection, protection and operation of the systems shall be provided. Premium efficient motors shall be provided. Unless otherwise specified for a particular application, electric motors shall have the following requirements.

b. Special Requirements:

(1) Where motor power requirements of equipment furnished deviate from power shown on plans, provide electrical service designed under the requirements of NFPA 70 at no additional cost or time to the Government.

(2) Assemblies of motors, starters, and controls and interlocks on factory assembled and wired devices shall be in accordance with the requirements of this specification.

(3) Wire and cable materials specified in the electrical division of the specifications shall be modified as follows:

(a) Wiring material located where temperatures can exceed 160° F shall be stranded copper with Teflon FEP insulation with jacket. This includes wiring on the boilers and water heaters.

(b) Other wiring at boilers and water heaters, and to control panels, shall be NFPA 70 designation THWN.

(c) Shielded conductors or wiring in separate conduits for all instrumentation and control systems shall be provided where

recommended by manufacturer of equipment.

(4) Motor sizes shall be selected so that the motors do not operate into the service factor at maximum required loads on the driven equipment. Motors on pumps shall be sized for non-overloading at all points on the pump performance curves.

(5) Motors utilized with variable frequency drives shall be rated “inverter-ready” per NEMA MG 1.

c. Motor Efficiency and Power Factor: All motors, when specified as “high efficiency or Premium Efficiency” by the project specifications on driven equipment, shall conform to efficiency and power factor requirements in Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT, with no consideration of annual service hours. Motor manufacturers generally define these efficiency requirements as “NEMA premium efficient” and the requirements generally exceed those of PL 109-58. Motors not specified as “high efficiency or premium efficient” shall comply with PL 109-58.

d. Single-phase Motors: Capacitor-start type for hard starting applications. Motors for centrifugal pumps may be split phase or permanent split capacitor (PSC).

e. Poly-phase Motors: NEMA Design B, Squirrel cage, induction type. Each two-speed motor shall have two separate windings. A time delay (20 seconds minimum) relay shall be provided for switching from high to low speed.

f. Rating: Rating shall be continuous duty at 100 percent capacity in an ambient temperature of 104° F; minimum horsepower as shown on drawings; maximum horsepower in normal operation shall not exceed nameplate rating without service factor.

g. Insulation Resistance: Not less than one-half meg-ohm between stator conductors and frame shall be measured at the time of final inspection.

2.7 VARIABLE SPEED MOTOR CONTROLLERS

a. Refer to Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS and Section 26 29 11, MOTOR CONTROLLERS for specifications.

b. The combination of controller and motor shall be provided by the respective pump manufacturer and shall be rated for 100 percent output performance. Multiple units of the same class of equipment, i.e. pumps, shall be product of a single manufacturer.

c. Motors shall be premium efficient type, “inverter duty”, and be approved by the motor controller manufacturer. The controller-motor combination shall be guaranteed to provide full motor nameplate horsepower in variable frequency operation. Both driving and driven motor sheaves shall be fixed pitch.

d. Controller shall not add any current or voltage transients to the input AC power distribution system, DDC controls, sensitive medical equipment, etc., nor shall be affected from other devices on the AC power system.

2.8 EQUIPMENT AND MATERIALS IDENTIFICATION

a. Use symbols, nomenclature and equipment numbers specified, shown in the drawings, or shown in the maintenance manuals. Coordinate equipment and valve identification with local VAMC shops. In addition, provide bar code identification nameplate for all equipment which will allow the equipment identification code to be scanned into the system for maintenance and inventory tracking. Identification for piping is specified in Section 09 91 00, PAINTING.

b. Interior (Indoor) Equipment: Engraved nameplates, with letters not less than 3/16 inch high of brass with black-filled letters, or rigid black plastic with white letters specified in Section 09 91 00, PAINTING shall be permanently fastened to the equipment. Unit components such as water heaters, tanks, coils, filters, etc. shall be identified.

c. Exterior (Outdoor) Equipment: Brass nameplates, with engraved black filled letters, not less than 3/16 inch high riveted or bolted to the equipment.

d. Control Items: All temperature, pressure, and controllers shall be labeled and the component’s function identified. Identify and label each item as they appear on the control diagrams.

e. Valve Tags and Lists:

(1) Plumbing: All valves shall be provided with valve tags and listed on a valve list (Fixture stops not included).

(2) Valve tags: Engraved black filled numbers and letters not less than 1/2 inch high for number designation, and not less than 1/4 inch for service designation on 19 gauge, 1-1/2 inches round brass disc, attached with brass "S" hook or brass chain.

(3) Valve lists: Valve lists shall be created using a word processing program and printed on plastic coated cards. The plastic-coated valve list card(s), sized 8-1/2 inches by 11 inches shall show valve tag number, valve function and area of control for each service or system. The valve list shall be in a punched 3-ring binder notebook. An additional copy of the valve list shall be mounted in picture frames for mounting to a wall. Contracting Officer shall instruct Contractor where frames shall be mounted.

(4) A detailed plan for each floor of the building indicating the location and valve number for each valve shall be provided in the 3-ring binder notebook. Each valve location shall be identified with a color-coded sticker or thumb tack in ceiling or access door.

2.9 FIRESTOPPING

Section 07 84 00, FIRESTOPPING specifies an effective barrier against the spread of fire, smoke and gases where penetrations occur for piping. Refer to Section 22 07 11, PLUMBING INSULATION, for pipe insulation.

2.10 GALVANIZED REPAIR COMPOUND

MIL-P-21035, paint.

2.11 PIPE AND EQUIPMENT SUPPORTS AND RESTRAINTS

a. In lieu of the paragraph which follows, suspended equipment support and restraints may be designed and installed in accordance with the International Building Code (IBC) Submittals based on the International Building Code (IBC) requirements, or the following paragraphs of this Section shall be stamped and signed by a professional engineer registered in the state where the project is located. The Support system of suspended equipment over 500 pounds shall be submitted for approval of the Contracting Officer in all cases. See the above specifications for lateral force design requirements.

b. Type Numbers Specified: For materials, design, manufacture, selection, application, and installation refer to MSS SP-58. Refer to Section 05 50 00, METAL FABRICATIONS, for miscellaneous metal support materials and prime coat painting.

c. For Attachment to Concrete Construction:

(1) Concrete insert: Type 18, MSS SP-58.

(2) Self-drilling expansion shields and machine bolt expansion anchors: Permitted in concrete not less than 4 inches thick when approved by the Contracting Officer for each job condition.

(3) Power-driven fasteners: Permitted in existing concrete or masonry not less than 4 inches thick when approved by the Contracting Officer for each job condition.

d. For Attachment to Steel Construction: MSS SP-58.

(1) Welded attachment: Type 22.

(2) Beam clamps: Types 20, 21, 28 or 29. Type 23 C-clamp may be used for individual copper tubing up to 7/8 inch outside diameter.

e. Attachment to Metal Pan or Deck: As required for materials specified in Section 05 31 00, STEEL DECKING. Section 05 36 00, COMPOSITE METAL DECKING.

f. For Attachment to Wood Construction: Wood screws or lag bolts.

g. Hanger Rods: Hot-rolled steel, ASTM A36/A36M or ASTM A575 for allowable load listed in MSS SP-58. For piping, provide adjustment means for controlling level or slope. Types 13 or 15 turn-buckles shall provide 1-1/2 inches minimum of adjustment and incorporate locknuts. All-thread rods are acceptable.

h. Multiple (Trapeze) Hangers: Galvanized, cold formed, lipped steel channel horizontal member, not less than 1-5/8 inches by 1-5/8 inches, No. 12 gauge, designed to accept special spring held, hardened steel nuts.

(1) Allowable hanger load: Manufacturers rating less 200 pounds.

(2) Guide individual pipes on the horizontal member of every other trapeze hanger with 1/4 inch U-bolt fabricated from steel rod. Provide Type 40 insulation shield, secured by two 1/2 inch

galvanized steel bands, or insulated calcium silicate shield for insulated piping at each hanger.

i. Pipe Hangers and Supports: (MSS SP-58), use hangers sized to encircle insulation on insulated piping. Refer to Section 22 07 11, PLUMBING INSULATION for insulation thickness. To protect insulation, provide Type 39 saddles for roller type supports or insulated calcium silicate shields. Provide Type 40 insulation shield or insulated calcium silicate shield at all other types of supports and hangers including those for insulated piping.

(1) General Types (MSS SP-58):

(a) Standard clevis hanger: Type 1; provide locknut.

(b) Riser clamps: Type 8.

(c) Wall brackets: Types 31, 32 or 33.

(d) Roller supports: Type 41, 43, 44 and 46.

(e) Saddle support: Type 36, 37 or 38.

(f) Turnbuckle: Types 13 or 15.

(g) U-bolt clamp: Type 24.

(h) Copper Tube:

1. Hangers, clamps and other support material in contact with tubing shall be painted with copper colored epoxy paint, copper-coated, plastic coated or taped with isolation tape to prevent electrolysis.

2. For vertical runs use epoxy painted, copper-coated or plastic coated riser clamps.

3. For supporting tube to strut: Provide epoxy painted pipe straps for copper tube or plastic inserted vibration isolation clamps.

4. Insulated Lines: Provide pre-insulated calcium silicate shields sized for copper tube.

(i) Supports for plastic or glass piping: As recommended by the pipe manufacturer with black rubber tape extending 1 inch beyond steel support or clamp. Spring Supports (Expansion and contraction of vertical piping):

1. Movement up to 3/4 inch: Type 51 or 52 variable spring unit with integral turn buckle and load indicator.

2. Movement more than 3/4 inch: Type 54 or 55 constant support unit with integral adjusting nut, turn buckle and travel position indicator.

(j) Spring hangers are required on all plumbing system pumps one horsepower and greater.

(2) Plumbing Piping (Other Than General Types):

(a) Horizontal piping: Type 1, 5, 7, 9, and 10.

(b) Chrome plated piping: Chrome plated supports.

(c) Hangers and supports in pipe chase: Prefabricated system ABS self-extinguishing material, not subject to electrolytic action, to hold piping, prevent vibration and compensate for all static and operational conditions.

(d) Blocking, stays and bracing: Angle iron or preformed metal channel shapes, 18 gauge minimum.

j. Pre-insulated Calcium Silicate Shields:

(1) Provide 360-degree water resistant high density 965 kPa 140 psig compressive strength calcium silicate shields encased in galvanized metal.

(2) Pre-insulated calcium silicate shields to be installed at the point of support during erection.

(3) Shield thickness shall match the pipe insulation.

(4) The type of shield is selected by the temperature of the pipe, the load it shall carry, and the type of support it will be used with.

(a) Shields for supporting cold water shall have insulation that extends a minimum of 1 inch past the sheet metal.

(b) The insulated calcium silicate shield shall support the maximum allowable water filled span as indicated in MSS SP-58. To support the load, the shields shall have one or more of the following features: structural inserts 600 psig compressive strength, an extra bottom metal shield, or formed structural steel (ASTM A36/A36M) wear plates welded to the bottom sheet metal jacket.

(5) Shields may be used on steel clevis hanger type supports, trapeze hangers, roller supports or flat surfaces.

2.12 PIPE PENETRATIONS

a. Pipe penetration sleeves shall be installed for all pipe other than rectangular blocked out floor openings for risers in mechanical bays.

b. Pipe penetration sleeve materials shall comply with all firestopping requirements for each penetration.

c. To prevent accidental liquid spills from passing to a lower level, provide the following:

(1) For sleeves: Extend sleeve 1 inch above finished floor and provide sealant for watertight joint.

(2) For blocked out floor openings: Provide 1-1/2 inch angle set in

silicone adhesive around opening.

(3) For drilled penetrations: Provide 1-1/2 inch angle ring or square set in silicone adhesive around penetration.

d. Penetrations are prohibited through beams or ribs, but may be installed in concrete beam flanges, with structural engineer prior approval. Any deviation from these requirements shall receive prior approval of Contracting Officer.

e. Sheet metal, plastic, or moisture resistant fiber sleeves shall be provided for pipe passing through floors, interior walls, and partitions, unless brass or steel pipe sleeves are specifically called for below.

f. Cast iron or zinc coated pipe sleeves shall be provided for pipe passing through exterior walls below grade. The space between the sleeve and pipe shall be made watertight with a modular or link rubber seal. The link seal shall be applied at both ends of the sleeve.

g. Galvanized steel or an alternate black iron pipe with asphalt coating sleeves shall be for pipe passing through concrete beam flanges, except where brass pipe sleeves are called for. A galvanized steel sleeve shall be provided for pipe passing through floor of mechanical rooms, laundry work rooms, and animal rooms above basement. Except in mechanical rooms, sleeves shall be connected with a floor plate.

h. Brass Pipe Sleeves shall be provided for pipe passing through quarry tile, terrazzo or ceramic tile floors. The sleeve shall be connected with a floor plate.

i. Sleeve clearance through floors, walls, partitions, and beam flanges shall be 1 inch greater in diameter than external diameter of pipe. Sleeve for pipe with insulation shall be large enough to accommodate the insulation plus 1 inch in diameter. Interior openings shall be caulked tight with firestopping material and sealant to prevent the spread of fire, smoke, water and gases.

j. Sealant and Adhesives: shall be as specified in Section 07 92 00, JOINT SEALANTS. Bio-based materials shall be utilized when possible.

k. Pipe passing through roof shall be installed through a 4.9 kg per square meter copper flashing with an integral skirt or flange. Skirt or flange shall extend not less than 8 inches from the pipe and set in a solid coating of bituminous cement. Extend flashing a minimum of 10 inches up the pipe. Pipe passing through a waterproofing membrane shall be provided with a clamping flange. The annular space between the sleeve and pipe shall be sealed watertight.

2.13 TOOLS AND LUBRICANTS

a. Furnish, and turn over to the Contracting Officer, special tools not readily available commercially, that are required for disassembly or adjustment of equipment and machinery furnished.

b. Grease Guns with Attachments for Applicable Fittings: One for each type of grease required for each motor or other equipment.

c. Tool Containers: metal, permanently identified for intended service

and mounted, or located, where directed by the Contracting Officer.

d. Lubricants: A minimum of 1 quart of oil, and 1 pound of grease, of equipment manufacturer's recommended grade and type, in unopened containers and properly identified as to use for each different application. Bio-based materials shall be utilized when possible.

2.14 WALL, FLOOR AND CEILING PLATES

a. Material and Type: Chrome plated brass or chrome plated steel, one piece or split type with concealed hinge, with set screw for fastening to pipe, or sleeve. Use plates that fit tight around pipes, cover openings around pipes and cover the entire pipe sleeve projection.

b. Thickness: Not less than 3/32 inch for floor plates. For wall and ceiling plates, not less than 0.025 inch for up to 3 inch pipe, 0.035 inch for larger pipe.

c. Locations: Use where pipe penetrates floors, walls and ceilings in exposed locations, in finished areas only. Wall plates shall be used where insulation ends on exposed water supply pipe drop from overhead. A watertight joint shall be provided in spaces where brass or steel pipe sleeves are specified.

2.15 ASBESTOS

Materials containing asbestos are prohibited.

PART 3 EXECUTION

3.1 ARRANGEMENT AND INSTALLATION OF EQUIPMENT AND PIPING

a. Location of piping, sleeves, inserts, hangers, and equipment, access provisions shall be coordinated with the work of all trades. Piping, sleeves, inserts, hangers, and equipment shall be located clear of windows, doors, openings, light outlets, and other services and utilities. Equipment layout drawings shall be prepared to coordinate proper location and personnel access of all facilities. The drawings shall be submitted for review.

b. Manufacturer's published recommendations shall be followed for installation methods not otherwise specified.

c. Operating Personnel Access and Observation Provisions: All equipment and systems shall be arranged to provide clear view and easy access, without use of portable ladders, for maintenance, testing and operation of all devices including, but not limited to: all equipment items, valves, backflow preventers, filters, strainers, transmitters, sensors, meters and control devices. All gauges and indicators shall be clearly visible by personnel standing on the floor or on permanent platforms. Maintenance and operating space and access provisions that are shown in the drawings shall not be changed nor reduced.

d. Structural systems necessary for pipe and equipment support shall be coordinated to permit proper installation.

e. Location of pipe sleeves, trenches and chases shall be accurately coordinated with equipment and piping locations.

f. Cutting Holes:

(1) Holes shall be located to avoid interference with structural members such as beams or grade beams. Holes shall be laid out in advance and drilling done only after approval by Contracting Officer. If the Contractor considers it necessary to drill through structural members, this matter shall be referred to Contracting Officer for approval.

(2) Waterproof membrane shall not be penetrated. Pipe floor penetration block outs shall be provided outside the extents of the waterproof membrane.

(3) Holes through concrete and masonry shall be cut by rotary core drill. Pneumatic hammer, impact electric, and hand or manual hammer type drill will not be allowed, except as permitted by Contracting Officer where working area space is limited.

g. Minor Piping: Generally, small diameter pipe runs from drips and drains, water cooling, and other services are not shown but shall be provided.

h. Protection and Cleaning:

(1) Equipment and materials shall be carefully handled, properly stored, and adequately protected to prevent damage before and during installation, in accordance with the manufacturer's recommendations and as approved by the Contracting Officer. Damaged or defective items in the opinion of the Contracting Officer, shall be replaced at no additional cost or time to the Government.

(2) Protect all finished parts of equipment, such as shafts and bearings where accessible, from rust prior to operation by means of protective grease coating and wrapping. Close pipe openings with caps or plugs during installation. Pipe openings, equipment, and plumbing fixtures shall be tightly covered against dirt or mechanical injury. At completion of all work thoroughly clean fixtures, exposed materials and equipment.

i. Concrete and Grout: Concrete and shrink compensating grout 3000 psig minimum, specified in Section 03 30 00, CAST-IN-PLACE CONCRETE, shall be used for all pad or floor mounted equipment.

j. Gauges, thermometers, valves and other devices shall be installed with due regard for ease in reading or operating and maintaining said devices. Thermometers and gauges shall be located and positioned to be easily read by operator or staff standing on floor or walkway provided. Servicing shall not require dismantling adjacent equipment or pipe work.

k. Interconnection of Controls and Instruments: Electrical interconnection is generally not shown but shall be provided. This includes interconnections of sensors, transmitters, transducers, control devices, control and instrumentation panels, alarms, instruments and computer workstations. Comply with NFPA 70.

l. Domestic cold and hot water systems interface with the HVAC control system for the temperature, pressure and flow monitoring requirements

to mitigate Legionella. See the HVAC control points list and Section 23 09 23, DIRECT DIGITAL CONTROL SYSTEM FOR HVAC.

m. Work in Existing Building:

(1) Perform as specified in Article, OPERATIONS AND STORAGE AREAS, Article, ALTERATIONS, and Article, RESTORATION of the Section 01 00 00, GENERAL REQUIREMENTS for relocation of existing equipment, alterations and restoration of existing building(s).

(2) As specified in Section 01 00 00, GENERAL REQUIREMENTS, Article, OPERATIONS AND STORAGE AREAS, make alterations to existing service piping at times that will cause the least interfere with normal operation of the facility.

n. Work in Animal Research Areas: Seal all pipe penetrations with silicone sealant to prevent entrance of insects.

o. Work in bathrooms, restrooms, housekeeping closets: All pipe penetrations behind escutcheons shall be sealed with plumbers’ putty.

p. Switchgear Drip Protection: Every effort shall be made to eliminate the installation of pipe above data equipment, and electrical and telephone switchgear. If this is not possible, encase pipe in a second pipe with a minimum of joints. Drain valve shall be provided in low point of casement pipe.

q. Inaccessible Equipment:

(1) Where the Government determines that the Contractor has installed equipment not conveniently accessible for operation and maintenance, equipment shall be removed and reinstalled or remedial action performed as directed at no additional cost or additional time to the Government.

(2) The term "conveniently accessible" is defined as capable of being reached without the use of ladders, or without climbing or crawling under or over obstacles such as electrical conduit, motors, fans, pumps, belt guards, transformers, high voltage lines, piping, and ductwork.

3.2 TEMPORARY PIPING AND EQUIPMENT

a. Continuity of operation of existing facilities may require temporary installation or relocation of equipment and piping. Temporary equipment or pipe installation or relocation shall be provided to maintain continuity of operation of existing facilities.

b. The Contractor shall provide all required facilities in accordance with the requirements of phased construction and maintenance of service. All piping and equipment shall be properly supported, sloped to drain, operate without excessive stress, and shall be insulated where injury can occur to personnel by contact with operating facilities. The requirements of paragraph 3.1 shall apply.

c. Temporary facilities and piping shall be completely removed back to the nearest active distribution branch or main pipe line and any openings in structures sealed. Dead legs are prohibited in potable water systems. Necessary blind flanges and caps shall be provided to

seal open piping remaining in service.

3.3 RIGGING

a. Openings in building structures shall be planned to accommodate design scheme.

b. Alternative methods of equipment delivery may be offered and will be considered by Government under specified restrictions of phasing and service requirements as well as structural integrity of the building.

c. All openings in the building shall be closed when not required for rigging operations to maintain proper environment in the facility for Government operation and maintenance of service.

d. Contractor shall provide all facilities required to deliver specified equipment and place on foundations. Attachments to structures for rigging purposes and support of equipment on structures shall be Contractor's full responsibility.

e. Contractor shall check all clearances, weight limitations and shall provide a rigging plan designed by a Registered Professional Engineer. All modifications to structures, including reinforcement thereof, shall be at Contractor's cost, time and responsibility.

f. Rigging plan and methods shall be referred to Contracting Officer for evaluation prior to actual work.

3.4 PIPE AND EQUIPMENT SUPPORTS

a. Where hanger spacing does not correspond with joist or rib spacing, use structural steel channels secured directly to joist and rib structure that will correspond to the required hanger spacing, and then suspend the equipment and piping from the channels. Holes shall be drilled or burned in structural steel ONLY with the prior written approval of the Contracting Officer.

b. The use of chain pipe supports, wire or strap hangers; wood for blocking, stays and bracing, or hangers suspended from piping above shall not be permitted. Rusty products shall be replaced.

c. Hanger rods shall be used that are straight and vertical. Turnbuckles for vertical adjustments may be omitted where limited space prevents use. A minimum of 1/2 inch clearance between pipe or piping covering and adjacent work shall be provided.

d. For horizontal and vertical plumbing pipe supports, refer to the IPC and these specifications.

e. Overhead Supports:

(1) The basic structural system of the building is designed to sustain the loads imposed by equipment and piping to be supported overhead.

(2) Provide steel structural members, in addition to those shown, of adequate capability to support the imposed loads, located in accordance with the final approved layout of equipment and piping.

(3) Tubing and capillary systems shall be supported in channel troughs.

f. Floor Supports:

(1) Provide concrete bases, concrete anchor blocks and pedestals, and structural steel systems for support of equipment and piping. Concrete bases and structural systems shall be anchored and doweled to resist forces under operating and seismic conditions (if applicable) without excessive displacement or structural failure.

(2) Bases and supports shall not be located and installed until equipment mounted thereon has been approved. Bases shall be sized to match equipment mounted thereon plus 2 inch excess on all edges. Structural drawings shall be reviewed for additional requirements. Bases shall be neatly finished and smoothed, shall have chamfered edges at the top, and shall be suitable for painting.

(3) All equipment shall be shimmed, leveled, firmly anchored, and grouted with epoxy grout. Anchor bolts shall be placed in sleeves, anchored to the bases. Fill the annular space between sleeves and bolts with a grout material to permit alignment and realignment.

3.5 LUBRICATION

a. All equipment and devices requiring lubrication shall be lubricated prior to initial operation. All devices and equipment shall be field checked for proper lubrication.

b. All devices and equipment shall be equipped with required lubrication fittings. A minimum of 1 quart of oil and 1 pound of grease of manufacturer's recommended grade and type for each different application shall be provided. All materials shall be delivered to Contracting Officer in unopened containers that are properly identified as to application.

c. A separate grease gun with attachments for applicable fittings shall be provided for each type of grease applied.

d. All lubrication points shall be accessible without disassembling equipment, except to remove access plates.

e. All lubrication points shall be extended to one side of the equipment.

3.6 PLUMBING SYSTEMS DEMOLITION

a. Rigging access, other than indicated in the drawings, shall be provided after approval for structural integrity by the Contracting Officer. Such access shall be provided at no additional cost or time to the Government. Where work is in an operating plant, approved protection from dust and debris shall be provided at all times for the safety of plant personnel and maintenance of plant operation and environment of the plant.

b. In an operating plant, cleanliness and safety shall be maintained. The plant shall be kept in an operating condition. Government

personnel will be carrying on their normal duties of operating, cleaning and maintaining equipment and plant operation. Work shall be confined to the immediate area concerned; maintain cleanliness and wet down demolished materials to eliminate dust. Dust and debris shall not be permitted to accumulate in the area to the detriment of plant operation. All flame cutting shall be performed to maintain the fire safety integrity of this plant. Adequate fire extinguishing facilities shall be available at all times. All work shall be performed in accordance with recognized fire protection standards including NFPA 51B. Inspections will be made by personnel of the VAMC, and the Contractor shall follow all directives of the Contracting Officer with regard to rigging, safety, fire safety, and maintenance of operations.

c. Unless specified otherwise, all piping, wiring, conduit, and other devices associated with the equipment not re-used in the new work shall be completely removed from Government property per Section 01 74 19, CONSTRUCTION WASTE MANAGEMENT. This includes all concrete equipment pads, pipe, valves, fittings, insulation, and all hangers including the top connection and any fastenings to building structural systems. All openings shall be sealed after removal of equipment, pipes, ducts, and other penetrations in roof, walls, floors, in an approved manner and in accordance with plans and specifications where specifically covered. Structural integrity of the building system shall be maintained. Reference shall also be made to the drawings and specifications of the other disciplines in the project for additional facilities to be demolished or handled.

d. All valves including gate, globe, ball, butterfly and check, all pressure gauges and thermometers with wells shall remain Government property and shall be removed and delivered to Contracting Officer and stored as directed. The Contractor shall remove all other material and equipment, devices and demolition debris under these plans and specifications. Such material shall be removed from Government property expeditiously and shall not be allowed to accumulate. Coordinate with the Contracting Officer and Infection Control.

3.7 CLEANING AND PAINTING

a. Prior to final inspection and acceptance of the plant and facilities for beneficial use by the Government, the plant facilities, equipment and systems shall be thoroughly cleaned and painted. Refer to Section 09 91 00, PAINTING.

b. In addition, the following special conditions apply:

(1) Cleaning shall be thorough. Solvents, cleaning materials and methods recommended by the manufacturers shall be used for the specific tasks. All rust shall be removed prior to painting and from surfaces to remain unpainted. Scratches, scuffs, and abrasions shall be repaired prior to applying prime and finish coats.

(2) The following Material and Equipment shall NOT be painted:

(a) Motors, controllers, control switches, and safety switches.

(b) Control and interlock devices.

(c) Regulators.

(d) Pressure reducing valves.

(e) Control valves and thermostatic elements.

(f) Lubrication devices and grease fittings.

(g) Copper, brass, aluminum, stainless steel and bronze surfaces.

(h) Valve stems and rotating shafts.

(i) Pressure gauges and thermometers.

(j) Glass.

(k) Name plates.

(3) Control and instrument panels shall be cleaned and damaged surfaces repaired. Touch-up painting shall be made with matching paint type and color obtained from manufacturer or computer matched.

(4) Pumps, motors, steel and cast-iron bases, and coupling guards shall be cleaned, and shall be touched-up with the same paint type and color as utilized by the pump manufacturer.

(5) Temporary Facilities: Apply paint to surfaces that do not have existing finish coats per Section 09 91 00, Painting.

(6) The final result shall be a smooth, even-colored, even-textured factory finish on all items. The entire piece of equipment shall be repainted, if necessary, to achieve this. Lead based paints shall not be used.

3.8 IDENTIFICATION SIGNS

a. Laminated plastic signs, with engraved lettering not less than 3/16 inch high, shall be provided that designates equipment function, for all equipment, switches, motor controllers, relays, meters, control devices, including automatic control valves. Nomenclature and identification symbols shall correspond to that used in maintenance manual, and in diagrams specified elsewhere. Attach by chain, adhesive, or screws.

b. Factory Built Equipment: Metal plate, securely attached, with name and address of manufacturer, serial number, model number, size, and performance data shall be placed on factory-built equipment.

c. Pipe Identification: Refer to Section 09 91 00, PAINTING.

3.9 STARTUP AND TEMPORARY OPERATION

a. Startup of equipment shall be performed as described in the equipment specifications. Vibration within specified tolerance shall be verified prior to extended operation. Temporary use of equipment is specified in Section 01 00 00, GENERAL REQUIREMENTS, Article, TEMPORARY USE OF MECHANICAL AND ELECTRICAL EQUIPMENT.

b. The CxA will observe startup and Contractor testing of selected equipment. Coordinate the startup and Contractor testing schedules with the Contracting Officer and CxA. Provide a minimum of

3.10 OPERATING AND PERFORMANCE TESTS

a. Prior to the final inspection, all required tests shall be performed as specified in Section 01 00 00, GENERAL REQUIREMENTS, Article, TESTS and submit the test reports and records to the Contracting Officer.

b. Should evidence of malfunction in any tested system, or piece of equipment or component part thereof, occur during or as a result of tests, make proper corrections, repairs or replacements, and repeat tests at no additional cost to the Government.

c. When completion of certain work or systems occurs at a time when final control settings and adjustments cannot be properly made to make performance tests, then conduct such performance tests and finalize control settings during the first actual seasonal use of the respective systems following completion of work. Rescheduling of these tests shall be requested in writing to Contracting Officer for approval.

d. Perform tests as required for commissioning provisions in accordance with Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS and Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

3.11 OPERATION AND MAINTENANCE MANUALS

a. All new and temporary equipment and all elements of each assembly shall be included.

b. Data sheet on each device listing model, size, capacity, pressure, speed, horsepower, impeller size, and other information shall be included.

c. Manufacturer’s installation, maintenance, repair, and operation instructions for each device shall be included. Assembly drawings and parts lists shall also be included. A summary of operating precautions and reasons for precautions shall be included in the Operations and Maintenance Manual.

d. Lubrication instructions, type and quantity of lubricant shall be included.

e. Schematic diagrams and wiring diagrams of all control systems corrected to include all field modifications shall be included.

f. Set points of all interlock devices shall be listed.

g. Trouble-shooting guide for the control system troubleshooting shall be inserted into the Operations and Maintenance Manual.

h. The control system sequence of operation corrected with submittal review comments shall be inserted into the Operations and Maintenance Manual.

i. Emergency procedures for shutdown and startup of equipment and systems.

3.12 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this Section of the specification will be tested as part of a larger system.

3.13 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for four hours to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

-- End of Section --

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SECTION 22 05 12
9/20

GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT

PART 1 GENERAL

1.1 DESCRIPTION

a. This Section describes the general motor requirements for plumbing equipment and applies to all Sections of Division 22.

b. A complete listing of common acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

f. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

g. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

h. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES.

j. Section 26 29 11, MOTOR CONTROLLERS: Starters, control and protection of motors.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and VHA standard will govern.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 112	(2017) Standard Test Procedure for Polyphase Induction Motors and Generators
IEEE 841	(2021) IEEE Standard for Petroleum and Chemical Industry-Premium-Efficiency, Severe-Duty Totally Enclosed Fan-Cooled (TEFC) Squirrel Cage Induction Motors--Up to and Including 370 kW (500 hp)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1	(2021) Motors and Generators
NEMA MG 2	(2014) Safety Standard for Construction and Guide for Selection, Installation and Use of Electric Motors and Generators
NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Certification; G, AE

SD-08 Manufacturer's Instructions

Training Plans; G, AE

Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this Section shall be marked "SUBMITTED UNDER SECTION 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT", with applicable paragraph identification.

c. Submit motor submittals with drive equipment.

d. Shop Drawings:

(1) Provide documentation to demonstrate compliance with Contract documents.

(2) Motor nameplate information shall be submitted including electrical ratings, efficiency, bearing data, power factor, frame size, dimensions, mounting details, materials, horsepower, voltage, phase, speed (RPM), enclosure, starting characteristics, torque characteristics, code letter, full load and locked rotor current, service factor, and lubrication method.

(3) Motor parameters required for the determination of the Reed Critical Frequency of vertical hollow shaft motors shall be submitted.

e. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

f. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

g. Certification: Two weeks prior to final inspection, unless otherwise noted, the following certification shall be submitted to the Contracting Officer.

Certification shall be submitted stating that the motors have been properly applied, installed, adjusted, lubricated, and tested.

h. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

i. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

a. Bio-Based Materials: For products designated by the USDA's bio-based Bio-Preferred Program, provide products that meet or exceed USDA

recommendations for bio-based content, so long as products meet all performance requirements in this specifications Section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

b. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in Paragraph “AS-BUILT DOCUMENTATION” of Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

PART 2 PRODUCTS

2.1 MOTORS

a. All material and equipment furnished and installation methods shall conform to the requirements of Section 26 29 11, MOTOR CONTROLLERS; and Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES. Provide all electrical wiring, conduit, and devices necessary for the proper connection, protection and operation of the systems. Provide premium efficiency type motors. Unless otherwise specified for a particular application, use electric motors with the following requirements.

b. For alternating current, fractional and integral horsepower motors, NEMA MG 1 and NEMA MG 2 shall apply.

c. For severe duty totally enclosed motors, IEEE 841 shall apply. For severe duty totally enclosed motors, IEEE 841 shall apply.

d. Single-phase Motors: Motors for centrifugal fans and pumps may be split phase or permanent split capacitor (PSC) type. Provide capacitor-start type for hard starting applications.

e. Poly-phase Motors: NEMA Design B, Squirrel cage, induction type.

Two Speed Motors: Each two-speed motor shall have two separate windings. Provide a time- delay (20 seconds minimum) relay for switching from high to low speed.

f. Voltage ratings shall be as follows:

(1) Single phase:

(a) Motors connected to 120-volt systems: 115 volts.

(b) Motors connected to 208-volt systems: 200 volts.

(c) Motors connected to 240-volt or 480-volt systems: 230/460 volts, dual connection.

(2) Three phase:

(a) Motors connected to 208-volt systems: 200 volts.

(b) Motors, less than 100 HP, connected to 240-volt or 480-volt systems: 230/460 volts, dual connection.

(c) Motors, 100 HP or greater, connected to 240-volt systems: 230 volts.

(d) Motors, 100 HP or greater, connected to 480-volt systems: 460 volts.

(e) Motors connected to high voltage systems (Over 600V): shall conform to NEMA MG 1 for connection to the nominal system voltage shown in the drawings.

g. Number of phases shall be as follows:

(1) Motors, less than 1/2 HP: Single phase.

(2) Motors, 1/2 HP and greater: Three phase.

(3) Exceptions:

(a) Hermetically sealed motors.

(b) Motors for equipment assemblies, less than 1 HP, may be single phase provided the manufacturer of the proposed assemblies cannot supply the assemblies with three phase motors.

h. Horsepower ratings shall be adequate for operating the connected loads continuously in the prevailing ambient temperatures in areas where the motors are installed, without exceeding the NEMA standard temperature rises for the motor insulation.

i. Motor designs, as indicated by the NEMA code letters, shall be coordinated with the connected loads to assure adequate starting, acceleration and running torque without exceeding nameplate ratings or considering service factor.

j. Motor Enclosures:

(1) shall be the NEMA types as specified and/or shown in the Contract Documents.

(2) Where the types of motor enclosures are not shown in the drawings, they shall be the NEMA types per NEMA 250, which are most suitable for the environmental conditions where the motors are being installed. Enclosure requirements for certain conditions are as follows:

(a) Motors located outdoors, indoors in wet or high humidity locations, or in unfiltered airstreams shall be totally enclosed type.

(b) Where motors are located in an NEC 511 classified area, provide TEFC explosion proof motor enclosures.

(c) Where motors are located in a corrosive environment, provide TEFC enclosures with corrosion resistant finish.

(3) Enclosures shall be primed and finish coated at the factory with manufacturer's prime coat and standard finish.

k. Electrical Design Requirements:

(1) Motors shall be continuous duty.

(2) The insulation system shall be rated minimum of Class B, 266 degrees F.

(3) The maximum temperature rise by resistance at rated power shall not exceed Class B limits, 144 degrees F.

(4) The speed/torque and speed/current characteristics shall comply with NEMA Design A or B, as specified.

(5) Motors shall be suitable for full voltage starting, unless otherwise noted. Coordinate motor features with applicable motor controllers.

(6) Motors for variable frequency drive applications shall adhere to NEMA MG 1, Part 30, Application Considerations for Constant Speed Motors Used on a Sinusoidal Bus with Harmonic Content and General Purpose Motors Used with Adjustable Voltage or Adjustable Frequency Controls, or both, or NEMA MG 1, Part 31, Definite Purpose Inverter Fed Polyphase Motors.

l. Mechanical Design Requirements:

(1) Bearings shall be rated for a minimum fatigue life of 26,280 hours for belt-driven loads and 100,000 hours for direct-drive loads based on L10 (Basic Rating Life) at full load direct coupled, except vertical high thrust motors which require a 40,000 hour rating. A minimum fatigue life of 40,000 hours is required for VFD drives.

(2) Vertical motors shall be capable of withstanding a momentary up thrust of at least 30 percent of normal down thrust.

(3) Grease lubricated bearings shall be designed for electric motor use. Grease shall be capable of the temperatures associated with electric motors and shall be compatible with Polyurea based greases.

(4) Grease fittings, if provided, shall be Alemite type or equivalent.

(5) Oil lubricated bearings, when specified, shall have an externally visible sight glass to view oil level.

(6) Vibration shall not exceed 0.15 inch per second, unfiltered peak.

(7) Noise level shall meet the requirements of the application.

(8) Motors on 180 frames and larger shall have provisions for lifting eyes or lugs capable of a safety factor of 5.

(9) All external fasteners shall be corrosion resistant.

(10) Condensation heaters, when specified, shall keep motor windings at least 9 degrees F above ambient temperature.

(11) Winding thermostats, when specified shall be normally closed,

connected in series.

(12) Grounding provisions shall be in the main terminal box.

m. Special Requirements:

(1) Where motor power requirements of equipment furnished deviate from power shown on plans, provide electrical service designed under the requirements of NFPA 70 without additional cost or time to the Government.

(2) Assemblies of motors, starters, controls and interlocks on factory assembled and wired devices shall be in accordance with the requirements of this specification.

(3) Wire and cable materials specified in the electrical division of the specifications shall be modified as follows:

(a) Wiring material located where temperatures can exceed 160 degrees F shall be stranded copper with Teflon FEP insulation with jacket. This includes wiring on the boilers.

(b) Other wiring at boilers and to control panels shall be NFPA 70 designation THWN.

(c) Provide shielded conductors or wiring in separate conduits for all instrumentation and control systems where recommended by manufacturer of equipment.

(4) Select motor sizes so that the motors do not operate into the service factor at maximum required loads on the driven equipment. Motors on pumps shall be sized for non-overloading at all points on the pump performance curves.

(5) Motors utilized with variable frequency drives shall be rated “inverter-duty” per NEMA MG 1, Part 31, Definite-Purpose Inverter-Fed Polyphase Motors. Provide motor shaft grounding apparatus that will protect bearings from damage from stray currents.

n. Additional requirements for specific motors, as indicated in other Sections, shall also apply.

o. NEMA Premium Efficiency Electric Motors, Motor Efficiencies: All permanently wired polyphase motors of 1 HP or greater shall meet the minimum full-load efficiencies as indicated in the following table, and as specified in this specification. Motors of 1 HP or greater with open, drip-proof or totally enclosed fan-cooled enclosures shall be NEMA premium efficiency type, unless otherwise indicated. Motors provided as an integral part of motor driven equipment are excluded from this requirement if a minimum seasonal or overall efficiency requirement is indicated for that equipment by the provisions of another Section.

Minimum Efficiencies Open Drip-Proof				Minimum Efficiencies Totally Enclosed Fan-Cooled			
Rating HP	1200 RPM	1800 RPM	3600 RPM	Rating HP	1200 RPM	1800 RPM	3600 RPM
1	82.5%	85.5%	77.0%	1	82.5%	85.5%	77.0%
1.5	86.5%	86.5%	84.0%	1.5	87.5%	86.5%	84.0%
2	87.5%	86.5%	85.5%	2	88.5%	86.5%	85.5%
3	88.5%	89.5%	85.5%	3	89.5%	89.5%	86.5%
5	89.5%	89.5%	86.5%	5	89.5%	89.5%	88.5%
7.5	90.2%	91.0%	88.5%	7.5	91.0%	91.7%	89.5%
10	91.7%	91.7%	89.5%	10	91.0%	91.7%	90.2%
15	91.7%	93.0%	90.2%	15	91.7%	92.4%	91.0%
20	92.4%	93.0%	91.0%	20	91.7%	93.0%	91.0%
25	93.0%	93.6%	91.7%	25	93.0%	93.6%	91.7%
30	93.6%	94.1%	91.7%	30	93.0%	93.6%	91.7%
40	94.1%	94.1%	92.4%	40	94.1%	94.1%	92.4%
50	94.1%	94.5%	93.0%	50	94.1%	94.5%	93.0%
60	94.5%	95.0%	93.6%	60	94.5%	95.0%	93.6%
75	94.5%	95.0%	93.6%	75	94.5%	95.4%	93.6%
100	95.0%	95.4%	93.6%	100	95.0%	95.4%	94.1%
125	95.0%	95.4%	94.1%	125	95.0%	95.4%	95.0%
150	95.4%	95.8%	94.1%	150	95.8%	95.8%	95.0%
200	95.4%	95.8%	95.0%	200	95.8%	96.2%	95.4%

p. Minimum Power Factor at Full Load and Rated Voltage: 90 percent at 1200 RPM, 1800 RPM and 3600 RPM. Power factor correction capacitors shall be provided unless the motor meets the 0.9 requirement without it or if the motor is controlled by a variable frequency drive. The power factor correction capacitors shall be able to withstand high voltage transients and power line variations without breakdown.

q. Energy Efficiency of Small Motors (Motor Efficiencies): All motors under 1 hp shall meet the requirements of the DOE Small Motor Regulation.

Polyphase Open MotorsAverage full load efficiency				**Capacitor-start capacitor-run and capacitor-start induction runopen motorsAverage full load efficiency**			
Rating hp	**6 poles**	**4 poles**	**2 poles**	**Rating hp**	**6 poles**	**4 poles**	**2 poles**
0.25	67.5	69.5	65.6	0.25	62.2	68.5	66.6
0.33	71.4	73.4	69.5	0.33	66.6	72.4	70.5
0.5	75.3	78.2	73.4	0.5	76.2	76.2	72.4
0.75	81.7	81.1	76.8	0.75	80.2	81.8	76.2

PART 3 EXECUTION

3.1 INSTALLATION

a. Install motors in accordance with manufacturer’s recommendations, the NEC, NEMA, as shown in the drawings and/or as required by other Sections of these specifications.

b. If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

3.2 FIELD TESTS

a. All tests shall be witnessed by the CxA or by the Contracting Officer.

b. Perform an electric insulation resistance Test using a megohmmeter on all motors after installation, before startup. All shall test free from grounds.

c. Perform Load test in accordance with IEEE 112, Test Method B, to determine freedom from electrical or mechanical defects and compliance with performance data.

d. Insulation Resistance: Not less than 1/2 meg-ohm between stator conductors and frame, to be determined at the time of final inspection.

e. All test data shall be complied into a report form for each motor and provided to the Contracting Officer or their representative.

3.3 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA will observe startup and Contractor testing of selected equipment. Coordinate the startup and Contractor testing schedules with Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

3.4 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this Section of the specification shall be tested as part of a larger system.

3.5 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for four hours to instruct each VA personnel responsible in operation and

maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

-- End of Section --

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SECTION 22 05 19

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3.3 STARTUP AND TESTING
3.4 COMMISSIONING
3.5 DEMONSTRATION AND TRAINING

-- End of Section Table of Contents --

SECTION 22 05 19
9/20

METERS AND GAUGES FOR PLUMBING PIPING

PART 1 GENERAL

1.1 DESCRIPTION

a. This Section describes the requirements for water meters and gauges primarily used for troubleshooting the system and to indicate system performance.

b. A complete listing of common acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

c. Components intended to be connected to BAS shall be furnished under Section 23 09 23 DIRECT DIGITAL CONTROL SYSTEMS FOR HVAC for installation under this Section.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

f. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS. Requirements for commissioning, systems readiness checklist, and training.

g. Section 23 09 23, DIRECT DIGITAL CONTROL SYSTEMS FOR HVAC.

h. Section 25 10 10, ADVANCED UTILITY METERING SYSTEM.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B40.100	(2022) Pressure Gauges and Gauge Attachments
ASME B40.200	(2008; R 2013) Thermometers, Direct Reading and Remote Reading

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C700	(2020) Cold-Water Meters - Displacement Type, Metal Alloy Main Case
AWWA C701	(2019) Cold-Water Meters - Turbine Type for Customer Service
AWWA C702	(2019) Cold-Water Meters - Compound Type
AWWA C707	(2022) Encoder-Type Remote-Registration Systems for Cold-Water Meters

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2	(2023) National Electrical Safety Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NSF INTERNATIONAL (NSF)

NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects
NSF 372	(2022) Drinking Water System Components - Lead Content

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Training Plans; G, AE

Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manual; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this Section shall be marked “SUBMITTED UNDER SECTION 22 05 19, METERS AND GAUGES FOR PLUMBING PIPING”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Water Meter.

(2) Pressure Gauges.

(3) Thermometers.

(4) Product certificates for each type of meter and gauge.

(5) BACnet communication protocol.

d. Complete operating and maintenance manual shall including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the system.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

e. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

f. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 AS-BUILT DOCUMENTATION

Comply with requirements in Paragraph “AS-BUILT DOCUMENTATION” of Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

PART 2 PRODUCTS

2.1 DISPLACEMENT WATER METER

a. For pipe sizes 2 inches and smaller, the water meter shall be displacement type, full size nutating disc, magnetic drive, sealed register, and fully conform to AWWA C700. Peak domestic flow shall be 34 gpm. The meter register shall indicate flow in U.S. gallons.

b. The water meter shall be rated for use at temperatures ranging from -40 degrees F and 158 degrees F and operate at a working pressure of 150 psig.

c. The meter case, bottom caps, and register box lids shall be constructed from cast bronze.

d. The meter shall register plus or minus 3 percent of the water actually passing through it at any rate of flow within the normal test flow limits specified in AWWA C700.

e. The water meter shall conform to NSF/ANSI 61 and NSF 372.

2.2 TURBINE WATER METER

a. The water meter shall be Turbine type, Class II, in-line, horizontal axis, and fully conform to AWWA C701. Peak domestic flow shall be as indicated on plans in gpm. The meter Register shall indicate flow in U.S. gallons.

b. The water meter shall be rated for use at temperatures ranging from -40 degrees F and 158 degrees F and operate at a working pressure of 150 psig.

c. The turbine case shall be constructed of cast bronze.

d. The register box rings and lid shall be made of cast copper alloy containing not less than 75 percent copper. Forged or die cast copper alloy containing not less than 75 percent copper or a suitable synthetic polymer.

e. The flow measuring turbine shall be made of a suitable synthetic polymer with specific gravity approximately equal to that of water. The measuring turbine shall have sufficient dimensional stability to retain operating clearances at the full range of working temperatures.

f. All external case closures, such as rings, clamps, screws, bolts, cap bolts, nuts and washers shall be designed for easy removal following lengthy service.

g. The turbine meter shall have flanged ends and supplied with companion flanges, gaskets, and with bolts and nuts. The companion flanges shall be made of cast iron.

h. The meter shall register plus or minus 3 percent of the water actually passing through it at any rate of flow within the normal test flow limits specified in AWWA C701.

i. The water meter shall conform to NSF/ANSI 61 and NSF 372.

2.3 COMPOUND WATER METER

a. The compound water meter shall be a combination of a main line meter of the turbine type and a meter of appropriate size for measuring low rates of flow. The compound meter shall have an automatic valve mechanism for diverting low rates of flow through the bypass meter. Both metering devices shall be provided with registers contained in the same case. The operating characteristics shall fully conform to AWWA C702. Peak domestic flow rate shall be as indicated on plan in

gpm. The bypass meter flow rate shall be as indicated on plans in gpm. Each Register shall indicate in U.S. gallons.

b. The water meter shall be rated for use at temperatures ranging from -40 degrees F and 158 degrees F and operate at a working pressure of 150 psig.

c. The main case shall be made of copper alloy containing no less than 75 percent copper.

d. The register box rings and lids shall be made of a cast copper alloy.

e. The measuring chambers shall be made of a copper alloy containing not less than 84 percent copper.

f. The measuring turbines shall be made of a suitable synthetic polymer with specific gravity approximately equal to that of water or stainless steel. The measuring turbines shall have sufficient dimensional stability to retain operating clearances at working temperatures.

g. The turbine meter shall have flanged ends and supplied with companion flanges, gaskets, and with bolts and nuts. The companion flanges shall be made of cast iron.

h. The meter shall register plus or minus 3 percent of the water actually passing through it at any rate of flow within the normal test flow limits specified in AWWA C702 except in the registration of flows within the changeover period from bypass meter to main meter.

i. The water meter shall conform to NSF/ANSI 61 and NSF 372.

2.4 WATER METER STRAINER

a. All meters shall be fitted with a factory installed integral strainer or bronze inlet strainer with top access. The strainer shall conform to AWWA C702.

b. The water meter strainer shall conform to NSF/ANSI 61 and NSF 372.

2.5 WATER METER PROGRAMMING

a. All meters or DN 50 2 inches and greater shall be programmable with software supplied by the meter manufacturer.

b. The software shall have a Microsoft based interface and operate on the latest Windows operating system. The software shall allow the user to configure the meter, troubleshoot the meter, query and display meter parameters, and configure data and stored values.

c. The meter firmware shall be upgradeable through one of the communication ports without removing the unit from service.

d. The meter shall include output for analog 4-20 milliamp signals and binary output.

e. The meter shall have two dry contact relays outputs for alarm or control functions.

2.6 WATER METER COMMUNICATION PROTOCOL

The meter shall use a native BACnet Ethernet communication protocol supporting HTTP SMTP Modbus. The communications shall be protected against surges induced on its communications channels.

2.7 REMOTE READOUT REGISTER

All meters shall be equipped with a remote readout register in accordance with AWWA C707.

2.8 PRESSURE GAUGES FOR WATER AND SEWAGE USAGE

a. ASME B40.100 all metal case 4-1/2 inches diameter, bottom connected throughout, graduated as required for service, and identity labeled. Range shall be 0 to 200 psig gauge.

b. The pressure element assembly shall be bourdon tube. The mechanical movement shall be lined to pressure element and connected to pointer.

c. The dial shall be non-reflective aluminum with permanently etched scale markings graduated in kPa and psig.

d. The pointer shall be dark colored metal.

e. The window shall be glass.

f. The ring shall be brass or stainless steel.

g. The accuracy shall be grade A, plus or minus 1 percent of middle half of scale range.

h. The pressure gauge for water domestic use shall conform to NSF/ANSI 61 and NSF 372.

2.9 THERMOMETERS

Thermometers shall be straight stem, metal case, red liquid-filled thermometer, approximately 7 inches high, 40 degrees F to 212 degrees F. Thermometers shall comply with ASME B40.200.

PART 3 EXECUTION

3.1 INSTALLATION

a. Direct mounted pressure gauges shall be installed in piping tees with pressure gauge located on pipe at the most readable position.

b. Valves and snubbers shall be installed in piping for each pressure gauge.

c. Test plugs shall be installed on the inlet and outlet pipes of all heat exchangers or water heaters serving more than one plumbing fixture.

d. Pressure gauges shall be installed where indicated in the drawings and at the following locations:

(1) Building water service entrance into building.

(2) Inlet and outlet of each pressure reducing valve.

(3) Suction and discharge of each domestic water pump or re-circulating hot water return pump.

e. Water meter installation shall conform to AWWA C700, AWWA C701, and AWWA C702. Electrical installations shall conform to IEEE C2, NFPA 70, and to the requirements specified herein. New materials shall be provided.

f. Remote readout register shall be mounted at the location indicated in the drawings or as directed by the Contracting Officer.

g. Thermometers shall be installed on the water heater inlet and outlet piping, thermostatic mixing valve outlet piping, and the hot water circulation pump inlet piping.

h. If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

i. Install portable water temperature, pressure and flow meters provided under Section 23 09 23, DIRECT DIGITAL CONTROL SYSTEMS FOR HVAC.

3.2 FIELD QUALITY CONTROL

The meter assembly shall be visually inspected and operationally tested. The correct multiplier placement on the face of the meter shall be verified.

3.3 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA will observe startup and Contractor testing of selected equipment. Coordinate the startup and Contractor testing schedules with the Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

3.4 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this Section of the specification will be tested as part of a larger system.

3.5 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for four

hours to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

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SECTION 22 05 23
11/22

GENERAL-DUTY VALVES FOR PLUMBING PIPING

PART 1 GENERAL

1.1 DESCRIPTION

a. This Section describes the requirements for general-duty valves for domestic water and sewer systems.

b. A complete listing of common acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

f. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A112.14.1	(2003; R 2017; R 2022) Backwater Valves

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

ASSE 1001	(2021) Performance Requirements for Atmospheric Type Vacuum Breakers
ASSE 1003	(2020) Performance Requirements for Water Pressure Reducing Valves for Domestic Water Distribution Systems - (ANSI approved 2010)
ASSE 1011	(2017) Performance Requirements for Hose Connection Vacuum Breakers
ASSE 1013	(2021) Performance Requirements for Reduced Pressure Principle Backflow

	Prevention Assemblies
ASSE 1015	(2021) Performance Requirements for Double Check Backflow Prevention Assemblies
ASSE 1017	(2023) Performance Requirements for Temperature Actuated Mixing Valves for Hot Water Distribution Systems - (ANSI approved 2010)
ASSE 1020	(2020) Performance Requirements for Pressure Vacuum Breaker Assemblies
ASSE 1035	(2020) Performance Requirements for Laboratory Faucet Backflow Preventers
ASSE 1069	(2020) Performance Requirements for Automatic Temperature Control Mixing Valves
ASSE 1070	(2020) Performance Requirements for Water Temperature Limiting Devices
ASSE 1017	(2023) Performance Requirements for Temperature Actuated Mixing Valves for Hot Water Distribution Systems - (ANSI approved 2010)

ASTM INTERNATIONAL (ASTM)

ASTM A126	(2004; R 2023) Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings
ASTM A276/A276M	(2024) Standard Specification for Stainless Steel Bars and Shapes
ASTM A536	(2024) Standard Specification for Ductile Iron Castings
ASTM B62	(2017) Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM B584	(2022) Standard Specification for Copper Alloy Sand Castings for General Applications

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-67	(2022) Butterfly Valves
MSS SP-70	(2011) Gray Iron Gate Valves, Flanged and Threaded Ends
MSS SP-71	(2018) Gray Iron Swing Check Valves, Flanged and Threaded Ends
MSS SP-80	(2019) Bronze Gate, Globe, Angle and Check Valves

MSS SP-85	(2011) Gray Iron Globe & Angle Valves Flanged and Threaded Ends
MSS SP-110	(2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends

NSF INTERNATIONAL (NSF)

NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects
NSF 372	(2022) Drinking Water System Components - Lead Content

FOUNDATION FOR CROSS-CONNECTION CONTROL AND HYDRAULIC RESEARCH (FCCCHR)

FCCCHR Manual	(10th Edition) Manual of Cross-Connection Control

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Test And Balance Reports For Balancing Valves; G, AE

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Training Plans; G, AE

Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this Section shall be marked “SUBMITTED UNDER SECTION 22 05 23, GENERAL-DUTY VALVES FOR PLUMBING PIPING”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description

and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Ball Valves.

(2) Gate Valves.

(3) Butterfly Valves.

(4) Balancing Valves.

(5) Check Valves.

(6) Globe Valves.

(7) Water Pressure Reducing Valves and Connections.

(8) Backwater Valves.

(9) Backflow Preventers.

(10) Chainwheels.

(11) Thermostatic Mixing Valves.

d. Test and Balance reports for balancing valves.

e. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

(4) Piping diagrams of thermostatic mixing valves to be installed.

f. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

g. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 DELIVERY, STORAGE, AND HANDLING

a. Valves shall be prepared for shipping as follows:

(1) Protect internal parts against rust and corrosion.

(2) Protect threads, flange faces, grooves, and weld ends.

(3) Set angle, gate, and globe valves closed to prevent rattling.

(4) Set ball and plug valves open to minimize exposure of functional surfaces.

(5) Set butterfly valves closed or slightly open.

(6) Block check valves in either closed or open position.

b. Valves shall be prepared for storage as follows:

(1) Maintain valve end protection.

(2) Store valves indoors and maintain at higher than ambient dew point temperature.

c. A sling shall be used for large valves. The sling shall be rigged to avoid damage to exposed parts. Hand wheels or stems shall not be used as lifting or rigging points.

1.7 AS BUILT DOCUMENTATION

Comply with requirements in Paragraph “AS-BUILT DOCUMENTATION” of Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

PART 2 PRODUCTS

2.1 VALVES, GENERAL

a. Asbestos packing and gaskets are prohibited.

b. Bronze valves shall be made with dezincification resistant materials. Bronze valves made with copper alloy (brass) containing greater than 15 percent zinc shall not be permitted.

c. Valves in insulated piping shall have or DN50 2 inch stem extensions and extended handles of non-thermal conductive material that allows operating the valve without breaking the vapor seal or disturbing the insulation. Memory stops shall be fully adjustable after insulation is applied.

d. Exposed Valves over or DN65 2-1/2 inches installed at an elevation over 8 feet shall have a chain-wheel attachment to valve hand-wheel, stem, or other actuator.

e. All valves used to supply potable water shall meet the requirements of NSF/ANSI 61 and NSF 372.

f. Bio-Based Materials: For products designated by the USDA’s bio-based Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications Section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

g. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

2.2 SHUT-OFF VALVES

a. Cold, Hot and Re-circulating Hot Water:

(1) or DN50 2 inches and smaller: Ball, MSS SP-110, Ball valve shall be full port three piece or two piece with a union design with adjustable stem package. Threaded stem designs are not allowed. The ball valve shall have a SWP rating of 150 psig and a CWP rating of 600 psig. The body material shall be Bronze ASTM B584, Alloy C844. The ends shall be non-lead solder.

(2) Less than DN100 4 inches: Butterfly shall have an iron body with EPDM seal and aluminum bronze disc. The butterfly valve shall meet MSS SP-67, type I standard. The butterfly valve shall have a SWP rating of 200 psig. The valve design shall be lug type suitable for bidirectional dead-end service at rated pressure. The body material shall meet ASTM A536, ductile iron.

(3) DN100 4 inches and greater:

(a) Class 125, OS&Y, Cast Iron Gate Valve. The gate valve shall meet MSS SP-70 type I standard. The gate valve shall have a CWP rating of 200 psig. The valve materials shall meet ASTM A126, grey iron with bolted bonnet, flanged ends, bronze trim, and positive-seal resilient solid wedge disc. The gate valve shall be gear operated for sizes under or DN200 8 inches and crank operated for sizes or DN200 8 inches and greater.

(b) Single flange, ductile iron butterfly valves: The single flanged butterfly valve shall meet the MSS SP-67 standard. The butterfly valve shall have a CWP rating of 200 psig. The butterfly valve shall be lug type, suitable for bidirectional dead-end service at rated pressure without use of downstream flange. The body material shall comply with ASTM A536 ductile iron. The seat shall be EPDM with stainless steel disc and stem.

(c) Grooved end, ductile iron butterfly valves. The grooved butterfly valve shall meet the MSS SP-67 standard. The grooved butterfly valve shall have a CWP rating of 200 psig. The valve materials shall be epoxy coated ductile iron conforming to ASTM A536 with two-piece stainless-steel stem, Buna-N EPDM encapsulated ductile iron disc, and EPDM seal. The butterfly valve shall be gear operated.

b. Reagent Grade Water: Valves for reagent grade, reverse osmosis, or deionized water service shall be ball type of same material as used for pipe.

2.3 MANUAL BALANCING VALVES

a. Hot Water Re-circulating, or DN75 3 inches and smaller manual balancing valve shall be of bronze body, brass ball construction with glass and carbon filled TFE seat rings and designed for positive shutoff. The manual balancing valve shall have differential pressure read-out ports across the valve seat area. The read out ports shall be fitted with internal EPT inserts and check valves. The valve body shall have or DN8 NPT 1/4 inch NPT) tapped drain and purge port. The valves shall have memory stops that allow the valve to close for service and then reopened to set point without disturbing the balance

position. All valves shall have calibrated nameplates to assure specific valve settings.

b. Greater than or DN75 3 inches: Manual balancing valves shall be of heavy duty cast iron flanged construction with 125 psig flange connections. The flanged manual balancing valves shall have either a brass ball with glass and carbon filled TFE seal rings or fitted with a bronze seat, replaceable bronze disc with EPDM seal insert and stainless steel stem. The design pressure shall be 175 psig at 250 degrees F.

2.4 THERMOSTATIC BALANCING VALVES

Thermostatic recirculation balancing valves for domestic hot water and domestic cold water application. Thermostatically controlled, spring actuated automatic balancing valve to very recirculation flow to maintain constant return temperatures. Stainless steel body, spring and brass or stainless steel thermal actuator and actuator carrier. Provide with threaded inlet, integral outlet union and stainless steel check valve. Direct acting to maintain return hot water temperature as indicated on plans.PTFE seat seal ring and EPDM body seal.

2.5 CHECK VALVES

a. or DN75 3 inches and smaller shall be Class 125, bronze swing check valves with non-metallic disc suitable for type of service. The check valve shall meet MSS SP-80 Type 4 standard. The check valve shall have a CWP rating of 200 psig. The check valve shall have a Y pattern horizontal body design with bronze body material conforming to ASTM B62, solder joints, and PTFE or TFE disc.

b. or DN100 4 inches and greater:

(1) Check valves shall be Class 125, iron swing check valve with lever and weight closure control. The check valve shall meet MSS SP-71 Type I standard. The check valve shall have a CWP rating of 200 psig. The check valve shall have a clear or full waterway body design with gray iron body material conforming to ASTM A126, bolted bonnet, flanged ends, bronze trim.

(2) All check valves on the discharge side of submersible sump pumps shall have factory installed exterior level and weight with sufficient weight to prevent the check valve from hammering against the seat when the sump pump stops.

2.6 GLOBE VALVES

a. or DN75 3 inches or smaller: Class 150, bronze globe valve with non-metallic disc. The globe valve shall meet MSS SP-80, Type 2 standard. The globe valve shall have a CWP rating of 300 psig. The valve material shall be bronze with integral seal and union ring bonnet conforming to ASTM B62 with solder ends, copper-silicon bronze stem, PTFE or TFE disc, and malleable iron hand wheel.

b. Greater than or DN75 3 inches: Similar to above, except with cast iron body and bronze trim, Class 125, iron globe valve. The globe valve shall meet MSS SP-85, Type 1 standard. The globe valve shall have a CWP rating of 200 psig. The valve material shall be gray iron with bolted bonnet conforming to ASTM A126 with flanged ends, bronze

trim, and malleable iron handwheel.

2.7 WATER PRESSURE REDUCING VALVE AND CONNECTIONS

a. or DN75 3 inches or smaller: The pressure reducing valve shall consist of a bronze body and bell housing, a separate access cover for the plunger, and a bolt to adjust the downstream pressure. The pressure reducing valve shall meet ASSE 1003. The bronze bell housing and access cap shall be threaded to the body and shall not require the use of ferrous screws. The assembly shall be of the balanced piston design and shall reduce pressure in both flow and no flow conditions. The assembly shall be accessible for maintenance without having to remove the body from the line.

b. or DN100 4 inches and greater: The pressure reducing valve shall consist of a flanged cast iron body and rated to 200 psig. The valve shall have a large elastomer diaphragm for sensitive response. The pressure reducing valve shall meet ASSE 1003.

c. The regulator shall have a tap for pressure gauge.

d. The regulator shall have a temperature rating of 212 degrees F for hot water or hot water return service. Pressure regulators shall have accurate pressure regulation to plus or minus 1 psig).

e. Setting: Entering water pressure, discharge pressure, capacity, size, and related measurements shall be as shown on the drawings.

f. Connections Valves and Strainers: Shut off valves shall be installed on each side of reducing valve and a bypass line equal in size to the regulator inlet pipe shall be installed with a normally closed globe valve. A strainer shall be installed on inlet side of, and same size as pressure reducing valve. A pressure gauge shall be installed on the inlet and outlet of the valve.

2.8 BACKWATER VALVE

a. The backwater valve shall have a cast iron body, automatic thermoplastic type valve seat and flapper suited for water service. The flapper shall be slightly open during periods of non-operation. The pressure reducing valve shall meet ASME A112.14.1. The cleanout shall be extended to the finish floor and fit with a threaded countersunk plug. A clamping device shall be included when the cleanout extends through the waterproofing membrane.

b. When the backwater valve is installed greater than 24 inches below the finish floor elevation, a pit or manhole large enough for a repair person can enter to service the backwater valve shall be installed.

2.9 BACKFLOW PREVENTERS (BFP)

a. A backflow prevention assembly shall be installed at any point in the plumbing system where the potable water supply comes in contact with a potential source of contamination. The backflow prevention assembly shall be approved by the University of Southern California Foundation for Cross Connection Control and Hydraulic Research (FCCCHR Manual).

b. The reduced pressure principle backflow prevention assembly shall be ASSE 1013 with full port OS&Y positive-seal resilient gate valves and

an integral relief monitor switch. The main body and access cover shall be epoxy coated ductile iron conforming to ASTM A536 grade 4. The seat ring and check valve shall be the thermoplastic type suited for water service. The stem shall be stainless steel conforming to ASTM A276/A276M. The seat disc shall be the elastomer type suited for water service. The checks and the relief valve shall be accessible for maintenance without removing the device from the line. An epoxy coated wye type strainer with flanged connections shall be installed on the inlet. Reduced pressure backflow preventers shall be installed in the following applications.

(1) Deionizers.

(2) Sterilizers.

(3) Dialysis, Deionized or Reverse Osmosis Water Systems.

(4) Water make up to heating systems, cooling tower, chilled water system, generators, and similar equipment consuming water.

(5) Water service entrance from loop system.

(6) Dental equipment.

(7) Power washer.

(8) Medical equipment.

(9) Process equipment.

(10) Autopsy, on each hot and cold water outlet at each table or sink.

(11) Reclaimed water systems.

c. The pipe applied or integral atmospheric vacuum breaker shall be ASSE 1001. The main body shall be cast bronze. The seat disc shall be the elastomer type suited for water service. The device shall be accessible for maintenance without removing the device from the service line. The installation shall not be in a concealed or inaccessible location or where the venting of water from the device during normal operation is deemed objectionable. Atmospheric vacuum breakers shall be installed in the following applications.

(1) Hose bibs and sinks with threaded outlets.

(2) Disposers.

(3) Showers (telephone/handheld type).

(4) Hydrotherapy units.

(5) All kitchen equipment, if not protected by air gap.

(6) Ventilating hoods with wash down system.

(7) Film processor.

(8) Detergent system.

(9) Fume hoods.

(10) Glassware washers.

(11) Service sinks (integral with faucet only).

(12) Laundry tubs (integral with faucet only).

(13) Sitz baths.

d. The hose connection vacuum breaker shall be ASSE 1011. The main body shall be cast brass with stainless steel working parts. The diaphragm and disc shall be the elastomer type suited for water service. The device shall permit the attachment of portable hoses to hose thread outlets. Hose connection vacuum breakers shall be installed in the following locations requiring non-continuous pressure:

Hose bibbs and wall hydrants.

e. The pressure vacuum breaker shall be ASSE 1020. The main body shall be brass. The disc and O-ring seal shall be the elastomer type. The valve seat and disc float shall be the thermoplastic type. Tee handle or lever handle shut-off ball valves. Test cocks for testing and draining where freezing conditions occur. All materials shall be suitable for water service. The device shall be accessible for maintenance without removing the device from the service line. The installation shall not be in a concealed or inaccessible location or where the venting of water from the device during normal operation is deemed objectionable. Pressure vacuum breakers shall be installed in the following locations requiring continuous pressure and no backpressure including equipment with submerged inlet connections:

Lawn Irrigation.

f. The laboratory faucet vacuum breaker shall be ASSE 1035. The main body shall be cast brass. Dual check valves with stainless steel working parts. The diaphragm and disc shall be the elastomer type suited for water service. The device shall permit the attachment of portable hoses to laboratory faucets for non-continuous pressure applications.

g. The double check backflow prevention assembly shall be ASSE 1015 and supply with full port, OS&Y, positive-seal, resilient gate valves. The main body and access cover shall be epoxy coated ductile iron conforming to ASTM A536 grade. The seat ring and check valve shall be the thermoplastic type suited for water service. The stem shall be stainless steel conforming to ASTM A276/A276M. The seat disc shall be the elastomer type suited for water service. The first and second check valve shall be accessible for maintenance without removing the device from the line. Double check valves shall be installed in the following location requiring continuous pressure subject to backpressure and backsiphonage conditions.

(1) Lawn Irrigation.

(2) Food Processing Equipment.

(3) Laundry equipment.

2.10 CHAINWHEELS

Valve chain wheel assembly with sprocket rim brackets and chain shall be constructed according to the following:

a. Brackets: Type, number, size, and fasteners required to mount actuator on valve.

b. Attachment: For connection to ball butterfly valve stem.

c. Sprocket rim with chain guides: Ductile or cast iron Aluminum Bronze of type and size required for valve with zinc coating.

d. Chain: Hot dipped galvanized steel Stainless steel of size required to fit sprocket rim.

2.11 THERMOSTATIC MIXING VALVES

a. Thermostatic Mixing Valves shall comply with the following general performance requirements:

(1) shall meet ASSE requirements for water temperature control.

(2) The body shall be cast bronze or brass with corrosion resistant internal parts preventing scale and biofilm build-up. Provide chrome-plated finish in exposed areas.

(3) No special tool shall be required for temperature adjustment, maintenance, replacing parts and disinfecting operations.

(4) Valve shall be able to be placed in various positions without making temperature adjustment or reading difficult.

(5) Valve finish shall be chrome plated in exposed areas.

(6) Valve shall allow easy temperature adjustments to allow hot water circulation. Internal parts shall be able to withstand disinfecting operations of chemical and thermal treatment of water temperatures up to 180 degrees F for 30 minutes or 50 ppm chlorine residual concentration for 24 hours.

(7) Parts shall be easily removed or replaced without dismantling the valves, for easy scale removal and disinfecting of parts.

(8) Valve shall have a manual adjustable temperature control with locking mechanism to prevent tampering by end user. Outlet temperature shall be visible to ensure outlet temperature does not exceed specified limits, particularly after thermal eradication procedures.

(9) Provide mixing valves with integral check valves with screens and stop valves.

b. Master Thermostatic Water Mixing Valves:

(1) Application: Tempered water distribution from hot water source.

(2) Standard: ASSE 1017.

(3) Pressure Rating: 125 psig.

(4) Type: Exposed-mounting or Cabinet-type, as indicated, thermostatically controlled water mixing valve.

(5) Connections: Flanged or threaded union inlets and outlet.

(6) Valve Finish: Chrome plated.

(7) Cabinet: Factory-fabricated, stainless steel, for recessed or surface mounting and with hinged, stainless-steel door.

(8) Thermometers shall be provided to indicate mixed water temperature.

(9) Provide a high temperature alarm device to detect mixing valve failure.

c. Hi-Lo Water-Mixing-Valve Assemblies:

(1) Application: Tempered water distribution from hot water source covering a wide range of flow.

(2) Description: Factory-fabricated, cabinet-type or exposed-mounting, thermostatically controlled, water-mixing-valve assembly in two-valve parallel arrangement including pressure regulators, pressure gauges and thermometer.

(3) Large-Flow Parallel: Master thermostatic water mixing valve and downstream pressure regulator with pressure gauges on inlet and outlet.

(4) Small-Flow Parallel: Master thermostatic water mixing valve.

(5) Master Thermostatic Mixing Valves: Comply with ASSE 1017.

(6) Water Regulator(s): Comply with ASSE 1003. Include pressure gauge on inlet and outlet.

(7) Component Pressure Ratings: 125 psig minimum, unless otherwise indicated.

(8) Cabinet: Factory-fabricated, stainless steel, for recessed or surface mounting and with hinged, stainless-steel door.

(9) Connections: Soldered or threaded union inlets and outlet.

(10) Thermometers shall be provided to indicate mixed water temperature.

(11) Provide a high temperature alarm device to detect mixing valve failure.

d. Automatic Water Temperature Control Mixing Valves:

(1) Application: Gang plumbing fixtures point-of-use when no other mixing at fixtures occurs.

(2) Standard: ASSE 1069.

(3) Pressure Rating: 125 psig.

(4) Type: Thermostatically controlled water mixing valve set at 110 degrees F.

(5) Connections: Threaded union or soldered inlets and outlet.

(6) Thermometers shall be provided to indicate mixed water temperature.

(7) Upon cold water supply failure the hot water flow shall automatically be reduced to 0.5 gpm maximum.

(8) Provide a high temperature alarm device to detect mixing valve failure.

e. Water Temperature Limiting Devices:

(1) Application: Single plumbing fixture point-of-use such as sinks or lavatories.

(2) Standard: ASSE 1070.

(3) Pressure Rating: 125 psig.

(4) Type: Thermostatically controlled water mixing valve set at 110 degrees F.

(5) Connections: Threaded union, compression or soldered inlets and outlet.

(6) Upon cold water supply failure the hot water flow shall automatically be reduced to 0.2 gpm maximum.

f. Temperature Activated Mixing Valves:

(1) Application: Emergency eye/face/drench shower equipment.

(2) Standard: ASSE 1017.

(3) Pressure Rating: 125 psig.

(4) Type: Thermostatically controlled water mixing valve set at 75-85 degrees F.

(5) Connections: Soldered or threaded union inlets and outlet.

(6) Cabinet: Factory-fabricated, stainless steel, for recessed or surface mounting and with hinged, stainless-steel door.

(7) Thermometers shall be provided to indicate mixed water temperature.

(8) Upon cold water supply failure the hot water flow shall automatically be reduced to 0.5 gpm maximum.

PART 3 EXECUTION

3.1 EXAMINATION

a. Valve interior shall be examined for cleanliness, freedom from foreign

matter, and corrosion. Special packing materials shall be removed, such as blocks, used to prevent disc movement during shipping and handling.

b. Valves shall be operated in positions from fully open to fully closed. Guides and seats shall be examined and made accessible by such operations.

c. Threads on valve and mating pipe shall be examined for form and cleanliness.

d. Mating flange faces shall be examined for conditions that might cause leakage. Bolting shall be checked for proper size, length, and material. Gaskets shall be verified for proper size and that its material composition is suitable for service and free from defects and damage.

e. Do not attempt to repair defective valves; replace with new valves.

3.2 INSTALLATION

a. Install valves with unions or flanges at each piece of equipment arranged to allow service, maintenance, and equipment removal without system shutdown.

b. Valves shall be located for easy access and shall be provide with separate support. Valves shall be accessible with access doors when installed inside partitions or above hard ceilings.

c. Valves shall be installed in horizontal piping with stem at or above center of pipe.

d. Valves shall be installed in a position to allow full stem movement.

e. Install chain wheels on operators for ball butterfly gate and globe valves NPS or DN65 2 1/2 inches and greater and installed greater than 8 feet above floor. Chains shall be extended to 60 inches above finished floor.

f. Check valves shall be installed for proper direction of flow and as follows:

Swing Check Valves: In horizontal position with hinge pin level and on top of valve.

g. Install backflow preventers in each water supply to mechanical equipment and systems and to other equipment and water systems that shall be sources of contamination. Comply with authorities having jurisdiction. Locate backflow preventers in same room as connected equipment or system.

Install drain for backflow preventers with atmospheric-vent drain connection with air-gap fitting, fixed air-gap fitting, or equivalent positive pipe separation of at least two pipe diameters in drain piping and pipe to floor drain. Locate air-gap device attached to or under backflow preventer. Simple air breaks are prohibited for this application.

h. Install pressure gauges on outlet of backflow preventers.

i. Do not install bypass piping around backflow preventers.

j. Install temperature-actuated water mixing valves with check stops or shutoff valves on inlets.

(1) Install thermometers if specified.

(2) Install cabinet-type units recessed in or surface mounted on wall as specified.

k. If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

l. Install thermostatic balancing valves with inlet strainer and inlet and outlet isolation valves.

3.3 LABELING AND IDENTIFYING

a. Equipment Nameplates and Signs: Install engraved plastic-laminate equipment nameplate or sign on or near each of the following:

(1) Calibrated balancing valves.

(2) Master, thermostatic, water mixing valves.

(3) Manifold, thermostatic, water-mixing-valve assemblies.

b. Distinguish among multiple units, inform operator of operational requirements, indicate safety and emergency precautions, and warn of hazards and improper operations, in addition to identifying unit.

3.4 ADJUSTING

a. Valve packing shall be adjusted or replaced after piping systems have been tested and put into service but before final adjusting and balancing. Valves shall be replaced if persistent leaking occurs.

b. Set field-adjustable flow set points of balancing valves and record data. Ensure recorded data represents actual measured or observed conditions. Permanently mark settings of valves and other adjustment devices allowing settings to be restored. Set and lock memory stops. After adjustment, take measurements to verify balance has not been disrupted or that such disruption has been rectified.

c. Set field-adjustable temperature set points of temperature-actuated water mixing valves.

d. Testing and adjusting of balancing valves shall be performed by an independent NEBB Accredited Test and Balance Contractor. A final settings and flow report shall be submitted to the VA Contracting Officer.

3.5 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the

various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA will observe startup and Contractor testing of selected equipment. Coordinate the startup and Contractor testing schedules with the Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

3.6 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this Section of the specification will be tested as part of a larger system.

3.7 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hours to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

-- End of Section --

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DIVISION 22 - PLUMBING

SECTION 22 05 33

HEAT TRACING FOR PLUMBING PIPING

9/20

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SECTION 22 05 33
9/20

HEAT TRACING FOR PLUMBING PIPING

PART 1 GENERAL

1.1 DESCRIPTION

a. This Section describes the requirement for supplying, installing, and testing of the electric heat tracing system of the plumbing piping. Freeze protection shall be utilized for domestic water piping in areas subject to freezing temperatures. Ice and snow melting shall be utilized for gutters, downspouts, roof drain bodies and roof drain leaders exposed to snow and ice accumulation.

b. A complete listing of common acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, GENERAL COMMISSIONING REQUIREMENTS.

e. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

f. Section 22 07 11, PLUMBING INSULATION.

g. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

h. Section 26 05 19, LOW VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 515	(2017) Standard for the Testing, Design, Installation, and Maintenance of Electrical Resistance Heat Tracing for Industrial Applications

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Manufacturer Qualifications; G, AE

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Field Quality Control Test Reports; G, AE

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Training Plans; G, AE

Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this Section shall be marked “SUBMITTED UNDER Section 22 05 33, HEAT TRACING FOR PLUMBING PIPING, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Rated capacity.

(2) Length of cable.

(3) Cable spacing.

(4) Electrical power requirements.

(5) Controls.

(6) Enclosures.

(7) Accessories.

d. The shop drawings shall include plans, Sections, details, wiring diagrams, and attachments to other work. The wiring diagrams shall include power, signal, and control wiring.

e. Field quality control test reports shall be submitted.

f. Complete operating and maintenance manuals including wiring diagram, technical data sheets, information for ordering replaceable parts, and troubleshoot guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

g. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

h. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

a. Manufacturer Qualifications: Ten years’ experience in design, engineering, manufacture and support of specified system and components.

b. Product Requirements:

(1) Pipe or tank tracing cable assembly be factory assembled, immersed in water for a minimum of 12 hours, and then tested for insulation resistance, high potential breakdown and continuity before leaving the factory.

(2) Factory Mutual approved heating cable that has the same wattage per lineal foot (power output), throughout its entire length.

(3) UL Listed, thermostat and contactor panel.

(4) UL Listed Control/Monitor Panel with gateway with protocol necessary, such as BACnet, to allow full integration with campus building automation system (BAS). Full integration will allow full control of the heat trace sysatm including setpoints, enabling, and alarms. Coordinate heat trace system with Mechanical Contractor, Controls Contractor, and specification section 23 21

13.

c. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

d. Bio-Based Materials: For products designated by the USDA's bio-based Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications Section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

e. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in Paragraph AS-BUILT DOCUMENTATION of Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

PART 2 PRODUCTS

2.1 ELECTRICAL HEAT TRACING SYSTEMS

a. The plastic insulated series resistance heating cables shall comply with IEEE 515 and NFPA 70.

b. Provide tracing for outdoor piping subject to freezing temperatures below 38 degrees F as follows:

(1) Water supply, transfer, and equilization piping.

(2) Overflow and drain.

c. Heat tracing shall be provided for all exposed piping.

d. Heating Cable: Flexible, parallel circuit construction consisting of a continuous self-limiting resistance, conductive inner core material between two parallel copper bus wires, designed for cut-to-length at the job site and for wrapping around valves and complex fittings. Unless not recommended or allowed by the manufacturer heat trace shall heat the full circumference of large piping and be wrapped around the piping or have multiple parallel cables on opposite sides of the pipe. Self regulation shall prevent overheating and burnouts even where the cable overlaps itself.

(1) Provide end seals at ends of circuits. Wire at the ends of the circuits is not to be tied together.

(2) Provide sufficient cable, as recommended by the manufacturer, to keep the pipe surface at 36 degrees F minimum during winter outdoor design temperature, but not less than the following:

(a) 3 inch pipe and smaller with 1 inch thick insulation: 4 watts per foot of pipe.

(b) 4 inch pipe and larger 1-1/2 inch thick insulation: 8 watts per feet of pipe.

2.2 CONTROLS

a. Control Panel shall be provided to connect and control heat trace system and connect to the building automation system for control, reporting, alarm, and recording of heat trace usage. System shall be provided with gateway necessary to allow BAS to have full access to all items the heat trace control panel can access. Coordinate control panel requirements with Controls, Electrical, and Plumbing Contractor. System shall allow each tank and tower piping system to be tracked and controlled separately from the other.

b. The pipe mounting thermostats, as needed, for Freeze protection shall have be a remote bulb unit with adjustable temperature range from minus 34 to 50 degrees F. The thermostat shall be snap action, open-on-rise, single pole switch with minimum current rating adequate for the connected cable. The thermostat shall be remote bulb on capillary, resistance temperature device, or thermistor for direct sensing of pipe wall temperature. The control enclosure shall be corrosion resistant and waterproof.

c. The enclosure shall be the NEMA 4X type.

d. A minimum 30 amp contactor shall be provided to energize cable or close other contactors. Provide relay with contacts to indicate operational status, on/off, and for interface with central energy management and control system.

2.3 ACCESSORIES

a. Cable Installation Accessories: Fiberglass tape, heat-conductive putty, cable ties, silicone end seals and splice kits, and installation clips all furnished by manufacturer, or as recommended in writing by manufacturer. Items shall include, but are not limited to, power supply connection fitting and stainless-steel mounting brackets. Provide stainless-steel worm gear clamp to fasten bracket to pipe. 1/2 inch wide fiberglass reinforced pressure sensitive cloth tape to fasten cable to pipe at 12 inch intervals. Pipe surface temperature control thermostat: Cast aluminum, NEMA 4 (watertight) enclosure, 1/2 inch NPT conduit hub, SPST switch rated 20 amps at 480 volts ac, with capillary and copper bulb sensor. Set thermostat to maintain pipe surface temperature at not less than (34 degrees F).

b. Signs: Manufacturer's standard (NFPA 70), stamped "ELECTRIC TRACED" located on the insulation jacket at 10 feet intervals along the pipe on alternating sides. Warning Labels: shall comply with NFPA 70.

c. Warning Tape: Continuously printed "Electrical Tracing"; vinyl, at least 3 mils thick, and with pressure-sensitive, permanent, waterproof, self-adhesive back.

(1) Width for Warning Tape on Pipes with Outside Dimension, Including Insulation, Less Than 6 inches: 3/4 inch minimum.

(2) Width for Warning Tape on Pipes with Outside Dimension, Including Insulation, 6 inches or Greater: 1-1/2 inches minimum.

PART 3 EXECUTION

3.1 GENERAL

a. Inspect surfaces and substrates of electric heating cables for compliance with requirements of this specification. Ensure surfaces and pipes in contact with electric heating cables are free of burrs and sharp protrusions.

b. Notify Contracting Officer if the existing substrate conditions are unsuitable for application of heating cables in accordance with manufacturer’s recommendations.

c. If the installation of the heat tracing is unsatisfactory, then the Contractor shall correct the installation at no additional cost or time to the Government.

3.2 INSTALLATION

a. Install tracing as recommended by the manufacturer. Unless not recommended or allowed by the manufacturer heat trace shall heat the full circumference of large piping and be wrapped around the piping or have multiple parallel cables on opposite sides of the pipe.Unless not recommended or allowed by the manufacturer heat trace shall be wrapped around the piping or have multiple parallel cables on opposite sides of the pipe. Electric heating cable be installed on all exposed piping subject to freezing temperatures below 38 degrees F including, but not limited to:

(1) Water supply, transfer, and equilization piping.

(2) Overflow and drain.

b. Electric heating cable shall be installed across expansion, construction, and control joints according to the manufacturer’s recommendations using cable protection conduit and slack cable to allow for movement without damage to cable.

c. The installation of electric heating cable for snow and ice melting on roofs, gutters and downspouts, and roof drain leaders shall be provided with clips furnished by the manufacturer that are compatible with roof, gutters and downspouts and roof drain leaders.

d. Electric heating cable for pipe freeze protection shall be installed according to the following:

(1) Electric heating cables shall be installed after piping has been tested and before insulation is installed.

(2) Electric heat cables shall be installed according to IEEE 515.

(3) Insulation shall be installed or applied over piping with electric cables. Refer to Section 22 07 11, PLUMBING INSULATION.

(4) Warning tape shall be installed on pipe insulation where piping is equipped with electric heating cables.

(5) Provide current transducer to monitor power in heating tape and monitor and alarm from BAS.

e. Field adjustable switches and circuit breaker trip ranges shall be set.

f. Heating cables including leads shall be protected from damage.

g. Equipment shall be grounded according to Section 26 05 19, LOW VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES.

h. Wiring shall be connected according to Section 26 05 19, LOW VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES.

3.3 TESTS

a. Manufacturer's Field Service: Engage a factory-authorized service representative to test and inspect components, assemblies, and equipment installations, including connections.

b. Perform the following tests and inspections with the assistance of a factory-authorized service representative:

(1) Perform tests after cable installation but before application of coverings such as insulation, wall or ceiling construction, or concrete.

(2) Test cables for electrical continuity and insulation integrity before energizing.

(3) Test cables to verify rating and power input. Energize and measure voltage and current simultaneously. Performance verification testing shall be completed prior to system acceptance. All installations shall be tested via thermal imaging during low ambient conditions or equivalent testing approved by the Contracting Officer..

c. Repeat tests for continuity, insulation resistance, and input power after applying thermal insulation on pipe-mounted cables.

d. If deficiency is found, Contractor shall correct all deficiencies at no addition cost or time to the Government.

e. Prepare test and inspection reports.

3.4 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this Section of the specification will be tested as part of a larger system.

3.5 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hours to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS

COMMISSIONING OF PLUMBING SYSTEMS.

-- End of Section --

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SECTION 22 07 11
9/15

PLUMBING INSULATION

PART 1 GENERAL

1.1 DESCRIPTION

a. Field applied insulation for thermal efficiency and condensation control for the following:

(1) Plumbing piping and equipment.

(2) Re-insulation of plumbing piping and equipment after asbestos abatement and or replacement of any part of existing insulation system (insulation, vapor retarder jacket, protective coverings/jacket) damaged during construction.

b. Definitions:

(1) ASJ: All Service Jacket, Kraft paper, white finish facing or jacket.

(2) Air conditioned space: Space having air temperature and/or humidity controlled by mechanical equipment.

(3) All insulation systems installed within supply, return, exhaust, relief and ventilation air plenums shall be limited to uninhabited crawl spaces, areas above a ceiling or below the floor, attic spaces, interiors of air conditioned or heating ducts, and mechanical equipment rooms shall be noncombustible or shall be listed and labeled as having a flame spread indexes of not more than 25 and a smoke-developed index of not more than 50 when tested in accordance with ASTM E84 or UL 723. Note: ICC IMC, Section 602.2.1.

(4) Cold: Equipment or piping handling media at design temperature of 60 degrees F or below.

(5) Concealed: Piping above ceilings and in chases,and pipe spaces.

(6) Exposed: Piping and equipment exposed to view in finished areas including mechanical equipment rooms or exposed to outdoor weather. Shafts, chases, unfinished attics, crawl spaces and pipe basements are not considered finished areas.

(7) FSK: Foil-scrim-Kraft facing.

(8) Hot: Plumbing equipment or piping handling media above 104 degrees F.

(9) Density: Pcf - pounds per cubic foot.

(10) Thermal conductance: Heat flow rate through materials.

(a) Flat surface: BTU per hour per square foot.

(b) Pipe or Cylinder: BTU per hour per linear foot for a given outside diameter.

(11) Thermal Conductivity (k): BTU - inch thickness, per hour, per square foot, per degree F temperature difference.

(12) Vapor Retarder (Vapor Barrier): A material which retards the transmission (migration) of water vapor. Performance of the vapor retarder is rated in terms of permeance (perms). For the purpose of this specification, vapor retarders/vapor barriers shall have a maximum published permeance of.02 perms.

(13) HWR: Hot water recirculating.

(14) CW: Cold water.

(15) HW: Hot water.

(17) PVDC: Polyvinylidene chloride vapor retarder jacketing, white.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS: Insulation material and insulation production method.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. .

f. .

g. Section 07 84 00, FIRESTOPPING: Mineral fiber and bond breaker behind sealant.

h. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING: General mechanical requirements and items, which are common to more than one section of Division 22.

i. Section 22 05 19, METERS AND GAGES FOR PLUMBING PIPING: Hot and cold water piping.

j. Section 22 05 23, GENERAL-DUTY VALVES FOR PLUMBING PIPING: Hot and cold water piping.

k. Section 22 05 33, HEAT TRACING FOR PLUMBING PIPING: Insulation over heating cables.

l. Section 22 08 00.01,DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS

m. Section 23 21 13, HYDRONIC PIPING: electrical heat tracing systems.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM C411	(2019) Standard Test Method for Hot-Surface Performance of High-Temperature Thermal Insulation
ASTM C449	(2007; R 2024) Standard Specification for Mineral Fiber Hydraulic-Setting Thermal Insulating and Finishing Cement
ASTM C450	(2008) Standard Practice for Fabrication of Thermal Insulating Fitting Covers for NPS Piping, and Vessel Lagging
ASTM C533	(2017; R 2023) Standard Specification for Calcium Silicate Block and Pipe Thermal Insulation
ASTM C534/C534M	(2024) Standard Specification for Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form
ASTM C547	(2022a) Standard Specification for Mineral Fiber Pipe Insulation
ASTM C552	(2022) Standard Specification for Cellular Glass Thermal Insulation
ASTM C591	(2022) Standard Specification for Unfaced Preformed Rigid Cellular Polyisocyanurate Thermal Insulation
ASTM C612	(2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation
ASTM C1126	(2024) Standard Specification for Faced or Unfaced Rigid Cellular Phenolic Thermal Insulation
ASTM C1136	(2023) Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation
ASTM C1710	(2011) Standard Guide for Installation of

	Flexible Closed Cell Preformed Insulation in Tube and Sheet Form
ASTM D1668/D1668M	(1997a; R 2014; E 2014) Glass Fabrics (Woven and Treated) for Roofing and Waterproofing
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials
ASTM E2231	(2021) Standard Practice for Specimen Preparation and Mounting of Pipe and Duct Insulation Materials to Assess Surface Burning Characteristics

U.S. DEPARTMENT OF DEFENSE (DOD)

FS L-P-535E	(2012; A 2022) Plastic Sheet (Sheeting): Plastic Strip; Poly (Vinyl Chloride) and Poly (Vinyl Chloride-Vinyl Acetate), Rigid
MIL-A-3316	(1987; Rev C; Am 2 1990) Adhesives, Fire-Resistant, Thermal Insulation
MIL-A-24179	(1969; Rev A; Am 2 1980; Notice 1 1987; Notice 2 2020) Adhesive, Flexible Unicellular-Plastic Thermal Insulation
MIL-PRF-19565	(1988; Rev C) Coating Compounds, Thermal Insulation, Fire- and Water-Resistant, Vapor-Barrier
MIL-C-20079	(1987) Cloth, Glass; Tape, Textile Glass; and Thread, Glass and Wire-Reinforced Glass

INTERNATIONAL CODE COUNCIL (ICC)

ICC IMC	(2024) International Mechanical Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A	(2024) Standard for the Installation of Air Conditioning and Ventilating Systems

UL SOLUTIONS (UL)

UL 723	(2023) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials
UL 1887	(2004; R 2021) Safety Fire Test of Plastic Sprinkler Pipe for Visible Flame and Smoke Characteristics

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control

approval. 01 81 13.02 SUSTAINABIE CERTIFICATION.Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Samples; G,AE

SD-02 Shop Drawings

Shop Drawings; G,AE

SD-03 Product Data

Manufacturer's Literature And Data; G,AE

SD-06 Test Reports

Completed System Readiness Checklist; G,AE

SD-07 Certificates

Certification Documentation; G,AE

SD-11 Closeout Submittals

As-Built Documentation; G,AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this Section shall be marked “SUBMITTED UNDER Section 22 07 11, PLUMBING INSULATION”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

d. Shop Drawings: All information, clearly presented, shall be included to determine compliance with drawings and specifications and ASTM Designation, Federal and Military specifications.

(1) Insulation materials: Specify each type used and state surface burning characteristics.

(2) Insulation facings and jackets: Each type used and state surface burning characteristics.

(3) Insulation accessory materials: Each type used.

(4) Manufacturer's installation and fitting fabrication instructions for flexible unicellular insulation shall follow the guidelines in accordance with ASTM C1710.

(5) Make reference to applicable specification paragraph numbers for coordination.

(6) All insulation fittings (exception flexible unicellular insulation) shall be fabricated in accordance with ASTM C450 and the referenced Adjunct to ASTM C450.

e. Samples:

(1) Each type of insulation: Minimum size 4 inches square for board/block/ blanket; 6 inches long, full diameter for round types.

(2) Each type of facing and jacket: Minimum size 4 inches square).

(3) Each accessory material: Minimum 4 ounce liquid container or 4 ounce dry weight for adhesives / cement / mastic.

f. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACEPROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

a. Refer to article QUALITY ASSURANCE, in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

b. Criteria:

(1) Comply with NFPA 90A, particularly paragraphs 4.3.3.1 through 4.3.3.6, 4.3.11.2.6, parts of which are quoted as follows:

(a) 4.3.3.1 Pipe and duct insulation and coverings, duct linings, vapor retarder facings, adhesives, fasteners, tapes, and supplementary materials added to air ducts, plenums, panels and duct silencers used in duct systems shall have, in the form in which they are used, a maximum flame spread index of 25 without evidence of continued progressive combustion and a maximum smoke developed index of 50 when tested in accordance with ASTM E84 and appropriate mounting practice, e.g. ASTM E2231.

(b) 4.3.3.3 Coverings and linings for air ducts, pipes, plenums andpanels including all pipe and duct insulation materials shall not flame, glow, smolder, or smoke when tested in accordance with a similar test for pipe covering, ASTM C411, Standard Test Method for Hot-Surface Performance of High-Temperature Thermal Insulation, at the temperature to which they are exposed in service. In no case shall the test temperature be below 250 degrees F.

(c) 4.3.11.2.6.3 Nonferrous fire sprinkler piping shall be listed as having a maximum peak optical density of 0.5 or less, an average optical density of 0.15 or less, and a maximum flame spread distance of 5 ft or less when tested in accordance with UL 1887.

(d) 4.3.11.2.6.8 Smoke detectors shall not be required to meet the provisions of Section 4.3.

(2) Test methods: ASTM E84, UL 723, and ASTM E2231.

(3) Specified k factors are at 75 degrees F mean temperature unless stated otherwise. Where optional thermal insulation material is used, select thickness to provide thermal conductance no greater than that for the specified material. For pipe, use insulation manufacturer's published heat flow tables. For domestic hot water supply and return, run out insulation and condensation control insulation, no thickness adjustment need be made.

(4) All materials shall be compatible and suitable for service temperature, and shall not contribute to corrosion or otherwise attack surface to which applied in either the wet or dry state.

c. Every package or standard container of insulation or accessories delivered to the job site for use shall have a manufacturer's stamp or label giving the name of the manufacturer, description of the material, and the production date or code.

d. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications Section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

1.7 AS-BUILT DOCUMENTATION

a. Submit manufacturer's literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, substitutions and construction revisions shall be in electronic version on compact disc or DVD inserted into a three ring binder. All aspects of system operation and maintenance procedures, including piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices such as damper and door closure interlocks shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing Contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. As-built drawings are to be provided, and a copy of them in REVIT version 2025 provided on compact disk or DVD. Should the installing Contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the 'third party testing company' requirement.

d. Certification documentation shall be provided prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and certification that all results of tests were within limits specified.

1.8 STORAGE AND HANDLING OF MATERIAL

Store materials in clean and dry environment, pipe insulation jackets shall be clean and unmarred. Place adhesives in original containers. Maintain ambient temperatures and conditions as required by printed instructions of manufacturers of adhesives, mastics and finishing cements.

PART 2 PRODUCTS

2.1 MINERAL FIBER OR FIBER GLASS

a. ASTM C547 (Pipe Fitting Insulation and Preformed Pipe Insulation), Class 1, k = 0.26 at 75 degrees F, for use at temperatures up to 446 degrees F with an all service vapor retarder jacket (ASJ) and with polyvinyl chloride (PVC) premolded fitting covering.

2.2 MINERAL WOOL OR REFRACTORY FIBER

Comply with Standard ASTM C612, Class 3, 842 degrees F.

2.3 RIGID CELLULAR PHENOLIC FOAM

a. Preformed (molded) pipe insulation, ASTM C1126, Type III, Grade 1, k = 0.15 at 50 degrees F, for use at temperatures up to 250 degrees F with vapor retarder and all service vapor retarder jacket (ASJ) and with PVC premolded fitting covering.

b. Equipment Insulation, ASTM C1126, Type II, grade 1, k = 0.15 at 50 degrees F, for use at temperatures up to 250 degrees F with rigid cellular phenolic insulation and covering, and all service vapor retarder jacket (ASJ).

2.4 CELLULAR GLASS CLOSED-CELL

a. Comply with Standard ASTM C552, density 7.5 pcf nominal, k = 0.29 at 75 degrees F.

b. Pipe insulation for use at process temperatures below ambient air to 900 degrees F with or without all service vapor retarder jacket (ASJ).

c. Pipe insulation for use at process temperatures for pipe and tube below ambient air temperatures or where condensation control is necessary are to be installed with a vapor retarder/barrier system of with or without all service vapor retarder sealed jacket (ASJ) system. Without ASJ shall require all longitudinal and circumferential joints to be vapor sealed with vapor barrier mastic.

d. Cellular glass thermal insulation intended for use on surfaces operating at temperatures between 450 and 900 degrees F. It is possible that special fabrication or techniques for pipe insulation, or both, shall be required for application in the temperature range from 250 to 800 degrees F.

2.5 POLYISOCYANURATE CLOSED-CELL RIGID

a. Preformed (fabricated) pipe insulation, ASTM C591, Type IV, K= 0.19 at 75 degrees F, flame spread not over 25, smoke developed not over 50, for use at temperatures up to 300 degree F with factory applied PVDC or all service vapor retarder jacket with PVC premolded fitting covers.

b. Equipment and duct insulation, ASTM C591, Type IV, K= 0.19 at 75 degrees F, for use at temperatures up to 300 degrees F with PVDC or all service jacket vapor retarder jacket.

2.6 FLEXIBLE ELASTOMERIC CELLULAR THERMAL

ASTM C534/C534M, k = 0.27 at 75 degrees F, flame spread not over 25, smoke developed not over 50, for temperatures from 40 degrees F to 199 degrees F. Under high humidity exposures for condensation control an external vapor retarder/barrier jacket is required. Consult ASTM C1710.

2.7 CALCIUM SILICATE

a. Preformed pipe Insulation: ASTM C533, Type I and Type II with indicator denoting asbestos-free material.

b. Premolded Pipe Fitting Insulation: ASTM C533, Type I and Type II with indicator denoting asbestos-free material.

c. Equipment Insulation: ASTM C533, Type I and Type II.

d. Characteristics:

Insulation Characteristics		
ITEMS	**TYPE I**	**TYPE II**
Surface Temperature, maximum degrees F	649 (1200)	927 (1700)
Density (dry), lb/ ft3	240 (15)	352 (22)
Thermal conductivity: MinBTU in/h ft2 degrees F@ mean temperature of 199 degrees F	0.065 (0.45)	0.078 (0.540)
Surface burning characteristics: Flame spread Index, Maximum	0	0
Smoke Density index, Maximum	0	0

2.8 INSULATION FACINGS AND JACKETS

a. Vapor Retarder, higher strength with low water permeance = 0.02 or less perm rating, Beach puncture 50 units for insulation facing on pipe insulation jackets. Facings and jackets shall be ASJ or PVDC Vapor Retarder jacketing.

b. ASJ shall be white finish (kraft paper) bonded to 1 mil thick aluminum foil, fiberglass reinforced, with pressure sensitive adhesive closure. Comply with ASTM C1136. Beach puncture is 50 units, suitable for painting without sizing. Jackets shall have minimum 1-1/2 inch lap on longitudinal joints and minimum 3 inch butt strip on end joints. Butt strip material shall be same as the jacket. Lap and butt strips shall be self-sealing type with factory-applied pressure sensitive adhesive.

c. Vapor Retarder medium strength with low water vapor permeance of 0.02 or less perm rating, Beach puncture 25 units: FSK or PVDC type for concealed ductwork and equipment.

d. Except for flexible elastomeric cellular thermal insulation (not for high humidity exposures), field applied vapor barrier jackets shall be provided, in addition to the specified facings and jackets, on all exterior piping as well as on interior piping conveying fluids below ambient temperature. The vapor barrier jacket shall consist of a multi-layer laminated cladding with a maximum water vapor permeance of 0.001 perms. The minimum puncture resistance shall be 30 inch-pounds for interior locations and 80 inch-pounds for exterior or exposed locations or where the insulation is subject to damage.

e. Except for cellular glass thermal insulation, when all longitudinal and circumferential joints are vapor sealed with a vapor barrier mastic or caulking, vapor barrier jackets may not be provided. For aesthetic and physical abuse applications, exterior jacketing is recommended. Otherwise field applied vapor barrier jackets shall be provided, in addition to the applicable specified facings and jackets, on all exterior piping as well as on interior piping conveying fluids below ambient temperature. The vapor barrier jacket shall consist of a multi-layer laminated cladding with a maximum water vapor permeance of 0.001 perms. The minimum puncture resistance shall be 30 inch-pounds for interior locations and 80 inch-pounds for exterior or exposed locations or where the insulation is subject to damage.

f. Glass Cloth Jackets: Presized, minimum 7.8 ounces per square yard, 300 psig bursting strength with integral vapor retarder where required or specified. Weather proof if utilized for outside service.

g. Pipe fitting insulation covering (jackets): Fitting covering shall be premolded to match shape of fitting and shall be PVC conforming to FS L-P-535E, composition A, Type II Grade GU, and Type III, minimum thickness 0.03 inches. Provide color matching vapor retarder pressure sensitive tape. Staples, tacks, or any other attachment that penetrates the PVC covering is not allowed on any form of a vapor barrier system in below ambient process temperature applications.

h. Aluminum Jacket-Piping systems and circular breeching and stacks: ASTM B209, 3003 alloy, H-14 temper, 0.023 inch minimum thickness with locking longitudinal joints. Jackets for elbows, tees and other fittings shall be factory-fabricated or with cut aluminum gores to match shape of fitting and of 0.024 inch minimum thickness aluminum. Aluminum fittings shall be of same construction with an internal moisture barrier as straight run jackets but need not be of the same alloy. Factory-fabricated stainless steel bands with wing seals shall be installed on all circumferential joints. Bands shall be 0.5 inch wide on 18 inch centers. System shall be weatherproof if utilized for outside service.

i. Aluminum jacket-Rectangular breeching: ASTM B209, 3003 alloy, H-14 temper, 0.020 inches thick with 1-1/4 inch corrugations or 0.032 inches thick with no corrugations. System shall be weatherproof if used for outside service.

2.9 PIPE COVERING PROTECTION SADDLES

a. Cold pipe support: Premolded pipe insulation 180 degrees (half-shells) on bottom half of pipe at supports. Material shall be cellular glass or high density Polyisocyanurate insulation of the same thickness as adjacent insulation. Density of Polyisocyanurate insulation shall be a minimum of 3.0 pcf.

Nominal Pipe Size and Accessories Material (Insert Blocks)	
Nominal Pipe Size inches	**Insert Blocks mm (inches)**
Up through 5	6 long
6	6 long
8, 10, 12	9 long
14, 16	12 long
18 through 24	14 long

b. Warm or hot pipe supports: Premolded pipe insulation (180 degree half-shells) on bottom half of pipe at supports. Material shall be high density Polyisocyanurate (for temperatures up to 300 degrees F), cellular glass or calcium silicate. Insulation at supports shall have same thickness as adjacent insulation. Density of Polyisocyanurate insulation shall be a minimum of 3.0 pcf.

2.10 ADHESIVE, MASTIC, CEMENT

a. MIL-A-3316, Class 1: Jacket and lap adhesive and protective finish coating for insulation.

b. MIL-A-3316, Class 2: Adhesive for laps and for adhering insulation to metal surfaces.

c. MIL-A-24179, Type II Class 1: Adhesive for installing flexible unicellular insulation and for laps and general use.

d. MIL-PRF-19565, Type I: Protective finish for outdoor use.

e. MIL-PRF-19565, Type I or Type II: Vapor barrier compound for indoor use.

f. ASTM C449: Mineral fiber hydraulic-setting thermal insulating and finishing cement.

g. Other: Insulation manufacturers' published recommendations.

2.11 MECHANICAL FASTENERS

a. Pins, anchors: Welded pins, or metal or nylon anchors with galvanized steel or fiber washer, or clips. Pin diameter shall be as recommended by the insulation manufacturer.

b. Staples: Outward clinching galvanized steel. Staples are not allowed for below ambient vapor barrier applications.

c. Wire: 18 gage thick soft annealed galvanized or 14 gage copper clad steel or nickel copper alloy or stainless steel.

d. Bands: 1/2 inch nominal width, brass, galvanized steel, aluminum or stainless steel.

e. Tacks, rivets, screws or any other attachment device capable of penetrating the vapor retarder shall NOT be used to attach/close the any type of vapor retarder jacketing. Thumb tacks sometimes used on PVC jacketing and preformed fitting covers closures are not allowed for below ambient vapor barrier applications.

2.12 REINFORCEMENT AND FINISHES

a. Glass fabric, open weave: ASTM D1668/D1668M, Type III (resin treated) and Type I (asphalt or white resin treated).

b. Glass fiber fitting tape: MIL-C-20079, Type II, Class 1.

c. Tape for Flexible Elastomeric Cellular Insulation: As recommended by the insulation manufacturer.

d. Hexagonal wire netting: one inch mesh, 22 gage thick galvanized steel.

e. Corner beads: 2 inch by 2 inch, 26 gage thick galvanized steel; or, 1 inch by 1 inch, 28 gage thick aluminum angle adhered to 2 inch by 2 inch Kraft paper.

f. PVC fitting cover: FS L-P-535E, Composition A, 11-86 Type II, Grade GU, with Form B Mineral Fiber insert, for media temperature 50 to 250 degrees F. Below 50 degrees F and above 250 degrees F provide mitered pipe insulation of the same type as insulating straight pipe. Provide double layer insert. Provide vapor barrier pressure sensitive tape matching the color of the PVC jacket.

2.13 FIRESTOPPING MATERIAL

Other than pipe insulation, refer to Section 07 84 00, FIRESTOPPING.

2.14 FLAME AND SMOKE

Unless shown otherwise all assembled systems shall meet flame spread 25 and smoke developed 50 rating as developed under ASTM and UL standards and specifications. See paragraph "Quality Assurance".

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

a. Required pressure tests of piping joints and connections shall be

completed and the work approved by the Contracting Officer for application of insulation. Surface shall be clean and dry with all foreign materials, such as dirt, oil, loose scale and rust removed.

b. Except for specific exceptions or as noted, insulate all specified equipment, and piping (pipe, fittings, valves, accessories). Insulate each pipe individually. Do not use scrap pieces of insulation where a full length Section will fit.

c.

d. Insulation materials shall be installed with smooth and even surfaces, with jackets and facings drawn tight and smoothly cemented down and sealed at all laps. Insulation shall be continuous through all sleeves and openings, except at fire dampers and duct heaters (NFPA 90A).

e. Vapor retarders shall be continuous and uninterrupted throughout systems with operating temperature 60 degrees F and below. Lap and seal vapor barrier over ends and exposed edges of insulation. Anchors, supports and other metal projections through insulation on cold surfaces shall be insulated and vapor sealed for a minimum length of 6 inches.

f. Install vapor stops with operating temperature 60 degrees F and below at all insulation terminations on either side of valves, pumps, fittings, and equipment and particularly in straight lengths every approx. 15 to 20 feet of pipe insulation. The annular space between the pipe and pipe insulation of approx. 1 inch in length at every vapor stop shall be sealed with appropriate vapor barrier sealant. Bio-based materials shall be utilized when possible.

g. Construct insulation on parts of equipment such as cold water pumps and heat exchangers that shall be opened periodically for maintenance or repair, so insulation can be removed and replaced without damage. Install insulation with bolted 20 gage thick galvanized steel or aluminum covers as complete units, or in Sections, with all necessary supports, and split to coincide with flange/split of the equipment. Do not insulate over equipment nameplate data.

h. Insulation on hot piping and equipment shall be terminated square at items not to be insulated, access openings and nameplates. Cover all exposed raw insulation with white sealer coating (caution about coating’s maximum temperature limit) or jacket material.

i. Protect all insulations outside of buildings with aluminum jacket using lock joint or other approved system for a continuous weather tight system. Access doors and other items requiring maintenance or access shall be removable and sealable.

j. Plumbing work not to be insulated unless otherwise noted:

(1) Piping and valves of fire protection system.

(2) Chromium plated brass piping.

(3) Water piping in contact with earth.

(4) Distilled water piping.

k. Apply insulation materials subject to the manufacturer's recommended temperature limits. Apply adhesives, mastic and coatings at the manufacturer's recommended minimum wet or dry film thickness. Bio-based materials shall be utilized when possible.

l. Elbows, flanges and other fittings shall be insulated with the same material as is used on the pipe straights. Use of polyurethane or polyisocyanurate spray-foam to fill a PVC elbow jacket is prohibited on cold applications.

m. Firestop Pipe insulation:

(1) Provide firestopping insulation at fire and smoke barriers through penetrations. Firestopping insulation shall be UL listed as defined in Section 07 84 00, FIRESTOPPING.

(2) Pipe penetrations requiring fire stop insulation including, but not limited to the following:

(a) Pipe risers through floors

(b) Pipe chase walls and floors

(c) Smoke partitions

(d) Fire partitions

(e) Hourly rated walls

n. Freeze protection of above grade outdoor piping (over heat tracing tape): 3/4 inch thick insulation, for all pipe sizes 3 inches and smaller and 1 inch thick insulation for larger pipes. Provide metal jackets for all pipe insulations. Provide freeze protection for cold water make-up piping and equipment where indicated on the drawings as described in Section 23 21 13, HYDRONIC PIPING (electrical heat tracing systems).

o. Provide vapor barrier systems as follows:

(1) All piping exposed to outdoor weather.

(2) All interior piping conveying fluids exposed to outdoor air (i.e. in attics, ventilated (not air conditioned) spaces, etc.) below ambient air temperature.

p. Provide metal jackets over insulation as follows:

(1) All plumbing piping exposed to outdoor weather.

(2) Piping exposed in building, within 6 feet of the floor, that connects to sterilizers, kitchen and laundry equipment. Jackets may be applied with pop rivets except for cold pipe or tubing applications. Provide aluminum angle ring escutcheons at wall, ceiling or floor penetrations.

(3) A 2 inch jacket overlap is required at longitudinal and circumferential joints with the overlap at the bottom.

q. Provide PVC jackets over insulation as follows:

(1) Piping exposed in building, within 6 feet of the floor, on piping that is not precluded in previous Sections.

(2) A 2 inch jacket overlap is required at longitudinal and circumferential joints with the overlap at the bottom.

3.2 INSULATION INSTALLATION

a. Mineral Fiber Board:

(1) Vapor retarder faced board: Apply board on pins spaced not more than 12 inches on center each way, and not less than 3 inches from each edge of board. In addition to pins, apply insulation bonding adhesive to entire underside of horizontal metal surfaces. (Bio-based materials shall be utilized when possible.) Butt insulation edges tightly and seal all joints with laps and butt strips. After applying speed clips cut pins off flush and apply vapor seal patches over clips.

(2) Plain unfaced board:

(a) Insulation shall be scored, beveled or mitered to provide tight joints and be secured to equipment with bands spaced 9 inches on center for irregular surfaces or with pins and clips on flat surfaces. Use corner beads to protect edges of insulation.

(b) For hot equipment: Stretch 1 inch mesh wire, with edges wire laced together, over insulation and finish with insulating and finishing cement applied in one coat, 1/4 inch thick, troweled to a smooth finish.

(c) For cold equipment: Apply meshed glass fabric in a tack coat 60 to 70 square feet per gallon of vapor mastic and finish with mastic at 12 to 15 square feet per gallon over the entire fabric surface.

(3) Cold equipment: 1-1/2inch thick insulation faced with vapor retarder ASJ or FSK. Seal all facings, laps, and termination points and do not use staples or other attachments that may puncture ASJ or FSK.

(a) Water filter, chemical feeder pot or tank.

(b) Pneumatic, cold storage water and surge tanks.

(4) Hot equipment: 1-1/2 inch thick insulation faced with unsealed ASJ or FSK.

(a) Domestic water heaters and hot water storage tanks (not factory insulated).

(b) Booster water heaters for dietetics dish and pot washers and for washdown grease-extracting hoods.

b. Molded Mineral Fiber Pipe and Tubing Covering:

(1) Fit insulation to pipe, aligning all longitudinal joints. Seal

longitudinal joint laps and circumferential butt strips by rubbing hard with a nylon sealing tool to assure a positive seal. Staples may be used to assist in securing insulation except for cold piping. Seal all vapor retarder penetrations on cold piping with a generous application of vapor barrier mastic. Provide cellar glass inserts and install with metal insulation shields at outside pipe supports. Install freeze protection insulation over heating cable.

(2) Contractor's options for fitting, flange and valve insulation:

(a) Insulating and finishing cement for sizes less than 4 inches operating at surface temperature of 60 degrees F or more.

(b) Factory premolded, one piece PVC covers with mineral fiber, (Form B), inserts surface temperature of above 40 degrees F to 250 degrees F. Provide mitered preformed insulation of the same type as the installed straight pipe insulation for pipe temperatures below 40 degrees F. Secure first layer of mineral fiber insulation with twine. Seal seam edges with vapor barrier mastic and secure with fitting tape.

(c) Factory preformed, ASTM C547 or fabricated mitered Sections, joined with adhesive or (hot only) wired in place. (Bio-based materials shall be utilized when possible.) For hot piping finish with a smoothing coat of finishing cement. For cold fittings, 60 degrees F or less, vapor seal with a layer of glass fitting tape imbedded between two 1/16 inch coats of vapor barrier mastic.

(d) Fitting tape shall extend over the adjacent pipe insulation and overlap on itself at least 2 inches.

(3) Nominal thickness in millimeters and inches specified in the schedule at the end of this Section.

c. Rigid Cellular Phenolic Foam:

(1) Rigid closed cell phenolic insulation may be provided, exterior only, for piping, ductwork and equipment for temperatures up to 250 degrees F.

(2) Note the ASTM E84 or UL 723 surface burning characteristics requirements of maximum 25/50 indexes in paragraph "Quality Assurance".

(3) Provide secure attachment facilities such as welding pins.

(4) Apply insulation with joints tightly drawn together.

(5) Apply adhesives, coverings, neatly finished at fittings, and valves.

(6) Final installation shall be smooth, tight, neatly finished at all edges.

(7) Minimum thickness in inches specified in the schedule at the end of this Section.

(8) Condensation control insulation: Minimum 1 inch thick for all

pipe sizes depending on high humidity exposures.

(a) Body of roof and overflow drains horizontal and vertical piping runs and offsets (including elbows) of interior downspout piping in all areas above grade.

(b) Waste piping from electric water coolers and icemakers to drainage system.

(c) Waste piping located above basement floor from ice making and film developing equipment and air handling units, from equipment (including trap) to main vertical waste pipe.

(d) Cold water piping, exterior only.

d. Cellular Glass Insulation:

(1) Pipe and tubing, covering nominal thickness in millimeters and inches as specified in the schedule at the end of this Section.

(2) Underground piping other than or in lieu of that specified in Section 22 11 00, FACILITY WATER DISTRIBUTION: Type II, factory jacketed with a 3 mm laminate jacketing consisting of 10 ft x 10 ft asphalt impregnated glass fabric, bituminous mastic and outside protective plastic film.

(a) 3 inches thick for hot water piping.

(b) As scheduled at the end of this Section for chilled water piping.

(c) Underground piping: Apply insulation with joints tightly butted. Seal longitudinal self-sealing lap. Use field fabricated or factory made fittings. Seal butt joints and fitting with jacketing as recommended by the insulation manufacturer. Use 4 inch wide strips to seal butt joints.

(d) Provide expansion chambers for pipe loops, anchors and wall penetrations as recommended by the insulation manufacturer.

(e) Underground insulation shall be inspected and approved by the Contracting Officer as follows:

1. Insulation in place before coating.

2. After coating.

(f) Sand bed and backfill: Minimum 3 inches all around insulated pipe or tank, applied after coating has dried.

(g) All piping up to 900 degrees F requiring protection from physical heavy contact/abuse including in mechanical rooms and exposures to the public.

(3) Cold equipment: 2 inch thick insulation faced with ASJ.

e. Polyisocyanurate Closed-Cell Rigid Insulation:

(1) Polyisocyanurate closed-cell rigid insulation (PIR) may be provided for exterior piping and equipment for temperature up to 300 degree F.

(2) Install insulation, vapor retarder and jacketing per manufacturer’s recommendations. Particular attention should be paid to recommendations for joint staggering, adhesive application, external hanger design, expansion/contraction joint design and spacing and vapor retarder integrity.

(3) Install insulation with all joints tightly butted (except expansion) joints in hot applications). Provide insulation contractions joints for very cold process temperatures.

(4) If insulation thickness exceeds 2-1/2 inches, install as a double layer system with longitudinal (lap) and butt joint staggering as recommended by manufacturer.

(5) For cold applications, vapor retarder shall be installed in a continuous manner. No staples, rivets, screws or any other attachment device capable of penetrating the vapor retarder shall be used to attach the vapor retarder or jacketing. No wire ties capable of penetrating the vapor retarder shall be used to hold the insulation in place. Stainless steel banding shall be used for cold applications to attach PVC or metal jacketing.

(6) Elbows, flanges and other fittings shall be insulated with the same material as is used on the pipe straights. The elbow/ fitting insulation shall be field-fabricated, mitered or factory prefabricated to the necessary size and shape to fit on the elbow/ fitting. Use of polyurethane or polyisocyanurate spray-foam to fill PVC elbow jacket is prohibited on cold applications.

(7) For cold applications, the vapor retarder on elbows/fittings shall be either mastic-fabric-mastic or 2 mil thick PVDC vapor retarder adhesive tape. Bio-based materials shall be utilized when possible.

(8) All PVC and metal jacketing shall be installed so as to naturally shed water. Joints shall point down and shall be sealed with either adhesive or caulking (except for periodic slip joints). Bio-based materials shall be utilized when possible.

(9) Note the NFPA 90A burning characteristic requirements of 25/50 in paragraph "Quality Assurance". Refer to paragraph “General Requirements” for items not to be insulated.

(10) Minimum thickness in millimeter (inches) specified in the schedule at the end of this Section.

f. Flexible Elastomeric Cellular Thermal Insulation:

(1) Apply insulation and fabricate fittings in accordance with the manufacturer's installation instructions and finish with two coats

of weather resistant finish as recommended by the insulation manufacturer. External vapor barrier jacketing may be required for expected or anticipated high humidity exposures. See ASTM C1710.

(2) Pipe and tubing insulation:

(a) Use proper size material. Do not stretch or strain insulation.

(b) To avoid undue compression of insulation, use supports as recommended by the elastomeric insulation manufacturer. Insulation shields are specified under Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

(c) Where possible, slip insulation over the pipe or tubing prior to connection, and seal the butt joints with adhesive. Where the slip-on technique is not possible, slit the insulation and apply it to the pipe sealing the seam and joints with contact adhesive. Optional tape sealing, as recommended by the manufacturer, may be employed. Bio-based materials shall be utilized when possible.

(3) Apply sheet insulation to flat or large curved surfaces with 100 percent adhesive coverage. For fittings and large pipe, apply adhesive to seams only.

(4) Pipe insulation: nominal thickness in millimeters (inches) as specified in the schedule at the end of this Section.

g. Calcium Silicate:

(1) Minimum thickness in inches specified below for piping other than in boiler plant.

Nominal Thickness Of Calcium Silicate Insulation(Non-Boiler Plant)				
Nominal Pipe Size Inches	Thru 1	1-1/4 to 3	4 to 8	8
199-500 degrees F(HPS, HPR)	4	5	6	6

(2) MRI Quench Vent Insulation: Type I, class D, 6 inch nominal thickness.

3.3 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS

b. Components provided under this Section of the specification will be tested as part of a larger system.

3.4 PIPE INSULATION SCHEDULE

Provide insulation for piping systems as scheduled below:

Insulation Thickness Inches					
		Nominal Pipe Size Inches			
Operating Temperature Range/Service	**Insulation Material**	**Less than 1**	**1 – 1¼**	**1½ - 3**	**4 and Greater**
100-140 degrees F (Domestic Hot Water Supply and Return)	Mineral Fiber (Above ground piping only)	1.5	1.5	2.0	2.0
100-140 degrees F (Domestic Hot Water Supply and Return)	Rigid Cellular Phenolic Foam (Above ground piping only) (exterior locations only)	1.5	1.5	2.0	2.0
100-140 degrees F (Domestic Hot Water Supply and Return)	Polyiso-cyanurate Closed-Cell Rigid (Exterior Locations only)	1.5	1.5	2.0	2.0
100-140 degrees F (Domestic Hot Water Supply and Return)	Flexible Elastomeric Cellular Thermal (Above ground piping only)	1.5	1.5	2.0	2.0
100-140 degrees F (Domestic Hot Water Supply and Return)	Cellular Glass Thermal	1.5	1.5	2.0	2.0
40-60 degrees F (Domestic Cold Water)	Rigid Cellular Phenolic Foam (Above ground piping only) (exterior locations only)	1.0	1.0	1.0	1.0
40-60 degrees F (Domestic Cold Water)	Polyiso-cyanurate Closed-Cell Rigid(Exterior Locations only)	1.0	1.0	1.0	1.0

40-60 degrees F (Domestic Cold Water)	Flexible Elastomeric Cellular Thermal (Above ground piping only)	1.0	1.0	1.0	1.0
40-60 degrees F (Domestic Cold Water)	Cellular Glass Thermal	1.5	1.5	1.5	1.5

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DIVISION 22 - PLUMBING

SECTION 22 08 00.01

DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS

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SECTION 22 08 00.01
8/19

DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

a. The requirements of this Section apply to all sections of Division 22.

b. This project will have selected building systems commissioned. The complete list of equipment and systems to be commissioned are specified in Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS. The commissioning process, which the Contractor is responsible to execute, is defined in Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS. A Commissioning Agent (CxA) appointed by the Government and Commissioning Manager (CxM) as indicated in Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS will manage the commissioning process.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Pre-Functional Checklists; G, DO

Contractors Tests; G, DO

Systems Functional Performance Testing; G, DO

SD-08 Manufacturer's Instructions

Training Of VA Personnel; G, DO

1.3 SUBMITTAL INSTRUCTIONS

a. The commissioning process requires review of selected Submittals. The CxA will provide a list of submittals that will be reviewed by the CxA and CxM. This list will be reviewed and approved by the DVA/USACE prior to forwarding to the Contractor. Refer to UFGS Section 01 33 00, SUBMITTAL PROCEDURES for further details.

b. The commissioning process requires Submittal review simultaneously with engineering review. Specific submittal requirements related to the commissioning process are specified in Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

1.4 RELATED WORK

a. UFGS GENERAL REQUIREMENTS sections.

b. Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

c. UFGS Section 01 33 00, SUBMITTAL PROCEDURES

1.5 SUMMARY

a. This Section includes requirements for commissioning plumbing systems, subsystems and equipment. This Section supplements the general requirements specified in Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

b. Refer to Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for more specifics regarding processes and procedures as well as roles and responsibilities for all Commissioning Team members.

1.6 DEFINITIONS

Refer to Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for definitions.

1.7 COMMISSIONED SYSTEMS

a. Commissioning of a system or systems specified in Division 22 is part of the construction process. Documentation and testing of these systems, as well as training of the VA's Operation and Maintenance personnel in accordance with the requirements of Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and of Division 22, is required in cooperation with the DVA/USACE and the CxA and CxM.

b. The Plumbing systems commissioning will include the systems listed in Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS:

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 CONSTRUCTION INSPECTIONS

Commissioning of the Building Plumbing Systems will require inspection of individual elements of the Plumbing construction throughout the construction period. The Contractor shall coordinate with the CxA and CxM in accordance with Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and the Commissioning Plan to schedule inspections as required to support the commissioning process.

3.2 PRE-FUNCTIONAL CHECKLISTS

The Contractor shall complete Pre-Functional Checklists to verify systems, subsystems, and equipment installation is complete and systems are ready for Systems Functional Performance Testing. The CxM will prepare Pre-Functional Checklists to be used to document equipment installation. The Contractor shall complete the checklists. Completed checklists shall be submitted to the DVA/USACE, CxA, and CxM for review. The CxM and CxA may spot check a sample of completed checklists. If the CxM and CxA determine that the information provided on the checklist is not accurate, the CxM will return the marked-up checklist to the Contractor for

correction and resubmission. If the CxM and CxA determine that a significant number of completed checklists for similar equipment are not accurate, the CxM in coordination with CxA will select a broader sample of checklists for review. If the CxM and CxA determine that a significant number of the broader sample of checklists is also inaccurate, all the checklists for the type of equipment will be returned to the Contractor for correction and resubmission. Refer to Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for submittal requirements for Pre-Functional Checklists, Equipment Startup Reports, and other commissioning documents.

3.3 CONTRACTORS TESTS

Contractor tests as required by other sections of Division 22 shall be scheduled and documented in accordance with UFGS GENERAL REQUIREMENTS sections. All testing shall be incorporated into the project schedule. Contractor shall provide no less than seven calendar days' notice of testing. The CxM will witness selected Contractor tests at the sole discretion of the CxM and CxA. The CxA and other Government representative(s) may choose to participate in some or all the testing procedures as observers. Contractor tests shall be completed prior to scheduling Systems Functional Performance Testing.

3.4 SYSTEMS FUNCTIONAL PERFORMANCE TESTING

The Commissioning Process includes Systems Functional Performance Testing that is intended to test systems functional performance under steady state conditions, to test system reaction to changes in operating conditions, and system performance under emergency conditions. The CxM will prepare final and detailed Systems Functional Performance Test procedures for review and approval by the CxA and Contracting Officer. The Contractor shall review and comment on the tests prior to approval. The Contractor shall provide the required labor, materials, and test equipment identified in the test procedure to perform the tests. The CxM will witness and document the testing. The Contractor shall sign the test reports to verify tests were performed. See Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS, for additional details.

3.5 TRAINING OF VA PERSONNEL

Training of the VA operation and maintenance personnel is required in cooperation with the Contracting Officer, VA Resident Engineer, and CxM. Provide competent, factory authorized personnel to provide instruction to operation and maintenance personnel concerning the location, operation, and troubleshooting of the installed systems. Contractor shall submit training agendas and trainer resumes in accordance with the requirements of Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS. The instruction shall be scheduled in coordination with the Contracting Officer and VA Resident Engineer after submission and approval of formal training plans. Refer to Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and Division 22 Sections for additional Contractor training requirements.

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SECTION 22 11 00
5/21

FACILITY WATER DISTRIBUTION

PART 1 GENERAL

1.1 DESCRIPTION

a. Domestic water systems, including piping, equipment and all necessary accessories as designated in this section.

b. A complete listing of all acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 07 84 00, FIRESTOPPING.

f. Section 07 92 00, JOINT SEALANTS.

g. Section 09 91 00, PAINTING.

h.

i. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

j. Section 22 07 11, PLUMBING INSULATION.

k. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.3 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.9	(2024) Factory-Made Wrought Buttwelding Fittings
ASME B16.15	(2024) Cast Copper Alloy Threaded Fittings Classes 125 and 250
ASME B16.18	(2021) Cast Copper Alloy Solder Joint Pressure Fittings

ASME B16.22	(2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings
ASME B16.24	(2022) Cast Copper Alloy Pipe Flanges, Flanged Fittings, and Valves Classes 150, 300, 600, 900, 1500, and 2500
ASME BPVC SEC IX	(2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

ASSE 1010	(2021) Performance Requirements for Water Hammer Arresters

ASTM INTERNATIONAL (ASTM)

ASTM A47/A47M	(1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings
ASTM A183	(2014; R 2020) Standard Specification for Carbon Steel Track Bolts and Nuts
ASTM A269/A269M	(2024) Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service
ASTM A312/A312M	(2022a) Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes
ASTM A403/A403M	(2025) Standard Specification for Wrought Austenitic Stainless Steel Piping Fittings
ASTM A536	(2024) Standard Specification for Ductile Iron Castings
ASTM B32	(2020) Standard Specification for Solder Metal
ASTM B43	(2020) Standard Specification for Seamless Red Brass Pipe, Standard Sizes
ASTM B75/B75M	(2020) Standard Specification for Seamless Copper Tube
ASTM B88	(2022) Standard Specification for Seamless Copper Water Tube
ASTM B584	(2022) Standard Specification for Copper Alloy Sand Castings for General Applications
ASTM B687	(1999; R 2023) Standard Specification for Brass, Copper, and Chromium-Plated Pipe Nipples

ASTM C919	(2024) Standard Practice for Use of Sealants in Acoustical Applications
ASTM D1785	(2021) Standard Specification for Poly(Vinyl Chloride) (PVC), Plastic Pipe, Schedules 40, 80, and 120
ASTM D2657	(2007; R 2015) Heat Fusion Joining Polyolefin Pipe and Fittings
ASTM D2855	(2020) Standard Practice for the Two-Step (Primer and Solvent Cement) Method of Joining Poly (Vinyl Chloride) (PVC) or Chlorinated Poly (Vinyl Chloride) (CPVC) Pipe and Piping Components with Tapered Sockets
ASTM E1120	(2021) Standard Specification for Liquid Chlorine
ASTM E1229	(2021) Standard Specification for Calcium Hypochlorite
ASTM F2389	(2024a) Standard Specification for Pressure-rated Polypropylene (PP) Piping Systems
ASTM F2620	(2024) Standard Practice for Heat Fusion Joining of Polyethylene Pipe and Fittings
ASTM F2769	(2022) Standard Specification for Polyethylene of Raised Temperature (PE-RT) Plastic Hot and Cold-Water Tubing and Distribution Systems

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C110/A21.10	(2021) Ductile-Iron and Gray-Iron Fittings
AWWA C151/A21.51	(2023) Ductile-Iron Pipe, Centrifugally Cast
AWWA C153/A21.53	(2019) Ductile-Iron Compact Fittings for Water Service
AWWA C213	(2022) Fusion-Bonded Epoxy Coatings and Linings for Steel Water Pipe and Fittings
AWWA C651	(2023) Standard for Disinfecting Water Mains

AMERICAN WELDING SOCIETY (AWS)

AWS A5.8/A5.8M	(2019) Specification for Filler Metals for Brazing and Braze Welding

INTERNATIONAL CODE COUNCIL (ICC)

ICC IPC	(2024) International Plumbing Code

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-58	(2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation
MSS SP-72	(2010a) Ball Valves with Flanged or Butt-Welding Ends for General Service
MSS SP-110	(2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends

NSF INTERNATIONAL (NSF)

NSF/ANSI 14	(2023) Plastics Piping System Components and Related Materials
NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects
NSF 372	(2022) Drinking Water System Components - Lead Content

PLUMBING AND DRAINAGE INSTITUTE (PDI)

PDI WH 201	(2010) Water Hammer Arresters Standard

U.S. DEPARTMENT OF VETERANS AFFAIRS (VA)

VA H-18-8	(2023) Seismic Design Requirements
VA PG-18-10	Design Manuals

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Welder’s Certification; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

Testing Plan; G, AE

SD-07 Certificates

Certification Documentation; G, AE

SD-08 Manufacturer's Instructions

Training Plans; G, AE

Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER Section 22 11 00, FACILITY WATER DISTRIBUTIONS”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity. All items listed in Part 2 - Products.

d. Complete operating and maintenance manuals including wiring diagrams, technical data sheets and information for ordering replacement parts:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

e. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

f. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

a. A certificate shall be submitted prior to welding of steel piping showing the Welder’s certification. The certificate shall be current and no more than one year old. Welder’s qualifications shall be in accordance with ASME BPVC SEC IX.

b. All grooved joint couplings, fittings, valves, and specialties shall be the products of a single manufacturer. Grooving tools shall be by the same manufacturer as the groove components.

c. All pipe, couplings, fittings, and specialties shall bear the identification of the manufacturer and any markings required by the applicable referenced standards.

d. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

1.7 SPARE PARTS

1.8 AS-BUILT DOCUMENTATION

a. Submit manufacturer's literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, substitutions and construction revisions shall be in electronic version on compact disc or DVD inserted into a three-ring binder. All aspects of system operation and maintenance procedures, including piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices shall be included. A list of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. As-built drawings are to be provided, and a copy of them in Revit version 2025 provided on compact disk or DVD. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the 'third party testing company' requirement.

d. Certification documentation shall be provided to Contracting Officer 10 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and certificate if applicable that all results of tests were within limits specified. If a certificate is not available, all documentation shall be on the Certifier's letterhead.

PART 2 PRODUCTS

2.1 MATERIALS

a. Material or equipment containing a weighted average of greater than 0.25 percent lead are prohibited in any potable water system intended for human consumption and shall be certified in accordance with NSF/ANSI 61 or NSF 372. Endpoint devices used to dispense water for drinking shall meet the requirements of NSF/ANSI 61, Section 9.

b. Plastic pipe, fittings, and solvent cement shall meet NSF/ANSI 14 and shall be NSF listed for the service intended.

2.2 UNDERGROUND WATER SERVICE CONNECTIONS TO BUILDINGS

a. From inside face of exterior wall to approximately 5 feet outside of building and underground inside building, material to be the same for the size specified inside the building.

b. 3 inch Diameter and Greater: Ductile iron, AWWA C151/A21.51, 350 psig pressure class, exterior bituminous coating, and cement lined. Bio-based materials shall be utilized when possible. Provide flanged and anchored connection to interior piping.

c. Under 3 inch Diameter: Copper tubing, ASTM B88, Type K, seamless, annealed. Fittings are as specified in paragraph “Above Ground (Interior) Water Piping”. Use brazing alloys, AWS A5.8/A5.8M, Classification BCuP.

d. Flexible Expansion Joint: Ductile iron with ball joints rated for 250 psig working pressure conforming to AWWA C153/A21.53, capable of deflecting a minimum of 20 degrees in each direction. Flexible expansion joint size shall match the pipe size it is connected to and shall have the expansion capability designed as an integral part of the ductile iron ball castings. Pressure containing parts shall be lined with a minimum of 15 mils of fusion bonded epoxy conforming to the applicable requirements of AWWA C213 and shall be factory tested with a 1500-volt spark test. Flexible expansion joint shall have flanged connections conforming to AWWA C110/A21.10. Bolts and nuts shall be 316 stainless steel and gaskets shall be neoprene. The flexible expansion fitting shall not expand or exert an axial thrust under internal water pressure. Provide piping joint restraints at each mechanical joint end connection and piping restraints at the penetration of the building wall. The restraints shall be provided to address the developed thrust at the change of piping direction.

2.3 ABOVE GROUND (INTERIOR) WATER PIPING

a. Pipe: Copper tube, ASTM B88, Type K or L, drawn. For pipe 6 inches and larger, stainless steel, ASTM A312/A312M, schedule 40 shall be used.

b. Fittings for Copper Tube:

(1) Wrought copper or bronze castings conforming to ASME B16.18 and ASME B16.22. Unions shall be bronze, MSS SP-72, MSS SP-110, solder or braze joints. Use 95/5 tin and antimony for all soldered joints.

(2) Grooved fittings, 2 to 6 inch wrought copper ASTM B75/B75M C12200, 5 to 6 inch bronze casting ASTM B584, C84400. Mechanical grooved couplings, 300 psig minimum ductile iron, ASTM A536 Grade 65-45-12, or malleable iron, ASTM A47/A47M Grade 32510 housing, with EPDM gasket, steel track head bolts, ASTM A183, coated with copper colored alkyd enamel.

(3) Mechanical press-connect fittings for copper pipe and tube **are prohibited**. See VA PG-18-10 for additional information.

(4) Mechanically formed tee connection: Form mechanically extracted collars in a continuous operation by drilling pilot hole and drawing out tube surface to form collar, having a height of not less than three times the thickness of tube wall. Adjustable collaring device shall ensure proper tolerance and complete uniformity of the joint. Notch and dimple joining branch tube in a single process to provide free flow where the branch tube penetrates the fitting. Braze joints.

(5) Flanged fittings, bronze, class 150, solder-joint ends conforming to ASME B16.24.

c. Fittings for Stainless Steel:

(1) Stainless steel butt-welded fittings, Type 316, Schedule 10, conforming to ASME B16.9.

(2) Grooved fittings, stainless steel, Type 316, Schedule 40, conforming to ASTM A403/A403M. Segmentally fabricated fittings are not allowed. Mechanical grooved couplings, ductile iron, 600 psig, ASTM A536 Grade 65-45-12, or malleable iron, ASTM A47/A47M Grade 32510 housing, with EPDM gasket, steel track head bolts, ASTM A183, coated with copper colored alkyd enamel.

d. Adapters: Provide adapters for joining pipe or tubing with dissimilar end connections.

e. Solder: ASTM B32 alloy type Sb5, HA or HB. Provide non-corrosive flux.

f. Brazing alloy: AWS A5.8/A5.8M, brazing filler metals shall be BCuP series for copper to copper joints and BAg series for copper to steel joints.

g. Re-agent Grade Water Piping and Dialysis Water Piping:

(1) Polypropylene, ASTM F2389, Schedule 80 pressure pipe without additions of modifiers, plasticizers, colorants, stabilizers or lubricants. Bio-based materials shall be utilized when possible. This virgin un-plasticized pipe and fittings shall transport 10 megohm water with no loss of purity. Provide socket or butt end fittings with ASTM D2657 heat fusion joints.

(2) Polyethylene, ASTM F2769, Schedule 80, food and medical grade, capable of transporting 10 megohm water with no loss of purity. Processed by continuous compression molding without the addition of fillers, polymer modifiers or processing aids. Uniform color with no cracks, flaws, blisters or other imperfections in appearance. Provide ASTM D2657 or ASTM F2620 heat fusion butt

welded joints. In accordance with manufacturer's recommendations, provide continuous channel support under all horizontal piping.

(3) Reverse Osmosis (RO) Water Piping:

(a) Low Pressure Feed, Reject and Recycle Piping: Less than or equal to 75 psig: ASTM D1785, Schedule 80 PVC, ASTM D2855 socket welded and flanged.

(b) RO Product Tubing from each Membrane Housing: ASTM D1785, Schedule 80 PVC, ASTM D2855 socket welded and flanged.

(c) Low Pressure Control and Pressure Gage Tubing: Polyethylene.

(d) High Pressure Reject and Recycle Piping: Greater than 75 psig: ASTM A269/A269M, Type 304 schedule 10 stainless steel with butt welded joints.

(e) High Pressure Control and Pressure Gage Tubing: 1000 psig burst nylon.

2.4 EXPOSED WATER PIPING

a. Finished Room: Use full iron pipe size chrome plated brass piping for exposed water piping connecting fixtures, casework, cabinets, equipment, and reagent racks when not concealed by apron including those furnished by the Government or specified in other sections.

(1) Pipe: ASTM B43, standard weight.

(2) Fittings: ASME B16.15 cast bronze threaded fittings with chrome finish.

(3) Nipples: ASTM B687, Chromium-plated.

(4) Unions: MSS SP-72, MSS SP-110, brass or bronze with chrome finish. Unions 2-1/2 inches and larger shall be flange type with approved gaskets.

b. Unfinished Rooms, Mechanical Rooms and Kitchens: Chrome-plated brass piping is not required. Paint piping systems as specified in Section 09 91 00, PAINTING.

2.5 ETHYLENE OXIDE (ETO) STERILIZER WATER SUPPLY PIPING

Stainless steel, ASTM A312/A312M, Schedule 10 with stainless-steel butt-welded fittings. Provide on sterilizer water supply.

2.6 TRAP PRIMER WATER PIPING

a. Pipe: Copper tube, ASTM B88, type K, hard drawn.

b. Fittings: Bronze castings conforming to ASME B16.18 Solder joints.

c. Solder: ASTM B32 alloy type Sb5. Provide non-corrosive flux.

2.7 STRAINERS

a. Provide on high pressure side of pressure reducing valves, on suction

side of pumps, on inlet side of indicating and control instruments and equipment subject to sediment damage and where shown on drawings. Strainer element shall be removable without disconnection of piping.

b. Water: Basket or "Y" type with easily removable cover and brass strainer basket.

c. Body: Less than 3 inches, brass or bronze; 3 inches and greater, cast iron or semi-steel.

2.8 DIELECTRIC FITTINGS

Provide dielectric couplings or unions between pipe of dissimilar metals.

2.9 STERILIZATION CHEMICALS

a. Hypochlorite: ASTM E1229.

b. Liquid Chlorine: ASTM E1120.

2.10 WATER HAMMER ARRESTER

a. Closed copper tube chamber with permanently sealed 60 psig air charge above a Double O-ring piston. Two high heat Buna-N 0-rings pressure packed and lubricated with FDA approved silicone compound. All units shall be designed in accordance with ASSE 1010. Access shall be provided where devices are concealed within partitions or above ceilings. Size and install in accordance with PDI WH 201 requirements. Provide water hammer arrestors at:

(1) All solenoid valves.

(2) All groups of two or more flush valves.

(3) All quick opening or closing valves.

(4) All medical washing equipment.

PART 3 EXECUTION

3.1 INSTALLATION

a. General: Comply with the International Plumbing Code and the following:

(1) Install branch piping for water from the piping system and connect to all fixtures, valves, cocks, outlets, casework, cabinets and equipment, including those furnished by the Government or specified in other sections.

(2) Pipe shall be round and straight. Cutting shall be done with proper tools. Pipe, except for plastic and glass, shall be reamed to remove burrs and a clean smooth finish restored to full pipe inside diameter.

(3) All pipe runs shall be laid out to avoid interference with other work/trades.

(4) Install union and shut-off valve on pressure piping at connections

to equipment.

(5) Pipe Hangers, Supports and Accessories:

(a) All piping shall be supported per the ICC IPC, VA H-18-8, MSS SP-58, and SMACNA as required.

(b) Shop Painting and Plating: Hangers, supports, rods, inserts and accessories used for pipe supports shall be shop coated with zinc chromate primer paint. Electroplated copper hanger rods, hangers and accessories may be used with copper tubing.

(c) Floor, Wall and Ceiling Plates, Supports, Hangers:

1. Solid or split un-plated cast iron.

2. All plates shall be provided with set screws.

3. Pipe Hangers: Height adjustable clevis type.

4. Adjustable Floor Rests and Base Flanges: Steel.

5. Concrete Inserts: "Universal" or continuous slotted type.

6. Hanger Rods: Mild, low carbon steel, fully threaded or Threaded at each end with two removable nuts at each end for positioning rod and hanger and locking each in place.

7. Pipe Hangers and Riser Clamps: Malleable iron or carbon steel. Pipe Hangers and riser clamps shall have a copper finish when supporting bare copper pipe or tubing.

8. Rollers: Cast iron.

9. Self-drilling type expansion shields shall be "Phillips" type, with case hardened steel expander plugs.

10. Hangers and supports utilized with insulated pipe and tubing shall have 180-degree (minimum) metal protection shield centered on and welded to the hanger and support. The shield thickness and length shall be engineered and sized for distribution of loads to preclude crushing of insulation without breaking the vapor barrier. The shield shall be sized for the insulation and have flared edges to protect vapor-retardant jacket facing. To prevent the shield from sliding out of the clevis hanger during pipe movement, center-ribbed shields shall be used.

11. Miscellaneous Materials: As specified, required, directed or as noted on the drawings for proper installation of hangers, supports and accessories. If the vertical distance exceeds 20 feet for cast iron pipe additional support shall be provided in the center of that span. Provide all necessary auxiliary steel to provide that support.

12. With the installation of each flexible expansion joint, provide piping restraints for the upstream and downstream section of the piping at the flexible expansion joint. Provide calculations supporting the restraint length design and type of selected restraints. Restraint calculations shall be based on the

criteria from the manufacturer regarding their restraint design.

(6) Install chrome plated cast brass escutcheon with set screw at each wall, floor and ceiling penetration in exposed finished locations and within cabinets and millwork.

(7) Penetrations:

(a) Firestopping: Where pipes pass through fire partitions, fire walls, smoke partitions, or floors, install a fire stop that provides an effective barrier against the spread of fire, smoke, and gases as specified in Section 07 84 00, FIRESTOPPING. Completely fill and seal clearances between raceways and openings with the firestopping materials.

(b) Waterproofing: At floor penetrations, completely seal clearances around the pipe and make watertight with sealant as specified in Section 07 92 00, JOINT SEALANTS. Bio-based materials shall be utilized when possible.

(c) Acoustical sealant: Where pipes pass through sound rated walls, seal around the pipe penetration with an acoustical sealant that is compliant with ASTM C919.

b. Domestic Water piping shall conform to the following:

(1) Grade all lines to facilitate drainage. Provide drain valves at bottom of risers and all low points in system. Design domestic hot and cold water circulating lines with no traps.

(2) Connect branch lines at bottom of main serving fixtures below and pitch down so that main may be drained through fixture. Connect branch lines to top of main serving only fixtures located on floor above.

3.2 TESTS

a. General: Test system either in its entirety or in sections. Submit testing plan to Contracting Officer 10 working days prior to test date.

b. Potable Water System: Test after installation of piping and domestic water heaters, but before piping is concealed, before covering is applied, and before plumbing fixtures are connected. Fill systems with water and maintain hydrostatic pressure of 150 psig gage for two hours. No decrease in pressure is allowed. Provide a pressure gage with a shutoff and bleeder valve at the highest point of the piping being tested. Pressure gauge shall have 1 psig increments.

c. Re-agent Grade Water Systems: Fill system with water and maintain hydrostatic pressure of 200 psig gage during inspection and prove tight.

d. All Other Piping Tests: Test new installed piping under 1-1/2 times actual operating conditions and prove tight.

e. The test pressure shall hold for the minimum time duration required by the applicable plumbing code or authority having jurisdiction.

3.3 STERILIZATION

a. After tests have been successfully completed, thoroughly flush and sterilize the interior domestic water distribution system in accordance with AWWA C651.

b. Use liquid chlorine or hypochlorite for sterilization.

3.4 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this Section of the specification will be tested as part of a larger system.

3.5 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for four hours to instruct VA Personnel in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

-- End of Section --

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DIVISION 22 - PLUMBING

SECTION 22 11 23

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-- End of Section Table of Contents --

SECTION 22 11 23
9/15

DOMESTIC WATER PUMPS

PART 1 GENERAL

1.1 DESCRIPTION

a. Hot water circulating pump, hot water recirculation pump and domestic water pressure booster system.

b. A complete listing of all acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e.

f. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

g. Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT.

h. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS: Requirements for commissioning, systems readiness checklist, and training.

i. Section 26 29 11, MOTOR CONTROLLERS.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC	(2010) Boiler and Pressure Vessels Code

NSF INTERNATIONAL (NSF)

NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects
NSF 372	(2022) Drinking Water System Components - Lead Content

UL SOLUTIONS (UL)

UL 778 (2016; Reprint Jul 2024) UL Standard for Safety Motor-Operated Water Pumps

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Certificate Of Shop Test For Domestic Water Booster System; G, AE

Certification Documentation; G, AE

SD-08 Manufacturer's Instructions

Training Plans; G, AE

Instructor Qualifications; G, AE

SD-09 Manufacturer's Field Reports

Factory And Construction Site Test Data; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER Section 22 11 23, DOMESTIC WATER PUMPS”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Pump:

(a) Manufacturer and model.

(b) Operating speed.

(c) Capacity.

(d) Characteristic performance curves.

(2) Motor:

(a) Manufacturer, frame and type.

(b) Speed.

(c) Current Characteristics.

(d) Efficiency.

(3) Tank:

(a) Manufacturer and model.

(b) Capacity

(4) Drive: Information in accordance with Section 26 29 11, MOTOR CONTROLLERS.

d. Certificate of shop test for domestic water booster system. Provide certified performance curves.

e. Certified copies of all the factory and construction site test data sheets and reports.

f. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

g. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

h. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

a. General:

(1) UL Compliance: Comply with UL 778 for motor-operated water pumps.

(2) Design Criteria:

(a) Pump sizes, capacities, pressures, operating characteristics and efficiency shall be as scheduled.

(b) Head-capacity curves shall slope up to maximum head at shut-off. Select pumps near the midrange of the curve, and near the point of maximum efficiency, without approaching the pump curve end point and possible cavitation and unstable operation. Select pumps for open systems so that required net positive suction head (NPSHR) does not exceed the net positive head available (NPSHA).

(c) Pump Driver: Furnish with pump. Size shall be non-overloading at any point on the head-capacity curve, including in a parallel or series pumping installation with one pump in operation.

(d) Provide all pumps with motors, impellers, drive assemblies, bearings, coupling guard and other accessories specified. Statically and dynamically balance all rotating parts.

(e) Furnish each pump and motor with a nameplate giving the manufacturers name, serial number of pump, capacity in GPM and head in feet at design condition, horsepower, voltage, frequency, speed and full load current and motor efficiency.

(f) Test all pumps before shipment. The manufacturer shall certify all pump ratings.

(g) After completion of balancing, provide replacement of impellers or trim impellers to provide specified flow at actual pumping head, as installed.

b. Hot Water Circulating and Recirculating Pumps: Components shall be assembled by a single manufacturer and the pump motor assembly shall be the standard cataloged product of the manufacturer.

c. Domestic Water Pressure Booster System:

(1) Components shall be furnished by a single manufacturer and the system shall be the standard cataloged product of the manufacturer.

(2) Shop Test: Water booster unit and its component parts shall undergo a thorough electric and hydraulic operating test prior to shipment. Tests shall include a system operating flow test from zero to 100 percent of design flow rate under specified suction and system pressure conditions. Certified performance curves shall be furnished.

d. Bio-Based Materials: For products designated by the USDA’s Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

1.7 AS-BUILT DOCUMENTATION

a. Submit manufacturer’s literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, substitutions and construction revisions shall be in electronic version on compact disc or DVD inserted into a three ring binder. All aspects of system operation and maintenance procedures, including piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices such as damper and door closure interlocks shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. As-built drawings are to be provided, and a copy of them in Revit version 2025 provided on compact disk or DVD. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the ‘third party testing company’ requirement.

d. Certification documentation shall be provided to Contracting Officer 10 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and certification that all results of tests were within limits specified.

PART 2 PRODUCTS

2.1 MATERIALS

Material or equipment containing a weighted average of greater than 0.25 percent lead shall be prohibited in any potable water system intended for human consumption, and shall be certified in accordance with NSF/ANSI 61 or NSF 372.

2.2 HOT WATER RECIRCULATING PUMP

a. General:

(1) Centrifugal, single stage, pump. Driver shall be electric motor, close coupled or connected by flexible or magnetic coupling. Pump for hot water system shall be designed for quiet, trouble-free operation at a minimum of 180 degrees F water service and 150 psig.

(2) Mounting shall be in-line, vertical or horizontal as indicated on drawing schedules.

(3) Stamped or engraved stainless steel nameplate.

(4) Motors: Maximum 104 degrees F ambient temperature rise, drip-proof, for operation with current, voltage, phase and cycle shown in schedule on Electrical drawings, conforming to NEMA Type 4. Motors shall be equipped with thermal overload protection. When motor has cooled down it shall re-start automatically if the operating control has been left on and the system requires pump to start.

(5) Pump shall operate continuously with on-off switch, or with an HOA switch for automatically controlled pumps, for manual shut down. In the inlet and outlet piping of the pump, shutoff valves shall be installed to permit service to the pump, strainer, and check valve without draining the system.

(6) A check valve shall be installed in the pump discharge piping immediately downstream of the pump. A strainer with drain valve and removable strainer screen or basket shall be installed immediately upstream of the pump. Flexible pipe connectors and isolation pipe hangers shall be installed to prevent pump vibration from being transferred to adjacent piping and the building structure.

b. Horizontal, Wet-Rotor Circulators:

(1) Maintenance free, close-coupled pump and motor with maximum 3,300 rpm rotational speed.

(2) Bronze Cast iron Stainless steel body construction with ceramic shaft, plastic impeller, fluid lubricated bearings, no mechanical seal, and flanged soldered joint connections. Pump shall be capable of pumping the capacity scheduled on drawings.

(3) Bearings: Carbon type.

c. Horizontal, Permanently Lubricated Circulators:

(1) Close-coupled pump and motor with maximum 3,500 rpm rotational speed.

(2) Bronze Cast iron body construction with solid steel shaft, plastic impeller, carbon/silicon carbide mechanical seal, and flanged soldered joint connections. Pump shall be capable of pumping the capacity scheduled on drawings.

(3) Bearings: Permanently oil lubricated and sealed, stainless steel ball bearings.

d. Horizontal, In-Line Pumps:

(1) Flexibly-coupled pump and motor with maximum 1,800 rpm rotational speed.

(2) Bronze Bronze-fitted cast iron body construction with solid steel shaft, and flanged connections. Pump shall be capable of pumping the capacity scheduled on drawings.

(3) Bearings: Permanently oil lubricated and sealed, stainless steel ball bearings.

(4) Impeller shall be high grade, cast brass or bronze, accurately machined and properly balanced.

(5) Seal: Mechanical, with carbon-steel rotating ring, stainless steel spring, ceramic seat, and rubber bellows and gasket. Include water slinger on shaft between motor and seal.

e. Vertical, In-Line Pumps:

(1) Close-coupled pump and motor with maximum 3,600 rpm rotational speed.

(2) Bronze Bronze-fitted cast iron body construction with solid steel shaft and flanged connections. Pump shall be suitable for mounting in a vertical or horizontal position, and shall be capable of pumping the capacity scheduled on drawings.

(3) Bearings: Permanently oil lubricated and sealed, stainless steel ball bearings.

(4) Impeller shall be high grade, cast brass or bronze, accurately machined and properly balanced.

(5) Seal: Mechanical, with carbon-steel rotating ring, stainless steel spring, ceramic seat, and rubber bellows and gasket. Include water slinger on shaft between motor and seal.

2.3 DOMESTIC WATER PRESSURE BOOSTER SYSTEM

(a) Pump lead/lag/stand-by assignments shall alternate to automatically equalize the run time in similarly sized pumps. (This excludes the normal assignment of lead pump responsibility to the uniquely-sized smallest pump in three-pump systems.)

a. General: Provide a factory prefabricated, prewired and pretested multi-stage pumping system including variable speed drive motors, pressure regulating valves with integral check valves, pressure transducers, vibration pads/isolators, emergency switches, duplex flow switches, power and control panels, suction and discharge manifolds, butterfly isolation valves, ball drain valves, bypass loops with appropriate valves and check valves, low pressure cut-off switches, hydropneumatic tanks and accessories. All components shall be furnished by a single manufacturer and the system shall be the standard cataloged product of the manufacturer. All components shall be factory installed on a common structural steel skid and shall be completely tested in the factory before shipment. Manufacturer shall assume “unit responsibility” to ensure that all components effectively interface to execute the operation of the designed system.

b. System Operation and Configuration:

(1) System shall automatically maintain constant system pressure as scheduled on drawingsat the outlet of the pressure control valve and hydropneumatic tank check valve at all times. Suction/inlet pressure variessee scheduled quantities. The multi-stage pumping system shall includethe scheduled quantity of pumps.

(2) Duplex pump systems shall include both pumps sized for 100 percent of the total capacity.

(3) Triplex pump systems shall include lead and lag pumps each sized for 50 percent of the total capacity and one stand-by pump of equal capacity

(4) Quadraplex pump systems shall include a lead pump sized for 33 percent of the total capacity, with one additional standby pump of equal cpacity, arrangedin a N+1 configuration.

d. Vertical Multistage Pump: Pumps shall be vertical multistage short-coupled pumps with variable speed drive motor, low pressure cut-off switches and bypass loops with ball, butterfly, check valves as indicated on the Contract Drawings.

(1) Impellers: AISI Type 304 stainless steel, enclosed radial flow type.

(2) Balancing of Impellers: Each impeller shall be statically and dynamically balanced prior to assembly in pump casing.

(3) Pump shaft: Stainless steel Type 416.

(4) Lubrication: Water lubricated type pumprequiring no external lubrication.

(5) Pump Chambers and Casing: Stainless steel construction suitable for the maximum operating pressure of the booster system.

(6) Pump Bearings: Factory-installed antifriction bearings designed for the expected service life of the pump.

(7) Pump Head: Fabricated steel with continuous bypass for low seal pressure. Cast iron heads are prohibited. Pump head shall be lined same as pump barrel.

(8) Seal: Cartridge-type mechanical seal suitable for potable water service and rated for the maximum operating pressure and temperature of the system.Seal shall be rated at 175 psig maximum.

(9) Adjustable Spacer Coupling: Removable type required so that pump seal can be replaced without disturbing motor.

(10) Motor: NEMA-rated premium efficiency motor with integrated variable frequency drive. Motors shall be capable of operating at any point on the pump performance curve without overloading and shall conform to NEMA standards.

(11) Controls: Factory-mounted Grundfos CU 352 controller or approved equal providing constant pressure control, automatic pump staging/destaging, lead-lag alternation, low suction pressure protection, alarm monitoring, and communication capability with the BAS as indicated on the Contract Drawings.

e. Pressure Control: System pressure shall be maintained by the integrated variable speed control system utilizing pressure transducer feedback and automatic pump speed modulation. Pressure regulating valves for normal system pressure control shall not be required.

System shall automatically stage and destage pumps to maintain the scheduled discharge pressure over the specified flow range.

f. Hydropneumatic Tank: Bladder type, hydropneumatic, designed and constructed in accordance with requirements of the ASME BPVC and stamped with appropriate symbol. Tank shall include pre-pressurized, sealed-in air cushion which shall accommodate pressure increases and expanded water volumes in the tank. Tank shall include butyl rubber or poly-propylene liner in lower, or water side of chamber. Minimum working pressure of tank shall be 175 psig. Unit shall be designed and manufactured for domestic water applications. Insulate tank as specified. Check valve at hydropneumatic tank shall include small orifice for undue loading.

g. Power and Control Panel: Class "A" shadow box double NEMA 1 enclosure, UL labeled, bonderized double prime coated with baked enamel finish:

(1) Fused disconnect switches with external operating handles.

(2) Magnetic contactor for each motor with HOA switch.

(3) Door interlock.

(4) Thermal overload protection relay for each motor, three leg type.

(5) Running light for each motor.

(6) Power light for each motor.

(7) Control transformer, switch, circuit breaker, light.

(8) Lead pump failure protection.

h. Motor and Starter: Maximum 104 degrees F ambient temperature rise, drip-proof type motor, ball bearings, voltage and phase as shown in schedule on Electrical drawings, conforming to NEMA Type 4. Motor shall be of such capacity that brake horsepower required by driven equipment at normal rated capacity will not exceed nameplate rating of the motor. Refer to Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT. Provide each motor with automatic, fully enclosed, magnetic starter of type specified in Section 26 29 11, MOTOR CONTROLLERS.

i. Instrumentation: All instrumentation shall be factory installed and shall include the following 4-1/2 inch dial gages with shut-off cock.

(1) Pump pressure gage for each pump.

(2) System pressure gage.

(3) Suction pressure gage.

j. Operating and Emergency Controls:

(1) The pump station shall receive a 4-20mA signal from each pressure transducer, as provided by the pumping station manufacturer. A pressure transducer signal shall be provided for each pump controller. The differential pressure transducers shall monitor

system discharge pressure versus suction line pressure and provide an analog signal 4-20mA to the pump control software, and allow the variable speed pump controller, to provide a variable Volts/Hz output to the motor. Once the pressure drops below the set system pressure, the pump shall start and provide system pressure (as determined by the station operator or program), if this pressure cannot be maintained by one pump, the next pump in sequence shall operate in a lead/lag stand-by capacity to provide the extra flow and pressure automatically without the use of additional panels or alternators. The sequence of the pumps shall be field adjustable, and completely automatic without additional panels or alternator controls. The variable speed pump controller shall be completely integrated with the VFD. Special type motors shall not be allowed. Refer to Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT. Pump assignments shall alternate as described below. All program settings shall be based on centrifugal pump language and centrifugal pumps. Program settings shall be field adjustable to provide onsite adjustments. When the system experiences low demand, the variable speed pump controllers shall reduce the speed of each pump, until demand has stopped. Pump controllers shall stop each pump at zero demand, without the use of external switches or controls.

(2) The pumping system includes multiple pumps as indicated above. In three-pump systems, the lead pump, lag pump, and stand-by pump normally satisfy demand in three stages. The first (lead) pump operates initially, until the water demand exceeds the lead pump’s capacity. The second (lag) pump starts and the lead pump stops, allowing the lag pump to operate until the water demand exceeds the lag pump’s capacity. At this point, the lead pump restarts, and operates concurrently with the lag pump to add capacity when needed. The third pump is a stand-by pump, ready for operation if the lead or lag pump is taken out of service and the system demand requires two pumps. The system is to be configured so that at least two pumps can be in service if any one pump is taken out of service. Pump assignments shall alternate to automatically equalize the run time in similarly sized pumps, which excludes uniquely-sized lead pumps.

(3) The pump logic controller shall provide the following standard user-selectable features:

(a) Low Suction Pressure Alarm and Cut Out

(b) High Suction Pressure Alarm and Cut Out

(c) Low System Pressure Alarm

(d) High System Pressure Alarm and Cut Out

(e) High Temperature Alarm and Cut Out

(f) Low Level Alarm and Cut Out

(g) No-Flow Shut Down

(h) Audible/Visible alarm with push to silence feature

(i) Overload Failure Alarm

(j) Pump Failure Alarm

(k) Pump operating order assignments

(l) Minimum run timers to prevent short cycle operation.

(4) Provide auxiliary contacts for remote communication with the BAS, including the following input/output points:

(a) Domestic water supply pressure (analog input to BAS)

(b) Alarm condition activated (binary input to BAS)

(c) Run status of lead pump (binary input to BAS)

(d) Run status of lag pump (binary input to BAS)

(e) Run status of stand-by pump (binary input to BAS)

k. Factory Test: The booster system and its component parts shall undergo a complete operation flow test from zero to 100 percent design flow rate under the specified suction and net system pressure conditions. The system certification shall include copies of the test and test data as performed in the factory prior to shipment. Performance test certifications should be placed inside the system control panel and two extra copies shall be provided to the Contracting Officer with the installation manual.

PART 3 EXECUTION

3.1 STARTUP AND TESTING

a. Make tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. System Test: After installation is completed provide an operational test of the completed system including flow rates, pressure compliance, alarms and all control functions.

c. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

d. The CxA will observe startup and Contractor testing of selected equipment. Coordinate the startup and Contractor testing schedules with the Contracting Officer and CxA. Contractor shall provide a minimum of 10 working days prior to startup and testing.

3.2 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.3 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for four hours to instruct VA Personnel in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

-- End of Section --

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SECTION 22 12 16
9/15

FACILITY ELEVATED, POTABLE-WATER STORAGE TANKS

PART 1 GENERAL

1.1 DESCRIPTION

a. An elevated steel water storage tank and tower, completed, including risers, piping, drains, materials, labor, equipment, painting, sterilization and all accessories as required for a complete working system.

b. A complete listing of all acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

f.

g. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

h. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

i. Section 26 41 00, FACILITY LIGHTNING PROTECTION: Lightning Protection.

j.

1.3 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 360	(2016) Specification for Structural Steel Buildings

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A112.6.3	(2022) Floor Drains
ASME B16.1	(2020) Gray Iron Pipe Flanges and Flanged

	Fittings Classes 25, 125, and 250
ASME BPVC SEC IX	(2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A123/A123M	(2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A153/A153M	(2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A234/A234M	(2024) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA B300	(2024) Hypochlorites
AWWA C110/A21.10	(2021) Ductile-Iron and Gray-Iron Fittings
AWWA C115/A21.15	(2020) Flanged Ductile-Iron Pipe With Ductile-Iron or Gray-Iron Threaded Flanges
AWWA C151/A21.51	(2023) Ductile-Iron Pipe, Centrifugally Cast
AWWA C652	(2019) Disinfection of Water-Storage Facilities
AWWA D100	(2021) Welded Steel Tanks for Water Storage
AWWA D102	(2024) Coating Steel Water-Storage Tanks

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel
AWS D1.3/D1.3M	(2018) Structural Welding Code - Sheet Steel
AWS D1.4/D1.4M	(2018) Structural Welding Code - Reinforcing Steel

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS O-C-114	(1974) Calcium Hypochlorite, Technical

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 22	(2023) Standard for Water Tanks for Private Fire Protection
NFPA 25	(2020; TIA 21-1) Standard for the Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems
NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NSF INTERNATIONAL (NSF)

NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects
NSF 372	(2022) Drinking Water System Components - Lead Content

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910	Occupational Safety and Health Standards

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Erection Procedure; G, AE

List Of Five Completed Projects; G, AE

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-05 Design Data

Design Calculations; G, AE

Power, Signal And Control Wiring; G, AE

Plans, Sections, Details, Elevations And Attachment To Other Work; G, AE

Foundation, Footings And Calculation, Designed And Stamped By A

Professional Structural Engineer; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Certification Documentation; G, AE

SD-08 Manufacturer's Instructions

Training Plans; G, AE

Instructor Qualifications; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER Section 22 12 16, FACILITY ELEVATED, POTABLE-WATER STORAGE TANKS”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Piping.

(2) Valves.

(3) Drain.

(4) Steel Tank and Accessories including P.E. sealed drawings and calculations.

(5) Alarm System.

(6) OSHA approved climbing equipment for maintenance personnel.

(7) Tank Heating Equipment.

(8) Lightning Protection.

(9) Paint.

d. Design Calculations and completely detailed erection procedure with shop drawings.

e. Foundation, footings and calculation, designed and stamped by a professional Structural Engineer.

f. Power, signal and control wiring.

g. Plans, sections, details, elevations and attachment to other work.

h. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

i. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

a. Criteria:

(1) Manufacturer regularly and presently manufactures the item submitted as one of their principal products.

(2) Installer, or supplier of a service, has technical qualifications, experience, trained personnel, and facilities to perform the specified work.

(3) Submit a list of five completed projects of similar type and size with references with current position, address, telephone number, and e-mail.

b. System Requirements: AWWA D100 for steel tanks plus the following:

(1) Water Tank Design Loads:

(a) Weight of Tank and Water.

(b) Wind Load.

(c) Seismic Load.

(d) Combined Loads: Design for the worst case:

1. Gravity and wind.

2. Gravity and seismic.

(e) Snow Load.

(f) Design Code: AISC 360.

(2) IBC and local building codes.

c. Tank welding qualification procedures and personnel shall be in accordance with AWS D1.1/D1.1M, AWS D1.3/D1.3M and AWS D1.4/D1.4M.

d. Pipe welding qualification procedures and personnel shall be in accordance with ASME BPVC SEC IX.

e. Electrical components, devices and accessories shall be listed and labeled as defined in NFPA 70, Article 100, by a testing agency

acceptable to authorities having jurisdiction, and marked for intended use.

f. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

1.7 AS-BUILT DOCUMENTATION

a. Submit manufacturer's literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, substitutions and construction revisions shall be in electronic version on compact disc or DVD inserted into a three ring binder. All aspects of system operation and maintenance procedures, including piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices such as damper and door closure interlocks shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. As-built drawings are to be provided, and a copy of them in Revit version 2025 provided on compact disk or DVD. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the 'third party testing company' requirement.

d. Certification documentation shall be provided to Contracting Officer 10 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and certification that all results of tests were within limits specified.

PART 2 PRODUCTS

2.1 MATERIALS

Material or equipment containing a weighted average of greater than 0.25 percent lead is prohibited in any potable water system intended for human consumption, and shall be certified in accordance with NSF/ANSI 61 or NSF 372.

2.2 TANK

a. Shall be in accordance with NFPA 22, AWWA D100 and ASTM A36/A36M. Tank drawings shall be submitted with calculations stamped by a P.E.

licensed in the state of Connecticut.

b. Circular style with roof and floor, comprised of thermoset epoxy powder coated (NSF/ANSI 61 and AWWA D103), field ereected, bolted flat panels with self-supported roof and flat bottom. Tanks shall have horizontal wind stiffeners if required to meet wind design requirements. Tanks shall meet NFPA-22 standards and be insulated. .

c. The tank shall be constructed of 1/4 inch minimum thick steel plus .01 inch for corrosion allowance or as required to meet latest edition of AWWA D103. Cathodic protection required if recommended by the manufacturer. Tank shall be gasketed and sealed throughout. Tanks shall be thoroughly tested per manufacturer's recommendations. Any items identified shall be corrected based upon manufacturer's recommendations to the satisfaction of the VA.

d. The entire structure shall withstand a 134 mph winds per AWWA.

e. Design the tank to withstand seismic earthquake zone per AWWA .

f. The balcony and ladder loading shall be as specified in AWWA D100, furnish OSHA approved climbing equipment.

2.3 FOUNDATION

a. The elevated water tower contractor shall be responsible for the soil borings, design and installation of the foundation and footings.

b. The soil borings shall be performed by an independent and established soils engineering firm.

2.4 ACCESSORIES

a. Comply with NFPA 22 and AWWA D100. Main riser and overflow shall be schedule 40 steel, ASTM A53/A53M with schedule 40, ASTM A234/A234M, butt welded fittings.

b. The main tank riser shall be as indicated on plans.

c. Overflow: Provide a (as indicated on plans) steel pipe extended to grade with an overflow that is visible/conspicuous before spilling into a storm sewer inlet. The overflow shall be provided with an insect screen and splash block.

d. Rigging Lugs for future maintenance: Provide 1/4 inch minimum diameter holes. Lugs shall be located to access all portions of the tank and shall be capable of sustaining a load of 992 pounds.

e.

f. Riser: Steel riser shall be as indicated on plans inch diameter. Supply pipe shall be as indicated on plans inch diameter. Riser manhole shall be 24 inches minimum.

g. Perimeter Safety Railing for Roof of Tank: Provide a circular steel railing around perimeter of the roof of the tank. Railing shall be 5 foot guardrail - 2 rail system with OSHA approved toe board and safety cage. Provide 30" roof hatch as indicated on drawings.

h. Vent: Vent shall be welded to the cover plate of the center manhole on the roof or as recommended by the manufacturer. Vent shall be tank manufacturer's standard type mushroom vent with aluminum bird screen. Screening for vent shall conform to Section 5.7.2 of AWWA D100. Ensure fail-safe operation in the event that screen frosts over and the bottom of the screen shall be sufficiently elevated for snow consideration in the area.

i. Ladders and Safety Devices: Ladders and safety devices shall be provided in accordance with Sections 7.4 and 7.5 of AWWA D100. Location of ladders shall be as shown on the drawings. Sections 7.4 and 7.5 of AWWA D100 represent the minimum requirement. In addition, safety cage, rest platforms, roof ladder handrails, and other safety devices shall be provided as required by federal or local laws or regulations. OSHA approved outside ladder inclusive of anti-scale door and saf-t-climb.

j. Additional: Level sensors, tank heater with building automation system (BAS) integration capability/connectivity, tank insulation, and submersible, electric water storage tank mixer with NEMA 4X control box for BAS integration capability/connectivity - start/stop conrol, monitoring for power draw, and alarms.

2.5 PIPING

a. Pipe:

(1) Ductile Iron (interior riser pipe to be PVC with carbon steel double FF nozzle and internal 45° ell):

(a) Underground: AWWA C151/A21.51, working pressure 150 psig, exterior bituminous coating and interior cement lining. Bio-based materials shall be utilized when possible.

(b) In Pits: AWWA C115/A21.15, flanged joints only, working pressure 150 psig.

(2) Steel (Above Ground Only): Galvanized, ASTM A53/A53M, standard weight, threaded flanges, ASME B16.1, class 125.

b. Fittings: Ductile-iron, AWWA C110/A21.10.

2.6 DRAIN

a. ASME A112.6.3, Cast iron with light duty grate.

b. Connects with overflow.

c. Side outlet.

2.7 HIGH AND LOW WATER LEVEL ALARM SYSTEMS

a. Provide high, intermediate (noted as "fire" on plans) and low level devices for alarm monitoring and an intermediate device for tank water level status.

(1) All three water levels shall be indicated by their respective pilot lights; green for high, amber for intermediate and red for low water levels, and a buzzer for low, intermediate, and high water levels.

(2) Buzzer and the respective pilot lights at high, intermediate, and low water levels shall be energized while the high or low water level pilot device is actuated.

(3) Depressing a silencing button shall silence the buzzer indicating the water level and shall remain in OFF condition. The pilot light shall remain energized.

(4) Resetting the pilot light shall de-energize the pilot light and release the buzzer from its sealed-off condition.

(5) Provide alarming contacts to a DDC (Direct Digital Controls) remotely to the BAS operator. Coordinate alarms with Controls Contractor and control sequences.

2.8 TANK HEATING

a. Provide tank heating to protect the tank from freezing in accordance with NFPA 22 Chapter 16 Water Tank Heating. Maintain water temperature at or above 42 degrees F at all times.

b. Electric Heat: Nickel chromium, electric-resistance elements, helix wound, factory-wired with time delay for element staging, and overcurrent and overheat protective devices.

c. Provide control panel capable of integrating with campus building automation system (BAS). Integration to allow the BAS to have the capability for tank heater on/off control, setpoint adjustment, monitoring, and alarms. Coordinate with Controls Contractor and control sequences.

PART 3 EXECUTION

3.1 INSTALLATION

a. Install tank and tower in accordance with NFPA 22.

b. Install safety ladders in accordance with NFPA 22 and 29 CFR 1910 requirements.

3.2 TANK INSPECTION

Comply with AWWA and NFPA 25 requirements.

3.3 LEAKAGE TEST

Comply with AWWA, NFPA 22 and NFPA 25 requirements.

3.4 LIGHTNING PROTECTION

Comply with NFPA 22.

3.5 FIELD PAINTING

After construction, all weld seams, unprimed margins, field bolts, lugs and any area where the factory primer has been damaged shall be blast cleaned and re-primed with the same factory primer. Apply paint in accordance with NFPA 22 and AWWA D102. The exterior paint color and the VAMC logo shall be as directed by the VAMC.

3.6 METALLIC COATINGS

a. Tower members, protection shield, ladders, ladder safety cages, rigging lugs, overflow pipe and riser shall be zinc-coated (galvanized) by the hot-dip method, ASTM A123/A123M. Bolts and similar fasteners shall be zinc-coated by the hot-dip method, ASTM A153/A153M. Hot-dip galvanized surfaces abraded or burned off by welding operations shall be repaired with zinc repair compound in stick or powder form, applied in accordance with the manufacturer's printed instructions.

b. All interior surfaces and all interior components of the tank shall be zinc metallized, and all exterior surfaces of the tank and any additional surfaces not previously designated shall be aluminum metallized after construction and testing.

c.

3.7 FLUSHING

After tests and painting , flush tank and connecting pipelines.

3.8 DISINFECTION/ STERILIZATION

a. After flushing, thoroughly disinfect and sterilize the tank and connecting piping with a chlorine and hypochlorite solution as required by AWWA C652. Following a contact period of 24 hours, flush the system with clean water until the residual chlorine content is not greater than that of the water supply. All valves in water line being disinfected shall be opened and closed several times during the 24 hour period. Hypochlorite material shall comply with FS O-C-114 grade B.

b. Hypochlorite material shall comply with AWWA B300.

c. Contracting Officer shall witness flush and disinfection/sterilization. Contractor shall coordinate schedules with the Contracting Officer and CxA. Contractor shall submit request to Contracting Officer a minimum of 10 working days prior to flushing, disinfection/sterilization, startup, and testing.

3.9 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.10 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for four hours to instruct VA Personnel in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

-- End of Section --

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DIVISION 22 - PLUMBING

SECTION 22 13 00

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SECTION 22 13 00
9/20

FACILITY SANITARY AND VENT PIPING

PART 1 GENERAL

1.1 DESCRIPTION

a. This section pertains to sanitary sewer and vent systems, including piping, equipment and all necessary accessories as designated in this section.

b. A complete listing of common acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e.

f. Section 07 84 00, FIRESTOPPING: Penetrations in rated enclosures.

g. Section 07 92 00, JOINT SEALANTS: Sealant products.

h. Section 09 91 00, PAINTING: Preparation and finish painting and identification of piping systems.

i. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING: Pipe Hangers and Supports, Materials Identification.

j. Section 22 07 11, PLUMBING INSULATION.

k. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS

l. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

m. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A112.36.2M	(2022) Cleanouts
ASME A112.6.3	(2022) Floor Drains
ASME B1.20.1	(2013; R 2018) Pipe Threads, General Purpose (Inch)
ASME B16.1	(2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250
ASME B16.4	(2021) Gray Iron Threaded Fittings; Classes 125 and 250
ASME B16.15	(2024) Cast Copper Alloy Threaded Fittings Classes 125 and 250
ASME B16.18	(2021) Cast Copper Alloy Solder Joint Pressure Fittings
ASME B16.21	(2021) Nonmetallic Flat Gaskets for Pipe Flanges
ASME B16.22	(2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings
ASME B16.23	(2021) Cast Copper Alloy Solder Joint Drainage Fittings - DWV
ASME B16.24	(2022) Cast Copper Alloy Pipe Flanges, Flanged Fittings, and Valves Classes 150, 300, 600, 900, 1500, and 2500
ASME B16.29	(2022) Wrought Copper and Wrought Copper Alloy Solder-Joint Drainage Fittings - DWV
ASME B16.39	(2020) Standard for Malleable Iron Threaded Pipe Unions; Classes 150, 250, and 300
ASME B18.2.1	(2012; R 2021) Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

ASSE 1044	(2023) Performance Requirements for Trap Seal Primer - Drainage Types and Electric Design Types
ASSE 1079	(2012) Performance Requirements for Dielectric Pipe Unions

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated,

	Welded and Seamless
ASTM A74	(2021) Standard Specification for Cast Iron Soil Pipe and Fittings
ASTM A888	(2024) Standard Specification for Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications
ASTM B32	(2020) Standard Specification for Solder Metal
ASTM B43	(2020) Standard Specification for Seamless Red Brass Pipe, Standard Sizes
ASTM B88	(2022) Standard Specification for Seamless Copper Water Tube
ASTM B306	(2020) Standard Specification for Copper Drainage Tube (DWV)
ASTM B687	(1999; R 2023) Standard Specification for Brass, Copper, and Chromium-Plated Pipe Nipples
ASTM B813	(2024) Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube
ASTM B828	(2023) Standard Practice for Making Capillary Joints by Soldering of Copper and Copper Alloy Tube and Fittings
ASTM C564	(2020a) Standard Specification for Rubber Gaskets for Cast Iron Soil Pipe and Fittings
ASTM F1545	(2015a; R 2021) Standard Specification for Plastic-Lined Ferrous Metal Pipe, Fittings and Flanges

CAST IRON SOIL PIPE INSTITUTE (CISPI)

CISPI HANDBOOK	(2006) Cast Iron Soil Pipe and Fittings Handbook
CISPI 301	(2018) Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications
CISPI 310	(2012) Coupling for Use in Connection with Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications

COPPER DEVELOPMENT ASSOCIATION (CDA)

CDA A4015	(2016; 14/17) Copper Tube Handbook

INTERNATIONAL CODE COUNCIL (ICC)

ICC IPC	(2024) International Plumbing Code

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-123	(2018) Non-Ferrous Threaded and Solder-Joint Unions for Use with Copper Water Tube

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawing; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Training Plans; G, AE

Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER Section 22 13 00, FACILITY SANITARY AND VENT PIPING”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Piping.

(2) Floor Drains.

(3) Cleanouts.

(4) Trap Seal Protection.

(5) Penetration Sleeves.

(6) Pipe Fittings.

(7) Traps.

(8) Exposed Piping and Fittings.

d. Detailed shop drawing of clamping device and extensions when required in connection with the waterproofing membrane or the floor drain.

e. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

f. Completed System Readiness Checklist provided by the CXA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

g. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

a. Bio-Based Materials: For products designated by the USDA’s bio-based Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

b. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in Paragraph “AS-BUILT DOCUMENTATION” of Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

PART 2 PRODUCTS

2.1 SANITARY WASTE, DRAIN, AND VENT PIPING

a. Cast iron waste, drain, and vent pipe and fittings.

(1) Cast iron waste, drain, and vent pipe and fittings shall be used for the following applications:

(a) Pipe buried in or in contact with earth.

(b) Sanitary pipe extensions to a distance of approximately 5 feet outside of the building.

(c) Interior waste and vent piping above grade.

(2) Cast iron Pipe shall be bell and spigot or hubless (plain end or no-hub or hubless).

(3) The material for all pipe and fittings shall be cast iron soil pipe and fittings and shall conform to the requirements of CISPI 301, ASTM A888, or ASTM A74.

(4) Cast iron pipe and fittings shall be made from a minimum of 95 percent post-consumer recycled material.

(5) Joints for hubless pipe and fittings shall conform to the manufacturer’s installation instructions. Couplings for hubless joints shall conform to CISPI 310. Joints for hub and spigot pipe shall be installed with compression gaskets conforming to the requirements of ASTM C564.

b. Copper Tube, (DWV):

(1) Copper DWV tube sanitary waste, drain and vent pipe may be used for piping above ground, except for urinal drains.

(2) The copper DWV tube shall be drainage type, drawn temper conforming to ASTM B306.

(3) The copper drainage fittings shall be cast copper or wrought copper conforming to ASME B16.23 or ASME B16.29.

(4) The joints shall be lead free, using a water flushable flux, and conforming to ASTM B32.

2.2 PUMP DISCHARGE PIPING

a. Galvanized steel pump discharge pipe and fittings:

(1) Galvanized steel pipe shall be Schedule 40 weight class conforming to ASTM A53/A53M, with square cut grooved or threaded ends to match joining method.

(2) Fittings shall be Class 125, gray-iron threaded fittings conforming to ASME B16.4.

(3) Unions shall be Class 150 hexagonal-stock body with ball and socket, metal to metal, bronze seating surface, malleable iron conforming to ASME B16.39 with female threaded ends.

(4) Flanges shall be Class 125 cast iron conforming to ASME B16.1.

(a) Flange gaskets shall be full face, flat nonmetallic, asbestos free conforming to ASME B16.21.

(b) Flange nuts and bolts shall be carbon steel conforming to ASME B18.2.1.

b. Copper pump discharge pipe and fittings:

(1) Copper tube shall be hard drawn Type L conforming to ASTM B88.

(2) Fittings shall be cast copper alloy conforming to ASME B16.18 or wrought copper conforming to ASME B16.22 with solder joint ends.

(3) Unions shall be copper alloy, hexagonal stock body with ball and socket, metal to metal seating surface conforming to MSS SP-123 with female solder-joint or threaded ends.

(4) Flanges shall be Class 150, cast copper conforming to ASME B16.24 with solder-joint end.

(a) Flange gaskets shall be full face, flat nonmetallic, asbestos free conforming to ASME B16.21.

(b) Flange nuts and bolts shall be carbon steel conforming to ASME B18.2.1.

(5) Solder shall be lead free, water flushable flux conforming to ASTM B32 and ASTM B813.

2.3 EXPOSED WASTE PIPING

a. Chrome plated brass piping of full iron pipe size shall be used in finished rooms for exposed waste piping connecting fixtures, casework, cabinets, equipment and reagent racks when not concealed by apron including those furnished by the Government or specified in other sections.

(1) The Pipe shall meet ASTM B43, regular weight.

(2) The Fittings shall conform to ASME B16.15.

(3) Nipples shall conform to ASTM B687, Chromium-plated.

(4) Unions shall be brass or bronze with chrome finish. Unions 2-1/2 inches and larger shall be flange type with approved gaskets.

b. In unfinished Rooms such as mechanical Rooms and Kitchens, Chrome-plated brass piping is not required. The pipe materials specified under the paragraph “Sanitary Waste, Drain, and Vent Piping” can be used. The sanitary pipe in unfinished rooms shall be painted as

specified in Section 09 91 00, PAINTING.

2.4 SPECIALTY PIPE FITTINGS

a. Transition pipe couplings shall join piping with small differences in outside diameters or different materials. End connections shall be of the same size and compatible with the pipes being joined. The transition coupling shall be elastomeric, sleeve type reducing or transition pattern and include shear and corrosion resistant metal, tension band and tightening mechanism on each end. The transition coupling sleeve coupling shall be of the following material:

(1) For cast iron soil pipes, the sleeve material shall be rubber conforming to ASTM C564.

(2) For dissimilar pipes, the sleeve material shall be PVC conforming to ASTM D5926, or other material compatible with the pipe materials being joined.

b. The dielectric fittings shall conform to ASSE 1079 with a pressure rating of 125 psig at a minimum temperature of 180 degrees F. The end connection shall be solder joint copper alloy and threaded ferrous.

c. Dielectric flange insulating kits shall be of non-conducting materials for field assembly of companion flanges with a pressure rating of 150 psig. The gasket shall be neoprene or phenolic. The bolt sleeves shall be phenolic or polyethylene. The washers shall be phenolic with steel backing washers.

d. The di-electric nipples shall be electroplated steel nipple complying with ASTM F1545 with a pressure rating of 300 psig at 225 degrees F. The end connection shall be male threaded. The lining shall be inert and noncorrosive propylene.

2.5 CLEANOUTS

a. Cleanouts shall be the same size as the pipe, up to 4 inches; and not less than 4 inches for larger pipe. Cleanouts shall be easily accessible and shall be gastight and watertight. Minimum clearance of 24 inches shall be provided for clearing a clogged sanitary line.

b. (FCO-A)Floor cleanouts shall be gray iron housing with clamping device and round, secured, scoriated, gray iron cover conforming to ASME A112.36.2M. A gray iron ferrule with hubless, socket, inside calk or spigot connection and counter sunk, taper-thread, brass or bronze closure plug shall be included. The frame and cover material and finish shall be nickel-bronze copper alloy with a square shape. The cleanout shall be vertically adjustable for a minimum of 2 inches. When a waterproof membrane is used in the floor system, clamping collars shall be provided on the cleanouts. Cleanouts shall consist of wye fittings and eighth bends with brass or bronze screw plugs. Cleanouts in the resilient tile floors, quarry tile and ceramic tile floors shall be provided with square top covers recessed for tile insertion. In the carpeted areas, carpet cleanout markers shall be provided. Two way cleanouts shall be provided where indicated in the contract document and at every building exit.(FCO-B)The loading classification for cleanouts in sidewalk areas or subject to vehicular traffic shall be heavy duty type.

c. (CO)Cleanouts shall be provided at or near the base of the vertical stacks with the cleanout plug located approximately 24 inches above the floor. If there are no fixtures installed on the lowest floor, the cleanout shall be installed at the base of the stack. The cleanouts shall be extended to the wall access cover. Cleanout shall consist of sanitary tees. Nickel-bronze square frame and stainless steel cover with minimum opening of 6 by 6 inches shall be furnished at each wall cleanout. Where the piping is concealed, a fixture trap or a fixture with integral trap, readily removable without disturbing concealed pipe, shall be accepted as a cleanout equivalent providing the opening to be used as a cleanout opening is the size required.

d. In horizontal runs above grade, cleanouts shall consist of cast brass tapered screw plug in fitting or caulked/hubless cast iron ferrule. Plain end (hubless) piping in interstitial space or above ceiling may use plain end (hubless) blind plug and clamp.

e. (YCO) Yard cleanouts shall be provided where indicated on the drawings. Cleanouts shall be round, dura-coated cast iron, size as listed on drawings. Cleanouts shall consist of double-flanged housing with heavy-duty, secured scoriated dura-coated cast iron cover complete with lifting device. Provide bronze cleanout plug with gas/water-tight seal. Yard cleanouts shall be installed flush with finished grade and set to provide accessibility for maintenance.

2.6 FLOOR DRAINS

a. General Data: floor drain shall comply with ASME A112.6.3. A caulking flange, inside gasket, or hubless connection shall be provided for connection to cast iron pipe, screwed or no hub outlets for connection to steel pipe. The drain connection shall be bottom outlet. A membrane clamp and extensions shall be provided, if required, where installed in connection with waterproof membrane. Puncturing membrane other than for drain opening shall not be permitted. Double drainage pattern floor drains shall have integral seepage pan for embedding into floor construction, and weep holes to provide adequate drainage from pan to drain pipe. For drains not installed in connection with a waterproof membrane, a flashing membrane, 24 inches square or another approved waterproof membrane shall be provided.

b. Type A (FD-A) medium duty (non-traffic) floor drain shall comply with ASME A112.6.3. The type A floor drain shall have a cast iron body, double drainage pattern, clamping device, light duty nickel bronze adjustable strainer with round or square grate of 6 inches width or diameter minimum for toilet rooms, showers and kitchens.

c. Type B (FD-B) floor drain shall comply with ASME A112.6.3. The type B floor drain shall have a heavy, cast iron body, double drainage pattern, heavy non-tilting ductile iron grate not less than 9 inches round, removable sediment bucket. Clearance between body and bucket shall be ample for free flow of waste water. For traffic use, an extra heavy duty load classification ductile iron grate shall be provided.

d. Type C (FD-C) floor drain shall comply with ASME A112.6.3. The type C floor drain shall have a heavy, cast iron body, double drainage pattern, heavy non-tilting ductile iron grate not less than 12 inches square, removable sediment bucket. Clearance between body and bucket shall be ample for free flow of waste water. For traffic use, an

extra heavy duty load classification ductile iron grate shall be provided.

e. Type D (FD-D) floor drain shall comply with ASME A112.6.3. The type D floor drain shall have a cast iron body, double drainage pattern and clamping device, less grate and sediment basket but with dome type secondary strainer. The drain shall be 8 inches in diameter or 8 inches square and approximately 6 inches deep. The interior and exposed exterior surfaces and rim shall have an acid resisting finish for indirect waste in sanitary areas.

f. Type E (FD-E) floor drain shall comply with ASME A112.6.3. The type E floor drain shall have a cast iron body, double drainage pattern and clamping device, less grate and sediment basket but with dome type secondary strainer. The drain shall be 8 inches in diameter or 8 inches square and approximately 6 inches deep. The interior and exposed exterior surfaces and rim shall have an acid resisting finish for indirect waste in sanitary areas.

g. Type A (FS-A) floor sink shall be constructed of cast iron body with acid-resistant coated interior, and shall be 16 inches square, and 12 inches deep. The floor sink shall be provided with a 6 inch bottom outlet connection. The rim and grate shall be nickel bronze, with a heavy-duty loose-set grate which shall be cut out to accommodate drain pipes to limit splashing. The receptor shall include a stainless steel mesh sediment bucket. The unit shall be provided with a seepage flange and clamping device.

h. Type A (TD-A) trench drain shall be a low-profile, pre-sloped shower trench system constructed of Type 304 stainless steel with adjustable height channel supports and a continuous internal slope. System shall include a nominal 2 inch diameter outlet connection and an integral waterproofing membrane clamping collar.

Trench drain lengths and configuration shall be as indicated on the Contract Drawings.

Grates shall be removable Type 304 stainless steel with a linear slotted pattern suitable for shower applications.

Prior to fabrication and ordering, manufacturer shall submit detailed layout drawings indicating trench sections, lengths, outlet locations, and dimensions coordinated with the project layout for approval.

h. Type B (TD-B) trench drain shall be modular, pre-sloped, heavy-duty polymer concrete, epoxy coated steel, or ductile iron system with a nominal 6 inch wide channel and a continuous 0.6% slope. System shall include ductile iron or steel integral rail as required to meet load class with integral catch basins with removable sediment buckets and 4 inch diameter outlets. Trench drain lengths and configuration shall be as indicated on the Contract Drawings.

Grates shall be ductile iron, E rated, locking type with longitudinal slotted ADA-compliant pattern.

Prior to fabrication and ordering, manufacturer shall submit detailed layout drawings indicating trench sections, lengths, outlet locations, catch basins, and dimensions coordinated with the project layout for approval.

2.7 TRAPS

Traps shall be provided on all sanitary branch waste connections from fixtures or equipment not provided with traps. Exposed brass shall be polished brass chromium plated with nipple and set screw escutcheons. Concealed traps may be rough cast brass or same material as the piping they are connected to. Slip joints are prohibited on sewer side of trap. Traps shall correspond to fittings on cast iron soil pipe or steel pipe respectively, and size shall be as required by connected service or fixture.

2.8 PRIMER VALVES AND TRAP SEAL PRIMER SYSTEMS

a. Trap Primer (TP-A and TP-B): The trap seal primer shall be an individual electronic trap priming device conforming to ASSE 1044.

(1) The trap primer shall be a self-contained, electronically controlled device designed to automatically deliver water to floor drain traps at adjustable or factory preset intervals. The unit shall not require a centralized or multi-zone controller.

(2) Each unit shall consist of a solenoid valve, integral timing module, and air gap assembly factory assembled as a single device. Body components shall be corrosion-resistant metal suitable for potable water service.

(3) The solenoid valve shall be normally closed, suitable for continuous duty, and rated for a minimum working pressure range of 20 psi to 100 psi.

(4) The unit shall operate on 120 VAC power with integral cord and plug connection or field-wired connection where required. Electrical components shall be UL listed and provided with integral overcurrent protection.

(5) The device shall incorporate an integral air gap to prevent backflow; separate vacuum breaker devices are not required unless indicated otherwise.

(6) Units shall be installed in accordance with manufacturer’s instructions and in compliance with NFPA 70 and applicable plumbing codes.

2.9 PENETRATION SLEEVES

A sleeve flashing device shall be provided at points where pipes pass through membrane waterproofed floors or walls. The sleeve flashing device shall be manufactured, cast iron fitting with clamping device that forms a sleeve for the pipe floor penetration of the floor membrane. A galvanized steel pipe extension shall be included in the top of the fitting that shall extend 2 inches above finished floor and galvanized steel pipe extension in the bottom of the fitting that shall extend through the floor slab. A waterproof caulked joint shall be provided at the top hub.

PART 3 EXECUTION

3.1 PIPE INSTALLATION

a. The pipe installation shall comply with the requirements of ICC IPC and these specifications.

b. Branch piping shall be installed for waste from the respective piping systems and connect to all fixtures, valves, cocks, outlets, casework, cabinets and equipment, including those furnished by the Government or specified in other sections.

c. Pipe shall be round and straight. Cutting shall be done with proper tools. Pipe shall be reamed to full size after cutting.

d. All pipe runs shall be laid out to avoid interference with other work.

e. The piping shall be installed above accessible ceilings where possible.

f. The piping shall be installed to permit valve servicing or operation.

g. The piping shall be installed free of sags and bends.

h. Seismic restraint shall be installed where required by code.

i. Changes in direction for soil and waste drainage and vent piping shall be made using appropriate branches, bends and long sweep bends. Sanitary tees and short sweep quarter bends may be used on vertical stacks if change in direction of flow is from horizontal to vertical. Long turn double wye branch and eighth bend fittings shall be used if two fixtures are installed back to back or side by side with common drain pipe. Straight tees, elbows, and crosses may be used on vent lines. Do not change direction of flow greater than 90 degrees. Proper size of standard increaser and reducers shall be used if pipes of different sizes are connected. Reducing size of drainage piping in direction of flow is prohibited.

j. Buried soil and waste drainage and vent piping shall be laid beginning at the low point of each system. Piping shall be installed true to grades and alignment indicated with unbroken continuity of invert. Hub ends shall be placed upstream. Required gaskets shall be installed according to manufacturer’s written instruction for use of lubricants, cements, and other installation requirements.

k. Cast iron piping shall be installed according to CISPI HANDBOOK, Chapter IV, “Installation of Cast Iron Soil Pipe and Fittings”

l. Aboveground copper tubing shall be installed according to CDA A4015.

m. If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

3.2 JOINT CONSTRUCTION

a. Hubless or No-hub, cast iron piping shall be joined in accordance with CISPI HANDBOOK for hubless piping coupling joints.

b. For threaded joints, thread pipe with tapered pipe threads according

to ASME B1.20.1. The threads shall be cut full and clean using sharp disc cutters. Threaded pipe ends shall be reamed to remove burrs and restored to full pipe inside diameter. Pipe fittings and valves shall be joined as follows:

(1) Apply appropriate tape or thread compound to external pipe threads unless dry seal threading is required by the pipe service.

(2) Pipe sections with damaged threads shall be replaced with new sections of pipe.

c. Copper tube and fittings with soldered joints shall be joined according to ASTM B828. A water flushable, lead free flux conforming to ASTM B813 and a lead-free alloy solder conforming to ASTM B32 shall be used.

3.3 SPECIALTY PIPE FITTINGS

a. Transition coupling shall be installed at pipe joints with small differences in pipe outside diameters.

b. Dielectric fittings shall be installed at connections of dissimilar metal piping and tubing.

3.4 PIPE HANGERS, SUPPORTS AND ACCESSORIES

a. All piping shall be supported according to ICC IPC, Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING, and these Specifications. Where conflicts arise between these the code and Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING the most restrictive or the requirement that specifies supports with highest loading or shortest spacing shall apply.

b. Hangers, supports, rods, inserts and accessories used for pipe supports shall be painted according to Section 09 91 00, PAINTING. Electroplated copper hanger rods, hangers and accessories may be used with copper tubing.

c. Horizontal piping and tubing shall be supported within 12 inches of each fitting or coupling.

d. Horizontal cast iron piping shall be supported with the following maximum horizontal spacing and minimum hanger rod diameters:

(1) NPS 1-1/2 inch to NPS 2 inch: 60 inches with 3/8 inch rod.

(2) NPS 3 inch: 60 inches with 1/2 inch rod.

(3) NPS 4 inch to NPS 5 inch: 60 inches with 5/8 inch rod.

(4) NPS 6 inch to NPS 8 inch: 60 inches with 3/4 inch rod.

(5) NPS 10 inch to NPS 12 inch: 60 inch with 7/8 inch rod.

e. Vertical piping and tubing shall be supported at the base, at each floor, and at intervals no greater than 15 feet.

f. In addition to the requirements in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING, Floor, Wall and Ceiling Plates, Supports, Hangers shall have the following characteristics:

(1) Solid or split unplated cast iron.

(2) All plates shall be provided with set screws.

(3) Height adjustable clevis type pipe hangers.

(4) Adjustable floor rests and base flanges shall be steel.

(5) Hanger rods shall be low carbon steel, fully threaded or threaded at each end with two removable nuts at each end for positioning rod and hanger and locking each in place.

(6) Riser clamps shall be malleable iron or steel.

(7) Rollers shall be cast iron.

(8) See Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING, for requirements on insulated pipe protective shields at hanger supports.

g. Miscellaneous materials shall be provided as specified, required, directed or as noted in the contract documents for proper installation of hangers, supports and accessories. If the vertical distance exceeds 20 feet for cast iron pipe additional support shall be provided in the center of that span. All necessary auxiliary steel shall be provided to provide that support.

h. Cast escutcheon with set screw shall be provided at each wall, floor and ceiling penetration in exposed finished locations and within cabinets and millwork.

i. Penetrations:

(1) Fire Stopping: Where pipes pass through fire partitions, fire walls, smoke partitions, or floors, a fire stop shall be installed that provides an effective barrier against the spread of fire, smoke and gases as specified in Section 07 84 00, FIRESTOPPING. Clearances between raceways and openings shall be completely filled and sealed with the fire stopping materials.

(2) Water proofing: At floor penetrations, clearances shall be completely sealed around the pipe and make watertight with sealant as specified in Section 07 92 00, JOINT SEALANTS.

j. Exhaust vents shall be extended separately through roof. Sanitary vents shall not connect to exhaust vents.

3.5 TESTS

a. Sanitary waste and drain systems shall be tested either in its entirety or in sections.

b. Waste System tests shall be conducted before trenches are backfilled or fixtures are connected. A water test or air test shall be conducted, as directed.

(1) If entire system is tested for a water test, tightly close all openings in pipes except highest opening, and fill system with water to point of overflow. If the waste system is tested in sections, tightly plug each opening except highest opening of section under test, fill each section with water and test with at least a 10 foot head of water. In testing successive sections, test at least upper 10 feet of next preceding section so that each joint or pipe except upper most 10 feet of system has been submitted to a test of at least a 10 foot head of water. Water shall be kept in the system, or in portion under test, for at least 15 minutes before inspection starts. System shall then be tight at all joints.

(2) For an air test, an air pressure of 5 psig gauge shall be maintained for at least 15 minutes without leakage. A force pump and mercury column gauge shall be used for the air test.

(3) After installing all fixtures and equipment, open water supply so that all p-traps can be observed. For 15 minutes of operation, all p-traps shall be inspected for leaks and any leaks found shall be corrected.

(4) Final Tests: Either one of the following tests may be used.

(a) Smoke Test: After fixtures are permanently connected and traps are filled with water, fill entire drainage and vent systems with smoke under pressure of 1 inch of water with a smoke machine. Chemical smoke is prohibited.

(b) Peppermint Test: Introduce 2 ounces of peppermint into each line or stack.

3.6 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this section of the specification shall be tested as part of a larger system.

3.7 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for four hours to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

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SECTION 22 14 00
9/15

FACILITY STORM DRAINAGE

PART 1 GENERAL

1.1 DESCRIPTION

a. This section describes the requirements for storm drainage systems, including piping and all necessary accessories as designated in this section.

b. A complete listing of all acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 07 84 00, FIRESTOPPING: Penetrations in rated enclosures.

f. Section 07 92 00, JOINT SEALANTS.

g. Section 09 91 00, PAINTING: Preparation and finish painting and identification of piping systems.

h.

i. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING: Pipe Hangers and Supports, Materials Identification.

j. Section 22 05 33, HEAT TRACING FOR PLUMBING PIPING.

k. Section 22 07 11, PLUMBING INSULATION.

l. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A112.6.4 (2022) Roof, Deck and Balcony Drains

ASME A112.36.2M (2022) Cleanouts

ASME B1.20.1	(2013; R 2018) Pipe Threads, General Purpose (Inch)
ASME B16.18	(2021) Cast Copper Alloy Solder Joint Pressure Fittings
ASME B16.22	(2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings
ASME B16.23	(2021) Cast Copper Alloy Solder Joint Drainage Fittings - DWV
ASME B16.29	(2022) Wrought Copper and Wrought Copper Alloy Solder-Joint Drainage Fittings - DWV

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

ASSE 1079	(2012) Performance Requirements for Dielectric Pipe Unions

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A74	(2021) Standard Specification for Cast Iron Soil Pipe and Fittings
ASTM A183	(2014; R 2020) Standard Specification for Carbon Steel Track Bolts and Nuts
ASTM A536	(2024) Standard Specification for Ductile Iron Castings
ASTM A888	(2024) Standard Specification for Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications
ASTM B32	(2020) Standard Specification for Solder Metal
ASTM B75/B75M	(2020) Standard Specification for Seamless Copper Tube
ASTM B88	(2022) Standard Specification for Seamless Copper Water Tube
ASTM B306	(2020) Standard Specification for Copper Drainage Tube (DWV)
ASTM B584	(2022) Standard Specification for Copper Alloy Sand Castings for General Applications
ASTM B828	(2023) Standard Practice for Making Capillary Joints by Soldering of Copper

and Copper Alloy Tube and Fittings

ASTM B813	(2024) Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube
ASTM C564	(2020a) Standard Specification for Rubber Gaskets for Cast Iron Soil Pipe and Fittings
ASTM C1173	(2022) Standard Specification for Flexible Transition Couplings for Underground Piping Systems
ASTM D2321	(2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications
ASTM D5926	(2015; R 2020) Standard Specification for Poly (Vinyl Chloride) (PVC) Gaskets for Drain, Waste, and Vent (DWV), Sewer, Sanitary, and Storm Plumbing Systems
ASTM F477	(2014; R 2021) Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe
ASTM F1545	(2015a; R 2021) Standard Specification for Plastic-Lined Ferrous Metal Pipe, Fittings and Flanges

COPPER DEVELOPMENT ASSOCIATION (CDA)

CDA A4015	(2016; 14/17) Copper Tube Handbook

CAST IRON SOIL PIPE INSTITUTE (CISPI)

CISPI 301	(2018) Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications
CISPI 310	(2012) Coupling for Use in Connection with Hubless Cast Iron Soil Pipe and Fittings for Sanitary and Storm Drain, Waste, and Vent Piping Applications
CISPI HANDBOOK	(2006) Cast Iron Soil Pipe and Fittings Handbook

INTERNATIONAL CODE COUNCIL (ICC)

ICC IPC	(2024) International Plumbing Code

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawing Of Clamping Device And Extensions; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Certification Documentation; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER Section 22 14 00, FACILITY STORM DRAINAGE”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Pipe and Fittings.

(2) Specialty Pipe Fittings.

(3) Cleanouts.

(4) Roof Drains.

(5) Expansion Joints.

(6) Downspout Nozzles.

(7) Sleeve Flashing Devices.

d. Detailed shop drawing of clamping device and extensions when required in connection with the waterproofing membrane.

e. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section

22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

f. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

1.7 AS-BUILT DOCUMENTATION

a. Submit manufacturer's literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, substitutions and construction revisions shall be in electronic version on compact disc or DVD inserted into a three ring binder. All aspects of system operation and maintenance procedures, including piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices such as damper and door closure interlocks shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. As-built drawings are to be provided, and a copy of them in the latest Auto-CADversion provided on compact disk or DVD. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the 'third party testing company' requirement.

d. Certification documentation shall be provided to Contracting Officer 10 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and certification that all results of tests were within limits specified.

PART 2 PRODUCTS

2.1 STORM WATER DRAIN PIPING

a. Cast Iron Storm Pipe and Fittings:

(1) Cast iron storm pipe and fittings shall be used for the following applications:

(a) Pipe buried in or in contact with earth.

(b) Extension of pipe to a distance of approximately 5 feet outside of building walls.

(c) Interior storm piping above grade.

(d) All mechanical equipment rooms or other areas containing mechanical air handling equipment.

(2) The cast iron storm pipe shall be hubless (plain end or no-hub) as required by selected jointing method.

(3) The material for all pipe and fittings shall be cast iron soil pipe and fittings and shall conform to the requirements of CISPI 301, ASTM A888, or ASTM A74.

(4) Joints for hubless pipe and fittings shall conform to the manufacturer’s installation instructions. Couplings for hubless joints shall conform to CISPI 310. Joints for hub and spigot pipe shall be installed with compression gaskets conforming to the requirements of ASTM C564.

b. Copper Tube, (DWV): May be used for piping above ground.

(1) The copper DWV tube shall be drainage type, drawn temper conforming to ASTM B306.

(2) The copper drainage fittings shall be cast copper or wrought copper conforming to ASME B16.23 or ASME B16.29.

(3) The joints shall be lead free, using a water flushable flux, and conforming to ASTM B32.

c. Roof drain piping and body of drain in locations where the outdoor conditions are subject to freezing shall be insulated.

2.2 PUMPED DRAIN PIPING

a. Pumped drain piping 3 inches and less shall be copper tube conforming to ASTM B88, type K or L. For pumped drain piping 4 inches and greater, galvanized steel conforming to ASTM A53/A53M, seamless, schedule 40 may be used.

b. Pumped drain pipe fittings shall comply with the following:

(1) Wrought copper or bronze castings for use with copper tube conforming to ASME B16.18 and ASME B16.22.

(2) Unions, for use with copper tube up to 2 inches shall be cast with bronze, conforming to ASME B16.18 and ASTM B584 with solder or braze joints.

(3) Grooved fittings, for use with copper tube 2-1/2 to 4 inch shall be wrought copper conforming to ASTM B75/B75M, alloy C12200, 5 to 6 inch bronze castings conforming to ASTM B584.

(4) Mechanical grooved couplings shall have a ductile iron housing conforming to ASTM A536 (Grade 65-45-12) elastomer gasket suitable for potable water service and process temperature and steel track head bolts conforming to ASTM A183, housing shall be coated with colored alkyd enamel paint.

c. Adapters shall be provided for joining pipe with different end connections.

d. The solder shall be lead free using a water flushable, non-corrosive flux conforming to ASTM B32.

e. Dielectric fittings and specialties shall be provided when joining pipe of dissimilar metals.

2.3 SPECIALTY PIPE FITTINGS

a. Transition pipe couplings shall join piping with small differences in outside diameters or be of different materials. End connections shall be of the same size and compatible with the pipes being joined. The transition coupling shall be unshielded, elastomeric, sleeve type reducing or transition pattern conforming with ASTM C1173 and include shear ring and corrosion resistant metal tension band and tightening mechanism on each end. The transition coupling sleeve coupling shall be of the following material:

(1) For cast iron soil pipes, the sleeve material shall be rubber conforming to ASTM C564.

(2) For PVC soil pipes, the sleeve material shall be elastomeric seal conforming to ASTM F477 or PVC conforming to ASTM D5926.

(3) dissimilar pipes, the sleeve material shall be PVC conforming to ASTM D5926, or other material compatible with the pipe materials being joined.

b. Dielectric fittings shall conform to ASSE 1079 with a pressure rating of 150 psig at a minimum temperature of 180 degrees F. The end connection shall be solder joint copper alloy and threaded ferrous.

c. Dielectric flanges shall conform to ASSE 1079 with a pressure rating of 150 psig. The flange shall be a factory fabricated, bolted, companion flange assembly. The end connection shall be threaded or solder-joint copper alloy and threaded ferrous.

d. Dielectric flange insulating kits shall be of non-conducting materials for field assembly of companion flanges with a pressure rating of 150 psig. The gasket shall be neoprene or phenolic. The bolt sleeves shall be phenolic or polyethylene. The washers shall be phenolic with steel backing washers.

e. Dielectric nipples shall be electroplated steel and shall conform with ASTM F1545 with a pressure ratings of 300 psig at 225 degrees F. The end connection shall be male threaded. The lining shall be inert and noncorrosive propylene. Bio-based materials shall be utilized when possible.

2.4 CLEANOUTS

a. Cleanouts shall be the same size as the pipe, up to 4 inches; not less than 4 inches for larger pipe. Cleanouts shall be easily accessible and shall be gastight and watertight. A minimum clearance of 24 inches shall be provided for clearing a clogged storm sewer line.

b. (FCO) Floor cleanouts shall be gray iron housing with clamping device and round, secured, scoriated, gray iron cover conforming to ASME A112.36.2M. A gray iron ferrule with hubless, socket, inside caulk or spigot connection and counter sunk, taper-thread, brass or bronze closure plug shall be included. The frame and cover material and finish shall be nickel-bronze copper alloy with a square shape. The cleanout shall be vertically adjustable for a minimum of 2 inches. When a waterproof membrane is used in the floor system, clamping collars shall be provided on the cleanouts. Cleanouts shall consist of wye fittings and eighth bends with brass or bronze screw plugs. Cleanouts in the resilient tile floors, quarry tile and ceramic tile floors shall be provided with square top covers recessed for tile insertion. In the carpeted areas, carpet cleanout markers shall be provided. Two way cleanouts shall be provided where indicated on the drawings and at each building exit. The loading classification for cleanouts in sidewalk areas or subject to vehicular traffic shall be heavy duty.

c. (CO) Cleanouts shall be provided at or near the base of the vertical stacks with the cleanout plug located approximately 24 inches above the floor. The cleanouts shall be extended to the wall access cover. Cleanout shall consist of sanitary tees. Nickel bronze square frame and stainless steel cover with minimum opening of 6 inch by 6 inch shall be provided at each wall cleanout.

d. In horizontal runs above grade, cleanouts shall consist of cast brass tapered screw plug in fitting or caulked/no hub cast iron ferrule. Plain end (no-hub) piping in interstitial space or above ceiling may use plain end (no-hub) blind plug and clamp.

e. (YCO) Yard cleanouts shall be provided where indicated on the drawings. Cleanouts shall be round, dura-coated cast iron, size as listed on drawings. Cleanouts shall consist of double-flanged housing with heavy-duty, secured scoriated dura-coated cast iron cover complete with lifting device. Provide bronze cleanout plug with gas/water-tight seal. Yard cleanouts shall be installed flush with finished grade and set to provide accessibility for maintenance.

2.5 ROOF DRAINS AND CONNECTIONS

a. Roof Drains: Roof Drains (RD) shall be cast iron with clamping device for making watertight connection and shall conform with ASME A112.6.4. Free openings through strainer shall be twice area of drain outlet. For roof drains not installed in connection with a waterproof membrane, a soft copper membrane shall be provided 12 inches in diameter greater than outside diameter of drain collar. An integral gravel stop shall be provided for drains installed on roofs having built up roofing covered with gravel or slag. Integral no-hub, soil pipe gasket or threaded outlet connection shall be provided.

(1) (RD-A)Flat Roofs: The roof drain shall have a beehive or dome shaped strainer with integral flange not less than 12 inches in

diameter. For an insulated roof, a roof drain with an adjustable drainage collar shall be provided, which can be raised or lowered to meet required insulation heights, sump receiver and deck clamp. The bottom section shall serve as roof drain during construction before insulation is installed.

(2)Protective Roof Membrane Insulation Assembly: The roof drain shall have a perforated stainless steel extension filter, non-puncturing clamp ring, large sump with extra wide roof flange and deck clamp.

(b) Pedestrian Roof: The roof drain shall have a bronze promenade top 14 inches square, set in square secured frame support collar.

(3)(OD-A)Roof Drains, Overflow or Secondary (Emergency): Roof Drains identified as overflow or secondary (emergency) drains shall have a 2 inch water dam integral to the drain body.

(4) Roof drains in areas subject to freezing shall have heat tape and shall be insulated.

b. Expansion Joints: Expansions joints shall be heavy cast iron with cast brass or PVC expansion sleeve having smooth bearing surface working freely against a packing ring held in place and under pressure of a bolted gland ring, forming a water and air tight flexible joint. Asbestos packing is prohibited.

c. Interior Downspouts: An expansion joint shall be provided, specified above, at top of run on straight, vertical runs of downspout piping 40 feet long or greater.

d. (DOC-A)Downspout Nozzle: The downspout nozzle fitting shall be of brass, unfinished, with internal pipe thread for connection to downspout.

e. (RDO-A)Downspout Nozzle: The downspout nozzle fitting shall be of brass, unfinished, with internal pipe thread for connection to downspout.

2.6 WATERPROOFING

A sleeve flashing device shall be provided at points where pipes pass through membrane waterproofed floors or walls. The sleeve flashing device shall be manufactured, cast iron fitting with clamping device that forms a sleeve for the pipe floor penetration of the floor membrane. A galvanized steel pipe extension shall be included in the top of the fitting that will extend 2 inches above finished floor and galvanized steel pipe extension in the bottom of the fitting that will extend through the floor slab. A waterproofed caulked joint shall be provided at the top hub.

PART 3 EXECUTION

3.1 PIPE INSTALLATION

a. The pipe installation shall comply with the requirements of the ICC IPC and these specifications.

b. Branch piping shall be installed from the piping system and connect to all drains and outlets.

c. Pipe shall be round and straight. Cutting shall be done with proper tools. Pipe, except for glass, shall be reamed to remove burrs and a clean smooth finish restored to full pipe inside diameter.

d. All pipe runs shall be laid out to avoid interference with other work/trades.

e. The piping shall be installed above accessible ceilings to allow for ceiling panel removal.

f. Unless otherwise stated on the documents, minimum horizontal slope shall be one inch for every 8 feet (1 percent slope) of pipe length.

g. The piping shall be installed free of sags and bends.

h. Seismic restraint shall be installed where required by code.

i. Changes in direction for storm drainage piping shall be made using appropriate branches, bends and long sweep bends. Sanitary tees and short sweep ¼ bends may be used on vertical stacks if change in direction of flow is from horizontal to vertical. Long turn double wye branch and 1/8 bend fittings shall be used if two drains are installed back to back or side by side with common drain pipe. Do not change direction of flow more than 90 degrees. Proper size of standard increaser and reducers shall be used if pipes of different sizes are connected. Reducing size of drainage piping in direction of flow is prohibited.

j. Buried storm drainage piping shall be laid beginning at the low point of each system. Piping shall be installed true to grades and alignment indicated with unbroken continuity of invert. Hub ends shall be placed upstream. Required gaskets shall be installed according to manufacturer’s written instruction for use of lubricants, cements, and other installation requirements. Bio-based materials shall be utilized when possible.

k. Cast iron piping shall be installed according to CISPI HANDBOOK Chapter IV, “Installation of Cast Iron Soil Pipe and Fittings”

l. Aboveground copper tubing shall be installed according to CDA A4015.

m. Underground PVC piping shall be installed according to ASTM D2321.

3.2 JOINT CONSTRUCTION

a. Hubless, cast iron piping shall be joined in accordance with CISPI HANDBOOK for hubless piping coupling joints.

b. For threaded joints, thread pipe with tapered pipe threads according to ASME B1.20.1. The threads shall be cut full and clean using sharp disc cutters. Threaded pipe ends shall be reamed to remove burrs and restored to full pipe inside diameter. Pipe fittings and valves shall be joined as follows:

(1) Apply appropriate tape or thread compound to external pipe threads

unless dry seal threading is required by the pipe service

(2) Pipe sections with damaged threads shall be replaced with new undamaged sections of pipe at no additional time or cost to Government.

c. Copper tube and fittings with soldered joints shall be joined according to ASTM B828. A water flushable, lead free flux conforming to ASTM B813 and a lead free alloy solder conforming to ASTM B32 shall be used.

3.3 SPECIALTY PIPE FITTINGS

a. Transition coupling shall be installed at pipe joints with small differences in pipe outside diameters.

b. Dielectric fittings shall be installed at connections of dissimilar metal piping and tubing.

3.4 PIPE HANGERS, SUPPORTS AND ACCESSORIES

a. All piping shall be supported according to the ICC IPC, Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING, and these specifications.

b. Hangers, supports, rods, inserts and accessories used for Pipe supports shall be shop coated with zinc Chromate primer paint. Electroplated copper hanger rods, hangers and accessories may be used with copper tubing.

c. Horizontal piping and tubing shall be supported within 12 inches of each fitting or coupling.

d. Horizontal cast iron piping shall be supported with the following maximum horizontal spacing and minimum hanger rod diameters:

(1) DN 40 to DN 50: 60 inches with 3/8 inch rod.

(2) DN 80: 60 inches with 1/2 inch rod.

(3) DN 100 to DN 125: 60 inches with 5/8 inch rod.

(4) DN 150 to DN 200: 60 inches with 3/4 inch rod.

(5) DN 250 to DN 300: 60 inches with 7/8 inch rod.

e. The maximum support spacing for horizontal plastic shall be 4 feet.

f. Vertical piping and tubing shall be supported at the base, at each floor, and at intervals no greater than 15 feet.

g. In addition to the requirements in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING, floor, wall and ceiling plates shall have the following characteristics:

(1) Solid or split unplated cast iron.

(2) All plates shall be provided with set screws.

(3) Height adjustable clevis type pipe hangers.

(4) Adjustable Floor Rests and Base Flanges shall be steel.

(5) Hanger Rods shall be low carbon steel, fully threaded or threaded at each end with two removable nuts at each end for positioning rod and hanger and locking each in place.

(6) Riser Clamps shall be malleable iron or steel.

(7) Roller shall be cast iron.

(8) Hangers and supports utilized with insulated pipe and tubing shall have 180 degree (minimum) metal protection shield centered on and welded to the hanger and support. The shield shall be 4 inches in length and be 16 gage steel. The shield shall be sized for the insulation.

h. Miscellaneous materials shall be provided as specified, required, directed or as noted on the drawings for proper installation of hangers, supports and accessories. If the vertical distance exceeds 20 feet for cast iron pipe additional support shall be provided in the center of that span. All necessary auxiliary steel shall be provided to provide that support.

i. Cast escutcheon with set screw shall be installed at each wall, floor and ceiling penetration in exposed finished locations and within cabinets and millwork.

j. Penetrations:

(1) Fire Stopping: Where pipes pass through fire partitions, fire walls, smoke partitions, or floors, a fire stop shall be installed that provides an effective barrier against the spread of fire, smoke and gases as specified in Section 07 84 00, FIRESTOPPING. Clearances between raceways and openings shall be completely filled and sealed with the fire stopping materials.

(2) Water proofing: At floor penetrations, Clearances around the pipe shall be completely sealed and made watertight with sealant as specified in Section 07 92 00, JOINT SEALANTS. Bio-based materials shall be utilized when possible.

3.5 INSULATION

Insulate horizontal sections and 2 feet past changes of direction to vertical sections for interior section of roof drains. Install insulation in accordance with the requirements of Section 22 07 11, PLUMBING INSULATION.

3.6 TESTS

a. Storm sewer system shall be tested either in its entirety or in sections.

b. Storm Water Drain tests shall be conducted before trenches are backfilled or fixtures are connected. A water test or air test shall be conducted, as directed.

(1) If entire system is tested with water, tightly close all openings in pipes except the highest opening, and fill system with water to

point of overflow. If system is tested in sections, tightly plug each opening except highest opening of section under test, fill each section with water and test with at least a 10 foot head of water. In testing successive sections, test at least upper 10 feet of next preceding section so that each joint or pipe except upper most 10 feet of system has been submitted to a test of at least a 10 foot head of water. Water shall be kept in the system, or in portion under test, for at least 15 minutes before inspection starts. System shall then be tight at all joints.

(2) For an air test, an air pressure of 5 psig gage shall be maintained for at least 15 minutes without leakage. A force pump and mercury column gage shall be used for the test.

(3) Final Tests: While either one of the following tests may be used, Contractor shall check with VA as to which test will be performed.

(a) Smoke Test: After fixtures are permanently connected and traps are filled with water, fill entire drainage and vent systems with smoke under pressure of 1 inch of water with a smoke machine. **Chemical smoke is prohibited.**

(b) Peppermint Test: Introduce 2 ounces of peppermint into each line or stack.

c. Contracting Officer shall witness all tests. Contractor shall coordinate schedules with the Contracting Officer and CxA. Contractor shall provide a minimum of 10 working days prior to flushing, disinfection/sterilization, startup, and testing.

3.7 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.8 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for one hours to instruct VA Personnel in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

-- End of Section --

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SUMP PUMPS

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SECTION 22 14 29
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SUMP PUMPS

PART 1 GENERAL

1.1 DESCRIPTION

a. Sump pumps. See schedule on Drawings for pump capacity and head.

b. A complete listing of all acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e.

f. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

g. Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT.

h. Section 22 05 23, GENERAL-DUTY VALVES FOR PLUMBING PIPING

i. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS. Requirements for commissioning, systems readiness checklist, and training.

j. Section 26 29 11, MOTOR CONTROLLERS.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A532/A532M	(2010; R 2023) Standard Specification for Abrasion-Resistant Cast Irons

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Certification Documentation; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-09 Manufacturer's Field Reports

Factory And Construction Site Test Data; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER Section 22 14 29, SUMP PUMPS”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Pump:

(a) Manufacturer and model.

(b) Operating speed (rpm).

(c) Capacity.

(d) Characteristic performance curves.

(2) Electric Motor:

(a) Manufacturer, frame and type.

(b) Speed.

(c) Current Characteristics and HP.

(d) Efficiency.

(3) Control panel.

(4) Sensors.

d. Certified copies of all the factory and construction site test data sheets and reports.

e. Complete operating and maintenance manuals including wiring diagrams, technical data sheets and information for ordering replacement parts:

(1) Include complete list which indicates all components of the system.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance, and troubleshooting.

f. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

g. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

For products designated by the USDA’s Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

1.7 AS-BUILT DOCUMENTATION

a. Submit manufacturer’s literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, substitutions and construction revisions shall be in electronic version on compact disc or DVD inserted into a three ring binder. All aspects of system operation and maintenance procedures, including piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices such as damper and door closure interlocks shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. As-built drawings are to be provided, and a copy of them in the latest Auto-CADD version provided on compact disk or DVD. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the 'third party testing company' requirement.

d. Certification documentation shall be provided to Contracting Officer 10 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and a certification that all results of tests were within limits specified.

PART 2 PRODUCTS

2.1 SUMP PUMP

a. Centrifugal, vertical, submersible pump and motor, designed for 140 degrees F maximum water service. Driver shall be electric motor. Support shall be rigid type. Provide perforated, suction strainer. Systems may include one, two, or more pumps with alternator as required by Contract Documents. Pump and/or Pumps shall be capable of continuous duty cycle.

(1) Pump housings may be cast iron, bronze, aluminum or stainless steel. Cast iron and aluminum housings for submersible pumps shall be epoxy coated. Bio-based materials shall be utilized when possible.

b. Impeller: Statically and dynamically balanced, keyed and secured to shaft,cast iron ASTM A532/A532M.

c. Shaft: Stainless steel or other approved corrosion-resisting metal.

d. Bearings: As required to hold shaft alignment, anti-friction type for thrust permanently lubricated. Bio-based materials shall be utilized when possible.

e. Seal: Mechanical.

f. Motor: Maximum 104 degrees F ambient temperature rise above the maximum fluid temperature being pumped, drip-proof hermitically sealed, lifting eye, capacitor start type, voltage and phase as shown in schedule on Electrical drawings conforming to NEMA Type 4X. Size the motor capacity to operate pump without overloading the motor at any point on the pump curve. Refer to Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT.

g. Starting Switch: Manually-operated, tumbler type, as specified in Section 26 29 11, MOTOR CONTROLLERS.

h. Automatic Control and Level Alarm: Furnish a control panel in a NEMA 1 enclosure for indoors or in a NEMA 4X enclosure for outdoors. The controls shall be suitable for operation with the electrical characteristics listed on the Electrical drawings. The control panel shall have a level control system with switches to start and stop

pumps automatically, and to activate a high water alarm. The level control system shall include sensors in the sump that detect the level of the liquid. The pump is also connected to a control which has the ability to prevent oil from being pumped. The same unit shall activate an alarm when oil is detected. The sensors may be float type switches, ultrasonic level sensors, or transducers. The high water alarm shall have a red beacon light at the control panel and a buzzer, horn, or bell. The alarm shall have a silencing switch. Provide auxiliary contacts for remote communication with, and alarm monitoring to, the BAS using a BACnet compatible open-protocol type interface to DDC Controls System.

(1) The circuitry of the control panel shall include:

(a) Power switch to turn on/off the automatic control mechanism

(b) HOA switches to manually override automatic control mechanism

(c) Run lights to indicate when pumps are powered up

(d) Level status lights to indicate when water in sump has reached the predetermined on/off and alarm levels

(e) Magnetic motor contactors

(f) Disconnect/breaker for each pump

(g) Automatic motor overload protection

(h) Wiring terminal block

(i) Dead front

(j) Auxiliary contacts

(k) Control circuit protection

(l) Fused control step down transformer

(2) Sensors that detect the level of water in the sump shall be so arranged as to allow the accumulation of enough volume of liquid below the normal on-level that the pump will run for a minimum cycle time as recommended by the pump manufacturer. Sensors shall be located to activate the alarm adequately before the water level rises to the inlet pipe.

(3) Provide two separate power supplies to the control panel, one for the control/alarm circuitry and one for power to the pump motors. Each power supply is to be fed from its own breaker so that if a pump overload trips a breaker, the alarm system shall still function. Each power supply is to be wired in its own conduit.

(4) Wiring from the sump to the control panel shall have separate conduits for the pump power and for the sensor switches. All conduits are to be sealed at the basin and at the control panel to prevent the intrusion of moisture and of flammable and/or corrosive gases.

i. Sump: Furnish fiberglass basin with gas tight covers. Cover shall

have 11 inch by 15 inch manhole with bolted cover, vent connection, openings for pumps and controls. Sump shall be sized to allow an adequate volume of water to accumulate for a minimum one minute cycle of pump operation.

j. Provide a check and ball valve in the discharge of each pump. Refer to Section 22 05 23, GENERAL-DUTY VALVES FOR PLUMBING PIPING.

k. Removal/Disconnect System: In a system utilizing a submersible pump, where sump depth, pump size, or other conditions make removal of the pump unusually difficult or unsafe, a manufacturer's removal/disconnect system shall be provided. The system shall consist of a discharge fitting mounted on vertical guide rails attached to the sump or quick connect pipe fitting connection to piping. The pump shall be fitted with an adapter fitting that easily connects to/disconnects from the discharge fitting as the pump is raised from or lowered into the sump. The discharge piping shall connect to the discharge fitting so that it is disconnected without workers entering the pit. Where the sump depth is greater than five feet or other conditions exist to make the removal of the pump difficult or hazardous, the system shall include a rail guided quick disconnect apparatus to allow the pump to be pulled up out of the sump.

PART 3 EXECUTION

3.1 STARTUP AND TESTING

a. Pump installation to comply with ANSI/HI 1.4 for sump pumps.

b. Leak Test: Charge piping system and test for leaks. Test until there are no leaks. Make tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

c. The tests shall include system capacity and all control and alarm functions.

d. When any defects are detected, correct defects and repeat test.

e. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Contractor shall provide a minimum of 10 working days prior to startup and testing.

3.2 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.3 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer's technical representative for one hours to instruct VA Personnel in operation and maintenance of units.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

-- End of Section --

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DIVISION 22 - PLUMBING

SECTION 22 14 36

PACKAGED, SUBMERSIBLE, DRAINAGE PUMP UNITS

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PART 3 EXECUTION

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3.3 DEMONSTRATION AND TRAINING

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SECTION 22 14 36
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PACKAGED, SUBMERSIBLE, DRAINAGE PUMP UNITS

PART 1 GENERAL

1.1 DESCRIPTION

a. Packaged submersible drainage pump units. See schedule on Drawings for pump capacity and head.

b. A complete listing of all acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e.

f. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

g. Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT.

h. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS: Requirements for commissioning, systems readiness checklist, and training.

i. Section 26 29 11, MOTOR CONTROLLERS.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

HYDRAULIC INSTITUTE (HI)

ANSI/HI 1.4	(2014) Rotodynamic Centrifugal Pumps for Manuals Describing Installation, Operation and Maintenance

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Factory And Construction Site Test Data; G, AE

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Certification Documentation; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 22 14 36, PACKAGED, SUBMERSIBLE, DRAINAGE PUMP UNITS”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Pump:

(a) Manufacturer and model.

(b) Operating speed (rpm).

(c) Capacity.

(d) Characteristic performance curves.

(2) Electric Motor:

(a) Manufacturer, frame and type.

(b) Speed.

(c) Current Characteristics and HP.

(d) Efficiency.

d. Certified copies of all the factory and construction site test data sheets and reports.

e. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance, and troubleshooting.

f. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

g. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

For products designated by the USDA’s Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

1.7 AS-BUILT DOCUMENTATION

a. Submit manufacturer’s literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments,substitutions and construction revisions shall be in electronic version on compact disc or DVDinserted into a three ring binder. All aspects of system operation and maintenance procedures, including piping isometrics,wiring diagrams of all circuits, a written description of system design,control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices such as damper and door closure interlocks shall be included. A List of recommended spare parts (manufacturer, model number, and quantity)shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. As-built drawings are to be provided, and a copy of them in the latest Auto-CAD version provided on compact disk or DVD. Should the installing contractor

engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the 'third party testing company' requirement.

d. Certification documentation shall be provided to Contracting Officer 10 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and certification that all results of tests were within limits specified.

PART 2 PRODUCTS

2.1 SUBMERSIBLE DRAINAGE PUMP

a. Centrifugal, submersible, designed for 140 degrees F maximum water temperature. Driver shall be electric motor with rigid type support. Provide perforated, nonferrous suction strainer: Systems may include one, two, or more pumps with alternator as required by Contract Documents:

(1) Pump housings may be cast iron, bronze, or stainless steel. Cast iron housings for submersible pumps shall be epoxy coated.

b. Impeller: Brass or bronze.

c. Shaft: Bronze, stainless steel or other VA approved corrosion-resisting metal.

d. Bearings: As required to hold shaft alignment, anti-friction type for thrust permanently lubricated. Bio-based materials shall be utilized when possible.

e. Motor: Maximum 104 degrees F ambient temperature rise, completely enclosed, voltage and phase as shown in schedule on Electrical drawings conforming to NEMA Type 6P. Size the motor capacity to operate the pump without overloading the motor at any point on the pump curve. Refer to Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT.

f. Starting Switch: Manually-operated, tumbler type, as specified in Section 26 29 11, MOTOR CONTROLLERS.

g. Automatic Control and Level Alarm: Furnish a control panel in a NEMA Type 1 enclosure for indoors and NEMA Type 4X enclosure for outdoors. The controls shall be suitable for operation with the electrical characteristics listed on the Electrical drawings. The control panel shall have a level control system with switches to start and stop pumps automatically, and to activate a high water alarm. The level control system shall include sensors in the sump that detect the level of the liquid. The sensors may be float type switches, ultrasonic level sensors, transducers, or other appropriate equipment. The high water alarm shall have a red beacon light at the control panel and a buzzer, horn, or bell. The alarm shall have a silencing switch. Provide auxiliary contacts for remote alarming to the Energy Control Center and BACnet compatible open-protocol type interface to DDC Controls System.

(1) The circuitry of the control panel shall include:

(a) Power switch to turn on/off the automatic control mechanism

(b) HOA switches to manually override automatic control mechanism

(c) Run lights to indicate when pumps are powered up

(d) Level status lights to indicate when water in sump has reached the predetermined on/off and alarm levels

(e) Magnetic motor contactors

(f) Disconnect/breaker for each pump

(g) Automatic motor overload protection

(h) Wiring terminal block

(i) Dead front

(j) Auxiliary contacts

(k) Control circuit protection

(l) Fused control step down transformer

(2) For a duplex system, provide an alternating relay to automatically alternate leadoff and standby duties of each pump of a duplex unit at the end of each pumping cycle. Standby pump shall start when water level in sump rises to a predetermined level that indicates excessive inflow or failure of the lead pump.

(3) Sensors that detect the level of water in the sump shall be so arranged as to allow the accumulation of enough volume of liquid below the normal on-level that the pump shall run for a minimum cycle as recommended by pump manufacturer to protect short cycling. Sensors shall be located to activate the alarm adequately before the water level rises to the inlet pipe.

(4) Provide two separate power supplies to the control panel, one for the control/alarm circuitry and one for power to the pump motors. Each power supply is to be fed from its own breaker so that if a pump overload trips a breaker, the alarm system shall still function. Each power supply is to be wired in its own conduit.

(5) Wiring from the sump to the control panel shall have separate conduits for the pump power and for the sensor switches. All conduits are to be sealed at the basin and at the control panel to prevent the intrusion of moisture and of flammable and/or corrosive gases.

h. Sump: Furnish cast iron, fiberglass or high density polyethylene basin with gas tight covers. Cover shall be sized so as to inspect and maintain pump installation manhole with bolted cover, vent connection, openings for pumps and controls. Sump shall be sized to allow an adequate volume of water to accumulate for a minimum one minute cycle of pump operation.

i. Provide a union, check and ball valve in the discharge of each pump

where indicated on drawings.

j. Removal/Disconnect System: Where indicated on drawings, a removal/ disconnect system shall be provided. The system shall consist of a discharge fitting mounted on a vertical guide rails attached to the sump. The pump shall be fitted with an adapter fitting that easily connects to/disconnects from the discharge fitting as the pump is raised from or lowered into the sump. The discharge piping shall connect to the discharge fitting so that it is not necessary to disconnect any piping in order to remove the pump. Where the sump depth is greater than five feet or other conditions exist to make the removal of the pump difficult or hazardous, the system shall include a rail guided quick disconnect apparatus to allow the pump to be pulled up out of the sump without workers entering the sump and without disconnecting the piping.

PART 3 EXECUTION

3.1 STARTUP AND TESTING

a. Pump installation to comply with ANSI/HI 1.4 for sump pumps.

b. Leak Test: Charge piping system and test for leaks. Test until there are no leaks. Make tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

c. System Test: After installation is completed provide an operational test of the completed system including flow rates, pressure compliance, alarms and all control functions.

d. When any defects are detected, correct defects and repeat test at no additional time or cost to Government.

e. The CxA will observe startup and Contractor testing of selected equipment. Coordinate the startup and Contractor testing schedules with the Contracting Officer and CxA. Contractor shall provide a minimum of 10 working days prior to startup and testing.

3.2 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.3 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for one hours to instruct VA Personnel in operation and maintenance of units.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

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DIVISION 22 - PLUMBING

SECTION 22 15 00

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SECTION 22 15 00
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GENERAL SERVICE COMPRESSED-AIR SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

a. This section describes the requirements for NFPA 99 Category 4 compressed air systems for non-medical air piping materials, including compressors, electric motors and starters, receiver, all necessary piping, fittings, valves, gauges, switches and all necessary accessories, connections and equipment. NFPA 99 Category 4 systems are non-medical systems of 100 psig or less in which failure of equipment would have no impact on patient care.

b. A complete listing of common acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00, GENERAL COMMISSIONING REQUIREMENTS.

e.

f. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

g. Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT.

h. Section 22 05 19, METERS AND GAGES FOR PLUMBING PIPING: Exposed piping and gauges.

i. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

j. Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standards will govern.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A13.1 (2023) Scheme for the Identification of Piping Systems

ASME B1.20.1	(2013; R 2018) Pipe Threads, General Purpose (Inch)
ASME B16.3	(2021) Malleable Iron Threaded Fittings, Classes 150 and 300
ASME B16.39	(2020) Standard for Malleable Iron Threaded Pipe Unions; Classes 150, 250, and 300
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2024) Factory-Made Wrought Buttwelding Fittings
ASME B16.21	(2021) Nonmetallic Flat Gaskets for Pipe Flanges
ASME B16.22	(2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings
ASME B16.24	(2022) Cast Copper Alloy Pipe Flanges, Flanged Fittings, and Valves Classes 150, 300, 600, 900, 1500, and 2500
ASME B18.2.1	(2012; R 2021) Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)
ASME B40.100	(2022) Pressure Gauges and Gauge Attachments
ASME BPVC SEC VIII	(2021) Boiler and Pressure Vessel Codes: Section VIII Rules for Construction of Pressure Vessel

ASTM INTERNATIONAL (ASTM)

ASTM A47/A47M	(1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings
ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A106/A106M	(2019a) Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM A126	(2004; R 2023) Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings
ASTM A135/A135M	(2021) Standard Specification for Electric-Resistance-Welded Steel Pipe

ASTM A536	(2024) Standard Specification for Ductile Iron Castings
ASTM A733	(2016; R 2022) Standard Specification for Welded and Seamless Carbon Steel and Austenitic Stainless Steel Pipe Nipples
ASTM B32	(2020) Standard Specification for Solder Metal
ASTM B62	(2017) Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM B88	(2022) Standard Specification for Seamless Copper Water Tube
ASTM B584	(2022) Standard Specification for Copper Alloy Sand Castings for General Applications
ASTM B813	(2024) Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube
ASTM B819	(2000; R 2011) Standard Specification for Seamless Copper Tube for Medical Gas Systems

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C606	(2022) Grooved and Shouldered Joints

AMERICAN WELDING SOCIETY (AWS)

AWS A5.8/A5.8M	(2019) Specification for Filler Metals for Brazing and Braze Welding

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-71	(2018) Gray Iron Swing Check Valves, Flanged and Threaded Ends
MSS SP-72	(2010a) Ball Valves with Flanged or Butt-Welding Ends for General Service
MSS SP-80	(2019) Bronze Gate, Globe, Angle and Check Valves
MSS SP-110	(2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends
MSS SP-123	(2018) Non-Ferrous Threaded and Solder-Joint Unions for Use with Copper Water Tube

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 99	(2024) Health Care Facilities Code

UL SOLUTIONS (UL)

UL 213	(2019) Rubber Gasketed Fittings for Fire-Protection Services

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Brazing And Welding Certificates; G, AE

Qualified Engineer Or Technician From The Compressor Manufacturer; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-05 Design Data

Seismic Restraint Design; G, AE

SD-06 Test Reports

Pneumatic Compressed Air System And Hydrostatic Drainage Piping Test Reports; G, AE

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Signed Certification; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 22 15 00, GENERAL SERVICE COMPRESSED-AIR SYSTEMS”, with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Aboveground Piping

(2) Underground Piping

(3) Supporting Elements

(4) Valves

(5) Pressure Gauges

(6) Air Pressure Reducing and Regulating Valves

(7) Automatic Drain Valves

(8) Filter Capacity and Operating Characteristics

(9) Vibration Isolation

(10) Quick Couplings

(11) Hose Assemblies

(12) Air Compressor System:

(a) Characteristic performance curves

(b) Efficiency

(c) Compressor; manufacturer and model

(d) Compressor operating speed

(e) Capacity (free air delivered at indicated pressure)

(f) Type of bearing in compressor

(g) Type of lubrication

(h) Capacity of receiver

(i) Unloader; manufacturer, type, and model

(j) Type and adjustment of drive

(k) Electrical motor; manufacturer, frame and model

(l) Speed of motor

(m) Current characteristics and HP of motor

(n) Air muffler filter; manufacture, type, and model

(o) After cooler; manufacturer, type, and model

d. Pneumatic compressed air system and hydrostatic drainage piping test reports shall be submitted.

e. Brazing and welding certificates shall be submitted.

f. For seismic restraint design the following shall be submitted:

(1) Dimensioned drawings of equipment identifying center of gravity and location and description of seismic mounting and anchorage systems.

g. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

h. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

i. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

a. The Contractor shall obtain the services of a qualified engineer or technician from the compressor manufacturer to review final installation and supervise startup and testing of the compressor. After satisfactory installation of the equipment, the engineer or technician shall provide a signed certification that the equipment is installed in accordance with the manufacturer's recommendations.

b. Bio-Based Materials: For products designated by the USDA’s Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit https://www.biopreferred.gov.

c. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for

additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in Paragraph AS-BUILT DOCUMENTATION of Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

PART 2 PRODUCTS

2.1 PIPES, TUBES AND FITTINGS

a. Pipe for general service compressed air system shall be drawn temper, Type "K" or "L" seamless copper tube, conforming to ASTM B88, ASTM B819 with wrought copper solder joint fittings conforming to ASME B16.22.

b. Copper unions shall conform to ASME B16.22 or MSS SP-123.

c. Cast copper alloy flanges shall be Class 300 conforming to ASME B16.24.

(1) Pipe-Flange Gasket Materials: ASME B16.21, nonmetallic, flat, asbestos free, 1/8 inch maximum thickness, full-face type.

(2) Flange Bolts and Nuts: ASME B18.2.1, carbon steel.

d. Solder filler metal shall consist of lead-free alloys conforming to ASTM B32 with water flushable flux conforming to ASTM B813.

e. Silver Brazing Filler metals shall be BCuP series, copper phosphorus alloys for general duty brazing conforming to AWS A5.8/A5.8M.

f. Schedule 40, Steel Pipe: ASTM A53/A53M, Type E or S, Grade B, black or hot-dip zinc coated with ends threaded according to ASME B1.20.1.

(1) Steel Nipples: ASTM A733, made of ASTM A53/A53M or ASTM A106/A106M, Schedule 40, galvanized seamless steel pipe. Include ends matching joining method.

(2) Malleable-Iron Fittings: ASME B16.3, Class 150 or 300, threaded.

(3) Malleable-Iron Unions: ASME B16.39, Class 150 or 300, threaded.

(4) Steel Flanges: ASME B16.5, Class 150 or 300, carbon steel, threaded.

(5) Wrought-Steel Butt-Welding Fittings: ASME B16.9, Schedule 40.

(6) Steel Flanges: ASME B16.5, Class 150 or 300, carbon steel.

(7) Grooved-End Fittings and Couplings:

(a) Grooved-End Fittings: ASTM A47/A47M, malleable-iron castings or ASTM A536, ductile-iron casting; with grooves according to AWWA C606 and dimensions matching steel pipe.

(b) Couplings: AWWA C606 or UL 213, for steel-pipe dimensions and rated for 300 psig minimum working pressure. Include ferrous housing sections, gasket suitable for compressed air, and bolts and nuts. Provide EDPM gaskets for oil-free compressed air. Provide NBR gaskets if compressed air contains oil or oil vapor.

g. Schedule 5, Steel Pipe: ASTM A135/A135M, carbon steel with plain ends and zinc-plated finish.

(1) Pressure-Seal Fittings: Listed and labeled by a qualified testing agency and FMG-approved, carbon-steel, pressure-seal housing with O-ring end seals suitable for compressed-air piping and rated for 300 psig minimum working pressure. Provide EDPM seals for oil-free compressed air. Provide NBR seals if compressed air contains oil or oil vapor.

h. Transition Couplings for Metal Piping: Metal coupling or other manufactured fitting same size as, with pressure rating at least equal to and ends compatible with, piping to be joined.

i. Pipe identification shall comply with ASME A13.1. Pipe identification labels shall be located as follows:

(1) At intervals of not more than 20 feet.

(2) At least visible once in or above every room.

(3) On both sides of walls or partitions penetrated by the piping.

(4) At least once in every story height traversed by risers.

2.2 VALVES

a. Ball:

(1) Ball valves NPS 3 inches and smaller shall be full port, two-piece or three-piece ball valve conforming to MSS SP-110. The ball valve shall have a SWP rating of 150 psig and a CWP rating of 600 psig. The body material shall be Bronze ASTM B584, Alloy C844. The ends shall be soldered.

(2) Ball valves NPS 4 inches and larger shall be flanged, Class 150, full port steel ball valve conforming to MSS SP-72. The body shall be split design. The CWP rating shall be 285 psig. The seals shall be PTFE or TFE. Ball and stem shall be stainless-steel.

b. Check:

(1) Check valves smaller than NPS 4 inches shall be Class 125, bronze swing check valves with non-metallic Buna-N disc. The check valve shall meet MSS SP-80 Type 4 standard. The check valve shall have a CWP rating of 200 psig. The check valve shall have a Y pattern horizontal body design with bronze body material conforming to ASTM B62, solder joints, and PTFE or TFE disc.

(2) Check valves NPS 4 inches and larger shall be Class 125, iron swing check valve with lever and weight closure control. The check valve shall meet MSS SP-71 Type I standard. The check valve shall have a CWP rating of 200 psig. The check valve shall have a clear or full waterway body design with gray iron body material conforming to ASTM A126, bolted bonnet, flanged ends, bronze trim.

2.3 DIELECTRIC FITTINGS

a. Fittings joining copper alloy and ferrous materials shall be isolated.

b. Dielectric unions shall be factory-fabricated union assemblies, rated at 250 psig minimum working pressure at 180 degrees F suitable for compressed air service.

c. Dielectric flanges shall be factory-fabricated companion flange assemblies, rated at 300 psig minimum working pressure at 180 degrees F suitable for compressed air service.

2.4 FLEXIBLE PIPE CONNECTORS

a. Stainless-steel hose flexible connectors shall be corrugated, stainless-steel tubing with stainless-steel wire braid covering and ends welded to inner tubing. The stainless-steel hose connectors shall be rated at 200 psig minimum. The end connections for NPS 2 inches and smaller shall be threaded steel pipe nipple. The end connections for NPS 2-1/2 inches and larger shall be flanged steel nipple.

b. Corrugated-bronze tubing with bronze wire-braid covering and ends brazed to inner tubing. The corrugated- bronze tubing shall be rated at 200 psig minimum.

2.5 SPECIALTIES

a. Pressure Gauges: Pressure gauges permanently installed in the system or used for testing purposes shall be listed for compressed air service and shall include a snubber or pulsation dampener and an isolation valve for maintenance access.

(1) For line pressure use adjacent to source equipment: ASME B40.100, pressure gauge, single, size 4-1/2 inches, for compressed air, accurate to within two percent, with metal case. Range shall be two times operating pressure. Dial graduations and figures shall be black on a white background, or white on a black background. Gage shall be labeled for appropriate service and marked "USE NO OIL".

(2) For all services downstream of main shutoff valve: Manufactured for compressed air use and marked "USE NO OIL", 1-1/2 inch diameter gauge with dial range 1 to 100 psig for air service.

b. Air Pressure Regulating Valves:

(1) Air pressure regulating valves under or DN80 (NPS 3 inches shall be pilot or diaphragm operated, bronze body and trim, direct acting, spring loaded manual pressure setting adjustment and rated for 200 psig inlet pressure. Delivered pressure shall not vary more than 1.5 psig variation in inlet pressure.

(2) Air pressure regulators NPS 3 inches and larger shall be pilot operated, bronze body, direct acting, spring loaded manual pressure setting adjustment and rated for 250 psig inlet pressure. Delivered pressure shall not vary more than 1.5 psig variation in inlet pressure.

c. Safety valves shall be constructed according to the ASME BPVC SEC VIII and be National Board Certified, labeled, and factory sealed. The safety valve shall be constructed of bronze body with poppet type safety valve for compressed air service.

d. The automatic drain valves shall have stainless-steel body and internal parts rated for 200 psig minimum working pressure. The automatic drain valve shall be capable of automatic discharge of collected condensate.

e. The coalescing filter shall be capable of removing water and oil aerosols, efficiency of 99.9 percent retention of particles 0.3 micrometer and smaller, with color change dye to indicate when carbon is saturated and warning light to indicate when selected maximum pressure drop has been exceeded. The coalescing filter shall include mounting brackets for wall mount application.

f. Air line lubricators shall come with a drip chamber and sight dome for observing oil drop entering air stream. The air line lubricator shall have oil feed adjustment screw and quick release collar for easy bowl removal. The Air line lubricators shall include mounting brackets for wall mount application. Lubricators shall be suitable for 200 psig at 160 degrees F.

2.6 QUICK CONNECT COUPLINGS

a. The quick connect coupling assemblies shall have a locking mechanism constructed to permit one-handed feature for quick connection and disconnection of compressed air hose and equipment. Furnish complete keyed indexing noninterchangeable coupling to prevent connection to medical compressed-air pressure outlets.

b. Automatic shutoff quick couplings shall be straight through brass body with O-ring or gasket seal and stainless-steel or nickel plated steel operating parts. The automatic shutoff quick connect coupling shall consist of socket or plug ends with one way valve and with barbed outlet or threaded hose fittings for attaching hose.

c. Valve less quick couplings shall be straight through brass body with O-ring or gasket seal and stainless-steel or nickel plated steel operating parts. The valve less quick connect coupling shall consist of socket or plug ends and with barbed outlet or threaded hose fittings for attaching hose.

2.7 HOSE ASSEMBLIES

a. Hose, clamps, couplings, splicers shall be suitable for compressed air service of nominal diameter indicated and rated for 300 psig minimum working pressure.

b. The hose shall be reinforced double wire braid, chloroprene reinforced covered hose.

c. Hose clamps shall be stainless-steel.

d. Hose couplings shall be two-piece straight through, threaded brass or stainless-steel O-ring or gasket seal swivel coupling with barbed ends for connecting two sections of hose.

e. Hose splicers shall be one piece, straight through brass or stainless-steel fitting with barbed ends.

2.8 PERFORMANCE REQUIREMENTS

a. Delegated Design: Engage a qualified professional engineer, as defined in Section 01 45 00 QUALITY CONTROL, to design compressed-air equipment mounting.

b.

(1)

2.9 AIR COMPRESSOR FOR SHOP AIR SYSTEM DESCRIPTION

a. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

b. ASME Compliance: Fabricate and label receivers to comply with ASME BPVC SEC VIII.

2.10 OILLESS, RECIPROCATING AIR COMPRESSORS

a. Compressor(s): Oilless (nonlubricated), reciprocating-piston type, with sealed oil-free bearings, that deliver air of quality equal to intake air.

(1) High discharge-air temperature switch.

(2) Belt guard totally enclosing pulleys and belts.

2.11 PRESSURE REGULATORS

For 100 psig regulator, provide duplex in parallel, valve for maintenance shut-down without service interruption. For additional pressures, locate regulators remote from compressor near point of use, and provide with isolation valves and valve bypass.

For systems 10 SCFM and below: Brass or bronze body and trim, reduced pressure range 25 to 123 psig adjustable, spring type, diaphragm operated, relieving. Delivered pressure shall vary not more than 0.15 psig for each 1.5 psig variation in inlet pressure.

2.12 INLET-AIR FILTERS

a. Description: Combination inlet-air filter-silencer, suitable for remote installation, for each air compressor.

(1) Construction: Weatherproof housing for replaceable, dry-type filter element, with silencer tubes or other method of sound reduction.

(2) Capacity: Match capacity of air compressor, with filter having collection efficiency of 99 percent retention of particles larger than 10 micrometers.

2.13 REFRIGERANT COMPRESSED-AIR DRYERSNoncycling, air-cooled, electric-motor-driven unit with steel enclosure and capability to deliver 35 degrees F, 100 psig air at dew point. Include automatic ejection of condensate from airstream, step-down transformers, disconnect switches, inlet and outlet pressure gauges, thermometers, automatic controls, and filters.2.14 MOTORS

a. Comply with NEMA designation, temperature rating, service factor, and efficiency requirements for motors specified in Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT.

(1) Enclosure:Totally enclosed, fan cooled .

(2) Motor Sizes: Minimum size as indicated. If not indicated, large enough so driven load does not require motor to operate in service factor range above 1.0.

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

b. Piping shall be installed concealed from view and protected from physical contact unless indicated to be exposed. Piping shall be installed exposed in mechanical rooms and service areas.

c. All pipe shall be installed at right angles or parallel to building walls. Diagonal runs are prohibited unless indicated.

d. Piping shall be installed above accessible ceilings, allowing for sufficient space for ceiling panel removal and to coordinate with other services occupying that that space.

e. Piping installed adjacent to equipment shall be located to allow for the required service clearances.

f. Air and drain piping shall be installed with a 1 percent slope downward in direction of flow.

g. Nipples, flanges, unions, transitions, and special fittings, and valves shall be installed with pressure ratings same as or higher than system pressure rating.

h. Cast copper alloy companion flange with gasket and brazed or soldered joints shall be used to connect equipment and specialties with flanged connections.

i. Flanged joints may be used instead of specified joint for any piping or tubing system.

j. Only eccentric reducers shall be installed where compressed air piping is reduced in direction of flow, with bottoms of both pipes and reducers fitting flush.

k. Branch connections shall be installed from the top of the main compressed air line. Drain legs and drain trap shall be installed at

the end of each main and branch and at all low points in the system.

l. Thermometers and pressure gauges shall be installed on discharge piping from each air compressor and on each receiver.

m. Valves shall be installed to permit servicing to all equipment.

n. Pipes shall be installed free of all sags and bends.

o. Seismic restraint shall be installed for all piping and equipment as required for location.

p. Piping shall be cut square and accurately with a tube cutter (sawing is prohibited) to measurements determined at place of installation and worked into place without springing or forcing the pipe. Tube shall bottom in each solder socket so there are no gaps between tube and fitting where solder can enter the inside of line. The tube shall be reamed to remove burrs, being careful not to expand tube and that no chips of copper remain in the line. Care shall be exercised in handling equipment and tools used in cutting or reaming of pipe to prevent oil or grease being introduced into piping.

q. Particular care shall be exercised, when flux is applied to avoid leaving any excess inside the completed joints. Thoroughly wash the outside of each joint with clean hot water after assembly to remove oxide coating.

r. Hanger spacing shall be based upon NFPA 99.

s. The Filtered Muffler shall be mounted to the air compressor outdoor intake line without the use of foundations or support frames. Silencer tubes shall be located between the filter and the housing.

t. Rigidly support valves and other equipment to prevent strain on tube or joints.

u. Compressor assembly shall have an equipment identification nameplate and data in accordance with 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

v. Suitably brace piping against sway and vibration. Bracing shall consist of brackets, anchor chairs, rods, and structural steel for vibration isolation.

3.2 PRELIMINARY STAGE TESTS

a. Preliminary tests shall be performed by the Contractor prior to testing witnessed by the Contracting Officer. Tests shall be pneumatic and shall use dry, oil-free compressed air, carbon dioxide or nitrogen in metallic systems.

b. Testing of any system for any purpose shall include preliminary testing by swabbing joints under test with standard soap solution and observing for bubbles at internal pressures not in excess of 5 psig.

c. When testing reveals system leakage, isolate and repair the leaks, replace defective materials where necessary, and retest the system until there is no loss of pressure. Remake leaking gaskets with new gaskets and new flange bolting, and discard used bolting and gaskets.

d. Drainage piping shall be hydrostatically tested to a pressure of 5 psig to ensure the piping does not leak. Repair all observed leaks and retest until all leaks have been corrected.

3.3 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part. Tests shall be pneumatic and shall use dry, oil-free compressed air, carbon dioxide or nitrogen in metallic systems.

b. The tests shall include initial piping purge test, initial pneumatic test for positive–pressure gas systems, initial cross-connection test, and initial standing positive-pressure gas piping tests, system capacity, control function, and alarm functions.

c. Pneumatic tests shall be performed utilizing a test pressure of 50 psig higher than the MAWP, minimum of 150 psig. Test pressure shall be maintained for a minimum period of four hours to ensure the temperature in the piping system stabilizes, then the pressure is refreshed and held for two hours with no loss of pressure. Pneumatic testing performance shall be in accordance with industry safety standards with the pressure gradually increased in increments of 25 percent of the MAWP until the required test pressure is reached. At each interval, the system pressure shall be held long enough for piping strains to stabilize. If leaks are observed, the leaks shall be identified, the system de-pressurized and repairs made before proceeding.

d. Other than standard piping flanges, plugs, caps and valves, only use commercially manufactured expandable elastomer plugs for sealing off piping for test purposes. Published safe test pressure rating of any plug used shall be not less than three times the actual test pressure being applied. During pneumatic testing evacuate personnel from areas where plugs are used.

e. Remove components that could be damaged by test pressure from piping systems to be tested.

f. Perform valve-operating tests and drainage tests to ensure valves do not leak when operating under pressure and are correctly labeled.

g. Check piping system components, such as valves, for proper operation under system test pressure.

h. No test media shall be added to a system during a test for a period specified or determined by the Contracting Officer.

i. Duration of a test will be determined by the Contracting Officer and will be for a minimum of 15 minutes with a maximum of 24 hours. Test may be terminated by direction of the Contracting Officer at any point after it has been determined that the pressure leak test has been satisfied.

j. Prepare and maintain test records of all piping systems tests. Records shall show Governmental and Contractor test personnel

responsibilities, dates, test gauge identification numbers, ambient temperatures, pressure ranges, rates of pressure drop, and leakage rates.

k. System verification and final testing shall be conducted comprising of a system verifier standing pressure test, verifier cross-connection test, verifier piping purge test, verifier final tie-in test, verifier operational pressure test, verifier piping particulate test, verifier piping purity test, labeling, and source equipment verification test.

l. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government. When testing reveals system leakage, isolate and repair the leaks, replace defective materials where necessary, and retest the system until there is no loss of pressure. Remake leaking gaskets with new gaskets and new flange bolting, and discard used bolting and gaskets.

m. The CxA will observe startup and Contractor testing of selected equipment. Coordinate the startup and Contractor testing schedules with the Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

3.4 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.5 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for one hour to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

-- End of Section --

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DIVISION 22 - PLUMBING

SECTION 22 33 00

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9/20

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-- End of Section Table of Contents --

SECTION 22 33 00
9/20

HEAT PUMP DOMESTIC WATER HEATER

PART 1 GENERAL

1.1 DESCRIPTION

a. This section describes the requirements for installing a complete electric domestic water heater system ready for operation including the water heaters, thermometers, and all necessary accessories, connections, and equipment.

b. A complete listing of common acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 03 30 00, CAST-IN-PLACE CONCRETE: Concrete and Grout.

f. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

g. Section 22 05 19, METERS AND GAGES FOR PLUMBING PIPING.

h. Section 22 05 23, GENERAL-DUTY VALVES FOR PLUMBING PIPING.

i. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standards will govern.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z21.22/CSA 4.4	(2015; R 2020) Relief Valves for Hot Water Supply Systems

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP	(2019) Energy Standard for Buildings

Except Low-Rise Residential Buildings

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC	(2010) Boiler and Pressure Vessels Code
ASME BPVC SEC VIII	(2021) Boiler and Pressure Vessel Codes: Section VIII Rules for Construction of Pressure Vessel
ASME B1.20.1	(2013; R 2018) Pipe Threads, General Purpose (Inch)
ASME B1.20.7	(1991; R 2013) Standard for Hose Coupling Screw Threads (Inch)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NSF INTERNATIONAL (NSF)

NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects
NSF 372	(2022) Drinking Water System Components - Lead Content

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-05 Design Data

Electric Domestic Hot Water Heater; G, AE

Seismic Qualification Certificates; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Documentation Indicating Compliance; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

Independent Testing Agency; G, AE

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 22 33 00, ELECTRIC DOMESTIC WATER HEATERS”, with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Water Heaters.

(2) Pressure and Temperature Relief Valves.

(3) Thermometers.

(4) Pressure Gauges.

(5) Vacuum Breakers.

(6) Expansion Tanks.

d. For each domestic hot water heater type and size, the following characteristics shall be submitted:

(1) Rated Capacities.

(2) Operating characteristics.

(3) Electrical characteristics.

(4) Furnished specialties and accessories.

(5) A Form U-1 or other documentation stating compliance with ASME BPVC.

e. Shop drawings shall include wiring diagrams for power, signal and control functions.f. Seismic qualification certificates shall be submitted that details equipment anchorage components, identifies equipment center of gravity with mounting and anchorage provisions, and whether the seismic qualification certificate is based on an actual test or calculations.

f. Submit documentation indicating compliance with applicable requirements with ASHRAE 90.1 - IP for Service Water Heating .

g. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

h. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

i. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

a. For commercial applications, comply with ASHRAE 90.1 - IP for efficiency performance.

b. Electrical components, devices and accessories shall be listed and labeled as defined in NFPA 70 by a qualified testing agency and marked for intended location and application.

c. ASME code construction shall be a vessel fabricated in compliance with the ASME BPVC SEC VIII.

d. Fabricate and label equipment components that will be in contact with potable water to comply with NSF/ANSI 61 and NSF 372.

e. The domestic water heater shall be certified and labeled by an independent testing agency.

f. Bio-Based Materials: For products designated by the USDA’s Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit https://www.biopreferred.gov.

g. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in Paragraph AS-BUILT DOCUMENTATION of Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING

PART 2 PRODUCTS

2.1 DOMESTIC WATER HEATER AND COMPONENTS

a. For each domestic hot water heater type and size, the following characteristics shall be submitted:

(1) Heat Pump Water Heater

(2) Expansion Tank

(3) Drain Pan

(4) Heat Traps

(5) Combination T&P Relief Valve

(6) Thermometers

(7) Supports

2.2 HEAT-PUMP DOMESTIC WATER HEATER

a. Provide an integrated air-source heat-pump water heater with factory-assembled storage tank. Tank construction shall be stainless-steel or glass-lined steel and rated for 150 psig. Tank shall comply with NSF/ANSI 61 and NSF 372.

b. Heat-pump assembly shall include hermetic compressor, evaporator, condenser coil, refrigerant charge, and controls for automatic operation. Electric resistance elements, where provided, shall function as supplemental or emergency backup heat only.

c. Provide factory-installed insulation meeting ASHRAE 90.1 and ENERGY STAR requirements.

d. Provide adjustable temperature controls. Minimum storage temperature: 140°F. If thermal eradication above is required, provide auxiliary electric heating or external means as scheduled.

e. Provide ASME-rated Combination Pressure and Temperature Relief Valve complying with ANSI Z21.22/CSA 4.4.

f. Provide corrosion-resistant drain valve, NPS 3/4 inch minimum, with ASME B1.20.7 hose threads.

g. A steel pressure rated tank constructed with welded joints and factory installed butyl rubber diaphragm shall be installed as scheduled. The air precharge shall be set to minimum system operating pressure at tank.

h. Provide all sensors, aquastats, controls, and piping specialties required for a complete operating system.

i. Provide condensate drain piping and neutralization (if required).

j. Heater capacities and electrical characteristics are scheduled on the drawings.

2.3 DOMESTIC HOT WATER EXPANSION TANKS

a. A steel pressure rated tank constructed with welded joints and factory installed butyl rubber diaphragm shall be installed as scheduled. The air precharge shall be set to minimum system operating pressure at tank.

b. The tappings shall be factory-fabricated steel, welded to the tank and include ASME B1.20.1 pipe thread.

c. The interior finish shall comply with NSF/ANSI 61 and NSF 372 for barrier materials for potable water tank linings and the liner shall extend into and through the tank fittings and outlets.

d. The air charging valve shall be factory installed.

2.4 WATER HEATER DRAIN PAN

A stainless-steel drain pan shall be provided that is 6 inches larger than the water heater and has a minimum of 6 inch sides. The drain pan shall include a drain outlet not less than NPS 3/4 inch with ASME B1.20.7 garden hose threads.

2.5 HEAT TRAPS

Heat traps shall be installed in accordance with ASHRAE 90.1 - IP unless provided integrally with the heater.

2.6 COMBINATION TEMPERATURE AND PRESSURE RELIEF VALVES

The combination pressure and temperature relief valve shall be ANSI Z21.22/CSA 4.4 and ASME rated and constructed of all brass or bronze with a self-closing reseating valve. The relief valves shall include a relieving capacity greater than the heat input and include a pressure setting less than the water heater’s working pressure rating. Sensing element shall extend into storage tank.

2.7 THERMOMETERS

Thermometers shall be rigid stem or remote sensing, scale or dial type with an aluminum, black metal, stainless-steel, or chromium plated brass case. The thermometer shall be back connected, red liquid (alcohol or organic-based) fill, vapor, bi-metal or gas actuated, with 9 inches high scale dial or circular dial 2 to 5 inches in diameter graduated from 40 to 212 degrees F, with two-degree graduations guaranteed accurate within one scale division. The socket shall be separable, double-seat, micrometer-fittings, with extension neck not less than 2-1/2 inches to clear tank or pipe covering. The thermometer shall be suitable for 3/4 inch pipe threads. Thermometers may be console-mounted with sensor installed in separate thermometer well.

2.8 SUPPORTS

a. Water heater stands shall be factory-fabricated steel for floor mounting capable of supporting water heater and water a minimum of 18 inches above the floor.

b. Wall brackets for wall mounted heaters shall be factory-fabricated steel capable of supporting water heater and water.

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the contractor shall correct the installation at no additional cost or time to the Government.

b. Water heaters shall be installed on concrete bases unless elevated above the floor. Refer to Section 03 30 00, CAST-IN-PLACE CONCRETE and Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

c. The water heaters shall be installed level and plumb and securely anchored.

d. The water heaters shall be installed and connected in accordance with manufacturer’s written instructions with manufacturer’s recommended clearances.

e. All pressure and temperature relief valves discharge shall be piped to nearby floor drains with air gap.

f. Thermometers shall be installed on the water heater inlet and outlet piping and shall be positioned such that they can be read by an operator or staff standing on floor or walkway.

g. The thermostatic control shall be set for a minimum setting of 140 degrees F for storage heaters.

h. Dielectric unions shall be provided if there are dissimilar metals between the water heater connections and the attached piping.

i. Provide vacuum breakers per ANSI Z21.22/CSA 4.4 on the inlet pipe if the water heater is bottom fed.

j. Shutoff valves shall be installed on the domestic water supply piping to the water heater and on the domestic hot water outlet piping.

k. All manufacturer’s required clearances shall be maintained.

l. The electric domestic water heaters shall be installed with seismic restraint devices.

m. A combination temperature and pressure relief valve shall be installed at the top portion of the storage tank in accordance with manufacturer’s recommendations. The sensing element shall extend into the tank. The relief valve outlet drain piping shall discharge by air gap into a floor drain.

n. Piping type heat traps shall be installed on the inlet and outlet piping of the electric domestic water heater storage tanks if not provided integrally with the tanks.

o. Water heater drain piping shall be installed as indirect waste to spill by air gap into open drains or over floor drains. Hose end drain valves shall be installed at low points in water piping for electric domestic water heaters without integral drains.

3.2 LEAKAGE TEST

Before piping connections are made, water heaters shall be tested with hydrostatic pressure of 200 psig and 240 psig for a unit with a MAWP of 160 psig. Any domestic water heater leaking water shall be replaced with a new unit at no additional cost or time to the Government.

3.3 PERFORMANCE TEST

Ensure that all the remote water outlets are always tested to a minimum of 110 degrees F and a maximum of 120 degrees F water flow.

3.4 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. The tests shall include system capacity, control function, and alarm functions.

c. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

d. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

3.5 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.6 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for four hour to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

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SECTION 22 40 00
5/24

PLUMBING FIXTURES

PART 1 GENERAL

1.1 DESCRIPTION

a. Plumbing fixtures, associated trim and fittings necessary to make a complete installation from wall or floor connections to rough piping, and certain accessories.

b. A complete listing of all acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 07 92 00, JOINT SEALANTS: Sealing between fixtures and other finish surfaces.

f. Section 08 31 13, ACCESS DOORS AND FRAMES: Flush panel access doors.

h. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

i. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS: Requirements for commissioning, systems readiness checklist, and training.

j. Section 22 13 00, FACILITY SANITARY AND VENT PIPING.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A112.6.1M	(1997; R 2017) Floor Affixed Supports for Off-the-Floor Plumbing Fixtures for Public Use
ASME A112.19.1/CSA B45.2	(2024) Enameled Cast Iron and Enameled Steel Plumbing Fixtures

NSF INTERNATIONAL (NSF)

NSF/ANSI 14	(2023) Plastics Piping System Components and Related Materials
NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects
NSF 372	(2022) Drinking Water System Components - Lead Content

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Submit Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating Instructions; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

Certification Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 22 40 00, PLUMBING FIXTURES”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, connections, and capacity.

d. Operating Instructions: Comply with requirements in Section 01 00 00, GENERAL REQUIREMENTS.

e. Completed System Readiness Checklist provided by the CxA and completed

by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

f. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

1.7 AS-BUILT DOCUMENTATION

a. Submit manufacturer's literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, substitutions and construction revisions shall bein electronic version on compact disc or DVDinserted into a three ring binder.All aspects of system operation and maintenance procedures, including piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices such as damper and door closure interlocks shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. As-built drawings are to be provided, and a copy of them in Revit version 2025 provided on compact disk or DVD. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the 'third party testing company' requirement.

d. Certification documentation shall be provided to Contracting Officer 10 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and certification that all results of tests were within limits specified.

PART 2 PRODUCTS

2.1 MATERIALS

a. Material or equipment containing a weighted average of greater than 0.25 percent lead is prohibited in any potable water system intended

for human consumption, and shall be certified in accordance with NSF/ANSI 61 or NSF 372. Endpoint devices used to dispense water for drinking shall meet the requirements of NSF/ANSI 61.

b. Plastic pipe, fittings, and solvent cement shall meet NSF/ANSI 14 and shall be NSF listed for the service intended.

2.2 STOPS

a. Provide lock-shield loose key or screw driver pattern angle stops, straight stops or stops integral with faucet, with each compression type faucet whether specifically called for or not, including sinks in solid-surface, wood and metal casework, laboratory furniture and pharmacy furniture. Locate stops centrally above or below fixture in accessible location.

b. Furnish keys for lock shield stops to the Contracting Officer.

c. Supply from stops not integral with faucet shall be chrome plated copper flexible tubing or flexible stainless steel with inner core of non-toxic polymer.

d. Supply pipe from wall to valve stop shall be rigid threaded IPS copper alloy pipe, i.e. red brass pipe nipple, chrome plated where exposed.

2.3 ESCUTCHEONS

Heavy type, chrome plated, with set screws. Provide for piping serving plumbing fixtures and at each wall, ceiling and floor penetrations in exposed finished locations and within cabinets and millwork.

2.4 LAMINAR FLOW CONTROL DEVICE

a. Smooth, bright stainless steel or satin finish, chrome plated metal laminar flow device shall provide non-aeration, clear, coherent laminar flow that will not splash in basin. Device shall also have a flow control restrictor and have vandal resistant housing. **Aerators are prohibited.**

b. Flow Control Restrictor:

(1) Capable of restricting flow from 0.25 gpm to 1.5 gpm for lavatories.

(2) Compensates for pressure fluctuation maintaining flow rate specified above within 10 percent between 25 psig and 80 psig.

(3) Operates by expansion and contraction, eliminates mineral/sediment build-up with self-cleaning action, and is capable of easy manual cleaning.

2.5 CARRIERS

a. ASME A112.6.1M, lavatory, concealed arm support. All lavatory chair carriers shall be capable of supporting the lavatory with a 250-pound vertical load applied at the front of the fixture.

2.6 WATER CLOSETS

a. (P-103) Water Closet (Wall Hung, ASME A112.19.1/CSA B45.2) office and industrial, elongated bowl, siphon jet dual flush oscillating bio-guard handle, 1.1 gallon/1.28 gallon per flush, wall outlet. Top of seat shall be between 16 inches and 17 inches above finished floor. Handicapped water closet shall have seat set 18 inches above finished floor.

(1) Seat: Institutional/Industrial, extra heavy duty, chemical resistant, solid plastic, open front less cover for elongated bowls, integrally molded bumpers, concealed check hinge with stainless steel post. Seat shall be posture contoured body design. Color shall be white.

(2) Fittings and Accessories: Gaskets-neoprene; bolts with chromium plated caps nuts and washers and carrier.

(3) Flush valve: Large chloramines resistant diaphragm, semi-red brass valve body, exposed chrome plated, dual flush non-hold open ADA approved side oscillating handle, water saver design per flush with maximum 10 percent variance 1 inch screwdriver back check angle stop with vandal resistant cap, adjustable tailpiece, a high back pressure vacuum breaker, spud coupling for 1-1/2 inches top spud, wall and spud flanges, solid-ring pipe support, and sweat solder adapter with cover tube and set screw wall flange. Valve body, cover, tailpiece and control stop shall be in conformance with ASTM alloy classification for semi-red brass. Seat bumpers shall be integral part of flush valve. Set centerline of inlet 11-1/2 inches above seat.

2.7 LAVATORIES

a. Dimensions for lavatories are specified, Length by width (distance from wall) and depth.

b. Brass components in contact with water shall contain no more than 0.25 percent lead content by dry weight. Faucet flow rates shall be 1.5 gpm for private lavatories and either 0.5 gpm or 0.25 gallons per cycle for public lavatories.

c. (P-402) Lavatory straight back, approximately 20 inches by 18 inches and a 4 inches maximum apron, first quality vitreous china. Punching for faucet on 4 inches centers. Set with rim 34 inches above finished floor.

(1) Faucet: Solid cast brass construction with washerless ceramic disc mixing cartridge type and centrally exposed rigid gooseneck spout with outlet 5-6 inches above rim. Provide laminar flow control device. 4 inches elbow handles on faucets shall be cast, formed or drop forged copper alloy. Faucet, wall and floor escutcheons shall be either copper alloy or CRS. Exposed metal parts, including exposed part under valve handle when in open position, shall have a smooth bright finish. Flow shall be limited to 0.35 gpm.

(2) Drain: Cast or wrought brass with flat grid strainer and offset tailpiece, chrome plated finish.

(3) Stops: Angle type, See paragraph “Stops”.

(4) Trap: Cast copper alloy, 1 1/2 inches by 1 1/4 inches P-trap. Adjustable with connected elbow and 17 gauge thick tubing extensions to wall. Exposed metal trap surfaces and connection hardware shall be chrome plated with a smooth bright finish. Set trap parallel to wall.

(5) Integral front shroud shall conceal piping, drain, stops, and trap and comply with 36 CFR 1191, A.D.A.H

2.8 SINKS

a. (P-502) Service Sink (Corner, Floor Mounted) stain resistant terrazzo, 28 inches by 28 inches by 12 inches with 6 inches drop front. Terrazzo, composed of marble chips and white Portland cement, shall develop compressive strength of 3000 psig seven days after casting. Provide extruded aluminum cap on front side.

(1) Faucet: Solid brass construction, 2.5 gpm combination faucet with replaceable Monel seat, removable replacement unit containing all parts subject to wear, integral check/stops, mounted on wall above sink. Spout shall have a pail hook, 3/4 inch hose coupling threads, vacuum breaker, and top or bottom brace to wall. Four-arm handles on faucets shall be cast, formed, or drop forged copper alloy. Escutcheons shall be either forged copper alloy or CRS. Exposed metal parts, including exposed part under valve handle when in open position, shall have a smooth bright finish. Provide 36 inches hose with wall hook. Centerline of rough in is 48 inches above finished floor.

(2) Drain: 3 inches cast brass drain with nickel bronze strainer.

(3) Trap: P-trap, drain through floor.

2.9 DISPENSER, DRINKING WATER

a. Standard rating conditions: 50 degrees F water with 80 degrees F inlet water temperature and 90 degrees F ambient air temperature.

b. (P-609) Electric Water Cooler: Mechanically cooled, self contained, wheel chair acessible, bubbler style, fully exposed dual height stainless steel fountain with integral bottle filling station., self-contained refrigeration system, fully exposed wall-mounted unit, , stainless steel grille, integral wall mounting frame or bracket, energy efficient cooling system consisting of a hermetically sealed reciprocating type compressor, 115v, 60 Hz, single phase, fan cooled condenser, permanently lubricated fan motor. Set highest bubbler 40 inches above finished floor. Provide with integral bottle filling station

2.10 SHOWER BATH FIXTURE

a. (P-704) Shower Bath Fixture (Wall Mounted, Concealed Supplies, Hose Spray):

(1) Shower Installation: Wall mounted showerhead connected to shower arm.

(2) Shower Heads: Chrome plated metal head, adjustable ball joint, self cleaning head with automatic flow control device to limit

discharge to not more than one point five (1.5) gpm. Body, internal parts of shower head and flow control fittings shall be copper alloy or CRS.Install showerhead 72 inches above finished floor.

(3) Valves: Type T/P combination temperature and pressure balancing, with chrome plated metal lever type operating with adjustment for rough-in variations handle and chrome plated metal or CRS face plate. Install diverter selector valve and elevated vacuum breaker to provide tempered water to shower head and hose spray. Valve body shall be any suitable copper alloy. Internal parts shall be copper nickel alloy, CRS or thermoplastic material. Valve inlet and outlet shall be 1/2 inch IPS. Provide external screwdriver check stops, and temperature limit stops. Set stops for a maximum temperature of 122 degrees F. All exposed fasteners shall be vandal resistant. Valve shall provide a minimum of 1.5 gpm at 45 psig pressure drop.

(4) Spray Assembly: Shall consist of a 60 inches length of rubber lined CRS, chrome plated metal flexible, or white vinyl reinforced hose with coupling for connection to 1/2 inch hose supply elbow protruding through wall. Spray shall consist of a self-closing, lever-handle, faucet with thumb control having open-shut positions and intermediate positions for regulating water flow and elevated pressure type vacuum breaker. Provide wall hook for faucet.

2.11 EMERGENCY FIXTURES

a. (P-707) Emergency Shower and Eye and Face Wash (Free Standing):

(1) Shower Head: Polished chrome plated, 8 inches in diameter, install head 84 inches above floor. Equip with stay-open ball valve, chrome plated. Operate valve with 24 inches stainless steel pull-rod with triangle handle. Pull-down opens valve; push-up closes valve. Flow rate shall be 20 gpm.(2) Emergency Eye and Face Wash: CRS receptor. Equipment with a 1/2 inch stay open ball valve operated by push flag handle. Mount eye and face wash spray heads 42 inches above finished floor. Flow rate shall be 3 gpm.

(2) Provide with thermostatic mixing valve, ASSE 1017 compliant, to provide tepid water from 75 to 85 degrees F. Refer to specification section 22 05 23 for additional mixing valve information.

(3) Shower head and emergency eye and face wash shall be mounted to stanchion with floor flange through floor waste connection and P-trap. Paint stanchion same color as room interior. Provide with signage to easily locate fixture.

(4) Provide with emergency alarm horn and light. Tie alarm to BAS.

(5) Unit shall be freezeless.

2.12 HYDRANT, HOSE BIBB AND MISCELLANEOUS DEVICES

a. (P-801) Wall Hydrant: Cast bronze non-freeze hydrant with detachable T-handle. Brass operating rod within casing of bronze pipe of sufficient length to extend through wall and place valve inside building. Brass valve with coupling and union elbow having

metal-to-metal seat. Valve rod and seat washer removable through face of hydrant; 3/4 inch hose thread on spout; 3/4 inch pipe thread on inlet. Finish may be rough; exposed surfaces shall be chrome plated. Set not less than 18 inches nor more than 36 inches above grade. On porches and platforms, set approximately 30 inches above finished floor. Provide integral vacuum breaker which automatically drains when shut off.

b. (P-804) Hose Bibb (Single Faucet, Wall Mounted to ExposedSupply Pipe): Cast or wrought copper alloy, single faucet with replaceable Monel seat, removable replacement unit containing all parts subject to wear, mounted on wall 36 inches above floor to concealed supply pipe. Provide faucet with 3/4 inch hose coupling thread on spout and vacuum breaker. Four-arm handle on faucet shall be cast, formed or drop forged copper alloy. Escutcheons shall be either forged copper alloy or CRS. Exposed metal parts, including exposed part under valve handle when in open position, shall have a bright finish.

c. (P-805) Roof Hydrant: Freezeless type. Shall be one-piece variable flow plunger type with rod guide, integral storage reservoir, and venturi system to evacuate water and provide drain-back operation for freeze-proof winterization without a drain connection. Hydrant shall be complete with cast iron flanged support, under-deck flange, and watertight seal between support and hydrant pipe with EPDM boot cover.

Provide hydrant with 3/4 inch threaded hose outlet and integral ASSE 1052 or ASSE 1057 approved, field-testable, double check valve backflow preventer.

Set hydrant not less than 18 inches nor more than 36 inches above roof, unless otherwise indicated.

PART 3 EXECUTION

3.1 INSTALLATION

a. Fixture Setting: Opening between fixture and floor and wall finish shall be sealed as specified under Section 07 92 00, JOINT SEALANTS. Bio-based materials shall be utilized when possible.

b. Supports and Fastening: Secure all fixtures, equipment and trimmings to partitions, walls and related finish surfaces. Exposed heads of bolts and nuts in finished rooms shall be hexagonal, polished chrome plated brass with rounded tops.

c. Not used.

d. Toggle Bolts: For hollow masonry units, finished or unfinished.

e. Expansion Bolts: For brick or concrete or other solid masonry. Shall be 1/4 inch diameter bolts, and to extend at least 3 inches into masonry and be fitted with loose tubing or sleeves extending into masonry. Wood plugs, fiber plugs, lead or other soft metal shields are prohibited.

f. Power Set Fasteners: May be used for concrete walls, shall be 1/4 inch threaded studs, and shall extend at least 1 1/4 inches into wall.

g. Tightly cover and protect fixtures and equipment against dirt, water and chemical or mechanical injury.

h. Where water closet waste pipe has to be offset due to beam interference, provide correct and additional piping necessary to eliminate relocation of water closet.

i. Aerators are prohibited on lavatories and sinks.

j. If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no cost or additional time to the Government.

3.2 CLEANING

At completion of all work, fixtures, exposed materials and equipment shall be thoroughly cleaned.

3.3 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.4 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for four hours to instruct VA Personnel in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

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VACUUM SYSTEMS FOR LABORATORY AND HEALTHCARE FACILITIES

PART 1 GENERAL

1.1 DESCRIPTION

a. Central Laboratory and Healthcare Vacuum Systems: This section describes the labor, equipment, and services necessary for and incidental to the installation of piped medical vacuum systems and medical vacuum and waste anesthesia gas disposal systems (WAGD). Medical vacuum and WAGD systems shall be installed started, tested, and ready for use. The scope of work shall include all necessary piping, fittings, valves, cabinets, station outlets and inlets, rough ins, ceiling services, gages, alarms including low voltage wiring, vacuum pumps, electric motors and starters, receivers, and all necessary parts, accessories, connections and equipment for a complete and operational system.

b. The Contractor shall provide all elements and accessories required for a complete system according to the most recent edition of NFPA 99, Gas and Vacuum Systems.

c. All necessary connections to owner furnished equipment shall be made as indicated on the Contract documents. A separate construction isolation valve shall be made at the point of connection to an existing vacuum system.

d. Electrical power and control wiring for vacuum pump(s), WAGD Producer(s), ceiling columns, alarms wiring from equipment to alarm panels, and modular accessories associated with the system(s) shall be included.

e. Pressure testing, cross connection testing and final testing per NFPA 99 shall be performed.

f. The Contractor shall retain a qualified third-party medical vacuum verifier acceptable to the engineer of record and VA to perform and attest to final verification of the systems. The Contractor shall make all corrections as determined by this third-party verifier, including additional testing if necessary to attain full and unqualified certification at no additional time or cost to the Government.

g. Coordinate with owner retained verifier for final verification of the systems. Make corrections as required, including additional testing if necessary to attain full certification.

h. A complete listing of all acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 07 84 00, FIRESTOPPING: Sealing around pipe penetrations to maintain the integrity of time rated construction.

f. Section 07 92 00, JOINT SEALANTS: Sealing around pipe penetrations through the floor to prevent moisture migration.

h. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING: General requirements and items common to more than one section of Division 22.

i. Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT: Electric motors.

j. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

k. Section 22 63 00, GAS SYSTEMS FOR LABORATORY AND HEALTHCARE FACILITIES: Laboratory and healthcare gases and vacuum alarms.

l. Section 22 63 00, GAS SYSTEMS FOR LABORATORY AND HEALTHCARE FACILITIES: Laboratory and healthcare gas piping and equipment.

m. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC: Alarm interface with BAS.

n. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Control wiring.

o. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduit.

p. Section 26 27 26, WIRING DEVICES: Electrical wiring and accessories.

q. Section 26 29 11, MOTOR CONTROLLERS: Motor starters.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the test by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A13.1	(2023) Scheme for the Identification of Piping Systems
ASME B16.22	(2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings
ASME B16.50	(2021) Wrought Copper and Copper Alloy

	Braze-Joint Pressure Fittings
ASME B40.100	(2022) Pressure Gauges and Gauge Attachments
ASME BPVC SEC VIII	(2021) Boiler and Pressure Vessel Codes: Section VIII Rules for Construction of Pressure Vessel

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

ASSE 6000 Series	(2021) Professional Qualification Standard for Medical Gas Systems Installers, Inspectors and Verifiers
ASSE 6030	(2018) Medical Gas Systems Verifiers

ASTM INTERNATIONAL (ASTM)

ASTM B819	(2000; R 2011) Standard Specification for Seamless Copper Tube for Medical Gas Systems

AMERICAN WELDING SOCIETY (AWS)

AWS A5.8/A5.8M	(2019) Specification for Filler Metals for Brazing and Braze Welding

COMPRESSED GAS ASSOCIATION (CGA)

CGA C-9	(2019) Standard Color Marking of Compressed Gas Containers for Medical Use
CGA P-9	(2008) The Inert Gases: Argon, Nitrogen, and Helium

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-72	(2010a) Ball Valves with Flanged or Butt-Welding Ends for General Service
MSS SP-110	(2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 6	(1993; R 2016) Industrial Control and Systems: Enclosures

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 99	(2024) Health Care Facilities Code

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control

approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Notarized Affidavit; G, AE

Equipment Installer; G, AE

Equipment Supplier; G, AE

Medical Gas System Testing Organization; G, AE

Testing Agency's Detailed Procedure; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-05 Design Data

Station Inlets; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

Final Inspection Documentation; G, AE

Vacuum Pump Test Results; G, AE

SD-07 Certificates

Certification; G, AE

Certification Documentation; G, AE

SD-08 Manufacturer's Instructions

Submit Training Plans And Instructor Qualifications; G, AE

Installation And Startup; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 22 62 00, VACUUM SYSTEMS FOR LABORATORY AND HEALTHCARE FACILITIES”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and

capacity.

(1) Complete specifications for the product intended to be installed, dimensional drawings, and wiring schematics.

(2) Package drawing indicating package style, dimensions when complete, method of disassembly and sizes of subsections for rigging and installation.

(3) Piping.

(4) Valves.

(5) Inlet and outlet cocks

(6) Valve cabinets.

(7) Gages.

(8) Station inlets, and rough in assemblies.

(9) Vacuum switches.

(10) Vacuum pump systems (Provide certified pump test data at startup):

(a) Pumps: Manufacturer and model.

(b) Pump performance curves.

(c) Pump operating speed (RPM).

(d) Capacity: Free air exhaust from 19 and 24 inches Hg gage vacuum (L/s) (SCFM).

(e) Capacity: Expanded air capacity at 19 and 24 inches Hg gage vacuum (L/s) (SCFM).

(f) Type of bearing in pump.

(g) Type of lubrication.

(h) Type and adjustment of drive.

(i) Electric motors: Manufacturer, frame and type.

(j) Speed of motors (RPM).

(k) Current characteristics and horsepower of motors.

(l) Receiver capacity and rating.

(m) Silencers: Manufacturer, type and model.

d. Station Inlets: A letter from manufacturer shall be submitted stating that inlets are designed and manufactured to comply with NFPA 99. Inlet shall bear label of approval as an assembly, of Underwriters Laboratories, Inc., or Associated Factory Mutual Research Corporation.

e. Certification: The completed systems have been installed, tested, purged and analyzed in accordance with the requirements of this specification. Certification shall be submitted to Contracting Officer.

f. A notarized affidavit from the verifier stating that the verifier undertakes to verify this project and thus agrees to disqualify themselves from supplying any equipment which shall be included in the scope of their verification. No verifier who supplies equipment shall be permitted to verify that equipment. Statement declaring that the vacuum system manufacturer has no fiduciary interest in the verifier and that the verifier is not an agent or representative of the vacuum system manufacturer. Statement declaring that the Contractor has no fiduciary interest in the third-party verifier and that the third-party verifier has no fiduciary interest in the Contractor.

g. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

h. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

1.6 QUALITY ASSURANCE

a. Contractor shall include with submittals an affidavit attesting to compliance with all relevant paragraphs of NFPA 99 most recent edition. Personnel assembling medical vacuum and WAGD system shall meet NFPA 99 5.1.10.11.10 “Qualification of Installers” and hold medical gas endorsements as under ASSE 6000 Series. The Contractor shall, on company letterhead, furnish documentation attesting that all installed piping materials were purchased cleaned and complied with the requirements of NFPA 99 5.1.10.1 and 5.1.10.2. Electrical Control systems and Medical vacuum Alarms are to be UL listed as assemblies with label affixed. Medical vacuum and WAGD controls are to be wired in accordance with NEC.

b. Equipment Installer: The equipment installer shall provide documentation proving that the personnel installing the equipment meet the standards set by ASSE 6000 Series. Show technical qualifications and previous experience in installing medical gas equipment on three similar projects. Submit names, phone numbers, and addresses of referenced projects. The equipment installer shall perform the following coordination functions:

(1) Coordinate with other trades to ensure timely installations and avoid conflicts and interferences.

(2) Coordinate and field verify with the metal stud partition installer and/or mason to ensure anchors, sleeves and similar items are provided in sufficient time to avoid delays; chases and openings are properly sized and prepared.

(3) Coordinate with VA to ensure medical vacuum inlets, whether owner supplied or Contractor supplied, in walls, ceiling and all equipment is provided by the same Medical Vacuum Equipment Manufacturer satisfactory to the owner.

(4) The Contractor shall coordinate with the Medical Vacuum System. Verifier to deliver a complete, operational, and tested medical gas installation ready for owner’s use.

c. Equipment Supplier: The Equipment supplier shall demonstrate evidence of installing equivalent product at three installations similar to this project that has been in satisfactory and efficient operation for three years. Names, phone numbers, and addresses where the product is installed shall be submitted for verification.

d. Medical Gas System Testing Organization: The Medical vacuum verifier shall show documentation proving that the medical gas verifier meets the standards set by ASSE 6000 Series . The testing shall be conducted by a party technically competent and experienced in the field of medical gas pipeline testing. Such testing shall be performed by a third-party testing company independent of the installing and general contractor.

e. Names of three projects where testing of vacuum systems has been performed by the testing agency shall be provided. The name of the project, names of such persons at that project who supervised the work for the project owner, or who accepted the report for the project owner, and a written statement that the projects listed required work of similar scope to that set forth in this specification shall be included in the documentation.

f. The testing agency's detailed procedure shall be followed in the testing of this project and submitted to Contracting Officer 10 working days prior to testing. In the testing agency’s procedure documentation, include details of the testing sequence, procedures for cross connection tests, outlet function tests, alarm tests, purity tests, etc., as required by this specification. For purity test procedures, data on test methods, types of equipment to be used, calibration sources and method references shall be submitted.

g. Installation and Startup: The manufacturer shall provide factory authorized representatives to review the installation and perform the initial startup of the system. The factory authorized representatives shall submit a report to the Contracting Officer and to the Contractor. The Contractor shall make all corrections identified by the factory authorized representative at no additional cost or time to the Government.

h. Certification: The Final inspection documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and certification that all results of tests were within limits allowed by this specification.

i. Bio-Based Materials: For products designated by the USDA’s Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the

Bio-Preferred Program, visit http://www.biopreferred.gov.

1.7 MAINTENANCE SUPPORT

The medical vacuum equipment manufacturer shall demonstrate a national factory direct service capability able to perform major overhauls. The medical vacuum equipment manufacturer shall provide factory direct preventative maintenance contract. The medical vacuum equipment manufacturer shall provide formal maintenance training courses. See paragraph “Demonstration and Training” for additional requirements for training. Servicer shall be no more than 100 miles away, be capable of responding within four hours, and provide certified personnel to perform all work.

1.8 AS-BUILT DOCUMENTATION

a. Submit manufacturer’s literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, substitutions and construction revisions shall be inserted into a three-ring binder. All aspects of system operation and maintenance procedures, including piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices such as damper and door closure interlocks shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner shall be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. As-built drawings are to be provided, and a copy of them in Revit version 2025 provided on compact disk or DVD. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the ‘third party testing company’ requirement.

d. Certification documentation shall be provided to Contracting Officer 10 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and certification that all results of tests were within limits specified.

PART 2 PRODUCTS

2.1 GENERAL PRODUCT REQUIREMENTS

A single Medical Vacuum Equipment Manufacturer shall supply the medical vacuum system(s) and equipment to include outlets, valves and gauges, valve boxes, alarm panels, manifolds, medical air, instrument air, vacuum and WAGD sources.

2.2 PIPING

a. Copper Tubing: Copper tubing shall be type "K", ASTM B819, seamless copper tube, hard drawn temper, with wrought copper fittings conforming to ASME B16.22 or brazing fittings complying with ASME B16.50. The copper tubing size designated reflects nominal inside diameter. All tubing and fittings shall be labeled "ACR/OXY", "OXY", "OXY/MED", "ACR/MED", or "MED".

b. Brazing Alloy: The brazing alloy shall comply with AWS A5.8/A5.8M, Classification BCuP, greater than 1000 degrees F melting temperature. Flux shall be strictly prohibited for copper-to-copper connections.

c. Screw Joints: Screw joints shall use polytetrafluoroethylene (Teflon) tape.

d. Use only copper or stainless-steel pipes for discharge from vacuum product (exhaust pipes).

e. Memory metal couplings shall have temperature and pressure ratings not less than that of a brazed joint.

f. Piping identification labels shall be applied at time of installation in accordance with NFPA 99. Supplementary color identification shall be in accordance with CGA C-9.

g. Special Fittings: The following special fittings shall be permitted to be used in lieu of brazed joints:

(1) Memory-metal couplings having temperature and pressure ratings joints not less than that of a brazed joint.

(2) Listed or approved metallic gas tube fittings that, when made up, provide a permanent joint having the mechanical, thermal, and sealing integrity of a brazed joint.

(3) Dielectric fittings where required by the manufacturer of special medical equipment to electrically isolate the equipment from the piping distribution system.

(4) Axially swaged, elastic strain preload fittings providing metal to metal seal having pressure and temperature ratings not less than that of a brazed joint and when complete are permanent and non-separable.

2.3 VALVES

a. Ball: Ball valves shall be in line, other than zone valves in cabinets.

(1) 2-1/2 inches and less: Ball valves shall be bronze/ brass body, MSS SP-72 and MSS SP-110, Type II, Class 150, Style 1, with tubing extensions for brazed connections, full ported, three piece or double union end connections, Teflon seat seals, full flow, 600 psig WOG minimum working pressure, with locking type handle.

(2) 3 to 4 inches: Ball valves shall be bronze/ brass body, MSS SP-72 and MSS SP-110, Type II, Class 150, Style 1 with tubing extensions brazed to flanges, full ported, three-piece, double

seal, Teflon seals, full flow, 600 psig WOG minimum working pressure, with locking type handle.

b. Check:

(1) 3 inches and less: Check valves shall be brass and bronze body, straight through design for minimum pressure drop, spring loaded, self-aligning with Teflon cone seat, vibration free, silent operation, supplied NPT female threads at each end with flow direction arrow permanently cast into body, 400 psig WOG minimum working pressure.

(2) 4 inches and greater: Check valves shall be iron body, bronze trim, swing type, vertical or horizontal installation, flange connection, 150 psig WSP with flow direction arrow permanently cast into body.

c. Zone valve in cabinet shall be ball valve with bronze/ brass body, double seal, three piece or double union end connections, replaceable Teflon seat seals, Teflon stem seal, 600 psig WOG, cold, non-shock gas working pressure or vacuum service to 29.5 inches Hg, blowout proof stem, one quarter turn of handle to completely open or close. Tubing extensions, factory brazed, pressure tested, cleaned for oxygen service shall be provided. A 1/8 inch NPT gauge port shall be provided for a 2 inch diameter monitoring gauge downstream of the shut-off valve. Zone valves shall be securely attached to the cabinet and provided with type “K” copper tube extensions for making connection to system piping outside the cabinet. Zone valves shall be products of one manufacturer, and uniform throughout in pattern, overall size and appearance. Trim with color coded plastic inserts or color-coded stick-on labels. Valves shall be in cabinets such that cover window cannot be in place when any valve is in the closed position. Color coding for identification plates and labels is as follows:

SERVICE LABEL	IDENTIFICATION COLORS	MFG. STD. CLR.
MEDICAL VACUUM	Black letters on white background	WHITE
Evacuation(Waste Gas)	White letters on violet background	VIOLET

2.4 GAGES

a. Vacuum Gages:

(1) For vacuum line adjacent to source equipment the vacuum gages shall comply with ASME B40.100, vacuum gage type, size 4-1/2 inches, gage listed for vacuum, accurate to within 2-1/2 percent, with metal case. The vacuum gage range shall be 0 to 29.5 inches Hg. Dial graduations and figures shall be black on a white background, or white on a black background. Label shall be for vacuum service. A gage cock shall be installed. Dual scale gages shall be installed for vacuum system.

(2) For vacuum service upstream of main shut-off valve: A 1-1/2 inches diameter gage shall be provided with steel case, bourdon tube and brass movement, dial range 0 to 29.5 inches Hg. Dual scale gages shall be provided for vacuum system.

2.5 MEDICAL VACUUM SYSTEMS

a. Provide a complete medical vacuum package, complying with NFPA 99 in all respects, as specified and scheduled in the Contract documents. All components shall be factory packaged (pre-wired and pre-piped), on a steel base, or tank mounted. All package piping shall be type “K” rigid copper. Provide discharge separator/silencer.

b. All components shall be duplex and valved (or check valved as permitted NFPA 99) to allow service to any component without interrupting vacuum service to the facility during any maintenance operation or any condition of single fault failure. The design load shall be met with the largest single unit out of service. Each pump exhaust shall be isolated by a union fitting permitting capping for service removal.

c. A complete and operational plant (source equipment) shall be furnished consisting of pumps, receiver and controls capable of providing the scheduled capacity with one pump out of service. All capacities shall be indicated in SCFM at 19 inches Hg and 24 inches Hg.

d. System shall be completely factory assembled, requiring only interconnection between modules on site. Systems requiring on site assembly other than interconnection are prohibited (replacement of components removed for shipping is permitted).

e. Motor and Starter: Maximum 104 degrees F ambient temperature rise, close coupled to a NEMA rated, High Efficiency, TEFC motor with a service factor of 1.15, ball bearings, for operation with current, voltage, phase and cycle specified in Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT. Motor shall be of such capacity that brake horsepower required by driver equipment at normal rated capacity will not exceed nameplate rating of the motor. Provide each motor with automatic, fully enclosed, magnetic starter of type specified in Section 26 29 11, MOTOR CONTROLLERS.

f. Each pump shall include inlet and outlet flex connectors supplied by the medical vacuum equipment manufacturer.

g. Programmable Logic Controllers (PLC) or Direct Digital Controls (DDC) can be used to implement operating logic. Controls shall have integral memory and EPROM backup. Controls shall control the automatic alternation lead-lag of the vacuum pumps with provisions for simultaneous operation if required, and automatic activation of reserve or lag unit if required. A lag alarm on control cabinet and contacts for the master alarm shall be provided.

h. The complete control system and all electrical components shall be NEMA ICS 6, Type 12 and UL labeled. The control system shall provide:

(1) Automatic lead/lag sequencing including self-adjusting minimum run timers which adaptively optimize the number of pump starts based on demand.

(2) Circuit breaker disconnects for each vacuum pump with external operators. Units with fuses instead of circuit breakers in motor circuit are prohibited. The control system shall include an automatic minimum run time adjustment to automatically adjust run time based on demand.

(3) Full voltage motor starters with overload protection.

(4) Redundant-120 volt control circuit transformers.

(5) Visual and audible reserve unit alarm with isolated contacts for remote alarms and audio cancel.

(6) Control cabinet shall have lighted HOA selector switches

(7) Panel mounted vacuum gauge, external visual lights indicating on/off status, audible and visual signals for thermal overload, oil discharge filter differential pressure or back pressure alarm, and alarm silence button.

(8) Contacts for external oil discharge filter differential pressure or back pressure alarm

(9) If silence has been pressed, audible and visual signal would be reactivated upon second alarm condition. Alarm shall be reset upon correction of original signal.

(10) Runtime hour-meter for each pump.

i. The medical vacuum pumps shall be oil free, single stage, positive displacement, and non-pulsating liquid ring type. The vacuum pump shall be fitted with mechanical seals. Each medical vacuum pump shall be all iron construction with bronze or stainless-steel rotor and carbon steel shaft. The medical vacuum pump design shall not require more than annual maintenance intervals. Fresh seal water shall be minimal under operation and shall include a reservoir of sufficient capacity for 96 hours of operation without a fresh water supply.

j. The vacuum pumps shall be individually tested and the vacuum pump test results shall be submitted to the Contracting Officer.

k. The medical vacuum pumps shall be non-contacting claw style rotary design. The internal construction shall be friction free and require no sealants. Each medical vacuum pump shall be air-cooled, and continuous duty rated. Each medical vacuum pump shall be provided with a single lubricated gearbox requiring an oil change not more often than 5,000 operating hours. Each medical vacuum pump shall be equipped with an exhaust silencer. Each medical vacuum pump shall be equipped with a high vacuum shutdown, a high temperature shutdown, a remote alarm at the BAS and local alarms. The lubricant supplied shall be inert with oxygen and the medical vacuum pump shall be factory cleaned for oxygen service.

l. The complete medical vacuum system and all electrical components shall be factory tested prior to shipment by the medical vacuum equipment manufacturer.

m. Waste Anesthetic Gas Disposal Pumps.

(1) Provide a complete WAGD source, complying with NFPA 99 5.1.3.7 in

all respects, as specified and scheduled.

(2) All components shall be at least duplexand valved to permit service to any component without interrupting WAGD supply to the facility during any maintenance operation or any condition of single fault failure.

(3) Furnish complete and operational plant consisting of producer and controls capable of providing the scheduled capacity with one producer out of service.

(4) System shall be completely factory assembled. Systems requiring site assembly are prohibited (removal of components for shipping is permitted).

(5) Each producer shall include inlet and outlet flex connectors supplied by the medical vacuum equipment manufacturer.

(6) A Programmable Logic Controller (PLC) or Direct Digital Controls (DDC) can be used to implement the operating sequence of operation. The controls shall have integral memory and EPROM memory backup. Alternating between the lead and lag WAGD pumps shall be made automatically by the programmable logic controller. The reserve unit shall be automatically activated as required to maintain uninterrupted service. A provision for simultaneous operation of two or more WAGD pumps shall be made when one operating WAGD pump is not meeting the demand. An alarm shall be activated whenever a reserve WAGD producer or lag WAGD producer is activated.

(7) The complete control system and all electrical components shall be NEMA 12 and UL labeled. The control system shall provide:

(a) Automatic lead/lag sequencing.

(b) Circuit breaker disconnects for each producer with external operators. Units with fuses instead of circuit breakers in motor circuit are prohibited.

(c) Full voltage motor starters with overload protection.

(d) Redundant 120-volt control circuit transformers.

(e) Visual and audible reserve unit alarm with isolated contacts for remote alarms and audio cancel.

(f) Control cabinet shall have lighted HOA selector switches

(g) Runtime hour-meter for each producer.

(7) Claw: A non-contacting claw style rotary pumps shall be provided. Internal construction is friction free and rotors are non-contacting. Air end is oil free and requires no sealants. Each pump is air cooled and continuous duty rated. Pump is provided with a single lubricated gearbox requiring lubricant change not more often than 5,000 operating hours. Pump is provided with exhaust silencer. Pumps shall be equipped with high

vacuum shutdown, high temperature shutdown, and visual and audible alarms per NFPA 99. Lubricant supplied shall be inert with oxygen. Pump shall be provided with vacuum modulated variable speed drive to control vacuum level at 7 inches Hg.

(6) The complete WAGD system and all electrical components shall be factory pretested prior to shipment.

n. Controls:

(1) Automatic: Adjustable, vacuum operated, automatic, electric switch to start and stop motor at receiver vacuum indicated. Provide heavy duty alternator, automatic, operating on a timed basis, to alternate the pumps by time forced alternation.

(2) Control panel: Housed in a NEMA ICS 6, Type 12, listed, dust proof enclosure; prewired to include all specified electrical, electronic and electro pneumatic devices. Include wiring diagrams and operating descriptions in the cabinet. Include the following:

(a) Circuit breakers for each control and motor circuit.

(b) Hand off automatic selector switch for each pump.

(c) Hour meter for each pump.

(d) Control circuit transformers.

(e) One magnetic motor starter for each pump.

(f) Provide panel with external visual (lights, red for running, green for off) and audible (horn/buzzer) signals. The signals provided include:

1. Pump in operation (visual only).

2. Thermal overload shutdown (visual and audible).

3. Oil discharge filter differential pressure or back pressure alarm (visual), with contacts for external signal. Wire to master alarm panel.

4. Cancel button, which shall silence an audible alarm, reactivate should a second alarm occur while the horn is silenced, and reset automatically upon correction of the original signal.

o. Receiver Tank: The receiver tank shall be welded galvanized steel, in compliance with ASME BPVC SEC VIII, 125 psig working pressure stamped and certified. The receiver tank shall be equipped with vacuum gage and gage glass. The receiver tank shall be of sufficient capacity to ensure practical on/off operation of pumps.

p. Bio-Hazard Safety Statement: "BIOHAZARD CAUTION: Fluid and waste material inside vacuum pipelines and vacuum equipment may be contaminated with blood and other potentially infectious material. Construction and service personnel shall use proper PERSONAL PROTECTIVE EQUIPMENT and practice UNIVERSAL PRECAUTIONS when opening or servicing vacuum systems."

PART 3 EXECUTION

3.1 INSTALLATION

a. All installation shall be performed in strict accordance with NFPA 99 5.1.10. Brazing procedures shall be as detailed in NFPA 99 5.1.10.4. Brazing shall be performed only by Brazers qualified under NFPA 99 5.1.10.11.10. Where piping runs underground, the installation shall be made in accordance with NFPA 99 5.1.10.11.5.

b. Contractor shall furnish 4 inches high concrete housekeeping pads. The Contractor shall furnish inertia bases in lieu of housekeeping pads where the equipment installed is not factory isolated by the manufacturer. Anchor bolts shall be cast into bases

c. Cast escutcheon shall be installed with set screw at each wall, floor and ceiling penetration in exposed finished locations and within cabinets and millwork.

d. Open ends of tube shall be capped or plugged at all times or otherwise sealed until final assembly to prevent infiltration of any foreign matter.

e. Piping shall be cut square and accurately with a tube cutter (**sawing is prohibited**) to measurements determined at place of installation. The tubing shall be reamed to remove burrs, being careful not to expand tube, and so no chips of copper remain in the tube. The tubing shall be worked into place without springing or forcing. The tubing shall be bottomed in socket so there are no gaps between tube and fitting. Care shall be exercised in handling equipment and tools used in cutting or reaming of tube to prevent oil or grease from being introduced into the tubing. Where contamination has occurred, material shall be no longer suitable for vacuum service and new, sealed tube sections used.

f. Piping shall be supported with pipe trays or hangers at intervals as shown on the Contract Drawings or as defined in NFPA 99. **Piping shall not be supported by other piping.** Isolation of copper piping from dissimilar metals shall be of a firm, positive nature. **Duct tape is prohibited as an isolation material.**

g. Valves and other equipment shall be rigidly supported to prevent strain on tube or joints.

h. Piping exposed to physical damage shall be protected.

i. During any brazing operation, the interior of the pipe shall be purged continuously with oil free, dry nitrogen NF, following the procedure in NFPA 99 5.1.10.4.5. At the completion of any section, all open pipe ends shall be capped using an EXTERNAL cap. The flow of purged gas shall be maintained until joint is cool to touch. The use of flux is prohibited when making of joints between copper-to-copper pipes and fittings.

j. Threaded joints in piping systems shall be avoided whenever possible. Where unavoidable, make up the male threads with polytetrafluoroethylene (such as Teflon) tape. Liquid sealants are prohibited.

k. Tubing shall not be bent. Fittings shall be used in all change of direction or angle.

l. After installation of the piping, but before installation of the outlet valves, blow lines clear using nitrogen NF per NFPA 99.

m. Two 1 inch minimum conduits shall be provided from ceiling column assembly to the adjacent corridor, one for mass spectrometer tubing and wiring and one for monitor wiring, and for connection to signal cabling network.

n. Pressure and vacuum switches, transmitter and gauges shall be installed to be easily accessed and provide access panel where installed above plaster ceiling. Pressure switch and sensors shall be installed with orifice nipple between the pipeline and switches/sensors.

o. Pipe labeling shall be applied during installation process and not after installation is completed. Size of legend letters shall be in accordance with ASME A13.1.

p. After initial leakage testing is completed, the piping shall be allowed to remain pressurized with testing gas until testing agency performs final tests.

q. Penetrations:

(1) Fire Stopping: Where pipes pass through fire partitions, fire walls, smoked partitions, or floors, fire stopping shall be installed that provides an effective barrier against the spread of fire, smoke and gases as specified in Section 07 84 00, FIRESTOPPING, Clearances between raceways and openings with the fire stopping material shall be completely filled and sealed.

(2) Water proofing: At floor penetrations, clearances shall be completely sealed around the pipe and made watertight with sealant as specified in Section 07 92 00, JOINT SEALANTS.

r. A vacuum gage 1-1/2 inch diameter shall be installed in line downstream of each valve located in a zone valve cabinet.

t.

s. Piping shall be labeled with name of service, identification color and direction of flow. Where non-standard pressures are piped, pressure shall be labeled. Labels shall be placed at least once every 20 feet of linear run or once in each story (whichever is more frequent). A label shall additionally be placed immediately on each side of all wall or floor penetrations. Pipe labels shall be self-adhesive vinyl type or other water-resistant material with permanent adhesive colored in accordance with NFPA 99 Table 5.1.11 and shall be visible on all sides of the pipe. Each master alarm signal shall be labeled for function after ring out. Each zone valve shall be labeled and each area alarm labeled for the area of control or surveillance after test. Labels shall be permanent and of a type approved by the VAMC.

t. Valves shall be labeled for service and areas monitored or

controlled. Coordinate with the VAMC for final room or area designations. Valves shall be labeled with name and identification color of the gas and direction of flow.

3.2 INSTALLER TESTING

a. Prior to declaring the lines ready for final verification, the installing Contractor shall strictly follow the procedures for verification as described in NFPA 99 5.1.12.2 and attest in writing over the notarized signature of an officer of the installing company the following:

(1) That all brazing was conducted by a Brazer qualified to ASSE 6000 Series and holding current medical gas endorsements.

(2) That all brazing was conducted with nitrogen purging. (Procedure per NFPA 99 5.1.10.4.5).

(3) That the lines have been blown clear of any construction debris using oil free dry nitrogen or air are clean and ready for use. (Procedure per NFPA 99 5.1.12.2.2).

(4) That the assembled piping, prior to the installation of any devices, maintained a test pressure 1 1/2 times the standard pressures listed in NFPA 99 Table 5.1.11 without leaks. (Procedure per NFPA 99 5.1.12.2.3).

(5) That after installation of all devices, the pipeline was proven leak free for 24 hours at a pressure 20 percent above the standard pressures listed in NFPA 99 Table 5.1.11. (Procedure per NFPA 99 5.1.12.2.6)

(6) That the systems have been checked for cross connections and none were found. (Procedure per NFPA 99 5.1.12.2.4)

(7) That the manufacturer has started up all medical air compressors, and medical vacuum pumpsystem(s) and that they are in operating order.

b. Four originals of the affidavit shall be distributed: two to the Contracting Officer, one to the general Contractor, and one to the verifier (www.mgpho.org).

3.3 VERIFIER TESTING

a. Prior to handing over the systems to VAMC, the Contractor shall retain a verifier acceptable to the engineer of record and VA who shall follow strictly the procedures for verification as described in NFPA 99 5.1.12.3 and provide a written report and certificate bearing the notarized signature of an officer of the verification company on company letterhead which contains at least the following:

(1) A current ACORD insurance certificate indicating professional liability coverage in the minimum amount of $1 Million per occurrence, and general aggregate liability in the minimum amount of $1 Million, valid and in force when the project is to be verified. General liability insurance alone is not acceptable.

(2) An affidavit bearing the notarized signature of an officer of the

verification company stating that the verification company is not the supplier of any equipment used on this project or tested in this report and that the verification contractor has no relationship to, or pecuniary interest in, the manufacturer, seller, or installer of any equipment used on this project or tested in this report.

(3) A listing of all tests performed, listing each source, outlet, valve and alarm included in the testing.

(4) An assertion that all tests were performed by a Medical Vacuum System Certified Medical Gas or vacuum Verifier or by individuals qualified to perform the work and holding valid qualifications to ASSE 6000 Series, ASSE 6030 and under the immediate supervision a Verifier. Include the names, credential numbers and expiration dates for all individuals working on the project.

(5) A statement that equipment used was calibrated at least within the last six months by a method traceable to a National Bureau of Standard Reference and enclosing certificates or other evidence of such calibration(s). Where outside laboratories are used in lieu of onsite equipment, those laboratories shall be named, and their original reports enclosed.

(6) A statement that where and when needed, equipment was re calibrated during the verification process and describing the method(s) used.

(7) A statement that the systems were tested and found to be free of debris to a procedure per NFPA 99 5.1.12.3.7.

(8) The flow from each outlet when tested to a procedure per NFPA 99 5.1.12.3.10.

(9) A statement that the systems were tested and found to have no cross-connections to a procedure per NFPA 99 5.1.12.3.3.

(10) A statement that the systems were tested and found to be free of contaminants to a procedure per NFPA 99 5.1.12.3.8 except that the purity standard shall be 2 ppm difference for halogenated hydrocarbons and 1 ppm total hydrocarbons (as methane).

(11) Statement that all local signals function as required under NFPA 99 5.1.3.5.8 and as per the relevant NFPA 99 sections relating to the sources.

rces include all alarms required by NFPA 99 Table A.5.1.9.5.

(12) The concentration of each component of NFPA 99 Table 5.1.12.3.11 in the medical air after 24 hours of operation of the medical air source.

b. Perform and document all cross-connection tests, labeling verification, supply system operation, and valve and alarm operation tests as required by, and in accordance with NFPA 99 and the

procedures set forth in pre-qualification documentation.

c. Verify that the systems, as installed, meet or exceed the requirements of NFPA 99, this specification, and that the systems operate as required.

d. Piping purge test: For each positive pressure gas system, verify cleanliness of piping system. Filter a minimum of 35 cubic feet of gas through a clean white 0.45-micron filter at a minimum velocity of 3.5 fpm. Filter shall show no discoloration and shall accrue no more than 0.1 mg of matter. Test each zone at the outlet most remote from the source. Perform test with the use of an inert gas as described in CGA P-9. Retest until all tests pass at no additional time or cost to the Government.

3.4 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this Section of the specification will be tested as part of a larger system.

3.5 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for four hours to instruct VA Personnel in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

-- End of Section --

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DIVISION 22 - PLUMBING

SECTION 22 63 00

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5/23

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SECTION 22 63 00
5/23

GAS SYSTEMS FOR LABORATORY AND HEALTHCARE FACILITIES

PART 1 GENERAL

1.1 DESCRIPTION

a. Central Laboratory and Healthcare Gas Systems: Consisting of compressed air services; complete, ready for operation, including all necessary piping, fittings, valves, gages, alarms including low voltage wiring, air compressors, electric motors and starters, air dryers, filters, pressure regulators, dew point monitor, carbon monoxide monitor and all necessary parts, accessories, connections and equipment.

b. Supply Lines Outside of Building (including PVC protective pipe): As specified in this Section.

c. A complete listing of all acronyms and abbreviations are included in Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 07 84 00, FIRESTOPPING: Sealing around pipe penetrations to maintain the integrity of time rated construction.

f. Section 07 92 00, JOINT SEALANTS: Sealing around pipe penetrations through the floor to prevent moisture migration.

i. Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING: General requirements and items common to more than one section of Division 22.

j. Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT: Electric motors.

k. Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS: Requirements for commissioning, systems readiness checklist, and training.

l. Section 22 62 00, VACUUM SYSTEMS FOR LABORATORY AND HEALTHCARE FACILITIES: Vacuum Piping and Equipment.

m. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC: Alarm interface with BAS.

n. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES:

Control wiring.

o. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduit.

p. Section 26 27 26, WIRING DEVICES: Electrical wiring and accessories.

q. Section 26 29 11, MOTOR CONTROLLERS: Motor starters.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A13.1	(2023) Scheme for the Identification of Piping Systems
ASME B16.22	(2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings
ASME B16.50	(2021) Wrought Copper and Copper Alloy Braze-Joint Pressure Fittings
ASME B40.100	(2022) Pressure Gauges and Gauge Attachments

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

ASSE 6000 Series	(2021) Professional Qualification Standard for Medical Gas Systems Installers, Inspectors and Verifiers

ASTM INTERNATIONAL (ASTM)

ASTM B819	(2000; R 2011) Standard Specification for Seamless Copper Tube for Medical Gas Systems
ASTM D1785	(2021) Standard Specification for Poly(Vinyl Chloride) (PVC), Plastic Pipe, Schedules 40, 80, and 120

AMERICAN WELDING SOCIETY (AWS)

AWS A5.8/A5.8M	(2019) Specification for Filler Metals for Brazing and Braze Welding

COMPRESSED GAS ASSOCIATION (CGA)

CGA C-9	(2019) Standard Color Marking of Compressed Gas Containers for Medical Use
CGA P-9	(2008) The Inert Gases: Argon, Nitrogen, and Helium

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-72	(2010a) Ball Valves with Flanged or Butt-Welding Ends for General Service
MSS SP-110	(2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 99	(2024) Health Care Facilities Code

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Equipment Installer; G,AE

Equipment Supplier; G,AE

Laboratory And Healthcare System Testing Organization; G,AE

Testing Agency's Detailed Procedure; G,AE

Medical Gas Partnering And Quality Control Plan; G,AE

SD-03 Product Data

Manufacturer's Literature And Data; G,AE

SD-04 Samples

Physical Sample Of A Typical Brazed Joint; G,AE

SD-05 Design Data

Piping; G,AE

SD-06 Test Reports

Completed System Readiness Checklist; G,AE

Documentation Of Conditions; G,AE

SD-07 Certificates

Certification; G,AE

Certification Documentation; G,AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G,AE

SD-10 Operation and Maintenance Data

Operation And Maintenance Data; G,AE

SD-11 Closeout Submittals

As-Built Documentation; G,AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Medical Gas Partnering and Quality Control Plan

(1) Identifies all stakeholders early in the project to partner together and develop a communication strategy that is productive and effective.

(2) Requires the 3 Phases of Quality Control

(a) Preparatory Phase - review contract requirements and necessary communication before execution of each portion of work.

(b) Initial Phase - Ensure the employees completing the work are qualified and executing as expected.

(c) Follow Up Phase - Check in on the work repeatedly to ensure continued compliance.

(3) Provides installer certifications for approval.

(4) Provides Verifiers Credentials for approval.

(5) Provides Certifiers Credentials for approval.

(6) Includes all Quality control processes, procedures and daily documentation to be used during installation.

(7) Outlines a process that clearly documents and demonstrates a nitrogen atmosphere is obtained and maintained within the piping throughout the installation process until final certification and activation.

(8) Outlines post installation/precertification blowdown procedures to ensure a sufficient volume and velocity of gas is expelled to carry any particulate through the entire system during the blowdown/white rag test. This test shall be witnessed by the Contracting Officer and appropriate stakeholders.

c. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 22 63 00, GAS SYSTEMS FOR LABORATORY AND HEALTHCARE FACILITIES”, with applicable paragraph identification.

d. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and

capacity.

(1) Piping.

(2) Valves.

(3) Inlet and outlet cocks

(4) Gages.

(5) Air compressor systems (Provide certified compressor test data at startup.):

(a) Compressors: Manufacturer and model.

(b) Characteristic performance curves.

(c) Compressor operating speed (RPM).

(d) Capacity: Free air delivered at indicated pressure (L/s) (SCFM).

(e) Type of bearing in compressor.

(f) Type of lubrication.

(g) Type and adjustment of drive.

(h) Electric motors: Manufacturer, frame and type.

(i) Speed of motors (RPM).

(j) Current characteristics and horsepower of motors.

(k) Receiver capacity and rating.

(l) Air silencer: Manufacturer, type and model.

(m) Air filters: Manufacturer, type, model and capacity.

(n) Pressure regulators: Manufacturer and capacity.

(o) Dew point monitor: Manufacturer, type and model.

(p) Air dryers: Manufacturer, type, model and capacity (L/s) (SCFM).

(q) Carbon monoxide monitor manufacturer, type and model.

(r) Aftercoolers.

e. Certification: The completed systems have been installed, tested, purged, analyzed and verified in accordance with the requirements of this specification. Certification shall be submitted to Contracting Officer.

f. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section

22 08 00.01, DVA/USACE PROJECTSCOMMISSIONING OF PLUMBING SYSTEMS.

g. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

h. Submit one physical sample of a typical brazed joint for each approved installer to be compared against their installed work as a Quality assurance measure.

i. Documentation of conditions while work is performed

(1) Daily reports, photos, uncapped pipingand all preapproved Pipefitter's names and all locations brazed by that individual each workday etc.

(2) All copies of verification reports documenting compliance with NFPA 99 during construction.

(3) Installation equipment calibration certificates.

1.6 QUALITY ASSURANCE

a. Materials and Installation: In accordance with NFPA 99 and as specified.

b. Equipment Installer: Show technical qualifications and previous experience in installing laboratory and healthcare equipment on three similar projects. Submit names, phone numbers, and addresses of referenced projects. Installers shall meet the qualifications of ASSE 6000 Series.

c. Equipment Supplier: Provide evidence of equivalent product installed at three installations similar to this project that has been in satisfactory and efficient operation for three years. Submit names, phone numbers, and addresses where the product is installed.

d. Laboratory and healthcare System Testing Organization: The testing shall be conducted by a party technically competent and experienced in the field of laboratory and healthcare pipeline testing. Testing and systems verification shall be performed by personnel meeting the qualifications of ASSE 6000 Series. Such testing shall be performed by a party other than the installing contractor.

e. Provide the names of three projects where testing of medical or laboratory gases systems has been performed by the testing agency. Include the name of the project, names of such persons at that project who supervised the work for the project owner, or who accepted the report for the project owner, and a written statement that the projects listed required work of similar scope to that set forth in this specification.

f. Submit the testing agency's detailed procedure which shall be followed in the testing of this project. Include details of the testing sequence, procedures for cross connection tests, outlet function tests, alarm tests, purity tests, etc., as required by this specification. For purity test procedures, include data on test methods, types of equipment to be used, calibration sources and method references.

g. Certification: Provide Contracting Officer documentation 10 working days prior to submitting request for final inspection to include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and certification that all results of tests were within limits allowed by this specification.

h. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

1.7 AS-BUILT DOCUMENTATION

a. Submit manufacturer's literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, substitutions and construction revisions shall be in electronic version on compact disc or DVD inserted into a three ring binder. All aspects of system operation and maintenance procedures, including piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices such as damper and door closure interlocks shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. As-built drawings are to be provided, and a copy of them in Revit version 2025 provided on compact disk or DVD. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the 'third party testing company' requirement.

d. Certification documentation shall be provided to Contracting Officer 10 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and certification that all results of tests were within limits specified.

PART 2 PRODUCTS

2.1 PIPING AND FITTINGS

a. Copper Tubing: Type "K", ASTM B819, seamless copper tube, hard drawn temper, with wrought copper fittings conforming to ASME B16.22 or brazing fittings complying with ASME B16.50. Size designated reflecting nominal inside diameter. All tubing and fittings shall be

labeled "ACR/OXY", "OXY", "OXY/MED", "ACR/MED", or "MED".

b. Brazing Alloy: AWS A5.8/A5.8M, Classification BCuP, greater than 1000 degrees F melting temperature. Flux is strictly prohibited for copper-to-copper connections.

c. Threaded Joints: Polytetrafluoroethylene (Teflon) tape.

d. Underground Protective Pipe: Polyvinyl Chloride (PVC), ASTM D1785, Schedule 80.

e. Memory metal couplings: Temperature and pressure rating shall not be less than that of a brazed joint in accordance with NFPA 99, paragraph 5.1.10.6.1.

f. Apply piping identification labels at the time of installation in accordance with NFPA 99. Apply supplementary color identification in accordance with CGA C-9.

g. Special Fittings: The following special fittings shall be permitted to be used in lieu of brazed joints:

(1) Memory-metal couplings having temperature and pressure ratings joints not less than that of a brazed joint.

(2) Listed or approved metallic gas tube fittings that, when made up, provide a permanent joint having the mechanical, thermal, and sealing integrity of a brazed joint.

(3) Dielectric fittings where required by the manufacturer of special medical equipment to electrically isolate the equipment from the piping distribution system.

(4) Axially swaged, elastic strain preload fittings providing metal to metal seal having pressure and temperature ratings not less than that of a brazed joint and when complete are permanent and non-separable.

2.2 VALVES

a. Ball: In-line, other than zone valves in cabinets:

(1) 3 inches and smaller: Bronze/ brass body, MSS SP-72, MSS SP-110, Type II, Class 150, Style 1, with tubing extensions for brazed connections, full port, three-piece or double union end connections, Teflon seat seals, full flow, 600 psig WOG minimum working pressure, with locking type handle, cleaned for oxygen use and labeled for intended service.

(2) 3 to 4 inches: Bronze/ brass body, MSS SP-72 MSS SP-110, Type II, Class 150, Style 1 with tubing extensions brazed to flanges, full port, three piece, double seal, Teflon seals, full flow, 600 psig WOG minimum working pressure, with locking type handle, cleaned for oxygen use and labeled for intended service.

b. Check:

(1) 3 inches and smaller: Bronze/brass body, straight through design for minimum pressure drop, spring loaded, self-aligning with

Teflon cone seat, vibration free, silent operation, supplied NPT female threads at each end with flow direction arrow permanently cast into, cleaned for oxygen use and labeled for intended service, 400 psig WOG minimum working pressure.

(2) 4 inches and larger: Iron body, bronze trim, swing type, vertical or horizontal installation, flange connection, with flow direction arrow permanently cast into, cleaned for oxygen use and labeled for intended service, 150 psig WSP.

SERVICE LABEL	IDENTIFICATION COLORS	MFG. STD. CLR.
MEDICAL AIR	Black letters on yellow background	YELLOW

2.3 GAGES

a. Pressure Gages: Includes gages temporarily supplied for testing purposes.

(1) For line pressure use adjacent to source equipment: ASME B40.100, pressure gage, single, size 4-1/2 inches, for compressed air accurate to within 2 percent, with metal case. Range shall be two times operating pressure. Dial graduations and figures shall be black on a white background, or white on a black background. Gage shall be cleaned for oxygen use, labeled for appropriate service, and marked "USE NO OIL". Install with gage cock.

2.4 AIR COMPRESSOR SYSTEMS

a. System Design: The laboratory air system shall be of a modular base mounted design consisting of triplex compressor, dryer/control, and an air receiver. Each unit shall be fully compliant with the latest edition of NFPA 99.

b. Compressors: Continuous duty rated “oil-less” type with permanently lubricated, sealed bearings. Single stage design, air cooled, reciprocating type with corrosion resistant reed type valves with stainless steel reeds. Both the compression rings and rider rings shall be made from a long life, fluororesin material designed for continuous duty operation. The crankshaft shall be constructed of a durable nodular graphite cast iron and designed to be fully supported on both ends by heavy duty ball bearings permanently lubricated and sealed. The crankcase shall be constructed of gray cast iron. Maximum heat dissipation shall be achieved through cast aluminum alloy cylinders treated for optimum corrosion and wear resistance. Cylinder sleeves shall not be required. Additionally, heat transmission from the piston wall to the piston pin needle bearing shall be minimized by an insulated “heat cut” piston pin. The connecting rod shall be of a one piece design for maximum reliability. Bio-based materials shall be utilized when possible.

c. Compressor Drive and Motor: V-belt driven through a combination flywheel/sheave and steel motor sheave with tapered bushing and protected by an OSHA approved, totally enclosed belt guard. Belt tensioning shall be achieved by a pivoting motor mounting base that is fully adjustable through twin adjusting screws. The motor shall be a

NEMA rated, open-drip-proof, 1800 RPM, with 1.15 service factor suitable for 208/230/460V electrical service, a specified in Section 22 05 12, GENERAL MOTOR REQUIREMENTS FOR PLUMBING EQUIPMENT and Section 26 29 11, MOTOR CONTROLLERS.

d. Intake Piping: Provide a pre-piped intake manifold with one inlet air filter with threaded opening for remote intake connection. Isolate filter housing from the intake manifold with a braided 304 stainless steel flex connector.

e. Discharge Piping: Provide an integral air cooled aftercooler designed for a maximum approach temperature of 12 degrees F complete with moisture separator and timed automatic solenoid drain valve with a manual drain value by-pass. Provide each cylinder head with a pre-wired high discharge air temperature shutdown switch. Include a flex connector, safety relief valve, and check valve. The compressor discharge line the piping shall be of ASTM B819 copper tubing, brass, and/or stainless steel. The discharge flex connector shall be braided 304 stainless steel, brass or bronze.

f. Isolation System: Isolate the compressor and monitor from the main compressor module base by means of a four point, heavy duty, spring isolation system for a minimum of 95 percent isolation efficiency.

g. Dryer/Control: The dryer/control shall include a NEMA 12, U.L. labeled control system, duplexed desiccant drying system, duplexed final line filters, duplexed final line regulators, and combination dew point/CO monitor. All of the above shall be pre-wired and pre-piped in accordance with NFPA 99 and include valving to allow complete air receiver by-pass, as well as air sampling port.

h. Dryer: Size each desiccant dryer for the peak calculated demand and capable of producing 10 degrees F pressure dew point. Dryer purge flow shall be minimized through an on-demand purge saving control system. Include a mounted prefilter rated for 0.01 micron with automatic drain and element change indicator on the inlet of each dryer.

i. Control System: Mounted and pre-wired control system shall be NEMA 12 and U.L. labeled. This control system shall provide automatic lead/lag sequencing with circuit breaker disconnects for each compressor with external operators, one non-fused main disconnect with external operators, full voltage motor magnetic starters with overload protection, redundant 120V control circuit transformers, visual and audible reserve unit alarm with isolated contacts for remote alarm, hand-off-auto (HOA) lighted selector switches, automatic alternation of both compressors with provisions for simultaneous operation if required, automatic activation of reserve unit if required, visual alarm indication for high discharge air temperature shutdown with isolated contacts for remote alarm, and duplexed run time hour meters.

j. Final Line Filters and Regulators: Fully duplexed final line filters rated for 0.01 micron with element change indicators shall be factory mounted and pre-piped, along with duplexed factory mounted and pre-piped final line regulators and duplex safety relief valves.

k. Dew Point Hygrometer/CO Monitor: Mounted, pre-piped and wired, combination dew point hygrometer/CO monitor shall be of the ceramic type with integral chemical type CO sensor. System accuracy shall be

plus or minus 2 degrees F for dew point and 2 PPM (at 10 PPM) for carbon monoxide. Dew point alarm shall be factory set at 40 degrees F) per NFPA 99, and the CO alarm shall be factory set at 10 PPM. Both set points shall be field adjustable.

l. Air Receiver: Vertical air receiver, galvanized, ASME Coded, National Board Certified, rated for minimum 150 psig design pressure and includes a sight gauge glass as well as a timed automatic solenoid drain valve. Provide three valve bypass on supply.

m. Example of an acceptable product and manufacturer: Beacon Medical Products “Lifeline Medical Air Systems”.

2.5 PRESSURE REGULATORS

a. For 100 psig regulator, provide duplex in parallel, valve for maintenance shut-down without service interruption. For additional pressures, locate regulators remote from compressor near point of use, and provide with isolation valves and valve bypass.

(1) For systems 10 SCFM and below: Brass or bronze body and trim, reduced pressure range 25 to 123 psig adjustable, spring type, diaphragm operated, relieving. Delivered pressure shall vary not more than 0.15 psig for each 1.5 psig variation in inlet pressure.

PART 3 EXECUTION

3.1 INSTALLATION

a. Install cast escutcheon with set screw at each wall, floor and ceiling penetration in exposed finished locations and within cabinets and millwork.

b. Open ends of tube shall be capped or plugged at all times or otherwise sealed until final assembly to prevent infiltration of any foreign matter.

c. Cut piping square and accurately with a tube cutter (sawing is prohibited) to measurements determined at place of installation. Ream tube to remove burrs, being careful not to expand tube, and so no chips of copper remain in the tube. Work into place without springing or forcing. Bottom tube in socket so there are no gaps between tube and fitting. Exercise care in handling equipment and tools used in cutting or reaming of tube to prevent oil or grease being introduced into tubing. Where contamination has occurred, material is no longer suitable for oxygen service.

d. Spacing of hangers: NFPA 99.

e. Rigidly support valves and other equipment to prevent strain on tube or joints.

f. While being brazed, joints shall be continuously purged with oil free nitrogen. The flow of purged gas shall be maintained until joint is cool to touch.

g. Do not bend tubing. Use fittings.

h. Support ceiling column assembly from heavy sub-mounting castings

furnished with the unit as part of roughing-in. Anchor with 1/2-inch diameter bolts attached to angle iron frame supported from structural ceiling, unless otherwise indicated.

i. Provide two 1 inch minimum conduits from ceiling column assembly to adjacent corridor, one for mass spectrometer tubing and wiring and one for monitor wiring, for connection to signal cabling network.

j. Install pressure switches, transmitter and gauges to be easily accessed, and provide access panel where installed above plaster ceiling. Install pressure switch and sensors with orifice nipple between the pipe line and switches/sensors.

k. Apply pipe labeling during installation process and not after installation is completed. Size of legend letters shall be in accordance with ASME A13.1.

l. Pipe compressor intake to a source of clean ambient air as indicated in NFPA 99.

m. After initial leakage testing is completed, allow piping to remain pressurized with testing gas until testing agency performs final tests.

n. Penetrations:

(1) Fire Stopping: Where pipes pass through fire partitions, fire walls, smoked partitions, or floors, install a fire stop that provides an effective barrier against the spread of fire, smoke and gases as specified in Section 07 84 00, FIRESTOPPING, with intumescent materials only. Completely fill and seal clearances between raceways and openings with the fire stopping material.

(2) Waterproofing: At floor penetrations, completely seal clearances around the pipe and make watertight with sealant as specified in Section 07 92 00, JOINT SEALANTS. Bio-based materials shall be utilized when possible.

3.2 STARTUP AND TESTING

a. Initial Tests: Blow down and high and low pressure leakage tests as required by NFPA 99 with documentation.

b. Laboratory and/or healthcare testing agency shall perform the following:

(1) Perform and document all cross connection tests, labeling verification, supply system operation, and valve and alarm operation tests as required by, and in accordance with NFPA 99 and the procedures set forth in pre-qualification documentation.

(2) Verify that the systems, as installed, meet or exceed the requirements of NFPA 99, this specification, and that the systems operate as required.

(3) Piping purge test: For each positive pressure gas system, verify cleanliness of piping system. Filter a minimum of 35 cubic feet of gas through a clean white 0.45 micron filter at a minimum velocity of 3.5 SCFM. Filter shall show no discoloration, and shall accrue no more than 0.0000035 ounces of matter. Test each

zone at the outlet most remote from the source. Perform test with the use of an inert gas as described in CGA P-9.

(4) Piping purity test: For each positive pressure system, verify purity of piping system. Test each zone at the most remote outlet for dew point, carbon monoxide, total hydrocarbons (as methane), and halogenated hydrocarbons, and compare with source gas. The two tests shall in no case exceed variation as specified in paragraph, “Maximum Allowable Variation”. Perform test with the use of an inert gas as described in CGA P-9.

(5) Source Contamination Test: Analyze each pressure gas source for concentration of contaminants, by volume. Take samples for air system test at the intake and at a point immediately downstream of the final filter outlet. The compared tests shall in no case exceed variation as specified in paragraph “Maximum Allowable Variation”. Allowable concentrations are below the following:

Dew point, air	40 degrees F pressure dew point at 100 psig
Carbon monoxide, air	10 mg/L (ppm)
Carbon dioxide, air	500 mg/L (ppm)
Gaseous hydrocarbons as methane, air	25 mg/L (ppm)
Halogenated hydrocarbons, air	2 mg/L (ppm)

(6) Analysis Test:

(a) Analyze each pressure gas source and outlet for concentration of gas, by volume.

(b) Make analysis with instruments designed to measure the specific gas dispensed.

(c) Allowable concentrations are within the following:

1. Laboratory air 19.5 percent to 23.5 percent oxygen.

Medical air	19.5% to 23.5% oxygen

(7) Maximum Allowable Variation: Between comparative test results required are as follows:

Dew point	35 degrees F
Carbon monoxide	2 mg/L (ppm)
Total hydrocarbons as methane	1 mg/L (ppm)
Halogenated hydrocarbons	2 mg/L (ppm)

c. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Contractor shall provide a minimum of 10 working days prior to startup and testing.

3.3 CONNECTION TO EXISTING LABORATORY GAS SYSTEM

a. Contactor shall test the existing system for hydrocarbons, dew point, etc. per NFPA 99. If problems are present, the Contracting Officer would notify the facility of the results. The facility would then make the necessary repairs and/or maintenance prior to connecting to new system.

b. Install shut-off valve at the connection of new line to existing line.

c. Time for shutdown of the existing laboratory and healthcare system shall be coordinated at least 10 work days prior to shutdown with the Contracting Officer and VA Medical Center.

d. Prior to any work being done, check the new pipeline for particulate or other forms of contamination per NFPA 99.

e. Ensure that the correct type of pipe tubing and fittings are being used.

f. Make a spot check of the existing pipelines in the facility to determine the level of cleanness present.

g. Reduce the pressure to zero and make the tie-in as quickly as possible. A nitrogen purge is not required since this would require another opening in the pipe.

h. After the tie-in is made and allowed to cool, slowly bleed the source gas back into the pipeline. Test the work area for leaks with soapy water and repair any leaks.

i. After all leaks, if any, are repaired and the line is fully recharged, perform blow down and testing. Open the zone that is closest to the main to the system, access the closest outlet to the work, and blow the main through the outlet. After the outlet blows clear into a white cloth, make an additional check at a zone most distant from the work. Perform all required NFPA 99 tests after connection.

3.4 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.5 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for four hours to instruct VA Personnel in operation and maintenance of units.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 22 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF PLUMBING SYSTEMS.

-- End of Section --

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SECTION 23 05 11
3/23

COMMON WORK RESULTS FOR HVAC

PART 1 GENERAL

1.1 DESCRIPTION

a. The requirements of this Section apply to all sections of Division 23, HEATING, VENTILATING, AND AIR CONDITIONING (HVAC).

b. Definitions:

(1) Exposed: Piping, ductwork, and equipment exposed to view in finished rooms.

(2) Exterior: Piping, ductwork, and equipment exposed to weather be it temperature, humidity, precipitation, wind, or solar radiation.

c. Abbreviations/Acronyms:

(1) ac: Alternating Current

(2) AC: Air Conditioning

(3) ACU: Air Conditioning Unit

(4) ACR: Air Conditioning and Refrigeration

(5) AI: Analog Input

(6) AISI: American Iron and Steel Institute

(7) AO: Analog Output

(8) ASJ: All Service Jacket

(9) AWG: American Wire Gauge

(10) BACnet: Building Automation and Control Networking Protocol

(11) BAg: Silver-Copper-Zinc Brazing Alloy

(12) BAS: Building Automation System

(13) BCuP: Silver-Copper-Phosphorus Brazing Alloy

(14) bhp: Brake Horsepower

(15) Btu: British Thermal Unit

(16) Btu/h: British Thermal Unit Per Hour

(17) CDA: Copper Development Association

(19) CD: Compact Disk

(20) CFM: Cubic Foot Per Minute

(21) CH: Chilled Water Supply

(22) CHR: Chilled Water Return

(23) CLR: Color

(24) CO: Carbon Monoxide

(25) CO2: Carbon Dioxide

(26) CPD: Condensate Pump Discharge

(27) CPM: Cycles Per Minute

(28) CPVC: Chlorinated Polyvinyl Chloride

(29) CRS: Corrosion Resistant Steel

(30) CTPD: Condensate Transfer Pump Discharge

(31) CTPS: Condensate Transfer Pump Suction

(32) CW: Cold Water

(33) CWP: Cold Working Pressure

(34) CxA: Commissioning Agent

(35) dB: Decibels

(36) dB(A): Decibels (A weighted)

(37) DDC: Direct Digital Control

(38) DI: Digital Input

(39) DO: Digital Output

(40) DVD: Digital Video Disc

(41) DN: Diameter Nominal

(42) DWV: Drainage, Waste and Vent

(43) EPDM: Ethylene Propylene Diene Monomer

(44) EPT: Ethylene Propylene Terpolymer

(45) ETO: Ethylene Oxide

(46) F: Fahrenheit

(47) FAR: Federal Acquisition Regulations

(48) FD: Floor Drain

(49) FED: Federal

(50) FG: Fiberglass

(51) FGR: Flue Gas Recirculation

(52) FOS: Fuel Oil Supply

(53) FOR: Fuel Oil Return

(54) FSK: Foil-Scrim-Kraft facing

(55) FWPD: Feedwater Pump Discharge

(56) FWPS: Feedwater Pump Suction

(57) GC: Chilled Glycol Water Supply

(58) GCR: Chilled Glycol Water Return

(59) GH: Hot Glycol Water Heating Supply

(60) GHR: Hot Glycol Water Heating Return

(61) gpm: Gallons Per Minute

(62) HDPE: High Density Polyethylene

(63) Hg: Mercury

(64) HOA: Hands-Off-Automatic

(65) hp: Horsepower

(66) HPS: High Pressure Steam 60 psig and above)

(67) HPR: High Pressure Steam Condensate Return

(68) HW: Hot Water

(69) HWH: Hot Water Heating Supply

(70) HWHR: Hot Water Heating Return

(71) Hz: Hertz

(72) ID: Inside Diameter

(73) IPS: Iron Pipe Size

(75) klb: 1000 lb

(77) lb: Pound

(78) lb/hr: Pounds Per Hour

(81) LPS: Low Pressure Steam ((15 psig) and below)

(82) LPR: Low Pressure Steam Condensate Gravity Return

(83) MAWP: Maximum Allowable Working Pressure

(84) MAX: Maximum

(85) MBtu/h: 1000 Btu/h

(86) MBtu: 1000 Btu

(87) MED: Medical

(89) MFG: Manufacturer

(92) MIN: Minimum

(96) MPS: Medium Pressure Steam ((16 psig) through (60 psig))

(97) MPR: Medium Pressure Steam Condensate Return

(98) MW: Megawatt

(99) NC: Normally Closed

(100) NF: Oil Free Dry (Nitrogen)

(101)

(102) NO: Normally Open

(103) NOx: Nitrous Oxide

(104) NPT: National Pipe Thread

(105) NPS: Nominal Pipe Size

(106) OD: Outside Diameter

(107) OSD: Open Sight Drain

(108) OS&Y: Outside Stem and Yoke

(109) PC: Pumped Condensate

(110) PID: Proportional-Integral-Differential

(111) PLC: Programmable Logic Controllers

(112) PP: Polypropylene

(113) PPE: Personal Protection Equipment

(114) ppb: Parts Per Billion

(115) ppm: Parts Per Million

(116) PRV: Pressure Reducing Valve

(117) PSIA: Pounds Per Square Inch Absolute

(118) psig: Pounds Per Square Inch Gauge

(119) PTFE: Polytetrafluoroethylene

(120) PVC: Polyvinyl Chloride

(121) PVDC: Polyvinylidene Chloride Vapor Retarder Jacketing, White

(122) PVDF: Polyvinylidene Fluoride

(123) rad: Radians

(124) RH: Relative Humidity

(125) RO: Reverse Osmosis

(126) rms: Root Mean Square

(127) RPM: Revolutions Per Minute

(128) RS: Refrigerant Suction

(129) RTD: Resistance Temperature Detectors

(130) RTRF: Reinforced Thermosetting Resin Fittings

(131) RTRP: Reinforced Thermosetting Resin Pipe

(132) SCFM: Standard Cubic Feet Per Minute

(133) SPEC: Specification

(134) SPS: Sterile Processing Services

(135) STD: Standard

(136) SDR: Standard Dimension Ratio

(137) SUS: Saybolt Universal Second

(138) SW: Soft water

(139) SWP: Steam Working Pressure

(140) TAB: Testing, Adjusting, and Balancing

(141) TDH: Total Dynamic Head

(142) TEFC: Totally Enclosed Fan-Cooled

(143) TFE: Tetrafluoroethylene

(144) THERM: 100,000 Btu

(145) THHN: Thermoplastic High-Heat Resistant Nylon Coated Wire

(146) THWN: Thermoplastic Heat & Water-Resistant Nylon Coated Wire

(147) T/P: Temperature and Pressure

(148) USDA: U.S. Department of Agriculture

(149) V: Volt

(150) VAC: Vacuum

(151) VA: Veterans Administration

(152) VAC: Voltage in Alternating Current

(153) VA CFM: VA Construction & Facilities Management

(154) VA CFM CSS: VA Construction & Facilities Management, Consulting Support Service

(155) VAMC: Veterans Administration Medical Center

(156) VHA OCAMES: Veterans Health Administration – Office of Capital Asset Management Engineering and Support

(157) VR: Vacuum condensate return

(158) WCB: Wrought Carbon Steel, Grade B

(159) WG: Water Gauge or Water Column

(160) WOG: Water, Oil, Gas

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITAL PROCEDURES.

c. Section 01 74 19, CONSTRUCTION WASTE MANAGEMENT.

d. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

e. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

f.

g. Section 03 30 00, CAST-IN-PLACE CONCRETE.

h. Section 05 31 00, STEEL DECKING.

i. Section 05 36 00, COMPOSITE METAL DECKING.

j. Section 05 50 00, METAL FABRICATIONS.

k. Section 07 84 00, FIRESTOPPING.

l. Section 07 92 00, JOINT SEALANTS.

m. Section 09 91 00, PAINTING.

n. Not used

o. Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC and STEAM GENERATION.

p. Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

q. Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, AND BALANCING FOR HVAC.

r. Section 23 07 11, HVAC AND BOILER PLANT INSULATION.

s. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

t.

u. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

v. Not used.

w. Section 23 82 00, CONVECTION HEATING AND COOLING UNITS.

x. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

y. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES.

z. Section 26 29 11, MOTOR CONTROLLERS.

1.3 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 410	(1996) Recommended Safety Practices for Users and Installers of Industrial and Commercial Fans

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B31.1	(2024) Power Piping
ASME B31.9	(2020) Building Services Piping
ASME BPVC	(2010) Boiler and Pressure Vessels Code

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A575	(2020) Standard Specification for Steel Bars, Carbon, Merchant Quality, M-Grades

RUBBER MANUFACTURERS ASSOCIATION (RMA)

RMA IP-20	(2007) Specifications for Drives Using Classical V-Belts and Sheaves. Specifications for A, B, C, and D Cross Sections
RMA IP-21	(2023) Specifications for Drives Using Double-V (Hexagonal) Belt
RMA IP-24	(2016) Specifications for Drives Using Synchronous Belts
RMA IP-27	(2015) Specifications for Drives Using Curvilinear Toothed Synchronous Belts

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-58	(2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation
MSS SP-127	(2014; Rev A) Bracing for Piping Systems: Seismic-Wind-Dynamic Design, Selection, and Application

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-P-21035	(1991; Rev B; Notice 2 2003; Notice 3 2021) Paint, High Zinc Dust Content, Galvanizing Repair (Metric)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

U.S. DEPARTMENT OF VETERANS AFFAIRS (VA)

VA H-18-8 (2023) Seismic Design Requirements

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Mock-Ups; G, AE

Rigging Plan; G, AE

SD-02 Shop Drawings

Coordination/Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-04 Samples

Samples; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Hvac Maintenance Data And Operating Instructions; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 05 11, COMMON WORK RESULTS FOR HVAC”, with applicable paragraph identification.

c. If the project is phased submit complete phasing plan/schedule with manpower levels prior to commencing work. The phasing plan shall be detailed enough to provide milestones in the process that can be verified.

d. Contractor shall make all necessary field measurements and investigations to assure that the equipment and assemblies will meet contract requirements, and all equipment that requires regular maintenance, calibration, etc are accessable from the floor or permanent work platform. It is the Contractor’s responsibility to ensure all submittals meet the VA specifications and requirements and it is assumed by the VA that all submittals do meet the VA specifications unless the Contractor has requested a variance in writing and approved by Contracting Officer prior to the submittal. If at any time during the project it is found that any item does not meet the VA specifications and there was no variance approval the Contractor shall correct at no additional cost or time to the Government even if a submittal was approved.

e. If equipment is submitted which differs in arrangement from that shown, provide documentation proving equivalent performance, design standards and drawings that show the rearrangement of all associated systems. Additionally, any impacts on ancillary equipment or services such as foundations, piping, and electrical shall be the Contractor’s responsibility to design, supply, and install at no additional cost or time to the Government. VA approval will be given only if all features of the equipment and associated systems, including accessibility, are equivalent to that required by the contract.

f. Prior to submitting shop drawings for approval, Contractor shall certify in writing that manufacturers of all major items of equipment have each reviewed contract documents, and have jointly coordinated and properly integrated their equipment and controls to provide a complete and efficient installation.

g. Submittals and shop drawings for interdependent items, containing applicable descriptive information, shall be furnished together. Coordinate and properly integrate materials and equipment to provide a completely compatible and efficient installation.

h. Samples: Samples will not be required, except for insulation or where materials offered differ from specification requirements. Samples shall be accompanied by full description of characteristics different from specification. The Government, at the Government’s expense, will perform evaluation and testing if necessary. The Contractor may submit samples of additional material at the Contractor's option; however, if additional samples of materials are submitted later, pursuant to Government request, adjustment in contract price and time will be made.

i. Coordination/Shop Drawings:

(1) Submit complete consolidated and coordinated shop drawings for all new systems, and for existing systems that are in the same areas.

(2) The coordination/shop drawings shall include plan views, elevations and sections of all systems and shall be on a scale of not less than 1: 32 3/8 inch equal to one foot). Clearly identify and dimension the proposed locations of the principal items of equipment. The drawings shall clearly show locations and adequate clearance for all equipment, piping, valves, control panels and other items. Show the access means for all items

requiring access for operations and maintenance. Provide detailed coordination/shop drawings of all piping and duct systems. The drawings should include all lockout/tagout points for all energy/hazard sources for each piece of equipment. Coordinate lockout/tagout procedures and practices with local VA requirements.

(3) Do not install equipment foundations, equipment or piping until coordination/shop drawings have been approved.

(4) In addition, for HVAC systems, provide details of the following:

(a) Mechanical equipment rooms.

(b) Interstitial space.

(c) Hangers, inserts, supports, and bracing.

(d) Pipe sleeves.

(e) Duct or equipment penetrations of floors, walls, ceilings, or roofs.

j. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity. Submit under the pertinent section rather than under this section.

(1) Submit belt drive with the driven equipment. Submit selection data for specific drives when requested by the Contracting Officer.

(2) Submit electric motor data and variable speed drive data with the driven equipment.

(3) Equipment and materials identification.

(4) Fire-stopping materials.

(5) Hangers, inserts, supports and bracing. Provide complete stress analysis for variable spring and constant support hangers.

(6) Wall, floor, and ceiling plates.

k. Rigging Plan: Provide documentation of the capacity and weight of the rigging and equipment intended to be used. The plan shall include the path of travel of the load, the staging area and intended access, and qualifications of the operator and signal person.

l. HVAC Maintenance Data and Operating Instructions:

(1) Maintenance and operating manuals in accordance with Section 01 00 00, GENERAL REQUIREMENTS, paragraph INSTRUCTIONS for systems and equipment.

(2) Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(a) Include complete list indicating all components of the systems.

(b) Include complete diagrams of the internal wiring for each item of equipment.

(c) Diagrams shall have their terminals identified to facilitate installation, operation, and maintenance.

(3) Provide a listing of recommended replacement parts for keeping in stock supply, including sources of supply for equipment. Include in the listing for belts: Belt manufacturer, model number, size and style, and distinguished whether of multiple belt sets.

m. Provide copies of approved HVAC equipment submittals to the TAB and Commissioning Subcontractor.

n. Completed System Readiness Checklist provided by the CxA and completed by the Contractor, signed by a qualified technician, and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

o. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.6 QUALITY ASSURANCE

a. Mechanical, electrical and associated systems shall be safe, reliable, efficient, durable, easily and safely operable and maintainable, easily and safely accessible, and in compliance with applicable codes as specified. The systems shall be comprised of high quality institutional-class and industrial-class products of manufacturers that are experienced specialists in the required product lines. All construction firms and personnel shall be experienced and qualified specialists in industrial and institutional HVAC.

b. Flow Rate Tolerance for HVAC Equipment: Section 23 05 93.01, DFVA/USACE PROJECTS TESTING, ADJUSTING, AND BALANCING FOR HVAC.

c. Equipment Vibration Tolerance:

(1) Refer to Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT. Equipment shall be factory-balanced to this tolerance and re-balanced on site, as necessary.

(2) After HVAC air balance work is completed and permanent drive sheaves are in place, perform field mechanical balancing and adjustments required to meet the specified vibration tolerance.

d. Products Criteria:

(1) Standard Products: Material and equipment shall be the standard products of a manufacturer regularly engaged in the manufacture of the products for at least 3 years (or longer as specified elsewhere). The design, model and size of each item shall have been in satisfactory and efficient operation on at least three installations for approximately three years. However, digital electronics devices, software and systems such as controls, instruments, computer work station, shall be the current generation of technology and basic design that has a proven

satisfactory service record of at least three years. See other specification sections for any exceptions and/or additional requirements.

(2) Refer to all other sections for quality assurance requirements for systems and equipment specified therein.

(3) All items furnished shall be free from defects that would adversely affect the performance, maintainability and appearance of individual components and overall assembly.

(4) The products and execution of work specified in Division 23 sections shall conform to the referenced codes and standards as required by the specifications. Local codes and amendments shall be enforced, along with requirements of local utility companies. The most stringent requirements of these specifications, local codes, or utility company requirements shall always apply. Any conflicts shall be brought to the attention of the Contracting Officer.

(5) Multiple Units: When two or more units of materials or equipment of the same type or class are required, these units shall be of the same manufacturer and model number, or if different models are required they shall be of the same manufacturer and identical to the greatest extent possible (i.e., same model series).

(6) Assembled Units: Performance and warranty of all components that make up an assembled unit shall be the responsibility of the manufacturer of the completed assembly.

(7) Nameplates: Nameplate bearing manufacturer's name or identifiable trademark shall be securely affixed in a conspicuous place on equipment, or name or trademark cast integrally with equipment, stamped or otherwise permanently marked on each item of equipment.

(8) Use of asbestos products or equipment or materials containing asbestos is prohibited.

e. HVAC Equipment Service Providers: Service providers shall be authorized and trained by the manufacturers of the equipment supplied. These providers shall be capable of responding onsite and provide acceptable service to restore equipment operations within 4 hours of receipt of notification by phone, e-mail or fax in event of an emergency, such as the shutdown of equipment; or within 24 hours in a non-emergency. Submit names, mail and e-mail addresses and phone numbers of service personnel and companies providing service under these conditions for (as applicable to the project): fans, air handling units, chillers, cooling towers, control systems, pumps, critical instrumentation, computer workstation and programming.

f. HVAC Mechanical Systems Welding: Before any welding is performed, Contractor shall submit a certificate certifying that welders comply with the following requirements:

(1) Qualify welding processes and operators for piping according to ASME BPVC Section IX. Provide proof of current certification.

(2) Comply with provisions of ASME B31 series "Code for Pressure Piping" (ASME B31.1 & ASME B31.9).

(3) Certify that each welder and welding operator has passed American Welding Society (AWS) qualification tests for the welding processes involved, and that certification is current.

(4) All welds shall be stamped according to the provisions of the AWS or ASME as required herein and by the associated code.

g. Manufacturer's Recommendations: Where installation procedures or any part thereof are required to be in accordance with the recommendations of the manufacturer of the material being installed, printed copies of these recommendations shall be furnished to the Contracting Officer with submittals. Installation of the item will not be allowed to proceed until the recommendations are received. Failure to furnish these recommendations can be cause for rejection of the material and removal by the Contractor and no additional cost or time to the Government.

h. Execution (Installation, Construction) Quality:

(1) Apply and install all items in accordance with manufacturer's written instructions. Refer conflicts between the manufacturer's instructions and the contract documents to the Contracting Officer for resolution. Provide written hard copies and computer files on CD or DVD of manufacturer’s installation instructions to the Contracting Officer with submittals prior to commencing installation of any item. Installation of the item will not be allowed to proceed until the recommendations are received and approved by the VA. Failure to furnish these recommendations is a cause for rejection of the material.

(2) All items that require access, such as for operating, cleaning, servicing, maintenance, and calibration, shall be easily and safely accessible by persons standing at floor level, or standing on permanent platforms, without the use of portable ladders. Examples of these items include, but are not limited to, all types of valves, filters and strainers, transmitters, control devices. Prior to commencing installation work, refer conflicts between this requirement and contract documents to the Contracting Officer for resolution. Failure of the Contractor to resolve, or point out any issues will result in the Contractor correcting at no additional cost or time to the Government.

(3) Complete coordination/shop drawings shall be required in accordance with paragraph, SUBMITTALS. Construction work shall not start on any system until the coordination/shop drawings have been approved by VA.

(4) Workmanship/craftsmanship will be of the highest quality and standards. The VA reserves the right to reject any work based on poor quality of workmanship this work shall be removed and done again at no additional cost or time to the Government.

i. Upon request by Government, provide lists of previous installations for selected items of equipment. Include contact persons who will serve as references, with current telephone numbers and e-mail addresses.

j. Guaranty: Warranty of Construction, FAR Clause 52.246-21.

k. Bio Based Materials: For products designated by the USDA's Bio Preferred Program, provide products that meet or exceed USDA recommendations for bio based content, so long as products meet all performance requirements in this specification section. For more information regarding the product categories covered by the Bio Preferred Program, visit http://www.biopreferred.gov.

l. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 DELIVERY, STORAGE AND HANDLING

a. Protection of Equipment:

(1) Equipment and material placed on the job site shall remain in the custody of the Contractor until phased acceptance, whether or not the Government has reimbursed the Contractor for the equipment and material. The Contractor is solely responsible for the protection of such equipment and material against any damage or theft.

(2) Large equipment such as boilers, chillers, cooling towers, fans, and air handling units if shipped on open trailer trucks shall be covered with shrink on plastics or water proof tarpaulins that provide protection from exposure to rain, road salts and other transit hazards. Protection shall be kept in place until equipment is moved into a building or installed as designed.

(3) Repair damaged equipment in first class, new operating condition and appearance; or, replace same as determined and directed by the Contracting Officer. Such repair or replacement shall be at no additional cost or time to the Government.

(4) Protect interiors of new equipment and piping systems against entry of foreign matter. Clean both inside and outside before painting or placing equipment in operation.

(5) Existing equipment and piping being worked on by the Contractor shall be under the custody and responsibility of the Contractor and shall be protected as required for new work.

(6) Protect plastic piping and tanks from ultraviolet light (sunlight).

b. Cleanliness of Piping and Equipment Systems:

(1) Exercise care in storage and handling of equipment and piping material to be incorporated in the work. Remove debris arising from cutting, threading and welding of piping.

(2) Piping systems shall be flushed, blown or pigged as necessary to deliver clean systems.

(3) Clean interior of all tanks prior to delivery for beneficial use by the Government.

(4) Boilers shall be left clean following final internal inspection by Government insurance representative or inspector.

(5) Contractor shall be fully responsible for all costs, damage, and

delay arising from failure to provide clean systems.

1.8 AS-BUILT DOCUMENTATION

a. Submit manufacturer’s literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, VA approved substitutions and construction revisions shall be four hard copied and one electronic copy of each inserted into a three-ring binder. All aspects of system operation and maintenance procedures, including applicable piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing Contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. Should the installing Contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the ‘third party testing company’ requirement. Provide record drawings as follows:

(1) Red-lined, hand-marked drawings are to be provided, with one paper copy and a scanned PDF version of the hand-marked drawings provided on CD or DVD.

d. The as-built drawings shall indicate the location and type of all lockout/tagout points for all energy sources for all equipment and pumps to include breaker location and numbers, valve tag numbers, etc. Coordinate lockout/tagout procedures and practices with local VA requirements.

e. Certification documentation shall be provided to Contracting Officer 21 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and provide documentation/certification that all results of tests were within limits specified. Test results shall contain written sequence of test procedure with written test results annotated at each step along with the expected outcome or setpoint. The results shall include all readings, including but not limited to data on device (make, model and performance characteristics), normal pressures, switch ranges, trip points, amp readings, and calibration data to include equipment serial numbers or individual identifications, etc.

1.9 JOB CONDITIONS – WORK IN EXISTING BUILDING

a. Building Operation: Government employees will be continuously

operating and managing all facilities, including temporary facilities that serve the VAMC.

b. Maintenance of Service: Schedule all work to permit continuous service as required by the VAMC.

c. Steam and Condensate Service Interruptions: Limited steam and condensate service interruptions, as required for interconnections of new and existing systems, will be permitted by the Contracting Officer during periods when the demands are not critical to the operation of the VAMC. These non-critical periods are limited to between 8 pm and 5 am in the appropriate off-season (if applicable). Provide at least 10 working days advance notice to the Contracting Officer. The request shall include a detailed plan on the proposed shutdown and the intended work to be done along with manpower levels. All equipment and materials shall be onsite and verified with plan 5 days prior to the shutdown or it will need to be rescheduled.

d. Phasing of Work: Comply with all requirements shown on contract documents. Contractor shall submit a complete detailed phasing plan/schedule with manpower levels prior to commencing work. The phasing plan shall be detailed enough to provide milestones in the process that can be verified.

e. Building Working Environment: Maintain the architectural and structural integrity of the building and the working environment at all times. Maintain the interior of building at (65 degrees F) minimum. Limit the opening of doors, windows or other access openings to brief periods as necessary for rigging purposes. Storm water or ground water leakage is prohibited. Provide daily clean-up of construction and demolition debris on all floor surfaces and on all equipment being operated by VA. Maintain all egress routes and safety systems/devices.

f. Acceptance of Work for Government Operation: As new equipment, systems and facilities are made available for operation and these items are deemed of beneficial use to the Government, inspections will be made and tests will be performed. Based on the inspections, a list of contract deficiencies will be issued to the Contractor. After correction of deficiencies as necessary for beneficial use, the Contracting Officer will process necessary acceptance and the equipment will then be under the control and operation of Government personnel.

PART 2 PRODUCTS

2.1 FACTORY-ASSEMBLED PRODUCTS

a. Provide maximum standardization of components to reduce spare part requirements.

b. Performance and warranty of all components that make up an assembled unit shall be the responsibility of the manufacturer of the completed assembly.

(1) All components of an assembled unit need not be products of same manufacturer.

(2) Constituent parts that are alike shall be products of a single manufacturer.

(3) Components shall be compatible with each other and with the total assembly for intended service.

(4) Contractor shall guarantee performance of assemblies of components, and shall repair or replace elements of the assemblies as required to deliver specified performance of the complete assembly.

c. Equipment and components of equipment shall bear manufacturer's name and trademark, model number, serial number and performance data on a nameplate securely affixed in a conspicuous place, or cast integral with, stamped or otherwise permanently marked upon the components of the equipment.

d. Major items of equipment, which serve the same function, shall be the same make and model. Exceptions must be approved by the VA, but may be permitted if performance requirements cannot be met.

2.2 COMPATIBILITY OF RELATED EQUIPMENT

Equipment and materials installed shall be compatible in all respects with other items being furnished and with existing items so that the result will be a complete and fully operational plant that conforms to contract requirements.

2.3 V-BELT DRIVES

a. Type: ARPM standard V-belts with proper motor pulley and driven sheave. Belts shall be constructed of reinforced cord and rubber.

b. Dimensions, rating and selection standards: RMA IP-20 and RMA IP-21.

c. Minimum Horsepower Rating: Motor horsepower plus recommended ARPM service factor (not less than 20 percent) in addition to the ARPM allowances for pitch diameter, center distance, and arc of contact.

d. Maximum Speed: (5000 feet per minute).

e. Adjustment Provisions: For alignment and ARPM standard allowances for installation and take-up.

f. Drives may utilize a single V-Belt (any cross section) when it is the manufacturer's standard.

g. Multiple Belts: Matched to ARPM specified limits by measurement on a belt measuring fixture. Seal matched sets together to prevent mixing or partial loss of sets. Replacement, when necessary, shall be an entire set of new matched belts.

h. Sheaves and Pulleys:

(1) Material: Pressed steel, or close-grained cast iron.

(2) Bore: Fixed or bushing type for securing to shaft with keys.

(3) Balanced: Statically and dynamically.

(4) Groove spacing for driving and driven pulleys shall be the same.

i. Drive Types, Based on ARI 435:

(1) Provide adjustable-pitch or fixed-pitch drive as follows:

(a) Fan speeds up to 1800 RPM: (10 horsepower) and smaller.

(b) Fan speeds over 1800 RPM: (3 horsepower) and smaller.

(2) Provide fixed-pitch drives for drives larger than those listed above.

(3) The final fan speeds required to just meet the system CFM and pressure requirements, without throttling the design air flow branch, shall be determined by adjustment of a temporary adjustable-pitch motor sheave or by fan law calculation if a fixed-pitch drive is used initially.

j. Final Drive Set: If adjustment is required beyond the capabilities of the factory drive set, the final drive set shall be provided as part of this contract at no additional cost or time to the Government.

2.4 SYNCHRONOUS BELT DRIVES

a. Type: ARPM synchronous belts with proper motor pulley and driven sheave. Belts shall be constructed of reinforced cord and rubber.

b. Dimensions, rating and selection standards: RMA IP-24 and RMA IP-27.

c. Minimum Horsepower Rating: Motor horsepower plus recommended ARPM service factor (not less than 20 percent) in addition to the ARPM allowances for pitch diameter, center distance, and arc of contact.

d. Maximum Speed: (5000 feet) per minute).

e. Adjustment Provisions: For alignment and ARPM standard allowances for installation and take-up.

f. Drives may utilize a single belt of manufacturer’s standard width for the application.

g. Multiple Belts: Matched to ARPM specified limits by measurement on a belt measuring fixture. Seal matched sets together to prevent mixing or partial loss of sets. Replacement, when necessary, shall be an entire set of new matched belts.

h. Sheaves and Pulleys:

(1) Material: Pressed steel, or close-grained cast iron.

(2) Bore: Fixed or bushing type for securing to shaft with keys.

(3) Balanced: Statically and dynamically.

i. Final Drive Set: The final fan speeds required to just meet the system CFM and pressure requirements, without throttling the design

air flow branch, shall be determined by fan law calculation. If adjustment is required beyond the capabilities of the factory drive set, the final drive set shall be provided as part of this contract at no additional cost or time to the Government.

2.5 DRIVE GUARDS

a. For machinery and equipment, provide guards as shown in AMCA 410 for belts, chains, couplings, pulleys, sheaves, shafts, gears and other moving parts regardless of height above the floor to prevent damage to equipment and injury to personnel. Drive guards may be excluded where motors and drives are inside factory-fabricated air handling unit casings.

b. Pump shafts and couplings shall be fully guarded by a sheet steel guard, covering coupling and shaft but not bearings. Material shall be minimum 16-gauge sheet steel; all edges shall be hemmed and ends shall be bent into flanges and the flanges shall be drilled and attached to pump base with minimum of four (1/4 inch) bolts. Reinforce guard as necessary to prevent side play forcing guard onto couplings.

c. V-belt and sheave assemblies shall be totally enclosed, firmly mounted, non-resonant. Guard shall be an assembly of minimum 22-gauge sheet steel and expanded or perforated metal to permit observation of belts. (1 inch) diameter hole shall be provided at each shaft centerline to permit speed measurement.

d. Materials: Sheet steel, expanded metal or wire mesh rigidly secured so as to be removable without disassembling pipe, duct, or electrical connections to equipment.

e. Access for Speed Measurement: (1 inch) diameter hole at each shaft center.

2.6 LIFTING ATTACHMENTS

Provide equipment with suitable lifting attachments to enable equipment to be lifted in its normal position. Lifting attachments shall withstand any handling conditions that might be encountered, without bending or distortion of shape, such as rapid lowering and braking of load.

2.7 ELECTRIC MOTORS

All material and equipment furnished and installation methods shall conform to the requirements of Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC AND STEAM GENERATION EQUIPMENT; Section 26 29 11, MOTOR CONTROLLERS; and, Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES. Provide all electrical wiring, conduit, and devices necessary for the proper connection, protection and operation of the systems. Provide special energy efficient premium efficiency type motors as scheduled.

2.8 VARIABLE SPEED MOTOR CONTROLLERS

a. Refer to Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS and Section 26 29 11, MOTOR CONTROLLERS for specifications.

b. Coordinate variable speed motor controller communication protocol with

Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

c. Provide variable speed motor controllers with or without a bypass contactor as indicated in contract drawings.

d. The combination of controller and motor shall be provided by the manufacturer of the driven equipment, such as pumps and fans, and shall be rated for 100 percent output performance. Multiple units of the same class of equipment, i.e. air handlers, fans, pumps, shall be product of a single manufacturer.

e. Motors shall be premium efficiency type and be approved by the motor controller manufacturer. The controller-motor combination shall be guaranteed to provide full motor nameplate horsepower in variable frequency operation. Both driving and driven motor/fan sheaves shall be fixed pitch.

f. Controller shall not add any current or voltage transients to the input ac power distribution system, DDC controls, sensitive medical equipment, etc., nor shall be affected from other devices on the ac power system.

2.9 EQUIPMENT AND MATERIALS IDENTIFICATION

a. Use symbols, nomenclature and equipment numbers specified, shown on the contract documents and shown in the maintenance manuals. Identification for piping is specified in Section 09 91 00, PAINTING.

b. Use symbols, nomenclature and equipment numbers specified, shown in the contract documents, and shown in the maintenance manuals. In addition, provide bar code identification nameplate for all equipment which will allow the equipment identification code to be scanned into the system for maintenance and inventory tracking. Identification for piping is specified in Section 09 91 00, PAINTING.

c. Interior (Indoor) Equipment: Engraved nameplates, with letters not less than (3/16 inch) high of brass with black-filled letters, or rigid black plastic with white letters specified in Section 09 91 00, PAINTING permanently fastened to the equipment. Identify unit components such as coils, filters, fans, etc.

d. Exterior (Outdoor) Equipment: Brass nameplates, with engraved black filled letters, not less than (3/16 inch) high riveted or bolted to the equipment.

e. Control Items: Label all instrumentation, temperature and humidity sensors, controllers and control dampers. Identify and label each item as they appear on the control diagrams.

f. Valve Tags and Lists:

(1) HVAC and Mechanical Rooms: Provide for all valves other than for equipment in Section 23 82 00, CONVECTION HEATING AND COOLING UNITS.

(2) Valve tags: Engraved black filled numbers and letters not less than (1/2 inch) high for number designation, and not less than (1/4 inch) for service designation on (19-gauge) (1-1/2 inches) round brass disc, attached with brass "S" hook or brass chain.

(3) Valve lists: Typed or printed plastic coated card(s), sized (8-1/2 inches) by (11 inches) showing tag number, valve function and area of control, for each service or system. Punch sheets for a 3-ring notebook.

(4) Provide detailed plan for each floor of the building indicating the location and valve number for each valve. Identify location of each valve with a color-coded thumb tack in ceiling.

g. Ceiling Grid Labels:

(1) (2 inch) long by (1/2 inch) wide by (1 mil) thick UV resistant metalized polyester label with red border color and black custom lettering on white background interior. Peel and stick adhesive backing. Label and adhesive manufactured specifically for use in equipment inventory tagging.

(2) Custom print labels with above ceiling HVAC equipment numbers.

2.10 FIRESTOPPING

Section 07 84 00, FIRESTOPPING specifies an effective barrier against the spread of fire, smoke and gases where penetrations occur for piping and ductwork. Refer to Section 23 07 11, HVAC AND BOILER PLANT INSULATION, for firestop pipe and duct insulation.

2.11 GALVANIZED REPAIR COMPOUND

MIL-P-21035, paint form.

2.12 HVAC PIPE AND EQUIPMENT SUPPORTS AND RESTRAINTS

a. Vibration Isolators: Refer to Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT.

b. Supports for Roof Mounted Items:

(1) Equipment: Equipment rails shall be galvanized steel, minimum (18 gauge), with integral baseplate, continuous welded corner seams, factory installed (2 by 4 inches) treated wood nailer, (18 gauge) galvanized steel counter flashing cap with screws, built-in cant strip, (except for gypsum or tectum deck), minimum height (11 inches). For surface insulated roof deck, provide raised cant strip to start at the upper surface of the insulation.

(2) Pipe/duct pedestals: Provide a galvanized Unistrut channel welded to U-shaped mounting brackets which are secured to side of rail with galvanized lag bolts.

c. Pipe Supports: Comply with MSS SP-58. Type Numbers specified refer to this standard. For selection and application comply with MSS SP-58. Refer to Section 05 50 00, METAL FABRICATIONS, for miscellaneous metal support materials and prime coat painting requirements.

d. Attachment to Concrete Building Construction:

(1) Concrete insert: MSS SP-58, Type 18.

(2) Self-drilling expansion shields and machine bolt expansion

anchors: Permitted in concrete not less than (4 inches) thick when approved by the COntracting Officer for each job condition.

(3) Power-driven fasteners: Permitted in existing concrete or masonry not less than (4 inches) thick when approved by the Contracting Officer for each job condition.

e. Attachment to Steel Building Construction:

(1) Welded attachment: MSS SP-58, Type 22.

(2) Beam clamps: MSS SP-58, Types 20, 21, 28 or 29. Type 23 C-clamp may be used for individual copper tubing up to (7/8 inch) outside diameter.

f. Attachment to Metal Pan or Deck: As required for materials specified in Section 05 31 00, STEEL DECKING Section 05 36 00, COMPOSITE METAL DECKING .

g. Attachment to existing structure: Support from existing floor/roof frame.

h. Attachment to Wood Construction: Wood screws or lag bolts.

i. Hanger Rods: Hot-rolled steel, ASTM A36/A36M or ASTM A575 for allowable load listed in MSS SP-58. For piping, provide adjustment means for controlling level or slope. Types 13 or 15 turn-buckles shall provide (1-1/2 inches) minimum of adjustment and incorporate locknuts. All-thread rods are acceptable.

j. Hangers Supporting Multiple Pipes (Trapeze Hangers): Galvanized, cold formed, lipped steel channel horizontal member, not less than (1-5/8 inches by 1-5/8 inches), (12 gauge), designed to accept special spring held, hardened steel nuts. Trapeze hangers are prohibited for use for steam supply and condensate piping.

(1) Allowable hanger load: Manufacturers rating less (200 pounds).

(2) Guide individual pipes on the horizontal member of every other trapeze hanger with (1/4 inch) U-bolt fabricated from steel rod. Provide Type 40 insulation shield, secured by two (1/2 inch) galvanized steel bands, or preinsulated calcium silicate shield for insulated piping at each hanger.

k. Supports for Piping Systems:

(1) Select hangers sized to encircle insulation on insulated piping. Refer to Section 23 07 11, HVAC AND BOILER PLANT INSULATION for insulation thickness. To protect insulation, provide Type 39 saddles for roller type supports or preinsulated calcium silicate shields. Provide Type 40 insulation shield or preinsulated calcium silicate shield at all other types of supports and hangers including those for preinsulated piping.

(2) Piping Systems except High and Medium Pressure Steam (MSS SP-58):

(a) Standard clevis hanger: Type 1; provide locknut.

(b) Riser clamps: Type 8.

(c) Wall brackets: Types 31, 32 or 33.

(d) Roller supports: Type 41, 43, 44 and 46.

(e) Saddle support: Type 36, 37 or 38.

(f) Turnbuckle: Types 13 or 15. Preinsulate.

(g) U-bolt clamp: Type 24.

(h) Copper Tube:

1. Hangers, clamps and other support material in contact with tubing shall be painted with copper colored epoxy paint, plastic coated or taped with non-adhesive isolation tape to prevent electrolysis.

2. For vertical runs use epoxy painted or plastic-coated riser clamps.

3. For supporting tube to strut: Provide epoxy painted pipe straps for copper tube or plastic inserted vibration isolation clamps.

4. Insulated Lines: Provide pre-insulated calcium silicate shields sized for copper tube.

(i) Supports for plastic piping: As recommended by the pipe manufacturer with black rubber tape extending one inch beyond steel support or clamp.

(3) High and Medium Pressure Steam (MSS SP-58):

(a) Provide eye rod or Type 17 eye nut near the upper attachment.

(b) Piping (2 inches) and greater: Type 43 roller hanger. For roller hangers requiring seismic bracing provide a Type 1 clevis hanger with Type 41 roller attached by flat side bars.

(c) Piping with Vertical Expansion and Contraction:

1. Movement up to (3/4 inch): Type 51 or 52 variable spring unit with integral turn buckle and load indicator.

2. Movement more than (3/4 inch): Type 54 or 55 constant support unit with integral adjusting nut, turn buckle and travel position indicator.

(4) Convertor and Expansion Tank Hangers: May be Type 1 sized for the shell diameter. Insulation where required will cover the hangers.

l. Pre-insulated Calcium Silicate Shields:

(1) Provide 360-degree water resistant high density (140 psig) compressive strength calcium silicate shields encased in galvanized metal.

(2) Pre-insulated calcium silicate shields to be installed at the

point of support during erection.

(3) Shield thickness shall match the pipe insulation.

(4) The type of shield is selected by the temperature of the pipe, the load it must carry, and the type of support it will be used with.

(a) Shields for supporting chilled or cold water shall have insulation that extends a minimum of 1 inch past the sheet metal. Provide for an adequate vapor barrier in chilled lines.

(b) The pre-insulated calcium silicate shield shall support the maximum allowable water filled span as indicated in MSS SP-58. To support the load, the shields may have one or more of the following features: structural inserts 600 psig compressive strength, an extra bottom metal shield, or formed structural steel (ASTM A36/A36M) wear plates welded to the bottom sheet metal jacket.

(5) Shields may be used on steel clevis hanger type supports, roller supports or flat surfaces.

m. Seismic Restraint and Bracing of Piping and Ductwork: Comply with MSS SP-127. Refer to Seismic Design Requirements VA H-18-8..

2.13 PIPE PENETRATIONS

a. Install sleeves during construction for other than blocked out floor openings for risers in mechanical bays.

b. To prevent accidental liquid spills from passing to a lower level, provide the following:

(1) For sleeves: Extend sleeve (1 inch) above finished floor and provide sealant for watertight joint.

(2) For blocked out floor openings: Provide (1-1/2 inch) angle set in silicone adhesive around opening.

(3) For drilled penetrations: Provide (1-1/2 inch) angle ring or square set in silicone adhesive around penetration.

c. Penetrations through beams or ribs are prohibited but may be installed in concrete beam flanges. Any deviation from these requirements shall receive prior approval of Contracting Officer.

d. Sheet Metal, Plastic, or Moisture-Resistant Fiber Sleeves: Provide for pipe passing through floors, interior walls, and partitions, unless brass or steel pipe sleeves are specifically called for below.

e. Cast Iron or Zinc Coated Pipe Sleeves: Provide for pipe passing through exterior walls below grade. Make space between sleeve and pipe watertight with a modular or link rubber seal. Seal shall be applied at both ends of sleeve.

f. Galvanized Steel or an Alternate Black Iron Pipe with Asphalt coating Sleeves: Provide for pipe passing through concrete beam flanges, except where brass pipe sleeves are called for. Provide sleeve for pipe passing through floor of mechanical rooms, laundry work rooms,

and animal rooms above basement. Except in mechanical rooms, connect sleeve with floor plate.

g. Brass Pipe Sleeves: Provide for pipe passing through quarry tile, terrazzo or ceramic tile floors. Connect sleeve with floor plate.

h. Sleeves are not required for wall hydrants for fire department connections or in drywall construction.

i. Sleeve Clearance: Sleeve through floors, walls, partitions, and beam flanges shall be one inch greater in diameter than external diameter of pipe. Sleeve for pipe with insulation shall be large enough to accommodate the insulation. Interior openings shall be caulked tight with fire stopping material and sealant to prevent the spread of fire, smoke, and gases.

j. Sealant and Adhesives: Shall be as specified in Section 07 92 00, JOINT SEALANTS.

2.14 DUCT PENETRATIONS

a. Provide curbs for roof mounted piping, ductwork and equipment. Curbs shall be (18 inches) high with continuously welded seams, built-in cant strip, interior baffle with acoustic insulation, curb bottom, hinged curb adapter.

b. Provide firestopping for openings through fire and smoke barriers, maintaining minimum required rating of floor, ceiling or wall assembly. See section 07 84 00, FIRESTOPPING.

2.15 SPECIAL TOOLS AND LUBRICANTS

a. Furnish, and turn over to the Contracting Officer, tools not readily available commercially, that are required for disassembly or adjustment of equipment and machinery furnished.

b. Grease Guns with Attachments for Applicable Fittings: One for each type of grease required for each motor or other equipment.

c. Refrigerant Tools: Provide system charging/Evacuation equipment, gauges, fittings, and tools required for maintenance of furnished equipment.

d. Tool Containers: Hardwood or metal, permanently identified for intended service and mounted, or located, where directed by the Contracting Officer.

e. Lubricants: A minimum of (1 quart) of oil, and (1 pound) of grease, of equipment manufacturer's recommended grade and type, in unopened containers and properly identified as to use for each different application.

2.16 WALL, FLOOR AND CEILING PLATES

a. Material and Type: Chrome plated brass or chrome plated steel, one piece or split type with concealed hinge, with set screw for fastening to pipe, or sleeve. Use plates that fit tight around pipes, cover openings around pipes and cover the entire pipe sleeve projection.

b. Thickness: Not less than (3/32 inch) for floor plates. For wall and

ceiling plates, not less than (0.025 inch) for up to (3-inch pipe), (0.035 inch) for larger pipe.

c. Locations: Use where pipe penetrates floors, walls and ceilings in exposed locations, in finished areas only. Provide a watertight joint in spaces where brass or steel pipe sleeves are specified.

2.17 ASBESTOS

Materials containing asbestos are prohibited.

PART 3 EXECUTION

3.1 GENERAL

If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

3.2 ARRANGEMENT AND INSTALLATION OF EQUIPMENT AND PIPING

a. Location of piping, sleeves, inserts, hangers, and equipment, access provisions shall be coordinated with the work of all trades. The coordination/shop drawings shall be submitted for review. Locate piping, sleeves, inserts, hangers, ductwork and equipment clear of windows, doors, openings, light outlets, and other services and utilities. Equipment coordination/shop drawings shall be prepared to coordinate proper location and personnel access of all facilities. The drawings shall be submitted for review. Follow manufacturer's published recommendations for installation methods not otherwise specified.

b. Operating Personnel Access and Observation Provisions: Select and arrange all equipment and systems to provide clear view and easy access, without use of portable ladders, for maintenance and operation of all devices including, but not limited to, all equipment items, valves, filters, strainers, transmitters, sensors, control devices. All gauges and indicators shall be clearly visible by personnel standing on the floor or on permanent platforms. Do not reduce or change maintenance and operating space and access provisions that are shown in the contract documents.

c. Equipment and Piping Support: Coordinate structural systems necessary for pipe and equipment support with pipe and equipment locations to permit proper installation.

d. Location of pipe sleeves, trenches and chases shall be accurately coordinated with equipment and piping locations.

e. Cutting Holes:

(1) Cut holes through concrete and masonry by rotary core drill. Pneumatic hammer, impact electric, and hand or manual hammer type drill is prohibited, except as permitted by Contracting Officer where working area space is limited.

(2) Locate holes to avoid interference with structural members such as slabs, columns, ribs, beams or reinforcing. Holes shall be laid out in advance and drilling done only after approval by

Contracting Officer. If the Contractor considers it necessary to drill through structural members, this matter shall be referred to Contracting Officer for approval.

(3) Do not penetrate membrane waterproofing.

f. Minor Piping: Generally, small diameter pipe runs from drips and drains, water cooling, and other service are not shown but must be provided.

g. Electrical Interconnection of Instrumentation or Controls: This generally not shown but shall be provided. This includes interconnections of sensors, transmitters, transducers, control devices, control and instrumentation panels, instruments and computer workstations. Devices shall be located so they are easily accessible for testing, maintenance, calibration, etc. The Contracting Officer has the final determination on what is accessible and what is not. Comply with NFPA 70.

h. Protection and Cleaning:

(1) Equipment and materials shall be carefully handled, properly stored, and adequately protected to prevent damage before and during installation, in accordance with the manufacturer's recommendations and as approved by the Contracting Officer. Damaged or defective items in the opinion of the Contracting Officer, shall be replaced.

(2) Protect all finished parts of equipment, such as shafts and bearings where accessible, from rust prior to operation by means of protective grease coating and wrapping. Close pipe openings with caps or plugs during installation. Tightly cover and protect fixtures and equipment against dirt, water chemical, or mechanical injury. At completion of all work thoroughly clean fixtures, exposed materials and equipment.

i. Concrete and Grout: Use concrete and non-shrink grout 3000 psig minimum, specified in Section 03 30 00, CAST-IN-PLACE CONCRETE.

j. Install gauges, thermometers, valves and other devices with due regard for ease in reading or operating and maintaining said devices. Locate and position thermometers and gauges to be easily read by operator or staff standing on floor or walkway provided. Servicing shall not require dismantling adjacent equipment or pipe work.

k. Install steam piping expansion joints as per manufacturer’s recommendations.

l. Work in Existing Building:

(1) Perform as specified in paragraph, OPERATIONS AND STORAGE AREAS, paragraph, ALTERATIONS, and paragraph, RESTORATION of the Section 01 00 00, GENERAL REQUIREMENTS for relocation of existing equipment, alterations and restoration of existing building(s).

(2) As specified in Section 01 00 00, GENERAL REQUIREMENTS, paragraph, OPERATIONS AND STORAGE AREAS, make alterations to existing service piping at times that will least interfere with normal operation of the facility.

m. Work in Animal Research Areas: Seal all pipe and duct penetrations with silicone sealant to prevent entrance of insects.

n. Switchgear/Electrical Equipment Drip Protection: Every effort shall be made to eliminate the installation of pipe above electrical and data/telephone switchgear. If this is not possible, encase pipe in a second pipe with a minimum of joints. Installation of piping, ductwork, leak protection apparatus or other installations foreign to the electrical installation shall not be located in the space equal to the width and depth of the equipment and extending from to a height of (6 feet) above the equipment or to ceiling structure, whichever is lower (NFPA 70).

o. Inaccessible Equipment:

(1) Where the Government determines that the Contractor has installed equipment not conveniently accessible for operation and maintenance or inspections, equipment shall be removed and reinstalled or remedial action performed as directed at no additional cost or time to the Government.

(2) The term "conveniently accessible" is defined as capable of being reached without the use of ladders, or without climbing or crawling under or over obstacles such as, but not limited to motors, fans, pumps, belt guards, transformers, high voltage lines, conduit and raceways, piping, hot surfaces, and ductwork. The Contracting Officer has final determination on whether an installation meets this requirement or not.

3.3 TEMPORARY PIPING AND EQUIPMENT

a. Continuity of operation of existing facilities will generally require temporary installation or relocation of equipment, ducts, and piping. B..

b. The Contractor shall provide all required facilities in accordance with the requirements of phased construction and maintenance of service. All piping and equipment shall be properly supported, sloped to drain, operate without excessive stress, and shall be insulated where injury can occur to personnel by contact with operating facilities. The requirements of paragraph, ARRANGEMENT AND INSTALLATION OF EQUIPMENT AND PIPING apply.

c. Temporary facilities and piping shall be completely removed and any openings in structures sealed. Provide necessary blind flanges and caps to seal open piping remaining in service.

3.4 RIGGING

a. Design is based on application of available equipment. Openings in building structures are planned to accommodate design scheme.

b. Alternative methods of equipment delivery may be offered by Contractor and will be considered by Government under specified restrictions of phasing and maintenance of service requirements as well as structural integrity of the building.

c. Close all openings in the building when not required for rigging

operations to maintain proper environment in the facility for Government operation and maintenance of service.

d. Contractor shall provide all facilities required to deliver specified equipment and place on foundations. Attachments to structures for rigging purposes and support of equipment on structures shall be Contractor's full responsibility. Upon request, the Government will check structure adequacy and advise Contractor of recommended restrictions.

e. Contractor shall check all clearances, weight limitations and shall offer a rigging plan designed by a Registered Professional Engineer. All modifications to structures, including reinforcement thereof, shall be at Contractor's cost, time and responsibility.

f. Follow approved rigging plan.

g. Restore building to original condition upon completion of rigging work.

3.5 PIPE AND EQUIPMENT SUPPORTS

a. Where hanger spacing does not correspond with joist or rib spacing, use structural steel channels designed by a structural engineer, secured directly to joist and rib structure that will correspond to the required hanger spacing, and then suspend the equipment and piping from the channels. Drill or burn holes in structural steel only with the prior approval of the Contracting Officer.

b. Use of chain pipe supports; wire or strap hangers; wood for blocking, stays and bracing; or, hangers suspended from piping above are prohibited. Replace or thoroughly clean rusty products and paint with zinc primer.

c. Hanger rods shall be used that are straight and vertical. Turnbuckles for vertical adjustments may be omitted where limited space prevents use. Provide a minimum of (1/2 inch) clearance between pipe or piping covering and adjacent work.

d. HVAC Horizontal Pipe Support Spacing: Refer to MSS SP-58. Provide additional supports at valves, strainers, inline pumps and other heavy components. Provide a support within one foot of each elbow.

e. HVAC Vertical Pipe Supports:

(1) Up to (6-inch pipe), (30 feet) long, bolt riser clamps to the pipe below couplings, or welded to the pipe and rests supports securely on the building structure.

(2) Vertical pipe larger than the foregoing, support on base elbows or tees, or substantial pipe legs extending to the building structure.

f. Overhead Supports:

(1) The basic structural system of the new building is designed to sustain the loads imposed by equipment and piping to be supported overhead.

(2) Provide steel structural members, in addition to those shown, of adequate capability to support the imposed loads, located in

accordance with the final approved layout of equipment and piping.

(3) Tubing and capillary systems shall be supported in channel troughs.

g. Floor Supports:

(1) Provide concrete bases, concrete anchor blocks and pedestals, and structural steel systems for support of equipment and piping. Concrete bases and structural systems shall be anchored and doweled to resist forces under operating and seismic conditions (if applicable) without excessive displacement or structural failure.

(2) Bases and supports shall not be located and installed until equipment mounted thereon has been approved. Bases shall be sized to match equipment mounted thereon plus (2 inch) excess on all edges unless noted otherwise on drawings. Chiller foundations shall have horizontal dimensions that exceed chiller base frame dimensions by at least (6 inches) on all sides. Structural contract documents shall be reviewed for additional requirements. Bases shall be neatly finished and smoothed, shall have chamfered edges at the top, and shall be suitable for painting.

(3) All equipment shall be shimmed, leveled, firmly anchored, and grouted with epoxy grout. Anchor bolts shall be placed in sleeves, anchored to the bases. Fill the annular space between sleeves and bolts with a granular material to permit alignment and realignment.

(4) Not used.

3.6 MECHANICAL DEMOLITION

a. Rigging access, other than indicated on the contract documents, shall be provided by the Contractor after approval for structural integrity by the Contracting Officer. Such access shall be provided without additional cost or time to the Government. Where work is in an operating plant, provide approved protection from dust and debris at all times for the safety of plant personnel and maintenance of plant operation and environment of the plant.

b. In an operating facility, maintain the operation, cleanliness and safety. Government personnel will be carrying on their normal duties of operating, cleaning and maintaining equipment and plant operation. Confine the work to the immediate area concerned; maintain cleanliness and wet down demolished materials to eliminate dust. Debris accumulated in the area to the detriment of plant operation is prohibited. Perform all flame cutting to maintain the fire safety integrity of this plant. Adequate fire extinguishing facilities shall be available at all times. Perform all work in accordance with recognized fire protection standards. Inspection will be made by personnel of the VAMC, and Contractor shall follow all directives of the Contracting Officer with regard to rigging, safety, fire safety, and maintenance of operations.

c. Unless specified otherwise, all piping, wiring, conduit, and other devices associated with the equipment not re-used in the new work shall be completely removed from Government property per Section 01 74 19, CONSTRUCTION WASTE MANAGEMENT. This includes all concrete

pads, pipe, valves, fittings, insulation, and all hangers including the top connection and any fastenings to building structural systems. All openings shall be sealed after removal of equipment, pipes, ducts, and other penetrations in roof, walls, floors, in an approved manner and in accordance with contract documents where specifically covered. Structural integrity of the building system shall be maintained. Reference shall also be made to the contract documents of the other disciplines in the project for additional facilities to be demolished or handled.

d. All indicated valves including gate, globe, ball, butterfly and check, all pressure gauges and thermometers with wells shall remain Government property and shall be removed and delivered to Contracting Officer and stored as directed. The Contractor shall remove all other material and equipment, devices and demolition debris under these contract documents. Such material shall be removed from Government property expeditiously and shall not be allowed to accumulate.

e.

3.7 CLEANING AND PAINTING

a. Prior to final inspection and acceptance of the plant and facilities for beneficial use by the Government, the plant facilities, equipment and systems shall be thoroughly cleaned and painted. Refer to Section 09 91 00, PAINTING.

b. In addition, the following special conditions apply:

(1) Cleaning shall be thorough. Solvents, cleaning materials and methods recommended by the manufacturers shall be used for the specific tasks. All rust shall be removed prior to painting and from surfaces to remain unpainted. Repair scratches, scuffs, and abrasions prior to applying prime and finish coats.

(2) The following material and equipment shall not be painted:

(a) Motors, controllers, control switches, and safety switches.

(b) Control and interlock devices.

(c) Regulators.

(d) Pressure reducing valves.

(e) Control valves and thermostatic elements.

(f) Lubrication devices and grease fittings.

(g) Copper, brass, aluminum, stainless steel and bronze surfaces.

(h) Valve stems and rotating shafts.

(i) Pressure gauges and thermometers.

(j) Glass.

(k) Nameplates.

(3) Control and instrument panels shall be cleaned, damaged surfaces repaired, and shall be touched-up with matching paint obtained from panel manufacturer.

(4) Pumps,fans, motors, steel and cast-iron bases, and coupling guards shall be cleaned, and shall be touched-up with the same paint type and color as utilized by the pump and fan manufacturer.

(5) Temporary Facilities: Apply paint to surfaces that do not have existing finish coats. This may include painting exposed metals where hangers were removed or where equipment was moved or removed.

(6) Paint shall withstand the following temperatures without peeling or discoloration:

(a) Condensate and Feedwater: (100 degrees F) on insulation jacket surface and (250 degrees F) on metal pipe surface.

(b) Steam: (125 degrees F) on insulation jacket surface and (374 degrees F) on metal pipe surface.

(7) Final result shall be smooth, even-colored, even-textured factory finish on all items. Completely repaint the entire piece of equipment if necessary to achieve this.

(8) Lead based paints are prohibited.

3.8 IDENTIFICATION SIGNS

a. Provide laminated plastic signs, with engraved lettering not less than 3/16 inch high, designating functions, for all equipment, switches, motor controllers, relays, meters, control devices, including automatic control valves. Nomenclature and identification symbols shall correspond to that used in maintenance manual, and in diagrams specified elsewhere. Attach by chain, adhesive, or screws.

b. Factory Built Equipment: Metal plate, securely attached, with name and address of manufacturer, serial number, model number, size, performance.

c. Pipe Identification: Refer to Section 09 91 00, PAINTING.

d. Attach ceiling grid label on ceiling grid location directly underneath above ceiling air terminal, control system component, valve, filter unit, fan etc.

3.9 MOTOR AND DRIVES

a. Use synchronous belt drives only on equipment controlled by soft starters or variable frequency drive motor controllers without a bypass contactor. Use V-belt drives on all other applications.

b. Alignment of V-Belt Drives: Set driving and driven shafts parallel and align so that the corresponding grooves are in the same plane.

c. Alignment of Synchronous Belt Drives: Set driving and driven shafts parallel and align so that the corresponding pulley flanges are in the same plane.

d. Alignment of Direct-Connect Drives: Securely mount motor in accurate alignment so that shafts are per coupling manufacturer’s tolerances when both motor and driven machine are operating at normal temperatures.

3.10 LUBRICATION

a. All equipment and devices requiring lubrication shall be lubricated prior to initial operation. Field-check all devices for proper lubrication.

b. All devices and equipment shall be equipped with required lubrication fittings or devices. A minimum of (1 quart) of oil and 1 pound of grease of manufacturer's recommended grade and type for each different application shall be provided; also provide 12 grease sticks for lubricated plug valves. Deliver all materials to Contracting Officer in unopened containers that are properly identified as to application.

c. All lubrication points shall be accessible without disassembling equipment, except to remove access plates.

d. All lubrication points shall be extended to one side of the equipment.

3.11 STARTUP, TEMPORARY OPERATION AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA will observe startup and Contractor testing of selected equipment. Coordinate the startup and Contractor testing schedules with Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

d. Startup of equipment shall be performed as described in equipment specifications. Vibration within specified tolerance shall be verified prior to extended operation. Temporary use of equipment is specified in Section 01 00 00, GENERAL REQUIREMENTS, paragraph, TEMPORARY USE OF MECHANICAL AND ELECTRICAL EQUIPMENT.

3.12 OPERATING AND PERFORMANCE TESTS

a. Prior to the final inspection, perform required tests as specified in Section 01 00 00, GENERAL REQUIREMENTS paragraph, TESTS, and in individual Division 23 specification sections and submit the test reports and records to the Contracting Officer.

b. Should evidence of malfunction in any tested system, or piece of equipment or component part thereof, occur during or as a result of tests, make proper corrections, repairs or replacements, and repeat tests at no additional cost or time to the Government.

c. When completion of certain work or system occurs at a time when final control settings and adjustments cannot be properly made to make

performance tests, then conduct such performance tests and finalize control settings for heating systems and for cooling systems respectively during first actual seasonal use of respective systems following completion of work. Rescheduling of these tests shall be requested in writing to Contracting Officer for approval.

d. No adjustments shall be made during the acceptance inspection. All adjustments shall have been made by this point.

e. Perform tests as required for commissioning provisions in accordance with Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS and Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

3.13 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.14 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hours to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

-- End of Section --

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

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SECTION 23 05 12
3/23

GENERAL MOTOR REQUIREMENTS FOR HVAC AND STEAM GENERATION EQUIPMENT

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies the furnishing, installation and connection of motors for HVAC and steam generation equipment.

b. A complete listing of common acronyms and abbreviations are included in Section 23 05 11 COMMON WORK RESULTS FOR HVAC.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Not used.

d. Section 01 91 00.15 10, TOTAL BUILDING COMMISSIONING.

e. Not used.

f. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

g. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

h. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

i. Not used.

j. Section 26 29 11, MOTOR CONTROLLERS.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)

ABMA 9	(2015) Load Ratings and Fatigue Life for Ball Bearings
ABMA 11	(2014; R 2020) Load Ratings and Fatigue Life for Roller Bearings

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 112	(2017) Standard Test Procedure for Polyphase Induction Motors and Generators

IEEE 841	(2021) IEEE Standard for Petroleum and Chemical Industry-Premium-Efficiency, Severe-Duty Totally Enclosed Fan-Cooled (TEFC) Squirrel Cage Induction Motors--Up to and Including 370 kW (500 hp)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Design A	Electrical Moter Design A - Squirrel-cage motor
NEMA Design B	Electrical Moter Design B - Squirrel-cage motor
NEMA MG 1	(2021) Motors and Generators
NEMA MG 2	(2014) Safety Standard for Construction and Guide for Selection, Installation and Use of Electric Motors and Generators
NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Motor Submittals With Driven Equipment; G, AE

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Certification; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Complete Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC AND STEAM GENERATION EQUIPMENT”, with applicable paragraph identification.

c. Submit motor submittals with driven equipment.

d. Shop Drawings:

(1) Provide documentation to demonstrate compliance with contract documents.

(2) Motor nameplate information shall be submitted including electrical ratings, efficiency, bearing data, power factor, frame size, dimensions, mounting details, materials, horsepower, voltage, phase, speed (RPM), enclosure, starting characteristics, torque characteristics, code letter, full load and locked rotor current, service factor, and lubrication method.

e. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

f. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

g. Certification: Two weeks prior to final inspection, unless otherwise noted, certification shall be submitted to the Contracting Officer stating that the motors have been properly applied, installed, adjusted, lubricated, and tested.

h. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

i. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS

COMMISSIONING OF HVAC SYSTEMS.

1.6 AS-BUILT DOCUMENTATION

a. Submit manufacturer’s literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, VA approved substitutions and construction revisions shall befour hard copies and one electronic copy of each inserted into a three-ring binder. All aspects of system operation and maintenance procedures, including applicable piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the ‘third party testing company’ requirement. Provide record drawings as follows:

(1) Red-lined, hand-marked drawings are to be provided, with one paper copy and a scanned PDF version of the hand-marked drawings provided on CD or DVD.

d. The as-built drawings shall indicate the location and type of all lockout/tagout points for all energy sources for all equipment and pumps to include breaker location and numbers, valve tag numbers, etc. Coordinate lockout/tagout procedures and practices with local VA requirements.

e. Certification documentation shall be provided to Contracting Officer 21 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and provide documentation/certification that all results of tests were within limits specified. Test results shall contain written sequence of test procedure with written test results annotated at each step along with the expected outcome or setpoint. The results shall include all readings, including but not limited to data on device (make, model and performance characteristics), normal pressures, switch ranges, trip points, amp readings, and calibration data to include equipment serial numbers or individual identifications, etc.

PART 2 PRODUCTS

2.1 MOTORS

a. For alternating current, fractional and integral horsepower motors, NEMA MG 1 and NEMA MG 2 shall apply.

b. For severe duty TEFC motors, IEEE 841 shall apply.

c. All material and equipment furnished and installation methods shall conform to the requirements of Section 26 29 11, MOTOR CONTROLLERS; and Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES. Provide all electrical wiring, conduit, and devices necessary for the proper connection, protection and operation of the systems. Provide premium efficiency type motors. Unless otherwise specified for a particular application, use electric motors with the following requirements.

d. Single-phase Motors: Motors for centrifugal fans and pumps may be split phase or permanent split capacitor (PSC) type. Provide capacitor-start type for hard starting applications.

e. Poly-phase Motors: NEMA Design B, Squirrel cage, induction type.

Two Speed Motors: Each two-speed motor shall have two separate windings. Provide a time- delay (20 seconds minimum) relay for switching from high to low speed.

f. Voltage ratings shall be as follows:

(1) Single phase:

(a) Motors connected to 120-volt systems: 115 volts.

(b) Motors connected to 208-volt systems: 200 volts.

(c) Motors connected to 240-volt or 480-volt systems: 230/460 volts, dual connection.

(2) Three phase:

(a) Motors connected to 208-volt systems: 200 volts.

(b) Motors, less than (100 hp), connected to 240-volt or 480-volt systems: 208-230/460 volts, dual connection.

(c) Motors, (100 hp) or larger, connected to 240-volt systems: 230 volts.

(d) Motors, (100 hp) or larger, connected to 480-volt systems: 460 volts.

(e) Motors connected to high voltage systems (Over 600V): Shall conform to NEMA MG 1 for connection to the nominal system voltage shown on the drawings.

g. Number of phases shall be as follows:

(1) Motors, less than (1/2 hp): Single phase.

(2) Motors, (1/2 hp) and larger: 3 phase.

(3) Exceptions:

(a) Hermetically sealed motors.

(b) Motors for equipment assemblies, less than (1 hp), may be single phase provided the manufacturer of the proposed assemblies cannot supply the assemblies with three phase motors.

h. Horsepower ratings shall be adequate for operating the connected loads continuously in the prevailing ambient temperatures in areas where the motors are installed, without exceeding the NEMA standard temperature rises for the motor insulation.

i. Motor designs, as indicated by the NEMA code letters, shall be coordinated with the connected loads to assure adequate starting, acceleration, and running torque without exceeding nameplate ratings or considering service factor.

j. Motor Enclosures:

(1) Shall be the NEMA types as specified and/or shown in the Contract Documents.

(2) Where the types of motor enclosures are not shown in the contract documents., they shall be the NEMA types per NEMA 250, which are most suitable for the environmental conditions where the motors are being installed. Enclosure requirements for certain conditions are as follows:

(a) Motors located outdoors, indoors in wet or high humidity locations, or in unfiltered airstreams shall be totally enclosed type.

(b) Where motors are located in an NEC 511 classified area, provide TEFC explosion proof motor enclosures.

(c) Where motors are located in a corrosive environment, provide TEFC enclosures with corrosion resistant finish.

(3) Enclosures shall be primed and finish coated at the factory with manufacturer's prime coat and standard finish.

k. Electrical Design Requirements:

(1) Motors shall be continuous duty.

(2) The insulation system shall be rated minimum of Class B, (266 degrees F).

(3) The maximum temperature rise by resistance at rated power shall not exceed Class B limits, (176 degrees F).

(4) The speed/torque and speed/current characteristics shall comply with NEMA Design A or NEMA Design B, as specified.

(5) Motors shall be suitable for full voltage starting, unless

otherwise noted. Coordinate motor features with applicable motor controllers.

(6) Motors for variable frequency drive applications shall adhere to NEMA MG 1, Part 30, Application Considerations for Constant Speed Motors Used on a Sinusoidal Bus with Harmonic Content and General-Purpose Motors Used with Adjustable-Voltage or Adjustable-Frequency Controls or Both, or NEMA MG 1, Part 31, Definite-Purpose Inverter-Fed Polyphase Motors.

l. Mechanical Design Requirements:

(1) Bearings shall be rated in accordance with ABMA 9 or ABMA 11 for a minimum fatigue life of 26,280 hours for belt-driven loads and 100,000 hours for direct-drive loads based on L10 (Basic Rating Life) at full load direct coupled, except vertical high thrust motors which require a 40,000 hours rating. A minimum fatigue life of 40,000 hours is required for VFD drives.

(2) Vertical motors shall be capable of withstanding a momentary up thrust of at least 30 percent of normal down thrust.

(3) Grease lubricated bearings shall be designed for electric motor use. Grease shall be capable of the temperatures associated with electric motors and shall be compatible with Polyurea based greases.

(4) Grease fittings, if provided, shall be Alemite type or equivalent.

(5) Oil lubricated bearings, when specified, shall have an externally visible sight glass to view oil level.

(6) Vibration shall not exceed0.15 inch per second, unfiltered peak.

(7) Noise level shall meet the requirements of the application.

(8) Motors on 180 frames and greater shall have provisions for lifting eyes or lugs capable of a safety factor of 5.

(9) All external fasteners shall be corrosion resistant.

(10) Condensation heaters, when specified, shall keep motor windings at least (9 degrees F) above ambient temperature.

(11) Winding thermostats, when specified shall be normally closed, connected in series.

(12) Grounding provisions shall be in the main terminal box.

m. Special Requirements:

(1) Where motor power requirements of equipment furnished deviate from power shown on plans, provide electrical service designed under the requirements of NFPA 70 without additional cost or time to the Government.

(2) Assemblies of motors, starters, controls and interlocks on factory assembled and wired devices shall be in accordance with the requirements of this specification.

(3) Wire and cable materials specified in the electrical division of the specifications shall be modified as follows:

(a) Wiring material located where temperatures can exceed (160 degrees F) shall be stranded copper with Teflon FEP insulation with jacket. This includes wiring on the boilers.

(b) Other wiring at boilers and to control panels shall be NFPA 70 designation THWN.

(c) Provide shielded conductors or wiring in separate conduits for all instrumentation and control systems where recommended by manufacturer of equipment.

(4) Select motor sizes so that the motors do not operate into the service factor at maximum required loads on the driven equipment. Motors on pumps shall be sized for non-overloading at all points on the pump performance curves.

(5) Motors utilized with variable frequency drives shall be rated “inverter-duty” per NEMA MG 1, Part 31, Definite-Purpose Inverter-Fed Polyphase Motors. Provide motor shaft grounding apparatus that shall protect bearings from damage from stray currents.

n. Additional requirements for specific motors, as indicated in the other sections listed in Article, RELATED SECTIONS shall also apply.

o. NEMA Premium Efficiency Electric Motors (Motor Efficiencies): All permanently wired polyphase motors of (1 hp) or more shall meet the minimum full-load efficiencies as indicated in the following table. Motors of (1 hp) or more with open, drip-proof, or TEFC enclosures shall be NEMA premium efficiency type, unless otherwise indicated. Motors provided as an integral part of motor driven equipment are excluded from this requirement if a minimum seasonal or overall efficiency requirement is indicated for that equipment by the provisions of another section.

Minimum Premium Efficiencies Open Drip-Proof				**Minimum Premium Efficiencies Totally Enclosed Fan-Cooled (TEFC)**			
Rating kW (hp)	**1200 RPM**	**1800 RPM**	**3600 RPM**	**Rating kW (hp)**	**1200 RPM**	**1800 RPM**	**3600 RPM**
(1)	82.5%	85.5%	77.0%	(1)	82.5%	85.5%	77.0%
(1.5)	86.5%	86.5%	84.0%	(1.5)	87.5%	86.5%	84.0%
(2)	87.5%	86.5%	85.5%	(2)	88.5%	86.5%	85.5%
(3)	88.5%	89.5%	85.5%	(3)	89.5%	89.5%	86.5%
(5)	89.5%	89.5%	86.5%	(5)	89.5%	89.5%	88.5%
(7.5)	90.2%	91.0%	88.5%	(7.5)	91.0%	91.7%	89.5%
(10)	91.7%	91.7%	89.5%	(10)	91.0%	91.7%	90.2%
(15)	91.7%	93.0%	90.2%	(15)	91.7%	92.4%	91.0%
(20)	92.4%	93.0%	91.0%	(20)	91.7%	93.0%	91.0%

(25)	93.0%	93.6%	91.7%	(25)	93.0%	93.6%	91.7%
(30)	93.6%	94.1%	91.7%	(30)	93.0%	93.6%	91.7%
(40)	94.1%	94.1%	92.4%	(40)	94.1%	94.1%	92.4%
(50)	94.1%	94.5%	93.0%	(50)	94.1%	94.5%	93.0%
(60)	94.5%	95.0%	93.6%	(60)	94.5%	95.0%	93.6%
(75)	94.5%	95.0%	93.6%	(75)	94.5%	95.4%	93.6%
(100)	95.0%	95.4%	93.6%	(100)	95.0%	95.4%	94.1%
(125)	95.0%	95.4%	94.1%	(125)	95.0%	95.4%	95.0%
(150)	95.4%	95.8%	94.1%	(150)	95.8%	95.8%	95.0%
(200)	95.4%	95.8%	95.0%	(200)	95.8%	96.2%	95.4%

p. Minimum Power Factor at Full Load and Rated Voltage: 90 percent at 1200 RPM, 1800 RPM, and 3600 RPM. Power factor correction capacitors shall be provided unless the motor meets the 0.90 requirement without it or if the motor is controlled by a variable frequency drive. The power factor correction capacitors shall be able to withstand high voltage transients and power line variations without breakdown.

q. Energy Efficiency of Small Motors (Motor Efficiencies): All motors under (1 hp) shall meet the requirements of the DOE Small Motor Regulation.

Polyphase Open MotorsAverage full load efficiency				**Capacitor-start capacitor-run and capacitor-start induction runopen motors Average full load efficiency**			
Rating (hp)	**6 poles**	**4 poles**	**2 poles**	**Rating kW (hp)**	**6 poles**	**4 poles**	**2 poles**
(0.25)	67.5	69.5	65.6	(0.25)	62.2	68.5	66.6
(0.33)	71.4	73.4	69.5	(0.33)	66.6	72.4	70.5
(0.5)	75.3	78.2	73.4	(0.5)	76.2	76.2	72.4
(0.75)	81.7	81.1	76.8	(0.75)	80.2	81.8	76.2

PART 3 EXECUTION

3.1 INSTALLATION

a. Install motors in accordance with manufacturer’s recommendations, the NEC, NEMA, as shown in the contract documents and/or as required by other sections of these specifications.

b. If in the substantiated evaluation of the Contracting Officer, the installation fails to meet the requirements of the construction documents with respect to function and maintainability, an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

3.2 FIELD TESTS

a. All tests shall be witnessed by the CxA or by the Contracting Officer.

b. Perform an electric insulation resistance Test using a megohmmeter on all motors after installation, before startup. All shall test free

from grounds.

c. Perform Load test in accordance with IEEE 112, Test Method B, to determine freedom from electrical or mechanical defects and compliance with performance data.

d. Insulation Resistance: Not less than one-half meg-ohm between stator conductors and frame, to be determined at the time of final inspection.

e. All test data shall be complied into a report form for each motor and provided to the contracting officer or their representative.

3.3 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA shall observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

3.4 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification shall be tested as part of a greater system.

3.5 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for one hour to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 05 41

NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT

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SECTION 23 05 41
3/23

NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies the application of noise control measures, seismic restraint for equipment and vibration control techniques to boiler plant rotating equipment and parts including chillers, cooling towers, boilers, pumps, fans, compressors, motors and steam turbines.

b. A complete listing of all common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

Noise criteria, seismic restraints for equipment, vibration tolerance and vibration isolation for HVAC and plumbing work.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Not used.

e. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

f. Section 23 31 00, HVAC DUCTS and CASINGS.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE HVAC APP SI HDBK	(2023) HVAC Applications Handbook, SI Edition

ASTM INTERNATIONAL (ASTM)

ASTM A307	(2023) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM D2240	(2015; E 2017) Standard Test Method for

Rubber Property - Durometer Hardness

U.S. DEPARTMENT OF VETERANS AFFAIRS (VA)

VA HVAC Design Manual (2023) VA HVAC Design Manual

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-05 Design Data

Load Calculations For Selection Of Isolators; G, AE

Anchors, Supports And Seismic Restraints; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 XX XX, SECTION TITLE”, with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Vibration isolators:

(a) Floor mountings

(b) Hangers

(c) Snubbers

(d) Thrust restraints

(2) Bases.

(3) Seismic restraint provisions and bolting.

(4) Acoustical enclosures.

d. Isolator manufacturer shall furnish with submittal load calculations for selection of isolators, including supplemental bases, based on

lowest operating speed of equipment supported.

e. Seismic Requirements: Submittals are required for all equipment anchors, supports and seismic restraints. Submittals shall include weights, dimensions, standard connections, and manufacturer's certification that all specified equipment will withstand seismic Lateral Force requirements as shown in the contract documents.

1.6 QUALITY ASSURANCE

a. Refer to article, QUALITY ASSURANCE in specification Section 23 05 11, COMMON WORK RESULTS FOR HVAC and STEAM GENERATION.

b. Noise Criteria:

Perform an acoustic analysis to demonstrate that the specified room noise levels are achieved in all octave bands for all air-handling units, heating and ventilating units, fans, chillers, boilers, generators, and outdoor noise producing equipment, such as cooling towers and chillers. Refer to VA HVAC Design Manual Chapter 6 Room Data Sheets for the required Noise Criteria (NC) levels. If the necessary room type is not listed in Chapter 6 of VA HVAC Design Manual then consult the ASHRAE Handbook - HVAC Applications (. The analysis shall consider both air duct borne noise and noise transmission through walls, floors and roofs and shall be completed for all duct systems and all HVAC equipment.

(2) For equipment which has no sound power ratings scheduled on the plans, the contractor shall select equipment such that the fore-going noise criteria, local ordinance noise levels, and OSHA requirements are not exceeded. Selection procedure shall be in accordance with ASHRAE HVAC APP SI HDBK, Chapter 8, Sound and Vibration.

(3) An allowance, not to exceed 5db, may be added to the measured value to compensate for the variation of the room attenuating effect between room test condition prior to occupancy and design condition after occupancy which may include the addition of sound absorbing material, such as, furniture. This allowance may not be taken after occupancy. The room attenuating effect is defined as the difference between sound power level emitted to room and sound pressure level in room.

(4) In absence of specified measurement requirements, measure equipment noise levels three feet from equipment and at an elevation of maximum noise generation.

c. Seismic Restraint Requirements:

(1) Equipment:

(a) All mechanical equipment not supported with isolators external to the unit shall be securely anchored to the structure. Such mechanical equipment shall be properly supported to resist a horizontal force of 20 percent of the weight of the equipment furnished.

(b) All mechanical equipment mounted on vibration isolators shall be provided with seismic restraints capable of resisting a

horizontal force of 100 percent of the weight of the equipment furnished.

(2) Piping: Refer to specification Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

(3) Ductwork: Refer to specification Section 23 31 00, HVAC DUCTS AND CASINGS.

d. Allowable Vibration Tolerances for Rotating, Non-reciprocating Equipment: Not to exceed a self-excited vibration maximum velocity of per second 0.20 inch per second) RMS, filter in, when measured with a vibration meter on bearing caps of machine in vertical, horizontal, and axial directions or measured at equipment mounting feet if bearings are concealed. Measurements for internally isolated fans and motors may be made at the mounting feet.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

a. Type of isolator, base, and minimum static deflection shall be as required for each specific equipment application as recommended by isolator or equipment manufacturer. Refer to ASHRAE HVAC APP SI HDBK - HVAC Applications Chapter 49, Noise and Vibration Control, Table 47 for selection guide for Vibration Isolation.

b. Elastometric Isolators shall comply with ASTM D2240 and be oil resistant neoprene with a maximum stiffness of 60 durometer and have a straight-line deflection curve.

c. Exposure to weather: Isolator housings to be either hot dipped galvanized or powder coated to ASTM B117 salt spray testing standards. Springs to be powder coated or electro galvanized. All hardware to be electro galvanized. In addition, provide limit stops to resist wind velocity. Velocity pressure established by wind shall be calculated in accordance with Section 1609 of the International Building Code. A minimum wind velocity of 75 mph shall be employed.

d. Uniform Loading: Select and locate isolators to produce uniform loading and deflection even when equipment weight is not evenly distributed.

e. Color code isolators by type and size for easy identification of capacity.

2.2 SEISMIC RESTRAINT REQUIREMENTS FOR EQUIPMENTS

a. Bolt pad mounted equipment, without vibration isolators, to the floor or other support using ASTM A307 standard bolting material.

b. Provide floor mounted equipment with Type SS vibration Isolators.. Where Type N isolators are used provide channel frame base horizontal restraints bolted to the floor, or other support, on all sides of the

equipment, Size and material required for the base shall be as recommended by the isolator manufacturer.

c. On all sides of suspended equipment, provide bracing for rigid supports and provide restraints for resiliently supported equipment.

2.3 VIBRATION ISOLATORS

a. Floor Mountings:

(1) Double Deflection Neoprene (Type N): Shall include neoprene covered steel support plated (top and bottom), friction pads, and necessary bolt holes.

(2) Spring Isolators (Type S): Shall be free-standing, laterally stable and include acoustical friction pads and leveling bolts. Isolators shall have a minimum ratio of spring diameter-to-operating spring height of 1.0 and an additional travel to solid equal to 50 percent of rated deflection.

(3) Captive Spring Mount for Seismic Restraint (Type SS):

(a) Design mounts to resiliently resist seismic forces in all directions. Snubbing shall take place in all modes with adjustment to limit upward, downward, and horizontal travel to a maximum of (1/4-inch) before contacting snubbers. Mountings shall have a minimum rating of one G coefficient of gravity as calculated and certified by a registered structural engineer.

(b) All mountings shall have leveling bolts that are rigidly bolted to the equipment. Spring diameters shall be no less than 0.8 of the compressed height of the spring at rated load. Springs shall have a minimum additional travel to solid equal to 50 percent of the rated deflection. Mountings shall have ports for spring inspection. Provide an all directional neoprene cushion collar around the equipment bolt.

(4) Spring Isolators with Vertical Limit Stops (Type SP): Similar to spring isolators noted above, except include a vertical limit stop to limit upward travel if weight is removed and also to reduce movement and spring extension due to wind loads. Provide clearance around restraining bolts to prevent mechanical short circuiting. Isolators shall have a minimum seismic rating of one G.

(5) Pads (Type D), Washers (Type W), and Bushings (Type L): Pads shall be natural rubber or neoprene waffle, neoprene and steel waffle, or reinforced duck and neoprene. Washers and bushings shall be reinforced duck and neoprene. Size pads for a maximum load of (50 pounds per square inch).

(6) Seismic Pad (Type DS): Pads shall be natural rubber / neoprene waffle with steel top plate and drilled for an anchor bolt. Washers and bushings shall be reinforced duck and neoprene. Size pads for a maximum load of (50 pounds per square inch).

b. Hangers: Shall be combination neoprene and springs unless otherwise noted and shall allow for expansion of pipe.

(1) Combination Neoprene and Spring (Type H): Vibration hanger shall contain a spring and double deflection neoprene element in series. Spring shall have a diameter not less than 0.8 of compressed operating spring height. Spring shall have a minimum additional travel of 50 percent between design height and solid height. Spring shall permit a 15 degree angular misalignment without rubbing on hanger box.

(2) Hanger supports for piping 50 mm (2 inches) and greater shall have a pointer and scale deflection indicator.

c. Snubbers: Each spring mounted base shall have a minimum of four all-directional or eight two directional (two per side) seismic snubbers that are double acting. Elastomeric materials shall be shock absorbent neoprene bridge quality bearing pads, maximum 60 durometer, replaceable and have a minimum thickness of (1/4 inch). Air gap between hard and resilient material shall be not less than 1/8 inch nor greater than (1/4 inch). Restraints shall be capable of withstanding design load without permanent deformation.

d. Thrust Restraints (Type THR): Restraints shall provide a spring element contained in a steel frame with neoprene pads at each end attachment. Restraints shall have factory preset thrust and be field adjustable to allow a maximum movement of 1/4 inch when the fan starts and stops. Restraint assemblies shall include rods, angle brackets and other hardware for field installation.

2.4 BASES

a. Rails (Type R): Design rails with isolator brackets to reduce mounting height of equipment and cradle machines having legs or bases that do not require a complete supplementary base. To assure adequate stiffness, height of members shall be a minimum of 1/12 of longest base dimension but not less than (4 inches). Where rails are used with neoprene mounts for small fans or close coupled pumps, extend rails to compensate overhang of housing.

c. Inertia Base (Type I): Base shall be a reinforced concrete inertia base. Pour concrete into a welded steel channel frame, incorporating prelocated equipment anchor bolts and pipe sleeves. Level the concrete to provide a smooth uniform bearing surface for equipment mounting. Provide grout under uneven supports. Channel depth shall be a minimum of 1/12 of longest dimension of base but not less than (six inches). Form shall include (1/2-inch) reinforcing bars welded in place on minimum of (eight inch) centers running both ways in a layer (1-1/2 inches) above bottom. Use height saving brackets in all mounting locations. Weight of inertia base shall be equal to or greater than weight of equipment supported to provide a maximum peak-to-peak displacement of (1/16 inch).

d. Curb Mounted Isolation Base (Type CB): Fabricate from aluminum to fit on top of standard curb with overlap to allow water run-off and have wind and water seals which shall not interfere with spring action. Provide resilient snubbers with 1/4 inch clearance for wind resistance. Top and bottom bearing surfaces shall have sponge type weather seals. Integral spring isolators shall comply with Spring Isolator (Type S) requirements.

2.5 SOUND ATTENUATING UNITS

Refer to specification Section 23 31 00, HVAC DUCTS and CASINGS.

2.6 ACOUSTICAL ENCLOSURES IN MECHANICAL ROOMS

Provide where shown in the contract documents. Enclosures shall be removable and sectional, of a size and weight that sections can be readily handled with typical lifting and moving equipment available in the equipment room. Enclosures shall contain access openings, observation ports, lights, and ventilation where required for normal operation, observation and servicing.

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the contractor shall correct the installation at no additional cost or time to the Government.

b. Vibration Isolation:

(1) No metal-to-metal contact will be permitted between fixed and floating parts.

(2) Connections to Equipment: Allow for deflections equal to or greater than equipment deflections. Electrical, drain, piping connections, and other items made to rotating or reciprocating equipment (pumps, compressors, etc.) which rests on vibration isolators, shall be isolated from building structure for first three hangers or supports with a deflection equal to that used on the corresponding equipment.

(3) Common Foundation: Mount each electric motor on same foundation as driven machine. Hold driving motor and driven machine in positive rigid alignment with provision for adjusting motor alignment and belt tension. Bases shall be level throughout length and width. Provide shims to facilitate pipe connections, leveling, and bolting.

(4) Provide heat shields where elastomers are subject to temperatures over (l00 degrees F).

(5) Extend bases for pipe elbow supports at discharge and suction connections at pumps. Pipe elbow supports shall not short circuit pump vibration to structure.

(6) Non-rotating equipment such as heat exchangers and convertors shall be mounted on isolation units having the same static deflection as the isolation hangers or support of the pipe connected to the equipment.

c. Inspection and Adjustments: Check for vibration and noise transmission through connections, piping, ductwork, foundations, and walls. Adjust, repair, or replace isolators as required to reduce vibration and noise transmissions to specified levels.

3.2 ADJUSTING

a. Adjust vibration isolators after piping systems are filled and equipment is at operating weight.

b. Adjust limit stops on restrained spring isolators to mount equipment at normal operating height. After equipment installation is complete, adjust limit stops so they are out of contact during normal operation.

c. Attach thrust limits at centerline of thrust and adjust to a maximum of 1/4inch movement during start and stop.

d. Adjust active height of spring isolators.

e. Adjust snubbers according to manufacturer's recommendations.

f. Adjust seismic restraints to permit free movement of equipment within normal mode of operation.

g. Torque anchor bolts according to equipment manufacturer's recommendations to resist seismic forces.

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 05 93.01

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SECTION 23 05 93.01
11/21

DVA/USACE PROJECTS TESTING, ADJUSTING, AND BALANCING FOR HVAC

PART 1 GENERAL

1.1 DESCRIPTION

a. Testing, adjusting, and balancing (TAB) of heating, ventilating and air conditioning (HVAC) systems. TAB includes the following:

(1) Planning systematic TAB procedures.

(2) Design Review Report.

(3) Systems Inspection report.

(4) Duct Air Leakage test report.

(5) Systems Readiness Report.

(6) Balancing air and water distribution systems; adjustment of total system to provide design performance; and testing performance of equipment and automatic controls.

(7) Vibration and sound measurements.

(8) Recording and reporting results.

b. Definitions:

(1) Basic TAB used in this Section: Chapter 39, "Testing, Adjusting and Balancing" of 2019 ASHRAE EQUIP IP HDBK, "HVAC Applications".

(2) TAB: Testing, Adjusting and Balancing; the process of checking and adjusting HVAC systems to meet design objectives.

(3) AABC: Associated Air Balance Council.

(4) NEBB: National Environmental Balancing Bureau.

(5) TABB: Testing Adjusting and Balancing Bureau

(6) SMACNA: Sheet Metal Contractors National Association

(7) Hydronic Systems: Includes chilled water,condenser water, heating hot waterand glycol-water systems.

(8) Air Systems: Includes all outside air, supply air, return air, exhaust air and relief air systems.

(9) Flow rate tolerance: The allowable percentage variation, minus to plus, of actual flow rate from values (design) in the contract documents.

1.2 RELATED WORK

a. UFGS Section 01 33 00 SUBMITTAL PROCEDURES

b. Not used.

c. Section 23 05 11, COMMON WORK RESULTS FOR HVAC

d. Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT

e. Section 23 07 11, HVAC, AND BOILER PLANT INSULATION

f. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS

g. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC

h. Section 23 31 00, HVAC DUCTS AND CASINGS

g. Not used.

i. Section 23 64 00, PACKAGED WATER CHILLERS

j. Section 23 65 00, COOLING TOWERS

1.3 QUALITY ASSURANCE

a. Refer to Articles, Quality Assurance and Submittals, in Section 23 05 11, COMMON WORK RESULTS FOR HVAC, and Section 23 08 00.01, DVA/USACE Projects COMMISSIONING OF HVAC SYSTEMS.

b. Qualifications:

(1) TAB Agency: The TAB agency shall be a subcontractor of the General Contractor and shall report to and be paid by the General Contractor.

(2) The TAB agency shall be either a certified member of AABC, TABB or certified by the NEBB to perform TAB service for HVAC, water balancing and vibrations and sound testing of equipment. The certification shall be maintained for the entire duration of duties specified herein. If, for any reason, the agency loses subject certification during this period, the General Contractor shall immediately notify the Contracting Officer and VA Resident Engineer and submit another qualified TAB firm for approval. Any agency that has been the subject of disciplinary action by either the AABC, TABB or NEBB within the five years preceding Contract Award shall not be eligible to perform any work related to the TAB. All work performed in this Section and in other related Sections by the TAB agency shall be considered invalid if the TAB agency loses its certification prior to Contract completion, and the successor agency’s review shows unsatisfactory work performed by the predecessor agency.

(3) TAB Specialist: The TAB specialist shall be either a member of AABC or TABB or an experienced technician of the Agency certified by NEBB. The certification shall be maintained for the entire duration of duties specified herein. If, for any reason, the Specialist loses subject certification during this period, the

General Contractor shall immediately notify the Contracting Officer and VA Resident Engineer and submit another TAB Specialist for approval. Any individual that has been the subject of disciplinary action by either the AABC or the NEBB within the five years preceding Contract Award shall not be eligible to perform any duties related to the HVAC systems, including TAB. All work specified in this Section and in other related Sections performed by the TAB specialist shall be considered invalid if the TAB Specialist loses its certification prior to Contract completion and must be performed by an approved successor.

(4) TAB Specialist shall be identified by the General Contractor within 60 days after the notice to proceed. The TAB specialist will be coordinating, scheduling and reporting all TAB work and related activities and will provide necessary information as required by the Contracting Officer and VA Resident Engineer. The responsibilities would specifically include:

(a) Shall directly supervise all TAB work.

(b) Shall sign the TAB reports that bear the seal of the TAB standard. The reports shall be accompanied by report forms and schematic drawings required by the TAB standard, AABC MN-1, TABB or NEBB MASV.

(c) Would follow all TAB work through its satisfactory completion.

(d) Shall provide final markings of settings of all HVAC adjustment devices.

(e) Permanently mark location of duct test ports.

(5) All TAB technicians performing actual TAB work shall be experienced and must have done satisfactory work on a minimum of 3 projects comparable in size and complexity to this project. Qualifications must be certified by the TAB agency in writing. The lead technician shall be certified by AABC, TABB or NEBB

c. Test Equipment Criteria: The instrumentation shall meet the accuracy/calibration requirements established by AABC MN-1 National Standards, TABB/SMACNA International Standards, or by NEBB PROCEDURAL STANDARDS for Testing, Adjusting and Balancing of Environmental Systems and instrument manufacturer. Provide calibration history of the instruments to be used for test and balance purpose.

d. TAB Criteria:

(1) One or more of the applicable AABC, NEBB, TABB or SMACNA publications, supplemented by ASHRAE EQUIP IP HDBK "2019 HVAC Applications" Chapter 39, and requirements stated herein shall be the basis for planning, procedures, and reports.

(2) Flow rate tolerance: Following tolerances are allowed. For tolerances not mentioned herein follow ASHRAE EQUIP IP HDBK HVAC Applications", Chapter 39, as a guideline. Air Filter resistance during tests, artificially imposed if necessary, shall be at least 100 percent of manufacturer recommended change over pressure drop values for pre-filters and after-filters.

(a) Air handling unit and all other fans, cubic meters/min (cubic feet per minute): Minus 0 percent to plus l0 percent.

(b) Air terminal units (maximum values): Minus 2 percent to plus l0 percent.

(c) Exhaust hoods/cabinets: 0 percent to plus l0 percent.

(d) Minimum outside air: 0 percent to plus 10 percent.

(e) Individual room air outlets and inlets, and air flow rates not mentioned above: Minus 5 percent to plus l0 percent except if the air to a space is 100 CFM or less the tolerance would be minus 5 to plus 5 percent.

(f) Heating hot water pumps and hot water coils: Minus 5 percent to plus 5 percent.

(g) Chilled water and condenser water pumps: Minus 0 percent to plus 5 percent.

(h) Chilled water coils: Minus 0 percent to plus 5 percent.

(3) Systems shall be adjusted for energy efficient operation as described in PART 3.

(4) Typical TAB procedures and results shall be demonstrated to the Resident Engineer for one air distribution system (including all fans, three terminal units, three rooms randomly selected by the Resident Engineer) and one hydronic system (pumps and three coils) as follows:

(a) When field TAB work begins.

(b) During each partial final inspection and the final inspection for the project if requested by VA.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications Of Tab Agency And Tab Specialists; G, AE

AABC, NEBB Or TABB Publications; G, AE

SD-03 Product Data

Plugs;

Insulation Repair Material; G, AE

SD-05 Design Data

Design Review Report; G, AE

SD-06 Test Reports

Systems Inspection Report; G, AE

Duct Air Leakage Test Report; G, AE

Systems Readiness Report; G, AE

Intermediate And Final Tab Reports; G, AE

Completed Test And Balance Report; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with UFGS Section 01 33 00 SUBMITTAL PROCEDURES.

b. Submit names and qualifications of TAB agency and TAB specialists within 60 days after the notice to proceed. Submit information on three recently completed projects and a list of proposed test equipment.

c. For use by the Resident Engineer staff, submit one complete set of applicable AABC, NEBB or TABB publications that will be the basis of TAB work.

d. Submit Following for Review and Approval:

(1) Design Review Report and within 60 days for design-build projects after the system layout on air and water side is completed by the Contractor.

(2) Systems inspection report on equipment and installation for conformance with design.

(3) Duct Air Leakage Test Report.

(4) Systems Readiness Report.

(5) Intermediate and Final TAB reports covering flow balance and adjustments, performance tests, vibration tests and sound tests.

(6) Include in final reports uncorrected installation deficiencies noted during TAB and applicable explanatory comments on test results that differ from design requirements.

e. Prior to request for Final or Partial Final inspection, submit completed Test and Balance report for the area.

1.6 REFERENCES

a. The following publications form a part of this specification to the extent indicated by the reference thereto. In text the publications are referenced to by the acronym of the organization.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE EQUIP IP HDBK	(2012) Handbook, HVAC Systems and Equipment (IP Edition)
ASHRAE EQUIP SI HDBK	(2012) Handbook, HVAC Systems and Equipment (SI Edition)

ASSOCIATED AIR BALANCE COUNCIL (AABC)

AABC MN-1	(2016, 7th ed) National Standards for Total System Balance

NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB)

NEBB PROCEDURAL STANDARDS	(2015) Procedural Standards for TAB (Testing, Adjusting and Balancing) Environmental Systems
NEBB MASV	(2006) Procedural Standards for Measurements and Assessment of Sound and Vibration
NEBB PROCEDURAL STANDARDS	(2015) Procedural Standards for TAB (Testing, Adjusting and Balancing) Environmental Systems

PART 2 PRODUCTS

2.1 PLUGS

Provide plastic plugs to seal holes drilled in ductwork for test purposes.

2.2 INSULATION REPAIR MATERIAL

See Section 23 07 11, HVAC and BOILER PLANT INSULATION Provide for repair of insulation removed or damaged for TAB work.

PART 3 EXECUTION

3.1 GENERAL

a. Refer to TAB Criteria in Article, Quality Assurance.

b. Obtain applicable contract documents and copies of approved submittals for HVAC equipment and automatic control systems.

3.2 DESIGN REVIEW REPORT

The TAB Specialist shall review the Contract Plans and specifications and advise the Resident Engineer of any design deficiencies that would prevent the HVAC systems from effectively operating in accordance with the sequence of operation specified or prevent the effective and accurate TAB of the system. The TAB Specialist shall provide a report individually listing each deficiency and the corresponding proposed corrective action necessary for proper system operation.

3.3 SYSTEMS INSPECTION REPORT

a. Inspect equipment and installation for conformance with design.

b. The inspection and report is to be done after air distribution equipment is on site and duct installation has begun, but well in advance of performance testing and balancing work. The purpose of the inspection is to identify and report deviations from design and ensure that systems will be ready for TAB at the appropriate time.

c. Reports: Follow check list format developed by AABC, NEBB or SMACNA(TABB), supplemented by narrative comments, with emphasis on air handling units and fans. Check for conformance with submittals. Verify that diffuser and register sizes are correct. Check air terminal unit installation including their duct sizes and routing.

3.4 DUCT AIR LEAKAGE TEST REPORT

TAB Agency shall perform the leakage test as outlined in "Duct leakage Tests and Repairs" in Section 23 31 00, HVAC DUCTS and CASINGS for TAB agency’s role and responsibilities in witnessing, recording and reporting of deficiencies.

3.5 SYSTEM READINESS REPORT

a. Inspect each System to ensure that it is complete including installation and operation of controls. Submit report to Contracting Officer and VA Resident Engineer in standard format and forms prepared and or approved by the CxM and CxA Commissioning Agents.

b. Verify that all items such as ductwork piping, dampers, valves, ports, terminals, connectors, etc., that is required for TAB are installed. Provide a report to the Contracting Officer and VA Resident Engineer.

3.6 TAB REPORTS

a. Submit an intermediate report for 25 percent of systems and equipment tested and balanced to establish satisfactory test results.

b. The TAB contractor shall provide raw data immediately in writing to the Contracting Officer and VA Resident Engineer if there is a problem in achieving intended results before submitting a formal report.

c. If over 20 percent of readings in the intermediate report fall outside the acceptable range, the TAB report shall be considered invalid and all contract TAB work shall be repeated after engineering and construction have been evaluated and re-submitted for approval at no additional cost to the owner.

d. Do not proceed with the remaining systems until intermediate report is approved by the Contracting Officer and VA Resident Engineer.

3.7 TAB PROCEDURES

a. TAB shall be performed in accordance with the requirement of the Standard under which TAB agency is certified by either AABC, TABB or NEBB. Balancing shall be done proportionally to all applicable

systems.

(1) At least one trunk damper shall be 100% open.

(2) At least one branch damper shall be 100%open per trunk.

(3) At least one terminal device duct be 100% open per branch.

b. General: During TAB all related system components shall be in full operation. Fan and pump rotation, motor loads and equipment vibration shall be checked and corrected as necessary before proceeding with TAB. Set controls and/or block off parts of distribution systems to simulate design operation of variable volume air or water systems for test and balance work.

c. Coordinate TAB procedures with existing systems and any phased construction completion requirements for the project. Provide TAB reports for each phase of the project prior to partial final inspections of each phase of the project.

d. Allow 30 days time in construction schedule for TAB and submission of all reports for an organized and timely correction of deficiencies.

e. Air Balance and Equipment Test: Include air handling units, fans, terminal units, fan coil units, room diffusers/outlets/inlets, computer room AC units, and laboratory fume hoods and biological safety cabinets.

(1) Artificially load air filters by partial blanking to produce air pressure drop of manufacturer’s recommended pressure drop.

(2) Adjust fan speeds to provide design air flow. V-belt drives, including fixed pitch pulley requirements, are specified in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

(3) Test and balance systems in all specified modes of operation, including variable volume, economizer, and fire emergency modes. Verify that dampers and other HVAC controls function properly.

(4) Variable air volume (VAV) systems:

(a) Coordinate TAB, including system volumetric controls, with Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

(b) Adjust operating pressure control setpoint to maintain the design flow to each space with the lowest setpoint.

(5) Record final measurements for air handling equipment performance data sheets.

f. Water Balance and Equipment Test: Include circulating pumps, convertors, coils, coolers and condensers:

(1) Coordinate water chiller flow balancing with Section 23 64 00, PACKAGED WATER CHILLERS.

(2) Adjust flow rates for equipment. Set coils and evaporator to

values on equipment submittals, if different from values on contract drawings.

(3) Primary-secondary (variable volume) systems: Coordinate TAB with Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC. Balance systems at design water flow and then verify that variable flow controls function as designed.

(4) Record final measurements for hydronic equipment on performance data sheets. Include entering and leaving water temperatures for heating and cooling coils, and for convertors. Include entering and leaving air temperatures (DB/WB for cooling coils) for air handling units and reheat coils. Make air and water temperature measurements at the same time.

3.8 VIBRATION TESTING

a. Furnish instruments and perform vibration measurements as specified in Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT. Field vibration balancing is specified in Section 23 05 11, COMMON WORK RESULTS FOR HVAC. Provide measurements for all rotating HVAC equipment of (1/2 horsepower) and larger, including centrifugal/screw compressors, cooling towers, pumps, fans and motors.

b. Record initial measurements for each unit of equipment on test forms and submit a report to the Contracting Officer and VA Resident Engineer. Where vibration readings exceed the allowable tolerance Contractor shall be directed to correct the problem. The TAB agency shall verify that the corrections are done and submit a final report to the Contracting Officer and VA Resident Engineer.

3.9 SOUND TESTING

a. Perform and record required sound measurements in accordance with Paragraph, QUALITY ASSURANCE in Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

(1) Take readings in rooms, approximately fifteen (15) percent of all rooms. The Contracting Officer and VA Resident Engineer may designate the specific rooms to be tested.

(2) Provide chiller and cooling tower sound measurements. Refer to Section 23 64 00, PACKAGED WATER CHILLERS and Section 23 65 00, COOLING TOWERS.

b. Take measurements with a calibrated sound level meter and octave band analyzer of the accuracy required by AABC, TABB or NEBB.

c. Sound reference levels, formulas and coefficients shall be according to 2019 ASHRAE EQUIP IP HDBK, "HVAC Applications", Chapter 49, SOUND AND VIBRATION CONTROL.

d. Determine compliance with specifications as follows:

(1) When sound pressure levels are specified, including the NC Criteria in Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT:

(a) Reduce the background noise as much as possible by shutting

off unrelated audible equipment.

(b) Measure octave band sound pressure levels with specified equipment "off."

(c) Measure octave band sound pressure levels with specified equipment "on."

(d) Use the DIFFERENCE in corresponding readings to determine the sound pressure due to equipment.

DIFFERENCE:	0	1	2	3	4	5 to 9	10 or More
FACTOR:	10	7	4	3	2	1	0

Sound pressure level due to equipment equals sound pressure level with equipment "on" minus FACTOR.

(e) Plot octave bands of sound pressure level due to equipment for typical rooms on a graph which also shows noise criteria (NC) curves.

(2) When sound power levels are specified:

(a) Perform steps 1.a. thru 1.d., as above.

(b) For indoor equipment: Determine room attenuating effect, i.e., difference between sound power level and sound pressure level. Determined sound power level will be the sum of sound pressure level due to equipment plus the room attenuating effect.

(c) For outdoor equipment: Use directivity factor and distance from noise source to determine distance factor, i.e., difference between sound power level and sound pressure level. Measured sound power level will be the sum of sound pressure level due to equipment plus the distance factor. Use 30 feet for sound level location.

(3) Where sound pressure levels are specified in terms of dB(A), as in Section 23 65 00, COOLING TOWERS, measure sound levels using the "A" scale of meter. Single value readings will be used instead of octave band analysis.

e. Where measured sound levels exceed specified level, the installing contractor or equipment manufacturer shall take remedial action approved by the Contracting Officer and VA Resident Engineer and the necessary sound tests shall be repeated.

f. Test readings for sound testing could go higher than 15 percent if determination is made by the Contracting Officer and VA Resident Engineer based on the recorded sound data.

3.10 MARKING OF SETTINGS

Following approval of Tab final Report, the setting of all HVAC adjustment devices including valves, splitters and dampers shall be permanently marked by the TAB Specialist so that adjustment can be restored if disturbed at any time. Style and colors used for markings shall be coordinated with the Contracting Officer and VA Resident Engineer.

3.11 IDENTIFICATION OF TEST PORTS

The TAB Specialist shall permanently and legibly identify the location points of duct test ports. If the ductwork has exterior insulation, the identification shall be made on the exterior side of the insulation. All penetrations through ductwork and ductwork insulation shall be sealed to prevent air leaks and maintain integrity of vapor barrier.

3.12 PHASING

a. Phased Projects: Testing and Balancing Work to follow project with areas shall be completed per the project phasing. Upon completion of the project all areas shall have been tested and balanced per the contract documents.

b. Existing Areas: Systems that serve areas outside of the project scope shall not be adversely affected. Measure existing parameters where shown to document system capacity.

3.13 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01 – DVA/USACE Projects COMMISSIONING OF HVAC SYSTEMS for all inspection, start up, and contractor testing required above and required by the System Readiness Checklist provided by the CxM and CxA Commissioning Agents.

b. Components provided under this section of the specification will be tested as part of a larger system. Refer to Section 23 08 00.01 – DVA/USACE Projects COMMISSIONING OF HVAC SYSTEMS and related sections for contractor responsibilities for system commissioning.

-- End of Section --

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SECTION 23 07 11
2/20

HVAC AND BOILER PLANT INSULATION

PART 1 GENERAL

1.1 DESCRIPTION

a. Field applied insulation for thermal efficiency and condensation control for

(1) HVAC piping, ductwork and equipment.

(2) Re-insulation of HVAC piping, ductwork and equipment, and boiler plant piping, breeching and stacks and equipment after asbestos abatement.

b. Definitions

(1) ASJ: All service jacket, white finish facing or jacket.

(2) Air conditioned space: Space having air temperature and/or humidity controlled by mechanical equipment.

(3) Cold: Equipment, ductwork or piping handling media at design temperature of (60 degrees F) or below.

(4) Concealed: Ductwork and piping above ceilings and in chases, interstitial space, and pipe spaces.

(5) Exposed: Piping, ductwork, and equipment exposed to view in finished areas including mechanical and electrical equipment rooms or exposed to outdoor weather. Attics and crawl spaces where air handling units are located are considered to be mechanical rooms. Shafts, chases, interstitial spaces, unfinished attics, crawl spaces and pipe basements are not considered finished areas.

(6) FSK: Foil-scrim-kraft facing.

(7) Hot: HVAC Ductwork handling air at design temperature above (60 degrees F);HVAC equipment or piping handling media above (105 degrees F).

(8) Density: (Pcf - pounds per cubic foot).

(9) Runouts: Branch pipe connections up to (one-inch) nominal size to fan coil units or reheat coils for terminal units.

(10) Thermal conductance: Heat flow rate through materials.

(a) Flat surface: (BTU per hour per square foot).

(b) Pipe or Cylinder: (BTU per hour per linear foot).

(11) Thermal Conductivity (k): (BTU per inch thickness, per hour, per square foot, per degree F temperature difference).

(12) Vapor Retarder (Vapor Barrier): A material which retards the transmission (migration) of water vapor. Performance of the vapor retarder is rated in terms of permeance (perms). For the purpose of this specification, vapor retarders shall have a maximum published permeance of 0.1 perms and vapor barriers shall have a maximum published permeance of 0.001 perms.

(13) HPS: High pressure steam (60 psig and above).

(14) HPR: High pressure steam condensate return.

(15) MPS: Medium pressure steam (thru 59 psig.

(16) MPR: Medium pressure steam condensate return.

(17) LPS: Low pressure steam (15 psig and below).

(18) LPR: Low pressure steam condensate gravity return.

(19) PC: Pumped condensate.

(20) HWH: Hot water heating supply.

(21) HWHR: Hot water heating return.

(22) GH: Hot glycol-water heating supply.

(23) GHR: Hot glycol-water heating return.

(24) FWPD: Feedwater pump discharge.

(25) FWPS: Feedwater pump suction.

(26) CTPD: Condensate transfer pump discharge.

(27) CTPS: Condensate transfer pump suction.

(28) VR: Vacuum condensate return.

(29) CPD: Condensate pump discharge.

(30) R: Pump recirculation.

(31) FOS: Fuel oil supply.

(32) FOR: Fuel oil return.

(33) CW: Cold water.

(34) SW: Soft water.

(35) HW: Hot water.

(36) CH: Chilled water supply.

(37) CHR: Chilled water return.

(38) GC: Chilled glycol-water supply.

(39) GCR: Chilled glycol-water return.

(40) RS: Refrigerant suction.

(41) PVDC: Polyvinylidene chloride vapor retarder jacketing, white.

1.2 RELATED WORK

A Section 01 33 00, SUBMITTAL PROCEDURES.

b. Not used.

c. Not used.

d. Section 07 84 00, FIRESTOPPING.

e. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

f. Section 23 08 00, COMMISSIONING OF HVAC SYSTEMS.

g. Section 23 21 13, HYDRONIC PIPING.

h. Section 23 22 13, STEAM and CONDENSATE HEATING PIPING

i. Section 23 22 23, STEAM CONDENSATE PUMPS

j. Section 23 50 11, BOILER PLANT MECHANICAL EQUIPMENT

1.3 QUALITY ASSURANCE

a. Refer to article QUALITY ASSURANCE, in Section 23 05 11, COMMON WORK RESULTS FOR HVAC .

b. Criteria:

(1) Comply with NFPA 90A, particularly paragraphs 4.3.3.1 through 4.3.3.6, 4.3.10.2.6, and 5.4.6.4, parts of which are quoted as follows:

4.3.3.1 Pipe insulation and coverings, duct coverings, duct linings, vapor retarder facings, adhesives, fasteners, tapes, and supplementary materials added to air ducts, plenums, panels, and duct silencers used in duct systems, unless otherwise provided for in 4.3.3.1.1 or 4.3.3.1.2., shall have, in the form in which they are used, a maximum flame spread index of 25 without evidence of continued progressive combustion and a maximum smoke developed index of 50 when tested in accordance with NFPA 255, {HL2}Standard Method of Test of Surface Burning Characteristics of Building Materials{/HL2}.
4.3.3.1.1 Where these products are to be applied with adhesives, they shall be tested with such adhesives applied, or the adhesives used shall have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when in the final dry state. *(See* 4.2.4.2*.)*

4.3.3.1.2 The flame spread and smoke developed index requirements of 4.3.3.1.1 shall not apply to air duct weatherproof coverings where they are located entirely outside f a building, do not penetrate a wall or roof, and do not create an exposure hazard.
4.3.3.2 Closure systems for use with rigid and flexible air ducts tested in accordance with UL 181, Standard for Safety Factory-Made Air Ducts and Air Connectors, shall have been tested, listed, and used in accordance with the conditions of their listings, in accordance with one of the following:
(1) UL 181A, Standard for Safety Closure Systems for Use with Rigid Air Ducts and Air Connectors
(2) UL 181B, Standard for Safety Closure Systems for Use with Flexible Air Ducts and Air Connectors
4.3.3.3 Air duct, panel, and plenum coverings and linings, and pipe insulation and coverings shall not flame, glow, smolder, or smoke when tested in accordance with a similar test for pipe covering, ASTM C411, Standard Test Method for Hot-Surface Performance of High-Temperature Thermal Insulation, at the temp rature to which they are exposed in service.
4.3.3.3.1 In no case shall the test temperature be below (250°F).
4.3.3.4 Air duct coverings shall not extend through walls or floors that are required to be fire stopped or required to have a fire resistance rating, unless such coverings meet the requirements of 5.4.6.4.
4.3.3.5* Air duct linings shall be interrupted at fire dampers to prevent interference with the operation of devices.
4.3.3.6 Air duct coverings shall not be installed so as to conceal or prevent the use of any service opening.
4.3.10.2.6 Materials exposed to the airflow shall be noncombustible or limited combustible and have a maximum smoke developed index of 50 or comply with the following.
4.3.10.2.6.1 Electrical wires and cables and optical fiber cables shall be listed as noncombustible or limited combustible and hav e a maximum smoke developed index of 50 or shall be listed as having a maximum peak optical density of 0.5 or less, an average optical density of 0.15 or less, and a maximum flame spread distance of (5 ft) or less when tested in accordance with NFPA 262, Standard Method of Test for Flame Travel and Smoke of Wires and Cables for Use in Air-Handling Spaces.

4.3.10.2.6.4 Optical-fiber and communication raceways shall be listed as having a maximum peak optical density of 0.5 or less, an average optical density of 0.15 or less, and a maximum flame spread distance of (5 ft) or less when tested in accordance with UL 2024, Standard for Safety Optical-Fiber Cable Raceway.
4.3.10.2.6.6 Supplementary materials for air distribution systems shall be permitted when complying with the provisions of 4.3.3.
5.4.6.4 Where air ducts pass through walls, floors, or partitions that are required to have a fire resistance rating and where fir e dampers are not required, the opening in the construction around the air duct shall be as follows:
(1) Not exceeding a (1 in.) average clearance on all sides
(2) Filled solid with an approved material capable of preventing the passage of flame and hot gases sufficient to ignite cotton waste when subjected to the time-temperature fire conditions required for fire barrier penetration as specified

in NFPA 251, {HL2}Standard Methods of Tests of Fire Endurance of Building Construction and Materials{/HL2}

(2) Test methods: ASTM E84, UL 723, or NFPA 255.

(3) Specified k factors are at (75 degrees F) mean temperature unless stated otherwise. Where optional thermal insulation material is used, select thickness to provide thermal conductance no greater than that for the specified material. For pipe, use insulation manufacturer's published heat flow tables. For domestic hot water supply and return, run out insulation and condensation control insulation, no thickness adjustment need be made.

(4) All materials shall be compatible and suitable for service temperature, and shall not contribute to corrosion or otherwise attack surface to which applied in either the wet or dry state.

c. Every package or standard container of insulation or accessories delivered to the job site for use must have a manufacturer's stamp or label giving the name of the manufacturer and description of the material.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Insulation Materials; G, AE

Insulation Facings And Jackets; G, AE

Insulation Accessory Materials; G, AE

SD-04 Samples

Samples; G, AE

SD-08 Manufacturer's Instructions

Manufacturer's Installation And Fitting Fabrication Instructions; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00, SUBMITTAL PROCEDURES..

b. Shop Drawings:

(1) All information, clearly presented, shall be included to determine compliance with drawings and specifications and ASTM, federal and military specifications.

(a) Insulation materials: Specify each type used and state surface burning characteristics.

(b) Insulation facings and jackets: Each type used. Make it clear that white finish will be furnished for exposed ductwork, casings and equipment.

(c) Insulation accessory materials: Each type used.

(d) Manufacturer's installation and fitting fabrication instructions for flexible unicellular insulation.

(e) Make reference to applicable specification paragraph numbers for coordination.

c. Samples:

(1) Each type of insulation: Minimum size 4 inches square for board/block/ blanket; 6 inches long, full diameter for round types.

(2) Each type of facing and jacket: Minimum size 4 inches square).

(3) Each accessory material: Minimum (4 ounce) liquid container or (4 ounce) dry weight for adhesives / cement / mastic.

1.6 STORAGE AND HANDLING OF MATERIAL

Store materials in clean and dry environment, pipe covering jackets shall be clean and unmarred. Place adhesives in original containers. Maintain ambient temperatures and conditions as required by printed instructions of manufacturers of adhesives, mastics and finishing cements.

1.7 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by basic designation only.

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-58	(2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation

U.S. DEPARTMENT OF DEFENSE (DOD)

FS L-P-535E	(2012; A 2022) Plastic Sheet (Sheeting): Plastic Strip; Poly (Vinyl Chloride) and Poly (Vinyl Chloride-Vinyl Acetate), Rigid

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-A-3316C	(1987) Adhesives, Fire-Resistant, Thermal Insulation
MIL-A-24179A	(2016) Adhesive, Flexible Unicellular-Plastic

MIL-C-19565C	(2016) Coating Compounds, Thermal Insulation, Fire-and Water-Resistant, Vapor-Barrier
MIL-C-20079H	(1987) Cloth, Glass; Tape, Textile Glass; and Thread, Glass and Wire-Reinforced Glass

ASTM INTERNATIONAL (ASTM)

ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM C177	(2019; E 2023) Standard Test Method for Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded-Hot-Plate Apparatus
ASTM C411	(2019) Standard Test Method for Hot-Surface Performance of High-Temperature Thermal Insulation
ASTM C449	(2007; R 2024) Standard Specification for Mineral Fiber Hydraulic-Setting Thermal Insulating and Finishing Cement
ASTM C518	(2021) Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus
ASTM C533	(2017; R 2023) Standard Specification for Calcium Silicate Block and Pipe Thermal Insulation
ASTM C547	(2022a) Standard Specification for Mineral Fiber Pipe Insulation
ASTM C553	(2024) Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications
ASTM C591	(2022) Standard Specification for Unfaced Preformed Rigid Cellular Polyisocyanurate Thermal Insulation
ASTM C612	(2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation
ASTM C1126	(2024) Standard Specification for Faced or Unfaced Rigid Cellular Phenolic Thermal Insulation
ASTM C1136	(2023) Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation
ASTM D1668/D1668M	(1997a; R 2014; E 2014) Glass Fabrics

	(Woven and Treated) for Roofing and Waterproofing
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A	(2024) Standard for the Installation of Air Conditioning and Ventilating Systems
NFPA 251	(2006) Standard Methods of Tests of Fire Resistance of Building Construction and Materials
NFPA 255	(2006; Errata 2006) Standard Method of Test of Surface Burning Characteristics of Building Materials
NFPA 262	(2023) Standard Method of Test for Flame Travel and Smoke of Wires and Cables for Use in Air-Handling Spaces

UL SOLUTIONS (UL)

UL 181	(2013; Reprint Dec 2021) UL Standard for Safety Factory-Made Air Ducts and Air Connectors
UL 181A	(2013; Reprint Dec 2021) Standard for Safety Closure Systems for Use with Rigid Air Ducts
UL 181B	(2013; Reprint Dec 2021) UL Standard for Safety Closure Systems for Use with Flexible Air Ducts and Air Connectors
UL 723	(2023) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-58	(2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation

PART 2 PRODUCTS

2.1 MINERAL FIBER OR FIBER GLASS

a. ASTM C612 (Board, Block), Class 1 or 2, density (3 pcf), (75 degrees F), external insulation for temperatures up to (400 degrees F) with foil scrim (FSK) facing.

b. ASTM C553 (Blanket, Flexible) Type I, Class B-3, Density (1 pcf), (0.31) at (75 degrees F), for use at temperatures up to (400 degrees

F) with foil scrim (FSK) facing.

c. ASTM C547 (Pipe Fitting Insulation and Preformed Pipe Insulation), Class 1, (0.26) at (75 degrees F), for use at temperatures up to (450 degrees F) with an all service vapor retarder jacket with polyvinyl chloride premolded fitting covering.

2.2 MINERAL WOOL OR REFRACTORY FIBER

Comply with Standard ASTM C612, Class 3, (850 degrees F).

2.3 RIGID CELLULAR PHENOLIC FOAM

a. Preformed (molded) pipe insulation, ASTM C1126, type III, grade 1, (0.15) at (50 degrees F), for use at temperatures up to (250 degrees F) with all service vapor retarder jacket with polyvinyl chloride premolded fitting covering.

b. Equipment and Duct Insulation, ASTM C1126, type II, grade 1, (0.15) at (50 degrees F), for use at temperatures up to (250 degrees F) with rigid cellular phenolic insulation and covering, and all service vapor retarder jacket.

2.4 CELLULAR GLASS CLOSED-CELL

a. Comply with Standard ASTM C177, ASTM C518, density (7.5 pcf) nominal, (0.29) at (75 degrees F).

b. Pipe insulation for use at temperatures up to (400 degrees F) with all service vapor retarder jacket.

2.5 POLYISOCYANURATE CLOSED-CELL RIGID

a. Preformed (fabricated) pipe insulation, ASTM C591, type IV, (0.19) at (75 degrees F), flame spread not over 25, smoke developed not over 50, for use at temperatures up to 149 degree C (300 degree F) with factory applied PVDC or all service vapor retarder jacket with polyvinyl chloride premolded fitting covers.

b. Equipment and duct insulation, ASTM C591, type IV, (0.19) at (75 degrees F), for use at temperatures up to (300 degrees F) with PVDC or all service jacket vapor retarder jacket.

2.6 FLEXIBLE ELASTOMERIC CELLULAR THERMAL

a. Pipe insulation, ASTM C177, ASTM C518, (0.27) at (75 degrees F), flame spread not over 25, smoke developed not over 50, for temperatures from minus (40 degrees F) to (200 degrees F). No jacket required.

2.7 CALCIUM SILICATE

a. Preformed pipe Insulation: ASTM C533, Type I and Type II with indicator denoting asbestos-free material.

b. Premolded Pipe Fitting Insulation: ASTM C533, Type I and Type II with indicator denoting asbestos-free material.

c. Equipment Insulation: ASTM C533, Type I and Type II

d. Characteristics:

Insulation Characteristics		
ITEMS	**TYPE I**	**TYPE II**
Temperature, maximum (degrees F)	(1200)	(1700)
Density (dry), (lb/ ft3)	(14.5)	(18)
Thermal conductivity: Min (Btu in/h ft2 degrees F)@ mean temperature of (200 degrees F)	(0.41)	(0.540)
Surface burning characteristics: Flame spread Index, Maximum	0	0
Smoke Density index, Maximum	0	0

2.8 INSULATION FACINGS AND JACKETS

a. Vapor Retarder, higher strength with low water permeance = 0.02 or less perm rating, Beach puncture 50 units for insulation facing on exposed ductwork, casings and equipment, and for pipe insulation jackets. Facings and jackets shall be all service type (ASJ) or PVDC Vapor Retarder jacketing.

b. ASJ jacket shall be white kraft bonded to (1 mil) thick aluminum foil, fiberglass reinforced, with pressure sensitive adhesive closure. Comply with ASTM C1136. Beach puncture 50 units, Suitable for painting without sizing. Jackets shall have minimum (1-1/2 inch) lap on longitudinal joints and minimum 3 inch butt strip on end joints. Butt strip material shall be same as the jacket. Lap and butt strips shall be self-sealing type with factory-applied pressure sensitive adhesive.

c. Vapor Retarder medium strength with low water vapor permeance of 0.02 or less perm rating), Beach puncture 25 units: Foil-Scrim-Kraft (FSK) or PVDC vapor retarder jacketing type for concealed ductwork and equipment.

d. Field applied vapor barrier jackets shall be provided, in addition to the specified facings and jackets, on all exterior piping and ductwork as well as on interior piping and ductwork conveying fluids below ambient temperature. The vapor barrier jacket shall consist of a multi-layer laminated cladding with a maximum water vapor permeance of 0.001 perms. The minimum puncture resistance shall be (30 inch-pounds) for interior locations and (80 inch-pounds) for exterior or exposed locations or where the insulation is subject to damage.

e. Glass Cloth Jackets: Presized, minimum (7.8 ounces per square yard), 300 psig bursting strength with integral vapor retarder where required or specified. Weather proof if utilized for outside service.

f. Factory composite materials may be used provided that they have been tested and certified by the manufacturer.

g. Pipe fitting insulation covering (jackets): Fitting covering shall be

premolded to match shape of fitting and shall be polyvinyl chloride (PVC) conforming to FS L-P-535E, composition A, Type II Grade GU, and Type III, minimum thickness (0.03 inches). Provide color matching vapor retarder pressure sensitive tape.

h. Aluminum Jacket-Piping systems: ASTM B209, 3003 alloy, H-14 temper, 0.023 inch minimum thickness with locking longitudinal joints. Jackets for elbows, tees and other fittings shall be factory-fabricated to match shape of fitting and of (0.024) inch minimum thickness aluminum. Fittings shall be of same construction as straight run jackets but need not be of the same alloy. Factory-fabricated stainless steel bands shall be installed on all circumferential joints. Bands shall be 0.5 inch wide on 18 inch centers. System shall be weatherproof if utilized for outside service.

i. Aluminum jacket-Rectangular breeching: ASTM B209, 3003 alloy, H-14 temper, 0.020 inches thick with (1-1/4 inch) corrugations or 0.032 inches thick with no corrugations. System shall be weatherproof if used for outside service.

2.9 REMOVABLE INSULATION JACKETS

a. Insulation and Jacket:

(1) Non-Asbestos Glass mat, type E needled fiber.

(2) Temperature maximum of 450°F, Maximum water vapor transmission of 0.00 perm, and maximum moisture absorption of 0.2 percent by volume.

(3) Jacket Material: Silicon/fiberglass and LFP 2109 pure PTFE.

(4) Construction: One piece jacket body with three-ply braided pure Teflon or Kevlar thread and insulation sewn as part of jacket. Belt fastened.

2.10 PIPE COVERING PROTECTION SADDLES

a. Cold pipe support: Premolded pipe insulation 180 degrees (half-shells) on bottom half of pipe at supports. Material shall be cellular glass or high density Polyisocyanurate insulation of the same thickness as adjacent insulation. Density of Polyisocyanurate insulation shall be a minimum of (3.0 pcf).

Nominal Pipe Size and Accessories Material (Insert Blocks)	
Nominal Pipe Size (inches)	Insert Blocks mm (inches)
Up through (5)	(6) long
(6)	(6) long
(8), (10), (12)	(9) long
(14), (16)	(12) long
(18)through 24)	(14) long

b. Warm or hot pipe supports: Premolded pipe insulation (180 degree half-shells) on bottom half of pipe at supports. Material shall be high density Polyisocyanurate (for temperatures up to 300 degrees F), cellular glass or calcium silicate. Insulation at supports shall have same thickness as adjacent insulation. Density of Polyisocyanurate

insulation shall be a minimum of (3.0 pcf).

c. Boiler Plant Pipe supports: MSS SP-58, Type 39. Apply at all pipe support points, except where MSS SP-58, Type 3 pipe clamps provided as part of the support system.

2.11 ADHESIVE, MASTIC, CEMENT

a. MIL-A-3316C, Class 1: Jacket and lap adhesive and protective finish coating for insulation.

b. MIL-A-3316C, Class 2: Adhesive for laps and for adhering insulation to metal surfaces.

c. MIL-A-24179A, Type II Class 1: Adhesive for installing flexible unicellular insulation and for laps and general use.

d. MIL-C-19565C, Type I: Protective finish for outdoor use.

e. MIL-C-19565C, Type I or Type II: Vapor barrier compound for indoor use.

f. ASTM C449: Mineral fiber hydraulic-setting thermal insulating and finishing cement.

g. Other: Insulation manufacturers' published recommendations.

2.12 MECHANICAL FASTENERS

a. Pins, anchors: Welded pins, or metal or nylon anchors with galvanized steel-coated or fiber washer, or clips. Pin diameter shall be as recommended by the insulation manufacturer.

b. Staples: Outward clinching galvanized steel.

c. Wire: thick (18 gage) soft annealed galvanized or (14 gage) copper clad steel or nickel copper alloy.

d. Bands: 0.5 inch nominal width, brass, galvanized steel, aluminum or stainless steel.

2.13 REINFORCEMENT AND FINISHES

a. Glass fabric, open weave: ASTM D1668/D1668M, Type III (resin treated) and Type I (asphalt treated).

b. Glass fiber fitting tape: MIL-C-20079H, Type II, Class 1.

c. Tape for Flexible Elastomeric Cellular Insulation: As recommended by the insulation manufacturer.

d. Hexagonal wire netting: (one inch) mesh, thick (22 gage) galvanized steel.

e. Corner beads: 2 inch by (2 inch), thick (26 gage) galvanized steel; or, 1 inch by (1 inch), thick (28 gage) aluminum angle adhered to 2 inch by 2 inch Kraft paper.

f. PVC fitting cover: FS L-P-535E, Composition A, 11-86 Type II, Grade

GU, with Form B Mineral Fiber insert, for media temperature (40 degrees F) to (250 degrees F). Below (40 degrees F) and above (250 degrees F). Provide double layer insert. Provide color matching vapor barrier pressure sensitive tape.

2.14 FIRESTOPPING MATERIAL

Other than pipe and duct insulation, refer to Section 07 84 00 FIRESTOPPING.

2.15 FLAME AND SMOKE

Unless shown otherwise all assembled systems shall meet flame spread 25 and smoke developed 50 rating as developed under ASTM, NFPA and UL standards and specifications. See paragraph 1.3 "Quality Assurance".

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

a. Required pressure tests of duct and piping joints and connections shall be completed and the work approved by the Resident Engineer for application of insulation. Surface shall be clean and dry with all foreign materials, such as dirt, oil, loose scale and rust removed.

b. Except for specific exceptions, insulate entire specified equipment, piping (pipe, fittings, valves, accessories), and duct systems. Insulate each pipe and duct individually. Do not use scrap pieces of insulation where a full length section will fit.

c.

d. Insulation materials shall be installed in a first class manner with smooth and even surfaces, with jackets and facings drawn tight and smoothly cemented down at all laps. Insulation shall be continuous through all sleeves and openings, except at fire dampers and duct heaters (NFPA 90A). Vapor retarders shall be continuous and uninterrupted throughout systems with operating temperature (60 degrees F) and below. Lap and seal vapor retarder over ends and exposed edges of insulation. Anchors, supports and other metal projections through insulation on cold surfaces shall be insulated and vapor sealed for a minimum length of (6 inches).

e. Install vapor stops at all insulation terminations on either side of valves, pumps and equipment and particularly in straight lengths of pipe insulation.

f. Construct insulation on parts of equipment such as chilled water pumps and heads of chillers, convertors and heat exchangers that must be opened periodically for maintenance or repair, so insulation can be removed and replaced without damage. Install insulation with bolted thick (20 gage) galvanized steel or aluminum covers as complete units, or in sections, with all necessary supports, and split to coincide with flange/split of the equipment.

g. Insulation on hot piping and equipment shall be terminated square at items not to be insulated, access openings and nameplates. Cover all exposed raw insulation with white sealer or jacket material.

h. Protect all insulations outside of buildings with aluminum jacket using lock joint or other approved system for a continuous weather tight system. Access doors and other items requiring maintenance or access shall be removable and sealable.

i. Insulate PRVs, flow meters, and steam traps.

i. HVAC work not to be insulated:

(1) Internally insulated ductwork and air handling units.

(2) Relief air ducts (Economizer cycle exhaust air).

(3) Exhaust air ducts and plenums, and ventilation exhaust air shafts.

(4) Equipment: Expansion tanks, flash tanks, hot water pumps, steam condensate pumps.

(5) In hot piping: Unions, flexible connectors, control valves, PRVs, safety valves and discharge vent piping, vacuum breakers, thermostatic vent valves, steam traps 3/4 inch and smaller, exposed piping through floor for convectors and radiators. Insulate piping to within approximately 3 inches of uninsulated items.

j. Boiler plant work not to be insulated(NI) or if insulated the insulation shall be removal jacket type (RJ):

(1) Pipes, valves and fittings:

(a) Gas fuel(NI)

(b) Oil unheated (NI)

(c) Compressed Air (NI)

(d) Flowmeter sensing piping and blowdown (NI)

(e) Level sensor piping and blowdown (NI)

(f) Tank drains (NI)

(g) Vents-tank, safety and back pressure valves except protective. (NI)

(h) Continuous blowdown and boiler water sampling except protective. (NI)

(i) Threaded valves (RJ)

(j) Check valves (RJ)

(k) Unions (RJ)

(l) Orifice flanges (RJ)

(m) Dielectric flanges and unions (RJ)

(n) Steam header drains (NI)

(o) Non-return stop and check valve drains (NI)

(p) Pneumatic controls (NI)

(q) Pressure transmission to gages (NI)

(r) Piping in control panels (NI)

(s) Tube cleaning piping (NI)

(t) Chemical feed from pump-type feeders (NI)

(u) Condensate piping from flash tank to condensate return pump (NI)

(2) Boilers:

(a) Water column, piping and blowdown (NI)

(b) Auxiliary low water cutoff, piping and blowdown(NI)

(c) Remote water level indicators and piping blowdown(NI)

(d) Steam gage piping(NI)

(e) Soot blower and piping(NI)

(f) Safety valves and drip pan ells(NI)

(g) Water level sensors and piping except where required by equipment manufacturer(NI)

(h) Control piping and devices or interlocks(NI)

(i) Drum heads (watertube boilers) (NI)

(3) Equipment:

(a) Condensate return pump units(NI)

(b) Vacuum return pump units(NI)

(c) Pumps-inlet to outlet(NI)

(d) Flash tanks(NI)

(e) Safety valves(NI)

(f) Water meters(NI)

(g) Oil meters(NI)

(h) Air compressors and tanks(NI)

(i) Refrigerated or desiccant air drier(NI)

(j) Chemical feeders(NI)

(k) Boiler and feedwater sampler(NI)

(l) All nameplates (NI)

(4) Specialties:

(a) Pressure reducing valves(RJ)

(b) Control valves-water and steam(NI)

(c) Level sensors-piping, valves and blowdown(NI)

(d) Back pressure regulators-oil and steam(NI)

(e) Strainers under (2-1/2 inch) pipe size(RJ)

(f) Expansion bellows(RJ)

(g) Flexible connectors(RJ)

(h) Ball joints except piping between joints(NI)

k. Apply insulation materials subject to the manufacturer's recommended temperature limits. Apply adhesives, mastic and coatings at the manufacturer's recommended minimum coverage.

l. Elbows, flanges and other fittings shall be insulated with the same material as is used on the pipe straights. The elbow/ fitting insulation shall be field-fabricated, mitered or factory prefabricated to the necessary size and shape to fit on the elbow/ fitting. Use of polyurethane spray-foam to fill a PVC elbow jacket is prohibited on cold applications.

m. Firestop Pipe and Duct insulation:

(1) Provide firestopping insulation at fire and smoke barriers through penetrations. Fire stopping insulation shall be UL listed as defines in Section 07 84 00, FIRESTOPPING.

(2) Pipe and duct penetrations requiring fire stop insulation including, but not limited to the following:

(a) Pipe risers through floors

(b) Pipe or duct chase walls and floors

(c) Smoke partitions

(d) Fire partitions

n. Freeze protection of above grade outdoor piping (over heat tracing tape): 10 inch thick insulation, for all pipe sizes (3 inches) and smaller and (1inch) thick insulation for larger pipes. Provide metal jackets for all pipes. Provide for cold water make-up to cooling towers and condenser water piping and chilled water piping as described in Section 23 21 13, HYDRONIC PIPING (electrical heat tracing systems).

o. Provide vapor barrier jackets over insulation as follows:

(1) All piping and ductwork exposed to outdoor weather.

(2) All interior piping and ducts conveying fluids below ambient air temperature .

p. Provide metal jackets over insulation as follows:

(1) All piping and ducts exposed to outdoor weather.

(2) Piping exposed in building, within 6 feet of the floor, that connects to sterilizers, kitchen and laundry equipment. Jackets may be applied with pop rivets. Provide aluminum angle ring escutcheons at wall, ceiling or floor penetrations.

(3) A 2 inch overlap is required at longitudinal and circumferential joints.

3.2 INSULATION INSTALLATION

a. Mineral Fiber Board:

(1) Faced board: Apply board on pins spaced not more than 12 inches on center each way, and not less than 3 inches from each edge of board. In addition to pins, apply insulation bonding adhesive to entire underside of horizontal metal surfaces. Butt insulation edges tightly and seal all joints with laps and butt strips. After applying speed clips cut pins off flush and apply vapor seal patches over clips.

(2) Plain board:

(a) Insulation shall be scored, beveled or mitered to provide tight joints and be secured to equipment with bands spaced 9 inches on center for irregular surfaces or with pins and clips on flat surfaces. Use corner beads to protect edges of insulation.

(b) For hot equipment: Stretch 1 inch mesh wire, with edges wire laced together, over insulation and finish with insulating and finishing cement applied in one coat, 1/4 inch thick, trowel led to a smooth finish.

(c) For cold equipment: Apply meshed glass fabric in a tack coat (60 to 70 square feet per gallon) of vapor mastic and finish with mastic at (12 to 15 square feet per gallon) over the entire fabric surface.

(d) Chilled water pumps: Insulate with removable and replaceable thick (20 gage) aluminum or galvanized steel covers lined with insulation. Seal closure joints/flanges of covers with gasket material. Fill void space in enclosure with flexible mineral fiber insulation.

(3) Exposed, unlined ductwork and equipment in unfinished areas, mechanical and electrical equipment rooms and attics, and duct work exposed to outdoor weather:

(a) (2 inch) thick insulation faced with ASJ (white all service jacket): Supply air duct and after filter housing.

(b) (2 inch) thick insulation faced with ASJ: Return air duct, mixed air plenums and prefilter housing.

(c) Outside air intake ducts: (one inch) thick insulation faced with ASJ.

(d) Exposed, unlined supply and return ductwork exposed to outdoor weather: 2 inch thick insulation faced with a reinforcing membrane and two coats of vapor barrier mastic or multi-layer vapor barrier with a maximum water vapor permeability of 0.001 perms.

(4) Supply air duct in the warehouse and in the laundry: (one inch) thick insulation faced with ASJ.

(5) Cold equipment: (1-1/2inch) thick insulation faced with ASJ.

(a) Chilled water pumps, water filter, chemical feeder pot or tank.

(b) Pneumatic, cold storage water and surge tanks.

(6) Hot equipment: (1-1/2 inch) thick insulation faced with ASJ.

(a) Convertors, air separators, steam condensate pump receivers.

(b) Reheat coil casing and separation chambers on steam humidifiers located above ceilings.

(c) Domestic water heaters and hot water storage tanks (not factory insulated).

b. Flexible Mineral Fiber Blanket:

(1) Adhere insulation to metal with 3 inch wide strips of insulation bonding adhesive at 8 inches on center all around duct. Additionally secure insulation to bottom of ducts exceeding 24 inches in width with pins welded or adhered on 18 inch centers. Secure washers on pins. Butt insulation edges and seal joints with laps and butt strips. Staples may be used to assist in securing insulation. Seal all vapor retarder penetrations with mastic. Sagging duct insulation will not be acceptable. Install firestop duct insulation where required.

(2) Supply air ductwork to be insulated includes main and branch ducts from AHU discharge to room supply outlets, and the bodies of ceiling outlets to prevent condensation. Insulate sound attenuator units, coil casings and damper frames. To prevent condensation insulate trapeze type supports and angle iron hangers for flat oval ducts that are in direct contact with metal duct.

(3) Concealed supply air ductwork.

(a) Above ceilings at a roof level, in attics, and duct work exposed to outdoor weather: 2 inch thick insulation faced with FSK.

(b) Above ceilings for other than roof level: (1 ½ inch) thick insulation faced with FSK.

(4) Concealed return air duct:

(a)

(b) Above ceilings at a roof level, unconditioned areas, and in chases with external wall or containing steam piping; (1-1/2 inch) thick, insulation faced with FSK.

(c) In interstitial spaces (where not subject to damage): (1-1/2 inch) thick insulation faced with FSK.

(d) Concealed return air ductwork in other locations need not be insulated.

(5) Concealed outside air duct: (1-1/2 inch) thick insulation faced with FSK.

(6) Exhaust air branch duct from autopsy refrigerator to main duct: (1-1/2 inch) thick insulation faced with FSK.

c. Molded Mineral Fiber Pipe and Tubing Covering:

(1) Fit insulation to pipe or duct, aligning longitudinal joints. Seal longitudinal joint laps and circumferential butt strips by rubbing hard with a nylon sealing tool to assure a positive seal. Staples may be used to assist in securing insulation. Seal all vapor retarder penetrations on cold piping with a generous application of vapor barrier mastic. Provide inserts and install with metal insulation shields at outside pipe supports. Install freeze protection insulation over heating cable.

(2) Contractor's options for fitting, flange and valve insulation:

(a) Insulating and finishing cement for sizes less than 4 inches operating at surface temperature of (61 degrees F) or more.

(b) Factory premolded, one piece PVC covers with mineral fiber, (Form B), inserts. Provide two insert layers for pipe temperatures below (40 degrees F), or above (250 degrees F). Secure first layer of insulation with twine. Seal seam edges with vapor barrier mastic and secure with fitting tape.

(c) Factory molded, ASTM C547 or field mitered sections, joined with adhesive or wired in place. For hot piping finish with a smoothing coat of finishing cement. For cold fittings, (60 degrees F) or less, vapor seal with a layer of glass fitting tape imbedded between two 1/16 inch coats of vapor barrier mastic.

(d) Fitting tape shall extend over the adjacent pipe insulation and overlap on itself at least (2 inches).

(3) Nominal thickness in inches specified in the schedule at the end of this section.

d. Rigid Cellular Phenolic Foam:

(1) Rigid closed cell phenolic insulation may be provided for piping, ductwork and equipment for temperatures up to (250 degrees F).

(2) Note the NFPA 90A burning characteristics requirements of 25/50 in paragraph 1.3.B

(3) Provide secure attachment facilities such as welding pins.

(4) Apply insulation with joints tightly drawn together

(5) Apply adhesives, coverings, neatly finished at fittings, and valves.

(6) Final installation shall be smooth, tight, neatly finished at all edges.

(8) Exposed, unlined supply and return ductwork exposed to outdoor weather: 2 inch thick insulation faced with a multi-layer vapor barrier with a maximum water vapor permeance of 0.00 perms.

(9) Condensation control insulation: Minimum 1.0 inch thick for all pipe sizes.

(a) HVAC: Cooling coil condensation piping to waste piping fixture or drain inlet. Omit insulation on plastic piping in mechanical rooms.

e. Cellular Glass Insulation:

(1) Pipe and tubing, covering nominal thickness in millimeters and inches as specified in the schedule at the end of this section.

(2) Underground Piping Other than or in lieu of that Specified in Section 23 21 13, HYDRONIC PIPING and Section 33 63 00, STEAM ENERGY DISTRIBUTION: Type II, factory jacketed with a laminate jacketing consisting of x (10 ft x 10 ft) asphalt impregnated glass fabric, bituminous mastic and outside protective plastic film.

(a) 3 inches thick for hot water piping.

(b) As scheduled at the end of this section for chilled water piping.

(c) Underground piping: Apply insulation with joints tightly butted. Seal longitudinal self-sealing lap. Use field fabricated or factory made fittings. Seal butt joints and fitting with jacketing as recommended by the insulation manufacturer. Use 4 inch wide strips to seal butt joints.

(d) Provide expansion chambers for pipe loops, anchors and wall penetrations as recommended by the insulation manufacturer.

(e) Underground insulation shall be inspected and approved by the Resident Engineer as follows:

1. Insulation in place before coating.

2. After coating.

(f) Sand bed and backfill: Minimum 3 inches all around insulated pipe or tank, applied after coating has dried.

(3) Cold equipment: 2 inch thick insulation faced with ASJ for chilled water pumps, water filters, chemical feeder pots or tanks, expansion tanks, air separators and air purgers.

(4) Exposed, unlined supply and return ductwork exposed to outdoor weather: 2 inch thick insulation faced with a reinforcing membrane and two coats of vapor barrier mastic or multi-layer vapor barrier with a water vapor permeability of 0.00 perms.

f. Polyisocyanurate Closed-Cell Rigid Insulation:

(1) Polyisocyanurate closed-cell rigid insulation (PIR) may be provided for exterior piping, equipment and ductwork for temperature up to (300 degree F).

(2) Install insulation, vapor barrier and jacketing per manufacturer’s recommendations. Particular attention should be paid to recommendations for joint staggering, adhesive application, external hanger design, expansion/contraction joint design and spacing and vapor barrier integrity.

(3) Install insulation with all joints tightly butted (except expansion) joints in hot applications).

(4) If insulation thickness exceeds (2.5 inches), install as a double layer system with longitudinal (lap) and butt joint staggering as recommended by manufacturer.

(5) For cold applications, vapor barrier shall be installed in a continuous manner. No staples, rivets, screws or any other attachment device capable of penetrating the vapor barrier shall be used to attach the vapor barrier or jacketing. No wire ties capable of penetrating the vapor barrier shall be used to hold the insulation in place. Banding shall be used to attach PVC or metal jacketing.

(6) Elbows, flanges and other fittings shall be insulated with the same material as is used on the pipe straights. The elbow/ fitting insulation shall be field-fabricated, mitered or factory prefabricated to the necessary size and shape to fit on the elbow/ fitting. Use of polyurethane spray-foam to fill PVC elbow jacket is prohibited on cold applications.

(7) For cold applications, the vapor barrier on elbows/fittings shall be either mastic-fabric-mastic or 2 mil thick PVDC vapor barrier adhesive tape.

(8) All PVC and metal jacketing shall be installed so as to naturally shed water. Joints shall point down and shall be sealed with either adhesive or caulking (except for periodic slip joints).

(9) Underground piping: Follow instructions for above ground piping but the vapor retarder jacketing shall be 6 mil thick PVDC or minimum 30 mil thick rubberized bituminous membrane. Sand bed and

backfill shall be a minimum of 6 inches all around insulated pipe.

(10) Exposed, unlined supply and return ductwork exposed to outdoor weather: 2 inch thick insulation faced with a multi-layer vapor barrier with a water vapor permeance of 0.00 perms.

(11) Note the NFPA 90A burning characteristic requirements of 25/50 in paragraph 1.3B. Refer to paragraph 3.1 for items not to be insulated.

(12) Minimum thickness in (inches) specified in the schedule at the end of this section.

g. Flexible Elastomeric Cellular Thermal Insulation:

(1) Apply insulation and fabricate fittings in accordance with the manufacturer's installation instructions and finish with two coats of weather resistant finish as recommended by the insulation manufacturer.

(2) Pipe and tubing insulation:

(a) Use proper size material. Do not stretch or strain insulation.

(b) To avoid undue compression of insulation, provide cork stoppers or wood inserts at supports as recommended by the insulation manufacturer. Insulation shields are specified under Section 23 05 11, COMMON WORK RESULTS FOR HVAC .

(c) Where possible, slip insulation over the pipe or tubing prior to connection, and seal the butt joints with adhesive. Where the slip-on technique is not possible, slit the insulation and apply it to the pipe sealing the seam and joints with contact adhesive. Optional tape sealing, as recommended by the manufacturer, may be employed. Make changes from mineral fiber insulation in a straight run of pipe, not at a fitting. Seal joint with tape.

(3) Apply sheet insulation to flat or large curved surfaces with 100 percent adhesive coverage. For fittings and large pipe, apply adhesive to seams only.

(4) Pipe insulation: nominal thickness in millimeters (inches as specified in the schedule at the end of this section.

(5) Minimum 0.75 inch thick insulation for pneumatic control lines for a minimum distance of 20 feet from discharge side of the refrigerated dryer.

(6) Use Class S (Sheet), 3/4 inch thick for the following:

(a) Chilled water pumps

(b) Bottom and sides of metal basins for winterized cooling towers (where basin water is heated).

(c) Chillers, insulate any cold chiller surfaces subject to condensation which has not been factory insulated.

(d) Piping inside refrigerators and freezers: Provide heat tape

under insulation.

(7) Exposed, unlined supply and return ductwork exposed to outdoor weather: 2 inch thick insulation faced with a multi-layer vapor barrier with a water vapor permeance of 0.00 perms.

h. Calcium Silicate:

(1) Minimum thickness in (inches) specified in the schedule at the end of this section for piping other than in boiler plant. See paragraphs 3.3 through 3.7 for Boiler Plant Applications.

3.3 APPLICATION –BOILER PLANT, PIPE, VALVES, STRAINERS AND FITTINGS

a. Temperature range (251 to 450 degrees F);

(1) Application; Steam service 16 psig nominal) and higher, high pressure condensate to trap assembly, boiler bottom blowoff from boiler to blowoff valve closest to boiler.

(2) Insulation and Jacket:

(a) Calcium silicate for piping from zero to 6 feet above boiler room floor, feedwater heater mezzanine floor or access platform and any floors or platforms on which tanks or pumps are located.

(b) Mineral fiber for remaining locations.

(c) ASJ with PVC premolded fitting coverings.

(d) Aluminum jacket from zero to 6 feet above floor on atomizing steam and condensate lines at boilers and burners.

(3) Thickness:

Nominal Thickness Of Calcium Silicate Insulation(Boiler Plant)

Pipe Diameter (in)	Insulation Thickness (in)
(1) and below)	(5)
(1-1/4 to 1-1/2)	(5)
(1-1/2) and above	(6)

b. Temperature range (211 to 250 degrees F):

(1) Application: Steam service 15 psig and below, trap assembly discharge piping, boiler feedwater from feedwater heater to boiler feed pump recirculation, feedwater heater overflow, heated oil from oil heater to burners.

(2) Insulation and Jacket:

(a) Calcium silicate for piping from zero to (0 to 6 feet above boiler room floor, feedwater heater mezzanine floor and access platform, and any floors or access platforms on which tanks or pumps are located.

(b) Mineral Fiber or rigid closed cell phenolic foam for remaining locations.

(c) ASJ with PVC premolded fitting coverings.

(d) Aluminum jacket from zero to 6 feet above floor on condensate lines at boilers and burners.

(3) Thickness-calcium silicate and mineral fiber insulation:

Nominal Thickness Of Insulation	
Pipe Diameter (in)	Insulation Thickness (in)
(1 and below)	(2)
(1-1/4 to 1-1/2)	(2)
(1-1/2) and above	(3)

(4) Thickness-rigid closed-cell phenolic foam insulation:

Nominal Thickness Of Insulation	
Pipe Diameter (in)	Insulation Thickness (in)
(1 and below)	(1.5)
(1-1/4 to 1-1/2)	(1.5)
(1-1/2) and above	(3)

c. Temperature range (90 to 211 degrees F):

(1) Application: Pumped condensate, vacuum heating return, gravity and pumped heating returns, condensate transfer, condensate transfer pump recirculation, heated oil system to heaters and returns from burners, condensate return from convertors and heated water storage tanks.

(2) Insulation Jacket:

(a) Calcium silicate for piping from zero to (six feet) above boiler room floor, feedwater heater mezzanine floor and access platform and any floor or access platform on which tanks or pumps are located.

(b) Mineral fiber or rigid closed-cell phenolic foam for remaining locations.

(c) ASJ with PVC premolded fitting coverings.

(3) Thickness-calcium silicate and mineral fiber insulation:

Nominal Thickness Of Insulation	
Pipe Diameter (in)	Insulation Thickness (in)
(1 and below)	(1.5)
(1-1/4 to 1-1/2)	(2)
(1-1/2) and above	(3)

(4) Thickness-rigid closed-cell phenolic foam insulation:

Nominal Thickness Of Insulation	
Pipe Diameter (in)	Insulation Thickness (in)
(1 and below)	(0.75)
(1-1/4 to 1-1/2)	(0.75)
(1-1/2) and above	(1)

d. Protective insulation to prevent personnel injury:

(1) Application: Piping from zero to 6 feet above all floors and access platforms including continuous blowoff, feedwater and boiler water sample, blowoff tank vent, flash tank vents and condensater tank vent, shot-type chemical feed, fire tube boiler bottom blowoff after valves, valve by-passes.

(2) Insulation thickness: (1 inch).

(3) Insulation and jacket: Calcium silicate with ASJ except provide aluminum jacket on piping at boilers within 6 feet of floor. Use PVC premolded fitting coverings when all service jacket is utilized.

e. Installation:

(1) At pipe supports, weld pipe covering protection saddles to pipe, except where MSS SP-58, type 3 pipe clamps are utilized.

(2) Insulation shall be firmly applied, joints butted tightly, mechanically fastened by stainless steel wires on 12 inch centers.

(3) At support points, fill and thoroughly pack space between pipe covering protective saddle bearing area.

(4) Terminate insulation and jacket hard and tight at anchor points.

(5) Terminate insulation at piping facilities not insulated with a 45 degree chamfered section of insulating and finishing cement covered with jacket.

(6) On calcium silicate, mineral fiber and rigid closed-cell phenolic foam systems, insulated flanged fittings, strainers and valves with sections of pipe insulation cut, fitted and arranged neatly and firmly wired in place. Fill all cracks, voids and coat outer surface with insulating cement. Install jacket. Provide similar construction on welded and threaded fittings on calcium silicate systems or use premolded fitting insulation.

(7) On mineral fiber systems, insulate welded and threaded fittings more than 2 inches in diameter with compressed blanket insulation (minimum 2/1) and finish with jacket or PVC cover.

(8) Insulate fittings 2 inches and smaller with mastic finishing material and cover with jacket.

(9) Insulate valve bonnet up to valve side of bonnet flange to permit bonnet flange removal without disturbing insulation.

(10) Install jacket smooth, tight and neatly finish all edges. Over wrap ASJ butt strips by 50 percent. Secure aluminum jacket with stainless steel bands 12 inches on center or aluminum screws on 4 inch centers.

(11) Do not insulate basket removal flanges on strainers.

3.4 Not Used.

3.5 APPLICATION ON HEATED OR TRACED OIL FACILITIES OUTSIDE OF BUILDING

a. Temperature range (85 to 250 degrees F).

(1) Application: Aboveground oil storage tank, oil and steam or hot water underground and aboveground piping systems.

(2) Insulation thickness:

(a) Tanks; (1-1/2 inches) plus finish.

(b) Oil suction and return piping: (1-1/2 inches plus finish).

(c) Steam or hot water piping: (1-1/2 inches) plus finish.

b. Insulation and jacket (aboveground tanks and piping): Calcium silicate with glass cloth or aluminum jacket, weatherproof jacket when used outside of building.

c. Insulation and Jacket (underground piping); Calcium silicate with fiberglass scrim jacket located within secondary containment. Allow space for heating cable (if provided) along bottom line of piping.

3.6 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of section 23 08 00 – COMMISSIONING OF HVAC SYSTEMS for all inspection, start up, and contractor testing required above and required by the System Readiness Checklist provided by the Commissioning Agent.

b. Components provided under this section of the specification will be tested as part of a larger system. Refer to section 23 08 00 – COMMISSIONING OF HVAC SYSTEMS and related sections for contractor responsibilities for system commissioning.

3.7 PIPE INSULATION SCHEDULE

Provide insulation for piping systems as scheduled below:

Insulation Wall Thickness (Inches)					
		Nominal Pipe Size (Inches)			
Operating Temperature Range/Service	Insulation Material	Less than 25 (1)	(1 – 1¼)	(1½ - 3)	(4) and Above
	Insulation Wall Thickness (Inches)				
(251-350 degrees F) (HPS, MPS)	Mineral Fiber (Above ground piping only)	(3)	(4)	(4.5)	(4.5)
(200-500 degrees F) (HPS, HPR)	Calcium Silicate	(4)	(5)	(6)	(6)
(212-250 degrees F) (HPR, MPR, LPS, vent piping from PRV Safety Valves, Condensate receivers and flash tanks)	Mineral Fiber (Above ground piping only)	(2.5)	(2.5)	(3.0)	(3.0)
(212-250 degrees F) (HPR, MPR, LPS, vent piping from PRV Safety Valves, Condensate receivers and flash tanks)	Rigid Cellular Phenolic Foam	(2.0)	(2.0)	(3.0)	(3.0)
(100-200 degrees F) (LPR, PC, HWH, HWHR, GH and GHR)	Mineral Fiber (Above ground piping only)	(1.5)	(1.5)	(2.0)	(2.0)
(100-211 degrees F) (LPR, PC, HWH, HWHR, GH and GHR)	Rigid Cellular Phenolic Foam	(1.5)	(1.5)	(2.0)	(2.0)
(100-211 degrees F) (LPR, PC, HWH, HWHR, GH and GHR)	Polyiso-cyanurate Closed-Cell Rigid (Exterior Locations only)	(1.5)	(1.5)	----	----

(100-200 degrees F) (LPR, PC, HWH, HWHR, GH and GHR)	Flexible Elastomeric Cellular Thermal (Above ground piping only)	(1.5)	(1.5)	----	----
(40-60 degrees F) (CH, CHR, GC, GCR and RS for DX refrigeration)	Rigid Cellular Phenolic Foam	(1.5)	(1.5)	(1.5)	(1.5)
(40-60 degrees F) (CH and CHR within chiller room and pipe chase and underground)	Cellular Glass Closed-Cell	(2.0)	(2.0)	(3.0)	(3.0)
(40-60 degrees F) (CH, CHR, GC, GCR and RS for DX refrigeration)	Cellular Glass Closed-Cell	(1.5)	(1.5)	(1.5)	(1.5)
(40-60 degrees F) (CH, CHR, GC and GCR (where underground)	Polyiso-cyanurate Closed-Cell Rigid	(1.5)	(1.5)	(2.0)	(2.0)
(40-60 degrees F) (CH, CHR, GC, GCR and RS for DX refrigeration)	Polyiso-cyanurate Closed-Cell Rigid (Exterior Locations only)	(1.5)	(1.5)	(1.5)	(1.5)
(40-60 degrees F) (CH, CHR, GC, GCR and RS for DX refrigeration)	Flexible Elastomeric Cellular Thermal (Above ground piping only)	(1.5)	(1.5)	(1.5)	(1.5)

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 08 00.01

DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS

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SECTION 23 08 00.01
8/19

DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

a. The requirements of this Section apply to all sections of Division 23.

b. This project will have selected building systems commissioned. The complete list of equipment and systems to be commissioned is specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS. The commissioning process, which the Contractor is responsible to execute, is defined in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS. A Commissioning Agent (CxA) appointed by the Government and Commissioning Manager (CxM) as indicated in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS will manage the commissioning process.

1.2 RELATED WORK

a. UFGS GENERAL REQUIREMENTS sections

b. Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS

c. UFGS Section 01 33 00 SUBMITTAL PROCEDURES

1.3 SUMMARY

a. This Section includes requirements for commissioning the Facility exterior closure, related subsystems and related equipment. This Section supplements the general requirements specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

b. Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for more details regarding processes and procedures as well as roles and responsibilities for all Commissioning Team members.

1.4 DEFINITIONS

Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for definitions.

1.5 COMMISSIONED SYSTEMS

a. Commissioning of a system or systems specified in Division 23 is part of the construction process. Documentation and testing of these systems, as well as training of the VA’s Operation and Maintenance personnel in accordance with the requirements of Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS and of Division 23, is required in cooperation with the DVA/USACE and the CxA and CxM..

b. The Facility exterior closure systems commissioning will include the systems listed in Section 01 91 00.01 DVA/USACE Projects GENERAL

COMMISSIONING REQUIREMENTS:

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Construction Inspections; G, DO

Pre-Functional Checklists; G, DO

Contractors Tests; G, DO

Systems Functional Performance Testing; G, DO

SD-08 Manufacturer's Instructions

Training Agendas And Trainer Resumes; G, DO

Formal Training Plans; G, DO

1.7 SUBMITTAL INSTRUCTIONS

a. The commissioning process requires review of selected Submittals that pertain to the systems to be commissioned. The CxA will provide a list of submittals that will be reviewed by the CxA and CxM. This list will be reviewed and approved by the VA prior to forwarding to the Contractor. Refer to UFGS Section 01 33 00 SUBMITTAL PROCEDURES for further details.

b. The commissioning process requires Submittal review simultaneously with engineering review. Specific submittal requirements related to the commissioning process are specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 CONSTRUCTION INSPECTIONS

Commissioning of HVAC systems will require inspection of individual elements of the HVAC systems construction throughout the construction period. The Contractor shall coordinate with the CxA and CxM in accordance with Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS and the Commissioning plan to schedule HVAC systems inspections as required to support the Commissioning Process.

3.2 PRE-FUNCTIONAL CHECKLISTS

The Contractor shall complete Pre-Functional Checklists to verify systems, subsystems, and equipment installation is complete and systems are ready for Systems Functional Performance Testing. The CxM will prepare Pre-Functional Checklists to be used to document equipment installation. The Contractor shall complete the checklists. Completed checklists shall

be submitted to the DVA/USACE, CxA, and CxM for review. The CxM and CxA may spot check a sample of completed checklists. If the CxM and CxA determine that the information provided on the checklist is not accurate, the CxM will return the marked-up checklist to the Contractor for correction and resubmission. If the CxM and CxA determine that a significant number of completed checklists for similar equipment are not accurate, the CxM in coordination with CxA will select a broader sample of checklists for review. If the CxM and CxA determine that a significant number of the broader sample of checklists is also inaccurate, all the checklists for the type of equipment will be returned to the Contractor for correction and resubmission. Refer to SECTION 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for submittal requirements for Pre-Functional Checklists, Equipment Startup Reports, and other commissioning documents.

3.3 CONTRACTORS TESTS

Contractor tests as required by other sections of Division 23 shall be scheduled and documented in accordance with UFGS GENERAL REQUIREMENTS sections. All testing shall be incorporated into the project schedule. Contractor shall provide no less than 7 calendar days' notice of testing. The CxM will witness selected Contractor tests at the sole discretion of the CxM and CxA. The CxA and other Government representative(s) may choose to participate in some or all the testing procedures as observers. Contractor tests shall be completed prior to scheduling Systems Functional Performance Testing.

3.4 SYSTEMS FUNCTIONAL PERFORMANCE TESTING

The Commissioning Process includes Systems Functional Performance Testing that is intended to test systems functional performance under steady state conditions, to test system reaction to changes in operating conditions, and system performance under emergency conditions. The CxM will prepare final and detailed Systems Functional Performance Test procedures for review and approval by the CxA and Contracting Officer. The Contractor shall review and comment on the tests prior to approval. The Contractor shall provide the required labor, materials, and test equipment identified in the test procedure to perform the tests. The CxM will witness and document the testing. The Contractor shall sign the test reports to verify tests were performed. See Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS, for additional details.

3.5 TRAINING OF VA PERSONNEL

Training of the VA operation and maintenance personnel is required in cooperation with the Contracting Officer, VA Resident Engineer, and CxM. Provide competent, factory authorized personnel to provide instruction to operation and maintenance personnel concerning the location, operation, and troubleshooting of the installed systems. Contractor shall submit training agendas and trainer resumes in accordance with the requirements of Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS. The instruction shall be scheduled in coordination with the Contracting Officer and VA Resident Engineer after submission and approval of formal training plans. Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and Division 23 Sections for additional Contractor training requirements.

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 08 00

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05/23, CHG 1: 08/24

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SECTION 23 08 00

COMMISSIONING OF MECHANICAL SYSTEMS
05/23, CHG 1: 08/24

PART 1 GENERAL

1.1 SEQUENCING AND SCHEDULING

Complete the following prior to starting Functional Performance Tests of mechanical systems:

a. All equipment and systems completed, cleaned, flushed, disinfected, calibrated, tested, and operate in accordance with contract documents and construction plans and specifications

b. Final DALT Report submitted and approved in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC

c. Pre-final Testing, Adjusting, and Balancing Report submitted in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC

e. The Certificate of Readiness submitted and approved in accordance with Section 01 91 00.15 BUILDING COMMISSIONING

f. Inspection and Testing in accordance with Section 22 33 30.00 10 SOLAR WATER HEATING EQUIPMENT

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Test Equipment; G

SD-06 Test Reports

Completed Pre-Functional Checklists; G

Seasonal Test Report; G

Full-Load Test Report; G

Post-Construction Trend Log Report; G

SD-07 Certificates

Certificate Of Readiness; G

1.3 ACCESSIBILITY REQUIREMENTS

Equipment, systems, and devices for commissioned systems must be accessible. Make necessary modifications if systems and devices are not

accessible for inspections and testing.

Assist commissioning team in testing by removing equipment covers, opening access panels, and other required activities that assist with visual oversight. Furnish ladders, flashlights, meters, gauges, or other inspection equipment as necessary.

1.4 COORDINATION

Refer to Section 01 91 00.15 BUILDING COMMISSIONING for requirements pertaining to coordination during the commissioning process. Coordinate with the in accordance with Section 01 91 00.15 and in accordance with the Commissioning Plan to schedule inspections as required to support the commissioning process. Furnish additional information requested by the. Coordinate scheduling of Functional Performance Testing with the commissioning team. Provide plans, reports, notes, and other documentation to the as specified in the commissioning plan, as it is completed.

1.5 CERTIFICATE OF READINESS

Submit Certificate of Readiness documentation in accordance with Section 01 91 00.15 BUILDING COMMISSIONING for all equipment and systems including start-up reports; completed Pre-Functional Checklists; Testing, Adjusting, and Balancing (TAB) Report; Issues Log; HVAC Controls Start-Up Reports. Do not schedule Functional Performance Tests for the system until the Certificate of Readiness for that system receives approval by the Government. The CxC, Quality Control Manager, and the Mechanical, Electrical, Controls, and TAB subcontractor representatives must sign and date the Certificate of Readiness.

PART 2 PRODUCTS

2.1 TEST EQUIPMENT

Provide all testing equipment required to perform testing for the systems to be commissioned, except for equipment specific to and used by TAB as required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Provide a sufficient quantity of two-way radios for each subcontractor. Submit list of Test Equipment and instrumentation to be used for testing including equipment/instrument identification number, equipment application or planned use, manufacturer, make, model, and serial number, and calibration history with certificates. Also list special equipment and proprietary tools specific to a piece of equipment required for testing.

2.1.1 Proprietary Equipment

Provide manufacturer's proprietary test equipment and software required by any equipment manufacturer for programming and start-up, whether specified or not. Provide manufacturer test equipment, demonstrate its use, and assist in the commissioning process as needed. Provide data logging equipment and software required to test equipment.

2.1.2 Calibration and Accuracy

Comply with equipment manufacturer's test equipment calibration procedures and intervals. Recalibrate test instruments immediately after instruments have been repaired resulting from being dropped or damaged. Affix

calibration tags to test instruments. Furnish calibration records to Contracting Officer upon request.

Provide all testing equipment of sufficient quality and accuracy to test and measure system performance with the tolerances specified. Unless otherwise noted, the following minimum requirements apply: Provide temperature sensors and digital thermometers with a certified calibration within the past year to an accuracy of 0.5 degrees F and a resolution of plus or minus 0.1 degrees F. Provide pressure sensors with an accuracy of plus or minus 2.0 percent of the value range being measured (not full range of meter) and calibrated within the last year.

PART 3 EXECUTION

3.1 MEETINGS

Attend all meetings in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

Provide timely updates on construction schedule changes to allow the to execute commissioning process efficiently. Notify Contracting Officer of anticipated construction delays to commissioning activities not yet performed or not yet scheduled.

3.2 COMMISSIONING CONSTRUCTION OBSERVATION CHECKLISTS

Commissioning construction observation checklists include Pre-Functional Checklists and Functional Performance Test Checklists. Provide commissioning construction observation checklists for the Interim and Final Construction Phase Commissioning Plan in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

3.2.1 Pre-Functional Checklists

The Pre-Functional Checklists must include items for physical inspection or testing that demonstrate that installation and start-up of equipment and systems is complete. Refer to paragraph PRE-FUNCTIONAL CHECKS. Pre-functional checklists must be tailored to verify the specific installation requirements and details of the construction documents and manufacturer's instructions.

3.2.2 Functional Performance Test Checklists

Functional Performance Test Checklists must include procedures that explain, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract. Refer to paragraph FUNCTIONAL PERFORMANCE AND INTEGRATED SYSTEMS TESTING. Include the following sections and details appropriate to the systems being tested in the Functional Performance Test Checklists:

a. Notable system features including information about controls to facilitate understanding of system operation and maintenance

b. Conclusions and recommendations. Conclusions must clearly indicate if system does or does not perform in accordance with contract requirements and specifications. Recommendation must clearly indicate that the system should or should not be approved by the Government.

c. Test conditions including date, beginning and ending time, and beginning and ending outdoor air conditions

d. Attendees

e. Identification of the equipment involved in the test

f. **System feature identification**

g. Point-to-point observations including demonstrating system flow meters and sensors have been calibrated and are correctly displayed on the Operator work station

h. Actuator operation observations demonstrating actuator responses to commands from the control system

i. As-found condition of the system operation

j. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.

k. Space for comments for each test item.

3.3 PRE-FUNCTIONAL CHECKS

Pre-Functional Checks are a type of Commissioning Inspection in accordance with Section 01 91 00.15 BUILDING COMMISSIONING. Complete one Pre-Functional Checklist for each individual item of equipment or system for each system required to be commissioned including, but not limited to, ductwork, piping, equipment, fixtures, and controls. Include manufacturer start-up checklists associated with equipment with the submission of the Pre-Functional Checklists. Provide manufacturer's installation manual for each type of unit. Indicate commissioning team member inspection and validation of each Pre-Functional Checklist item by initials. Validation of each Pre-Functional Checklist item by each team member indicates that item conforms to the contract documents and validated design in their area of responsibility. Commissioning Specialist validation of each Pre-Functional Checklist item indicates that each item has been installed correctly and in accordance with contract documents. Required commissioning team members for Pre-Functional Checks includes the CxC, Mechanical Commissioning Specialist, Quality Control Manager, sub-contractor representative for each trade responsible for construction/installation of the checklist item, and the TAB representative (for items impacting TAB). Submit the initialed and Completed Pre-Functional Checklists no later than 7 calendar days after completion of inspection of all checklists items for each system.

3.4 STARTUP AND INITIAL CHECKOUT

Document start-up and initial testing procedures, and include in the Completed Pre-Functional Checklists submittal, including:

a. Startup tests and factory testing reports.

b. Manufacturer's representative start-up, operating, troubleshooting and maintenance procedures.

c. Additional documentation necessary for third party certification programs.

d. Perform and clearly document system operational checks and quality control checks as they are completed, and providing a copy to the commissioning team.

e. Correct deficiencies and sign the Certificate of Readiness for each system before functional performance testing

3.5 DUCT AIR LEAKAGE TEST (DALT) REPORT REVIEW AND VERIFICATION

3.5.1 Duct Air Leakage Test (DALT) Report Review

The Mechanical System Technical Commissioning Specialist must review the pre-final DALT Report required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. All deficiencies must be resolved prior to DALT Report approval.

3.5.2 Duct Air Leakage Test (DALT) Report Verification

The Mechanical System Technical Commissioning Specialist must witness the DALT Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. All deficiencies must be resolved prior to DALT Report approval.

3.6 TESTING, ADJUSTING, AND BALANCING REPORT REVIEW AND VERIFICATION

3.6.1 Testing, Adjusting, and Balancing (TAB) Report Review

The Mechanical System Technical Commissioning Specialist must review the pre-final TAB Report required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. All deficiencies must be resolved prior to TAB Report approval.

3.6.2 Testing, Adjusting, and Balancing (TAB) Report Verification

The Mechanical System Technical Commissioning Specialist must witness the TAB Field Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. All deficiencies must be resolved prior to TAB Report approval.

3.7 HVAC CONTROLS START-UP AND PERFORMANCE VERIFICATION TEST REVIEW

The Mechanical System Technical Commissioning Specialist must review the Start-Up Testing Report, PVT Procedures and PVT Reports, including endurance testing trend submittals, required by Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. The Mechanical System Technical Commissioning Specialist must review each submittal and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel and include in the issues log. All

deficiencies must be resolved prior to final acceptance.

3.8 FUNCTIONAL PERFORMANCE TESTING

Functional Performance Testing are a type of Commissioning Testing in accordance with Section 01 91 00.15 BUILDING COMMISSIONING. Conduct Functional Performance Testing in accordance with Section 01 91 00.15 BUILDING COMMISSIONING and requirements in this Section. Prior to Functional Performance Testing, complete all prerequisites in accordance with paragraph SEQUENCING AND SCHEDULING. Demonstrate that all system components have been installed, that each control device and item of equipment operates, and that the systems operate and perform, including interactive operation between systems, in accordance with contract documents.

3.8.1 Test Scheduling and Coordination

Schedule and conduct Initial Functional Performance Tests. Develop and implement means of artificial loading to demonstrate, to a reasonable level of confidence, the ability of the HVAC systems to handle peak seasonal loads. Schedule Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. Correct all deficiencies identified through any prior review, inspection, or test activity before the start of Functional Performance Tests.

Functional Performance Tests must be performed with the CxC present. Government reserves the right to witness all tests. Coordinate test schedule with Government representatives.

3.8.2 Preparation

Put equipment and systems into operation and continue operation during each working day of functional performance testing, as required. Verify temperature and pressure taps in accordance with Contract Documents. Provide a pressure/temperature plug at each water sensor which is an input point to control system.

Perform minor adjustments to equipment and systems during Functional Performance Tests as deemed necessary by the commissioning team. Where calibrated DDC sensors cannot be used to record test data, provide measuring instruments, logging devices, and data acquisition equipment to record data for the complete range of test data for the required test period.

3.8.3 Testing Procedures

Provide all necessary materials and system modifications to produce the necessary flows, pressures, temperatures, and other conditions necessary to execute the test according to the specified conditions. At completion of the test, return the affected building equipment and systems to their pre-test condition.

Follow the Functional Performance Test from the approved Final Construction Phase Commissioning Plan. Perform Functional Performance Tests for each item of equipment and each system required to be commissioned. Verify all sensor calibrations, control responses, safeties, interlocks, operating modes, sequences of operation, capacities, and all other performance requirements comply with contract, regardless of

the specific items listed within the checklists provided. In general, testing must progress from equipment or components to subsystems to systems to interlocks and connections between systems. Commissioning Specialists are responsible for determining the order of components and systems to be tested.

Indicate validation of each item of equipment and systems tested by signature of each commissioning team member for each test. The Quality Control Representative, Commissioning Specialists, and Contracting Officer's Representative, if present, must indicate validation after the equipment and systems are free of deficiencies.

3.8.4 Simulating Conditions

Functional performance testing is conducted by simulating conditions at control devices to initiate a control system response. Before testing, calibrate all sensors, transducers and devices. Over-writing control input values through the control system is not acceptable unless approved by the Contracting Officer. Perform each test under conditions that simulate actual conditions as close as is practically possible. Specific examples of simulating conditions are provided below. Do not simulate conditions when damage to the system or building may result.

a. When varying static pressures inside ductwork cannot be simulated within the duct, and where a sensor signals the controls system to initiate sequences at various duct static pressures, it is acceptable to simulate the various pressures with a Pneumatic Squeeze-Bulb Type Signaling Device with gauge temporarily attached to the sensing tube leading to the transmitter. It is not acceptable to reset the various set-points, nor to simulate an electric analog signal (unless approved as noted above).

b. Dirty filter pressure drops can be simulated by partially blocking filter face.

c. Freeze-stat safeties can be simulated by packing portion of sensor with ice.

d. High outside air temperatures can be simulated with a hair blower.

e. Raising entering cooling coil temperatures by activating a heating/preheat coil can be used to simulate entering cooling coil conditions.

f. Do not use signal generators to simulate sensor signals unless approved by the Contracting Officer, as noted above, for special cases.

g. Control set points can be altered. For example, to see the air conditioning compressor lockout work at an outside air temperature below 55 degrees F, when the outside air temperature is above 55 degrees F, temporarily change the lockout set point to be 0 degrees F above the current outside air temperature. Caution: Set points are not to be raised or lowered to a point to cause damage to the components, systems, or the building structure and contents.

h. Test duct mounted smoke detectors in accordance with the manufacturer's recommendations. Perform the tests with air system at minimum airflow condition.

i. Test current sensing relays used for fan and pump status signals to control system to indicate unit failure and run status by resetting the set point on the relay to simulate a lost belt or unit failure while the unit is running. Confirm that the failure alarm was generated and received at the control system. After the test is conducted, return the set point to its original set-point or a set-point as indicated by the Contracting Officer.

3.8.5 Sample Strategy

Refer to Section 01 91 00.15 BUILDING COMMISSIONING for sample strategy. Complete a Functional Performance Test Checklist for each item of equipment or system to be tested. During testing, Government representatives may select the specific equipment or system to be tested for sample sizes less than 100 percent.

3.8.5.1 100 Percent Sample Procedures

Systems or equipment for which 100 percent sample size are tested fail if one or more of the test procedures results in discovery of a deficiency and the deficiency cannot be resolved within 5 minutes during the test.

Re-test to the extent necessary to confirm that the deficiencies have been corrected without negatively impacting the performance of the rest of the system.

3.8.5.2 Less than 100 Percent Sample Procedures

Randomly test each sample group of identical equipment. If 10 percent of the units in the first sample fail the functional performance tests, test a second sample group, the same size as the first sample group. The second sample must not include any units from the first sample group.

If 10 percent of the units in the second sample fail, test all remaining units. If at any point frequent failures occur the CxC may stop the testing and require the contractor to perform and document a checkout of the remaining units prior to continuing functional testing.

3.9 RETESTING REQUIREMENTS

Abort tests if any deficiency prevents successful completion of the test or if any required commissioning team member is not present for the test. Re-test only after all deficiencies identified during the original tests have been corrected.

Contracting Officer may withhold payment equivalent to lost time, re-testing, and aborted tests. These costs may include salary, travel costs, and per diem for Government team members. Correct deficiencies as identified by the commissioning team and retest the systems to be commissioned.

3.10 SYSTEM ACCEPTANCE

Systems may be partially accepted prior to seasonal testing if they comply with all construction contract and accepted design requirements that can be tested during initial Functional Performance Tests. All test procedures must be successful completed prior to full systems acceptance.

3.11 SEASONAL TESTS

Perform Initial Functional Performance Tests as soon as all contract work is completed, but prior to facility turnover, regardless of the season.

In addition to the Initial Functional Performance Tests, perform Functional Performance Tests of HVAC systems during peak heating and cooling seasons during outdoor air condition design extremes. Schedule Seasonal Functional Performance Tests in coordination with the Contracting Officer. Submit Seasonal Test Report within 14 days of test completion. Include seasonal test report in the Updated Final Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING

Execute seasonal functional performance testing, witnessed by the Contracting Officer. Correct deficiencies and make adjustments to O&M manuals and as-built drawings for applicable issues identified in any seasonal testing.

3.12 TRAINING

The Mechanical Systems Technical Commissioning Specialist must review the training plan required by Section 01 78 00 OPERATION AND MAINTENANCE DATA and identify any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel.

3.13 COMMISSIONING REPORT

Include all completed Pre-Functional Checklists and Functional Performance Checklists in the Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING. Include the approved TAB Report.

3.14 POST-CONSTRUCTION ENDURANCE TEST

Perform an Endurance Test in accordance with the paragraph in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC once during the peak heating season and once during the peak cooling season during outdoor air condition extremes with the exception that network bandwidth usage measurement and recording is not required.

The Mechanical System Commissioning Specialists must review the trend logs from the Endurance Tests to ensure that the systems have stable operation and operate as required by the construction contract, the accepted design. The Commissioning Specialists must provide a Post-Construction Trend Log Report that identifies any deficiencies noted in operation, recommendations for correction, and includes a graphical representation of the trends. Provide one Trend Log Report for the peak cooling season and one Trend Log Report for the peak heating season. Submit the Post-Construction Trend Log Reports no later than 14 calendar days following receipt of the trend log data by the Commissioning Specialist. Include the Post-Construction Trend Log Reports in the Updated Final Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING

-- End of Section --

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SECTION 23 09 23
3/23

DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC

PART 1 – GENERAL

1.1 DESCRIPTION

a. General Contractor shall provide direct-digital control system(s) as indicated on the project documents, point list, interoperability tables, drawings and as described in these specifications. Include a complete and working direct-digital control system. Include all engineering, programming, configuration/setup hardware and software, controls and installation materials, installation labor, commissioning and startup, training, final project documentation and warranty. .

(1) The direct-digital control system(s) shall consist of high-speed, peer-to-peer network of DDC controllers, a control system server, all configuration and setup software and hardware devices, and an Engineering Control Center. Provide a remote user using Building Controllers to access the control system graphics and change adjustable setpoints with the proper password.

(2) All new building controllers shall be native BACnet. All new BACnet workstations, controllers, devices and components shall be listed by BACnet Testing Laboratories. All new BACnet workstations, controllers, devices and components shall be accessible using a HTML5 Web browser interface. Browsers shall not require the use of an extension or add on software in order to access aforementioned workstations, controllers, devices, and components.

(a) If used, gateways shall be BTL listed.

(b) If used, gateways shall provide all object properties and read/write services shown on VA-approved interoperability schedules.

(3) The work administered by this Section of the technical specifications shall include all labor, materials, special tools, equipment, enclosures, power supplies, software, software licenses, Project specific software configurations and database entries, interfaces, wiring, tubing, installation, labeling, engineering, calibration, documentation, submittals, testing, verification, training services, permits and licenses, transportation, shipping, handling, administration, supervision, management, insurance, Warranty, specified services and any other items required for a complete and fully functional Controls System.

(4) The control systems shall be designed such that each mechanical system shall operate under stand-alone mode. The A/E shall designate what each “mechanical systems” is composed of. The contractor administered by this Section of the technical

specifications shall provide controllers for each mechanical system. In the event of a network communication failure, or the loss of any other controller, the control system shall continue to operate independently. Failure of the ECC shall have no effect on the field controllers, including those involved with global strategies.

(5) The control system shall accommodate 1 Engineering Control Center(s) and the control system shall accommodate 5 web-based Users simultaneously, and the access to the system should be limited only by operator password.

b. Some products are furnished but not installed by the contractor administered by this Section of the technical specifications. The contractor administered by this Section of the technical specifications shall formally coordinate in writing and receive from other contractors formal acknowledgements in writing prior to submission the installation of the products. These products include but are not limited to the following:

(1) Control valves.

(2) Flow switches.

(3) Flow meters.

(4) Sensor wells and sockets in piping.

(5) Terminal unit controllers.

c. Some products are installed but not furnished by the contractor administered by this Section of the technical specifications. The contractor administered by this Section of the technical specifications shall formally coordinate in writing and receive from other contractors formal acknowledgements in writing prior to submission the procurement of the products. These products include but are not limited to the following:

(1) Refrigerant leak detection system.

(2) Factory-furnished accessory thermostats and sensors furnished with unitary equipment.

d. Some products are not provided by, but are nevertheless integrated with the work executed by, the contractor administered by this Section of the technical specifications. These products include but are not limited to the following:

(1) Fire alarm systems. If zoned fire alarm is required by the project-specific requirements, this interface shall require multiple relays, which are provided and installed by the fire alarm system contractor, to be monitored.

(2) Advanced utility metering systems. These systems may take information from the control system or its component meters and sensors.

(3) Cchiller controls. These controls, if not native BACnet, will require a BACnet Gateway.

(4) Terminal units' velocity sensors

(5) Condenser water quality systems: condenser water high- and low-parts hydrogen (pH) alarms.

(6) Unitary HVAC equipment (air conditioning units, split systems, packaged pumping stations) controls. These include:

(a) Discharge temperature control.

(b) Economizer control.

(c) Flowrate control.

(d) Setpoint reset.

(e) Time of day indexing.

(f) Status alarm.

(7) Steam trap monitoring system: Monitoring and alarms shall be connected to the ECC via BACnet.

8. Variable frequency drives. These controls, if not native BACnet, will require a BACnet Gateway.

9. The following systems have limited control (as individually noted below) from the ECC:

(a) Medical and dental vacuum systems: high pressure and status alarms.

(b) Medical and dental compressed air systems: low pressure and status alarms.

(c) Domestic water heating systems: low temperature, high temperature and status alarms.

(d) Elevators: status alarms.

(e) Building lighting systems: on/off and scene control.

(f) Stormwater removal pumps: status alarm.

(g) Sanitary sewage pumps: status alarm.

e. Responsibility Table:

Work/Item/System	Furnish	Install	Low Voltage Wiring	Line Power
Control system low voltage and communication wiring	23 09 23	23 09 23	23 09 23	N/A
Terminal units	23	23	N/A	26
Controllers for terminal units	23 09 23	23	23 09 23	16
LAN conduits and raceway	23 09 23	23 09 23	N/A	N/A
Automatic dampers (not furnished with equipment)	23 09 23	23	N/A	N/A
Automatic damper actuators	23 09 23	23 09 23	23 09 23	23 09 23
Manual valves	23	23	N/A	N/A
Automatic valves	23 09 23	23	23 09 23	23 09 23
Pipe insertion devices and taps, flow and pressure stations.	23	23	N/A	N/A
Thermowells	23 09 23	23	N/A	N/A
Current Switches	23 09 23	23 09 23	23 09 23	N/A
Control Relays	23 09 23	23 09 23	23 09 23	N/A
Power distribution system monitoring interfaces	23 09 23	23 09 23	23 09 23	26
Interface with chiller/boiler controls	23 09 23	23 09 23	23 09 23	26
Chiller/boiler controls interface with control system	23	23	23 09 23	26
All control system nodes, equipment, housings, enclosures and panels.	23 09 23	23 09 23	23 09 23	26
Smoke detectors	28 31 00	28 31 00	28 31 00	28 31 00
Fire/Smoke Dampers	23	23	28 31 00	28 31 00
Smoke Dampers	23	23	28 31 00	28 31 00
Fire Dampers	23	23	N/A	N/A
Chiller/starter interlock wiring	N/A	N/A	26	26
Chiller Flow Switches	23	23	23	N/A
Boiler interlock wiring	23	23	23	26
Boiler Flow Switches	23	23	23	N/A
Water treatment system	23	23	23	26
VFDs	23	26	23 09 23	26
Refrigerant monitors	23	23 09 23	23 09 23	26
Laboratory Environmental Controls	23 09 23	23 09 23	23 09 23	26
Fume hood controls	23 09 23	23 09 23	23 09 23	26
Medical gas panels	23	23	26	26
Laboratory Air Valves	23	23	23 09 23	N/A
Computer Room A/C Unit field-mounted controls	23	23	26	26
Control system interface with CRU A/C controls	23 09 23	23 09 23	23 09 23	26
CRU A/C unit controls interface with control system	23	23 09 23	23 09 23	26
Fire Alarm shutdown relay interlock wiring	28	28	28	26

Work/Item/System	Furnish	Install	Low Voltage Wiring	Line Power
Control system monitoring of fire alarm smoke control relay	28	28	23 09 23	28
Fire-fighter’s smoke control station (FSCS	28	28	28	28
Fan Coil Unit controls (not furnished with equipment)	23 09 23	23 09 23	23 09 23	26
Unit Heater controls (not furnished with equipment)	23 09 23	23 09 23	23 09 23	26
Packaged RTU space-mounted controls (not furnished with equipment)	23 09 23	23 09 23	23 09 23	26
Packaged RTU unit-mounted controls (not furnished with equipment)	23 09 23	23 09 23	23 09 23	26
Cooling Tower Vibration Switches	23	23	23 09 23	23 09 23
Cooling Tower Level Control Devices	23	23	23 09 23	23 09 23
Cooling Tower makeup water control devices	23	23	23 09 23	23 09 23
Starters, HOA switches	23	23	N/A	26
Steam Trap Monitoring in Boiler Plant.	23 21 11	23 21 11	26	26
Steam Trap Monitoring not in Boiler Plant.	23 22 13	23 22 13	26	26

f. This facility’s existing direct-digital control (DDC) system is manufactured by Siemens , and its ECC is located in Building 19 . The existing system’s top-end communications is via BACnet . The existing system’s ECC and controllers were installed in the year of 2026 . The contractor administered by this Section of the technical specifications shall observe the capabilities, communication network, services, spare capacity of the existing control system and its ECC prior to beginning work. The contractor shall connect the new ECC system, either JCI or Siemens, located in Building 17 to the existing campus ECC located in Building 19.

(1)

(2) Provide a new BACnet JCI or Siemens ECC for Building 17, communications network, and controllers. Provide a programmable internetworking gateway allowing for real-time communication between the existing direct-digital control system and the new BACnet control system. Real-time communication shall provide all object properties and read/write services shown on VA-approved interoperability schedules. The contractor administered by this Section of the technical specifications shall provide all necessary investigation and site-specific programming to execute

the interoperability schedules.

(a) The combined system shall operate and function as one complete system including one database of control point objects and global control logic capabilities. Facility operators shall have complete operations and control capability over all systems, new and existing including; monitoring, trending, graphing, scheduling, alarm management, global point sharing, global strategy deployment, graphical operations interface and custom reporting as specified.

(3) Upgrade the existing direct-digital control system's ECC to include all properties and services required by an ASHRAE 135 BACnet B-AWS Profile. The upgraded ECC shall continue to communicate with the existing direct-digital control system's devices. The upgraded ECC shall communicate directly with the new native-BACnet devices over the existing control system's communications network without the use of a gateway. Provide programming converting the existing non-BACnet devices, objects and services to ASHRAE 135 BACnet-complaint BIBBs. The contractor administered by this Section of the technical specifications shall provide all necessary investigation and site-specific programming to execute the interoperability schedules.

(b) The combined system shall operate and function as one complete system including one database of control point objects and global control logic capabilities. Facility operators shall have limited operations and control capability over the legacy systems as described in the VA-approved interoperability schedule.

g. This campus has standardized on an existing standard ASHRAE 135, BACnet/IP Control System supported by a preselected controls service company. This entity is referred to as the "Control System Integrator" in this Section of the technical specifications. The Control system integrator is responsible for ECC system graphics and expansion. It also prescribes control system-specific commissioning/ verification procedures to the contractor administered by this Section of the technical specification. It lastly provides limited assistance to the contractor administered by this Section of the technical specification in its commissioning/verification work.

(1) The General Contractor of this project shall directly hire the Control System Integrator in a contract separate from the contract procuring the controls contractor administered by this Section of the technical specifications.

(2) The contractor administered by this Section of the technical specifications shall coordinate all work with the Control System Integrator. The contractor administered by this Section of the technical specifications shall integrate the ASHRAE 135, BACnet/IP control network(s) with the Control System Integrator's B-AWS through an Ethernet connection provided by either the Control System Integrator or VA.

(3) The contractor administered by this Section of the technical specifications shall provide a peer-to-peer networked, stand-alone, distributed control system. This direct digital control (DDC) system at least shall include one portable operator terminal - laptop, one digital display unit, microprocessor-based controllers, instrumentation, end control devices, wiring, piping, software, and related systems. This contractor is responsible for all device mounting and wiring.

(4) Responsibility Table:

Item/Task	**Section 23 09 23 contactor**	**Control system integrator**	**VA**
ECC expansion		X	
ECC programming		X	
Devices, controllers, control panels and equipment	X		
Point addressing: all hardware and software points including setpoint, calculated point, data point(analog/ binary), and reset schedule point	X		
Point mapping		X	
Network Programming	X		
ECC Graphics		X	
Controller programming and sequences	X		
Integrity of LAN communications			X
Electrical wiring	X		
Operator system training		X	
LAN connections to devices			X
LAN connections to ECC			X
IP addresses			X
Overall system verification (Cx)		X	
Controller and system verification	X		

h. Unitary standalone systems including Unit Heaters, Cabinet Unit Heaters, Fan Coil Units, Base Board Heaters, thermal comfort ventilation fans, and similar units for control of room environment conditions may be equipped with integral controls furnished and installed by the equipment manufacturer or field mounted. Refer to equipment specifications and as indicated in project documents. Application of standalone unitary controls is limited to at least those systems wherein remote monitoring, alarm and start-up are not necessary. Examples of such systems include:

(1) Light-switch-operated toilet exhaust

(2) Vestibule heater

(3) Exterior stair heater

(4) Attic heating and ventilation

(5) Mechanical or electrical room heating and ventilation.

i. The direct-digital control system shall start and stop equipment, move (position) damper actuators and valve actuators, and vary speed of equipment to execute the mission of the control system. Use electricity as the motive force for all damper and valve actuators, unless use of pneumatics as motive force is specifically granted in writing by the VA.

1.2 RELATED WORK

a. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

b. Not used.

c. Not used.

d. Section 23 21 13, Hydronic Piping.

e. Section 23 22 13, Steam and Condensate Heating Piping.

f. Section 23 31 00, HVAC Ducts and Casings.

g. Not used.

h. Not used.

i. Not used.

j. Not used.

k. Section 23 64 00, Packaged Water Chillers.

l. Section 23 73 00, Indoor Central-Station Air-Handling Units.

m. Not used.

n. Section 23 81 00, Decentralized Unitary HVAC Equipment.

o. Not used.

p. Section 23 81 43, Air-Source Unitary Heat Pumps.

q. Not used.

r. Not used.

s. Section 25 10 10, Advanced Utility Metering System.

t. Section 26 05 11, Requirements for Electrical Installations.

u. Not used.

v. Section 26 05 26, Grounding and Bonding for Electrical Systems.

w. Section 26 05 33, Raceway and Boxes for Electrical Systems.

x. Section 26 09 23, Lighting Controls.

y. Not used.

z. Section 26 27 26, Wiring Devices.

aa. Section 26 29 11, Motor Starters.

bb. Not used.

cc. Section 27 15 00, Communications Horizontal Cabling

dd. Section 28 31 00, Fire Detection and Alarm.

1.3 DEFINITION

a. Algorithm: A logical procedure for solving a recurrent mathematical problem; A prescribed set of well-defined rules or processes for the solution of a problem in a finite number of steps.

b. Analog: A continuously varying signal value (e.g., temperature, current, velocity etc.

c. BACnet: A Data Communication Protocol for Building Automation and Control Networks -as defined by ASHRAE 135. This communications protocol allows diverse building automation devices to communicate data and services over a network.

d. BACnet/IP: Annex J of Standard 135. It defines and allows for using a reserved UDP socket to transmit BACnet messages over IP networks. A BACnet/IP network is a collection of one or more IP sub-networks that share the same BACnet network number.

e. BACnet Internetwork: Two or more BACnet networks connected with routers. The two networks may use different LAN technologies.

f. BACnet Network: One or more BACnet segments that have the same network address and are interconnected by bridges at the physical and data link layers.

g. BACnet Segment: One or more physical segments of BACnet devices on a BACnet network, connected at the physical layer by repeaters.

h. BACnet Broadcast Management Device (BBMD): A communications device which broadcasts BACnet messages to all BACnet/IP devices and other BBMDs connected to the same BACnet/IP network.

i. BACnet Interoperability Building Blocks (BIBBs): BACnet Interoperability Building Blocks (BIBBs) are collections of one or more BACnet services. These are prescribed in terms of an "A" and a "B" device. Both of these devices are nodes on a BACnet internetwork.

j. BACnet Testing Laboratories (BTL). The organization responsible for testing products for compliance with the BACnet standard, operated under the direction of BACnet International.

k. Baud: It is a signal change in a communication link. One signal change can represent one or more bits of information depending on type of transmission scheme. Simple peripheral communication is normally one bit per Baud. (e.g., Baud rate = 78,000 Baud/sec is 78,000 bits/sec, if one signal change = 1 bit).

l. Binary: A two-state system where a high signal level represents an "ON" condition and an "OFF" condition is represented by a low signal

level.

m. BMP or bmp: Suffix, computerized image file, used after the period in a DOS-based computer file to show that the file is an image stored as a series of pixels.

n. Bus Topology: A network topology that physically interconnects workstations and network devices in parallel on a network segment.

o. Control Unit (CU): Generic term for any controlling unit, stand-alone, microprocessor based, digital controller residing on secondary LAN or Primary LAN, used for local controls or global controls

p. Deadband: A temperature range over which no heating or cooling is supplied, i.e., (72-78 degrees F), as opposed to a single point change over or overlap).

q. Device: a control system component that contains a BACnet Device Object and uses BACnet to communicate with other devices.

r. Device Object: Every BACnet device requires one Device Object, whose properties represent the network visible properties of that device. Every Device Object requires a unique Object Identifier number on the BACnet internetwork. This number is often referred to as the device instance.

s. Device Profile: A specific group of services describing BACnet capabilities of a device, as defined in ASHRAE 135-2008, Annex L. Standard device profiles include BACnet Operator Workstations (B-OWS), BACnet Building Controllers (B-BC), BACnet Advanced Application Controllers (B-AAC), BACnet Application Specific Controllers (B-ASC), BACnet Smart Actuator (B-SA), and BACnet Smart Sensor (B-SS). Each device used in new construction is required to have a PICS statement listing which service and BIBBs are supported by the device.

t. Diagnostic Program: A software test program, which is used to detect and report system or peripheral malfunctions and failures. Generally, this system is performed at the initial startup of the system.

u. Direct Digital Control (DDC): Microprocessor based control including Analog/Digital conversion and program logic. A control loop or subsystem in which digital and analog information is received and processed by a microprocessor, and digital control signals are generated based on control algorithms and transmitted to field devices in order to achieve a set of predefined conditions.

v. Distributed Control System: A system in which the processing of system data is decentralized and control decisions can and are made at the subsystem level. System operational programs and information are provided to the remote subsystems and status is reported back to the Engineering Control Center. Upon the loss of communication with the Engineering Control center, the subsystems shall be capable of operating in a stand-alone mode using the last best available data.

w. Download: The electronic transfer of programs and data files from a central computer or operation workstation with secondary memory devices to remote computers in a network (distributed) system.

x. DXF: An AutoCAD 2-D graphics file format. Many CAD systems import and export the DXF format for graphics interchange.

y. Electrical Control: A control circuit that operates on line or low voltage and uses a mechanical means, such as a temperature sensitive bimetal or bellows, to perform control functions, such as actuating a switch or positioning a potentiometer.

z. Electronic Control: A control circuit that operates on low voltage and uses a solid-state components to amplify input signals and perform control functions, such as operating a relay or providing an output signal to position an actuator.

aa. Engineering Control Center (ECC): The centralized control point for the intelligent control network. The ECC comprises of personal computer and connected devices to form a single workstation.

bb. Ethernet: A trademark for a system for exchanging messages between computers on a local area network using coaxial, fiber optic, or twisted-pair cables.

cc. Firmware: Firmware is software programmed into read only memory (ROM) chips. Software may not be changed without physically altering the chip.

dd. Gateway: Communication hardware connecting two or more different protocols. It translates one protocol into equivalent concepts for the other protocol. In BACnet applications, a gateway has BACnet on one side and non-BACnet (usually proprietary) protocols on the other side.

ee. GIF: Abbreviation of Graphic interchange format.

ff. Graphic Program (GP): Program used to produce images of air handler systems, fans, chillers, pumps, and building spaces. These images can be animated and/or color-coded to indicate operation of the equipment.

gg. Graphic Sequence of Operation: It is a graphical representation of the sequence of operation, showing all inputs and output logical blocks.

hh. I/O Unit: The section of a digital control system through which information is received and transmitted. I/O refers to analog input (AI, digital input (DI), analog output (AO) and digital output (DO). Analog signals are continuous and represent temperature, pressure, flow rate etc, whereas digital signals convert electronic signals to digital pulses (values), represent motor status, filter status, on-off equipment etc.

ii. I/P: a method for conveying and routing packets of information over LAN paths. User Datagram Protocol (UDP) conveys information to “sockets” without confirmation of receipt. Transmission Control Protocol (TCP) establishes "sessions", which have end-to-end confirmation and guaranteed sequence of delivery.

jj. JPEG: A standardized image compression mechanism stands for Joint Photographic Experts Group, the original name of the committee that wrote the standard.

kk. Local Area Network (LAN): A communication bus that interconnects operator workstation and digital controllers for peer-to-peer communications, sharing resources and exchanging information.

ll. Network Repeater: A device that receives data packet from one network and rebroadcasts to another network. No routing information is added to the protocol.

mm. MS/TP: Master-slave/token-passing (ISO 8802-3). It uses twisted-pair wiring for relatively low speed and low cost communication.

nn. Native BACnet Device: A device that uses BACnet as its primary method of communication with other BACnet devices without intermediary gateways. A system that uses native BACnet devices at all levels is a native BACnet system.

oo. Network Number: A site-specific number assigned to each network segment to identify for routing. This network number must be unique throughout the BACnet internetwork.

pp. Object: The concept of organizing BACnet information into standard components with various associated properties. Examples include analog input objects and binary output objects.

qq. Object Identifier: An object property used to identify the object, including object type and instance. Object Identifiers must be unique within a device.

rr. Object Properties: Attributes of an object. Examples include present value and high limit properties of an analog input object. Properties are defined in ASHRAE 135; some are optional and some are required. Objects are controlled by reading from and writing to object properties.

ss. Operating system (OS): Software, which controls the execution of computer application programs.

tt. PCX: File type for an image file. When photographs are scanned onto a personal computer they can be saved as PCX files and viewed or changed by a special application program as Photo Shop.

uu. Peripheral: Different components that make the control system function as one unit. Peripherals include monitor, printer, and I/O unit.

vv. Peer-to-Peer: A networking architecture that treats all network stations as equal partners- any device can initiate and respond to communication with other devices.

ww. PICS: Protocol Implementation Conformance Statement, describing the BACnet capabilities of a device. All BACnet devices have published PICS.

xx. PID: Proportional, integral, and derivative control, used to control modulating equipment to maintain a setpoint.

yy. Repeater: A network component that connects two or more physical segments at the physical layer.

zz. Router: a component that joins together two or more networks using different LAN technologies. Examples include joining a BACnet Ethernet LAN to a BACnet MS/TP LAN.

AAA. Sensors: devices measuring state points or flows, which are then transmitted back to the DDC system.

BBB. Thermostats : devices measuring temperatures, which are used in control of standalone or unitary systems and equipment not attached to the DDC system.

1.4 QUALITY ASSURANCE

a. Criteria:

(1) Single Source Responsibility of subcontractor: Either the DDC Contractor or the System Integrator shall obtain hardware and software supplied under this Section and delegate the responsibility to a single source controls installation subcontractor. The Integration subcontractor shall be responsible for the complete design, installation, integration, and commissioning of the system. The controls subcontractor shall be in the business of design, installation and service of such building automation control systems similar in size and complexity.

(2) Equipment and Materials: Equipment and materials shall be cataloged products of manufacturers regularly engaged in production and installation of HVAC control systems. Products shall be manufacturer’s latest standard design and have been tested and proven in actual use.

(3) The controls subcontractor shall provide a list of no less than five similar projects which have building control systems as specified in this Section. These projects must be on-line and functional such that the Department of Veterans Affairs (VA) representative could observe the control systems in full operation.

(4) The controls subcontractor shall have an in-place facility within 50 miles with technical staff, spare parts inventory for the next five (5) years, and necessary test and diagnostic equipment to support the control systems.

(5) The controls subcontractor shall have minimum of three years of experience in design and installation of building automation systems similar in performance to those specified in this Section.

(6) Provide a competent and experienced Project Manager employed by the Controls Contractor. The Project Manager shall be supported as necessary by other Contractor employees in order to provide professional engineering, technical and management service for the work. The Project Manager shall attend scheduled Project Meetings as required and shall be empowered to make technical, scheduling and related decisions on behalf of the Controls Contractor.

b. Codes and Standards:

(1) All work shall conform to the applicable Codes and Standards.

(2) Electronic equipment shall conform to the requirements of FCC Part 15, Governing Radio Frequency Electromagnetic Interference, and be so labeled.

1.5 PERFORMANCE

a. The system shall conform to the following:

(1) Graphic Display: The system shall display up to four (4) graphics on a single screen with a minimum of twenty (20) dynamic points per graphic. All current data shall be displayed within ten (10) seconds of the request.

(2) Graphic Refresh: The system shall update all dynamic points with current data within eight (8) seconds. Data refresh shall be automatic, without operator intervention.

(3) Object Command: The maximum time between the command of a binary object by the operator and the reaction by the device shall be two(2) seconds. Analog objects shall start to adjust within two (2) seconds.

(4) Object Scan: All changes of state and change of analog values shall be transmitted over the high-speed network such that any data used or displayed at a controller or work-station will be current, within the prior six (6) seconds.

(5) Alarm Response Time: The maximum time from when an object goes into alarm to when it is annunciated at the workstation shall not exceed (10) seconds.

(6) Program Execution Frequency: Custom and standard applications shall be capable of running as often as once every (5) seconds. The Contractor shall be responsible for selecting execution times consistent with the mechanical process under control.

(7) Multiple Alarm Annunciations: All workstations on the network shall receive alarms within five (5) seconds of each other.

(8) Performance: Programmable Controllers shall be able to execute DDC PID control loops at a selectable frequency from at least once every one (1) second. The controller shall scan and update the process value and output generated by this calculation at this same frequency.

(9) Reporting Accuracy: Listed below are minimum acceptable reporting end-to-end accuracies for all values reported by the specified system:

Measured Variable	**Reported Accuracy**
Space temperature	±0.5 degrees C (±1 degrees F)
Ducted air temperature	±1 degrees F
Outdoor air temperature	±2 degrees F
Dew Point	±3 degrees F
Water temperature	±1 degrees F
Relative humidity	±2 percent RH

Water flow	±1 percent of reading
Air flow (terminal)	±10 percent of reading
Air flow (measuring stations)	±5 percent of reading
Carbon Monoxide (CO)	±5 percent of reading
Carbon Dioxide (CO 2)	±50 ppm
Air pressure (ducts)	±25 Pa ±0.1 inch w.c.
Air pressure (space)	±0.3 Pa ±0.001 inch w.c.
Water pressure	±2 percent of full scale *Note 1
Electrical Power	±0.5 percent of reading

Note 1: for both absolute and differential pressure

10. Control stability and accuracy: Control sequences shall maintain measured variable at setpoint within the following tolerances:

Controlled Variable	**Control Accuracy**	**Range of Medium**
Air Pressure	(±0.2 inch wg)	(0–6 inch wg)
Air Pressure	±3 Pa (±0.01 in. wg)	(-0.1 to 0.1 inch wg)
Airflow	±10 percent of full scale	
Space Temperature	(±2.0 degrees F)	
Duct Temperature	(±3 degrees F)	
Humidity	±5 percent RH	MRI, SPS, PHARMACY
Fluid Pressure	(±1.5 psig)	(1–150 psig)
Fluid Pressure	(±1.0 inch wg)	(0–50 inch wg) differential

11. Extent of direct digital control: control design shall allow for at least the points indicated on the points lists on the drawings.

1.6 WARRANTY

a. Labor and materials for control systems shall be warranted for a period as specified under Warranty in FAR clause 52.246-21.

b. Control system failures during the warranty period shall be adjusted, repaired, or replaced at no cost or reduction in service to the owner. The system includes all computer equipment, transmission equipment, and all sensors and control devices.

c. The on-line support service shall allow the Controls supplier to dial out over telephone lines to or connect via (through password-limited access) VPN through the internet to monitor and control the facility’s building automation system. This remote connection to the facility shall be within two (2) hours of the time that the problem is

reported. This coverage shall include normal business hours, after business hours, weekend and holidays. If the problem cannot be resolved with on-line support services, the Controls supplier shall dispatch the qualified personnel to the job site to resolve the problem within 8 hours after the problem is reported.

d. Controls subcontractor shall be responsible for temporary operations and maintenance of the control systems during the construction period until final commissioning, training of facility operators and acceptance of the project by VA.

1.7 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer’s Literature And Data; G, AE

SD-05 Design Data

Licenses; G, AE

SD-06 Test Reports

Validation Test Plan; G, AE

Performance Report; G, AE

SD-07 Certificates

Product Certificates; G, AE

SD-10 Operation and Maintenance Data

Operation And Maintenance (O/M) Manuals); G, AE

SD-11 Closeout Submittals

As Built Control Drawings; G, AE

1.8 SUBMITTAL INSTRUCTIONS

a. Submit shop drawings in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC”, with applicable paragraph identification.

c. Manufacturer’s Literature and Data Including:

(1) A wiring diagram for each type of input device and output device including DDC controllers, modems, repeaters, etc. Diagram shall show how the device is wired and powered, showing typical connections at the digital controllers and each power supply, as

well as the device itself. Show for all field connected devices, including but not limited to, control relays, motor starters, electric or electronic actuators, and temperature pressure, flow and humidity sensors and transmitters.

(2) A diagram of each terminal strip, including digital controller terminal strips, terminal strip location, termination numbers and the associated point names.

(3) Control dampers and control valves schedule, including the size and pressure drop.

(4) Control air-supply components, and computations for sizing compressors, receivers and main air-piping, if pneumatic controls are furnished.

(5) Catalog cut sheets of all equipment used. This includes, but is not limited to software (by manufacturer and by third parties), DDC controllers, panels, peripherals, airflow measuring stations and associated components, and auxiliary control devices such as sensors, actuators, and control dampers. When manufacturer’s cut sheets apply to a product series rather than a specific product, the data specifically applicable to the project shall be highlighted. Each submitted piece of literature and drawings should clearly reference the specification and/or drawings that it supposed to represent.

(6) Sequence of operations for each system and the associated control diagrams. Equipment and control labels shall correspond to those shown on the drawings.

(7) Color prints of proposed graphics with a list of points for display.

(8) Furnish a BACnet Protocol Implementation Conformance Statement (PICS) for each BACnet-compliant device.

(9) Schematic wiring diagrams for all control, communication and power wiring. Provide a schematic drawing of the central system installation. Label all cables and ports with computer manufacturers’ model numbers and functions. Show all interface wiring to the control system.

(10) An instrumentation list for each controlled system. Each element of the controlled system shall be listed in table format. The table shall show element name, type of device, manufacturer, model number, and product data sheet number.

(11) Riser diagrams of wiring between central control unit (CCU) and all control panels.

(12) Plan drawings showing routing of LAN and locations of control panels, controllers, routers, gateways, ECC, and larger controlled devices.

(13) Construction details for all installed conduit, cabling, raceway, cabinets, and similar. Construction details of all penetrations and their protection.

(14) Quantities of submitted items may be reviewed but it is the responsibility of the contractor administered by this Section of the technical specifications to provide sufficient quantities for a complete and working system.

d. Product Certificates: Compliance with Article, QUALITY ASSURANCE.

e. Validation Test Plan: Along with manufacturer's literature, product certificates, wiring and functional diagrams, and sequence of operations, submit for review and approval a Validation Test Plan which is specific to the work of this project and which references the specific controls component nomenclatures found in the control contractor's wiring and functional diagrams and the specific sequences of controls for this project and which describes how the contractor will implement the controls system validation and demonstration as specified in paragraph 3.2 SYSTEM VALIDATION AND DEMONSTRATION found in this section.

f. Licenses: Provide licenses for all software residing on and used by the Controls Systems, ECC, and portable OWS and transfer these licenses to the Owner prior to completion.

g. As Built Control Drawings:

(1) Furnish three (3) copies of as-built drawings for each control system. The documents shall be submitted for approval prior to final completion.

(2) Furnish one (1) set of applicable control system prints for each mechanical system for wall mounting. The documents shall be submitted for approval prior to final completion.

(3) Furnish one (1) CD-ROM in CAD DWG and/or.DXF format for the drawings noted in subparagraphs above.

h. Operation and Maintenance (O/M) Manuals):

(1) Submit in accordance with Article, INSTRUCTIONS, in Specification Section 01 00 00, GENERAL REQUIREMENTS.

(2) Include the following documentation:

(a) General description and specifications for all components, including logging on/off, alarm handling, producing trend reports, overriding computer control, and changing set points and other variables.

(b) Detailed illustrations of all the control systems specified for ease of maintenance and repair/replacement procedures, and complete calibration procedures.

(c) One copy of the final version of all software provided including operating systems, programming language, operator workstation software, and graphics software.

(d) Complete troubleshooting procedures and guidelines for all systems.

(e) Complete operating instructions for all systems.

(f) Recommended preventive maintenance procedures for all system components including a schedule of tasks for inspection, cleaning and calibration. Provide a list of recommended spare parts needed to minimize downtime.

(g) Training Manuals: Submit the course outline and training material to the Owner for approval three (3) weeks prior to the training to VA facility personnel. These persons will be responsible for maintaining and the operation of the control systems, including programming. The Owner reserves the right to modify any or all of the course outline and training material.

(h) Licenses, guaranty, and other pertaining documents for all equipment and systems.

i. Submit Performance Report to Contracting Officer prior to final inspection.

1.9 INSTRUCTIONS

a. Instructions to VA operations personnel: Perform in accordance with Article, INSTRUCTIONS, in Specification Section 01 00 00, GENERAL REQUIREMENTS, and as noted below.

(1) First Phase: Formal instructions to the VA facilities personnel for a total of 16 32 hours, given in multiple training sessions (each no longer than four hours in length), conducted sometime between the completed installation and prior to the performance test period of the control system, at a time mutually agreeable to the Contractor and the VA.

(2) Second Phase: This phase of training shall comprise of on the job training during startup, checkout period, and performance test period. VA facilities personnel will work with the Contractor’s installation and test personnel on a daily basis during startup and checkout period. During the performance test period, controls subcontractor will provide 8 hours of instructions, given in multiple training sessions (each no longer than four hours in length), to the VA facilities personnel.

(3) The O/M Manuals shall contain approved submittals as outlined in Article 1.7, SUBMITTALS. The Controls subcontractor will review the manual contents with VA facilities personnel during second phase of training.

(4) Training shall be given by direct employees of the controls system subcontractor.

1.10 PROJECT CONDITIONS (ENVIRONMENTAL CONDITIONS OF OPERATION)

a. The ECC and peripheral devices and system support equipment shall be designed to operate in ambient condition of 20 to 35 degrees C (65 to 90 degrees F) at a relative humidity of 20 to 80 percent non-condensing.

b. The Controllers used outdoors shall be mounted in NEMA 4 waterproof enclosures, and shall be rated for operation at –40 to 65 degrees C

(-40 to 150 degrees F).

c. All electronic equipment shall operate properly with power fluctuations of plus 10 percent to minus 15 percent of nominal supply voltage.

d. Sensors and controlling devices shall be designed to operate in the environment, which they are sensing or controlling.

1.11 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI INCITS 154	(1988; R 2004) Office Machines and Supplies - Alphanumeric Machines - Keyboard Arrangement

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 135	(2020; Interpretation 1-8 2021; Errata 1-2 2021; Addenda CD 2021; Addenda BV-CE 2022; Interpretation 9-12 2022; Interpretation 13-24 2023; Addenda BV-CF 2023; Errata 3 2023) BACnet—A Data Communication Protocol for Building Automation and Control Networks
ASHRAE 147	(2019) Reducing the Release of Halogenated Refrigerants from Refrigerating and Air-Conditioning Equipment and Systems

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-485	(1998a; R 2012) Electrical Characteristics of Generators and Receivers for Use in Balanced Digital Multipoint Systems
TIA-568.0	(2020e) Generic Telecommunications Cabling for Customer Premises

U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.3	(2022; CORR 2024) Ethernet

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 8802-3	(2021) Telecommunications and exchange between information technology systems — Requirements for local and metropolitan area networks — Part 3: Standard for Ethernet

U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

FCC Part 68	Connection of Terminal Equipment to the Telephone Network (47 CFR 68)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 262	(2023) Standard Method of Test for Flame Travel and Smoke of Wires and Cables for Use in Air-Handling Spaces

PART 2 – PRODUCS

2.1 MATERIALS

Use new products that the manufacturer is currently manufacturing and that have been installed in a minimum of **25** installations. Spare parts shall be available for at least **five** years after completion of this contract.

2.2 CONTROLS SYSTEM ARCHITECTURE

a. General

(1) The Controls Systems shall consist of multiple Nodes and associated equipment connected by industry standard digital and communication network arrangements.

(2) The ECC, building controllers and principal communications network equipment shall be standard products of recognized major manufacturers available through normal PC and computer vendor channels – not "Clones" assembled by a third-party subcontractor.

(3) The networks shall, at minimum, comprise, as necessary, the following:

(a) A fixed ECC and a portable operator’s terminal.

(b) Network computer processing, data storage and BACnet-compliant communication equipment including Servers and digital data processors.

(c) BACnet-compliant routers, bridges, switches, hubs, modems, gateways, interfaces and similar communication equipment.

(d) Active processing BACnet-compliant building controllers connected to other BACNet-compliant controllers together with their power supplies and associated equipment.

(e) Addressable elements, sensors, transducers and end devices.

(f) Third-party equipment interfaces and gateways as described and required by the Contract Documents.

(g) Other components required for a complete and working Control Systems as specified.

b. The Specifications for the individual elements and component subsystems shall be minimum requirements and shall be augmented as necessary by the Contractor to achieve both compliance with all applicable codes, standards, and to meet all requirements of the Contract Documents.

c. Network Architecture

(1) The Controls communication network shall utilize BACnet communications protocol operating over a standard Ethernet LAN and operate at a minimum speed of 100 Mb/sec.

(2) The networks shall utilize only copper and optical fiber communication media as appropriate and shall comply with applicable codes, ordinances and regulations.They may also utilize digital wireless technologies as appropriate to the application and if approved by the VA.

(3) All necessary telephone lines, ISDN lines and internet Service Provider services and connections will be provided by the VA.

d. Third Party Interfaces:

(1) The contractor administered by this Section of the technical specifications shall include necessary hardware, equipment, software and programming to allow data communications between the controls systems and building systems supplied by other trades.

(2) Other manufacturers and contractors supplying other associated systems and equipment shall provide their necessary hardware, software and start-up at their cost and shall cooperate fully with the contractor administered by this Section of the technical specifications in a timely manner and at their cost to ensure complete functional integration.

e. Servers:

(1) Provide data storage server(s) to archive historical data including trends, alarm and event histories and transaction logs.

(2) Equip these server(s) with the same software tool set that is located in the BACnet building controllers for system configuration and custom logic definition and color graphic configuration.

(3) Access to all information on the data storage server(s) shall be through the same browser functionality used to access individual nodes. When logged onto a server the operator will be able to also interact with any other controller on the control system as required for the functional operation of the controls systems. The contractor administered by this Section of the technical specifications shall provide all necessary digital processor programmable data storage server(s).

(4) These server(s) shall be utilized for controls systems application configuration, for archiving, reporting and trending of data, for operator transaction archiving and reporting, for network information management, for alarm annunciation, for operator interface tasks, for controls application management and similar.

These server(s) shall utilize IT industry standard data base platforms which utilize a database declarative language designed for managing data in relational database management systems (RDBMS) such as SQL.

2.3 COMMUNICATION

a. Control products, communication media, connectors, repeaters, hubs, and routers shall comprise a BACnet internetwork. Controller and operator interface communication shall conform to ASHRAE 135, BACnet.

(1) The Data link / physical layer protocol between the ECC and all B-BC's (for communication) acceptable to the VA throughout its facilities is Ethernet (ISO 8802-3) and BACnet/IP.

(3) The MS/TP data link / physical layer protocol is prohibited to the VA in any new BACnet network or sub-network in its healthcare or lab facilities.

b. Each controller shall have a communication port for connection to an operator interface.

c. Project drawings indicate remote buildings or sites to be connected by a nominal 56,000 baud modem over voice-grade telephone lines. In each remote location a modem and field device connection shall allow communication with each controller on the internetwork as specified in Paragraph D.

d. Internetwork operator interface and value passing shall be transparent to internetwork architecture.

(1) An operator interface connected to a controller shall allow the operator to interface with each internetwork controller as if directly connected. Controller information such as data, status, reports, system software, and custom programs shall be viewable and editable from each internet controller.

(2) Inputs, outputs, and control variables used to integrate control strategies across multiple controllers shall be readable by each controller on the internetwork. Program and test all cross-controller links required to execute specified control system operation. An authorized operator shall be able to edit cross-controller links by typing a standard object address.

e. System shall be expandable to at least twice the required input and output objects with additional controllers, associated devices, and wiring. Expansion shall not require operator interface hardware additions or software revisions.

f. ECCs and Controllers with real-time clocks shall use the BACnet Time Synchronization service. The system shall automatically synchronize system clocks daily from an operator-designated device via the internetwork. The system shall automatically adjust for daylight savings and standard time as applicable.

2.4 ENGINEERING CONTROL CENTER (ECC)

a. The ECC shall reside on a high-speed network with controllers as shown on system drawings. The ECC and each standard browser connected to server shall be able to access all system information.

b. ECC and controllers shall communicate using BACnet protocol. ECC and control network backbone shall communicate using ISO 8802-3 (Ethernet) Data Link/Physical layer protocol and BACnet/IP addressing as specified in ASHRAE 135, BACnet Annex J.

c. Hardware: ECC shall conform to the BACnet Advanced Workstation (B-AWS) Profile and shall be BTL-Listed as a B-AWS device.

(1) ECC shall be based on commercially available server grade hardware. Computers based on desktop architectures shall not be permitted. ECC shall have remote management capabilities.

(2) Processor(s):

(a) Processors shall be either Intel Xeon or AMD EPYC chips sets designed specifically for server use. Desktop processors will not be allowed.

(b) Minimum core count shall be 16 cores. Each Core shall be capable of executing 2 threads simultaneously.

Minimum base clock speed shall be 3.0 GHz

(3) Memory:

(a) Engineering Control Center shall be equipped with a minimum of 32G of DDR4 Error Correcting Code (ECC) memory. After installation of required 32G of RAM, the motherboard must still have a minimum of 2 DIMM slots open for expansion. Motherboard shall be capable of minimum of 64GB of ECC memory.

(b) Minimum speed shall be 2133MHz for memory

(4) Storage:

(a) ECC shall be equipped with a RAID capable drive controller capable of handling at least 8 internal, hot swappable drives.

(b) All installed drives shall be “Enterprise Class” drives designed specifically for server use.

(c) Minimum configuration shall consist of 6 drives in 2 separate RAID arrays:

1. The operating system shall be stored on a RAID 1 array compromising of 2 drives with each drive having a minimum capacity of 1TB each.

2 The B-AWS software and all its related databases shall be stored in RAID 5 array consisting of 4 drives with each drive have a minimum of 4TB capacity for a minimum storage capacit of 12TB’s.

3. An alternative configuration for the 2 RAID arrays described above is 8 drives arrayed in a RAID 10 configuration. In this case the OS RAID array would consist of 4 drives of 1TB minimum each, and the B-AWS RAID array consisting of 4 drives of 16TB minimum each. This configuration would provide faster write times and much quicker rebuild times in the event of a drive failure.

(d) ECC will include an 16X DVD R/W drive

(5) Case:

(a) Case shall have space for a minimum of 8 hot swap 3.5 inch hard drives and one internal optical drive

(b) Real-time clock:

1. Accuracy: Plus or minus 1 minute per month.

2. Time Keeping Format: 24-hour time format including seconds, minutes, hours, date, day, and month; automatic reset by software.

3. Clock shall function for one year without power.

4. Provide automatic time correction once every 24 hours by synchronizing clock with the Time Service Department of the U.S. Naval Observatory.

(c) Serial ports: Four USB ports and two RS-232-F serial ports for general use, with additional ports as required. Data transmission rates shall be selectable under program control.

(d) Parallel port: Enhanced.

(e) Sound card: For playback and recording of digital WAV sound files associated with audible warning and alarm functions.

(f) Color monitor: PC compatible, not less than 22 inches, LCD type, with a minimum resolution of 1280 by 1024 pixels, non-interlaced, and a maximum dot pitch of 0.28 mm.

(g) Keyboard: Minimum of 64 characters, standard ASCII character set based on ANSI INCITS 154.

(h) Mouse: Standard, compatible with installed software.

(i) Removable disk storage: Include the following, each with appropriate controller:

1)Minimum 1 TB removable hard disk, maximum average access time of 10 ms.

(j) Network interface card (NIC): integrated 10-100-1000 Base-TX Ethernet NIC with an RJ45 connector or a 100Base-FX Ethernet NIC with an SC/ST connector.

(6) Cable modem: 42.88 MBit/s, DOCSIS 2.0 Certified, also backwards compatible with DOCSIS 1.1/1.0 standards. Provide Ethernet or USB

connectivity.

(7) Optical modem: full duplex link, for use on 10 GBase-R single-mode and multi-mode fiber with a XENPAK module.

(8) Auto-dial modem: 56,600 bps, full duplex for asynchronous communications. With error detection, auto answer/autodial, and call-in-progress detection. Modem shall comply with requirements in ITU-T V.34, ITU-T V.42, ITU-T V.42 Appendix VI for error correction, and ITU-T V.42 BIS for data compression standards; and shall be suitable for operating on unconditioned voice-grade telephone lines complying with FCC Part 68.

(9) Audible Alarm: Manufacturer's standard.

(10) Printers:

(a) Provide a dedicated, minimum resolution 600 dpi, color laser printer, connected to the ECC through a USB interface.

1. If a network printer is used instead of this dedicated printer, it shall have a 100Base-T interface with an RJ45 connection and shall have a firmware print spooler compatible with the Operating System print spooler.

2. RAM: 512 MB, minimum.

3. Printing Speed: Minimum twenty-six pages per minute (color); minimum 30 pages per minute (black/white).

4. Paper Handling: Automatic sheet feeder with 250-sheet x 8.5 inch x 11 inch paper cassette and with automatic feed.

(b) Provide a dedicated black/white tractor-feed dot matrix printer for status/alarm message printing, minimum 10 characters per inch, minimum 160 characters per second, connected to the ECC through a USB interface.

1. Paper: One box of 2000 sheets of 8-1/2 x 11 multi-fold type printer paper.

(11) RS-232 ASCII Interface

(a) ASCII interface shall allow RS-232 connections to be made between a meter or circuit monitor operating as the host PC and any equipment that will accept RS-232 ASCII command strings, such as local display panels, dial-up modems, and alarm transmitters.

(b) Pager System Interface: Alarms shall be able to activate a pager system with customized message for each input alarm.

(c) Alarm System Interface: RS-232 output shall be capable of transmitting alarms from other monitoring and alarm systems to workstation software.

(d) RS-232 output shall be capable of connection to a pager interface that can be used to call a paging system or service and send a signal to a portable pager. System shall allow an individual alphanumeric message per alarm input to be sent to

paging system. This interface shall support both numeric and alphanumeric pagers.

(e) Cables: provide Plenum-Type, RS-232 Cable: Paired, 2 pairs, No. 22 AWG, stranded (7x30) tinned copper conductors, plastic insulation, and individual aluminum foil-polyester tape shielded pairs with 100 percent shield coverage; plastic jacket. Pairs are cabled on common axis with No. 24 AWG, stranded (7x32) tinned copper drain wire.

1. NFPA 70, Type CMP.

2. Flame Resistance: NFPA 262, Flame Test.

(12) Self-contained uninterruptible power supply (UPS):

(a) Size: Provide a minimum of six hours of operation of ECC equipment, including two hours of alarm printer operation.

(b) Batteries: Sealed, valve regulated, recombinant, lead calcium.

(c) Accessories:

1. Transient voltage suppression.

2. Input-harmonics reduction.

3. Rectifier/charger.

4. Battery disconnect device.

5. Static bypass transfer switch.

6. Internal maintenance bypass/isolation switch.

7. External maintenance bypass/isolation switch.

8. Output isolation transformer.

9. Remote UPS monitoring.

10. Battery monitoring.

11. Remote battery monitoring.

d. ECC Software:

(1) Provide for automatic system database save and restore on the ECC’s hard disk a copy of the current database of each Controller. This database shall be updated whenever a change is made in any system panel. In the event of a database loss in a building management panel, the ECC shall automatically restore the database for that panel. This capability may be disabled by the operator.

(2) Provide for manual database save and restore. An operator with proper clearance shall be able to save the database from any system panel. The operator also shall be able to clear a panel database and manually initiate a download of a specified database

to any panel in the system.

(3) Provide a method of configuring the system. This shall allow for future system changes or additions by users with proper clearance.

(4) Operating System. Furnish a concurrent multi-tasking operating system. The operating system also shall support the use of other common software applications. Acceptable operating systems are Windows Server 2019, Linux, and UNIX.

(5) System Graphics. The operator workstation software shall be graphically oriented. The system shall allow display of up to 10 graphic screens at once for comparison and monitoring of system status. Provide a method for the operator to easily move between graphic displays and change the size and location of graphic displays on the screen. The system graphics shall be able to be modified while on-line. An operator with the proper password level shall be able to add, delete, or change dynamic objects on a graphic. Dynamic objects shall include analog and binary values, dynamic text, static text, and animation files. Graphics shall have the ability to show animation by shifting image files based on the status of the object.

(6) Custom Graphics. Custom graphic files shall be created with the use of a graphics generation package furnished with the system. The graphics generation package shall be a graphically based system that uses the mouse to create and modify graphics that are saved in industry standard formats such as PCX, TIFF, and GEM. The graphics generation package also shall provide the capability of capturing or converting graphics from other programs such as Designer or AutoCAD.

(7) Graphics Library. Furnish a complete library of standard HVAC equipment graphics such as chillers, boilers, air handlers, terminals, fan coils, and unit ventilators. This library also shall include standard symbols for other equipment including fans, pumps, coils, valves, piping, dampers, and ductwork. The library shall be furnished in a file format compatible with the graphics generation package program.

(8) The Controls Systems Operator Interfaces shall be user friendly, readily understood and shall make maximum use of colors, graphics, icons, embedded images, animation, text based information and data visualization techniques to enhance and simplify the use and understanding of the displays by authorized users at the ECC. The operating system shall be Windows 10 latest version, or better, and shall support the third party software.

(9) Provide graphical user software, which shall minimize the use of keyboard through the use of the mouse and "point and click" approach to menu selection.

(10) The software shall provide a multi-tasking type environment that will allow the user to run several applications simultaneously. The mouse or Alt-Tab keys shall be used to quickly select and switch between multiple applications. The operator shall be able automatically export data to and work in Microsoft Word, Excel, and other Windows based software programs, while concurrently on-line system alarms and monitoring information.

(11) On-Line Help. Provide a context-sensitive, on-line help system to assist the operator in operating and editing the system. On-line help shall be available for all applications and shall provide the relevant data for that particular screen. Additional help information shall be available through the use of hypertext.

(12) User access shall be protected by a flexible and Owner re-definable software-based password access protection. Password protection shall be multi-level and partition able to accommodate the varied access requirements of the different user groups to which individual users may be assigned. Provide the means to define unique access privileges for each individual authorized user. Provide the means to on-line manage password access control under the control of a project specific Master Password. Provide an audit trail of all user activity on the Controls Systems including all actions and changes.

(13) The system shall be completely field-programmable from the common operator’s keyboard thus allowing hard disk storage of all data automatically. All programs for the CUs shall be able to be downloaded from the hard disk. The software shall provide the following functionality as a minimum:

(a) Point database editing, storage and downloading of controller databases.

(b) Scheduling and override of building environmental control systems.

(c) Collection and analysis of historical data.

(d) Alarm reporting, routing, messaging, and acknowledgement.

(e) Definition and construction of dynamic color graphic displays.

(f) Real-time graphical viewing and control of environment.

(g) Scheduling trend reports.

(h) Program editing.

(i) Operating activity log and system security.

(j) Transfer data to third party software.

(14) Provide functionality such that using the least number of steps to initiate the desired event may perform any of the following simultaneously:

(a) Dynamic color graphics and graphic control.

(b) Alarm management.

(c) Event scheduling.

(d) Dynamic trend definition and presentation.

(e) Program and database editing.

(f) Each operator shall be required to log on to the system with a username and password to view, edit or delete the data. System security shall be selectable for each operator, and the password shall be able to restrict the operator's access for viewing and changing the system programs. Each operator shall automatically be logged off the system if no keyboard or mouse activity is detected for a selected time.

(15) Graphic Displays:

(a) The workstation shall allow the operator to access various system schematics and floor plans via a graphical penetration scheme, menu selection, or text-based commands. Graphic software shall permit the importing of AutoCAD or scanned pictures in the industry standard format (such as PCX, BMP, GIF, and JPEG) for use in the system.

(b) System Graphics shall be project specific and schematically correct for each system. (i.e.: coils, fans, dampers located per equipment supplied with project.) Standard system graphics that do not match equipment or system configurations are prohibited. Operator shall have capability to manually operate the entire system from each graphic screen at the ECC. Each system graphic shall include a button/tab to a display of the applicable sequence of operation.

(c) Dynamic temperature values, humidity values, flow rates, and status indication shall be shown in their locations and shall automatically update to represent current conditions without operator intervention and without pre-defined screen refresh values.

(d) Color shall be used to indicate status and change in status of the equipment. The state colors shall be user definable.

(e) A clipart library of HVAC equipment, such as chillers, boilers, air handling units, fans, terminal units, pumps, coils, standard ductwork, piping, valves and laboratory symbols shall be provided in the system. The operator shall have the ability to add custom symbols to the clipart library.

(f) A dynamic display of the site-specific architecture showing status of the controllers, the ECC and network shall be provided.

(g) The windowing environment of the workstation shall allow the user to simultaneously view several applications at a time to analyze total building operation or to allow the display of graphic associated with an alarm to be viewed without interrupting work in progress. The graphic system software shall also have the capability to split screen, half portion of the screen with graphical representation and the other half with sequence of operation of the same HVAC system.

(16) Trend reports shall be generated on demand or pre-defined schedule and directed to monitor display, printers or disk. As a minimum, the system shall allow the operator to easily obtain the following types of reports:

(a) A general list of all selected points in the network.

(b) List of all points in the alarm.

(c) List of all points in the override status.

(d) List of all disabled points.

(e) List of all points currently locked out.

(f) List of user accounts and password access levels.

(g) List of weekly schedules.

(h) List of holiday programming.

(i) List of limits and dead bands.

(j) Custom reports.

(k) System diagnostic reports, including, list of digital controllers on the network.

(l) List of programs.

(17) ASHRAE 147 Report: Provide a daily report that shows the operating condition of each chiller as recommended by ASHRAE 147. At a minimum, this report shall include:

(a) Chilled water (or other secondary coolant) inlet and outlet temperature

(b) Chilled water (or other secondary coolant) flow

(c) Chilled water (or other secondary coolant) inlet and outlet pressures

(d) Evaporator refrigerant pressure and temperature

(e) Condenser refrigerant pressure and liquid temperature

(f) Condenser water inlet and outlet temperatures

(g) Condenser water flow

(h) Refrigerant levels

(i) Oil pressure and temperature

(j) Oil level

(k) Compressor refrigerant discharge temperature

(l) Compressor refrigerant suction temperature

(m) Addition of refrigerant

(n) Addition of oil

(o) Vibration levels or observation that vibration is not excessive

(p) Motor amperes per phase

(q) Motor volts per phase

(r) PPM refrigerant monitor level

(s) Purge exhaust time or discharge count

(t) Ambient temperature (dry-bulb and wet-bulb)

(u) Date and time logged

(18) Electrical, Gas, and Weather Reports

(a) Electrical Meter Report: Provide a monthly report showing the daily electrical consumption and peak electrical demand with time and date stamp for each building meter.

(b) Provide an annual (12-month) summary report showing the monthly electrical consumption and peak demand with time and date stamp for each meter.

(c) Gas Meter Report: Provide a monthly report showing the daily natural gas consumption for each meter. Provide an annual (12-month) report that shows the monthly consumption for each meter.

(d) Weather Data Report: Provide a monthly report showing the daily minimum, maximum, and average outdoor air temperature, as well as the number of heating and cooling degree-days for each day. Provide an annual (12-month) report showing the minimum, maximum, and average outdoor air temperature for the month, as well as the number of heating and cooling degree-days for the month.

(19) Scheduling and Override:

(a) Provide override access through menu selection from the graphical interface and through a function key.

(b) Provide a calendar type format for time-of-day scheduling and overrides of building control systems. Schedules reside in the ECC. The digital controllers shall ensure equipment time scheduling when the ECC is off-line. The ECC shall not be required to execute time scheduling. Provide the following spreadsheet graphics as a minimum:

1. Weekly schedules.

2. Zone schedules, minimum of 100 zones.

3. Scheduling up to 365 days in advance.

4. Scheduled reports to print at workstation.

(20) Collection and Analysis of Historical Data:

(a) Provide trending capabilities that will allow the operator to monitor and store records of system activity over an extended period. Points may be trended automatically on time-based intervals or change of value, both of which shall be user definable. The trend interval could be five (5) minutes to 120 hours. Trend data may be stored on hard disk for future diagnostic and reporting. Additionally trend data may be archived to network drives or removable disk media for off-site retrieval.

(b) Reports may be customized to include individual points or predefined groups of at least six points. Provide additional functionality to allow pre-defined groups of up to 250 trended points to be easily accessible by other industry standard word processing and spreadsheet packages. The reports shall be time and date stamped and shall contain a report title and the name of the facility.

(c) System shall have the set up to generate spreadsheet reports to track energy usage and cost based on weekly or monthly interval, equipment run times, equipment efficiency, and/or building environmental conditions.

(d) Provide additional functionality that will allow the operator to view real time trend data on trend graph displays. A minimum of 20 points may be graphed regardless of whether they have been predefined for trending. In addition, the user may pause the graph and take snapshots of the screens to be stored on the workstation disk for future reference and trend analysis. Exact point values may be viewed, and the graph may be printed. Operator shall be able to command points directly on the trend plot by double clicking on the point.

(21) Alarm Management:

(a) Alarm routing shall allow the operator to send alarm notification to selected printers or operator workstation based on time of day, alarm severity, or point type.

(b) Alarm notification shall be provided via two alarm icons, to distinguish between routine, maintenance type alarms and critical alarms. The critical alarms shall display on the screen at the time of its occurrence, while others shall display by clicking on their icon.

(c) Alarm display shall list the alarms with highest priority at the top of the display. The alarm display shall provide selector buttons for display of the associated point graphic and message in English language. The operator shall be able to sort out the alarms.

(d) Alarm messages shall be customized for each point to display detailed instructions to the operator regarding actions to take in the event of an alarm.

(e) An operator with proper security level access may acknowledge and clear the alarm. All that have not been cleared shall be archived at workstation disk.

(22) Remote Communications: The system shall have the ability to dial

out in the event of an alarm. Receivers shall include operator workstations, e-mail addresses, and alpha-numeric pagers. The alarm message shall include the name of the calling location, the device that generated the alarm, and the alarm message itself.

(23) System Configuration:

(a) Network control strategies shall not be restricted to a single digital controller but shall be able to include data from all other network devices to allow the development of global control strategies.

(b) Provide automatic backup and restore of all digital controller databases on the workstation hard disk. In addition to all backup data, all databases shall be performed while the workstation is on-line without disturbing other system operations.

2.5 PORTABLE OPERATOR'S TERMINAL (POT)

a. Provide a portable operator's terminal (POT) that shall be capable of accessing all system data. POT may be connected to any point on the system network or may be connected directly to any controller for programming, setup, and troubleshooting. POT shall communicate using BACnet protocol. POT may be connected to any point on the system network or it may be connected directly to controllers using the BACnet PTP (Point-To-Point) Data Link/ Physical layer protocol. The terminal shall use the Read (Initiate) and Write (Execute) BACnet Services. POT shall be an IBM-compatible notebook-style PC including all software and hardware required.

b. Hardware: POT shall conform to the BACnet Advanced Workstation (B-AWS) Profile and shall be BTL-Listed as a B-AWS device.

(1) POT shall be commercial standard with supporting 32- or 64-bit hardware (as limited by the direct-digital control system software) and software enterprise server. Internet Explorer v6.0 SP1 or higher, Windows Script Hosting version 5.6 or higher, Windows Message Queuing, Windows Internet Information Services (IIS) v5.0 or higher, minimum 2.8 GHz processor, minimum 500 GB 7200 rpm SATA hard drive with 16 MB cache, minimum 2GB DDR3 SDRAM (minimum 1333 Mhz) memory, 512 MB video card, minimum 16 inch (diagonal) screen, 10-100-1000 Base-TX Ethernet NIC with an RJ45 connector or a 100Base-FX Ethernet NIC with an SC/ST connector, 56,600 bps modem, an ASCII RS-232 interface, and a 16 speed high density DVD-RW+/- optical drive.

c. Software: POT shall include software equal to the software on the ECC.

2.6 BACnet PROTOCOL ANALYZER

For ease of troubleshooting and maintenance, provide a BACnet protocol analyzer. Provide its associated fittings, cables and appurtenances, for connection to the communications network. The BACnet protocol analyzer shall be able to, at a minimum: capture and store to a file all data traffic on all network levels; measure bandwidth usage; filter out (ignore) selected traffic.

2.7 NETWORK AND DEVICE NAMING CONVENTION

a. Network Numbers

(1) BACnet network numbers shall be based on a "facility code, network" concept. The "facility code" is the VAMC's or VA campus' assigned numeric value assigned to a specific facility or building. The "network" typically corresponds to a "floor" or other logical configuration within the building. BACnet allows 65535 network numbers per BACnet internet work.

(2) The network numbers are thus formed as follows: "Net #" = "FFFNN" where:

(a) FFF = Facility code (see below)

(b) NN = 00-99 This allows up to 100 networks per facility or building

b. Device Instances

(1) BACnet allows 4194305 unique device instances per BACnet internet work. Using Agency's unique device instances are formed as follows: "Dev #" = "FFFNNDD" where

(a) FFF and N are as above and

(b) DD = 00-99, this allows up to 100 devices per network.

(2) Note Special cases, where the network architecture of limiting device numbering to DD causes excessive subnet works. The device number can be expanded to DDD and the network number N can become a single digit. In NO case shall the network number N and the device number D exceed 4 digits.

(3) Facility code assignments:

(4) 000-400 Building/facility number

(5) Note that some facilities have a facility code with an alphabetic suffix to denote wings, related structures, etc. The suffix will be ignored. Network numbers for facility codes greater than 400 will be assigned in the range 000-399.

c. Device Names

(1) Name the control devices based on facility name, location within a facility, the system or systems that the device monitors and/or controls, or the area served. The intent of the device naming is to be easily recognized. Names can be up to 254 characters in length, without embedded spaces. Provide the shortest descriptive, but unambiguous, name. For example, in building #123 prefix the number with a "B" followed by the building number, if there is only one chilled water pump "CHWP-1", a valid name would be "B123.CHWP. 1.STARTSTOP". If there are two pumps designated "CHWP-1", one in a basement mechanical room (Room 0001) and one in a penthouse mechanical room (Room PH01), the names could be "B123.R0001.CHWP.1. STARTSTOP" or "B123.RPH01.CHWP.1.STARTSTOP". In the case of unitary controllers, for example a VAV box

controller, a name might be "B123.R101.VAV". These names should be used for the value of the "Object_Name" property of the BACnet Device objects of the controllers involved so that the BACnet name and the EMCS name are the same.

2.8 BACnet DEVICES

a. All BACnet Devices – controllers, gateways, routers, actuators, Operator Displays, and sensors shall conform to BACnet Device Profiles and shall be BACnet Testing Laboratories (BTL) -Listed as conforming to those Device Profiles. Protocol Implementation Conformance Statements (PICSs), describing the BACnet capabilities of the Devices shall be published and available for the Devices through links in the BTL website.

(1) BACnet Building Controllers, shall conform to the BACnet B-BC Device Profile, and shall be BTL-Listed as conforming to the B-BC Device Profile. The Device’s PICS shall be submitted.

(2) BACnet Advanced Application Controllers shall conform to the BACnet B-AAC Device Profile and shall be BTL-Listed as conforming to the B-AAC Device Profile. The Device’s PICS shall be submitted.

(3) BACnet Application Specific Controllers shall conform to the BACnet B-ASC Device Profile and shall be BTL-Listed as conforming to the B-ASC Device Profile. The Device’s PICS shall be submitted.

(4) BACnet Smart Actuators shall conform to the BACnet B-SA Device Profile and shall be BTL-Listed as conforming to the B-SA Device Profile. The Device’s PICS shall be submitted.

(5) BACnet Smart Sensors shall conform to the BACnet B-SS Device Profile and shall be BTL-Listed as conforming to the B-SS Device Profile. The Device’s PICS shall be submitted.

(6) BACnet routers and gateways shall conform to the BACnet B-OTH Device Profile, and shall be BTL-Listed as conforming to the B-OTH Device Profile. The Device’s PICS shall be submitted.

2.9 CONTROLLERS

a. General. Provide an adequate number of BTL listed B-BC building controllers, BTL listed B-AAC, BTL listed B-ASC, BTL listed B-SA, and BTL listed B-SS’s to achieve the performance specified in the Part 1 Article on “System Performance.” Each of these controllers shall meet the following requirements.

(1) Communication.

(a) Each B-BC controller shall reside on a BACnet network using the ISO 8802-3 (Ethernet) Data Link/Physical layer protocol for its communications.

(b) Each B-BC controller shall provide a service communication port using BACnet Data Link/Physical layer protocol for connection to a portable operator’s terminal. If this port is not available built into the controller, contractor is to install a 4 port unmanaged switch inside the B-BC control cabinet.

(2) Keypad. A local keypad and display shall be provided for each controller. The keypad shall be provided for interrogating and editing data. Provide a system security password to prevent unauthorized use of the keypad and display.

(3) Serviceability. Provide diagnostic LEDs for power, communication, and processor. All wiring connections shall be made to field-removable, modular terminal strips or to a termination card connected by a ribbon cable.

(4) Memory. The controller shall maintain all BIOS and programming information in the event of a power loss for at least 72 hours.

(5) The controller shall be able to operate at 90 percent to 110 percent of nominal voltage rating and shall perform an orderly shutdown below 80 percent nominal voltage. Controller operation shall be protected against electrical noise of 5 to 120 Hz and from keyed radios up to 5 W at (3 ft).

(6) Transformer. Power supply for the ASC must be rated at a minimum of 125 percent of B-ASC power consumption and shall be of the fused or current limiting type.

b. Provide BTL-Listed B-ASC application specific controllers for each piece of equipment for which they are constructed. Application specific controllers shall communicate with other BACnet devices on the internetwork using the BACnet Read (Execute) Property service.

(1) Each B-ASC shall be capable of stand-alone operation and shall continue to provide control functions without being connected to the network.

(2) Each B-ASC will contain sufficient I/O capacity to control the target system.

(3) Communication.

(a) Each controller shall reside on a BACnet network using the ISO 8802-3 (Ethernet) Data Link/Physical layer protocol for its communications. Each building controller also shall perform BACnet routing if connected to a network of custom application and application specific controllers.

(b) Each controller shall have a BACnet Data Link/Physical layer compatible connection for a laptop computer or a portable operator’s tool. This connection shall be extended to a space temperature sensor port where shown.

4.Serviceability. Provide diagnostic LEDs for power, communication, and processor. All wiring connections shall be made to field-removable, modular terminal strips or to a termination card connected by a ribbon cable.

(5) Memory. The application specific controller shall use nonvolatile memory and maintain all BIOS and programming information in the event of a power loss.

(6) Immunity to power and noise. Controllers shall be able to operate at 90 percent to 110 percent of nominal voltage rating and shall

perform an orderly shutdown below 80 percent. Operation shall be protected against electrical noise of 5-120 Hz and from keyed radios up to 5 W at (3 ft).

c. Direct Digital Controller Software

(1) The software programs specified in this section shall be commercially available, concurrent, multi-tasking operating system and support the use of software application that operates under DOS, Linuxor Microsoft Windows.

(2) All points shall be identified by up to 30-character point name and 16-character point descriptor. The same names shall be used at the ECC.

(3) All control functions shall execute within the stand-alone control units. All new controllers installed will also include all software and/or hardware required to program, commission, or alter the sequence of operation of said controller(s). Controllers requiring software or hardware that is not commercially available will not be allowed. Installation of software and/or hardware for controller configuration will be the responsibility of the DDC contractor. Contracting Officer will direct to install said hardware and/or software on either the B-AWS or portable operator terminal. The VA shall be able to customize control strategies and sequences of operations defining the appropriate control loop algorithms and choosing the optimum loop parameters without requiring the services of a DDC contractor.

(4) All controllers shall be capable of being programmed to utilize stored default values for assured fail-safe operation of critical processes. Default values shall be invoked upon sensor failure or, if the primary value is normally provided by the central or another CU, or by loss of bus communication. Individual application software packages shall be structured to assume a fail-safe condition upon loss of input sensors. Loss of an input sensor shall result in output of a sensor-failed message at the ECC. Each ACU and RCU shall have capability for local readouts of all functions. The UCUs shall be read remotely.

(5) All DDC control loops shall be able to utilize any of the following control modes:

(a) Two position (on-off, slow-fast) control.

(b) Proportional control.

(c) Proportional plus integral (PI) control.

(d) Proportional plus integral plus derivative (PID) control. All PID programs shall automatically invoke integral wind up prevention routines whenever the controlled unit is off, under manual control of an automation system or time-initiated program.

(e) Automatic tuning of control loops.

(6) System Security: Operator access shall be secured using individual password and operator’s name. Passwords shall restrict the operator to the level of object, applications, and system

functions assigned to him. A minimum of three (3) or a maximum of six (6) levels of security for operator access shall be provided.

(7) Application Software: The controllers shall provide the following programs as a minimum for the purpose of optimizing energy consumption while maintaining comfortable environment for occupants. All application software shall reside and run in the system digital controllers. Editing of the application shall occur at the ECC or via a portable operator's terminal, when it is necessary, to access directly the programmable unit.

(a) Power Demand Limiting (PDL): Power demand limiting program shall monitor the building power consumption and limit the consumption of electricity to prevent peak demand charges. PDL shall continuously track the electricity consumption from a pulse input generated at the kilowatt-hour/demand electric meter. PDL shall sample the meter data to continuously forecast the electric demand likely to be used during successive time intervals. If the forecast demand indicates that electricity usage will likely to exceed a user preset maximum allowable level, then PDL shall automatically shed electrical loads. Once the demand load has met, loads that have been shed shall be restored and returned to normal mode. Control system shall be capable of demand limiting by resetting the HVAC system set points to reduce load while maintaining indoor air quality.

(b) Economizer: An economizer program shall be provided for VAV systems. This program shall control the position of air handler relief, return, and outdoor dampers. If the outdoor air dry bulb temperature falls below changeover set point the energy control center will modulate the dampers to provide 100 percent outdoor air. The operator shall be able to override the economizer cycle and return to minimum outdoor air operation at any time.

(c) Night Setback/Morning Warm up Control: The system shall provide the ability to automatically adjust set points for this mode of operation.

(d) Optimum Start/Stop (OSS): Optimum start/stop program shall automatically be coordinated with event scheduling. The OSS program shall start HVAC equipment at the latest possible time that will allow the equipment to achieve the desired zone condition by the time of occupancy, and it shall also shut down HVAC equipment at the earliest possible time before the end of the occupancy period and still maintain desired comfort conditions. The OSS program shall consider both outside weather conditions and inside zone conditions. The program shall automatically assign longer lead times for weekend and holiday shutdowns. The program shall poll all zones served by the associated AHU and shall select the warmest and coolest zones. These shall be used in the start time calculation. It shall be possible to assign occupancy start times on a per air handler unit basis. The program shall meet the local code requirements for minimum outdoor air while the building is occupied. Modification of assigned occupancy start/stop times shall be possible via the ECC.

(e) Event Scheduling: Provide a comprehensive menu driven program to automatically start and stop designated points or a group of points according to a stored time. This program shall provide the

capability to individually command a point or group of points. When points are assigned to one common load group it shall be possible to assign variable time advances/delays between each successive start or stop within that group. Scheduling shall be calendar based and advance schedules may be defined up to one year in advance. Advance schedule shall override the day-to-day schedule. The operator shall be able to define the following information:

1. Time, day.

2. Commands such as on, off, auto.

3. Time delays between successive commands.

4. Manual overriding of each schedule.

5)Allow operator intervention.

(f) Alarm Reporting: The operator shall be able to determine the action to be taken in the event of an alarm. Alarms shall be routed to the ECC based on time and events. An alarm shall be able to start programs, login the event, print and display the messages. The system shall allow the operator to prioritize the alarms to minimize nuisance reporting and to speed operator's response to critical alarms. A minimum of six (6) priority levels of alarms shall be provided for each point.

(g) Remote Communications: The system shall have the ability to dial out in the event of an alarm to the ECC and alpha-numeric pagers. The alarm message shall include the name of the calling location, the device that generated the alarm, and the alarm message itself. The operator shall be able to remotely access and operate the system using dial up communications. Remote access shall allow the operator to function the same as local access.

(h) Maintenance Management (PM): The program shall monitor equipment status and generate maintenance messages based upon the operators defined equipment run time, starts, and/or calendar date limits. A preventative maintenance alarm shall be printed indicating maintenance requirements based on pre-defined run time. Each preventive message shall include point description, limit criteria and preventative maintenance instruction assigned to that limit. A minimum of 480-character PM shall be provided for each component of units such as air handling units.

(i) Chilled water Plant Operation: This program shall have the ability to sequence the multiple chillers to minimize energy consumption. The program shall provide sequence of operation as described on the drawings and include the following as a minimum:

1. Automatic start/stop of chillers and auxiliaries in accordance with the sequence of operation shown on the drawings, while incorporating requirements and restraints, such as starting frequency of the equipment imposed by equipment manufacturers.

2. Secondary chilled water pumps and controls.

3. Generate chilled water plant load profiles for different

seasons for use in forecasting efficient operating schedule.

4. Cooling Tower Operation Program: The objective of cooling tower control is to optimize chiller/tower energy use within the equipment restraints and minimum condenser water temperature limit recommended by the equipment manufacturer. Maintain chilled water plant performance records and print reports at intervals selected by the operator. It shall be possible for the operator to change the set points and the operating schedule.

5. The chilled water plant program shall display the following as a minimum:

a) Secondary chilled flow rate.

b) Secondary chilled water supply and return temperature.

c) Condenser water supply and return temperature.

d) Outdoor air dry bulb temperature.

e) Outdoor air wet bulb temperature.

f) Ton-hours of chilled water per day/month/year.

g) On-off status for each chiller.

h) Chilled water flow rate.

i) Chilled water supply and return temperature.

j) Operating set points-temperature and pressure.

k) Kilowatts and power factor.

l) Current limit set point.

m) Date and time.

n) Operating or alarm status.

o) Operating hours.

2.10 SENSORS (AIR, WATER AND STEAM)

a. Sensors' measurements shall be read back to the DDC system, and shall be visible by the ECC.

b. Temperature and Humidity Sensors shall be electronic, vibration and corrosion resistant for wall, immersion, and/or duct mounting. Provide all remote sensors as required for the systems.

(1) Temperature Sensors: thermistor type for terminal units and Resistance Temperature Device (RTD) with an integral 4-20 mA or 0-10 VDCtransmitter type for all other sensors.

(a) Duct sensors shall be rigid or averaging type as shown on

drawings. Averaging sensor shall be a minimum of 1 linear foot of sensing element for each square foot of cooling/heating coil face area.

(b) Immersion sensors shall be provided with a separable well made of stainless steel, bronze or monel material. Pressure rating of well is to be consistent with the system pressure in which it is to be installed. Temperature well shall be filled with a thermal compound compatible with installed sensor.

(c) All space sensors shall be equipped with in-space User set-point adjustment, override switch, numerical temperature display on sensor cover, and BACnet communication port. Match room thermostats. Provide a tooled-access cover.

1. Public space sensor: setpoint adjustment shall be only through the ECC or through the DDC system’s diagnostic device/laptop. Do not provide in-space User set-point adjustment. Provide an opaque keyed-entry cover if needed to restrict in-space User set-point adjustment.

(d) Outdoor air temperature sensors shall have watertight inlet fittings and be shielded from direct sunlight.

(e) Room security sensors shall have stainless steel cover plate with insulated back and security screws.

(f) Wire: Twisted, shielded-pair cable.

(g) Output Signal: 4-20 mA or 0-10 VDC..

(2) Humidity Sensors: Bulk polymer sensing element type.

(a) Duct and room sensors shall have a sensing range of 20 to 80 percent with accuracy of ± 2 to ± 5 percent RH, including hysteresis, linearity, and repeatability.

(b) Outdoor humidity sensors shall be furnished with element guard and mounting plate and have a sensing range of 0 to 100 percent RH.

(c) Continuous Output Signal: 4-20 mA 0-10 VDC.

c. Static Pressure Sensors: Non-directional, temperature compensated.

(1) 4-20 mA or 0-10 VDC output signal.

(2) 0 to 5 inches wg for duct static pressure range.

(3) 0 to 0.25 inch wg for Building static pressure range.

d. Vortex Water flow sensors:

(1) Type: Insertion vortex type with retractable probe assembly and 2 inch full port gate valve.

(a) Pipe size: up to 24 inches.

(b) Retractor: ASME threaded, non-rising stem type with hand wheel.

(c) Mounting connection: 2 inch 150 PSI flange.

(d) Sensor assembly: Design for expected water flow and pipe size.

(e) Seal: Teflon (PTFE).

(2) Controller:

(a) Integral to unit.

(b) Locally display flow rate and total.

(c) Output flow signal to BAS/EMS/BES/BMCS: Digital pulse or BACNet type.

(3) Performance:

(a) Turndown: 20:1

(b) Response time: Adjustable from 1 to 100 seconds.

(c) Power: 24 volt DC

(4) Install flow meters according to manufacturer’s recommendations. Where recommended by manufacturer because of mounting conditions, provide flow rectifier.

e. Turbine Water Flow Sensors: shall be insertion turbine type with turbine element, retractor and preamplifier/transmitter mounted on a two-inch full port isolation valve; assembly easily removed or installed as a single unit under line pressure through the isolation valve without interference with process flow; calibrated scale shall allow precise positioning of the flow element to the required insertion depth within plus or minute (0.05 inch); wetted parts shall be constructed of stainless steel. Operating power shall be nominal 24 VDC . Local instantaneous flow indicator shall be LED type in NEMA 4 enclosure with 3-1/2 digital display, for wall or panel mounting.

(1) Performance characteristics:

(a) Ambient conditions: (-40 degrees F to 140 degrees F), 5 to 100 percent humidity.

(b) Operating conditions: 850 kPa (125 psig), (30 degrees F to 250 degrees F), per second (0.5 to 40 feet per second) velocity.

(c) Nominal range (turn down ratio): 10 to 1.

(d) Preamplifier mounted on meter shall provide 4-20 mA or 0-10 VDC, a divided pulse output or switch closure signal for units of volume or mass per a time base. Signal transmission distance shall be a minimum of (6,000 feet). .

(e) Pressure Loss: Maximum 1 percent of the line pressure in line sizes above (4 inches).

(f) Ambient temperature effects, less than 0.005 percent calibrated span per (degrees F) temperature change.

(g) RFI effect - flow meter shall not be affected by RFI.

(h) Power supply effect less than 0.02 percent of span for a variation of plus or minus 10 percent power supply.

f. Steam Flow Sensor/Transmitter:

(1) Sensor: Vortex shedder incorporating wing type sensor and amplification technology for high signal-to-noise ratio, carbon steel body with 316 stainless steel working parts, 24 VDC power, NEMA 4 enclosure.

(a) Ambient conditions, (-40 degrees F to 175 degrees F).

(b) Process conditions, 125 psig saturated steam.

(c) Turn down ratio, 20 to 1.

(d) Output signal, 4-20 ma DC 0-10 VDC DC.

(e) Processor/Transmitter, NEMA 4 enclosure with keypad program selector and six digit LCD output display of instantaneous flow rate or totalized flow, solid state switch closure signal shall be provided to the nearest DDC panel for totalization.

1. Ambient conditions, (0 degrees F-120 degrees F), 0 95 percent non-condensing RH.

2. Power supply, 120 VAC, 60 hertz or 24 VDC.

3. Internal battery, provided for 24-month retention of RAM contents when all other power sources are removed.

(f) Sensor on all steam lines shall be protected by pigtail siphons installed between the sensor and the line, and shall have an isolation valve installed between the sensor and pressure source.

g. Flow switches:

(1) Shall be either paddle or differential pressure type.

(a) Paddle-type switches (liquid service only) shall be UL Listed, SPDT snap-acting, adjustable sensitivity with NEMA 4 enclosure.

(b) Differential pressure type switches (air or water service) shall be UL listed, SPDT snap acting, NEMA 4 enclosure, with scale range and differential suitable for specified application.

h. Current Switches: Current operated switches shall be self powered, solid state with adjustable trip current as well as status, power, and relay command status LED indication. The switches shall be selected to match the current of the application and output requirements of the DDC systems.

i. Water Overflow Indicators:

1. Float Switches: Float switch manufactured to close a normally open electrical contact when water level rises at the bottom of a drain pan or sump. All parts that come in contact with water shall be plastic, brass, or stainless steel. Select switch configuration for reliable operation in the specific location it is to be used (primary drain pan, primary humidifier drain pan, secondary pan, pump sump etc.) Contact rating 4A at 24 VAC maximum and 100 mA at 5 VDC minimum. For installation the switch shall be clamped to the side of the fan or sump. Switches that required drilling through the side of the pan or sump shall be prohibited.

2. Water Sensors: 24 VAC powered solid state single pole – double throw contacts that closes upon the sensor’s detection of water at the bottom of pan, sump, floor etc. Sensor designed to be placed at the bottom of the pan, floor etc. where it will detect moisture. Sensor design shall incorporate water sensor array inside a hydrophilic pad designed to absorb and alarm with only a small amount of water. Contact rating 4A at 24 VAC maximum and 100 mA at 5 VDC minimum.

2.11 CONTROL CABLES

a. General:

(1) Ground cable shields, drain conductors, and equipment to eliminate shock hazard and to minimize ground loops, common-mode returns, noise pickup, cross talk, and other impairments. Comply with Sections 27 05 26 and 26 05 26.

(2) Cable conductors to provide protection against induction in circuits. Crosstalk attenuation within the System shall be in excess of -80 dB throughout the frequency ranges specified.

(3) Minimize the radiation of RF noise generated by the System equipment so as not to interfere with any audio, video, data, computer main distribution frame (MDF), telephone customer service unit (CSU), and electronic private branch exchange (EPBX) equipment the System may service.

(4) The as-installed drawings shall identify each cable as labeled, used cable, and bad cable pairs.

(5) Label system’s cables on each end. Test and certify cables in writing to the VA before conducting proof-of-performance testing. Minimum cable test requirements are for impedance compliance, inductance, capacitance, signal level compliance, opens, shorts, cross talk, noise, and distortion, and split pairs on all cables in the frequency ranges used. Make available all cable installation and test records at demonstration to the VA. All changes (used pair, failed pair, etc.) shall be posted in these records as the change occurs.

(6) Power wiring shall not be run in conduit with communications trunk wiring or signal or control wiring operating at 100 volts or less.

b. Analogue control cabling shall be not less than No. 18 AWG solid or stranded, with thermoplastic insulated conductors.

c. Copper digital communication cable between the ECC and the B-BC and

B-AAC controllers shall be 100BASE-TX Ethernet, Category 5e or 6, not less than minimum 24 American Wire Gauge (AWG) solid, Shielded Twisted Pair (STP) or Unshielded Twisted Pair (UTP), with thermoplastic insulated conductors, enclosed in a thermoplastic outer jacket, as specified in Section 27 15 00.

(1) Other types of media commonly used within IEEE 802.3 LANs (e.g., 10Base-T and 10Base-2) shall be used only in cases to interconnect with existing media.

d. All MS/TP communications cables for devices utilizing the TIA-485 standard must be listed for use on TIA-485 networks by the manufacturer of the cable. This requirement overrides any cable recommendation by the controller manufacturer. The use of TIA-485 communication cables shall not affect the warranty from the installing DDC contractor. Cables shall have the following characteristic:

(1) Nominal Impedance: 100-130 Ohms

(2) Twisted/shielded construction of 1, 1.5, or 2 pairs depending on controller requirements.

(3) Be plenum rated when required

(4) Cables designated for use by the cable manufacturer for use in PA or Speaker systems shall not be allowed, regardless of recommendations by the controller manufacturer.

e. Optical digital communication fiber, if used, shall be Multimode or Single mode fiber, 62.5/125 micron for multimode or 10/125 micron for single mode micron with SC or ST connectors as specified in TIA-568.0 Terminations, patch panels, and other hardware shall be compatible with the specified fiber and shall be as specified in Section 27 15 00. Fiber-optic cable shall be suitable for use with the 100Base-FX or the 100Base-SX standard (as applicable) as defined in IEEE 802.3.

2.12 THERMOSTATS AND HUMIDISTATS

a. Room thermostats controlling unitary standalone heating and cooling devices not connected to the DDC system shall have three modes of operation (heating - null or dead band - cooling). Thermostats for patient bedrooms shall have capability of being adjusted to eliminate null or dead band. Wall mounted thermostats shall have satin chrome manufacturer's recommendation finish, setpoint range and temperature display and external adjustment:

(1) Electronic Thermostats: Solid-state, microprocessor based, programmable to daily, weekend, and holiday schedules.

(a) Public Space Thermostat: Public space thermostat shall have a thermistor sensor and shall not have a visible means of set point adjustment. Adjustment shall be via the digital controller to which it is connected.

(b) Patient Room Thermostats: thermistor with in-space User set point adjustment and an on-casing room temperature numerical temperature display.

(c) Psychiatric Patient Room Sensors: Electronic duct sensor as

noted under Article 2.4.

(d) Battery replacement without program loss.

b. Strap-on aquastat shall be enclosed in a dirt-and-moisture proof housing with adjustable temperature switching point and minimumsingle pole, double throw switch. Variable setpoint and variable differential ranges selected for the application. Automatic reset.

c. Freezestats shall have a minimum of (one linear foot) of sensing element for each (one square foot) of coil area. A freezing condition at any increment of (one foot) anywhere along the sensing element shall be sufficient to operate the thermostatic element. Freezestats shall be manually-reset.

d. Room Humidistats: Provide fully proportioning humidistat with adjustable throttling range for accuracy of settings and conservation. The humidistat shall have set point scales shown in percent of relative humidity located on the instrument. Systems showing moist/dry or high/low are prohibited.

e. Low Signal Selector High Limit Duct Humidistat: Duct mounted high limit humidistat mounted downstream of the humidifier steam injection grid to protect the ductwork and other components from moisture in the event of humidifier loop problems. Typically set at 70 percent to 80 percent RH the device provides proportional signal back to steam control equipment as duct humidity approaches high limit setpoint thus providing more accurate control. Duct mounted device with polymer sensor:
1. Output 4 to 20 milliamp or 0-10 VDC.
2. Setpoint range 60 percent RH to 95 percent RH.
3. Proportional Band range 5 percent to 30 percent.
4. Auxiliary contacts for alarms. 5. Accuracy 2 percent RH and drift less than 0.5 percent per year.

2.13 FINAL CONTROL ELEMENTS AND OPERATORS

a. Fail Safe Operation: Control valves and dampers shall provide "fail safe" operation in either the normally open or normally closed position as required for freeze, moisture, and smoke or fire protection.

b. Spring Ranges: Range as required for system sequencing and to provide tight shut-off.

c. Power Operated Control Dampers (other than VAV Boxes): Factory fabricated, balanced type dampers. All modulating dampers shall be opposed blade type and gasketed. Blades for two-position, duct-mounted dampers shall be parallel, airfoil (streamlined) type for minimum noise generation and pressure drop.

(1) Leakage: Except as specified in subparagraph 2 below, maximum leakage in closed position shall not exceed (15 CFMs) differential pressure for outside air and exhaust dampers and (40 CFM/sq. ft.) at 2 inches differential pressure for other dampers.

(2) Frame shall be galvanized steel channel with seals as required to meet leakage criteria.

(3) Blades shall be galvanized steel or aluminum, 8 inch maximum width, with edges sealed as required.

(4) Bearing shall be nylon, bronze sleeve or ball type.

(5) Hardware shall be zinc-plated steel. Connected rods and linkage shall be non-slip. Working parts of joints shall be brass, bronze, nylon or stainless steel.

(6) Maximum air velocity and pressure drop through free area of the dampers:

(a) Smoke damper in air handling unit: (1000 fpm).

(b) Duct mounted damper: (2000 fpm).

(c) Maximum static pressure loss: 0.20 inches water gage).

d. Smoke Dampers and Combination Fire/Smoke Dampers: Dampers and operators are specified in Section 23 31 00, HVAC DUCTS AND CASINGS. Control of these dampers is specified under this Section.

e. Control Valves:

(1) Valves shall be rated for a minimum of 150 percent of system operating pressure at the valve location but not less than (125 psig).

(2) Valves 2 inches and smaller shall be bronze body with threaded or flare connections.

(3) Valves (2 1/2 inches and greater shall be bronze or iron body with flanged connections.

(4) Brass or bronze seats except for valves controlling media greater than (210 degrees F), which shall have stainless steel seats.

(5) Flow characteristics:

(a) Three way modulating valves shall be globe pattern. Position versus flow relation shall be linear relation for steam or equal percentage for water flow control.

(b) Two-way modulating valves shall be globe pattern. Position versus flow relation shall be linear for steam and equal percentage for water flow control.

(c) Two-way 2-position valves shall be ball, gate or butterfly type.

(6) Maximum pressure drop:

(a) Two position steam control: 20 percent of inlet gauge pressure.

(b) Modulating Steam Control: 80 percent of inlet gauge pressure (acoustic velocity limitation).

(c) Modulating water flow control, greater of 10 feet of water or

the pressure drop through the apparatus.

(7) Two position water valves shall be line size.

f. Damper and Valve Operators and Relays:

(1) Electric operator shall provide full modulating control of dampers and valves. For dampers a linkage and pushrod shall be furnished for mounting the actuator on the damper frame internally in the duct, externally in the duct, externally on the duct wall, or shall be furnished with a direct-coupled design. Metal parts shall be aluminum, mill finish galvanized steel, or zinc plated steel or stainless steel. Provide actuator heads which allow for electrical conduit attachment. The motor(s) shall have sufficient closure torque to allow for complete closure of valve or damper under pressure. Provide multiple motors as required to achieve sufficient close-off torque.

(a) Minimum valve close-off pressure shall be equal to the system pump’s dead-head pressure, minimum 50 psig for valves smaller than 4 inches.

(2) Electronic damper operators: Metal parts shall be aluminum, mill finish galvanized steel, or zinc plated steel or stainless steel. Provide actuator heads which allow for electrical conduit attachment. The motors shall have sufficient closure torque to allow for complete closure of valve or damper under pressure. Provide multiple motors as required to achieve sufficient close-off torque.

(4) See and coordinate drawings for required control operation.

2.14 AIR FLOW CONTROL

a. Airflow and static pressure shall be controlled via digital controllers with inputs from airflow control measuring stations and static pressure inputs as specified. Controller outputs shall be analog or pulse width modulating output signals. The controllers shall include the capability to control via simple proportional (P) control, proportional plus integral (PI), proportional plus integral plus derivative (PID), and on-off. The airflow control programs shall be factory-tested programs that are documented in the literature of the control manufacturer.

c. Air Flow Measuring Station -- Electronic Thermal Type:

(1) Air Flow Sensor Probe:

(a) Each air flow sensor shall contain two individual thermal sensing elements. One element shall determine the velocity of the air stream while the other element shall compensate for changes in temperature. Each thermal flow sensor and its associated control circuit and signal conditioning circuit shall be factory calibrated and be interchangeable to allow replacement of a sensor

without recalibration of the entire flow station. The sensor in the array shall be located at the center of equal area segment of the duct and the number of sensors shall be adequate to accommodate the expected velocity profile and variation in flow and temperature. The airflow station shall be of the insertion type in which sensor support structures are inserted from the outside of the ducts to make up the complete electronic velocity array.

(b) Thermal flow sensor shall be constructed of hermetically sealed thermistors or nickel chromium or reference grade platinum wire, wound over an epoxy, stainless steel or ceramic mandrel and coated with a material suitable for the conditions to be encountered. Each dual sensor shall be mounted in an extruded aluminum alloy strut.

(2) Air Flow Sensor Grid Array:

(a) Each sensor grid shall consist of a lattice network of temperature sensors and linear integral controllers (ICs) situated inside an aluminum casing suitable for mounting in a duct . Each sensor shall be mounted within a strut facing downstream of the airflow and located so that it is protected on the upstream side. All wiring shall be encased (out of the air stream) to protect against mechanical damage.

(b) The casing shall be made of welded aluminum of sufficient strength to prevent structural bending and bowing. Steel or iron composite shall not be acceptable in the casing material.

(c) Pressure drop through the flow station shall not exceed (0.015 inch W.G.) at (3,000 FPM).

(3) Electronics Panel:

(a) Electronics Panel shall consist of a surface mounted enclosure complete with solid-state microprocessor and software.

(b) Electronics Panel shall be A/C powered 24 VAC and shall have the capability to transmit signals of 4-20 ma or 0-10 VDC type or PWM type for use in control of the HVAC Systems. The electronic panel shall have the capability to accept user defined scaling parameters for all output signals.

(c) Electronics Panel shall have the capability to digitally display airflow in CFM LPS and temperature in degrees F. The displays shall be provided as an integral part of the electronics panel. The electronic panel shall have the capability to totalize the output flow in CFM for two or more systems, as required. A single output signal shall be provided which will equal the sum of the systems totalized. Output signals shall be provided for temperature and airflow. Provide remote mounted air flow or temperature displays where indicated on the plans.

(d) Electronics Panel shall have the following:

1. Minimum of 12-bit A/D conversion.

2. Field adjustable digital primary output offset and gain.

3. Airflow analog output scaling of 100 to 10,000 FPM.

4. Temperature analog output scaling from (-50 degrees F to 160 degrees F).

5. Analog output resolution (full scale output) of 0.025 percent.

(e) All readings shall be in I.P. and S.I. units.

(4) Thermal flow sensors and its electronics shall be installed as per manufacturer's instructions. The required probe sensor density shall be as follows:

Probe Sensor Density	
Area (square feet)	Qty. Sensors
<=1	2
>1 to <4	4
4 to <8	6
8 to <12	8
12 to <16	12
>=16	16

(a) Complete installation shall not exhibit more than ± 2.0 percent error in airflow measurement output for variations in the angle of flow of up to 10 percent in any direction from its calibrated orientation. Repeatability of readings shall be within ± 0.25 percent.

d. Static Pressure Measuring Station: shall consist of one or more static pressure sensors and transmitters along with relays or auxiliary devices as required for a complete functional system. The span of the transmitter shall not exceed two times the design static pressure at the point of measurement. The output of the transmitter shall be true representation of the input pressure with plus or minus (0.1 inch) W.G. of the required input pressure:

(1) Static pressure sensors shall have the same requirements as Airflow Measuring Devices except that total pressure sensors are optional, and only multiple static pressure sensors positioned on an equal area basis connected to a network of headers are required.

(2) For systems with multiple major or main trunk supply ducts, furnish a static pressure transmitter for each trunk duct. The transmitter signal representing the lowest static pressure shall be selected and this shall be the input signal to the controller.

(3) The controller shall receive the static pressure transmitter signal and Control Unit (CU) shall provide a control output signal to the supply fan capacity control device. The control mode shall be proportional plus integral (PI) (automatic reset) and where required shall also include derivative mode.

(4) In systems with multiple static pressure transmitters, provide a switch located near the fan discharge to prevent excessive

pressure during abnormal operating conditions. High-limit switches shall be manually reset.

e. Constant Volume Control Systems shall consist of an air flow measuring station along with such relays and auxiliary devices as required to produce a complete functional system. The transmitter shall receive its air flow signal or static pressure signal from the flow measuring station and shall have a span not exceeding three times the design flow rate. The CU shall receive the transmitter signal and shall provide an output to the fan volume control device to maintain a constant flow rate. The CU shall provide proportional plus integral (PI) (automatic reset) control mode and where required also inverse derivative mode. Overall system accuracy shall be plus or minus the equivalent of (0.008 inch) velocity pressure as measured by the flow station.

f. Airflow Synchronization:

(1) Systems shall consist of an air flow measuring station for each main supply and return duct, the CU and such relays, as required to provide a complete functional system that will maintain a constant flow rate difference between supply and return air to an accuracy of ±10 percent. In systems where there is no suitable location for a flow measuring station that will sense total supply or return flow, provide multiple flow stations with a differential pressure transmitter for each station. Signals from the multiple transmitters shall be added through the CU such that the resultant signal is a true representation of total flow.

(2) The total flow signals from supply and return air shall be the input signals to the CU. This CU shall track the return air fan capacity in proportion to the supply air flow under all conditions.

2.15 SAFETY

Provide hard-wired interlocked connections for such all safety devices, such as freeze stats, smoke detectors, smoke dampers, and refrigerant leak detection devices. All safety devises shall be provided with additional dry contacts and shall be connected to the DDC system for monitoring and sequencing.

PART 3 EXECUTION

3.1 INSTALLATION

a. General:

(1) Examine project plans for control devices and equipment locations; and report any discrepancies, conflicts, or omissions to Contracting Officer for resolution before proceeding for installation.

(2) Install equipment, piping, wiring /conduit parallel to or at right angles to building lines.

(3) Install all equipment and piping in readily accessible locations. Do not run tubing and conduit concealed under insulation or inside ducts.

(4) Mount control devices, tubing and conduit located on ducts and apparatus with external insulation on standoff support to avoid interference with insulation.

(5) Provide sufficient slack and flexible connections to allow for vibration of piping and equipment.

(6) Run tubing and wire connecting devices on or in control cabinets parallel with the sides of the cabinet neatly racked to permit tracing.

(7) Install equipment level and plumb.

b. Electrical Wiring Installation:

(1) All wiring and cabling shall be installed in conduits. Install conduits and wiring in accordance with Specification Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS. Conduits carrying control wiring and cabling shall be dedicated to the control wiring and cabling: these conduits shall not carry power wiring. Provide plastic end sleeves at all conduit terminations to protect wiring from burrs.

(2) Install analog signal and communication cables in conduit and in accordance with Specification Division 27 – COMMINICATIONS. Install digital communication cables in conduit and in accordance with Specification Section 27 15 00, COMMINICATIONS STRUCTURED CABLING.

(3) Install conduit and wiring between operator workstation(s), digital controllers, electrical panels, indicating devices, instrumentation, miscellaneous alarm points, thermostats, and relays as shown on the drawings or as required under this section.

(4) Install all electrical work required for a fully functional system and not shown on electrical plans or required by electrical specifications. Where low voltage (less than 50 volt) power is required, provide suitable Class B transformers.

(5) Install all system components in accordance with local Building Code and NFPA 70.

(a) Splices: Splices in shielded and coaxial cables shall consist of terminations and the use of shielded cable couplers. Terminations shall be in accessible locations. Cables shall be harnessed with cable ties.

(b) Equipment: Fit all equipment contained in cabinets or panels with service loops, each loop being at least (12 inches) long. Equipment for fiber optics system shall be rack mounted, as applicable, in ventilated, self-supporting, code gauge steel enclosure. Cables shall be supported for minimum sag.

(c) Cable Runs: Keep cable runs as short as possible. Allow extra length for connecting to the terminal board. Do not bend flexible coaxial cables in a radius less than ten times the cable outside diameter.

(d) Use vinyl tape, sleeves, or grommets to protect cables from

vibration at points where they pass around sharp corners, through walls, panel cabinets, etc.

(6) Conceal cables, except in mechanical rooms and areas where other conduits and piping are exposed.

(7) Permanently label or code each point of all field terminal strips to show the instrument or item served. Color-coded cable with cable diagrams may be used to accomplish cable identification.

(8) Grounding: ground electrical systems per manufacturer’s written requirements for proper and safe operation.

c. Install Sensors and Controls:

(1) Temperature Sensors:

(a) Install all sensors and instrumentation according to manufacturer’s written instructions. Temperature sensor locations shall be readily accessible, permitting quick replacement and servicing of them without special skills and tools.

(b) Calibrate sensors to accuracy specified, if not factory calibrated.

(c) Use of sensors shall be limited to its duty, e.g., duct sensor shall not be used in lieu of room sensor.

(d) Install room sensors permanently supported on wall frame. They shall be mounted at (5.0 feet) above the finished floor unless otherwise noted on the plans or drawings.

(e) Mount sensors rigidly and adequately for the environment within which the sensor operates. Separate extended-bulb sensors form contact with metal casings and coils using insulated standoffs.

(f) Sensors used in mixing plenum, and hot and cold decks shall be of the averaging of type. Averaging sensors shall be installed in a serpentine manner horizontally across duct. Each bend shall be supported with a capillary clip.

(g) All pipe mounted temperature sensors shall be installed in wells.

(h) All wires attached to sensors shall be air sealed in their conduits or in the wall to stop air transmitted from other areas affecting sensor reading.

(i) Permanently mark terminal blocks for identification. Protect all circuits to avoid interruption of service due to short-circuiting or other conditions. Line-protect all wiring that comes from external sources to the site from lightning and static electricity.

(2) Pressure Sensors:

(a) Install duct static pressure sensor tips facing directly downstream of airflow.

(b) Install high-pressure side of the differential switch between the pump discharge and the check valve.

(c) Install snubbers and isolation valves on steam pressure sensing devices.

(3) Actuators:

(a) Mount and link damper and valve actuators according to manufacturer’s written instructions.

(b) Check operation of damper/actuator combination to confirm that actuator modulates damper smoothly throughout stroke to both open and closed position.

(c) Check operation of valve/actuator combination to confirm that actuator modulates valve smoothly in both open and closed position.

(4) Flow Switches:

(a) Install flow switch according to manufacturer’s written instructions.

(b) Mount flow switch a minimum of 5 pipe diameters up stream and 5 pipe diameters downstream or 2 feet whichever is greater, from fittings and other obstructions.

(c) Assure correct flow direction and alignment.

(d) Mount in horizontal piping-flow switch on top of the pipe.

d. Installation of network:

(1) Ethernet:

(a) The network shall employ Ethernet LAN architecture, as defined by IEEE 802.3. The Network Interface shall be fully Internet Protocol (IP) compliant allowing connection to currently installed IEEE 802.3, Compliant Ethernet Networks.

(b) The network shall directly support connectivity to a variety of cabling types. As a minimum provide the following connectivity: 100 Base TX (Category 5e cabling) for the communications between the ECC and the B-BC and the B-AAC controllers.

(2) Third party interfaces: Contractor shall integrate real-time data from building systems by other trades and databases originating from other manufacturers as specified and required to make the system work as one system.

e. Installation of digital controllers and programming:

(1) Provide a separate digital control panel for each major piece of equipment, such as air handling unit, chiller, pumping unit etc. Points used for control loop reset such as outdoor air, outdoor humidity, or space temperature could be located on any of the remote control units.

(2) Provide sufficient internal memory for the specified control sequences and trend logging. There shall be a minimum of 25 percent of available memory free for future use.

(3) System point names shall be human readable, permitting easy operator interface without the use of a written point index.

(4) Provide software programming for the applications intended for the systems specified, and adhere to the strategy algorithms provided.

(5) Provide graphics for each piece of equipment and floor plan in the building. This includes each chiller, cooling tower, air handling unit, fan, terminal unit, boiler, pumping unit etc. These graphics shall show all points dynamically as specified in the point list.

3.2 SYSTEM VALIDATION AND DEMONSTRATION

a. As part of final system acceptance, a system demonstration is required (see below). Prior to start of this demonstration, the contractor is to perform a complete validation of all aspects of the controls and instrumentation system.

b. Validation

(1) Prepare and submit for approval a validation test plan including test procedures for the performance verification tests. Test Plan shall be submitted for review with the control system submittal as indicated in paragraph 1.7 Submittals. Test Plan, address all specified functions of the ECC and all specified sequences of operation. Explain in detail actions and expected results used to demonstrate compliance with the requirements of this specification. Explain the method for simulating the necessary conditions of operation used to demonstrate performance of the system. Test plan shall include a test check list to be used by the Installer’s agent to check and initial that each test has been successfully completed. Deliver test plan documentation for the performance verification tests to the owner’s representative 30 days prior to start of performance verification tests. Provide draft copy of operation and maintenance manual with performance verification test.

(2) After approval of the validation test plan, installer shall carry out all tests and procedures therein. Installer shall completely check out, calibrate, and test all connected hardware and software to ensure that system performs in accordance with approved specifications and sequences of operation submitted. Installer shall complete and submit Test Check List.

c. Demonstration

(1) System operation and calibration to be demonstrated by the installer in the presence of the Architect, CxA or Contracting Officer on random samples of equipment as dictated by the Contracting Officer. Should random sampling indicate improper work, the owner reserves the right to subsequently witness complete calibration of the system at no addition cost to the VA.

(2) Make accessible, personnel to provide necessary adjustments and corrections to systems as directed by balancing agency.

(3) The following witnessed demonstrations of field control equipment shall be included:

(a) Observe HVAC systems in shut down condition. Check dampers and valves for normal position.

(b) Test application software for its ability to communicate with digital controllers, operator workstation, and uploading and downloading of control programs.

(c) Demonstrate the software ability to edit the control program off-line.

(d) Demonstrate reporting of alarm conditions for each alarm and ensure that these alarms are received at the assigned location, including operator workstations.

(e) Demonstrate ability of software program to function for the intended applications-trend reports, change in status etc.

(f) Demonstrate via graphed trends to show the sequence of operation is executed in correct manner, and that the HVAC systems operate properly through the complete sequence of operation, e.g., seasonal change, occupied/unoccupied mode, and warm-up condition.

(g) Demonstrate hardware interlocks and safeties functions, and that the control systems perform the correct sequence of operation after power loss and resumption of power loss.

(h) Prepare and deliver to the VA graphed trends of all control loops to demonstrate that each control loop is stable and the set points are maintained.

(i) Demonstrate that each control loop responds to set point adjustment and stabilizes within one (1) minute(s). Control loop trend data shall be instantaneous and the time between data points shall not be greater than one (1) minute.

(4) Witnessed demonstration of ECC functions shall consist of:

(a) Running each specified report.

(b) Display and demonstrate each data entry to show site specific customizing capability. Demonstrate parameter changes.

(c) Step through penetration tree, display all graphics, demonstrate dynamic update, and direct access to graphics.

(d) Execute digital and analog commands in graphic mode.

(e) Demonstrate DDC loop precision and stability via trend logs of inputs and outputs (6 loops minimum).

(f) Demonstrate Energy Management System (EMS) performance via

trend logs and command trace.

(g) Demonstrate scan, update, and alarm responsiveness.

(h) Demonstrate spreadsheet/curve plot software, and its integration with database.

(i) Demonstrate on-line user guide, and help function and mail facility.

(j) Demonstrate digital system configuration graphics with interactive upline and downline load, and demonstrate specified diagnostics.

(k) Demonstrate multitasking by showing dynamic curve plot, and graphic construction operating simultaneously via split screen.

(l) Demonstrate class programming with point options of beep duration, beep rate, alarm archiving, and color banding.

3.3 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and Commissioning Agent. Provide a minimum notice of 10 working days prior to startup and testing.

3.4 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.5 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4

-- hours to instruct each VA personnel responsible in the operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

3.6 CONSTRUCTION WASTE MANAGEMENT

a. General: Comply with Contractor’s Waste Management Plan and Section

01 74 19, CONSTRUCTION WASTE MANAGEMENT.

b. To the greatest extent possible, separate reusable and recyclable products from contaminated waste and debris in accordance with the Contractor’s Waste Management Plan. Place recyclable and reusable products in designated containers and protect from moisture and contamination.

-- End of Section --

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-- End of Section Table of Contents --

SECTION 23 10 00
4/20

FACILITY FUEL OIL SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

a. Diesel fuel oil burner fuel oil tanks, piping, and accessories located outside, aboveground as shown on contract drawings. Refer to contract drawings for type of fuel and for tank capacities.

b. Tank fluid level monitoring and alarm systems.

c. Leak detection system for tanks and underground piping.

d. Fuel oil quality maintenance system (water and particulate removal).

e. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, AUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 03 30 00, CAST-IN-PLACE CONCRETE.

f. Section 05 50 00, METAL FABRICATIONS.

g. Section 09 91 00, PAINTING.

h. Not used.

j. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

k. Section 23 07 11, HVAC AND BOILER PLANT INSULATION.

l. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

m. Not used.

n. Section 23 21 11, BOILER PLANT PIPING SYSTEMS.

o. Section 23 50 11, BOILER PLANT MECHANICAL EQUIPMENT.

p. Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION.

q. Not used.

r. Section 31 20 00, EARTHWORK.

1.3 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.

AMERICAN PETROLEUM INSTITUTE (API)

API RP 1631	(2001) Interior Lining and Periodic Inspection of Underground Storage Tanks

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 /Inch Standard
ASME B16.9	(2024) Factory-Made Wrought Buttwelding Fittings
ASME B16.11	(2022) Forged Fittings, Socket-Welding and Threaded

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A105/A105M	(2023) Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A106/A106M	(2019a) Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM A126	(2004; R 2023) Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings
ASTM A234/A234M	(2024) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
ASTM B62	(2017) Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM D2996	(2023) Standard Specification for Filament-Wound "Fiberglass" (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

NACE SP0169	(2024) Control of External Corrosion on Underground or Submerged Metallic Piping Systems
SSPC SP 6/NACE No.3	(2007) Commercial Blast Cleaning
SSPC 7/NACE No.4	(2007) Brush-Off Blast Cleaning

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 30	(2024; TIA 24-1) Flammable and Combustible Liquids Code
NFPA 31	(2024; TIA 23-1) Standard for the Installation of Oil-Burning Equipment
NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

UL SOLUTIONS (UL)

UL 142	(2006; Reprint Jan 2021) UL Standard for Safety Steel Aboveground Tanks for Flammable and Combustible Liquids
UL 971	(1995; Reprint Mar 2006) UL Standard for Safety Nonmetallic Underground Piping for Flammable Liquids

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Aboveground Steel Tanks; G, AE

Leak Detection System; G, AE

Tank Fluid Level Monitoring Instrumentation System; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Fuel Piping; G, AE

Pipe Fittings, Unions, Flanges; G, AE

Check Valves, Overfill Prevention Valves; G, AE

Secondary Containment System For Fuel Piping; G, AE

Tank And Piping Accessories; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 10 00, FACILITY FUEL OIL SYSTEMS”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

e. Aboveground Steel Tanks, Including Vault-type Tanks:

(1) Drawings of tanks, supports, ladders, platforms, heating coils, tank manholes, emergency relief vents and all accessories. Include overall dimensions and dimensional locations and sizes of pipe connections, and access openings.

(2) Recommended tank support locations.

(3) Weight of entire tank assembly, empty and flooded.

(4) Design and construction of primary tanks, insulation, secondary containment, supports, pipe connections, platforms.

(5) Application and performance data on coatings from manufacturer of coatings.

(6) Data certifying tanks are designed for surcharge loads of platforms shown.

(7) Certification of compliance with specified standards.

(8) Certification that steel tank manufacturer participates in Steel Tank Institute (STI) Quality Assurance Program.

(9) Design, construction, performance, dimensions of emergency relief vents.

(10) Not used.

f. Fuel Piping:

(1) ASTM and UL compliance.

(2) Grade, class or type, schedule number.

(3) Manufacturer.

g. Pipe Fittings, Unions, Flanges:

(1) ASTM and UL compliance.

(2) ASTM standards number.

(3) Catalog cuts.

(4) Pressure and temperature rating.

h. Check Valves, Overfill Prevention Valves, External Emergency Valves:

(1) Catalog cuts showing design and construction.

(2) Pressure and temperature ratings.

(3) Pressure loss and flow rate data.

(4) Materials of construction.

(5) Accessories.

i. Secondary Containment System for Fuel Piping:

(1) Sizes, materials, construction of containment system including end seals, sumps, coatings and pipe supports.

(2) Layout of system.

(3) Installation instructions.

(4) Design of cathodic protection system (steel casing).

j. Leak Detection System:

(1) Drawings, description and performance data on sensors, control units.

(2) Description of operation.

(3) Layout of system.

(4) Installation and operating instructions.

(5) Data on interconnecting wiring systems to be furnished.

k. Tank Fluid Level Monitoring Instrumentation System:

(1) Drawings showing instruments and in-tank sensing units, with dimensions.

(2) Design and construction of all elements of system.

(3) Installation instructions.

l. Tank and Piping Accessories: Design, construction, and dimensions of vent caps, fill boxes, fill caps, spill containers and other accessories.

m. Fuel Oil Polishing System:

(1) Drawings and description of all components and arrangement of system.

(2) Design and performance of pumps, filters.

(3) Catalog data and operation of control system.

(4) Installation instructions.

n. Fuel Oil Heating System:

(1) Drawings and description of all components and arrangement of system.

(2) Design and performance of pumps, heaters.

(3) Catalog data and operation of control system.

(4) Installation instructions.

o. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

p. Completed System Readiness Checklist provided by the Commissioning Agent and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

q . Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.6 QUALITY ASSURANCE

a. Approval by Contracting Officer is required of products or services of proposed manufacturers, suppliers and installers, and will be based on Contractor’s certification that:

(1) Manufacturers regularly and currently manufacture tanks, tank and piping accessories, tank fluid level monitoring and leak detection systems, and fuel quality management systems.

(2) Manufacturers of steel tanks participate in the Quality Assurance Program of the Steel Tank Institute (STI).

(3) The design and size of each item of equipment provided for this project is of current production and has been in satisfactory operation on at least three installations for approximately three years. Current models of fluid level and leak detection systems with less than three years’ service experience are acceptable if similar previous models from the same manufacturer have at least three years’ service experience.

b. Apply and install materials, equipment and specialties in accordance with manufacturer’s written instructions. Conflicts between the manufacturer’s instructions and the contract drawings and specifications shall be referred to the Contracting Officer for resolution. Provide copies of installation instructions to the Contracting Officer two weeks prior to commencing installation of any item.

c. All equipment shall be free from defects that would adversely affect the performance, maintainability and appearance of individual components or overall assembly.

d. Tanks, Secondary Containment Systems for Piping, Plastic Piping and Containment Systems, Tank Level Monitoring Systems, Leak Detection Systems, Fuel Quality Management Systems, Cathodic Protection Systems: Authorized manufacturer’s representatives shall provide onsite training of installers and supervision of the installation and testing of the equipment and systems to assure conformance to written instructions of manufacturers.

e. Tank and piping installation contractor shall be certified as acceptable by local and state pollution control authorities.

f. Entire installation shall conform to requirements of local and state pollution control authorities.

g. Pipe Welding: Conform to requirements of ASME B31.1. Welders shall show evidence of qualification. Welders shall utilize a stamp to identify their work. Unqualified personnel will be rejected.

h. Assembly of Glass Fiber Reinforced Plastic Piping: Installation personnel shall have been trained, tested and certified under a procedure approved by the manufacturer of the piping. Proof of

certification, in writing, shall be provided to the Contracting Officer.

i. Where specified codes or standards conflict, consult the Contracting Officer.

j. Label of Conformance (definition): Labels of accredited testing laboratories showing conformance to the standards specified.

k. Equipment and materials installed shall be compatible in all respects with other items being furnished and with existing items so that the result will be a safe, complete and fully operational system which conforms to contract requirements and in which no item is subject to conditions beyond its design capabilities.

1.7 AS-BUILT DOCUMENTATION

a. Submit manufacturer’s literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, VA approved substitutions and construction revisions shall be in electronic version on CD or DVD inserted into a three-ring binder. All aspects of system operation and maintenance procedures, including applicable piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the ‘third party testing company’ requirement. Provide record drawings as follows:

(1) Red-lined, hand-marked drawings are to be provided, with one paper copy and a scanned PDF version of the hand-marked drawings provided on CD or DVD.

d. The as-built drawings shall indicate the location and type of all lockout/tagout points for all energy sources for all equipment and pumps to include breaker location and numbers, valve tag numbers, etc. Coordinate lockout/tagout procedures and practices with local VA requirements.

e. Certification documentation shall be provided to Contracting Officer 21 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and provide

documentation/certification that all results of tests were within limits specified. Test results shall contain written sequence of test procedure with written test results annotated at each step along with the expected outcome or setpoint. The results shall include all readings, including but not limited to data on device (make, model and performance characteristics), normal pressures, switch ranges, trip points, amp readings, and calibration data to include equipment serial numbers or individual identifications, etc.

1.8 PERMITS

Contractor shall obtain and complete all tank permit and registration forms required by governmental authorities.

PART 2 PRODUCTS

2.1 ABOVEGROUND STEEL TANKS

a. Type: Factory-fabricated all welded steel, vertical cylindrical configuration, insulated, atmospheric pressure, internal and external corrosion protection as specified. Inner primary tank dimensions shall be approximately 13'-0" diameter and 49'-10" vertical height having approximately 49,500 gallons in total volume. In addition to specified requirements, tanks shall be fabricated in accordance with Steel Tank Institute (STI) design standards by manufacturer that participates in STI Quality Assurance Program.

b. Construction:

(1) ASTM A36/A36M steel, conform to UL 142. Inner 1/4-inch-thick primary tank and secondary 7-gauge steel tank shall both conform. Provide label of conformance.

(2) Conform to NFPA 30 or NFPA 31 as applicable.

(3) Double-wall, insulated tank with 2-inch-thick poured urethane foam between secondary 7-gauge steel tank and outer 10-gauge steel skin.

(7) Design for surcharge load produced by tank-mounted platforms and platform loadings shown.

(8) Leaks and abrasions are prohibited. Maximum permissible out-of-roundness of cylindrical shells is one percent of the diameter.

(9) Provide lifting lugs for rigging tanks.

(10) Make provisions for leak detectors to be installed at lowest part of interstitial space between walls of double-wall tanks.

c. Platforms, Stairs, Ladders and Handrails: Provide welded steel assemblies as shown, conforming to OSHA requirements. Ladders shall be equipped with a personal fall arrest system (PFAS) or a ladder safety system suitable for access the exterior top of the tank and the

interior tank bottom. Provide welded steel tank attachments designed to support platform framing, stairs, ladders and live and dead loadings. Clean and coat all surfaces as specified for tank and steel dike exterior. Galvanizing is an acceptable alternative.

d. Factory Cleaning: Clean interior and exterior of tanks. Remove mill scale, dirt, rust, oil, welding debris, loose coatings and coatings incompatible with fuel stored or protective coating. Sandblast exterior in accordance with NACE 3.

e. Factory Coating: Provide tanks with exterior coat of rust resistant metal primer, specified under Section 09 91 00, PAINTING. Coat interior from bottom of tank to 3 feet above bottom in compliance with API RP 1631.

g. Pipe Connections to Tanks:

(1) Conform to UL 142.

(2) Pipe sizes 2 inches and smaller, threaded. Pipe sizes 2-1/2 inches and larger, flanged, 150 psig ASME rating.

(3) Welded joints required on steel piping located inside tanks.

(4) Provide and coordinate tank connection quantities, sizes and types with requirements of tank level gauge unit; sounding rod; vent, fill, supply and return pipes; and other pipes as shown.

(5) On double-wall tanks, provide valved drain of interstitial space.

h. Tank Manholes: Provide quantity shown. Bolted cover type, gasketed.

i. External Ladder: Provide steel climbing ladder as shown with anodized aluminum rail attached to ladder with climbing trolley.

j. Internal Ladder: Provide steel climbing ladder as shown with 2 inch x 1/4 inch sides and 3/4 inch diameter rungs at 12 inches on center. Attach anodized aluminum rail to ladder with climbing trolley for access to tank bottom. Provide slide supports to allow for tank movement.

k. Wear (Striker) Plates: Provide 12 inch square, 1/4 inch thick steel plates welded to tank bottom directly under the sounding opening, the fuel return discharge, and the fill discharge.

l. Lifting Lugs: Provide for rigging tanks.

m. Emergency Relief Vents for Fire Exposure: Venting capacity shall conform to NFPA 30 or NFPA 31 as applicable. Standard product of a manufacturer, designed to automatically open at tank pressure of 2.5 psig gauge. Aluminum or cast-iron construction with Teflon seating surface. Provide separate vents for primary and secondary tanks.

n. Provide fittings for grounding per NFPA 70.

o. Not used.

2.2 TANK AND PIPING ACCESSORIES

a. Vent Caps: Galvanized cast iron or cast aluminum with brass or bronze screens, arranged to permit full venting and to prevent entry of foreign material into the vent line. Same pipe size as vent pipe.

b. Remote Fill Box:

(1) Remote fill container enclosure mounted on adjustable stainless-steel legs to elevate camlock hose connector fill connection centerline between minimum 32 inches to maximum 46 inches above grade.

(2) Spill-container type enclosing a fill cap assembly with stainless-steel camlock hose connector with closure coordinated with fittings used by fuel supplier. Provide 3-inch diameter stainless-steel manually operated ball valve in fill box downstream of hose connector.

(3) NEMA-4 rated remote fill box enclosure and hinged door fabricated from minimum 14-gauge Type 304 stainless-steel shall include a spill containment area with minimum 10-gallon capacity. Remote fill box enclosure shall be a vented assembly with the hinged door being lockable with a padlock. Padlock to be furnished by the Owner. Fill box bottom container shall slope to one side toward drain and pump mounting location. Fill box shall be equipped with a manually operated hand pump assembly of stainless-steel construction with PTFE seals and FKM O-rings. Hand pump assembly shall discharge through stainless-steel check and manual shutoff ball valves to the aboveground tank fill pipe.

(4) Fill connection outlet shall have a 3-inch NPT threaded connection exiting the top of the fill box.

d. Ladders, Access Platforms, and Catwalks.

(1) Design criteria:

(a) In addition to the dead loads, design fabrications to support the following live loads unless otherwise specified.

(b) Ladders and Rungs: 120 kg (250 pounds) at any point.

(c) Railings and handrails:

1. Handrail:

a. Linear load: 50 pounds per linear foot.

b. Concentrated load: 200 pounds in any direction at any point.

2. Intermediate rail:

a. Concentrated load: 50 pounds in any direction at any point.

(d) Floor Plates, Gratings, Covers, Catwalks, and Platforms: 100 pounds per square foot. Use 300-pound force for concentrated loads

applied on an area of 4 square inch.

(2) Design criteria:

(a) Structural Steel: ASTM A36.

(b) Stainless Steel: ASTM A240, Type 302 or 304.

(c) Aluminum, Extruded: ASTM B221, Alloy 6063-T5 unless otherwise specified. For structural shapes use alloy 6061-T6 and alloy 6061-T4511.

(d) Grating with serrated walking surface:

1. Steel ASTM A786.

2. Aluminum: ASTM B632.

(e) Cast-Iron: ASTM A48, Class 30, commercial pattern.

(f) Malleable Iron Castings: A47.

(g) Primer Paint: As specified in Section 09 91 00, PAINTING.

(h) Stainless Steel Tubing: ASTM A269, type 302 or 304.

(i) Modular Channel Units:

1. Factory fabricated, channel shaped, cold formed sheet steel shapes, complete with fittings bolts and nuts required for assembly.

2. Form channel within turned pyramid shaped clamping ridges on each side.

3. Provide case hardened steel nuts with serrated grooves in the top edges designed to be inserted in the channel at any point and be given a quarter turn so as to engage the channel clamping ridges. Provide each nut with a spring designed to hold the nut in place.

4. Factory finish channels and parts with oven baked primer when exposed to view. Channels fabricated of ASTM A525, G90 galvanized steel may have primer omitted in concealed locations. Finish screws and nuts with zinc coating.

5. Fabricate snap-in closure plates to fit and close exposed channel openings of not more than 0.0125 inch thick stainless steel.

(3) Hardware

(a) Rough Hardware:

1. Furnish rough hardware with a standard plating, applied after punching, forming and assembly of parts; galvanized, cadmium plated, or zinc-coated by electro-galvanizing process. Galvanized G-90 where specified.

2. Use G-90 galvanized coating on ferrous metal for exterior work unless non-ferrous metal or stainless is used.

(b) Fasteners:

1. Bolts with nuts.

a. ASME B18.2.2.

b. ASTM A307 for 60,000 psi tensile strength bolts.

c. ASTM F468 for nonferrous bolts.

d. d) ASTM F593 for stainless steel.

2. Washers: ASTM F436, type to suit material and anchorage.

(4) Supports

(a) General:

1. Fabricate ASTM A36 structural steel shapes as shown.

2. Use clip angles or make provisions for welding hangers and braces to overhead construction.

3. Field connections may be welded or bolted.

(5) Grating

(a) Fabricate gratings to support live loads specified and a concentrated load as specified.

(b) Provide clearance at all sides to permit easy removal of grating.

(c) Make cutouts in gratings with 1/4 inch minimum to one inch maximum clearance for penetrations or passage of pipes and ducts. Edge band cutouts.

(d) Fabricate in sections not to exceed 25 square feet in area and 200 pounds in weight.

(e) Fabricate sections of grating with end-banding bars.

(f) Fabricate angle frames and supports, including anchorage as shown.

1. Fabricate intermediate supporting members from "T's" or angles.

2. Locate intermediate supports to support grating section edges.

3. Fabricate frame to finish flush with top of grating.

4. Locate anchors at ends and not over 24 inches O.C.

5. Butt or miter, and weld angle frame at corners.

(g) Steel Bar Gratings:

1. Fabricate grating using steel bars, frames, supports and other members shown in accordance with Metal Bar Grating Manual.

2. Galvanize steel members after fabrication in accordance with ASTM A123, G-90 for exterior gratings.

3. Use serrated bars for exterior gratings.

(h) Plank Gratings:

1. Conform to Fed. Spec. RR-G-1602.

2. Manufacturers standard widths, lengths and side channels to meet live load requirements.

3. Galvanize exterior steel gratings ASTM A123, G-90 after fabrication.

(6) Ladders

(a) Steel Ladders:

1. Fixed-rail type with steel rungs shouldered and headed into and welded to rails.

2. Fabricate angle brackets of 2-inch wide by 1/2-inch) thick steel; brackets spaced maximum of 4-feet apart and of length to hold ladder 7-inches from vertical surface to center of rungs. Provide turned ends or clips for anchoring.

3. Provide holes for anchoring with expansion bolts through turned ends and brackets.

4. Where shown, fabricate side rails curved, twisted and formed into a gooseneck.

5. Galvanize exterior ladders after fabrication, ASTM A123, G-90.

(b) Aluminum Ladders:

1. Fixed-rail type, constructed of structural aluminum, with mill finish.

2. Fabricate side rails and rungs of size and design shown, with the rungs shouldered and headed into and welded to the rails.

3. Where shown fabrication side rails curved, twisted and formed into gooseneck.

4. Fabricate angle brackets at top and bottom and intermediate brackets where shown. Drill for bolting.

(7) Railings

(a) In addition to the dead load design railing assembly to

support live load specified.

(b) Fabrication General:

1. Provide continuous welded joints, dressed smooth and flush.

2. Standard flush fittings, designed to be welded, may be used.

3. Exposed threads will not be approved.

4. Form handrail brackets to size and design shown.

(c) Handrails:

1. Close free ends of rail with flush metal caps welded in place except where flanges are secured using bolts with lock washers.

2. Make provisions for attaching handrail brackets to posts and handrail as shown.

(d) Steel Pipe Railings:

1. Fabricate of steel pipe with welded joints.

2. Number and space of rails as shown.

3. Space posts for railings not over 6 feet on centers between end posts.

4. Form handrail brackets from malleable iron.

5. Fabricate removable sections with posts at end of section.

6. Removable Rails:

a. Provide "U" shape brackets at each end for holding removable rails. Use for top and bottom horizontal rail when rails are joined together with vertical members.

b. Secure rail to brackets with 9 mm (3/8 inch) stainless steel through bolts and nuts at top rail only when rails joined with vertical members.

c. Continuously weld brackets to post.

d. Provide slotted bolt holes in rail bracket.

e. Weld bolt heads flush with top of rail.

f. Weld flanged fitting to post where posts are installed in sleeves.

(e) Opening Guard Rails:

1. Fabricate rails with flanged fitting at each end to fit between wall opening jambs.

2. Design flange fittings for fastening with machine screws to

steel plate anchored to jambs.

3. Fabricate rails for floor openings for anchorage in sleeves.

(f) Gates:

1. Fabricate from steel pipe as specified for railings.

2. Fabricate gate fittings from either malleable iron or wrought steel.

3. Hang each gate on suitable spring hinges of clamp on or through bolted type. Use bronze hinges for exterior gates.

4. Provide suitable stops, so that gate will swing as shown.

(8) Catwalks

(a) Fabricate catwalks including platforms, railings, ladders, supports and hangers, and arrangement of members as shown on drawings.

(b) Fabricate stairs as specified in Section 05 51 00, METAL STAIRS.

(c) Fabricate steel ladders as specified under paragraph LADDERS unless shown otherwise.

(d) Fabricate steel pipe railings as specified under paragraph "Railings".

(e) Catwalk and platforms floor surfaces as shown.

1. Steel gratings as specified under paragraph gratings, either bar or plank type.

e. Support horizontal portion of pipes located inside tank every 7 feet maximum.

f. Furnish gauging chart, versus and gallons versus inches depth.

g. Furnish sounding rod for each tank size. Mark rods in increments representing five percent of tank capacity. Rods shall be graduated in gallons.

2.3 PIPING, VALVES, FITTINGS

a. Fuel supply and return, tank fill, vents, sounding, and pump out.

b. Steel Pipe and Fittings:

(1) Piping: Steel, seamless or electric resistance welded (ERW), ASTM A53/A53M Grade B or ASTM A106/A106M Grade B, Schedule 40. Aboveground, outdoor, uninsulated piping shall be painted. Refer to Section 09 91 00, PAINTING.

(2) Joints: Socket or butt-welded. Threaded joints are prohibited except at valves, unions and tank connections.

(3) Fittings:

(a) Butt-welded joints: Steel, ASTM A234/A234M, Grade B, ASME B16.9, same schedule as adjoining pipe.

(b) Socket-welded joints: Forged steel, ASME B16.11, 2000 psig class.

(4) Unions: Malleable iron, 300 psig class.

(5) Companion flanges: Flanges and bolting, ASME B16.5.

(6) Welding flanges: Weld neck, ASME B16.5, forged steel ASTM A105/A105M, 150 psig.

c. Glass Fiber Reinforced Plastic (FRP) Pipe and Fittings:

(1) Conform to UL 971 and ASTM D2996 using a filament-winding process and epoxy or vinyl ester resins.

(2) Design pipe, fittings and joining system for required fuel service, 150 degrees F, 150 psig pressure, 20 inches Hg vacuum.

(3) Provide an integral resin-rich liner, 0.020 inches minimum thickness to enhance the corrosion resistance. Outer layer shall include ultra-violet inhibitors. Joining adhesive shall be designed for the pipe furnished and shall be supplied by the pipe manufacturer.

(4) Plastic piping allowed in underground use only.

d. Check Valves, Overfirll Prevention Valves, Ecternal Emergency Valves, Valves, Extractor Fittings, and Overfill Prevention.

(1) Pipe Sizes 2 inches and under: Rated for 200 psig water-oil-gas, swing-type, threaded ends, ASTM B62 bronze body. Provide union adjacent to valve.

(2) Pipe Sizes 2-1/2 inches and above: Rated for 200 psig water-oil-gas, swing-type, 125 pounds ASME flanged ends, ASTM A126 class B cast iron body.

e. Extractor Fittings: Arranged to permit removal of overfill prevention valves and other devices that are located in the aboveground storage tank. Provide extractor wrench.

f. Overfill Prevention Valve: Aluminum automatic valve designed for aboveground tanks. Locate valve near the top of the tank in the fill pipe. Pprovide single-stage valve, rated for fill flow and pressure, which stops flow completely at 95 percent of tank capacity. Valve shall include method for draining oil trapped above the valve into the tank.

g. External Emergency Valves:

(1) Normally open valve designed to shut off the flow of liquid in the event of a fire or excessive temperature. The fusible link shall hold a spring-actuated poppet in the open position.

(2) Body, cap, and poppet shall be fabricated from ductile iron and shall have non-shock C.W.P. rating of minimum 200 PSI W.O.G. Flanged models shall conform to ANSI B16.42 specification for class 150# raised face flanges. Cap gasket shall be fabricated from polytetrafluoroethylene (PTFE) with the stem seal fabricated from polytetrafluoroethylene encapsulated fluorocarbon elastomer.

(3) Provide with 165°F UL Listed fusible link. Valves 1-1/2-inch and smaller shall have threaded connections; valves 2-inch and larger shall have flanged connections.

h. Ball Valves:

(1) Valves 1-1/2-inch diameter and smaller:

(a) Forged brass valve body shall have threaded connections and shall have a non-shock C.W.P. rating of minimum 600 PSI W.O.G. Brass full port ball shall be chrome plated. Seals shall be polytetrafluoroethylene (PTFE).

(b) Valve shall be UL 842 listed for flammable liquids.

(2) Valves 2-inch diameter and larger:

(a) CF8M stainless steel valve body shall have 150# raised flange connections and shall have a non-shock C.W.P. rating of minimum 275 PSI W.O.G. Valve cap and full port ball shall be fabricated from CF8M stainless steel. Valve seat shall be molecularly enhanced polytetrafluoroethylene (ME-PTFE).

(b) Valve shall have API 607 4th edition fire safe approval.

i. Swing Check Valves:

(1) Valves 2-inch diameter and smaller:

(a) Body and cap shall be fabricated from ductile iron with threaded connections and shall have non-shock C.W.P. rating of minimum 200 PSI W.O.G. Internal components shall be fabricated from either ductile iron or stainless steel and shall have metal-to-metal seats. Cap gasket shall be fabricated from polytetrafluoroethylene (PTFE).

(b) Provide valve with thermal expansion relief valve having 50 psi setting and fabricated from stainless-steel and fluoroelastomer (FKM) components.

(2) Valves 3-inch diameter and larger:

(a) Body and cap shall be fabricated from ductile iron with 150# raised face flange connections and shall have non-shock C.W.P. rating of minimum 200 PSI W.O.G. Internal components shall be fabricated from either ductile iron or stainless steel and shall have metal-to-metal seats. Cap gasket shall be fabricated from polytetrafluoroethylene (PTFE).

(b) Provide valve with thermal expansion relief valve having 50 psi setting and fabricated from stainless-steel and fluoroelastomer (FKM) components.

2.4 SECONDARY CONTAINMENT FOR UNDERGROUND FUEL PIPING SYSTEMS AND FUEL PIPING ROUTED WITHIN SHALLOW CONCRETE TRENCHES

a. Enclose the fuel supply, return and fill pipes in factory-engineered and fabricated secondary containment conduit systems. The systems shall be complete with end seals, with 1 inches minimum continuous annular space, 1-1/2 inches between carrier pipes, which shall contain all leakage and which has provisions for leak detection system as specified.

b. Where piping is installed with less than 36-inches of earth cover, including where installed inside shallow concrete trenches, inner fuel oil carrier pipe shall have minimum 1-inch thick polyisocyanurate or phenolic insulation. Annular spacing between insulated carrier pipe and secondary containment conduit shall be maintained as described in paragraph above.

c. Steel Conduit with Fusion-Bonded Epoxy Coating and Cathodic Protection:

(1) Galvanized carbon steel pipe, ASTM A53/A53M, Grade B, Schedule 40 for diameters through 5 inches, 0.134 inch thick for diameters greater than 5 inches up through 26 inches. All welded construction.

(2) Sand blast exterior per SSPC SP 6/NACE No.3.

(3) Coat exterior with 20 mils thick fusion-bonded epoxy.

(4) Provide cathodic protection designed by corrosion specialist and consisting of galvanic anodes, test stations, interconnecting wiring in conformance with UL 1746 and NACE SP0169. Electrical isolation required between all connecting systems in manholes and buildings.

d. Steel Conduit with Fiberglass Reinforced Plastic (FRP) Coating:

(1) Carbon steel pipe, ASTM A53/A53M, Grade B, Schedule 40 for diameters through 5 inches, 0.134 inch thick for diameters greater than 5 inches up thru 26 inches. All welded construction.

(2) Blast clean exterior per SSPC 7/NACE No.4.

(3) Apply fiberglass reinforced polyester (FRP) external cladding at least 0.10 inches thick with ultra-violet inhibitor. Cladding on field joints shall be equivalent to factory-applied cladding applied on remainder of system.

(4) Test entire system for holidays using a 35,000-volt holiday detector.

e. Glass Fiber Reinforced Plastic (FRP) Conduit:

(1) Conform to UL 971 and ASTM D2996 using a filament-winding process and epoxy or vinyl ester resins.

(2) Design pipe, fittings and joining system for carrier pipe fuel service, 150 degrees F, 150 psig pressure, 20 inches Hg vacuum.

(3) Provide an integral resin-rich liner, minimum thickness 0.010 inch. Outer layer shall include ultra-violet inhibitors.

(4) Minimum total wall thickness 0.07 inch for diameters below 8 inches, 0.11 inch for diameters 8 inches and 10 inches, 3/16 inch for diameters 10 inches through 20 inches, and 1/4 inch for diameters above 20 inches.

f. Pipe Supports: Provide supports within conduit for fuel carrier pipes spaced 7 feet apart except 10 feet apart for carrier pipe size 2 inches through 4 inches. Support design shall permit differential movement of pipes, allow drainage of leakage to sumps, and maintain alignment of carrier pipes.

g. Conduit End Seals: Same material and coating as conduit; leak tight.

h. Leak Detector Sensor Locations: On each piping system, provide sumps at the low points with water-tight openings above grade for access to leak detector sensors. Design sumps to intercept all potential leakage. Maximum spacing between sumps, 10 feet.

2.5 LEAK DETECTION SYSTEMS

a. Automatic digital continuous monitoring systems responsive to the presence of water and hydrocarbons in the interstitial space of the double-wall tanks, in the tank manhole access enclosures, and in the secondary containment of fuel piping systems. System shall distinguish between hydrocarbon and water and identify location of leak as to individual tank and piping system. System may be combined with tank fluid level monitor and alarm system specified in paragraph, TANK FILL CONTROL, ACTIVE TANK SELECTION, FLUID LEVEL MONITOR AND ALARM SYSTEM.

b. Functions and Arrangement:

(1) Single control station to monitor all sensing probes.

(2) Visual indicator to monitor and identify leaks as water or hydrocarbon and location.

(3) Indicators showing system status including faults and alarms.

(4) On board printer that provides complete reports of all system functions upon command.

(5) Panel circuit test button.

(6) 95 dB audible alarm with silencing control to sound when leak is detected.

(7) Eight-hour memory backup system with battery.

(8) NEMA 250 Type 4 cabinet.

(9) UL or other accredited testing laboratory listing.

(10) RS232 Modbus communications with engineering control system to indicate system in service and alarm conditions.

c. Sensors:

(1) Designed for required locations including: Insertion between walls of double-wall tanks, in sumps in double-wall piping systems and in tank manhole enclosures. Sensing points shall be at lowest point of each tank or sump. Intrinsically safe design.

(2) Sensing units shall detect presence of water and a minimum 1/8 inch thick layer of hydrocarbon on surface of water and minimum 2 inch thickness of hydrocarbon in area that has no water present.

(3) Sensors shall be arranged to allow replacement of individual sensors without disturbing other portions of leak detection system or fuel storage and piping system. Underground sensors shall be accessed through caps at grade.

(4) Materials of construction shall be non-corroding.

(5) Transmit status signal to control unit.

d. Components:

(1) Provide manholes at grade for each sensor cap. Manholes shall be cast iron, quick-opening cover, watertight, minimum size necessary to accommodate sensor caps. Provide identification plates, similar to those specified for fill points, labeled “MONITORING/OBSERVATION WELL-DO NOT FILL”. Provide special tools if necessary for opening covers.

(2) Sensor housings from tank and piping to grade shall be Schedule 40 PVC, or stainless steel.

(3) Underground wiring between probes and control unit: Place in water-tight corrosion-resistant conduit system conforming to Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION.

2.6 TANK FILL CONTROL, ACTIVE TANK SELECTION, FLUID LEVEL MONITOR AND ALARM SYSTEMS

a. Digital systems for central monitoring and control tank filling, selecting the active fuel oil tank, and monitoring of fuel and water levels in all fuel oil storage tanks in the project. High and low level visual and audible alarms. Volumetric tank-tightness testing. Complete with all transducing, transmitting, and receiving devices. On board printer to provide complete report of all system functions upon command. System may be combined with leak detection system specified in paragraph, LEAK DETECTION SYSTEMS.

b. Fill Control:

(1) Digital readout, showing positions of all five (5) FOF 2-position fill valves, and which tank is selected for filling aboveground fuel oil tank. Programming shall allow only one fill valve to open at a time. Fuel oil supplier shall be able to connect a tanker truck to a common fill port at the remote aboveground tank fill box and select a single aboveground tank to pump fuel oil into.

(2) Fill control shall indicate an "full tank" condition when tank being filled reaches 90% of its volumetric capacity. Fill control

shall indicate an "alarm" condition when tank being filled reaches 95% of its volumetric capacity.

c. Active Tank Selection:

(1) Digital readout, showing positions of all five (5) FOR 2-position return valves, and all five (5) FOS 2-position supply valves, and which tank is selected as the "active" tank. Programming shall require one "active" tank at all times. The "active" tank shall have both 2-position FOS and FOR valves in the open position. The "active" tank will be the tank supply fuel oil to the equipment connected to the fuel oil piping system. Programming shall allow only a single tank to be "active" at a time.

(2) Active tank controls shall indicate an "alarm" condition when the fuel oil level drops below 10% (adjustable) of the tank's volumetric capacity.

b. Fluid Level Monitor:

(1) Digital continuous readout, showing tank oil and water levels in gallons, smallest reading one gallon. Provide identification of product measured, measuring units, and the tank number.

(2) Tank and fuel characteristics contained in preprogrammed non-volatile field-replaceable databases. Protected power supply.

c. High and Low Fluid Level Alarm System:

(1) Automatic continuous on-line monitoring of all tanks.

(2) Visual and audible indicators combined with fluid level monitor. Identify the tank that is in alarm condition.

(3) Manual alarm test and silencing controls.

(4) Low level alarm actuation adjustable 0-25 percent of tank capacity. High level alarm actuation adjustable 75-100 percent of tank capacity.

d. Locate all indicators, selector switches, alarms on face of wall-mounted NEMA 250, Type 4 panel.

e. Remote Alarm Annunciator:

(1) Visual and audible high-level alarms adjacent to tank fill box locations. Locate in NEMA 250 Type 4X weatherproof exterior wall or pole-mounted panels.

(2) Alarm shall include flashing red light with 180-degree visibility for each tank and 95 dB horn or 4 inch diameter bell. Provide alarm silence control.

(3) Provide identification sign: "WHEN ALARM SOUNDS - FUEL TANK FILLED TO CAPACITY - DO NOT OVERFILL".

f. Modbus communication to engineering control system to indicate tank fluid level and alarm conditions. Telephone modem communication capability.

g. System Performance: Accuracy plus or minus 0.10 inch of fluid height in inventory mode and 0.01 inch in leak detection mode. Automatic compensation for fluid temperature changes. Volumetric tank tightness sensitivity of 0.1 gph.

h. Sensors:

(1) Provide sensor types such as magnetostrictive, capacitance, float, hydrostatic and other types as necessary for the applications.

(2) Apply in accordance with manufacturer's instructions with provisions for easy future replacement without need for excavation.

(3) Provide for each hydrostatic sensor a constant flow differential pressure regulator and transmitter protected from fuel contamination. Air supply shall include filter and over-pressure protection. Provide desiccant-type dryer on air supply designed for removal of water vapor. Dryer rating, minimum 10 SCFM. Provide moisture indicator. Dryer may be deleted if air supply source has a refrigerated dryer.

(4) Float-type units shall be designed for installation and removal through a 4 inch diameter vertical pipe mounted in the top of the tank.

i. Underground Wiring and Piping: Enclose in water-tight corrosion-resistant conduit system sized and arranged as recommended by system manufacturer and conforming to Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION.

j. Code Conformance: NFPA 70.

2.7 FUEL OIL POLISHING SYSTEMS

a. Complete factory-assembled automatic particulate filtration and dewatering system to maintain the purity of No. 2 fuel oil in storage. The system shall circulate the oil from the storage tank, through the system, and back to the storage tank. Provide quantity and capacity of systems to serve tanks as shown, connected to the tank suction and return pipes. Refer to drawings for the quantity of aboveground storage tanks being served by the fuel oil polishing system. Smaller systems without large water storage tanks and without fuel additive injection shall be enclosed in a NEMA 4 enclosure suitable for mounting on Unistrut support system. Digital controls.

b. Performance: Design for nearly 100 percent water removal. Provide 2-micron particulate filtration. Each system shall have capacity to turn over the largest connected full tank one time within 22 hours maximum. System shall be designed to allow continuous operation with brief interruptions to manually change filters and clean strainers.

c. Components:

(1) Strainer: 20 mesh wye strainer with perforated stainless steel element. 125 psig design pressure.

(2) Water Separation Unit: Two stage, designed to reduce water content of fuel to less than 10 ppm. Centrifugal separator for

removal of large droplets and renewable resin-impregnated cellulose water coalescing elements. Water removed shall flow to water holding sump in the unit. Water sensing probe to alert the operator when water level in bowl has reached capacity.

(3) Filter: Two filters in series as follows:

(a) 25-micron particulate filtration with 96 percent removal efficiency, valved manual drain. Replaceable elements.

(b) 3-micron particulate filtration with 96 percent removal efficiency, valved manual drain. Replaceable elements.

(4) Pressure gauges across the following system components:

(a) Water separator.

(b) Pump.

(c) 25-micron particulate filter.

(d) 3-micron particulate filter.

(5) Magnitic conditioner for removing ferrous particles.

(6) Heaters: Provide the following:

(a) Enclosure heater to prevent the build-up of condensation.

(b) Fuel heater to prevent gelling during low ambient temperatures.

(7) Leak Detector: shall be provided to indicate an alarm condition upon detection of hydrocarbon liquids in bottom of enclosure.

(8) Filtration Pump: Positive displacement spur gear pump with cast iron or bronze housing, for circulating the oil from the storage tank, through the water separation and filter units and back to the storage tank. Pump shall have carbon bushings, stainless steel shaft and Teflon mechanical seal, ODP motor. Refer to Pump Schedule on drawings for pump sizing, power, and flow requirements.

(9) Controls:

(a) Digital PLC electronic controls for all system control and alarm functions. Relay logic not acceptable.

(b) Control panel with selector for modes of operation, indicators to show system status, and visual and audible alarms to signal the need for operator intervention. Operator interface shall be 2 x 20 LCD and keypad.

(c) Controls shall include:

1. Control power “on-off”.

2. “Cycle Start”.

3. “Cycle Cancel”.

4. “Hand-off-Auto” for filtration pump.

5. Pump cycle timer set function.

6. Cycle duration selector.

7. Vacuum switch at pump suction for low pressure system alarm and pump shutdown safety.

8. Pressure transmitters across 3-micron and 25-micron filter elements for monitoring and alarming of system pressure levels.

9. Flow transmitter for indicating and monitoring of polishing system flow rate.

(d) Indications on control panel cover shall include:

1. “System Power On”.

2. “Pump Run”.

3. “High Water” alarm.

4. “Basin Leak” alarm.

5. “High Pressure” alarm.

6. “Low Flow” alarm.

7. “High Vacuum” alarm.

(e) Filter and strainer differential pressure gauges, differential pressure transmitters and control. Provide indication when filters should be changed.

(f) Over pressure switch and control to shut down pump if filter inlet pressure exceeds limits.

(g) All primary wiring exiting the enclosure shall be encased in conduit.

(h) Magnetic motor starters with overload protection.

(i) Circuit breakers.

(j) Control enclosure shall be NEMA 4, stainless-steel, fully gasketed doors with 3 point lockable latching. Interior shall have white gloss finish. All controls and indicating devices shall be mounted on front of enclosure and labeled with black Phenolic labels with white lettering. Refer to E2/M503 for control panel cover lights and switches required on control panel face.

(k) Modbus communication to engineering control system for alarms and system status.

d. Enclosure – Wall Mounted Units (mounted from Unistrut support members):

14-gauge Type 304 stainless steel, NEMA Type 4 enclosure, continuously welded, framed cabinet. Provide doors for complete access to all equipment. Doors shall have a turned edge, piano hinges, three-point locking mechanisms. No penetrations shall be made through the enclosure bottom to prevent the possibility of fuel oil leaks.

e. Sequence of Operation: Program controls to automatically polish the aboveground fuel oil tanks as follows

(a) Fuel oil polisher control panel shall be programmed to automatically polish a single aboveground fuel oil storage tank at a time by opening one group of fuel oil supply and return valves. Refer to "Fuel Oil Polisher Controls (FPC-A)" on the drawings for additional information.

(b) Once a group of fuel oil supply and return valves serving a single tank have proven open, the fuel oil polisher pump shall be energized for 24-hours (adjustable).

(c) After the fuel in a single tank has been polished the pump shall be de-energized and the fuel oil supply and return valves shall be closed. Controls programming shall then switch to the next tank to be polished, and the cycle shall repeat until all five (5) tanks have been polished.

(d) Controls programming shall automatically polish all five tanks on a once every four-week period, starting on a Monday and concluding on Friday (adjustable) basis.

(e) Controls programming shall allow for any aboveground fuel oil tank as being "out-of-service" which shall prevent automatic fuel oil polishing for that tank until it is released from that mode.

f. Water Drainage System: Sealed bowl (bottle) with high level alarm system. Water collected in filters shall drain to a sealed bowl that can be easily removed and emptied. Provide electric heater within enclosure as needed to prevent collected water from freezing during periods of low ambient temperatures.

h. Piping: Refer to Section 23 21 11, BOILER PLANT PIPING SYSTEMS.

i. Pressure Gauges: Refer to Section 23 21 11, BOILER PLANT PIPING SYSTEMS.

2.8 BURIED UTILITY WARNING TRACING TAPE

Tape shall be 0.004 inch thick, 6 inches wide, yellow polyethylene with a metallic core, acid and alkali-resistant and shall have a minimum strength of 1740 psig lengthwise and 1500 psig crosswise with an elongation factor of 350 percent. Provide bold black letters on the tape identifying the type of system. Insulating and labeling shall be unaffected by moisture and other substances contained in the backfill material.

2.9 ELECTRIC FUEL OIL HEATER FOR OUTDOOR, ABOVEGROUND TANKS

a. Complete factory-assembled and programmed automatic fuel oil heating system to maintain fuel oil temperature above 36°F (adjustable). Provide each aboveground fuel oil tank with dedicated heater mounted adjacent to tank as shown on drawings. Digital controls programmed per

sequence of operation as shown on drawings.

b. Components:

(1) Strainer: 20 mesh wye strainer with perforated stainless steel element. 125 psig design pressure.

(2) Pump: Positive displacement spur gear pump with cast iron or bronze housing, for circulating the oil from the storage tank, through the fuel oil heater and back to the storage tank. Pump shall have carbon bushings, stainless steel shaft and Teflon mechanical seal, ODP motor. Refer to pump schedule on drawing for sizing requirements. 125 psig design pressure.

(3) Electric Heater: UL Listed electric fuel oil heater shall be equipped with manufacturer furnished thermostat that will energize heater element when fuel oil entering heater drops below an adjustable set point. A current switch shall disable electric heater when pump motor is not energized. Refer to fuel oil heater schedule on drawing for sizing requirements.

(4) Vacuum Switch (VS): A vacuum switch mounted at fuel oil pump suction shall disable fuel oil pump (manual reset) when pressure drops below 5 inches of mercury (adjustable).

(5) Pressure Transmitter (PT): A pressure transmitter mounted at fuel oil pump discharge shall disable fuel oil pump (manual reset) when pump discharge exceeds 75 psig for more than 5 seconds (adjustable). Digital controller shall display fuel oil pressure at pump discharge when pump is energized. Programming shall prevent nuisance pressure trips when pump motor is first energized.

(6) Thermometer: Provide fluid-filled thermometers, with 0°F to 160°F scale, in fuel oil piping upstream and downstream of electric fuel oil heater.

(7) Fuel Oil Temperature Transmitter: Provide temperature transmitters in fuel oil piping upstream and downstream of electric fuel oil heater.

(8) Outside Air Temperature Transmitter: Provide outside air temperature transmitters for sensing outside air temperature for use as described in the sequence of operation (refer to drawings). Outside air temperature sensor shall penetrate the enclosure via a weather-tight rubber grommet.

(9) Safety Valve: Provide safety valve at heater to allow excess pressure to bypass discharge isolation valve (refer to flow diagram on drawings).

(10) Hydrocarbon Detection Cable: Provide hydrocarbon leak detection cable in bottom of enclosure to indicate an alarm at the control panel upon fuel oil detection.

(11) Controls:

(a) Digital PLC electronic controls for all system control and alarm functions. Relay logic not acceptable.

(b) Control panel with selector for modes of operation, indicators to show system status, and visual and audible alarms to signal the need for operator intervention. Operator interface shall be 2 x 20 LCD and keypad.

(c) Controls shall include:

1. Control power “on-off”.

2. “Hand-off-Auto” for heater pump.

(d) Indications shall include:

1. “Control Power On”.

2. “Pump Run”.

3. “Pump Failure”.

4. “Low Pump Suction Pressure”.

5. “High Pump Discharge Pressure”.

6. “Rupture Basin Leak” alarm.

7. “High Pressure” alarm and automatic shutdown.

(e) Strainer differential pressure gauges, differential pressure switches and control.

(f) Over pressure switch and control to shut down pump if discharge pressure exceeds 75 psig.

(g) All primary wiring exiting the enclosure shall be encased in conduit.

(h) Magnetic motor starters with overload protection.

(i) Circuit breakers.

(j) Control enclosure shall be NEMA 4, stainless-steel, fully gasketed doors with 3 point lockable latching. Interior shall have white gloss finish. All controls and indicating devices shall be mounted on front of enclosure and labeled with black Phenolic labels with white lettering. No penetrations shall be made through the enclosure bottom to prevent the possibility of fuel oil leaks.

(k) Modbus communication to engineering control system for alarms and system status.

c. Provide electric fuel oil heating system with single point power connection sized for 460V 3-phase power. Provide all required transformers, circuit breakers/fuses, and relays as required to comply with National Electric Code requirements. Provide wiring diagram and corresponding labeled terminal blocks.

d. Refer to drawings for fuel oil heater schedule, flow diagrams, and

control diagrams.

PART 3 EXECUTION

3.1 GENERAL

If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

3.2 INSTALLATION AND TESTING, ABOVEGROUND TANKS

a. Conform to NFPA 30 or NFPA 31 as applicable.

b. Support tanks on steel saddles welded to the tanks. Anchor to concrete foundations. Provide molded neoprene isolation pads between the steel supports and the concrete foundation.

c. After tanks are installed, test steel tanks with air pressure of 3 to 5 psig, using soapsuds to locate leaks. Repair leaks by chipping to bare metal and rewelding. Retest until all leaks are repaired. Repair all damaged areas of prime coat on tanks and steel dikes (if furnished). Test interstitial area between steel tank walls with air at pressure recommended by tank manufacturer. Tests shall be witnessed by the Contracting Officer.

d. For steel tanks storing heated oil, field-applied insulation requirements are specified under Section 23 07 11, HVAC AND BOILER PLANT INSULATION.

e. Surface finish coating for tanks and steel dikes (if furnished) is specified under Section 09 91 00, PAINTING.

f. Provide electrical grounding in accordance with NFPA 70.

3.3 INSTALLATION AND TESTING, UNDERGROUND PIPING SYSTEMS

a. Leak Detection System: Arrange fuel and tracing media carrier piping, enclosed in secondary containment piping, to accommodate leak detection system. Slope piping down toward tanks and leak detectors at 1 inch in 40 feet.

b. Steel Fuel and Tracing Media Carrier Piping: All joints butt or socket welding. Threaded piping is prohibited. Piping ends shall be accurately cut, true, and beveled for welding.

c. Glass Fiber Reinforced Plastic (FRP) Fuel Carrier Piping and Secondary Containment Piping: Install in accordance with printed instructions of pipe manufacturer. Installation personnel trained in accordance with paragraph, QUALITY ASSURANCE. Plastic piping is prohibited in the same secondary containment system with steam or condensate piping.

d. Secondary Containment Piping:

(1) Provide sand bedding and backfill material for steel piping and pea gravel for FRP piping.

(2) Top of system 18 inches minimum below grade.

(3) Design and locate leak detector sumps to intercept all potential leakage. Maximum spacing along each system, 10 feet.

(4) Seal all building and manhole wall penetrations with a modular, watertight flexible penetration seal system. The modular penetration seal shall have a nitrile rubber seal, or if a fire separation is required, a high temperature silicone fire seal.

(5) After placing system, prior to backfill, repair all damage, including coatings, as recommended in printed instructions of system manufacturer. Perform 10,000-volt holiday test on coated steel systems.

(6) Fuel oil piping is prohibited in the same secondary containment system as steam or condensate piping.

(7) On steel systems that do not have FRP cladding, install cathodic protection system.

e. Leak Test: Test carrier pipes with air pressure at 100 psig, and test the containment piping with air pressure at 8 psig. Systems shall hold the pressure for 30 minutes. Repair all leaks and retest.

f. Coatings for Steel Piping not in Secondary Containment System: Provide urethane coating and cathodic protection.

3.4 INSTALLATION, FILL BOXES

Provide for tank fill, tank sounding, leak detector sensors, and extractor fittings. Refer to fill box detail.

3.5 INSTALLATION AND TESTING, LEAK DETECTOR SYSTEMS FOR TANKS AND PIPING

a. Wiring shall conform to NFPA 70.

b. Locate control monitor panels 5 feet above the floor on inside wall of boiler room, generator room or garage, depending on type of fuel tank served, unless shown otherwise.

c. Test operation of each probe, and monitoring system with fuel and water. If type of probe utilized is damaged by exposure to fuel, provide temporary probe for testing monitoring system.

3.6 INSTALLATION, TANK FLUID LEVEL INDICATOR AND ALARM SYSTEM

a. Wiring shall conform to NFPA 70.

b. Locate level indicator and alarm panel 5 feet above grade near the remote fill box.

c. Locate remote high-level alarm on exterior wall or pole in view of tank fill point, 8 feet above grade.

3.7 INSTALLATION, BURIED UTILITY WARNING TRACING TAPE

Install tracer wire in the trench approximately 18 inches above the non-metallic pipe. The tracer wire shall be taped approximately every 10 feet to the pipe, where practical. The tracer wire shall be installed so that electrical continuity is maintained throughout the pipe system. As

few connections as possible shall be made in the tracer wire. The wire shall be contiguous except at test stations, valve boxes, and where splicing is required. All splices shall be encased. Connections will be made by stripping the insulation back one inch and joining the two ends using an approved mechanical connector and a split bolt connector. Twisting of copper wire is prohibited. To complete this connection, wrap all exposed wire thoroughly with electrical tape. A minimum 5 foot of additional tracer wire will be coiled, buried and terminate aboveground at the ends of the pipeline.

3.8 INSTALLATION, FUEL OIL QUALITY MAINTENANCE SYSTEMS

a. Locate systems within easy reach of persons standing on floor.

b. Connect to tank suction and return piping systems with isolation valves. Provide compound pressure gauges at suction and discharge piping connections. Refer to Section 23 21 11, BOILER PLANT PIPING SYSTEMS for gauge requirements.

3.9 TANK MANHOLE ENCLOSURES

All pipe penetrations shall be leak tight permitting no groundwater into enclosure.

3.10 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The Commissioning Agent will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with Contracting Officer and Commissioning Agent. Provide a minimum notice of 10 working days prior to startup and testing.

3.11 INSTALLATION, CATHODIC PROTECTION TEST STATIONS

Provide separate station for each piping system, anchor firmly, locate so that terminal board is 2 feet minimum above grade. Connect wiring from all anodes and protected structures to the test stations.

3.12 TESTING, CATHODIC PROTECTION

a. Testing performed by NACE-certified corrosion specialist; witnessed by Contracting Officer.

b. Test Instruments:

(1) Volt-Ammeter.

(2) Saturated copper-copper sulfate reference electrode.

(3) Other instruments as required.

c. Procedures: Conform to NACE SP0169.

d. Test Results Required for Acceptance:

(1) Potential of minus 0.85 volt between protected structure and reference electrode.

(2) Minimum shift of minus 300 millivolts upon application of protective current. Voltage measured between protected structure and reference electrode.

(3) Minimum shift of minus 100 millivolts upon interruption of protective current. Voltage measured between protected structure and reference electrode.

e. Test Report: Provide complete report to Contracting Officer showing all test measurements, calculations, list of instruments used.

3.13 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.14 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 8 hours to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

-- End of Section --

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SECTION 23 21 11
3/24

BOILER PLANT PIPING SYSTEMS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2024) Factory-Made Wrought Buttwelding Fittings
ASME B16.11	(2022) Forged Fittings, Socket-Welding and Threaded
ASME B16.34	(2021) Valves - Flanged, Threaded and Welding End
ASME B31.1	(2024) Power Piping
ASME B31.9	(2020) Building Services Piping
ASME B40.100	(2022) Pressure Gauges and Gauge Attachments
ASME BPVC	(2010) Boiler and Pressure Vessels Code
ASME BPVC SEC I	(2017) BPVC Section I-Rules for Construction of Power Boilers
ASME BPVC SEC IX	(2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications
ASME BPVC SEC VIII	(2021) Boiler and Pressure Vessel Codes: Section VIII Rules for Construction of Pressure Vessel

AMERICAN WELDING SOCIETY (AWS)

AWS B2.1/B2.1M	(2021) Specification for Welding Procedure and Performance Qualification
AWS Z49.1	(2021) Safety in Welding, Cutting and Allied Processes

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A105/A105M	(2023) Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A106/A106M	(2019a) Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM A216/A216M	(2021) Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service
ASTM A234/A234M	(2024) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-45	(2020) Bypass and Drain Connections
MSS SP-58	(2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation
MSS SP-80	(2019) Bronze Gate, Globe, Angle and Check Valves
MSS SP-127	(2014; Rev A) Bracing for Piping Systems: Seismic-Wind-Dynamic Design, Selection, and Application
MSS SP-97	(2019) Integrally Reinforced Forged Branch Outlet Fittings: Socket Welding, Threaded, and Buttwelding Ends

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 30	(2024; TIA 24-1) Flammable and Combustible Liquids Code
NFPA 31	(2024; TIA 23-1) Standard for the Installation of Oil-Burning Equipment

PIPE FABRICATION INSTITUTE (PFI)

PFI ES 24	(2015) PIPE BENDING METHODS, TOLERANCES, PROCESS AND MATERIAL REQUIREMENTS

1.2 DESCRIPTION

a. All boiler plant piping systems, except plumbing and sanitary, including piping supports. Piping located outside of the boiler plant

building is not included except for gas regulator and meter stations.

b. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

1.3 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 09 91 00, PAINTING.

f. Not used.

g. Section 22 11 00 FACILITY WATER DISTRIBUTION.

l. Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, AND BALANCING FOR HVAC.

m. Section 23 07 11, HVAC AND BOILER PLANT INSULATION.

n. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

r. Section 23 50 11, BOILER PLANT MECHANICAL EQUIPMENT.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Welding Qualifications; G, AE

SD-02 Shop Drawings

Pipe Support Systems; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 21 11, BOILER PLANT PIPING SYSTEMS”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Piping:

(a) ASTM material specification number.

(b) Grade, class or type, schedule number.

(c) Manufacturer.

(d) Intended service.

(2) Pipe Fittings, Unions, Flanges:

(a) ASTM material specification number.

(b) ASME standards number.

(c) Catalog cuts.

(d) Pressure and temperature ratings.

(e) Intended service.

(3) Valves - Gate, Globe, Check, Plug, Butterfly, Ball:

(a) Catalog cuts showing design and construction.

(b) Pressure and temperature ratings.

(c) Materials of construction.

(d) Accessories.

(e) Intended service.

(4) Sight Flow Indicators:

(a) Catalog cuts showing design and construction.

(b) Pressure and temperature ratings.

(c) Materials of construction.

(d) Intended service.

(5) Quick - Couple Hose Connectors:

(a) Catalog cuts showing design and construction.

(b) Pressure and temperature ratings.

(c) Materials of construction.

(d) Type of seal between couplings.

(6) Pressure Reducing and Regulating Valves, Safety Valves, Relief Valves:

(a) Catalog cuts showing design and construction.

(b) Service limitations (type of fluid, maximum pressure, and temperatures).

(c) Materials of construction.

(d) Flow capacity at required set pressure or differential pressure.

(7) Strainers:

(a) Catalog cuts showing design and construction.

(b) Pressure and temperature ratings.

(c) Materials of construction.

(d) Strainer basket or liner mesh size.

(e) Pressure loss and flow rate data.

(f) Intended service.

(8) Flexible Connectors:

(a) Catalog cuts showing design and construction.

(b) Pressure and temperature ratings.

(c) Materials of construction.

(d) Maximum allowable lateral and axial movements.

(e) Description of type of movement permitted intermittent offset or continuous vibration.

(f) Intended service.

(9) Pipe Support Systems: The contractor shall provide the following with submissions.

(a) Credentials of technical personnel who will design the support systems.

(b) Description of computer program for pipe support selection indicating its capability and limitations. Provide documentation showing both program verification and validation results. List of projects where it was employed successfully.

(c) Input and output data for pipe support selection program for all piping systems with pipe sizes 2-1/2 inches and above.

(d) Hanger load calculation methods and results for piping systems with pipe sizes 2 inches and below.

(e) Piping layouts showing location and type of each hanger support and anchors with unique identifiers that can be referenced back to the calculation results.

(f) Catalog cuts showing design and construction of each hanger support and anchor and their conformance to MSS standards.

(g) Drawings showing arrangement and sizes of all components comprising each spring-type hanger and support assembly.

(h) Load rating and movement tables for all spring hangers, and seismic shock absorbing devices.

(j) Stress analyses on the boiler plant piping systems under all possible load conditions as part of the design. Once all piping is completed another stress analysis is required on the as built systems. Documentation results shall flag locations/components requiring recommended revision/modification to obtain acceptable stress levels.

d. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation, and maintenance.

e. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

f. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.6 QUALITY ASSURANCE

a. The products and execution of work specified in this section shall conform to the referenced codes and standards as required by the specifications. Local codes and amendments shall be enforced, along with requirements of local utility companies. The most stringent requirements of these specifications, local codes, or utility company requirements shall always apply. Any conflicts shall be brought to the attention of the Contracting Officer.

b. Welding Qualifications: Before any welding is performed, contractor shall submit a certificate certifying that welders comply with the following requirements:

(1) Qualify welding processes and operators for piping according to ASME BPVC SEC IX, AWS Z49.1 and AWS B2.1/B2.1M.

(2) Comply with provisions in ASME B31.9.

(3) Certify that each welder and welding operator has passed AWS qualification tests for welding processes involved and that certification is current and recent. Submit documentation to the Contracting Officer.

(4) All welds shall be stamped according to the provisions of the American Welding Society.

c. ASME Compliance: Comply with ASME B31.9 for materials, products, and installation. Safety valves and pressure vessels shall bear appropriate ASME labels.

1.7 DELIVERY, STORAGE AND HANDLING

All piping shall be stored and kept free of foreign material and shall be internally and externally cleaned of all oil, dirt, rust, and foreign material. Deliver and store valves and pipe hangers in sealed shipping containers with labeling in place. Storage shall be in dry, protected location.

1.8 INFORMATION ON PRESSURE-TEMPERATURE DESIGN OF PIPING SYSTEMS

a. Fuel oil system pressures are determined by the requirements of the burners and fuel trains. No. 2 oil systems are designed for maximum temperatures of 130 degrees F, and pressures of 150 psig.

1.9 AS-BUILT DOCUMENTATION

Comply with requirements in Paragraph “AS-BUILT DOCUMENTATION” in Section 23 05 11 COMMON WORK RESULTS FOR HVAC.

PART 2 PRODUCTS

2.1 FUEL PIPING

a. Fuel oil (No. 2) for main burner and igniter (pilot) fuels. Comply with ASME B31.1.

b. Piping: Carbon steel, ASTM A53/A53M Grade B or ASTM A106/A106M Grade B, seamless or ERW, Schedule 40.

c. Joints:

(1) Pipe sizes 2-l/2 inches and above: Butt-welded, no other types allowed.

(2) Pipe sizes 2 inches and below: Socket-welded or butt-welded.

d. Fittings:

(1) Butt-welded joints: Steel, ASTM A234/A234M, Grade B, ASME B16.9, same schedule as adjoining pipe.

(2) Socket-welded joints: Forged steel, ASME B16.11, 2000 psig class.

e. Unions on piping 2 inches and under: Forged steel, 2000 psig class or 3000 psig class.

f. Flanges: Forged steel weld neck, ASME B16.5, ASTM A105/A105M, 150 psig.

g. Companion flanges: Flanges and bolting shall conform to ASME B16.5.

2.2 VALVES; GATE, GLOBE, PLUG, CHECK, BALL, BUTTERFLY, VENT COCKS

a. Valves for particular services are generally specified as Type Numbers. The Type Numbers are defined below. All valves of the same type shall be the products of a single manufacturer. Comply with MSS SP-45, MSS SP-80, and ASME B31.1. Design valves for the service fluids and conditions. Pressure-temperature ratings listed are minimum requirements. Packing and gaskets shall not contain asbestos.

b. Valve Type Designations:

(1) Gate Valves:

(a) Type 101: Cast steel body ASTM A216/A216M WCB, rated for at 150 psig at 500 degrees F, ll.5 to l3 percent chromium stainless steel flexible wedge and hard faced (stellite) or nickel copper alloy seats, 150 psig ASME flanged ends, OS&Y, rising stem, bolted bonnet.

1. Provide factory installed globe-valved warm-up bypass when main valve is 3 inch pipe size or greater and serves steam main longer than 20 feet. Conform to MSS SP-45.

2. Drill and tap bosses for connection of drains if valve is in steam service. Conform to MSS SP-45.

(b) Type 102: Cast steel body ASTM A216/A216M WCB, Class 300,

ll.5 to l3 percent chromium stainless steel flexible wedge and hard faced (stellite) alloy seats, ASME flanged ends, OS&Y, rising stem, and bolted bonnet.

1. Provide factory installed globe-valved bypass when main valve is 3 inch pipe size or greater and serves steam main longer than 20 feet. Conform to MSS SP-45.

2. Drill and tap bosses for connection of drains if valve is in steam service. Conform to MSS SP-45.

(c) Type 103: Cast steel body ASTM A216/A216M WCB, Class 300, ll.5 to l3 percent chromium stainless steel flexible wedge and hard faced (stellite) alloy seats, ASME flanged ends, OS&Y, rising stem, and bolted bonnet.

1. Provide factory installed globe-valved bypass when main valve is 3 inch pipe size or greater and serves steam main longer than 20 feet. Conform to MSS SP-45.

2. Drill and tap bosses for connection of drains if valve is in steam service. Conform to MSS SP-45.

(d) Type 105: Forged steel body ASTM A105/A105M, rated for at 300 psig at 420 degrees F minimum, Class 600 psig or Class 800 psig, hardened stainless steel or stellite wedge and seats, threaded ends, OS&Y, rising stem, bolted bonnet.

(2) Globe Valves:

(a) Type 201: Cast steel body ASTM A216/A216M WCB, rated for at 150 psig at 500 degrees F, 11-1/2 to 13 percent chromium stainless steel or stellite disc and seat, 150 psig ASME flanged ends, OS&Y, rising stem, bolted bonnet, renewable seat rings. Drill and tap bosses for connection of drains where shown. Conform to MSS SP-45.

(b) Type 205: Forged steel body ASTM A105/A105M, rated for at 300 psig at 420 degrees F minimum, Class 600 psig or Class 800 psig, stainless steel disc, stellite seat, threaded ends, OS&Y, rising stem, bolted bonnet.

(3) Plug Valves: Cast steel body ASME B16.5 Class 150, one-fourth turn to open. 125 psig ASME flanged ends for pipe sizes above 2 inches, threaded ends for pipe sizes 2 inches and under. All components designed for service to which applied: natural gas, LP gas (propane), or fuel oil. Furnish lever handle for each valve.

(a) Type 301: Two-way valves up through 4 inches pipe size. Eccentric action, non-lubricated plug with resilient seal molded into groove on plug face providing bubble-tight shut off. O-ring stem seal, corrosion-resistant bearings, corrosion-resistant seat coating, seal materials as recommended by valve manufacturer for the service. Valves on natural gas service AGA approved.

(b) Type 302: Two-way valves 5 inches pipe size and above, all sizes of three-way valves. Lubricated full-port plug type with lubricant for intended service. Reinforced Teflon stem seal, valve plug floated on Teflon surfaces, lubricant injection system

that has sufficient pressure to fully lubricate all sealing surfaces. Provide laminated plastic label attached to each valve stating, “Lubricate with Sealant once a year”.

(4) Check Valves:

(a) Type 401: Not used.

(b) Type 402: Swing-type, cast steel body ASME B16.34, rated for 250 psig saturated steam, 500 psig WOG, bronze or bronze-faced disc and seat, 250 psig ASME flanged ends, bolted cover, renewable disc, and seat.

(c) Type 403: Swing-type, cast steel body ASME B16.34, rated for l25 psig saturated steam, 200 psig WOG, bronze or bronze-faced disc and seat, l25 psig ASME flanged ends, bolted cover, renewable disc, and seat.

(d) Type 405: Lift-type, forged steel body ASTM A105/A105M, rated for 300 psig at 420 degrees F minimum Class 600 psig or 800 psig, hardened stainless steel disc, hard faced seat, bolted cover, threaded ends.

(e) Type 406: Swing-type, Type 316 stainless steel body, disc, and hanger, rated for 250 psig at 360 degrees F minimum.

(f) Type 408: Silent spring-loaded wafer type, cast steel ASTM A216/A216M WCB body, rated for 300 psig water, 250 degrees F, stainless steel trim.

(5) Ball Valves: Reduced port permitted for bypass (throttling) service; full port required for all other services, one-fourth turn to open.

(a) Type 501: Type 316 stainless steel body, ball, and stem, rated for at 150 psig at 365 degrees F, 600 psig at 200 degrees F; reinforced TFE seat, stem seal and thrust washer; end entry, threaded ends.

(b) Type 502: Steel body, rated for 150 psig at 365 degrees F, 250 psig at 250 degrees F, reinforced TFE seat, stem seal and thrust washer; end entry, threaded ends, UL-listed for natural or LP gas shut off service when used on those services.

(c) Type 503: Carbon steel body, steam service, rated for 200 psig at 392 degrees F, stainless steel ball and stem, Polyfil seat, live-loaded or adjustable stem seal, threaded ends.

(d) Type 504: Carbon steel body, saturated steam service, rated for 150 psig, stainless steel ball and stem, Polyfil seat, live-loaded stem seal, ASME flanged ends.

c. Fuel Oil: Discharge side of pumps. Conform to NFPA 30 and NFPA 31.

(1) Gate valves, 2 inches and under: Type 105.

(2) Gate Valves, 2-l/2 inches and above: Type 101 or 102.

(3) Globe valves, 2 inches and under: Type 205.

(4) Plug valves, 4 inches and under: Type 301. (Tank isolating valve on return line.)

(5) Check valves, 2 inches and under: Type 405 or 408.

(6) Check valves, 2-l/2 inches and above: Type 402 or 408.

(7) Ball valves, 2 inches and under: Type 502.

d. Fuel Oil: Suction side of pumps. Conform to NFPA 30 and NFPA 31.

(1) Gate valves, 2 inches and under: Type 105.

(2) Gate valves, 2-l/2 inches and above: Type 103.

(3) Plug valves, 4 inches and under: Type 301.

(4) Check valves, 2 inches and under: Type 405.

(5) Check valves, 2-l/2 inches and above: Type 403.

(6) Ball valves, 2 inches and under: Type 502.

e. Fuel Oil: Tank fill lines where tank is above fill point.

(1) Gate valves, 2 inches and under: Type 105.

(2) Gate valves, 2-1/2 inches and above: Type 103.

(3) Check valves, all sizes: Type 408.

f. Instrumentation and Control Piping: Ball valves, 2 inches and under: Type 502.

2.3 GAUGES, PRESSURE AND COMPOUND

a. ASME B40.100, Accuracy Grade 1A, (pressure, vacuum, or compound), initial mid-scale accuracy 1 percent of scale (Qualify grade), metal or phenolic case, 4-1/2 inches in diameter, 1/4 inch NPT bottom connection, white dial with black graduations and pointer, clear glass or acrylic plastic window, suitable for board mounting. Provide red "set hand" to indicate normal working pressure.

b. Provide steel, lever handle union cock. Provide steel or stainless-steel pressure snubber for gauges in water service. Provide steel pigtail syphon for steam gauges.

c. Pressure gauge ranges shall be selected such that the normal operating pressure for each gauge is displayed near the midpoint of each gauge’s range. Gauges with ranges selected such that the normal pressure is displayed at less than 30 percent or more than 70 percent of the gauge’s range are prohibited. The units of pressure shall be psig.

2.4 SIGHT FLOW INDICATORS

a. Provide, where shown, to allow observation of flow in piping systems.

b. Type: In line, dual portholes on opposite sides, with safety shield,

with or without rotor as shown in contract documents. Where provided, rotor shall have minimum of three vanes.

c. Construction: Carbon steel body, tempered borosilicate window, PTFE seals (except Buna-N on oil service), threaded ends on pipe sizes under 2-1/2 inches, flanged ends on sizes 2-1/2 inches and above. Pressure and temperature ratings shall be equivalent to requirements for valves on the same pipelines.

d. Safety Shield: Transparent wrap-around overlap covering entire sight flow indicator, designed to protect personnel from failure of indicator. Shield shall fit the indicator tightly and be suitable for 150 psig, 302 degrees F.

2.5 QUICK-COUPLE HOSE CONNECTORS

a. Provide on all Y-strainer drains and as shown to allow quick connection of length of hose to piping drain or blowoff so that discharge fluid (water or steam) can be conveyed to a drainage system.

b. Type: Straight through, plug and socket, screw type or cam locking connections, all units 3/4 inch pipe size. Integral shut-off devices not required.

c. Service: Design for water and steam at 15 psig, 309 degrees F.

d. Spare Parts: Furnish one socket and one plug.

e. Accessories: Furnish two hoses 20 feet long, 3/4 inch inside diameter, rated for steam service at 100 psig, 300 degrees F. Hose shall be sufficiently flexible to be placed in 4 foot diameter coil. Provide connector on one end of each hose to mate with connectors on drains. Provide hose rack for holding both hoses. Securely mount rack in location selected by Contracting Officer.

2.6 SAFETY VALVES, RELIEF VALVES, SAFETY RELIEF VALVES AND ACCESSORIES

a. Provide valves and accessories to protect piping systems and pressure vessels from over-pressure. All valves shall comply with ASME BPVC SEC I and ASME BPVC SEC VIII). Flow capacities shall be certified by National Board of Boiler and Pressure Vessel Inspectors (NB).

b. Fuel Oil Service: Refer to Section 23 50 11, BOILER PLANT MECHANICAL EQUIPMENT.

2.7 STRAINERS, SIMPLEX BASKET TYPE

a. Provide on condensate lines where shown. Refer to Section 23 50 11, BOILER PLANT MECHANICAL EQUIPMENT, for duplex basket strainers at oil pumps.

b. Type: Simplex cylindrical basket type, clamp cover, closed-bottom, removable basket, drain at bottom with threaded plug.

c. Service: Water at 212 degrees F, 15 psig maximum pressure.

d. Construction:

(1) Body: Cast steel rated for 125 psig ASME flanged ends, flow arrows cast on side.

(2) Basket: Stainless steel, 1/8 inch perforations. Ratio of screen open area to cross section of pipe; four to one minimum.

2.8 STRAINERS, Y-TYPE

a. Provide as shown on steam, water, and compressed air piping systems.

b. Type: Open-end removable cylindrical screen. Threaded blow-off connection.

c. Construction:

(1) Steam Service 61 to 150 psig: Cast steel rated for 150 psig saturated steam with 150 psig ASME flanged ends, or forged steel with 250 psig ASME flanged ends, for pipe sizes above 2 inches. Cast steel rated for saturated steam at 150 psig threaded ends, for pipe sizes 2 inches and under.

(2) Steam Service 60 psig and under, water (except boiler feed between feedwater pumps and boilers), compressed air: Cast steel rated for 125 psig saturated steam, 175 psig WOG, with 125 psig ASME flanged ends, for pipe sizes above 2 inches. Cast steel, threaded ends, rated for 125 psig saturated steam, 175 psig WOG, for pipe sizes 2 inches and under.

(3) Boiler Feed between Feedwater Pumps and Boilers: Cast steel rated for 250 psig at 450 degrees F with 300 psig ASME flanged ends or cast steel with 250 psig ASME flanged ends, for pipe sizes above 2 inches. Cast steel, threaded ends, rated for at 232 degrees F 250 psig at 450 degrees F for pipe sizes 2 inches and under.

d. Screen: Monel or stainless steel, free area not less than 2-1/2 times flow area of pipe. For strainers 3 inch pipe size and smaller, diameter of openings shall be 0.032 inch or less on steam service, 0.05 inch or less on water service, 0.01 inch or less on compressed air service. For strainers 4 inch pipe size and greater, diameter of openings shall be 0.05 inch on steam service, 1/8 inch on water service. Provide 80 mesh stainless steel screen liner on all strainers installed upstream of water meters or control valves.

e. Accessories: Gate or ball valve and quick-couple hose connection on all blowoff connections. These items are specified elsewhere in this section.

2.9 PIPING SUPPORT SYSTEMS

a. Provide an engineered piping support system with all hangers, supports and anchors designed and located by experienced technical pipe support specialists, utilizing piping system design and analysis software. The system design shall be completely documented and submitted for review.

b. All pipe hangers and supports, and selection and installation shall comply with MSS SP-58 and MSS SP-127.

c. All pipe hanger and support devices shall be in compliance with

specified MSS SP-58 type numbers, have published load ratings, and be products of engineered pipe support manufacturers.

d. All pipe stresses and forces and moments on connecting equipment and structures shall be within the allowances of the ASME B31.1, applicable building codes, and equipment manufacturer’s design limits.

e. Piping that expands and contracts horizontally including steam, steam condensate, boiler feed, condensate transfer, shall be supported by roller or sliding type hangers and supports except when long vertical hanger rods permit sufficient horizontal movement with the vertical angles of the rods less than 4 degrees.

f. Piping that expands and contracts vertically including steam, steam condensate, boiler feed, condensate transfer, shall be supported by engineered variable spring and spring cushion hangers. Utilize MSS SP-58 selection requirements and guidelines. Vibration isolator hanger types are prohibited.

g. Seismic braces and shock absorbers shall be provided. Comply with MSS SP-127 design requirements and guidelines. Piping shall remain fully connected and supported under the design seismic events. Piping and connected equipment shall not be overstressed beyond code limits during seismic events.

h. Piping system anchors shall be engineered and located to control movement of piping that is subject to thermal expansion.

i. Prior to construction, submit complete engineering calculation methods and results, descriptions of all devices with MSS numbers, sizes, load capabilities and locations. Submit calculations on all moments and forces at anchors and guides, all hanger loads, all pipe stresses that are within 20 percent of the code allowable or exceed ASME B31.1 code allowable, all pipe movements at supports.

j. Detailed Design Requirements:

(1) Piping system design and analysis software shall be current state of the art that performs ASME B31.1 code analyses and shall be utilized to analyze pipe movement and deflection, pipe stresses, pipe support forces and moments, and for selection of pipe support types and sizes. Seismic restraint calculations shall utilize the applicable shock spectra for the type of building structure, type of supported system, and the locality. Comply with MSS SP-127.

(2) Each support for piping 2-1/2 inches and above shall be completely engineered to include location, type and size, hot and cold loads, and movement. Submit layout drawings showing precise support locations and submit individual drawings for each support assembly showing all components, sizes, loadings.

(3) Supports for piping 2 inches and below shall be engineered in general terms with approximate locations, typical support types and sizes, approximate movements. Submit layout drawings showing general locations and support types and sizes.

(4) Obtain permissible loadings (forces and moments) for equipment nozzles (pipe connections) from the manufacturer of the boilers, the feedwater deaerator, and any other equipment as necessary.

Professional structural engineer shall verify capability of building structure to handle piping loads.

(5) The project drawings shall show locations and types of resilient supports including rollers and springs, and shall also show special supports including anchors, guides, and braces. Comply with the drawing requirements unless it is determined that piping shall be overstressed or supports overloaded. Refer conflicts to the Contracting Officer.

(6) Variable spring hangers conforming the MSS SP-58, Type 51, shall support all piping that expands vertically from thermal effects which shall include connected equipment, such as boilers. Spring rates shall be selected to avoid excessive load transfer to the connected equipment as the piping expands vertically. Vibration-type spring isolators are prohibited. Light duty spring hangers, MSS SP-58, Type 48, shall be utilized on loads of 200 pounds or less, and vertical movement of 1/8 inches or less. Spring cushion hangers, MSS SP-58, Type 49, shall be utilized for vertical movement of 1/8 inches or less.

(7) Locate supports to permit removal of valves and strainers from pipelines without disturbing supports.

(8) If equipment and piping arrangement differs from that shown in contract documents, support locations and types shall be revised at no cost or time to the Government. The Government will also require a complete stress analysis of the system as-built at no additional cost or time to the Government.

k. Hangers and Supports - Products:

(1) Factory-built products of a manufacturer specializing in engineered pipe supports. All components shall have published load ratings. All spring type supports shall have published spring rates and movement limits. All support assemblies shall include threaded connections that permit vertical position adjustment. Supports shall comply with MSS SP-58 Type Numbers as listed below.

(2) Upper Attachments to Building Structure: Types 18, 20, 21, 22, 23, 29, and 30.

(3) Roller Supports: Types 41, 43, and 46. Provide vertical adjustment for Type 41 with threaded studs and nuts adjacent to the roller.

(4) Variable Spring Hanger Assembly:

(a) Type 51 variable spring, with Type 3 pipe clamp or Type 1 clevis. Type 53 variable spring trapeze shall also be used. Locate Type 51 variable spring within 1 foot above pipe attachment. Attach rod to top of variable spring with Type 14 clevis.

(b) Typical features of variable spring hangers include spring rates under 150 lb/in, enclosed spring, load and travel indicator, sizes available with load capabilities ranging from 50 lb to multiples of 10,000 lb.

(5) Spring Cushion Hanger Assembly: Double Rod: Type 41 and 49.

(6) Light Duty Spring Hanger Assembly: Type 48 light duty spring, with Type 3 pipe clamp or Type 1 clevis. Locate Type 48 light duty spring within 1 foot above pipe attachment.

(7) Clevis Hangers: Type 1.

(8) Wall Brackets: Type 31, 32, and 33.

(9) Pipe Stands: Type 38.

(10) Riser Clamps: Type 42.

(11) Roller Guides: Type 44. Construct guides to restrain movement perpendicular to the long axis of the piping. All members shall be welded steel.

(12) Trapeze Supports: Shall be used where pipes are close together and parallel. Construct with structural steel channels or angles. Bolt roller supports to steel to support piping subject to horizontal thermal expansion. Attach other piping with U-bolts.

(13) Pipe Covering Protection Saddles: Type 39. Provide at all support points on insulated pipe except where Type 3 pipe clamp is provided. Insulation shields are prohibited. Refer to Section 23 07 11, HVAC AND BOILER PLANT INSULATION.

(14) Sliding Supports: Type 35. Welded steel attachments to pipe and building structure with Teflon or graphite sliding surfaces bonded to the attachments. Provide steel guides, except at expansion bends, to prevent lateral movement of the pipe.

(15) Piping Anchors: Provide engineered designs to accommodate the calculated loads. All ferrous material shall be painted in accordance with Section 09 91 00, PAINTING.

(16) Seismic Restraints:

(a) Comply with MSS SP-127.

(b) Bracing: Provide as determined by engineering calculations.

(c) Shock Absorbers: Type 50. Mechanical or hydraulic type rated for shock loads. Pipe attachments shall be Type 3.

2.10 PIPE AND VALVE FLANGE GASKETS

Non-asbestos, designed for the service conditions. On steam service utilize 1/8 inch thick Class 300 spiral-wound with Type 304 stainless steel and mica/graphite filler and carbon steel gauge ring.

2.11 THREAD SEALANTS

As recommended by the sealant manufacturer for the service. Teflon tape of any type is prohibited on fuel oil systems.

2.12 PIPE SLEEVES

a. Service: For pipes passing through floors, walls, partitions.

b. Construction: Steel pipe, schedule 10 minimum.

c. Sleeve Diameter: Not less than 1 inch larger than the diameter of the enclosed pipe and thermal insulation, vapor barrier, and protective covering for insulated pipe; sleeves for un-insulated pipe shall be not less than 1 inch larger than the diameter of the enclosed pipe.

PART 3 EXECUTION

3.1 GENERAL

a. If in the substantiated evaluation of the Contracting Officer, the installation fails to meet the requirements of the construction documents with respect to function and maintainability, an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

3.2 ARRANGEMENT OF PIPING

a. The piping arrangement shown is a design based on currently available equipment. The plans show typical equipment to scale and show practical arrangement. Modification will be necessary during construction, at no additional cost or time to the Government, to adapt the equipment layout and piping plans to the precise equipment purchased by the Contractor. Accessibility for operation and maintenance shall be maintained.

b. All piping shall be installed parallel to walls and column centerlines (unless shown otherwise). Fully coordinate work of each trade to provide the designed systems without interference between systems. All piping shall be accurately cut, true, and beveled for welding. Threaded piping shall be accurately cut, reamed, and threaded with sharp dies. Copper piping work shall be performed in accordance with best practices requiring accurately cut clean joints and soldering in accordance with the recommended practices for the material and solder employed. Compression type fittings are prohibited.

c. All piping shall be pitched for drainage at a constant slope of in 1 inch in 40 feet. Steam, condensate, trap discharge, drip, drain, air, gas, and blowdown piping shall pitch down in direction of flow. Service water, pumped condensate, pumped boiler feedwater, oil, shall pitch up in direction of flow. Provide valved air vents at top of rise and valved drains at low points. Gas piping shall be run level as it is presumed to be dry, but dirt pockets shall be provided at base of risers.

d. Valves shall be located and stems oriented to permit proper and easy operation and access to valve bonnet for maintenance of packing, seat and disc. Valve stems shall not be below centerline of pipe. Refer to plans for stem orientation. Where valves are more than 7 feet above the floor or platform, stems shall be horizontal unless shown otherwise. Gate and globe valves more than 10 feet above floor or platform, shall be accessed using additional permanent work platforms to be provided by contractor. Provide hammer-blow wheel on any valve

that cannot be opened or tightly closed by one person. Steam line gate type isolation valves 3 inch pipe size and above shall have factory or field-fabricated or 3/4 inch or 1 inch globe-valved warm-up bypasses if the steam line length is 20 feet or longer.

e. Provide union adjacent to all threaded end valves.

f. Provide valves as necessary to permit maintenance of a device or sub-system without discontinuing service to other elements of that service or system.

g. Do not install any piping within 2 feet of water tube boiler side or top casings.

3.3 WELDING

a. The contractor is entirely responsible for the quality of the welding and shall:

(1) Conduct tests of the welding procedures used on the project, verify the suitability of the procedures used, verify that the welds made will meet the required tests, and also verify that the welding operators have the ability to make sound welds under standard conditions.

(2) Perform all welding operations required for construction and installation of the piping systems.

b. Qualification of Welders: Rules of procedure for qualification of all welders and general requirements for fusion welding shall conform with the applicable portions of ASME B31.1, AWS B2.1/B2.1M, AWS Z49.1, and also as outlined below.

c. Examining Welder: Examine each welder at job site, in the presence of the Contracting Officer, to determine the ability of the welder to meet the qualifications required. Test welders for piping for all positions, including welds with the axis horizontal (not rolled) and with the axis vertical. Each welder shall be allowed to weld only in the position in which he has qualified and shall be required to identify his welds with his specific code marking signifying his name and number assigned.

d. Examination Results: Provide the Contracting Officer with a list of names and corresponding code markings. Retest welders who fail to meet the prescribed welding qualifications. Disqualify welders, who fail the second test, for work on the project.

e. Beveling: Field bevels and shop bevels shall be done by mechanical means or by flame cutting. Where beveling is done by flame cutting, surfaces shall be thoroughly cleaned of scale and oxidation just prior to welding. Conform to specified standards.

f. Alignment: Provide approved welding method for joints on all pipes greater than 2 inches to assure proper alignment, complete weld penetration, and prevention of weld spatter reaching the interior of the pipe.

g. Erection: Piping shall not be split, bent, flattened, or otherwise damaged before, during, or after installation. If the pipe

temperature falls to 32 degrees F or lower, the pipe shall be heated to approximately 100 degrees F for a distance of 1 foot on each side of the weld before welding, and the weld shall be finished before the pipe cools to 32 degrees F.

h. Non-Destructive Examination of Piping Welds:

(1) Perform radiographic examination of 50 percent of the first 10 welds made by each welder, and 10 percent of all additional welds made by each welder. The Contracting Officer reserves the right to identify individual welds for which the radiographic examination shall be performed. All welds shall be visually inspected by the Contracting Officer, or their representative. The VA reserves the right to require testing on additional welds up to 100 percent if more than 25 percent of the examined welds fail the inspection.

(2) An approved independent testing firm regularly engaged in radiographic testing shall perform the radiographic examination of pipe joint welds. All radiographs shall be reviewed and interpreted by an ASNT Certified Level III radiographer, employed by the testing firm, who shall sign the reading report.

(3) Comply with ASME B31.1. Furnish a set of films showing each weld inspected, a reading report evaluating the quality of each weld, and a location plan showing the physical location where each weld is to be found in the completed project. The Contracting Officer and the CxA shall be given a copy of all reports to be maintained as part of the project records and shall review all inspection records.

i. Defective Welds: Replace and reinspect defective welds. Repairing defective welds by adding weld material over the defect or by peening are prohibited. Welders responsible for defective welds shall be requalified prior to resuming work on the project.

j. Electrodes: Electrodes shall be stored in a dry heated area and be kept free of moisture and dampness during the fabrication operations. Discard electrodes that have lost part of their coating.

3.4 PIPING JOINTS

a. All butt-welded piping shall be welded at circumferential joints, flanges shall be weld neck type; slip-on flanges, screwed flanges shall be applied only with written approval of the Contracting Officer.

b. Companion flanges at equipment or valves shall match flange construction of equipment or valve. Raised face shall be removed at all companion flanges when attached to flanges equipped for flat face construction.

c. Gaskets and bolting shall be applied in accordance with the recommendations of the gasket manufacturer and bolting standards of ASME B31.1. Strains shall be evenly applied without overstress of bolts. Gaskets shall cover entire area of mating faces of flanges.

d. Screw threads shall be made up with Teflon tape except gas and oil piping joints shall utilize specified joint compound.

3.5 BRANCH INTERSECTION CONNECTIONS

a. Factory-built reinforced tees and laterals are required.

b. Factory-built integrally-reinforced forged steel branch outlet fittings shall be used on reduced size connections upon approval of Contracting Officer. They shall comply with MSS SP-97.

3.6 EXPANSION AND FLEXIBILITY

The design includes provision for piping expansion due to pressure, thermal, weight and seismic (where applicable) effects. It is the Contractor's responsibility to avoid reduction in flexibility and increase in stress in piping systems. Major deviation will be shown by submittal for review of scale working drawings and stress calculations for the piping systems. Contractor shall provide any necessary additional construction and materials to limit stresses to safe values as directed by the Contracting Officer and at no additional cost or time to the Government.

3.7 PIPE BENDING

a. Pipe bending shall be in accordance with the recommended practices of PFI ES 24. Only ASTM A106/A106M seamless pipe shall be bent. Sizes below 2 inches shall be bent in field; sizes 2-1/2 inches and larger shall have factory-fabricated bends. Minimum radii and tangent lengths for field bent piping are shown in the following table:

Size	Minimum Radius	Minimum Tangent
1/2 inch	2-1/2 inches	1-1/2 inches
3/4 inch	2-3/4 inches	1-3/4 inches
1 inch	5 inches	2 inches
1-1/4 inches	6-1/4 inches	2 inches
1-1/2 inches	7-1/2 inches	2-1/2 inches

3.8 SIZE CHANGES

Piping size changes shall be accomplished by use of line reducers, reducing ell, reducing tee. Apply eccentric reduction in all piping requiring continuous drainage, steam, condensate, vacuum, blowdown. Concentric reduction shall be applied in run of piping involving pressure water systems except at pump inlets. Use concentric increasers where flow is in direction of increased size. Eccentric reduction, top flat, at all pump connections.

3.9 MINOR PIPING

Minor piping associated with instrumentation and control is generally not shown. Interconnection of sensors, transducers, control devices, instrumentation panels, combustion control panel, burner control panels is the responsibility of the contractor. Small piping associated with water cooling, drips, drains and other minor piping shall not be shown to avoid confusion in the plan presentation but shall be provided as part of contract work.

3.10 INSTALLATION - PRESSURE AND TEMPERATURE REGULATORS, CONTROL VALVES, SAFETY SHUT-OFF VALVES

a. Provide sufficient clearance on all sides of valve to permit replacement of working parts without removing valve from pipeline.

b. Maintain access.

3.11 INSTALLATION – PIPE SUPPORT SYSTEMS

a. Coordinate support locations with building structure prior to erection of piping. Also refer to approved shop drawings of equipment and approved piping layout and hanger layout drawings when locating hangers. Arrangement of supports shall facilitate operating, servicing and removal of valves, strainers, and piping specialties. Hanger parts shall be marked at the factory with a numbering system keyed to hanger layout drawings. Layout drawings shall be available at the site.

b. Upper attachments to Building Structure:

(1) New Reinforced Concrete Construction: Concrete inserts.

(2) Existing Reinforced Concrete Construction: Upper attachment welded or clamped to steel clip angles (or other construction shown in contract documents) which are expansion-bolted to the concrete. Expansion bolting shall be located so that loads place bolts in shear.

(3) Steel Deck and Structural Framing: Upper attachments welded or clamped to structural steel members.

c. Expansion Fasteners and Power Set Fasteners: In existing concrete floor, ceiling and wall construction, expansion fasteners shall be used for hanger loads up to 1/3 the manufacturer's rated strength of the expansion fastener. Power set fasteners shall be used for loads up to 1/4 of rated load. When greater hanger loads are encountered, additional fasteners shall be used and interconnected with steel members combining to support the hanger.

d. Special Supports:

(1) Secure horizontal pipes where necessary to prevent vibration or excess sway.

(2) Where hangers cannot be adequately secured as specified, (for example, support for flow metering sensing lines, control piping) special provisions shall be made for hanging and supporting pipe as directed by the Contracting Officer.

(3) Pipe supports, hangers, clamps or anchors shall not be attached to equipment unless specifically permitted by the specifications for that equipment or unless Contracting Officer gives written permission. Attachments to boiler casings are prohibited.

e. Spring Hangers: Locate spring units within one foot of the pipe, breeching or stack attachment except in locations where spring assemblies interfere with pipe insulation. Adjust springs to loads calculated by hanger manufacturer.

f. Do not insulate piping within one foot of device until device has been inspected by Contracting Officer.

3.12 CLEANING OF PIPING AFTER INSTALLATION

Flush all piping sufficiently to remove all dirt and debris. Fill piping completely. Velocity shall be equivalent to that experienced during normal plant operation at maximum loads. During flushing, all control valves, steam traps and pumps shall be disconnected from the system. After cleaning is complete, remove, clean, and replace all strainer baskets and elements. Reconnect all equipment. Provide safe points of discharge for debris blown from pipes. Flushing of the systems shall be witnessed by the Contracting Officer or their representative. Provide documentation of system flow rates to flushing flow rates and for how long was the flow maintained.

3.13 TESTING

a. Testing of piping components is not required prior to installation. Valves and fittings shall be capable of withstanding hydrostatic shell test equal to twice the primary design service pressure except as modified by specifications on fittings, ASME B16.5. This test capability is a statement of quality of material. Tests of individual items of pipe, fittings or equipment shall be required only on instruction of Contracting Officer and at Government cost, except where required by the specifications for specific equipment such as the boilers.

b. After erection, all piping systems shall be capable of withstanding a hydrostatic test pressure of 1.5 times design pressure, as stipulated in ASME B31.1. Hydrostatic tests will be required only on boiler external steam piping, utilizing water as the test medium. Hydrostatic tests will be required on other piping when operating tests described are unsatisfactory, or when inspection of welds shows poor workmanship and is subject to question by the Contracting Officer. When hydrostatic tests show leaks, the Contracting Officer will require necessary welding repairs, in accordance with ASME B31.1, at the Contractor's cost.

c. Perform operating test as follows:

(1) Test oil systems under service conditions at pressure equal to highest setting of safety and relief valves in the individual systems.

(5) Make corrections and retests to establish systems that have no leaks. Replace or recut any defective fittings or defective threads. Soldered material shall be thoroughly cleaned prior to resoldering. Back welding of threads is prohibited.

d. Prepare and submit test and inspection reports to the Contracting Officer within 5 working days of test completion and prior to covering the pipe.

e. All tests shall be witnessed by the Contracting Officer, their representative, or the CxA and be documented by each section tested, date tested, and list or personnel present.

h. Safety, Safety-Relief, Relief Valves: After installation, test under pressure in presence of Contracting Officer. Test operation, including set pressure, flow, and blowdown in accordance with ASME BPVC. Install relief valve set at pressure no more than 1/3 higher than test pressure and replace safety valves of the appropriate pressure. Reset pressure setpoint of all relief valves to the appropriate pressures and replace safety valves after all tests have been completed. Contractor to provide written report of the reset with date and time stamp for each relief valve and replacement of the safety valves. Any deficiencies shall be corrected and retest performed.

3.14 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA shall observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

3.15 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification shall be tested as part of a larger system.

3.16 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hours to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

-- End of Section --

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SECTION 23 21 13
06/25

HYDRONIC PIPING

PART 1 GENERAL

1.1 DESCRIPTION

a. Water piping to connect HVAC equipment, including the following:

(1) Chilled water, condenser water, heating hot water and drain piping.

(2) Extension of domestic water make-up piping for HVAC systems.

(3) Glycol-water piping.

b. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

f. Not used.

h. Section 23 05 11, COMMON WORK RESULTS FOR HVAC: General mechanical requirements and items, which are common to more than one section of Division 23.

i. Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT.

j. Section 23 07 11, HVAC AND BOILER PLANT INSULATION: Piping insulation.

k. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

l. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC: Temperature and pressure sensors and valve operators.

m. Section 23 21 23, HYDRONIC PUMPS: Pumps.

n. Section 23 22 13, STEAM AND CONDENSATE HEATING PIPING.

o. Section 23 25 00, HVAC WATER TREATMENT: Water treatment for open and closed systems.

p. Section 23 82 00, CONVECTION HEATING AND COOLING UNITS: Induction units, fan coil units, Unit Heaters and radiant ceiling panels.

q. Section 31 20 00, EARTHWORK: Excavation and backfill.

1.3 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.20.1	(2013; R 2018) Pipe Threads, General Purpose (Inch)
ASME B16.3	(2021) Malleable Iron Threaded Fittings, Classes 150 and 300
ASME B16.4	(2021) Gray Iron Threaded Fittings; Classes 125 and 250
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2024) Factory-Made Wrought Buttwelding Fittings
ASME B16.11	(2022) Forged Fittings, Socket-Welding and Threaded
ASME B16.18	(2021) Cast Copper Alloy Solder Joint Pressure Fittings
ASME B16.22	(2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings
ASME B16.24	(2022) Cast Copper Alloy Pipe Flanges, Flanged Fittings, and Valves Classes 150, 300, 600, 900, 1500, and 2500
ASME B16.39	(2020) Standard for Malleable Iron Threaded Pipe Unions; Classes 150, 250, and 300
ASME B16.42	(2021) Ductile Iron Pipe Flanges and Flanged Fittings, Classes 150 and 300
ASME B31.9	(2020) Building Services Piping
ASME B40.100	(2022) Pressure Gauges and Gauge Attachments
ASME BPVC	(2010) Boiler and Pressure Vessels Code
ASME BPVC SEC VIII	(2021) Boiler and Pressure Vessel Codes: Section VIII Rules for Construction of

Pressure Vessel

ASTM INTERNATIONAL (ASTM)

ASTM A47/A47M	(1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings
ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A126	(2004; R 2023) Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings
ASTM A216/A216M	(2021) Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service
ASTM A307	(2023) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM A536	(2024) Standard Specification for Ductile Iron Castings
ASTM B62	(2017) Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM B88	(2022) Standard Specification for Seamless Copper Water Tube
ASTM F439	(2024) Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80
ASTM F441/F441M	(2023) Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe, Schedules 40 and 80

AMERICAN WELDING SOCIETY (AWS)

AWS B2.1/B2.1M	(2021) Specification for Welding Procedure and Performance Qualification

EXPANSION JOINT MANUFACTURERS ASSOCIATION (EJMA)

EJMA Stds	(2015) (10th Ed) EJMA Standards

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-67	(2022) Butterfly Valves
MSS SP-70	(2011) Gray Iron Gate Valves, Flanged and Threaded Ends
MSS SP-71	(2018) Gray Iron Swing Check Valves,

	Flanged and Threaded Ends
MSS SP-80	(2019) Bronze Gate, Globe, Angle and Check Valves
MSS SP-85	(2011) Gray Iron Globe & Angle Valves Flanged and Threaded Ends
MSS SP-110	(2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends
MSS SP-125	(2010) Gray Iron and Ductile Iron In-Line, Spring-Loaded, Center-Guided Check Valves

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

TUBULAR EXCHANGER MANUFACTURERS ASSOCIATION (TEMA)

TEMA Stds	(2007) Standards of the Tubular Exchange Manufacturers Association (TEMA)

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Welder’s Qualifications; G, AE

SD-02 Shop Drawings

Coordination Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Form No. U-1, For Asme Pressure Vessels; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Complete Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Piping Diagrams; G, AE

Spare Parts; G, AE

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 21 13, HYDRONIC PIPING”, with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Pipe and equipment supports. Submit calculations for variable spring and constant support hangers.

(2) Pipe and tubing, with specification, class or type, and schedule.

(3) Pipe fittings, including miscellaneous adapters and special fittings.

(4) Flanges, gaskets and bolting.

(5) Couplings and fittings.

(6) Valves of all types.

(7) Strainers.

(8) Flexible connectors for water service.

(9) Pipe alignment guides.

(10) Expansion joints.

(11) Expansion compensators.

(12) All specified hydronic system components.

(13) Water flow measuring devices.

(14) Gauges.

(15) Thermometers and test wells.

(16) Electric heat tracing systems.

(17) Seismic bracing details for piping.

d. Manufacturer's certified data report, Form No. U-1, for ASME pressure vessels:

(1) Heat exchangers (Water to Water).

(2) Air separators.

(3) Expansion tanks.

(4) Buffer tanks.

e. Submit the welder’s qualifications in the form of a current (less than 1-year old) and formal certificate.

f. Coordination Drawings: Refer to paragraph, SUBMITTALS of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

g. As-Built Piping Diagrams: Provide drawings as follows for chilled water, condenser water, and heating hot water system and other piping systems and equipment.

(1) 1wall-mounted stick file with complete set of prints. Mount stick file in the chiller plant or control room along with control diagram stick file.

(2) 1 complete set of reproducible drawings.

(3) 1 complete set of drawings in electronic AutoCAD and pdf format.

h. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

i. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician, and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

j. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

QUALITY ASSURANCE

a. Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC, which includes welding qualifications.

b. Submit prior to welding of steel piping a certificate of Welder’s certification. The certificate shall be current and not more than 1-year old.

c. All couplings, fittings, valves, and specialties shall be the products of a single manufacturer.

(1) All castings used for coupling housings, fittings, valve bodies, etc., shall be date stamped for quality assurance and traceability.

d. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

e. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.6 QUALITY ASSURANCE

a. Section 23 05 11, COMMON WORK RESULTS FOR HVAC, which includes welding qualifications.

b. Submit prior to welding of steel piping a certificate of Welder's certification. The certificate shall be current and not more than 1-year old.

c. All couplings, fittings, valves, and specialties shall be the products of a single manufacturer.

(1) All castings used for coupling housings, fittings, valve bodies, etc., shall be date stamped for quality assurance and traceability.

d. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit:

e. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

1.8 SPARE PARTS

For mechanical pressed sealed fittings provide tools required for each pipe size used at the facility.

PART 2 PRODUCTS

2.1 PIPE AND EQUIPMENT SUPPORTS, PIPE SLEEVES, AND WALL AND CEILING PLATES

Provide in accordance with Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

2.2 PIPE AND TUBING

a. Chilled Water, Condenser Water, Heating Hot Water, and Glycol-Water ,

and Vent Piping:

(1) Steel: ASTM A53/A53M Grade B, seamless or ERW, Schedule 40.

(2) Copper water tube option: ASTM B88, Type K or L, hard drawn.

b. Extension of Domestic Water Make-up Piping: ASTM B88, Type K or L, hard drawn copper tubing.

c. Cooling Coil Condensate Drain Piping:

(1) From air handling units: Copper water tube, ASTM B88, Type M, or Schedule 40 PVC plastic piping. Use of PVC piping is prohibited in exterior applications.

(2) From fan coil or other terminal units: Copper water tube, ASTM B88, Type M for runouts and Type L for mains.

d. Chemical Feed Piping for Condenser Water Treatment: CPVC, Schedule 80, ASTM F441/F441M.

e. Pipe supports, including insulation shields, for above ground piping: Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

2.2.1 PREINSULATED CONDUIT SYSTEM (UNDERGROUND OUTSIDE - DIRECT BURY)

a. Design Pressure 125 psig (865 kPa gauge), Maximum Design Temperature: 210°F (99°C).

b. Conduit System; Preinsulated; HDPE Jacket:

(1) Black Steel; Standard Weight; Welded Joints.

(a) Carrier Pipe: Standard weight black steel, beveled ends, ASTM A53, Type E or S, Grade B.

(b) Joints: Butt-welded.

(c) Fittings: Standard weight wrought steel, butt-welding type, ASTM A234, ASME B16.9. Pipe and fittings shall be a system provided by the same manufacturer.

(d) Where possible, straight sections shall be supplied in 40 foot random lengths, with piping exposed at each end for field joint fabrication.

(2) Accessories

(a) Elbows, tees, reducers, anchors, field joints and end seals shall be designed and factory fabricated to prevent the ingress of moisture into the system.

(b) Fitting insulation and covers shall match the pipe insulation.

(c) Install gland seal terminations at all building and manhole entrancers. Gland seals shall allow the inside pipe to move freely to allow for pipe expansion.

(3) Carrier Pipe Insulation: Polyurenthane foam with 2 lb/ft3 minimum

density, 90% minimum closed cell content and maximum initial thermal conductivity of 0.18 BTU-in/hr-ft2-F. The insulation shall completly fill the annular space between the service pipe and jacket and shall be bonded to both. Systems using open cell insulation or a nonbonded design shall not be allowed. The insulation shall be provided to the minmum thickness specified below:

Pipe Size (in)	Minimum Insulation Thickness (in)	
	Chilled Water	Hot Water
1 to 8	1	1
10 to 12	1	1.5
14 to 36	1.5	2

(4) Insulation Jacket: The outer protective insulation jacket shall be seamless high denisty polyethylene (HDPE) in accordance with ASTM D3350 minimum cell classification PE 345444 C. PVC or tape materials are not allowed. The minimum thickness of the HDPE jacket shall be as follows:

Jacket OD (in)	Minimum Jacket Thickness (in)
OA <= 12	0.080
12 < OA <= 24	0.120
OD > 24	0.160

(5) Field Joints: The carrier pipe shall be hydrostatically tested to 150 psig or 1.5 times the design pressure whichever is greater. Insulation shall then be poured in place into the field joint area. All field applied insulation shall be placed only in straight sections of pipe. Field insulation of fittings is not acceptable. The installer shall seal the field joint area with a heat shrinkable adhesive backed sleeve. Backfilling shall not begin until the heat shrink sleeve has cooled.

(6) Backfill: A 4 inch layer of sand or fine gravel shall be placed and tamped in the trench, to provde a uniform bedding for the pipe. The entire trench width shall be evenly backfilled with a similar material as the bedding in 6 inch compacted layers to a minimum height of 6 inches above the top of the insulated pipe. The remaining trench shall be evenly and continuously backfilled and compacted in uniform layers with suitable excavated soil.

2.3 FITTINGS FOR STEEL PIPE

a. 2 inches and Smaller: Screwed or welded joints.

(1) Butt welding: ASME B16.9 with same wall thickness as connecting piping.

(2) Forged steel, socket welding or threaded: ASME B16.11.

(3) Screwed: 150-pound malleable iron, ASME B16.3. 125-pound cast iron, ASME B16.4, may be used in lieu of malleable iron. Bushing reduction of a single pipe size, or use of close nipples, is not acceptable.

(4) Unions: ASME B16.39.

(5) Water hose connection adapter: Brass, pipe thread to 3/4 inch garden hose thread, with hose cap nut.

b. 2-1/2 inches and Larger: Welded or flanged joints.

(1) Butt welding fittings: ASME B16.9 with same wall thickness as connecting piping. Elbows shall be long radius type, unless otherwise noted.

(2) Welding flanges and bolting: ASME B16.5:

(a) Water service: Weld neck or slip-on, plain face, with 1/8 inch thick full-face neoprene gasket suitable for 220 degrees F.

1. Contractor's option: Convoluted, cold formed 150-pound steel flanges, with Teflon gaskets, may be used for water service.

(b) Flange bolting: Carbon steel machine bolts or studs and nuts, ASTM A307, Grade B.

c. Welded Branch and Tap Connections: Forged steel weldolets, or branchlets and threadolets shall be used for branch connections up to 1 pipe size smaller than the main. Forged steel half-couplings, ASME B16.11 shall be used for drain, vent, and gauge connections.

2.4 FITTINGS FOR COPPER TUBING

a. Joints:

Solder Joints: Joints shall be made up in accordance with recommended practices of the materials applied. Apply 95/5 tin and antimony on all copper piping.

b. Bronze Flanges and Flanged Fittings: ASME B16.24.

c. Fittings: ASME B16.18 cast copper or ASME B16.22 solder wrought copper.

2.5 FITTINGS FOR PLASTIC PIPING

a. Schedule 40, socket type for solvent welding.

b. Schedule 40 PVC drain piping: Drainage pattern.

c. Chemical feed piping for condenser water treatment: CPVC, Schedule 80, ASTM F439.

2.6 DIELECTRIC FITTINGS

a. Provide where copper tubing and ferrous metal pipe are joined.

b. 2 inches and Smaller: Threaded dielectric union, ASME B16.39.

c. 2-1/2 inches and Larger: Flange union with dielectric gasket and bolt sleeves, ASME B16.42. Dielectric gasket material shall be compatible with hydronic medium.

d. Temperature Rating, 210 degrees F.

e. Contractor’s option: On pipe sizes 2 inch and smaller, screwed end brass ball valves or dielectric nipples may be used in lieu of dielectric unions.

2.7 SCREWED JOINTS

a. Pipe Thread: ASME B1.20.1.

b. Lubricant or Sealant: Oil and graphite or other compound approved for the intended service.

2.8 VALVES

a. Asbestos packing is prohibited.

b. All valves of the same type shall be products of a single manufacturer.

c. Provide chain operators for valves 2 1/2 inches and larger when the centerline is located 8 feet or more above the floor or operating platform.

d. Shut-Off Valves:

(1) Ball Valves (Pipe sizes 2 inch and smaller): MSS SP-110, screwed or solder connections, brass or bronze body with chrome-plated ball with full port and Teflon seat at 400 psig working pressure rating. Provide stem extension to allow operation without interfering with pipe insulation.

(2) Butterfly Valves (Pipe Sizes 2-1/2 inch and larger): Provide stem extension to allow 2 inches of pipe insulation without interfering with valve operation. MSS SP-67, flange lug type rated 175 psig working pressure at 200 degrees F. Valves shall be ANSI Leakage Class VI and rated for bubble tight shut-off to full valve pressure rating. Valve shall be rated for dead end service and bi-directional flow capability to full rated pressure. Butterfly valves are prohibited for direct buried pipe applications.

(a) Body: Cast iron, ASTM A126, Class B. Malleable iron, ASTM A47/A47M electro-plated, or ductile iron, ASTM A536, Grade 65-45-12 electro-plated.

(b) Trim: Bronze, aluminum bronze, or 300 series stainless-steel disc, bronze bearings, 316 stainless-steel shaft and manufacturer's recommended resilient seat. Resilient seat shall be field replaceable, and fully line the body to completely isolate the body from the product. A phosphate coated steel shaft or stem is acceptable, if the stem is completely isolated from the product.

(3) Gate Valves:
a. 2 inches and Smaller: MSS SP-80, Bronze, 150 psig, wedge disc, rising stem, union bonnet. b. 2-1/2 inches and Larger: Flanged, outside screw and yoke. MSS SP-70, iron body, bronze mounted, 125 psig wedge disc.

(4) Field Interchangeable Actuators: Valves for balancing service shall have adjustable memory stop to limit open position.

(a) Valves 6 inches and Smaller: Lever actuator with minimum of seven locking positions, except where chain wheel is required.

(b) Valves 8 inches and Larger: Enclosed worm gear with handwheel, and where required, chain-wheel operator.

e. Globe and Angle Valves:

(1) Globe Valves:

(a) 2 inches and smaller: MSS SP-80, bronze, 150 psig Globe valves shall be union bonnet with metal plug type disc.

(b) 2-1/2 inches and larger: 125 psig, flanged, iron body, bronze trim, MSS SP-85 for globe valves.

(2) Angle Valves:

(a) 2 inches and smaller: MSS SP-80, bronze, 150 psig Angle valves shall be union bonnet with metal plug type disc.

(b) 2-1/2 inches and larger: 125 psig, flanged, iron body, bronze trim, MSS SP-85 for angle.

f. Check Valves:

(1) Swing Check Valves:

(a) 2 inches and smaller: MSS SP-80, bronze, 150 psig, 45-degree swing disc.

(b) 2-1/2 inches and larger: 125 psig, flanged, iron body, bronze trim, MSS SP-71 for check valves.

(2) Non-Slam or Silent Check Valve: Spring loaded double disc swing check or internally guided flat disc lift type check for bubble tight shut-off. Provide where check valves are shown in chilled water and hot water piping. Check valves incorporating a balancing feature may be used.

(a) Body: MSS SP-125 cast iron, ASTM A126, Class B, or steel, ASTM A216/A216M, Class WCB, or ductile iron, ASTM A536, flanged or wafer type.

(b) Seat, disc and spring: 18-8 stainless steel, or bronze, ASTM B62. Seats may be elastomer material.

g. Water Flow Balancing Valves: For flow regulation. Provide separate

isolation valve.. Valves shall be sized to provide the needed pressure drop in the middle of their throttling range.

(1) Ball or Globe style valve.

(2) A dual-purpose flow balancing valve and adjustable flow meter, with bronze or cast-iron body, calibrated position pointer, valved pressure taps or quick disconnects with integral check valves and preformed polyurethane insulating enclosure.

(3) Provide a readout kit including flow meter, readout probes, hoses, flow charts or calculator, and carrying case.

h. Automatic Balancing Control Valves: Factory calibrated to maintain constant flow (plus or minus five percent) over system pressure fluctuations of 4 to 57 psig. Provide standard pressure taps and 4 sets of capacity charts. Valves shall be line size and be one of the following designs:

(1) Gray iron ASTM A126 or brass body rated 175 psig at 200 degrees F, with stainless-steel piston and spring.

(2) Brass or ferrous body designed for 300 psig service at 250 degrees F, with corrosion resistant, tamper proof, self-cleaning piston/spring assembly that is easily removable for inspection or replacement.

(3) Combination assemblies containing ball type shut-off valves, unions, flow regulators, strainers with blowdown valves and pressure temperature ports shall be acceptable.

(4) Provide a readout kit including flow meter, probes, hoses, flow charts and carrying case.

i. Manual Radiator/Convector Valves: Brass, packless, with position indicator.

2.9 WATER FLOW MEASURING DEVICES

a. Minimum overall accuracy plus or minus 3 percent over a range of 70 to 110 percent of design flow. Select devices for not less than 110 percent of design flow rate.

b. Venturi Type: Bronze, steel, or cast-iron with bronze throat, with valved pressure sensing taps upstream and at the throat.

c. Wafer Type Circuit Sensor: Cast-iron wafer-type flow meter equipped with readout valves to facilitate the connecting of a differential pressure meter. Each readout valve shall be fitted with an integral check valve designed to minimize system fluid loss during the monitoring process.

d. Self-Averaging Annular Sensor Type: Brass or stainless-steel metering tube, shutoff valves and quick-coupling pressure connections. Metering tube shall be rotatable so all sensing ports may be pointed down-stream when unit is not in use.

e. Insertion Turbine Type Sensor: Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

f. Flow Measuring Device Identification:

(1) Metal tag attached by chain to the device.

(2) Include meter or equipment number, manufacturer's name, meter model, flow rate factor and design flow rate in gpm.

g. Portable Water Flow Indicating Meters:

(1) Minimum 6 inch diameter dial, forged brass body, beryllium-copper bellows designed for 175 psig working pressure at 250 degrees F.

(2) Bleed and equalizing valves.

(3) Vent and drain hose and two 10 feet lengths of hose with quick disconnect connections.

(4) Factory-fabricated carrying case with hose compartment and a bound set of capacity curves showing flow rate versus pressure differential.

(5) Provide one portable meter for each range of differential pressure required for the installed flow devices.

h. Permanently Mounted Water Flow Indicating Meters: Minimum 6 inch diameter, or 18 inch long scale, for 120 percent of design flow rate, direct reading in gpm, with 3 valve manifold and 2 shut-off valves.

2.10 STRAINERS

a. Y Type.

(1) Screens: Bronze, Monel metal or 18-8 stainless-steel, free area not less than 2-1/2 times pipe area, with perforations as follows: 0.045 inch diameter perforations for 4 inches and larger: 1/8 inch diameter perforations.

b. Suction Diffusers: Specified in Section 23 21 23, HYDRONIC PUMPS.

2.11 FLEXIBLE CONNECTORS FOR WATER SERVICE

a. Flanged Spool Connector:

(1) Single arch or multiple arch type. Tube and cover shall be constructed of chlorobutyl elastomer with full faced integral flanges to provide a tight seal without gaskets. Connectors shall be internally reinforced with high strength synthetic fibers impregnated with rubber or synthetic compounds as recommended by connector manufacturer, and steel reinforcing rings.

(2) Working pressures and temperatures shall be as follows:

(a) Connector sizes 2 inches to 4 inches, 165 psig at 250 degrees F.

(b) Connector sizes 5 inches to 12 inches, 140 psig at 250 degrees F.

(3) Provide ductile iron retaining rings and control units.

b. Braided Hose: Corrugated-metal inner hose and braided outer shell.

1. Piping Up To 2 inches: Bronze hoses and single-braid bronze sheaths with 450 psig at 70 degrees F and 340 psig at 450 degrees F ratings.

2. Piping From 2-1/2 inches To 4 inches: Stainless steel hoses and single-braid, stainless-steel sheaths with 300 psig at 70 degrees F and 225 psig at 450 degrees F ratings.

3. Piping From 150 mm 6 inches and Above: Stainless-steel hoses and single-braid, stainless-steel sheaths with 200 psig at 70 degrees F and 145 psig at 600 degrees F ratings.

2.12 EXPANSION JOINTS

a. Factory built devices, inserted in the pipe lines, designed to absorb axial cyclical pipe movement which results from thermal expansion and contraction. This includes factory-built or field-fabricated guides located along the pipe lines to restrain lateral pipe motion and direct the axial pipe movement into the expansion joints.

b. Manufacturing Quality Assurance: Conform to Expansion Joints Manufacturers Association (EJMA Stds) Standards.

c. Bellows - Internally Pressurized Type:

(1) Multiple corrugations of Type 304 or Type A240-321 stainless steel.

(2) Internal stainless-steel sleeve entire length of bellows.

(3) External cast iron equalizing rings for services exceeding (50 psig).

(4) Welded ends.

(5) Design shall conform to standards of EJMA Stds and ASME B31.9.

(6) External tie rods designed to withstand pressure thrust force upon anchor failure if one or both anchors for the joint are at change in direction of pipeline.

(7) Integral external cover.

d. Bellows - Externally Pressurized Type:

(1) Multiple corrugations of Type 304 stainless-steel.

(2) Internal and external guide integral with joint.

(3) Design for external pressurization of bellows to eliminate squirm.

(4) Welded ends.

(5) Conform to the standards of EJMA Stds and ASME B31.9.

(6) Threaded connection at bottom, 1 inch minimum, for drain or drip point.

(7) Integral external cover and internal sleeve.

e. Expansion Compensators:

(1) Corrugated bellows, externally pressurized, stainless-steel or bronze.

(2) Internal guides and anti-torque devices.

(3) Threaded ends.

(4) External shroud.

(5) Conform to standards of EJMA Stds.

f. Expansion Joint (Contractor’s Option): 350 psig maximum working pressure, steel pipe fitting consisting of telescoping body and slip-pipe sections, PTFE modified polyphenylene sulfide coated slide section, with welded or flanged ends, suitable for axial end movement to 3 inch.

g. Expansion Joint Identification: Provide stamped brass or stainless-steel nameplate on each expansion joint listing the manufacturer, the allowable movement, flow direction, design pressure and temperature, date of manufacture, and identifying the expansion joint by the identification number in the contract drawings.

h. Guides: Provide factory-built guides along the pipe line to permit axial movement only and to restrain lateral and angular movement. Guides shall be designed to withstand a minimum of 15 percent of the axial force which will be imposed on the expansion joints and anchors. Field-built guides shall be used if detailed in the contract documents.

i. Supports: Provide saddle supports and frame or hangers for heat exchanger. Mounting height shall be adjusted to facilitate gravity return of steam condensate. Construct supports from steel, weld joints.

2.13 HYDRONIC SYSTEM COMPONENTS

a. Heat Exchanger (Water to Water): Shell and tube type, U-bend removable tube bundle, heating fluid in shell, heated fluid in tubes, equipped with support cradles.

(1) Maximum tube velocity: 7.5 f/s.

(2) Tube fouling factor: TEMA Stds, but not less than 0.001.

(3) Materials:

(a) Shell: Steel.

(b) Tube sheet and tube supports: Steel or brass.

(c) Tubes: 3/4 inch OD copper.

(d) Head or bonnet: Cast iron or steel.

(4) Construction: In accordance with ASME BPVC SEC VIII for 125 psig working pressure for shell and tubes. Provide manufacturer's certified data report, Form No. U-1.

b. Plate and Frame Heat Exchanger:

(1) Fixed frame with bolted removable corrugated channel plate assembly, ASME code stamped for 150 psig working pressure.

(2) Corrugated channel plates shall be type 316 or 304 stainless-steel.

(3) Channel plate ports to be double gasketed to prevent mixing or cross-contamination of hot side and cold side fluids. Gaskets to be EPPM.

(4) Channel plate carrying bars to be carbon steel with zinc yellow chromate finish.

(5) Fixed frame plates and moveable pressure plates to be corrosion resistant epoxy painted carbon steel.

(6) Piping connections 2 inch and smaller to be carbon steel NPT tappings. Piping connections 4 inch and larger to be studded port design to accept ANSI flange connections. Connection ports to be integral to the frame or pressure plate.

(7) Finished units to be provided with OSHA required, formed aluminum splash guards to enclose exterior channel plate and gasket surfaces.

(8) Provide 2 sets of replacement gaskets and provide 1 set of wrenches for disassembly of plate type heat exchangers.

(9) Performance: As scheduled on drawings.

c. Optional Heat Transfer Package: In lieu of field erected individual components, the Contractor may provide a factory or shop assembled package of converters, pumps, and other components, pre-piped and pre-wired supported on a welded steel frame or skid. Refer to Section 23 22 13, STEAM AND CONDENSATE HEATING PIPING, for additional requirements.

d. Air Purger: Cast iron or fabricated steel, 125 psig water working pressure, for in-line installation.

e. Tangential Air Separator: ASME BPVC SEC VIII construction for 125 psig working pressure, flanged tangential inlet and outlet connection, internal perforated stainless-steel air collector tube designed to direct released air into expansion tank, bottom blowdown connection. Provide Form No. U-1. If scheduled on the in the contract documents, provide a removable stainless-steel strainer element having 3/16 inch perforations and free area of not less than five times the cross-sectional area of connecting piping.

f. Diaphragm Type Pre-Pressurized Expansion Tank: ASME BPVC SEC VIII construction for 125 psig working pressure, welded steel shell, rustproof coated, with a flexible elastomeric diaphragm suitable for a maximum operating temperature of 240 degrees F. Provide Form No. U-1. Tank shall be equipped with system connection, drain connection, standard air fill valve and be factory pre-charged to a minimum of 12 psig.

(1) Vertical floor-mounted expansion tank: Provide gauge glass, system or drain connection (bottom) and air charging (top) tappings. Provide gate valve and necessary adapters for charging system. Tank support shall consist of floor mounted base ring with drain access opening or four angle iron legs with base plates.

h. Pressure Reducing Valve (Water): Diaphragm or bellows operated, spring loaded type, with minimum adjustable range of 4 psig above and below set point. Bronze, brass or iron body and bronze, brass or stainless-steel trim, rated 125 psig working pressure at 225 degrees F.

i. Pressure Relief Valve: Bronze or iron body and bronze or stainless-steel trim, with testing lever. Comply with ASME BPVC Section VIII and bear ASME stamp.

j. Automatic Air Vent Valves (where shown on drawings): Cast iron or semi-steel body, 150 psig working pressure, stainless steel float, valve, valve seat and mechanism, minimum 1/2 inch water connection and 1/4 inch air outlet. Air outlet shall be piped to the nearest floor drain.

k. Buffer Tank: Buffer tank shall be constructed with a built-in baffle to allow mixing of the fluid inside the tank. Tank shall be constructed in accordance with ASME BPVC Section VIII requirements and stamped and registered with the National Board of Boiler and Pressure Vessel Inspectors. Tank shall have a working pressure of 125 psig and shall come equipped with a base ring for installing the buffer tank directly on a level surface. The tank shall be furnished with two flanged connections, tappings for air vent, relief valve and drain. Buffer tank shall have a capacity as indicated in the contract documents.

2.14 WATER FILTERS AND POT CHEMICAL FEEDERS

See Section 23 25 00, HVAC WATER TREATMENT, paragraph, CHEMICAL TREATMENT FOR CLOSED LOOP SYSTEMS.

2.15 GAUGES, PRESSURE AND COMPOUND

a. ASME B40.100, Accuracy Grade 1A, (pressure, vacuum, or compound for air, oil or water), initial mid-scale accuracy 1 percent of scale (Qualify grade), metal or phenolic case, 4-1/2 inches in diameter, 1/4 inch NPT bottom connection, white dial with black graduations and pointer, clear glass or acrylic plastic window, suitable for board mounting. Provide red "set hand" to indicate normal working pressure.

b. Provide brass lever handle union cock. Provide brass/bronze pressure snubber for gauges in water service.

c. Range of Gauges: Provide range equal to at least 130 percent of

normal operating range.

(1) For condenser water suction (compound): 30 inches Hg) to 100 psig.

2.16 PRESSURE/TEMPERATURE TEST PROVISIONS

a. Pete's Plug: 1/4 inch MPT by 3 inches long, brass body and cap, with retained safety cap, Nordel self-closing valve cores, permanently installed in piping where shown, or in lieu of pressure gauge test connections shown on the drawings.

b. Provide one each of the following test items to the Contracting Officer:

(1) 1/4 inch FPT by 1/8 inch diameter stainless-steel pressure gauge adapter probe for extra-long test plug.

(2) 3-1/2 inch diameter, one percent accuracy, compound gauge, 29.5 inches Hg) to 100 psig range.

(3) 32 to 220 degrees F pocket thermometer 1/2 degree accuracy, 1 inch dial, 5 inch long stainless-steel stem, plastic case.

2.17 THERMOMETERS

a. Mercury or organic liquid filled type, red or blue column, clear plastic window, with 6 inch brass stem, straight, fixed or adjustable angle as required for each in reading.

b. Case: Chrome plated brass or aluminum with enamel finish.

c. Scale: Not less than 9 inches, range as described below, 2-degree graduations.

d. Separable Socket (Well): Brass, extension neck type to clear pipe insulation.

e. Scale ranges:

(1) Chilled Water and Glycol-Water: 32 to 100 degrees F.

(2) Hot Water and Glycol-Water: 100 to 200 degrees F.

2.18 FIRESTOPPING MATERIAL

Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

2.19 ELECTRICAL HEAT TRACING SYSTEMS

a. Systems shall meet requirements of NFPA 70.

b. Provide tracing for outdoor piping subject to freezing temperatures below 38 degrees F as follows:

(1) Condenser water, including equalizer and bypass piping, piping for cooling towers.

(2) Basin sweeper piping.

(3) Hot water piping at units and all other areas exposed to the

weather.

c. Heat tracing shall be provided for all exposed piping.

d. Heating Cable: Flexible, parallel circuit construction consisting of a continuous self-limiting resistance, conductive inner core material between two parallel copper bus wires, designed for cut-to-length at the job site and for wrapping around valves and complex fittings. Unless not recommended or allowed by the manufacturer heat trace shall heat the full circumference of large piping and be wrapped around the piping or have multiple parallel cables on opposite sides of the pipe. Self regulation shall prevent overheating and burnouts even where the cable overlaps itself.

(1) Provide end seals at ends of circuits. Wire at the ends of the circuits is not to be tied together.

(2) Provide sufficient cable, as recommended by the manufacturer, to keep the pipe surface at 36 degrees F minimum during winter outdoor design temperature, but not less than the following:

(a) 3 inch pipe and smaller with 1 inch thick insulation: 4 watts per foot of pipe.

(b) 4 inch pipe and larger 1-1/2 inch thick insulation: 8 watts per feet of pipe.

e. Electrical Heating Tracing Accessories:

(1) Control Panel shall be provided to connect and control heat trace system and connect to the building automation system for control, reporting, alarm, and recording of heat trace usage. System shall be provided with gateway necessary to allow BAS to have full access to all items the heat trace control panel can access. Coordinate control panel requirements with Controls, Electrical, and Plumbing Contractor. System shall allow each tank and tower piping system to be tracked and controlled separately from the other.

(2) Power supply connection fitting and stainless-steel mounting brackets. Provide stainless-steel worm gear clamp to fasten bracket to pipe.

(3) 1/2 inch wide fiberglass reinforced pressure sensitive cloth tape to fasten cable to pipe at 12 inch intervals.

(4) Pipe surface temperature control thermostat: Cast aluminum, NEMA 4 (watertight) enclosure, 1/2 inch NPT conduit hub, SPST switch rated 20 amps at 480 volts ac, with capillary and copper bulb sensor. Set thermostat to maintain pipe surface temperature at not less than (34 degrees F).

(5) Signs: Manufacturer's standard (NFPA 70), stamped "ELECTRIC TRACED" located on the insulation jacket at 10 feet intervals along the pipe on alternating sides.

PART 3 EXECUTION

3.1 GENERAL

a. If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

b. The contract documents show the general arrangement of pipe and equipment but do not show all required fittings and offsets that may be necessary to connect pipes to equipment, fan-coils, coils, radiators, etc., and to coordinate with other trades. Provide all necessary fittings, offsets and pipe runs based on field measurements and at no additional cost or time to the Government. Coordinate with other trades for space available and relative location of HVAC equipment and accessories to be connected on ceiling grid. Pipe location in the contract documents shall be altered by contractor where necessary to avoid interferences and clearance difficulties.

c. Store materials to avoid excessive exposure to weather or foreign materials. Keep inside of piping relatively clean during installation and protect open ends when work is not in progress.

d. Support piping securely. Refer to PART 3, Section 23 05 11, COMMON WORK RESULTS FOR HVAC. Install heat exchangers at height sufficient to provide gravity flow of condensate to the flash tank and condensate pump.

e. Install piping generally parallel to walls and column center lines, unless shown otherwise in the contract documents. Space piping, including insulation, to provide 1 inch minimum clearance between adjacent piping or other surface. Unless shown otherwise, slope drain piping down in the direction of flow not less than 1 inch in (40 feet). Provide eccentric reducers to keep bottom of sloped piping flat.

f. Locate and orient valves to permit proper operation and access for maintenance of packing, seat and disc. Generally, locate valve stems in overhead piping in horizontal position. Provide a union adjacent to one end of all threaded end valves. Control valves usually require reducers to connect to pipe sizes shown in the contract documents. Install butterfly valves with the valve open as recommended by the manufacturer to prevent binding of the disc in the seat.

g. Offset equipment connections to allow valving off for maintenance and repair with minimal removal of piping. Provide flexibility in equipment connections and branch line take-offs with 3-elbow swing joints where noted in the contract documents.

h. Tee water piping runouts or branches into the side of mains or other branches. Avoid bull-head tees, which are two return lines entering opposite ends of a tee and exiting out the common side.

i. Provide manual or automatic air vent at all piping system high points and drain valves at all low points. Install piping to floor drains from all automatic air vents.

j. Connect piping to equipment as shown in the contract documents. Install components furnished by others, mechanical contractor shall coordinate with chemical treatment vendor and controls contractor on

the below items, such as:

(1) Water treatment pot feeders and condenser water treatment systems furnished by the chemical treatment vendor.

(2) Flow elements (orifice unions), control valve bodies, flow switches, pressure taps with valve, and wells for sensors furnished by the controls contractor.

k. Thermometer Wells: In pipes 2-1/2 inches and smaller increase the pipe size to provide free area equal to the upstream pipe area.

l. Firestopping: Fill openings around uninsulated piping penetrating floors or fire walls, with firestop material. For firestopping insulated piping refer to Section 23 07 11, HVAC AND BOILER PLANT INSULATION.

m. Where copper piping is connected to steel piping, provide dielectric connections.

3.2 UNDERGROUND PIPING

a. Lay all underground piping in trenches. Provide and operate pumping equipment to keep trenches free of water.

b. Install thrust blocking and restraints on all underground piping at elbows and other changes in pipe direction and as directed by manufacturer.

c. Refer to manufacturer'sinstructions for additional requirements. Install direct bury system per manufacturer's recommendations.

3.3 PIPE JOINTS

a. Welded: Beveling, spacing and other details shall conform to ASME B31.9 and AWS B2.1/B2.1M. See Welder’s qualification requirements under "Quality Assurance" in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Screwed: Threads shall conform to ASME B1.20.1; joint compound shall be applied to male threads only and joints made up so no more than 3 threads show. Coat exposed threads on steel pipe with joint compound, or red lead paint for corrosion protection.

c. 125 Pound Cast-Iron Flange (Plain Face): Mating flange shall have raised face, if any, removed to avoid overstressing the cast iron flange.

d. Solvent Welded Joints: As recommended by the manufacturer.

3.4 EXPANSION JOINTS (BELLOWS AND SLIP TYPE)

a. Anchors and Guides: Provide type, quantity and spacing as recommended by manufacturer of expansion joint and as shown. A professional engineer shall verify in writing that anchors and guides are properly designed for forces and moments which will be imposed.

b. Cold Set: Provide setting of joint travel at installation as recommended by the manufacturer for the ambient temperature during the

installation.

c. Preparation for Service: Remove all apparatus provided to restrain joint during shipping or installation. Representative of manufacturer shall visit the site and verify that installation is proper.

d. Access: Expansion joints shall be located in readily accessible space. Locate joints to permit access without removing piping or other devices. Allow clear space to permit replacement of joints and to permit access to devices for inspection of all surfaces and for adding.

3.5 SEISMIC BRACING ABOVEGROUND PIPING

Not used.

3.6 LEAK TESTING ABOVE GROUND PIPING

a. Inspect all joints and connections for leaks and workmanship and make corrections as necessary, to the satisfaction of the Contracting Officer. Tests shall either of those below, or a combination, as approved by the Contracting Officer.

b. An operating test at design pressure, and for hot systems, design maximum temperature.

c. A hydrostatic test at 1.5 times design pressure. For water systems, the design maximum pressure would usually be the static head, or expansion tank maximum pressure, plus pump head. Factory tested equipment (convertors, exchangers, coils, etc.) need not be field tested. Isolate equipment where necessary to avoid excessive pressure on mechanical seals and safety devices.

3.7 FLUSHING AND CLEANING PIPING SYSTEMS

a. Water Piping: Clean systems as recommended by the suppliers of chemicals specified in Section 23 25 00, HVAC WATER TREATMENT.

b. Initial Flushing: Remove loose dirt, mill scale, metal chips, weld beads, rust, and like deleterious substances without damage to any system component. Provide temporary piping or hose to bypass coils, control valves, exchangers and other factory cleaned equipment unless acceptable means of protection are provided and subsequent inspection of hide-out areas takes place. Isolate or protect clean system components, including pumps and pressure vessels, and remove any component which may be damaged. Open all valves, drains, vents and strainers at all system levels. Remove plugs, caps, spool pieces, and components to facilitate early debris discharge from system. Sectionalize system to obtain debris carrying velocity of (5.9 f/s), if possible. Connect dead-end supply and return headers as necessary. Flush bottoms of risers. Install temporary strainers where necessary to protect down-stream equipment. Supply and remove flushing water and drainage by various type hose, temporary and permanent piping and Contractor's booster pumps. Flush until clean as approved by the Contracting Officer.

c. Cleaning: Using products supplied in Section 23 25 00, HVAC WATER TREATMENT, circulate systems at normal temperature to remove adherent organic soil, hydrocarbons, flux, pipe mill varnish, pipe joint

compounds, iron oxide, and like deleterious substances not removed by flushing, without chemical or mechanical damage to any system component. Removal of tightly adherent mill scale is not required. Keep isolated equipment which is "clean" and where dead-end debris accumulation cannot occur. Sectionalize system if possible, to circulate at velocities not less than 5.9 f/s. Circulate each section for not less than 4 hours. Blow-down all strainers or remove and clean as frequently as necessary. Drain and prepare for final flushing.

d. Final Flushing: Return systems to conditions required by initial flushing after all cleaning solution has been displaced by clean make-up. Flush all dead ends and isolated clean equipment. Gently operate all valves to dislodge any debris in valve body by throttling velocity. Flush for not less than one hour.

3.8 WATER TREATMENT

a. Install water treatment equipment and provide water treatment system piping.

b. Close and fill system as soon as possible after final flushing to minimize corrosion.

c. Charge systems with chemicals specified in Section 23 25 00, HVAC WATER TREATMENT.

d. Utilize this activity, by arrangement with the Contracting Officer, for instructing VA operating personnel.

3.9 ELECTRIC HEAT TRACING

a. Install tracing as recommended by the manufacturer. Unless not recommended or allowed by the manufacturer heat trace shall heat the full circumference of large piping and be wrapped around the piping or have multiple parallel cables on opposite sides of the pipe.Unless not recommended or allowed by the manufacturer heat trace shall be wrapped around the piping or have multiple parallel cables on opposite sides of the pipe.

b. Coordinate electrical connections.

c. Performance verification testing shall be completed prior to system acceptance. All installations shall be tested via thermal imaging during low ambient conditions or equivalent testing approved by the Contracting Officer.

3.10 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA will observe startup and contractor testing of selected

equipment. Coordinate the startup and contractor testing schedules with Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

d. Adjust red set hand on pressure gauges to normal working pressure.

3.11 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.12 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hours to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

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SECTION 23 21 23
3/23

HYDRONIC PUMPS

PART 1 GENERAL

1.1 DESCRIPTION

a. Hydronic pumps for Heating, Ventilating and Air Conditioning.

b. Definitions:

(1) Capacity: (Gallons per minute (gpm)) of the fluid pumped.

(2) Head: Total dynamic head in (feet) of the fluid pumped.

(3) Flat head-capacity curve: Where the shutoff head is less than 1.16 times the head at the best efficiency point.

c. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Not used.

f. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

g. Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC AND STEAM GENERATION EQUIPMENT.

h. Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT.

i. Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, AND BALANCING FOR HVAC.

j. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

k. Section 23 21 13, HYDRONIC PIPING.

l. Section 26 29 11, MOTOR CONTROLLERS.

1.3 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.

ASTM INTERNATIONAL (ASTM)

ASTM A48/A48M	(2022) Standard Specification for Gray Iron Castings
ASTM B62	(2017) Standard Specification for Composition Bronze or Ounce Metal Castings

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Bio-Based Materials; S, G, AE

SD-06 Test Reports

Characteristic Curves; G, AE

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION XX XX XX, SECTION TITLE”, with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and

capacity.

(1) Pumps and accessories.

(2) Motors and drives.

(3) Variable speed motor controllers.

d. Characteristic Curves: Head-capacity, efficiency-capacity, brake horsepower-capacity, and NPSHR-capacity for each pump and for combined pumps in parallel or series service. Identify pump and show fluid pumped, specific gravity, pump speed and curves plotted from zero flow to maximum for the impeller being furnished and at least the maximum diameter impeller that can be used with the casing.

e. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

f. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician, and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

g. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.6 QUALITY ASSURANCE

a. Design Criteria:

(1) Pumps design and manufacturer shall conform to Hydraulic Institute Standards.

(2) Pump sizes, capacities, pressures, operating characteristics and efficiency shall be as scheduled.

(3) Head-capacity curves shall slope up to maximum head at shut-off. Curves shall be relatively flat for closed systems. Select pumps near the midrange of the curve, so the design capacity falls to the left of the best efficiency point, to allow a cushion for the usual drift to the right in operation, without approaching the pump curve end point and possible cavitation and unstable operation. Select pumps for open systems so that required net positive suction head (NPSHR) does not exceed the net positive head available (NPSHA).

(4) Pump Driver: Furnish with pump. Size shall be non-overloading at any point on the head-capacity curve, including in a parallel or series pumping installation with one pump in operation.

(5) Provide all pumps with motors, impellers, drive assemblies, bearings, coupling guard and other accessories specified. Statically and dynamically balance all rotating parts.

(6) Furnish each pump and motor with a nameplate giving the manufacturers name, serial number of pump, capacity in gpm and head in feet at design condition, horsepower, voltage, frequency, speed and full load current and motor efficiency. Name plates shall be visible when pumps are insulated.

(7) Test all pumps before shipment. The manufacturer shall certify all pump ratings.

(8) After completion of balancing, provide replacement of impellers or trim impellers to provide specified flow at actual pumping head, as installed.

b. Allowable Vibration Tolerance for Pump Units: Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT.

c. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this section. For more information regarding the product catagories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

d. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

1.8 SPARE MATERIALS

Furnish one spare seal and casing gasket for each pump to the Contracting Officer.

PART 2 PRODUCTS

2.1 CENTRIFUGAL PUMPS, BRONZE FITTED

a. General:

(1) Provide pumps that will operate continuously without overheating bearings or motors at every condition of operation on the pump curve, or produce noise audible outside the room or space in which installed.

(2) Provide pumps of size, type and capacity as indicated, complete with electric motor and drive assembly, unless otherwise indicated. Design pump casings for the indicated working pressure and factory test at 1-1/2 times the designed pressure.

(3) Provide pumps of the same type, the product of a single manufacturer, with pump parts of the same size and type

interchangeable.

(4) General Construction Requirements

(a) Balance: Rotating parts, statically and dynamically.

(b) Construction: To permit servicing without breaking piping or motor connections.

(c) Pump Motors: Provide high efficiency motors, inverter duty for variable speed service. Refer to Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC AND STEAM GENERATION EQUIPMENT. Motors shall be open drip proof and operate at 1750 RPM unless noted otherwise.

(d) Heating pumps shall be suitable for handling water to 225 degrees F.

(e) Provide coupling guards that meet OSHA requirements.

(f) Pump Connections: Flanged.

(g) Pump shall be factory tested.

(h) Performance: As scheduled on the Contract Drawings.

(5) Variable Speed Pumps:

(a) The pumps shall be the type shown in the contract documents and specified herein flex coupled to an open drip proof motor.

(b) Variable Speed Motor Controllers: Refer to Section 26 29 11, MOTOR CONTROLLERS and to Section 23 05 11, COMMON WORK RESULTS FOR HVAC paragraph, VARIABLE SPEED MOTOR CONTROLLERS. Furnish controllers with pumps and motors.

(c) Pump operation and speed control shall be as shown in the contract documents.

(d) Direct drive pumps with integrated variable frequency drive (VFD) utilizing the design pump curve programmed on board the built-in controller (also known as sensor-less, or self-sensing). Pump to comply with paragraphs in this section. VFD and motor to comply with Section 26 29 11, MOTOR CONTROLLERS and Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC AND STEAM GENERATION EQUIPMENT.

b. In-Line Type, Base Mounted End Suction or Double Suction Type:

(1) Casing and Bearing Housing: Close-grained cast-iron, ASTM A48/A48M.

(2) Casing Wear Rings: Bronze.

(3) Suction and Discharge: Plain face flange, 125 psig, ASME B16.1 Union End.

(4) Casing Vent: Manual brass cock at high point.

(5) Casing Drain and Gauge Taps: 1/2 inch plugged connections minimum

size.

(6) Impeller: Bronze, ASTM B62, enclosed type, keyed to shaft.

(7) Shaft: Steel, Type 1045 or stainless-steel.

(8) Shaft Seal: Mechanical seal consisting of carbon rotating ring against a ceramic seat held by a stainless-steel spring, and EPDM bellows and gasket.

(9) Shaft Sleeve: Bronze or stainless-steel.

(10) Motor: Furnish with pump. Refer to Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC AND STEAM GENERATION EQUIPMENT.

(11) Base Mounted Pumps:

(a) Designed for disassembling for service or repair without disturbing the piping or removing the motor.

(b) Impeller Wear Rings: Bronze.

(c) Shaft Coupling: Non-lubricated steel flexible type or spacer type with coupling guard, bolted to the baseplate.

(d) Bearings (Double-Suction pumps): Regreaseable ball or roller type.

(e) Provide lip seal and slinger outboard of each bearing.

(f) Base: Cast-iron or fabricated steel for common mounting to a concrete base.

(12) Provide line sized shut-off valve and suction strainer, maintain manufacturer recommended straight pipe length on pump suction (with blowdown valve). Contractor option: Provide suction diffuser as follows:

(a) Body: Cast-iron with steel inlet vanes and combination diffuser-strainer-orifice cylinder with 3/16 inch diameter openings for pump protection. Provide taps for strainer blowdown and gauge connections.

(b) Provide adjustable foot support for suction piping.

(c) Strainer free area: Not less than 5 times the suction piping.

(d) Provide disposable startup strainer.

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or

time to the Government.

b. Follow manufacturer's written instructions for pump mounting and startup. Access/Service space around pumps shall not be less than minimum space recommended by pumps manufacturer.

c. Provide drains for bases and seals for base mounted pumps, piped to and discharging into floor drains.

d. Coordinate location of thermometer and pressure gauges as per Section 23 21 13, HYDRONIC PIPING.

3.2 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

d. Verify that the piping system has been flushed, cleaned and filled.

e. Lubricate pumps before startup.

f. Prime the pump, vent all air from the casing and verify that the rotation is correct. To avoid damage to mechanical seals, never start or run the pump in dry condition.

g. Verify that correct size heaters-motor over-load devices are installed for each pump controller unit.

h. Field modifications to the bearings and or impeller (including trimming) are prohibited. If the pump does not meet the specified vibration tolerance, send the pump back to the manufacturer for a replacement pump. All modifications to the pump shall be performed at the factory.

i. Ensure the disposable strainer is free of debris prior to testing and balancing of the hydronic system.

j. After several days of operation, replace the disposable startup strainer with a regular strainer in the suction diffuser.

3.3 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.4 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hours to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

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SECTION 23 22 13
3/24

STEAM AND CONDENSATE HEATING PIPING

PART 1 GENERAL

1.1 DESCRIPTION

a. Steam, condensate, and vent piping inside buildings.

b. Boiler plant and outside steam distribution piping is covered in specification Section 33 63 00, STEAM ENERGY DISTRIBUTION and Section 23 21 11, BOILER PLANT PIPING SYSTEMS.

c. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 09 91 00, PAINTING.

f. Not used.

g. Not used.

h. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

h. Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, AND BALANCING FOR HVAC.

i. Section 23 07 11, HVAC AND BOILER PLANT INSULATION.

j. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

k. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

l. Section 23 22 23, STEAM CONDENSATE PUMPS.

m. Section 23 25 00, HVAC WATER TREATMENT.

1.3 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standards shall govern.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.20.1	(2013; R 2018) Pipe Threads, General Purpose (Inch)
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2024) Factory-Made Wrought Buttwelding Fittings
ASME B16.11	(2022) Forged Fittings, Socket-Welding and Threaded
ASME B16.42	(2021) Ductile Iron Pipe Flanges and Flanged Fittings, Classes 150 and 300
ASME B31.1	(2024) Power Piping
ASME B40.100	(2022) Pressure Gauges and Gauge Attachments
ASME BPVC	(2010) Boiler and Pressure Vessels Code
ASME BPVC SEC II-A	(2017) BPVC Section II-Materials-Part A-Ferrous Materials Specifications
ASME BPVC SEC VIII	(2021) Boiler and Pressure Vessel Codes: Section VIII Rules for Construction of Pressure Vessel
ASME BPVC SEC IX	(2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A106/A106M	(2019a) Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM A216/A216M	(2021) Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service
ASTM A285/A285M	(2024) Standard Specification for Pressure Vessel Plates, Carbon Steel, Low- and Intermediate-Tensile Strength
ASTM A516/A516M	(2017) Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service
ASTM A536	(2024) Standard Specification for Ductile Iron Castings

AMERICAN WELDING SOCIETY (AWS)

AWS B2.1/B2.1M	(2021) Specification for Welding Procedure and Performance Qualification
AWS Z49.1	(2021) Safety in Welding, Cutting and Allied Processes

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-S-901	Military Specification: Shock Tests. H.I. (High-Impact) Shipboard Machinery, Equipment, And Systems

TUBULAR EXCHANGER MANUFACTURERS ASSOCIATION (TEMA)

TEMA Stds	(2007) Standards of the Tubular Exchange Manufacturers Association (TEMA)

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Welding Qualifications; G, AE

SD-02 Shop Drawings

Coordination Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Manufacturer's Certified Data Report; G, AE

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Piping Diagrams; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 22 13, STEAM AND CONDENSATE HEATING PIPING”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Pipe and equipment supports.Submit calculations for variable spring and constant support hangers.

(2) Pipe and tubing, with specification, class or type, and schedule.

(3) Pipe fittings, including miscellaneous adapters and special fittings.

(4) Flanges, gaskets and bolting.

(5) Valves of all types.

(6) Steam traps.

(7) Steam trap monitoring system.

(8) Strainers.

(9) Pipe alignment guides.

(10) Expansion joints.

(11) Expansion compensators.

(12) Flexible ball joints: Catalog sheets, performance charts, schematic drawings, specifications and installation instructions.

(13) All specified steam system components.

(14) Gauges.

(15) Thermometers and test wells.

(16) Electric heat tracing systems.

(17) Seismic bracing details for piping.

d. Manufacturer's certified data report, Form No. U-1, for ASME pressure vessels:

(1) Heat Exchangers (Steam-to-Hot Water).

(2) Flash tanks.

e. Coordination Drawings: Refer to paragraph, SUBMITTALS of Section

23 05 11, COMMON WORK RESULTS FOR HVAC.

f. As-Built Piping Diagrams: Provide drawing as follows for steam and steam condensate piping and other central plant equipment.

(1) One wall-mounted stick file for prints. Mount stick file in the chiller plant or adjacent control room along with control diagram stick file.

(2) One set of reproducible drawings.

g. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

h. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

i. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.6 QUALITY ASSURANCE

a. Section 23 05 11, COMMON WORK RESULTS FOR HVAC, which includes welding qualifications.

b. The products and execution of work specified in this section shall conform to the referenced codes and standards as required by the specifications. Local codes and amendments shall be enforced, along with requirements of local utility companies. The most stringent requirements of these specifications, local codes, or utility company requirements shall always apply. Any conflicts shall be brought to the attention of the Contracting Officer.

c. Welding Qualifications: Before any welding is performed, contractor shall submit a certificate certifying that welders comply with the following requirements:

(1) Qualify welding processes and operators for piping according to ASME BPVC SEC IX, AWS Z49.1 and AWS B2.1/B2.1M.

(2) Comply with provisions in ASME B31.9 or ASME B31.1.

(3) Certify that each welder and welding operator has passed AWS qualification tests for welding processes involved and that certification is current and recent. Submit documentation to the Contracting Officer.

(4) All welds shall be stamped according to the provisions of the American Welding Society.

d. ASME Compliance: Comply with ASME B31.9 or ASME B31.1 for materials, products, and installation. Safety valves and pressure vessels shall bear appropriate ASME labels.

1.7 AS-BUILT DOCUMENTATION

a. Submit manufacturer’s literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, VA approved substitutions and construction revisions shall be four hard copies and one electronic copy of each inserted into a three-ring binder. All aspects of system operation and maintenance procedures, including applicable piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the ‘third party testing company’ requirement. Provide record drawings as follows:

(1) Red-lined, hand-marked drawings are to be provided, with one paper copy and a scanned PDF version of the hand-marked drawings provided on CD or DVD.

d. The as-built drawings shall indicate the location and type of all lockout/tagout points for all energy sources for all equipment and pumps to include breaker location and numbers, valve tag numbers, etc. Coordinate lockout/tagout procedures and practices with local VA requirements.

e. Certification documentation shall be provided to Contracting Officer 21 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and provide documentation/certification that all results of tests were within limits specified. Test results shall contain written sequence of test procedure with written test results annotated at each step along with the expected outcome or setpoint. The results shall include all readings, including but not limited to data on device (make, model and performance characteristics), normal pressures, switch ranges, trip points, amp readings, and calibration data to include equipment serial

numbers or individual identifications, etc.

PART 2 PRODUCTS

2.1 PIPE AND EQUIPMENT SUPPORTS, PIPE SLEEVES, AND WALL AND CEILING PLATES

Provide in accordance with Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

2.2 PIPE AND TUBING

a. Steam Piping: Steel, ASTM A53/A53M, Grade B, seamless or ERW; ASTM A106/A106M Grade B, seamless; Schedule 40.

b. Steam Condensate and Pumped Condensate Piping: Steel, ASTM A53/A53M, Grade B, seamless or ERW; or ASTM A106/A106M Grade B, seamless, Schedule 80.

c. Vent Piping: Steel, ASTM A53/A53M, Grade B, seamless or ERW; ASTM A106/A106M Grade B, seamless; Schedule 40, galvanized.

2.3 FITTINGS FOR STEEL PIPE

a. 2 inches and Smaller: Screwed or welded.

(1) Cast iron fittings or piping is prohibited for steam and steam condensate piping. Bushing reduction or use of close nipples is prohibited.

(2) Forged steel, socket welding or threaded: ASME B16.11, 2000 psig class with ASME B1.20.1 threads. Use Schedule 80 pipe and fittings for threaded joints. Lubricant or sealant shall be oil and graphite, or other compound approved for the intended service.

(3) Unions: Forged steel, 2000 psig class or 3000 psig class on piping 2 inches and less.

(4) Steam line drip station and strainer quick-couple blowdown hose connection: Straight through, plug and socket, screw or cam locking type for 1/2 inch ID hose. No integral shut-off is required.

b. 2-1/2 inches and Larger: Welded or flanged joints.

(1) Cast iron fittings or piping is prohibited for steam and steam condensate piping.

(2) Butt welding fittings: ASME B16.9 with same wall thickness as connecting piping. Elbows shall be long radius type, unless otherwise noted.

(3) Welding flanges and bolting: ASME B16.5:

(a) Steam service: Weld neck or slip-on, raised face, with non-asbestos gasket. Non-asbestos gasket shall either be stainless steel spiral wound strip with flexible graphite filler or compressed inorganic fiber, such as carbon fiber, with nitrile binder rated for saturated and superheated steam service 750 degrees F and 1500 psig.

(b) Flange bolting: Bolts shall be high strength steel ASTM A193/A193M, Grade B7. Nuts shall be ASTM A194/A194M.

c. Welded Branch and Tap Connections: Forged steel weldolets, or branchlets and threadolets may be used for branch connections up to one pipe size smaller than the main. Forged steel half-couplings, ASME B16.11 may be used for drain, vent and gauge connections.

2.4 DIELECTRIC FITTINGS

a. Provide where dissimilar metal pipe are joined.

b. 2 inches and Smaller: Threaded dielectric union.

c. 2-1/2 inches and Larger: Flange union with dielectric gasket and bolt sleeves, ASME B16.42.

d. Temperature Rating, 250 degrees F for steam condensate and as required for steam service.

e. Contractor’s option: On pipe sizes 2 inches and smaller, screwed end steel gate valves or dielectric nipples may be used in lieu of dielectric unions.

2.5 VALVES

a. Asbestos packing is prohibited.

b. All valves of the same type shall be products of a single manufacturer.

c. Provide chain operators for valves 6 inches and larger when the centerline is located 7 feet or greater above the floor or operating platform.

d. Shut-Off Valves:

(1) Ball Valves:

(a) 2 inches and smaller: Forged steel body, rated for 200 psig saturated steam, 400 psig WOG, bronze wedges and Monel or stainless-steel seats, threaded ends, rising stem, and union bonnet.

(b) 2-1/2 inches and larger: Flanged, outside screw and yoke.

1. High pressure steam 110 kPa 16 psig and above system: Cast steel body, ASTM A216/A216M grade WCB, 150 psig at 500 degrees F, 11-1/2 to 13 percent chrome stainless steel solid disc and seats. Provide 1 inch factory installed bypass with globe valve on valves 4 inches and larger.

2. All other services: Forged steel body, Class B, rated for 123 psig saturated steam, 200 psig WOG, bronze or bronze face wedge and seats, 123 psig ASME flanged ends, OS&Y, rising stem, bolted bonnet, and renewable seat rings.

e. Ball Valves: Reduced port permitted for bypass (throttling) service; full port required for all other services, one-fourth turn to open.

1. Type 501: Type 316 stainless steel body, ball and stem, rated for 150 psig at 365 degrees F, 600 psig at 200 degrees F; reinforced TFE seat, stem seal and thrust washer; end entry, threaded ends.

2. Type 502: Steel body, rated for 150 psig at 365 degrees F, 250 psig at 250 degrees F, reinforced TFE seat, stem seal and thrust washer; end entry, threaded ends, UL-listed for natural or LP gas shut off service when used on those services.

3. Type 503: Carbon steel body, steam service, rated for 200 psig at 392 degrees F, stainless steel ball and stem, Polyfil seat, live-loaded or adjustable stem seal, threaded ends.

f. Globe and Angle Valves:

(1) Globe Valves:

(a) 2 inches and smaller: Forged steel body, rated for 200 psig saturated steam, 400 psig WOG, hardened stainless steel disc and seat, threaded ends, rising stem, union bonnet, and renewable seat rings.

(b) 2-1/2 inches and larger:

1. Globe valves for high pressure steam 16 psig: Cast steel body, ASTM A216/A216M grade WCB, flanged, OS&Y, 150 psig at 500 degrees F, 11-1/2 to 13 percent chrome stainless steel disc and renewable seat rings.

2. All other services: Steel body, rated for 850 kPa 123 psig saturated steam, 1380 kPa 200 psig WOG, bronze or bronze-faced disc (Teflon or composition facing permitted) and seat, 123 psig ASME flanged ends, OS&Y, rising stem, bolted bonnet, and renewable seat rings.

(2) Angle Valves:

(a) 2 inches and smaller: Cast steel 150 psig, union bonnet with metal plug type disc.

(b) 2-1/2 inches and larger:

1. Angle valves for high pressure steam 16 psig: Cast steel body, ASTM A216/A216M grade WCB, flanged, OS&Y, 150 psig at 500 degrees F, 11-1/2 to 13 percent chrome stainless steel disc and renewable seat rings.

2. All other services: 125 psig, flanged, cast steel body, and bronze trim.

g. Swing Check Valves:

(1) 2 inches and smaller: Cast steel, 150 psig, 45-degree swing disc.

(2) 2-1/2 inches and Larger:

(a) Check valves for high pressure steam 16 psig and above system: Cast steel body, ASTM A216/A216M grade WCB, flanged, OS&Y, 150 psig at 500 degrees F, 11-1/2 to 13 percent chrome

stainless steel disc and renewable seat rings.

(b) All other services: 125 psig, flanged, cast steel body, and bronze trim.

h. Manual Radiator/Convector Valves: Brass, packless, with position indicator.

2.6 STRAINERS

a. Y Type. Tee type is acceptable for gravity flow and pumped steam condensate service.

b. High Pressure Steam: Rated 150 psig saturated steam.

(1) 2 inches and smaller: Cast steel, rated for saturated steam at 150 psig threaded ends.

(2) 2-1/2 inches and larger: Cast steel rated for 150 psig saturated steam with 150 psig ASME flanged ends or forged steel with 250 psig ASME flanged ends.

c. All Other Services: Rated 125 psig saturated steam.

(1) 2 inches and smaller: Cast steel body.

(2) 2-1/2 inches and larger: Flanged, cast steel body.

d. Screens: Bronze, Monel metal or 18-8 stainless steel, free area not less than 2-1/2 times pipe area, with perforations as follows:

(1) 3 inches and smaller: 20 mesh for steam and 0.045 inch diameter perforations for liquids.

(2) 4 inches and larger: 0.045 inch diameter perforations for steam and 1/8 inch diameter perforations for liquids.

2.7 PIPE ALIGNMENT

Provide factory-built guides along the pipe line to permit axial movement only and to restrain lateral and angular movement. Guides shall be designed to withstand a minimum of 15 percent of the axial force which will be imposed on the expansion joints and anchors. Field-built guides may be used if detailed in the contract documents.

2.8 EXPANSION JOINTS

a. Factory built devices, inserted in the pipe lines, designed to absorb axial cyclical pipe movement which results from thermal expansion and contraction. This includes factory-built or field-fabricated guides located along the pipe lines to restrain lateral pipe motion and direct the axial pipe movement into the expansion joints.

b. Minimum Service Requirements:

(1) Pressure Containment:

(a) Steam Service 5-29 psig: Rated 50 psig at 298 degrees F.

(b) Steam Service 31-123 psig: Rated 150 psig at 366 degrees F.

(c) Steam Service 126-150 psig: Rated 200 psig at 381 degrees F.

(d) Condensate Service: Rated 100 psig at 309 degrees F.

(2) Number of Full Reverse Cycles without failure: Minimum 1000.

(3) Movement: As shown on contract documents plus recommended safety factor of manufacturer.

c. Manufacturing Quality Assurance: Conform to Expansion Joints Manufacturers Association Standards.

d. Bellows - Internally Pressurized Type:

(1) Multiple corrugations of Type 304 or Type A240-321 stainless steel.

(2) Internal stainless-steel sleeve entire length of bellows.

(3) External cast iron equalizing rings for services exceeding 50 psig.

(4) Welded ends.

(5) Design shall conform to standards of EJMA and ASME B31.1.

(6) External tie rods designed to withstand pressure thrust force upon anchor failure if one or both anchors for the joint are at change in direction of pipeline.

(7) Integral external cover.

e. Bellows - Externally Pressurized Type:

(1) Multiple corrugations of Type 304 stainless steel.

(2) Internal and external guide integral with joint.

(3) Design for external pressurization of bellows to eliminate squirm.

(4) Welded ends.

(5) Conform to the standards of EJMA and ASME B31.1.

(6) Threaded connection at bottom, 1 inch minimum, for drain or drip point.

(7) Integral external cover and internal sleeve.

f. Expansion Joint Identification: Provide stamped brass or stainless-steel nameplate on each expansion joint listing the manufacturer, the allowable movement, flow direction, design pressure and temperature, date of manufacture, and identifying the expansion joint by the identification number in the contract documents.

2.9 FLEXIBLE BALL JOINTS

a. Design and Fabrication: One-piece component construction, fabricated from steel with welded ends, designed for a working steam pressure of

250 psig and a temperature of 450 degrees F. Each joint shall provide for 360 degrees rotation in addition to a minimum angular flexible movement of 30 degrees for sizes 1/4 inch to 6 inch inclusive, and 15 degrees for sizes 2-1/2 inches to 30 inches. Joints through 14 inches shall have forged pressure retaining members, while size 16 inches through 30 inches shall be of one-piece construction.

b. Material:

(1) Cast or forged steel pressure containing parts and bolting in accordance with ASME BPVC SEC II-A or ASME B31.1. Retainer may be ductile iron ASTM A536, Grade 65-45-12, or ASME BPVC SEC II-A SA 515, Grade 70.

(2) Gaskets: Steam pressure molded composition design for a temperature range of from minus 50 degrees F to plus 525 degrees F.

c. Certificates: Submit qualifications of ball joints in accordance with the following test data:

(1) Low pressure leakage test: 6 psig saturated steam for 60 days.

(2) Flex cycling: 800 Flex cycles at 500 psig saturated steam.

(3) Thermal cycling: 100 saturated steam pressure cycles from atmospheric pressure to operating pressure and back to atmospheric pressure.

(4) Environmental shock tests: Forward certificate from a recognized test laboratory, that ball joints of the type submitted has passed shock testing in accordance with MIL-S-901.

(5) Vibration: 170 hours on each of three mutually perpendicular axes at 25 to 125 Hz; to 0.05 inch to 0.10 inch double amplitude on a single ball joint and 3 ball joint off set.

2.10 STEAM SYSTEM COMPONENTS

a. Heat Exchanger (Steam to Hot Water): Shell and tube type, U-bend removable tube bundle, steam in shell, water in tubes, equipped with support cradles.

(1) Maximum tube velocity: 7.5 feet/second.

(2) Tube fouling factor: TEMA Stds, but not less than 0.001 square foot/hour degrees F/Btu.

(3) Materials:

(a) Shell: Steel.

(b) Tube sheet and tube supports: Steel or brass.

(c) Tubes: 3/4 inch OD copper.

(d) Head or bonnet: Steel.

(4) Construction: In accordance with ASME BPVC for 125 psig working

pressure for shell and tubes. Provide manufacturer's certified data report, Form No. U-1.

b. Optional Heat Transfer Package: In lieu of field erected individual components, the Contractor may provide a factory or shop assembled package of heat exchangers, pumps, and other components, pre-piped and pre-wired and supported on a welded steel frame or skid.

c. Steam Pressure Reducing Valves in PRV Stations:

(1) Type: Single-seated, diaphragm operated, spring-loaded, external or internal steam pilot-controlled, normally closed, adjustable set pressure. Pilot shall sense controlled pressure downstream of main valve.

(2) Service: Provide controlled reduced pressure to steam piping systems.

(3) Pressure control shall be smooth and continuous with maximum drop of 10 percent deviation from set pressure. Maximum flow capacity of each valve shall not exceed capacity of downstream safety valve(s).

(4) Main valve and pilot valve shall have replaceable valve plug and seat of stainless steel, Monel, or similar durable material.

(a) Pressure rating for high pressure steam: Not less than 150 psig saturated steam.

(b) Connections: Flanged for valves 2-1/2 inches and larger; flanged or threaded ends for smaller valves.

(5) Select pressure reducing valves to develop less than 85 dBA at 5 feet elevation above adjacent floor, and 5 feet distance in any direction. Inlet and outlet piping for steam pressure reducing valves shall be Schedule 80 minimum for required distance to achieve required levels or sound attenuators shall be applied.

d. Safety Valves and Accessories: Comply with ASME BPVC SEC VIII. Capacities shall be certified by National Board of Boiler and Pressure Vessel Inspectors, maximum accumulation 10 percent. Provide lifting lever. Provide drip pan elbow where shown. Valve shall have stainless steel seats and trim.

e. Steam PRV for Individual Equipment: Cast steel body, screwed or flanged ends, rated 125 psig, or 20 percent above the working pressure, whichever is greater. Single-seated, diaphragm operated, spring loaded, adjustable range, all parts renewable.

f. Moisture Separator: Shall be of the high efficiency internal baffle type, with a pressure drop that does not exceed an equivalent length of straight pipe. Body shall be iron, steel or cast steel with screwed, socket weld or flanged connections as detailed on drawing schedule. Steel construction shall be in accordance with ASME BPVC SEC VIII, Division 1 of the ASME Boiler and Pressure Vessel Code with ASME Class 150, 300 or 600 RF flange connections. A bottom drain connection shall be provided for the installation of a trap to

discharge any accumulated liquid. Separators shall be installed before steam PRV and Control Valves.

g. Flash Tanks: Horizontal or vertical vortex type, constructed of copper bearing steel, ASTM A516/A516M or ASTM A285/A285M, for a steam working pressure of 125 psig to comply with ASME Code for Unfired Pressure Vessels and stamped with "U" symbol. Perforated pipe inside tank shall be ASTM A53/A53M Grade B, seamless or ERW, or ASTM A106/A106M Grade B seamless, Schedule 80. Corrosion allowance of 1/16 inch may be provided in lieu of the copper bearing requirement. Provide data Form No. U-1.

h. Steam Trap: Each type of trap shall be the product of a single manufacturer. Provide trap sets at all low points and at 200 feet intervals on the horizontal main lines.

(1) Floats and linkages shall provide sufficient force to open trap valve over full operating pressure range available to the system. Unless otherwise indicated in the contract documents, traps shall be sized for capacities indicated at minimum pressure drop as follows:

(a) For equipment with modulating control valve: (1/4 psig), based on a condensate leg of 12 inches at the trap inlet and gravity flow to the receiver.

(b) For main line drip trap sets and other trap sets at steam pressure: Up to 70 percent of design differential pressure. Condensate may be lifted to the return line.

(2) Trap bodies: Steel and stainless-steel, constructed to permit ease of removal and servicing working parts without disturbing connecting piping. The use of raised face flange is required on pipe sizes 1-1/2 inch and greater. The use of unions is acceptable for pipe sizes less than 1-1/2 inches. For systems without relief valve traps shall be rated for the pressure upstream of the steam supplying the system.

(3) Balanced pressure thermostatic elements: Phosphor bronze, stainless steel or Monel metal.

(4) Valves and seats: Suitable hardened corrosion resistant alloy.

(5) Mechanism: Brass, stainless steel or corrosion resistant alloy.

(6) Floats: Stainless steel.

(7) Inverted bucket traps: Provide bi-metallic thermostatic element for rapid release of non-condensables.

i. Thermostatic Air Vent (Steam): Steel body, balanced pressure bellows, stainless steel (renewable) valve and seat, rated 125 psig working pressure, 3/4 inch screwed connections. Air vents shall be balanced pressure type that responds to steam pressure-temperature curve and vents air at any pressure.

j. Steam Flow Meter/Recorder: Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

2.11 GAUGES, PRESSURE AND COMPOUND

a. ASME B40.100, Accuracy Grade 1A, (pressure, vacuum, or compound), initial mid-scale accuracy 1 percent of scale (Qualify grade), metal or phenolic case, 4-1/2 inches in diameter, 1/4 inch NPT bottom connection, white dial with black graduations and pointer, clear glass or acrylic plastic window, suitable for board mounting. Provide red "set hand" to indicate normal working pressure.

b. Provide steel, lever handle union cock. Provide steel or stainless-steel pressure snubber for gauges in water service. Provide steel pigtail syphon for steam gauges.

c. Pressure gauge ranges shall be selected such that the normal operating pressure for each gauge is displayed near the midpoint of each gauge’s range. Gauges with ranges selected such that the normal pressure is displayed at less than 30 percent or greater than 70 percent of the gauge’s range are prohibited. The units of pressure shall be psig.

2.12 PRESSURE/TEMPERATURE TEST PROVISIONS

a. Provide one each of the following test items to the Contracting Officer:

(1) 1/4 inch FPT by 1/8 inch diameter stainless steel pressure gauge adapter probe for extra-long test plug. Pressure/temperature plug is an example.

(2) 3-1/2 inch diameter, one percent accuracy, compound gauge, 30 inches Hg to 100 psig range.

(3) 32 to 220 degrees F pocket thermometer one-half degree accuracy, 1 inch dial, 5 inch long stainless-steel stem, plastic case.

2.13 FIRESTOPPING MATERIAL

Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 3 EXECUTION

3.1 GENERAL

a. If in the substantiated evaluation of the Contracting Officer, the installation fails to meet the requirements of the construction documents with respect to function and maintainability, an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

b. The contract documents show the general arrangement of pipe and equipment but do not show all required fittings and offsets that may be necessary to connect pipes to equipment, fan-coils, coils, radiators, etc., and to coordinate with other trades. Provide all necessary fittings, offsets and pipe runs based on field measurements and at no additional cost or time to the Government. Coordinate with other trades for space available and relative location of HVAC

equipment and accessories to be connected on ceiling grid. Pipe location in the contract documents shall be altered by contractor where necessary to avoid interferences and clearance difficulties.

c. Store materials to avoid excessive exposure to weather or foreign materials. Keep inside of piping relatively clean during installation and protect open ends when work is not in progress.

d. Support piping securely. Refer to PART 3, Section 23 05 11, COMMON WORK RESULTS FOR HVAC. Install convertors and other heat exchangers at height sufficient to provide gravity flow of condensate to the flash tank and condensate pump.

e. Install piping generally parallel to walls and column center lines, unless shown otherwise in the contract documents. Space piping, including insulation, to provide 1 inch minimum clearance between adjacent piping and another surface. Unless shown otherwise, slope steam, condensate and drain piping down in the direction of flow not less than 1 inch in 40 feet. Provide eccentric reducers to keep bottom of sloped piping flat.

f. Locate and orient valves to permit proper operation and access for maintenance of packing, seat and disc. Generally, locate valve stems in overhead piping in horizontal position. Provide a union adjacent to one end of all threaded end valves. Control valves usually require reducers to connect to pipe sizes shown in the contract documents.

g. Offset equipment connections to allow valving off for maintenance and repair with minimal removal of piping. Provide flexibility in equipment connections and branch line take-offs with 3-elbow swing joints where noted in the contract documents.

h. Tee water piping runouts or branches into the side of mains or other branches. Avoid bull-head tees, which are two return lines entering opposite ends of a tee and exiting out the common side.

i. Connect piping to equipment as shown in the contract documents. Install components furnished by controls contractor such as flow elements (orifice unions), control valve bodies, flow switches, pressure taps with valve, and wells for sensors.

j. Firestopping: Fill openings around uninsulated piping penetrating floors or fire walls, with firestop material. For firestopping insulated piping refer to Section 23 07 11, HVAC AND BOILER PLANT INSULATION.

k. Pipe vents to the exterior. Where a combined vent is provided, the cross-sectional area of the combined vent shall be equal to sum of individual vent areas. Slope vent piping 1 inch in 40 feet 0.25 percent in direction of flow. Provide a drip pan elbow on relief valve outlets if the vent rises to prevent backpressure. Terminate vent minimum 12 inches above the roof or through the wall minimum 8 feet above grade with down turned elbow.

l.

m. Contract documents do not identify all locations where steam trap sets are required. Contractor shall provide at no additional cost as many trap-sets as are necessary for the proper, reliable operation of the

system. Provide trap-sets at all low points and at 200 feet intervals on all horizontal main lines.

3.2 WELDING

a. The contractor is entirely responsible for the quality of the welding and shall:

(1) Conduct tests of the welding procedures used on the project, verify the suitability of the procedures used, verify that the welds made will meet the required tests, and also verify that the welding operators have the ability to make sound welds under standard conditions.

(2) Perform all welding operations required for construction and installation of the piping systems.

b. Qualification of Welders: Rules of procedure for qualification of all welders and general requirements for fusion welding shall conform with the applicable portions of ASME B31.1, AWS B2.1/B2.1M, AWS Z49.1, and also as outlined below.

c. Examining Welder: Examine each welder at job site, in the presence of the Contracting Officer, to determine the ability of the welder to meet the qualifications required. Test welders for piping for all positions, including welds with the axis horizontal (not rolled) and with the axis vertical. Each welder shall be allowed to weld only in the position in which he has qualified and shall be required to identify his welds with his specific code marking signifying his name and number assigned.

d. Examination Results: Provide the Contracting Officer with a list of names and corresponding code markings. Retest welders who fail to meet the prescribed welding qualifications. Disqualify welders, who fail the second test, for work on the project.

e. Beveling: Field bevels and shop bevels shall be done by mechanical means or by flame cutting. Where beveling is done by flame cutting, surfaces shall be thoroughly cleaned of scale and oxidation just prior to welding. Conform to specified standards.

f. Alignment: Provide approved welding method for joints on all pipes greater than 2 inches to assure proper alignment, complete weld penetration, and prevention of weld spatter reaching the interior of the pipe.

g. Erection: Piping shall not be split, bent, flattened, or otherwise damaged before, during, or after installation. If the pipe temperature falls to 32 degrees F or lower, the pipe shall be heated to approximately 100 degrees F for a distance of 1 foot on each side of the weld before welding, and the weld shall be finished before the pipe cools to 32 degrees F.

h. Non-Destructive Examination of Piping Welds:

(1) Perform radiographic examination of 50 percent of the first 10 welds made and 10 percent of all additional welds made. The Contracting Officer reserves the right to identify individual welds for which the radiographic examination shall be performed.

All welds shall be visually inspected by the Contracting Officer. The VA reserves the right to require testing on additional welds up to 100 percent if greater than 25 percent of the examined welds fail the inspection.

(2) An approved independent testing firm regularly engaged in radiographic testing shall perform the radiographic examination of pipe joint welds. All radiographs shall be reviewed and interpreted by an ASNT Certified Level III radiographer, employed by the testing firm, who shall sign the reading report.

(3) Comply with ASME B31.1. Furnish a set of films showing each weld inspected, a reading report evaluating the quality of each weld, and a location plan showing the physical location where each weld is to be found in the completed project. The Contracting Officer and the CxA shall be given a copy of all reports to be maintained as part of the project records and shall review all inspection records.

i. Defective Welds: Replace and reinspect defective welds. Repairing defective welds by adding weld material over the defect or by peening are prohibited. Welders responsible for defective welds shall be requalified prior to resuming work on the project.

j. Electrodes: Electrodes shall be stored in a dry heated area and be kept free of moisture and dampness during the fabrication operations. Discard electrodes that have lost part of their coating.

3.3 PIPE JOINTS

a. Welded: Beveling, spacing and other details shall conform to ASME B31.1 and AWS B2.1/B2.1M. See Welder’s qualification requirements under "Quality Assurance" in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Screwed: Threads shall conform to ASME B1.20.1; joint compound shall be applied to male threads only and joints made up so no more than three threads show. Coat exposed threads on steel pipe with joint compound, or red lead paint for corrosion protection.

c. 125 Pound Cast Steel Flange (Plain Face): Mating flange shall have raised face, if any, removed to avoid overstressing the cast steel flange.

3.4 EXPANSION JOINTS (BELLOWS AND SLIP TYPE)

a. Anchors and Guides: Provide type, quantity and spacing as recommended by manufacturer of expansion joint and as shown. A professional engineer shall verify in writing that anchors and guides are properly designed for forces and moments which will be imposed.

b. Cold Set: Provide setting of joint travel at installation as recommended by the manufacturer for the ambient temperature during the installation.

c. Preparation for Service: Remove all apparatus provided to restrain joint during shipping or installation. Representative of manufacturer shall visit the site and verify that installation is proper.

d. Access: Expansion joints shall be located in readily accessible space. Locate joints to permit access without removing piping or other devices. Allow clear space to permit replacement of joints and to permit access to devices for inspection of all surfaces and for adding packing.

3.5 STEAM TRAP PIPING

a. Install to permit gravity flow to the trap. Provide gravity flow (avoid lifting condensate) from the trap where modulating control valves are used. Support traps weighing over 24 pounds independently of connecting piping.

(1) On pipe size 1-1/2 inch and greater a raised face flange is required to allow for removal of the steam trap without disturbing surrounding piping.

(2) On pipe size less than 1-1/2inch raised face flanges or unions may be used to allow for removal of the traps.

3.6 SEISMIC BRACING

Not used.

3.7 LEAK TESTING

a. Inspect all joints and connections for leaks and workmanship and make corrections as necessary to the satisfaction of the Contracting Officer in accordance with the specified requirements. Testing shall be performed in accordance with the specification requirements.

b. An operating test at design pressure, and for hot systems, design maximum temperature.

c. A hydrostatic test at 1.5 times design pressure. For water systems, the design maximum pressure would usually be the static head, or expansion tank maximum pressure, plus pump head. Factory tested equipment (convertors, exchangers, coils, etc.) need not be field tested. Avoid excessive pressure on mechanical seals and safety devices.

d. Prepare and submit test and inspection reports to the Contracting Officer within 5 working days of test completion and prior to covering the pipe.

e. All tests shall be witnessed by the Contracting Officer, their representative, or the CxA and be documented by each section tested, date tested, and list or personnel present.

3.8 FLUSHING AND CLEANING PIPING SYSTEMS

The piping system shall be flushed clean prior to equipment connection. Cleaning includes pulling all strainer screens and cleaning all scale/dirt legs during startup operation. Contractor shall be responsible for damage caused by inadequately cleaned/flushed systems.

3.9 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA shall observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

d. Adjust red set hand on pressure gauges to normal working pressure.

3.10 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification shall be tested as part of a larger system.

3.11 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hour s to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

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SECTION 23 22 23
4/20

STEAM CONDENSATE PUMPS

PART 1 GENERAL

1.1 DESCRIPTION

a. Steam condensate pumps for Heating, Ventilating and Air Conditioning.

b. Definitions:

(1) Capacity: Liters per second Gallons per minute (gpm) of the fluid pumped.

(2) Head: Total dynamic head in feet of the fluid pumped.

c. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Not used.

g. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

h. Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC AND STEAM GENERATION EQUIPMENT.

i. Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT.

j. Section 23 08 00.01, DVA/USCAE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

k. Section 23 22 13, STEAM AND CONDENSATE HEATING PIPING.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Characteristic Curves; G, AE

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.4 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 22 23, STEAM CONDENSATE PUMPS”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Pumps and accessories.

(2) Motors and drives.

d. Characteristic Curves: Head-capacity, efficiency-capacity, brake horsepower-capacity, and NPSHR-capacity for each pump .

e. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

f. Completed System Readiness Checklist provided by the Commissioning Agent and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING

OF HVAC SYSTEMS.

g. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.5 QUALITY ASSURANCE

a. Design Criteria:

(1) Pumps design and manufacturer shall conform to Hydraulic Institute Standards.

(2) Pump sizes, capacities, pressures, operating characteristics and efficiency shall be as scheduled.

(3) Select pumps so that required net positive suction head (NPSHR) does not exceed the net positive head available (NPSHA).

(4) Pump Driver: Furnish with pump. Size shall be non-overloading at any point on the head-capacity curve including one pump operation in a parallel or series pumping installation.

(5) Provide all electric-powered pumps with motors, impellers, drive assemblies, bearings, coupling guard and other accessories specified. Statically and dynamically balance all rotating parts.

(6) Furnish each pump and motor with a nameplate giving the manufacturers name, serial number of pump, capacity in gpm and head in feet at design condition, horsepower, voltage, frequency, speed and full load current and motor efficiency.

(7) Test all pumps before shipment. The manufacturer shall certify all pump ratings.

(8) After completion of balancing, provide replacement of impellers or trim impellers to provide specified flow at actual pumping head, as installed.

(9) Furnish one spare seal and casing gasket for each pump to the Contracting Officer.

b. Allowable Vibration Tolerance for Pump Units: Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT.

1.6 AS-BUILT DOCUMENTATION

a. Submit manufacturer’s literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, VA approved substitutions and construction revisions shall be in electronic version on CD or DVD inserted into a three-ring binder. All aspects of system operation and maintenance procedures, including applicable piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency

situations. Notes on all special systems or devices shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the 'third party testing company' requirement. Provide record drawings as follows:

(1) Red-lined, hand-marked drawings are to be provided, with one paper copy and a scanned PDF version of the hand-marked drawings provided on CD or DVD.

d. The as-built drawings shall indicate the location and type of all lockout/tagout points for all energy sources for all equipment and pumps to include breaker location and numbers, valve tag numbers, etc. Coordinate lockout/tagout procedures and practices with local VA requirements.

e. Certification documentation shall be provided to Contracting Officer 21 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and provide documentation/certification that all results of tests were within limits specified. Test results shall contain written sequence of test procedure with written test results annotated at each step along with the expected outcome or setpoint. The results shall include all readings, including but not limited to data on device (make, model and performance characteristics), normal pressures, switch ranges, trip points, amp readings, and calibration data to include equipment serial numbers or individual identifications, etc.

PART 2 PRODUCTS

2.1 CONDENSATE PUMP, PAD-MOUNTED

a. General: Factory assembled unit consisting of vented receiver tank, motor-driven pumps, interconnecting piping and wiring, motor controls (including starters, if necessary) and accessories, designed to receive, store, and pump steam condensate up to 212°F without cavitation.

b. Receiver Tank: Cast iron with threaded openings for connection of piping and accessories and facilities for mounting float switches. Receivers for simplex pumps shall include all facilities for future mounting of additional pump and controls.

c. Furnish seals for condensate pump with a minimum temperature rating of 250 degrees F.

d. Centrifugal Pumps: Bronze fitted with mechanical shaft seals.

(1) Designed to allow removal of rotating elements without disturbing connecting piping or pump casing mounting.

(2) Shafts: Stainless steel, Type 416 or alloy steel with bronze shaft sleeves.

(3) Bearings: Regreaseable ball or roller type.

(4) Casing wearing rings: Bronze.

e. Motors: Refer to Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC AND STEAM GENERATION EQUIPMENT.

f. Pump Operation:

(1) Float Switches: NEMA 4, mounted on receiver tank, to start and stop pumps in response to changes in the water level in the receiver and adjustable to permit the controlled water levels to be changed. Floats and connecting rods shall be copper, bronze or stainless steel.

(2) Alternator: Provide for duplex units to automatically start the second pump when the first pump fails in keeping the receiver water level from rising and to alternate the order of starting the pumps to equalize wear. For units 1/3 hp and smaller, the alternator may be the mechanical type for use in lieu of float switches.

g. Control Cabinet for 3 Phase (1/2 hp and larger) Units: NEMA 4, UL approved, factory wired, enclosing all controls, with indicating lights, manual switches and resets mounted on the outside of the panel. Unit shall be capable of connecting to BAS for control and alarms. Attach cabinet to the pump set with rigid steel framework, unless remote mounting is noted on the pump schedule.

(1) Motor starters: Magnetic contact types with circuit breakers or combination fusible disconnect switches. Provide low voltage control circuits (120-volt maximum) and HOA switches for each pump.

(2) Indicating lights for each pump: Green to show that power is on, red to show that the pump is running.

h. Electric Wiring: Suitable for 200 degrees F service; enclosed in liquid-tight flexible metal conduit where located outside of control cabinet.

i. Receiver Accessories:

(1) Thermometer: 100 to 420 degrees F, mounted below minimum water level.

(2) Water level gauge glass: Brass with gauge cocks which automatically stop the flow of water when the glass is broken. Provide drain on the lower gauge cock and protection rods for the glass.

2.2 INSULATION

b. Removable Insulation Jacket:

(1) The insulation jacket should be of sewn construction with Velcro fasteners and have openings for inlet, outlet, drain, and gauge glass.

(2) Materials:

(a) Liner and jacket shall be silicone impregnated heavy duty glass fiber rated for a maximum temperature of 500 degrees F.

(b) Insulation shall be 1 inch minimum thickness, Type E needled glass fiber mat rated for a maximum temperature of 1200 F.

(c) Jacket shall be sewn with Nomex thread with a UV inhibitor.

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

b. Follow manufacturer's written instructions for pump mounting and startup. Access/Service space around pumps shall not be less than minimum space recommended by pumps manufacturer.

c. Sequence of installation for base-mounted pumps:

(1) Level and shim the unit base and grout to the concrete pad.

(2) Shim the driver and realign the pump and driver. Correct axial, angular or parallel misalignment of the shafts.

(3) Connect properly aligned and independently supported piping.

(4) Recheck alignment.

d. Pad-mounted Condensate Pump : Level, shim, bolt, and grout the unit base onto the concrete pad.

e. Sump Type Condensate Pump: Apply two coats of asphalt or bituminous compound on the exterior of the receiver tank, and mount level and flush in the floor with waterproofing gaskets and grouting to prevent ground water from entering the building from around the receiver.

f. Coordinate location of thermometer and pressure gauges as per Section 23 22 13, STEAM AND CONDENSATE HEATING PIPING.

3.2 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The Commissioning Agent will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with Contrating Officer and Commissioning Agent. Provide a minimum notice of 10 working days prior to startup and testing.

d. Verify that the piping system has been flushed, cleaned and filled.

e. Lubricate pumps before startup.

f. Prime the pump, vent all air from the casing and verify that the rotation is correct. To avoid damage to mechanical seals, never start or run the pump in dry condition.

g. Verify that correct size heaters-motor over-load devices are installed for each pump controller unit.

h. Field modifications to the bearings and or impeller (including trimming) are prohibited. If the pump does not meet the specified vibration tolerance send the pump back to the manufacturer for a replacement pump. All modifications to the pump shall be performed at the factory.

3.3 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.4 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hour s to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

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SECTION 23 25 00

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SECTION 23 25 00
3/23

HVAC WATER TREATMENT

PART 1 GENERAL

1.1 DESCRIPTION

a. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. This section specifies cleaning and treatment of circulating HVAC water systems, including the following.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

(1) Cleaning compounds.

(2) Chemical treatment for closed loop heat transfer systems.

(3) Chemical treatment for open loop systems.

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1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

d. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

e. Section 23 21 13, HYDRONIC PIPING.

f. Section 23 22 13, STEAM and CONDENSATE HEATING PIPING.

g. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.3 REFERENCES

The publication listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standards will govern.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC (2010) Boiler and Pressure Vessels Code

ASTM INTERNATIONAL (ASTM)

ASTM A666	(2023) Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate and Flat Bar
ASTM F441/F441M	(2023) Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe, Schedules 40 and 80

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, E

Materials Safety Data Sheet; G, AE

Bio-Based Materials; S, G, AE

SD-06 Test Reports

Water Analysis Verification; G, AE

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Certified Laboratory Reports; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

Log Forms; G, AE

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked "SUBMITTED UNDER SECTION XX XX XX, SECTION TITLE", with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Cleaning compounds and recommended procedures for their use.

(2) Chemical treatment for closed systems, including installation and operating instructions.

(3) Chemical treatment for open loop systems, including installation and operating instructions.

(4) Glycol-water system materials, equipment, and installation.

d. Water analysis verification.

e. Materials Safety Data Sheet for all proposed chemical compounds, based on U.S. Department of Labor Form No. L5B-005-4.

f. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation, and maintenance.

g. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician, and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

h. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING of HVAC SYSTEMS.

1.6 QUALITY ASSURANCE

a. Refer to paragraph QUALITY ASSURANCE in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Technical Services: Provide the services of an experienced water treatment chemical engineer or technical representative to direct flushing, cleaning, pre-treatment, training, debugging, and acceptance testing operations; direct and perform chemical limit control during construction period and monitor systems for a period of 12 months after acceptance, including not less than 6 service calls and written status reports. Emergency calls are not included. During this period perform monthly tests of the cooling tower for Legionella pneumophila and Heterotrophic Plate Count (HPC). Submit reports stating Legionella

bacteria count per millimeter. These tests shall be conducted in a certified laboratory and not by a technician in the field. Minimum service during construction/startup shall be 6 hours.

c. Field Quality Control and Certified Laboratory Reports: During the 1-year guarantee period, the water treatment laboratory shall provide not less than 12 reports based on onsite periodic visits, as stated in above paragraph Technical Services, sample taking and testing, and review with VA personnel, of water treatment control for the previous period. In addition to field tests, the water treatment laboratory shall provide certified laboratory test reports. These monitoring reports shall assess chemical treatment accuracy, scale formation, fouling and corrosion control, and shall contain instructions for the correction of any out-of-control condition.

d. Log Forms: Provide 1-year supply of preprinted water treatment test log forms.

e. Chemicals: Chemicals shall be non-toxic approved by local authorities and meeting applicable EPA requirements.

f. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specification section. For more information regarding the product categories covered by the Bio-Preferred Program, visit.

g. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

a. Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 2 PRODUCTS

2.1 CLEANING COMPOUNDS

a. Alkaline phosphate or non-phosphate detergent/surfactant/specific to remove organic soil, hydrocarbons, flux, pipe mill varnish, pipe compounds, iron oxide, and like deleterious substances, with or without inhibitor, suitable for system wetted metals without deleterious effects.

b. All chemicals to be acceptable for discharge to sanitary sewer.

c. Refer to Section 23 21 13, HYDRONIC PIPING and Section 23 22 13, STEAM and CONDENSATE HEATING PIPING, PART 3, for flushing and cleaning procedures.

2.2 CHEMICAL TREATMENT FOR CLOSED LOOP SYSTEMS

a. Inhibitor: Provide sodium nitrite/borate, molybdate-based inhibitor or other approved compound suitable for make-up quality and make-up rate and which will cause or enhance bacteria/corrosion problems or mechanical seal failure due to excessive total dissolved solids. Shot feed manually. Maintain inhibitor residual as determined by water

treatment laboratory, taking into consideration residual and temperature effect on pump mechanical seals.

b. pH Control: Inhibitor formulation shall include adequate buffer to maintain pH range of 8.0 to 10.5.

c. Performance: Protect various wetted, coupled, materials of construction including ferrous, and red and yellow metals. Maintain system essentially free of scale, corrosion, and fouling. Corrosion rate of following metals shall not exceed specified mills per year penetration; ferrous, 0-2; brass, 0-1; copper, 0-1. Inhibitor shall be stable at equipment skin surface temperatures and bulk water temperatures of not less than 250 degrees F and 125 degrees Fahrenheit respectively. Heat exchanger fouling and capacity reduction shall not exceed that allowed by fouling factor 0.0005.

d. Pot Feeder: Bypass type, complete with necessary shutoff valves, drain and air release valves, and system connections, for introducing chemicals into system, cast-iron or steel tank with funnel or large opening on top for easy chemical addition. Feeders shall be 5 gallon minimum capacity at 125 psig minimum working pressure.

e. Side stream Water Filter for Closed Loop Systems: Stainless-steel housing, and polypropylene filter media with stainless-steel core. Filter media shall be compatible with antifreeze and water treatment chemicals used in the system. Replaceable filter cartridges for sediment removal service with minimum 20 micrometer particulate at 98 percent efficiency for approximately 5 percent of system design flow rate. Filter cartridge shall have a maximum pressure drop of 2 psig at design flow rate when clean, and maximum pressure drop of 25 psig when dirty. A constant flow rate valve shall be provided in the piping to the filter. Inlet and outlet pressure gauges shall be provided to monitor filter condition.

2.3 CHEMICAL TREATMENT FOR OPEN LOOP SYSTEM(S)

a. General:

(1) A factory-fabricated and tested packaged, self-contained, chemical feed/blow-down monitoring, controlling, recording and alarming system, containing all except specified or indicated remote components, and requiring only terminal sample stream and chemical piping/tubing connections, remote component electrical connection and power supply.

(2) System shall be suitable for a broad spectrum make-up water supply and chemical treatment program. Components, except those specified or indicated otherwise, shall be housed in one or more joined or divided steel enclosures.

(3) System shall be integrated into the campus Direct-Digital Control System for HVAC for trending and alarms. Refer to Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

b. System Functions:

(1) Automatically maintain a predetermined, selectable, total dissolved solids concentration through a continuously monitoring conductivity controller, maintain a predetermined, selectable,

scale/corrosion inhibitor and dispersant residual, through a continuously make-up monitoring meter/counter/timer and inhibitor/dispersant ratio controller; achieve a predetermined, selectable, peak concentration of 1 or 2 microbiocides as needed on an alternating basis, through a programmable timer controller. De-energize controller or stagger feed chemicals that would degrade or could be incompatible if fed simultaneously.

(2) Automatically maintain a predetermined, selectable, pH level through a continuously monitoring pH controller. For systems with make-up water alkalinity in excess of 125 PPM or hardness above 300 PPM, provide acid feed limit timer and audible/visual alarm actuated on low pH.

(3) Monitoring and Data Collection:

a. The water treatment system shall monitor the following conditions and report to the building control system:

1) Biocide levels.

2) Water chemistry including pH and conductivity.

3) Corrosion inhibitor levels.

b. All conditions shall be trended on an hourly basis.

c. Main control panel and accessories:

(1) Housed in a NEMA Type 4X enclosure:

(a) Hinged key lock door with viewing window.

(b) Hard wire connected to power source.

(c) Provide minimum of three (3) 115V, 1 Ph, 60 Hz receptacles located on enclosure for electrical connection and control of chemical pumps.

(d) Prewired for ease of installation.

(2) Provide an external combination mounted flow switch with transparent sight tube.

(a) Disable control outputs upon loss of water flow to prevent chemical feeding.

(b) Provide complete with 3/4 inch connections and combination conductivity and temperature electrode.

(3) Keypad or remote control: Access all measurements and set points through chemical resistant key pad or remote.

(a) Security code to prevent unauthorized access.

(4) Utilize microprocessor technology.

(5) Menu driver programs.

(6) Liquid crystal display (LCD).

(7) Provide temperature corrected measurements by reading water temperature and adjusting conductivity values according to known temperature curve.

(a) Range: (32-212 degrees F with an adjustable high alarm.

(8) Provide real-time clock.

(9) Conductivity monitor:

(a) Provide linear measurements of full range.

(b) Provide 2 scales for selection of high and low in field to assure accurate measurements.

(c) Provide increments of 1 micro ohm/cm with adjustable hysteresis.

(d) Provide bleed-off control in following manner:

1. Standard operation-controller actuates a bleed-off solenoid valve when dissolved solids level is exceeded by trip point.

2. Provide an adjustable bleed limit timer to prevent excessive bleed-off.

3. An alarm contact shall close when timer has timed out.

(10) Biocide operation:

(a) Provide a programmable 28-day biocide timer for accurate addition of algaecide.

(b) Provide a secondary bleed-off timer to lower conductivity in system prior to biocide feed.

(c) Lockout cooling water bleed-off during biocide feed period.

(11) Chemical feed control: Provide 3 timers that are capable of operating in one of following field programmable modes.

(a) Counter-timer-chemical feed proportioned to make-up water rate.

1. Controller shall send low voltage signal to a contacting head water meter.

2. Low voltage signal will ensure long contact life.

3. Water meter shall read in gallons.

(12) UV Biocide Equipment: a. Target Irradiation: Minimum 5,000 micro Wxs/sq. cm. b. Light Source Vessels:

1. ASTM A666, Type 304 stainless-steel.

2. Construct for minimum 150 psig at 150 degrees F in accordance with ASME BPVC Section VIII and equipped with pressure-relief valve.

3. Light Source Sleeve: Quartz, with EPDM O-ring seals.

4. Light Source: Replaceable UV lamp producing minimum target irradiation of 254-nm wavelength light.

c. Controls: Interlock with pumps to operate when water is circulating.

(13) Alarms: Provide alarm LEDs with silence button for high and low conductivity, 10 -60 minute bleed-off, chemical feed limit timers, and chemical drum level. Provide remote output relay to indicate alarm condition to Building Control System specified under Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

(14) Controller Operating Data History:

a. Retain in memory all operating data for following parameters:

1) Standard memory shall allow acquisition and storage of all analog inputs for a 1 week period.

2) A 3-hour minimum, maximum average of all conditions shall be stored for a 1 week period.

3) A minute-by-minute account of operating conditions shall be available for latest 3-hour period.

(15) Electrode: Combination temperature and conductivity type.

a. Quick disconnect.

b. Supplied in flow switch assembly.

(16) pH Monitor:

a. Sensor for monitoring purposes only.

b. Acid shall not be used to control pH.

(17) Remote Communication: Provide open protocol BACnet/IP interface to perform the following functions:

a. Access real-time system values.

b. Change operating parameters.

d. Impulse water meter:

(1) General:

(a) Measure in gallons.

(b) Sized to meter peak make up rates.

(c) Equipped with an electrical contacting register.

(d) Totalize flow at main control panel.

(2) Provide at following locations:

(a) Cooling tower make up line.

(b) Cooling tower bleed off line.

e. Provide CPVC injection nozzles, ASTM F441/F441M with corporation stop to inject chemical into main circulating water line.

(1) Pressure rating: 100 psig

(2) Size: 3/4 inch NPT.

(3) Quantity: 3.

f. Provide chemical feed pumps operated by a 115V, 60 cycle, single PH motor.

(1) Provide separate stroke and stroke frequency setting capabilities.

(2) Positive Displacement Type Pump

(a) Provide with anti-siphon/pressure relief valve installed on pump head which provides anti-siphon protection and aids in priming under pressure.

(b) Capacity: As determined by Water Treatment Vendor.

(c) Complete with discharge check valves, foot valves, polyethylene suction and discharge tubing.

(3) Quantity: Provide 1 pump for each chemical provided.

g. Bleed-off piping assembly:

(1) Inlet shutoff valve.

(2) Wye strainer.

(3) Strainer blowdown valve.

(4) Throttling valve.

(5) Brass solenoid valve compatible with main control panel.

(6) Assembly shall be sized by Water Treatment Vendor.

h. Secondary containment spill pallets for chemical drums:

(1) Material: Polyethylene.

(2) Capacity: 66 gallons each.

(3) Dimensions each: 53 inches length x 29 inches wide X 17 inches high.

(4) Provide each pallet with grating and drain plug.

(5) Provide 1 portable loading ramp.

(6) Quantity: 2.

i. Provide liquid level switch assemblies with a CPVC bung hole adapter, ASTM F441/F441M, to mount directly into 55 gallons chemical drum bung hole.

(1) Interface with main control panel.

(2) Quantity: 3.

j. Corrosion monitor rack:

(1) Materials: Corrosion resistant.

(2) Construction: ASME standards.

(3) Number of coupons: 4.

(4) Coupon holders: quick disconnect type.

k. Provide test kits for monitoring inhibitor levels, total dissolved solids, chlorides, alkalinity and closed system inhibitors.

m. Provide 1 year's supply of chemical treatment including quantity of chemicals necessary to chemically treat system to control scale, and corrosion and biological fouling. Provide water treatment products that perform the following:

(1) Inhibitor to protect against corrosion and scale formation.

(2) 2 liquid biocides for prevention of slime, bacteria and algae.

(3) Chromate based chemical are prohibited.

(4) Water treatment chemicals to remain stable throughout operating temperature range.

(5) Compatible with pump seals and other elements in the systems.

(6) Maintain required pH balance to prevent precipitation and/or breakdown of circulating fluid.

(7) Where analysis justifies addition of pH control, provide alteration of chemical formulation.

n. Chemicals: Except for acid, provide sufficient chemicals for startup and testing and 12 months operation from date of project acceptance.

(1) Scale/corrosion inhibitor: Provide a concentrated liquid organic corrosion/scale/fouling inhibiting formation without phosphates, chromates, zinc and other materials in excess of allowable, local, effluent limits. Feed automatically. Maintain residual as determined by water treatment laboratory.

(2) Dispersant: Provide a concentrated liquid organic/polyelectrolyte formulation. Feed automatically. Maintain residual as determined by water treatment laboratory.

(3) pH Control: Depending upon local water conditions, provide 60 or 66 degree Baume technical grade, concentrated sulfuric acid for acidic treatment or sodium hydroxide (NaOH) for basic treatment to maintain pH in the range of 7.0 to 8.0 automatically. Provide 1 initial 12.5 gallon carboy of acid or base and 1 spare carboy of acid or base, if required.

(4) Microbiocides: Provide 2 different, 1 oxidizing and 1 non-oxidizing, concentrated algaecide-biocide formations containing no heavy metals and which are effective at maximum encountered pH. Alternate solutions as needed to effectuate selective kill without buildup of immunity. Period treatment with a chlorine releasing agent is permissible within allowable, local, effluent limits. Feed automatically. Develop peak concentration and maintain for minimum period as determined by water treatment laboratory.

(5) All chemicals to be acceptable for discharge to sanitary sewer.

o. Water Analysis: Confirm raw water analysis or provide analysis if none is furnished:

1. Description Year (Avg.)

2. Silica (SiO2) _____

3. Iron & Aluminum _____

4. Calcium (Ca) _____

5. Magnesium (Mg) _____

6. Sodium (Na) & Potassium (K) _____

7. Carbonate (C03) _____

8. Bicarbonate (HC03) _____

9. Sulfate (S04) _____

10. Chloride (C1) _____

11. Nitrate (N03) _____

12. Turbidity _____

13. pH _____

14. Residual Chlorine _____

15. Total Alkalinity _____

16. Non Carbonate Hardness _____

17. Total Hardness _____

18. Dissolved Solids _____

p. Conduct performance test to prove capacity and performance of treatment system.

(1) Raw water total hardness, PPM

(2) Concentration cycles

(3) Raw water, pH

(4) System water, pH

(5) Chemical solution used

(6) Acid solution used

(7) Quantity or chemical solution injected into system per cycle

(8) Quantity of acid injected into system per cycle

(9) Make up water required

(10) Waste to drain requirement

q. Centrifugal Solid Separator:

(1) Material: The separator shall be fabricated of carbon steel with shell material and head material of 0.135 inch wall or heavier. Maximum operating pressure shall be 150 psi, unless specified otherwise.

(2) Finish: Paint coating shall be acrylic urethane, sprayon, and royal blue.

(3) Performance: The removal of solids from a pumped/pressurized liquid system shall be accomplished with a centrifugal-action vortex separator. Solids removal efficiency is principally predicated on the difference in specific gravity between the solids and the liquid. Single pass test performance shall be less than 95 percent removal of solids 74 microns and larger. Pressure loss shall be between 5-12 psi.

(4) Purging: Evacuation of separated solids shall be accomplished automatically, employing a timer-activated motorized ball valve. Straight-through valve design, with bronze valve body and stainless-steel ball in a Teflon seat. NEMA 4 housing for indoor and outdoor installation. Valve size: 2 inch.

(5) Provide a differential pressure sensor interface with DDC system.

r. Chemical Treatment System Piping and Valves

(1) Schedule 80 CPVC and fittings. Pipe size shall be 1 inch unless otherwise shown.

(2) Ball Valves: CPVC type.

2.4 EQUIPMENT AND MATERIALS IDENTIFICATION

Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the contractor shall correct the installation at no additional cost or time to the Government.

b. Delivery and Storage: Deliver all chemicals in manufacturer's sealed shipping containers. Store in designated space and protect from deleterious exposure and hazardous spills.

c. Install equipment furnished by the chemical treatment supplier and charge systems according to the manufacturer's instructions and as directed by the Technical Representative.

d. Refer to Section 23 21 13, HYDRONIC PIPING for chemical treatment piping, installed as follows.

e. Before adding cleaning chemical to the closed system, all air handling coils and fan coil units shall be isolated by closing the inlet and outlet valves and opening the bypass valves. This is done to prevent dirt and solids from lodging the coils.

1. Provide a bypass line around water meters and bleed off piping assembly. Provide ball valves to allow for bypassing, isolation, and servicing of components.

2. Bleed off water piping with bleed off piping assembly shall be piped from pressure side of circulating water piping to a convenient drain. Bleed off connection to main circulating water piping shall be upstream of chemical injection nozzles.

3. Provide piping for the flow assembly piping to the main control panel and accessories.

a. The inlet piping shall connect to the discharge side of the circulating water pump.

b. The outlet piping shall connect to the water piping serving the cooling tower downstream of the heat source.

c. Provide inlet Y-strainer and ball valves to isolate and service

4. Install injection nozzles with corporation stops in the water piping serving the cooling tower downstream of the heat source.

5. Provide piping for corrosion monitor rack per manufacturer's installation instructions. Provide ball valves to isolate and service rack.

6. Provide piping for erosion chemical feeder per manufacturer's installation instructions. Provide ball valves to isolate and service

feeder.

7. Provide installation supervision, startup and operating instruction by manufacturer's technical representative.

f. UV-Irradiation Unit Installation: Install UV-Irradiation units on concrete bases level and plumb. Maintain manufacturer's recommended clearances. Arrange units, so controls and devices that require servicing are accessible. Anchor mineral and brine tanks and floor-mounting accessories to substrate.

g. Do not valve in or operate system pumps until after system has been cleaned.

h. After chemical cleaning is satisfactorily completed, open the inlet and outlet valves to each coil and close the bypass valves. Also, clean all strainers.

i. Perform tests and report results in accordance with Section 01 00 00, GENERAL REQUIREMENTS.

j. After cleaning is complete, and water pH is acceptable to manufacturer of water treatment chemical, add manufacturer recommended amount of chemicals to systems.

3.2 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

3.3 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.4 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer's technical representative for 4 hour s to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS

COMMISSIONING OF HVAC SYSTEMS.

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

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SECTION 23 31 00
3/23

HVAC DUCTS AND CASINGS

PART 1 GENERAL

1.1 DESCRIPTION

a. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Ductwork and accessories for HVAC including the following:

(1) Supply air, return air, outside air, exhaust, make-up air, and relief systems.

(2) Exhaust duct with HEPA filters .

(3) Exhaust duct for “wet exhaust” ducts.

c. Definitions:

(1) Seal or Sealing: Use of liquid or mastic sealant, with or without compatible tape overlay, or gasketing of flanged joints, to keep air leakage at duct joints, seams and connections to an acceptable minimum.

(2) Duct Pressure Classification: SMACNA 1922 HVAC Duct Construction Standards, Metal and Flexible.

(3) Exposed Duct: Exposed to view in a finished room, exposed to weather.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 07 84 00, FIRESTOPPING.

e. Section 08 90 00, LOUVERS and VENTS.

f. Section 22 11 00, FACILITY WATER DISTRIBUTION.

g. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

h. Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

i. Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, and BALANCING FOR HVAC.

j. Section 23 07 11, HVAC and BOILER PLANT INSULATION.

k. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

l. Section 23 34 00, HVAC FANS.

m. Section 23 40 00, HVAC AIR CLEANING DEVICES.

n. Section 23 73 00, INDOOR CENTRAL-STATION AIR-HANDLING UNITS.

o. Section 23 82 00, CONVECTION HEATING and COOLING UNITS.

p. Section 23 82 16, AIR COILS.

q. Section 28 31 00, FIRE DETECTION and ALARM.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.Where conflicts occur these specifications and the VHA standards will govern.

ASTM INTERNATIONAL (ASTM)

ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM C1071	(2019) Standard Specification for Fibrous Glass Duct Lining Insulation (Thermal and Sound Absorbing Material)
ASTM E1	(2014) Standard Specification for ASTM Liquid-in-Glass Thermometers
ASTM E77	(2021) Standard Test Method for Inspection and Verification of Thermometers

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A	(2024) Standard for the Installation of Air Conditioning and Ventilating Systems

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1922	(2004) Rectangular Industrial Duct Construction Standards, 2nd Edition
SMACNA 1972 CD	(2012) HVAC Air Duct Leakage Test Manual - 2nd Edition

UL SOLUTIONS (UL)

UL 181	(2013; Reprint Dec 2021) UL Standard for

	Safety Factory-Made Air Ducts and Air Connectors
UL 555	(2006; Reprint Aug 2016) UL Standard for Safety Fire Dampers

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Coordination Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Bio-Based Materials; S, G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 XX XX, SECTION TITLE”, with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Rectangular Ducts:

(a) Schedules of duct systems, materials and selected SMACNA construction alternatives for joints, sealing, gauge and reinforcement.

(b) Duct liner.

(c) Sealants and gaskets.

(d) Access doors.

(2) Round and Flat Oval DuctS:

(a) Manufacturer's details for duct fittings.

(b) Sealants and gaskets.

(c) Access sections.

(d) Installation instructions.

(3) Volume dampers, back draft dampers.

(4) Upper hanger attachments.

(5) Fire dampers, fire doors, and smoke dampers with installation instructions.

(6) Sound attenuators, including pressure drop and acoustic performance.

(7) Flexible ducts and clamps, with manufacturer's installation instructions.

(8) Flexible connections.

(9) Instrument test fittings.

(10) Details and design analysis of alternate or optional duct systems.

d. Coordination Drawings: Refer to paragraph SUBMITTALS in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

e. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation, and maintenance.

1.6 QUALITY ASSURANCE

a. Bio-Based Materials: For products designated by the USDA’s Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

b. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

c. Fire Safety Code: Comply with NFPA 90A.

d. Duct System Construction and Installation: Referenced SMACNA Standards are the minimum acceptable quality.

e. Duct Sealing, Air Leakage Criteria, and Air Leakage Tests: Ducts shall be sealed as per duct sealing requirements of SMACNA 1972 CD HVAC Air Duct Leakage Test Manual for duct pressure classes shown in the contract documents.

f. Duct accessories exposed to the air stream, such as dampers of all types (except smoke dampers) and access openings, shall be of the same material as he duct or provide at least the same level of corrosion resistance.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 2 PRODUCTS

2.1 DUCT MATERIALS AND SEALANTS

a. General: Except for systems specified otherwise, construct ducts, casings, and accessories of galvanized sheet steel, ASTM A653/A653M, coating G90; or, aluminum sheet, ASTM B209, alloy 1100, 3003 or 5052.

b. Specified Corrosion Resistant Systems: Stainless-steel sheet, Class 302 or 304, Condition A (annealed) Finish No. 4 for exposed ducts and Finish No. 2B for concealed duct or ducts located in mechanical rooms.

c. Joint Sealing: Refer to SMACNA 1922 HVAC Duct Construction Standards, Metal and Flexible.

(1) Sealant: Elastomeric compound, gun or brush grade, maximum 25 flame spread, and 50 smoke developed (dry state) compounded specifically for sealing ductwork as recommended by the manufacturer. Generally, provide liquid sealant, with or without compatible tape, for low clearance slip joints and heavy, permanently elastic, mastic type where clearances are larger. Oil base caulking and glazing compounds are prohibited because they do not retain elasticity and bond.

(2) Tape: Use only tape specifically designated by the sealant manufacturer and apply only over wet sealant. Pressure sensitive tape shall not be used on bare metal or on dry sealant.

(3) Gaskets in Flanged Joints: Soft neoprene.

d. Approved factory-made joints may be used.

2.2 DUCT CONSTRUCTION AND INSTALLATION

a. Regardless of the pressure classifications outlined in the SMACNA Standards, fabricate and seal the ductwork in accordance with the following pressure classifications:

b. Duct Pressure Classification:

(1) 0 to 1/2 inch

(2) to 1/2 inch to 1 inch

(3) to 1 inch to 2 inch

(4) to 2 inch to 3 inch

(5) to 3 inch to 4 inch

c. Seal Class: All ductwork shall receive Class A Seal.

d. Wet Air Exhaust Ducts and Accessories: Ducts for dishwashers, scullery hood, cart washers, manual cart washers, cage washers, steam sterilizer hoods and orthotics hoods shall be 18 gauge stainless-steel made liquid tight with continuous external weld for all seams and joints. Provide neoprene gaskets at flanged connections. Where ducts are not self-draining back to the equipment, provide low point drain pocket with copper drain pipe to sanitary sewer. Provide access door in side of duct at drain pockets.

e. Duct for Negative Pressure Up to 3 inch WG: Provide for exhaust duct between HEPA filters and exhaust fan inlet .

(1) Round Duct: Galvanized steel, spiral lock seam construction with standard slip joints.

(2) Rectangular Duct: Galvanized steel, minimum 20 gauge, lock seam, companion angle joints 1-1/4 by 1/8 inch minimum at no greater than 8 feet spacing. Approved pre-manufactured joints are acceptable in lieu of companion angles.

f. Round and Flat Oval Ducts: Furnish double wall duct and fittings made by the same manufacturer to ensure good fit of slip joints. When submitted and approved in advance, round and flat oval duct, with size converted on the basis of equal pressure drop, may be furnished in lieu of rectangular duct design shown in the contract documents.

(1) Elbows: Diameters 3 through 8 inches shall be two sections die stamped, all others shall be gored construction, maximum 18 degree angle, with all seams continuously welded or standing seam. Coat galvanized areas of fittings damaged by welding with corrosion resistant aluminum paint or galvanized repair compound.

(2) Provide bell mouth, conical tees or taps, laterals, reducers, and other low loss fittings as shown in SMACNA 1922 HVAC Duct Construction Standards, Metal and Flexible.

(3) Ribbed Duct Option: Lighter gauge round/oval duct and fittings may be furnished provided certified tests indicating that the rigidity and performance is equivalent to SMACNA standard gauge ducts are submitted.

(a) Ducts: Manufacturer's published standard gauge, G90 coating, spiral lock seam construction with an intermediate standing rib.

(b) Fittings: May be manufacturer's standard as shown in published catalogs, fabricated by spot welding and bonding with neoprene base cement or machine formed seam in lieu of continuous welded seams.

(4) Provide flat side reinforcement of oval ducts as recommended by the manufacturer and SMACNA 1922 HVAC Duct Construction Standards, Metal and Flexble. Because of high pressure loss, do not use internal tie-rod reinforcement unless approved by the Contracting Officer.

g. Casings and Plenums: Construct in accordance with SMACNA 1922 HVAC Duct Construction Standards, Metal and Flexible, including curbs, access doors, pipe penetrations, eliminators and drain pans. Access doors shall be hollow metal, insulated, with latches and door pulls, 20 inches wide by (48 to 54 inches high. Provide view port in the doors where shown. Provide drain for outside air louver plenum. Outside air plenum shall have exterior insulation. Drain piping shall be routed to the nearest floor drain.

h. Volume Dampers: Single blade or opposed blade, multi-louver type as detailed in SMACNA Standards. Refer to SMACNA for Single Blade and Multi-blade Volume Dampers.

i. Duct Hangers and Supports: Refer to SMACNA Standards. Avoid use of trapeze hangers for round duct.

2.3 DUCT LINER (WHERE INDICATED IN THE CONTRACT DOCUMENTS)

a. Duct sizes shown in the contract documents for lined duct are clear opening inside lining.

b. Duct liner is only permitted to be used for return, relief and general exhaust ducts. Duct liner is prohibited for outside air ducts, supply air ducts or any other positive pressure ductwork (provide exterior insulation only).

c. Rectangular Duct or Casing Liner: ASTM C1071, Type I (flexible), or Type II (board), 1 inch minimum thickness, applied with mechanical fasteners and 100 percent coverage of adhesive in conformance with SMACNA.

2.4 DUCT ACCESS DOORS, PANELS AND SECTIONS

a. Provide access doors, sized and located for maintenance work, upstream, in the following locations:

(1) Each duct mounted coil and humidifier.

(2) Each fire damper (for link service), smoke damper and automatic control damper.

(3) Each duct mounted smoke detector.

(4) For cleaning operating room supply air duct and kitchen hood exhaust duct, locate access doors at 20 feet intervals and at each

change in duct direction.

b. Openings shall be as large as feasible in small ducts, by 12 inch by 12 inch minimum where possible. Access sections in insulated ducts shall be double-wall, insulated. Transparent shatterproof covers are preferred for uninsulated ducts.

(1) For Rectangular Ducts: Refer to SMACNA 1922 HVAC Duct Construction Standards, Metal and Flexible.

(2) For Round and Flat Oval Ducts: Refer to SMACNA 1922 HVAC Duct Construction Standards, Metal and Flexible.

2.5 FIRE DAMPERS

a. Galvanized steel, interlocking blade type, UL listing and label, 1-1/2-hour rating, 158 degrees F fusible line, 100 percent free opening with no part of the blade stack or damper frame in the air stream.

b. Fire dampers in wet air exhaust shall be of stainless-steel construction, all others may be galvanized steel.

c. Minimum Requirements for Fire Dampers:

(1) The damper frame may be of design and length as to function as the mounting sleeve, thus eliminating the need for a separate sleeve, as allowed by UL 555. Otherwise provide sleeves and mounting angles, minimum 14 gauge, required to provide installation equivalent to the damper manufacturer's UL test installation.

(2) Submit manufacturer's installation instructions conforming to UL rating test.

2.6 FIRE DOORS

Galvanized steel, interlocking blade type, UL listing and label, 160 degrees F fusible link, 3-hour rating and approved for openings in Class A fire walls with rating up to 4 hours, 100 percent free opening with no part of the blade stack or damper frame in the air stream.

2.7 FLEXIBLE AIR DUCT

a. General: Factory-fabricated, complying with NFPA 90A for connectors not passing through floors of buildings. Flexible ducts shall not penetrate any fire or smoke barrier which is required to have a fire resistance rating of one hour or more. Flexible duct length shall not exceed (5 feet). Provide insulated acoustical air duct connectors in supply air duct systems and elsewhere as shown.

b. Flexible ducts shall be listed by Underwriters Laboratories, Inc., complying with UL 181. Ducts greater than 8 inches in diameter shall be Class 1. Ducts less than 8 inches in diameter shall be Class 1 or Class 2.

c. Insulated Flexible Air Duct: Factory-fabricated including mineral fiber insulation with maximum C factor of 0.25 at 75 degrees F mean temperature, encased with a low permeability moisture barrier outer jacket, having a puncture resistance of not less than 50 Beach Units.

Acoustic insertion loss shall not be less than 3 dB per foot of straight duct, at 500 Hz, based on 6 inch duct, of 2500 fpm.

d. Application Criteria:

(1) Temperature Range: 0 to 200 degrees F internal.

(2) Maximum Working Velocity: 4000 fpm.

(3) Minimum Working Pressure, Inches of Water Gauge: 10 inches WG positive, 2 inches WG negative.

e. Duct Clamps: 100 percent nylon strap, 176 pounds minimum loop tensile strength manufactured for this purpose or stainless-steel strap with cadmium plated worm gear tightening device. Apply clamps with sealant and as approved for UL 181, Class 1 installation.

f. Flexible air duct is prohibited in operating room suites, SPS processing spaces, Pharmacy compounding spaces, Protective Environment spaces, Isolation Rooms, Clean Rooms, Bio-containment Laboratories, Research Laboratories, and Clinical Laboratories.

2.8 FLEXIBLE DUCT CONNECTIONS

Where duct connections are made to fans, air terminal units, and air handling units, install a non-combustible flexible connection of 29 ounces neoprene coated fiberglass fabric approximately 6 inches wide. For connections exposed to sun and weather provide hypalon coating in lieu of neoprene. Burning characteristics shall conform to NFPA 90A. Securely fasten flexible connections to round ducts with stainless-steel or zinc-coated iron draw bands with worm gear fastener. For rectangular connections, crimp fabric to sheet metal and fasten sheet metal to ducts by screws 2 inches on center. Fabric shall not be stressed other than by air pressure. Allow at least one inch slack to ensure that no vibration is transmitted.

2.9 SOUND ATTENUATING UNITS

a. Casing, not less than 20 gauge galvanized sheet steel, or18 gauge aluminum fitted with suitable flanges to make clean airtight connections to ductwork. Sound-absorbent material faced with glass fiber cloth and covered with not less than 24 gauge or heavier galvanized perforated sheet steel, or 22 gauge or heavier perforated aluminum. Perforations shall not exceed 5/32 inch diameter, approximately 25 percent free area. Sound absorbent material shall be long glass fiber acoustic blanket meeting requirements of NFPA 90A.

b. Entire unit shall be completely air tight and free of vibration and buckling at internal static pressures up to (8 inches WG) at operating velocities.

c. Pressure drop through each unit: Not to exceed indicated value at design air quantities indicated.

d. Submit complete independent laboratory test data showing pressure drop and acoustical performance.

e. Cap open ends of attenuators at factory with plastic, heavy duty paper, cardboard, or other appropriate material to prevent entrance of

dirt, water, or any other foreign matter to inside of attenuator. Caps shall not be removed until attenuator is installed in duct system.

2.10 PREFABRICATED ROOF CURBS

Galvanized steel or extruded aluminum 12 inches above finish roof service, continuous welded corner seams, treated wood nailer, 1-1/2 inch thick, 3 pound/cubic feet density rigid mineral fiberboard insulation with metal liner, built-in cant strip (except for gypsum or tectum decks). For surface insulated roof deck, provide raised cant strip (recessed mounting flange) to start at the upper surface of the insulation. Curbs shall be constructed for pitched roof or ridge mounting as required to keep top of curb level.

2.11 FIRESTOPPING MATERIAL

Refer to Section 07 84 00, FIRESTOPPING.

2.12 SEISMIC RESTRAINT FOR DUCTWORK

Not used.

2.13 DUCT MOUNTED THERMOMETER (AIR)

a. Stem Type Thermometers: ASTM E1, 7 inch scale, red appearing mercury, lens front tube, cast aluminum case with enamel finish and clear glass or polycarbonate window, brass stem, 2 percent of scale accuracy to ASTM E77 scale calibrated in degrees Fahrenheit.

b. Thermometer Supports:

(1) Socket: Brass separable sockets for thermometer stems with or without extensions as required, and with cap and chain.

(2) Flange: 3 inch outside diameter reversible flange, designed to fasten to sheet metal air ducts, with brass perforated stem.

2.14 DUCT MOUNTED TEMPERATURE SENSOR (AIR)

Refer to Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

2.15 INSTRUMENT TEST FITTINGS

a. Manufactured type with a minimum 2 inch length for insulated duct, and a minimum 1 inch length for duct not insulated. Test hole shall have a flat gasket for rectangular ducts and a concave gasket for round ducts at the base, and a screw cap to prevent air leakage.

b. Provide instrument test holes at each duct or casing mounted temperature sensor or transmitter, and at entering and leaving side of each heating coil, cooling coil, and heat recovery unit.

2.16 AIR FLOW CONTROL VALVES (AFCV)

Refer to Section 23 82 00, CONVECTION HEATING and COOLING UNITS.

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the contractor shall correct the installation at no additional cost or time to the Government.

b. Comply with provisions of Section 23 05 11, COMMON WORK RESULTS FOR HVAC, particularly regarding coordination with other trades and work in existing buildings.

c. Fabricate and install ductwork and accessories in accordance with referenced SMACNA Standards:

(1) Contract documents show the general layout of ductwork and accessories but do not show all required fittings and offsets that may be necessary to connect ducts to equipment, boxes, diffusers, grilles, etc., and to coordinate with other trades. Fabricate ductwork based on field measurements. Provide all necessary fittings and offsets at no additional cost to the Government. Coordinate with other trades for space available and relative location of HVAC equipment and accessories on ceiling grid. Duct sizes in the contract documents are inside dimensions which shall be altered by Contractor to other dimensions with the same air handling characteristics where necessary to avoid interferences and clearance difficulties.

(2) Provide duct transitions, offsets and connections to dampers, coils, and other equipment in accordance with SMACNA Standards. Provide streamliner, when an obstruction cannot be avoided and must be taken in by a duct. Repair galvanized areas with galvanizing repair compound.

(3) Provide bolted construction and tie-rod reinforcement in accordance with SMACNA Standards.

(4) Construct casings, eliminators, and pipe penetrations in accordance with SMACNA Standards. Design casing access doors to swing against air pressure so that pressure helps to maintain a tight seal.

d. Install duct hangers and supports in accordance with SMACNA Standards.

e. Install fire dampers and smoke dampers in accordance with the manufacturer's instructions to conform to the installation used for the rating test. Install fire dampers and smoke dampers at locations indicated and where ducts penetrate fire rated and/or smoke rated walls, shafts and where required by the Contracting Officer. Install with required perimeter mounting angles, sleeves, breakaway duct connections, corrosion resistant springs, bearings, bushings and hinges per UL and NFPA. Demonstrate resetting of fire dampers and operation of smoke dampers to the Contracting Officer.

f. Seal openings around duct penetrations of floors and fire rated partitions with fire stop material as required by NFPA 90A.

g. Flexible Duct Installation: Refer to SMACNA Standards. Ducts shall be continuous, single pieces not over 5 feet long per NFPA 90A, as

straight and short as feasible, adequately supported. Centerline radius of bends shall be not less than two duct diameters. Make connections with clamps as recommended by SMACNA. Clamp per SMACNA with one clamp on the core duct and one on the insulation jacket. Flexible ducts shall not penetrate floors, or any chase or partition designated as a fire or smoke barrier, including corridor partitions fire rated one hour or two hours. Support ducts SMACNA Standards.

h. Where diffusers, registers and grilles cannot be installed to avoid seeing inside the duct, paint the inside of the duct with flat black paint to reduce visibility.

i. Control Damper Installation:

(1) Provide necessary blank-off plates required to install dampers that are smaller than duct size. Provide necessary transitions required to install dampers larger than duct size.

(2) Assemble multiple sections dampers with required interconnecting linkage and extend required number of shafts through duct for external mounting of damper motors.

(3) Provide necessary sheet metal baffle plates to eliminate stratification and provide air volumes specified. Locate baffles by experimentation, and affix and seal permanently in place, only after stratification problem has been eliminated.

(4) Install all damper control/adjustment devices on stand-offs to allow complete coverage of insulation.

j. Air Flow Measuring Devices: Install units with minimum straight run distances, upstream and downstream as recommended by the manufacturer.

k. Low Pressure Duct Liner: Install in accordance with SMACNA Standards.

l. Protection and Cleaning: Adequately protect equipment and materials against physical damage. Place equipment in first class operating condition or return to source of supply for repair or replacement, as determined by Contracting Officer. Protect equipment and ducts during construction against entry of foreign matter to the inside and clean both inside and outside before operation and painting. When new ducts are connected to existing ductwork, clean both new and existing ductwork by mopping and vacuum cleaning inside and outside before operation.

3.2 DUCT LEAKAGE TESTS AND REPAIR

a. Ductwork leakage testing shall be performed by the Testing and Balancing Contractor directly contracted by the General Contractor and independent of the Sheet Metal Contractor.

b. Ductwork leakage testing shall be performed for the entire air distribution system (including all supply, return, exhaust and relief ductwork), section by section, including fans, coils and filter sections. Based upon satisfactory initial duct leakage test results, the scope of the testing may be reduced by the Contracting Officer on ductwork constructed to the 2 inches WG duct pressure classification. In no case shall the leakage testing of ductwork constructed above the 2 inches WG) duct pressure classification or

ductwork located in shafts or other inaccessible areas be eliminated.

c. Test procedure, apparatus and report shall conform to SMACNA 1972 CD HVAC Air Duck Leakage Test Manual.

d. All ductwork shall be leak tested first before enclosed in a shaft or covered in other inaccessible areas.

e. All tests shall be performed in the presence of the Contracting Officer and the Test and Balance agency. The Test and Balance agency shall measure and record duct leakage and report to the Contracting Officer and identify leakage source with excessive leakage.

f. If any portion of the duct system tested fails to meet the permissible leakage level, the Contractor shall rectify sealing of ductwork to bring it into compliance and shall retest it until acceptable leakage is demonstrated to the Contracting Officer.

g. All tests and necessary repairs shall be completed prior to insulation or concealment of ductwork.

h. Make sure all openings used for testing flow and temperatures by TAB Contractor are sealed properly.

3.3 DUCTWORK EXPOSED TO WIND VELOCITY

Provide additional support and bracing to all exposed ductwork installed on the roof or outside the building to withstand wind velocity of (90mph).

3.4 TESTING, ADJUSTING, AND BALANCING (TAB)

Refer to Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, and BALANCING FOR HVAC.

3.5 OPERATING AND PERFORMANCE TESTS

Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

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SECTION 23 34 00
3/23

HVAC FANS

PART 1 GENERAL

1.1 DESCRIPTION

a. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Fans for heating, ventilating and air conditioning.

c. Product Definitions: AMCA 99.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Not used.

f. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

g. Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC and STEAM GENERATION EQUIPMENT.

h. Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

i. Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, and BALANCING FOR HVAC.

j. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

k. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

l. Section 23 73 00, INDOOR CENTRAL-STATION AIR-HANDLING UNITS.

m. Section 23 82 16, AIR COILS.

n. Section 26 29 11, MOTOR CONTROLLERS.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standards will govern.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 99	(2016) Standards Handbook
AMCA 210	(2016) Laboratory Methods of Testing Fans for Aerodynamic Performance Rating
AMCA 300	(2014) Reverberant Room Method for Sound Testing of Fans

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 51	(2016) Laboratory Methods of Testing Fans for Aerodynamic Performance Rating

ASTM INTERNATIONAL (ASTM)

ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM D1735	(2014) Standard Practice for Testing Water Resistance of Coatings Using Water Fog Apparatus
ASTM D3359	(2017) Standard Test Methods for Rating Adhesion by Tape Test
ASTM G152	(2013; R 2021) Standard Practice for Operating Open Flame Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials
ASTM G153	(2013; R 2021) Standard Practice for Operating Enclosed Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 96	(2024) Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturers Literature And Data; G, AE

Bio-Based Materials; S, G, AE

SD-05 Design Data

Motor Ratings, Types, Electrical Characteristics, And Accessories; G, AE

SD-06 Test Reports

Certified Sound Power Levels; G, AE

Certified Fan Performance Curves; G, AE

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 XX XX, SECTION TITLE”, with applicable paragraph identification.

c. Manufacturers Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Plenum fan, motors, and drives.

(2) Centrifugal fans, motors, drives, accessories, and coatings.

(a) Inline centrifugal fans.

(b) Tubular Centrifugal Fans.

(c) Up-blast kitchen hood exhaust fans.

(d) Industrial fans.

(e) Utility fans and vent sets.

(3) Power roof and wall ventilators.

(4) Centrifugal ceiling fans.

(5) Propeller fans.

(6) Packaged hood make-up air units.

(7) Vane axial fans.

(8) Tube-axial fans.

(9) Air curtain units.

(10) Belt guards.

d. Certified sound power levels for each fan.

e. Motor ratings, types, electrical characteristics, and accessories.

f. Certified fan performance curves for each fan showing cubic feet per minute (CFM) versus static pressure, efficiency, and horsepower for design point of operation.

g. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation, and maintenance.

h. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician, and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

i. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.6 QUALITY ASSURANCE

a. Fans and power ventilators shall bear the AMCA performance seal.

b. Operating Limits for Centrifugal Fans: AMCA 99 (Class I, II, and III).

c. Fans and power ventilators shall comply with the following standards:

(1) Testing and Rating: AMCA 210.

(2) Sound Rating: AMCA 300.

d. Vibration Tolerance for Fans and Power Ventilators: Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

e. Performance Criteria:

(1) The fan schedule shall show the design air volume and static pressure. Select the fan motor hp by increasing the fan bhp by 10 percent to account for the drive losses and field conditions.

(2) Select the fan operating point as follows:

(a) Forward Curve and Axial Flow Fans: Right hand side of peak pressure point.

(b) Air Foil, Backward Inclined, or Tubular: At or near the peak static efficiency.

f. Safety Criteria: Provide manufacturer's standard screen on fan inlet and discharge where exposed to operating and maintenance personnel.

g. Corrosion Protection:

(1) Except for fans in fume hood exhaust service, all steel shall be mill-galvanized, or phosphatized and coated with minimum two coats, corrosion resistant enamel paint. Manufacturer’s paint and paint system shall meet the minimum requirements of ASTM D1735 water fog, ASTM B117 salt spray, ASTM D3359 adhesion, and ASTM G152 and ASTM G153 for carbon arc light apparatus for exposure of non-metallic material.

(2) Fans for general purpose fume hoods, or chemical hoods, and radioisotope hoods shall be constructed of materials compatible with the chemicals being transported in the air through the fan.

h. Spark resistant construction: If flammable gas, vapor or combustible dust is present in concentrations above 20 percent of the Lower Explosive Limit (LEL), the fan construction shall be as recommended by AMCA 99 Classification for Spark Resistant Construction. Drive set shall be comprised of non-static belts for use in an explosive.

i. Bio-Based Materials: For products designated by the USDA’s Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

j. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

1.8 EXTRA MATERIALS

Provide one additional set of belts for all belt-driven fans.

PART 2 PRODUCTS

2.1 PLENUM FANS

a. Plenum Fans – Single and/or Multiple Fans in an Array:

(1) General: Fans shall be Class II (minimum) construction with single inlet, aluminum wheel and stamped air-foil aluminum bladed. The fan wheel shall be mounted on the directly-driven

motor shaft in AMCA 99 Arrangement 4. Fans shall be dynamically balanced and internally isolated to minimize the vibrations. Provide a steel inlet cone for each wheel to match with the fan inlet. Locate fan in the air stream to assure proper flow. The fan performance shall be rated in accordance with AMCA 210 or ASHRAE 51.

(2) Allowable vibration tolerances for fan shall not exceed a self-excited vibration maximum velocity of 0.20 inch per second) RMS, filter in, when measured with a vibration meter on bearing caps of machine in vertical, horizontal and axial directions or measured at equipment mounting feet if bearings are concealed. After field installation, compliance to this requirement shall be demonstrated with field test in accordance with Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT and Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, AND BALANCING FOR HVAC. Following fan assembly, the complete fan assembly balance shall be tested using an electronic balance analyzer with a tunable filter and stroboscope. Vibration measurements shall be taken on each motor bearing housing in the vertical, horizontal, and axial planes (5 total measurements, 2 each motor bearing and 1 axial).

(3) The plenum fans shall be driven by variable speed drives with at least one back-up drive as shown in the design documents. Use of a drive with bypass is not permitted.

(4) Multiple fans shall be installed in a pre-engineered structural frame to facilitate fan stacking. All fans shall modulate in unison, above or below the synchronous speed within the limits specified by the manufacturer, by a common control sequence. Staging of the fans is not permitted. Redundancy requirement shall be met by all operating fans in an array and without the provision of an idle standby fan.

(5) Fan Accessories

(a) Fan Isolation: Provide a manual blank off plate to isolate the fan not in operation due to failure.

(b) Fan Airflow Measurement: Provide an airflow measuring device integral to the fan to measure air volume within +/- 5 percent accuracy. The probing device shall not be placed in the airflow path to stay clear of turbulence and avoid loss of performance.

2.2 CENTRIFUGAL FANS

a. Record factory vibration test results on the fan or furnish to the Contractor.

b. Fan arrangement, unless noted or approved otherwise:

(1) DWDl fans: Arrangement 3.

(2) SWSl fans: Arrangement 1, 3, 9 or 10.

c. Construction: Wheel diameters and outlet areas shall be in accordance with AMCA standards.

(1) Housing: Low carbon steel, arc welded throughout, braced and supported by structural channel or angle iron to prevent vibration or pulsation, flanged outlet, inlet fully streamlined. Provide lifting clips, and casing drain. Provide manufacturer's standard access door. Provide 1/2 inches wire mesh screens for fan inlets without duct connections.

(2) Wheel: Steel plate with die formed blades welded or riveted in place, factory balanced statically and dynamically.

(3) Shaft: Designed to operate at no more than 70 percent of the first critical speed at the top of the speed range of the fans class.

(4) Bearings: Heavy duty ball or roller type sized to produce a B10 life of not less than 50,000 hours, and an average fatigue life of 200,000 hours. Extend filled lubrication tubes for interior bearings or ducted units to outside of housing.

(5) Belts: Oil resistant, non-sparking and non-static.

(6) Belt Drives: Factory installed with final alignment belt adjustment made after installation.

(7) Motors and Fan Wheel Pulleys: Adjustable pitch for use with motors through 15 hp, fixed pitch for use with motors greater than 15 hp. Select pulleys so that pitch adjustment is at the middle of the adjustment range at fan design conditions.

(8) Motor, adjustable motor base, drive and guard: Furnish from factory with fan. Provide protective sheet metal enclosure for fans located outdoors. Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC for specifications.

(9) Furnish variable speed fan motor controllers where shown in the contract documents. Refer to Section 26 29 11, MOTOR CONTROLLERS. Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC for controller/motor combination requirements.

d. Inline Centrifugal Fans: Provide minimum 18 Gauge galvanized steel housing with inlet and outlet flanges, backward inclined aluminum centrifugal fan wheel, bolted access door and supports as required. Motors shall be factory pre-wired to an external junction box. Provide factory wired disconnect switch.

e. Tubular Centrifugal Fans: Provide hot rolled steel, one-piece design housing incorporating integral guide vanes, motor mounts, bolted access hatch and end flanges. Provide spun inlet bell and screen for unducted inlet and screen for unducted outlet. Provide welded steel, flanged inlet and outlet cones for ducted connection. Provide mounting legs or suspension brackets as required for support. Guide vanes shall straighten the discharge air pattern to provide linear flow.

f. Industrial Fans: Use where scheduled or in lieu of centrifugal fans for low volume high static service. Provide material handling flat blade type fan wheel.

g. Utility Fans, Vent Sets and Small Capacity Fans: Class l design, arc

welded housing, spun intake cone. Requirement for AMCA seal is waived for wheel diameters less than l2 inches and housings may be cast-iron.

h. Spark Resistant/Explosion Proof Fans: If flammable gas, vapor, or combustible dust is present in concentrations above 20 percent of the Lower Explosive Limit (LEL), provide AMCA construction option: A, B or C as indicated. Drive set shall be comprised of non-static belts for use in an explosive atmosphere. Motor shall be explosion proof type if located in air stream.

2.3 POWER ROOF VENTILATOR

a. Type: Centrifugal fan, backward inclined blades. Provide down-blast or up-blast type as indicated.

b. Construction: Steel or aluminum, completely weatherproof, for curb mounting, exhaust cowl or entire drive assembly readily removable for servicing, aluminum bird screen on discharge, UL approved safety disconnect switch, conduit for wiring, vibration isolators for wheel, motor and drive assembly. Provide self-acting back draft damper. Provide electric motor operated damper where indicated.

c. Motor and Drive: Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC. Bearings shall be pillow block ball type with a minimum L-50 life of 200,000 hours. Motor shall be located out of air stream.

d. Up-blast Type: Top discharge exhauster, motor out of air stream. For kitchen hood exhaust applications, provide grease trough on base and threaded drain. The mounting height of the kitchen up-blast exhaust fan shall be in compliance with NFPA 96. (Provide vented curb extension if required to maintain required clearances.)

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the contractor shall correct the installation at no additional cost or time to the Government.

b. Install fan, motor and drive in accordance with manufacturer's instructions.

c. Align fan and motor sheaves to allow belts to run true and straight.

d. Bolt equipment to curbs with galvanized lag bolts.

e. Install vibration control devices as shown in the contract documents and specified in Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

3.2 PRE-OPERATION MAINTENANCE

a. Lubricate bearings, pulleys, belts and other moving parts with manufacturer recommended lubricants.

b. Rotate impeller by hand and check for shifting during shipment and check all bolts, collars, and other parts for tightness.

c. Clean fan interiors to remove foreign material and construction dirt and dust.

3.3 STARTUP AND TESTING

a. Verify operation of motor, drive system and fan wheel according to the contract documents.

b. Check vibration and correct as necessary for air balance work.

c. After air balancing is complete and permanent sheaves are in place perform necessary field mechanical balancing to meet vibration tolerance in Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

d. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

e. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

f. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

3.4 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.5 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hour s to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01,DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 37 00

AIR OUTLETS AND INLETS

3/23

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SECTION 23 37 00
3/23

AIR OUTLETS AND INLETS

PART 1 GENERAL

1.1 DESCRIPTION

a. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Air Outlets and Inlets: Diffusers, registers, and grilles.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 08 90 00, LOUVERS and VENTS.

e. Not used.

f. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

g. Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

h. Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, and BALANCING FOR HVAC.

1.3 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standards will govern.

AIR DUCT COUNCIL (ADC)

ADC Standards Manual	(2008; 5th Edition) Flexible Duct Performance Installation Standards

ASTM INTERNATIONAL (ASTM)

ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 90A	(2024) Standard for the Installation of Air Conditioning and Ventilating Systems

UL SOLUTIONS (UL)

UL 181	(2013; Reprint Dec 2021) UL Standard for Safety Factory-Made Air Ducts and Air Connectors

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Bio-Based Materials; S, G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 XX XX, SECTION TITLE”, with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Air intake/exhaust hoods.

(2) Diffusers, registers, grilles, and accessories.

d. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation, and maintenance.

1.6 QUALITY ASSURANCE

a. Fire Safety Code: Comply with NFPA 90A.

b. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

c. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 2 PRODUCTS

2.1 GRAVITY INTAKE/EXHAUST VENTILATORS (ROOF MOUNTED)

a. Aluminum, ASTM B209/B209M, louvered, spun, or fabricated using panel sections with roll-formed edges, 1/2 inch mesh aluminum welded wire bird screen, with gravity or motorized dampers where shown, accessible interior, designed for wind velocity specified in paragraph Intake/Exhaust Hoods Exposed to Wind Velocity in this section.

(1) Spun Intake/Exhaust Ventilators: Spun aluminum structural components shall be constructed of minimum 16 Gauge marine alloy aluminum, bolted to a rigid aluminum support structure. The aluminum base shall have continuously welded curb cap corners for maximum leak protection. The spun aluminum baffle shall have a rolled bead for added strength.

(2) Louvered Intake/Exhaust Hoods: Louvered hood constructed from 0.003 inch extruded aluminum tiers welded to a minimum 8 Gauge aluminum support structure. The aluminum hood shall be constructed of a minimum 0.025 inch marine alloy aluminum and provided with a layer of anti-condensate coating. The aluminum base shall have continuously welded curb cap corners for maximum leak protection.

(3) Low Silhouette Intake/Exhaust Ventilator: The unit shall be of bolted and welded construction utilizing corrosion resistant fasteners. The aluminum hood shall be constructed of minimum 14 Gauge marine alloy aluminum, bolted to a minimum 8 Gauge aluminum support structure. The aluminum base shall have continuously welded curb cap corners for maximum leak protection. Birdscreen constructed of 1/2 inch mesh shall be mounted across the relief opening.

b. See ventilator schedule on the contract drawings. Sizes shown on the contract drawings designate throat size. Area of ventilator perimeter

opening shall be not less than the throat area.

c. Dampers for Gravity Ventilators without Duct Connection: Construct damper of the same material as the ventilator and of the design to completely close opening or remain wide open. Hold damper in closed position by a brass chain and catch. Extend chains 12 inches below and engage catch when damper is closed.

d. See paragraph Intake/Exhaust Hoods Exposed to Wind Velocity in this section for exposure to high wind velocities.

e. Provide roof curb by unit manufacturer. Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC for additional requirements.

2.2 EQUIPMENT SUPPORTS

Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

2.3 AIR OUTLETS AND INLETS

a. Materials:

(1) Steel or aluminum . Provide manufacturer's standard gasket.

(2) Exposed Fastenings: The same material as the respective inlet or outlet. Fasteners for aluminum may be stainless-steel.

(3) Contractor shall review all ceiling drawings and details and provide all ceiling mounted devices with appropriate dimensions and trim for the specific locations.

b. Performance Test Data: In accordance with Air Duct Council's ADC Standards Manual “Flexible Duct Performance & Installation Standards (Manual), Green Book”. Refer to Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT for NC criteria.

c. Air Supply Outlets:

(1) Ceiling Diffusers: Suitable for surface mounting, exposed T-bar or special tile ceilings, off-white finish, square or round neck connection as shown on the contract drawings. Provide plaster frame for units in plaster ceilings.

(a) Square, Louver, Fully Adjustable Pattern: Round neck, surface mounting unless shown otherwise on the contract drawings. Provide equalizing or control grid and volume control damper.

(b) Louver Face Type: Square or rectangular, removable core for 1, 2, 3, or 4 way directional pattern. Provide equalizing or control grid and opposed blade damper.

(c) Perforated Face Type: Manual adjustment for one-, two-, three-, or four-way horizontal air distribution pattern without change of air volume or pressure. Provide equalizing or control grid and opposed blade over overlapping blade damper. Perforated face diffusers for VAV systems shall have the pattern controller on the inner face, rather than in the neck and designed to discharge air horizontally at the ceiling maintaining a Coanda effect.

(d) Slot diffuser/plenum:

1. Diffuser: Frame and support bars shall be constructed of heavy gauge extruded aluminum. Form slots or use adjustable pattern controllers, to provide stable, horizontal air flow pattern over a wide range of operating conditions.

2. Galvanized steel boot lined with 1/2 inch thick fiberglass conforming to NFPA 90A and complying with UL 181 for erosion. The internal lining shall be factory-fabricated, anti-microbial, and non-friable.

3. Provide inlet connection diameter equal to duct diameter shown on the contract drawings or provide transition coupling if necessary. Inlet duct and plenum size shall be as recommended by the manufacturer.

4. Maximum pressure drop at design flow rate: 0.15 inch WG)

(f) Linear Bar Grilles and Diffusers: Extruded aluminum, manufacturer's standard finish, and positive holding concealed fasteners.

(1) Margin Frame: Flat, 3/4 inch wide.

(2) Bars: Minimum 3/16 inch wide by 3/4 inch deep, zero deflection unless otherwise shown. Bar spacing shall be a minimum of 1/8 inch on center. Reinforce bars on 18 inch center for sidewall units and on 6 inch center for units installed in floor or sills.

(3) Provide opposed blade damper and equalizing or control grid where shown.

(3) Supply Registers: Double deflection type with horizontal face bars and opposed blade damper with removable key operator.

(a) Margin: Flat, 1-1/4 inches wide.

(b) Bar Spacing: 3/4 inch maximum.

(c) Finish: Off white baked enamel for ceiling mounted units. Wall units shall have a prime coat for field painting or shall be extruded with manufacturer's standard finish.

(4) Supply Grilles: Same as registers but without the opposed blade damper.

d. Return and Exhaust Registers and Grilles: Provide opposed blade damper without removable key operator for registers.

(1) Finish: Off-white baked enamel for ceiling mounted units. Wall units shall have a prime coat for field painting or shall be

extruded aluminum with manufacturer's standard aluminum finish.

(2) Standard Type: Fixed horizontal face bars set at 30 to 45 degrees, approximately 1-1/4 inch margin.

(3) Perforated Face Type: To match supply units.

(4) Grid Core Type: 1/2 inch by 1/2 inch core with 1-1/4 inch margin.

(5) Linear Type: To match supply units.

(6) Door Grilles: Are furnished with the doors.

(7) Egg Crate Grilles: Aluminum or Painted Steel 1/2 inch by 1/2 inch by 1/2 inch grid providing 90 percent free area.

(a) Heavy extruded aluminum frame shall have countersunk screw mounting. Unless otherwise indicated, register blades and frame shall have factory applied white finish.

(b) Grille shall be suitable for duct or surface mounting as indicated on the contract drawings. All necessary appurtenances shall be provided to allow for mounting.

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the contractor shall correct the installation at no additional cost or time to the Government.

b. Comply with provisions of Section 23 05 11, COMMON WORK RESULTS FOR HVAC, particularly regarding coordination with other trades and work in existing buildings.

c. Protection and Cleaning: Protect equipment and materials against physical damage. Place equipment in first class operating condition or return to source of supply for repair or replacement, as determined by Contracting Officer. Protect equipment during construction against entry of foreign matter to the inside and clean both inside and outside before operation and painting.

3.2 INTAKE/EXHAUST HOODS EXPOSED TO WIND VELOCITY

Provide additional support and bracing to all exposed ductwork installed on the roof or outside the building to withstand wind velocity of 90 mph or, in coastal areas.

3.3 TESTING, ADJUSTING, AND BALANCING (TAB)

Refer to Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, and BALANCING FOR HVAC.

3.4 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

-- End of Section --

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 40 00

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3.2 STARTUP AND TESTING

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SECTION 23 40 00
3/23

HVAC AIR CLEANING DEVICES

PART 1 GENERAL

1.1 DESCRIPTION

a. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Air filters for heating, ventilating and air conditioning.

c. Definitions: Refer to ASHRAE 52.2 for definitions of face velocity, net effective filtering area, media velocity, initial resistance (pressure drop), MERV (Minimum Efficiency Reporting Value), PSE (Particle Size Efficiency), particle size ranges for each MERV number, dust holding capacity and explanation of electrostatic media based filtration products versus mechanical filtration products. Refer to ASHRAE 52.2 for definition of MERV-A.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Not used.

e. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

f. Section 23 37 00, AIR OUTLETS AND INLETS.

g. Section 23 73 00, INDOOR CENTRAL-STATION AIR-HANDLING UNITS

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by basic designation only. Where conflicts occur these specifications and the VHA standards will govern.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 52.2	(2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC SEC IX	(2017; Errata 2018) BPVC Section

IX-Welding, Brazing and Fusing Qualifications

AMERICAN WELDING SOCIETY (AWS)

AWS D9.1/D9.1M (2018) Sheet Metal Welding Code

INSTITUTE OF ENVIRONMENTAL SCIENCES AND TECHNOLOGY (IEST)

IEST RP-CC-001 (2016; Rev 6) HEPA and ULPA Filters

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 9001 (2015) Quality Management Systems-Requirements

UL SOLUTIONS (UL)

UL 586 (2009; Reprint Sep 2022) UL Standard for Safety High-Efficiency Particulate, Air Filter Units

UL 900 (2015; Reprint Aug 2022) UL Standard for SafetyStandard for Air Filter Units

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Bio-Based Materials; S, G, AE

SD-06 Test Reports

Air Filter Performance Reports; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

Suppliers Warranty; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 XX XX, SECTION TITLE”, with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Extended surface filters.

(2) Holding frames. Identify locations.

(3) Side access housings. Identify locations, verify insulated doors.

(4) HEPA filters.

(5) Magnehelic gauges.

(6) Carbon Filters.

d. Air Filter performance reports.

e. Suppliers warranty.

f. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation, and maintenance.

1.6 QUALITY ASSURANCE

a. Air Filter Performance Report for Extended Surface Filters:

(1) Submit a test report for each Grade of filter being offered. The report shall not be more than three (3) years old and prepared by using test equipment, method and duct section as specified by ASHRAE 52.2 for type filter under test and acceptable to Contracting Officer, indicating that filters comply with the requirements of this specification. Filters utilizing partial or complete synthetic media will be tested in compliance with pre-conditioning steps as stated in ASHRAE 52.2. All testing is to be conducted on filters with a nominal by 24 inch by 24 inch face dimension. Test for 500 fpm will be accepted for lower velocity rated filters provided the test report of an independent testing laboratory complies with all the requirements of this specification.

(3) Guarantee Performance: The manufacturer shall supply ASHRAE 52.2 test reports on each filter type submitted. Any filter supplied will be required to maintain the minimum efficiency shown on the ASHRAE 52.2 report throughout the time the filter is in service. Within the first 6 to 12 weeks of service a filter may be pulled

out of service and sent to an independent laboratory for ASHRAE 52.2 testing for initial efficiency only. If this filter fails to meet the minimum level of efficiency shown in the previously submitted reports, the filter manufacturer/distributor shall take back all filters and refund the owner all monies paid for the filters, cost of installation, cost of freight and cost of testing.

b. Filter Warranty for Extended Surface Filters: Guarantee the filters against leakage, blow-outs, and other deficiencies during their normal useful life, up to the time that the filter reaches the final pressure drop. Defective filters shall be replaced at no additional cost or time to the Government.

c. Comply with UL 900 for flame test.

d. Nameplates: Each filter shall bear a label or name plate indicating manufacturer's name, filter size, rated efficiency, UL classification .

e. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

f. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 2 PRODUCTS

2.1 REPLACEMENT FILTER ELEMENTS TO BE FURNISHED

a. To allow temporary use of HVAC systems for testing and in accordance with Paragraph, TEMPORARY USE OF MECHANICAL AND ELECTRICAL SYSTEMS in Section 01 00 00, GENERAL REQUIREMENTS. Provide one complete set of additional filters to the Contracting Officer.

b. The Contracting Officer will direct whether these additional filters will either be installed as replacements for dirty units or turned over to VA for future use as replacements.

2.2 EXTENDED SURFACE AIR FILTERS

a. Use factory assembled air filters of the extended surface type with supported or non-supported cartridges for removal of particulate matter in air conditioning, heating and ventilating systems. Filter units shall be of the extended surface type fabricated for disposal when the contaminant load limit is reached as indicated by maximum (final) pressure drop.

b. Filter Classification: UL listed and approved conforming to UL 900.

c. HVAC Filter Types:

HVAC Filter Types				
MERV Value ASHRAE 52.2		Application	Particle Size	Thickness /Type
8		Pre-Filter	3 to 10 Microns	2 inch Throwaway
11		After-Filter	1 to 3 Microns	6 inch or 12 inch Rigid Cartridge
13		(Alternate) After-Filter	0.3 to 1 Microns	6 inch or 12 inch Rigid Cartridge
14		After-Filter	0.3 to 1 Microns	6 inch or 12 inch Rigid Cartridge

d. HEPA Filters:

HEPA Filters				
Efficiency at 0.3 Micron	Application	Initial Resistance (inches WG)	Rated cfm	Construction
99.97	Final Filter	1.35	1100	Galvanized Frame X-Body
99.97	Final Filter	1.00	2000	Aluminum Frame V-Bank

2.3 MEDIUM EFFICIENCY PLEATED PANEL PRE-FILTERS

a. Construction: MEDIUM EFFICIENCY PLEATED PANEL PRE-FILTERS 50 MM 2 INCH; MERV 8; UL 900 CLASS 2 Air filters shall be medium efficiency ASHRAE pleated panels consisting of cotton and synthetic or 100 percent virgin synthetic media, self-supporting media with required media stabilizers, and beverage board enclosing frame. Filter media shall be lofted to a uniform depth and formed into a uniform radial pleat. The media stabilizers shall be bonded to the downstream side of the media to maintain radial pleats and prevent media oscillation. An enclosing frame of no less than 28-point high wet-strength beverage board shall provide a rigid and durable enclosure. The frame shall be bonded to the media on all sides to prevent air bypass. Integral diagonal support members on the air entering and air exiting side shall be bonded to the apex of each pleat to maintain uniform pleat spacing in varying airflows.

b. Performance: The filter shall have a Minimum Efficiency Reporting Value of MERV 8 when evaluated under the guidelines of ASHRAE 52.2. The media shall maintain or increase in efficiency over the life of the filter. All testing is to be conducted on filters with a nominal 24 inch by 24 inch face dimension.

Minimum Efficiency Reporting (MERV)	8

Dust Holding Capacity (Grams)	105
Nominal Size (Inches Width x Height x Depth)	24x24x2
Rated Air Flow Capacity (cfm)	2,000
Rated Air Flow Rate (fpm)	500
Final Resistance (Inches WG) Maximum Recommended Change-Out Resistance (Inches WG)	1.0 0.66
Rated Initial Resistance (Inches WG)	0.33

c. The filters shall be approved and listed by Underwriters Laboratories, Inc. as Class 2 when tested according to UL 900.

2.4 HIGH EFFICIENCY EXTENDED SURFACE (INTERMEDIATE/AFTER (FINAL)) CARTRIDGE FILTERS)

a. HIGH EFFICIENCY EXTENDED SURFACE (INTERMEDIATE/AFTER (FINAL)) CARTRIDGE FILTERS (12 INCH; MERV 14/13/11; UL 900 CLASS 2)Construction: Air filters shall consist of 8 pleated media packs assembled into 4 V-banks within a totally plastic frame. The filters shall be capable of operating at temperatures up to 176 degrees F. The filters shall either fit without modification or be adaptable to the existing holding frames. The molded end panels are to be made of high impact polystyrene plastic. The center support members shall be made of plastic. No metal components are to be used.

b. Media: The media shall be made of micro glass fibers with a water repellent binder. The media shall be a dual density construction, with coarser fibers on the air entering side and finer fibers on the air leaving side. The media shall be pleated using separators made of continuous beads of low profile thermoplastic material. The media packs shall be bonded to the structural support members at all points of contact, this improves the rigidity as well as eliminates potential air bypass in the filter.

c. Performance: Filters of the size, air flow capacity and nominal efficiency (MERV) shall meet the following rated performance specifications based on the ASHRAE 52.2 test method. All testing is to be conducted on filters with a nominal 24 inch by 24 inch header dimension.

Minimum Efficiency Reporting Value (MERV)	14	13	11
Gross Media Area (sf) Dust Holding Capacity (Grams)	197 486	197 430	197 465
Nominal Size (Inches Width x Height x Depth)	24x24x12	24x24x12	24x24x12
Rated Air Flow Capacity (cfm)	2,000	2,000	2,000
Rated Air Flow Rate (fpm)	500	500	500
Final Resistance (Inches WG) Maximum Recommended Change-Out Resistance (Inches WG)	2.0 0.74	2.0 0.68	2.0 0.54

Rated Initial Resistance (Inches WG)	0.37	0.34	0.27

2.5 HIGH EFFICIENCY PARTICULATE AIR (HEPA) FILTERS STANDARD CAPACITY (FINAL FILTER APPLICATION)

a. Air filters shall be HEPA grade standard capacity air filters with waterproof micro glass fiber media, corrugated aluminum separators, urethane sealant, 16 gauge steel enclosing frame and fluid sealing gasket. Sizes shall be as noted on enclosed drawings or other supporting materials.

b. Construction: Filter media shall be one continuous pleating of microfine glass fiber media. Pleats shall be uniformly separated by corrugated aluminum separators incorporating a hemmed edge to prevent damage to the media. The media pack shall be potted into the enclosing frame with a fire-retardant urethane sealant. The enclosing frame shall be of 16 gauge steel, with a zinc aluminum alloy finish, and shall be bonded to the media pack to form a rugged and durable enclosure. The filter shall be assembled without the use of fasteners to ensure no frame penetrations. Overall dimensional tolerance shall be correct within -1/8 inch, +0 inch, and square within 1/8 inch. A poured-in-place seamless sealing gasket shall be included on the downstream side of the enclosing frame to form a positive seal upon installation.

c. Performance: The filter shall have a tested efficiency of 99.97 percent when evaluated according to IEST RP-CC-001 Recommended Practice. Initial resistance to airflow shall not exceed 1 inch WG at rated capacity. Filter shall be listed by Underwriters Laboratories as UL 900. The filter shall be capable of withstanding 10 inches WG without failure of the media pack. Manufacturer shall provide evidence of facility certification to ISO 9001.

HEPA Performance (Standard Capacity)		
Nominal Size (Inches Height x Width x Depth)	Airflow Capacity (cfm)	Media Area (sf)
24x24x12	1080 at 1 Inch WG	153
24x12x12	500 at 1 Inch WG	33
Follow manufacturer’s recommendation for change out resistance, typically double the initial.		

d. Supporting Data: The filter shall be labeled as to tested efficiency, rated/tested cfm, pressure drop and shall be serialized for identification. The manufacturer shall supply a Certificate of Conformance for each HEPA filter supplied to the facility.

2.6 HEPA FILTERS HIGH CAPACITY V-BANK HIGH CAPACITY FILTERS (FINAL FILTER APPLICATION)

a. Air filters shall be absolute grade HEPA filters consisting of pleated media packs assembled in a V-bank configuration, polyurethane sealant, anodized aluminum enclosure and seamless fluid sealing gasket. Sizes

shall be as noted on enclosed drawings or other supporting materials.

b. Construction: Filter media shall be micro fiber glass formed into mini-pleat pleat-in-pleat V-bank design. The media packs shall be potted into the enclosing frame with fire retardant polyurethane sealant. An enclosing frame of anodized extruded aluminum shall form a rugged and durable enclosure. A seamless sealing gasket shall be included on the downstream side of the filter to form a positive seal upon installation.

c. Performance: Filter efficiency at 0.3 micron shall be 99.99 percent when evaluated according to the IEST RP-CC-001 Recommended Practice for applicable type. Each filter shall be labeled as to tested performance. Initial resistance target shall not exceed 1 inch WG) at rated airflow.

HEPA Performance V-Bank Style (High Capacity)		
Nominal Size (Inches Height x Width x Depth)	Airflow Capacity (cfm)	Media Area (sf)
24x24x12	2000 at 1 Inch WG	390
24x12x12	900 at 1 Inch WG	174
Follow manufacturer’s recommendation for change out resistance, typically double the initial.		

d. Supporting Data: The filter shall be labeled as to tested efficiency, rated/tested cfm, pressure drop and shall be serialized for identification. The manufacturer shall supply a Certificate of Conformance for each HEPA filter supplied to the facility.

e. Filter shall be listed as UL 586 and UL 900 per Underwriters Laboratories. Manufacturer shall provide evidence of facility certification to ISO 9001.

2.7 FILTER HOUSINGS/SUPPORT FRAMES

a. Side Servicing Housings (HVAC Grade):

(1) Filter housing shall be two-stage filter system consisting of 16 gauge galvanized steel enclosure, aluminum filter mounting track, universal filter holding frame, insulated dual-access doors, static pressure tap, filter gaskets and seals. Inline housing depth shall not exceed 21 inches. Sizes shall be as noted on enclosed drawings or other supporting materials.

(2) Construction: The housing shall be constructed of 16 gauge galvanized steel with pre-drilled standing flanges to facilitate attachment to other system components. Corner posts of Z-channel construction shall ensure dimensional adherence. The housing shall incorporate the capability of two stages of filtration without modification to the housing. A filter track, of aluminum construction shall be an integral component of housing construction. The track shall accommodate a 2 inch deep prefilter, a 6 inch or 12 inch deep rigid final filter, or a pocket filter with header. Insulated dual access doors, swing-open type, shall include high-memory sponge neoprene gasket

to facilitate a door-to-filter seal. Each door shall be equipped with adjustable and replaceable positive sealing UV-resistant star-style knobs and replaceable door hinges. A universal holding frame constructed of 18 gauge galvanized steel, equipped with centering dimples, multiple fastener lances, and polyurethane filter sealing gasket, shall be included to facilitate installation of high-efficiency filters. The housing shall include a pneumatic pressure fitting to allow the installation of a static pressure gauge to evaluate pressure drop across a single filter or any combination of installed filters.

(3) Performance: Leakage at rated airflow, upstream to downstream of filter, holding frame, and slide mechanism shall be less than 1 percent at 3 inches WG. Leakage in to or out of the housing shall be less than one half of 1 percent at 3 inches WG). Accuracy of pneumatic pressure fitting, when to evaluate a single-stage, or multiple filter stages, shall be accurate within ± 3 percent at 0.6 inch WG.

(4) Manufacturer shall provide evidence of facility certification to ISO 9001.

b. Holding Frame System (HVAC Grade):

(1) Air filter-holding frames shall be 16 gauge galvanized steel with filter sealing flange, centering dimples, sealing gasket and lances for appropriate air filter fasteners. Sizes shall be as noted on enclosed drawings or other supporting materials.

(2) Construction: Filter holding frame shall be constructed of (16 gauge) galvanized steel. The frame shall be assembled from two corner sections and welded to assure a rigid and durable frame assembly. The frame shall include a variety of pre-punched lances for filter fastener attachment. Fastener shall be capable of being installed without the use of tools, nuts or bolts. Lance penetrations shall be upstream of filter flange to assure leak-free integrity. The frame shall include filter-centering dimples on each frame wall to facilitate ease of filter installation and assure filter centering against filter sealing flange. A 3/4 inch filter-sealing flange shall be an integral component of the holding frame. All corners shall be flush mitered and a permanently mounted polyurethane foam gasket shall be mounted on the sealing flange to assure filter to frame sealing integrity.

(3) Manufacturer shall provide evidence of facility certification to ISO 9001.

c. Side-Access Housing (HEPA Grade):

(1) Filter housing shall be two-stage filter system consisting of 14 gauge galvanized steel enclosure, spring-loaded crank-type sealing assembly for gasket seal type final filters, insulated dual-access doors with gasketing and positive sealing doorknobs. Inline housing depth shall not exceed 25 inches. Sizes shall be as noted on enclosed drawings or other supporting materials.

(2) Construction: The housing shall be constructed of 14 gauge galvanized steel with mating flanges to facilitate attachment to

other system components. All pressure boundaries shall be of all welded construction. The housing shall be weatherproof and suitable for rooftop/outdoor installation. A prefilter track to accommodate nominal 2 inch deep prefilters, shall be an integral component of the housing. The housing shall incorporate a spring-loaded crank-type final filter sealing mechanism. The mechanism shall be geared to exert 700 pounds of pressure against each filter. The clamping frame shall have a continuous flat surface seal to compress all four downstream gasketed surfaces of the downstream seal filter. The final filter locking mechanism shall include a 3/4 inch socket adapter to facilitate opening or closing the mechanism. Insulated dual access doors shall include high-memory sponge neoprene gasket to facilitate a door-to-filter seal. Each door shall be equipped with adjustable and replaceable UV-resistant positive sealing knobs. The access doors shall be both hinged for swing open operation or designed to be completely removable. The housing shall include static pressure ports 1/8 inch NPT male) to facilitate pressure drop measurements across prefilter, final filter, or combination thereof.

(3) Performance: Manufacturer shall provide evidence of facility certification to ISO 9001.

d. Built-Up Bank HEPA Holding Frames:

(1) Holding frames shall be constructed of 14 guage galvanized steel. Frames shall be welded and include centering dimples, pre-drilled mounting holes, filter sealing flange and swing bolt assemblies. An appropriate number of swing bolts to match air filters shall also be included. Sizes shall be as noted on enclosed drawings or other supporting materials.

(2) Construction: Filter frame shall be all-welded construction of 14 guage galvanized steel. The frame shall include pre-drilled mounting holes to align frame-to-frame and ensure built-up bank support. Annular based centering dimples shall be an integral component to assist in proper seating of filter gasket to filter sealing flange. Assembly holes shall be within dimples to recess assembly bolts. Filter securing swing bolt assemblies, of the same construction as the frame, shall be offset to facilitate multiple filter installations. The assembly shall include appropriate swing bolts to match filter depth and equi-bearing clamps to allow uniform filter gasket sealing.

(3) Performance: The sealing assembly shall be capable of sealing each element with 30 inch/lbs of torque to 50 percent filter gasket compression. Manufacturer shall provide evidence of facility certification to ISO 9001.

e. Bag-In/Bag-Out HEPA Air Filter Housing:

(1) Housing shall be side-access bag-in/bag-out, fluid seal housing. The housing shall be adequately reinforced to withstand a negative or positive pressure of 15 inches WG. Housing design and filter arrangement shall allow air to enter and exit housing without changing direction. The housing shall accommodate standard size filters that do not require any special attachments or devices to function properly in the housing. The housing shall accommodate fluid seal filters which require a penetrating knife edge

installed on all filter sealing surfaces. The knife edge shall insert into the fluid filled perimeter channel located on the face of the filter. By engaging the filter seal/release mechanism the filter shall move (push) the fluid filled channel to the sealed position. For removal of the air filters the filter seal/release mechanism shall remove (pull) the filter free of the blade type knife edge. This entire process is performed from inside the filter change out bag.

(2) Construction:

(a) Housing shall be constructed of 14 gauge and 11 gauge T-304 stainless-steel metal. All pressure retaining joints and seams shall be continuously welded with no porosities. Joints and seams requiring intermittent welds, such as reinforcement members, shall be intermittently welded. Housing shall be free of burrs and sharp edges. All weld joints and seams that are a portion of any gasket setting surface, and duct connection flanges, shall be ground smooth and flush with adjacent base metals. All welded joints and seams shall be wire brushed to remove heat discoloration. The housing shall be reinforced to withstand a positive or negative pressure of 15 inches WG. The upstream and downstream ductwork connections shall have 1-1/2 inch outward-turned flanges.

(b) The housing shall have a bagging ring around each filter access port that is sealed by a gasketed filter access door. The filter access door gasket shall be silicone and shall be replaceable, if necessary. The bagging ring shall have two (2) continuous formed raised ridges to secure the PVC change-out bag. The bagging ring shall be hemmed on the outer edge to prevent the change-out bag from tearing.

(c) Ancillary hardware including filter seal/release mechanism, door handles, door studs and labels shall be 300 series stainless-steel. Filter access door knobs shall be cast aluminum and designed to prevent galling of threads.

(d) One (1) PVC change-out bag shall be furnished with each filter access port. Change-out bags shall be 8-mil. thick with a yellow translucent, non-sticking, matte finish. It shall include a 1/4 inch diameter elastic shock cord hemmed into the opening of the bag so when stretched around the housing bagging ring flange, a secure fit is created. The bag shall include three (3) integral glove ports to assist in filter change-out. One (1) nylon security strap shall be included per filter access port to prevent the bag from sliding off the bagging flange during the change-out process. Design of components shall be such that all change-out operations shall be within the bag so there is a barrier between the worker and the filter at all times.

(3) Performance: All welding procedures, welders, and welder operators shall be qualified in accordance with ASME BPVC SEC IX. All production welds shall be visually inspected by qualified personnel, incorporating the workmanship acceptance criteria described in AWS D9.1/D9.1M.

(4) The filter housing shall be factory tested for filter fit, alignment of filter sealing knife edge and operation of filter

clamping mechanism. The filter sealing surface shall be tested at +10 inches WG and have a maximum leak rate of 0.0005 cfm per cubic foot of housing volume. The overall system pressure boundary shall be leak tested at +15 inches WG and have a maximum leak rate of 0.0005 cfm per cubic foot of housing volume.

(5) Filter bags shall be capable of continuous operating to temperature extremes of 0 degrees F to 150 degrees F.

(6) Multi-wide housing shall be equipped with a filter removal rod to pull the filters to the change-out position. The removal rod shall operate from the inside of the filter change out bag.

(7) Manufacturer shall provide evidence of facility certification to ISO 9001.

f. Equipment Identification: Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

2.8 INSTRUMENTATION

a. Magnehelic Differential Pressure Filter Gauges: Nominal 4 inch diameter, 0 to 2 inches WG, 3 inch for HEPA range, except for MERV 17 HEPA final filters, where the range shall be 0 to 3 inches WG. Gauges shall be flush-mounted in aluminum panel board, complete with static tips, copper or aluminum tubing, and accessory items to provide zero adjustment.

b. DDC Static (Differential) Air Pressure Measuring Station: Refer to Section 23 09 23, DIRECT DIGITAL CONTROL SYSTEM FOR HVAC.

c. Provide one DDC sensor across each extended surface filter. Provide Petcocks for each gauge or sensor.

d. Provide one common filter gauge for two-stage filter banks with isolation valves to allow differential pressure measurement.

2.9 HVAC EQUIPMENT FACTORY FILTERS

a. Manufacturer standard filters within fabricated packaged equipment should be specified with the equipment and should adhere to industry standard.

b. Cleanable filters are prohibited.

c. Automatic roll type filters are prohibited.

2.10 FILTER RETURN GRILLES

Refer to Section 23 37 00, AIR OUTLETS AND INLETS.

PART 3 – EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the contractor shall correct the installation at no additional cost or time to the Government.

b. Install supports, filters, and gauges in accordance with manufacturer's instructions.

c. Label clearly with words "Contaminated Air" on exhaust ducts leading to the HEPA filter housing.

3.2 STARTUP AND TESTING

a. Clean and vacuum air handling units and plenums prior to starting air handling systems.

b. Replace pre-filters and install clean filter units prior to final inspection as directed by the Contracting Officer.

c. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

d. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

e. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

-- End of Section --

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SECTION 23 50 11

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SECTION 23 50 11
3/22

BOILER PLANT MECHANICAL EQUIPMENT

PART 1 GENERAL

1.1 DESCRIPTION

a. Fuel oil pumping that supports the operation of the boilers.

b. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 09 91 00, PAINTING.

f. Not used.

l. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.g. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM OR HVAC.

h. Section 23 21 11, BOILER PLANT PIPING SYSTEMS.

i. Section 26 29 11, MOTOR CONTROLLERS.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC SEC VIII D1	(2023) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

UL SOLUTIONS (UL)

UL 574	(2003; Reprint Oct 2024) UL Standard for Safety Electric Oil Heaters

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

No. 2 Fuel Oil Temperature Control System; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

SD-06 Test Reports

Acceptance Tests; G, AE

Completed System Readiness Checklist; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 50 11, BOILER PLANT MECHANICAL EQUIPMENT”, with applicable paragraph identification.

c. Manufacturer's Literature and Data including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

d. No. 2 Fuel Oil Temperature Control System:

(1) Drawing with dimensions and arrangement of pumps, motors, heaters, relief valves and accessories.

(2) Catalog data and specification sheets on the design and construction of pumps, motors, heaters and controls.

(3) Pressure and temperature limitations of pumps, heaters, valves, fittings, strainers and relief valves.

(4) Material (ASTM No.) and pressure rating of pipe and fittings.

(5) Performance data on oil pumps including discharge head, flow, suction lift and motor power required at viscosity range shown. Refer to drawings for requirements.

(6) Performance data on relief valves.

e. Test Data – Acceptance Tests, on-site: Four copies all specified tests.

f. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

g. Completed System Readiness Checklist provided by the Commissioning Agent and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

h. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.6 AS-BUILT DOCUMENTATION

a. Submit manufacturer’s literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, VA approved substitutions and construction revisions shall be in electronic version on CD or DVD inserted into a three-ring binder. All aspects of system operation and maintenance procedures, including applicable piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices shall be included. A List of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the ‘third party testing company’ requirement. Provide

record drawings as follows:

(1) Red-lined, hand-marked drawings are to be provided, with one paper copy and a scanned PDF version of the hand-marked drawings provided on CD or DVD.

d. The as-built drawings shall indicate the location and type of all lockout/tagout points for all energy sources for all equipment and pumps to include breaker location and numbers, valve tag numbers, etc. Coordinate lockout/tagout procedures and practices with local VA requirements.

e. Certification documentation shall be provided to Contracting Officer 21 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and provide documentation/certification that all results of tests were within limits specified. Test results shall contain written sequence of test procedure with written test results annotated at each step along with the expected outcome or setpoint. The results shall include all readings, including but not limited to data on device (make, model and performance characteristics), normal pressures, switch ranges, trip points, amp readings, and calibration data to include equipment serial numbers or individual identifications, etc.

PART 2 PRODUCTS

2.1 GENERAL

Electric motor control cabinets/enclosures including VFDs in the boiler plant shall be minimum NEMA 250 NEMA 4 or better. The design AE shall determine at the design stage based on the environmental condition and location. This shall also be indicated on the drawings.

2.2 NO. 2 FUEL OIL TEMPERATURE CONTROL SYSTEM

a. General: Provide for each aboveground fuel oil tank system that stores No. 2 fuel oil. Consisting of an oil pump, an electric oil heater, controls, valves, and piping connected to the fuel oil tank supply and return lines. The purpose is to maintain oil tank temperature of approximately (30 degrees F) to control the oil viscosity and to keep the oil tank temperature above the pour point of the oil.

b. Oil Pump: Electric motor-driven, rotary gear-type, mechanical shaft seal, hardened steel gears and shafts. Pump shall be close-coupled, motor-mounted. Shaft couplings shall have no organic material. Pump performance shall be as shown on the drawings.

c. Electric Oil Heater: Shell-type with immersion-type electric resistance heating elements, designed for fuel oil heating. Design unit for maximum oil pressure of (200 psig). Materials and fabrication of shell and heads shall be in accordance with the ASME BPVC SEC VIII D1. Heating elements shall have electrical ratings in accordance with drawing requirements and shall be removable. Comply with UL 574. Provide pipe connections shown.

d. Controls: Provide locally mounted control panel consisting of manual

start-stop controls for the oil pump, thermostatically controlled contactors for the oil heater, red indicating lights for "pump running", "heater on", and green for "power on". Also include in panel, fusible disconnect switches or circuit breakers and control transformer (120 volt) for heater thermostat and indicating lights. Thermostat shall have minimum adjustment range of (10 to 61 degrees F). Provide devices to signal computer workstation that system is on or off.

e. Pressure Relief Valves: Provide on the shell of the oil heater and on the oil pump discharge line where shown. Valves shall be sized to relieve the maximum combined pumping capability of all oil pumps, at 140 psig set pressure plus 25 percent accumulation. Pressure settings shall be adjustable. Route relief discharge pipe back to oil tank.

f. Pipe, Valves, Fittings, Miscellaneous Piping Specialties, Pressure Gauges and Thermometers: Refer to specification Section 23 21 11, BOILER PLANT PIPING SYSTEMS.

g. Arrangement: Heaters, pumps, controls and interconnecting piping shall be mounted within a NEMA-4 cabinet as shown on the drawings.

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

b. Fuel Oil Pumping Equipment and Fuel Oil Heaters and Accessories: Locate equipment to permit access to all valves and controls, and to permit removal and replacement of system components.

3.2 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The Commissioning Agent will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and Commissioning Agent. Provide a minimum notice of 10 working days prior to startup and testing.

3.3 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.4 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hours to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 64 00

PACKAGED WATER CHILLERS

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SECTION 23 64 00
3/24

PACKAGED WATER CHILLERS

PART 1 GENERAL

1.1 DESCRIPTION

a. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. This section covers Centrifugal water-cooled chillers, complete with accessories.

1.2 DEFINITIONS

a. Engineering Control Center (ECC): The centralized control point for the intelligent control network. The ECC comprises of personal computer and connected devices to form a single workstation.

b. BACNET: Building Automation Control Network Protocol, ASHRAE 135.

c. Ethernet: A trademark for a system for exchanging messages between computers on a local area network using coaxial, fiber optic, or twisted-pair cables.

d. FTT-10: Echelon Transmitter-Free Topology Transceiver.

1.3 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 03 30 00, CAST-IN-PLACE CONCRETE.

f. Not used.

g. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

h. Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC and STEAM GENERATION EQUIPMENT.

i. Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

j. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

k. Section 23 21 13, HYDRONIC PIPING.

l. Section 23 21 23, HYDRONIC PUMPS.

m. Section 23 31 00, HVAC DUCTS and CASINGS

n. Section 23 81 00, DECENTRALIZED UNITARY HVAC EQUIPMENT.

o. Section 26 29 11, MOTOR CONTROLLERS.

1.4 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standards will govern.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 550/590 I-P	(2023) Performance Rating Of Water-Chilling and Heat Pump Water-Heating Packages Using the Vapor Compression Cycle
AHRI 575	(2017; R 2020) Method of Measuring Machinery Sound Within an Equipment Space

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)

ABMA 9	(2015) Load Ratings and Fatigue Life for Ball Bearings
ABMA 11	(2014; R 2020) Load Ratings and Fatigue Life for Roller Bearings

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 135	(2020; Interpretation 1-8 2021; Errata 1-2 2021; Addenda CD 2021; Addenda BV-CE 2022; Interpretation 9-12 2022; Interpretation 13-24 2023; Addenda BV-CF 2023; Errata 3 2023) BACnet—A Data Communication Protocol for Building Automation and Control Networks
ASHRAE 15 & 34	(2022) ASHRAE Standard 34-2016 Safety Standard for Refrigeration Systems/ASHRAE Standard 34-2016 Designation and Safety Classification of Refrigerants-ASHRAE Standard 34-2016

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC SEC VIII	(2021) Boiler and Pressure Vessel Codes: Section VIII Rules for Construction of Pressure Vessel

ASTM INTERNATIONAL (ASTM)

ASTM B117	(2019) Standard Practice for Operating

	Salt Spray (Fog) Apparatus
ASTM C534/C534M	(2024) Standard Specification for Preformed Flexible Elastomeric Cellular Thermal Insulation in Sheet and Tubular Form
ASTM D4587	(2011; R 2019; E 2019) Standard Practice for Fluorescent UV-Condensation Exposures of Paint and Related Coatings

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.95	Occupational Noise Exposure

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Seismic Restraints; G, AE

Bio-Based Materials; S, G, AE

SD-06 Test Reports

Run Test Report; G, AE

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Product Certificate; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.6 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 XX XX, SECTION TITLE”, with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Centrifugal water chillers, including motor starters, control panels, and vibration isolators, and remote condenser data shall include the following:

(a) Rated capacity.

(b) Pressure drop.

(c) Efficiency at full load and part load WITHOUT applying any tolerance indicated in the AHRI 550/590 I-P.

(d) Refrigerant

(e) Fan performance (Air-Cooled Chillers only.)

(f) Accessories.

(g) Installation instructions and certifications.

(h) Startup procedures.

(i) Wiring diagrams, including factory-installed and field-installed wiring.

(j) Sound/Noise data report. Manufacturer shall provide sound ratings. Noise warning labels shall be posted on equipment.

(k) Refrigerant vapor detectors and monitors.

d. Run test report for all chillers.

e. Product Certificate: Signed by chiller manufacturer certifying that chillers furnished comply with AHRI requirements. The test report shall include calibrated curves, calibration records, and data sheets for the instrumentation used in factory tests.

f. Provide IBC certification and seismic restraints for refrigeration equipment to withstand seismic forces.

g. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation, and maintenance.

h. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician, and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

i. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.7 QUALITY ASSURANCE

a. Comply with AHRI requirements for testing and certification of the chillers.

b. Refer to paragraph, WARRANTY, Section 01 00 00, GENERAL REQUIREMENTS, except as noted below:

(1) Provide a 5-year motor, transmission, and compressor warranty to include materials, parts and labor.

c. Refer to OSHA 29 CFR 1910.95(a) and (b) for Occupational Noise Exposure Standard.

d. Refer to ASHRAE 15 & 34 for refrigerant vapor detectors and monitor.

e. Bio-Based Materials: For products designated by the USDA’s Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

f. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.8 AS-BUILT DOCUMENTATION

Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 2 PRODUCTS

2.1 CENTRIFUGAL WATER-COOLED WATER CHILLERS

a. General: Chiller shall be factory assembled and factory tested, complete with evaporator, condenser, marine water boxes for condenser and evaporator, compressor, motor, starter, oil heater and cooler, economizer or intercooler, purge system, refrigerant piping, instrumentation and control piping, operating and safety controls mounted on the chiller, and other auxiliaries necessary for safe and proper operation of the unit. Chiller operation shall be fully automatic. Make provision for space and design piping layout to suit the marine water boxes.

b. Performance: Provide the capacity as shown on the contract drawings. Part load and full load efficiency ratings of the chiller shall not exceed those shown on the contract drawings.

c. Capacity of a single water-cooled chiller shall not exceed 1,250 Tons (standard AHRI conditions).

d. Applicable Standard: Chillers shall be rated and certified in accordance with AHRI 550/590 I-P. Chillers shall be AHRI stamped. Chiller efficiency shall comply with FEMP (Federal Energy Management Progress) requirements.

e. Acoustics: Sound pressure levels shall not exceed the following specified levels. The manufacturer shall provide sound treatment if required to comply with the specified maximum levels. Testing shall be in accordance with AHRI 575.

OCTAVE BAND								Overall
63	125	250	500	1000	2000	4000	8000	dB(A)

f. Hermetic or Open: Chillers shall be open or hermetically sealed, using refrigerant type R-514A. EPA approved refrigerants are listed at https://www.epa.gove/snap/. Submit proposed refrigerant for Government approval.

g. Compressor (Centrifugal Type): Single or multistage, having statically and dynamically balanced impeller, either direct or gear driven. Impeller shaft shall be heat-treated carbon steel of sufficient rigidity to prevent whip or vibration at operating speed. Shaft main bearings shall be of journal type with bronze or babbitt line steel cartridge, aluminum alloy one-piece insert type, or rolling element typewith an ABMA L10 life of a minimum of 200,000 hours. Rolling element bearings shall be rated in accordance with ABMA 9 or ABMA 11 as applicable. Casing shall be cast-iron or steel plate with split sections gasketed and bolted together. Lubrication System shall be forced-feed type and shall provide oil at proper temperature to all parts requiring lubrication. Make provisions to ensure lubrication of bearings prior to starting and of shaft seal both on stopping and starting, or bearings and shaft seal shall be submerged in oil. On units providing for forced-feed lubrication prior to starting, a differential oil pressure cutout interlocked with compressor starting equipment shall allow compressor to operate only when required oil pressure is provided to bearings. Capacity control shall be by means of variable inlet guide vanes in the compressor suction to modulate the chiller capacity from 100 to 10 percent of full unit rated capacity without unstable compressor operation. The inlet guide vanes shall be electrically operated upon the actuation of temperature or pressure sensor.

j. Evaporator: Shell-and-tube type, constructed and tested and stamped in accordance with ASME BPVC SEC VIII where applicable for working pressure produced by refrigerant used and water system installed, but not less than 150 psig waterside working pressure. Shell shall be fabricated of carbon steel and shall have carbon steel tube sheets; drilled and reamed to accommodate the tubes. Tubes shall be externally and internally enhanced individually replaceable and shall

be expanded full diameter into tube sheets, providing a leak proof seal. Intermediate tube supports sheets shall be provided as recommended by the manufacturer to minimize tube vibration, stress, and wear. Performance shall be based on a water velocity not less than 3 fps nor greater than 12 fps, and fouling factor of 0.0001 hr. sq. ft. degrees F/Btu. Removable marine water box shall be constructed of steel. Design working pressure shall be 150 psig ; pressure tested at 130 percent of working pressure. Water nozzle connections shall be flanged.

k. Condenser: Shell-and-tube type, constructed, tested, and stamped in accordance with applicable portions of ASME BPVC SEC VIII, where applicable for working pressure produced by the refrigerant used and water system installed, but not less than 150 psig. Shell shall be fabricated of carbon steel and shall have carbon steel tube sheets; drilled and reamed to accommodate the tubes. Tubes shall be nonferrous metal, externally enhanced, and internally enhanced, individually replaceable, and shall be expanded full diameter into tube sheets, providing a leak proof seal. Intermediate tube support sheets shall be provided as recommended by the manufacturer to minimize tube vibration, stress and wear. Tubes shall fit tightly in the supports to prevent chafing due to vibration or pulsation. Performance of condenser shall be based on a water velocity not less than 3 fps nor greater than 12 fps, and a fouling factor of 0.00025 hr. sq. ft. degrees F/Btu. Removable marine water box shall be constructed of steel. Design working pressure shall be 150 psig ; pressure tested at 130 percent of working pressure. Water nozzle connections shall be flanged.

l. Insulation: Evaporator, suction piping, compressor, and all other parts subject to condensation shall be insulated with 1.5 inch minimum thickness of flexible-elastomeric thermal insulation, complying with ASTM C534/C534M.

m. Economizer: Provide if required by manufacturer. Flash gas shall be piped from economizer to inlet of intermediate stage impeller wheel. In case of rotary compressor flash gas shall be piped from economizer to the intermediate compressor point. Provide a refrigerant flow control system (float valve or variable/multiple orifice system) to automatically regulate flow of liquid refrigerant through economizer. If external-type economizer is used, such economizer shall be constructed and tested in accordance with ASME BPVC SEC VIII for working pressures produced by refrigerant used.

n. Motor Load Limiter: Provide a sensing and control system, which will limit maximum load current of compressor motor to a manually selectable percentage of 40 percent to 100 percent of full load current. System shall sense compressor motor current and limit it by modulating inlet guide vanes at the compressor, overriding other controls in their ability to increase loading, but not overriding their ability to reduce loading.

o. Purge System: Chillers utilizing HCFC-123 and chillers using refrigerants with vapor pressure less than 14.5 psig, or considered low pressure refrigerants, shall be supplied with Purge System. Purge unit shall be factory mounted, complete with necessary, piping, operating and safety controls and refrigerant service valves to isolate the unit from the chilling unit. Purge unit shall be air, water, or refrigerant cooled. When in operation, purge system shall

function automatically to remove, water vapor, and condensable gases from refrigeration system and to condense, separate, and return to system any refrigerant present therein. Purge system shall be manually or automatically started and stopped, and shall be assembled as a compact unit. As an option, a fully automatic purge system that operates continuously while main unit is operating may be furnished. Such purge system shall provide a means to signal operator of occurrence of excessive purging indicating abnormal air leakage into unit. The purge system shall be of high efficiency in recapturing the refrigerant at all load and head conditions and with capability to operate when the chiller is off. The purge unit shall be UL listed.

p. Isolation Pads: Manufacturer’s standard.

q. Spring Isolators: Per Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

r. Refrigerant and Oil:

(1) Provide sufficient volume of dehydrated refrigerant and lubricating oil to permit maximum unit capacity operation before and during tests. Refrigerant charge lost during the warranty period due to equipment failure shall be replaced without cost to the Government.

(2) The manufacturer shall certify that chiller components, such as seals, O-ring, motor windings, etc, are fully compatible with the specified refrigerants.

s. Chillers utilizing HCFC-123 shall be supplied with a vacuum prevention system to maintain the chiller at positive pressure during non-operational cycles.

t. Chillers utilizing HCFC-123 shall be supplied withfrangible carbon rupture discall metal, non-fragmented with reverse buckling design rupture disc and a safety relief valve downstream of the rupture disc Low pressure refrigerant chillers shall be supplied with safety relief valve in addition to high efficiency refrigerant purge system. .

u. Service valves shall be provided to facilitate refrigerant reclaim/removal required during maintenance.

v. Controls: Chiller shall be furnished with unit mounted, stand-alone, microprocessor-based controls in NEMA 250 type 12 enclosure, hinged and lockable, factory wired with a single point power connection and separate control circuit. The control panel provide chiller operation, including monitoring of sensors and actuators, and shall be furnished with light emitting diodes or liquid-crystal display keypad.

(1) The following functions shall display as a minimum:

(a) Date and Time.

(b) Outdoor air temperature.

(c) Operating set point temperature and pressure.

(d) Operating hours.

(e) Operating or alarm status.

(f) Chilled water temperature-entering and leaving.

(g) Condenser water temperature-entering and leaving.

(h) Refrigerant pressure-condenser and evaporator.

(i) Low oil pump pressure.

(j) High oil supply pressure.

(k) Chiller diagnostic codes.

(l) Current limit set point.

(m) Number of compressor starts.

(n) Purge suction temperature, if refrigerant HCFC-123 or any low pressure refrigerant is used.

(o) Purge elapsed time, if refrigerant HCFC-123 or any low pressure refrigerant is used.

(2) Control Functions:

(a) Manual or automatic startup and shutdown time schedule.

(b) Control set points for entering and leaving chilled temperatures.

(c) Condenser water temperature.

(d) Current/demand limit.

(e) Motor load limit.

(3) Safety Controls: Following conditions shall shut down the chiller and require manual reset to start:

(a) High condenser pressure.

(b) High oil temperature.

(c) High or low oil pressure.

(d) Loss of flow-condenser or chilled water.

(e) Low chilled water temperature.

(f) Low evaporator refrigerant temperature.

(g) Sensor malfunctions.

(h) Power fault.

(i) Extended compressor surge.

(j) Communication loss between the chiller and its control panel.

A signal shall be transmitted to Energy Control Center, if provided, for this communication loss and for any abnormal.

(4) The chiller control panel shall provide a relay output to initiate system changeover to free cooling. This relay shall be energized upon initiation of free cooling at the chiller control panel.

(5) Leaving chilled water temperature reset, where specified in the control sequence, shall be based on return water temperature 4-20 MA or 0-10 VDC signal from a building automation system.

(6) Chillers shall be pre-wired to terminal strips for interlocked to other equipment.

(7) Provide contacts for remote start/stop, alarm for abnormal operation or shut down, and for Engineering Control Center (ECC) interface.

(8) Chiller control panel shall reside on the "BACnet network", provide data using open protocol network variable types and configuration properties, and BACnet interworking using ARCNET or MS/TP physical data link layer protocol for communication with building automation control system.

(9) Auxiliary hydronic system and the chiller(s) shall be electronically interlocked to provide time delay and starting sequence as indicated on control drawings.

(10) The chiller control panel shall utilize the following components to automatically take action to prevent unit shutdown due to abnormal operating conditions which will perform as follows.

(a) High pressure switch that is set to 20 psig (adjustable setting) lower than factory pressure switch that will automatically unload the compressor to help prevent a high pressure condenser control trip. One switch is required for each compressor and indicating light shall also be provided.

(b) Motor surge pressure that is set at 95 percent of compressor RLA that will automatically unload the compressor to prevent an over current trip. One protector is required for each compressor and indicating light shall also be provided.

(c) Low pressure switch that is set at 5 psig above the factory low pressure switch that will automatically unload the compressor to help prevent a low evaporator temperature trip. One switch is required for each compressor and indicating light shall also be provided.

(d) In all the above cases, the chiller will continue to run, in an unloaded state and will continue to produce some chilled water in an attempt to meet the cooling load. However, if the chiller reaches the trip-out limits, the chiller controls will take the chiller off line for protection, and a manual reset is required. Once the "near trip" condition is corrected, the chiller will return to normal operation and can then produce full load cooling.

(11) With variation of +/-10 percent of design flow per minute, chiller shall be able to maintain +/-0.5 degrees F leaving water

temperature control. The chiller shall be able to withstand a +/- 30 percent change in flow rate per minute without unit trip. Variations in the primary flow allow for optimal system efficiency, but the chiller shall be able to maintain temperature control to help ensure occupant comfort.

(12) The chiller control panel shall provide +/-0.5 degrees F leaving water temperature control during normal operation. The chiller shall provide multiple steps leaving chilled water temperature controller to minimize part load energy use and optimize leaving chilled water temperature control. If manufacturer is unable to provide at least several steps of unloading, hot gas bypass shall be required to minimize loss of leaving water temperature control.

(13) The chiller control panel shall provide a 2-minute stop-to-start and 5 minute start-to-start solid state timer. If the anti-recycle timers are longer than 5 minutes, then hot-gas bypass shall be provided to limit loss of leaving chilled water temperature control in low-load conditions.

w. Motor: Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC. Compressor motor furnished with the chiller shall be in accordance with the chiller manufacturer and Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC and STEAM GENERATION EQUIPMENT. Starting torque of the motor shall be suitable for the driven chiller machine.

x. Motor Starter: Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC. Provide a starter for each centrifugal or rotary-screw chiller in NEMA 250 type 1 enclosure, designed for floor or unit mounting. For floor mounted starter provide wiring from starter to chiller. Starter shall be a autotransformer type as noted on drawings. Unit mounted starter should be used to meet IBC certification. Provide starter with the following features in addition to the ones specified in Section 26 29 11, MOTOR CONTROLLERS.

(1) Starter shall include incoming line provision for the number and size cables shown on the contract drawings. Incoming line lugs shall be copper mechanical type.

(2) Terminals connection pads shall be provided to which customers supply lugs can be attached.

(3) Starters shall be coordinated with chiller packages(s) making certain all terminals are properly marked according to the chiller manufacturer’s wiring diagram.

(4) Contactors shall be sized per NEMA requirements to the chillers for full load currents.

(5) Ammeter(s) shall be provided, capable of displaying current to all three phases. Ammeter shall be calibrated so that inrush current can be indicated.

(6) Chiller starter shall include an advanced motor protection system incorporating electronic three phase overloads and current transformers. This electronic motor protection system shall monitor and protect against the following conditions:

(a) Three phase loss with under and over voltage protection.

(b) Phase imbalance.

(c) Phase reversal.

(d) Motor overload.

(e) Motor overload protection incorrectly set.

(f) Momentary power loss protection with auto restart consisting of three phase current sensing device that monitor the status of the current.

(g) Starter contactor fault protection.

(h) Starter transition failure.

(i) Distribution fault protection.

(7) When a motor driven oil pump is furnished, provide a 120-volt control circuit, mounted within starter enclosure. When an oil pump starter is provided at the refrigeration machine, provide fused disconnect in star delta starter for oil pump.

(8) The starter shall be equipped with pilot relays to initiate the start sequence of compressor. These relays shall be a self-monitoring safety circuit, which shall indicate improper operation (slow operation, welding of contacts, etc) and shall cause the chiller unit to be shut down and a fault trip indicator be displayed. The "starter circuit fault" indicator shall be located in the door of the enclosure and shall require manual reset.

(9) A lockout transition safety circuit shall be provided to prevent damage from prolonged energization due to malfunction of the transistor contactor. Malfunction shall cause the chiller unit to shut down and the "starter circuit fault" indicator be displayed.

(10) A permanent nameplate shall be provided and mounted on the starter panel. It shall identify the manufacturer, serial or model number identifying the date of manufacturing and component replacement parts, and all current and voltage rating, and as built wiring schematic showing all items provided.

(11) Provide non-fused main power disconnect switch.

2.2 CONDENSING UNITS FOR AIR CONDITIONING SERVICE

Refer to Section 23 81 00, DECENTRALIZED UNITARY HVAC EQUIPMENT.

2.3 REFRIGERANT MONITORING AND SAFETY EQUIPMENT

a. General: Provide refrigerant monitoring sensor/alarm system and safety equipment as specified here. Refrigerant sensor and alarm system shall comply with ASHRAE 15 & 34. The refrigerant monitoring system will be provided by the chiller manufacturer and shall be interfaced with the DDC control system.

b. Refrigerant monitor shall continuously display the specific gas

(refrigerant used) concentration; shall be capable of indicating, alarming and shutting down equipment; and automatically activating ventilation system. On leak detection by refrigerant sensor(s), the following shall occur:

(1) Activate machinery (chiller) room ventilation.

(2) Activate visual and audio alarm inside and outside of machinery room, with beacon light(s) and horn sounds equipment room and outside equipment room door(s). Shut down combustion process where combustion equipment is employed in the machinery room.

(3) Notify Engineering Control Center (ECC) of the alarm condition.

c. Refrigerant monitor shall be capable of detecting concentration of 1 part per million (ppm) for low-level detection and for ensuring the safety of operators. It shall be supplied factory-calibrated for the apparent refrigerant.

d. Monitor design and construction shall be compatible with temperature, humidity, barometric pressure, and voltage fluctuations of the machinery room operating environment.

2.4 CORROSION PROTECTION

a. Remote Outdoor Condenser Coils (Epoxy Immersion Coating – Electrically Deposited): The multi-stage corrosion-resistant coating application comprises of cleaning (heated alkaline immersion bath) and reverse-osmosis immersion rinse prior to the start of the coating process. The coating thickness shall be maintained between 0.6-mil and 1.2-mil. Before the coils are subjected to high-temperature oven cure, they are treated to permeate immersion rinse and spray. Where the coils are subject to UV exposure, UV protection spray treatment comprising of UV-resistant urethane mastic topcoat shall be applied. Provide complete coating process traceability for each coil and minimum five years of limited warranty. The coating process shall be such that uniform coating thickness is maintained at the fin edges. The quality control shall be maintained by ensuring compliance to the applicable ASTM Standards for the following:

(1) Salt Spray Resistance (Minimum 6,000 Hours)

(2) Humidity Resistance (Minimum 1,000 Hours)

(3) Water Immersion (Minimum 260 Hours)

(4) Cross-Hatch Adhesion (Minimum 4B-5B Rating)

(5) Impact Resistance (Up to 160 Inch/Pound)

b. Exposed Outdoor Cabinet: Casing Surfaces (Exterior and Interior): All exposed and accessible metal surfaces shall be protected with a water-reducible acrylic with stainless-steel pigment spray-applied over the manufacturer’s standard finish. The spray coating thickness shall be 2-4 mils and provide minimum salt-spray resistance of 1,000 hours in accordance with ASTM B117 and 500 hours UV resistance in accordance with ASTM D4587.

PART 3 – EXECUTION

3.1 EXAMINATION

a. If an installation is unsatisfactory to the Contracting Officer, the contractor shall correct the installation at no additional cost or time to the Government.

b. Examine roughing-in for concrete equipment bases, anchor-bolt sizes and locations, piping and electrical to verify actual locations and sizes before chiller installation and other conditions that might affect chiller performance, maintenance, and operation. Equipment locations shown on the contract drawings are approximate. Determine exact locations before proceeding with installation.

3.2 EQUIPMENT INSTALLATION

a. Install chiller on concrete base with isolation pads or vibration isolators.

(1) Concrete base is specified in Section 03 30 00, CAST-IN-PLACE CONCRETE.

(2) Vibration isolator types and installation requirements are specified in Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

(3) Anchor chiller to concrete base according to manufacturer’s written instructions and for seismic restraint on vibration isolators.

(4) Charge the chiller with refrigerant, if not factory charged.

(5) Install accessories and any other equipment furnished loose by the manufacturer, including remote starter, remote control panel, and remote flow switches, according to the manufacturer written instructions and electrical requirements.

(6) Chillers shall be installed in a manner as to provide easy access for tube pull and removal of compressor and motors etc.

b. Install refrigerant monitoring and safety equipment in accordance with ASHRAE 15 & 34.

c. Install refrigerant piping as specified in ASHRAE 15 & 34.

d. Install thermometers and gauges as recommended by the manufacturer and/or as shown on the contract drawings.

e. Piping Connections:

(1) Make piping connections to the chiller for chilled water, condenser water, and automatic tube brush cleaning system and other connections as necessary for proper operation and maintenance of the equipment.

(2) Make equipment connections with flanges and couplings for easy removal and replacement of equipment from the equipment room.

(3) Extend vent piping from the relief valve and purge system to the outside.

3.3 STARTUP AND TESTING

a. Engage manufacturer's factory-trained representative to perform startup and testing service.

b. Inspect, equipment installation, including field-assembled components, and piping and electrical connections.

c. After complete installation startup checks, according to the manufacturers written instructions, do the following to demonstrate to the Contracting Officer that the equipment operate and perform as intended.

(1) Check refrigerant charge is sufficient and chiller has been tested for refrigerant leak.

(2) Check bearing lubrication and oil levels.

(3) Verify proper motor rotation.

(4) Verify pumps associated with chillers are installed and operational.

(5) Verify thermometers and gauges are installed.

(6) Verify purge system, if installed, is functional and relief piping is routed outdoor.

(7) Operate chiller for run-in-period in accordance with the manufacturer’s instruction and observe its performance.

(8) Check and record refrigerant pressure, water flow, water temperature, and power consumption of the chiller.

(9) Test and adjust all controls and safeties. Replace or correct all malfunctioning controls, safeties and equipment as soon as possible to avoid any delay in the use of the equipment.

(10) Prepare a written report outlining the results of tests and inspections, and submit it to the Contracting Officer.

d. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

e. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

f. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

3.4 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.5 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hour s to instruct each VA personnel responsible in operation and maintenance of the system. Coordinate this training with that of the cooling tower, if furnished together.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

-- End of Section --

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SECTION 23 65 00
3/23

COOLING TOWERS

PART 1 GENERAL

1.1 DESCRIPTION

a. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. This section covers packaged, induced draft open circuit cooling tower complete with fill, fan, inlet louvers and associated accessories and equipment.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 03 30 00, CAST-IN-PLACE CONCRETE.

f. Not used.

g. Section 22 11 00, FACILITY WATER DISTRIBUTION.

h. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

i. Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC and STEAM GENERATION EQUIPMENT.

j. Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

k. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

l. Section 23 21 13, HYDRONIC PIPING.

m. Section 23 25 00, HVAC WATER TREATMENT.

n. Section 23 31 00, HVAC DUCTS and CASINGS.

o. Section 26 29 11, MOTOR CONTROLLERS.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standards will govern.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASSE A10.18	(2007(R2012))Safety Requirements for Temporary Roof and Floor Holes, Wall Openings, Stairways, and Other Unprotected Edges in Construction and Demolition Operations

ASTM INTERNATIONAL (ASTM)

ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A666	(2023) Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate and Flat Bar
ASTM E84	(2024) Standard Test Method for Surface Burning Characteristics of Building Materials

COOLING TECHNOLOGY INSTITUTE (CTI)

CTI ATC 105	(2011; R2022) Acceptance Test Code for Closed Circuit Cooling Towers
CTI STD 201 OM	(2019) Operations Manual for Thermal Performance Certification of Evaporative Heat Rejection Equipment

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1	(2021) Motors and Generators
NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

U.S DEPARTMENT OF VETERANS AFFAIRS (VA)

VHA ES-2019-01	(2021) Engineering Standard Cooling Tower Water Systems

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Bio-Based Materials; S, G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Certification; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 XX XX, SECTION TITLE”, with applicable paragraph identification.

c. Shop Drawings:

(1) Sufficient information, clearly presented, shall be included to determine compliance with the contract documents.

(2) Include rated capacities, pressure drop, fan performance and rating curves, dimensions, weights, mounting details, front view, side view, and equipment and device arrangement.

(3) Include electrical rating, detail wiring for power, signals, and controls.

(5) Sound curves and characteristics of sound attenuators if required to meet the noise criteria.

d. Certification:

(1) Submit 4 copies of performance curves, for CTI certified cooling towers, showing compliance with actual conditions specified, to

the Contracting Officer 2 weeks prior to delivery of the equipment.

(2) Two weeks prior to final inspection, submit 4 copies of the following to the Contracting Officer:

(a) Certification from the manufacturer that the cooling tower(s), accessories, and components are suitable for seismic design category C installations and that the unit will be fully operational after the seismic event at the project site.

(b) Certification by the manufacturer that the cooling towers conform to the requirements of the contract documents.

(c) Certification by the Contractor that the cooling towers have been installed, adjusted, and tested.

e. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

f. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation, and maintenance.

g. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician, and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

h. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.6 QUALITY ASSURANCE

a. Refer to paragraph QUALITY ASSURANCE of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Design Criteria:

(1) Design to withstand (60 psf) wind load.

(2) Free water drift loss shall not be greater than five tenths of one percent (0.05) or 0.005% of the water circulated to tower. Efficient drift eliminators that reduce drift to maximum of 0.002% of recirculated water volume for counterflow towers and 0.005% of recirculated water flow for cross-flow towers.

(3) Sound levels at 5 feet and 55 feet in any direction from the tower shall not exceed 40 dB (A) and 48 dB (A), respectively. Select

"low Noise" model cooling towers, where available. Provide sound attenuators if necessary to meet the noise criteria.

c. Performance Criteria:

(1) Manufacturer shall certify that performance of cooling towers will meet contract requirements, stating entering air wet bulb temperature, entering and leaving condenser water temperatures, water flow rates, fan kW (horsepower)and pump head at base of tower. Certification shall be made at the time of submittal.

(2) Cooling Technology Institute (CTI) Certified Towers: These towers shall have been tested, rated, and certified in accordance with CTI STD 201 OM, and shall bear the CTI certification label, and shall be listed in the CTI directory of certified cooling towers.

(3) The alignment and balancing of the fans, motors and drive shaft as installed shall operate within the vibration tolerance specified in Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

d. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

e. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 2 PRODUCTS

2.1 INDUCED DRAFT OPEN CIRCUIT COOLING TOWER

a. Cooling tower shall be a factory assembled, induced draft, cross-flow type with a vertical discharge configuration. Unit shall be Factory Mutual approved and IBC certified. Unit shall not exceed dimensions, in particular height, and weight noted on schedule.

b. Casing: Heavy gauge (minimum 16 gauge) galvanized steelwith stainless-steel basins.

(1) Galvanized Steel: Hot-dip galvanized steel complying with ASTM A653/A653M, and having G235 (Z700) coating.

(2) Stainless-Steel: ASTM A666, Type 304.

(4) Fasteners: Zinc or cadmium coated bolts or tapping screws for assembly. Use stainless-steel washers with neoprene backing where required for preventing leaks.

(5) Joints and Seams: Sealed watertight.

(6) Welded Connections: Continuous and watertight.

c. Framing: Rolled structural steel shapes, hot-dip galvanized after fabrication or structural shapes cold formed from galvanized steel sheets or plates, complying with ASTM A653/A653M, and having G235 (Z700) coating.

d. Louvers:

(1) Spaced to minimize air resistance and prevent splash out. Louver materials shall be similar to the casings or may be polyvinyl chloride (PVC) if formed integral with the fill material.

(2) 1/2 inch inlet screen, stainless-steel. Attach the screen securely to air intakes.

e. Fill: PVC resistant to rot, decay and biological attack; with a maximum flame spread rating of five per ASTM E84 and fabricated, formed and installed by manufacturer to ensure that water breaks up into droplets.

f. Drift Eliminators: Two stage drift eliminator system consisting of integral first stage and separate second stage to prevent water carryover into the plenum. Same as fill material. Effectively trap water droplets entrained in discharge air stream and limit drift loss to less than 0.005 percent of the total water circulated. Efficient drift eliminators that reduce drift to maximum of 0.002% of recirculated water volume for counterflow towers and 0.005% of recirculated water flow for cross-flow towers. Sections shall be assembled into easily removable racks of the same material as the casing. Eliminators can be PVC neoprene honeycomb type.

g. Hot Water Distribution System: Open basin,constructed of Type 304 Stainless Steel, flume and troughs, or a pipe system with nozzles spaced for even distribution of water over fill material. Provide access door. System shall be self-draining and non-clogging. Spray nozzles, if used, shall be cleanable stainless-steel, bronze or high impact plastic, non-clog, removable type properly spaced for even distribution. Provide cover for entire nozzle area or flume/trough area. Provide manufacturer's standard pre-strainer assembly and butterfly or globe valve, for cross flow tower, to balance the water flow to each basin.

h. Cold Water Collection Basin: Casing panels and upper structural members shall be constructed of heavy gauge, zinc-coated or hot-dip galvanized steel. Cold water basin shall be constructed of TYpe 304 Stainless Steel panels and structural members . Overflow drain not less than 2 inches, and a 304 stainless-steel strainer assembly with openings smaller than nozzle orifices and with built-in vortex baffling to prevent cavitation and air entrainment in the water basin circulating pump.

i. Accessories: Bypass, overflow, drain, and equalizer connections (multiple cooling tower systems) .Provide units with air intake screens, perimeter handrails per linear feet, internal walkway, access doors and panels, fan motor space heaters, basin heaters, ladder with safety cage, and sump sweeper piping (high flow eductors). Basin

section shall have two oversized hinged access doors for access into the plenum section.

j. Water Level Control: Electronic operated, 5-probe, with slow closing 120V solenoid valve and NEMA MG 1, Type 4X enclosure. Solid state controls with stainless-steel electrode probes and relays factory wired to a terminal strip to provide control of makeup valve, low and high level alarms and output for shutoff of pump on low level.

k. Fans: Heavy duty, Super Low Sound Fans, axial flow type, belt driven and balanced at the factory after assembly, with FRP blades. Fans shall be driven by variable speed motor. The fan drive and moving parts shall be completely enclosed by removable hot-dip galvanized screens and panels complying with OSHA regulations. Fan shaft bearings of the self-aligning, grease-lubricated ball or roller bearings with moisture proof seals and premium, moisture-resistant grease suitable for temperatures between (minus 20 and plus 300 degrees F). Bearings designed for an L-10 life of 100,000 hours and with extended lubrication lines to an easily accessible location outside of the wet air stream. Provide access doors for inspection and cleaning and vibration cutout switch.

l. Motors and Drives:

(1) The alignment and balancing of the fans, motors and drive shaft as installed shall operate within the vibration limits specified in Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

(2) In addition to the requirements of Section 23 05 11, COMMON WORK RESULTS FOR HVAC, the following shall apply:

(a) Motors: Totally enclosed or epoxy encapsulated NEMA MG 1. Protect fan, bearings, and appurtenances from damage by weather, corrosion, water spray and grit. Provide motors with severe duty rating with the rotor and stator protected with corrosion-inhibiting epoxy resin, double shielded, vacuum-degassed bearings lubricated with premium moisture-resistant grease suitable for temperatures between minus 20 and plus 300 degrees F, and an internal heater automatically energized when motor is de-energized. Provide an adjustable motor base or other suitable provision for adjusting belt tension.

(b) Fans for towers of 100 tons and less shall be belt driven. For towers larger than 100 tons, fan shall be driven through a gear reducer, or driven by a V-belt.

1. Gear Reducer Drive: Specially designed for cooling tower operation, with dynamically balanced drive shaft assembly or shock absorbent flexible coupling requiring no lubrication, cast-iron case with readily accessible oil drum and fill, and self-contained oil reservoir sealed against water entrance.

2. V-Belt Drive: Fan shall be driven by a one-piece, multi-groove, neoprene/polyester belt, where this is the manufacturer’s standard. Belt drives shall be "V" type as specified in Section 23 05 11, COMMON WORK RESULTS FOR HVAC. Belt driven fan and motor shafts shall have taper-lock sheaves fabricated from corrosion resistant material.

(c) Motor Controllers: Provide variable speed motors and controllers, if shown in the contract documents for cooling tower fans. See Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

(d) Lubrication fittings shall be readily accessible outside the wet air stream. Provide access doors for inspection and cleaning.

(e) The alignment and balancing of the fans, motors and drive shaft as installed shall operate within the vibration tolerance specified in Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

m. Fans over 60 inches in diameter include a vibration cutout switch located in a protected position to effectively monitor fan vibration. Vibration switch shall be solid-state with adjustable time delay in NEMA 250, Type 4 enclosure. It shall stop fan motor under excessive fan vibration. Interface the vibration cut-out switch with the DDC control system to provide an alarm in the event the fans stop due to excessive vibration.

n. Safety: Provide fan guards, ladders, handrails, and platform in conformance with the ANSI/ASSE A10.18 as follows:

(1) Fan Guard: Removable fan discharge with a rigid framed screen guard, installed over the fan cylinder.

(2) Ladders: Vertical hot-dip galvanized steel or aluminum ladder for each tower located outdoors. Ladders higher than 12 feet shall have safety cage. Ladders shall extend to within 1 foot of the grade or the roof deck surface.

(3) Hand Railing: Steel or aluminum hand railings not less than 42 inches high around perimeter of each fan-deck, or working surface 12 feet or more above ground, roof or other supporting construction. Handrails shall meet OSHA Standards.

(4) Platform: Galvanized steel with a bar grating floor.

o. Electric Basin Heater: Furnish stainless-steel electric immersion heater installed in a threaded coupling on the side of the basin and with watertight junction boxes mounted in the basin with sufficient capacity to maintain plus 40 degrees F water in the basin at 32 degrees C (0 degrees F) ambient. Provide a NEMA 250, Type 4 mounted on the side of each cooling tower cell with magnetic contactors controlled by a temperature sensor/controller to maintain collection basin water-temperature set point. Provide a water-level probe to monitor cooling tower water level and de-energize the heater when the water reaches low-level set point. Provide a control-circuit transformer with primary and secondary side fuses, terminal blocks with numbered and color-coded wiring to match wiring diagram, Single-point, field-power connection to a fused disconnect switch and heater branch circuiting complying with NFPA 70. Provide a Metal raceway for factory-installed wiring outside of enclosures, except make connections to each electric basin heater with liquid tight conduit.

p. Electric Heat Tracing: Provide in Section 23 21 13, HYDRONIC PIPING.

2.2 COOLING TOWER WATER SYSTEM

Mandatory design requirements and operational standards for the management of cooling towers water systems associated with VHA owned and/or operated facilities/buildings designed as recommended per VHA Engineering Standard VHA ES-2019-01.

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the contractor shall correct the installation at no additional cost or time to the Government.

b. Install cooling tower according to equipment manufacturer’s written instruction.

c. Install cooling towers plumb, level and anchored on structure provided. Coordinate steel structure with cooling tower mounting requirements. If installed on concrete base, refer to Section 03 30 00, CAST-IN-PLACE CONCRETE for concrete materials and installation requirements.

d. Install vibration controls according to manufacturer’s recommendations.

e. Install anchor bolts to elevations required for proper attachment to supported equipment.

f. Maintain manufacturer’s recommended clearances for service and maintenance.

g. Piping:

(1) Install piping, including flanges or union adjacent to cooling towers to allow for service and maintenance.

(2) Install flexible pipe connectors at connections to cooling towers mounted on vibration isolators.

(3) Install shutoff/balancing valves at cooling tower inlet connections.

(4) Install piping adjacent to cooling towers to allow service and maintenance.

(5) Provide drain piping with valve at cooling tower drain connections and at low points in piping.

(6) Connect cooling tower overflows and drains, and piping drains to sanitary sewage system.

(7) Equipment, piping and distribution system components are flushed of debris and disinfected prior to being placed into service.

(8) Documentation of flushing and disinfection shall be maintained.

(9) System piping shall be installed to avoid stagnation or dead legs.

(10) Domestic Water Piping: Comply with applicable requirements in Section 22 11 00, FACILITY WATER DISTRIBUTION. Connect to water-level control with shutoff valve and union, flange, or mechanical coupling at each connection.

(11) Supply and Return Piping: Comply with applicable requirements in Section 23 21 13, HYDRONIC PIPING. Connect to entering cooling tower connections with shutoff valve, balancing valve, thermometer, plugged tee with pressure gauge,flow meter and drain connection with valve. Connect to leaving cooling tower connection with shutoff valve. Make connections to cooling tower with a flange .

(12) Equalizer Piping: Piping requirements to match supply and return piping. Connect an equalizer pipe, full size of cooling tower connection, between tower cells. Connect to cooling tower with shutoff valve.

h. Not used.

i. Electrical Wiring: Install electrical devices, components and accessories furnished loose by manufacturer, including remote flow switches and variable frequency drives. Refer to Section 26 29 11, MOTOR CONTROLLERS.

3.2 WATER TREATMENT SYSTEM INSTALLATION

Biocide-based cooling tower water treatment systems review, design, installation, and operation shall be maintained current. Refer to paragraph CHEMICAL TREATMENT FOR OPEN LOOP SYSTEM(S) of Section 23 25 00, HVAC WATER TREATMENT for biocide concentration levels in the cooling tower water distribution systems.

3.3 FIELD QUALITY CONTROL

a. Provide the services of an independent testing and inspection agency to perform the field tests and inspections of non-CTI certified cooling towers, 200 tons and larger, according to CTI ATC 105 for Cooling Towers. Submit qualification of the independent testing agency to the Contracting Officer 2 weeks prior to the inspection for approval.

b. If the cooling tower does not meet the specified performance, the Contractor shall make the tower corrections necessary to bring the tower into compliance with the specified performance including replacing the tower if necessary. Additional tests will be required until the tower meets the specified performance. Costs for the tower corrections or replacement, and tests shall be borne by the Contractor.

3.4 STARTUP AND TESTING

a. Provide the services of a factory-authorized and qualified representative to perform startup service.

b. Clean entire unit including basin.

c. Inspect field-assembled components and equipment installation, including piping and electrical connections.

d. Verify that accessories are properly installed.

e. Obtain and review performance curves and tables.

f. Perform startup checks, according to manufacturer's written instructions, and as noted below:

(1) Check clearances for airflow and tower servicing.

(2) Check for vibration isolation and structural support.

(3) Verify fan rotation for correct direction and for vibration or binding and correct problems.

(4) Adjust belts to proper alignment and tension.

(5) Lubricate rotating parts and bearings.

(6) Verify proper oil level in gear-drive housing. Fill with oil to proper level.

(7) Operate variable-speed fans through entire operating range and check for harmonic vibration imbalance. Set motor controller to skip speeds resulting in abnormal vibration.

(8) Check vibration switch setting. Verify operation.

(9) Verify operation of basin heater and control.

(10) Operate equipment controls and safeties.

(11) Verify that tower discharge is high enough and it does not recirculate into HVAC air intakes. Recommend corrective action.

g. Adjust water level for operating level and balance condenser water flow to each tower inlet.

h. Check water treatment water system, including blow down for proper operation of the tower. Refer to Section 23 25 00, HVAC WATER TREATMENT. Check makeup water-level control and valve.

i. Start cooling tower, including condenser water pumps and verify the tower operation.

j. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

k. Prepare and submit a written report of startup and inspection service to the Contracting Officer.

l. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

m. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

3.5 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.6 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hour s to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

-- End of Section --

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SECTION 23 73 00
3/23

INDOOR CENTRAL-STATION AIR-HANDLING UNITS

PART 1 GENERAL

1.1 DESCRIPTION

a. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Air handling units including integral components specified herein.

c. Definition of Air Handling Unit (AHU): A factory-fabricated and tested assembly of modular sections consisting of single or multiple plenum fans with direct-drive, coils, filters, and other necessary equipment to perform one or more of the following functions of circulating, cleaning, heating, cooling, humidifying, dehumidifying, and mixing of air. Design capacities of units shall be as scheduled in the contract documents.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Section 09 91 00, PAINTING.

f. Not used.

g. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

h. Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC AND STEAM GENERATION EQUIPMENT.

i. Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT.

j. Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, AND BALANCING FOR HVAC.

k. Section 23 07 11, HVAC AND BOILER PLANT INSULATION.

l. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

m. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

n. Section 23 21 13, HYDRONIC PIPING.

o. Section 23 22 13, STEAM AND CONDENSATE HEATING PIPING.

p. Section 23 31 00, HVAC DUCTS and CASINGS.

q. Section 23 34 00, HVAC FANS.

r. Section 23 40 00, HVAC AIR CLEANING DEVICES.

s. Section 23 82 16, AIR COILS.

t. Section 26 29 11, MOTOR CONTROLLERS.

1.3 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standards will govern.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 260	(2012) Sound Rating of Ducted Air Moving and Conditioning Equipment
AHRI 410	(2001; Addendum 1 2002; Addendum 2 2005; Addendum 3 2011) Forced-Circulation Air-Cooling and Air-Heating Coils
AHRI 430	(2009) Central-Station Air-Handling Units

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 99	(2016) Standards Handbook
AMCA 210	(2016) Laboratory Methods of Testing Fans for Aerodynamic Performance Rating

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 51	(2016) Laboratory Methods of Testing Fans for Aerodynamic Performance Rating

ASTM INTERNATIONAL (ASTM)

ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1972 CD	(2012) HVAC Air Duct Leakage Test Manual - 2nd Edition

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-P-21035	(1991; Rev B; Notice 2 2003; Notice 3 2021) Paint, High Zinc Dust Content,

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

PL 109-58 Energy Policy Act of 2005 (EPAct05)

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submittals with an "S" are for inclusion in the Sustainability Notebook. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Shipping Information; G, AE

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Bio-Based Materials; S, G, AE

As-Built Documentation; S, G, AE

SD-06 Test Reports

Written Test Procedures; G, AE

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Air Handling Units Certification; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 XX XX, SECTION TITLE”, with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Submittals for AHUs shall include fans, drives, motors, coils,

sound attenuators, mixing box with outside/return air dampers, filter housings, blender sections, and all other related accessories. The contractor shall provide custom drawings showing total air handling unit assembly including dimensions, operating weight, access sections, flexible connections, door swings, controls penetrations, electrical disconnect, lights, duplex receptacles, switches, wiring, utility connection points, unit support system, vibration isolators, drain pan, pressure drops through each component (filter, coil etc).

(2) Submittal drawings of section or component only will not be acceptable. Contractor shall also submit performance data including performance test results, charts, curves, or certified computer selection data; data sheets; fabrication and insulation details. If the unit cannot be shipped in one piece, the contractor shall indicate the number of pieces that each unit will have to be broken into to meet shipping and job site rigging requirements.

(3) Submit sound power levels in each octave band for the inlet and discharge of the fan and at entrance and discharge of AHUs at scheduled conditions. In absence of sound power ratings refer to Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT.

(4) Provide fan curves showing Liters/Second (cubic feet per minute), static pressure, efficiency, and horsepower for design point of operation and at maximum design Liters/Second (cubic feet per minute).

(5) Submit total fan static pressure, external static pressure, for AHU including total, inlet and discharge pressures, and itemized specified internal losses and unspecified internal losses. Refer to air handling unit schedule in the contract documents.

d. Submit written test procedures two weeks prior to factory testing. Submit written results of factory tests for approval prior to shipping.

e. Submit shipping information that clearly indicates how the units will be shipped in compliance with the descriptions below.

(1) Units shall be shipped in one (1) piece where possible and in shrink wrapping to protect the unit from dirt, moisture and/or road salt.

(2) If not shipped in one (1) piece, provide manufacturer approved shipping splits where required for installation or to meet shipping and/or job site rigging requirements in modular sections. Indicate clearly that the shipping splits shown in the submittals have been verified to accommodate the construction constraints for rigging as required to complete installation and removal of any section for replacement through available access without adversely affecting other sections.

(3) If shipping splits are provided, each component shall be individually shrink wrapped to protect the unit and all necessary hardware (e.g., bolts, gaskets etc.) will be included to assemble unit onsite.

(4) Lifting lugs will be provided to facilitate rigging on shipping splits and joining of segments. If the unit cannot be shipped in one piece, the contractor shall indicate the number of pieces that each unit will have to be broken into to meet shipping and job site rigging requirements.

f. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include instructions for lubrication, filter replacement, motor and drive replacement, and spare parts list.

(3) Include complete diagrams of the internal wiring for each item of equipment.

(4) Diagrams shall have their terminals identified to facilitate installation, operation, and maintenance.

g. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician, and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

h. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.6 QUALITY ASSURANCE

a. Comply with requirements in paragraph QUALITY ASSURANCE of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Air Handling Units Certification:

(1) Air Handling Units with Plenum Fans:

(a) Air Handling Units with a single Plenum Fan shall be certified in accordance with AHRI 430 and tested/rated in accordance with AHRI 260.

(b) Air handling Units with Multiple Fans in an Array shall be tested and rated in accordance with AHRI 430 and AHRI 260.

c. Heating, Cooling, and Air Handling Capacity and Performance Standards: AHRI 430, AHRI 410, ASHRAE 51, and AMCA 210.

d. Performance Criteria:

(1) The fan BHP shall include all system effects for all fans and v-belt drive losses for housed centrifugal fans.

(2) The fan motor shall be selected within the rated nameplate capacity, without relying upon NEMA Standard Service Factor.

(3) Select the fan operating point as follows:

(a) Forward Curve and Axial Flow Fans: Right hand side of peak pressure point.

(b) Air Foil, Backward Inclined, or Tubular Fans Including Plenum Fans: At or near the peak static efficiency but at an appropriate distance from the stall line.

(4) Operating Limits: AMCA 99 and manufacturer's recommendations.

e. Units shall be factory-fabricated, assembled, and tested by a manufacturer, in business of manufacturing similar air-handling units for at least five years.

f. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

g. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 2 – PRODUCTS

2.1 AIR HANDLING UNITS

a. General:

(1) AHUs shall be fabricated from insulated, solid double-wall galvanized steel without any perforations in draw-through configuration. Galvanizing shall be hot dipped conforming to ASTM A653/A653M and shall provide a minimum of 0.97 oz. of zinc per square meter (G90). Aluminum constructed units, subject to VA approval, may be used in place of galvanized steel. The unit manufacturer shall provide published documentation confirming that the structural rigidity of aluminum air-handling units is equal or greater than the specified galvanized steel.

(2) The contractor and the AHU manufacturer shall be responsible for ensuring that the unit will not exceed the allocated space shown in the contract documents, including required clearances for service and future overhaul or removal of unit components. All structural, piping, wiring, and ductwork alterations of units, which are dimensionally different than those specified, shall be the responsibility of the contractor at no additional cost or time to the Government.

(3) AHUs shall be fully assembled by the manufacturer in the factory in accordance with the arrangement shown in the contract documents. The unit shall be assembled into the largest sections possible subject to shipping and rigging restrictions. The

correct fit of all components and casing sections shall be verified in the factory for all units prior to shipment. All units shall be fully assembled, tested, and then split to accommodate shipment and job site rigging. On units not shipped fully assembled, the manufacturer shall tag each section and include air flow direction to facilitate assembly at the job site. Lifting lugs or shipping skids shall be provided for each section to allow for field rigging and final placement of unit.

(4) The AHU manufacturer shall provide the necessary gasketing, caulking, and all screws, nuts, and bolts required for assembly. The manufacturer shall provide a factory-trained and qualified local representative at the job site to supervise the assembly and to assure that the units are assembled to meet manufacturer's recommendations and requirements noted in the contract documents. Provide documentation to the Contracting Officer that the local representative has provided services of similar magnitude and complexity on jobs of comparable size. If a local representative cannot be provided, the manufacturer shall provide a factory representative.

(5) Gaskets: All door and casing and panel gaskets and gaskets between air handling unit components, if joined in the field, shall be high quality which seal airtight and retain their structural integrity and sealing capability after repeated assembly and disassembly of bolted panels and opening and closing of hinged components. Bolted sections may use a more permanent gasketing method provided they are not disassembled.

(6) Structural Rigidity: Provide structural reinforcement when required by span or loading so that the deflection of the assembled structure shall not exceed 1/200 of the span based on a differential static pressure of 8 inch WG or higher.

b. Base:

(1) Provide a heavy duty steel base for supporting all major AHU components. Bases shall be constructed of wide-flange steel I-beams, channels, or minimum 5 inch high10 Gauge steel base rails. Welded or bolted cross members shall be provided as required for lateral stability. Contractor shall provide supplemental steel supports as required to obtain proper operation heights for cooling coil condensate drain trap as shown in the contract documents.

(2) AHUs shall be completely self-supporting for installation on concrete housekeeping pad, steel support pedestals, or suspended as shown in the contract documents.

(3) The AHU bases not constructed of galvanized steel shall be cleaned, primed with a rust inhibiting primer, and finished with rust inhibiting exterior enamel.

c. Casing (Including Wall, Floor, and Roof):

(1) General: AHU casing shall be constructed as solid double wall, galvanized steel insulated panels without any perforations, integral of or attached to a structural frame. The thickness of insulation, mode of application and thermal breaks shall be such

that there is no visible condensation on the exterior panels of the AHU located in the non-conditioned spaces.

(2) Casing Construction:

Outer Panel	22 Gauge Minimum
Inner Panel	22 Gauge Minimum
Insulation Thickness Density	Foam 2 inch Minimum 33.0 lb/ft3 Minimum
Total R Value	13.0 ft2.OF.hr/Btu Minimum

Not used.

(4) Blank-Off: Provide blank-offs as required to prevent air bypass between the AHU sections, around coils, and filters.

(5) Casing panels shall be secured to the support structure with stainless-steel or zinc-chromate plated screws and gaskets installed around the panel perimeter. Panels shall be completely removable to allow removal of fan, coils, and other internal components for future maintenance, repair, or modifications. Welded exterior panels are prohibited.

(6) Access Doors: Provide in each access section and where shown in the contract documents. Show single-sided and double-sided access doors with door swings on the floor plans. Doors shall be a minimum of2 inch thick with same double wall construction as the unit casing. Doors shall be a minimum of24 inches wide, unless shown of different size in the contract documents, and shall be the full casing height up to a maximum of 6 feet. Doors shall be gasketed, hinged, and latched to provide an airtight seal. The access doors for fan section, mixing box, coil section shall include a minimum 6 inch x 6 inch double thickness, with air space between the glass panes tightly sealed, reinforced glass or Plexiglas window in a gasketed frame.

(a) Hinges: Manufacturer’s standard, designed for door size, weight and pressure classifications. Hinges shall hold door completely rigid with minimum100 lb weight hung on latch side of door.

(b) Latches: Non-corrosive alloy construction, with operating levers for positive cam action, operable from either inside or outside. Doors that do not open against unit operating pressure shall allow the door to ajar and then require approximately 45 degrees further movement of the handle for complete opening. Latch shall be capable of restraining explosive opening of door with a force not less than 8 inch WG.

(c) Gaskets: Neoprene, continuous around door, positioned for direct compression with no sliding action between the door and gasket. Secure with high quality mastic to eliminate possibility of gasket slipping or coming loose.

(7) Provide sealed sleeves, metal or plastic escutcheons or grommets for penetrations through casing for power and temperature control wiring and pneumatic tubing. Coordinate with electrical and temperature control subcontractors for number and location of penetrations. Coordinate lights, switches, and duplex receptacles and disconnect switch location and mounting. All penetrations and equipment mounting may be provided in the factory or in the field. All field penetrations shall be performed neatly by drilling or saw cutting. No cutting by torches will be allowed. Neatly seal all openings airtight.

e. Floor:

(1) Unit floor shall be level without offset space or gap and designed to support a minimum of 91 lbs per square foot distributed load without permanent deformation or crushing of internal insulation. Provide adequate structural base members beneath floor in service access sections to support typical service foot traffic and to prevent damage to unit floor or internal insulation. Unit floors in casing sections, which may contain water or condensate, shall be watertight with drain pan.

(2) Where indicated, furnish and install floor drains flush with the floor, with nonferrous grate cover and stub through floor for external connection.

f. Condensate Drain Pan: Drain pan shall be designed to extend entire length of cooling coils including headers and return bends. Depth of drain pan shall be at least1.7 inches and shall handle all condensate without overflowing. Drain pan shall be double-wall, double sloping type, and fabricated from stainless-steel (304) with at least2 inch thick insulation sandwiched between the inner and outer surfaces. Drain pan shall be continuous metal or welded watertight. No mastic sealing of joints exposed to water will be permitted. Drain pan shall be placed on top of casing floor or integrated into casing floor assembly. Drain pan shall be pitched in all directions to drain line.

(1) An intermediate, stainless-steel (304) condensate drip pan with copper downspouts shall be provided on stacked cooling coils. Use of intermediate condensate drain channel on upper casing of lower coil is permissible provided it is readily cleanable. Design of intermediate condensate drain shall prevent upper coil condensate from flowing across face of lower coil.

(2) Drain pan shall be piped to the exterior of the unit. Drain pan shall be readily cleanable.

(3) Installation, including frame, shall be designed and sealed to prevent blow-by.

i. Plenum Fans – Single and/or Multiple Fans in an Array:

(1) General: Fans shall be Class II (minimum) construction with single inlet, aluminum wheel and stamped air-foil aluminum bladed. The fan wheel shall be mounted on the directly-driven

motor shaft in AMCA Arrangement 4. Fans shall be dynamically balanced and internally isolated to minimize the vibrations. Provide a steel inlet cone for each wheel to match with the fan inlet. Locate fan in the air stream to assure proper flow. The fan performance shall be rated in accordance with AMCA 210 or ASHRAE 51.

(2) Allowable vibration tolerances for fan shall not exceed a self-excited vibration maximum velocity of 0.20 inch per second) RMS, filter in, when measured with a vibration meter on bearing caps of machine in vertical, horizontal, and axial directions or measured at equipment mounting feet if bearings are concealed. After field installation, compliance to this requirement shall be demonstrated with field test in accordance with Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT and Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, AND BALANCING FOR HVAC. Following fan assembly, the complete fan assembly balance shall be tested using an electronic balance analyzer with a tunable filter and stroboscope. Vibration measurements shall be taken on each motor bearing housing in the vertical, horizontal, and axial planes (5 total measurements, 2 each motor bearing and 1 axial).

(3) The plenum fans shall be driven by variable speed drives. Use of a drive with bypass is prohibited.

(4) Multiple fans shall be installed in a pre-engineered structural frame to facilitate fan stacking. All fans shall modulate in unison, above or below the synchronous speed within the limits specified by the manufacturer, by a common control sequence. Staging of the fans is prohibited. Redundancy requirement shall be met by all operating fans in an array and without the provision of an idle standby fan.

(5) Fan Accessories:

(a) Fan Isolation: Provide a backdraft damper to isolate the fan not in operation due to failure or as required in the control sequence.

(b) Fan Airflow Measurement: Provide an airflow measuring device integral to the fan to measure air volume within +/- 5 percent accuracy. The probing device shall not be placed in the airflow path to stay clear of turbulence and avoid loss of performance.

j. Fan Motor and Drive (Plenum Fans): Motors shall be premium energy efficient type, as mandated by the PL 109-58 Energy Policy Act of 2005, with efficiencies as shown in Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC AND STEAM GENERATION EQUIPMENT, in the contract documents and suitable for use in variable frequency drive applications. Refer to Section 23 05 11, COMMON WORK RESULTS FOR HVAC, for additional motor and drive specifications. Refer to Section 26 29 11, MOTOR CONTROLLERS.

l. Mixing Boxes: Mixing box shall consist of casing and outdoor air and return air dampers in opposed blade arrangement with damper linkage for automatic operation. Unit serving the control room does not require a mixing box. Coordinate damper operator with Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC. Dampers shall be of ultra-low

leak design with metal compressible bronze jamb seals and extruded vinyl edge seals on all blades or lowest leak design offered by manufacturer. Blades shall rotate on stainless-steel sleeve bearings or bronze bushings. Leakage rate shall not exceed 5 CFM per square foot at 1 inch WG and 9 CFM per square foot at 4 inch WG. operators shall be furnished and mounted in an accessible and easily serviceable location by the air handling unit manufacturer at the factory. Damper operators shall be of same manufacturer as controls furnished under Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

m. Not used.

n. Filter Section: Refer to Section 23 40 00, HVAC AIR CLEANING DEVICES, for filter requirements.

(1) Filters including one complete set for temporary use at site shall be provided independent of the AHU. The AHU manufacturer shall install filter housings and racks in filter section compatible with filters furnished. The AHU manufacturer shall be responsible for furnishing temporary filters (pre-filters and after-filters, as shown in the contract documents) required for AHU testing.

(2) Factory-fabricated filter section shall be of the same construction and finish as the AHU casing including filter racks and hinged double wall access doors. Filter housings shall be constructed in accordance with side service or holding frame housing requirements in Section 23 40 00, HVAC AIR CLEANING DEVICES.

p. Coils: Coils shall be mounted on hot dipped galvanized steel supports to assure proper anchoring of coil and future maintenance. Coils shall be face or side removable for future replacement thru the access doors or removable panels. Each coil shall be removable without disturbing adjacent coil. Cooling coils shall be designed and installed to ensure no condensate carry over. Provide factory-installed extended supply, return, drain, and vent piping connections. All coils in the high humidity areas, shall be epoxy-coated as shown below. Refer to the contract documents and Section 23 82 16, AIR COILS for additional coil requirements.

(1) Epoxy Immersion Coating – Electrically Deposited: The multi-stage corrosion-resistant coating application comprises of cleaning (heated alkaline immersion bath) and reverse-osmosis immersion rinse prior to the start of the coating process. The coating thickness shall be maintained between 0.6-mil and 1.2-mil. Before the coils are subjected to high-temperature oven cure, they are treated to permeate immersion rinse and spray. Where the coils are subject to UV exposure, UV protection spray treatment comprising of UV-resistant urethane mastic topcoat shall be applied. Provide complete coating process traceability for each coil and minimum five years of limited warranty.

(2) The coating process shall such that uniform coating thickness is maintained at the fin edges. The quality control shall be maintained by ensuring compliance to the applicable ASTM Standards for the following:

(a) Salt Spray Resistance (Minimum 6,000 Hours)

(b) Humidity Resistance (Minimum 1,000 Hours)

(c) Water Immersion (Minimum 260 Hours)

(d) Cross-Hatch Adhesion (Minimum 4B-5B Rating)

(e) Impact Resistance (Up to 160 Inch/Pound)

(3) Water coils .

q. Humidifier: When included in design, coordinate the humidification requirements with Section 23 22 13 STEAM AND CONDENSATE HEATING PIPING. Provide air-handling unit-mounted humidification section with stainless-steel drain pan of adequate length to allow complete absorption of water vapor. Provide stainless-steel dispersion panel or distributors as indicated, with stainless-steel supports and hardware.

r. Not used.

s. Discharge Section: Provide aerodynamically designed framed discharge openings or spun bellmouth fittings to minimize pressure loss.

t. Electrical and Lighting: Wiring and equipment specifications shall conform to Division 26, ELECTRICAL.

(1) Vapor-proof lights using cast aluminum base style with glass globe and cast aluminum guard shall be installed in access sections for fan, mixing box, and any section over 12 inch wide. A switch shall control the lights in each compartment with pilot light mounted outside the respective compartment access door. Wiring between switches and lights shall be factory-installed. All wiring shall run in neatly installed electrical conduits and terminate in a junction box for field connection to the building system. Provide single point 115 volt - one phase connection at junction box.

(2) Install compatible LED bulb in each light fixture.

(3) Provide a convenience duplex weatherproof receptacle next to the light switch.

(4) Disconnect switch and power wiring: Provide factory or field mounted disconnect switch. Coordinate with Division 26, ELECTRICAL.

(5) Provide single point power connection. Coordinate with Division 26, ELECTRICAL.

PART 3 – EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the contractor shall correct the installation at no additional cost or time to the Government.

b. Install air handling unit in conformance with AHRI 430.

c. Assemble air handling unit components following manufacturer's instructions for handling, testing and operation. Repair damaged galvanized areas with paint in accordance with MIL-P-21035. Repair painted units by touch up of all scratches with finish paint material. Vacuum the interior of air handling units clean prior to operation.

d. Install seismic restraints for roof top units.

e. Leakage and test requirements for air handling units shall be the same as specified for ductwork in Section 23 31 00, HVAC DUCTS AND CASINGS except leakage shall not exceed Leakage Class (CL) 12 listed in SMACNA 1972 CD HVAC Air Duct Leakage Test Manual when tested at 1.5 times the design static pressure. Repair casing air leaks that can be heard or felt during normal operation and to meet test requirements.

f. Perform field mechanical (vibration) balancing in accordance with Section 23 05 41, NOISE AND VIBRATION CONTROL FOR HVAC PIPING AND EQUIPMENT.

g. Seal and/or fill all openings between the casing and AHU components and utility connections to prevent air leakage or bypass.

3.2 STARTUP AND TESTING

a. The air handling unit shall not be operated for any purpose, temporary or permanent, until ductwork is clean, filters are in place, bearings are lubricated, and the fan has been test run under observation.

b. After the air handling unit is installed and tested, provide startup and operating instructions to VA personnel.

c. An authorized factory representative should startup, test, and certify the final installation and application specific calibration of control components. Items to be verified include fan performance over entire operating range, noise and vibration testing, verification of proper alignment, overall inspection of the installation, Owner/Operator training, etc.

d. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

e. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

f. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Provide a minimum notice of 10 working days prior to startup and testing.

3.3 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING

OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.4 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hour s to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

-- End of Section --

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SECTION 23 81 00
3/24

DECENTRALIZED UNITARY HVAC EQUIPMENT

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies self-contained, split-systems.

b. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

c. Definitions:

(1) Energy Efficiency Ratio (EER): The ratio of net cooling capacity is Btu/h to total rate of electricity input in watts under designated operating conditions (Btu hour/Watt).

(2) Seasonal Energy Efficiency Ratio (EER): The ratio of the total cooling output of an air conditioner during its normal annual usage period for cooling in Btu/h divided by total electric energy input in watts during the same period (Btu hour/).

(3) Unitary: A Unitary Air Conditioner consists of one or more factory-made assemblies which normally include an evaporator or cooling coil, a compressor and condenser combination, and may include a heating function as well.

(4) Where such equipment is provided in greater than one assembly the separated assemblies are to be designed to be used together and the requirements of rating are based upon use of matched assemblies.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 03 30 00, CAST-IN-PLACE CONCRETE.

e. Section 07 72 00, ROOF ACCESSORIES.

f. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

g. Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC and STEAM GENERATION EQUIPMENT.

h. Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

i. Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, and BALANCING FOR HVAC.

j. Section 23 07 11, HVAC and BOILER PLANT INSULATION.

k. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

l. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

m. Section 23 31 00, HVAC DUCTS and CASINGS.

n. Section 23 73 00, INDOOR CENTRAL-STATION AIR-HANDLING UNITS.

o. Section 23 82 16, AIR COILS.

p. Section 28 31 00, FIRE DETECTION and ALARM.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer’s Literature And Data; G, AE

SD-05 Design Data

Performance Rating; G, AE

SD-06 Test Reports

Completed System Readiness Checklists; G, AE

SD-07 Certificates

Certification; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manual; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.4 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with specification Section 01 33 00, SUBMITTAL PROCEDURES

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 81 00, DECENTRALIZED UNITARY HVAC EQUIPMENT”, with applicable paragraph identification.

c. Manufacturer’s literature and data Including:

(1) Sufficient information, including capacities, pressure drops, and piping connections clearly presented, shall be included to

determine compliance with contract documents for units noted below:

(a) Unitary air conditioners:

1. Self-contained units

2. Split systems

(2) Unit Dimensions required clearances, operating weights accessories and start-up instructions.

(3) Electrical requirements, wiring diagrams, interlocking and control wiring showing factory installed and portions to be field installed.

(4) Mounting and flashing of the roof curb to the roofing structure with coordinating requirements for the roof membrane system.

d. Certification: Submit proof of specified ARI Certification.

e. Performance Rating: Submit catalog selection data showing equipment ratings and compliance with required sensible-to-heat-ratio, integrated energy efficiency ratio (IEER), and coefficient of performance (COP).

f. Operating and Maintenance Manual: Submit three copies of Operating and Maintenance manual to Contracting Officer three weeks prior to final inspection.

g. Completed System Readiness Checklists provided by the CxA and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.5 QUALITY ASSURANCE

a. Refer to specification Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Safety Standards: ANSI/ASHRAE 15 & 34, Safety Code for Mechanical Refrigeration.

1.6 REFERENCES

(1) Make material requirements agree with requirements specified in the referenced Applicable Publications. Verify and update the publication list to that which applies to the project, unless the reference applies to all mechanical systems Publications that apply to all mechanical systems may not be specifically referenced in the body of the specification but shall form a part of this specification.

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standards will govern.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

ANSI/AHRI 210/240	(2008; Add 1 2011; Add 2 2012) Performance Rating of Unitary Air-Conditioning & Air-Source Heat Pump Equipment
AHRI 340/360	(I-P 2019)Performance Rating of Commercial and Industrial Unitary Air Conditioning and Heat Pump Equipment

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 410	(1996) Recommended Safety Practices for Users and Installers of Industrial and Commercial Fans

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 90.1 - SI	(2019) Energy Standard for Buildings Except Low-Rise Residential Buildings
ANSI/ASHRAE 15 & 34	(2022) ASHRAE Standard 15-Safety Standard for Refrigeration Systems and ANSI/ASHRAE Standard 34-Designation and Safety Classification of Refrigerants
ASHRAE 52.2	(2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size
ASHRAE 62.1	(2016) Ventilation for Acceptable Indoor Air Quality

ASTM INTERNATIONAL (ASTM)

ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in Paragraph “AS-BUILT DOCUMENTATION” in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 2 PRODUCTS

2.1 UNITARY AIR CONDITIONERS - GENERAL

a. Applicable ARI Standards:

(1) Cooling Capacity 135,000 Btu/hour and greater: AHRI 340/360.

(2) Cooling Capacity Less Than 135,000 Btu/hour: ANSI/AHRI 210/240. Units shall be listed in the ARI Directory of Certified Unitary Air-Conditioners.

b. Performance Rating: Cooling capacity of units shall meet the sensible heat and total heat requirements shown in the contract documents. In selecting unit size, make true allowance for "sensible to total heat

ratio" to satisfy required sensible cooling capacity.

c. Machinery Guards: Provide guards as shown in AMCA 410 for belts, chains, couplings, pulleys, sheaves, shafts, gears and other moving parts regardless of height above the floor. Drive guards may be excluded where motors and drives are inside factory fabricated casings.

d. Corrosion Prevention: Unless specified otherwise, equipment fabricated from ferrous metals that do not have a zinc coating or a duplex coating of zinc and paint shall be treated for prevention of rust with a factory coating or paint system that will withstand 125 hours in a salt-spray fog test, except that equipment located outdoors shall be tested for 500 hours. The salt-spray fog test shall be in accordance with ASTM B117 using a 20 percent sodium chloride solution. Immediately after completion of the test, the coating shall show no signs of blistering, wrinkling or cracking, no loss of adhesion, and the specimen shall show no signs of rust beyond 1/8-inch on both sides from the scratch mark. Provide factory-coated coils for protection from corrosion by using multiple stage electro-deposition coating process.

2.2 SPLIT-SYSTEM AIR CONDITIONERS

a. Description: Factory assembled and tested, wall-mounted unit, with an air- cooled remote condensing unit, and field-installed refrigeration piping. Unit shall include a heating coil.

d. Wall-Mounting, Evaporator-Fan Components:

(1) Cabinet: Enameled steel with removable panels on front and ends in color selected by Architect, and discharge drain pans with drain connection.

(2) Airstream Surfaces: Surfaces in contact with the airstream shall comply with requirements in ASHRAE 62.1.

(3) Drain Pan and Drain Connection: Comply with ASHRAE 62.1-2007.

(4) Refrigerant Coil: Copper tube, with mechanically bonded aluminum fins, complying with ANSI/AHRI 210/240, and with thermal-expansion valve.

(5) Electric-Resistance Heating Coil: Helical, nickel-chrome, resistance-wire heating elements with refractory ceramic support bushings; automatic-reset thermal cutout; built-in magnetic contactors; manual-reset thermal cutout; airflow proving device; and one-time fuses in terminal box for overcurrent protection.

(6) Fan: Direct drive, centrifugal fan.

(7) Fan Motors: Comply with requirements in Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC and STEAM GENERATION EQUIPMENT for multi-tapped, multi-speed motors with internal thermal protection and permanent lubrication.

(8) Filters: Disposable, with MERV rating of 7 or higher according to ASHRAE 52.2.

e. Ceiling-Mounting, Evaporator-Fan Components:

(1) Cabinet: Enameled steel with removable panels on front and ends in color selected by Architect, and discharge drain pans with drain connection.

(2) Airstream Surfaces: Surfaces in contact with the airstream shall comply with requirements in ASHRAE 62.1.

(3) Drain Pan and Drain Connection: Comply with ASHRAE 62.1.

(4) Refrigerant Coil: Copper tube, with mechanically bonded aluminum fins, complying with ANSI/AHRI 210/240, and with thermal-expansion valve.

(5) Electric-Resistance Heating Coil: Helical, nickel-chrome, resistance-wire heating elements with refractory ceramic support bushings; automatic-reset thermal cutout; built-in magnetic contactors; manual-reset thermal cutout; airflow proving device; and one-time fuses in terminal box for overcurrent protection.

(6) Fan: Direct drive, centrifugal fan, and integral condensate pump.

(7) Fan Motors: Comply with requirements in Section 23 05 12, GENERAL MOTOR REQUIREMENTS FOR HVAC and STEAM GENERATION EQUIPMENT for multi-tapped, multi-speed motors with internal thermal protection and permanent lubrication.

(8) Filters: Disposable, with MERV rating of 7 or higher according to ASHRAE 52.2.

f. Air-Cooled, Compressor-Condenser Components:

(1) Casing: Steel, finished with baked enamel in color selected by Architect, with removable panels for access to controls, weep holes for water drainage, and mounting holes in base. Service valves, fittings, and gauge ports shall be brass and located outside of the casing.

(2) Compressor: Hermetically sealed reciprocating scroll with crankcase heater and mounted on vibration isolation. Compressor motor shall have thermal- and current-sensitive overload devices, start capacitor, relay, and contactor.

(3) Compressor motor with manual-reset, high-pressure switch and automatic-reset, low-pressure switch.

(4) Refrigerant: Use refrigerant type R-454B as listed in https://www.epa.gov/snap/. Submit selected refrigerant for government approval.

(5) Refrigerant Coil: Copper tube, with mechanically bonded aluminum fins, complying with ANSI/AHRI 210/240, and with liquid subcooler.

(6) Fan: Aluminum, propeller type, directly connected to motor.

(7) Motor: Permanently lubricated, with integral thermal-overload protection.

(8) Low Ambient Kit: Permit operation down to minus 0 degrees F.

(9) Mounting Base: Polyethylene.

(10) Minimum Energy Efficiency: Comply with ASHRAE 90.1 - SI "Energy Standard for Buildings except Low-Rise Residential Buildings."

PART 3 EXECUTION

3.1 INSTALLATION

a.

b. Rooftop Unit Support: Install unit level on structural curbs . Coordinate wall penetrations and flashing with wall construction. Secure rooftop units to structural support with anchor bolts.

c. Install wind and seismic restraints according to manufacturer's written instructions. Wind and seismically restrained vibration isolation roof-curb rails are specified in Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

d. Install units level and plumb maintaining manufacturer’s recommended clearances and tolerances.

e. Install water-cooled units with thermometer and pressure gauge at the water supply and return connection.

f. Install vibration spring isolators under base of self-contained unit, with minimum static deflection of 1 inch unless otherwise indicated. Refer to Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT

g. Install ground-mounting, compressor-condenser components on4-inch thick, reinforced concrete base; 4 inches greater on each side than unit. Concrete, reinforcement, and formwork are specified in Section 03 30 00, CAST-IN-PLACE CONCRETE. Coordinate anchor installation with concrete base.

h. Install ground-mounting, compressor-condenser components on polyethylene mounting base.

i. Install roof-mounting compressor-condenser components on equipment supports specified in Section 07 72 00, ROOF ACCESSORIES. Anchor units to supports with removable, cadmium-plated fasteners.

j. Install seismic restraints.

k. Install compressor-condenser components on restrained, spring isolators with a minimum static deflection of 1 inch unless otherwise indicated. Refer to Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

l. Install and connect precharged refrigerant tubing to component's

quick-connect fittings. Install tubing to allow access to unit.

m. Install wall sleeves in finished wall assembly and weatherproof. Install and anchor wall sleeves to withstand, without damage seismic forces as required by code.

3.2 CONNECTIONS

a. Verify condensate drainage requirements.

b. Install condensate drain, minimum connection size, with trap and indirect connection to nearest roof drain or area drain.

c. Install piping adjacent to units to allow service and maintenance.

d.

e. Install ducts to termination at top of roof curb. Cut roof decking only as required for passage of ducts. Do not cut out decking under entire roof curb.

f. Connect supply ducts to units with flexible duct connectors specified in Section 23 31 00, HVAC DUCTS and CASINGS.

g. Install return-air duct continuously through roof structure.

h. Install normal-weight, 3000-psi, compressive strength (28-day) concrete mix inside roof curb, 4 inches thick.

i. Ground equipment and install power wiring, switches, and controls for self-contained and split systems.

j. Connect refrigerant piping to coils with shutoff valves on the suction and liquid lines at the coil and a union or flange at each connection at the coil and condenser.

k. Install ducts to the units with flexible duct connections.

l. Connect piping with shutoff duty valves on the supply and return side of the coil and unions at all connections and with a throttling valve on the return piping near the coil.

m. Connect piping with shutoff duty valves on the supply and return side of the water-cooled condenser and unions at all connections and with a throttling valve on the return piping near the condenser

n. Connect piping with shutoff duty valves and unions on the steam supply and condensate side of the steam coil. On the condensate line near the coil provide a strainer, trap and shutoff valve.

3.3 FIELD QUALITY CONTROL

a. Perform tests and inspections and prepare test reports.

b. Tests and Inspections: After installing units and after electrical circuitry has been energized, test units for compliance with requirements. Inspect for and remove shipping bolts, blocks, and tie-down straps. After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation. Test and

adjust controls and safeties. Replace damaged and malfunctioning controls and equipment. Remove and replace malfunctioning units and retest as specified above.

3.4 STARTUP AND TESTING

a. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Provide a minimum of seven days prior notice.

b. Provide services of manufacturer’s technical representative for four hours to instruct VA personnel in operation and maintenance of units.

3.5 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS for all inspection, start up, and contractor testing required above and required by the System Readiness Checklist provided by the CxA.

b. Components provided under this section of the specification will be tested as part of a larger system. Refer to Section 23 08 00.01, DVA/USACE PROJEC COMMISSIONING OF HVAC SYSTEMS and related sections for contractor responsibilities for system commissioning.

3.6 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hours to instruct VA personnel in operation and maintenance of units.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

-- End of Section --

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 81 43

AIR-SOURCE UNITARY HEAT PUMPS

3/23

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SECTION 23 81 43
3/23

AIR-SOURCE UNITARY HEAT PUMPS

PART 1 GENERAL

1.1 DESCRIPTION

a. This Section specifies electrically operated air-source unitary heat pumps.

b. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

c. Definitions:

(1) Coefficient of Performance (COP) - Cooling: The ratio of the rate of heat removed to the rate of energy input in consistent units, for a complete refrigerating system or some specific portion of that system under designated operating conditions.

(2) Coefficient of Performance (COP) - Heating: The ratio of the rate of heat delivered to the rate of energy input is consistent units for a complete heat pump system, including the compressor and, if applicable, auxiliary heat under designated operating conditions.

(3) Energy Efficiency Ratio (EER): The ratio of net cooling capacity is Btu/h to total rate of electricity input in watts under designated operating conditions.

(4) Heating Seasonal Performance Factor (HSPF) - Total heating output of heat pump during its normal annual usage period for heating in Btu/h divided by total electric energy input in watts during the same period.

(5) Seasonal Energy Efficiency Ratio (SEER) - Total cooling output of an air conditioner during its normal annual usage period for cooling in Btu/h divided by total electric energy input in watts during the same period.

(6) Air-Source Unitary Heat Pump: One or more factory made assemblies that normally include an indoor conditioning coil, compressor(s) and an outdoor refrigerant-to-air coil. These units provide both heating and cooling functions.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01 – DVA/USACE PROJECTS GENERAL COMMISSIONING

REQUIREMENTS.

e. Not used.

f. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

g. Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, AND BALANCING FOR HVAC.

h. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

i. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

j. Not used.

k. Section 23 31 00, HVAC DUCTS AND CASINGS.

1.3 QUALITY ASSURANCE

a. Refer to specification Section 23 05 11, COMMON WORK RESULTS FOR HVAC

b. Comply with ANSI/ASHRAE 15 & 34, Safety Code for Mechanical Refrigeration.

c. Comply with ASHRAE 90.1 - SI, Energy Standard for Buildings except Low-Rise Residential Buildings for cooling and heating performance requirements when tested in accordance with ANSI/AHRI 340/360 and UL 1995.

d. Heating Performance shall conform to ASHRAE requirements when tested in accordance with ANSI/AHRI 340/360 and UL 1995.

e. Mot used.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer’s Literature And Data; G, AE

SD-05 Design Data

Performance Rating; G, AE

SD-06 Test Reports

Completed System Readiness Checklists; G, AE

SD-07 Certificates

Certification; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with specification Section 01 33 00, SUBMITTAL PROCEDURES.

b. Manufacturer’s Literature and Data Including:

(1) Air-Source Unitary Heat pump:

(a) Packaged units

(b) Split system

c. Certification: Submit, simultaneously with shop drawings, a proof of certification that this product has been certified by AHRI.

d. Performance Rating: Submit catalog selection data showing equipment ratings and compliance with required cooling and heating capacities EER and COP values as applicable.

e. Completed System Readiness Checklists provided by the CxA and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.6 REFERENCES

(1) Make material requirements agree with requirements specified in the referenced Applicable Publications. Verify and update the publication list to that which applies to the project, unless the reference applies to all mechanical systems. Publications that apply to all mechanical systems may not be specifically referenced in the body of the specification, but, shall form a part of this specification.

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standards will govern.

U.S. DEPARTMENT OF DEFENSE (DOD)

FS A-A-50502	(1990) Air Conditioner (Unitary Heat Pump), Air to Air (3000 TO 300,000 BTU)

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

ANSI/AHRI 210/240	(2008; Add 1 2011; Add 2 2012) Performance Rating of Unitary Air-Conditioning & Air-Source Heat Pump Equipment
ANSI/AHRI 340/360	(2007; Addendum 1 2010; Addendum 2 2011) Performance Rating of Commercial and Industrial Unitary Air-Conditioning and Heat Pump Equipment

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ANSI/ASHRAE 15 & 34	(2022) ASHRAE Standard 15-Safety Standard for Refrigeration Systems and ANSI/ASHRAE Standard 34-Designation and Safety Classification of Refrigerants
ASHRAE 62.1	(2016) Ventilation for Acceptable Indoor Air Quality
ASHRAE 90.1 - SI	(2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

ASTM INTERNATIONAL (ASTM)

ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM D4587	(2011; R 2019; E 2019) Standard Practice for Fluorescent UV-Condensation Exposures of Paint and Related Coatings

UL SOLUTIONS (UL)

UL 1995	(2015; Reprint Aug 2022) UL Standard for Safety Heating and Cooling Equipment

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in Paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 2 PRODUCTS

2.1 UNITARY HEAT PUMPS, AIR TO AIR

a. Units shall comply with FS A-A-50502Type I, having factory assembled refrigerant circuit or circuits (Packaged Unit), Class 1, "Department of Energy" (DOE) covered products (units with cooling capacity up to 65000 Btu/hr .

(1) Unitary heat pumps shall bear the United States Environmental Protection Agency, Energy Star label and shall have a minimum Heating Season Performance Factor (HSPF) of 8 (Type I – Packaged Unit),8.2 (Type II – Split System),and a minimum Seasonal Energy Efficiency Ratio (SEER) of14 (Type I – Packaged Unit),.

b. Applicable AHRI Standards: Units shall be listed in the corresponding ARI Directory of Certified products shown in paragraph, REFERENCES:

(1) Air Source Unitary heat pumps with capacity less than 19 KW (65,000 Btu/hr), Comply with ANSI/AHRI 210/240.

(2) Air Source heat pump with capacity above 19KW (above 65,000 Btu/hr, Comply with ANSI/AHRI 340/360.

c. Casing: Unit shall be constructed of zinc coated, 1-inch matt-faced, fire-resistant, odorless, glass fiber material. Surfaces in contact

with the airstream shall comply with requirements in ASHRAE 62.1. The base of the unit shall have provisions for forklift and crane lifting.

d. Filters: One inch, MERV 7, throwaway filter shall be standard on all units below 6 Tons. Filter rack can be converted to two-inch capability. Two inch, MERV 8, throwaway filters shall be factory supplied on all units above 6 Tons.

e. Compressors: Compressors shall be direct-drive, hermetic scroll type with centrifugal type oil pumps. Motor shall be suction gas-cooled. Internal overloads and crankcase heaters shall be utilized with all compressors.

f. Refrigerant Circuit: A minimum of two circuits is required. Each refrigerant circuit shall have independent fixed orifice or thermostatic expansion devices, service pressure ports, and refrigerant line filter driers factory installed as standard. An area shall be provided for replacement suction line driers.

g. Evaporator and Condenser Coils: Internally finned, DN 10 (NPS 3/8) copper tubes mechanically bonded to a configured aluminum plate fin shall be standard. The evaporated coil and condenser coil shall be leak tested at the factory to 200 psig and pressure tested to 400 psig. All dual compressor units shall have intermingled evaporator coils. Sloped condensate drain pans shall be provided.

h. Outdoor fans: Direct driven, statically and dynamically balanced, draw-through in the vertical discharge position. The fan motors shall be permanently lubricated and shall have built-in thermal overload protection.

j. Defrost Controls: A time initiated, temperature terminated defrost system shall ship with a setting of 70-minute cycle, with a choice of 50- or 90-minute cycle. Timed override limits defrost cycle to 10 minutes shall be available on units from 10 to 20 tons. Adaptive demand defrost shall be provided on units below 10 Tons.

k. Unit Electrical

(1) Provide single point unit power connection.

(2) Unit control box shall be located within the unit and shall contain controls for compressor, reversing valve and fan motor operation and shall have a 50 VA 24-volt control circuit transformer and a terminal block for low voltage field wiring connections.

(3) Safety Controls - High pressure, low temperature, and low-pressure safety switches shall be wired through a latching lockout circuit to hold the conditioner off until it is reset electrically be interrupting the power supply to the conditioner. All safety switches shall be normally closed, opening upon fault detection.

l. Operating Controls

(1) Provide unit with low voltage electric controls factory supplied DDC control system.

(2) Low voltage, adjustable room thermostat to control heating and cooling in sequence with delay between stages, compressor and supply fan to maintain temperature setting. Include system selector switch and fan control switch (auto-on).

(3) Unit DDC Controller:

(a) Unit controller shall include input, output and self-contained programming as needed for complete control of unit.

(b) Unit controller shall be BAC net compliant and utilize BAC net operating protocol.

(c) Control system shall seamlessly interface with temperature control system as specified in Section 23 09 23, DIRECT DIGITAL CONTROL SYSTEMS FOR HVAC without requiring gateways or any other interface devices.

(d) All program sequences shall be stored on board in EEPROM. No batteries shall be needed to retain logic program. All program sequences shall be executed by controller 10 times per second and shall be capable of multiple PID loops for control of multiple devices. Programming of logic controller shall be completely modifiable in the field over installed BACnet LANs.

(e) Temperature Control System Interface: Points shall be available from the unit controller for service access and display and/or control.

(f) Space Temperature Sensor: The wall mounted sensor shall include occupied and unoccupied setpoint control, pushbutton unoccupied override, space temperature offset and space temperature indication. Refer to Section 23 09 23, DIRECT DIGITAL CONTROL SYSTEM FOR HVAC for additional requirements.

m. Accessories:

(1) Electric Heater: Constructed of heavy-duty nickel chromium elements. Staging shall be achieved through the unit control processor. Each heater shall have automatically reset high limit control. Heaters shall be individually fused from the factory and shall comply with NEC and CEC requirements. Power assemblies shall provide single point connection. Electric heat modules shall be listed and labeled by a national recognized testing laboratory acceptable to authorities having jurisdiction. Electric heater controls shall confirm the supply fan is operating before electric elements are energized. Operate electric heater in 2 stages when outdoor ambient is too low to maintain space thermostat setting with compressor operation.

(2) Economizer: Horizontal flow; field installed; and shall include fully modulating 0-100 percent motor and dampers, barometric relief, minimum position setting and fixed dry bulb.Solid state enthalpy and differential enthalpy control shall be field-installed.

(3) Oversized Motors: Factory installed over sized motor shall be available for high-static application.

2.2 CORROSION PROTECTION

a. Remote Outdoor Condenser Coils:

(1) Epoxy Immersion Coating – Electrically Deposited: The multi-stage corrosion-resistant coating application comprises of cleaning (heated alkaline immersion bath) and reverse-osmosis immersion rinse prior to the start of the coating process. The coating thickness shall be maintained between 0.6-mil and 1.2-mil. Before the coils are subjected to high-temperature oven cure, they are treated to permeate immersion rinse and spray. Where the coils are subject to UV exposure, UV protection spray treatment comprising of UV-resistant urethane mastic topcoat shall be applied. Provide complete coating process traceability for each coil and minimum five years of limited warranty. The coating process shall be such that uniform coating thickness is maintained at the fin edges. The quality control shall be maintained by ensuring compliance to the applicable ASTM Standards for the following:

(a) Salt Spray Resistance (Minimum 6,000 Hours)

(b) Humidity Resistance (Minimum 1,000 Hours)

(c) Water Immersion (Minimum 260 Hours)

(d) Cross-Hatch Adhesion (Minimum 4B-5B Rating)

(e) Impact Resistance (Up to 160 Inch/Pound)

b. Exposed Outdoor Cabinet:

(1) Casing Surfaces (Exterior and Interior): All exposed and accessible metal surfaces shall be protected with a water-reducible acrylic with stainless steel pigment spray-applied over the manufacturer’s standard finish. The spray coating thickness shall be 2-4 mils and provide minimum salt-spray resistance of 1,000 hours (ASTM B117) AND 500 hours UV resistance ASTM D4587-11(2019)e.

PART 3 EXECUTION

3.1 INSTALLATION

a. If in the substantiated evaluation of the Contracting Officer, the installation fails to meet the requirements of the construction documents with respect to function and maintainability, an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

b. Install heat pumps according to manufacturers printed instructions.

c. Install electrical and control devices furnished by the manufacturer but not specified to be factory mounted. All electrical work shall comply with Division 26 Sections.

d. Ductwork: Comply with requirements in Section 23 31 00, HVAC DUCTS AND CASINGS.

e. Not used.

3.2 STARTUP AND TESTING

a. Perform startup checks according to manufacturer’s written instructions.

b. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Provide a minimum notice of 7 working days prior to startup and testing.

c. Test controls and demonstrate its compliance with project requirements. Replace damaged or malfunctioning controls and equipment and retest the equipment to the satisfaction of the Contracting Officer.

d. Furnish test reports to the Contracting Officer in accordance with specification Section 01 00 00, GENERAL REQUIREMENTS.

3.3 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS for all inspection, start up, and contractor testing required above and required by the System Readiness Checklist provided by the CxA.

b. Components provided under this section of the specification will be tested as part of a larger system. Refer to Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS and related sections for contractor responsibilities for system commissioning.

3.4 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hours to instruct VA personnel in operation and maintenance of units.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

-- End of Section --

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 82 00

CONVECTION HEATING AND COOLING UNITS

3/23

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SECTION 23 82 00
3/23

CONVECTION HEATING AND COOLING UNITS

PART 1 GENERAL

1.1 DESCRIPTION

a. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. This section specifies unit heaters, and cabinet unit heaters.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Not used.

f. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

g. Section 23 05 41, NOISE and VIBRATION CONTROL FOR HVAC PIPING and EQUIPMENT.

h. Section 23 05 93.01, DVA/USACE PROJECTS TESTING, ADJUSTING, and BALANCING FOR HVAC.

i. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

j. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

k. Section 23 21 13, HYDRONIC PIPING.

l. Section 23 31 00, HVAC DUCTS and CASINGS.

m. Section 23 82 16, AIR COILS.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Bio-Based Materials; S

SD-06 Test Reports

Completed System Readiness Checklists; G, AE

SD-07 Certificates

Certificates; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.4 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 XX XX, SECTION TITLE”, with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Unit heaters.

(2) Wall Mounted Unit Heaters.

d. Certificates:

(1) Compliance with paragraph QUALITY ASSURANCE.

(2) Compliance with specified standards.

e. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate

installation, operation, and maintenance.

f. Completed System Readiness Checklists provided by the CxA and completed by the contractor, signed by a qualified technician, and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

g. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.5 QUALITY ASSURANCE

a. Refer to paragraph QUALITY ASSURANCE of Section 23 05 11, COMMON WORK RESULTS FOR HVAC. Provide guarantee in accordance with FAR clause 52.246-21.

b. Bio-Based Materials: For products designated by the USDA's Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

c. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.6 AS-BUILT DOCUMENTATION

Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 2 PRODUCTS

2.1 UNIT HEATERS

a. General: Horizontal or vertical discharge type for hot water or electric heating medium, as indicated.

b. Casing: Steel sheet, phosphatized to resist rust and finished in baked enamel. Provide hanger supports.

c. Fan: Propeller type, direct driven by manufacturer's standard electric motor. Provide resilient mounting. Provide fan guard for horizontal discharge units.

d. Discharge Air Control:

(1) Horizontal Discharge: Horizontal, adjustable louvers.

(2) Vertical Discharge: Radial louver diffuser.

e. Hot Water Coil: Aluminum fins bonded to seamless copper tubing by mechanical expansion of the tubing, designed for 75 psig working pressure.

f. Electric Units: UL listed, factory wired to terminal strips for field connection of power and control wiring.

(1) Heating Elements: Nickel chromium alloy resistance wire embedded in a magnesium oxide insulating refractory and sealed in corrosion resisting metallic sheath with fins. Three phase heaters shall have balanced phases.

(2) Thermal Cutout: Manual reset type, which disconnects elements, and motor in the event normal operating temperatures are exceeded.

(3) Magnetic Contactor: Factory-installed with low voltage relay for remote pilot duty thermostat operation.

g. Controls: Provide field installed remote wall mounted line voltage electric space thermostats or unit mounted return air thermostats, where shown or specified Provide an aquastat on hot water units to prevent fan operation when the heating system is off.

2.2 WALL-MOUNTED UNIT HEATERS

a. General: Electric heat, fan driven, thermostatic control, UL listed.

b. Enclosure:

(1) Wall box: Not less than l8 gauge steel, recessed type.

(2) Ribbed l6 gauge steel front cover.

(3) Closely spaced discharge louvers.

(4) Concealed screws for locking trim frame to front cover.

(5) Finished in baked enamel of manufacturer’s standard color with satin finish anodized aluminum trim frame.

c. Heating Elements: Steel sheath enclosed finned-tube type.

d. Integral Controls:

(l) Two-pole terminal block.

(2) Built-in fan delay switch.

(3) Automatic reset line voltage internal thermal overheats protection.

(4) Built-in thermostat comfort control with adjustment range between 30 to 90 degrees F, and manually set “No Heat” position; tamper resistant adjustment by inserting screwdriver through front cover louvers.

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the contractor shall correct the installation at no additional cost or time to the Government.

b. Work shall be installed as shown and according to the manufacturer’s diagrams and recommendations.

c. Handle and install units in accordance with manufacturer's written instructions.

d. Support units rigidly so they always remain stationary. Cross-bracing or other means of stiffening shall be provided as necessary. Method of support shall be such that distortion and malfunction of units cannot occur.

e. Install fiberglass blanket insulation with a minimum R value of 8 above hydronic radiant panels.

3.2 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Provide a minimum notice of 10 working days prior o startup and testing.

3.3 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.4 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hour s to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

-- End of Section --

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DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 82 16

AIR COILS

3/24

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SECTION 23 82 16
3/24

AIR COILS

PART 1 GENERAL

1.1 DESCRIPTION

a. A complete listing of common acronyms and abbreviations are included in Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Heating and cooling coils for air handling units and duct applications.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 00, SUBMITTAL PROCEDURES.

c. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

d. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

e. Not used.

f. Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

g. Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

h. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEMS for HVAC.

i. Section 23 31 00, HVAC DUCTS AND CASINGS.

j. Section 23 73 00, INDOOR CENTRAL-STATION AIR-HANDLING UNITS.

m. Section 23 82 00, CONVECTION HEATING AND COOLING UNITS.

1.3 REFERENCES

a. The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and the VHA standards will govern.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 410 (2001; Addendum 1 2002; Addendum 2 2005; Addendum 3 2011) Forced-Circulation Air-Cooling and Air-Heating Coils

ASTM INTERNATIONAL (ASTM)

ASTM B75/B75M (2020) Standard Specification for Seamless Copper Tube

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Manufacturer's Literature And Data; G, AE

Coils; G

Bio-Based Materials; S

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

SD-07 Certificates

Certification Compliance; G, AE

SD-08 Manufacturer's Instructions

Installation, Operating And Maintenance Instructions; G, AE

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSTRUCTIONS

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 00, SUBMITTAL PROCEDURES.

b. Information and material submitted under this section shall be marked “SUBMITTED UNDER SECTION 23 XX XX, SECTION TITLE”, with applicable paragraph identification.

c. Manufacturer's Literature and Data Including: Full item description and optional features and accessories. Include dimensions, weights, materials, applications, standard compliance, model numbers, size, and capacity.

(1) Submit type, size, arrangements, and performance details. Present application ratings in the form of tables, charts, or curves.

d. Provide installation, operating and maintenance instructions.

e. Certification Compliance: Evidence of listing in current AHRI Directory of Certified Applied Air Conditioning Products.

f. Coils may be submitted with Section 23 73 00, INDOOR CENTRAL-STATION AIR-HANDLING UNITS, , or Section 23 82 00, CONVECTION HEATING AND COOLING UNITS.

g. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replaceable parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation, and maintenance.

h. Completed System Readiness Checklist provided by the CxA and completed by the contractor, signed by a qualified technician, and dated on the date of completion, in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

i. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

1.6 QUALITY ASSURANCE

a. Refer to paragraph QUALITY ASSURANCE of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

b. Unless specifically exempted by these specifications, heating and cooling coils shall be tested, rated, and certified in accordance with AHRI 410 and shall bear the AHRI certification label.

c. Bio-Based Materials: For products designated by the USDA’s Bio-Preferred Program, provide products that meet or exceed USDA recommendations for bio-based content, so long as products meet all performance requirements in this specifications section. For more information regarding the product categories covered by the Bio-Preferred Program, visit http://www.biopreferred.gov.

d. Refer to Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for additional sustainable design requirements.

1.7 AS-BUILT DOCUMENTATION

Comply with requirements in paragraph AS-BUILT DOCUMENTATION of Section 23 05 11, COMMON WORK RESULTS FOR HVAC.

PART 2 PRODUCTS

2.1 HEATING AND COOLING COILS

a. Conform to ASTM B75/B75M and AHRI 410.

b. Tubes: Minimum 0.625 inch tube diameter; Seamless copper tubing.

c. Fins: 0.0055 inch aluminum or 0.0045 inch copper mechanically bonded or soldered or helically wound around tubing.

d. Headers: Copper, welded steel or cast-iron. Provide seamless copper tubing or resistance welded steel tube for volatile refrigerant coils.

e. "U" Bends, Where Used: Machine die-formed, silver brazed to tube ends.

f. Coil Casing: l6 gauge galvanized steel with tube supports at48 inch maximum spacing. Construct casing to eliminate air bypass and moisture carry-over. Provide duct connection flanges.

g. Pressures kPa (PSIG):

Pressure	Water Coil		Refrigerant Coil	
Test	(300)		(300)	
Working	(200)		(250)	

h. Protection: Unless protected by the coil casing, provide cardboard, plywood, or plastic material at the factory to protect tube and finned surfaces during shipping and construction activities.

i. Vents and Drain: Coils that are not vented or drainable by the piping system shall have capped vent/drain connections extended through coil casing.

PART 3 EXECUTION

3.1 INSTALLATION

a. If an installation is unsatisfactory to the Contracting Officer, the contractor shall correct the installation at no additional cost or time to the Government.

b. Follow coil manufacturer's instructions for handling, cleaning, installation, and piping connections.

c. Comb fins, if damaged. Eliminate air bypass or leakage at coil sections.

3.2 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The CxA will observe startup and contractor testing of selected

equipment. Coordinate the startup and contractor testing schedules with the Contracting Officer and CxA. Provide a minimum of 10 working days prior to startup and testing.

3.3 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.4 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hours to instruct VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 23 08 00.01, DVA/USACE PROJECTS COMMISSIONING OF HVAC SYSTEMS.

-- End of Section --

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SECTION 25 05 11.01

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08/24

cybersecurity for LOW and MODERATE impact facility-related control systems

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SECTION 25 05 11.01

CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - BUILDING AUTOMATION SYSTEMS (L-M-M)
08/24

PART 1 GENERAL

Many subparts in this Section contain text in curly braces ("{" and "}") indicating which cybersecurity control and control correlation identifier (CCI) the requirements of the subpart relate to. The text inside these curly braces is for Government reference only and enables coordination of the requirements of this Section with the RMF process throughout the design and construction process. Text in curly braces are not contractor requirements.

This Section refers to Security Requirements Guide (SRGs) and Security Technical Implementation Guide (STIGs). STIGs and SRGs are available online at the Information Assurance Support Environment (IASE) website at https://public.cyber.mil/stigs/downloads/ and an SRG/STIG Applicability Guide and Collection Tool is available at https://public.cyber.mil/stigs/SCAP/. Not all control system components have applicable STIGs or SRGs. The "Control Systems SRG" does not apply to work performed under this Section; all requirements within this section to apply applicable SRGs DO NOT include the "Control Systems SRG".

1.1 CONTROL SYSTEM APPLICABILITY

There are multiple versions of this Section associated with this project. Different versions have requirements applicable to different control systems. This specific Section applies only to the following control systems: Building Automation Systems.

1.2 RELATED REQUIREMENTS

This section does not contain sufficient requirements to procure a control system and must be used in conjunction with other Sections which specify control systems. This Section adds cybersecurity requirements to the control systems specified in other Sections, and as these requirements are conditioned on the control system being provided, there may be requirements in this Section that will not apply to this project. All Sections containing facility-related control systems or control system components are related to the requirements of this Section. Review all specification sections to determine related requirements.

In cases where a requirement is specified in both this Section and in another Section, the more stringent requirement must be met. In cases where a requirement in this Section conflicts with the requirements of another Section such that both requirements cannot be met at the same time, request direction from the Contracting Officer to determine which requirement applies to the project.

Systems that are to be connected to the VA network shall have an Enterprise Risk Assessment (ERA) performed. ERAs shall conform to the medical center's standards.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 135	(2024; INT 1-3 2025; Errata 1 2025; INT 4-6 2025; Addenda CU 2025) BACnet—A Data Communication Protocol for Building Automation and Control Networks

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.1x	(2010) Local and Metropolitan Area Networks - Port Based Network Access Control

INTERNET ENGINEERING TASK FORCE (IETF)

IETF RFC 2819	(2000) Remote Network Monitoring (RMON) Management Information Base (MIB)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST FIPS 140-2	(2001) Security Requirements for Cryptographic Modules

U.S. DEPARTMENT OF DEFENSE (DOD)

DODI 8551.01	(2014) Ports, Protocols, and Services Management (PPSM)
DTM 08-060	(2008) Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement

1.4 DEFINITIONS

1.4.1 Administrator Account

An administrator account is an account with full permissions to a device, application, or operating system, including the ability to create and modify other user accounts.

Note that the operating system Administrator Account may be different than Administrator Accounts for applications hosted on that operating system. Also, most controllers will not have any support for accounts and will therefore not have an 'Administrator Account'.

1.4.2 Computer

A computer is one of the following:

a. a device running a non-embedded desktop or server version of Microsoft Windows

b. a device running a non-embedded version of MacOS

c. a device running a non-embedded version of Linux

d. a device running a version or derivative of the Android Operating System, where Android is considered separate from Linux

e. a device running a version of Apple iOS

Unless otherwise indicated or clear from context use of the word "device" in this Section includes computers.

1.4.3 Controller

A device other than a computer or Ethernet switch.

1.4.4 Mission Space

A device or media is in mission space if physical access to the device or media is controlled by the organization served by the device. For example, a variable air volume (VAV) box controller in a suspended ceiling is in mission space if the VAV box serves that room; an electrical switchgear in an electrical room or an air handling unit (AHU) in a mechanical room or on a rooftop may still be considered to be in mission space if the organization (mission) served by that switchgear or AHU controls access to the electrical room, mechanical room or rooftop.

1.4.5 Network

A network is a group of two or more devices that can communicate using a network protocol. Network protocols must provide a method for addressing devices on the network; a communication method that does not provide an addressing scheme is not a networked form of communication. Devices that communicate using a method of communication that does not support device addressing are not using a network. Addresses may be other than IP addresses, and addressing may be at either Open Systems Interconnection (OSI) layer 2 or layer 3.

1.4.6 Network Connected

A component is network connected (or "connected to a network") only when the device has a network transceiver which is directly connected to the network and implements the network protocol. A device lacking a network transceiver (and accompanying protocol implementation) can never be considered network connected. Note that (unlike many IT definitions of "Network Connected") a device connected to a non-IP network is still considered network connected (an IP connection or IP address is not required for a device to be network connected).

1.4.6.1 Wireless Network Connected

Any device that supports wireless network communication is network connected to a wireless network, regardless of whether the device is communicating using wireless. Unless physically disabled, devices with wireless transceivers support wireless, it is not sufficient to disable the wireless in software.

1.4.7 Network Media

The thing that provides the communication channel between the devices on a network. Typically wire, but might include wireless, fiber optic, or even power line (some network protocols allow sending network signals over power wiring).

1.4.8 User Account Support Levels

The support for user accounts is categorized in this Section as one of three levels:

1.4.8.1 FULLY Supported

Device supports configurable individual accounts. Accounts can be created, deleted, modified, etc. Privileges can be assigned to accounts. These devices support user-based (as opposed to role-based) authentication.

1.4.8.2 MINIMALLY Supported

Device supports a small, fixed number of accounts (perhaps only one). Accounts cannot be modified. A device with only a "User" and an "Administrator" account would fit this category. Similarly, a device with two PINs for logon - one for restricted and one for unrestricted rights would fit here (in other words, the accounts do not have to be the traditional "username and password" structure). These devices typically only support role-based authentication.

Examples of devices which MINIMALLY support accounts are a) a variable frequency drive with a single account which requires a PIN for access to configuration; and b) a room lighting control touchpad interface that has a single account.

1.4.8.3 NOT Supported

Device does not support any Access Enforcement therefore the whole concept of "account" is meaningless.

1.4.9 Manual Local Input

Manual Local Inputs are system analog or binary inputs that are adjustable by a person but are, by intrinsic hardware design, very limited in potential capabilities. Manual Local Inputs do not have touch screens or full keyboards, but may have a few buttons or dials to allow input. Manual Local Inputs do not have full graphic screens or dot-matrix displays, but may have simple lights (LEDs) or 7-segment displays. Manual Local Inputs do not have any sort of menu structure, each button has a single well-defined function.

Examples of Manual Local Inputs are H-O-A switches, simple thermostats, and disconnect switches.

1.4.10 Card Reader

A card reader is an input/output device whose primary function is to assist in two-factor authentication. A card reader must have an interface to read data from a card and may be able to write data to a card. A card reader may have a means (such as buttons, keypad, touchscreen, etc.) for a user to input a PIN or password, as well as a limited display.

1.4.11 User Interface

A User Interface (UI) is something other than a Manual Local Input or Card Reader that allows a person to interact with the system or device. Note that while a Card Reader is not by itself a User Interface, a User Interface may contain a Card Reader in order for it to authenticate its user. Within control systems, there are a wide range of User Interfaces.

Two important distinctions are 1) whether the user interface is Local or Remote, and 2) the effective capabilities of the User Interface to alter data, which is the "privilege" of the user interface (where effective privilege available to a specific user at a specific user interface is the combination of the greatest privilege offered by the user interface and the specific account the user is logged into).

1.4.11.1 Local User Interface

A Local User Interface is a user interface where the physical hardware the user interacts with (keyboard, buttons, display, etc.) is physically part of the device being affected. All of the relevant characteristics of the user interface are embodied within a single device.

Note that a Local UI may be able to access data in a different device, Local versus Remote in this context refers to the user interface itself; the capability to access data in a different device is covered under "Full User Interface".

1.4.11.2 Remote User Interface

A Remote User Interface implements a Client/Server model where the physical hardware the user interacts with (Client) is physically distinct from the device being affected (Server). Most or all of the security and functionality characteristics of the user interface are defined by the Server, not the Client. The Client and Server communicate via a network connection. A common example of a remote user interface is a web-based interface where the browser (client) is generally on different hardware than the web server (server). A Remote UI remains a Remote UI even if the user happens to be at a Client on the same hardware as the Server. What is important is that a) the Client may be on different hardware than the Server and b) the majority of the security and functional characteristics of the interface are defined at the Server.

Note that this definition of "remote" is consistent with that generally used in the control industry but is not aligned with the NIST 800-53 definition of "Remote", which refers to "outside the system". The term "Remote" here better aligns with the NIST 800-53 definition of "Network" (remote from within the system) Access.

1.4.11.3 Types of User Interface (by capability)

User interfaces are also categorized by their capabilities as being Read Only, Limited, or Full.

1.4.11.3.1 Read-Only User Interface

A Read Only User Interface (also referred to as a View-Only User Interface) is a user interface that only allows for reading data, it does not allow (have the capability to) modify data. A Read Only User

Interface may be either Local or Remote. A User Interface that is configured to be Read Only (by some other means than the interface itself, such as using configuration software on a laptop) is a Read-Only Interface. Note a Read Only User Interface may have buttons (or touch screen, etc.) allowing the user to navigate through the presentation of data.

Examples of a Read Only User Interfaces are a) a publicly viewable "energy dashboard" showing weather data and energy usage within a building and b) digital wayfinding signage.

1.4.11.3.2 Limited User Interface

A Limited User Interface is a user interface that - by design - can only alter information local to the user interface. Note that the determination of "alter" includes only direct interactions, it explicitly excludes interactions that might occur as secondary effects. For example, an interface changing the flow setpoint in a pump controller is a direct interaction, the subsequent change in flow (as well as any subsequent downstream changes in valve position) are not direct interactions.

Two examples of LIMITED UIs are: a) a variable speed drive has a Limited Local User Interface which allows the user to change properties within the drive, but does not allow affecting things outside the drive; and b) a typical home WiFi Router has a Limited Remote User Interface which allows configuration of the Router, but does not allow direct interaction with other devices.

1.4.11.3.3 Full User Interface

A Full User Interface can alter information in devices outside the device with the user interface. For example, a typical Local Display Panel is a Full Local User Interface while a browser-based front end is a Full Remote User Interface.

1.4.11.3.4 View-Only User Interface

See Read-Only User Interface

1.4.11.4 Other User Interface Terminology

In addition to defining whether a user interface is a Hardware Limited, Read-Only, Limited or Full, and whether it is Local or Remote, user interfaces are classified by whether they are writable or privileged.

1.4.11.4.1 Writable User Interface

Any User Interface that is not Read-Only is Writable. (Limited User Interfaces and Full User Interfaces are both writable user interfaces (as they are capable of changing a value)).

1.4.11.4.2 Privileged User Interface

A Privileged UI is a UI that has sufficient capabilities or functionality that it requires specific cybersecurity measures to be put in place to limit its unauthorized use. Ultimately, whether a specific user interface is considered a Privileged User Interface must be determined by usage. Unless otherwise specified, user interfaces can be determined to be privileged or not using the following:

a. Read-Only User Interfaces are not privileged user interfaces.

b. Full User Interfaces are privileged user interfaces.

c. User interfaces that allow for configuration of auditing or allows for modification or deletion of audit logs are privileged user interface.

d. User interfaces that allow for reprogramming a network connected device is a privileged user interface.

e. Except as specified above, a Limited User Interface must be determined to be privileged or not based on the specific capabilities and use case of the user interface. In general however, user interfaces that do not offer significant capabilities above and beyond those available at that location via other means (e.g. such as a disconnect switch, breaker, or hand-off-auto switch, or physical attack) are not privileged.

1.4.12 Wireless Network

Any network that communicates without using wires or fiber optics as the communication media. Wireless networks include: WiFi, Bluetooth, ZigBee, cellular, satellite, long and short wave radio, 2.4 GHz, free space optical, point-to-point laser, microwave, and IR.

1.4.13 Wired Broadcast Network

Wired Broadcast Networks are any network, such as powerline carrier networks and modem (wired telephony), that use wire-based technologies where there is not a clearly defined boundary for signal propagation.

1.5 ADMINISTRATIVE REQUIREMENTS

1.5.1 Points of Contact

Coordinate with the following Points of Contact as indicated in this Section and as required. Not all projects will require coordination with all Points of Contact. When coordination is required and no Point of Contact is indicated, coordinate with the Contracting Officer (KO).

a. Government Computer Access Point of Contact: Contracting Office (KO)

b. HTTPS Certificate Point of Contact: Contracting Officer (KO)

c. Email Address Point of Contact: Contracting Officer (KO)

d. Password Point of Contact: Contracting Officer (KO)

1.5.2 Coordination

Coordinate the execution of this Section with the execution of all other Sections related to control systems as indicated in the paragraph RELATED REQUIREMENTS. Items that must be considered when coordinating project efforts include but are not limited to:

a. If requesting permission for alternate account lock permissions, the Device Account Lock Exception Request must be approved prior to control system device selection and installation.

b. If requesting permission for the use of a device with multiple physical connections to IP networks, the Multiple IP Connection Device Request must be approved prior to control system device selection and installation.

c. If the Device Audit Record Upload Software is to be installed on a computer not being provided as part of the control system, coordination is required to identify the computer on which to install the software.

d. The Cybersecurity Interconnection Schedule must be coordinated with other work that will be interconnected to, and interconnections must be approved by the Government before relying on them for system functionality.

e. Cybersecurity testing support must be coordinated across control systems and with the Government cybersecurity testing schedule.

f. Passwords must be coordinated with the indicated contact for the project site.

g. If applicable, HTTPS web server certificates must be obtained from the indicated HTTPS Certificate Point of Contact.

h. Contractor Computer Cybersecurity Compliance Statements must be provided for each contractor using contractor owned computers.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Wireless and Wired Broadcast Communication Request; GDevice Account Lock Exception Request; G

Multiple Ethernet Connection Device Request; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

Cybersecurity Interconnection Schedule ; G

Protection of Information At Rest Proposal; G

Proposed STIG and SRG Applicability Report; G

Pre-Construction Control System Inventory Report ; G

SD-02 Shop Drawings

Network Communication, Ports, Protocols and Services Report; G

Cybersecurity Network (Riser) Diagram ; G

SD-03 Product Data

Control System Cybersecurity Documentation; G

SD-06 Test Reports

Control System Cybersecurity Testing Procedures; G

Control System Cybersecurity Testing Report; G

SD-11 Closeout Submittals

Password Change Summary Report; G

Enclosure Keys; G

Software and Configuration Backups; G

System Maintenance Tool Software; G

Control System Scanning Tools; G

STIG, SRG and Vendor Guide Compliance Result Report; G

Final Control System Inventory Report; G

1.7 QUALITY CONTROL

1.7.1 Regulatory Requirements

No regulatory requirements are enforced for the Building Automation System.

]1.8 CYBERSECURITY DOCUMENTATION

{For Government Reference Only: This subpart (and its subparts) relates

to PL-7; CCI-003071}

1.8.1 Proposed STIG and SRG Applicability Report

For each model of network connected or network infrastructure device, use the DISA SRG/STIG Applicability Guide and Collection Tool (available at https://public.cyber.mil/stigs/SCAP/to identify applicable STIGs or SRGs and provide a report indicating applicable STIGs and SRGs for each model. Provide the Proposed STIG and SRG Applicability Report concurrently with the Pre-Construction Control System Inventory Report.

1.8.2 Cybersecurity Interconnection Schedule

{For Government Reference Only: This subpart relates to CA-3(b), PL-8, SC-7(9), SC-7(11); CCI-000258, CCI-003072, CCI-003073, CCI-003075, CCI-002398, CCI-002399, CCI-002401, CCI-002402, CCI-002403. For MODERATE Impact systems, this subpart also relates to SC-7; CCI-001126, CCI-001109}

Provide a completed Cybersecurity Interconnection Schedule documenting network connections between the installed system and other systems. Provide the following information for each device directly communicating between systems: Device Identifier, Device Manufacturer, Device Description, Transport layer Protocol, Network Address, Port (if applicable), MAC (Layer 2) address (if applicable), Media, Application Protocol, Service (if applicable), Descriptive Purpose of communication. For MODERATE Impact Systems: Also describe the impact of loss of the connection on the control system. If other control system Sections used on this project include submittals documenting this information, provide copies of those submittals to meet this requirement.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Cybersecurity Interconnection Schedule as an editable Microsoft Excel file (a template Cybersecurity Interconnection Schedule in Excel format is available athttps://www.wbdg.org/dod/ufgs/ufgs-25-05-11.)

1.8.3 Network Communication, Ports, Protocols and Services Report

{For Government Reference Only: This subpart (and its subparts) relates to CA-9, PL-8; CCI-002102, CCI-002103, CCI-002104, CCI-002105, CCI-003072, CCI-003073, CCI-003075 and also the submittal requirements associated with CM-6, CM-7, including CM-7(3), CCI-000388.}

Provide a Network Communication, Ports, Protocols and Services Report. For each networked device, document device identifier and the communication characteristics of the device including communication protocols, services used, encryption employed, and a general description of what information is communicated over the network. For each device using IP communication, document all TCP and UDP ports used. For each device using non-IP communication, document communication protocol and media used. If other control system Sections used on this project include submittals documenting this information, provide copies of those submittals to meet this requirement.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Network Communication, Ports, Protocols and Services Report as an editable Microsoft Excel file.

1.8.4 Control System Inventory Reports

{For Government Reference Only: This subpart (and its subparts) relates to CM-8(a), SI-17, IA-3; CCI-000389, CCI-000392, CCI-000398, CCI-002773, CCI-002774, CCI-002775, CCI-000777, CCI-000778, CCI-001958}

Provide a Pre-Construction Control System Inventory Report report and a Final Control System Inventory Report, using the Inventory Spreadsheet listed under this Section at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. In the Control System Inventory Reports, document all networked devices, including network infrastructure devices. For each device provide all applicable information for which there is a field on the spreadsheet in accordance with the instructions on the spreadsheet.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Control System Inventory Reports as editable Microsoft Excel files. Provide the Pre-Construction Control System Inventory Report concurrently with the Proposed STIG and SRG Applicability Report

1.8.5 Software and Configuration Backups

{For Government Reference Only: This subpart (and its subparts) relates to CP-10; CCI-000550, CCI-000551, CCI-000552}

For each computer on which software is installed under this project, provide a recovery image of the final as-built computer . This image must allow for bare-metal restore such that restoration of the image is sufficient to restore system operation to the imaged state without the need for re-installation of software. If additional user permissions are required to meet this requirement, coordinate the creation of the image with the identified Government Computer Access Point of Contact.

For all Ethernet switches provide a backup of the switch configuration. For all controllers, provide a backup of the controller configuration and the source code for all loaded application programs (all software that is not common to every controller of the same manufacturer and model).

If any or all of these are provided under another Section, provide documentation indicating this and referencing those submittals.

1.8.6 Cybersecurity Network (Riser) Diagram

{For Government Reference Only: This subpart (and its subparts) relates to PL-2(a), PL-8; CCI-003051, CCI-003053, CCI-003072, CCI-003073, CCI-003075}

Provide a cybersecurity network (riser) diagram of the complete control system including all network and device hardware. For each device, include the device identifier, device type, and manufacturer. If the control system specifications require a riser diagram submittal, provide a copy of that submittal as the cybersecurity riser diagram. Otherwise, provide a riser diagram in one-line format.

1.8.7 STIG, SRG and Vendor Guide Compliance Result Report

For every component (device or software) with an applicable STIG or SRG in the Proposed STIG and SRG Applicability Report, document compliance with

the STIG or SRG requirements.

a. For components which are scannable by the SCAP (security content automation protocol) tool (available online at https://public.cyber.mil/stigs/scap), include the SCAP report and raw scan results in addition to the final, manually reviewed and revised, documentation of compliance with STIG and SRG requirements. Checklist files should not contain any findings with a Not Reviewed (NR) status after manual reviews.

b. For components which do not support automated scanning, a manual review using the General Purpose STIG option should be done. A completed Checklist file should not contain any findings with a Not Reviewed (NR) status after manual reviews

For every component (device or software) with manufacturer provided cybersecurity documentation, procedure, or method for secure configuration or installation, provide a report documenting how the component was configured and any deviation from the manufacturer instructions.

1.8.8 Control System Cybersecurity Documentation

{For Government Reference Only: This subpart (and its subparts) relates to SA-5(a),SA-5(b),SA-5(c), SA-22(b); CCIs: CCI-003124, CCI-003125, CCI-003126, CCI-003127, CCI-003128, CCI-003129, CCI-003130, CCI-003131, CCI-003374}

Provide a Control System Cybersecurity Documentation submittal containing the indicated information for each device and software application.

1.8.8.1 Software Applications

For all software applications running on computers provide:

a. administrator documentation that describes secure configuration of the software {For Government Reference Only: relates to CCI-003124}

b. administrator documentation that describes secure installation of the software and software updates. {For Government Reference Only: relates to CCI-003125}

c. administrator documentation that describes secure operation of the software {For Government Reference Only: relates to CCI-003124}

d. administrator documentation that describes effective use and maintenance of security functions or mechanisms for the software {For Government Reference Only: relates to CCI-003127}

e. administrator documentation that describes known vulnerabilities regarding configuration and use of administrative (i.e. privileged) functions for the software {For Government Reference Only: relates to CCI-003128}

f. user documentation that describes user-accessible security functions or mechanisms in the software and how to effectively use those security functions or mechanisms {For Government Reference Only: relates to CCI-003129}

g. user documentation that describes methods for user interaction which

enables individuals to use the software in a more secure manner {For Government Reference Only: relates to CCI-003130}

h. user documentation that describes user responsibilities in maintaining the security of the software {For Government Reference Only: relates to CCI-003131}

1.8.8.2 For HVAC Control System Devices

1.8.8.2.1 HVAC Control System Devices FULLY Supporting User Accounts

For all HVAC Control System Devices which FULLY support user accounts, provide:

a. Documentation that describes secure configuration of the device {For Government Reference Only: relates to CCI-003124}

b. Documentation that describes secure operation of the device {For Government Reference Only: relates to CCI-003124}

c. Documentation that describes effective use and maintenance of security functions or mechanisms for the device {For Government Reference Only: relates to CCI-003127}

d. Documentation that describes known vulnerabilities regarding configuration and use of administrative (i.e. privileged) functions for the device {For Government Reference Only: relates to CCI-003128}

e. Documentation that describes user-accessible security functions or mechanisms in the device and how to effectively use those security functions or mechanisms; or a specific indication that there are no user-accessible security functions or mechanisms in the device {For Government Reference Only: relates to CCI-003129}

f. Documentation that describes methods for user interaction which enables individuals to use the device in a more secure manner {For Government Reference Only: relates to CCI-003130}

1.8.8.2.2 All Other HVAC Control System Devices

For all HVAC Control System Devices which do not FULLY support user accounts, provide:

a. Documentation that describes secure configuration of the device; or a specific indication that there are no secure configuration steps that apply {For Government Reference Only: relates to CCI-003124}

b. Documentation that describes effective use and maintenance of security functions or mechanisms for the device; or a specific indication that there are no security functions or mechanisms in the device {For Government Reference Only: relates to CCI-003127}

c. For devices which include a user interface, documentation that describes methods for user interaction which enables individuals to use the device in a more secure manner {For Government Reference Only: relates to CCI-003130}

1.9 CYBERSECURITY DURING CONSTRUCTION

{For Government Reference Only: This subpart (and its subparts) relates to AC-18, CA-3; CCI-000258}

In addition to the control system cybersecurity requirements indicated in this section, meet following requirement throughout the construction process.

1.9.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. Contractor computers connected to the control system, control system network, or a control system component at any point during construction must meet the following requirements:

1.9.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.9.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. Computers used on this project must perform a full antivirus scan at least once per day. Computers which have connected to any other network since the last full antivirus scan must perform a full antivirus scan prior to connection to the control system network or to the temporary contractor-installed IP network. Perform an antivirus scan on all removable media (e.g., external hard drives, CDs, DVDs, USB flash drives) prior to connecting to the control system environment.

1.9.1.3 Passwords and Passphrases

The passwords and passphrases for computers, applications, and web-based applications supporting passwords must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.9.1.4 User-Based Authentication

Each user must have a unique account; sharing of a single account between multiple users is prohibited.

1.9.1.5 Demonstration of Compliance

The Government has the right to require demonstration of computer compliance with these requirements at any time during the project.

1.9.1.6 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor

owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. Each Statement must be signed by a cybersecurity representative for the relevant company.

1.9.2 Government Access to Network

Government personnel must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification.

1.9.3 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.9.4 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

1.10 CYBERSECURITY DURING WARRANTY PERIOD

All work performed on the control system after acceptance must be performed using Government Furnished Equipment or equipment specifically and individually approved by the Government.

PART 2 PRODUCTS

All products used on this project must meet the indicated requirements, but not all products specified here will be required by every project.

2.1 ETHERNET SWITCH

Provide Open Systems Interconnection (OSI) Layer 2 Ethernet switches with the following capabilities, and with an interface to support switch configuration for these capabilities:

2.1.1 Required Functionality

Switches must:

a. Copper Ethernet ports must auto negotiate for 10, 100 and 1000 megabits-per-second links.

b. Be capable of implementing port level access control by MAC address and limit the number of MAC addresses to one MAC address per port.

c. For MODERATE Impact Systems, be capable of implementing per-port access control lists (ACLs) where the list can be filtered by source and destination IP addresses, and by source and destination UDP or TCP ports.

d. Support Remote Network Monitoring (RMON) Port Analysis in accordance with IETF RFC 2819

e. Configure target port and analysis port such that switch clones all target port traffic to analysis port.

2.1.2 Configuration Requirements

Switches must:

a. Support configuration save and restore.

b. Support both manual IP address assignment and acquisition of a dynamic IP address via Dynamic Host Configuration Protocol (DHCP).

c. Be capable of limiting access for configuration to one or more of: a web interface using HTTPS, a command line interface using SSH, or an SNMP connection using SNMP version 3 or later.

2.2 DAISY CHAIN IP CONTROLLERS

Controllers used as Daisy Chain IP Controllers must be IP controllers with exactly two Ethernet network connections and basic built-in switch capabilities to allow implementation of an Ethernet network in a daisy chain architecture. Switches incorporated by Daisy Chain IP Controllers are not required to meet the requirements for Ethernet Switches as defined in this Section.

2.3 DATABASE AND WEB SERVER SOFTWARE FOR MODERATE IMPACT SYSTEMS

{For Government Reference Only: This subpart (and its subparts) relate to RA-5(1), RA-5(5); CCI-001062, CCI-001067, CCI-001645, CCI-002906}

All computer-based databases must use Microsoft SQL Server or Oracle or MySQL. All computer-based web interfaces must use Internet Information Services (IIS) or Apache as the web server.

PART 3 EXECUTION

3.1 CYBERSECURITY HARDENING AND CONFIGURATION GUIDES

Install, configure, and harden all hardware and software furnished on this project in accordance with manufacturer provided documentation, procedures, or methods for secure configuration or installation. Do not implement specific hardening actions if that action would conflict with required functionality or another requirement of this Section.

3.2 NETWORK REQUIREMENTS

3.2.1 Information Flow Enforcement In MODERATE Impact Systems

{For Government Reference Only: This subpart (and its subparts) relate to AC-4; CCI-001368, CCI-001414, CCI-001548, CCI-001549, CCI-001550, CCI-001551}

Install and configure Ethernet switches to block all traffic on all ports not required by the control protocol.

3.2.2 Wireless and Wired Broadcast Communication

{For Government Reference Only: This subpart (and its subparts) relates to AC-18, AC-18(3); CCI-001438, CCI-001439, CCI-002323, CCI-001441, CCI-001449}

Unless explicitly authorized by the Government, do not use any wireless or wired broadcast communication.

3.2.2.1 Wireless and Wired Broadcast IP Communications

Do not install wireless or wired broadcast IP networks, including: do not install a wireless access point; do not install or configure an ad-hoc wireless network; do not install or configure a WiFi Direct communication.

When explicitly authorized by the Government, wireless IP communication may be used to communicate with an existing wireless network.

3.2.2.2 Non-IP Wireless Communication

For MODERATE Impact Systems: When non-IP wireless communication is explicitly authorized by the Government, the radios must meet NIST FIPS 140-2 Level 2.

3.2.2.3 Wireless and Wired Broadcast Communication Request

Provide a report documenting the proposed use of wireless or wired broadcast communication prior to device selection using the Wireless and Wired Broadcast Communication Request Schedule at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. If there is no proposed use of wireless or wired broadcast communication, provide a document indicating this instead of the Request Schedule.

For each device proposed to use wireless or wired broadcast communication show: the device identifier, a description of the device, the location of the device, the device identifiers of other devices communicating with the device, the protocol used for communication, encryption type and strength. For wireless communication, also show: RF Frequency, Radiated Power in dBm (decibel with a milliwatt reference), free-space range, and the expected as-installed range.

3.2.3 Non-IP Control Networks

When control system specifications require particular communication protocols, use only those communication protocols and only as specified. Do not implement any other communication protocol.

When control system specifications do not indicate requirements for communication protocols, use only those protocols required for operation of the system as specified.

3.2.4 IP Control Networks

{For Government Reference Only: This subpart relates to CM-6(a), CM-7(a), CM-7(b), CM-7(1)(b), SC-41; CCI-001588, CCI-000381, CCI-000380, CCI-000381, CCI-000382, CCI-001761, CCI-001762, CCI-002544, CCI-002545,

CCI-002546. For Moderate Impact Systems, this subpart (and its subparts) also relates to SC-5(1), SC-5(2); CCI-001094 CCI-001095}

IP Networks must be Ethernet networks and must use switches which are Ethernet Switches or Daisy Chain IP Controllers as defined in this Section. Do not use nonsecure functions, ports, protocols and services as defined in DODI 8551.01 unless those ports, protocols and services are specifically required by the control system specifications or otherwise specifically authorized by the Government. Do not use ports, protocols and services that are not specified in the control system specifications or required for operation of the control system.

For MODERATE Impact Systems, unless explicitly authorized, do not use IP networks if the same control functionality is available through the use of non-IP networks.

3.2.4.1 IP Devices With Multiple Ethernet Connections

Except for Ethernet Switches and Daisy Chain IP Controllers, devices must not have more than one Ethernet connection to IP networks unless doing so is required by the project specifications and the specific application is approved. If a device with Multiple Ethernet Connections to IP networks is required, provide a Multiple Ethernet Connection Device Request using the Multiple Ethernet Connection Device Request Template at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11 to request approval for each device. If a device with Multiple Ethernet Connections to IP networks is not required, instead provide a document stating that no approval is being requested.

3.2.5 Cryptographic Protection

{For Government Reference Only: This subpart relates to IA-2(9), IA-3(1), SC-8, SC-13, SC-23(1), SC-23(3); CCI-001942, CCI-001959, CCI-001967, CCI-002418, CCI-002449, CCI-002450, CCI-001185, CCI-001188, CCI-001664.}

All remote user interfaces must use HTTPS for all traffic between the user interface client and user interface server.

3.2.6 Device Identification and Authentication

{For Government Reference Only: This subpart (and its subparts) relates to IA-3; CCI-000777, CCI-000778, CCI-001958. For MODERATE Impact systems, this subpart (and its subparts) also relates to SC-23, SC-23(5); CCI-001184, CCI-002470.}

All computers must support IEEE 802.1x for device authentication to the network.

3.2.6.1 For HVAC Control System Devices

Devices using HTTP as a control protocol must use HTTPS instead. Devices using BACnet must support network security as specified for BACnet Secure Connect in ASHRAE 135.

3.2.7 Cryptographic Module Authentication

{For Government Reference Only: This subpart (and its subparts) relates to IA-7; CCI-000803}

For devices (including but not limited to NIST FIPS 140-2 compliant radios) that have STIG/SRGs related to cryptographic module authentication (CCI-000803), comply with the requirements of those STIG/SRGs.

3.3 ACCESS CONTROL REQUIREMENTS

3.3.1 User Accounts

{For Government Reference Only: This subpart (and its subparts) relate to AC-2(a), AC-3, AC-6(1), AC-6(10), AC-6(2), AC-6(9), CM-11(2), and IA-2; CCI-002110, CCI-000213, CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-002235, CCI-000039, CCI-001419, CCI-002234, CCI-001812, and CCI-000764. For MODERATE Impact systems, this subpart (and its subparts) also relate to AC-2 (2), AC-2(3), AC-2(4), AC-6(1), and CM-5(1); CCI-001361, CCI-000017, CCI-000217, CCI-000018, CCI-001403, CCI-001404, CCI-001405, CCI-002130, CCI-001683, CCI-001684, CCI-001685, CCI-001686, CCI-002132, CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-001813.}

Any user interface supporting user accounts (either FULLY or MINIMALLY) must limit access according to specified limitations for each account. Install and configure any device having a STIG or SRG in accordance with that STIG or SRG.

All user interfaces FULLY supporting accounts must implement user-based authentication where each account is uniquely assigned to a specific user. User interfaces FULLY supporting accounts must implement at least three (3) levels of user account privilege including: 1) an account with read-only permissions 2) an account with full permissions including account creation and modification and 3) an account with greater permissions than read-only but without account creation and modification. Disable any unnecessary or unused accounts. Disable any "guest-level" accounts that are created on the system by default.

3.3.1.1 Computers

All computer operating systems must FULLY support user accounts and implement accounts for access. Each control system software application not supporting accounts and running on a computer must be installed such that use of the software is restricted by the computer operating system to specific users.

Applications running on computers must not require the user be logged in to a computer operating system administrator account for normal operation. It is permissible to require the computer operating system administrator account for initial application installation and configuration.

3.3.1.2 Controllers

For user interfaces provided by controllers, provide access control in accordance with the User Interface Requirements table for the applicable control system and user interface type.

a. For table entries of "NA": NA means Not Applicable, there are no interfaces in this category.

b. For table entries of "None Required": The user interface is not required to support user accounts.

c. For table entries of "MINIMALLY": The user interface must at least MINIMALLY support user accounts.

d. For table entries of "FULLY": The user interface must at FULLY support user accounts.

e. For table entries of "KEY": The user interface must have physical security in the form of either a key lock on the interface itself or be furnished inside a lockable enclosure. Where this is required for a read only interface, the lockable enclosure must prevent viewing of data on the interface; for other interfaces, this lockable enclosure must prevent using the interface to alter data.

f. For table entries of "Physical Security": For Local FULL interfaces, the interface must be located inside mission space. For Local Limited (not FULL) interfaces, the user interface must either a) be located within mission space or b) be protected by physical security at least as good as the control devices (and equipment controlled by the control devices) affected by the interface. For purposes of this requirement, 'affected' includes controllers with data that can be directly altered by the interface, as well as mechanical and/or electrical equipment directly controlled by those controllers, but does not include other interactions.

g. Entries of the form "X and Y" must meet both the requirement indicated for X and the requirement indicated for Y. For example, an entry of "MINIMALLY and Physical Security" indicates the user interface must both MINIMALLY support accounts and have physical security.

h. Entries of the form "X or Y" must meet either the requirement indicated for X or the requirement indicated for Y.

3.3.1.2.1 HVAC Control Systems

User Interface Requirements for MODERATE Impact HVAC Control Systems	
User Interface Type	**Access Control Requirement** (See note 3)
Local Read Only (see note 1)	None Required
Local Limited, Non-privileged	None Required
Local Limited, Privileged	MINIMALLY and Physical Security
Local Full	MINIMALLY and Physical Security
Remote Read Only	None Required
Remote Limited, Non-Privileged	FULLY
Remote Limited, Privileged AND Remote Full (see note 2)	FULLY

User Interface Requirements for MODERATE Impact HVAC Control Systems	
User Interface Type	**Access Control Requirement** (See note 3)
Notes: 1)Local Read Only User Interfaces are always Non-Privileged 2)Remote Full User Interfaces are always Privileged 3)Devices outside mission space require physical security protections as indicated (in "PHYSICAL SECURITY IN MODERATE IMPACT SYSTEMS")	

3.3.1.3 Additional User Account Expiration Requirements In MODERATE Impact Systems:

In addition to other user account requirements, user account expiration and auditing must be configured as indicated.

3.3.1.3.1 For Control System Applications Running on Computers

If temporary accounts are supported, expire temporary accounts 72 hours after creation. Expire all other accounts after 35 days of inactivity.

3.3.1.3.2 For Other Control System Devices FULLY Supporting Accounts

If temporary accounts are supported, expire temporary accounts 72 hours after creation. Expire all other accounts after 365 days of inactivity.

3.3.2 Unsuccessful Logon Attempts

{For Government Reference Only: This subpart (and its subparts) relate to AC-7 (a), AC-7 (b); CCI-000043, CCI-000044, CCI-001423, CCI-002236, CCI-002237, CCI-002238}

Except for high availability user interfaces indicated as exempt, devices must meet the indicated requirements for handling unsuccessful logon attempts. If a device cannot meet these requirements, document device capabilities to protect from subsequent logon attempts and propose alternate protections in a Device Account Lock Exception Request submittal. Do not implement alternate protection measures in lieu of the indicated requirements without explicit permission from the Government. If no Device Account Lock Exceptions are requested, provide a document stating that no approval is being requested as the Device Account Lock Exception Request.

3.3.2.1 Devices MINIMALLY Supporting Accounts

For MODERATE Impact Systems: Devices which MINIMALLY (but not FULLY) support accounts must lock the user accountafter five consecutive failed login attempts and must unlock the user account after 60 minutes have elapsed without an unsuccessful login attempt or by a successful login to a separate administrator account.

3.3.2.2 Devices FULLY Supporting Accounts

Devices which FULLY support accounts must meet the following requirements.

a. It must lock the user account when three unsuccessful logon attempts occur within a 15 minute interval.

b. Once an account is locked, the account must stay locked until unlocked by an administrator. If the account being locked is the sole administrator account on the device, the account must stay locked for 1 hour and then automatically unlock.

c. Once the indicated number of unsuccessful logon attempts occurs, delay further logon prompts by 5 seconds.

3.3.2.3 High Availability Interfaces Exempt from Unsuccessful Logon Attempts Requirements

There are no high availability interfaces which are exempt from unsuccessful logon attempts requirements.

3.3.3 System Use Notification

{For Government Reference Only: This subpart (and its subparts) relates to AC-8; CCI-000048, CCI-002247, CCI-002243, CCI-002244, CCI-002245, CCI-002246, CCI-000050, CCI-002248}

3.3.3.1 System Use Notification for Remote User Interfaces

Remote user interfaces must display a warning banner meeting the requirements of DTM 08-060 on screen.

3.3.3.2 System Use Notification for Local User Interfaces

Devices which are connected to a network and have a local user interface must display a warning banner meeting the requirements of DTM 08-060 on the user interface screen if capable of doing so and must have a permanently affixed label with an approved banner from DTM 08-060 if unable to display the warning banner on the screen. Where it is impractical (perhaps due to device size) to affix the label to the device, affix the label to the device enclosure.

Labels must be machine printed or engraved, plastic or metal, designed for permanent installation, must use a font no smaller than 14 point, and must provide a high contrast between font and background colors.

3.3.4 Session Lock and Session Termination Requirements In MODERATE Impact Systems:

{For Government Reference Only: This subpart (and its subparts) relates to AC-11(a), AC-11(b), AC-11(1), AC-12, SC-10; AC-10; CCI-000058, CCI-000059, CCI-000056, CCI-000057, CCI-000060, CCI-002360, CCI-002361, CCI-001133, CCI-001134, CCI-000054, CCI-000055, CCI-002252}

3.3.4.1 Session Termination

When session termination is required for a User Interface, the User Interface must implement session termination a) based on manual initiation, or b) based on lack of activity, or c) based on either manual

initiation or lack of activity, as indicated.

Session Termination must result in logging out the user. A logged out User Interface may only perform actions as indicated in the "Permitted Actions Without Identification or Authentication" subpart of this Section or display a publicly viewable image or blank screen. User Interfaces must remain logged out (session terminated) until a user enters correct authentication information, which must initiate a new session. All User Interfaces running on computers and all Remote User Interfaces must also terminate network connections as part of session termination.

3.3.4.2 Session Lock

When session lock is required for a User Interface, the User Interface must implement session lock a) based on manual initiation, or b) based on lack of activity, or c) based on either manual initiation or lack of activity, as indicated.

Session lock must result in the User Interface being suspended and the user interface must display a publicly viewable image or blank screen. No interaction with the user interface must be possible until either a) the same user enters valid authentication information, in which case that session must be continued, or b) until a different user enters valid authentication information at which point the first session must be terminated and a new session initiated for the new user.

3.3.4.3 Session Lock and Termination for Computers

User Interface sessions provided by computer operating systems must support the requirement for both Session Lock and Session Termination. Session Lock and Session Termination must be capable of being initiated by the user and must also be initiated by lack of activity. Session Lock must occur after 15 minutes of inactivity, and Session Termination must occur after 30 minutes total of inactivity (including, not in addition to, the time for Session Lock). When a user initiates a new session, terminate existing sessions if necessary to limit the total number of concurrent sessions to 1.

Other User Interface sessions running on computers (for local user interfaces) or hosted on a computer (for remote user interfaces) and supporting accounts must support user initiation of Session Termination. In addition, remote User Interface sessions must also initiate Session Termination after 30 minutes of inactivity .

3.3.4.4 Session Lock and Termination for Controllers

Writable Remote User Interfaces must support requirements for Session Termination, and must both be capable of being initiated by the user and initiated by lack of activity. Session Termination must initiate after 30 minutes of inactivity.

Local User Interfaces supporting accounts must support manual initiation of Session Termination. Privileged Local User Interfaces must also support timed initiation of Session Termination, with Session Termination initiated at 30 minutes of inactivity.

3.3.5 Permitted Actions Without Identification or Authentication

{For Government Reference Only: This subpart (and its subparts) relates

to AC-14; CCI-000061, CCI-000232}

The control system must require identification and authentication before allowing any actions except read-only actions by a user acting from a user interface which MINIMALLY or FULLY supports accounts.

3.3.6 Physical Security in MODERATE Impact Systems

{For Government Reference Only: This subpart relates to PE-3(1), PE-4, PE-5, SC-7(a), SC-7(c), SC-8, SC-8(1); CCI-000928, CCI-002926, CCI-000936, CCI-002930, CCI-002931, CCI-000937, CCI-001097, CCI-001109, CCI-002418, CCI-002419, CCI-002421.}

3.3.6.1 Physical Security for Media

3.3.6.1.1 Physical Security for Media Inside Mission Space

Install all non-IP network media located inside of the mission space in conduit. Install all IP network media located inside of the mission space in intermediate metallic conduit.

3.3.6.1.2 Physical Security for Media Outside Mission Space

Install all network media (both IP and non-IP) located outside of the mission space in rigid metallic conduit.

3.3.6.2 Physical Security for Devices

Install all devices (computers and controllers) which are located outside of mission space in lockable enclosures. (Recall that per definition of mission space, a room controlled by the mission is mission space regardless of whether it is contiguous with other mission space.)

Install all controllers connected to an IP network in lockable enclosures (both inside and outside of mission space).

3.3.6.3 Physical Security for User Interfaces

Physical security requirements for User Interfaces are specified in the preceding paragraphs of this Section.

3.3.7 Enclosures

Prior to final acceptance of the system, lock all lockable enclosures. Submit an Enclosure Keys submittal with all copies of keys for all enclosures and a key inventory list documenting all keys. Label each key with the matching enclosure identifier.

3.4 USER IDENTIFICATION AND AUTHENTICATION

{For Government Reference Only: This subpart (and its subparts) relates to IA-2, IA-2(1),IA-2(12), IA-5 IA-5(b), IA-5(c), IA-5(e), IA-5(g), IA-5(1), IA-5(11); CCI-000764, CCI-000765, CCI-001953, CCI-001954, CCI-001544, CCI-001989, CCI-000182, CCI-001610, CCI-000192, CCI-000193, CCI-000194, CCI-000205, CCI-001619, CCI-001611, CCI-001612, CCI-001613, CCI-001614, CCI-000195, CCI-001615, CCI-000196, CCI-000197, CCI-000199, CCI-000198, CCI-001616, CCI-001617, CCI-000200, CCI-001618, CCI-002041, CCI-002002, CCI-002003. For MODERATE Impact systems, this subpart also relates to AC-6 (1), AC-6(10), AC-6(2), AC-6(9)-IA-2(4), IA-5(13);

CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-002235, CCI-000039, CCI-001419, CCI-002234, CCI-000768, CCI-002007.}

This subpart indicates requirements for specific methods of identification and authentication for users and user accounts. Where these requirements conflict apply the following order of precedence: 1) If present, Device Specific Requirements take precedence over any other requirements; and then 2) multifactor authentication requirements take precedence over password requirements.

3.4.1 User Identification and Authentication for All System Types

Unless otherwise indicated, all user interfaces supporting accounts (either FULLY or MINIMALLY) must implement Identification and Authorization via passwords.

For MODERATE Impact Systems: Software running on computers and computer operating systems must manage cached authenticators in accordance with the relevant STIGs. All other devices and software must not use cached authenticators.

3.4.2 User Identification and Authentication for Specific System Types

System specific requirements are in addition to and supersede those indicated for all system types. When no additional requirements are indicated for a specific system type the requirements for all systems still apply to that system type.

3.4.2.1 HVAC Control Systems Devices

No additional system specific requirements apply.

3.4.3 User Identification and Authentication for Specific Devices

There are no additional device specific user interface requirements

3.4.4 Implementation of Identification and Authorization Requirements

Identification and Authorization must be met by one of the following methods:

a. Direct implementation in the user interface.

b. For user interfaces on a computer: inheriting the Identification and Authorization from the computer operating system, either by the operating system limiting access to specific applications by user, or by the application itself having permissions based on the user logged into the computer.

c. For remote interfaces: an implementation shared between the remote user interface server and the remote user interface client. For example, a requirement for PIV authentication may be met on a remote user interface by a PIV reader on a web browser client which sends the authentication information via HTTPS to the remote server.

3.4.5 Password-Based Authentication Requirements

3.4.5.1 Passwords for Software and Applications Running on Computers

All software and applications running on computers supporting password-based authentication must enforce the following requirements:

a. Minimum password length of 12 characters

b. Password must contain at least one uppercase character.

c. Password must contain at least one lowercase character.

d. Password must contain at least one numeric character.

e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.

f. Password must have a minimum lifetime of 24 hours.

g. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.

h. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters (where location is significant, a character may be reused if it is in a different position).

i. Passwords must be cryptographically protected during storage and transmission.

3.4.5.2 Passwords for Controllers FULLY Supporting Accounts

All controllers FULLY supporting accounts and supporting password-based authentication must enforce the following requirements:

a. Minimum password length of 12 characters

b. Password must contain at least one uppercase character.

c. Password must contain at least one lowercase character.

d. Password must contain at least one numeric character.

e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.

f. Password must have a maximum lifetime of sixty (60) days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.

g. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters.

h. Passwords must be cryptographically protected during storage and transmission.

3.4.5.3 Passwords for Remote Interfaces

Passwords for connecting to a Remote User Interface supporting password-based authentication must enforce the following requirements:

a. Minimum password length of 12 characters

b. Password must contain at least one uppercase character.

c. Password must contain at least one lowercase character.

d. Password must contain at least one numeric character.

e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.

f. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.

g. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters (where location is significant, a character may be reused if it is in a different position).

h. Passwords must be cryptographically protected during storage and transmission.

3.4.5.4 Passwords for Devices Minimally Supporting Accounts

Devices MINIMALLY supporting accounts must support passwords with a minimum length of four characters.

3.4.5.5 Password Configuration and Reporting

For all devices with a password, coordinate the changing of passwords with the project site following testing of the system but prior to turnover to the Government. Coordinate with Password Point of Contact to determine appropriate project site personnel to complete password changes. Accompany identified personnel to each device with a password and instruct personnel on the process of changing password. Record the time, date and personnel present when each device's password is changed and submit a Password Change Summary Report documenting this information.

Provide the Password Summary Report electronically in both PDF and Microsoft Excel.

3.4.6 Authenticator Feedback

{For Government Reference Only: This subpart relates to IA-6; CCI-000206}

Devices must never show authentication information, including passwords, on a display. Devices that momentarily display a character as it is entered, and then obscure the character, are acceptable. For devices that have STIGs or SRGs related to obscuring of authenticator feedback (CCI-000206), comply with the requirements of those STIGS/SRGs.

3.5 CYBERSECURITY AUDITING

Where an auditing requirement exists for email notification, notify via email the application administrator and Information System Security Officer (ISSO) of the event. Coordinate with the Email Address Point of Contact for email addresses. If outgoing email is not available to the system, disable email notifications.

3.5.1 Audit Events, Content of Audit Records, and Audit Generation

{For Government Reference Only: This subpart (and its subparts) relates to AU-2(a), AU-2(c), AU-2(d), AU-3, AU-10, AU-12, AU-14(b), AU-14(1), AU-14(2), AU-14(3), CM-5(1), SC-7 (9); CCI-000123, CCI-001571, CCI-000125, CCI-001485, CCI-000130, CCI-000131, CCI-000132, CCI-001230, CCI-000133, CCI-000134, CCI-001487, CCI-000166, CCI-001899, CCI-000169, CCI-001459, CCI-000171, CCI-000172, CCI-001910, CCI-001914, CCI-001919, CCI-001464, CCI-001462, CCI-001920, CCI-001814, CCI-002400. For MODERATE Impact systems, this subpart (and its subparts) also relates to AU-3 (1); CCI-000135, CCI-001488}

For devices that have STIG/SRGs related to audit events, content of audit records or audit generation, comply with the requirements of those STIG/SRGs.

If auditing requirements can be met using existing control system alarm or event capabilities, those existing capabilities may be used to meet these requirements.

3.5.1.1 Computers

For each computer, provide the capability to select audited events and the content of audit logs. Configure computers to audit the indicated events, and to record the indicated information for each auditable event. Send logs to a syslog server that is only accessible via accounts on the system with privileged level access.

3.5.1.1.1 Audited Events

Configure each computer to audit the following events:

a. Successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels)

b. Successful and unsuccessful logon attempts

c. Successful logouts

d. Privileged activities or other system level access

e. Concurrent logons from different workstations

f. Successful and unsuccessful accesses to objects

g. Program initiations

h. Direct access to the information system

i. Account creations, modifications, disabling, and terminations. For MODERATE Impact Systems, also provide email notification when these audit events occur.

j. Kernel module load, unload, and restart

3.5.1.1.2 Audit Event Information To Record

Configure each computer to record, for each auditable event, the following information (where applicable to the event):

a. What type of event occurred

b. When the event occurred

c. Where the event occurred

d. The source of the event

e. The outcome of the event

f. The identity of any individuals or subjects associated with the event

g. For MODERATE Impact Systems: For all privileged commands, full-text recording of the executed command and the user executing the command

For MODERATE Impact Systems: Audit records must provide sufficient detail to reconstruct events to determine cause of compromise and magnitude of damage, malfunction, or security violation.

3.5.1.2 For HVAC Control System Controllers

3.5.1.2.1 HVAC Control System Controllers FULLY Supporting User Accounts

For each controller which FULLY supports accounts, provide the capability to select audited events and the content of audit logs. Configure controllers to audit the indicated events, and to record the indicated information for each auditable event.

3.5.1.2.1.1 Audited Events

Configure each controller to audit the following events:

a. Successful and unsuccessful logon attempts to the controller

b. Successful logouts

c. All account creations, modifications, disabling, and terminations. For MODERATE Impact Systems, also provide email notification when these audit events occur.

d. All controller shutdown and startup

e. For privileged user interfaces in MODERATE Impact Systems: All user commands.

3.5.1.2.1.2 Audit Event Information To Record

Configure each controller to record, for each auditable event, the following information (where applicable to the event):

a. what type of event occurred

b. when the event occurred

c. the identity of any individuals or subjects associated with the event

d. For privileged user interfaces in MODERATE Impact Systems: Full text recording of the executed command and the user executing the command.

For MODERATE Impact Systems: Audit records must provide sufficient detail to reconstruct events to determine cause of compromise and magnitude of damage, malfunction, or security violation.

3.5.1.2.2 Other HVAC Control System Controllers

There are no requirements to perform auditing at HVAC field controllers that do not FULLY support accounts.

3.5.2 Audit Time Stamps

{For Government Reference Only: This subpart (and its subparts) relates to AU-8; CCI-000159, CCI-001889, CCI-001890. For MODERATE Impact systems, this subpart (and its subparts) also relates to AU-8 (1); CCI-001891, CCI-001892, CCI-002046.}

Any device (computer or controller) generating audit records must have an internal clock capable of providing time with a resolution of one second. Clocks must not drift more than 10 seconds per day. Configure the system so that each device (computer or controller) generating audit records maintains accurate time to within 1 second. Note that if the control system specifications include requirement for clocks, the most stringent requirement applies.

3.5.3 Response to Audit Processing Failures

{For Government Reference Only: This subpart (and its subparts) relates to AU-5; CCI-000139, CCI-000140, CCI-001490.}

In the case of a failure in the auditing system, computers associated with auditing must provide email notification. For MODERATE Impact systems, the computer must also notify the associated auditing front end software. In case of an audit failure, if possible, continue to collect audit records by overwriting existing audit records.

For MODERATE Impact Systems: In the case of an audit failure at a controller performing auditing, the device must notify the associated auditing front end software of the audit failure if able, and must continue to collect audit records by overwriting existing audit records if able. The auditing front end software must provide notification as indicated, treating the notification of failure from the device as a failure in the auditing system.

3.6 REQUIREMENTS FOR LEAST FUNCTIONALITY

{For Government Reference Only: This subpart (and its subparts), along with the Network Communication, Ports, Protocols and Services Report submittal specified elsewhere in this section, relates to CM-6(a), CM-6(c), CM-7, CM-7(1)(b), SC-41; CCI-000363, CCI-000364, CCI-000365, CCI-001588, CCI-001755, CCI-000381, CCI-000380, CCI-000382, CCI-001761, CCI-001762, CCI-002544, CCI-002545, CCI-002546. For MODERATE Impact systems, this subpart (and its subparts) also relates to CM-7(2), CM-7(5)(a), CM-7(5)(b); CCI-000381, CCI-000380, CCI-000382, CCI-001761, CCI-001762}

For devices that have a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port and device I/O access for least functionality), install and configure the device in accordance with that STIG or SRGs.

3.6.1 Device Capabilities

For HVAC Control Systems: Do not provide devices with remote user interfaces or with full user interfaces where one was not required. Do not use a networked sensor or actuator where a non-networked sensor or actuator would suffice.

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For all MODERATE Impact Systems: Unless specifically required by the government, do not provide a capability to update device firmware over the network.

3.6.2 Software

For software that has a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port access for least functionality), install and configure the software in accordance with that STIG or SRG.

Do not install software that is not specifically required to meet a contract requirement. Remove any previously installed that is not specifically required to meet a contract requirement. Do not implement functionality within software
that is not specifically required to meet contract requirements.

3.7 SYSTEM AND COMMUNICATION PROTECTION

3.7.1 Collaborative Computing

{For Government Reference Only: This subpart relates to SC-15(a), SC-15(b); CCI-001150, CCI-001152.}

Without explicit approval from the project site, control systems must not use collaborative computing technologies.

3.7.2 Denial of Service Protection and Application Partitioning In MODERATE Impact Systems:

{For Government Reference Only: This subpart relates to SC-5, SC-12, SC-7(a); CCI-001093, CCI-002385, CCI-002386, CCI-002430, CCI-001097. For MODERATE Impact systems, this subpart also relates to SC-2; CCI-001082.}

To the greatest extent practical, implement control logic without reliance on the network. Except when required to meet the requirements of the control system Section (where the requirement can only be met using computer hardware), do not implement control logic in computers. For MODERATE Impact systems, do not implement control logic in a device providing (i.e. acting as a server for) a Full Remote User Interface.

3.7.2.1 Network Reliance in MODERATE Impact HVAC Control Systems

Except for networked input and outputs on input-output buses specifically designed to provide high reliability or redundancy, sensors and actuators must not rely on the network to exchange data with the controller executing the sequence of operation which uses the sensor value or determines the actuator command.

Sensor values required by multiple devices may be shared over the network provided they are connected to a controller requiring the value for execution of the sequence and that controller shares the value on the network.

3.7.3 Mobile Code In MODERATE Impact Systems:

{For Government Reference Only: This subpart relates to SC-18(a), SC-18(b),
SC-18(c), SC-18(1), SC-18(3), SC-18(4); CCI-001160, CCI-001161, CCI-001162, CCI-001163, CCI-001164, CCI-001165, CCI-001166, CCI-001662, CCI-002457, CCI-002458, CCI-001169, CCI-001695, CCI-001170, CCI-002469}

Devices with STIGs/SRGs related to Mobile Code and to Security Control SC-18 must be installed in accordance with the relevant STIGs/SRGs. All remote user interfaces must meet the requirements of the "Web Browsers and Application SRG".

Mobile code may only be downloaded from a specifically authorized mobile code repository. Coordinate with the Mobile Code Point of Contact for the location of a repository.

3.7.4 Protection of Information at Rest In MODERATE Impact Systems:

{For Government Reference Only: This subpart relates to SC-28, SC-28(1); CCI-001199, CCI-002472, CCI-002475, CCI-002476}

Computers must protect information at rest in accordance with applicable STIGs.

Any control system device storing personally identifiable information (PII), controlled unclassified information (CUI), or classified information must be protected by an Information At Rest encryption solution or by a physical security solution. Provide a Protection of Information At Rest Proposal indicating each device storing PII, CUI, or classified information and the encryption or physical security solution proposed for that device for government approval. If no devices stores PII, CUI, or classified information, provide a document stating this as the Protection of Information At Rest Proposal submittal. Do proceed with device selection and installation until the Protection of Information At Rest Proposal is approved. Once approved, implement approved Information At Rest protections.

3.8 SAFE MODE AND FAIL SAFE OPERATION

{For Government Reference Only: This subpart (and its subparts) relates to CP-12, SI-10(3), SI-17; CCI-002855, CCI-002856, CCI-002857, CCI-002754, CCI-002773, CCI-002774, CCI-002775}

For all control system components with an applicable STIG or SRG, configure the component in accordance with all applicable STIGs and SRGs.

3.9 SYSTEM MAINTENANCE TOOL SOFTWARE

{For Government Reference Only: This subpart (and its subparts) relates to MA-3; CCI-000865.}

Submit and license to the Government all software required to operate, maintain and modify the control system such the Government or their agents are able to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer. Submit hard copies of user manuals for each software with the software submittal. Provide any hardware keys or dongles, software keys, license numbers, and other information required to enable the Government to access or change the system.

For software provided and licensed to the Government under the requirements of another Section, submit a statement indicating the Section and Submittal under which the software was provided. For software provided to meet the requirements of this Section and not provided and licensed under another Section, submit software and software user manuals on DVD or CD as a Technical Data Package and submit one hard copy of the software user manual for each piece of software.

3.10 DEVICE POWER

{For Government Reference Only: This subpart (and its subparts) relates to PE-11, PE-11(1); CCI-002955, CCI-000961. For MODERATE Impact systems, this subpart (and its subparts) also relates to PE-9, PE-9(1); CCI-000952, CCI-002953, CCI-002954.}

For MODERATE Impact Systems: Provide control system with power supply meeting or exceeding the reliability of the controlled equipment. Powering control system devices using the same power source as the equipment controlled by the device is a permissible method of meeting this requirement. Without explicit approval from the government, do not install local uninterruptible power supplies (UPSes) as a source of device power.

3.10.1 Device Behavior on Loss of Power In MODERATE Impact Systems:

Application programs and configuration settings must be stored in devices in manner such that a loss of power does not result in a loss of the application program or configuration settings: Loss of power must never result in the loss of application programs, regardless of the length of time power is lost; and loss of power for less than 2,500 hours must not result in the loss of configured settings.

In the event of a loss of power, when power is restored, controllers and computers executing control logic (and the underlying equipment) must recover and resume their normal sequences of operation. Note that the sequence of operation may require specific actions (e.g. startup

sequences) upon recovery from loss of power.

3.11 VULNERABILITY SCANNING

{For Government Reference Only: This subpart (and its subparts) relates to RA-5 RA-5(a),RA-5(b),RA-5(c),RA-5(d); CCI-001054, CCI-001055, CCI-000156, CCI-001641, CCI-001643, CCI-001057, CCI-001058, CCI-001059. For MODERATE Impact systems, this subpart (and its subparts) also relates to RA-5(1), RA-5(5); CCI-001062, CCI-001067, CCI-001645, CCI-002906.}

All IP devices must be scannable, such that the device can be scanned by industry standard IP network scanning utilities without harm to the device, application, or functionality.

3.11.1 Computers and Software Running on Computers

Computers and applications running on computers must meet relevant vulnerability scanning STIGs/SRGs and respond to approved DoD vulnerability scanning tools.

3.11.2 Controllers

Provide controllers that are scannable by standard control system discovery tools or control system browsers and return meaningful status information including the network inputs and outputs for the controller. This information must contain sufficient detail to detect vulnerabilities or exploits of the controller.

Provide all software needed to scan the control system as the Control System Scanning Tools submittal. If the software required to scan the system is already installed at the project site or is provided under a separate section instead provide a statement indicating this.

3.12 SYSTEM AND INTEGRATION INTEGRITY

3.12.1 Malicious Code Protection

{For Government Reference Only: This subpart (and its subparts) relates to SI-3(c); CCI-001241, CCI-002623}

For all computers installed under this project, provide malware protection software media, provide licenses, and install and configure malware protection software as indicated. Coordinate with the Government Computer Access Point of Contact as required.

a. Malware protection software licenses will be Government furnished.

b. Malware protection software media will be Government furnished.

c. Malware protection software will be Government installed.

3.13 CONTROL SYSTEM CYBERSECURITY TESTING

{For Government Reference Only: For MODERATE Impact systems, this subpart (and its subparts) relates to SA-11(a), SA-11(b), SA-11(c), SA-11(d), SA-11(e); CCI-003171, CCI-003172, CCI-003173, CCI-003174, CCI-003175, CCI-003176, CCI-003177, CCI-003178.}

3.13.1 Control System Cybersecurity Testing Procedures

Prepare and submit Control System Cybersecurity Testing Procedures explaining step-by-step, the actions and expected results that will demonstrate that the control system meets the requirements of this Section. The Control System Cybersecurity Testing Procedures may be submitted as a Technical Data Package.

3.13.2 Control System Cybersecurity Testing Execution

Using the Control System Cybersecurity Testing Procedures verify that the control system meets the requirements of this Section. UNLESS GOVERNMENT WITNESSING OF A TEST IS SPECIFICALLY WAIVED BY THE GOVERNMENT, PERFORM ALL TESTS WITH A GOVERNMENT WITNESS. If testing reveals deficiencies in the system, correct the deficiency and retest until successful.

3.13.3 Control System Cybersecurity Testing Report

Prepare and submit a Control System Cybersecurity Testing Report documenting all tests performed and their results. Include all tests in the Control System Cybersecurity Testing Procedures and any additional tests performed during testing. Document test failures and repairs conducted with the test results. The Control System Cybersecurity Testing Report may be submitted as a Technical Data Package

3.14 FIELD QUALITY CONTROL, CYBERSECURITY VALIDATION SUPPORT

In addition to testing and testing support required by other Sections, provide a minimum of eight hours of technical support for cybersecurity testing of control systems to support the DoD Risk Management Framework process Cybersecurity assessment of the control system. This support is independent of (and in addition to) the Control System Cybersecurity Testing specified in this section.

3.15 CYBERSECURITY TRAINING

Provide eight hours of classroom training for six Government personnel on the cybersecurity operation and maintenance of the control system provided. This training is in addition to and must be coordinated with control system training specified in other Sections.

The Government will provide the training location. Training must cover, at a minimum: (a) applying software and firmware updates, (b) user account creation, modification and deletion, (c) audit log upload procedures and (d) identification of privileged user interfaces and system impact of those interfaces. Training session must include a question and answer period during which government staff questions about cybersecurity aspects of the control system are answered.

-- End of Section --

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08/24

cybersecurity for LOW impact facility-related control systems

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SECTION 25 05 11.02

CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - UTILITY MONITORING AND CONTROL SYSTEMS (L-L-L)
08/24

PART 1 GENERAL

Many subparts in this Section contain text in curly braces ("{" and "}") indicating which cybersecurity control and control correlation identifier (CCI) the requirements of the subpart relate to. The text inside these curly braces is for Government reference only and enables coordination of the requirements of this Section with the RMF process throughout the design and construction process. Text in curly braces are not contractor requirements.

This Section refers to Security Requirements Guide (SRGs) and Security Technical Implementation Guide (STIGs). STIGs and SRGs are available online at the Information Assurance Support Environment (IASE) website at https://public.cyber.mil/stigs/downloads/ and an SRG/STIG Applicability Guide and Collection Tool is available at https://public.cyber.mil/stigs/SCAP/. Not all control system components have applicable STIGs or SRGs. The "Control Systems SRG" does not apply to work performed under this Section; all requirements within this section to apply applicable SRGs DO NOT include the "Control Systems SRG".

1.1 CONTROL SYSTEM APPLICABILITY

There are multiple versions of this Section associated with this project. Different versions have requirements applicable to different control systems. This specific Section applies only to the following control systems: Utility Monitoring and Control Systems, SCADA.

1.2 RELATED REQUIREMENTS

This section does not contain sufficient requirements to procure a control system and must be used in conjunction with other Sections which specify control systems. This Section adds cybersecurity requirements to the control systems specified in other Sections, and as these requirements are conditioned on the control system being provided, there may be requirements in this Section that will not apply to this project. All Sections containing facility-related control systems or control system components are related to the requirements of this Section. Review all specification sections to determine related requirements.

In cases where a requirement is specified in both this Section and in another Section, the more stringent requirement must be met. In cases where a requirement in this Section conflicts with the requirements of another Section such that both requirements cannot be met at the same time, request direction from the Contracting Officer to determine which requirement applies to the project.

Systems that are to be connected to the VA network shall have an Enterprise Risk Assessment (ERA) performed. ERAs shall conform to the medical center's standards.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.1x	(2010) Local and Metropolitan Area Networks - Port Based Network Access Control

INTERNET ENGINEERING TASK FORCE (IETF)

IETF RFC 2819	(2000) Remote Network Monitoring (RMON) Management Information Base (MIB)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST FIPS 140-2	(2001) Security Requirements for Cryptographic Modules

U.S. DEPARTMENT OF DEFENSE (DOD)

DODI 8551.01	(2014) Ports, Protocols, and Services Management (PPSM)
DTM 08-060	(2008) Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement

1.4 DEFINITIONS

1.4.1 Administrator Account

An administrator account is an account with full permissions to a device, application, or operating system, including the ability to create and modify other user accounts.

Note that the operating system Administrator Account may be different than Administrator Accounts for applications hosted on that operating system. Also, most controllers will not have any support for accounts and will therefore not have an 'Administrator Account'.

1.4.2 Computer

A computer is one of the following:

a. a device running a non-embedded desktop or server version of Microsoft Windows

b. a device running a non-embedded version of MacOS

c. a device running a non-embedded version of Linux

d. a device running a version or derivative of the Android Operating System, where Android is considered separate from Linux

e. a device running a version of Apple iOS

Unless otherwise indicated or clear from context use of the word "device" in this Section includes computers.

1.4.3 Controller

A device other than a computer or Ethernet switch.

1.4.4 Mission Space

A device or media is in mission space if physical access to the device or media is controlled by the organization served by the device. For example, a variable air volume (VAV) box controller in a suspended ceiling is in mission space if the VAV box serves that room; an electrical switchgear in an electrical room or an air handling unit (AHU) in a mechanical room or on a rooftop may still be considered to be in mission space if the organization (mission) served by that switchgear or AHU controls access to the electrical room, mechanical room or rooftop.

1.4.5 Network

A network is a group of two or more devices that can communicate using a network protocol. Network protocols must provide a method for addressing devices on the network; a communication method that does not provide an addressing scheme is not a networked form of communication. Devices that communicate using a method of communication that does not support device addressing are not using a network. Addresses may be other than IP addresses, and addressing may be at either Open Systems Interconnection (OSI) layer 2 or layer 3.

1.4.6 Network Connected

A component is network connected (or "connected to a network") only when the device has a network transceiver which is directly connected to the network and implements the network protocol. A device lacking a network transceiver (and accompanying protocol implementation) can never be considered network connected. Note that (unlike many IT definitions of "Network Connected") a device connected to a non-IP network is still considered network connected (an IP connection or IP address is not required for a device to be network connected).

1.4.6.1 Wireless Network Connected

Any device that supports wireless network communication is network connected to a wireless network, regardless of whether the device is communicating using wireless. Unless physically disabled, devices with wireless transceivers support wireless, it is not sufficient to disable the wireless in software.

1.4.7 Network Media

The thing that provides the communication channel between the devices on a network. Typically wire, but might include wireless, fiber optic, or even power line (some network protocols allow sending network signals over power wiring).

1.4.8 User Account Support Levels

The support for user accounts is categorized in this Section as one of

three levels:

1.4.8.1 FULLY Supported

Device supports configurable individual accounts. Accounts can be created, deleted, modified, etc. Privileges can be assigned to accounts. These devices support user-based (as opposed to role-based) authentication.

1.4.8.2 MINIMALLY Supported

Device supports a small, fixed number of accounts (perhaps only one). Accounts cannot be modified. A device with only a "User" and an "Administrator" account would fit this category. Similarly, a device with two PINs for logon - one for restricted and one for unrestricted rights would fit here (in other words, the accounts do not have to be the traditional "username and password" structure). These devices typically only support role-based authentication.

Examples of devices which MINIMALLY support accounts are a) a variable frequency drive with a single account which requires a PIN for access to configuration; and b) a room lighting control touchpad interface that has a single account.

1.4.8.3 NOT Supported

Device does not support any Access Enforcement therefore the whole concept of "account" is meaningless.

1.4.9 Manual Local Input

Manual Local Inputs are system analog or binary inputs that are adjustable by a person but are, by intrinsic hardware design, very limited in potential capabilities. Manual Local Inputs do not have touch screens or full keyboards, but may have a few buttons or dials to allow input. Manual Local Inputs do not have full graphic screens or dot-matrix displays, but may have simple lights (LEDs) or 7-segment displays. Manual Local Inputs do not have any sort of menu structure, each button has a single well-defined function.

Examples of Manual Local Inputs are H-O-A switches, simple thermostats, and disconnect switches.

1.4.10 Card Reader

A card reader is an input/output device whose primary function is to assist in two-factor authentication. A card reader must have an interface to read data from a card and may be able to write data to a card. A card reader may have a means (such as buttons, keypad, touchscreen, etc.) for a user to input a PIN or password, as well as a limited display.

1.4.11 User Interface

A User Interface (UI) is something other than a Manual Local Input or Card Reader that allows a person to interact with the system or device. Note that while a Card Reader is not by itself a User Interface, a User Interface may contain a Card Reader in order for it to authenticate its user. Within control systems, there are a wide range of User Interfaces.

Two important distinctions are 1) whether the user interface is Local or

Remote, and 2) the effective capabilities of the User Interface to alter data, which is the "privilege" of the user interface (where effective privilege available to a specific user at a specific user interface is the combination of the greatest privilege offered by the user interface and the specific account the user is logged into).

1.4.11.1 Local User Interface

A Local User Interface is a user interface where the physical hardware the user interacts with (keyboard, buttons, display, etc.) is physically part of the device being affected. All of the relevant characteristics of the user interface are embodied within a single device.

Note that a Local UI may be able to access data in a different device, Local versus Remote in this context refers to the user interface itself; the capability to access data in a different device is covered under "Full User Interface".

1.4.11.2 Remote User Interface

A Remote User Interface implements a Client/Server model where the physical hardware the user interacts with (Client) is physically distinct from the device being affected (Server). Most or all of the security and functionality characteristics of the user interface are defined by the Server, not the Client. The Client and Server communicate via a network connection. A common example of a remote user interface is a web-based interface where the browser (client) is generally on different hardware than the web server (server). A Remote UI remains a Remote UI even if the user happens to be at a Client on the same hardware as the Server. What is important is that a) the Client may be on different hardware than the Server and b) the majority of the security and functional characteristics of the interface are defined at the Server.

Note that this definition of "remote" is consistent with that generally used in the control industry but is not aligned with the NIST 800-53 definition of "Remote", which refers to "outside the system". The term "Remote" here better aligns with the NIST 800-53 definition of "Network" (remote from within the system) Access.

1.4.11.3 Types of User Interface (by capability)

User interfaces are also categorized by their capabilities as being Read Only, Limited, or Full.

1.4.11.3.1 Read-Only User Interface

A Read Only User Interface (also referred to as a View-Only User Interface) is a user interface that only allows for reading data, it does not allow (have the capability to) modify data. A Read Only User Interface may be either Local or Remote. A User Interface that is configured to be Read Only (by some other means than the interface itself, such as using configuration software on a laptop) is a Read-Only Interface. Note a Read Only User Interface may have buttons (or touch screen, etc.) allowing the user to navigate through the presentation of data.

Examples of a Read Only User Interfaces are a) a publicly viewable "energy dashboard" showing weather data and energy usage within a building and b) digital wayfinding signage.

1.4.11.3.2 Limited User Interface

A Limited User Interface is a user interface that - by design - can only alter information local to the user interface. Note that the determination of "alter" includes only direct interactions, it explicitly excludes interactions that might occur as secondary effects. For example, an interface changing the flow setpoint in a pump controller is a direct interaction, the subsequent change in flow (as well as any subsequent downstream changes in valve position) are not direct interactions.

Two examples of LIMITED UIs are: a) a variable speed drive has a Limited Local User Interface which allows the user to change properties within the drive, but does not allow affecting things outside the drive; and b) a typical home WiFi Router has a Limited Remote User Interface which allows configuration of the Router, but does not allow direct interaction with other devices.

1.4.11.3.3 Full User Interface

A Full User Interface can alter information in devices outside the device with the user interface. For example, a typical Local Display Panel is a Full Local User Interface while a browser-based front end is a Full Remote User Interface.

1.4.11.3.4 View-Only User Interface

See Read-Only User Interface

1.4.11.4 Other User Interface Terminology

In addition to defining whether a user interface is a Hardware Limited, Read-Only, Limited or Full, and whether it is Local or Remote, user interfaces are classified by whether they are writable or privileged.

1.4.11.4.1 Writable User Interface

Any User Interface that is not Read-Only is Writable. (Limited User Interfaces and Full User Interfaces are both writable user interfaces (as they are capable of changing a value)).

1.4.11.4.2 Privileged User Interface

A Privileged UI is a UI that has sufficient capabilities or functionality that it requires specific cybersecurity measures to be put in place to limit its unauthorized use. Ultimately, whether a specific user interface is considered a Privileged User Interface must be determined by usage. Unless otherwise specified, user interfaces can be determined to be privileged or not using the following:

a. Read-Only User Interfaces are not privileged user interfaces.

b. Full User Interfaces are privileged user interfaces.

c. User interfaces that allow for configuration of auditing or allows for modification or deletion of audit logs are privileged user interface.

d. User interfaces that allow for reprogramming a network connected device is a privileged user interface.

e. Except as specified above, a Limited User Interface must be determined to be privileged or not based on the specific capabilities and use case of the user interface. In general however, user interfaces that do not offer significant capabilities above and beyond those available at that location via other means (e.g. such as a disconnect switch, breaker, or hand-off-auto switch, or physical attack) are not privileged.

1.4.12 Wireless Network

Any network that communicates without using wires or fiber optics as the communication media. Wireless networks include: WiFi, Bluetooth, ZigBee, cellular, satellite, long and short wave radio, 2.4 GHz, free space optical, point-to-point laser, microwave, and IR.

1.4.13 Wired Broadcast Network

Wired Broadcast Networks are any network, such as powerline carrier networks and modem (wired telephony), that use wire-based technologies where there is not a clearly defined boundary for signal propagation.

1.5 ADMINISTRATIVE REQUIREMENTS

1.5.1 Points of Contact

Coordinate with the following Points of Contact as indicated in this Section and as required. Not all projects will require coordination with all Points of Contact. When coordination is required and no Point of Contact is indicated, coordinate with the Contracting Officer (KO).

a. Government Computer Access Point of Contact: Contracting Office (KO)

b. HTTPS Certificate Point of Contact: Contracting Officer (KO)

c. Email Address Point of Contact: Contracting Officer (KO)

d. Password Point of Contact: Contracting Officer (KO)

1.5.2 Coordination

Coordinate the execution of this Section with the execution of all other Sections related to control systems as indicated in the paragraph RELATED REQUIREMENTS. Items that must be considered when coordinating project efforts include but are not limited to:

a. If requesting permission for wireless or wired broadcast communication, the Wireless and Wired Broadcast Communication Request submittal must be approved prior to control system device selection and installation.

b. If requesting permission for alternate account lock permissions, the Device Account Lock Exception Request must be approved prior to control system device selection and installation.

c. If requesting permission for the use of a device with multiple physical connections to IP networks, the Multiple IP Connection Device

Request must be approved prior to control system device selection and installation.

d. Wireless testing may be required as part of the control system testing. See requirements for the Wireless Communication Test Report submittal.

e. If the Device Audit Record Upload Software is to be installed on a computer not being provided as part of the control system, coordination is required to identify the computer on which to install the software.

f. The Cybersecurity Interconnection Schedule must be coordinated with other work that will be interconnected to, and interconnections must be approved by the Government before relying on them for system functionality.

g. Cybersecurity testing support must be coordinated across control systems and with the Government cybersecurity testing schedule.

h. Passwords must be coordinated with the indicated contact for the project site.

i. If applicable, HTTPS web server certificates must be obtained from the indicated HTTPS Certificate Point of Contact.

j. Contractor Computer Cybersecurity Compliance Statements must be provided for each contractor using contractor owned computers.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Wireless and Wired Broadcast Communication Request; GDevice Account Lock Exception Request; G

Multiple Ethernet Connection Device Request; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

Cybersecurity Interconnection Schedule; G

Proposed STIG and SRG Applicability Report; G

Pre-Construction Control System Inventory Report; G

SD-02 Shop Drawings

Network Communication, Ports, Protocols and Services Report; G

Cybersecurity Network (Riser) Diagram; G

SD-03 Product Data

Control System Cybersecurity Documentation; G

SD-06 Test Reports
Wireless Communication Test Report; G
Control System Cybersecurity Testing Procedures; G

Control System Cybersecurity Testing Report; G

SD-11 Closeout Submittals

Password Change Summary Report; G

Enclosure Keys; G

Software and Configuration Backups; G

System Maintenance Tool Software; G,

Control System Scanning Tools; G,

STIG, SRG and Vendor Guide Compliance Result Report; G

1.7 QUALITY CONTROL

1.7.1 Regulatory Requirements

No regulatory requirements are enforced for the Utility Monitoring and Control System.

1.8 CYBERSECURITY DOCUMENTATION

{For Government Reference Only: This subpart (and its subparts) relates to PL-7; CCI-003071}

1.8.1 Proposed STIG and SRG Applicability Report

For each model of network connected or network infrastructure device, use the DISA SRG/STIG Applicability Guide and Collection Tool (available at https://public.cyber.mil/stigs/SCAP/to identify applicable STIGs or SRGs and provide a report indicating applicable STIGs and SRGs for each model. Provide the Proposed STIG and SRG Applicability Report concurrently with

the Pre-Construction Control System Inventory Report.

1.8.2 Cybersecurity Interconnection Schedule

{For Government Reference Only: This subpart relates to CA-3(b), PL-8, SC-7(9), SC-7(11); CCI-000258, CCI-003072, CCI-003073, CCI-003075, CCI-002398, CCI-002399, CCI-002401, CCI-002402, CCI-002403.}

Provide a completed Cybersecurity Interconnection Schedule documenting network connections between the installed system and other systems. Provide the following information for each device directly communicating between systems: Device Identifier, Device Manufacturer, Device Description, Transport layer Protocol, Network Address, Port (if applicable), MAC (Layer 2) address (if applicable), Media, Application Protocol, Service (if applicable), Descriptive Purpose of communication.

If other control system Sections used on this project include submittals documenting this information, provide copies of those submittals to meet this requirement.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Cybersecurity Interconnection Schedule as an editable Microsoft Excel file (a template Cybersecurity Interconnection Schedule in Excel format is available athttps://www.wbdg.org/dod/ufgs/ufgs-25-05-11.)

1.8.3 Network Communication, Ports, Protocols and Services Report

{For Government Reference Only: This subpart (and its subparts) relates to CA-9, PL-8; CCI-002102, CCI-002103, CCI-002104, CCI-002105, CCI-003072, CCI-003073, CCI-003075 and also the submittal requirements associated with CM-6, CM-7, including CM-7(3), CCI-000388.}

Provide a Network Communication, Ports, Protocols and Services Report. For each networked device, document device identifier and the communication characteristics of the device including communication protocols, services used, encryption employed, and a general description of what information is communicated over the network. For each device using IP communication, document all TCP and UDP ports used. For each device using non-IP communication, document communication protocol and media used. If other control system Sections used on this project include submittals documenting this information, provide copies of those submittals to meet this requirement.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Network Communication, Ports, Protocols and Services Report as an editable Microsoft Excel file.

1.8.4 Control System Inventory Reports

{For Government Reference Only: This subpart (and its subparts) relates to CM-8(a), SI-17, IA-3; CCI-000389, CCI-000392, CCI-000398, CCI-002773, CCI-002774, CCI-002775, CCI-000777, CCI-000778, CCI-001958}

Provide a Pre-Construction Control System Inventory Report report and a Final Control System Inventory Report, using the Inventory Spreadsheet listed under this Section at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. In the Control System Inventory Reports, document all networked devices, including network infrastructure devices. For each device provide all applicable information for which there is a field on the spreadsheet in accordance with the instructions on the spreadsheet.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Control System Inventory Reports as editable Microsoft Excel files. Provide the Pre-Construction Control System Inventory Report concurrently with the Proposed STIG and SRG Applicability Report

1.8.5 Software and Configuration Backups

{For Government Reference Only: This subpart (and its subparts) relates to CP-10; CCI-000550, CCI-000551, CCI-000552}

For each computer on which software is installed under this project, provide a recovery image of the final as-built computer . This image must allow for bare-metal restore such that restoration of the image is sufficient to restore system operation to the imaged state without the need for re-installation of software. If additional user permissions are required to meet this requirement, coordinate the creation of the image with the identified Government Computer Access Point of Contact.

For all Ethernet switches provide a backup of the switch configuration. For all controllers, provide a backup of the controller configuration and the source code for all loaded application programs (all software that is not common to every controller of the same manufacturer and model).

If any or all of these are provided under another Section, provide documentation indicating this and referencing those submittals.

1.8.6 Cybersecurity Network (Riser) Diagram

{For Government Reference Only: This subpart (and its subparts) relates to PL-2(a), PL-8; CCI-003051, CCI-003053, CCI-003072, CCI-003073, CCI-003075}

Provide a cybersecurity network (riser) diagram of the complete control system including all network and device hardware. For each device, include the device identifier, device type, and manufacturer. If the control system specifications require a riser diagram submittal, provide a copy of that submittal as the cybersecurity riser diagram. Otherwise, provide a riser diagram in one-line format.

1.8.7 STIG, SRG and Vendor Guide Compliance Result Report

For every component (device or software) with an applicable STIG or SRG in the Proposed STIG and SRG Applicability Report, document compliance with the STIG or SRG requirements.

a. For components which are scannable by the SCAP (security content automation protocol) tool (available online at https://public.cyber.mil/stigs/scap), include the SCAP report and raw scan results in addition to the final, manually reviewed and revised, documentation of compliance with STIG and SRG requirements. Checklist files should not contain any findings with a Not Reviewed (NR) status after manual reviews.

b. For components which do not support automated scanning, a manual review using the General Purpose STIG option should be done. A completed Checklist file should not contain any findings with a Not

Reviewed (NR) status after manual reviews

For every component (device or software) with manufacturer provided cybersecurity documentation, procedure, or method for secure configuration or installation, provide a report documenting how the component was configured and any deviation from the manufacturer instructions.

1.8.8 Control System Cybersecurity Documentation

{For Government Reference Only: This subpart (and its subparts) relates to SA-5(a),SA-5(b),SA-5(c), SA-22(b); CCIs: CCI-003124, CCI-003125, CCI-003126, CCI-003127, CCI-003128, CCI-003129, CCI-003130, CCI-003131, CCI-003374}

Provide a Control System Cybersecurity Documentation submittal containing the indicated information for each device and software application.

1.8.8.1 Software Applications

For all software applications running on computers provide:

a. administrator documentation that describes secure configuration of the software {For Government Reference Only: relates to CCI-003124}

b. administrator documentation that describes secure installation of the software and software updates. {For Government Reference Only: relates to CCI-003125}

c. administrator documentation that describes secure operation of the software {For Government Reference Only: relates to CCI-003124}

d. administrator documentation that describes effective use and maintenance of security functions or mechanisms for the software {For Government Reference Only: relates to CCI-003127}

e. administrator documentation that describes known vulnerabilities regarding configuration and use of administrative (i.e. privileged) functions for the software {For Government Reference Only: relates to CCI-003128}

f. user documentation that describes user-accessible security functions or mechanisms in the software and how to effectively use those security functions or mechanisms {For Government Reference Only: relates to CCI-003129}

g. user documentation that describes methods for user interaction which enables individuals to use the software in a more secure manner {For Government Reference Only: relates to CCI-003130}

h. user documentation that describes user responsibilities in maintaining the security of the software {For Government Reference Only: relates to CCI-003131}

1.8.8.2 Default Requirements for Control System Devices

For control system devices where Control System Cybersecurity Documentation requirements are not otherwise indicated in this Section, provide:

a. Documentation that describes secure configuration of the device {For Government Reference Only: relates to CCI-003124}

b. Documentation that describes secure installation of the device {For Government Reference Only: relates to CCI-003125}

c. Documentation that describes secure operation of the device {For Government Reference Only: relates to CCI-003124}

d. Documentation that describes effective use and maintenance of security functions or mechanisms for the device {For Government Reference Only: relates to CCI-003127}

e. Documentation that describes known vulnerabilities regarding configuration and use of administrative (i.e. privileged) functions for the device {For Government Reference Only: relates to CCI-003128}

f. Documentation that describes user-accessible security functions or mechanisms in the device and how to effectively use those security functions or mechanisms {For Government Reference Only: relates to CCI-003129}

g. Documentation that describes methods for user interaction which enables individuals to use the device in a more secure manner {For Government Reference Only: relates to CCI-003130}

h. Documentation that describes user responsibilities in maintaining the security of the device {For Government Reference Only: relates to CCI-003131}

i. Documentation of the published last date of support by the manufacturer or indication that a published date is not available. {For Government Reference Only: relates to CCI-003374}

1.9 CYBERSECURITY DURING CONSTRUCTION

{For Government Reference Only: This subpart (and its subparts) relates to AC-18, CA-3; CCI-000258}

In addition to the control system cybersecurity requirements indicated in this section, meet following requirement throughout the construction process.

1.9.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. Contractor computers connected to the control system, control system network, or a control system component at any point during construction must meet the following requirements:

1.9.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.9.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. Computers used on this project must perform a full antivirus scan at least once per day. Computers which have connected to any other network since the last full antivirus scan must perform a full antivirus scan prior to connection to the control system network or to the temporary contractor-installed IP network. Perform an antivirus scan on all removable media (e.g., external hard drives, CDs, DVDs, USB flash drives) prior to connecting to the control system environment.

1.9.1.3 Passwords and Passphrases

The passwords and passphrases for computers, applications, and web-based applications supporting passwords must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.9.1.4 User-Based Authentication

Each user must have a unique account; sharing of a single account between multiple users is prohibited.

1.9.1.5 Demonstration of Compliance

The Government has the right to require demonstration of computer compliance with these requirements at any time during the project.

1.9.1.6 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. Each Statement must be signed by a cybersecurity representative for the relevant company.

1.9.2 Government Access to Network

Government personnel must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification.

1.9.3 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.9.4 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity

Compliance Statements must use the template published at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

1.10 CYBERSECURITY DURING WARRANTY PERIOD

All work performed on the control system after acceptance must be performed using Government Furnished Equipment or equipment specifically and individually approved by the Government.

PART 2 PRODUCTS

All products used on this project must meet the indicated requirements, but not all products specified here will be required by every project.

2.1 ETHERNET SWITCH

Provide Open Systems Interconnection (OSI) Layer 2 Ethernet switches with the following capabilities, and with an interface to support switch configuration for these capabilities:

2.1.1 Required Functionality

Switches must:

a. Copper Ethernet ports must auto negotiate for 10, 100 and 1000 megabits-per-second links.

b. Be capable of implementing port level access control by MAC address and limit the number of MAC addresses to one MAC address per port.

c. For LOW Impact Systems, be capable of implementing per-port access control lists (ACLs) where the list can be filtered by source and destination IP addresses, and by source and destination UDP or TCP ports.

d. Support Remote Network Monitoring (RMON) Port Analysis in accordance with IETF RFC 2819

e. Configure target port and analysis port such that switch clones all target port traffic to analysis port.

2.1.2 Configuration Requirements

Switches must:

a. Support configuration save and restore.

b. Support both manual IP address assignment and acquisition of a dynamic IP address via Dynamic Host Configuration Protocol (DHCP).

c. Be capable of limiting access for configuration to one or more of: a web interface using HTTPS, a command line interface using SSH, or an SNMP connection using SNMP version 3 or later.

2.2 DAISY CHAIN IP CONTROLLERS

Controllers used as Daisy Chain IP Controllers must be IP controllers with exactly two Ethernet network connections and basic built-in switch capabilities to allow implementation of an Ethernet network in a daisy chain architecture. Switches incorporated by Daisy Chain IP Controllers are not required to meet the requirements for Ethernet Switches as defined in this Section.

PART 3 EXECUTION

3.1 CYBERSECURITY HARDENING AND CONFIGURATION GUIDES

Install, configure, and harden all hardware and software furnished on this project in accordance with manufacturer provided documentation, procedures, or methods for secure configuration or installation. Do not implement specific hardening actions if that action would conflict with required functionality or another requirement of this Section.

3.2 NETWORK REQUIREMENTS

3.2.1 Wireless and Wired Broadcast Communication

{For Government Reference Only: This subpart (and its subparts) relates to AC-18, AC-18(3); CCI-001438, CCI-001439, CCI-002323, CCI-001441, CCI-001449}

Unless explicitly authorized by the Government, do not use any wireless or wired broadcast communication.

3.2.1.1 Wireless and Wired Broadcast IP Communications

Do not install wireless or wired broadcast IP networks, including: do not install a wireless access point; do not install or configure an ad-hoc wireless network; do not install or configure a WiFi Direct communication.

When explicitly authorized by the Government, wireless IP communication may be used to communicate with an existing wireless network.

3.2.1.2 Non-IP Wireless Communication

For LOW Impact Systems: When non-IP wireless communication is explicitly authorized by the Government, use the maximum level of encryption supported by the specific protocol employed and select signal strength and radiated power to the minimum necessary for reliable communication.

3.2.1.3 Wireless and Wired Broadcast Communication Request

Provide a report documenting the proposed use of wireless or wired broadcast communication prior to device selection using the Wireless and Wired Broadcast Communication Request Schedule at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. If there is no proposed use of wireless or wired broadcast communication, provide a document indicating this instead of the Request Schedule.

For each device proposed to use wireless or wired broadcast communication show: the device identifier, a description of the device, the location of

the device, the device identifiers of other devices communicating with the device, the protocol used for communication, encryption type and strength. For wireless communication, also show: RF Frequency, Radiated Power in dBm (decibel with a milliwatt reference), free-space range, and the expected as-installed range.

3.2.2 Non-IP Control Networks

When control system specifications require particular communication protocols, use only those communication protocols and only as specified. Do not implement any other communication protocol.

When control system specifications do not indicate requirements for communication protocols, use only those protocols required for operation of the system as specified.

3.2.3 IP Control Networks

{For Government Reference Only: This subpart relates to CM-6(a), CM-7(a), CM-7(b), CM-7(1)(b), SC-41; CCI-001588, CCI-000381, CCI-000380, CCI-000381, CCI-000382, CCI-001761, CCI-001762, CCI-002544, CCI-002545, CCI-002546.}

IP Networks must be Ethernet networks and must use switches which are Ethernet Switches or Daisy Chain IP Controllers as defined in this Section. Do not use nonsecure functions, ports, protocols and services as defined in DODI 8551.01 unless those ports, protocols and services are specifically required by the control system specifications or otherwise specifically authorized by the Government. Do not use ports, protocols and services that are not specified in the control system specifications or required for operation of the control system.

3.2.3.1 IP Devices With Multiple Ethernet Connections

Except for Ethernet Switches and Daisy Chain IP Controllers, devices must not have more than one Ethernet connection to IP networks unless doing so is required by the project specifications and the specific application is approved. If a device with Multiple Ethernet Connections to IP networks is required, provide a Multiple Ethernet Connection Device Request using the Multiple Ethernet Connection Device Request Template at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11 to request approval for each device. If a device with Multiple Ethernet Connections to IP networks is not required, instead provide a document stating that no approval is being requested.

3.2.4 Cryptographic Protection

{For Government Reference Only: This subpart relates to IA-2(9), IA-3(1), SC-8, SC-13, SC-23(1), SC-23(3); CCI-001942, CCI-001959, CCI-001967, CCI-002418, CCI-002449, CCI-002450, CCI-001185, CCI-001188, CCI-001664.}

All remote user interfaces must use HTTPS for all traffic between the user interface client and user interface server.

3.2.5 Device Identification and Authentication

{For Government Reference Only: This subpart (and its subparts) relates to IA-3; CCI-000777, CCI-000778, CCI-001958.}

All computers must support IEEE 802.1x for device authentication to the network.

3.2.5.1 Default Requirements for Control System Devices

For control system devices where Device Identification and Authentication requirements are not otherwise indicated in this Section: Devices using HTTP as a control protocol must use HTTPS instead.

3.2.6 Cryptographic Module Authentication

{For Government Reference Only: This subpart (and its subparts) relates to IA-7; CCI-000803}

For devices (including but not limited to NIST FIPS 140-2 compliant radios) that have STIG/SRGs related to cryptographic module authentication (CCI-000803), comply with the requirements of those STIG/SRGs.

3.3 ACCESS CONTROL REQUIREMENTS

3.3.1 User Accounts

{For Government Reference Only: This subpart (and its subparts) relate to AC-2(a), AC-3, AC-6(1), AC-6(10), AC-6(2), AC-6(9), CM-11(2), and IA-2; CCI-002110, CCI-000213, CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-002235, CCI-000039, CCI-001419, CCI-002234, CCI-001812, and CCI-000764. }

Any user interface supporting user accounts (either FULLY or MINIMALLY) must limit access according to specified limitations for each account. Install and configure any device having a STIG or SRG in accordance with that STIG or SRG.

All user interfaces FULLY supporting accounts must implement user-based authentication where each account is uniquely assigned to a specific user. User interfaces FULLY supporting accounts must implement at least three (3) levels of user account privilege including: 1) an account with read-only permissions 2) an account with full permissions including account creation and modification and 3) an account with greater permissions than read-only but without account creation and modification. Disable any unnecessary or unused accounts. Disable any "guest-level" accounts that are created on the system by default.

3.3.1.1 Computers

All computer operating systems must FULLY support user accounts and implement accounts for access. Each control system software application not supporting accounts and running on a computer must be installed such that use of the software is restricted by the computer operating system to specific users.

Applications running on computers must not require the user be logged in to a computer operating system administrator account for normal operation. It is permissible to require the computer operating system administrator account for initial application installation and configuration.

3.3.1.2 Controllers

For user interfaces provided by controllers, provide access control in accordance with the User Interface Requirements table for the applicable control system and user interface type.

a. For table entries of "NA": NA means Not Applicable, there are no interfaces in this category.

b. For table entries of "None Required": The user interface is not required to support user accounts.

c. For table entries of "MINIMALLY": The user interface must at least MINIMALLY support user accounts.

d. For table entries of "FULLY": The user interface must at FULLY support user accounts.

e. For table entries of "KEY": The user interface must have physical security in the form of either a key lock on the interface itself or be furnished inside a lockable enclosure. Where this is required for a read only interface, the lockable enclosure must prevent viewing of data on the interface; for other interfaces, this lockable enclosure must prevent using the interface to alter data.

f. For table entries of "Physical Security": For Local FULL interfaces, the interface must be located inside mission space. For Local Limited (not FULL) interfaces, the user interface must either a) be located within mission space or b) be protected by physical security at least as good as the control devices (and equipment controlled by the control devices) affected by the interface. For purposes of this requirement, 'affected' includes controllers with data that can be directly altered by the interface, as well as mechanical and/or electrical equipment directly controlled by those controllers, but does not include other interactions.

g. Entries of the form "X and Y" must meet both the requirement indicated for X and the requirement indicated for Y. For example, an entry of "MINIMALLY and Physical Security" indicates the user interface must both MINIMALLY support accounts and have physical security.

h. Entries of the form "X or Y" must meet either the requirement indicated for X or the requirement indicated for Y.

3.3.1.2.1 Utility Montioring and Control Systems

User Interface Requirements for LOW Impact Utility Monitoring and Control Systems	
User Interface Type	**Access Control Requirement**
Local Read Only (see note 1)	None Required
Local Limited, Non-privileged	None Required
Local Limited, Privileged	MINIMALLY

User Interface Requirements for LOW Impact Utility Monitoring and Control Systems	
User Interface Type	**Access Control Requirement**
Local Full	MINIMALLY
Remote Read Only	None Required
Remote Limited, Non-Privileged	MINIMALLY
Remote Limited, Privileged AND Remote Full (see note 2)	
Notes: 1)Local Read Only User Interfaces are always Non-Privileged 2)Remote Full User Interfaces are always Privileged	

3.3.1.2.2 Default Requirements for Other Control Systems

For control system devices where User Interface Requirements are not otherwise indicated in this Section, use the Default User Interface Requirements tables.

Default User Interface Requirements for LOW Impact Control Systems	
User Interface Type	**Access Control Requirement**
Local Read Only (see note 1)	None Required
Local Limited, Non-privileged	None Required
Local Limited, Privileged	MINIMALLY
Local Full	MINIMALLY
Remote Read Only	None Required
Remote Limited, Non-Privileged	MINIMALLY
Remote Limited, Privileged AND Remote Full (see note 2)	FULLY
Notes: 1)Local Read Only User Interfaces are always Non-Privileged 2)Remote Full User Interfaces are always Privileged	

3.3.2 Unsuccessful Logon Attempts

{For Government Reference Only: This subpart (and its subparts) relate to AC-7 (a), AC-7 (b); CCI-000043, CCI-000044, CCI-001423, CCI-002236, CCI-002237, CCI-002238}

Except for high availability user interfaces indicated as exempt, devices must meet the indicated requirements for handling unsuccessful logon attempts. If a device cannot meet these requirements, document device capabilities to protect from subsequent logon attempts and propose alternate protections in a Device Account Lock Exception Request submittal. Do not implement alternate protection measures in lieu of the indicated requirements without explicit permission from the Government. If no Device Account Lock Exceptions are requested, provide a document stating that no approval is being requested as the Device Account Lock Exception Request.

3.3.2.1 Devices MINIMALLY Supporting Accounts

For LOW Impact Systems: Devices which MINIMALLY (but not FULLY) support accounts are not required to lock based on unsuccessful logon attempts.

3.3.2.2 Devices FULLY Supporting Accounts

Devices which FULLY support accounts must meet the following requirements.

a. It must lock the user account when three unsuccessful logon attempts occur within a 15 minute interval.

b. Once an account is locked, the account must stay locked until unlocked by an administrator. If the account being locked is the sole administrator account on the device, the account must stay locked for 1 hour and then automatically unlock.

c. Once the indicated number of unsuccessful logon attempts occurs, delay further logon prompts by 5 seconds.

3.3.2.3 High Availability Interfaces Exempt from Unsuccessful Logon Attempts Requirements

There are no high availability interfaces which are exempt from unsuccessful logon attempts requirements.

3.3.3 System Use Notification

{For Government Reference Only: This subpart (and its subparts) relates to AC-8; CCI-000048, CCI-002247, CCI-002243, CCI-002244, CCI-002245, CCI-002246, CCI-000050, CCI-002248}

3.3.3.1 System Use Notification for Remote User Interfaces

Remote user interfaces must display a warning banner meeting the requirements of DTM 08-060 on screen.

3.3.3.2 System Use Notification for Local User Interfaces

Devices which are connected to a network and have a local user interface must display a warning banner meeting the requirements of DTM 08-060 on the user interface screen if capable of doing so and must have a permanently affixed label with an approved banner from DTM 08-060 if unable to display the warning banner on the screen. Where it is impractical (perhaps due to device size) to affix the label to the device, affix the label to the device enclosure.

Labels must be machine printed or engraved, plastic or metal, designed for permanent installation, must use a font no smaller than 14 point, and must provide a high contrast between font and background colors.

3.3.4 Permitted Actions Without Identification or Authentication

{For Government Reference Only: This subpart (and its subparts) relates to AC-14; CCI-000061, CCI-000232}

The control system must require identification and authentication before allowing any actions except read-only actions by a user acting from a user interface which MINIMALLY or FULLY supports accounts.

3.3.5 Enclosures

Prior to final acceptance of the system, lock all lockable enclosures. Submit an Enclosure Keys submittal with all copies of keys for all enclosures and a key inventory list documenting all keys. Label each key with the matching enclosure identifier.

3.4 USER IDENTIFICATION AND AUTHENTICATION

{For Government Reference Only: This subpart (and its subparts) relates to IA-2, IA-2(1),IA-2(12), IA-5 IA-5(b), IA-5(c), IA-5(e), IA-5(g), IA-5(1), IA-5(11); CCI-000764, CCI-000765, CCI-001953, CCI-001954, CCI-001544, CCI-001989, CCI-000182, CCI-001610, CCI-000192, CCI-000193, CCI-000194, CCI-000205, CCI-001619, CCI-001611, CCI-001612, CCI-001613, CCI-001614, CCI-000195, CCI-001615, CCI-000196, CCI-000197, CCI-000199, CCI-000198, CCI-001616, CCI-001617, CCI-000200, CCI-001618, CCI-002041, CCI-002002, CCI-002003. }

This subpart indicates requirements for specific methods of identification and authentication for users and user accounts. Where these requirements conflict apply the following order of precedence: 1) If present, Device Specific Requirements take precedence over any other requirements; and then 2) multifactor authentication requirements take precedence over password requirements.

3.4.1 User Identification and Authentication for All System Types

Unless otherwise indicated, all user interfaces supporting accounts (either FULLY or MINIMALLY) must implement Identification and Authorization via passwords.

3.4.2 User Identification and Authentication for Specific System Types

System specific requirements are in addition to and supersede those indicated for all system types. When no additional requirements are indicated for a specific system type the requirements for all systems still apply to that system type.

3.4.2.1 Utility Monitoring and Control System Devices

No additional system specific requirements apply.

3.4.3 User Identification and Authentication for Specific Devices

There are no additional device specific user interface requirements

3.4.4 Implementation of Identification and Authorization Requirements

Identification and Authorization must be met by one of the following methods:

a. Direct implementation in the user interface.

b. For user interfaces on a computer: inheriting the Identification and Authorization from the computer operating system, either by the operating system limiting access to specific applications by user, or by the application itself having permissions based on the user logged into the computer.

c. For remote interfaces: an implementation shared between the remote user interface server and the remote user interface client. For example, a requirement for PIV authentication may be met on a remote user interface by a PIV reader on a web browser client which sends the authentication information via HTTPS to the remote server.

3.4.5 Password-Based Authentication Requirements

3.4.5.1 Passwords for Software and Applications Running on Computers

All software and applications running on computers supporting password-based authentication must enforce the following requirements:

a. Minimum password length of 12 characters

b. Password must contain at least one uppercase character.

c. Password must contain at least one lowercase character.

d. Password must contain at least one numeric character.

e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.

f. Password must have a minimum lifetime of 24 hours.

g. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.

h. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters (where location is significant, a character may be reused if it is in a different position).

i. Passwords must be cryptographically protected during storage and transmission.

3.4.5.2 Passwords for Controllers FULLY Supporting Accounts

All controllers FULLY supporting accounts and supporting password-based authentication must enforce the following requirements:

a. Minimum password length of 12 characters

b. Password must contain at least one uppercase character.

c. Password must contain at least one lowercase character.

d. Password must contain at least one numeric character.

e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.

f. Password must have a maximum lifetime of sixty (60) days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.

g. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters.

h. Passwords must be cryptographically protected during storage and transmission.

3.4.5.3 Passwords for Remote Interfaces

Passwords for connecting to a Remote User Interface supporting password-based authentication must enforce the following requirements:

a. Minimum password length of 12 characters

b. Password must contain at least one uppercase character.

c. Password must contain at least one lowercase character.

d. Password must contain at least one numeric character.

e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.

f. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.

g. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters (where location is significant, a character may be reused if it is in a different position).

h. Passwords must be cryptographically protected during storage and transmission.

3.4.5.4 Passwords for Devices Minimally Supporting Accounts

Devices MINIMALLY supporting accounts must support passwords with a

minimum length of four characters.

3.4.5.5 Password Configuration and Reporting

For all devices with a password, coordinate the changing of passwords with the project site following testing of the system but prior to turnover to the Government. Coordinate with Password Point of Contact to determine appropriate project site personnel to complete password changes. Accompany identified personnel to each device with a password and instruct personnel on the process of changing password. Record the time, date and personnel present when each device's password is changed and submit a Password Change Summary Report documenting this information.

Provide the Password Summary Report electronically in both PDF and Microsoft Excel.

3.4.6 Authenticator Feedback

{For Government Reference Only: This subpart relates to IA-6; CCI-000206}

Devices must never show authentication information, including passwords, on a display. Devices that momentarily display a character as it is entered, and then obscure the character, are acceptable. For devices that have STIGs or SRGs related to obscuring of authenticator feedback (CCI-000206), comply with the requirements of those STIGS/SRGs.

3.5 CYBERSECURITY AUDITING

Where an auditing requirement exists for email notification, notify via email the application administrator and Information System Security Officer (ISSO) of the event. Coordinate with the Email Address Point of Contact for email addresses. If outgoing email is not available to the system, disable email notifications.

3.5.1 Audit Events, Content of Audit Records, and Audit Generation

{For Government Reference Only: This subpart (and its subparts) relates to AU-2(a), AU-2(c), AU-2(d), AU-3, AU-10, AU-12, AU-14(b), AU-14(1), AU-14(2), AU-14(3), CM-5(1), SC-7 (9); CCI-000123, CCI-001571, CCI-000125, CCI-001485, CCI-000130, CCI-000131, CCI-000132, CCI-001230, CCI-000133, CCI-000134, CCI-001487, CCI-000166, CCI-001899, CCI-000169, CCI-001459, CCI-000171, CCI-000172, CCI-001910, CCI-001914, CCI-001919, CCI-001464, CCI-001462, CCI-001920, CCI-001814, CCI-002400. }

For devices that have STIG/SRGs related to audit events, content of audit records or audit generation, comply with the requirements of those STIG/SRGs.

If auditing requirements can be met using existing control system alarm or event capabilities, those existing capabilities may be used to meet these requirements.

3.5.1.1 Computers

For each computer, provide the capability to select audited events and the content of audit logs. Configure computers to audit the indicated events, and to record the indicated information for each auditable event. Send

logs to a syslog server that is only accessible via accounts on the system with privileged level access.

3.5.1.1.1 Audited Events

Configure each computer to audit the following events:

a. Successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels)

b. Successful and unsuccessful logon attempts

c. Successful logouts

d. Privileged activities or other system level access

e. Concurrent logons from different workstations

f. Successful and unsuccessful accesses to objects

g. Program initiations

h. Direct access to the information system

i. Account creations, modifications, disabling, and terminations.

j. Kernel module load, unload, and restart

3.5.1.1.2 Audit Event Information To Record

Configure each computer to record, for each auditable event, the following information (where applicable to the event):

a. What type of event occurred

b. When the event occurred

c. Where the event occurred

d. The source of the event

e. The outcome of the event

f. The identity of any individuals or subjects associated with the event

3.5.1.2 Default Requirements for Control System Controllers

For control system controllers where Audit Events, Content of Audit Records, and Audit Generation are not otherwise indicated in this Section:

3.5.1.2.1 Controllers Which FULLY Support Accounts

For each controller which FULLY supports accounts, provide the capability to select audited events and the content of audit logs. Configure controllers to audit the indicated events, and to record the indicated information for each auditable event.

3.5.1.2.1.1 Audited Events

Configure each controller to audit the following events:

a. Successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels)

b. Successful and unsuccessful logon attempts

c. Successful logouts

d. Concurrent logons from different workstations

e. All account creations, modifications, disabling, and terminations.

f. All kernel module load, unload, and restart

3.5.1.2.1.2 Audit Event Information To Record

Configure each controller to record, for each auditable event, the following information (where applicable to the event):

a. what type of event occurred

b. when the event occurred

c. where the event occurred

d. the source of the event

e. the outcome of the event

f. the identity of any individuals or subjects associated with the event

3.5.1.2.2 Controllers Which Do Not FULLY Support Accounts

For each controller which does not FULLY support accounts configure the controller to audit all controller shutdown and startup events and to record for each event the type of event and when the event occurred.

3.5.2 Audit Time Stamps

{For Government Reference Only: This subpart (and its subparts) relates to AU-8; CCI-000159, CCI-001889, CCI-001890.}

Any device (computer or controller) generating audit records must have an internal clock capable of providing time with a resolution of one second. Clocks must not drift more than 10 seconds per day. Configure the system so that each device (computer or controller) generating audit records maintains accurate time to within 1 second. Note that if the control system specifications include requirement for clocks, the most stringent requirement applies.

3.5.3 Response to Audit Processing Failures

{For Government Reference Only: This subpart (and its subparts) relates to AU-5; CCI-000139, CCI-000140, CCI-001490.}

In the case of a failure in the auditing system, computers associated with auditing must provide email notification. In case of an audit failure, if possible, continue to collect audit records by overwriting existing audit records.

3.6 REQUIREMENTS FOR LEAST FUNCTIONALITY

{For Government Reference Only: This subpart (and its subparts), along with the Network Communication, Ports, Protocols and Services Report submittal specified elsewhere in this section, relates to CM-6(a), CM-6(c), CM-7, CM-7(1)(b), SC-41; CCI-000363, CCI-000364, CCI-000365, CCI-001588, CCI-001755, CCI-000381, CCI-000380, CCI-000382, CCI-001761, CCI-001762, CCI-002544, CCI-002545, CCI-002546. }

For devices that have a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port and device I/O access for least functionality), install and configure the device in accordance with that STIG or SRGs.

3.6.1 Device Capabilities

Do not provide devices with remote user interfaces or with full user interfaces where one was not required.

3.6.2 Software

For software that has a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port access for least functionality), install and configure the software in accordance with that STIG or SRG.

Do not install software that is not specifically required to meet a contract requirement. Remove any previously installed that is not specifically required to meet a contract requirement. Do not implement functionality within software
that is not specifically required to meet contract requirements.

3.7 SYSTEM AND COMMUNICATION PROTECTION

3.7.1 Collaborative Computing

{For Government Reference Only: This subpart relates to SC-15(a), SC-15(b); CCI-001150, CCI-001152.}

Without explicit approval from the project site, control systems must not use collaborative computing technologies.

3.7.2 Denial of Service Protection

{For Government Reference Only: This subpart relates to SC-5, SC-12, SC-7(a); CCI-001093, CCI-002385, CCI-002386, CCI-002430, CCI-001097. }

To the greatest extent practical, implement control logic without reliance on the network. Except when required to meet the requirements of the control system Section (where the requirement can only be met using computer hardware), do not implement control logic in computers.

3.8 SAFE MODE AND FAIL SAFE OPERATION

{For Government Reference Only: This subpart (and its subparts) relates to CP-12, SI-10(3), SI-17; CCI-002855, CCI-002856, CCI-002857, CCI-002754, CCI-002773, CCI-002774, CCI-002775}

For all control system components with an applicable STIG or SRG, configure the component in accordance with all applicable STIGs and SRGs.

3.9 SYSTEM MAINTENANCE TOOL SOFTWARE

{For Government Reference Only: This subpart (and its subparts) relates to MA-3; CCI-000865.}

Submit and license to the Government all software required to operate, maintain and modify the control system such the Government or their agents are able to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer. Submit hard copies of user manuals for each software with the software submittal. Provide any hardware keys or dongles, software keys, license numbers, and other information required to enable the Government to access or change the system.

For software provided and licensed to the Government under the requirements of another Section, submit a statement indicating the Section and Submittal under which the software was provided. For software provided to meet the requirements of this Section and not provided and licensed under another Section, submit software and software user manuals on DVD or CD as a Technical Data Package and submit one hard copy of the software user manual for each piece of software.

3.10 VULNERABILITY SCANNING

{For Government Reference Only: This subpart (and its subparts) relates to RA-5 RA-5(a),RA-5(b),RA-5(c),RA-5(d); CCI-001054, CCI-001055, CCI-000156, CCI-001641, CCI-001643, CCI-001057, CCI-001058, CCI-001059. }

All IP devices must be scannable, such that the device can be scanned by industry standard IP network scanning utilities without harm to the device, application, or functionality.

3.10.1 Computers and Software Running on Computers

Computers and applications running on computers must meet relevant vulnerability scanning STIGs/SRGs and respond to approved DoD vulnerability scanning tools.

3.10.2 Controllers

Provide controllers that are scannable by standard control system discovery tools or control system browsers and return meaningful status information including the network inputs and outputs for the controller. This information must contain sufficient detail to detect vulnerabilities or exploits of the controller.

Provide all software needed to scan the control system as the Control System Scanning Tools submittal. If the software required to scan the system is already installed at the project site or is provided under a

separate section instead provide a statement indicating this.

3.11 SYSTEM AND INTEGRATION INTEGRITY

3.11.1 Malicious Code Protection

{For Government Reference Only: This subpart (and its subparts) relates to SI-3(c); CCI-001241, CCI-002623}

For all computers installed under this project, provide malware protection software media, provide licenses, and install and configure malware protection software as indicated. Coordinate with the Government Computer Access Point of Contact as required.

a. Malware protection software licenses will be Government furnished.

b. Malware protection software media will be Government furnished.

c. Malware protection software will be Government installed.

3.12 CONTROL SYSTEM CYBERSECURITY TESTING

3.12.1 Control System Cybersecurity Testing Procedures

Prepare and submit Control System Cybersecurity Testing Procedures explaining step-by-step, the actions and expected results that will demonstrate that the control system meets the requirements of this Section. The Control System Cybersecurity Testing Procedures may be submitted as a Technical Data Package.

3.12.2 Control System Cybersecurity Testing Execution

Using the Control System Cybersecurity Testing Procedures verify that the control system meets the requirements of this Section. UNLESS GOVERNMENT WITNESSING OF A TEST IS SPECIFICALLY WAIVED BY THE GOVERNMENT, PERFORM ALL TESTS WITH A GOVERNMENT WITNESS. If testing reveals deficiencies in the system, correct the deficiency and retest until successful.

3.12.3 Control System Cybersecurity Testing Report

Prepare and submit a Control System Cybersecurity Testing Report documenting all tests performed and their results. Include all tests in the Control System Cybersecurity Testing Procedures and any additional tests performed during testing. Document test failures and repairs conducted with the test results. The Control System Cybersecurity Testing Report may be submitted as a Technical Data Package

3.13 FIELD QUALITY CONTROL, CYBERSECURITY VALIDATION SUPPORT

In addition to testing and testing support required by other Sections, provide a minimum of eight hours of technical support for cybersecurity testing of control systems to support the DoD Risk Management Framework process Cybersecurity assessment of the control system. This support is independent of (and in addition to) the Control System Cybersecurity Testing specified in this section.

3.14 CYBERSECURITY TRAINING

Provide eight hours of classroom and hands-on training for six Government personnel on the cybersecurity operation and maintenance of the control system provided. This training is in addition to and must be coordinated with control system training specified in other Sections.

The Government will provide the training location. Training must cover, at a minimum: (a) applying software and firmware updates, (b) user account creation, modification and deletion, (c) audit log upload procedures and (d) identification of privileged user interfaces and system impact of those interfaces. Training session must include a question and answer period during which government staff questions about cybersecurity aspects of the control system are answered.

-- End of Section --

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08/24

cybersecurity for LOW and MODERATE impact facility-related control systems

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SECTION 25 05 11.03

CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - ELECTRONIC SECURITY SYSTEMS (M-M-M)
08/24

PART 1 GENERAL

Many subparts in this Section contain text in curly braces ("{" and "}") indicating which cybersecurity control and control correlation identifier (CCI) the requirements of the subpart relate to. The text inside these curly braces is for Government reference only and enables coordination of the requirements of this Section with the RMF process throughout the design and construction process. Text in curly braces are not contractor requirements.

This Section refers to Security Requirements Guide (SRGs) and Security Technical Implementation Guide (STIGs). STIGs and SRGs are available online at the Information Assurance Support Environment (IASE) website at https://public.cyber.mil/stigs/downloads/ and an SRG/STIG Applicability Guide and Collection Tool is available at https://public.cyber.mil/stigs/SCAP/. Not all control system components have applicable STIGs or SRGs. The "Control Systems SRG" does not apply to work performed under this Section; all requirements within this section to apply applicable SRGs DO NOT include the "Control Systems SRG".

1.1 CONTROL SYSTEM APPLICABILITY

There are multiple versions of this Section associated with this project. Different versions have requirements applicable to different control systems. This specific Section applies only to the following control systems: Electronic Security Systems (ESS).

1.2 RELATED REQUIREMENTS

This section does not contain sufficient requirements to procure a control system and must be used in conjunction with other Sections which specify control systems. This Section adds cybersecurity requirements to the control systems specified in other Sections, and as these requirements are conditioned on the control system being provided, there may be requirements in this Section that will not apply to this project. All Sections containing facility-related control systems or control system components are related to the requirements of this Section. Review all specification sections to determine related requirements.

In cases where a requirement is specified in both this Section and in another Section, the more stringent requirement must be met. In cases where a requirement in this Section conflicts with the requirements of another Section such that both requirements cannot be met at the same time, request direction from the Contracting Officer to determine which requirement applies to the project.

Systems that are to be connected to the VA network shall have an Enterprise Risk Assessment (ERA) performed. ERAs shall conform to the medical center's standards.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.1x	(2010) Local and Metropolitan Area Networks - Port Based Network Access Control

INTERNET ENGINEERING TASK FORCE (IETF)

IETF RFC 2819	(2000) Remote Network Monitoring (RMON) Management Information Base (MIB)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST FIPS 140-2	(2001) Security Requirements for Cryptographic Modules
NIST FIPS 201-2	(2013) Personal Identity Verification (PIV) of Federal Employees and Contractors

U.S. DEPARTMENT OF DEFENSE (DOD)

DODI 8551.01	(2014) Ports, Protocols, and Services Management (PPSM)
DTM 08-060	(2008) Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement

1.4 DEFINITIONS

1.4.1 Administrator Account

An administrator account is an account with full permissions to a device, application, or operating system, including the ability to create and modify other user accounts.

Note that the operating system Administrator Account may be different than Administrator Accounts for applications hosted on that operating system. Also, most controllers will not have any support for accounts and will therefore not have an 'Administrator Account'.

1.4.2 Computer

A computer is one of the following:

a. a device running a non-embedded desktop or server version of Microsoft Windows

b. a device running a non-embedded version of MacOS

c. a device running a non-embedded version of Linux

d. a device running a version or derivative of the Android Operating

System, where Android is considered separate from Linux

e. a device running a version of Apple iOS

Unless otherwise indicated or clear from context use of the word "device" in this Section includes computers.

1.4.3 Controller

A device other than a computer or Ethernet switch.

1.4.4 Mission Space

A device or media is in mission space if physical access to the device or media is controlled by the organization served by the device. For example, a variable air volume (VAV) box controller in a suspended ceiling is in mission space if the VAV box serves that room; an electrical switchgear in an electrical room or an air handling unit (AHU) in a mechanical room or on a rooftop may still be considered to be in mission space if the organization (mission) served by that switchgear or AHU controls access to the electrical room, mechanical room or rooftop.

1.4.5 Network

A network is a group of two or more devices that can communicate using a network protocol. Network protocols must provide a method for addressing devices on the network; a communication method that does not provide an addressing scheme is not a networked form of communication. Devices that communicate using a method of communication that does not support device addressing are not using a network. Addresses may be other than IP addresses, and addressing may be at either Open Systems Interconnection (OSI) layer 2 or layer 3.

1.4.6 Network Connected

A component is network connected (or "connected to a network") only when the device has a network transceiver which is directly connected to the network and implements the network protocol. A device lacking a network transceiver (and accompanying protocol implementation) can never be considered network connected. Note that (unlike many IT definitions of "Network Connected") a device connected to a non-IP network is still considered network connected (an IP connection or IP address is not required for a device to be network connected).

1.4.6.1 Wireless Network Connected

Any device that supports wireless network communication is network connected to a wireless network, regardless of whether the device is communicating using wireless. Unless physically disabled, devices with wireless transceivers support wireless, it is not sufficient to disable the wireless in software.

1.4.7 Network Media

The thing that provides the communication channel between the devices on a network. Typically wire, but might include wireless, fiber optic, or even power line (some network protocols allow sending network signals over power wiring).

1.4.8 User Account Support Levels

The support for user accounts is categorized in this Section as one of three levels:

1.4.8.1 FULLY Supported

Device supports configurable individual accounts. Accounts can be created, deleted, modified, etc. Privileges can be assigned to accounts. These devices support user-based (as opposed to role-based) authentication.

1.4.8.2 MINIMALLY Supported

Device supports a small, fixed number of accounts (perhaps only one). Accounts cannot be modified. A device with only a "User" and an "Administrator" account would fit this category. Similarly, a device with two PINs for logon - one for restricted and one for unrestricted rights would fit here (in other words, the accounts do not have to be the traditional "username and password" structure). These devices typically only support role-based authentication.

Examples of devices which MINIMALLY support accounts are a) a variable frequency drive with a single account which requires a PIN for access to configuration; and b) a room lighting control touchpad interface that has a single account.

1.4.8.3 NOT Supported

Device does not support any Access Enforcement therefore the whole concept of "account" is meaningless.

1.4.9 Manual Local Input

Manual Local Inputs are system analog or binary inputs that are adjustable by a person but are, by intrinsic hardware design, very limited in potential capabilities. Manual Local Inputs do not have touch screens or full keyboards, but may have a few buttons or dials to allow input. Manual Local Inputs do not have full graphic screens or dot-matrix displays, but may have simple lights (LEDs) or 7-segment displays. Manual Local Inputs do not have any sort of menu structure, each button has a single well-defined function.

Examples of Manual Local Inputs are H-O-A switches, simple thermostats, and disconnect switches.

1.4.10 Card Reader

A card reader is an input/output device whose primary function is to assist in two-factor authentication. A card reader must have an interface to read data from a card and may be able to write data to a card. A card reader may have a means (such as buttons, keypad, touchscreen, etc.) for a user to input a PIN or password, as well as a limited display.

1.4.11 User Interface

A User Interface (UI) is something other than a Manual Local Input or Card Reader that allows a person to interact with the system or device. Note that while a Card Reader is not by itself a User Interface, a User Interface may contain a Card Reader in order for it to authenticate its

user. Within control systems, there are a wide range of User Interfaces.

Two important distinctions are 1) whether the user interface is Local or Remote, and 2) the effective capabilities of the User Interface to alter data, which is the "privilege" of the user interface (where effective privilege available to a specific user at a specific user interface is the combination of the greatest privilege offered by the user interface and the specific account the user is logged into).

1.4.11.1 Local User Interface

A Local User Interface is a user interface where the physical hardware the user interacts with (keyboard, buttons, display, etc.) is physically part of the device being affected. All of the relevant characteristics of the user interface are embodied within a single device.

Note that a Local UI may be able to access data in a different device, Local versus Remote in this context refers to the user interface itself; the capability to access data in a different device is covered under "Full User Interface".

1.4.11.2 Remote User Interface

A Remote User Interface implements a Client/Server model where the physical hardware the user interacts with (Client) is physically distinct from the device being affected (Server). Most or all of the security and functionality characteristics of the user interface are defined by the Server, not the Client. The Client and Server communicate via a network connection. A common example of a remote user interface is a web-based interface where the browser (client) is generally on different hardware than the web server (server). A Remote UI remains a Remote UI even if the user happens to be at a Client on the same hardware as the Server. What is important is that a) the Client may be on different hardware than the Server and b) the majority of the security and functional characteristics of the interface are defined at the Server.

Note that this definition of "remote" is consistent with that generally used in the control industry but is not aligned with the NIST 800-53 definition of "Remote", which refers to "outside the system". The term "Remote" here better aligns with the NIST 800-53 definition of "Network" (remote from within the system) Access.

1.4.11.3 Types of User Interface (by capability)

User interfaces are also categorized by their capabilities as being Read Only, Limited, or Full.

1.4.11.3.1 Read-Only User Interface

A Read Only User Interface (also referred to as a View-Only User Interface) is a user interface that only allows for reading data, it does not allow (have the capability to) modify data. A Read Only User Interface may be either Local or Remote. A User Interface that is configured to be Read Only (by some other means than the interface itself, such as using configuration software on a laptop) is a Read-Only Interface. Note a Read Only User Interface may have buttons (or touch screen, etc.) allowing the user to navigate through the presentation of data.

Examples of a Read Only User Interfaces are a) a publicly viewable "energy dashboard" showing weather data and energy usage within a building and b) digital wayfinding signage.

1.4.11.3.2 Limited User Interface

A Limited User Interface is a user interface that - by design - can only alter information local to the user interface. Note that the determination of "alter" includes only direct interactions, it explicitly excludes interactions that might occur as secondary effects. For example, an interface changing the flow setpoint in a pump controller is a direct interaction, the subsequent change in flow (as well as any subsequent downstream changes in valve position) are not direct interactions.

Two examples of LIMITED UIs are: a) a variable speed drive has a Limited Local User Interface which allows the user to change properties within the drive, but does not allow affecting things outside the drive; and b) a typical home WiFi Router has a Limited Remote User Interface which allows configuration of the Router, but does not allow direct interaction with other devices.

1.4.11.3.3 Full User Interface

A Full User Interface can alter information in devices outside the device with the user interface. For example, a typical Local Display Panel is a Full Local User Interface while a browser-based front end is a Full Remote User Interface.

1.4.11.3.4 View-Only User Interface

See Read-Only User Interface

1.4.11.4 Other User Interface Terminology

In addition to defining whether a user interface is a Hardware Limited, Read-Only, Limited or Full, and whether it is Local or Remote, user interfaces are classified by whether they are writable or privileged.

1.4.11.4.1 Writable User Interface

Any User Interface that is not Read-Only is Writable. (Limited User Interfaces and Full User Interfaces are both writable user interfaces (as they are capable of changing a value)).

1.4.11.4.2 Privileged User Interface

A Privileged UI is a UI that has sufficient capabilities or functionality that it requires specific cybersecurity measures to be put in place to limit its unauthorized use. Ultimately, whether a specific user interface is considered a Privileged User Interface must be determined by usage. Unless otherwise specified, user interfaces can be determined to be privileged or not using the following:

a. Read-Only User Interfaces are not privileged user interfaces.

b. Full User Interfaces are privileged user interfaces.

c. User interfaces that allow for configuration of auditing or allows for modification or deletion of audit logs are privileged user interface.

d. User interfaces that allow for reprogramming a network connected device is a privileged user interface.

e. Writeable User Interfaces in Electronic Security Systems (ESS) are privileged user interfaces.

f. Except as specified above, a Limited User Interface must be determined to be privileged or not based on the specific capabilities and use case of the user interface. In general however, user interfaces that do not offer significant capabilities above and beyond those available at that location via other means (e.g. such as a disconnect switch, breaker, or hand-off-auto switch, or physical attack) are not privileged.

1.4.12 Wireless Network

Any network that communicates without using wires or fiber optics as the communication media. Wireless networks include: WiFi, Bluetooth, ZigBee, cellular, satellite, long and short wave radio, 2.4 GHz, free space optical, point-to-point laser, microwave, and IR.

1.4.13 Wired Broadcast Network

Wired Broadcast Networks are any network, such as powerline carrier networks and modem (wired telephony), that use wire-based technologies where there is not a clearly defined boundary for signal propagation.

1.5 ADMINISTRATIVE REQUIREMENTS

1.5.1 Points of Contact

Coordinate with the following Points of Contact as indicated in this Section and as required. Not all projects will require coordination with all Points of Contact. When coordination is required and no Point of Contact is indicated, coordinate with the Contracting Officer (KO).

a. Government Computer Access Point of Contact: Contracting Office (KO)

b. HTTPS Certificate Point of Contact: Contracting Officer (KO)

c. Email Address Point of Contact: Contracting Officer (KO)

d. Password Point of Contact: Contracting Officer (KO)

1.5.2 Coordination

Coordinate the execution of this Section with the execution of all other Sections related to control systems as indicated in the paragraph RELATED REQUIREMENTS. Items that must be considered when coordinating project efforts include but are not limited to:

a. If requesting permission for alternate account lock permissions, the Device Account Lock Exception Request must be approved prior to control system device selection and installation.

b. If requesting permission for the use of a device with multiple physical connections to IP networks, the Multiple IP Connection Device Request must be approved prior to control system device selection and installation.

c. If the Device Audit Record Upload Software is to be installed on a computer not being provided as part of the control system, coordination is required to identify the computer on which to install the software.

d. The Cybersecurity Interconnection Schedule must be coordinated with other work that will be interconnected to, and interconnections must be approved by the Government before relying on them for system functionality.

e. Cybersecurity testing support must be coordinated across control systems and with the Government cybersecurity testing schedule.

f. Passwords must be coordinated with the indicated contact for the project site.

g. If applicable, HTTPS web server certificates must be obtained from the indicated HTTPS Certificate Point of Contact.

h. Contractor Computer Cybersecurity Compliance Statements must be provided for each contractor using contractor owned computers.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Wireless and Wired Broadcast Communication Request; GDevice Account Lock Exception Request; G

Multiple Ethernet Connection Device Request; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

Cybersecurity Interconnection Schedule ; G

Protection of Information At Rest Proposal; G

Proposed STIG and SRG Applicability Report; G

Pre-Construction Control System Inventory Report ; G

SD-02 Shop Drawings

Network Communication, Ports, Protocols and Services Report; G

Cybersecurity Network (Riser) Diagram ; G

SD-03 Product Data

Control System Cybersecurity Documentation; G

SD-06 Test Reports

Control System Cybersecurity Testing Procedures; G

Control System Cybersecurity Testing Report; G

SD-11 Closeout Submittals

Password Change Summary Report; G

Enclosure Keys; G

Software and Configuration Backups; G

System Maintenance Tool Software; G

Control System Scanning Tools; G

STIG, SRG and Vendor Guide Compliance Result Report; G

Final Control System Inventory Report; G

1.7 QUALITY CONTROL

1.7.1 Regulatory Requirements

No regulatory requirements are enforced for the Electronic Security System.

1.8 CYBERSECURITY DOCUMENTATION

{For Government Reference Only: This subpart (and its subparts) relates to PL-7; CCI-003071}

1.8.1 Proposed STIG and SRG Applicability Report

For each model of network connected or network infrastructure device, use

the DISA SRG/STIG Applicability Guide and Collection Tool (available at https://public.cyber.mil/stigs/SCAP/to identify applicable STIGs or SRGs and provide a report indicating applicable STIGs and SRGs for each model. Provide the Proposed STIG and SRG Applicability Report concurrently with the Pre-Construction Control System Inventory Report.

1.8.2 Cybersecurity Interconnection Schedule

{For Government Reference Only: This subpart relates to CA-3(b), PL-8, SC-7(9), SC-7(11); CCI-000258, CCI-003072, CCI-003073, CCI-003075, CCI-002398, CCI-002399, CCI-002401, CCI-002402, CCI-002403. For MODERATE Impact systems, this subpart also relates to SC-7; CCI-001126, CCI-001109}

Provide a completed Cybersecurity Interconnection Schedule documenting network connections between the installed system and other systems. Provide the following information for each device directly communicating between systems: Device Identifier, Device Manufacturer, Device Description, Transport layer Protocol, Network Address, Port (if applicable), MAC (Layer 2) address (if applicable), Media, Application Protocol, Service (if applicable), Descriptive Purpose of communication. For MODERATE Impact Systems: Also describe the impact of loss of the connection on the control system. If other control system Sections used on this project include submittals documenting this information, provide copies of those submittals to meet this requirement.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Cybersecurity Interconnection Schedule as an editable Microsoft Excel file (a template Cybersecurity Interconnection Schedule in Excel format is available athttps://www.wbdg.org/dod/ufgs/ufgs-25-05-11.)

1.8.3 Network Communication, Ports, Protocols and Services Report

{For Government Reference Only: This subpart (and its subparts) relates to CA-9, PL-8; CCI-002102, CCI-002103, CCI-002104, CCI-002105, CCI-003072, CCI-003073, CCI-003075 and also the submittal requirements associated with CM-6, CM-7, including CM-7(3), CCI-000388.}

Provide a Network Communication, Ports, Protocols and Services Report. For each networked device, document device identifier and the communication characteristics of the device including communication protocols, services used, encryption employed, and a general description of what information is communicated over the network. For each device using IP communication, document all TCP and UDP ports used. For each device using non-IP communication, document communication protocol and media used. If other control system Sections used on this project include submittals documenting this information, provide copies of those submittals to meet this requirement.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Network Communication, Ports, Protocols and Services Report as an editable Microsoft Excel file.

1.8.4 Control System Inventory Reports

{For Government Reference Only: This subpart (and its subparts) relates to CM-8(a), SI-17, IA-3; CCI-000389, CCI-000392, CCI-000398, CCI-002773, CCI-002774, CCI-002775, CCI-000777, CCI-000778, CCI-001958}

Provide a Pre-Construction Control System Inventory Report report and a

Final Control System Inventory Report, using the Inventory Spreadsheet listed under this Section at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. In the Control System Inventory Reports, document all networked devices, including network infrastructure devices. For each device provide all applicable information for which there is a field on the spreadsheet in accordance with the instructions on the spreadsheet.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Control System Inventory Reports as editable Microsoft Excel files. Provide the Pre-Construction Control System Inventory Report concurrently with the Proposed STIG and SRG Applicability Report

1.8.5 Software and Configuration Backups

{For Government Reference Only: This subpart (and its subparts) relates to CP-10; CCI-000550, CCI-000551, CCI-000552}

For each computer on which software is installed under this project, provide a recovery image of the final as-built computer . This image must allow for bare-metal restore such that restoration of the image is sufficient to restore system operation to the imaged state without the need for re-installation of software. If additional user permissions are required to meet this requirement, coordinate the creation of the image with the identified Government Computer Access Point of Contact.

For all Ethernet switches provide a backup of the switch configuration. For all controllers, provide a backup of the controller configuration and the source code for all loaded application programs (all software that is not common to every controller of the same manufacturer and model).

If any or all of these are provided under another Section, provide documentation indicating this and referencing those submittals.

1.8.6 Cybersecurity Network (Riser) Diagram

{For Government Reference Only: This subpart (and its subparts) relates to PL-2(a), PL-8; CCI-003051, CCI-003053, CCI-003072, CCI-003073, CCI-003075}

Provide a cybersecurity network (riser) diagram of the complete control system including all network and device hardware. For each device, include the device identifier, device type, and manufacturer. If the control system specifications require a riser diagram submittal, provide a copy of that submittal as the cybersecurity riser diagram. Otherwise, provide a riser diagram in one-line format.

1.8.7 STIG, SRG and Vendor Guide Compliance Result Report

For every component (device or software) with an applicable STIG or SRG in the Proposed STIG and SRG Applicability Report, document compliance with the STIG or SRG requirements.

a. For components which are scannable by the SCAP (security content automation protocol) tool (available online at https://public.cyber.mil/stigs/scap), include the SCAP report and raw scan results in addition to the final, manually reviewed and revised, documentation of compliance with STIG and SRG requirements. Checklist

files should not contain any findings with a Not Reviewed (NR) status after manual reviews.

b. For components which do not support automated scanning, a manual review using the General Purpose STIG option should be done. A completed Checklist file should not contain any findings with a Not Reviewed (NR) status after manual reviews

For every component (device or software) with manufacturer provided cybersecurity documentation, procedure, or method for secure configuration or installation, provide a report documenting how the component was configured and any deviation from the manufacturer instructions.

1.8.8 Control System Cybersecurity Documentation

{For Government Reference Only: This subpart (and its subparts) relates to SA-5(a),SA-5(b),SA-5(c), SA-22(b); CCIs: CCI-003124, CCI-003125, CCI-003126, CCI-003127, CCI-003128, CCI-003129, CCI-003130, CCI-003131, CCI-003374}

Provide a Control System Cybersecurity Documentation submittal containing the indicated information for each device and software application.

1.8.8.1 Software Applications

For all software applications running on computers provide:

a. administrator documentation that describes secure configuration of the software {For Government Reference Only: relates to CCI-003124}

b. administrator documentation that describes secure installation of the software and software updates. {For Government Reference Only: relates to CCI-003125}

c. administrator documentation that describes secure operation of the software {For Government Reference Only: relates to CCI-003124}

d. administrator documentation that describes effective use and maintenance of security functions or mechanisms for the software {For Government Reference Only: relates to CCI-003127}

e. administrator documentation that describes known vulnerabilities regarding configuration and use of administrative (i.e. privileged) functions for the software {For Government Reference Only: relates to CCI-003128}

f. user documentation that describes user-accessible security functions or mechanisms in the software and how to effectively use those security functions or mechanisms {For Government Reference Only: relates to CCI-003129}

g. user documentation that describes methods for user interaction which enables individuals to use the software in a more secure manner {For Government Reference Only: relates to CCI-003130}

h. user documentation that describes user responsibilities in maintaining the security of the software {For Government Reference Only: relates to CCI-003131}

1.8.8.2 Default Requirements for Control System Devices

For control system devices where Control System Cybersecurity Documentation requirements are not otherwise indicated in this Section, provide:

a. Documentation that describes secure configuration of the device {For Government Reference Only: relates to CCI-003124}

b. Documentation that describes secure installation of the device {For Government Reference Only: relates to CCI-003125}

c. Documentation that describes secure operation of the device {For Government Reference Only: relates to CCI-003124}

d. Documentation that describes effective use and maintenance of security functions or mechanisms for the device {For Government Reference Only: relates to CCI-003127}

e. Documentation that describes known vulnerabilities regarding configuration and use of administrative (i.e. privileged) functions for the device {For Government Reference Only: relates to CCI-003128}

f. Documentation that describes user-accessible security functions or mechanisms in the device and how to effectively use those security functions or mechanisms {For Government Reference Only: relates to CCI-003129}

g. Documentation that describes methods for user interaction which enables individuals to use the device in a more secure manner {For Government Reference Only: relates to CCI-003130}

h. Documentation that describes user responsibilities in maintaining the security of the device {For Government Reference Only: relates to CCI-003131}

i. Documentation of the published last date of support by the manufacturer or indication that a published date is not available. {For Government Reference Only: relates to CCI-003374}

1.9 CYBERSECURITY DURING CONSTRUCTION

{For Government Reference Only: This subpart (and its subparts) relates to AC-18, CA-3; CCI-000258}

In addition to the control system cybersecurity requirements indicated in this section, meet following requirement throughout the construction process.

1.9.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. Contractor computers connected to the control system, control system network, or a control system component at any point during construction must meet the following requirements:

1.9.1.1 Operating System

The operating system must be an operating system currently supported by

the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.9.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. Computers used on this project must perform a full antivirus scan at least once per day. Computers which have connected to any other network since the last full antivirus scan must perform a full antivirus scan prior to connection to the control system network or to the temporary contractor-installed IP network. Perform an antivirus scan on all removable media (e.g., external hard drives, CDs, DVDs, USB flash drives) prior to connecting to the control system environment.

1.9.1.3 Passwords and Passphrases

The passwords and passphrases for computers, applications, and web-based applications supporting passwords must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.9.1.4 User-Based Authentication

Each user must have a unique account; sharing of a single account between multiple users is prohibited.

1.9.1.5 Demonstration of Compliance

The Government has the right to require demonstration of computer compliance with these requirements at any time during the project.

1.9.1.6 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. Each Statement must be signed by a cybersecurity representative for the relevant company.

1.9.2 Government Access to Network

Government personnel must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification.

1.9.3 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.9.4 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

1.10 CYBERSECURITY DURING WARRANTY PERIOD

All work performed on the control system after acceptance must be performed using Government Furnished Equipment or equipment specifically and individually approved by the Government.

PART 2 PRODUCTS

All products used on this project must meet the indicated requirements, but not all products specified here will be required by every project.

2.1 ETHERNET SWITCH

Provide Open Systems Interconnection (OSI) Layer 2 Ethernet switches with the following capabilities, and with an interface to support switch configuration for these capabilities:

2.1.1 Required Functionality

Switches must:

a. Copper Ethernet ports must auto negotiate for 10, 100 and 1000 megabits-per-second links.

b. Be capable of implementing port level access control by MAC address and limit the number of MAC addresses to one MAC address per port.

c. For MODERATE Impact Systems, be capable of implementing per-port access control lists (ACLs) where the list can be filtered by source and destination IP addresses, and by source and destination UDP or TCP ports.

d. Support Remote Network Monitoring (RMON) Port Analysis in accordance with IETF RFC 2819

e. Configure target port and analysis port such that switch clones all target port traffic to analysis port.

2.1.2 Configuration Requirements

Switches must:

a. Support configuration save and restore.

b. Support both manual IP address assignment and acquisition of a dynamic IP address via Dynamic Host Configuration Protocol (DHCP).

c. Be capable of limiting access for configuration to one or more of: a

web interface using HTTPS, a command line interface using SSH, or an SNMP connection using SNMP version 3 or later.

2.2 DAISY CHAIN IP CONTROLLERS

Controllers used as Daisy Chain IP Controllers must be IP controllers with exactly two Ethernet network connections and basic built-in switch capabilities to allow implementation of an Ethernet network in a daisy chain architecture. Switches incorporated by Daisy Chain IP Controllers are not required to meet the requirements for Ethernet Switches as defined in this Section.

2.3 DATABASE AND WEB SERVER SOFTWARE FOR MODERATE IMPACT SYSTEMS

{For Government Reference Only: This subpart (and its subparts) relate to RA-5(1), RA-5(5); CCI-001062, CCI-001067, CCI-001645, CCI-002906}

All computer-based databases must use Microsoft SQL Server or Oracle or MySQL. All computer-based web interfaces must use Internet Information Services (IIS) or Apache as the web server.

PART 3 EXECUTION

3.1 CYBERSECURITY HARDENING AND CONFIGURATION GUIDES

Install, configure, and harden all hardware and software furnished on this project in accordance with manufacturer provided documentation, procedures, or methods for secure configuration or installation. Do not implement specific hardening actions if that action would conflict with required functionality or another requirement of this Section.

3.2 NETWORK REQUIREMENTS

3.2.1 Information Flow Enforcement In MODERATE Impact Systems

{For Government Reference Only: This subpart (and its subparts) relate to AC-4; CCI-001368, CCI-001414, CCI-001548, CCI-001549, CCI-001550, CCI-001551}

Install and configure Ethernet switches to block all traffic on all ports not required by the control protocol.

3.2.2 Wireless and Wired Broadcast Communication

{For Government Reference Only: This subpart (and its subparts) relates to AC-18, AC-18(3); CCI-001438, CCI-001439, CCI-002323, CCI-001441, CCI-001449}

Unless explicitly authorized by the Government, do not use any wireless or wired broadcast communication.

3.2.2.1 Wireless and Wired Broadcast IP Communications

Do not install wireless or wired broadcast IP networks, including: do not install a wireless access point; do not install or configure an ad-hoc wireless network; do not install or configure a WiFi Direct communication.

When explicitly authorized by the Government, wireless IP communication may be used to communicate with an existing wireless network.

3.2.2.2 Non-IP Wireless Communication

For MODERATE Impact Systems: When non-IP wireless communication is explicitly authorized by the Government, the radios must meet NIST FIPS 140-2 Level 2.

3.2.2.3 Wireless and Wired Broadcast Communication Request

Provide a report documenting the proposed use of wireless or wired broadcast communication prior to device selection using the Wireless and Wired Broadcast Communication Request Schedule at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. If there is no proposed use of wireless or wired broadcast communication, provide a document indicating this instead of the Request Schedule.

For each device proposed to use wireless or wired broadcast communication show: the device identifier, a description of the device, the location of the device, the device identifiers of other devices communicating with the device, the protocol used for communication, encryption type and strength. For wireless communication, also show: RF Frequency, Radiated Power in dBm (decibel with a milliwatt reference), free-space range, and the expected as-installed range.

3.2.3 Non-IP Control Networks

When control system specifications require particular communication protocols, use only those communication protocols and only as specified. Do not implement any other communication protocol.

When control system specifications do not indicate requirements for communication protocols, use only those protocols required for operation of the system as specified.

3.2.4 IP Control Networks

{For Government Reference Only: This subpart relates to CM-6(a), CM-7(a), CM-7(b), CM-7(1)(b), SC-41; CCI-001588, CCI-000381, CCI-000380, CCI-000381, CCI-000382, CCI-001761, CCI-001762, CCI-002544, CCI-002545, CCI-002546. For Moderate Impact Systems, this subpart (and its subparts) also relates to SC-5(1), SC-5(2); CCI-001094 CCI-001095}

IP Networks must be Ethernet networks and must use switches which are Ethernet Switches or Daisy Chain IP Controllers as defined in this Section. Do not use nonsecure functions, ports, protocols and services as defined in DODI 8551.01 unless those ports, protocols and services are specifically required by the control system specifications or otherwise specifically authorized by the Government. Do not use ports, protocols and services that are not specified in the control system specifications or required for operation of the control system.

For MODERATE Impact Systems, unless explicitly authorized, do not use IP networks if the same control functionality is available through the use of non-IP networks.

3.2.4.1 IP Devices With Multiple Ethernet Connections

Except for Ethernet Switches and Daisy Chain IP Controllers, devices must not have more than one Ethernet connection to IP networks unless doing so is required by the project specifications and the specific application is approved. If a device with Multiple Ethernet Connections to IP networks is required, provide a Multiple Ethernet Connection Device Request using the Multiple Ethernet Connection Device Request Template at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11 to request approval for each device. If a device with Multiple Ethernet Connections to IP networks is not required, instead provide a document stating that no approval is being requested.

3.2.5 Cryptographic Protection

{For Government Reference Only: This subpart relates to IA-2(9), IA-3(1), SC-8, SC-13, SC-23(1), SC-23(3); CCI-001942, CCI-001959, CCI-001967, CCI-002418, CCI-002449, CCI-002450, CCI-001185, CCI-001188, CCI-001664.}

All remote user interfaces must use HTTPS for all traffic between the user interface client and user interface server.

3.2.6 Device Identification and Authentication

{For Government Reference Only: This subpart (and its subparts) relates to IA-3; CCI-000777, CCI-000778, CCI-001958. For MODERATE Impact systems, this subpart (and its subparts) also relates to SC-23, SC-23(5); CCI-001184, CCI-002470.}

All computers must support IEEE 802.1x for device authentication to the network.

3.2.6.1 Default Requirements for Control System Devices

For control system devices where Device Identification and Authentication requirements are not otherwise indicated in this Section: Devices using HTTP as a control protocol must use HTTPS instead.

3.2.7 Cryptographic Module Authentication

{For Government Reference Only: This subpart (and its subparts) relates to IA-7; CCI-000803}

For devices (including but not limited to NIST FIPS 140-2 compliant radios) that have STIG/SRGs related to cryptographic module authentication (CCI-000803), comply with the requirements of those STIG/SRGs.

3.3 ACCESS CONTROL REQUIREMENTS

3.3.1 User Accounts

{For Government Reference Only: This subpart (and its subparts) relate to AC-2(a), AC-3, AC-6(1), AC-6(10), AC-6(2), AC-6(9), CM-11(2), and IA-2; CCI-002110, CCI-000213, CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-002235, CCI-000039, CCI-001419, CCI-002234, CCI-001812, and CCI-000764. For MODERATE Impact systems, this subpart (and its subparts) also relate to AC-2 (2), AC-2(3), AC-2(4), AC-6(1), and CM-5(1); CCI-001361, CCI-000017, CCI-000217, CCI-000018, CCI-001403, CCI-001404, CCI-001405, CCI-002130, CCI-001683, CCI-001684, CCI-001685, CCI-001686, CCI-002132,

CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-001813.}

Any user interface supporting user accounts (either FULLY or MINIMALLY) must limit access according to specified limitations for each account. Install and configure any device having a STIG or SRG in accordance with that STIG or SRG.

All user interfaces FULLY supporting accounts must implement user-based authentication where each account is uniquely assigned to a specific user. User interfaces FULLY supporting accounts must implement at least three (3) levels of user account privilege including: 1) an account with read-only permissions 2) an account with full permissions including account creation and modification and 3) an account with greater permissions than read-only but without account creation and modification. Disable any unnecessary or unused accounts. Disable any "guest-level" accounts that are created on the system by default.

3.3.1.1 Computers

All computer operating systems must FULLY support user accounts and implement accounts for access. Each control system software application not supporting accounts and running on a computer must be installed such that use of the software is restricted by the computer operating system to specific users.

Applications running on computers must not require the user be logged in to a computer operating system administrator account for normal operation. It is permissible to require the computer operating system administrator account for initial application installation and configuration.

3.3.1.2 Controllers

For user interfaces provided by controllers, provide access control in accordance with the User Interface Requirements table for the applicable control system and user interface type.

a. For table entries of "NA": NA means Not Applicable, there are no interfaces in this category.

b. For table entries of "None Required": The user interface is not required to support user accounts.

c. For table entries of "MINIMALLY": The user interface must at least MINIMALLY support user accounts.

d. For table entries of "FULLY": The user interface must at FULLY support user accounts.

e. For table entries of "KEY": The user interface must have physical security in the form of either a key lock on the interface itself or be furnished inside a lockable enclosure. Where this is required for a read only interface, the lockable enclosure must prevent viewing of data on the interface; for other interfaces, this lockable enclosure must prevent using the interface to alter data.

f. For table entries of "Physical Security": For Local FULL interfaces, the interface must be located inside mission space. For Local Limited (not FULL) interfaces, the user interface must either a) be located

within mission space or b) be protected by physical security at least as good as the control devices (and equipment controlled by the control devices) affected by the interface. For purposes of this requirement, 'affected' includes controllers with data that can be directly altered by the interface, as well as mechanical and/or electrical equipment directly controlled by those controllers, but does not include other interactions.

g. Entries of the form "X and Y" must meet both the requirement indicated for X and the requirement indicated for Y. For example, an entry of "MINIMALLY and Physical Security" indicates the user interface must both MINIMALLY support accounts and have physical security.

h. Entries of the form "X or Y" must meet either the requirement indicated for X or the requirement indicated for Y.

3.3.1.2.1 Electronic Security Systems (ESS)

User Interface Requirements for MODERATE Impact Electronic Security Systems	
User Interface Type	**Access Control Requirement** (See note 3)
Local Read Only (see note 1)	MINIMALLY
Local Limited, Non-privileged	NA
Local Limited, Privileged	FULLY
Local Full	FULLY and Physical Security
Remote Read Only	None Required
Remote Limited, Non-Privileged	NA
Remote Limited, Privileged AND Remote Full (see note 2)	FULLY
Notes: 1)Local Read Only User Interfaces are always Non-Privileged 2)Remote Full User Interfaces are always Privileged 3)Devices outside mission space require physical security protections as indicated (in "PHYSICAL SECURITY IN MODERATE IMPACT SYSTEMS")	

3.3.1.3 Additional User Account Expiration Requirements In MODERATE Impact Systems:

In addition to other user account requirements, user account expiration and auditing must be configured as indicated.

3.3.1.3.1 For Control System Applications Running on Computers

If temporary accounts are supported, expire temporary accounts 72 hours after creation. Expire all other accounts after 35 days of inactivity.

3.3.1.3.2 For Other Control System Devices FULLY Supporting Accounts

If temporary accounts are supported, expire temporary accounts 72 hours after creation. Expire all other accounts after 365 days of inactivity.

3.3.2 Unsuccessful Logon Attempts

{For Government Reference Only: This subpart (and its subparts) relate to AC-7 (a), AC-7 (b); CCI-000043, CCI-000044, CCI-001423, CCI-002236, CCI-002237, CCI-002238}

Except for high availability user interfaces indicated as exempt, devices must meet the indicated requirements for handling unsuccessful logon attempts. If a device cannot meet these requirements, document device capabilities to protect from subsequent logon attempts and propose alternate protections in a Device Account Lock Exception Request submittal. Do not implement alternate protection measures in lieu of the indicated requirements without explicit permission from the Government. If no Device Account Lock Exceptions are requested, provide a document stating that no approval is being requested as the Device Account Lock Exception Request.

3.3.2.1 Devices MINIMALLY Supporting Accounts

For MODERATE Impact Systems: Devices which MINIMALLY (but not FULLY) support accounts must lock the user accountafter five consecutive failed login attempts and must unlock the user account after 60 minutes have elapsed without an unsuccessful login attempt or by a successful login to a separate administrator account.

3.3.2.2 Devices FULLY Supporting Accounts

Devices which FULLY support accounts must meet the following requirements.

a. It must lock the user account when three unsuccessful logon attempts occur within a 15 minute interval.

b. Once an account is locked, the account must stay locked until unlocked by an administrator. If the account being locked is the sole administrator account on the device, the account must stay locked for 1 hour and then automatically unlock.

c. Once the indicated number of unsuccessful logon attempts occurs, delay further logon prompts by 5 seconds.

3.3.2.3 High Availability Interfaces Exempt from Unsuccessful Logon Attempts Requirements

There are no high availability interfaces which are exempt from unsuccessful logon attempts requirements.

3.3.3 System Use Notification

{For Government Reference Only: This subpart (and its subparts) relates to AC-8; CCI-000048, CCI-002247, CCI-002243, CCI-002244, CCI-002245, CCI-002246, CCI-000050, CCI-002248}

3.3.3.1 System Use Notification for Remote User Interfaces

Remote user interfaces must display a warning banner meeting the requirements of DTM 08-060 on screen.

3.3.3.2 System Use Notification for Local User Interfaces

Devices which are connected to a network and have a local user interface must display a warning banner meeting the requirements of DTM 08-060 on the user interface screen if capable of doing so and must have a permanently affixed label with an approved banner from DTM 08-060 if unable to display the warning banner on the screen. Where it is impractical (perhaps due to device size) to affix the label to the device, affix the label to the device enclosure.

Labels must be machine printed or engraved, plastic or metal, designed for permanent installation, must use a font no smaller than 14 point, and must provide a high contrast between font and background colors.

3.3.4 Session Lock and Session Termination Requirements In MODERATE Impact Systems:

{For Government Reference Only: This subpart (and its subparts) relates to AC-11(a), AC-11(b), AC-11(1), AC-12, SC-10; AC-10; CCI-000058, CCI-000059, CCI-000056, CCI-000057, CCI-000060, CCI-002360, CCI-002361, CCI-001133, CCI-001134, CCI-000054, CCI-000055, CCI-002252}

3.3.4.1 Session Termination

When session termination is required for a User Interface, the User Interface must implement session termination a) based on manual initiation, or b) based on lack of activity, or c) based on either manual initiation or lack of activity, as indicated.

Session Termination must result in logging out the user. A logged out User Interface may only perform actions as indicated in the "Permitted Actions Without Identification or Authentication" subpart of this Section or display a publicly viewable image or blank screen. User Interfaces must remain logged out (session terminated) until a user enters correct authentication information, which must initiate a new session. All User Interfaces running on computers and all Remote User Interfaces must also terminate network connections as part of session termination.

3.3.4.2 Session Lock

When session lock is required for a User Interface, the User Interface must implement session lock a) based on manual initiation, or b) based on lack of activity, or c) based on either manual initiation or lack of activity, as indicated.

Session lock must result in the User Interface being suspended and the user interface must display a publicly viewable image or blank screen. No interaction with the user interface must be possible until either a) the same user enters valid authentication information, in which case that session must be continued, or b) until a different user enters valid authentication information at which point the first session must be terminated and a new session initiated for the new user.

3.3.4.3 Session Lock and Termination for Computers

User Interface sessions provided by computer operating systems must support the requirement for both Session Lock and Session Termination. Session Lock and Session Termination must be capable of being initiated by the user and must also be initiated by lack of activity. Session Lock must occur after 15 minutes of inactivity, and Session Termination must occur after 30 minutes total of inactivity (including, not in addition to, the time for Session Lock). When a user initiates a new session, terminate existing sessions if necessary to limit the total number of concurrent sessions to 1.

Other User Interface sessions running on computers (for local user interfaces) or hosted on a computer (for remote user interfaces) and supporting accounts must support user initiation of Session Termination. In addition, remote User Interface sessions must also initiate Session Termination after 30 minutes of inactivity .

3.3.4.4 Session Lock and Termination for Controllers

Writable Remote User Interfaces must support requirements for Session Termination, and must both be capable of being initiated by the user and initiated by lack of activity. Session Termination must initiate after 30 minutes of inactivity.

Local User Interfaces supporting accounts must support manual initiation of Session Termination. Privileged Local User Interfaces must also support timed initiation of Session Termination, with Session Termination initiated at 30 minutes of inactivity.

3.3.5 Permitted Actions Without Identification or Authentication

{For Government Reference Only: This subpart (and its subparts) relates to AC-14; CCI-000061, CCI-000232}

The control system must require identification and authentication before allowing any actions except read-only actions by a user acting from a user interface which MINIMALLY or FULLY supports accounts.

3.3.6 Physical Security in MODERATE Impact Systems

{For Government Reference Only: This subpart relates to PE-3(1), PE-4, PE-5, SC-7(a), SC-7(c), SC-8, SC-8(1); CCI-000928, CCI-002926, CCI-000936, CCI-002930, CCI-002931, CCI-000937, CCI-001097, CCI-001109, CCI-002418, CCI-002419, CCI-002421.}

3.3.6.1 Physical Security for Media

3.3.6.1.1 Physical Security for Media Inside Mission Space

Install all non-IP network media located inside of the mission space in conduit. Install all IP network media located inside of the mission space in intermediate metallic conduit.

3.3.6.1.2 Physical Security for Media Outside Mission Space

Install all network media (both IP and non-IP) located outside of the mission space in rigid metallic conduit.

3.3.6.2 Physical Security for Devices

Install all devices (computers and controllers) which are located outside of mission space in lockable enclosures. (Recall that per definition of mission space, a room controlled by the mission is mission space regardless of whether it is contiguous with other mission space.)

Install all controllers connected to an IP network in lockable enclosures (both inside and outside of mission space).

3.3.6.3 Physical Security for User Interfaces

Physical security requirements for User Interfaces are specified in the preceding paragraphs of this Section.

3.3.7 Enclosures

Prior to final acceptance of the system, lock all lockable enclosures. Submit an Enclosure Keys submittal with all copies of keys for all enclosures and a key inventory list documenting all keys. Label each key with the matching enclosure identifier.

3.4 USER IDENTIFICATION AND AUTHENTICATION

{For Government Reference Only: This subpart (and its subparts) relates to IA-2, IA-2(1),IA-2(12), IA-5 IA-5(b), IA-5(c), IA-5(e), IA-5(g), IA-5(1), IA-5(11); CCI-000764, CCI-000765, CCI-001953, CCI-001954, CCI-001544, CCI-001989, CCI-000182, CCI-001610, CCI-000192, CCI-000193, CCI-000194, CCI-000205, CCI-001619, CCI-001611, CCI-001612, CCI-001613, CCI-001614, CCI-000195, CCI-001615, CCI-000196, CCI-000197, CCI-000199, CCI-000198, CCI-001616, CCI-001617, CCI-000200, CCI-001618, CCI-002041, CCI-002002, CCI-002003. For MODERATE Impact systems, this subpart also relates to AC-6 (1), AC-6(10), AC-6(2), AC-6(9)-IA-2(4), IA-5(13); CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-002235, CCI-000039, CCI-001419, CCI-002234, CCI-000768, CCI-002007.}

This subpart indicates requirements for specific methods of identification and authentication for users and user accounts. Where these requirements conflict apply the following order of precedence: 1) If present, Device Specific Requirements take precedence over any other requirements; and then 2) multifactor authentication requirements take precedence over password requirements.

3.4.1 User Identification and Authentication for All System Types

Unless otherwise indicated, all user interfaces supporting accounts (either FULLY or MINIMALLY) must implement Identification and Authorization via passwords.

For MODERATE Impact Systems: Software running on computers and computer operating systems must manage cached authenticators in accordance with the relevant STIGs. All other devices and software must not use cached authenticators.

3.4.2 User Identification and Authentication for Specific System Types

System specific requirements are in addition to and supersede those

indicated for all system types. When no additional requirements are indicated for a specific system type the requirements for all systems still apply to that system type.

3.4.2.1 Electronic Security System Devices

User interfaces which FULLY support accounts and which run on a computer must use multifactor authentication via PIV. User interfaces which MINIMALLY support accounts must use either passwords or multifactor authentication via PIV.

3.4.3 User Identification and Authentication for Specific Devices

There are no additional device specific user interface requirements

3.4.4 Implementation of Identification and Authorization Requirements

Identification and Authorization must be met by one of the following methods:

a. Direct implementation in the user interface.

b. For user interfaces on a computer: inheriting the Identification and Authorization from the computer operating system, either by the operating system limiting access to specific applications by user, or by the application itself having permissions based on the user logged into the computer.

c. For remote interfaces: an implementation shared between the remote user interface server and the remote user interface client. For example, a requirement for PIV authentication may be met on a remote user interface by a PIV reader on a web browser client which sends the authentication information via HTTPS to the remote server.

3.4.5 Password-Based Authentication Requirements

3.4.5.1 Passwords for Software and Applications Running on Computers

All software and applications running on computers supporting password-based authentication must enforce the following requirements:

a. Minimum password length of 12 characters

b. Password must contain at least one uppercase character.

c. Password must contain at least one lowercase character.

d. Password must contain at least one numeric character.

e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.

f. Password must have a minimum lifetime of 24 hours.

g. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.

h. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters (where location is significant, a character may be reused if it is in a different position).

i. Passwords must be cryptographically protected during storage and transmission.

3.4.5.2 Passwords for Controllers FULLY Supporting Accounts

All controllers FULLY supporting accounts and supporting password-based authentication must enforce the following requirements:

a. Minimum password length of 12 characters

b. Password must contain at least one uppercase character.

c. Password must contain at least one lowercase character.

d. Password must contain at least one numeric character.

e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.

f. Password must have a maximum lifetime of sixty (60) days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.

g. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters.

h. Passwords must be cryptographically protected during storage and transmission.

3.4.5.3 Passwords for Remote Interfaces

Passwords for connecting to a Remote User Interface supporting password-based authentication must enforce the following requirements:

a. Minimum password length of 12 characters

b. Password must contain at least one uppercase character.

c. Password must contain at least one lowercase character.

d. Password must contain at least one numeric character.

e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.

f. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.

g. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters (where

location is significant, a character may be reused if it is in a different position).

h. Passwords must be cryptographically protected during storage and transmission.

3.4.5.4 Passwords for Devices Minimally Supporting Accounts

Devices MINIMALLY supporting accounts must support passwords with a minimum length of eight characters.

3.4.5.5 Password Configuration and Reporting

For all devices with a password, coordinate the changing of passwords with the project site following testing of the system but prior to turnover to the Government. Coordinate with Password Point of Contact to determine appropriate project site personnel to complete password changes. Accompany identified personnel to each device with a password and instruct personnel on the process of changing password. Record the time, date and personnel present when each device's password is changed and submit a Password Change Summary Report documenting this information.

Provide the Password Summary Report electronically in both PDF and Microsoft Excel.

3.4.6 Authenticator Feedback

{For Government Reference Only: This subpart relates to IA-6; CCI-000206}

Devices must never show authentication information, including passwords, on a display. Devices that momentarily display a character as it is entered, and then obscure the character, are acceptable. For devices that have STIGs or SRGs related to obscuring of authenticator feedback (CCI-000206), comply with the requirements of those STIGS/SRGs.

3.5 CYBERSECURITY AUDITING

Where an auditing requirement exists for email notification, notify via email the application administrator and Information System Security Officer (ISSO) of the event. Coordinate with the Email Address Point of Contact for email addresses. If outgoing email is not available to the system, disable email notifications.

3.5.1 Audit Events, Content of Audit Records, and Audit Generation

{For Government Reference Only: This subpart (and its subparts) relates to AU-2(a), AU-2(c), AU-2(d), AU-3, AU-10, AU-12, AU-14(b), AU-14(1), AU-14(2), AU-14(3), CM-5(1), SC-7 (9); CCI-000123, CCI-001571, CCI-000125, CCI-001485, CCI-000130, CCI-000131, CCI-000132, CCI-001230, CCI-000133, CCI-000134, CCI-001487, CCI-000166, CCI-001899, CCI-000169, CCI-001459, CCI-000171, CCI-000172, CCI-001910, CCI-001914, CCI-001919, CCI-001464, CCI-001462, CCI-001920, CCI-001814, CCI-002400. For MODERATE Impact systems, this subpart (and its subparts) also relates to AU-3 (1); CCI-000135, CCI-001488}

For devices that have STIG/SRGs related to audit events, content of audit records or audit generation, comply with the requirements of those

STIG/SRGs.

If auditing requirements can be met using existing control system alarm or event capabilities, those existing capabilities may be used to meet these requirements.

3.5.1.1 Computers

For each computer, provide the capability to select audited events and the content of audit logs. Configure computers to audit the indicated events, and to record the indicated information for each auditable event. Send logs to a syslog server that is only accessible via accounts on the system with privileged level access.

3.5.1.1.1 Audited Events

Configure each computer to audit the following events:

a. Successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels)

b. Successful and unsuccessful logon attempts

c. Successful logouts

d. Privileged activities or other system level access

e. Concurrent logons from different workstations

f. Successful and unsuccessful accesses to objects

g. Program initiations

h. Direct access to the information system

i. Account creations, modifications, disabling, and terminations. For MODERATE Impact Systems, also provide email notification when these audit events occur.

j. Kernel module load, unload, and restart

3.5.1.1.2 Audit Event Information To Record

Configure each computer to record, for each auditable event, the following information (where applicable to the event):

a. What type of event occurred

b. When the event occurred

c. Where the event occurred

d. The source of the event

e. The outcome of the event

f. The identity of any individuals or subjects associated with the event

g. For MODERATE Impact Systems: For all privileged commands, full-text recording of the executed command and the user executing the command

For MODERATE Impact Systems: Audit records must provide sufficient detail to reconstruct events to determine cause of compromise and magnitude of damage, malfunction, or security violation.

3.5.1.2 Default Requirements for Control System Controllers

For control system controllers where Audit Events, Content of Audit Records, and Audit Generation are not otherwise indicated in this Section:

3.5.1.2.1 Controllers Which FULLY Support Accounts

For each controller which FULLY supports accounts, provide the capability to select audited events and the content of audit logs. Configure controllers to audit the indicated events, and to record the indicated information for each auditable event.

3.5.1.2.1.1 Audited Events

Configure each controller to audit the following events:

a. Successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels)

b. Successful and unsuccessful logon attempts

c. Successful logouts

d. Concurrent logons from different workstations

e. All account creations, modifications, disabling, and terminations. For MODERATE Impact Systems, also provide email notification when these audit events occur.

f. All kernel module load, unload, and restart

g. For privileged user interfaces in MODERATE Impact Systems: All user commands.

3.5.1.2.1.2 Audit Event Information To Record

Configure each controller to record, for each auditable event, the following information (where applicable to the event):

a. what type of event occurred

b. when the event occurred

c. where the event occurred

d. the source of the event

e. the outcome of the event

f. the identity of any individuals or subjects associated with the event

g. For privileged user interfaces in MODERATE Impact Systems: Full text recording of the executed command and the user executing the command.

For MODERATE Impact Systems: Audit records must provide sufficient detail to reconstruct events to determine cause of compromise and magnitude of damage, malfunction, or security violation

3.5.1.2.2 Controllers Which Do Not FULLY Support Accounts

For each controller which does not FULLY support accounts configure the controller to audit all controller shutdown and startup events and to record for each event the type of event and when the event occurred.

3.5.2 Audit Time Stamps

{For Government Reference Only: This subpart (and its subparts) relates to AU-8; CCI-000159, CCI-001889, CCI-001890. For MODERATE Impact systems, this subpart (and its subparts) also relates to AU-8 (1); CCI-001891, CCI-001892, CCI-002046.}

Any device (computer or controller) generating audit records must have an internal clock capable of providing time with a resolution of one second. Clocks must not drift more than 10 seconds per day. Configure the system so that each device (computer or controller) generating audit records maintains accurate time to within 1 second. Note that if the control system specifications include requirement for clocks, the most stringent requirement applies.

3.5.3 Response to Audit Processing Failures

{For Government Reference Only: This subpart (and its subparts) relates to AU-5; CCI-000139, CCI-000140, CCI-001490.}

In the case of a failure in the auditing system, computers associated with auditing must provide email notification. For MODERATE Impact systems, the computer must also notify the associated auditing front end software. In case of an audit failure, if possible, continue to collect audit records by overwriting existing audit records.

For MODERATE Impact Systems: In the case of an audit failure at a controller performing auditing, the device must notify the associated auditing front end software of the audit failure if able, and must continue to collect audit records by overwriting existing audit records if able. The auditing front end software must provide notification as indicated, treating the notification of failure from the device as a failure in the auditing system.

3.6 REQUIREMENTS FOR LEAST FUNCTIONALITY

{For Government Reference Only: This subpart (and its subparts), along with the Network Communication, Ports, Protocols and Services Report submittal specified elsewhere in this section, relates to CM-6(a), CM-6(c), CM-7, CM-7(1)(b), SC-41; CCI-000363, CCI-000364, CCI-000365, CCI-001588, CCI-001755, CCI-000381, CCI-000380, CCI-000382, CCI-001761, CCI-001762, CCI-002544, CCI-002545, CCI-002546. For MODERATE Impact systems, this subpart (and its subparts) also relates to CM-7(2), CM-7(5)(a), CM-7(5)(b); CCI-000381, CCI-000380, CCI-000382, CCI-001761, CCI-001762}

For devices that have a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port and device I/O access for least functionality), install and configure the device in accordance with that STIG or SRGs.

3.6.1 Device Capabilities

For MODERATE Impact Systems: Do not provide devices with remote user interfaces or full user interfaces where one was not required. Do not use a networked sensor or actuator where a non-networked sensor or actuator would suffice.

For all MODERATE Impact Systems: Unless specifically required by the government, do not provide a capability to update device firmware over the network.

3.6.2 Software

For software that has a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port access for least functionality), install and configure the software in accordance with that STIG or SRG.

Do not install software that is not specifically required to meet a contract requirement. Remove any previously installed that is not specifically required to meet a contract requirement. Do not implement functionality within software
that is not specifically required to meet contract requirements.

3.7 SYSTEM AND COMMUNICATION PROTECTION

3.7.1 Collaborative Computing

{For Government Reference Only: This subpart relates to SC-15(a), SC-15(b); CCI-001150, CCI-001152.}

Without explicit approval from the project site, control systems must not use collaborative computing technologies.

3.7.2 Denial of Service Protection and Application Partitioning In MODERATE Impact Systems:

{For Government Reference Only: This subpart relates to SC-5, SC-12, SC-7(a); CCI-001093, CCI-002385, CCI-002386, CCI-002430, CCI-001097. For MODERATE Impact systems, this subpart also relates to SC-2; CCI-001082.}

To the greatest extent practical, implement control logic without reliance on the network. Except when required to meet the requirements of the control system Section (where the requirement can only be met using computer hardware), do not implement control logic in computers. For MODERATE Impact systems, do not implement control logic in a device providing (i.e. acting as a server for) a Full Remote User Interface.

3.7.2.1 Default Requirements for MODERATE Impact Control Systems

Except for networked input and outputs on input-output buses specifically designed to provide high reliability or redundancy, sensors and actuators

must not rely on the network to exchange data with the controller executing the sequence of operation which uses the sensor value or determines the actuator command.

Sensor values required by multiple devices may be shared over the network provided they are connected to a controller requiring the value for execution of the sequence and that controller shares the value on the network.

3.7.3 Mobile Code In MODERATE Impact Systems:

{For Government Reference Only: This subpart relates to SC-18(a), SC-18(b), SC-18(c), SC-18(1), SC-18(3), SC-18(4); CCI-001160, CCI-001161, CCI-001162, CCI-001163, CCI-001164, CCI-001165, CCI-001166, CCI-001662, CCI-002457, CCI-002458, CCI-001169, CCI-001695, CCI-001170, CCI-002469}

Devices with STIGs/SRGs related to Mobile Code and to Security Control SC-18 must be installed in accordance with the relevant STIGs/SRGs. All remote user interfaces must meet the requirements of the "Web Browsers and Application SRG".

Mobile code may only be downloaded from a specifically authorized mobile code repository. Coordinate with the Mobile Code Point of Contact for the location of a repository.

3.7.4 Protection of Information at Rest In MODERATE Impact Systems:

{For Government Reference Only: This subpart relates to SC-28, SC-28(1); CCI-001199, CCI-002472, CCI-002475, CCI-002476}

Computers must protect information at rest in accordance with applicable STIGs.

Any control system device storing personally identifiable information (PII), controlled unclassified information (CUI), or classified information must be protected by an Information At Rest encryption solution or by a physical security solution. Provide a Protection of Information At Rest Proposal indicating each device storing PII, CUI, or classified information and the encryption or physical security solution proposed for that device for government approval. If no devices stores PII, CUI, or classified information, provide a document stating this as the Protection of Information At Rest Proposal submittal. Do proceed with device selection and installation until the Protection of Information At Rest Proposal is approved. Once approved, implement approved Information At Rest protections.

3.8 SAFE MODE AND FAIL SAFE OPERATION

{For Government Reference Only: This subpart (and its subparts) relates to CP-12, SI-10(3), SI-17; CCI-002855, CCI-002856, CCI-002857, CCI-002754, CCI-002773, CCI-002774, CCI-002775}

For all control system components with an applicable STIG or SRG, configure the component in accordance with all applicable STIGs and SRGs.

3.9 SYSTEM MAINTENANCE TOOL SOFTWARE

{For Government Reference Only: This subpart (and its subparts) relates

to MA-3; CCI-000865.}

Submit and license to the Government all software required to operate, maintain and modify the control system such the Government or their agents are able to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer. Submit hard copies of user manuals for each software with the software submittal. Provide any hardware keys or dongles, software keys, license numbers, and other information required to enable the Government to access or change the system.

For software provided and licensed to the Government under the requirements of another Section, submit a statement indicating the Section and Submittal under which the software was provided. For software provided to meet the requirements of this Section and not provided and licensed under another Section, submit software and software user manuals on DVD or CD as a Technical Data Package and submit one hard copy of the software user manual for each piece of software.

3.10 DEVICE POWER

{For Government Reference Only: This subpart (and its subparts) relates to PE-11, PE-11(1); CCI-002955, CCI-000961. For MODERATE Impact systems, this subpart (and its subparts) also relates to PE-9, PE-9(1); CCI-000952, CCI-002953, CCI-002954.}

For MODERATE Impact Systems: Provide control system with power supply meeting or exceeding the reliability of the controlled equipment. Powering control system devices using the same power source as the equipment controlled by the device is a permissible method of meeting this requirement. Without explicit approval from the government, do not install local uninterruptible power supplies (UPSes) as a source of device power.

3.10.1 Device Behavior on Loss of Power In MODERATE Impact Systems:

Application programs and configuration settings must be stored in devices in manner such that a loss of power does not result in a loss of the application program or configuration settings: Loss of power must never result in the loss of application programs, regardless of the length of time power is lost; and loss of power for less than 2,500 hours must not result in the loss of configured settings.

In the event of a loss of power, when power is restored, controllers and computers executing control logic (and the underlying equipment) must recover and resume their normal sequences of operation. Note that the sequence of operation may require specific actions (e.g. startup sequences) upon recovery from loss of power.

3.11 VULNERABILITY SCANNING

{For Government Reference Only: This subpart (and its subparts) relates to RA-5 RA-5(a),RA-5(b),RA-5(c),RA-5(d); CCI-001054, CCI-001055, CCI-000156, CCI-001641, CCI-001643, CCI-001057, CCI-001058, CCI-001059. For MODERATE Impact systems, this subpart (and its subparts) also relates to RA-5(1), RA-5(5); CCI-001062, CCI-001067, CCI-001645, CCI-002906.}

All IP devices must be scannable, such that the device can be scanned by industry standard IP network scanning utilities without harm to the

device, application, or functionality.

3.11.1 Computers and Software Running on Computers

Computers and applications running on computers must meet relevant vulnerability scanning STIGs/SRGs and respond to approved DoD vulnerability scanning tools.

3.11.2 Controllers

Provide controllers that are scannable by standard control system discovery tools or control system browsers and return meaningful status information including the network inputs and outputs for the controller. This information must contain sufficient detail to detect vulnerabilities or exploits of the controller.

Provide all software needed to scan the control system as the Control System Scanning Tools submittal. If the software required to scan the system is already installed at the project site or is provided under a separate section instead provide a statement indicating this.

3.12 FIPS 201-2 REQUIREMENT

{For Government Reference Only: This subpart (and its subparts) relates to SA-4 (10); CCI-003116}

Devices in the following systems which implement PIV must be on the NIST FIPS 201-2 approved product list (https://www.idmanagement.gov/approved-products-list/): electronic security systems(ESS).

3.13 SYSTEM AND INTEGRATION INTEGRITY

3.13.1 Malicious Code Protection

{For Government Reference Only: This subpart (and its subparts) relates to SI-3(c); CCI-001241, CCI-002623}

For all computers installed under this project, provide malware protection software media, provide licenses, and install and configure malware protection software as indicated. Coordinate with the Government Computer Access Point of Contact as required.

a. Malware protection software licenses will be Government furnished.

b. Malware protection software media will be Government furnished.

c. Malware protection software will be Government installed.

3.14 CONTROL SYSTEM CYBERSECURITY TESTING

{For Government Reference Only: For MODERATE Impact systems, this subpart (and its subparts) relates to SA-11(a), SA-11(b), SA-11(c), SA-11(d), SA-11(e); CCI-003171, CCI-003172, CCI-003173, CCI-003174, CCI-003175, CCI-003176, CCI-003177, CCI-003178.}

3.14.1 Control System Cybersecurity Testing Procedures

Prepare and submit Control System Cybersecurity Testing Procedures explaining step-by-step, the actions and expected results that will

demonstrate that the control system meets the requirements of this Section. The Control System Cybersecurity Testing Procedures may be submitted as a Technical Data Package.

3.14.2 Control System Cybersecurity Testing Execution

Using the Control System Cybersecurity Testing Procedures verify that the control system meets the requirements of this Section. UNLESS GOVERNMENT WITNESSING OF A TEST IS SPECIFICALLY WAIVED BY THE GOVERNMENT, PERFORM ALL TESTS WITH A GOVERNMENT WITNESS. If testing reveals deficiencies in the system, correct the deficiency and retest until successful.

3.14.3 Control System Cybersecurity Testing Report

Prepare and submit a Control System Cybersecurity Testing Report documenting all tests performed and their results. Include all tests in the Control System Cybersecurity Testing Procedures and any additional tests performed during testing. Document test failures and repairs conducted with the test results. The Control System Cybersecurity Testing Report may be submitted as a Technical Data Package

3.15 FIELD QUALITY CONTROL, CYBERSECURITY VALIDATION SUPPORT

In addition to testing and testing support required by other Sections, provide a minimum of eight hours of technical support for cybersecurity testing of control systems to support the DoD Risk Management Framework process Cybersecurity assessment of the control system. This support is independent of (and in addition to) the Control System Cybersecurity Testing specified in this section.

3.16 CYBERSECURITY TRAINING

Provide eight hours of classroom training for six Government personnel on the cybersecurity operation and maintenance of the control system provided. This training is in addition to and must be coordinated with control system training specified in other Sections.

The Government will provide the training location. Training must cover, at a minimum: (a) applying software and firmware updates, (b) user account creation, modification and deletion, (c) audit log upload procedures and (d) identification of privileged user interfaces and system impact of those interfaces. Training session must include a question and answer period during which government staff questions about cybersecurity aspects of the control system are answered.

-- End of Section --

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08/24

cybersecurity for LOW impact facility-related control systems

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SECTION 25 05 11.04

CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS - FIRE AND LIFE SAFETY SYSTEMS (L-L-L)
08/24

PART 1 GENERAL

Many subparts in this Section contain text in curly braces ("{" and "}") indicating which cybersecurity control and control correlation identifier (CCI) the requirements of the subpart relate to. The text inside these curly braces is for Government reference only and enables coordination of the requirements of this Section with the RMF process throughout the design and construction process. Text in curly braces are not contractor requirements.

This Section refers to Security Requirements Guide (SRGs) and Security Technical Implementation Guide (STIGs). STIGs and SRGs are available online at the Information Assurance Support Environment (IASE) website at https://public.cyber.mil/stigs/downloads/ and an SRG/STIG Applicability Guide and Collection Tool is available at https://public.cyber.mil/stigs/SCAP/. Not all control system components have applicable STIGs or SRGs. The "Control Systems SRG" does not apply to work performed under this Section; all requirements within this section to apply applicable SRGs DO NOT include the "Control Systems SRG".

1.1 CONTROL SYSTEM APPLICABILITY

There are multiple versions of this Section associated with this project. Different versions have requirements applicable to different control systems. This specific Section applies only to the following control systems: Fire Alarm Systems, Fire Protection Systems, Life Safety Systems.

1.2 RELATED REQUIREMENTS

This section does not contain sufficient requirements to procure a control system and must be used in conjunction with other Sections which specify control systems. This Section adds cybersecurity requirements to the control systems specified in other Sections, and as these requirements are conditioned on the control system being provided, there may be requirements in this Section that will not apply to this project. All Sections containing facility-related control systems or control system components are related to the requirements of this Section. Review all specification sections to determine related requirements.

In cases where a requirement is specified in both this Section and in another Section, the more stringent requirement must be met. In cases where a requirement in this Section conflicts with the requirements of another Section such that both requirements cannot be met at the same time, request direction from the Contracting Officer to determine which requirement applies to the project.

Systems that are to be connected to the VA network shall have an Enterprise Risk Assessment (ERA) performed. ERAs shall conform to the medical center's standards.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.1x	(2010) Local and Metropolitan Area Networks - Port Based Network Access Control

INTERNET ENGINEERING TASK FORCE (IETF)

IETF RFC 2819	(2000) Remote Network Monitoring (RMON) Management Information Base (MIB)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST FIPS 140-2	(2001) Security Requirements for Cryptographic Modules

U.S. DEPARTMENT OF DEFENSE (DOD)

DODI 8551.01	(2014) Ports, Protocols, and Services Management (PPSM)
DTM 08-060	(2008) Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement

1.4 DEFINITIONS

1.4.1 Administrator Account

An administrator account is an account with full permissions to a device, application, or operating system, including the ability to create and modify other user accounts.

Note that the operating system Administrator Account may be different than Administrator Accounts for applications hosted on that operating system. Also, most controllers will not have any support for accounts and will therefore not have an 'Administrator Account'.

1.4.2 Computer

A computer is one of the following:

a. a device running a non-embedded desktop or server version of Microsoft Windows

b. a device running a non-embedded version of MacOS

c. a device running a non-embedded version of Linux

d. a device running a version or derivative of the Android Operating System, where Android is considered separate from Linux

e. a device running a version of Apple iOS

Unless otherwise indicated or clear from context use of the word "device" in this Section includes computers.

1.4.3 Controller

A device other than a computer or Ethernet switch. For Fire Protection systems this includes fire alarm control panels, remote operating consoles, and remote annunciators.

1.4.4 Mission Space

A device or media is in mission space if physical access to the device or media is controlled by the organization served by the device. For example, a variable air volume (VAV) box controller in a suspended ceiling is in mission space if the VAV box serves that room; an electrical switchgear in an electrical room or an air handling unit (AHU) in a mechanical room or on a rooftop may still be considered to be in mission space if the organization (mission) served by that switchgear or AHU controls access to the electrical room, mechanical room or rooftop.

1.4.5 Network

A network is a group of two or more devices that can communicate using a network protocol. Network protocols must provide a method for addressing devices on the network; a communication method that does not provide an addressing scheme is not a networked form of communication. Devices that communicate using a method of communication that does not support device addressing are not using a network. Addresses may be other than IP addresses, and addressing may be at either Open Systems Interconnection (OSI) layer 2 or layer 3.

1.4.6 Network Connected

A component is network connected (or "connected to a network") only when the device has a network transceiver which is directly connected to the network and implements the network protocol. A device lacking a network transceiver (and accompanying protocol implementation) can never be considered network connected. Note that (unlike many IT definitions of "Network Connected") a device connected to a non-IP network is still considered network connected (an IP connection or IP address is not required for a device to be network connected).

1.4.6.1 Wireless Network Connected

Any device that supports wireless network communication is network connected to a wireless network, regardless of whether the device is communicating using wireless. Unless physically disabled, devices with wireless transceivers support wireless, it is not sufficient to disable the wireless in software.

1.4.7 Network Media

The thing that provides the communication channel between the devices on a network. Typically wire, but might include wireless, fiber optic, or even power line (some network protocols allow sending network signals over power wiring).

1.4.8 User Account Support Levels

The support for user accounts is categorized in this Section as one of three levels:

1.4.8.1 FULLY Supported

Device supports configurable individual accounts. Accounts can be created, deleted, modified, etc. Privileges can be assigned to accounts. These devices support user-based (as opposed to role-based) authentication.

1.4.8.2 MINIMALLY Supported

Device supports a small, fixed number of accounts (perhaps only one). Accounts cannot be modified. A device with only a "User" and an "Administrator" account would fit this category. Similarly, a device with two PINs for logon - one for restricted and one for unrestricted rights would fit here (in other words, the accounts do not have to be the traditional "username and password" structure). These devices typically only support role-based authentication.

Examples of devices which MINIMALLY support accounts are a) a variable frequency drive with a single account which requires a PIN for access to configuration; and b) a room lighting control touchpad interface that has a single account.

1.4.8.3 NOT Supported

Device does not support any Access Enforcement therefore the whole concept of "account" is meaningless.

1.4.9 Manual Local Input

Manual Local Inputs are system analog or binary inputs that are adjustable by a person but are, by intrinsic hardware design, very limited in potential capabilities. Manual Local Inputs do not have touch screens or full keyboards, but may have a few buttons or dials to allow input. Manual Local Inputs do not have full graphic screens or dot-matrix displays, but may have simple lights (LEDs) or 7-segment displays. Manual Local Inputs do not have any sort of menu structure, each button has a single well-defined function.

Examples of Manual Local Inputs are H-O-A switches, simple thermostats, and disconnect switches.

1.4.10 Card Reader

A card reader is an input/output device whose primary function is to assist in two-factor authentication. A card reader must have an interface to read data from a card and may be able to write data to a card. A card reader may have a means (such as buttons, keypad, touchscreen, etc.) for a user to input a PIN or password, as well as a limited display.

1.4.11 User Interface

A User Interface (UI) is something other than a Manual Local Input or Card Reader that allows a person to interact with the system or device. Note that while a Card Reader is not by itself a User Interface, a User Interface may contain a Card Reader in order for it to authenticate its

user. Within control systems, there are a wide range of User Interfaces.

Two important distinctions are 1) whether the user interface is Local or Remote, and 2) the effective capabilities of the User Interface to alter data, which is the "privilege" of the user interface (where effective privilege available to a specific user at a specific user interface is the combination of the greatest privilege offered by the user interface and the specific account the user is logged into).

1.4.11.1 Local User Interface

A Local User Interface is a user interface where the physical hardware the user interacts with (keyboard, buttons, display, etc.) is physically part of the device being affected. All of the relevant characteristics of the user interface are embodied within a single device.

Note that a Local UI may be able to access data in a different device, Local versus Remote in this context refers to the user interface itself; the capability to access data in a different device is covered under "Full User Interface".

1.4.11.2 Remote User Interface

A Remote User Interface implements a Client/Server model where the physical hardware the user interacts with (Client) is physically distinct from the device being affected (Server). Most or all of the security and functionality characteristics of the user interface are defined by the Server, not the Client. The Client and Server communicate via a network connection. A common example of a remote user interface is a web-based interface where the browser (client) is generally on different hardware than the web server (server). A Remote UI remains a Remote UI even if the user happens to be at a Client on the same hardware as the Server. What is important is that a) the Client may be on different hardware than the Server and b) the majority of the security and functional characteristics of the interface are defined at the Server.

Note that this definition of "remote" is consistent with that generally used in the control industry but is not aligned with the NIST 800-53 definition of "Remote", which refers to "outside the system". The term "Remote" here better aligns with the NIST 800-53 definition of "Network" (remote from within the system) Access.

1.4.11.3 Types of User Interface (by capability)

User interfaces are also categorized by their capabilities as being Read Only, Limited, or Full.

1.4.11.3.1 Read-Only User Interface

A Read Only User Interface (also referred to as a View-Only User Interface) is a user interface that only allows for reading data, it does not allow (have the capability to) modify data. A Read Only User Interface may be either Local or Remote. A User Interface that is configured to be Read Only (by some other means than the interface itself, such as using configuration software on a laptop) is a Read-Only Interface. Note a Read Only User Interface may have buttons (or touch screen, etc.) allowing the user to navigate through the presentation of data.

Examples of a Read Only User Interfaces are a) a publicly viewable "energy dashboard" showing weather data and energy usage within a building and b) digital wayfinding signage.

1.4.11.3.2 Limited User Interface

A Limited User Interface is a user interface that - by design - can only alter information local to the user interface. Note that the determination of "alter" includes only direct interactions, it explicitly excludes interactions that might occur as secondary effects. For example, an interface changing the flow setpoint in a pump controller is a direct interaction, the subsequent change in flow (as well as any subsequent downstream changes in valve position) are not direct interactions.

Two examples of LIMITED UIs are: a) a variable speed drive has a Limited Local User Interface which allows the user to change properties within the drive, but does not allow affecting things outside the drive; and b) a typical home WiFi Router has a Limited Remote User Interface which allows configuration of the Router, but does not allow direct interaction with other devices.

1.4.11.3.3 Full User Interface

A Full User Interface can alter information in devices outside the device with the user interface. For example, a typical Local Display Panel is a Full Local User Interface while a browser-based front end is a Full Remote User Interface.

1.4.11.3.4 View-Only User Interface

See Read-Only User Interface

1.4.11.4 Other User Interface Terminology

In addition to defining whether a user interface is a Hardware Limited, Read-Only, Limited or Full, and whether it is Local or Remote, user interfaces are classified by whether they are writable or privileged.

1.4.11.4.1 Writable User Interface

Any User Interface that is not Read-Only is Writable. (Limited User Interfaces and Full User Interfaces are both writable user interfaces (as they are capable of changing a value)).

1.4.11.4.2 Privileged User Interface

A Privileged UI is a UI that has sufficient capabilities or functionality that it requires specific cybersecurity measures to be put in place to limit its unauthorized use. Ultimately, whether a specific user interface is considered a Privileged User Interface must be determined by usage. Unless otherwise specified, user interfaces can be determined to be privileged or not using the following:

a. Read-Only User Interfaces are not privileged user interfaces.

b. User Interfaces that can inhibit or force the activation of a fire suppression system (e.g. such as for a pre-action or deluge system) are always privileged user interfaces. Other Full User interfaces for Fire Alarm Systems are privileged user interfaces as indicated and

shown, or when another requirement of this Section establishes they are privileged. For all other systems, Full User Interfaces are privileged user interfaces.

c. User interfaces that allow for configuration of auditing or allows for modification or deletion of audit logs are privileged user interface.

d. User interfaces that allow for reprogramming a network connected device is a privileged user interface.

e. Except as specified above, a Limited User Interface must be determined to be privileged or not based on the specific capabilities and use case of the user interface. In general however, user interfaces that do not offer significant capabilities above and beyond those available at that location via other means (e.g. such as a disconnect switch, breaker, or hand-off-auto switch, or physical attack) are not privileged.

1.4.12 Wireless Network

Any network that communicates without using wires or fiber optics as the communication media. Wireless networks include: WiFi, Bluetooth, ZigBee, cellular, satellite, long and short wave radio, 2.4 GHz, free space optical, point-to-point laser, microwave, and IR.

1.4.13 Wired Broadcast Network

Wired Broadcast Networks are any network, such as powerline carrier networks and modem (wired telephony), that use wire-based technologies where there is not a clearly defined boundary for signal propagation.

1.5 ADMINISTRATIVE REQUIREMENTS

1.5.1 Points of Contact

Coordinate with the following Points of Contact as indicated in this Section and as required. Not all projects will require coordination with all Points of Contact. When coordination is required and no Point of Contact is indicated, coordinate with the Contracting Officer (KO).

a. Government Computer Access Point of Contact: Contracting Office (KO)

b. HTTPS Certificate Point of Contact: Contracting Officer (KO)

c. Email Address Point of Contact: Contracting Officer (KO)

d. Password Point of Contact: Contracting Officer (KO)

1.5.2 Coordination

Coordinate the execution of this Section with the execution of all other Sections related to control systems as indicated in the paragraph RELATED REQUIREMENTS. Items that must be considered when coordinating project efforts include but are not limited to:

a. If requesting permission for wireless or wired broadcast communication, the Wireless and Wired Broadcast Communication Request

submittal must be approved prior to control system device selection and installation.

b. If requesting permission for alternate account lock permissions, the Device Account Lock Exception Request must be approved prior to control system device selection and installation.

c. If requesting permission for the use of a device with multiple physical connections to IP networks, the Multiple IP Connection Device Request must be approved prior to control system device selection and installation.

d. Wireless testing may be required as part of the control system testing. See requirements for the Wireless Communication Test Report submittal.

e. If the Device Audit Record Upload Software is to be installed on a computer not being provided as part of the control system, coordination is required to identify the computer on which to install the software.

f. The Cybersecurity Interconnection Schedule must be coordinated with other work that will be interconnected to, and interconnections must be approved by the Government before relying on them for system functionality.

g. Cybersecurity testing support must be coordinated across control systems and with the Government cybersecurity testing schedule.

h. Passwords must be coordinated with the indicated contact for the project site.

i. If applicable, HTTPS web server certificates must be obtained from the indicated HTTPS Certificate Point of Contact.

j. Contractor Computer Cybersecurity Compliance Statements must be provided for each contractor using contractor owned computers.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Device Account Lock Exception Request; G

Multiple Ethernet Connection Device Request; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

Cybersecurity Interconnection Schedule ; G

Proposed STIG and SRG Applicability Report; G

Pre-Construction Control System Inventory Report ; G

SD-02 Shop Drawings

Network Communication, Ports, Protocols and Services Report; G

Cybersecurity Network (Riser) Diagram ; G

SD-03 Product Data

Control System Cybersecurity Documentation; G

SD-06 Test Reports

Control System Cybersecurity Testing Procedures; G

Control System Cybersecurity Testing Report; G

SD-11 Closeout Submittals

Password Change Summary Report; G

Enclosure Keys; G

Software and Configuration Backups; G

System Maintenance Tool Software; G

Control System Scanning Tools; G

STIG, SRG and Vendor Guide Compliance Result Report; G

1.7 QUALITY CONTROL

1.7.1 Regulatory Requirements

No regulatory requirements are enforced for the Fire and Life Safety Systems.

1.8 CYBERSECURITY DOCUMENTATION

{For Government Reference Only: This subpart (and its subparts) relates to PL-7; CCI-003071}

1.8.1 Proposed STIG and SRG Applicability Report

For each model of network connected or network infrastructure device, use the DISA SRG/STIG Applicability Guide and Collection Tool (available at https://public.cyber.mil/stigs/SCAP/to identify applicable STIGs or SRGs and provide a report indicating applicable STIGs and SRGs for each model. Provide the Proposed STIG and SRG Applicability Report concurrently with the Pre-Construction Control System Inventory Report.

1.8.2 Cybersecurity Interconnection Schedule

{For Government Reference Only: This subpart relates to CA-3(b), PL-8, SC-7(9), SC-7(11); CCI-000258, CCI-003072, CCI-003073, CCI-003075, CCI-002398, CCI-002399, CCI-002401, CCI-002402, CCI-002403.}

Provide a completed Cybersecurity Interconnection Schedule documenting network connections between the installed system and other systems. Provide the following information for each device directly communicating between systems: Device Identifier, Device Manufacturer, Device Description, Transport layer Protocol, Network Address, Port (if applicable), MAC (Layer 2) address (if applicable), Media, Application Protocol, Service (if applicable), Descriptive Purpose of communication.
If other control system Sections used on this project include submittals documenting this information, provide copies of those submittals to meet this requirement.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Cybersecurity Interconnection Schedule as an editable Microsoft Excel file (a template Cybersecurity Interconnection Schedule in Excel format is available athttps://www.wbdg.org/dod/ufgs/ufgs-25-05-11.)

1.8.3 Network Communication, Ports, Protocols and Services Report

{For Government Reference Only: This subpart (and its subparts) relates to CA-9, PL-8; CCI-002102, CCI-002103, CCI-002104, CCI-002105, CCI-003072, CCI-003073, CCI-003075 and also the submittal requirements associated with CM-6, CM-7, including CM-7(3), CCI-000388.}

Provide a Network Communication, Ports, Protocols and Services Report. For each networked device, document device identifier and the communication characteristics of the device including communication protocols, services used, encryption employed, and a general description of what information is communicated over the network. For each device using IP communication, document all TCP and UDP ports used. For each device using non-IP communication, document communication protocol and media used. If other control system Sections used on this project include submittals documenting this information, provide copies of those submittals to meet this requirement.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Network Communication, Ports, Protocols and Services Report as an editable Microsoft Excel file.

1.8.4 Control System Inventory Reports

{For Government Reference Only: This subpart (and its subparts) relates to CM-8(a), SI-17, IA-3; CCI-000389, CCI-000392, CCI-000398, CCI-002773, CCI-002774, CCI-002775, CCI-000777, CCI-000778, CCI-001958}

Provide a Pre-Construction Control System Inventory Report report and a Final Control System Inventory Report, using the Inventory Spreadsheet listed under this Section at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. In the Control System Inventory Reports, document all networked devices, including network infrastructure devices. For each device provide all applicable information for which there is a field on the spreadsheet in accordance with the instructions on the spreadsheet.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Control System Inventory Reports as editable Microsoft Excel files. Provide the Pre-Construction Control System Inventory Report concurrently with the Proposed STIG and SRG Applicability Report

1.8.5 Software and Configuration Backups

{For Government Reference Only: This subpart (and its subparts) relates to CP-10; CCI-000550, CCI-000551, CCI-000552}

For each computer on which software is installed under this project, provide a recovery image of the final as-built computer . This image must allow for bare-metal restore such that restoration of the image is sufficient to restore system operation to the imaged state without the need for re-installation of software. If additional user permissions are required to meet this requirement, coordinate the creation of the image with the identified Government Computer Access Point of Contact.

For all Ethernet switches provide a backup of the switch configuration. For all controllers, provide a backup of the controller configuration and the source code for all loaded application programs (all software that is not common to every controller of the same manufacturer and model).

If any or all of these are provided under another Section, provide documentation indicating this and referencing those submittals.

1.8.6 Cybersecurity Network (Riser) Diagram

{For Government Reference Only: This subpart (and its subparts) relates to PL-2(a), PL-8; CCI-003051, CCI-003053, CCI-003072, CCI-003073, CCI-003075}

Provide a cybersecurity network (riser) diagram of the complete control system including all network and device hardware. For each device, include the device identifier, device type, and manufacturer. If the control system specifications require a riser diagram submittal, provide a copy of that submittal as the cybersecurity riser diagram. Otherwise, provide a riser diagram in one-line format.

1.8.7 STIG, SRG and Vendor Guide Compliance Result Report

For every component (device or software) with an applicable STIG or SRG in the Proposed STIG and SRG Applicability Report, document compliance with

the STIG or SRG requirements.

a. For components which are scannable by the SCAP (security content automation protocol) tool (available online at https://public.cyber.mil/stigs/scap), include the SCAP report and raw scan results in addition to the final, manually reviewed and revised, documentation of compliance with STIG and SRG requirements. Checklist files should not contain any findings with a Not Reviewed (NR) status after manual reviews.

b. For components which do not support automated scanning, a manual review using the General Purpose STIG option should be done. A completed Checklist file should not contain any findings with a Not Reviewed (NR) status after manual reviews

For every component (device or software) with manufacturer provided cybersecurity documentation, procedure, or method for secure configuration or installation, provide a report documenting how the component was configured and any deviation from the manufacturer instructions.

1.8.8 Control System Cybersecurity Documentation

{For Government Reference Only: This subpart (and its subparts) relates to SA-5(a),SA-5(b),SA-5(c), SA-22(b); CCIs: CCI-003124, CCI-003125, CCI-003126, CCI-003127, CCI-003128, CCI-003129, CCI-003130, CCI-003131, CCI-003374}

Provide a Control System Cybersecurity Documentation submittal containing the indicated information for each device and software application.

1.8.8.1 Software Applications

For all software applications running on computers provide:

a. administrator documentation that describes secure configuration of the software {For Government Reference Only: relates to CCI-003124}

b. administrator documentation that describes secure installation of the software and software updates. {For Government Reference Only: relates to CCI-003125}

c. administrator documentation that describes secure operation of the software {For Government Reference Only: relates to CCI-003124}

d. administrator documentation that describes effective use and maintenance of security functions or mechanisms for the software {For Government Reference Only: relates to CCI-003127}

e. administrator documentation that describes known vulnerabilities regarding configuration and use of administrative (i.e. privileged) functions for the software {For Government Reference Only: relates to CCI-003128}

f. user documentation that describes user-accessible security functions or mechanisms in the software and how to effectively use those security functions or mechanisms {For Government Reference Only: relates to CCI-003129}

g. user documentation that describes methods for user interaction which

enables individuals to use the software in a more secure manner {For Government Reference Only: relates to CCI-003130}

h. user documentation that describes user responsibilities in maintaining the security of the software {For Government Reference Only: relates to CCI-003131}

1.8.8.2 Default Requirements for Control System Devices

For control system devices where Control System Cybersecurity Documentation requirements are not otherwise indicated in this Section, provide:

a. Documentation that describes secure configuration of the device {For Government Reference Only: relates to CCI-003124}

b. Documentation that describes secure installation of the device {For Government Reference Only: relates to CCI-003125}

c. Documentation that describes secure operation of the device {For Government Reference Only: relates to CCI-003124}

d. Documentation that describes effective use and maintenance of security functions or mechanisms for the device {For Government Reference Only: relates to CCI-003127}

e. Documentation that describes known vulnerabilities regarding configuration and use of administrative (i.e. privileged) functions for the device {For Government Reference Only: relates to CCI-003128}

f. Documentation that describes user-accessible security functions or mechanisms in the device and how to effectively use those security functions or mechanisms {For Government Reference Only: relates to CCI-003129}

g. Documentation that describes methods for user interaction which enables individuals to use the device in a more secure manner {For Government Reference Only: relates to CCI-003130}

h. Documentation that describes user responsibilities in maintaining the security of the device {For Government Reference Only: relates to CCI-003131}

i. Documentation of the published last date of support by the manufacturer or indication that a published date is not available. {For Government Reference Only: relates to CCI-003374}

1.9 CYBERSECURITY DURING CONSTRUCTION

{For Government Reference Only: This subpart (and its subparts) relates to AC-18, CA-3; CCI-000258}

In addition to the control system cybersecurity requirements indicated in this section, meet following requirement throughout the construction process.

1.9.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. Contractor

computers connected to the control system, control system network, or a control system component at any point during construction must meet the following requirements:

1.9.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.9.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. Computers used on this project must perform a full antivirus scan at least once per day. Computers which have connected to any other network since the last full antivirus scan must perform a full antivirus scan prior to connection to the control system network or to the temporary contractor-installed IP network. Perform an antivirus scan on all removable media (e.g., external hard drives, CDs, DVDs, USB flash drives) prior to connecting to the control system environment.

1.9.1.3 Passwords and Passphrases

The passwords and passphrases for computers, applications, and web-based applications supporting passwords must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.9.1.4 User-Based Authentication

Each user must have a unique account; sharing of a single account between multiple users is prohibited.

1.9.1.5 Demonstration of Compliance

The Government has the right to require demonstration of computer compliance with these requirements at any time during the project.

1.9.1.6 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. Each Statement must be signed by a cybersecurity representative for the relevant company.

1.9.2 Government Access to Network

Government personnel must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification.

1.9.3 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.9.4 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

1.10 CYBERSECURITY DURING WARRANTY PERIOD

All work performed on the control system after acceptance must be performed using Government Furnished Equipment or equipment specifically and individually approved by the Government.

PART 2 PRODUCTS

All products used on this project must meet the indicated requirements, but not all products specified here will be required by every project.

2.1 ETHERNET SWITCH

Provide Open Systems Interconnection (OSI) Layer 2 Ethernet switches with the following capabilities, and with an interface to support switch configuration for these capabilities:

2.1.1 Required Functionality

Switches must:

a. Copper Ethernet ports must auto negotiate for 10, 100 and 1000 megabits-per-second links.

b. Be capable of implementing port level access control by MAC address and limit the number of MAC addresses to one MAC address per port.

c. For LOW Impact Systems, be capable of implementing per-port access control lists (ACLs) where the list can be filtered by source and destination IP addresses, and by source and destination UDP or TCP ports.

d. Support Remote Network Monitoring (RMON) Port Analysis in accordance with IETF RFC 2819

e. Configure target port and analysis port such that switch clones all target port traffic to analysis port.

2.1.2 Configuration Requirements

Switches must:

a. Support configuration save and restore.

b. Support both manual IP address assignment and acquisition of a dynamic IP address via Dynamic Host Configuration Protocol (DHCP).

c. Be capable of limiting access for configuration to one or more of: a web interface using HTTPS, a command line interface using SSH, or an SNMP connection using SNMP version 3 or later.

2.2 DAISY CHAIN IP CONTROLLERS

Controllers used as Daisy Chain IP Controllers must be IP controllers with exactly two Ethernet network connections and basic built-in switch capabilities to allow implementation of an Ethernet network in a daisy chain architecture. Switches incorporated by Daisy Chain IP Controllers are not required to meet the requirements for Ethernet Switches as defined in this Section.

PART 3 EXECUTION

3.1 CYBERSECURITY HARDENING AND CONFIGURATION GUIDES

Install, configure, and harden all hardware and software furnished on this project in accordance with manufacturer provided documentation, procedures, or methods for secure configuration or installation. Do not implement specific hardening actions if that action would conflict with required functionality or another requirement of this Section.

3.2 NETWORK REQUIREMENTS

3.2.1 Wireless and Wired Broadcast Communication for Fire Protection Systems

The use of wireless and wired broadcast communication for fire protection systems within a facility is prohibited. Wireless communication may be used to provide communication from the fire protection system in a facility to the central monitoring station.

3.2.2 Non-IP Control Networks

When control system specifications require particular communication protocols, use only those communication protocols and only as specified. Do not implement any other communication protocol.

When control system specifications do not indicate requirements for communication protocols, use only those protocols required for operation of the system as specified.

3.2.3 IP Control Networks

{For Government Reference Only: This subpart relates to CM-6(a), CM-7(a), CM-7(b), CM-7(1)(b), SC-41; CCI-001588, CCI-000381, CCI-000380, CCI-000381, CCI-000382, CCI-001761, CCI-001762, CCI-002544, CCI-002545, CCI-002546.}

IP Networks must be Ethernet networks and must use switches which are Ethernet Switches or Daisy Chain IP Controllers as defined in this Section. Do not use nonsecure functions, ports, protocols and services as defined in DODI 8551.01 unless those ports, protocols and services are specifically required by the control system specifications or otherwise specifically authorized by the Government. Do not use ports, protocols and services that are not specified in the control system specifications or required for operation of the control system.

3.2.3.1 IP Devices With Multiple Ethernet Connections

Except for Ethernet Switches and Daisy Chain IP Controllers, devices must not have more than one Ethernet connection to IP networks unless doing so is required by the project specifications and the specific application is approved. If a device with Multiple Ethernet Connections to IP networks is required, provide a Multiple Ethernet Connection Device Request using the Multiple Ethernet Connection Device Request Template at https://www.wbdg.org/dod/ufgs/ufgs-25-05-11 to request approval for each device. If a device with Multiple Ethernet Connections to IP networks is not required, instead provide a document stating that no approval is being requested.

3.2.4 Cryptographic Protection

{For Government Reference Only: This subpart relates to IA-2(9), IA-3(1), SC-8, SC-13, SC-23(1), SC-23(3); CCI-001942, CCI-001959, CCI-001967, CCI-002418, CCI-002449, CCI-002450, CCI-001185, CCI-001188, CCI-001664.}

All remote user interfaces must use HTTPS for all traffic between the user interface client and user interface server.

3.2.5 Device Identification and Authentication

{For Government Reference Only: This subpart (and its subparts) relates to IA-3; CCI-000777, CCI-000778, CCI-001958.}

All computers must support IEEE 802.1x for device authentication to the network.

3.2.5.1 Default Requirements for Control System Devices

For control system devices where Device Identification and Authentication requirements are not otherwise indicated in this Section: Devices using HTTP as a control protocol must use HTTPS instead.

3.2.6 Cryptographic Module Authentication

{For Government Reference Only: This subpart (and its subparts) relates to IA-7; CCI-000803}

For devices (including but not limited to NIST FIPS 140-2 compliant radios) that have STIG/SRGs related to cryptographic module authentication (CCI-000803), comply with the requirements of those STIG/SRGs.

3.3 ACCESS CONTROL REQUIREMENTS

3.3.1 User Accounts

{For Government Reference Only: This subpart (and its subparts) relate to AC-2(a), AC-3, AC-6(1), AC-6(10), AC-6(2), AC-6(9), CM-11(2), and IA-2; CCI-002110, CCI-000213, CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-002235, CCI-000039, CCI-001419, CCI-002234, CCI-001812, and CCI-000764. }

Any user interface supporting user accounts (either FULLY or MINIMALLY) must limit access according to specified limitations for each account. Install and configure any device having a STIG or SRG in accordance with that STIG or SRG.

All user interfaces FULLY supporting accounts must implement user-based authentication where each account is uniquely assigned to a specific user. User interfaces FULLY supporting accounts must implement at least three (3) levels of user account privilege including: 1) an account with read-only permissions 2) an account with full permissions including account creation and modification and 3) an account with greater permissions than read-only but without account creation and modification. Disable any unnecessary or unused accounts. Disable any "guest-level" accounts that are created on the system by default.

3.3.1.1 Computers

All computer operating systems must FULLY support user accounts and implement accounts for access. Each control system software application not supporting accounts and running on a computer must be installed such that use of the software is restricted by the computer operating system to specific users.

Applications running on computers must not require the user be logged in to a computer operating system administrator account for normal operation. It is permissible to require the computer operating system administrator account for initial application installation and configuration.

3.3.1.2 Controllers

For user interfaces provided by controllers, provide access control in accordance with the User Interface Requirements table for the applicable control system and user interface type.

a. For table entries of "NA": NA means Not Applicable, there are no interfaces in this category.

b. For table entries of "None Required": The user interface is not required to support user accounts.

c. For table entries of "MINIMALLY": The user interface must at least MINIMALLY support user accounts.

d. For table entries of "FULLY": The user interface must at FULLY support user accounts.

e. For table entries of "KEY": The user interface must have physical security in the form of either a key lock on the interface itself or

be furnished inside a lockable enclosure. Where this is required for a read only interface, the lockable enclosure must prevent viewing of data on the interface; for other interfaces, this lockable enclosure must prevent using the interface to alter data.

f. For table entries of "Physical Security": For Local FULL interfaces, the interface must be located inside mission space. For Local Limited (not FULL) interfaces, the user interface must either a) be located within mission space or b) be protected by physical security at least as good as the control devices (and equipment controlled by the control devices) affected by the interface. For purposes of this requirement, 'affected' includes controllers with data that can be directly altered by the interface, as well as mechanical and/or electrical equipment directly controlled by those controllers, but does not include other interactions.

g. Entries of the form "X and Y" must meet both the requirement indicated for X and the requirement indicated for Y. For example, an entry of "MINIMALLY and Physical Security" indicates the user interface must both MINIMALLY support accounts and have physical security.

h. Entries of the form "X or Y" must meet either the requirement indicated for X or the requirement indicated for Y.

3.3.1.2.1 Fire Protection Systems

User Interface Requirements for LOW Impact Fire Protection Systems	
User Interface Type	**Access Control Requirement**
Local Read Only (see note 1)	None Required
Local Limited, Non-privileged	None Required
Local Limited, Privileged	KEY and Physical Security
Local Full	KEY
Remote Read Only	None Required
Remote Limited, Non-Privileged	MINIMALLY
Remote Limited, Privileged AND Remote Full	FULLY
Notes: 1)Local Read Only User Interfaces are always Non-Privileged	

3.3.1.2.2 Default Requirements for Other Control Systems

For control system devices where User Interface Requirements are not otherwise indicated in this Section, use the Default User Interface Requirements tables.

Default User Interface Requirements for LOW Impact Control Systems	
User Interface Type	**Access Control Requirement**
Local Read Only (see note 1)	None Required
Local Limited, Non-privileged	None Required
Local Limited, Privileged	MINIMALLY
Local Full	MINIMALLY
Remote Read Only	None Required
Remote Limited, Non-Privileged	MINIMALLY
Remote Limited, Privileged AND Remote Full (see note 2)	FULLY
Notes: 1)Local Read Only User Interfaces are always Non-Privileged 2)Remote Full User Interfaces are always Privileged	

3.3.2 Unsuccessful Logon Attempts

{For Government Reference Only: This subpart (and its subparts) relate to AC-7 (a), AC-7 (b); CCI-000043, CCI-000044, CCI-001423, CCI-002236, CCI-002237, CCI-002238}

Except for high availability user interfaces indicated as exempt, devices must meet the indicated requirements for handling unsuccessful logon attempts. If a device cannot meet these requirements, document device capabilities to protect from subsequent logon attempts and propose alternate protections in a Device Account Lock Exception Request submittal. Do not implement alternate protection measures in lieu of the indicated requirements without explicit permission from the Government. If no Device Account Lock Exceptions are requested, provide a document stating that no approval is being requested as the Device Account Lock Exception Request.

3.3.2.1 Devices MINIMALLY Supporting Accounts

For LOW Impact Systems: Devices which MINIMALLY (but not FULLY) support accounts are not required to lock based on unsuccessful logon attempts.

3.3.2.2 Devices FULLY Supporting Accounts

Devices which FULLY support accounts must meet the following requirements.

a. It must lock the user account when three unsuccessful logon attempts occur within a 15 minute interval.

b. Once an account is locked, the account must stay locked until unlocked by an administrator. If the account being locked is the sole

administrator account on the device, the account must stay locked for 1 hour and then automatically unlock.

c. Once the indicated number of unsuccessful logon attempts occurs, delay further logon prompts by 5 seconds.

3.3.2.3 High Availability Interfaces Exempt from Unsuccessful Logon Attempts Requirements

There are no high availability interfaces which are exempt from unsuccessful logon attempts requirements.

3.3.3 System Use Notification

{For Government Reference Only: This subpart (and its subparts) relates to AC-8; CCI-000048, CCI-002247, CCI-002243, CCI-002244, CCI-002245, CCI-002246, CCI-000050, CCI-002248}

3.3.3.1 System Use Notification for Remote User Interfaces

Remote user interfaces must display a warning banner meeting the requirements of DTM 08-060 on screen.

3.3.3.2 System Use Notification for Local User Interfaces

Devices which are connected to a network and have a local user interface must display a warning banner meeting the requirements of DTM 08-060 on the user interface screen if capable of doing so and must have a permanently affixed label with an approved banner from DTM 08-060 if unable to display the warning banner on the screen. Where it is impractical (perhaps due to device size) to affix the label to the device, affix the label to the device enclosure.

Labels must be machine printed or engraved, plastic or metal, designed for permanent installation, must use a font no smaller than 14 point, and must provide a high contrast between font and background colors.

3.3.4 Permitted Actions Without Identification or Authentication

{For Government Reference Only: This subpart (and its subparts) relates to AC-14; CCI-000061, CCI-000232}

The control system must require identification and authentication before allowing any actions except read-only actions by a user acting from a user interface which MINIMALLY or FULLY supports accounts.

3.3.5 Enclosures

Prior to final acceptance of the system, lock all lockable enclosures. Submit an Enclosure Keys submittal with all copies of keys for all enclosures and a key inventory list documenting all keys. Label each key with the matching enclosure identifier.

3.4 USER IDENTIFICATION AND AUTHENTICATION

{For Government Reference Only: This subpart (and its subparts) relates to IA-2, IA-2(1),IA-2(12), IA-5 IA-5(b), IA-5(c), IA-5(e), IA-5(g), IA-5(1), IA-5(11); CCI-000764, CCI-000765, CCI-001953, CCI-001954, CCI-001544, CCI-001989, CCI-000182, CCI-001610, CCI-000192, CCI-000193,

CCI-000194, CCI-000205, CCI-001619, CCI-001611, CCI-001612, CCI-001613, CCI-001614, CCI-000195, CCI-001615, CCI-000196, CCI-000197, CCI-000199, CCI-000198, CCI-001616, CCI-001617, CCI-000200, CCI-001618, CCI-002041, CCI-002002, CCI-002003. }

This subpart indicates requirements for specific methods of identification and authentication for users and user accounts. Where these requirements conflict apply the following order of precedence: 1) If present, Device Specific Requirements take precedence over any other requirements; and then 2) multifactor authentication requirements take precedence over password requirements.

3.4.1 User Identification and Authentication for All System Types

Unless otherwise indicated, all user interfaces supporting accounts (either FULLY or MINIMALLY) must implement Identification and Authorization via passwords.

3.4.2 User Identification and Authentication for Specific System Types

System specific requirements are in addition to and supersede those indicated for all system types. When no additional requirements are indicated for a specific system type the requirements for all systems still apply to that system type.

3.4.2.1 Fire and Life Safety Control System Devices

No additional system specific requirements apply.

3.4.3 User Identification and Authentication for Specific Devices

There are no additional device specific user interface requirements

3.4.4 Implementation of Identification and Authorization Requirements

Identification and Authorization must be met by one of the following methods:

a. Direct implementation in the user interface.

b. For user interfaces on a computer: inheriting the Identification and Authorization from the computer operating system, either by the operating system limiting access to specific applications by user, or by the application itself having permissions based on the user logged into the computer.

c. For remote interfaces: an implementation shared between the remote user interface server and the remote user interface client. For example, a requirement for PIV authentication may be met on a remote user interface by a PIV reader on a web browser client which sends the authentication information via HTTPS to the remote server.

3.4.5 Password-Based Authentication Requirements

3.4.5.1 Passwords for Software and Applications Running on Computers

All software and applications running on computers supporting password-based authentication must enforce the following requirements:

a. Minimum password length of 12 characters

b. Password must contain at least one uppercase character.

c. Password must contain at least one lowercase character.

d. Password must contain at least one numeric character.

e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.

f. Password must have a minimum lifetime of 24 hours.

g. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.

h. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters (where location is significant, a character may be reused if it is in a different position).

i. Passwords must be cryptographically protected during storage and transmission.

3.4.5.2 Passwords for Controllers FULLY Supporting Accounts

All controllers FULLY supporting accounts and supporting password-based authentication must enforce the following requirements:

a. Minimum password length of 12 characters

b. Password must contain at least one uppercase character.

c. Password must contain at least one lowercase character.

d. Password must contain at least one numeric character.

e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.

f. Password must have a maximum lifetime of sixty (60) days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.

g. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters.

h. Passwords must be cryptographically protected during storage and transmission.

3.4.5.3 Passwords for Remote Interfaces

Passwords for connecting to a Remote User Interface supporting password-based authentication must enforce the following requirements:

a. Minimum password length of 12 characters

b. Password must contain at least one uppercase character.

c. Password must contain at least one lowercase character.

d. Password must contain at least one numeric character.

e. Password must contain at least one special character. The list of supported special characters must include at least 4 separate characters.

f. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.

g. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters (where location is significant, a character may be reused if it is in a different position).

h. Passwords must be cryptographically protected during storage and transmission.

3.4.5.4 Passwords for Devices Minimally Supporting Accounts

Devices MINIMALLY supporting accounts must support passwords with a minimum length of four characters.

3.4.5.5 Password Configuration and Reporting

For all devices with a password, coordinate the changing of passwords with the project site following testing of the system but prior to turnover to the Government. Coordinate with Password Point of Contact to determine appropriate project site personnel to complete password changes. Accompany identified personnel to each device with a password and instruct personnel on the process of changing password. Record the time, date and personnel present when each device's password is changed and submit a Password Change Summary Report documenting this information.

Provide the Password Summary Report electronically in both PDF and Microsoft Excel.

3.4.6 Authenticator Feedback

{For Government Reference Only: This subpart relates to IA-6; CCI-000206}

Devices must never show authentication information, including passwords, on a display. Devices that momentarily display a character as it is entered, and then obscure the character, are acceptable. For devices that have STIGs or SRGs related to obscuring of authenticator feedback (CCI-000206), comply with the requirements of those STIGS/SRGs.

3.5 CYBERSECURITY AUDITING

Where an auditing requirement exists for email notification, notify via email the application administrator and Information System Security Officer (ISSO) of the event. Coordinate with the Email Address Point of Contact for email addresses. If outgoing email is not available to the system, disable email notifications.

3.5.1 Audit Events, Content of Audit Records, and Audit Generation

{For Government Reference Only: This subpart (and its subparts) relates to AU-2(a), AU-2(c), AU-2(d), AU-3, AU-10, AU-12, AU-14(b), AU-14(1), AU-14(2), AU-14(3), CM-5(1), SC-7 (9); CCI-000123, CCI-001571, CCI-000125, CCI-001485, CCI-000130, CCI-000131, CCI-000132, CCI-001230, CCI-000133, CCI-000134, CCI-001487, CCI-000166, CCI-001899, CCI-000169, CCI-001459, CCI-000171, CCI-000172, CCI-001910, CCI-001914, CCI-001919, CCI-001464, CCI-001462, CCI-001920, CCI-001814, CCI-002400. }

For devices that have STIG/SRGs related to audit events, content of audit records or audit generation, comply with the requirements of those STIG/SRGs.

If auditing requirements can be met using existing control system alarm or event capabilities, those existing capabilities may be used to meet these requirements.

3.5.1.1 Computers

For each computer, provide the capability to select audited events and the content of audit logs. Configure computers to audit the indicated events, and to record the indicated information for each auditable event. Send logs to a syslog server that is only accessible via accounts on the system with privileged level access.

3.5.1.1.1 Audited Events

Configure each computer to audit the following events:

a. Successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels)

b. Successful and unsuccessful logon attempts

c. Successful logouts

d. Privileged activities or other system level access

e. Concurrent logons from different workstations

f. Successful and unsuccessful accesses to objects

g. Program initiations

h. Direct access to the information system

i. Account creations, modifications, disabling, and terminations.

j. Kernel module load, unload, and restart

3.5.1.1.2 Audit Event Information To Record

Configure each computer to record, for each auditable event, the following information (where applicable to the event):

a. What type of event occurred

b. When the event occurred

c. Where the event occurred

d. The source of the event

e. The outcome of the event

f. The identity of any individuals or subjects associated with the event

3.5.1.2 Default Requirements for Control System Controllers

For control system controllers where Audit Events, Content of Audit Records, and Audit Generation are not otherwise indicated in this Section:

3.5.1.2.1 Controllers Which FULLY Support Accounts

For each controller which FULLY supports accounts, provide the capability to select audited events and the content of audit logs. Configure controllers to audit the indicated events, and to record the indicated information for each auditable event.

3.5.1.2.1.1 Audited Events

Configure each controller to audit the following events:

a. Successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels)

b. Successful and unsuccessful logon attempts

c. Successful logouts

d. Concurrent logons from different workstations

e. All account creations, modifications, disabling, and terminations.

f. All kernel module load, unload, and restart

3.5.1.2.1.2 Audit Event Information To Record

Configure each controller to record, for each auditable event, the following information (where applicable to the event):

a. what type of event occurred

b. when the event occurred

c. where the event occurred

d. the source of the event

e. the outcome of the event

f. the identity of any individuals or subjects associated with the event

3.5.1.2.2 Controllers Which Do Not FULLY Support Accounts

For each controller which does not FULLY support accounts configure the controller to audit all controller shutdown and startup events and to record for each event the type of event and when the event occurred.

3.5.2 Audit Time Stamps

{For Government Reference Only: This subpart (and its subparts) relates to AU-8; CCI-000159, CCI-001889, CCI-001890.}

Any device (computer or controller) generating audit records must have an internal clock capable of providing time with a resolution of one second. Clocks must not drift more than 10 seconds per day. Configure the system so that each device (computer or controller) generating audit records maintains accurate time to within 1 second. Note that if the control system specifications include requirement for clocks, the most stringent requirement applies.

3.5.3 Response to Audit Processing Failures

{For Government Reference Only: This subpart (and its subparts) relates to AU-5; CCI-000139, CCI-000140, CCI-001490.}

In the case of a failure in the auditing system, computers associated with auditing must provide email notification. In case of an audit failure, if possible, continue to collect audit records by overwriting existing audit records.

3.6 REQUIREMENTS FOR LEAST FUNCTIONALITY

{For Government Reference Only: This subpart (and its subparts), along with the Network Communication, Ports, Protocols and Services Report submittal specified elsewhere in this section, relates to CM-6(a), CM-6(c), CM-7, CM-7(1)(b), SC-41; CCI-000363, CCI-000364, CCI-000365, CCI-001588, CCI-001755, CCI-000381, CCI-000380, CCI-000382, CCI-001761, CCI-001762, CCI-002544, CCI-002545, CCI-002546. }

For devices that have a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port and device I/O access for least functionality), install and configure the device in accordance with that STIG or SRGs.

3.6.1 Device Capabilities

Do not provide devices with remote user interfaces or with full user interfaces where one was not required.

3.6.2 Software

For software that has a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port access for least functionality), install and configure the software in accordance with that STIG or SRG.

Do not install software that is not specifically required to meet a contract requirement. Remove any previously installed that is not specifically required to meet a contract requirement. Do not implement functionality within software
that is not specifically required to meet contract requirements.

3.7 SYSTEM AND COMMUNICATION PROTECTION

3.7.1 Collaborative Computing

{For Government Reference Only: This subpart relates to SC-15(a), SC-15(b); CCI-001150, CCI-001152.}

Without explicit approval from the project site, control systems must not use collaborative computing technologies.

3.7.2 Denial of Service Protection

{For Government Reference Only: This subpart relates to SC-5, SC-12, SC-7(a); CCI-001093, CCI-002385, CCI-002386, CCI-002430, CCI-001097. }

To the greatest extent practical, implement control logic without reliance on the network. Except when required to meet the requirements of the control system Section (where the requirement can only be met using computer hardware), do not implement control logic in computers.

3.7.3 Process Isolation and Boundary Protection in Moderate Impact Fire Protection Systems

{For Government Reference Only: This subpart relates to SC-7(a), SC-7(c), SC-7(4)(a), SC-7(4)(c), SC-7(5), SC-7(7), SC-7(9)(a), SC-7(11), SC-7(13), SC-7(13), SC-7(18); CCI-001097, CCI-001098, CCI-001102, CCI-002396, CCI-001109, CCI-002397, CCI-002398, CCI-002399, CCI-002403, CCI-001120, CCI-001119, CCI-001126}

3.7.3.1 Radio Interfaces for Fire Protection Systems

When radios interfacing a local fire protection system to a supervisory system are not NIST FIPS 140-2 validated, use a relay panel interface between the local fire protection system and the radio. Install and configure the relay panel to prohibit initiating any action within the local fire protection system other than causing the system to play a pre-recorded message or causing the system to play a live audio message.

3.7.3.2 Fire Suppression System Network Isolation

For fire suppression systems including a release panel, any network used in these systems must be dedicated to these systems and must be isolated from any other network, including other components of the Fire Alarm and Fire Suppression systems. Use only dry contacts and relays to transfer signals from these systems to any other systems.

3.8 SAFE MODE AND FAIL SAFE OPERATION

{For Government Reference Only: This subpart (and its subparts) relates to CP-12, SI-10(3), SI-17; CCI-002855, CCI-002856, CCI-002857, CCI-002754, CCI-002773, CCI-002774, CCI-002775}

For all control system components with an applicable STIG or SRG, configure the component in accordance with all applicable STIGs and SRGs.

3.9 SYSTEM MAINTENANCE TOOL SOFTWARE

{For Government Reference Only: This subpart (and its subparts) relates to MA-3; CCI-000865.}

Submit and license to the Government all software required to operate, maintain and modify the control system such the Government or their agents are able to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer. Submit hard copies of user manuals for each software with the software submittal. Provide any hardware keys or dongles, software keys, license numbers, and other information required to enable the Government to access or change the system.

For software provided and licensed to the Government under the requirements of another Section, submit a statement indicating the Section and Submittal under which the software was provided. For software provided to meet the requirements of this Section and not provided and licensed under another Section, submit software and software user manuals on DVD or CD as a Technical Data Package and submit one hard copy of the software user manual for each piece of software.

3.10 VULNERABILITY SCANNING

{For Government Reference Only: This subpart (and its subparts) relates to RA-5 RA-5(a),RA-5(b),RA-5(c),RA-5(d); CCI-001054, CCI-001055, CCI-000156, CCI-001641, CCI-001643, CCI-001057, CCI-001058, CCI-001059. }

All IP devices must be scannable, such that the device can be scanned by industry standard IP network scanning utilities without harm to the device, application, or functionality.

3.10.1 Computers and Software Running on Computers

Computers and applications running on computers must meet relevant vulnerability scanning STIGs/SRGs and respond to approved DoD vulnerability scanning tools.

3.10.2 Controllers

Provide controllers that are scannable by standard control system discovery tools or control system browsers and return meaningful status information including the network inputs and outputs for the controller. This information must contain sufficient detail to detect vulnerabilities or exploits of the controller.

Provide all software needed to scan the control system as the Control System Scanning Tools submittal. If the software required to scan the system is already installed at the project site or is provided under a

separate section instead provide a statement indicating this.

3.11 SYSTEM AND INTEGRATION INTEGRITY

3.11.1 Malicious Code Protection

{For Government Reference Only: This subpart (and its subparts) relates to SI-3(c); CCI-001241, CCI-002623}

For all computers installed under this project, provide malware protection software media, provide licenses, and install and configure malware protection software as indicated. Coordinate with the Government Computer Access Point of Contact as required.

a. Malware protection software licenses will be Government furnished.

b. Malware protection software media will be Government furnished.

c. Malware protection software will be Government installed.

]3.12 CONTROL SYSTEM CYBERSECURITY TESTING

3.12.1 Control System Cybersecurity Testing Procedures

Prepare and submit Control System Cybersecurity Testing Procedures explaining step-by-step, the actions and expected results that will demonstrate that the control system meets the requirements of this Section. The Control System Cybersecurity Testing Procedures may be submitted as a Technical Data Package.

3.12.2 Control System Cybersecurity Testing Execution

Using the Control System Cybersecurity Testing Procedures verify that the control system meets the requirements of this Section. UNLESS GOVERNMENT WITNESSING OF A TEST IS SPECIFICALLY WAIVED BY THE GOVERNMENT, PERFORM ALL TESTS WITH A GOVERNMENT WITNESS. If testing reveals deficiencies in the system, correct the deficiency and retest until successful.

3.12.3 Control System Cybersecurity Testing Report

Prepare and submit a Control System Cybersecurity Testing Report documenting all tests performed and their results. Include all tests in the Control System Cybersecurity Testing Procedures and any additional tests performed during testing. Document test failures and repairs conducted with the test results. The Control System Cybersecurity Testing Report may be submitted as a Technical Data Package

3.13 FIELD QUALITY CONTROL, CYBERSECURITY VALIDATION SUPPORT

In addition to testing and testing support required by other Sections, provide a minimum of eight hours of technical support for cybersecurity testing of control systems to support the DoD Risk Management Framework process Cybersecurity assessment of the control system. This support is independent of (and in addition to) the Control System Cybersecurity Testing specified in this section.

3.14 CYBERSECURITY TRAINING

Provide eight hours of classroom training for six Government personnel on the cybersecurity operation and maintenance of the control system provided. This training is in addition to and must be coordinated with control system training specified in other Sections.

The Government will provide the training location. Training must cover, at a minimum: (a) applying software and firmware updates, (b) user account creation, modification and deletion, (c) audit log upload procedures and (d) identification of privileged user interfaces and system impact of those interfaces. Training session must include a question and answer period during which government staff questions about cybersecurity aspects of the control system are answered.

-- End of Section --

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ADVANCED UTILITY METERING SYSTEM

PART 1 GENERAL

1.1 DESCRIPTION

This Section includes the following for the advanced metering of the systems of the facility. The metered systems include the electrical power, natural gas distribution, fuel gas and fuel oil, steam, steam condensate, chilled water, heating water, domestic water, recovered water and makeup water systems. The metering systems in each facility are part of a Corporate-Wide utility metering system, rendering the VA accurate and automated metering of its facilities' energy and water flows. Metering systems are comprised of:

(1) PC-based workstation(s) or server(s) and software.

(2) Communication network and interface modules for RS-232, RS-485, Modbus TCP/IP, IEEE 802.3 data transmission protocols.

(3) Electric meters.

(4) Volumetric flowmeters, temperature sensors and pressure transducers.

(5) Mass flowmeters.

1.2 RELATED WORK

a. Not used.

b. Section 22 05 19 METERS AND GAGES FOR PLUMBING PIPING: meters and gages.

c. Section 22 33 00 ELECTRIC DOMESTIC WATER HEATERS: references meters.

d. Section 23 05 11, COMMON WORK RESULTS FOR HVAC AND STEAM GENERATION: General mechanical requirements, common to more than one section in mechanical.

e. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC: Flowmeters and communications

f. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: General electrical requirements and items that are common to more than one section of Division 26.

g. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

h. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduits.

i. Section 26 11 16, SECONDARY UNIT SUBSTATIONS: Unit secondary substation.

j. Not used.

k. Section 26 23 00, LOW-VOLTAGE SWITCHGEAR: Secondary distribution switchgear.

l. Not used.

m. Section 33 10 00 WATER UTILITIES: references meters.

n. Section 33 63 00 STEAM ENERGY DISTRIBUTION: references meters.

1.3 DEFINITIONS

a. AMR: Automatic meter reading is the technology of automatically collecting consumption, diagnostic, and status data from water and energy metering devices (water, gas, electric, steam) and transferring that data to a central database for billing, troubleshooting, and analyzing.

b. AUMS: Advanced Utility Metering System: the system described by this Section.

c. BACnet: BACnet is a Data Communications Protocol for Building Automation and Control Networks. It is defined by ASHRAE 135 (ISO 16484-5) standard protocol.

d. Data Over Cable Service Interface Specification (DOCSIS): an international standard defining communications and operation support interface requirements for a data over cable system, by the Cable Television Laboratories, Inc. consortium

e. Data Head (on meters): converts analog and pulse signals to digital signals for transmission to the Site Data Aggregation Device. Also provides for limited storage of the digital signals.

f. Device Accuracy: accuracy in this section is based on actual flow, not full scale or full range. Device accuracy measures the conversion of flow information to analog or pulse signals.

g. Ethernet: Local area network, based on IEEE 802.3 standards.

h. Firmware: Software (programs or data) that has been written onto read-only memory (ROM). Firmware is a combination of software and hardware. Storage media with ROMs that have data or programs recorded on them are firmware.

i. Gateway: Bi-directional protocol translator connecting control systems that use different communication protocols.

j. GB: gigabyte. When used to describe data storage, "GB" represents 1024 megabytes.

k. HTML: Hypertext markup language.

l. I/O: Input/output.

m. KB: Short for kilobyte. When used to describe data storage, "KB" represents 1024 bytes.

n. KY Pulse: A term used by the metering industry to describe a method of measuring consumption of electricity that is based on a relay changing status in response to the rotation of the disk in the meter.

o. LAN: Local area network. Sometimes plural as "LANs."

p. LCD: Liquid crystal display.

q. LonMark: An association comprising of suppliers and installers of LonTalk products. The Association provides guidelines for the implementation of the LonTalk protocol to ensure interoperability through Standard implementation.

r. LonTalk: An open standard protocol developed by the Echelon Corporation that uses a “Neuron Chip” for communication.

s. LonWorks: Network technology developed by the Echelon Corporation.

t. Low Voltage: As defined in NFPA 70 for circuits and equipment operating at less that 50 V or remote-control, signaling and power-limited circuits.

u. MB: megabyte. When used to describe data storage, "MB" represents 1024 kilobytes.

v. Mbps: Megabytes per second, equal to 8 megabits per second

w. Modbus TCP/IP: An open protocol for exchange of process data.

x. Monitoring: Acquisition, processing, communication, and display of equipment status data, metered electrical parameter values, power quality evaluation data, event and alarm signals, tabulated reports, and event logs.

y. OTDR: Optical Time Domain Reflectometer. A test instrument that analyzes the light loss in an optical fiber. Used to find faults, splices and bends in the line, it works by sending out a light pulse and measuring its reflection. Such devices can measure fiber lines that are longer than 150 miles

z. PC: Personal computer

AA. PICS, Protocol Implementation Conformance Statement: A written document that identifies the particular options specified by BACnet that are implemented in a device.

BB. REO: Resident Engineer Office: the VA office administering the construction contract.

CC. Reporting Accuracy: this is the root-mean-square sum of all of the metering devices' inaccuracies: measurement inaccuracy, mechanical inaccuracy, analog-to-digital or pulse integration inaccuracy, etc., up to the meter's data head.

DD. rms: Root-mean-square value of alternating voltage, which is the square root of the mean value of the square of the voltage values during a complete cycle.

EE. Router: A device that connects two or more networks at the network layer.

FF. RS-232: A Telecommunications Industry Association standard for asynchronous serial data communications between terminal devices.

GG. RS-485: A Telecommunications Industry Association standard for multipoint communications using two twisted-pairs.\

HH. TB: terrabyte. When used to describe data storage, "TB" represents 1024 gigabytes.

II. TCP/IP: Transport control protocol/internet protocol.

JJ. Turn-down: the maximum flow divided by the minimum flow through a meter; used along with accuracy requirements. For example, a meter shall be accurate to within 2% of actual flow with throughout a 20:1 turndown

KK. THD: Total harmonic distortion.

LL. UPS: Uninterruptible power supply; used both in singular and plural context.

MM. UTP: Unshielded twisted pair cabling, used to limit crosstalk and electromagnetic interference from the environment

NN. WAN: Wide area network.

1.4 QUALITY ASSURANCE

a. Installer Qualifications: Manufacturer's authorized representative who is trained and approved for installation of units required for this Project.

b. Manufacturer Qualifications: A firm experienced at least three years in manufacturing and installing power monitoring and control equipment similar to that indicated for this Project and with a record of successful in-service performance.

c. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100, by a testing agency, and marked for intended use.

d. System Modifications: Make recommendations for system modification in writing to the VA. No system modifications shall be made without prior written approval of the VA. Any modifications made to the system shall be incorporated into the Operations and Maintenance Instructions, and other documentation affected. Provide to the VA software updates for all software furnished under this specification during this contract's construction and verification periods and for the first two years after government acceptance. All updated software shall be verified as part of this contract.

1.5 PERFORMANCE

The advanced utility metering system shall conform to the following:

(1) Site Data Aggregation Device Graphic Display: The system shall display up to 4 graphics on a single screen with a minimum of (20) dynamic points per graphic. All current data shall be displayed within (10) seconds of the request.

(2) Site Data Aggregation Device Graphic Refresh: The system shall update all dynamic points with current data within ten seconds. Data refresh shall be automatic, without operator intervention.

(3) Meter Scan: All changes of metered values shall be transmitted over the high-speed network such that any data used or displayed at a controller or Site Data Aggregation Device will be current, within the prior ten seconds.

(4) Alarm Response Time: The maximum time from when meter goes into alarm to when it is annunciated at the workstation shall not exceed ten seconds.

(5) Reporting Accuracy: Listed below are minimum acceptable reporting accuracies for all values within the below minimum turn-down envelope reported by the meters:

Measured Variable	**Units Measured**	**Minimum Turn-Down of Meter**	**Reporting Accuracy (Note 1)**
Electricity	V, A, W, etc.	n/a	±0.5% of measured value
Natural Gas	l/s (CFH)	10:1	±2%
Liquefied Petroleum Gas	l/s (CFH)	10:1	±2%
Steam	kW (MBH)	20:1	±2%
Condensate	kW (MBH)	20:1	±2%
Domestic Water flow	l/s (GPH)	20:1	±2%
Reclaimed Water flow	l/s (GPH)	20:1	±2%
Make-up Water to Boilers flow	l/s (GPH)	10:1	±2%
Make-up Water to Cooling Towers flow	l/s (GPH)	10:1	±2%
No. 2 Heating Oil	l/s (GPH)	10:1	±2%
No. 6 Heating Oil	l/s (GPH)	10:1	±2%
Heating Water	kW (MBH)	20:1	±2%
Chilled Water	kW (MBH)	20:1	±2%

Outside Air Temperature	°C (°F)	n/a	±2%
Outside Air Relative Humidity	% rh	n/a	±2.5%

Table 1.5: Meter Performance Criteria

Table Notes:

(1) This table shows reporting accuracy, not merely the meter's accuracy. Reporting accuracy includes meter accuracy and data conversion accuracy. See Article 1.3 in this Section for definition. Accuracy is shown against the measured value, not against the full range of the meter.

(2) l/s: liter per second

CFH: cubic feet per hour
kW: kilowatt
MBH: 1000's British Thermal Units per hour
GPH: gallons per hour

1.6 WARRANTY

a. Labor and materials for advanced utility metering systems shall be warranted for a period as specified under Warranty in FAR clause 52.246-21.

b. Advance utility metering system failures during the warranty period shall be adjusted, repaired, or replaced at no cost or reduction in service to the owner. The system includes all computer equipment, transmission equipment, and all sensors and metering devices.

1.7 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualification Data; G, AE

SD-02 Shop Drawings

Shop Drawings; G, AE

SD-03 Product Data

Product Data; G, AE

Software Licenses And Upgrades; G, AE

SD-05 Design Data

Design Drawing; G, AE

Licensing Agreement; G, AE

SD-06 Test Reports

Software And Firmware Operational Documentation; G, AE

Field Quality-Control Test Reports; G, AE

SD-10 Operation and Maintenance Data

Informational Submittals; G, AE

Operation And Maintenance Data; G, AE

SD-11 Closeout Submittals

Warranty; G, AE

Software Upgrade Kit; G, AE

Firmware Upgrade Kit; G, AE

1.8 SUBMITTAL INSTRUCTIONS

a. Product Data: for each type of product indicated, Attach copies of approved Product Data submittals for products (such as flowmeters, temperature sensors and pressure transmitters, switchboards and switchgear) that describe advance utility metering features to illustrate coordination among related equipment and utility metering and control.

b. Shop Drawings: include plans, elevations, sections, details, and attachments to other work.

(1) Outline Drawings: Indicate arrangement of meters, components and clearance and access requirements. Clearly identify system components, internal connections, and all field connections.

(2) Block Diagram: Show interconnections between components specified in this Section and devices furnished with power distribution system components. Indicate data communication paths and identify networks, data buses, data gateways, concentrators, and other devices to be used. Describe characteristics of network and other data communication lines.

(3) Detail equipment assemblies and indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.

(4) Wiring Diagrams: Power, signal, and communications wiring. Coordinate nomenclature and presentation with a block diagram. Show all communications network components and include a communications single-line diagram indicating device interconnection and addressing information for all system devices. Identify terminal blocks used for interconnections and wire type to be used.

(5) UPS sizing calculations for workstation.

c. Software and Firmware Operational Documentation:

(1) Self-study guide describing the process for setting equipment's network address; setting Owner's options; procedures to ensure data access from any PC on the network, using a standard Web browser; and recommended firewall setup.

(2) Software operating and upgrade manuals.

(3) Software Backup: On a compact disc, complete with Owner-selected options.

(4) Device address list and the set point of each device and operator option, as set in applications software.

(5) Graphic file and printout of graphic screens and related icons, with legend.

(6) "Quick-Start" guide to describe a simple, three-step commissioning process for setting the equipment’s Ethernet address, and ensuring trouble-free data access from any PC on the network, using a standard web browser.

d. Software Upgrade Kit: For Owner to use in modifying software to suit future utility metering system revisions.

e. Firmware Upgrade Kit: For Owner to use in modifying firmware to suit future power system revisions or advanced utility metering system revisions. Firmware updates, and necessary software tools for firmware updates, shall be downloadable from the internet. VA shall be able to update firmware, in equipment, without removing device from the equipment. VA shall be capable of updating firmware over the utility metering communication network or through local communication ports on the device.

f. Software licenses and upgrades required by and installed for operating and programming digital and analog devices.

g. Qualification Data: For installer and manufacturer

h. Other Informational Submittals:

(1) System installation and setup guides, with data forms to plan and record options and setup decisions.

i. Revise and update the Contract Drawings to include details of the system design. Drawings shall be on 17 by 11 inches sheets. Details to be shown on the Design Drawing include:

(1) Details on logical structure of the network. This includes logical location of all network hardware.

(2) Manufacturer and model number for each piece of computer and network hardware.

(3) Physical location for each piece of network or computer hardware.

(4) Physical routing of LAN cabling.

(5) Physical and qualitative descriptions of connectivities.

1.9 CLOSEOUT SUBMITTAL INSTRUCTIONS

a. Operation and Maintenance Data: For advanced utility metering system components and meters, to include in emergency, operation, and maintenance manuals. Include the following:

(1) Operating and applications software documentation.

(2) Software licenses.

(3) Software service agreement.

(4) PC installation and operating documentation, manuals, and software for the PC and all installed peripherals. Software shall include system restore, emergency boot compact disks, and drivers for all installed hardware. Provide separately for each PC.

(5) Hard copies of manufacturer's specification sheets, operating specifications, design guides, user's guides for software and hardware, and PDF files on CD-ROM of the hard-copy submittal.

(6) In addition to the copies required by 01 00 00, provide 5 bound paper copies of the Operation and Maintenance Data and two compact disks (CD), with all Instructions as Acrobat PDF files. The pdf files shall identical to the paper copies and shall Acrobat navigation tools including Bookmarks for each Chapter.

(7) The advanced utility metering system Operation and Maintenance Instructions shall include:

(a) Procedures for the AUMS system start-up, operation and shut-down.

(b) Final As-Built drawings, including actual LAN cabling routing shown on architectural backgrounds.

1. IP address(es) as applicable for each piece of network hardware.

2. IP address for each computer server, workstation and networked printer.

3. Network identifier (name) for each printer, computer server and computer workstation.

4. CEA-709.1B address (domain, subnet, node address) for each CEA-709.1B TP/FT-10 to IP Router.

(c) Routine maintenance checklist, rendered in a Microsoft Excel format. The routine maintenance checklist shall be arranged in a columnar format. The first column shall list all installed devices, the second column shall list each device’s node identifier/address, the third column shall describe each device’s physical location, the fourth column shall state the maintenance activity or state no maintenance required, the fifth column shall state the frequency of the maintenance activity, frequency of

calibration and the sixth column for additional comments or reference.

(d) Qualified service organization list.

(e) In addition to the requirements in Section 01 33 00, the submittal shall include manufacturer Installation Requirements.

(f) Include complete instructions for calibration of each meter type and model.

(g) Start-Up and Start-Up Testing Report.

(h) Performance verification test procedures and reports.

(i) Preventive Maintenance Work Plan.

(j) In addition to factory-trained manufacturers' representatives requirements in 01 00 00, provide signed letter by factory-trained manufacturers' representatives stating that the system and components are installed in strict accordance with the manufacturers' recommendations.

b. Field quality-control test reports.

1.10 LICENSING AGREEMENT

a. Licenses procured as part of this work become the property of the government upon acceptance of the work. Licenses shall have no expiration.

b. Technical Support: Beginning with Government Acceptance, provide software support for one years.

c. Upgrade Service: Update software to latest version at Project completion. Install and program software upgrades that become available within two years from date of Government Acceptance. Upgrading software shall include the operating systems. Upgrade shall include new or revised licenses for use of software.

(1) Provide 30-day notice to Owner to allow scheduling and access to system and to allow Owner to upgrade computer equipment if necessary.

1.11 MAINTENANCE AND SERVICE

a. Preventive Maintenance Requirements: provide a preventative maintenance plan with attached procedures indicated by meter and component manufacturers. Perform maintenance procedures for a period of 1 year after government acceptance, at frequencies and using procedures required by the meter and component manufacturers. At a minimum and if the manufacturer is silent on its preventative maintenance requirements, frequencies, deliverables and activities shall comply with the following:

(1) Preventive Maintenance Work Plan: prepare a Preventive Maintenance Work Plan to schedule all required preventive maintenance. VA approval of the Work Plan shall be obtained. Adhere to the approved work plan to facilitate VA verification of

work. If the Contractor finds it necessary to reschedule maintenance, a written request shall be made to the VA detailing the reasons for the proposed change at least five days prior to the originally scheduled date. Scheduled dates shall be changed only with the prior written approval of the REO.

(2) Semiannual Maintenance: perform the following Semiannual Maintenance as specified:

(a) Perform data backups on all Server Hardware.

(b) Run system diagnostics and correct diagnosed problems.

(c) Perform fan checks and filter changes for AUMS hardware.

(d) Perform all necessary adjustments on printers.

(e) Resolve all outstanding problems.

(f) Install new ribbons, ink cartridges and toner cartridges into printers, and ensure that there is at least one spare ribbon or cartridge located at each printer.

(3) Maintenance Procedures

(a) Maintenance Coordination: Any scheduled maintenance event by Contractor that will result in component downtime shall be coordinated with the VA as follows. Time periods shall be measured as actual elapsed time from beginning of equipment off-line period, including working and non-working hours.

1. For non-redundant computer server hardware, provide 14 days notice, components shall be off-line for no more than 8 hours.

2. For redundant computer server hardware, provide 7 days notice, components shall be off-line for no more than 36 hours.

3. For active (powered) network hardware, provide 14 days notice, components shall be off-line for no more than 6 hours.

4. For cabling and other passive network hardware, provide 21 days notice, components shall be off-line for no more than 12 hours.

(b) Software/Firmware: Software/firmware maintenance shall include operating systems, application programs, and files required for the proper operation of the advanced utility metering system regardless of storage medium. User- (project site-) developed software is not covered by this contract, except that the advanced utility metering system software/firmware shall be maintained to allow user creation, modification, deletion, and proper execution of such user-developed software as specified. Perform diagnostics and corrective reprogramming as required to maintain total advanced utility metering system operations as specified. Back up software before performing any computer hardware and software maintenance. Do not modify any parameters without approval from the VA. Any approved changes and additions shall be properly documented, and the appropriate manuals shall be updated.

(c) Network: Network maintenance shall include testing transmission media and equipment to verify signal levels, system data rates, errors and overall system performance.

b. Service Call Reception

(1) A VA representative will advise the Contractor by phone or in person of all maintenance and service requests, as well as the classification of each based on the definitions specified. A description of the problem or requested work, date and time notified, location, classification, and other appropriate information will be placed on a Service Call Work Authorization Form by the VA.

(2) The Contractor shall have procedures for receiving and responding to service calls during regular working hours. A single telephone number shall be provided for receipt of service calls during regular working hours. Service calls shall be considered received by the Contractor at the time and date the telephone call is placed by the VA.

(3) Separately record each service call request, as received on the Service Call Work Authorization form.Complete the Service Call Work Authorization form for each service call. The completed form shall include the serial number identifying the component involved, its location, date and time the call was received, nature of trouble, names of the service personnel assigned to the task, instructions describing what has to be done, the amount and nature of the materials to be used, the time and date work started, and the time and date of completion.

(4) Respond to each service call request within two working hours. The status of any item of work must be provided within four hours of the inquiry during regular working hours, and within sixteen hours after regular working hours or as needed to repair equipment.

1.12 SPARE PARTS

a. Furnish spare parts described below that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

(1) Addressable Relays: One for every ten installed. Furnish at least one of each type.

(2) Data Line Surge Suppressors: One for every ten of each type installed. Furnish at least one of each type.

b. Furnish spare parts shall not be used for any warranty-required remediation.

1.13 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced, unless otherwise noted. Publications are referenced in the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B40.200	(2008; R 2013) Thermometers, Direct Reading and Remote Reading

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 135	(2020; Interpretation 1-8 2021; Errata 1-2 2021; Addenda CD 2021; Addenda BV-CE 2022; Interpretation 9-12 2022; Interpretation 13-24 2023; Addenda BV-CF 2023; Errata 3 2023) BACnet—A Data Communication Protocol for Building Automation and Control Networks

CONSUMER ELECTRONICS ASSOCIATION (CEA)

CEA-852-C	(2014) Tunneling Device Area Network Protocols Over Internet Protocol Channels

U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

FCC EMC	(2002) FCC Electromagnetic Compliance Requirements

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.1x	(2010) Local and Metropolitan Area Networks - Port Based Network Access Control
IEEE 802.3	(2022; CORR 2024) Ethernet

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 62053-22	(2020) Electricity Metering Equipment (A.C.) - Particular Requirements - Part 22: Static Meters for Active Energy (Classes 0,2 S and 0,5 S)

INTERNATIONAL TELECOMMUNICATION UNION (ITU)

ITU-T	ITU-T Recommendation Series

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)
NEMA 1	Enclosures
NEMA 4	Enclosures for outdoor use where extra protection against moisture and wind driven rain is required
ANSI C12.20	(2015; E 2018) Electricity Meters - 0.1, 0.2, and 0.5 Accuracy Classes

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 262	(2023) Standard Method of Test for Flame Travel and Smoke of Wires and Cables for Use in Air-Handling Spaces
NFPA 780	(2023) Standard for the Installation of Lightning Protection Systems

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-232	(1997f; R 2012) Interface Between Data Terminal Equipment and Data Circuit-Terminating Equipment Employing Serial Binary Data Interchange
EIA/TIA 455-61	Measurement of Fiber or Cable Attenuation Using on ODR
TIA-485	(1998a; R 2012) Electrical Characteristics of Generators and Receivers for Use in Balanced Digital Multipoint Systems
TIA-526-14	(2023d) OFSTP-14A Optical Power Loss Measurements of Installed Multimode Fiber Cable Plant
TIA-568.1	(2020e) Commercial Building Telecommunications Infrastructure Standard
TIA-568.2	(2018d) Balanced Twisted-Pair Telecommunications Cabling and Components Standards
TIA-568.3	(2022e) Optical Fiber Cabling Components Standard
TIA-526-7	(2015a; R 2022) Measurement of Optical Power Loss of Installed Single-Mode Fiber Cable Plant, Adoption of IEC 61280-4-2 edition 2: Fibre-Optic Communications Subsystem Test Procedures - Part 4-2: Installed Cable Plant - Single-Mode Attenuation and Optical Return Loss Measurement
TIA-606	(2021d) Administration Standard for Telecommunications Infrastructure

UL SOLUTIONS (UL)

UL 1449	(2021; Reprint Dec 2022) UL Standard for Safety Surge Protective Devices

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 8802-3	(2021) Telecommunications and exchange between information technology systems — Requirements for local and metropolitan area networks — Part 3: Standard for Ethernet
ISO 16484-5	Data communication protocol

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST FIPS 140-2	(2001) Security Requirements for Cryptographic Modules

INTERNET ENGINEERING TASK FORCE (IETF)

RFC 1812	Requirements for IP Version 4 Routers

PART 2 PRODUCTS

2.1 ADVANCED UTILITY METERING SYSTEM

a. Functional Description

(1) Meter and record load profiles. Chart energy and water consumption patterns.

(a) Calculate and record the following:

1. Load factor.

2. Peak demand periods.

3. Consumption correlated with facility activities.

(b) Measure and record metering data for the following:

1. Electricity.

2. Steam and condensate

3. Domestic water.

4. Natural gas.

5. Oil.

6. Liquefied Petroleum Gas.

7. Used, Boiled/Evaporated, Reclaimed and Recovered water.

8. Chilled water

9. Heating water

(c) Software: calculate allocation of utility costs.

(d) Electric Power Quality Monitoring: Identify power system anomalies and measure, display, capture waveforms, and record trends and alarms of the following power quality parameters:

1. Voltage regulation and unbalance.

2. Continuous three-phase rms voltage.

3. Periodic max./min./avg. samples.

4. Harmonics.

5. Voltage excursions.

6.

(e) Emergency Load Shedding. Preserve critical loads or avoid total shutdown due to unforeseen loss of power sources according to the following logic:

1. Determine system topology.

2. Evaluate remaining loads and sources.

3. Shed loads in less than 100 ms.

4.

(f) Demand Management:

1. Peaking or co-generator control.

2. Load interlocking.

3. Load shedding.

4. Load trimming.

5.

(g) System: Report equipment status and power system control.

b. Communications Components and Networks

(1) Site Data Aggregation Device and its networked meters shall communicate using BACNet protocol. Backbone shall communicate using ISO 8802-3 (Ethernet) Data Link/Physical layer protocol and BACnet/IP addressing as specified in ASHRAE 135, BACnet Annex J.

(a) Control products, communication media, connectors, repeaters, hubs, and routers shall comprise a BACnet internetwork. Controller and operator interface communication shall conform to ASHRAE 135, BACnet.

(b) Each controller shall have a communication port for connection to an operator interface.

(2) Network Configuration: High-speed, multi-access, open nonproprietary, industry standard LAN and WAN and Internetworked

LAN.

(3) Communication protocol; LANs complying with RS-485 or RS-485 accessed through Ethernet, 100 Base-TX Ethernet, and Modbus TCP/IP.

(4) Network Hardware

(a) Building Point of Connection Hardware

1. Active equipment and communication interfaces.

2. Switches, hubs, bridges, routers and servers.

(b) IP Network Hardware

1. Wire and Cables, copper connectivity devices.

2. Fiber Optic Patch Panel.

3. Fiber Optic Media Converter

4. Ethernet Switch

5. IP Router

(5) Communication Security

(a) Remote teleworking and remote access of the network shall be through a firewall, at the Site Data Aggregation Device, complying with the requirements associated with Level 1 security in the NIST FIPS 140-2 (2002), Security Requirements for Cryptographic Modules.

(b) Direct access to network shall be restricted as described in

2.2 SITE DATA AGGREGATION DEVICE – PERSONAL COMPUTER WORKSTATION

a. Hardware

(1) Workstation Hardware

(a) Environmental Conditions: System components shall be capable of withstanding Indoor installation in spaces that have environmental controls to maintain ambient conditions of 36 to 140 deg F dry bulb temperature and 20 to 95% relative humidity, noncondensing environmental conditions without mechanical or electrical damage or degradation of operating capability.

(b) Computer: Commercial standard with supporting 32- or 64-bit hardware (as limited by the advanced utility metering system software) and software enterprise server. Internet Explorer v6.0 SP1 or higher, Windows Script Hosting version 5.6 or higher, Windows Message Queuing, Windows Internet Information Services (IIS) v5.0 or higher, minimum 2.8 GHz processor, minimum 4GB DDR3 SDRAM (minimum 1333 Mhz) memory, minimum 1 TB 7200 rpm SATA hard drive with 16 MB cache, 512 MB video card, and 16 speed high density DVD-RW+/- optical drive.

(c) Real-Time Clock:

1. Accuracy: Plus or minus 1 minute per month.

2. Time Keeping Format: 24-hour time format including seconds, minutes, hours, date, day, and month; automatic reset by software.

3. Clock shall function for one year without power.

4. Provide automatic time correction once every 24 hours by synchronizing clock with the Time Service Department of the U.S. Naval Observatory.

(d) Serial Ports: Four USB ports and two RS-232-F serial ports for general use, with additional ports as required. Data transmission rates shall be selectable under program control.

(e) Parallel Port: Enhanced.

(f) Sound Card: For playback and recording of digital WAV sound files associated with audible warning and alarm functions.

(g) Color Monitor: PC compatible, not less than 22 inches, LCD type, with a minimum resolution of 1280 by 1024 pixels, noninterlaced, and a maximum dot pitch of 0.28 mm.

(h) Keyboard: Minimum of 64 characters, standard ASCII character set based on ANSI INCITS 154.

(i) Mouse: Standard, compatible with installed software.

(j) Removable Disk Storage: Include the following, each with appropriate controller:

1. Minimum 1 TB removable hard disk, maximum average access time of 10 ms.

(k) Network Interface Card (NIC): integrated 10-100-1000 Base-TX Ethernet NIC with an RJ45 connector or a 100Base-FX Ethernet NIC with an SC/ST connector.

(l) Cable Modem: 42.88 Mbps, DOCSIS 3.0 Certified, also backwards compatible with DCOSIS 2.0 and DOCSIS 1.1/1.0 standards. Provide Ethernet or USB connectivity.

(m) Optical Modem: full duplex link, for use on 10 GBase-R single-mode and multi-mode fiber with a XENPAK module.

(n) Modem: 56,600 bits per second, full duplex for asynchronous communications. With error detection, auto answer/autodial, and call-in-progress detection. Modem shall comply with requirements in ITU-T v.34, ITU-T v.42, ITU-T v.42 Appendix VI for error correction, and ITU-T v.42 BIS for data compression standards; and shall be suitable for operating on unconditioned voice-grade telephone lines complying with FCC EMC 47 CFR 68.

(o) Audible Alarm: Manufacturer's standard.

(2) Printers: provide a dedicated, minimum resolution 600 dpi, color

laser printer, connected to the Site Data Aggregation Device through a USB interface.

(a) If a network printer is used instead of this dedicated printer, it shall have a 100Base-T interface with an RJ45 connection and shall have a firmware print spooler compatible with the Operating System print spooler.

(b) RAM: 512 MB, minimum.

(c) Printing Speed: Minimum twenty six pages per minute (color); minimum 30 pages per minute (black/white).

(d) Paper Handling: Automatic sheet feeder with 250-sheet x 8.5 inch x 11 inch paper cassette and with automatic feed.

(3) TIA-232 (RS-232) ASCII Interface

(a) ASCII interface shall allow TIA-232 connections to be made between a meter or circuit monitor operating as the host PC and any equipment that will accept RS-232 ASCII command strings, such as local display panels, dial-up modems, and alarm transmitters.

(b) Pager System Interface: Alarms shall be able to activate a pager system with customized message for each input alarm.

(c) TIA-232 output shall be capable of connection to a pager interface that can be used to call a paging system or service and send a signal to a portable pager. System shall allow an individual alphanumeric message per alarm input to be sent to paging system. This interface shall support both numeric and alphanumeric pagers.

(d) Alarm System Interface: TIA-232 output shall be capable of transmitting alarms from other monitoring and alarm systems to workstation software.

(e) Cables: provide Plenum-Type, TIA-232 Cable: Paired, 2 pairs, No. 22 AWG, stranded (7x30) tinned copper conductors, plastic insulation, and individual aluminum foil-polyester tape shielded pairs with 100 percent shield coverage; plastic jacket. Pairs are cabled on common axis with No. 24 AWG, stranded (7x32) tinned copper drain wire.

1. NFPA 70, Type CMP.

2. Flame Resistance: NFPA 262, Flame Test.

(4) Rack-Mounted Server Hardware

(a) Environmental Conditions: System components shall be capable of withstanding Indoor installation in spaces that have environmental controls to maintain ambient conditions of 36 to 140 deg F dry bulb temperature and 20 to 95% relative humidity, noncondensing environmental conditions without mechanical or electrical damage or degradation of operating capability.

(b) Computer: Commercial rack-mounted with supporting 32- or 64-bit hardware (as limited by the advanced utility metering

system software) and software enterprise server. Internet Explorer v6.0 SP1 or higher, Windows Script Hosting version 5.6 or higher, Windows Message Queuing, Windows Internet Information Services (IIS) v5.0 or higher, minimum 2.8 GHz processor, minimum 4GB DDR3 SDRAM (minimum 1333 Mhz) memory, minimum 1 TB 7200 rpm SATA hard drive with 16 MB cache, and 16 speed high density DVD-RW+/- optical drive.

(c) Real-Time Clock:

1. Accuracy: Plus or minus 1 minute per month.

2. Time Keeping Format: 24-hour time format including seconds, minutes, hours, date, day, and month; automatic reset by software.

3. Clock shall function for one year without power.

4. Provide automatic time correction once every 24 hours by synchronizing clock with the Time Service Department of the U.S. Naval Observatory.

(d) Serial Ports: Four USB ports and two RS-232-F serial ports for general use, with additional ports as required. Data transmission rates shall be selectable under program control.

(e) Parallel Port: Enhanced.

(f) Removable Disk Storage: Include minimum 1 TB removable hard disk, maximum average access time of 10 ms, with appropriate controller:

(g) Network Interface Card (NIC): integrated 10-100-1000 Base-TX Ethernet NIC with an RJ45 connector or a 100Base-FX Ethernet NIC with an SC/ST connector.

(h) Cable Modem: 42.88 Mbps, DOCSIS 2.0 Certified, also backwards compatible with DOCSIS 2.0 and DOCSIS 1.1/1.0 standards. Provide Ethernet or USB connectivity.

(i) Optical Modem: full duplex link, for use on 10 GBase-R single-mode and multi-mode fiber with a XENPAK module.

(j) Modem: 56,600 bits per second, full duplex for asynchronous communications. With error detection, auto answer/autodial, and call-in-progress detection. Modem shall comply with requirements in ITU-T v.34, ITU-T v.42, ITU-T v.42 Appendix VI for error correction, and ITU-T v.42 BIS for data compression standards; and shall be suitable for operating on unconditioned voice-grade telephone lines complying with FCC EMC47 CFR 68.

(k) Audible Alarm: Manufacturer's standard.

(5) TIA-232 (RS-232) ASCII Interface

(a) ASCII interface shall allow TIA-232 connections to be made between a meter or circuit monitor operating as the host PC and any equipment that will accept TIA-232 ASCII command strings, such as local display panels, dial-up modems, and alarm transmitters.

(b) Pager System Interface: Alarms shall be able to activate a pager system with customized message for each input alarm.

(c) TIA-232 output shall be capable of connection to a pager interface that can be used to call a paging system or service and send a signal to a portable pager. System shall allow an individual alphanumeric message per alarm input to be sent to paging system. This interface shall support both numeric and alphanumeric pagers.

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(e) Cables: provide Plenum-Type, TIA-232 Cable: Paired, 2 pairs, No. 22 AWG, stranded (7x30) tinned copper conductors, plastic insulation, and individual aluminum foil-polyester tape shielded pairs with 100 percent shield coverage; plastic jacket. Pairs are cabled on common axis with No. 24 AWG, stranded (7x32) tinned copper drain wire.

1. NFPA 70, Type CMP.

2. Flame Resistance: NFPA 262, Flame Test.

b. Software

(1) Operating System (OS)

(a) For a Site Data Aggregation Device connected to multiple utility meters, software shall reside on the Workstation or Server PC connected to a network able to poll and support over 1000 utility metering devices; software shall be web-enabled with the option to add custom graphics displays and additional web-enabled clients. BACNet, Ethernet, Modbus TCP/IP, TIA-232, and TIA-485 digital communications.

(b) Operating System Software: Based on 32- or 64-bit, Microsoft Windows operating system, as required by the metering and database software. Software shall have the following features:

1. Multiuser and multitasking to allow independent activities and monitoring to occur simultaneously at different workstations.

2. Graphical user interface to show pull-down menus and a menu tree format.

3. Capability for future additions within the indicated system size limits.

(2) Office Automation Software shall consist of the e-mail, spreadsheet and word processing portions of the project site's standard office automation software.

(3) Virus Protection Software shall consist of the project site's standard virus protection software complete with a virus definition update subscription.

(4) Configuration server shall meet the requirements of CEA-852-C.

(5) Network configuration tool shall meet the following minimum requirements:

(a) It shall allow configuration of the network while off-line such that an operator may set up changes to the network while disconnected from the network, and then execute all of them once connected.

(b) It shall have a graphics-based user interface, and be able to display and print a graphical representation of the control network.

(c) It shall be capable of generating and printing a table containing domain/subnet/node address and node identifier for the entire network or any subset thereof, selected by the User.

(d) It shall be capable of merging two existing standard databases into a single standard database.

(6) Metering Software

(a) Basic Requirements:

1. Fully compatible with and based on the approved operating system.

2. Password-protected operator login and access; three levels, minimum.

3. Password-protected setup functions.

4. Context sensitive on-line help.

5. Capability of creating, deleting, and copying files; and automatically maintaining a directory of all files, including size and location of each sequential and random-ordered record.

6. Capability for importing custom icons into graphic views to represent alarms and I/O devices.

7. Automatic and encrypted backups for database and history; automatically stored at the Site Data Aggregation Device and encrypted with a nine-character alphanumeric password, which must be used to restore or read data contained in backup.

8. Operator audit trail for recording and reporting all changes made to user-defined system options.

(b) Workstation and Server Functions:

1. Support other client PCs on the LAN and WAN.

2. Maintain recorded data in databases accessible from other PCs on the LAN and WAN.

(c) Data Formats:

1. User-programmable export and import of data to and from commonly used Microsoft Windows spreadsheet, database, billing, and other applications; using dynamic data exchange technology.

2. Option to convert reports and graphics to HTML format.

3. Interactive graphics.

4. Option to send preprogrammed or operator designed e-mail reports.

5. Option to serve information to third-party applications via Object Linking and Embedding for Process Control using open standards.

(d) Metered data: Display metered values in real time with a rigid time-stamp. Couple all metered data with measured outside air conditions at the relevant facility.

(e) Metered Data alarms: Provide generic alarm modules to notify Users and highlight metered data gaps, data spikes outside of range, and data timestamp errors.

1. Customize the generic alarm modules to the application.

2. Modules shall allow for user adjustment of alarm criteria.

3. Alarm notices shall be shown via hyperlinks on the graphical User interface, and shall also be shown by flags within the data set.

(f) Automatic Data Scrubbing: Provide tools for User-programming of rules to scrub the data of the followings errors: data gaps, data spikes outside of range, and data timestamp errors. Use these rules to scrub the raw metered data. Flag all data which has been so scrubbed.

(g) Remote control:

1. (for electrical load control) Display circuit-breaker status and allow breaker control.

2. User defined with load-shedding automatically initiated and executed schemes responding to programmed time schedules, set points of metered demands, utility contracted load shedding, or combinations of these.

(h) Equipment Documentation: Database for recording of equipment ratings and characteristics; with capability for graphic display on monitors.

(i) User-Defined Events: Display and record with date and time stamps accurate to 0.1 second, and including the following:

1. Operator log on/off.

2. Attempted operator log on/off.

3. All alarms.

4. Equipment operation counters.

5. Out-of-limit, pickup, trip, and no-response events.

(j) (for electrical power monitoring) Waveform Data: Display and record waveforms on demand or automatically on an alarm or programmed event; include the graphic displays of the following, based on user-specified criteria:

1. Phase voltages, phase currents, and residual current.

2. Overlay of three-phase currents, and overlay each phase voltage and current.

3. Waveforms ranging in length from 2 cycles to 5 minutes.

4. Disturbance and steady-state waveforms up to 512 points per cycle.

5. Transient waveforms up to 83,333 points per cycle on 60-Hz base.

6. Calculated waveform on a minimum of four cycles of data of the following:

a) THD.

b) rms magnitudes.

c) Peak values.

d) Crest factors.

e) Magnitude of individual harmonics.

(k) Data Sharing: Allow export of recorded displays and tabular data to third-party applications software on the local server.

(l) Activity Tracking Software:

1. Automatically compute and prepare activity demand and energy-use statements based on metering of energy use and peak demand integrated over user-defined interval.

2. Intervals shall be same as used by electric utilities, including current vendor.

3. Import metered data from saved records that were generated by metering and monitoring software.

4. Maintain separate directory for each activity's historical billing information.

5. Prepare summary reports in user-defined formats and time intervals.

(m) Passwords

(n) Protocol Drivers

(o) System Graphic Displays: provide interactive color-graphics platform with pull-down menus and mouse-driven generation of power system graphics, in formats widely used for such drafting; to include the following:

1. Site plan.

2. Floor plans.

3. Equipment elevations.

4. Single-line diagrams.

5. Custom graphic screens configured, not programmed, using drag-and-drop tools available within the software.

(p) Alarms: display and record alarm messages from discrete input and controls outputs, according to user programmable protocol.

1. Functions requiring user acknowledgment shall run in background during computer use for other applications and override other presentations when they occur.

(q) Trending: display and record data acquired in real-time from different meters or devices, in historical format over user-defined time; unlimited as to interval, duration, or quantity of trends.

1. Spreadsheet functions of sum, delta, percent, average, mean, standard deviation, and related functions applied to recorded data.

2. Charting, statistical, and display functions of standard Windows-based spreadsheet.

(r) Report Generation: User commands initiate the reporting of a list of current alarm, supervisory, and trouble conditions in system or a log of past events.

1. Print a record of user-defined alarm, supervisory, and trouble events on workstation printer.

a) Sort and report by device name and by function.

b) Report type of signal (alarm, supervisory, or trouble), description, date, and time of occurrence.

c) Differentiate alarm signals from other indications.

d) When system is reset, report reset event with same information concerning device, location, date, and time.

(7) BACnet: Site Data Aggregation Device shall have demonstrated interoperability during at least one BMA Interoperability Workshop and shall substantially conform to BACnet Operator Workstation (B-OWS) device profile as specified in ASHRAE 135, BACnet Annex L

(8) Site Data Aggregation Device shall periodically upload metered data to the VA Corporate-wide server:

(a) The metering software shall provide periodic upload (adjustable interval, initially set on 15-minute intervals) of the scrubbed and collected data.

(b) The VA's Corporate wide server accepts the following data structures:

1. Information structured using the 2005 and 2008 SQL server database engine.

2. The following data stores are acceptable:

a) Databases: SQL Server, DB2, Oracle, Access, Sybase, MySQL.

b) Flat files:.CSV,.XLS,.TXT,.XML,.PQDIF

(c) The minimum data to be uploaded (per meter) includes:

1. A time stamp

2. A device identifier

3. A flow (power or water flow) value

4. A flow order of magnitude

5. Description of the flow's units

6. The outside air drybulb temperature at the time stamp

7. The outside air wetbulb temperature at the time stamp

8. A "scrubbed data" flag

9. An irregular data alarm stamp

c. Self-contained uninterruptible power supply (UPS):

(1) Size: Provide a minimum of six hours of operation of workstation station equipment, including two hours of alarm printer operation.

(2) Batteries: Sealed, valve regulated, recombinant, lead calcium.

(3) Accessories:

(a) Transient voltage suppression.

(b) Input-harmonics reduction.

(c) Rectifier/charger.

(d) Battery disconnect device.

(e) Static bypass transfer switch.

(f) First six subparagraphs below are optional accessories.

(g) Internal maintenance bypass/isolation switch.

(h) External maintenance bypass/isolation switch.

(i) Output isolation transformer.

(j) Remote UPS monitoring.

(k) Battery monitoring.

(l) Remote battery monitoring.

2.3 CABLE SYSTEMS - TWISTED PAIR AND FIBER OPTIC

a. General:

(1) All metallic cable sheaths, etc. (i.e.: risers, underground, station wiring, etc. shall be grounded.

(2) Install temporary cable and wire pairs so as to not present a pedestrian safety hazard. Provide for all associated work for any temporary installation and for removal when no longer necessary. Temporary cable installations are not required to meet Industry Standards; but, must be reviewed and approved by the VA prior to installation.

(3) Cable conductors to provide protection against induction in circuits. Crosstalk attenuation within the System shall be in excess of -80 dB throughout the frequency ranges specified.

(4) Minimize the radiation of RF noise generated by the System equipment so as not to interfere with audio, video, data, computer main distribution frame (MDF), telephone customer service unit (CSU), and electronic private branch exchange (EPBX) equipment the System may service.

(5) The as-installed drawings shall identify each cable as labeled, used cable, and bad cable pairs.

(6) Label system’s cables on each end. Test and certify cables in writing to the VA before conducting proof-of-performance testing. Minimum cable test requirements are for impedance compliance, inductance, capacitance, signal level compliance, opens, shorts, cross talk, noise, and distortion, and split pairs on all cables in the frequency ranges specified. The cable tests shall demonstrate the operation of this cable at not less than 10 mega (m) Hertz (Hz) full bandwidth, fully channel loaded and a Bit Error Rate of a minimum of 10-6 at the maximum rate of speed. Make available all cable installation and test records at acceptance testing by the VA and shall thereafter be maintained in the Facility’s Telephone Switch Room. All changes (used pair, failed pair, etc.) shall be posted in these records as the change occurs.

(7) Coordinate with the Electrical Contractor to install the telephone entrance cable to the nearest point of entry into the Facility and as shown on the drawings. Coordinate with the VA and the

Electrical Contractor to provide all cable pairs/circuits from the Facility point of entry to the Telephone Switch Room all telephone, FTS, DHCP, ATM, Frame Relay, data, pay stations, patient phones, and any low voltage circuits as described herein.

(8) Provide all cable pairs/circuits from the Server Room and establish circuits throughout the Facility for all cabling as described herein.

(9) Provide proper test equipment to demonstrate that cable pairs meet each OEM's standard transmission requirements, and guarantee the cable will carry data transmissions at the required speeds, frequencies, and fully loaded bandwidth.

b. LAN COPPER CABLES

(1) Comply with Section 27 15 00 "Communications Horizontal Cabling."

(2) RS-485 Cable:

(a) PVC-Jacketed, RS-485 Cable: Paired, 2 pairs, twisted, No. 22 AWG, stranded (7x30) tinned copper conductors, PVC insulation, unshielded, PVC jacket, and NFPA 70, Type CMG.

(3) Unshielded Twisted Pair Cables: Category 5e or 6 as specified for horizontal cable for data service in Section 27 15 00 "Communications Horizontal Cabling."

(4) Cabling products shall be tested and certified for use at data speeds up to at least 100 Mbps. Other types of media commonly used within IEEE 802.3 LANs (e.g., 10Base-T and 10Base-2) shall be used only in cases to interconnect with existing media. Short lengths of media and transceivers may be used in these applications. Provide separately orderable media, taps and connectors.

(5) Ethernet Switch shall be IEEE 802.3 bridges which shall function as the center of a distributed-star architecture and shall be "learning" bridges with spanning tree algorithms in accordance with IEEE 802.1x. The switch shall support the connected media types and shall have a minimum of 150% the required ports and no fewer than 4 ports. One port shall be switch selectable as an uplink port.

(6) Provide IP router network equipment. The routers shall be fully configurable for protocol types, security, and routing selection of sub-networks. The router shall meet all requirements of RFC 1812.

c. LAN FIBER OPTICAL CABLES

(1) Interior Fiber Optic Cable: Interior Fiber Optic Cable shall be Multimode or Singlemode fiber, 62.5/125 micron for multimode or 10/125 micron for singlemode micron with SC or ST connectors as specified in TIA-568.1. Terminations, patch panels, and other hardware shall be compatible with the specified fiber and shall be as specified in Section 27 15 00 "Communications Horizontal Cabling." The data communications equipment shall use the 850-nm range of multimode or 1310-nm range of singlemode fiber-optic

cable. Fiber-optic cable shall be suitable for use with the 100Base-FX standard as defined in IEEE 802.3.

(2) Exterior Fiber Optic Cable: Exterior Fiber Optic Cable shall be Multimode or Singlemode Fiber, 62.5/125 micron for multimode or 10/125 micron for singlemode micron with SC or ST connectors as specified in TIA-568.1. Terminations, patch panels, and other hardware shall be compatible with the specified fiber and shall be as specified in Section 27 15 00 "Communications Horizontal Cabling.". The data communications equipment shall use the 850-nm range of multimode or 1310-nm range of singlemode fiber-optic cable. Fiber-optic cable shall be suitable for use with the 100Base-FX standard as defined in IEEE 802.3.

(3) Fiber Optic Patch Panels shall be wall or rack mountable and designed to provide termination facilities for up to 24 fibers. Unit shall also have capability to be equipped with spliced trays, six packs (for adapters), and blank panels for easy termination of the fiber bundles and tube cables. Fiber-optic terminating equipment shall provide for mounting of ST or SC connectors on an optical patch panel. Provide fiber-cable management and cable-routing hardware to assure conformance to minimum fiber and cable bend radii. Connectors on the patch panel shall be ST or SC feed through. Provide access to both sides of the panel. The patch panel for the connectors shall be mounted to facilitate rearrangement and identification. Each apparatus shall have cabling and connection instructions associated with it.

(4) Fiber Optic media converter shall provide media conversion between layer 1 copper and fiber media to support data rates equal to the greater of the physical layer or 100 Mbps as specified in IEEE 802.3.

d. LOW-VOLTAGE WIRING

(1) Low-Voltage Control Cable: Multiple conductor, color-coded, No. 20 AWG copper, minimum.

(a) Sheath: PVC; except in plenum-type spaces, use sheath listed for plenums.

(b) Ordinary Switching Circuits: Three conductors, unless otherwise indicated.

(c) Switching Circuits with Pilot Lights or Locator Feature: Five conductors, unless otherwise indicated.

e. WIRELESS MODEMS

Provide wireless modems for high speed, point-to-point Ethernet communications between sites. Transceivers shall be single integral units and may be mounted within the building in a NEMA 1 enclosure or weatherproof with integral antenna and pole mounted. System shall have the following features as a minimum:

(1) 4.9x GHz Licensed Frequency or 902-928Mhz Industrial/ Scientific/Medical (ISM) or 2.4GHz Industrial/Scientific/Medical (ISM) or 5.8GHz Industrial/Scientific/Medical (ISM) band.

(2) Obtain FCC license on behalf of the VA for each licensed frequency.

(3) Security protocol shall utilize a minimum of 128-bit data encryption. Provide Simple Network Management Protocol (SNMP) for network diagnostics and management. Transceiver shall have status indicators for power, ethernet link status and RF link status.

(4) Operating Conditions: , 85% relative humidity (32 degrees F to 131 degrees F, 85% relative humidity).

(5) Transmitter/Receiver/Antenna combination shall provide less than 0.005% frame error rate at 10Mbps data rate between sites.

(6) Antennas may be omni-directional or directional as required for system gain. Antennas and supports shall withstand a combined load of ½" ice and 125mph wind loading.

(7) Provide heavy-duty antenna masks and wall and roof mask mount to support antennas. All hardware shall be stainless steel. Ground antenna mast per NFPA 780.

(8) Coaxial cable shall be 0.200 diameter minimum for lengths below 50' and 0.400 diameter or greater for length greater than 50'.

(9) Surge suppressors for coaxial cables shall be rated for the frequency of operation, utilize gas tube technology and have a maximum let thru of 1mJ. Provide UL 1449 listed, Type 1, 50kA, 120V, surge protective device for each power circuit.

2.4 GROUNDING

Ground cable shields, drain conductors, and equipment to eliminate shock hazard and to minimize ground loops, common-mode returns, noise pickup, cross talk, and other impairments. Comply with VA 27 05 26 Grounding and Bonding for Communications Systems and with VA 26 05 26 Grounding and Bonding for Electrical Systems.

2.5 METER COMMUNICATION

a. Provide a BACNet network allowing communication from the meters' data heads to the Site Data Aggregation Device.

b. Provide data heads at each meter, converting analog and pulsed information to digital information. Data heads shall allow for up to 24 hours of data storage (including time stamp, measured value, and scaling factor).

(1) Each data head shall reside on a BACnet network using the MS/TP Data Link/Physical layer protocol. Each data head shall have a communication port for connection to an operator interface.

(2) Environment: Data Head hardware shall be suitable for the conditions ranging from (-20°F to 140°F). Data Heads used outdoors and/or in wet ambient conditions shall be mounted within waterproof enclosures and shall be rated for operation at conditions ranging from (-20°F to 140°F).

(3) Provide a local keypad and display for interrogating and editing

data. An optional system security password shall be available to prevent unauthorized use of the keypad and display.

(4) Serviceability. Provide diagnostic LEDs for power, communication, and processor. All wiring connections shall be made to field-removable, modular terminal strips or to a termination card connected by a ribbon cable.

(5) Memory. The building controller shall maintain all BIOS and data in the event of a power loss for at least 72 hours.

(6) Immunity to power and noise. Controller shall be able to operate at 90% to 110% of nominal voltage rating and shall perform an orderly shutdown below 80% nominal voltage. Operation shall be protected against electrical noise of 5 to 120 Hz and from keyed radios up to 5 W at (3 ft).

2.6 ELECTRICAL POWER METERS AND SUB-METERS

a. ELECTRICAL METER APPLICATIONS

(1) Energy meters in the advanced utility metering system shall have models available for amperage ranges of 100-2400 amperes.

(a) The RS-485 communications shall provide communications links up to 10,000 feet long.

(2) Power meters shall be installed as part of the advanced utility metering system.

(a) All setup parameters required by the power meter shall be stored in nonvolatile memory and retained in the event of a control power interruption.

(b) The power meter may be applied in three-phase, three- or four-wire systems.

(c) The power meter shall be capable of being applied without modification at nominal frequencies of 50, 60, or 400 Hz.

(d) The power meter shall provide for onboard data logging, able to log data, alarms, waveforms and events.

b. Physical and Common Requirements

(1) Electrical power meters shall be separately mounted, and enclosed in a NEMA 250, Type 1 enclosure. Environmental Conditions: System components shall be capable of withstanding the following environmental conditions without mechanical or electrical damage or degradation of operating capability:

(a) Ambient conditions of 0 to 140 deg F dry bulb and 20 to 95 percent relative humidity, noncondensing.

c. Current and voltage ratings:

(1) Designed for use with current inputs from standard instrument current transformers with 5-A secondary and shall have a metering range of 0-10 A.

(2) Withstand ratings shall be not less than 15 A, continuous; 50 A, lasting over 10 seconds, no more frequently than once per hour; 500 A, lasting 1 second, no more frequently than once per hour.

(3) Voltage inputs from standard instrument potential transformers with 120 volt secondary output. The power meter shall support PT primaries through 3.2 MV.

(4) The power meter shall operate properly over a wide range of control power including 90-457 VAC or 100-300 VDC.

d. Electrical measurements and calculated values

(1) Power meters shall include the following rms Real-Time Measurements:

(a) Current: Each phase, neutral, average of three phases, percent unbalance.

(b) Voltage: Line-to-line each phase, line-to-line average of three phases, line-to-neutral each phase, line-to-neutral average of three phases, line-to-neutral percent unbalance.

(c) Power: Per phase and three-phase total.

(d) Reactive Power: Per phase and three-phase total.

(e) Apparent Power: Per phase and three-phase total.

(f) True Power Factor: Per phase and three-phase total.

(g) Displacement Power Factor: Per phase and three-phase total.

(h) Frequency.

(i) THD: Current and voltage.

(j) Accumulated Energy: Real kWh, reactive kVARh, apparent kVAh (signed/absolute).

(k) Incremental Energy: Real kWh, reactive kVARh, apparent kVAh (signed/absolute).

(l) Conditional Energy: Real kWh, reactive kVARh, apparent kVAh (signed/absolute).

(2) Power meters shall perform the following demand current calculations, per phase, three-phase average and neutral:

(a) Present.

(b) Running average.

(c) Last completed interval.

(d) Peak.

(3) Power meters shall perform the following demand real power

calculations, three-phase total:

(a) Present.

(b) Running average.

(c) Last completed interval.

(d) Predicted.

(e) Peak.

(f) Coincident with peak kVA demand.

(g) Coincident with kVAR demand.

(4) Power meters shall perform the following demand reactive power calculations, three-phase total:

(a) Present.

(b) Running average.

(c) Last completed interval.

(d) Predicted.

(e) Peak.

(f) Coincident with peak kVA demand.

(g) Coincident with kVAR demand.

(5) Power meters shall perform the following demand apparent power calculations, three-phase total:

(a) Present.

(b) Running average.

(c) Last completed interval.

(d) Predicted.

(e) Peak.

(f) Coincident with peak kVA demand.

(g) Coincident with kVAR demand.

(6) Power meters shall perform the following average true power factor calculations, demand coincident, three-phase total:

(a) Last completed interval.

(b) Coincident with kW peak.

(c) Coincident with kVAR peak.

(d) Coincident with kVA peak.

(7) Power Analysis Values:

(a) THD, Voltage and Current: Per phase, three phase, and neutral.

(b) Displacement Power Factor: Per phase, three phase.

(c) Fundamental Voltage, Magnitude and Angle: Per phase.

(d) Fundamental Currents, Magnitude and Angle: Per phase.

(e) Fundamental Real Power: Per phase, three phase.

(f) Fundamental Reactive Power: Per phase.

(g) Harmonic Power: Per phase, three phase.

(h) Phase rotation.

i Unbalance: Current and voltage.

(j) Harmonic Magnitudes and Angles for Current and Voltages: Per phase, up to 31st harmonic.

(8) Power meters shall perform one of the following demand calculations, selectable by the User; meters shall be capable of performance of all of the following demand calculations.

(a) Block interval with optional subintervals: Adjustable for 1-minute intervals, from 1 to 60 minutes. User-defined parameters for the following block intervals:

1. Sliding block that calculates demand every second, with intervals less than 15 minutes, and every 15 seconds with an interval between 15 and 60 minutes.

2. Fixed block that calculates demand at end of the interval.

3. Rolling block subinterval that calculates demand at end of each subinterval and displays it at end of the interval.

(b) Demand calculations initiated by a Utility-furnished synchronization signal:

1. Signal is a pulse from an external source. Demand period begins with every pulse. Calculation shall be configurable as either a block or rolling block calculation.

2. Signal is a communication signal. Calculation shall be configurable as either a block or rolling block calculation.

3. Demand can be synchronized with clock in the power meter.

(c) Minimum and maximum values: Record monthly minimum and maximum values, including date and time of record. For three-phase measurements, identify phase of recorded value. Record the following parameters:

1. Line-to-line voltage.

2. Line-to-neutral voltage.

3. Current per phase.

4. Line-to-line voltage unbalance.

5. Line-to-neutral voltage unbalance.

6. Power factor.

7. Displacement power factor.

8. Total power.

9. Total reactive power.

10)Total apparent power.
11)THD voltage L-L.
12)THD voltage L-N.
13)THD current.
14)Frequency.

(d) Harmonic calculation: display and record the following:

1. Harmonic magnitudes and angles for each phase voltage and current through 31st harmonic. Calculate for all three phases, current and voltage, and residual current. Current and voltage information for all phases shall be obtained simultaneously from same cycle.

2. Harmonic magnitude reported as a percentage of the fundamental or as a percentage of rms values, as selected by the VA.

e. Waveform Capture:

(1) Capture and store steady-state waveforms of voltage and current channels; initiated manually. Each capture shall be for 3 cycles, 128 data points for each cycle, allowing resolution of harmonics to 31st harmonic of basic 60 Hz.

(2) Capture and store disturbance waveform captures of voltage and current channels, initiated automatically based on an alarm event. Each capture shall be fully configurable for duration with resolution of at least 128 data points per cycle, for all channels simultaneously. Waveform shall be configurable to capture pre-event cycles for analysis.

(3) Store captured waveforms in internal nonvolatile memory; available for PC display, archiving, and analysis.

f. Meter accuracy:

(1) Comply with ANSI C12.20, Class 0.5; and IEC 62053-22, Class 0.5 for revenue meters.

(2) Accuracy from Light to Full Rating:

(a) Power: Accurate to 0.5 percent of reading.

(b) Voltage and Current: Accurate to 0.5 percent of reading.

(c) Power Factor: Plus or minus 0.005, from 0.5 leading to 0.5 lagging.

(d) Frequency: Plus or minus 0.01 Hz at 45 to 67 Hz.

g. Meter input, sampling, display, output, recording and reading Capabilities

(1) Input: One digital input signal.

(a) Normal mode for on/off signal.

(b) Demand interval synchronization pulse, accepting a demand synchronization pulse from a utility demand meter.

(c) Conditional energy signal to control conditional energy accumulation.

(d) GPS time synchronization.

(2) Sampling:

(a) Current and voltage shall be digitally sampled at a rate high enough to provide accuracy to 63rd harmonic of 60-Hz fundamental.

(b) Power monitor shall provide continuous sampling at a rate of 128 samples per cycle on all voltage and current channels in the meter.

(3) Display Monitor:

(a) Backlighted LCD to display metered data with touch-screen or touch-pad selecting device.

(b) Touch-screen display shall be a minimum 12-inch diagonal, resolution of 800 by 600 RGB pixels, 256 colors; NEMA 250, Type 1 display enclosure.

(c) Display four values on one screen at same time.

1. Coordinate list below with meter capabilities specified in subparagraphs above.

2. Current, per phase rms, three-phase average and neutral.

3. Voltage, phase to phase, phase to neutral, and three-phase averages of phase to phase and phase to neutral.

4. Real power, per phase and three-phase total.

5. Reactive power, per phase and three-phase total.

6. Apparent power, per phase and three-phase total.

7. Power factor, per phase and three-phase total.

8. Frequency.

9. Demand current, per phase and three-phase average.

10. Demand real power, three-phase total.

11. Demand apparent power, three-phase total.

12. Accumulated energy (MWh and MVARh).

13. THD, current and voltage, per phase.

(d) Reset: Allow reset of the following parameters at the display:

1. Peak demand current.

2. Peak demand power (kW) and peak demand apparent power (kVA).

3. Energy (MWh) and reactive energy (MVARh).

(4) Outputs:

(a) Operated either by user command sent via communication link, or set to operate in response to user-defined alarm or event.

(b) Closed in either a momentary or latched mode as defined by user.

(c) Each output relay used in a momentary contact mode shall have an independent timer that can be set by user.

(d) One digital KY pulse to a user-definable increment of energy measurement. Output ratings shall be up to 120-V ac, 300-V dc, 50 mA, and provide 3500-V rms isolation.

(e) One relay output module, providing a load voltage range from 20- to 240-V ac or from 20- to 30-V dc, supporting a load current of 2 A.

(f) Output Relay Control:

1. Relay outputs shall operate either by user command sent via communication link or in response to user-defined alarm or event.

2. Normally open and normally closed contacts, field configured to operate as follows:

a) Normal contact closure where contacts change state for as long as signal exists.

b) Latched mode when contacts change state on receipts of a pickup signal; changed state is held until a dropout signal is received.

c) Timed mode when contacts change state on receipt of a

pickup signal; changed state is held for a preprogrammed duration.

d) End of power demand interval when relay operates as synchronization pulse for other devices.

e) Energy Pulse Output: Relay pulses quantities used for absolute kWh, absolute kVARh, kVAh, kWh In, kVARh In, kWh Out, and kVARh Out.

f) Output controlled by multiple alarms using Boolean-type logic.

(5) Onboard Data Logging:

(a) Store logged data, alarms, events, and waveforms in 2 MB of onboard nonvolatile memory.

(b) Stored Data:

1. Billing Log: User configurable; data shall be recorded every 15 minutes, identified by month, day, and 15-minute interval. Accumulate 24 months of monthly data, 32 days of daily data, and between 2 to 52 days of 15-minute interval data, depending on number of quantities selected.

2. Custom Data Logs: three user-defined log(s) holding up to 96 parameters. Date and time stamp each entry to the second and include the following user definitions:

a) Schedule interval.

b) Event definition.

c) Configured as "fill-and-hold" or "circular, first-in first-out."

3. Alarm Log: Include time, date, event information, and coincident information for each defined alarm or event.

4. Waveform Log: Store captured waveforms configured as "fill-and-hold" or "circular, first-in first-out."

(c) Default values for all logs shall be initially set at factory, with logging to begin on device power up.

(6) Alarms.

(a) User Options:

1. Define pickup, dropout, and delay.

2. Assign one of four severity levels to make it easier for user to respond to the most important events first.

3. Allow for combining up to four alarms using Boolean-type logic statements for outputting a single alarm.

(b) Alarm Events:

1. Over/undercurrent.
2. Over/undervoltage.
3. Current imbalance.
4. Phase loss, current.
5. Phase loss, voltage.
6. Voltage imbalance.
7. Over kW demand.
8. Phase reversal.
9. Digital input off/on.
10. End of incremental energy interval.
11. End of demand interval.

2.7 WATER, OIL, GAS METER DEVICES

a. Water, oil and gas meter applications:

(1) Steam Meters: provide vortex-shedding flowmeters, along with temperature sensors and pressure transducers to develop the energy flow.

(2) Steam Condensate Meters: provide a magnetic flowmeter in new installations; provide an ultrasonic or vortex-shedding flowmeter in existing installations which service interruption is not allowed. Provide temperature and pressure transducers to develop the energy flow.

(3) Natural Gas Meters: provide vortex-shedding flowmeters with pressure sensors.

(4) Liquefied Petroleum Gas Meters: provide vortex-shedding flowmeters with pressure sensors.

(5) Potable (Domestic) Water: provide a magnetic flowmeter in new installations; provide an ultrasonic or vortex-shedding flowmeter with pressure sensor in existing installations which service interruption is not allowed.

(6) Reclaimed (storm or gray): provide a magnetic flowmeter in new installations; provide a vortex-shedding flowmeter with pressure sensor in existing installations which service interruption is not allowed.

(7) Make-up Water Meters to Cooling Towers, Evaporative Cooling Systems and Boiler Systems: provide a magnetic flowmeter in new installations; provide a vortex-shedding flowmeter with pressure sensor in existing installations which service interruption is not allowed.

(8) Oil Meters

(a) No. 2 Oils and other Light Oils (less than 60 SSU viscosity at (100 degrees F)): use screw-type or positive-displacement (such as rotating piston or oscillating piston) flowmeters.

(b) No. 6 and Other Heavy Oils (more than 90 SSU viscosity at (100 degrees F)): use positive-displacement flowmeters(such as rotating piston or oscillating piston).

(9) HVAC Hydronic System Water Meters

(a) Chilled Water Systems: provide vortex-shedding flowmeters with pressure and temperature sensors to determine energy flow.

(b) Heating Water Systems: provide vortex-shedding flowmeters with pressure and temperature sensors to determine energy flow.

b. Associated Devices (to provide outside air conditions as well as energy metering, not merely flow metering):

(1) Temperature Sensors: Resistance Temperature Device (RTD) with an integral transmitter type.

(a) Immersion sensors shall be provided with a separable thermowell. Pressure rating of well is to be consistent with the system pressure in which it is to be installed.

(b) Outdoor air temperature sensors shall have watertight inlet fittings and be shielded from direct sunlight.

(c) Output Signal: 4-20 ma or digital.

(2) Humidity Sensors: Bulk polymer sensing element type.

(a) Outdoor humidity sensors shall be furnished with element guard and mounting plate and have a sensing range of 0 to 100 percent RH.

(b) Output Signal: 4-20 ma continuous output signal.

(3) Pressure sensors.

(a) Gas Pressure Transmitter: Nondirectional sensor with suitable range for expected input, and temperature compensated.

(b) Water Pressure Transmitters: Stainless-steel diaphragm construction, suitable for service; minimum 150-psig operating pressure and tested to 300-psig; linear output 4 to 20 mA.

(4) Thermowells.

(a) Description: Pressure-tight, socket-type fitting made for insertion into piping tee fitting. Stepped shank unless straight or tapered shank is indicated. ASME B40.200. Bore diameter required to match thermometer bulb or stem. Insertion length required to match thermometer bulb or stem. Provide a lagging extension on thermowells for insulated piping and tubing. Provide bushings. Use a mixture of graphite and glycerin for the thermowell’s heat transfer medium.

1. Material for Use with Copper Tubing: copper nickel (90-10).

2. Material for Use with Steel Piping: stainless steel.

c. Turbine flowmeters (natural gas duty).

(1) Flowmeter shall be as specified below. Provide data head on meter as specified in this section.

(2) Meter shall be designed for 125 psi. Meter's pipe connection flanges shall be ANSI Class 125 or Class 150 . All meter bearings and gearing shall be in areas sealed from metered fluid and contaminants. Metering transducers shall be operated through magnetic coupling. The measuring devices shall be contained within a module that can be removed from the meter body for service and calibration without breaking the system piping connections. Meter shall be constructed for measured fluid's chemical characteristics. Construct meter of corrosion-resistant materials, or provide a corrosion-resistant coating.

(3) Provide a data head on the meter.

(4) Straightening Vanes: Provide as recommended by the meter manufacturer for the actual installation arrangement.

(5) Performance:

(a) Transmitted signal from flowmeter and its transmitter shall have a total (rms) accuracy plus or minus 0.5% of flow rate.

(b) Flowmeter accuracy shall be no more than plus or minus 0.1%. Flowmeter repeatability shall be no more than 0.3% of actual flow rate. Meter shall be designed to minimize vibration effect and to provide elimination of this effect.

(c) Minimum turndown capability shall be 10: 1.

(d) Pressure drop shall not exceed 5 inches WC).

(6) Calibration: Factory calibrated. Furnish three-point curve spanning required flow range on actual meter furnished.

(a) This meter requires annual re-calibration.

(7) Accessories:

(a) Filter: Shall have replaceable glass-fiber or cellulose cartridge with ten micron or smaller particle retention. Filter enclosure shall be the pipe size of the meter or larger as required by pressure drop considerations. Static pressure capability shall be at least twice lockup pressure of service supply regulators. Maximum pressure loss 5 inches WC) at maximum design flow rate of meter. Plug all drains or instrumentation outlets. Provide vent with cock for relieving pressure in filter.

d. Turbine flowmeters (water duty).

(1) Flowmeter shall be as specified in Section 23 09 23, in the "water

flow sensors" paragraph. Provide data head on meter as specified in this section.

(2) Sensor shall be insertion turbine type with turbine element, retractor and preamplifier/transmitter mounted on a two-inch full port isolation valve; assembly easily removed or installed as a single unit under line pressure through the isolation valve without interference with process flow; calibrated scale shall allow precise positioning of the flow element to the required insertion depth within plus or minute 0.05 inch; wetted parts shall be constructed of stainless steel. Operating power shall be nominal 24 VDC. Local instantaneous flow indicator shall be LED type in NEMA 4 enclosure with 3-1/2 digit display, for wall or panel mounting.

(a) Ambient conditions: (-40 to 140 degrees F), 5 to 100 percent humidity

(b) Operating conditions: (125 psig), (30 to 250 degrees F), (0.5 to 40 feet per second) velocity.

(3) Performance:

(a) Transmitted signal from flowmeter and its transmitter shall have a total (rms) accuracy plus or minus 1% of actual flow rate.

(b) Flowmeter accuracy shall be no more than plus or minus 0.1% of actual flow rate. Flowmeter repeatability shall be no more than 0.3% of actual flow rate. Meter shall be designed to minimize vibration effect and to provide elimination of this effect.

(c) Minimum turndown capability shall be 20: 1.

(d) Pressure drop shall be as scheduled, maximum 1% of line pressure in lines sized 4 inches and larger.

(e) Ambient temperature effects, less than 0.005 percent calibrated span per (degree F) temperature change.

(f) RFI effect - flow meter shall not be affected by RFI.

(g) Power supply effect less than 0.02 percent of actual flow rate for a variation of plus or minus 10 percent power supply.

(4) Provide a data head on the meter.

(a) Preamplifier mounted on meter shall provide 4-20 ma divided pulse output or switch closure signal for units of volume or mass per a time base. Signal transmission distance shall be a minimum of (6,000 feet)..

(5) Straightening Vanes: Provide as recommended by the meter manufacturer for the actual installation arrangement.

e. Screw type flowmeters (No. 2 oil duty)

(1) Meters shall have cast iron cases, nitrided steel spindles, Viton seals, threaded pipe connections, designed for pressure exceeding set pressure, plus 25 percent, of nearest upstream relief valve.

Meters shall be rated for (250 degrees F) if utilized for heated oil.

(2) Provide a meter data head.

(a) Meter Registers: Hermetically sealed flow computer with digital flow rate readout and digital register for totalizer with at least five digits located at meter, positioned for easy viewing. Provide a data head on the meter.

(3) Performance:

(a) Transmitted signal from flowmeter and its transmitter shall have a total (rms) accuracy plus or minus 1% of flow rate.

(b) Flowmeter accuracy shall be no more than plus or minus 0.1%, over the required flow range. Flowmeter repeatability shall be no more than 0.2% of actual flow rate. Meter shall be designed to minimize vibration effect and to provide elimination of this effect.

(c) Minimum turndown capability shall be 10: 1. Its maximum fluid pressure drop through meter shall be as scheduled.

f. Vortex-shedding flowmeters.

(1) Meter shall have an all-welded flanged 316 stainless steel meter body with no seals. No sensor parts shall be exposed to the flow stream. Provide a 316 stainless steel trapezoidal shedder bar, sensing by detecting stresses in the shedder bar caused by vortices, and dual piezoelectric crystals located outside the process flow sense the shed vortices (dual crystal alignment cancels effects of noise and vibration). Design meter for Schedule 40 piping.

(a) Meter shall be suitable for 25% warmer than the fluid operating temperature and for 25% higher than either the fluid’s operating pressure or 25% higher than the piping system’s safety valve set pressure, whichever is higher.

(b) Meter flanges shall be Class 300 or higher, if required by the piping system’s temperature and pressure Class.

(c) Meter shall be suitable for installation in ambient conditions ranging from (-20 to 140 degrees F).

(2) Provide meter data head.

(a) Meters shall have digital readout of pressure-compensated flow rate and totalization located at transmitter and transmit flow rate and totalization digital signals to the Site Data Aggregation Device and recorders. As an option, pressure compensation and the compensated flow rate may be performed and displayed by the Site Data Aggregation Device receiving signals from the flow meter and from a pressure transmitter.

(b) Provide programmable microprocessor electronics with on-board programming. Output signals shall be immune to ambient temperature swings. Processor shall include continuous

self-diagnostic routines that identify electronics problems and provide a warning. Electronics shall be replaceable in the field without affecting metering accuracy. Provide power supply as recommended by meter manufacturer. Mount electronics in a NEMA 4 enclosure separate from meter body in position accessible from platform or floor without the use of a portable ladder.

1. Power supply to meter and transmitter shall be 120V/60hz. Provide a Class 2 control voltage transformer for 24VDC power to meter as needed.

2. Provide an internal battery, provided for 24-month retention of RAM contents when all other power sources are removed.

(3) Performance:

(a) Transmitted signal from flowmeter and its transmitter shall have a total (rms) accuracy plus or minus 1.5% of flow rate.

(b) Flowmeter accuracy shall be no more than plus or minus 1% of span for gasses and plus or minus 0.7% of span for liquids. Flowmeter repeatability shall be no more than 0.2% of actual flow rate. Meter shall be designed to minimize vibration effect and to provide elimination of this effect.

(c) Minimum turndown ratio shall be 20: 1 for gasses and liquids. Maximum fluid pressure drop shall be as scheduled.

g. Ultrasonic (Doppler and time of travel) flowmeters.

(1) Provide a clamp-on flowmeter precluding the requirement of penetrating into the process pipe. The flowmeter shall be completely microprocessor based utilizing the transit-time flow measurement technique. The flowmeter shall employ the phase detection multiple pulse transmit principle in conjunction with multiple frequency axial beam transducer technology to insure operation on liquids with solids and or bubbles. In addition, the flowmeter shall incorporate an alternate Doppler method measurement mode for highly aerated or heavy solid bearing liquids.

(2) Provide a meter data head.

(a) The flowmeter shall provide automatic transducer spacing for clamp-on transducers utilizing a prefabricated mounting frame or mounting track (ruler scales shall not be acceptable), the meter shall also support in-line transducers. The meter shall also provide automatic Reynolds Number and liquid sonic velocity variation compensation and live zero flow measurement.

1. By use of either transit-time or Doppler modes of operation, the flowmeter shall be capable of measuring all liquids in full sonically conductive pipes.

(b) The flowmeter shall have the ability to indicate flow rate, flow velocity, total flow, signal strength, liquid sonic velocity, Reynolds Number and liquid aeration level.

(c) The flowmeter shall be equipped with an integral front panel keypad and multifunction 240 X 128 pixel LCD display. In

addition, the flowmeter shall provide self and application diagnostics to isolate any fault conditions to either equipment failure or abnormal process conditions.

(d) The flowmeter shall have full HELP menu routines corresponding to all levels of programming and operation.

(e) The flowmeter electronics shall be housed in a NEMA 4 enclosure and powered by 90-240VAC, 50-60Hz. Two isolated 4 to 20 maDC and two 0 to 5000 Hz pulse outputs proportional to flow shall be provided. The current outputs must be capable of driving a 1000-ohm resistive load. In addition, the unit shall provide two 0 to 10 volt outputs and four SPDT alarm relays assignable to flow velocity, liquid sonic velocity, signal strength or liquid aeration.

(f) Provide an internal 1 MB data logger shall be provided to allow storage of all measured and calculated variables and alarms in intervals of 10 minutes.

(g) Two each bi-directional communications ports shall be provided.

1. One each RS-485 with Modbus RTU or BACnet protocol.

(3) Performance:

(a) The flowmeter shall have an accuracy of plus or minus 1% of flow over span. Repeatability shall be 0.25% of flow.

(b) Meter shall have a flow sensitivity of 0.001 fps at any flow rate including no flow conditions.

h. Magnetic flowmeters.

(1) Meter shall have an all-welded flanged 316 stainless steel engineered flow tube with no seals. No sensor parts shall be exposed to the flow stream. Design meter for mating with Schedule 40 piping.

(a) Meter shall be suitable for 25% warmer than the fluid operating temperature and for 25% higher than either the fluid’s operating pressure or 25% higher than the piping system’s safety valve set pressure, whichever is higher.

(b) Meter flanges shall be Class 150, Class 300 or higher, if required by the piping system’s temperature and pressure Class.

(c) Meter shall be suitable for installation in ambient conditions ranging from (-20 to 140 degrees F).

(2) Provide meter data head.

(a) Meters shall have digital readout of pressure-compensated flow rate and totalization located at transmitter and transmit flow rate and totalization digital signals to the Site Data Aggregation Device and recorders. As an option, pressure compensation and the compensated flow rate may be performed and displayed by the Site Data Aggregation Device receiving signals from the flow meter and from a pressure transmitter.

(b) Provide programmable microprocessor electronics with on-board programming. Output signals shall be immune to ambient temperature swings. Processor shall include continuous self-diagnostic routines that identify electronics problems and provide a warning. Electronics shall be replaceable in the field without affecting metering accuracy. Provide power supply as recommended by meter manufacturer. Mount electronics in a NEMA 4 enclosure separate from meter body in position accessible from platform or floor without the use of a portable ladder.

1. Power supply to meter and transmitter shall be 120V/60hz. Provide a Class 2 control voltage transformer for 24VDC power to meter as needed.

(3) Performance:

(a) Transmitted signal from flowmeter and its transmitter shall have a total (rms) accuracy plus or minus 1.5% of flow rate.

(b) Flowmeter accuracy shall be no more than plus or minus 1.5% of actual flow rate for gasses and plus or minus 1% of actual flow rate for liquids. Flowmeter repeatability shall be no more than 0.2% of actual flow rate. Meter shall be designed to minimize vibration effect and to provide elimination of this effect.

(c) Minimum turndown ratio shall be 20: 1 for gasses and liquids. Maximum fluid pressure drop shall be as scheduled.

i. Positive-displacement oil flowmeters.

(1) Meter shall be a rotating- or oscillating-piston meter with a and aluminum piston, cast bronze body, safety filter, viton o-rings and Class 150 flanges mating with Schedule 40 piping. Meter shall transmit pulse signals to meter’s data head.

(2) Provide meter data head.

(a) Meters shall have digital readout flow rate and totalization located at transmitter and transmit flow rate and totalization digital signals to the Site Data Aggregation Device and recorders. As an option,

(b) Provide programmable microprocessor electronics with on-board programming. Output signals shall be immune to ambient temperature swings. Processor shall include continuous self-diagnostic routines that identify electronics problems and provide a warning. Electronics shall be replaceable in the field without affecting metering accuracy. Provide power supply as recommended by meter manufacturer. Mount electronics in a NEMA 4 enclosure separate from meter body in position accessible from platform or floor without the use of a portable ladder.

1. Power supply to meter and transmitter shall be 120V/60hz. Provide a Class 2 control voltage transformer for 24VDC power to meter as needed.

(3) Performance:

(a) Flowmeter accuracy shall be no more than plus or minus 1.5% of actual flow rate.

(b) Minimum turndown ratio shall be 10: 1.

PART 3 EXECUTION

3.1 INSTALLATION REQUIREMENTS

a. Cabling

(1) Install Category 5e UTP, Category 6 UTP, and optical fiber cabling system as detailed in TIA-568.1, TIA-568.2, or TIA-568.3.

(2) Screw terminals shall not be used except where specifically indicated on plans.

(3) Use an approved insulation displacement connection (IDC) tool kit for copper cable terminations.

(4) Do not untwist Category 5e, Category 6 UTP cables more than 1/2 inch from the point of termination to maintain cable geometry.

(5) Provide service loop on each end of the cable, 10 feet at the server rack and 12 inches at the meter.

(6) Do not exceed manufacturers' cable pull tensions for copper and optical fiber cables.

(7) Provide a device to monitor cable pull tensions. Do not exceed (25 pounds) pull tension for four pair copper cables.

(8) Do not chafe or damage outer jacket materials.

(9) Use only lubricants approved by cable manufacturer.

(10) Do not over cinch cables, or crush cables with staples.

(11) For UTP cable, bend radii shall not be less than four times the cable diameter.

(12) Cables shall be terminated; no cable shall contain unterminated elements.

(13) Cables shall not be spliced.

(14) Label cabling in accordance with paragraph Labeling in this section.

b. Labeling

(1) Labels: Provide labeling in accordance with TIA-606. Handwritten labeling is unacceptable. Stenciled lettering for all circuits shall be provided using laser printer.

(2) Cables: Cables shall be labeled using color labels on both ends with identifiers in accordance with TIA-606.

c. Grounding: ground exposed, non-current-carrying metallic parts of

electrical equipment, metallic raceway systems, grounding conductor in metallic and nonmetallic raceways, telecommunications system grounds, and grounding conductor of nonmetallic sheathed cables, as well as equipment to eliminate shock hazard and to minimize ground loops, common-mode returns, noise pickup, cross talk, and other impairments. Comply with VA 27 05 26 GROUNDING AND BONDING FOR COMMUNICATIONS SYSTEMS and with VA 26 05 26 GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS.

d. Surge Protection

(1) Provide surge protective devices on all metallic cables entering and leaving an interior environment to an exterior environment or vice versa, i.e. surge protective device at each interior location of a penetration to the exterior environment.

e. Network Hardware

(1) System components and appurtenances shall be installed in accordance with the manufacturer's instructions and as shown. Necessary interconnections, services, and adjustments required for a complete and operable wired or wireless data transmission system shall be provided and shall be fully integrated with the configured network chosen for the project.

f. Computer Hardware

(1) Provide the server(s) or personal computer workstation(s) where shown on the plans or indicated diagrammatically.

g. Computer Software

(1) User friendly software shall be suitable for operation on computer workstations which serve as site data aggregation devices by monitoring the meters in the system, recording events, indicating alarm conditions, and logging and displaying system reports.

(2) The software shall be developed by the manufacturer of the monitoring devices, and shall be designed specifically for energy, power monitoring and control. Additional utilities, i.e. water, air gas, electric and steam shall also be easily integrated.

(3) The software shall be configured, not programmed. All software shall be configured by the vendor and delivered ready to use. This configuration shall include preparation of all graphics, displays, and interactive one-line diagrams required as a part of this project.

(a) Configuration shall be to the point that when monitoring devices are required to be added, the user shall only convey to the software the communications address and type of device.

(4) The software shall be a standard product offering with no customization required and clients shall interface with the server or computer workstation via Internet Explorer browser.

(a) The web-enabled interactive graphics client shall only reside on the server PC, client PC not required to host any application software other than Internet Explorer 6.0 SP1 or higher browser to

become a fully functional system.

h. Electrical Meters

(1) Power monitoring and control components shall all be factory installed, wired and tested prior to shipment to the job site.

(2) All control power, CT, PT and data communications wire shall be factory wired and harnessed within the equipment enclosure.

(3) Where external circuit connections are required, terminal blocks shall be provided and the manufacturer’s drawings must clearly identify the interconnection requirements including wire type to be used.

(4) All wiring required to externally connect separate equipment lineups shall be furnished and installed at the site as part of the contractor’s responsibility.

(5) Contractor interconnection wiring requirements shall be clearly identified on the power monitoring and control system shop drawings.

i. Water, Oil and Gas Meters

(1) Thermowells

(a) Install thermowells with socket extending a minimum of 2 inches into fluid or one-third of pipe diameter and in vertical position in piping tees.

(b) Install thermowells of sizes required to match temperature sensor connectors. Include bushings if required to match sizes.

(c) Install thermowells with extension on insulated piping.

(d) Fill thermowells with heat-transfer medium.

(2) Provide a test plug beside each temperature sensor.

(3) Flow meters, general

(a) Install meters and gages adjacent to machines and equipment to allow service and maintenance of meters, gages, machines, and equipment.

(b) Connect flowmeter-system elements to meters, connect flowmeter transmitters to meters, and connect thermal-energy meter transmitters to meters.

(c) Assemble and install connections, tubing, and accessories between flow-measuring elements and flowmeters according to manufacturer's written instructions.

(d) Install flowmeter elements in accessible positions in piping systems.

(e) Install flowmeter, with minimum 20 x pipe diameter straight lengths of pipe upstream and minimum 10 x pipe diameter straight

lengths of pipe downstream from flowmeter unless otherwise indicated by manufacturer's written instructions.

(f) Mount thermal-energy meters on wall if accessible; if not, provide brackets to support meters.

3.2 ADJUSTING AND IDENTIFICATION

a. Install a permanent wire marker on each wire at each termination.

b. Identifying numbers and letters on the wire markers shall correspond to those on the wiring diagrams used for installing the systems.

c. Wire markers shall retain their markings after cleaning.

3.3 FIELD QUALITY CONTROL

a. The power monitoring and control system vendor must be able to provide development, integration and installation services required to complete and turn over a fully functional system including:

(1) Project management to coordinate personnel, information and on-site supervision for the various levels and functions of suppliers required for completion of the project.

(2) All technical coordination, installation, integration, and testing of all components.

(3) Detailed system design and system drawings.

b. Cabling, equipment and hardware manufacturers shall have a minimum of 5 years experience in the manufacturing, assembly, and factory testing of components which comply with EIA TIA-568.1, EIA TIA-568.2 and EIA TIA-568.3.

c. The network cabling contractor shall be a firm which is regularly and professionally engaged in the business of the applications, installation, and testing of the specified network cabling systems and equipment. The contractor shall demonstrate experience in providing successful systems within the past 3 years. Submit documentation for a minimum of three and a maximum of five successful network cabling system installations.

(1) Supervisors and installers assigned to the installation of this system or any of its components shall be Building Industry Consulting Services International (BICSI) Registered Cabling Installers, Technician Level. Submit documentation of current BICSI certification for each of the key personnel.

3.4 ACCEPTANCE TESTING

a. Develop testing procedures to address all specified functions and components of the Advanced Utility Metering System (AUMS). Testing shall demonstrate proper and anticipated responses to normal and abnormal operating conditions.

(1) Provide skilled technicians to start and operate equipment.

(2) Coordinate with equipment manufacturers to determine specific

requirements to maintain the validity of the warranty.

(3) Correct deficiencies and make necessary adjustments to O&M manuals and as-built drawings for issues identified in testing.

(4) Provide all tools to start, check-out and functionally test equipment and systems.

(5) Correct deficiencies and make necessary adjustments to O&M manuals and as-built drawings for issues identified in any testing

(6) Review test procedures, testing and results with Government.

b. Testing checklists: Develop project-specific checklists to document the systems and all components are installed in accordance with the manufacturers recommendation and the Contract Documents.

c. Before testing, the following prerequisite items must be completed.

(1) All related equipment has been started and start-up reports and checklists submitted and approved as ready for testing:

(2) All associated system functions for all interlocking systems are programmed and operable per contract documents.

(3) All punchlist items for the AUMS and equipment are corrected.

(4) The test procedures reviewed and approved.

(5) Safeties and operating ranges reviewed.

d. The following testing shall be included:

(1) Demonstrate reporting of data and alarm conditions for each point and ensure that alarms are received at the assigned location, including Site Data Collection Device.

(2) Demonstrate ability of software program to function for the intended application.

(3) Demonstrate via graphed trends to show the reports are executed in correct manner.

(4) Demonstrate that the meter readings are accurate using portable NIST traceable portable devices and calibrated valves in the piping system

(5) Demonstrate that the systems perform during power loss and resumption of power.

e. Copper cables: Contractor shall provide all necessary testing equipment to test all copper network circuit cables. Tests shall conform to TIA-568.1 Permanent Link testing criteria. All testers are to be TIA-568.2, Level IIe compliant. The primary field test parameters are:

(1) Wire map: The wire map test is intended to verify pair to pin termination at each end and check for installation connectivity errors. For each of the conductors in the cable, the wire map

indicates:

(a) Continuity to the remote end

(b) Shorts between any two or more conductors

(c) Crossed pairs

(d) Reversed pairs

(e) Split pairs

(f) Any other mis-wiring

(2) Length requirements: The maximum physical length of the basic link shall be 94 meters (including test equipment cords).

(3) Insertion Loss: Worst case insertion loss relative to the maximum insertion loss allowed shall be reported.

(4) Near-end crosstalk (NEXT) loss: Field tests of NEXT shall be performed at both ends of the test configuration.

(5) Power sum near-end crosstalk (PSNEXT) loss

(6) Equal-level far-end crosstalk (ELFEXT: Field tests of ELFEXT shall be performed at both ends of the test configuration

(7) Power sum equal-level far-end crosstalk (PSELFEXT): Must be determined from both ends of the cable. Power sum Near End Crosstalk is not a category 3 parameter. For all frequencies from 1 to 100 MHz, the category 5e PSELFEXT of the cabling shall be measured in accordance with annex E of TIA-568.2 and shall meet the values determined using equations (12) and (13) for the permanent link. PSELFEXT is not a required category 3 measurement parameter.

(8) Return loss: Includes all the components of the link. The limits are based on the category of components and cable lengths. Return loss must be tested at both ends of the cable. Cabling return loss is not a required measurement for category 3 cabling.

(9) Propagation delay and delay skew: Propagation delay is the time it takes for a signal to propagate from one end to the other. Propagation delay shall be measured in accordance with annex D of TIA-568.2. The maximum propagation delay for all category permanent link configurations shall not exceed 498 ns measured at 10 MHz. Delay skew is a measurement of the signaling delay difference from the fastest pair to the slowest. Delay skew shall be measured in accordance with annex D of TIA-568.2. The maximum delay skew for all category permanent link configurations shall not exceed 44 ns.

(10) Administration: In addition to Pass/Fail indications, measured values of test parameters should be recorded in the administration system. Any reconfiguration of link components after testing may change the performance of the link and thus invalidates previous test results. Such links shall require retesting to regain conformance.

(11) Test equipment connectors and cords: Adapter cords that are qualified and determined by the test equipment manufacturer to be suitable for permanent link measurements shall be used to attach the field tester to the permanent link under consideration.

(12) Test setup: The permanent link test configuration is to be used by installers and users of data telecommunications systems to verify the performance of permanently installed cabling. A schematic representation of the permanent link is illustrated in figure 1. The permanent link consists of up to (295 ft) of horizontal cabling and one connection at each end and may also include an optional transition/consolidation point connection. The permanent link excludes both the cable portion of the field test instrument cord and the connection to the field test instrument.

(13) Replace or repair and cables, connectors, and/or terminations found to be defective.

(14) Repair, replace, and/or re-work any or all defective components to achieve cabling tests which meet or exceed TIA-568.1 permanent link requirements prior to acceptance of the installation or payment for services.

f. Optical Fiber cables: Contractor shall provide all necessary testing equipment to test all optical fiber cables.

(1) Attenuation Testing:

(a) Singlemode testing shall conform to TIA-526-7 Method A.1 single jumper reference and TIA-568.1 requirements for link segment testing.

(b) Multimode testing shall conform to TIA-526-14 Method B single jumper reference and TIA-568.1 requirements for link segment testing.

(c) Attenuation testing shall be performed in one direction at each operating wavelength.

(d) Testing of backbone fiber optic cabling shall be performed from main telecommunications room to each telecommunications room.

(e) Testing of horizontal fiber optic cabling shall be performed from telecommunications room to station outlet location.

(f) Tester shall be capable of recording and reporting test reading in an electronic format.

(2) OTDR Testing:

(a) OTDR testing is required on all backbone fiber optic cables

(b) The test shall be preformed as per the EIA/TIA 455-61.

(c) Multimode testing shall be performed with a minimum 80 meter launch cable.

(d) Singlemode testing shall be performed with a minimum of 500

meter launch cable.

(e) Tests shall be performed on each fiber in each direction at both operating wavelengths.

(3) Test report data shall reference cables by cable labeling standards. Tests shall be submitted on a 1.5mb, 3.5" DOS formatted floppy disk. Contractor shall provide tests in the native file format of the tester. Contractor shall provide all software needed to view, print, and edit tests.

(4) Replace or repair and defective cables, connectors, terminations, etc.

(5) Mated connector pairs shall have no more than 0.5dB loss. Fusion splices shall have no more than.15dB loss per splice. Cable attenuation shall be no more than 2% more than the attenuation of the cable on the reel as certified at the factory. Repair, replace, and/or rework any or all defective components to achieve specified test results prior to acceptance of the installation or payment for services.

g. Wireless Modems: Test system by sending 100,000 commands. Frame error rate shall not be greater than 5 out 100,000 commands.

3.5 DEMONSTRATION AND INSTRUCTION

a. Furnish the services of a factory-trained engineer or technician for a total of two four-hour classes to instruct designated Facility Information Technologies personnel. Instruction shall include cross connection, corrective, and preventive maintenance of the wired network system and connectivity equipment.

b. Before the System can be accepted by the VA, this training must be provided and executed. Training will be scheduled at the convenience of the Facilities Contracting Officer and Chief of Engineering Service.

c. On-site start-up and training of the advanced utility metering system shall include a complete working demonstration of the system with simulation of possible operating conditions that may be encountered.

(1) Include any documentation and hands-on exercises necessary to enable electrical and mechanical operations personnel to assume full operating responsibility for the advanced utility monitoring system after completion of the training period.

d. Include 6 days on-site start-up assistance and 3 days on-site training in two sessions separated by minimum 1 month.

e. Regularly schedule and make available factory training for VA staff training on all aspects of advanced utility metering system including:

(1) Comprehensive software and hardware setup, configuration, and operation.

(2) Advanced monitoring and data reporting.

(3) Advanced power quality and disturbance monitoring.

f. Before the system is accepted by the VA, the contractor shall walk-through the installation with the VA's representative and the design engineer to verify proper installation. The contractor may be requested to open enclosures and terminal compartments to verify cable labeling and/or installation compliance.

g. As-built drawings shall be provided noting the exact cable path and cable labeling information. Drawings in.DWG format will be available to the contractor. As-builts shall be submitted to the VA on disk saved as.DXF or.DWG files. Redline hardcopies shall be provided as well. CAD generated as-built information shall be shown on a new layer named AS_BUILT.

-- End of Section --

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SECTION 26 05 11
11/22

REQUIREMENTS FOR ELECTRICAL INSTALLATIONS

PART 1 GENERAL

1.1 DESCRIPTION

a. This Section applies to all Sections of Division 26.

b. Furnish and install electrical systems, materials, equipment, and accessories in accordance with the specifications and drawings. Capacities and ratings of motors, transformers, conductors and cable, switchboards, switchgear, panelboards, motor control centers, generators, automatic transfer switches, and other items and arrangements for the specified items are shown on the drawings.

c. Electrical service entrance equipment and arrangements for temporary and permanent connections to the electric utility company’s system shall conform to the electric utility company's requirements. Coordinate fuses, circuit breakers and relays with the electric utility company’s system, and obtain electric utility company approval for sizes and settings of these devices.

d. Conductor ampacities specified or shown on the drawings are based on copper conductors, with the conduit and raceways sized per NFPA 70. Aluminum conductors are prohibited.

1.2 MINIMUM REQUIREMENTS

a. The latest International Building Code (IBC), Underwriters Laboratories, Inc. (UL), Institute of Electrical and Electronics Engineers (IEEE), and National Fire Protection Association (NFPA) codes and standards are the minimum requirements for materials and installation.

b. The drawings and specifications shall govern in those instances where requirements are greater than those stated in the above codes and standards.

1.3 TEST STANDARDS

a. All materials and equipment shall be listed, labeled, or certified by a Nationally Recognized Testing Laboratory (NRTL) to meet Underwriters Laboratories, Inc. (UL), standards where test standards have been established. Materials and equipment which are not covered by UL standards will be accepted, providing that materials and equipment are listed, labeled, certified or otherwise determined to meet the safety requirements of a NRTL. Materials and equipment which no NRTL accepts, certifies, lists, labels, or determines to be safe, will be considered if inspected or tested in accordance with national industrial standards, such as ANSI, NEMA, and NETA. Evidence of compliance shall include certified test reports and definitive shop drawings.

b. Definitions:

(1) Listed: Materials and equipment included in a list published by an organization that is acceptable to the Authority Having Jurisdiction and concerned with evaluation of products or services, that maintains periodic inspection of production or listed materials and equipment or periodic evaluation of services, and whose listing states that the materials and equipment either meets appropriate designated standards or has been tested and found suitable for a specified purpose.

(2) Labeled: Materials and equipment to which has been attached a label, symbol, or other identifying mark of an organization that is acceptable to the Authority Having Jurisdiction and concerned with product evaluation, that maintains periodic inspection of production of labeled materials and equipment, and by whose labeling the manufacturer indicates compliance with appropriate standards or performance in a specified manner.

(3) Certified: Materials and equipment which:

(a) Have been tested and found by a NRTL to meet nationally recognized standards or to be safe for use in a specified manner.

(b) Are periodically inspected by a NRTL.

(c) Bear a label, tag, or other record of certification.

(4) Nationally Recognized Testing Laboratory: Testing laboratory which is recognized and approved by the Secretary of Labor in accordance with OSHA regulations.

1.4 QUALIFICATIONS (PRODUCTS AND SERVICES)

a. Manufacturer’s Qualifications: The manufacturer shall regularly and currently produce, as one of the manufacturer's principal products, the materials and equipment specified for this project, and shall have manufactured the materials and equipment for at least three years.

b. Product Qualification:

(1) Manufacturer's materials and equipment shall have been in satisfactory operation, on three installations of similar size and type as this project, for at least three years.

(2) The Government reserves the right to require the Contractor to submit a list of installations where the materials and equipment have been in operation before approval.

c. Service Qualifications: There shall be a permanent service organization maintained or trained by the manufacturer which will render satisfactory service to this installation within four hours of receipt of notification that service is needed. Submit name and address of service organizations.

1.5 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only. Where conflicts occur these specifications and

the VHA standards will govern.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 70B	(2023) Recommended Practice for Electrical Equipment Maintenance
NFPA 70E	(2024) Standard for Electrical Safety in the Workplace
NFPA 99	(2024) Health Care Facilities Code
NFPA 110	(2025) Standard for Emergency and Standby Power Systems

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910-SUBPART J	General Environmenal Controls
29 CFR 1910-SUBPART K	Medical and First Aid
29 CFR 1910-SUBPART S	Electrical

1.6 MANUFACTURED PRODUCTS

a. Materials and equipment furnished shall be of current production by manufacturers regularly engaged in the manufacture of such items, and for which replacement parts shall be available. Materials and equipment furnished shall be new, and shall have superior quality and freshness.

b. When more than one unit of the same class or type of materials and equipment is required, such units shall be the product of a single manufacturer.

c. Equipment Assemblies and Components:

(1) Components of an assembled unit need not be products of the same manufacturer.

(2) Manufacturers of equipment assemblies, which include components made by others, shall assume complete responsibility for the final assembled unit.

(3) Components shall be compatible with each other and with the total assembly for the intended service.

(4) Constituent parts which are similar shall be the product of a single manufacturer.

d. Factory wiring and terminals shall be identified on the equipment being furnished and on all wiring diagrams.

e. When Factory Tests are specified, Factory Tests shall be performed in the factory by the equipment manufacturer, and witnessed by the contractor. In addition, the following requirements shall be complied

with:

(1) The Government shall have the option of witnessing factory tests. The Contractor shall notify the Government through the Contracting Officer a minimum of 30 days prior to the manufacturer's performing of the factory tests.

(2) When factory tests are successful, contractor shall furnish four copies of the equipment manufacturer's certified test reports to the Contracting Officer 14 days prior to shipment of the equipment, and not more than 90 days after completion of the factory tests.

(3) When factory tests are not successful, factory tests shall be repeated in the factory by the equipment manufacturer, and witnessed by the Contractor. The Contractor shall be liable for all additional expenses for the Government to witness factory re-testing.

1.7 VARIATIONS FROM CONTRACT REQUIREMENTS

Where the Government or the Contractor requests variations from the contract requirements, the connecting work and related components shall include, but not be limited to additions or changes to branch circuits, circuit protective devices, conduits, wire, feeders, controls, panels and installation methods.

1.8 MATERIALS AND EQUIPMENT PROTECTION

Materials and equipment shall be protected during shipment and storage against physical damage, vermin, dirt, corrosive substances, fumes, moisture, cold and rain.

a. Store materials and equipment indoors in clean dry space with uniform temperature to prevent condensation.

b. During installation, equipment shall be protected against entry of foreign matter, and be vacuum-cleaned both inside and outside before testing and operating. Compressed air shall not be used to clean equipment. Remove loose packing and flammable materials from inside equipment.

c. Damaged equipment shall be repaired or replaced, as determined by the Contracting Officer.

d. Painted surfaces shall be protected with factory installed removable heavy kraft paper, sheet vinyl or equal.

e. Damaged paint on equipment shall be refinished with the same quality of paint and workmanship as used by the manufacturer so repaired areas are not obvious.

1.9 WORK PERFORMANCE

a. All electrical work shall comply with requirements of the latest NFPA 70, NFPA 70B, NFPA 70E, NFPA 99, NFPA 110, 29 CFR 1910-SUBPART J, 29 CFR 1910-SUBPART K, and 29 CFR 1910-SUBPART S, in addition to other references required by Contract.

b. Job site safety and worker safety is the responsibility of the Contractor.

c. Electrical work shall be accomplished with all affected circuits or equipment de-energized. However, energized electrical work may be performed only for the non-destructive and non-invasive diagnostic testing(s), or when scheduled outage poses an imminent hazard to patient care, safety, or physical security. In such case, all aspects of energized electrical work, such as the availability of appropriate/correct personal protective equipment (PPE) and the use of PPE, shall comply with the latest NFPA 70E, as well as the following requirements:

(1) Only Qualified Person(s) shall perform energized electrical work. Supervisor of Qualified Person(s) shall witness the work of its entirety to ensure compliance with safety requirements and approved work plan.

(2) At least two weeks before initiating any energized electrical work, the Contractor and the Qualified Person(s) who is designated to perform the work shall visually inspect, verify and confirm that the work area and electrical equipment can safely accommodate the work involved.

(3) At least two weeks before initiating any energized electrical work, the Contractor shall develop and submit a job specific work plan, and energized electrical work request to the Contracting Officer, and Medical Center’s Chief Engineer or his/her designee. At the minimum, the work plan must include relevant information such as proposed work schedule, area of work, description of work, name(s) of Supervisor and Qualified Person(s) performing the work, equipment to be used, procedures to be used on and near the live electrical equipment, barriers to be installed, safety equipment to be used, and exit pathways.

(4) Energized electrical work shall begin only after the Contractor has obtained written approval of the work plan, and the energized electrical work request from the Contracting Officer, and Medical Center’s Chief Engineer or his/her designee. The Contractor shall make these approved documents present and available at the time and place of energized electrical work.

(5) Energized electrical work shall begin only after the Contractor has invited and received acknowledgment from the Contracting Officer, and Medical Center’s Chief Engineer or his/her designee to witness the work.

d. For work that affects existing electrical systems, arrange, phase and perform work to assure minimal interference with normal functioning of the facility. Refer to Article OPERATIONS AND STORAGE AREAS under Section 01 00 00, GENERAL REQUIREMENTS.

e. New work shall be installed and connected to existing work neatly, safely and professionally. Disturbed or damaged work shall be replaced or repaired to its prior conditions, as required by Section 01 00 00, GENERAL REQUIREMENTS.

f. Coordinate location of equipment and conduit with other trades to minimize interference.

1.10 EQUIPMENT INSTALLATION AND REQUIREMENTS

a. Equipment location shall be as close as practical to locations shown on the drawings.

b. Working clearances shall not be less than specified in the NFPA 70.

c. Inaccessible Equipment:

(1) Where the Government determines that the Contractor has installed equipment not readily accessible for operation and maintenance, the equipment shall be removed and reinstalled as directed at no additional cost to the Government.

(2) "Readily accessible" is defined as being capable of being reached quickly for operation, maintenance, or inspections without the use of ladders, or without climbing or crawling under or over obstacles such as, but not limited to, motors, pumps, belt guards, transformers, piping, ductwork, conduit and raceways.

d. Electrical service entrance equipment and arrangements for temporary and permanent connections to the electric utility company's system shall conform to the electric utility company's requirements. Coordinate fuses, circuit breakers and relays with the electric utility company's system, and obtain electric utility company approval for sizes and settings of these devices.

1.11 EQUIPMENT IDENTIFICATION

a. In addition to the requirements of NFPA 70, install an identification sign which clearly indicates information required for use and maintenance of items such as switchboards and switchgear, panelboards, cabinets, motor controllers, fused and non-fused safety switches, generators, automatic transfer switches, separately enclosed circuit breakers, individual breakers and controllers in switchboards, switchgear and motor control assemblies, control devices and other significant equipment.

b. Identification signs for Normal Power System equipment shall be laminated black phenolic resin with a white core with engraved lettering. Identification signs for Essential Electrical System (EES) equipment, as defined in NFPA 70, shall be laminated red phenolic resin with a white core with engraved lettering. Lettering shall be a minimum of 1/2 inch high. Identification signs shall indicate equipment designation, rated bus amperage, voltage, number of phases, number of wires, and type of EES power branch as applicable. Secure nameplates with screws.

c. Install adhesive arc flash warning labels on all equipment as required by the latest NFPA 70E. Label shall show specific and correct information for specific equipment based on its arc flash calculations. Label shall show the followings:

(1) Nominal system voltage.

(2) Equipment/bus name, date prepared, and manufacturer name and address.

(3) Arc flash boundary.

(4) Available arc flash incident energy and the corresponding working distance.

(5) Minimum arc rating of clothing.

(6) Site-specific level of PPE.

1.12 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Interconnection Wiring Diagrams For Communication And Signal Systems; G

SD-10 Operation and Maintenance Data

Maintenance And Operation Manuals; G

Parts List; G

1.13 SUBMITTAL INSTRUCTIONS

a. Submit to the Contracting Officer .

b. The Government's approval shall be obtained for all materials and equipment before delivery to the job site. Delivery, storage or installation of materials and equipment which has not had prior approval will not be permitted.

c. All submittals shall include six copies of adequate descriptive literature, catalog cuts, shop drawings, test reports, certifications, samples, and other data necessary for the Government to ascertain that the proposed materials and equipment comply with drawing and specification requirements. Catalog cuts submitted for approval shall be legible and clearly identify specific materials and equipment being submitted.

d. Submittals for individual systems and equipment assemblies which consist of more than one item or component shall be made for the system or assembly as a whole. Partial submittals will not be considered for approval.

(1) Mark the submittals, "SUBMITTED UNDER SECTION_____".

(2) Submittals shall be marked to show specification reference including the section and paragraph numbers.

(3) Submit each section separately.

e. The submittals shall include the following:

(1) Information that confirms compliance with contract requirements.

Include the manufacturer's name, model or catalog numbers, catalog information, technical data sheets, shop drawings, manuals, pictures, nameplate data, and test reports as required.

(2) Elementary and interconnection wiring diagrams for communication and signal systems, control systems, and equipment assemblies. All terminal points and wiring shall be identified on wiring diagrams.

(3) Parts list which shall include information for replacement parts and ordering instructions, as recommended by the equipment manufacturer.

f. Maintenance and Operation Manuals:

(1) Submit as required for systems and equipment specified in the technical sections. Furnish in hardcover binders or an approved equivalent.

(2) Inscribe the following identification on the cover: the words "MAINTENANCE AND OPERATION MANUAL," the name and location of the system, material, equipment, building, name of Contractor, and contract name and number. Include in the manual the names, addresses, and telephone numbers of each subcontractor installing the system or equipment and the local representatives for the material or equipment.

(3) Provide a table of contents and assemble the manual to conform to the table of contents, with tab sheets placed before instructions covering the subject. The instructions shall be legible and easily read, with large sheets of drawings folded in.

(4) The manuals shall include:

(a) Internal and interconnecting wiring and control diagrams with data to explain detailed operation and control of the equipment.

(b) A control sequence describing start-up, operation, and shutdown.

(c) Description of the function of each principal item of equipment.

(d) Installation instructions.

(e) Safety precautions for operation and maintenance.

(f) Diagrams and illustrations.

(g) Periodic maintenance and testing procedures and frequencies, including replacement parts numbers.

(h) Performance data.

(i) Pictorial "exploded" parts list with part numbers. Emphasis shall be placed on the use of special tools and instruments. The list shall indicate sources of supply, recommended spare and replacement parts, and name of servicing organization.

(j) List of factory approved or qualified permanent servicing organizations for equipment repair and periodic testing and maintenance, including addresses and factory certification qualifications.

g. Approvals will be based on complete submission of shop drawings, manuals, test reports, certifications, and samples as applicable.

h. After approval and prior to installation, furnish the Contracting Officer with one sample of each of the following:

(1) A minimum 12 inches length of each type and size of wire and cable along with the tag from the coils or reels from which the sample was taken. The length of the sample shall be sufficient to show all markings provided by the manufacturer.

(2) Each type of conduit coupling, bushing, and termination fitting.

(3) Conduit hangers, clamps, and supports.

(4) Duct sealing compound.

(5) Each type of receptacle, toggle switch, lighting control sensor, outlet box, manual motor starter, device wall plate, engraved nameplate, wire and cable splicing and terminating material, and branch circuit single pole molded case circuit breaker.

1.14 SINGULAR NUMBER

Where any device or part of equipment is referred to in these specifications in the singular number (e.g., "the switch"), this reference shall be deemed to apply to as many such devices as are required to complete the installation as shown on the drawings.

1.15 POLYCHLORINATED BIPHENYL (PCB) EQUIPMENT

a. This project requires the removal, transport, and disposal of electrical equipment containing Polychlorinated Biphenyls (PCB) in accordance with the Federal Toxic Substances Control Act (TSCA).

b. The equipment to be removed is shown on the drawings.

c. The selective demolition shall be in accordance with Section 02 41 00, DEMOLITION.

1.16 ACCEPTANCE CHECKS AND TESTS

a. The Contractor shall furnish the instruments, materials, and labor for tests.

b. Where systems are comprised of components specified in more than one section of Division 26, the Contractor shall coordinate the installation, testing, and adjustment of all components between various manufacturer’s representatives and technicians so that a complete, functional, and operational system is delivered to the Government.

c. When test results indicate any defects, the Contractor shall repair or replace the defective materials or equipment, and repeat the tests for

the equipment. Repair, replacement, and re-testing shall be accomplished at no additional cost to the Government.

1.17 WARRANTY

All work performed and all equipment and material furnished under this Division shall be free from defects and shall remain so for a period of one year from the date of acceptance of the entire installation by the Contracting Officer for the Government.

1.18 INSTRUCTION

a. Instruction to designated Government personnel shall be provided for the particular equipment or system as required in each associated technical specification section.

b. Furnish the services of competent and factory-trained instructors to give full instruction in the adjustment, operation, and maintenance of the specified equipment and system, including pertinent safety requirements. Instructors shall be thoroughly familiar with all aspects of the installation, and shall be factory-trained in operating theory as well as practical operation and maintenance procedures.

c. A training schedule shall be developed and submitted by the Contractor and approved by the Contracting Officer at least 30 days prior to the planned training.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

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SECTION 26 05 13
06/25

MEDIUM-VOLTAGE CABLES

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, and connection of medium-voltage cables, indicated as cable or cables in this section, and medium-voltage cable splices and terminations.

1.2 RELATED WORK

a. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

b. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

c. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduits for medium-voltage cables.

d. Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION: Manholes and ducts for medium-voltage cables.

e. Section 26 12 19, PAD-MOUNTED, MEDIUM-VOLTAGE TRANSFORMERS: Medium-voltage cable terminations for use in pad-mounted, medium-voltage transformers.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 FACTORY TESTS

a. Factory Tests shall be required.

b. Factory Tests shall be in accordance with Paragraph, MANUFACTURED PRODUCTS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirement:

(1) A representative sample of Medium-voltage cables from each lot shall be factory tested per NEMA WC 74/ICEA S-93-639 to ensure that there are no electrical defects in that specific lot of cable.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualified Worker Approval; G,

Electric Utility Company Approval; G,

SD-02 Shop Drawings

Shop Drawings; G,

SD-04 Samples

Samples; G,

SD-06 Test Reports

Factory Tests; G,

Field Test Reports; G

SD-07 Certificates

Certifications; G,

1.6 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

(1) Shop Drawings:

(a) Submit sufficient information to demonstrate compliance with drawings and specifications.

(b) Submit the following data for approval:

1. Complete electrical ratings.

2. Installation instructions.

(2) Samples: After approval of submittal and prior to installation, Contractor shall furnish sample in accordance with Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

(3) Certifications:

(a) Factory Test Reports: Submit certified factory production test reports for approval.

(b) Field Test Reports: Submit field test reports for approval.

(c) Compatibility: Submit a certificate from the cable manufacturer that the splices and terminations are approved for use with the cable.

(d) Two weeks prior to final inspection, submit the following.

1. Certification by the manufacturer that the cables,

splices, and terminations conform to the requirements of the drawings and specifications.

2. Certification by the Contractor that the cables, splices, and terminations have been properly installed and tested.

3. Certification by the Contractor that each splice and each termination were completely installed in a single continuous work period by the same qualified worker or qualified installation crew without any overnight interruption.

(4) Qualified Worker Approval:

(a) Qualified workers who install cables, splices, and terminations shall have a minimum of five years of experience splicing and terminating cables, including experience with the materials in the approved splices and terminations. Qualified workers who perform cable testing shall have a minimum of five year s of experience performing electrical testing of medium-voltage cables, including the ability to understand, interpret results and develop test reports.

(b) Furnish satisfactory proof of such experience for each qualified worker who tests, splices, or terminates the cables.

(5) Electric Utility Company Approval: Prior to construction, obtain written approval from the electric utility company (United Illuminating Company) for the following items:

(a) Service entrance cables, splices, and terminations, defined as connections to equipment owned and operated by the utility company.

(b) A list of qualified workers who will install, splice, and terminate the service entrance cables.

1.7 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

ASTM INTERNATIONAL (ASTM)

ASTM B3	(2013; R 2024) Standard Specification for Soft or Annealed Copper Wire
ASTM B8	(2023) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 48	(2020) Test Procedures and Requirements for Alternating-Current Cable Terminations Used on Shielded Cables Having Laminated Insulation Rated 2.5 kV through 765 kV or Extruded Insulation Rated 2.5 kV through 500 kV

IEEE 386	(2016) Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV
IEEE 400	(2012) Guide for Field Testing and Evaluation of the Insulation of Shielded Power Cable Systems Rated 5 kV and Above
IEEE 400.2	(2024) Guide for Field Testing of Shielded Power Cable Systems Using Very Low Frequency (VLF)
IEEE 404	(2012) Standard for Extruded and Laminated Dielectric Shielded Cable Joints Rated 2500 V to 500,000 V

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA WC 74/ICEA S-93-639	(2022) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

1.8 SHIPMENT AND STORAGE

a. Cable shall be shipped on reels such that it is protected against physical, mechanical and environmental damage. Each end of each length of cable shall be hermetically sealed with manufacturer’s end caps and securely attached to the reel.

b. Cable stored and/or cut on site shall have the ends turned down, and sealed with cable manufacturer’s standard cable end seals, or field-installed heat-shrink cable end seals.

PART 2 PRODUCTS

2.1 CABLE

a. Cable shall be in accordance with ASTM, IEEE, NEC, NEMA and UL, and as shown on the drawings.

b. Single conductor Class B concentric-laystranded copper conforming to ASTM B8 and ASTM B3.

c. Voltage Rating:

(1) 15,000 V cable shall be used on all distribution systems with voltages ranging from 4,800 V to 15,000 V.

d. Insulation:

(1) Insulation level shall be 133 percent.

(2) Types of insulation:

(a) Cable type abbreviation, EPR: Ethylene propylene rubber insulation shall be thermosetting, light and heat stabilized.

e. Insulation shield shall be semi-conducting. Conductor shield shall be semi-conducting.

f. Insulation shall be wrapped with copper shielding tape, helically-applied over semi-conducting insulation shield.

g. Heavy duty, overall protective polyvinyl chloride jacket shall enclose every cable. The manufacturer's name, cable type and size, and other pertinent information shall be marked or molded clearly on the overall protective jacket.

h. Cable temperature ratings for continuous operation, emergency overload operation, and short circuit operation shall be not less than the NFPA 70, , /ICEA S-96-659, or NEMA WC 74/ICEA S-93-639 standard for the respective cable.

2.2 SPLICES AND TERMINATIONS

a. Materials shall be compatible with the cables being spliced and terminated, and shall be suitable for the prevailing environmental conditions.

b. In locations where moisture might be present, the splices shall be watertight. In manholes and pullboxes, the splices shall be submersible.

c. Splices:

(1) Shall comply with IEEE 404. Include all components required for complete splice, with detailed instructions.

d. Terminations:

(1) Shall comply with IEEE 48. Include shield ground strap for shielded cable terminations.

(2) Class 1 terminations for indoor use: Kit with stress-relief tube, molded-silicone rubber insulator modules, and compression-type connector.

(3) Class 3 terminations for outdoor use: Kit with stress cone and compression-type connector.

(4) Dead-break terminations for indoor and outdoor use: 600 A deadbreak premolded rubber elbow connectors with bushing inserts, suitable for submersible applications. Separable connectors shall comply with the requirements of IEEE 386, and shall be interchangeable between suppliers. Allow sufficient slack in medium-voltage cable, ground, and drain wires to permit elbow connectors to be moved to their respective parking stands.

(5) Ground metallic cable shields with a device designed for that purpose, consisting of a solderless connector enclosed in watertight rubber housing covering the entire assembly.

(6) Provide insulated cable supports to relieve any strain imposed by

cable weight or movement. Ground cable supports to the grounding system.

2.3 FIREPROOFING TAPE

Fireproofing tape shall be flexible, non-corrosive, self-extinguishing, arcproof, and fireproof intumescent elastomer. Securing tape shall be glass cloth electrical tape not less than 7 mils thick, and 0.75 inch wide.

PART 3 EXECUTION

3.1 GENERAL

a. Installation shall be in accordance with NFPA 70, as shown on the drawings, and manufacturer’s instructions.

b. Cable shall be installed in conduit above grade and duct bank below grade.

c. All cables in the same racewayshall be pulled simultaneously.

d. Conductors of different systems (e.g., 5kV and 15kV) shall not be installed in the same raceway.

e. Splice the cables only in manholes for underground feeders or in pullboxes for above ground feeders.

f. Ground shields in accordance with Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS.

g. Cable maximum pull length, maximum pulling tension, and minimum bend radius shall conform with the recommendations of the manufacturer. Provide calculations showing maximum pulling tension, sidewall pressure, jam ration, bend radius, pulling direction, and anticipated pulling tensions for each pull greater than 50 feet or more than two bends.

h. Use suitable lubricating compounds on the cables to prevent pulling damage. Provide compounds that are not injurious to the cable jacket and do not harden or become adhesive.

i. Seal the cable ends prior to pulling, to prevent the entry of moisture or lubricant.

3.2 PROTECTION DURING SPLICING OPERATIONS

Blowers shall be provided to force fresh air into manholes where free movement or circulation of air is obstructed. Waterproof protective coverings shall be available on the work site to provide protection against moisture while a splice is being made. Pumps shall be used to keep manholes dry during splicing operations. Under no conditions shall a splice or termination be made that exposes the interior of a cable to moisture. A manhole ring at least 6 inches above ground shall be used around the manhole entrance to keep surface water from entering the manhole. Unused ducts shall be plugged and water seepage through ducts in use shall be stopped before splicing.

3.3 PULLING CABLES IN DUCTS AND MANHOLES

a. Cables shall be pulled into ducts with equipment designed for this purpose, including power-driven winches, cable-feeding flexible tube guides, cable grips, pulling eyes, and lubricants. A sufficient number of qualified workers and equipment shall be employed to ensure the careful and proper installation of the cable.

b. Cable reels shall be set up at the side of the manhole opening and above the duct or hatch level, allowing cables to enter through the opening without reverse bending. Flexible tube guides shall be installed through the opening in a manner that will prevent cables from rubbing on the edges of any structural member.

c. Cable shall be unreeled from the top of the reel. Pay-out shall be carefully controlled. Cables to be pulled shall be attached through a swivel to the main pulling wire by means of a suitable cable grip and pulling eye.

d. Woven-wire cable grips shall be used to grip the cable end when pulling small cables and short straight lengths of heavier cables.

e. Pulling eyes shall be attached to the cable conductors to prevent damage to the cable structure, pulling eyes are required for all manhole pulls.

f. Cables shall be liberally coated with a suitable lubricant as they enter the tube guide or duct. Rollers, sheaves, or tube guides around which the cable is pulled shall conform to the minimum bending radius of the cable.

g. Cables shall be pulled into ducts at a reasonable speed. Cable pulling using a vehicle shall not be permitted. Pulling operations shall be stopped immediately at any indication of binding or obstruction, and shall not be resumed until the potential for damage to the cable is corrected. Sufficient slack shall be provided for free movement of cable due to expansion or contraction.

h. Splices in manholes shall be firmly supported on cable racks. Cable ends shall overlap at the ends of a section to provide sufficient undamaged cable for splicing.

i. Cables cut in the field shall have the cut ends immediately sealed to prevent entrance of moisture.

3.4 SPLICES AND TERMINATIONS

a. Install the materials as recommended by the manufacturer, including precautions pertaining to air temperature and humidity during installation.

b. Installation shall be executed by qualified person trained to perform medium-voltage equipment installations. Tools shall be as recommended or provided by the manufacturer. Installation shall comply with manufacturer’s instructions.

c. Splices for underground feeders shall only occur in manholes and only where expressly identified on the drawings. Splices in manholes shall be located midway between cable racks on walls of manholes, and

supported with cable arms at approximately the same elevation as the enclosing duct.

d. Where the Government determines that unsatisfactory splices and terminations have been installed, the Contractor shall replace the unsatisfactory splices and terminations with approved material at no additional cost to the Government.

3.5 FIREPROOFING

a. Cover all cable segments exposed in manholes and pullboxes with fireproofing tape.

b. Apply the tape in a single layer, wrapped in a half-lap manner, or as recommended by the manufacturer. Extend the tape not less than 1 inch into each duct.

c. At each end of a taped cable section, secure the fireproof tape in place with glass cloth tape.

3.6 CIRCUIT IDENTIFICATION OF FEEDERS

In each manhole and pullbox, install permanent identification tags on each circuit's cables to clearly designate the circuit identification and voltage. The tags shall be the embossed stainless steel type, 1.5 inches in diameter and minimum 40 mils thick. Attach tags with plastic ties. Position the tags so they will be easy to read after the fireproofing tape is installed.

3.7 ACCEPTANCE CHECKS AND TESTS

a. General:

(1) Perform tests in accordance with the latest IEEE 400 and IEEE 400.2, manufacturer's recommendations, and as specified in this specification.

(2) Contractor shall make arrangements to have tests witnessed by the Contracting Officer. Contractor shall proceed with tests only after obtaining approval from the Contracting Officer.

b. Visual Inspection: Perform visual inspection prior to electrical tests.

(1) Inspect exposed sections of cables for physical damage.

(2) Inspect shield grounding, cable supports, splices, and terminations.

(3) Verify that visible cable bends meet manufacturer’s minimum bending radius requirement.

(4) Verify installation of fireproofing tape and identification tags.

(5) At the time of final acceptance, Contractor shall provide the Contracting Officer visual field inspection notes, findings, and photographs detailing accessible inspection locations.

c. Electrical Tests - New Cables: Perform preparation and tests in order

shown below:

(1) Preparation Prior to Testing: Splices and terminations applied to new cables shall be completed prior to testing. For renovation installation, ends of new cables intended to be spliced to existing service-aged cables shall be prepared (cut back) to allow testing without flashover or tracking. Cables shall not be connected to other equipment while under test.

(2) Perform Insulation-Resistance Test. Test all cables with respect to ground and adjacent cables. All adjacent cables shall be grounded during testing.

(a) Apply test voltage for a period sufficient to stabilize output voltage and insulation resistance measurement.

(b) Test data shall include megohm, applied test voltage, and leakage current readings.

(c) Further testing shall not continue unless the insulation resistance test results meet or exceed the values listed below. Test voltages and minimum acceptable resistance values shall be:

Voltage Class	Test Voltage	Min. Insulation Resistance
5kV	2,500 VDC	1,000 megohms
15kV	2,500 VDC	5,000 megohms
25kV	5,000 VDC	20,000 megohms
35kV	15,000 VDC	100,000 megohms

(3) Perform Tan Delta test. Review test readings with the Contracting Officer prior to proceeding with the Very Low Frequency (VLF) Withstand test

(4) Perform Very Low Frequency (VLF) Withstand test. Utilize test voltages in accordance with IEEE 400.2.

d. Electrical Tests - Service-Aged Cables: Tests shall be performed for serviced-age cables before inter-connecting to new cables. Perform tests in order shown below:

(1) Preparation Prior to Testing: Splices and terminations applied to cables shall be completed prior to testing. Ends of cables intended to be spliced to existing service-aged cables shall be prepared (cut back) to allow testing without flashover or tracking. Cables shall not be connected to other equipment while under test.

(2) Perform Insulation-Resistance Test. Test all cables with respect to ground and adjacent cables. All adjacent cables shall be grounded during testing.

(a) Apply test voltage for a period sufficient to stabilize output voltage and insulation resistance measurement.

(b) Test data shall include megohm, applied test voltage, and leakage current readings.

(c) Further testing shall not continue unless the insulation resistance test results meet or exceed the values listed below. Test voltages and minimum acceptable resistance values shall be:

Voltage Class	Test Voltage	Min. Insulation Resistance
5kV	2,500 VDC	1,000 megohms
15kV	2,500 VDC	5,000 megohms
25kV	5,000 VDC	20,000 megohms
35kV	15,000 VDC	100,000 megohms

(3) Perform Tan Delta test. Review test readings with the Contracting Officer prior to proceeding with the VLF Withstand test.

(4) Perform VLF Withstand test. Utilize test voltages in accordance with IEEE 400.2.

e. Electrical Tests – Inter-connected New Cables and Service-Aged Cables: After successful Tan Delta and VLF Withstand testing of new cables and service-aged cables, perform final splicing inter-connecting between new and service-aged cables. Once new and service-aged cables are completely inter-connected, conduct Tan Delta and VLF Withstand tests for the entire inter-connected cable. Utilize maintenance test voltage for VLF Withstand testing.

f. Field Test Report: Submit a field test report to the Contracting Officer that includes the following information:

(1) Project Name, Location, Test Date.

(2) Name of Technician and Company performing the test.

(3) Ambient temperature and humidity at time of test.

(4) Name, Model Number and Description of Test Equipment used.

(5) Circuit identification, cable length, cable type and size, insulation type, cable manufacturer, service age (if any), voltage rating, description of splices or terminations.

(6) Visual field inspection notes, findings, and photographs.

(7) Insulation Resistance Test results:

(a) Test voltage.

(b) Measurement in Megohms.

(c) Leakage current.

(8) Tan Delta results:

(a) Test voltage.

(b) Waveform (sinusoidal or cosine-rectangular).

(c) Mean Tan Delta at V0.

(d) Stability measured by Standard Deviation at V0.

(e) Differential Tan Delta.

(f) IEEE Condition Assessment Rating.

(9) VLF Withstand results:

1. Test voltage.

2. Waveform (sinusoidal or cosine-rectangular).

3. Pass/Fail Rating.

(10) Conclusions. If any deficiency is discovered based on test results, provide recommendations for corrective action.

g. Final Acceptance: Final acceptance shall depend upon the satisfactory performance of the cables under test. No cable shall be put into service until all tests are successfully passed, and field test reports have been approved by the Contracting Officer.

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SECTION 26 05 19
11/22

LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, connection, and testing of the electrical conductors and cables for use in electrical systems rated 600 V and below, indicated as cable(s), conductor(s), wire, or wiring in this section.

1.2 RELATED WORK

a. Section 07 84 00, FIRESTOPPING: Sealing around penetrations to maintain the integrity of fire-resistant rated construction.

b. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

c. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

d. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduits for conductors and cables.

e. Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION: Installation of conductors and cables in manholes and ducts.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-07 Certificates

Certifications; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Submit the following data for approval:

(a) Electrical ratings and insulation type for each conductor and cable.

(b) Splicing materials and pulling lubricant.

b. Certifications: Two weeks prior to final inspection, submit the following.

(1) Certification by the manufacturer that the conductors and cables conform to the requirements of the drawings and specifications.

(2) Certification by the Contractor that the conductors and cables have been properly installed, adjusted, and tested.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements and errata) form a part of this specification to the extent referenced. Publications are reference in the text by designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D2304	(2018) Standard Test Method for Thermal Endurance of Rigid Electrical Insulating Materials

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

PART 2 PRODUCTS

2.1 CONDUCTORS AND CABLES

a. Conductors and cables shall be in accordance with ASTM, NEMA, NFPA, UL, as specified herein, and as shown on the drawings.

b. All conductors shall be copper.

c. Single Conductor and Cable:

(1) No. 12 AWG: Minimum size, except where smaller sizes are specified herein or shown on the drawings.

(2) No. 8 AWG and larger: Stranded.

(3) No. 10 AWG and smaller: Solid; except shall be stranded for final connection to motors, transformers, and vibrating equipment.

(4) Insulation: THHN-THWN and XHHW-2. XHHW-2 shall be used for

isolated power systems.

d. Color Code:

(1) No. 10 AWG and smaller: Solid color insulation or solid color coating.

(2) No. 8 AWG and larger: Color-coded using one of the following methods:

(a) Solid color insulation or solid color coating.

(b) Stripes, bands, or hash marks of color specified.

(c) Color using 0.75 inches wide tape.

(4) For modifications and additions to existing wiring systems, color coding shall conform to the existing wiring system.

(5) Conductors shall be color-coded as follows:

208/120 V	Phase	480/277 V
Black	A	Brown
Red	B	
Blue	C	Yellow
White *	Neutral	Gray *
* neutral conductors shall have a color tracer that matches associated phase for single phase circuits.		

(6) Lighting circuit “switch legs”, and 3-way and 4-way switch “traveling wires,” shall have color coding that is unique and distinct (e.g., pink and purple) from the color coding indicated above. The unique color codes shall be solid and in accordance with NFPA 70. Coordinate color coding in the field with the Contracting Officer/.

(7) Color code for isolated power system wiring shall be in accordance with NFPA 70.

2.2 SPLICES

a. Splices shall be in accordance with NFPA 70 and UL. Splices shall only occur where required to terminate fixture or device wiring, or where branch circuits split.

b. Above Ground Splices for No. 10 AWG and Smaller:

(1) Solderless, screw-on, reusable pressure cable type, with integral insulation, approved for copper and aluminum conductors.

(2) The integral insulator shall have a skirt to completely cover the stripped conductors.

(3) The number, size, and combination of conductors used with the connector, as listed on the manufacturer's packaging, shall be strictly followed.

c. Above Ground Splices for No. 8 AWG to No. 4/0 AWG shall be avoided unless expressly identified on the documents. Where identified, splices shall consist of:

(1) Compression, hex screw, or bolt clamp-type of high conductivity and corrosion-resistant material, listed for use with copper and aluminum conductors.

(2) Insulate with materials approved for the particular use, location, voltage, and temperature. Insulation level shall be not less than the insulation level of the conductors being joined.

(3) Splice and insulation shall be product of the same manufacturer.

(4) All bolts, nuts, and washers used with splices shall be zinc-plated steel.

d. Above Ground Splices for 250 kcmil and Larger shall be avoided unless expressly identified on the documents. Where identified, splices shall consist of:

(1) Long barrel “butt-splice” or “sleeve” type compression connectors, with minimum of two compression indents per wire, listed for use with copper and aluminum conductors.

(2) Insulate with materials approved for the particular use, location, voltage, and temperature. Insulation level shall be not less than the insulation level of the conductors being joined.

(3) Splice and insulation shall be product of the same manufacturer.

e. Underground Splices for No. 10 AWG and Smaller shall be avoided unless expressly identified on the documents. Where identified, splices shall consist of:

(1) Solderless, screw-on, reusable pressure cable type, with integral insulation. Listed for wet locations, and approved for copper and aluminum conductors.

(2) The integral insulator shall have a skirt to completely cover the stripped conductors.

(3) The number, size, and combination of conductors used with the connector, as listed on the manufacturer's packaging, shall be strictly followed.

f. Underground Splices for No. 8 AWG and Larger shall be avoided unless expressly identified on the documents. Where identified, splices shall consist of:

(1) Mechanical type, of high conductivity and corrosion-resistant material. Listed for wet locations, and approved for copper and aluminum conductors.

(2) Insulate with materials approved for the particular use, location, voltage, and temperature. Insulation level shall be not less than the insulation level of the conductors being joined.

(3) Splice and insulation shall be product of the same manufacturer.

g. Plastic electrical insulating tape: Per ASTM D2304, flame-retardant, cold and weather resistant.

2.3 CONNECTORS AND TERMINATIONS

a. Mechanical type of high conductivity and corrosion-resistant material, listed for use with copper and aluminum conductors.

b. Long barrel compression type of high conductivity and corrosion-resistant material, with minimum of two compression indents per wire, listed for use with copper and aluminum conductors.

c. All bolts, nuts, and washers used to connect connections and terminations to bus bars or other termination points shall be zinc-plated steel.

2.4 CONTROL WIRING

a. Unless otherwise specified elsewhere in these specifications, control wiring shall be as specified herein, except that the minimum size shall be not less than No. 14 AWG.

b. Control wiring shall be sized such that the voltage drop under in-rush conditions does not adversely affect operation of the controls.

2.5 WIRE LUBRICATING COMPOUND

a. Lubricating compound shall be suitable for the wire insulation and conduit, and shall not harden or become adhesive.

b. Shall not be used on conductors for isolated power systems.

PART 3 EXECUTION

3.1 GENERAL

a. Installation shall be in accordance with NFPA 70, as shown on the drawings, and manufacturer’s instructions.

b. Install all conductors in raceway systems.

c. Splice conductors only in outlet boxes, junction boxes, pullboxes, manholes, or handholes.

d. Conductors of different systems (e.g., 120 V and 277 V) shall not be installed in the same raceway.

e. Install cable supports for all vertical feeders in accordance with NFPA 70. Provide split wedge type which firmly clamps each individual cable and tightens due to cable weight.

f. In panelboards, cabinets, wireways, switches, enclosures, and equipment assemblies, neatly form, train, and tie the conductors with non-metallic ties.

g. For connections to motors, transformers, and vibrating equipment, stranded conductors shall be used only from the last fixed point of connection to the motors, transformers, or vibrating equipment.

h. Use expanding foam or non-hardening duct-seal to seal conduits entering a building, after installation of conductors.

i. Conductor and Cable Pulling:

(1) Provide installation equipment that will prevent the cutting or abrasion of insulation during pulling. Use lubricants approved for the cable.

(2) Use nonmetallic pull ropes.

(3) Attach pull ropes by means of either woven basket grips or pulling eyes attached directly to the conductors.

(4) All conductors in a single conduit shall be pulled simultaneously.

(5) Do not exceed manufacturer’s recommended maximum pulling tensions and sidewall pressure values.

j. No more than three branch circuits shall be installed in any one conduit.

k. When stripping stranded conductors, use a tool that does not damage the conductor or remove conductor strands.

3.2 INSTALLATION IN MANHOLES

a. Train the cables around the manhole walls, but do not bend to a radius less than six times the overall cable diameter.

b. Fireproofing:

(1) Install fireproofing on low-voltage conductors where the low-voltage conductors are installed in the same manholes with medium-voltage conductors.

(2) Use fireproofing tape as specified in Section 26 05 13, MEDIUM-VOLTAGE CABLES, and apply the tape in a single layer, half-lapped, or as recommended by the manufacturer. Install the tape with the coated side towards the cable and extend it not less than 1 inch into each duct.

(3) Secure the fireproofing tape in place by a random wrap of glass cloth tape.

3.3 SPLICE AND TERMINATION INSTALLATION

a. Splices and terminations shall be mechanically and electrically secure, and tightened to manufacturer’s published torque values using a torque screwdriver or wrench.

b. Where the Government determines that unsatisfactory splices or terminations have been installed, replace the splices or terminations at no additional cost to the Government.

3.4 CONDUCTOR IDENTIFICATION

When using colored tape to identify phase, neutral, and ground conductors

larger than No. 8 AWG, apply tape in half-overlapping turns for a minimum of 3 inches from terminal points, and in junction boxes, pullboxes, and manholes. Apply the last two laps of tape with no tension to prevent possible unwinding. Where cable markings are covered by tape, apply tags to cable, stating size and insulation type.

3.5 FEEDER CONDUCTOR IDENTIFICATION

In each interior pullbox and each underground manhole and handhole, install brass tags on all feeder conductors to clearly designate their circuit identification and voltage. The tags shall be the embossed type, 1-1/2 inches in diameter and 40 mils thick. Attach tags with plastic ties.

3.6 EXISTING CONDUCTORS

Unless specifically indicated on the plans, existing conductors shall not be reused.

3.7 CONTROL WIRING INSTALLATION

a. Unless otherwise specified in other sections, install control wiring and connect to equipment to perform the required functions as specified or as shown on the drawings.

b. Install a separate power supply circuit for each system, except where otherwise shown on the drawings.

3.8 CONTROL WIRING IDENTIFICATION

a. Install a permanent wire marker on each wire at each termination.

b. Identifying numbers and letters on the wire markers shall correspond to those on the wiring diagrams used for installing the systems.

c. Wire markers shall retain their markings after cleaning.

d. In each manhole and handhole, install embossed brass tags to identify the system served and function.

3.9 ACCEPTANCE CHECKS AND TESTS

a. Perform in accordance with the manufacturer's recommendations. In addition, include the following:

(1) Visual Inspection and Tests: Inspect physical condition.

(2) Electrical tests:

(a) After installation but before connection to utilization devices, such as fixtures, motors, or appliances, test conductors phase-to-phase and phase-to-ground resistance with an insulation resistance tester. Existing conductors to be reused shall also be tested.

(b) Applied voltage shall be 500 V DC for 300 V rated cable, and 1000 for 600 V rated cable. Apply test for one minute or until reading is constant for 15 seconds, whichever is longer. Minimum insulation resistance values shall not be less than 25 megohms for 300 V rated cable and 100 megohms for 600 V rated cable.

(c) Perform phase rotation test on all three-phase circuits.

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SECTION 26 05 26
11/22

GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies the furnishing, installation, connection, and testing of grounding and bonding equipment, indicated as grounding equipment in this section.

b. “Grounding electrode system” refers to grounding electrode conductors and all electrodes required or allowed by NFPA 70, as well as made, supplementary, and lightning protection system grounding electrodes.

c. The terms “connect” and “bond” are used interchangeably in this section and have the same meaning.

1.2 RELATED WORK

a. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

b. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Low-voltage conductors.

c. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduit and boxes.

d. Section 26 12 19, PAD-MOUNTED, MEDIUM-VOLTAGE TRANSFORMERS: pad-mounted, medium-voltage transformers.

e. Section 26 13 13, MEDIUM-VOLTAGE CIRCUIT BREAKER SWITCHGEAR: Medium-voltage circuit breaker switchgear.

f. Section 26 22 00, LOW-VOLTAGE TRANSFORMERS: Low-voltage transformers.

g. Section 26 23 00, LOW-VOLTAGE SWITCHGEAR: Low-voltage switchgear.

h. Section 26 24 13, DISTRIBUTION SWITCHBOARDS: Low-voltage distribution switchboards.

i. Section 26 24 16, PANELBOARDS: Low-voltage panelboards.

j. Section 26 36 23, AUTOMATIC TRANSFER SWITCHES: Automatic transfer switches.

k. Section 26 41 00, FACILITY LIGHTNING PROTECTION: Lightning protection.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-06 Test Reports

Test Reports; G,

SD-07 Certificates

Certifications; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Submit plans showing the location of system grounding electrodes and connections, and the routing of aboveground and underground grounding electrode conductors.

b. Test Reports: Two weeks prior to the final inspection, submit ground resistance field test reports to the Contracting Officer.

c. Certifications: Certification by the Contractor that the grounding equipment has been properly installed and tested.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 81	(2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

PART 2 PRODUCTS

2.1 GROUNDING AND BONDING CONDUCTORS

a. Equipment grounding conductors shall be insulated stranded copper, except that sizes No. 10 AWG and smaller shall be solid copper. Insulation color shall be continuous green for all equipment grounding conductors, except that wire sizes No. 4 AWG and larger shall be identified per NFPA 70.

b. Bonding conductors shall be bare stranded copper, except that sizes No. 10 AWG and smaller shall be bare solid copper. Bonding conductors shall be stranded for final connection to motors, transformers, and vibrating equipment.

c. Conductor sizes shall not be less than shown on the drawings, or not less than required by NFPA 70, whichever is greater.

d. Insulation: THHN-THWN.

2.2 GROUND RODS

a. Copper clad steel , 0.75 inch diameter by 10 feet long.

b. Quantity of rods shall be as shown on the drawings, and as required to obtain the specified ground resistance.

2.3 CONCRETE ENCASED ELECTRODE

Concrete encased electrode shall be No. 4 AWG bare copper wire, installed per NFPA 70.

2.4 GROUND CONNECTIONS

a. Below Grade and Inaccessible Locations: Exothermic-welded type connectors.

b. Above Grade:

(1) Bonding Jumpers: Listed for use with copper conductors. For wire sizes No. 8 AWG and larger, use compression-type connectors. For wire sizes smaller than No. 8 AWG, use mechanical type lugs. Connectors or lugs shall use zinc-plated steel bolts, nuts, and washers. Bolts shall be torqued to the values recommended by the manufacturer.

(2) Connection to Building Steel: Exothermic-welded type connectors.

(3) Connection to Grounding Bus Bars: Listed for use with copper conductors. Use mechanical type lugs, with zinc-plated steel bolts, nuts, and washers. Bolts shall be torqued to the values recommended by the manufacturer.

(4) Connection to Equipment Rack and Cabinet Ground Bars: Listed for use with copper conductors. Use mechanical type lugs, with zinc-plated steel bolts, nuts, and washers. Bolts shall be torqued to the values recommended by the manufacturer.

2.5 EQUIPMENT RACK AND CABINET GROUND BARS

Provide solid copper ground bars designed for mounting on the framework of open or cabinet-enclosed equipment racks. Ground bars shall have minimum dimensions of 0.25 inch thick x 0.75 inch wide, with length as required or as shown on the drawings. Provide insulators and mounting brackets.

2.6 GROUND TERMINAL BLOCKS

At any equipment mounting location (e.g., backboards and hinged cover enclosures) where rack-type ground bars cannot be mounted, provide mechanical type lugs, with zinc-plated steel bolts, nuts, and washers. Bolts shall be torqued to the values recommended by the manufacturer.

2.7 GROUNDING BUS BAR

Pre-drilled rectangular copper bar with stand-off insulators, minimum 0.25 inch thick x 4 inches high in cross-section, length as shown on the drawings, with hole size, quantity, and spacing per detail shown on the drawings. Provide insulators and mounting brackets.

PART 3 EXECUTION

3.1 GENERAL

a. Installation shall be in accordance with NFPA 70, as shown on the drawings, and manufacturer’s instructions.

b. System Grounding:

(1) Secondary service neutrals: Ground at the supply side of the secondary disconnecting means or at the related transformer.

(2) Separately derived systems (transformers downstream from the service entrance): Ground the secondary neutral.

c. Equipment Grounding: Metallic piping, building structural steel, electrical enclosures, raceways, junction boxes, outlet boxes, cabinets, machine frames, and other conductive items in close proximity with electrical circuits, shall be bonded and grounded.

3.2 INACCESSIBLE GROUNDING CONNECTIONS

Make grounding connections, which are normally buried or otherwise inaccessible, by exothermic weld.

3.3 MEDIUM-VOLTAGE EQUIPMENT AND CIRCUITS

a. Switchgear: Provide a bare grounding electrode conductor from the switchgear ground bus to the grounding electrode system.

b. Duct Banks and Manholes: Provide an insulated equipment grounding conductor in each duct containing medium-voltage conductors, sized per NFPA 70 except that minimum size shall be No. 2 AWG. Bond the equipment grounding conductors to the switchgear ground bus, to all manhole grounding provisions and hardware, to the cable shield grounding provisions of medium-voltage cable splices and terminations, and to equipment enclosures.

c. Outdoor Switchgear:

(1) Provide a driven ground rod and bond with a grounding electrode conductor to the transformer grounding pad.

d. Lightning Arresters: Connect lightning arresters to the equipment ground bus or ground rods as applicable.

3.4 SECONDARY VOLTAGE EQUIPMENT AND CIRCUITS

a. Main Bonding Jumper: Bond the secondary service neutral to the ground bus in the service equipment.

b. Metallic Piping, Building Structural Steel, and Supplemental Electrode(s):

(1) Provide a grounding electrode conductor sized per NFPA 70 between the service equipment ground bus and all metallic water pipe systems, building structural steel, and supplemental or made electrodes. Provide jumpers across insulating joints in the metallic piping.

(2) Provide a supplemental ground electrode as shown on the drawings and bond to the grounding electrode system.

c. Switchgear, Switchboards, Unit Substations, Panelboards, Motor Control Centers, Engine-Generators, Automatic Transfer Switches, and other electrical equipment:

(1) Connect the equipment grounding conductors to the ground bus.

(2) Connect metallic conduits by grounding bushings and equipment grounding conductor to the equipment ground bus.

d. Transformers:

(1) Exterior: Exterior transformers supplying interior service equipment shall have the neutral grounded at the transformer secondary. Provide a grounding electrode at the transformer.

(2) Separately derived systems (transformers downstream from service equipment): Ground the secondary neutral at the transformer. Provide a grounding electrode conductor from the transformer to the nearest component of the grounding electrode system .

3.5 RACEWAY

a. Conduit Systems:

(1) Ground all metallic conduit systems. All metallic conduit systems shall contain an equipment grounding conductor.

(2) Non-metallic conduit systems, except non-metallic feeder conduits that carry a grounded conductor from exterior transformers to interior or building-mounted service entrance equipment, shall contain an equipment grounding conductor.

(3) Metallic conduit that only contains a grounding conductor, and is

provided for its mechanical protection, shall be bonded to that conductor at the entrance and exit from the conduit.

(4) Metallic conduits which terminate without mechanical connection to an electrical equipment housing by means of locknut and bushings or adapters, shall be provided with grounding bushings. Connect bushings with a equipment grounding conductor to the equipment ground bus.

b. Feeders and Branch Circuits: Install equipment grounding conductors with all feeders, and power and lighting branch circuits.

c. Boxes, Cabinets, Enclosures, and Panelboards:

(1) Bond the equipment grounding conductor to each pullbox, junction box, outlet box, device box, cabinets, and other enclosures through which the conductor passes (except for special grounding systems for intensive care units and other critical units shown).

(2) Provide lugs in each box and enclosure for equipment grounding conductor termination.

d. Wireway Systems:

(1) Bond the metallic structures of wireway to provide electrical continuity throughout the wireway system, by connecting a No. 6 AWG bonding jumper at all intermediate metallic enclosures and across all section junctions.

(2) Install insulated No. 6 AWG bonding jumpers between the wireway system, bonded as required above, and the closest building ground at each end and approximately every 50 feet.

(3) Use insulated No. 6 AWG bonding jumpers to ground or bond metallic wireway at each end for all intermediate metallic enclosures and across all section junctions.

(4) Use insulated No. 6 AWG bonding jumpers to ground cable tray to column-mounted building ground plates (pads) at each end and approximately every 49 feet.

e. Receptacles shall not be grounded through their mounting screws. Ground receptacles with a jumper from the receptacle green ground terminal to the device box ground screw and a jumper to the branch circuit equipment grounding conductor.

f. Ground lighting fixtures to the equipment grounding conductor of the wiring system. Fixtures connected with flexible conduit shall have a green ground wire included with the power wires from the fixture through the flexible conduit to the first outlet box.

g. Fixed electrical appliances and equipment shall be provided with a ground lug for termination of the equipment grounding conductor.

3.6 CORROSION INHIBITORS

When making grounding and bonding connections, apply a corrosion inhibitor to all contact surfaces. Use corrosion inhibitor appropriate for

protecting a connection between the metals used.

3.7 CONDUCTIVE PIPING

a. Bond all conductive piping systems, interior and exterior, to the grounding electrode system. Bonding connections shall be made as close as practical to the equipment ground bus.

b. In operating rooms and at intensive care and coronary care type beds, bond the medical gas piping and medical vacuum piping at the outlets directly to the patient ground bus.

3.8 LIGHTNING PROTECTION SYSTEM

Bond the lightning protection system to the electrical grounding electrode system.

3.9 MAIN ELECTRICAL ROOM GROUNDING

Provide ground bus bar and mounting hardware at each main electrical room where incoming feeders are terminated, as shown on the drawings. Connect to pigtail extensions of the building grounding ring, as shown on the drawings.

3.10 EXTERIOR LIGHT POLES

Provide 20 feet of No. 4 AWG bare copper coiled at bottom of pole base excavation prior to pour, plus additional un-spliced length in and above foundation as required to reach pole ground stud.

3.11 GROUND RESISTANCE

a. Grounding system resistance to ground shall not exceed 5 ohms. Make any modifications or additions to the grounding electrode system necessary for compliance without additional cost to the Government. Final tests shall ensure that this requirement is met.

b. Grounding system resistance shall comply with the electric utility company ground resistance requirements.

3.12 GROUND ROD INSTALLATION

a. For outdoor installations, drive each rod vertically in the earth, until top of rod is 24 inches below final grade.

b. For indoor installations, leave 4 inches of each rod exposed, locate outside of travel paths to prevent trip hazards..

c. Where buried or permanently concealed ground connections are required, make the connections by the exothermic process, to form solid metal joints. Accessible ground connections may be madewith mechanical pressure-type ground connectors.

d. Where rock or impenetrable soil prevents the driving of vertical ground rods, install angled ground rods or grounding electrodes in horizontal trenches to achieve the specified ground resistance.

3.13 ACCEPTANCE CHECKS AND TESTS

a. Resistance of the grounding electrode system shall be measured using a four-terminal fall-of-potential method as defined in IEEE 81. Ground resistance measurements shall be made before the electrical distribution system is energized or connected to the electric utility company ground system, and shall be made in normally dry conditions not fewer than 48 hours after the last rainfall.

b. Resistance measurements of separate grounding electrode systems shall be made before the systems are bonded together. The combined resistance of separate systems may be used to meet the required resistance, but the specified number of electrodes must still be provided.

c. Below-grade connections shall be visually inspected by the Contracting Officer prior to backfilling. The Contractor shall notify the Contracting Officer 24 hours before the connections are ready for inspection.

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SECTION 26 05 33
11/22

RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

a. This Section specifies the furnishing, installation, and connection of conduit, fittings, and boxes, to form complete, coordinated, grounded raceway systems. Raceways are required for all wiring unless shown or specified otherwise.

b. Definitions: The term conduit, as used in this specification, shall mean any or all of the raceway types specified.

1.2 RELATED WORK

a. Section 06 10 00, ROUGH CARPENTRY: Mounting board for telephone closets.

b. Section 07 60 00, FLASHING AND SHEET METAL: Fabrications for the deflection of water away from the building envelope at penetrations.

c. Section 07 84 00, FIRESTOPPING: Sealing around penetrations to maintain the integrity of fire rated construction.

d. Section 07 92 00, JOINT SEALANTS: Sealing around conduit penetrations through the building envelope to prevent moisture migration into the building.

e. Section 09 91 00, PAINTING: Identification and painting of conduit and other devices.

f. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: General electrical requirements and items that are common to more than one section of Division 26.

g. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

h. Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION: Underground conduits.

i. Section 31 20 00, EARTHWORK: Bedding of conduits.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation;

submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-07 Certificates

Certifications; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Size and location of main feeders.

(2) Size and location of panels and pull-boxes.

(3) Layout of required conduit penetrations through structural elements.

(4) Submit the following data for approval:

(a) Raceway types and sizes.

(b) Conduit bodies, connectors and fittings.

(c) Junction and pull boxes, types and sizes.

b. Certifications: Two weeks prior to final inspection, submit the following:

(1) Certification by the manufacturer that raceways, conduits, conduit bodies, connectors, fittings, junction and pull boxes, and all related equipment conform to the requirements of the drawings and specifications.

(2) Certification by the Contractor that raceways, conduits, conduit bodies, connectors, fittings, junction and pull boxes, and all related equipment have been properly installed.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C80.1	(2020) American National Standard for Electrical Rigid Steel Conduit (ERSC)
ANSI C80.3	(2020) American National Standard for Electrical Metallic Tubing (EMT)

ANSI C80.6	(2018) Electrical Intermediate Metal Conduit (ELMC)
NEMA FB 1	(2014) Standard for Fittings, Cast Metal Boxes, and Conduit Bodies for Conduit, Electrical Metallic Tubing, and Cable
NEMA TC 3	(2021) Polyvinyl Chloride (PVC) Fittings for Use With Rigid PVC Conduit and Tubing

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

UL SOLUTIONS (UL)

UL 1	(2005; Reprint Jan 2022) UL Standard for Safety Flexible Metal Conduit
UL 5	(2016; Reprint Jul 2022) UL Standard for Safety Surface Metal Raceways and Fittings
UL 6	(2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel
UL 50	(2024) UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations
UL 360	(2013; Reprint Jan 2024) UL Standard for Safety Liquid-Tight Flexible Metal Conduit
UL 467	(2022) UL Standard for Safety Grounding and Bonding Equipment
UL 514A	(2024) UL Standard for Safety Metallic Outlet Boxes
UL 514B	(2012; Reprint Mar 2024) UL Standard for Safety Conduit, Tubing and Cable Fittings
UL 514C	(2014; Reprint Apr 2024) UL Standard for Safety Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers
UL 651	(2011; Reprint May 2022) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
UL 651A	(2023; Reprint Jan 2024) UL Standard for Safety Schedule 40 and 80 High Density Polyethylene (HDPE) Conduit
UL 797	(2007; Reprint Apr 2023) UL Standard for Safety Electrical Metallic Tubing -- Steel
UL 1242	(2006; Reprint Apr 2022) UL Standard for

PART 2 PRODUCTS

2.1 MATERIAL

a. Conduit Size: In accordance with NFPA 70, but not less than 0.5-inch unless otherwise shown. Where permitted by NFPA 70, 0.5-inch flexible conduit may be used for tap connections to recessed lighting fixtures.

b. Conduit:

(1) Size: In accordance with NFPA 70, but not less than 0.5-inch.

(2) Rigid Steel Conduit (RMC): Shall conform to UL 6 and ANSI C80.1.

(4) Rigid Intermediate Steel Conduit (IMC): Shall conform to UL 1242 and ANSI C80.6.

(5) Electrical Metallic Tubing (EMT): Shall conform to UL 797 and ANSI C80.3. Maximum size not to exceed 4 inches and shall be permitted only with cable rated 600 V or less.

(6) Flexible Metal Conduit: Shall conform to UL 1.

(7) Liquid-tight Flexible Metal Conduit: Shall conform to UL 360.

(8) Direct Burial Plastic Conduit: Shall conform to UL 651 and UL 651A, heavy wall PVC or high density polyethylene (PE).

(9) Surface Metal Raceway: Shall conform to UL 5.

c. Conduit Fittings:

(1) Rigid Steel and Intermediate Metallic Conduit Fittings:

(a) Fittings shall meet the requirements of UL 514B and NEMA FB 1.

(b) Standard threaded couplings, locknuts, bushings, conduit bodies, and elbows: Only steel or malleable iron materials are acceptable. Integral retractable type IMC couplings are also acceptable.

(c) Locknuts: Bonding type with sharp edges for digging into the metal wall of an enclosure.

(d) Bushings: Metallic insulating type, consisting of an insulating insert, molded or locked into the metallic body of the fitting. Bushings made entirely of metal or nonmetallic material are not permitted.

(e) Erickson (Union-Type) and Set Screw Type Couplings: Approved for use in concrete are permitted for use to complete a conduit run where conduit is installed in concrete. Use set screws of case-hardened steel with hex head and cup point to firmly seat in conduit wall for positive ground. Tightening of set screws with pliers is prohibited.

(f) Sealing Fittings: Threaded cast iron type. Use continuous drain-type sealing fittings to prevent passage of water vapor. In concealed work, install fittings in flush steel boxes with blank cover plates having the same finishes as that of other electrical plates in the room.

(2) Rigid Aluminum Conduit Fittings:

(a) Standard threaded couplings, locknuts, bushings, conduit bodies, and elbows: Malleable iron, steel or aluminum alloy materials; Zinc or cadmium plate iron or steel fittings. Aluminum fittings containing more than 0.4 percent copper are prohibited.

(b) Locknuts and Bushings: As specified for rigid steel and IMC conduit.

(c) Set Screw Fittings: Not permitted for use with aluminum conduit.

(3) Electrical Metallic Tubing Fittings:

(a) Fittings and conduit bodies shall meet the requirements of UL 514B, ANSI C80.3, and NEMA FB 1.

(b) Only steel or malleable iron materials are acceptable.

(c) Compression Couplings and Connectors: Concrete-tight and rain-tight, with connectors having insulated throats.

(c) Indent-type connectors or couplings are prohibited.

(d) Die-cast or pressure-cast zinc-alloy fittings or fittings made of "pot metal" are prohibited.

(4) Flexible Metal Conduit Fittings:

(a) Conform to UL 514B. Only steel or malleable iron materials are acceptable.

(b) Clamp-type, with insulated throat.

(5) Liquid-tight Flexible Metal Conduit Fittings:

(a) Fittings shall meet the requirements of UL 514B and NEMA FB 1.

(b) Only steel or malleable iron materials are acceptable.

(c) Fittings must incorporate a threaded grounding cone, a steel or plastic compression ring, and a gland for tightening. Connectors shall have insulated throats.

(6) Direct Burial Plastic Conduit Fittings: Fittings shall meet the requirements of UL 514C and NEMA TC 3.

(7) Surface Metal Raceway Fittings: As recommended by the raceway manufacturer. Include couplings, offsets, elbows, expansion joints, adapters, hold-down straps, end caps, conduit entry fittings, accessories, and other fittings as required for complete system.

(8) Expansion and Deflection Couplings:

(a) Conform to UL 467 and UL 514B.

(b) Accommodate a 0.75-inch deflection, expansion, or contraction in any direction, and allow 30 degree angular deflections.

(c) Include internal flexible metal braid, sized to guarantee conduit ground continuity and a low-impedance path for fault currents, in accordance with UL 467 and NFPA 70 tables for equipment grounding conductors.

(d) Jacket: Flexible, corrosion-resistant, watertight, moisture and heat-resistant molded rubber material with stainless steel jacket clamps.

d. Conduit Supports:

(1) Parts and Hardware: Zinc-coat or provide equivalent corrosion protection.

(2) Individual Conduit Hangers: Designed for the purpose, having a pre-assembled closure bolt and nut, and provisions for receiving a hanger rod.

(3) Multiple Conduit (Trapeze) Hangers: Not less than 1.5 x 1.5 inches, 12-gauge steel, cold-formed, lipped channels; with not less than 0.375-inch diameter steel hanger rods.

(4) Solid Masonry and Concrete Anchors: Self-drilling expansion shields, or machine bolt expansion.

e. Outlet, Junction, and Pull Boxes:

(1) Comply with UL 50 and UL 514A.

(2) Rustproof cast metal where required by NFPA 70 or shown on drawings.

(3) Sheet Metal Boxes: Galvanized steel, except where shown on drawings.

f. Metal Wireways: Equip with hinged covers, except as shown on drawings. Include couplings, offsets, elbows, expansion joints, adapters, hold-down straps, end caps, and other fittings to match and mate with wireways as required for a complete system.

PART 3 EXECUTION

3.1 PENETRATIONS

a. Cutting or Holes:

(1) Cut holes in advance where they should be placed in the structural elements, such as ribs or beams. Obtain the approval of the Contracting Officer prior to drilling through structural elements.

(2) Cut holes through concrete and masonry in new and existing

structures with a diamond core drill or concrete saw. Pneumatic hammers, impact electric, hand, or manual hammer-type drills are not allowed, except when permitted by the Contracting Officer where working space is limited.

b. Firestop: Where conduits, wireways, and other electrical raceways pass through fire partitions, fire walls, smoke partitions, or floors, install a fire stop that provides an effective barrier against the spread of fire, smoke and gases as specified in Section 07 84 00, FIRESTOPPING.

c. Waterproofing: At floor, exterior wall, and roof conduit penetrations, completely seal the gap around conduit to render it watertight, as specified in Section 07 92 00, JOINT SEALANTS.

3.2 INSTALLATION, GENERAL

a. In accordance with NFPA 70, NEMA, UL, as shown on drawings, and as specified herein.

b. Raceway systems used for Essential Electrical Systems (EES) shall be entirely independent of other raceway systems.

c. Install conduit as follows:

(1) In complete mechanically and electrically continuous runs before pulling in cables or wires.

(2) Unless otherwise indicated on the drawings or specified herein, installation of all conduits shall be concealed within finished walls, floors, and ceilings.

(3) Flattened, dented, or deformed conduit is not permitted. Remove and replace the damaged conduits with new conduits.

(4) Assure conduit installation does not encroach into the ceiling height head room, walkways, or doorways.

(5) Cut conduits square, ream, remove burrs, and draw up tight.

(6) Independently support conduit at a maximum of 8 feet on centers with specified materials and as shown on drawings and specifications. Contractor shall submit hangar load calculations when more than four conduits are supported by a single horizontal support.

(7) Do not use suspended ceilings, suspended ceiling supporting members, lighting fixtures, other conduits, cable tray, boxes, piping, or ducts to support conduits and conduit runs.

(8) Support within 12 inches of changes of direction, and within 12 inches of each enclosure to which connected.

(9) Close ends of empty conduits with plugs or caps at the rough-in stage until wires are pulled in, to prevent entry of debris.

(10) Conduit installations under fume and vent hoods are prohibited.

(11) Secure conduits to cabinets, junction boxes, pull-boxes, and

outlet boxes with bonding type locknuts. For rigid steel and IMC conduit installations, provide a locknut on the inside of the enclosure, made up wrench tight. Do not make conduit connections to junction box covers.

(12) Flashing of penetrations of the roof membrane is specified in Section 07 60 00, FLASHING AND SHEET METAL.

(13) Conduit bodies shall only be used for changes in direction, and shall not contain splices.

d. Conduit Bends:

(1) Make bends with standard conduit bending machines.

(2) Conduit hickey may be used for slight offsets and for straightening stubbed out conduits.

(3) Bending of conduits with a pipe tee or vise is prohibited.

e. Layout and Homeruns:

(1) Install conduit with wiring, including homeruns, as shown on drawings.

(2) Deviations: Make only where necessary to avoid interferences and only after drawings showing the proposed deviations have been submitted and approved by the Contracting Officer.

3.3 CONCEALED WORK INSTALLATION

a. In Concrete:

(1) Conduit: Rigid steel or IMC. Do not install EMT in concrete slabs.

(2) Align and run conduit in direct lines.

(3) Install conduit through concrete beams only:

(a) Where shown on the structural drawings.

(b) As approved by the Contracting Officer prior to construction, and after submittal of drawing showing location, size, and position of each penetration.

(4) Installation of conduit in concrete that is less than 3 inches thick is prohibited.

(a) Conduit outside diameter larger than one-third of the slab thickness is prohibited.

(b) Space between conduits in slabs: Approximately six conduit diameters apart, and one conduit diameter at conduit crossings.

(c) Install conduits approximately in the center of the slab so that there will be a minimum of 0.75-inch of concrete around the conduits.

(5) Make couplings and connections watertight. Use thread compounds

that are UL approved conductive type to ensure low resistance ground continuity through the conduits. Tightening setscrews with pliers is prohibited.

b. Above Furred or Suspended Ceilings and in Walls:

(1) Conduit for Conductors Above 600 V: Rigid steel. Mixing different types of conduits in the same system is prohibited.

(2) Conduit for Conductors 600 V and Below: Rigid steel, IMC, or EMT. Mixing different types of conduits in the same system is prohibited.

(3) Align and run conduit parallel or perpendicular to the building lines.

(4) Connect recessed lighting fixtures to conduit runs with maximum 6 feet of flexible metal conduit extending from a junction box to the fixture.

(5) Tightening set screws with pliers is prohibited.

(6) For conduits running through metal studs, limit field cut holes to no more than 70 percent of web depth. Spacing between holes shall be at least 18 inches. Cuts or notches in flanges or return lips shall not be permitted.

3.4 EXPOSED WORK INSTALLATION

a. Unless otherwise indicated on drawings, exposed conduit is only permitted in mechanical and electrical rooms.

b. Conduit for Conductors Above 600 V: Rigid steel. Mixing different types of conduits in the system is prohibited.

c. Conduit for Conductors 600 V and Below: Rigid steel, IMC, or EMT. Mixing different types of conduits in the system is prohibited.

d. Align and run conduit parallel or perpendicular to the building lines.

e. Install horizontal runs close to the ceiling or beams and secure with conduit straps.

f. Support horizontal or vertical runs at not over 8 feet intervals.

g. Surface Metal Raceways: Use only where shown on drawings.

h. Painting:

(1) Paint exposed conduit as specified in Section 09 91 00, PAINTING.

(2) Paint all conduits containing cables rated over 600 V safety orange. Refer to Section 09 91 00, PAINTING for preparation, paint type, and exact color. In addition, paint legends, using 2 inch high black numerals and letters, showing the cable voltage rating. Provide legends where conduits pass through walls and floors and at maximum 20 feet intervals in between.

3.5 DIRECT BURIAL INSTALLATION

Refer to Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION.

3.6 HAZARDOUS LOCATIONS

a. Use rigid steel conduit only.

b. Install UL approved sealing fittings that prevent passage of explosive vapors in hazardous areas equipped with explosion-proof lighting fixtures, switches, and receptacles, as required by NFPA 70.

3.7 WET OR DAMP LOCATIONS

a. Use rigid steel conduit only.

b. Provide sealing fittings to prevent passage of water vapor where conduits pass from warm to cold locations, i.e., refrigerated spaces, constant-temperature rooms, air-conditioned spaces, building exterior walls, roofs, or similar spaces.

c. Use rigid steel conduit within 5 feet of the exterior and below concrete building slabs in contact with soil, gravel, or vapor barriers, unless as shown on drawings. Conduit shall be half-lapped with 10 mil PVC tape before installation. After installation, completely recoat or retape any damaged areas of coating.

d. Conduits run on roof shall be supported with integral galvanized lipped steel channel, attached to UV-inhibited polycarbonate or polypropylene blocks every 8 feet with 3/8-inch galvanized threaded rods, square washer and locknut. Conduits shall be attached to steel channel with conduit clamps.

3.8 MOTORS AND VIBRATING EQUIPMENT

a. Use flexible metal conduit for connections to motors and other electrical equipment subject to movement, vibration, misalignment, cramped quarters, or noise transmission.

b. Use liquid-tight flexible metal conduit for installation in exterior locations, moisture or humidity laden atmosphere, corrosive atmosphere, water or spray wash-down operations, inside airstream of HVAC units, and locations subject to seepage or dripping of oil, grease, or water.

c. Provide a green equipment grounding conductor with flexible and liquid-tight flexible metal conduit.

3.9 EXPANSION JOINTS

a. Conduits 3 inch and larger that are secured to the building structure on opposite sides of a building expansion joint require expansion and deflection couplings. Install the couplings in accordance with the manufacturer's recommendations.

b. Provide conduits smaller than 3 inch with junction boxes on both sides of the expansion joint. Connect flexible metal conduits to junction boxes with sufficient slack to produce a 5 inch vertical drop midway between the ends of the flexible metal conduit. Flexible metal

conduit shall have a green insulated copper bonding jumper installed. In lieu of this flexible metal conduit, expansion and deflection couplings as specified above are acceptable.

c. Install expansion and deflection couplings where shown.

3.10 CONDUIT SUPPORTS

a. Safe working load shall not exceed one-quarter of proof test load of fastening devices.

b. Use pipe straps or individual conduit hangers for supporting individual conduits.

c. Support multiple conduit runs with trapeze hangers. Use trapeze hangers that are designed to support a load equal to or greater than the sum of the weights of the conduits, wires, hanger itself, and an additional 200 lbs. Attach each conduit with U-bolts or other approved fasteners. Submit calculations for hangars supporting more than four conduits.

d. Support conduit independently of junction boxes, pull-boxes, fixtures, suspended ceiling T-bars, angle supports, and similar items.

e. Fasteners and Supports in Solid Masonry and Concrete:

(1) New Construction: Use steel or malleable iron concrete inserts set in place prior to placing the concrete.

(2) Existing Construction:

(a) Steel expansion anchors not less than 0.25-inch bolt size and not less than 1.125 inch in embedment.

(b) Power set fasteners not less than 0.25-inch diameter with depth of penetration not less than 3 inch.

(c) Use vibration and shock-resistant anchors and fasteners for attaching to concrete ceilings.

f. Hollow Masonry: Toggle bolts.

g. Bolts supported only by plaster or gypsum wallboard are not acceptable.

h. Metal Structures: Use machine screw fasteners or other devices specifically designed and approved for the application.

i. Attachment by wood plugs, rawl plug, plastic, lead or soft metal anchors, or wood blocking and bolts supported only by plaster is prohibited.

j. Chain, wire, or perforated strap shall not be used to support or fasten conduit.

k. Spring steel type supports or fasteners are prohibited for all uses except horizontal and vertical supports/fasteners within walls.

l. Vertical Supports: Vertical conduit runs shall have riser clamps and

supports in accordance with NFPA 70 and as shown. Provide supports for cable and wire with fittings that include internal wedges and retaining collars.

3.11 BOX INSTALLATION

a. Boxes for Concealed Conduits:

(1) Flush-mounted.

(2) Provide raised covers for boxes to suit the wall or ceiling, construction, and finish.

b. In addition to boxes shown, install additional boxes where needed to prevent damage to cables and wires during pulling-in operations or where more than the equivalent of 4-90 degree bends are necessary.

c. Locate pullboxes so that covers are accessible and easily removed. Coordinate locations with piping and ductwork where installed above ceilings.

d. Remove only knockouts as required. Plug unused openings. Use threaded plugs for cast metal boxes and snap-in metal covers for sheet metal boxes.

e. Outlet boxes mounted back-to-back in the same wall are prohibited. A minimum 24 inch center-to-center lateral spacing shall be maintained between boxes.

f. Flush-mounted wall or ceiling boxes shall be installed with raised covers so that the front face of raised cover is flush with the wall. Surface-mounted wall or ceiling boxes shall be installed with surface-style flat or raised covers.

g. Minimum size of outlet boxes for ground fault circuit interrupter (GFCI) receptacles is 4 inches square x 2.125 inches deep, with device covers for the wall material and thickness involved.

h. Stencil or install phenolic nameplates on covers of the boxes identified on riser diagrams; for example "SIG-FA JB No. 1."

i. On all branch circuit junction box covers, identify the circuits with black marker.

-- End of Section --

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06/25

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SECTION 26 05 41
06/25

UNDERGROUND ELECTRICAL CONSTRUCTION

PART 1 GENERAL

1.1 DESCRIPTION

a. This Section specifies the furnishing, installation, and connection of underground ducts and raceways, and precast manholes and pullboxes to form a complete underground electrical raceway system for 600V and below cables and feeders. .

b. The terms “duct” and “conduit” are used interchangeably in this section.

1.2 RELATED WORK

a. Section 07 92 00, JOINT SEALANTS: Sealing of conduit penetrations.

b. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

c. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

d. Section 31 20 00, EARTH MOVING : Trenching, backfill, and compaction.

1.3 QUALITY ASSURANCE

a. Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

b. Coordinate layout and installation of ducts, manholes, and pullboxes with final arrangement of other utilities, site grading, and surface features.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-07 Certificates

Certifications; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Submit information on manholes, pullboxes, ducts, and hardware. Submit manhole plan and elevation drawings, showing openings, pulling irons, cable supports, cover, ladder, sump, and other accessories.

(3) Proposed deviations from the drawings shall be clearly marked on the submittals. If it is necessary to locate manholes, pullboxes, or duct banks at locations other than shown on the drawings, show the proposed locations accurately on scaled site drawings, and submit to the Contracting Officer for approval prior to construction.

b. Certifications: Two weeks prior to the final inspection, submit the following.

(1) Certification by the manufacturer that the materials conform to the requirements of the drawings and specifications.

(2) Certification by the Contractor that the materials have been properly installed, connected, and tested.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

UL SOLUTIONS (UL)

UL 651	(2011; Reprint May 2022) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
UL 651A	(2023; Reprint Jan 2024) UL Standard for Safety Schedule 40 and 80 High Density Polyethylene (HDPE) Conduit

PART 2 PRODUCTS

2.1 PRE-CAST CONCRETE MANHOLES AND HARDWARE

a. Structure: Factory-fabricated, reinforced-concrete, monolithically-poured walls and bottom. Frame and cover shall form top of manhole.

b. Cable Supports:

(1) Cable stanchions shall be hot-rolled, heavy duty, hot-dipped galvanized "T" section steel, 2.25 inches x 0.25 inch in size, and

punched with 14 holes on 1.5 inches centers for attaching cable arms.

(2) Cable arms shall be 0.1875 inch gauge, hot-rolled, hot-dipped galvanized sheet steel, pressed to channel shape. Arms shall be approximately 2.5 inches wide x 14 inches long.

(3) Insulators for cable supports shall be porcelain, and shall be saddle type or type that completely encircles the cable.

(4) Equip each cable stanchion with one spare cable arm, with three spare insulators for future use.

c. Ladder: Provide Fiberglass ladder with a maximum rung spacing of 14 inches - as measured between center line of the rungs, and a minimum rung clear width of 12 inches. Provide securely-mounted ladder for every manhole over 4 feet deep.

d. Ground Rod Sleeve: Provide a 3 inches PVC sleeve in manhole floors so that a driven ground rod may be installed.

e. Sump: Provide 12 inches x 12 inches covered sump frame and grated cover.

2.2 DUCTS

a. Number and sizes shall be as shown on the drawings.

b. Ducts (concrete-encased):

(1) Plastic Duct:

(a) UL 651 and UL 651A Schedule 40 PVC conduit.

(b) Duct shall be suitable for use with 194 degrees F rated conductors.

(2) Conduit Spacers: Prefabricated plastic.

2.3 GROUNDING

Per Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS.

2.4 WARNING TAPE

4-mil polyethylene 3 inches wide detectable tape, red with black letters, imprinted with “CAUTION - BURIED ELECTRIC CABLE BELOW” or similar.

2.5 PULL ROPE FOR SPARE DUCTS

Plastic with 200 lb minimum tensile strength.

PART 3 EXECUTION

3.1 MANHOLE AND PULLBOX INSTALLATION

a. Assembly and installation shall be per the requirements of the manufacturer.

(1) Install manholes and pullboxes level and plumb.

(2) Units shall be installed on a 12 inches thick level bed of 90 percent compacted granular fill, well-graded from the 1 inch sieve to the No. 4 sieve. Granular fill shall be compacted with a minimum of four passes with a plate compactor.

b. Access: Ensure the top of frames and covers are flush with finished grade.

c. Grounding in Manholes:

(1) Ground Rods in Manholes: Drive a ground rod into the earth, through the floor sleeve, after the manhole is set in place. Fill the sleeve with sealant to make a watertight seal. Rods shall protrude approximately 4 inches above the manhole floor.

(2) Install a No. 3/0 AWG bare copper ring grounding conductor around the inside perimeter of the manhole and anchor to the walls with metallic cable clips.

(3) Connect the ring grounding conductor to the ground rod by an exothermic welding process.

(4) Bond the ring grounding conductor to the duct bank equipment grounding conductors, the exposed non-current carrying metal parts of racks, sump covers, and like items in the manholes with a minimum No. 6 AWG bare copper jumper using an exothermic welding process.

3.2 TRENCHING

a. Refer to Section 31 20 00, EARTH MOVING for trenching, backfilling, and compaction.

b. Before performing trenching work at existing facilities, a Ground Penetrating Radar Survey shall be carefully performed by a certified technician to reveal all existing underground ducts, conduits, cables, and other utility systems.

c. Work with extreme care near existing ducts, conduits, and other utilities to avoid damaging them.

d. Cut the trenches neatly and uniformly.

e. For Concrete-Encased Ducts:

(1) After excavation of the trench, stakes shall be driven in the bottom of the trench at 4 feet intervals to establish the grade and route of the duct bank.

(2) Pitch the trenches uniformly toward manholes or both ways from high points between manholes for the required duct line drainage. Avoid pitching the ducts toward buildings wherever possible.

(3) The walls of the trench may be used to form the side walls of the duct bank, provided that the soil is self-supporting and that the concrete envelope can be poured without soil inclusions. Forms are required where the soil is not self-supporting.

(4) After the concrete-encased duct has sufficiently cured, the trench shall be backfilled to grade with earth, and appropriate warning tape installed.

f. Individual conduits to be installed under existing paved areas and roads that cannot be disturbed shall be jacked into place using rigid metal conduit, or bored using plastic utilities duct or PVC conduit, as approved by the Contracting Officer.

3.3 DUCT INSTALLATION

a. General Requirements:

(1) Ducts shall be in accordance with the NEC, as shown on the drawings, and as specified.

(2) Join and terminate ducts with fittings recommended by the manufacturer.

(3) Slope ducts to drain towards manholes and pullboxes, and away from building and equipment entrances. Pitch not less than 4 inches in 100 feet.

(4) Underground conduit stub-ups and sweeps to equipment inside of buildings shall be galvanized rigid metal conduit half-lap wrapped with PVC tape, and shall extend a minimum of 5 feet outside the building foundation. Tops of conduits below building slab shall be minimum 24 inches below bottom of slab.

(5) Stub-ups and sweeps to equipment mounted on outdoor concrete slabs shall be galvanized rigid metal conduit half-lap wrapped with PVC tape, and shall extend a minimum of 5 feet away from the edge of slab.

(6) Install insulated grounding bushings on the conduit terminations.

(7) Radius for sweeps shall be sufficient to accomplish pulls without damage. Minimum radius shall be six times conduit diameter.

(8) All multiple conduit runs shall have conduit spacers. Spacers shall securely support and maintain uniform spacing of the duct assembly a minimum of 3 inches above the bottom of the trench during the concrete pour. Spacer spacing shall not exceed 5 feet. Secure spacers to ducts and earth to prevent floating during concrete pour. Provide nonferrous tie wires to prevent displacement of the ducts during concrete pour. Tie wires shall not act as substitute for spacers.

(9) Duct lines shall be installed no less than 12 inches from other utility systems, such as water, sewer, chilled water.

(10) Clearances between individual ducts:

(a) For similar services, not less than 3 inches.

(b) For power and signal services, not less than 6 inches.

(11) Duct lines shall terminate at window openings in manhole walls as

shown on the drawings. All ducts shall be fitted with end bells.

(12) Couple the ducts with proper couplings. Stagger couplings in rows and layers to ensure maximum strength and rigidity of the duct bank.

(13) Keep ducts clean of earth, sand, or gravel, and seal with tapered plugs upon completion of each portion of the work.

(14) Spare Ducts: Where spare ducts are shown, they shall have a nylon pull rope installed. They shall be capped at each end and labeled as to location of the other end.

(15) Duct Identification: Place continuous strip of warning tape approximately 12 inches above ducts before backfilling trenches. Warning tape shall be preprinted with proper identification.

(16) Duct Sealing: Seal ducts, including spare ducts, at building entrances and at outdoor terminations for equipment, with a suitable non-hardening compound to prevent the entrance of foreign objects and material, moisture, and gases.

(17) Use plastic ties to secure cables to insulators on cable arms. Use minimum two ties per cable per insulator.

b. Concrete-Encased Ducts:

(1) Install concrete-encased ducts for medium-voltage systems, low-voltage systems, and signal systems, unless otherwise shown on the drawings.

(2) Duct banks shall be single or multiple duct assemblies encased in concrete. Ducts shall be uniform in size and material throughout the installation.

(3) Tops of concrete-encased ducts shall be:

(a) Not less than 24 inches and not less than shown on the drawings, below finished grade.

(b) Not less than 30 inches and not less than shown on the drawings, below roads and other paved surfaces.

(c) Additional burial depth shall be required in order to accomplish NEC-required minimum bend radius of ducts.

(d) Conduits crossing under grade slab construction joints shall be installed a minimum of 4 feet below slab.

(4) Extend the concrete envelope encasing the ducts not less than 3 inches beyond the outside walls of the outer ducts.

(5) Within 10 feet of building and manhole wall penetrations, install reinforcing steel bars at the top and bottom of each concrete envelope to provide protection against vertical shearing.

(6) Install reinforcing steel bars at the top and bottom of each concrete envelope of all ducts underneath roadways and parking areas.

(7) Where new ducts and concrete envelopes are to be joined to existing manholes, pullboxes, ducts, and concrete envelopes, make the joints with the proper fittings and fabricate the concrete envelopes to ensure smooth durable transitions.

(8) Duct joints in concrete may be placed side by side horizontally, but shall be staggered at least 6 inches vertically.

(9) Pour each run of concrete envelope between manholes or other terminations in one continuous pour. If more than one pour is necessary, terminate each pour in a vertical plane and install 0.75 inch reinforcing rod dowels extending 18 inches into concrete on both sides of joint near corners of envelope.

(10) Pour concrete so that open spaces are uniformly filled. Do not agitate with power equipment unless approved by Contracting Officer.

c. Direct-Burial Ducts:

(1) Install direct-burial ducts only where shown on the drawings. Provide direct-burial ducts only for low-voltage power and lighting branch circuits.

(2) Tops of ducts shall be:

(a) Not less than 24 inches and not less than shown on the drawings, below finished grade.

(b) Not less than 30 inches and not less than shown on the drawings, below roads and other paved surfaces.

(c) Additional burial depth shall be required in order to accomplish NEC-required minimum bend radius of ducts.

(3) Do not kink the ducts. Compaction shall not deform the ducts.

d. Connections to Manholes: Ducts connecting to manholes shall be flared to have an enlarged cross-section to provide additional shear strength. Dimensions of the flared cross-section shall be larger than the corresponding manhole opening dimensions by no less than 12 inches in each direction. Perimeter of the duct bank opening in the manhole shall be flared toward the inside or keyed to provide a positive interlock between the duct and the wall of the manhole. Use vibrators when this portion of the encasement is poured to ensure a seal between the envelope and the wall of the structure.

e. Connections to Existing Manholes: For duct connections to existing manholes, break the structure wall out to the dimensions required and preserve the steel in the structure wall. Cut steel and extend into the duct bank envelope. Chip the perimeter surface of the duct bank opening to form a key or flared surface, providing a positive connection with the duct bank envelope.

f. Connections to Existing Ducts: Where connections to existing ducts are indicated, excavate around the ducts as necessary. Cut off the ducts and remove loose concrete from inside before installing new ducts. Provide a reinforced-concrete collar, poured monolithically

with the new ducts, to take the shear at the joint of the duct banks.

g. Partially-Completed Ducts: During construction, wherever a construction joint is necessary in a duct bank, prevent debris such as mud and dirt from entering ducts by providing suitable plugs. Fit concrete envelope of a partially completed ducts with reinforcing steel extending a minimum of 2 feet back into the envelope and a minimum of 2 feet beyond the end of the envelope. Provide one No. 4 bar in each corner, 3 inches from the edge of the envelope. Secure corner bars with two No. 3 ties, spaced approximately 12 inches apart. Restrain reinforcing assembly from moving during pouring of concrete.

3.4 ACCEPTANCE CHECKS AND TESTS

Duct Testing and Cleaning:

a. Upon completion of the duct installation, a standard flexible mandrel shall be pulled through each duct to loosen particles of earth, sand, or foreign material left in the duct, and to test for out-of-round conditions.

b. The mandrel shall be not less than 12 inches long, and shall have a diameter not less than 0.5 inch less than the inside diameter of the duct. A brush with stiff bristles shall then be pulled through each duct to remove the loosened particles. The diameter of the brush shall be the same as, or slightly larger than, the diameter of the duct.

c. If testing reveals obstructions or out-of-round conditions, the Contractor shall replace affected section(s) of duct and retest to the satisfaction of the Contracting Officer.

d. Mandrel pulls shall be witnessed by the Contracting Officer.

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SECTION 26 05 73

OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY

11/22

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PART 3 EXECUTION (NOT USED)

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SECTION 26 05 73
11/22

OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies the overcurrent protective device coordination study, related calculations and analysis, indicated as the study in this section.

b. A short-circuit and selective coordination study, and arc flash calculations and analysis shall be prepared for the electrical overcurrent devices to be installed under this project.

c. The study shall present a well-coordinated time-current analysis of each overcurrent protective device from the individual device up to the utility sourceand Generators provided with Boiler Plant project..

1.2 RELATED WORK

a. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: General electrical requirements that are common to more than one section of Division 26.

b. Section 26 13 13, MEDIUM-VOLTAGE CIRCUIT BREAKER SWITCHGEAR: Medium-voltage circuit breaker switchgear.

c. Section 26 23 00, LOW-VOLTAGE SWITCHGEAR: Low-voltage switchgear.

d. Section 26 24 13, DISTRIBUTION SWITCHBOARDS: Low-voltage distribution switchboards.

e. Section 26 24 16, PANELBOARDS: Low-voltage panelboards.

f. Section 26 36 23, AUTOMATIC TRANSFER SWITCHES: Automatic transfer switches.

1.3 QUALITY ASSURANCE

a. Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

b. The study shall be prepared by the equipment manufacturer, and performed by the equipment manufacturer’s licensed electrical engineer.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Equipment Manufacturer’s Licensed Electrical Engineer; G,

Equipment Manufacturer; G,

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Product Data; G,

SD-05 Design Data

One Line Diagram; G,

Short-Circuit Study; G,

Arc Flash Calculations And Analysis; G,

SD-06 Test Reports

Complete Study; G,

Coordination Study; G,

SD-07 Certificates

Certifications; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Product data on the software program to be used for the study. Software shall be in mainstream use in the industry, shall provide device settings and ratings, and shall show selective coordination by time-current drawings.

b. Complete study as described in paragraph 1.7. Submittal of the study shall be well-coordinated with submittals of the shop drawings for equipment in related specification sections.

c. Certifications: Two weeks prior to final inspection, submit the following: Certification by the Contractor that the overcurrent protective devices have been set in accordance with the approved study.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1584 (2018; E 2019) Guide for Performing Arc-Flash Hazard Calculations

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70E	(2024) Standard for Electrical Safety in the Workplace

1.7 STUDY REQUIREMENTS

a. The study shall be in accordance with IEEE and NFPA standards.

b. The study shall include one line diagram, short-circuit and ground fault analysis, protective coordination plots for all overcurrent protective devices, and arc flash calculations and analysis.

c. One Line Diagram:

(1) Show all electrical equipment and wiring to be protected by the overcurrent devices.

(2) Show the following specific information:

(a) Calculated fault impedance, X/R ratios, and short-circuit values at each feeder and branch circuit bus.

(b) Relay, circuit breaker, and fuse ratings.

(c) Generator kW/kVA and transformer kVA and voltage ratings, percent impedance, X/R ratios, and wiring connections.

(d) Voltage at each bus.

(e) Identification of each bus, matching the identification on the drawings.

(f) Conduit, conductor, and busway material, size, length, and X/R ratios.

d. Short-Circuit Study:

(1) The study shall be performed using computer software designed for this purpose. Pertinent data and the rationale employed in developing the calculations shall be described in the introductory remarks of the study.

(2) Calculate the fault impedance to determine the available short-circuit and ground fault currents at each bus. Incorporate applicable motor and/or generator contribution in determining the momentary and interrupting ratings of the overcurrent protective devices.

(3) Present the results of the short-circuit study in a table. Include the following:

(a) Device identification.

(b) Operating voltage.

(c) Overcurrent protective device type and rating.

(d) Calculated short-circuit current.

e. Coordination Study:

(1) Prepare the coordination curves to determine the required settings of overcurrent protective devices to demonstrate selective coordination. Graphically illustrate on log-log paper that adequate time separation exists between devices, including the utility company upstream device if applicable. Plot the specific time-current characteristics of each overcurrent protective device in such a manner that all devices are clearly depicted.

(2) The following specific information shall also be shown on the coordination curves:

(a) Device identification.

(b) Potential transformer and current transformer ratios.

(c) Three-phase and single-phase ANSI damage points or curves for each cable, transformer, or generator.

(d) Applicable circuit breaker or protective relay characteristic curves.

(e) No-damage, melting, and clearing curves for fuses.

(f) Transformer in-rush points.

(3) Develop a table to summarize the settings selected for the overcurrent protective devices. Include the following in the table:

(a) Device identification.

(b) Protective relay or circuit breaker potential and current transformer ratios, sensor rating, and available and suggested pickup and delay settings for each available trip characteristic.

(c) Fuse rating and type.

f. Arc Flash Calculations and Analysis:

(1) Arc flash warning labels shall comply with Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

(2) Arc flash calculations shall be based on actual over-current protective device clearing time. Maximum clearing time shall be in accordance with IEEE 1584.

(3) Arc flash analysis shall be based on the lowest clearing time setting of the over-current protective device to minimize the incident energy level without compromising selective coordination.

(4) Arc flash boundary and available arc flash incident energy at the corresponding working distance shall be calculated for all electrical power distribution equipment specified in the project, and as shown on the drawings.

(5) Required arc-rated clothing and other PPE shall be selected and specified in accordance with NFPA 70E.

1.8 ANALYSIS

Analyze the short-circuit calculations, and highlight any equipment determined to be underrated as specified. Propose solutions to effectively protect the underrated equipment.

1.9 ADJUSTMENTS, SETTINGS, AND MODIFICATIONS

Final field settings and minor modifications of the overcurrent protective devices shall be made to conform with the study, without additional cost to the Government.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

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DVA/USACE PROJECTS COMMISSIONING OF ELECTRICAL SYSTEMS

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SECTION 26 08 00.01
8/19

DVA/USACE PROJECTS COMMISSIONING OF ELECTRICAL SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

a. The requirements of this Section apply to all sections of Division 26.

b. This project will have selected building systems commissioned. The complete list of equipment and systems to be commissioned is specified in Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS. The commissioning process, which the Contractor is responsible to execute, is defined in Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIRMENTS. A Commissioning Agent (CxA) appointed by the Government and Commissioning Manager (CxM) as indicated in Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIRMENTS will manage the commissioning process.

1.2 RELATED WORK

a. UFGS GENERAL REQUIREMENTS sections

b. Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS

c. UFGS Section 01 33 00, SUBMITTAL PROCEDURES

1.3 SUMMARY

a. This Section includes requirements for commissioning the Facility electrical systems, related subsystems and related equipment. This Section supplements the general requirements specified in Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING.

b. Refer to Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS for more details regarding processes and procedures as well as roles and responsibilities for all Commissioning Team members.

1.4 DEFINITIONS

Refer to Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS for definitions.

1.5 COMMISSIONED SYSTEMS

a. Commissioning of a system or systems specified in Division 26 is part of the construction process. Documentation and testing of these systems, as well as training of the VA’s Operation and Maintenance personnel in accordance with the requirements of Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIRMENTS and of Division 26, is required in cooperation with the DVA/USACE and the CxA and CxM.

b. The Facility electrical systems commissioning will include the systems listed in Section 01 91 00.01, DVA/USACE PROJECTS GENERAL

COMMISSIONING REQUIREMENTS:

1.6 SUBMITTAL INSTRUCTIONS

a. The commissioning process requires review of selected Submittals that pertain to the systems to be commissioned. The CxA will provide a list of submittals that will be reviewed by the CxA and CxM. This list will be reviewed and approved by the DVA/USACE prior to forwarding to the Contractor. Refer to UFGS Section 01 33 00, SUBMITTAL PROCEDURES for further details.

b. The commissioning process requires Submittal review simultaneously with engineering review. Specific submittal requirements related to the commissioning process are specified in Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS.

1.7 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Pre-Functional Checklists; G, DO

Systems Functional Performance Testing; G, DO

SD-06 Test Reports

Contractors Tests; G, DO

SD-08 Manufacturer's Instructions

Training Agendas And Trainer Resumes; G, DO

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 CONSTRUCTION INSPECTIONS

Commissioning of Electrical systems will require inspection of individual elements of the electrical systems construction throughout the construction period. The Contractor shall coordinate with the CxA and CxM in accordance with Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIRMENTS and the Commissioning plan to schedule electrical systems inspections as required to support the Commissioning Process.

3.2 PRE-FUNCTIONAL CHECKLISTS

The Contractor shall complete Pre-Functional Checklists to verify systems, subsystems, and equipment installation is complete and systems are ready for Systems Functional Performance Testing. The CxM will prepare Pre-Functional Checklists to be used to document equipment installation. The Contractor shall complete the checklists. Completed checklists shall be submitted to the DVA/USACE, CxA, and CxM for review. The CxM and CxA

may spot check a sample of completed checklists. If the CxM and CxA determine that the information provided on the checklist is not accurate, the CxM will return the marked-up checklist to the Contractor for correction and resubmission. If the CxM and CxA determine that a significant number of completed checklists for similar equipment are not accurate, the CCxM in coordination with CxA will select a broader sample of checklists for review. If the CxM and CxA determine that a significant number of the broader sample of checklists is also inaccurate, all the checklists for the type of equipment will be returned to the Contractor for correction and resubmission. Refer to Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS for submittal requirements for Pre-Functional Checklists, Equipment Startup Reports, and other commissioning documents.

3.3 CONTRACTORS TESTS

Contractor tests as required by other sections of Division 26 shall be scheduled and documented in accordance with UFGS GENERAL REQUIREMENTS sections. All testing shall be incorporated into the project schedule. Contractor shall provide no less than seven calendar days' notice of testing. The CxM will witness selected Contractor tests at the sole discretion of the CxM and CxA. The CxA and other Government representative(s) may choose to participate in some or all the testing procedures as observers. Contractor tests shall be completed prior to scheduling Systems Functional Performance Testing.

3.4 SYSTEMS FUNCTIONAL PERFORMANCE TESTING

The Commissioning Process includes Systems Functional Performance Testing that is intended to test systems functional performance under steady state conditions, to test system reaction to changes in operating conditions, and system performance under emergency conditions. The CxM will prepare final and detailed Systems Functional Performance Test procedures for review and approval by the CxA and Contracting Officer. The Contractor shall review and comment on the tests prior to approval. The Contractor shall provide the required labor, materials, and test equipment identified in the test procedure to perform the tests. The CxM will witness and document the testing. The Contractor shall sign the test reports to verify tests were performed. See Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS, for additional details.

3.5 TRAINING OF VA PERSONNEL

Training of the VA operation and maintenance personnel is required in cooperation with the Contracting Officer, VA Resident Engineer, and CxM. Provide competent, factory authorized personnel to provide instruction to operation and maintenance personnel concerning the location, operation, and troubleshooting of the installed systems. Contractor shall submit training agendas and trainer resumes in accordance with the requirements of Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIRMENTS. The instruction shall be scheduled in coordination with the Contracting Officer and VA Resident Engineer after submission and approval of formal training plans. Refer to Section 01 91 00.01, DVA/USACE PROJECTS GENERAL COMMISSIONING REQUIREMENTS and Division 26 Sections for additional Contractor training requirements.

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LIGHTING CONTROLS

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SECTION 26 09 23
11/22

LIGHTING CONTROLS

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation and connection of the lighting controls.

1.2 RELATED WORK

a. Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC: Interface of lighting controls with HVAC control systems.

b. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: General requirements that are common to more than one section of Division 26.

c. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES (600 VOLTS AND BELOW): Cables and wiring.

d. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path to ground for possible ground fault currents.

e. Section 26 24 16, PANELBOARDS: Panelboard enclosure and interior bussing used for lighting control panels.

f. Section 26 27 26, WIRING DEVICES: Wiring devices used for control of the lighting systems.

g. Section 26 51 00, INTERIOR LIGHTING: Luminaire ballast and drivers used in control of lighting systems.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit the following information for each type of lighting controls.

(2) Material and construction details.

(3) Physical dimensions and description.

(4) Wiring schematic and connection diagram.

(5) Installation details.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, complete maintenance and operating manuals, including technical data sheets, wiring diagrams, and information for ordering replacement parts.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

c. Certifications: Two weeks prior to final inspection, submit the following: Certification by the Contractor that the lighting control systems have been properly installed and tested.

1.6 REFERENCES

a. Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

UL SOLUTIONS (UL)

UL 773A	(2016; Reprint Jan 2024) UL Standard for Safety Nonindustrial Photoelectric Switches for Lighting Control
UL 916	(2015; Reprint Oct 2021) UL Standard for Safety Energy Management Equipment
UL 917	(2006; Reprint Aug 2013) UL Standard for Safety Clock-Operated Switches

PART 2 PRODUCTS

2.1 ELECTRONIC TIME SWITCHES

a. Electronic, solid-state programmable units with alphanumeric display; complying with UL 916 and or UL 917.

(1) Contact Configuration: DPST .

(2) Contact Rating: 30-A inductive or resistive120-277 volt .

(3) Astronomical Clock: Capable of switching a load on at sunset and off at sunrise, and automatically changing the settings each day in accordance with seasonal changes of sunset and sunrise. Additionally, it shall be programmable to a fixed on/off weekly schedule.

(4) Power Backup: Battery or capacitor for schedules and time clock.

2.2 ELECTROMECHANICAL-DIAL TIME SWITCHES

a. Electromechanical-dial time switches; complying with UL 917.

(1) Contact Configuration: DPST .

(2) Contact Rating: 30-A inductive or resistive, 120-277 volt .

(3) Wound-spring reserve carryover mechanism to keep time during power failures.

2.3 OUTDOOR PHOTOELECTRIC SWITCHES

a. Solid state, with DPST dry contacts rated for 1800 VA tungsten or 1000 VA inductive, complying with UL 773A.

(1) Light-Level Monitoring Range: 1.5 to 10 fc, with adjustable turn-on and turn-off levels.

(2) Time Delay: 15-second minimum.

(3) Surge Protection: Metal-oxide varistor.

(4) Mounting: Twist lock, with base-and-stem mounting or stem-and-swivel mounting accessories as required.

2.4 TIMER SWITCHES

a. Digital switches with backlit LCD display, 120/277 volt rated, fitting as a replacement for standard wall switches.

(1) Compatibility: Compatible with all ballasts.

(2) Warning: Audible warning to sound during the last minute of “on” operation.

(3) Time-out: Adjustable from five minutes to 12 hours.

(4) Faceplate: Refer to wall plate material and color requirements for toggle switches, as specified in Section 26 27 26, WIRING

DEVICES.

2.5 CEILING-MOUNTED PHOTOELECTRIC SWITCHES

a. Solid-state, light-level sensor unit, with separate relay unit.

(1) Sensor Output: Contacts rated to operate the associated relay. Sensor shall be powered from the relay unit.

(2) Relay Unit: Dry contacts rated for 20A ballast load at 120 volt and 277 volt, for 13A tungsten at 120 volt, and for 1 hp at 120 volt.

(3) Monitoring Range: 10 to 200 fc , with an adjustment for turn-on and turn-off levels.

(4) Time Delay: Adjustable from 5 to 300 seconds, with deadband adjustment.

(5) Indicator: Two LEDs to indicate the beginning of on-off cycles.

2.6 INDOOR OCCUPANCY SENSORS

a. Wall- or ceiling-mounting, solid-state units with a power supply and relay unit, suitable for the environmental conditions in which installed.

(1) Operation: Unless otherwise indicated, turn lights on when covered area is occupied and off when unoccupied; with a 1 to 15 minute adjustable time delay for turning lights off.

(2) Sensor Output: Contacts rated to operate the connected relay. Sensor shall be powered from the relay unit.

(3) Relay Unit: Dry contacts rated for 20A ballast load at 120 volt and 277 volt, for 13A tungsten at 120 volt, and for 1 hp at 120 volt.

(4) Mounting:

(a) Sensor: Suitable for mounting in any position on a standard outlet box.

(b) Time-Delay and Sensitivity Adjustments: Recessed and concealed behind hinged door.

(5) Indicator: LED, to show when motion is being detected during testing and normal operation of the sensor.

(6) Bypass Switch: Override the on function in case of sensor failure.

(7) Manual/automatic selector switch.

(8) Automatic Light-Level Sensor: Adjustable from 2 to 200 fc; keep lighting off when selected lighting level is present.

(9) Faceplate for Wall-Switch Replacement Type: Refer to wall plate material and color requirements for toggle switches, as specified in Section 26 27 26, WIRING DEVICES.

b. Dual-technology Type: Ceiling mounting; combination PIR and ultrasonic detection methods, field-selectable.

(1) Sensitivity Adjustment: Separate for each sensing technology.

(2) Detector Sensitivity: Detect occurrences of 6-inch minimum movement of any portion of a human body that presents a target of not less than 36 sq. in, and detect a person of average size and weight moving not less than 12 inches in either a horizontal or a vertical manner at an approximate speed of 305 mm/s (12 inches/s).

c. Detection Coverage: Shall be sufficient to provide coverage as required by sensor locations shown on drawing.

2.7 INDOOR VACANCY SENSOR SWITCH

a. Wall mounting, solid-state units with integral sensor and switch.

(1) Operation: Manually turn lights on with switch and sensor detects vacancy to turn lights off.

(2) Switch Rating: 120/277 volt, 1200 watts at 277 volt, 800 watts at 120 volt unit.

(3) Mounting:

(a) Sensor: Suitable for mounting in a standard switch box.

(b) Time-Delay and Sensitivity Adjustments: Integral with switch and accessible for reprogramming without removing switch.

(4) Indicator: LED, to show when motion is being detected during testing and normal operation of the sensor.

(5) Switch: Manual operation to turn lights on and override lights off.

(6) Faceplate: Refer to wall plate material and color requirements for toggle switches, as specified in Section 26 27 26, WIRING DEVICES.

2.8 OUTDOOR MOTION SENSOR (PIR)

a. Suitable for operation in ambient temperatures ranging from minus 40 to plus 130 degrees F.

(1) Operation: Turn lights on when sensing infrared energy changes between background and moving body in area of coverage; with a 1 to 15 minute adjustable time delay for turning lights off.

(2) Mounting:

(a) Sensor: Suitable for mounting in any position on a standard outdoor junction box.

(b) Relay: Internally mounted in a standard weatherproof electrical enclosure.

(c) Time-Delay and Sensitivity Adjustments: Recessed and concealed behind hinged door.

(3) Bypass Switch: Override the on function in case of sensor failure.

(4) Automatic Light-Level Sensor: Adjustable from 1 to 20 fc; keep lighting off during daylight hours.

b. Detector Sensitivity: Detect occurrences of 6-inch minimum movement of any portion of a human body that presents a target of not less than 36 sq. in.

c. Detection Coverage: Shall be sufficient to provide coverage as required by sensor locations shown on drawing.

d. Individually Mounted Sensor: Contacts rated to operate the connected relay, complying with UL 773A. Sensor shall be powered from the relay unit.

(1) Relay Unit: Dry contacts rated for 20A ballast load at 120 volt and 277 volt, for 13A tungsten at 120 volt, and for 1 hp at 120 volt.

(2) Indicator: LED, to show when motion is being detected during testing and normal operation of the sensor.

2.9 LIGHTING CONTROL SYSTEM – DISTIBUTIVE RELAY TYPE

a. System Description:

(1) The lighting control system shall be a network of remote relay modules connected to a digital network via network hubs and controlled through a system server / central station. Lighting control devices connect to the relay modules and communicate via the digital network with the system server. System includes all associated network interfaces and wiring, hubs, relay modules, relays, photocells, switches, dimmers, time clock, and occupancy sensors. System shall utilize distributed relays modules, allowing these relay modules to be located above accessible ceilings in or adjacent to rooms they are controlling.

(2) System shall include server / central station with operating software, data network, and BACnet IP communication with other systems as described. System communication protocol shall be compatible with the building automation system specified in Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

(3) System server / central station shall provide programmable operation of lights connected via system relays and controlled with system devices. System software shall provide control of relays and control devices, time and sequence scheduling, timed out and blink light operation and monitoring and reporting of system events and components. Initial programming shall be as shown on plans and schedules.

b. Server / Central Control Station: Lighting control system manufacturer shall be responsible to assure coordination between relay modules, network hubs and control system server/ central station such that system performs as described. Server / central control station

shall have a minimum 80 GB hard drive, 8 GB RAM, 3 GHz speed minimum, three Ethernet ports, 1024 x 768 resolution graphic card, and 3 USB 2.0 ports. Server shall be provided with monitor, keyboard and mouse, and plugged into a receptacle connected to an equipment emergency circuit as a minimum.

c. Network Hub: Network Hub shall contain processor and astronomic time clock for control and monitoring of lighting. Network Hub shall be fed from an equipment emergency circuit at a minimum.

d. Relay Modules: Mounted in NEMA enclosure with physically separate 120/277 volt wiring compartment from low voltage control wiring. Provide low voltage digital communication to control devices as shown on drawings and schedules. Supplemental power packs shall be provided as required for multiple control devices. This equipment shall be identified in shop drawing submission. Dimmable relay modules shall be provided where indicated. Relay modules shall contain up to four relays. Relay modules shall be labeled with room number that relays control lighting within.

e. Single-Pole Relays: Mechanically held unless otherwise indicated; split-coil, momentary-pulsed type, rated 20 A, 125 volt AC for tungsten filaments and 20 A, 277 volt AC for electronic ballasts, 50,000 cycles at rated capacity.

f. Control Devices: All occupancy sensors (Ultrasonic, IR and Dual Technology type), photocells, switches and timers shall be provided with system and designed to operate on system network. Supplemental power packs shall be provided as required for multiple control devices. This equipment shall be identified in shop drawing submission.

2.10 LIGHTING CONTROL SYSTEM – DIGITAL ADDRESSABLE LIGHTING INTERFACE (DALI)

a. System Description:

(1) The lighting control system shall consist of digital lighting control network connecting DALI compliant digital addressable ballasts, control modules and lighting control devices directly with a system server / central control station. Individually addressable electronic ballasts, control modules, and control devices are operated from signals received through DALI-compliant bus from variety of DALI compliant digital controllers and interfaces and programmed through the system server / central control station. System includes all associated network bus and wiring, DALI controllers and interfaces, panels, photocells, switches, dimmers, time clock, and occupancy sensors. System shall utilize DALI compliant ballast and dimming modules provided with light fixtures.

(2) System shall include server / central station with DALI operating software, data network, and BACnet IP communication with other systems as described. System communication protocol shall be compatible with the building automation system specified in Section 23 09 23, DIRECT-DIGITAL CONTROL SYSTEM FOR HVAC.

(3) System server / central station shall provide programmable operation of lights connected via system bus and controlled with

system devices. System software shall provide control of DALI ballast, control modules and control devices, time and sequence scheduling, timed out and blink light operation and monitoring and reporting of system events and components. Initial programming shall be as shown on plans and schedules.

b. Server / Central Control Station: Lighting control system manufacturer shall be responsible to assure coordination between relay modules, network hubs and control system server/ central station such that system performs as described. Server / central control station shall have a minimum 80 GB hard drive, 8 GB RAM, 3 GHz speed minimum, three Ethernet ports, 1024 x 768 resolution graphic card, and 3 USB 2.0 ports. Server shall be provided with monitor, keyboard and mouse, and plugged into a receptacle connected to an equipment emergency circuit as a minimum.

c. Control Devices: All occupancy sensors (Ultrasonic, IR and Dual Technology type), photocells, switches and timers shall be provided with system and be DALI compliant. Devices shall be designed to operate on system network. Supplemental DALI compliant signal repeaters and controllers shall be provided as required. This equipment shall be identified in shop drawing submission.

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation shall be in accordance with the NEC, manufacturer's instructions, as shown on the drawings, and as specified.

b. Aim outdoor photoelectric sensor according to manufacturer's recommendations. Set adjustable window slide for 1 footcandle turn-on.

c. Aiming for wall-mounted and ceiling-mounted motion sensor switches shall be per manufacturer’s recommendations.

d. Set occupancy sensor "on" duration to 15 minutes.

e. Locate photoelectric sensors as indicated and in accordance with the manufacturer's recommendations. Adjust sensor for the available light level at the typical work plane for that area.

f. Label time switches and contactors with a unique designation.

g. Program lighting control panels per schedule on drawings.

3.2 ACCEPTANCE CHECKS AND TESTS

a. Perform in accordance with the manufacturer's recommendations.

b. Upon completion of installation, conduct an operating test to show that equipment operates in accordance with requirements of this section.

c. Test for full range of dimming ballast and dimming controls capability. Observe for visually detectable flicker over full dimming range.

d. Test occupancy sensors for proper operation. Observe for light

control over entire area being covered.

e. Upon completion of the installation, the system shall be commissioned by the manufacturer’s factory-authorized technician who will verify all adjustments and sensor placements.

3.3 FOLLOW-UP VERIFICATION

Upon completion of acceptance checks and tests, the Contractor shall show by demonstration in service that the lighting control devices are in good operating condition and properly performing the intended function in the presence of Contracting Officer.

3.4 INSTRUCTION

a. Furnish the services of a factory-trained technician for one 8-hour training period for instructing personnel in the maintenance and operation of the lighting control system on the dates requested by the Contracting Officer.

b. Contractor shall submit written instructions on training and maintenance as reviewed in training session.

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SECTION 26 11 16
11/22

SECONDARY UNIT SUBSTATIONS

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, connection, and testing of the secondary unit substations, referred to as substation(s) in this section.

1.2 RELATED WORK

a. Section 03 30 00, CAST-IN-PLACE CONCRETE: Requirements for concrete equipment pads.

c. Section 25 10 10, ADVANCED UTILITY METERING: Electric meters installed in substations.

d. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

e. Section 26 05 13, MEDIUM-VOLTAGE CABLES: Medium-voltage cables.

f. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Low-voltage conductors.

g. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

h. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduits.

i. Section 26 05 73, OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY: Short circuit, coordination and arc-flash study, and requirements for a coordinated electrical system.

j. Section 26 13 13, MEDIUM-VOLTAGE CIRCUIT BREAKER SWITCHGEAR: Medium-voltage circuit breaker switchgear for use in secondary unit substations.

k. Section 26 23 00, LOW-VOLTAGE SWITCHGEAR: Switchgear for use in secondary unit substations.

l. Section 26 24 13, DISTRIBUTION SWITCHBOARDS: Distribution switchboards for use in secondary unit substations.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 FACTORY TESTS

a. Factory Tests shall be required.

b. Factory Tests shall be in accordance with Paragraph, MANUFACTURED PRODUCTS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirement:

(1) Substations shall be thoroughly tested at the factory to assure that there are no electrical or mechanical defects. Tests shall be conducted per UL and ANSI Standards. Factory tests shall be certified. The following tests shall be performed:

(a) Medium-Voltage Section: Refer to Section 26 13 13, MEDIUM-VOLTAGE CIRCUIT BREAKER SWITCHGEAR.

(b) Transformer Section:

1. Perform insulation-resistance tests winding-to-winding and each winding-to-ground.

2. Perform turns-ratio tests at all tap positions.

(c) Low-Voltage Section: Refer to Section 26 23 00, LOW-VOLTAGE SWITCHGEAR Section 26 24 13, DISTRIBUTION SWITCHBOARDS.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Product Data; G,

SD-06 Test Reports

Factory Tests; G,

Certified Factory Design And Production Test Reports; G,

SD-07 Certificates

Certifications; G,

SD-09 Manufacturer's Field Reports

Certified Field Test Reports; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.6 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Prior to fabrication of substations, submit the following product data for approval:

(a) Complete electrical ratings, including primary and secondary voltage, decibel rating, temperature rise, nominal impedance, voltage regulation, and no load and full load losses.

(b) Nameplate data.

(c) Elementary and interconnection wiring diagrams.

(d) Technical data for each component.

(e) Dimensioned exterior views of the substations.

(f) Dimensioned section views of the substations.

(g) Floor plan of the substations.

(h) Foundation plan for the substations.

(i) Provisions and required locations for external conduit and wiring entrances.

(j) Approximate design weights.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, complete maintenance and operating manuals, including technical data sheets, wiring diagrams, and information for ordering replacement parts.

(a) Include three-line diagrams showing device terminal numbers.

(b) Include schematic signal and control diagrams, with all terminals identified, matching terminal identification in the substation.

(c) Include information for testing, repair, troubleshooting, assembly, and disassembly.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

c. Test Reports:

(1) Submit certified factory design and production test reports for approval.

(2) Two weeks prior to the final inspection, submit certified field test reports.

d. Certifications: Two weeks prior to final inspection, submit the following:

(1) Certification by the manufacturer that substations conform to the requirements of the drawings and specifications.

(2) Certification by the Contractor that substations have been properly installed, adjusted, and tested.

1.7 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata), form a part of this specification to the extent referenced. Publications are referenced in the text by basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C57.12.01	(2020) General Requirements for Dry-Type Distribution and Power Transformers Including Those with Solid-Cast and/or Resin-Encapsulated Windings
IEEE C57.12.50	(1981; R 1998) Ventilated Dry-Type Distribution Transformers, 1 to 500 kVA, Single-Phase, and 15 to 500 kVA, Three-Phase, with High-Volt 601 to 34,500 Volts
IEEE C57.12.51	(2019) IEEE Guide for Mechanical Interchangeability of Ventilated Dry-Type Transformers

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ST 20	(2014) Dry-Type Transformers for General Applications

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 431	Energy Efficiency Program for Certain Commercial and Industrial Equipment

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

a. Unless otherwise specified, substations shall be in accordance with ANSI, ASTM, IEEE, NFPA, UL, 10 CFR 431, and as shown on the drawings.

b. Substations shall be a unitized integral assemblies, complete, grounded, continuous-duty, metal-clad, dead-front, dead-rear, with dry-type transformer.

c. Substations shall be designed, manufactured, and rated for indoor installation and service, with ventilation openings . External doors shall have provisions for padlocking.

d. Substation ratings shall be not less than required by NFPA 70, and not less than shown on the drawings. Short circuit current ratings shall be not less than the available maximum short circuit currents as shown on the drawings.

e. Substations shall conform to the arrangements and details shown on the drawings, and to the space designated for installation.

f. Incorporate key-type mechanical interlock systems as required in Section 26 13 13, MEDIUM-VOLTAGE CIRCUIT BREAKER SWITCHGEAR, , and Section 26 23 00, LOW-VOLTAGE SWITCHGEAR , as shown on the drawings.

g. Substations shall be assembled and prewired by the manufacturer at the factory. Substations shall be sub-assembled and shipped in complete sections ready for connection at the site. Where practical, a substation shall be shipped as one unit.

h. Substations shall be thoroughly cleaned, phosphate treated, and painted at the factory with light gray rust-inhibiting paint or baked enamel.

i. Provide advanced utility metering devices, instrument transformers (C.T. and P.T.), communication interfaces, wiring and raceway, and accessories as specified in Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS). Meters shall be multifunction electronic type with local display, remote communications, data logging, power quality monitoring, and alarming capabilities. All new meters shall be fully compatible with, and integrated into, the existing facility SCADA system for real-time monitoring, trending, alarming, and data archiving. Coordinate communication protocol (e.g., Modbus, TCP/IP, BACnet/IP, or facility-standard), point mapping/registers, and programming with the existing SCADA system integrator/vendor. Provide complete point lists, Modbus/BACnet object mappings, and support for HMI screen updates, alarm configuration, and trend setup. Dedicated communication cabling (Ethernet CAT6 or fiber as required by distance and facility standard) shall connect new meters to the existing SCADA network/data aggregation devices; do not share with IT or BAS networks. Meters and integration shall comply with VA Electrical Design Manual (PG-18-10) Section 5.4 POWER MONITORING AND METERING. Contractor shall perform integration testing, verify data accuracy against calibrated reference meters, and demonstrate full SCADA functionality (display, alarms, trends) prior to acceptance, refer to Section 25 08 10 UTILITY MONITORING AND CONTROL SYSTEM TESTING.

2.2 MEDIUM-VOLTAGE SECTION

a. Medium-Voltage Circuit Breaker(s): Refer to Section 26 13 13, MEDIUM-VOLTAGE CIRCUIT BREAKER SWITCHGEAR.

b. Interrupting ratings shall be not less than the maximum short circuit

current available, as shown on the drawings.

2.3 DRY-TYPE TRANSFORMERS

a. Shall comply with IEEE C57.12.01, and IEEE C57.12.50 for dry-type transformers rated up to 500 kVA, and IEEE C57.12.51 for dry-type transformers rated 501 kVA and larger.

b. Provide a cast coil type transformer with primary and secondary windings individually cast in epoxy. Transformer shall have an insulation system rated 185 degrees C, with an 80 degree C average winding temperature rise above a 40 degrees C maximum ambient.

c. Transformer shall be rated kVA rating as shown on the drawings, 95 kV BIL primary and 10 kV BIL secondary.

d. Primary and secondary windings:

(1) Windings shall be copper.

(2) Primary windings shall be delta-connected.

(3) Secondary windings shall be wye-connected except where otherwise shown on the drawings.

(4) Secondary windings shall have neutral bushings for transformers with wye-connected secondary windings.

(5) Terminals shall be the most suitable clamp or blade type as required for the circuit connections.

e. Provide four 2-1/2 percent full capacity taps, two above and two below rated primary voltage. Locate tap adjustments on the face of the medium voltage coil. Adjustments shall be accessible by removing the front panel and shall be made when the transformer is de-energized.

f. Features and accessories shall include the following:

(1) Winding temperature indicator.

(2) Auxiliary cooling equipment and controls.

(a) Transformer shall have provisions for future addition of automatically controlled fans for forced-air-cooling.

g. Transformer energy efficiency shall comply with the United States Department of Energy’s 10 CFR 431.

2.4 LOW VOLTAGE SECTION

a. Refer to Section 26 23 00, LOW-VOLTAGE SWITCHGEAR.

2.5 AUXILIARIES

a. Install additional components as shown on the drawings or otherwise required for the substations.

b. Provide 120-volt heaters inincoming section, dry-type transformer section, and outgoing section. Heaters shall be of sufficient

capacity to control moisture condensation in the compartments, shall be 250 watts minimum, and shall be controlled by a thermostat and humidistat located in each section. Thermostat shall be industrial type, high limit, to maintain compartments within the range of 60 to 90 degrees F. Humidistat shall have a range of 30 to 60 percent relative humidity. If heater voltage is different than substation secondary voltage, provide transformer rated to carry 125 percent of heater full load rating. Transformer shall have 220 degrees C insulation system with a temperature rise not exceeding 115 degrees C and shall conform to NEMA ST 20. Energize electric heaters while the substation is in storage or in place prior to being placed in service. Provide method for easy connection of heater to external power source.

PART 3 – EXECUTION

3.1 INSTALLATION

a. Install substations in accordance with NFPA 70, as shown on the drawings, and as recommended by the manufacturer.

b. Coordinate the components of the substations and their arrangements electrically and mechanically. Coordinate all circuit entrances into the substations, including methods of entrance and connections.

c. Anchor substations with rustproof bolts, nuts, and washers not less than 1/2 inch diameter, in accordance with manufacturer’s instructions, and as shown on the drawings.

d. Exterior Location. Mount substations on concrete slab. Unless otherwise indicated, the slab shall be at least 8 inches thick, reinforced with a 150 by 6 inches by 6 inches No. 6 mesh placed uniformly 4 inches from the top of the slab. Slab shall be placed on a 6 inches thick, well-compacted gravel base. The top of the concrete slab shall be approximately 4 inches above the finished grade. Edges above grade shall have 1/2 inch chamfer. The slab shall be of adequate size to project at least 8 inches beyond the equipment. Provide conduit turnups and cable entrance space required by the equipment to be mounted. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant. Cut off and bush conduits 3 inches above slab surface. Concrete work shall be as specified in Section 03 30 00, CAST-IN-PLACE CONCRETE.

e. Interior Location. Mount substations on concrete slab. Unless otherwise indicated, the slab shall be at least 4 inches thick. The top of the concrete slab shall be approximately 4 inches above finished floor. Edges above floor shall have 1/2 inch chamfer. The slab shall be of adequate size to project at least 8 inches beyond the equipment. Provide conduit turnups and cable entrance space required by the equipment to be mounted. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant. Cut off and bush conduits 3 inches above slab surface. Concrete work shall be as specified in Section 03 30 00, CAST-IN-PLACE CONCRETE.

f. Substation Grounding:

(1) Provide bare copper cable not smaller than No. 4/0 AWG, and not less than 24 inches below grade, interconnecting with the ground rods.

(2) Surge arresters (if applicable) and neutral shall be bonded directly to the transformer enclosure, and then to the grounding electrode system with bare copper conductors, sized as shown on the drawings. Lead lengths shall be kept as short as practical with no kinks or sharp bends.

3.2 ACCEPTANCE CHECKS AND TESTS

Perform tests in accordance with the manufacturer's recommendations. In addition, include the following:

a. Medium-Voltage Section Tests: Refer to Section 26 13 13, MEDIUM-VOLTAGE CIRCUIT BREAKER SWITCHGEAR.

b. Transformer Inspection and Tests:

(1) Compare equipment nameplate data with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition. Check for damaged or cracked insulators.

(3) Inspect all field-installed bolted electrical connections, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey after energization under load.

(4) Perform specific inspections and mechanical tests as recommended by manufacturer.

(5) Verify correct equipment grounding.

(6) Verify that the tap-changer is set at specified ratio.

(7) Verify proper secondary voltage phase-to-phase and phase-to-neutral after energization and prior to loading.

c. Low-Voltage Section Tests: Refer to Section 26 23 00, LOW VOLTAGE SWITCHGEAR .

3.3 FOLLOW-UP VERIFICATION

Upon completion of acceptance checks, settings, and tests, the Contractor shall demonstrate that the substations are in good operating condition and properly performing the intended function.

3.4 TEMPORARY HEATING

Apply temporary heat to substations, according to manufacturer's written instructions, throughout periods when the environment is not controlled for temperature and humidity within manufacturer's stipulated service conditions.

3.5 ONE LINE DIAGRAM AND SEQUENCE OF OPERATION

a. At final inspection, an as-built one line diagram shall be laminated or mounted under acrylic glass, and installed in a frame mounted in the substation room or in the outdoor substation enclosure.

b. Furnish a written sequence of operation for the substation and connected line side/load side electrical distribution equipment. The sequence of operation shall be laminated or mounted under acrylic glass, and installed in a frame mounted in the substation room or in the outdoor substation enclosure.

c. Deliver an additional four copies of the as-built one line diagram and sequence of operation to the Contracting Officer.

3.6 INSTRUCTION

Furnish the services of a factory-trained technician for one 4-hour training period for instructing personnel in the maintenance and operation of the substations, on the dates requested by the Contracting Officer.

-- End of Section --

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SECTION 26 12 19
11/22

PAD-MOUNTED, MEDIUM-VOLTAGE TRANSFORMERS

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, connection, and testing of platform-mounted / pad-mounted, **cast-coil dry-type**, medium-voltage transformers. **Oil-filled or liquid-immersed transformers are strictly prohibited.** Transformers shall be cast-coil (cast resin) dry-type and suitable for installation on structural steel dunnage on a building roof in an environment exposed to cooling tower chemical plumes and drift. Transformers shall include a vibration isolation system coordinated with the structural support system.

1.2 RELATED WORK

a. Section 26 05 73, OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY: External primary and secondary overcurrent protection is provided under other sections. Transformer impedance and ratings shall be coordinated with the study.

b. Section <SRF>25 10 10</SRF>, UTILITY MONITORING AND CONTROL SYSTEM (UMCS) FRONT END AND INTEGRATION: SCADA/monitoring integration requirements for transformer instrumentation.

c. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

d. Section 26 05 13, MEDIUM-VOLTAGE CABLES: Medium-voltage cables.

e. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path to ground for possible ground currents.

f. Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION: Manholes, pull-boxes, and ducts for underground raceway systems.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 FACTORY TESTS

a. Factory Tests shall be required.

b. Factory Tests shall be in accordance with Paragraph, MANUFACTURED PRODUCTS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirement: Transformers shall be thoroughly tested at the factory to ensure that there are no electrical or mechanical defects. Tests shall be conducted as per IEEE Standards. Factory tests shall be certified. The following

tests shall be performed:

(1) Perform insulation-resistance tests, winding-to-winding and each winding-to-ground.

(2) Perform turns-ratio tests at all tap positions.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-06 Test Reports

Factory Tests; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.6 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Paragraph, SUBMITTALS in Section26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

(1) Shop Drawings:

(a) Submit sufficient information to demonstrate compliance with drawings and specifications.

(b) Include electrical ratings, nameplate data, impedance, outline drawing with dimensions and front, top, and side views, weight, mounting details, decibel rating, termination information, temperature rise, no-load and full-load losses, regulation, overcurrent protection, connection diagrams, and accessories.

(c) Complete nameplate data, including manufacturer’s name and catalog number.

(2) Manuals:

(a) When submitting the shop drawings, submit companion copies of complete maintenance and operating manuals, including technical data sheets, wiring diagrams, and information for ordering replacement parts.

1. Identify terminals on wiring diagrams to facilitate installation, maintenance, and operation.

2. Indicate on wiring diagrams the internal wiring for each piece of equipment and interconnections between the pieces of equipment.

3. Approvals will be based on complete submissions of manuals, together with shop drawings.

(b) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

1. Update the manual to include any information necessitated by shop drawing approval.

2. Show all terminal identification.

3. Include information for testing, repair, troubleshooting, assembly, disassembly, and recommended maintenance intervals.

4. Provide a replacement parts list with current prices. Include a list of recommended spare parts, tools, and instruments for testing and maintenance purposes.

b. Certifications: Two weeks prior to the final inspection, submit the following certifications.

(1) Certification by the manufacturer that the transformers conform to the requirements of the drawings and specifications.

(2) Certification by the Contractor that the transformers have been properly installed, connected, and tested.

1.7 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 386	(2016) Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV
IEEE C62.11	(2020) Standard for Metal-Oxide Surge Arresters for Alternating Current Power Circuits (>1kV)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ST 20	(2014) Dry-Type Transformers for General Applications

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 431	Energy Efficiency Program for Certain Commercial and Industrial Equipment

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

a. Transformers shall be in accordance with ASTM, IEEE, NFPA, UL, as shown on the drawings, and as specified herein. Each transformer shall be assembled as an integral unit by a single manufacturer.

b. Transformers shall be cast-coil (cast resin) dry-type only. Oil-filled, liquid-immersed, or any transformer containing dielectric fluid is strictly prohibited. All transformers shall be designed, manufactured, and tested in accordance with IEEE C57.12.00, IEEE C57.12.01, IEEE C57.12.90, and DOE energy efficiency standards. Transformers shall be suitable for outdoor installation on a roof in close proximity to cooling towers.

c. Ratings shall not be less than shown on the drawings and these specifications.

d. Completely fabricate transformers at the factory so that only the external cable connections are required at the project site.

e. Thoroughly clean, phosphatize, and finish all the metal surfaces at the factory with a rust-resistant primer and dark green enamel finish coat, except where a different color is specified. All surfaces of the transformer that will be in contact with the concrete pad shall be treated with corrosion-resistant compounds and epoxy resin or a rubberized sealing compound.

2.2 COMPARTMENTS

a. Construction:

(1) Transformers shall be of cast-coil (cast resin) construction with aluminum windings. The core shall be high-grade silicon steel. The complete core and coil assembly shall be vacuum cast in epoxy resin. Transformers shall be self-extinguishing and shall not produce toxic gases in the event of fire (C2-E2-F1 classification).

(2) The enclosure shall be NEMA 3R / IP23 rated for exterior use with stainless steel hardware. The enclosure and all exposed components shall be suitable for exposure to cooling tower chemical plumes and drift. Enhanced corrosion protection is required.

(3) The compartments shall be constructed of sheet steel (gauge to meet ANSI requirements) with bracing and with reinforcing gussets using jig welds to assure rectangular rigidity.

(4) All bolts, nuts, and washers shall be zinc-plated steel.

(5) Sufficient space shall be provided for equipment, cabling, and terminations within the compartments.

(6) Affix transformer nameplate permanently within the low-voltage compartment. Voltage and kVA rating, connection configuration, impedance, date of manufacture, and serial number shall be shown on the nameplate.

b. Doors:

(1) Provide a separate door for each compartment with provisions for a single padlock to secure all doors. Provide each compartment door with open-position doorstops and corrosion-resistant tamperproof hinges welded in place. The medium-voltage compartment door shall be mechanically prevented from opening unless the low-voltage compartment door is open.

(2) The secondary compartment door shall have a one-piece steel handle and incorporate three-point locking mechanisms.

(3) Provide a 2 inches size padlock for each assembly, as approved by the Contracting Officer. Padlocks shall be keyed to the Contracting Officer'sestablished key set. Firmly attach the padlock to the door assembly by a chain.

2.3 RATINGs

Transformers shall be rated as follows:

a. Capacity: 5,000 kVA base rating, self-cooled (AN).

b. Function: Step-down.

c. Winding Material: Aluminum.

d. Winding Method: Cast-coil (cast resin) construction.

e. Average Winding Temperature Rise: 125 °C over 30 °C ambient.

f. Insulation Class: Class H (180 °C).

g. Vector Group: Dyn11.

h. Percent Impedance: 5.75 % (tolerance ±7.5 % per IEEE C57.12).

i. Primary Voltage: 13,800 V delta connected, 95 kV BIL.

j. Secondary Voltage: 4,160 V wye connected, 60 kV BIL. Secondary neutral (X0) shall be isolated from the casing and provided with a removable ground strap.

k. Cooling Class: AN (Air Natural) self-cooled. Provisions for future conversion to AN/AF (forced-air) cooling shall be included so that the temperature control system can activate fans without modification to the transformer.

l. Fire Classification: C2-E2-F1 (cast-coil) per applicable standards. Transformers shall be self-extinguishing and shall not produce toxic gases in the event of fire.

m. K-Factor Rating: K13 (suitable for the harmonic content of multiple 4,160 V VFD-driven centrifugal chiller loads).

n. Maximum In-Rush Current: 5 times rated current.

o. Voltage Adjustment Taps: Seven (7) taps — two (2) above and four (4) below nominal voltage, each ±2.5 %, de-energized tap changer, externally operable.

p. Efficiency: DOE 2016 compliant (or latest applicable edition).

2.4 PRIMARY CONNECTIONS

a. Primary connections shall be 600 A deadbreak wells and inserts for cable sizes shown on the drawings.

b. Surge Arresters: Distribution class, one for each primary phase, complying with IEEE C62.11, supported from tank wall.

2.5 MEDIUM-VOLTAGE TERMINATIONS

a. Terminate the medium-voltage cables in the primary compartment with 600 A deadbreak premolded rubber elbow connectors, suitable for submersible applications. Elbow connectors shall have a semi-conductive shield material covering the housing. The separable connector system shall include the loadbreak elbow, the bushing insert, and the bushing well. Separable connectors shall comply with the requirements of IEEE 386, and shall be interchangeable between suppliers. Allow sufficient slack in medium-voltage cable, ground, and drain wires to permit elbow connectors to be moved to their respective parking stands.

b. Ground metallic cable shield with a cable shield grounding adapter, consisting of a solderless connector enclosed in watertight rubber housing covering the entire assembly, bleeder wire, and ground braid.

2.6 LOW-VOLTAGE EQUIPMENT

a. Mount the low-voltage bushings, and hot stick in the low-voltage compartment.

b. The low-voltage leads shall be brought out of the tank by epoxy pressure tight bushings, and shall be standard arrangement.

c. Tin-plate the low-voltage neutral terminal and isolate from the transformer tank. Provide a removable ground strap sized in accordance with NFPA 70 and connect between the secondary neutral and ground pad.

2.7 TRANSFORMERS

a. Transformers shall be of cast-coil (cast resin) dry-type construction only. Oil-filled or liquid-immersed designs are prohibited. Windings shall be aluminum, vacuum cast in epoxy resin. The core shall be high-grade silicon steel (standard, non-amorphous unless otherwise approved). Transformers shall be suitable for outdoor installation exposed to cooling tower chemical plumes and drift. Enhanced corrosion protection suitable for this environment is required.

b. Enclosure shall be NEMA 3R / IP23 exterior rated with stainless steel hardware. The enclosure shall protect against rain, sleet, snow, and external ice formation while allowing ventilation for AN cooling. All hardware and exposed metal shall be stainless steel or equivalently corrosion resistant.

c. Transformers shall be suitable for installation on structural steel dunnage on a building roof, including reinforced base and vibration

isolation system coordinated with the structural support system.

d. Primary and Secondary Windings for Three-Phase Transformers:

(1) Primary windings shall be delta-connected.

(2) Secondary windings shall be wye-connected, except where otherwise indicated on the drawings. Provide isolated neutral bushings for secondary wye-connected transformers.

(3) Secondary leads shall be brought out through pressure-tight epoxy bushings.

e. Core and Coil Assemblies:

(1) Cores shall be grain-oriented, non-aging, silicon steel to minimize losses.

(2) Core and coil assemblies shall be rigidly braced to withstand the stresses caused by rough handling during shipment, and stresses caused by any possible short-circuit currents.

(3) Coils shall be continuous-winding type without splices except for taps. Material shall be copper.

(4) Coil and core losses shall be optimum for efficient operation.

(5) Primary, secondary, and tap connections shall be brazed or pressure type.

(6) Provide end fillers or tie-downs for coil windings.

f. The transformer casing, cover, and radiator gauge thickness shall not be less than that required by ANSI.

g. Accessories:

Transformers shall be provided with the following accessories as a minimum:

a. Temperature Control and Monitoring System: PTC sensors in the low-voltage windings at the hottest spot. The system shall provide continuous hot-spot temperature display, alarm, and trip functions. It shall be capable of controlling future forced-air cooling fans. The controller shall support remote monitoring via RS-485 or hardwired contacts to the UMCS.

b. Surge Arresters: Distribution-class surge arresters on the primary (one per phase), mounted adjacent to the HV bushings.

c. Grounding: Removable neutral ground strap on the secondary neutral (X0). Ground pads in both HV and LV compartments.

d. Nameplate: Stainless steel or anodized aluminum, permanently affixed in the low-voltage compartment, containing all required data per IEEE C57.12.00 plus project-specific ratings.

e. Lifting Lugs, Grounding Pads, and NEMA Safety Labels: As required for safe handling and installation on roof dunnage.

f. Electrostatic Shielding: Not required.

g. Provide 120-volt heaters inincoming section, dry-type transformer section, and outgoing section. Heaters shall be of sufficient capacity to control moisture condensation in the compartments, shall be 250 watts minimum, and shall be controlled by a thermostat and humidistat located in each section. Thermostat shall be industrial type, high limit, to maintain compartments within the range of 60 to 90 degrees F. Humidistat shall have a range of 30 to 60 percent relative humidity. If heater voltage is different than substation secondary voltage, provide transformer rated to carry 125 percent of heater full load rating. Transformer shall have 220 degrees C insulation system with a temperature rise not exceeding 115 degrees C and shall conform to NEMA ST 20. Energize electric heaters while the transformer is in storage or in place prior to being placed in service. Provide method for easy connection of heater to external power source. Circuit to a 20A/1P breaker in Equipment Branch panelboard 17-1LP-EQA with 2#10 & 1#10 GND in 3/4" C.

(2) The accessories shall be made accessible within the compartments without disassembling trims and covers.

h. Transformers shall meet the energy conservation standards for transformers per the United States Department of Energy 10 CFR 431.

2.8 VIBRATION ISOLATION

a. Provide a complete vibration isolation system between the transformer base and structural steel dunnage. The system shall consist of spring-type isolators with minimum 25 mm (1 inch) static deflection (or approved elastomer pads) with seismic and wind restraints. Isolators shall be selected for the actual transformer weight. Coordinate exact type, deflection, and attachment with the structural engineer. The system shall reduce structure-borne vibration and audible hum into the building.

2.9 INSTRUMENTATION AND SCADA PROVISIONS

a. Provide the following instrumentation with auxiliary contacts for remote indication and alarming to the UMCS. All contacts shall be wired to terminal blocks located in an external NEMA 4 (or NEMA 4X where required) control cabinet mounted on the low-voltage side of the transformer. The control cabinet shall be accessible for maintenance without opening the high-voltage or low-voltage compartment doors. Provide sufficient spare terminals for future use.

1. Dial-type thermometer (top-oil temperature) with auxiliary contacts and maximum reading pointer.

b. Provide provisions for future installation of current transformers and potential transformers for UMCS metering as indicated on the drawings.

PART 3 EXECUTION

3.1 INSTALLATION

a. Install transformers outdoors, as shown on the drawings, in accordance with NFPA 70, and as recommended by the manufacturer.

b. Anchor transformers with rustproof bolts, nuts, and washers not less than 1/2 inch diameter, in accordance with manufacturer’s instructions, and as shown on drawings.

c. Transformers shall be suitable for installation on structural steel dunnage on a building roof, including vibration isolation system coordinated with the structural support system.

d. Grounding:

(1) Ground each transformer in accordance with the requirements of NFPA 70. Install ground rods per the requirements of Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS, to maintain a maximum resistance of 5 ohms to ground.

(2) Connect the ground rod to the ground pads in the medium- and low-voltage compartments.

(3) Install and connect the cable shield grounding adapter per the manufacturer’s instructions. Connect the bleeder wire of the cable shield grounding adapter to the loadbreak or deadbreak elbow grounding point with minimum No. 14 AWG wire, and connect the ground braid to the grounding system with minimum No. 6 AWG bare copper wire. Use soldered or mechanical grounding connectors listed for this purpose.

3.2 ACCEPTANCE CHECKS AND TESTS

Perform manufacturer’s required field tests in accordance with the manufacturer's recommendations. In addition, include the following: Visual Inspection and Tests:

a. Compare equipment nameplate data with specifications and approved shop drawings.

b. Inspect physical and mechanical condition. Check for damaged or cracked bushings or coil damage.

c. Verify that control and alarm settings on temperature indicators are as specified.

d. Inspect all field-installed bolted electrical connections, using the calibrated torque-wrench method to verify tightness of accessible bolted electrical connections, and perform thermographic survey after energization under load.

e. Vacuum-clean transformer interior. Clean transformer enclosure exterior.

f. Verify transformer arrangement allows for proper drainage of water..

g. Verify correct equipment grounding per the requirements of Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS.

h. Verify the presence and connection of transformer surge arresters, if provided.

i. Verify that the tap-changer is set at rated system voltage.

3.3 FOLLOW-UP VERIFICATION

Upon completion of acceptance checks, settings, and tests, the Contractor shall demonstrate that the transformers are in good operating condition and properly performing the intended function. Verify metering information is reported as required in Section 25 10 10, UTILITY MONITORING AND CONTROL SYSTEM (UMCS) FRONT END AND INTEGRATION: SCADA/monitoring integration requirements for transformer instrumentation.

3.4 SPARE PARTS

Deliver the following spare parts for the project to the Contracting Officer two weeks prior to final inspection:

a. Six insulated protective caps.

b. One spare set of medium-voltage fuses for each size and type of fuse used in the project.

3.5 INSTRUCTION

The Contractor shall instruct maintenance personnel, for not less than one 2-hour period, on the maintenance and operation of the equipment on the date requested by the Contracting Officer.

-- End of Section --

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SECTION 26 13 01

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05/24

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SECTION 26 13 01

PAD-MOUNTED DEAD-FRONT AIR INSULATED SWITCHGEAR
05/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A240/A240M	(2025a) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM D1535	(2014; R 2018) Standard Practice for Specifying Color by the Munsell System

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 386	(2016) Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV
IEEE C2	(2023) National Electrical Safety Code
IEEE C37.74	(2014) Standard Requirements for Subsurface, Vault, and Pad-Mounted Load-Interrupter Switchgear and Fused Load-Interrupter Switchgear for Alternating Current Systems Up to 38 kV
IEEE C37.75	(2023) Standard for Pad-Mounted, Pole-Mounted, and Submersible Switchgear Enclosures and Associated Control Enclosures - Coastal and Non-Coastal Environmental Integrity
IEEE C62.11	(2020) Standard for Metal-Oxide Surge Arresters for Alternating Current Power Circuits (>1kV)

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS	(2025) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NFPA 70B (2023) Recommended Practice for Electrical Equipment Maintenance

UL SOLUTIONS (UL)

UL 467 (2022) UL Standard for Safety Grounding and Bonding Equipment

1.2 RELATED REQUIREMENTS

1.3 DEFINITIONS

1.3.1 Switched Way

A switched way is considered a three-phase circuit entrance to the bus through a switch.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Switchgear Drawings; G

SD-03 Product Data

Fuse Time-Current Characteristic Curves; G

Air Insulated Pad-mounted Switchgear; G

Insulated High-Voltage Connectors; G

Surge Arresters; G

SD-06 Test Reports

Acceptance Checks and Tests; G

SD-07 Certificates

Paint Coating System; G

SD-09 Manufacturer's Field Reports

Switchgear Design and Production Tests; G

SD-10 Operation and Maintenance Data

Air Insulated Pad-Mounted Switchgear Operation and Maintenance, Data Package 5; G

1.5 QUALITY ASSURANCE

1.5.1 Switchgear Drawings

Furnish drawings that include, but are not limited to, the following:

a. Overall dimensions, weights, plan view, and front view

b. Ratings

c. Single-line diagram.

1.5.2 Paint Coating System

Submit IEEE C37.75 paint coating system performance requirement tests.

1.5.3 Fuse Time-Current Characteristic Curves

Provide time-current characteristic curves in PDF format and in electronic format suitable for import or updating into the SKM PowerTools for Windows computer program.

1.6 MAINTENANCE

1.6.1 Air Insulated Pad-mounted Switchgear Operation and Maintenance Data

Submit Operation and Maintenance Manuals in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

PART 2 PRODUCTS

2.1 AIR INSULATED PAD-MOUNTED SWITCHGEAR

IEEE C37.74. Submit manufacturer's information on switches and each component, device, and accessory provided with the equipment.

2.1.1 Ratings and Test Requirements

Provide switchgear with a nominal voltage rating of 14.4 kV and the following corresponding ratings:

Rated Maximum Voltage, kV	17
Rated Withstand Impulse Voltage, kV BIL	95
Continuous and Load Interrupting Current, A	600
Short-Circuit Current, kA rms Sym	25
Power Fuses, Amperes, Maximum Interrupting, kA rms Sym	500E

Provide switched ways rated for the required continuous and load interrupting current.

2.1.2 Switchgear Construction

Provide switchgear with the following construction and configuration:

a. Switch contacts and cable entrance terminations contained in an enclosed, steel compartment.

b. Configured with load interrupting and fused switched ways as indicated.

c. Accessible terminations suitable for cables entering from below.

d. Switch contact positions for switched ways visible through viewing windows in the switchgear termination compartment.

e. Each switched way with two position switch; Open, Closed and provisions for grounding.

2.1.2.1 Pad-mounting Provisions

Provide enclosed switchgear suitable for installation on a concrete pad, including the following:

a. Fabricate switchgear enclosure and enclosure base of ASTM A240/A240M type 304 or 304L stainless steel.

b. Enclosure base includes any part of the switchgear enclosure that is within 3 inches of concrete pad.

c. Paint enclosure including base ASTM D1535 Munsell 7GY3.29/1.5 green .

d. Comply with IEEE C37.75 for the paint coating system regardless of equipment material.

2.1.3 Load Interrupting Switched Ways

Provide the following for load interrupter switched ways:

a. Three-pole group operated switching.

b. Interrupter switches operated by means of an externally accessible switch-operating hub.

c. Switch-operating hub located within a recessed stainless steel pocket mounted on the side of the pad-mounted gear enclosure.

d. Padlockable stainless steel access cover and hood on the switch-operating-hub pocket to protect the padlock shackle from tampering.

e. Stops on the switch-operating hub to prevent overtravel.

f. Labels in the switch-operating-hub pocket to indicate switch position.

g. Folding switch-operating handle for each interrupter switch.

2.1.4 Fused Ways

Provide the following:

a. Fuse mountings enclosed in an inner steel compartment.

b. Each fuse mounting installed as an integral part of a fuse handling mechanism that does not allow access to the fuse until the elbow for that fuse has been disconnected and a mechanical interlock to the fuse-access panel has been actuated.

c. The opening into the component compartment covered by the fuse-access panel in both the open and closed positions to prevent access to high voltage.

d. Blown-fuse indicators for fused ways visible through viewing windows in the termination compartment.

2.1.4.1 Fuses

Provide fuses in accordance with the following:

a. Fuse ratings as indicated on the drawings.

b. Helically coiled fuses if rated 10 amperes or larger.

c. Solid-material power fuses capable of detecting and interrupting all faults under all realistic conditions of circuitry, with line-to-line or line-to-ground voltage across the fuse, and capable of handling the full range of transient recovery voltage severity associated with these faults.

d. All arcing accompanying operation of the fuse contained within the fuse, and all arc products and gases evolved effectively contained within the exhaust control device during fuse operation.

e. Fusible elements nonaging and nondamagable with melting time-current characteristics that are permanently accurate to within a maximum tolerance of 10 percent in terms of current.

f. Equipped with a blown-fuse indicator that provides visible evidence of fuse operation while installed in the fuse mounting.

2.1.5 Dead-Front High-Voltage Bushings

IEEE 386. 15 kV, 95 kV BIL. Provide 600 ampere one-piece deadbreak apparatus bushings for each switched way. Provide connectors with 200 ampere bushing interface.

a. Parking stands: Provide a parking stand near each dead-front bushing.

2.2 Insulated High-Voltage Connectors

IEEE 386. Provide corresponding connector for each switched way; provide connectors with a grounding eye and test point.

a. 600 Ampere deadbreak connector ratings: Voltage: 15 kV, 95 kV BIL. Short time rating: 25,000 rms symmetrical amperes. Provide 200 ampere bushing interface for surge arresters.

2.3 Surge Arresters

IEEE C62.11, rated15kV, fully shielded, dead-front, metal-oxide-varistor, elbow type with resistance-graded gap, suitable for plugging into inserts. Provide arresters on switched ways as indicated.

2.4 Grounding Provisions

Provide a ground-connection pad in each termination compartment.

2.5 SOURCE QUALITY CONTROL

2.5.1 Switchgear Design and Production Tests

Furnish reports which include results of design and production tests performed according to IEEE C37.74. Perform production tests by the manufacturer on each switchgear assembly to ensure that design performance is maintained in production.

PART 3 EXECUTION

3.1 INSTALLATION

Conform to IEEE C2, NFPA 70, and to the requirements specified herein.

3.2 GROUNDING

Conform to NFPA 70 and IEEE C2, except provide grounding systems with a resistance to solid earth ground not exceeding 25 ohms. The provision of the contract covering "Changes" applies when work is directed to obtain the specified ground resistance in addition to that indicated or specified.

3.2.1 Grounding Electrodes

Provide driven ground rods as specified in at each corner of switchgear pad.

3.2.2 Switchgear Grounding

Connect #4/0 bare copper conductor ground loop, not less than 24 inches below grade, to the upper end of the ground rods by exothermic welds or compression connectors. Provide #4/0 bare copper conductors connecting the switchgear grounding provisions to two different ground rods.

3.2.3 Connections

Make joints in grounding conductors and ground loop by exothermic weld or compression connector. Install exothermic welds and compression connectors as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

3.2.4 Grounding and Bonding Equipment

Conform to UL 467, except as indicated or specified otherwise.

3.3 FOUNDATION FOR EQUIPMENT AND ASSEMBLIES

Mount switch on concrete slab, including the following:

a. Provide slab of size at least 12 inches thick, reinforced with a 6 by

6 - W2.9 by W2.9 mesh, placed uniformly 4 inches from the top of the slab.

b. Place Slab on a 6 inch thick, well-compacted gravel base.

c. Install top of concrete slab approximately 4 inches above finished grade. Provide edges above grade 1/2 inch chamfer.

d. Provide slab of adequate size to project at least 8 inches beyond equipment.

Stub up conduits, with bushings, 2 inches into cable wells in the concrete pad. Coordinate dimensions of cable wells with switch cable training areas. Provide concrete work as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.4 FIELD QUALITY CONTROL

3.4.1 Performance of Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, NFPA 70B, NETA ATS, and referenced ANSI standards. Submit reports, including acceptance criteria and limits for each test in accordance with NETA ATS "Test Values".

Include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

3.4.1.1 Switchgear

a. Visual and Mechanical Inspection

(1) Compare equipment nameplate information with specifications and approved shop drawings.

(2) Inspect physical and mechanical condition.

(3) Check for proper anchorage, alignment, required area clearances, and grounding.

(4) Perform mechanical operator tests in accordance with manufacturer's instructions.

(5) Inspect all indicating devices for proper operation.

(6) Test interlock systems for proper operation and sequencing.

b. Electrical Tests

(1) Perform contact-resistance tests.

(2) Perform insulation-resistance tests.

(3) Perform an over-potential test on each switched way pole with the switched way in the closed position in accordance with the manufacturer's instructions.

3.4.1.2 Grounding System

a. Visual and Mechanical Inspection

(1) Inspect ground system for compliance with contract plans and specifications.

b. Electrical Tests

(1) Perform ground impedance measurements utilizing the fall-of-potential method. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. Use an instrument equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test.

(2) Submit the measured ground resistance of each ground rod and grounding system, indicating the location of the rod and grounding system. Include the test method and test setup (i.e., pin location) used to determine ground resistance and soil conditions at the time the measurements were made.

3.4.2 Follow-Up Verification

Upon completion of acceptance checks and tests, show by demonstration in service that devices are in good operating condition and properly performing the intended function. Perform each item not less than three times to demonstrate its function. As an exception to requirements stated elsewhere in the contract, notify the Contracting Officer 5 working days in advance of the dates and times for checks and tests.

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SECTION 26 13 13

MEDIUM-VOLTAGE CIRCUIT BREAKER SWITCHGEAR

06/25

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SECTION 26 13 13
06/25

MEDIUM-VOLTAGE CIRCUIT BREAKER SWITCHGEAR

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, connection, and testing of medium-voltage circuit breaker switchgear, indicated as switchgear in this section.

1.2 RELATED WORK

a. Section 03 30 00, CAST-IN-PLACE CONCRETE: Requirements for concrete equipment pads.

b. Section 25 10 10, ADVANCED UTILITY METERING: Electric meters installed in switchgear.

c. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

d. Section 26 05 13, MEDIUM-VOLTAGE CABLES: Medium-voltage cables and terminations.

e. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Low-voltage conductors.

f. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path to ground for possible ground fault currents.

g. Section 26 05 73, OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY: Short circuit and coordination study, and requirements for a coordinated electrical system.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 FACTORY TESTS

a. Factory Tests shall be required.

b. Factory Tests shall be in accordance with Paragraph, MANUFACTURED PRODUCTS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

(1) Switchgear shall be tested, with the circuit breakers in the connected position in their cubicles. Tests shall be in accordance with NEMA/ANSI C37.54 and NEMA/ANSI C37.55, and IEEE C37.09. Factory tests shall be certified, and shall include

the following tests:

(a) Design tests.

(b) Production tests.

(c) Conformance tests.

(2) The following additional tests shall be performed:

(a) Verify that circuit breaker sizes and types correspond to drawings, and the Overcurrent Protective Device Coordination Study.

(b) Verify that current and voltage transformer ratios correspond to drawings.

(c) Verify tightness of bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer’s published data.

(d) Verify correct barrier and shutter installation and operation.

(e) Exercise all active components.

(f) Inspect indicating devices for correct operation.

(g) Perform an insulation-resistance test, phase to ground, on each bus section, with phases not under test grounded, in accordance with manufacturer’s published data.

(h) Perform insulation-resistance tests on control wiring with respect to ground. Applied potential shall be 500 V DC for 300-volt rated cable and 1000 V DC for 600-volt rated cable, or as required if solid-state components or control devices cannot tolerate the applied voltage.

(i) If applicable, verify correct function of control transfer relays located in the switchgear with multiple control power sources.

(j) Perform phasing checks on double-ended or dual-source switchgear to insure correct bus phasing from each source.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-06 Test Reports

Test Reports; G,

Factory Tests; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.6 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Switchgear shop drawings shall be submitted simultaneously with or after the Overcurrent Protective Device Coordination Study.

(2) Submit sufficient information to demonstrate compliance with drawings and specifications.

(3) Prior to fabrication of switchgear, submit the following data for approval:

(a) Complete electrical ratings.

(b) Circuit breaker sizes.

(c) Interrupting ratings.

(d) Safety features.

(e) Accessories and nameplate data.

(f) Switchgear one line diagram, showing ampere rating, number of bars per phase and neutral in each bus run (horizontal and vertical), bus spacing, equipment ground bus, and bus material.

(g) Elementary and interconnection wiring diagrams.

(h) Technical data for each component.

(i) Dimensioned exterior views of the switchgear.

(j) Dimensioned section views of the switchgear.

(k) Floor plan of the switchgear.

(l) Foundation plan for the switchgear.

(m) Provisions and required locations for external conduit and wiring entrances.

(n) Approximate design weights.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, companion copies of complete maintenance and operating manuals, including technical data sheets, wiring diagrams, and information for ordering replacement parts.

(a) Three-line diagrams showing device terminal numbers.

(b) Schematic signal and control diagrams, with all terminals identified, matching terminal identification in the switchgear.

(c) Include information for testing, repair, troubleshooting, assembly, disassembly, and factory recommended/required periodic maintenance procedures and frequency.

(d) Provide a replacement and spare parts list. Include a list of tools and instruments for testing and maintenance purposes.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

c. Test Reports:

(1) Submit certified factory design and production test reports for approval.

(2) Two weeks prior to the final inspection, submit certified field test reports and data sheets.

d. Certifications: Two weeks prior to final inspection, submit four copies of the following.

(1) Certification by the manufacturer that switchgear conforms to the requirements of the drawings and specifications.

(2) Certification by the Contractor that switchgear has been properly installed, adjusted, and tested.

1.7 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C37.09	(2018; Errata 2019; Corr 2021) Standard Test Procedure for AC High-Voltage Circuit Breakers Rated on a Symmetrical Current Basis
IEEE C37.2	(2022 Standard for Electrical Power System Device Function Numbers, Acronyms and Contact Designations
IEEE C37.23	(2015) Metal-Enclosed Bus
IEEE C37.90	(2005; R 2011) Standard for Relays and

	Relay Systems Associated With Electric Power Apparatus
IEEE C62.11	(2020) Standard for Metal-Oxide Surge Arresters for Alternating Current Power Circuits (>1kV)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA/ANSI C37.54	(2002) Indoor Alternating Current High-Voltage Circuit Breakers Applied as Removable Elements in Metal-Enclosed Switchgear-–Conformance Test Procedures
NEMA/ANSI C37.55	(2020) Switchgear-Medium Voltage Metal-Clad Assemblies-Conformance Test Procedures

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

Switchgear shall be in accordance with IEEE, NEMA, NFPA, as shown on the drawings, and have the following features:

a. Switchgear shall be a complete, grounded, continuous-duty, integral assembly, metal clad, dead-front, dead-rear, self-supporting, indoor type switchgear assembly, . Incorporate devices shown on the drawings and all related components required to fulfill operational and functional requirements.

b. Ratings shall not be less than shown on the drawings. Short circuit ratings shall not be less than 250 MVA.

c. Switchgear shall conform to the arrangements and details shown on the drawings.

d. Key-type mechanical interlocks for multiple circuit breakers shall be provided as shown on the drawings.

e. Switchgear shall be assembled, connected, and wired at the factory so that only external circuit connections are required at the construction site. Split the structure only as required for shipping and installation. Circuit breakers and accessories shall be packaged and shipped separately. Packaging shall provide adequate protection against rough handling during shipment.

f. All non-current-carrying parts shall be grounded per Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS, for additional requirements.

h. Provide advanced utility metering devices, instrument transformers (C.T. and P.T.), communication interfaces, wiring and raceway, and accessories as specified in Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS). Meters shall be multifunction electronic type

with local display, remote communications, data logging, power quality monitoring, and alarming capabilities. All new meters shall be fully compatible with, and integrated into, the existing facility SCADA system for real-time monitoring, trending, alarming, and data archiving. Coordinate communication protocol (e.g., Modbus, TCP/IP, BACnet/IP, or facility-standard), point mapping/registers, and programming with the existing SCADA system integrator/vendor. Provide complete point lists, Modbus/BACnet object mappings, and support for HMI screen updates, alarm configuration, and trend setup. Dedicated communication cabling (Ethernet CAT6 or fiber as required by distance and facility standard) shall connect new meters to the existing SCADA network/data aggregation devices; do not share with IT or BAS networks. Meters and integration shall comply with VA Electrical Design Manual (PG-18-10) Section 5.4 POWER MONITORING AND METERING. Contractor shall perform integration testing, verify data accuracy against calibrated reference meters, and demonstrate full SCADA functionality (display, alarms, trends) prior to acceptance, refer to Section 25 08 10 UTILITY MONITORING AND CONTROL SYSTEM TESTING.

2.2 HOUSING

a. Shall have the following features:

(1) Frames and enclosures:

(a) The assembly shall be braced with reinforcing gussets using bolted connections to assure rectangular rigidity.

(b) The enclosure shall be steel, leveled, and not less than the gauge required by applicable publications.

(c) Die-pierce the holes for connecting adjacent structures to insure proper alignment, and to allow for future additions.

(d) All bolts, nuts, and washers shall be zinc-plated steel.

(2) Cubicles:

(a) An individual cubicle shall be supplied for each circuit breaker and each future circuit breaker as shown on the drawings. Cubicles shall be provided with isolated wireways for control wiring between devices.

1. Compartment each cubicle so that the circuit breaker, buses, and cable terminations are in separate compartments with steel partitions or barriers of approved and properly installed insulation.

2. Each cubicle furnished with a circuit breaker (active or spare) shall be fully equipped as noted on drawings and specified below.

3. Each cubicle noted as space for future circuit breaker shall be fully equipped for positioning and connecting the breakers. Provide all equipment required to implement the future breaker installation, except the relays and meters on the cubicle doors and the associated current transformers.

(b) Conveniently locate test blocks within each cubicle for

circuit breaker wiring connections.

(3) Auxiliary compartments:

(a) Cubicles shall be provided for auxiliaries, metering, and transition or termination sections as required by the manufacturer, and as shown on drawings. Cubicles shall be provided with isolated wireways for control wiring between devices.

(4) Cubicle doors:

(a) The doors shall permit convenient removal and interchanging of the circuit breakers between cubicles. The doors shall be capable of a swing approaching 180 degrees and shall be provided with intermediate doorstops.

(b) Each door shall include suitable handles and padlocking provisions. Concealed or semi-concealed hinges shall be provided to attach the doors. Weld the hinges to the equipment structure and to the cubicle doors.

(c) The following equipment shall be mounted on the door of circuit breaker cubicles:

1. A breaker control switch.

2. Breaker-position-indicator lamps.

3. Protective relays and/or metering as indicated on the drawings or other sections of the specifications.

4. Any additional components indicated on the drawings.

c. Finish:

(1) All metal surfaces shall be thoroughly cleaned, phosphatized and factory primed prior to applying baked enamel or lacquer finish.

(2) Provide a light gray finish for indoor switchgear.

(3) Outdoor switchgear:

(a) Interior finish shall be light gray.

(b)

(c) The underside of the switchgear and enclosure shall be treated with corrosion resistant compounds, epoxy resin, or rubberized sealing compound.

2.3 BUS

a. Bus Bars and Interconnections:

(1) Provide copper buses, fully rated for the amperage shown on the drawings for entire length of the switchgear.

(2) Fully insulate and totally enclose the buses within the bus compartment of switchgear cubicles.

(3) Mount the buses on appropriately spaced insulators and brace to withstand the available short circuit currents.

(4) The bus and bus compartment shall be designed so that the acceptable NEMA standard temperature rises are not exceeded.

(5) Install a copper ground bus the full length of the switchgear assembly.

(6) All bolts, nuts, and washers shall be zinc-plated steel. Bolts shall be torqued to the values recommended by the manufacturer.

(7) Make provisions for future bus extensions by means of bolt holes or other approved method.

b. Insulation: The insulation shall be a high flame-retardant, self extinguishing, high track-resistant material that complies with the IEEE C37.23 149 degree F) temperature rise.

c. Control Bus: Extend the control buses to all of the circuit breaker cubicles including spare and spaces for future circuit breakers.

2.4 CIRCUIT BREAKERS

a. Breakers that have the same ratings shall be interchangeable with other breakers in that line-up.

b. Circuit breakers shall have the following features:

(1) Outdoor: vacuum interrupter type.

(a) Sulfur Hexafloride (SF6):

1. Three independently sealed SF6 interrupters.

2. Protect the interrupter contacts from moisture and contaminated atmospheres.

3. Arc interruption based on SF6 single-pressure puffer principle.

4. Low pressure; normal operating pressure of 2.5 bar gauge for the SF6.

5. Provide a low-pressure alarm on each interrupter.

6. Readily accessible contact wear indicator for each interrupter.

7. Provisions for slow closing (testing).

8. Breaker total interrupting time of five cycles.

9. Maintenance free interrupter.

(b) Indoor: Vacuum:

1. Three independent sealed high-vacuum interrupters.

2. Protect the interrupter contacts from moisture and contaminated atmospheres.

3. Readily accessible contact wear indicator for each interrupter.

4. Breaker total interrupting time of three cycles.

5. Maintenance free interrupter.

6. Contact surfaces to be of special alloys (such as copper chrome) to reduce effect of chopping.

(2) Operating mechanism:

(a) The mechanism shall operate in a quick-make, quick-break manner and shall be charged by a small universal motor to provide stored-energy for breaker operation. Breaker tripping, closing, and indicating lamps shall be DC operated.

(b) The speed of the contacts during the operation shall be independent of the control voltage and the operator's movements.

(c) Equip the mechanism for manual opening and closing of the contacts during loss of normal control power.

(3) Relays: Comply with IEEE C37.90, integrated digital type with test blocks and plugs. Provide relay functions per the IEEE C37.2, and as shown on the drawings.

(4) Drawout rails:

(a) Design the rails to guide the breakers to their disconnected, test, and connected positions. Provide a positive stop at each of the positions by a levering mechanism.

(b) The breaker shall maintain contact with ground in all positions through flexible connections and ground shoes.

(c) Make provisions for padlocking the breaker in the test and disconnected position.

(5) Power line and load disconnecting contact fingers and springs:

(a) The contact fingers shall be silver-plated, full-floating, self-aligning, self-coupling, and designed for cleaning action during engaging and disengaging movements.

(b) Provide adequate flexibility between stationary and movable components to assure proper meeting of the contact fingers, while also providing adequate pressure on the contact surfaces.

(6) The stationary contacts for the line and load breaker contact fingers shall be isolated from the breaker compartment by shutters when the breaker is removed from the connected position.

(7) The control and auxiliary contacts of the breaker shall be silver plated, multi-contact, self-coupling, plug and socket type. The

contacts shall connect the circuits through terminal blocks that shall be conveniently mounted on the breaker for visual inspection.

(8) Mechanical interlocks:

(a) Shall prevent the breaker from movement, except when the breaker contacts are in the open position.

(b) Shall prevent the breaker from closing the contacts while in the connected position, except when the power line and load disconnecting contacts are completely connected.

(c) Key interlock shall be installed on the front cover of the breaker.

c. The interrupting ratings of the breakers shall be not less than 250 MVA.

2.5 CURRENT TRANSFORMERS

a. Provide encapsulated type current transformers or approved equal. The transformers shall have a mechanical and one-second thermal rating in RMS amperes of not less than the momentary and interrupting rating of the breaker at rated voltage.

b. Provide transformer ratios as shown on the drawings. Accuracies shall be coordinated with the associated relays by the switchgear manufacturer to assure proper operation at the selected pick-up and operating current ratings.

2.6 POTENTIAL TRANSFORMERS

a. The potential transformers shall be encapsulated, drawout, disconnecting type, and shall be properly protected by primary current-limiting fuses.

b. When the transformers are withdrawn from the compartment the primary terminals shall be grounded.

c. The transformer ratios and accuracies shall be coordinated, with the associated relays by the switchgear manufacturer.

2.7 CONTROL POWER TRANSFORMERS

a. The control power transformers shall be encapsulated, drawout, disconnecting type and shall be properly protected by primary current-limiting fuses.

b. The ratings of the transformer shall be as indicated on the drawings.

c. Refer to the drawings for rating and capacity of the circuit breaker equipped panelboard served by the control power transformer.

d. Equip the control power transformer compartment door with indicating lights and nameplates to indicate when the control power is energized.

e. Dual Control Power Supplies:

(1) For each of the incoming feeders, provide a separate control power

transformer.

(2) An automatic transfer switch shall transfer the secondary connected load as follows:

(a) While the preferred incoming feeder is energized, the load shall be connected to the transformer energized by the feeder.

(b) While the preferred incoming feeder is de-energized and the other incoming feeder is energized, the load shall be transferred to the energized incoming feeder.

2.8 SYSTEM

a. Batteries:

(1) Provide high discharge rate type maintenance-free nickel-cadmium batteries. voltage shall be 125 volts nominal. Calculate the battery capacity based on the lowest ambient temperature in the room where it is to be installed. Include a safety margin of 50 percent for reserve capacity.

(a) Provide sufficient battery capacity to carry all continuous loads (lamps, relays, etc.) for 8 hours and then perform the greater of the following duties, with the charger de-energized.

1. Trip all circuit breakers simultaneously or,

2. Close the largest breaker in a line-up of four or less breakers, or close the two largest breakers simultaneously in a line-up of more than four breakers. Breaker closing current shall include both the spring release coil current and the starting current of the spring charging motor.

(2) Provide battery connector covers for protection against external short circuits.

(3) Provide corrosion-resistant steel battery racks.

b. Charger:

(1) Provide a charger of the full-wave rectifier type utilizing silicon controlled rectifiers as the power-control elements. Construction shall be modular with plug-in control units for easy replacement.

(2) The charger shall maintain 1/2 of one percent voltage regulation from no load to full load for line voltage variation of 10 percent, and frequency variation of 3 Hz from 60 Hz.

(3) The charger shall maintain a nominal float voltage of 1.4 vpc, and a nominal equalizing voltage of 1.5 vpc.

(4) The charger shall be capable of continuous operation in an ambient temperature of 104 degrees F without derating. The charger shall be installed in a convection cooled NEMA Type 1 ventilated enclosure. The housing is to have a hinged front door with all equipment accessible from the front.

(5) Provide both AC and DC transient protection. Charger shall be able to recharge a fully discharged battery without tripping AC protective devices. AC circuit breaker shall not trip under any DC load condition, including short circuit on output terminals.

(6) The charger shall be capable of supplying the following demand simultaneously:

(a) Recharging a fully discharged battery in 12 hours.

(b) Supervisory panel and control panel.

(c) Steady loads (indicating lamps, relays, etc.).

(7) The charger shall have fused AC input and DC output protection.

(8) The charger shall not discharge the batteries when AC power fails.

(9) The charger shall have the following accessories:

(a) On-off control switch with pilot light.

(b) AC power failure alarm light.

(c) High DC voltage alarm light.

(d) Low DC voltage alarm light.

(e) Ground detection switch and alarm light.

(f) DC ammeter - 2 percent accuracy.

(g) DC voltmeter - 2 percent accuracy: Float/equalize voltage marked in red on voltmeter.

(h) Provisions for activation of remote annunciation of trouble for the above conditions.

2.9 METERING

a. Refer to Section 25 10 10, ADVANCED UTILITY METERING. Refer to drawings for meter locations.

b. As necessary, provide vertical structure with a front hinged door to provide safe isolated access to meters and all associated terminal and fuse blocks for maintenance, calibration or testing.

c. Provide current transformers for each meter. Current transformers shall be wired to shorting-type terminal blocks.

d. Provide voltage transformers including primary fuses and secondary protective devices for metering as shown on the drawings.

2.10 OTHER EQUIPMENT

a. Furnish tools and accessories required for circuit breaker and switchgear test, inspection, maintenance, and proper operation.

b. Cable terminations:

(1) Cable terminations shall conform to the requirements in Section 26 05 13, MEDIUM-VOLTAGE CABLES.

(2) Coordinate cable terminations with the switchgear being furnished.

c. Medium-voltage surge arresters:

(1) Distribution class, metal-oxide-varistor type. Comply with IEEE C62.11.

(2) Provide each ungrounded conductor of each incoming circuit with an appropriate arrester for the application voltage.

(3) Provide each phase of each circuit breaker with appropriate surge arrester for application voltage.

d. Panelboards: Requirements for panelboards shown to be installed in the switchgear shall be as shown on the drawings and in Section 26 24 16, PANELBOARDS.

e. Circuit breaker removal equipment: Furnish a permanent circuit breaker removal device mounted on top of enclosure for installation and removal of circuit breakers.

2.11 CONTROL WIRING

Switchgear control wiring shall not be less than No. 14 AWG copper 600 volt rated. Install wiring complete at the factory, adequately bundled and protected. Provide separate control circuit fuses in each breaker compartment and locate for ease of access and maintenance.

2.12 NAMEPLATES AND MIMIC BUS

a. Nameplates: For Normal Power system, provide laminated black phenolic resin with white core with 1/2 inch engraved lettered nameplates next to each circuit breaker. For Essential Electrical System, provide laminated red phenolic resin with white core with 1/2 inch engraved lettered nameplates next to each circuit breaker. Nameplates shall indicate equipment served, spaces, or spares in accordance with one line diagram shown on drawings. Nameplates shall be mounted with plated screws on front of breakers or on equipment enclosure next to breakers.

b. Mimic Bus: Provide an approved mimic bus on front of each switchgear assembly. Color shall be black for the Normal Power system and red for the Essential Electrical System, either factory-painted plastic or metal strips. Plastic tape shall not be used. Use symbols similar to one line diagram shown on drawings. Plastic or metal strips shall be mounted with plated screws.

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation shall be in accordance with NFPA 70, as shown on the drawings, and manufacturer’s instructions.

b. Anchor switchgear with rustproof bolts, nuts, and washers not less

than 1/2 inch diameter, in accordance with manufacturer’s instructions, and as shown on drawings.

b. Exterior Location. Mount switchgear on concrete slab. Unless otherwise indicated, the slab shall be at least 8 inches thick, reinforced with a 6 by 6 inches No. 6 mesh placed uniformly 4 inches from the top of the slab. Slab shall be placed on a 6 inches thick, well-compacted gravel base. The top of the concrete slab shall be approximately 4 inches above the finished grade. Edges above grade shall have 1/2 inch chamfer. The slab shall be of adequate size to project at least 8 inches beyond the equipment. Provide conduit turnups and cable entrance space required by the equipment to be mounted. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant. Cut off and bush conduits 3 inches above slab surface. Concrete work shall be as specified in Section 03 30 00, CAST-IN-PLACE CONCRETE.

c. Interior Location. Mount switchgear on concrete slab. Unless otherwise indicated, the slab shall be at least 4 inches thick. The top of the concrete slab shall be approximately 4 inches above finished floor. Edges above floor shall have 1/2 inch chamfer. The slab shall be of adequate size to project at least 8 inches beyond the equipment. Provide conduit turnups and cable entrance space required by the equipment to be mounted. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant. Cut off and bush conduits 3 inches above slab surface. Concrete work shall be as specified in Section 03 30 00, CAST-IN-PLACE CONCRETE.

3.2 ACCEPTANCE CHECKS AND TESTS

a. An authorized representative of the switchgear manufacturer shall technically supervise and participate during all of the field adjustments and tests. Major adjustments and field tests shall be witnessed by the Contracting Officer. The manufacturer’s representative shall certify in writing that the equipment has been installed, adjusted and tested in accordance with the manufacturer’s recommendations.

b. Prior to the final inspection for acceptance, a technical representative from the electric utility company shall witness the testing of the equipment to assure the proper operation of the individual components, and to confirm proper operation/coordination with electric utility company’s equipment.

c. Perform manufacturer’s required field tests in accordance with the manufacturer's recommendations. In addition, include the following:

(1) Visual Inspection and Tests:

(a) Compare equipment nameplate data with specifications and approved shop drawings.

(b) Inspect physical, electrical, and mechanical condition.

(c) Confirm correct application of manufacturer's recommended lubricants.

(d) Verify appropriate anchorage, required area clearances, and correct alignment.

(e) Verify that circuit breaker sizes and types correspond to approved shop drawings.

(f) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey after energization.

(g) Verify appropriate equipment grounding.

(h) Confirm correct operation and sequencing of key-type mechanical interlock systems.

(i) Vacuum-clean enclosure interior. Clean enclosure exterior.

(j) Inspect insulators for evidence of physical damage or contaminated surfaces.

(k) Verify correct shutter installation and operation.

(l) Exercise all active components.

(m) Verify the correct operation of all sensing devices, alarms, and indicating devices.

(n) Verify that vents are clear.

(o) Inspect control power transformers.

(2) Electrical tests:

(a) Perform insulation-resistance tests on each bus section.

(b) Perform overpotential tests.

(c) Perform insulation-resistance test on control wiring; do not perform this test on wiring connected to solid-state components.

(d) Perform phasing check on double-ended switchgear to ensure correct bus phasing from each source.

(e) Circuit breakers shall be tripped by operation of each protective device.

3.3 FOLLOW-UP VERIFICATION

Upon completion of acceptance checks and tests, the Contractor shall show by demonstration in service that the medium-voltage circuit breaker switchgear is in good operating condition and properly performing the intended function.

3.4 TEMPORARY HEATING

Apply temporary heat to switchgear, according to manufacturer's written instructions, throughout periods when switchgear environment is not controlled for temperature and humidity within manufacturer's stipulated service conditions.

3.5 WARNING SIGN

Mount on each entrance door of the switchgear room, approximately 5 feet above grade or floor, a clearly lettered warning sign for warning personnel. The sign shall be attached with rustproof metal screws.

3.6 ONE LINE DIAGRAM AND SEQUENCE OF OPERATION

a. At final inspection, an as-built one line diagram shall be laminated or mounted under acrylic glass, and installed in a frame mounted in the switchgear room or in the outdoor switchgear enclosure.

b. Furnish a written sequence of operation for the switchgear and connected line side/load side electrical distribution equipment. The sequence of operation shall be laminated or mounted under acrylic glass, and installed in a frame mounted in the switchgear room or in the outdoor switchgear enclosure.

c. Deliver an additional four copies of the as-built one line diagram and sequence of operation to the Contracting Officer.

3.7 AS-LEFT RELAY SETTINGS, FUSE RATINGS FOR CONTROL EQUIPMENT

a. The relay settings shall be set in the field by an authorized representative of the switchgear manufacturer per the approved Overcurrent Protective Device Coordination Study in accordance with Section 26 05 73, OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY.

b. The relay settings of the main breaker(s) shall be reviewed by the electric utility company to assure coordination with the electric utility company primary fusing. Prior to switchgear activation, provide written verification of this review to the Contracting Officer.

c. Post a durable copy of the "as-left" relay settings, and fuse ratings for control equipment in a convenient location in the switchgear room . Deliver four additional copies of the settings and fuse ratings to the Contracting Officer. Furnish this information prior to the activation of the switchgear.

3.8 INSTRUCTION

Furnish the services of a factory-trained technician for one 4-hour training period for instructing personnel in the maintenance and operation of the switchgear, on the dates requested by the Contracting Officer.

-- End of Section --

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SECTION 26 22 00
11/22

LOW-VOLTAGE TRANSFORMERS

PART 1 – GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, connection, and testing of low-voltage dry-type general-purpose transformers, indicated as transformers in this section.

1.2 RELATED WORK

a. Section 03 30 00, CAST-IN-PLACE CONCRETE: Requirements for concrete equipment pads.

b. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

c. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Low-voltage conductors.

d. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

e. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduit.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Transformers; G,

Nonlinear Transformers; G,

Energy Saving Harmonic Cancellation Transformers; G,

Buck-Boost Transformers; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Include electrical ratings, dimensions, mounting details, materials, required clearances, terminations, weight, temperature rise, wiring and connection diagrams, plan, front, side, and rear elevations, accessories, and device nameplate data.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, companion copies of complete maintenance and operating manuals including technical data sheets and wiring diagrams.

(a) Schematic signal and control diagrams, with all terminals identified, matching terminal identification in the transformers.

(b) Include information for testing, repair, troubleshooting, assembly, disassembly, and factory recommended/required periodic maintenance procedures and frequency.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

c. Certifications: Two weeks prior to final inspection, submit the following.

(1) Certification by the manufacturer that the transformers conform to the requirements of the drawings and specifications.

(2) Certification by the Contractor that the transformers have been properly installed, adjusted, and tested.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 431 Energy Efficiency Program for Certain Commercial and Industrial Equipment

PART 2 PRODUCTS

2.1 TRANSFORMERS

a. Unless otherwise specified, transformers shall be in accordance with NEMA, NFPA, UL and as shown on the drawings.

b. Transformers shall have the following features:

(1) Self-cooled by natural convection, isolating windings, indoor dry-type. Autotransformers will not be accepted, except as specifically allowed for buck-boost applications.

(2) Rating and winding connections shall be as shown on the drawings.

(3) Ratings shown on the drawings are for continuous duty without the use of cooling fans.

(4) Copper windings.

(5) Insulation systems:

(a) Transformers 30 kVA and larger: UL rated 428 degrees F) system with an average maximum rise by resistance of 302 degrees F) in a maximum ambient of 104 degrees F.

(b) Transformers below 30 kVA: Same as for 30 kVA and larger or UL rated 365 degrees F) system with an average maximum rise by resistance of 239 degrees F in a maximum ambient of 104 degrees F.

(6) Core and coil assemblies:

(a) Rigidly braced to withstand the stresses caused by short-circuit currents and rough handling during shipment.

(b) Cores shall be grain-oriented, non-aging, and silicon steel.

(c) Coils shall be continuous windings without splices except for taps.

(d) Coil loss and core loss shall be minimized for efficient operation.

(e) Primary and secondary tap connections shall be brazed or pressure type.

(f) Coil windings shall have end filters or tie-downs for maximum strength.

(7) Average audible sound levels shall comply with NEMA.

(8) If not shown on drawings, nominal impedance shall be as permitted by NEMA.

(9) Single phase transformers rated 15 kVA through 25 kVA shall have two 5 percent full capacity taps below normal rated primary voltage. All transformers rated 30 kVA and larger shall have two 2.5 percent full capacity taps above, and four 2.5 percent full

capacity taps below normal rated primary voltage.

(10) Core assemblies shall be grounded to their enclosures with adequate flexible ground straps.

(11) Enclosures:

(a) Comprised of not less than code gauge steel.

(b) Outdoor enclosures shall be NEMA 3R.

(c) Temperature rise at hottest spot shall conform to NEMA Standards, and shall not bake and peel off the enclosure paint after the transformer has been placed in service.

(d) Ventilation openings shall prevent accidental access to live components.

(e) The enclosure at the factory shall be thoroughly cleaned and painted with manufacturer's prime coat and standard finish.

(12) Standard NEMA features and accessories, including ground pad, lifting provisions, and nameplate with the wiring diagram and sound level indicated.

(13) Dimensions and configurations shall conform to the spaces designated for their installations.

(14) Transformers shall meet the energy conservation standards for transformers per the United States Department of Energy’s 10 CFR 431.

2.2 NONLINEAR TRANSFORMERS

a. Nonlinear transformers shall be as specified in Paragraph 2.1, with additional features as specified below.

b. Transformers shall be designed to withstand the overheating effects caused by harmonics resulting from non-linear (non-sinusoidal) loads.

c. Neutral rating shall be 200 percent of rated secondary phase current.

d. Transformers with K factor of 13 shall be provided, if K factor is not shown on contract drawings.

2.3 ENERGY SAVING HARMONIC CANCELLATION TRANSFORMERS

a. Shall be as specified in Paragraph 2.1, with additional features as specified below.

b. Zero sequence currents shall be treated via flux cancellation in the secondary windings.

c. Each winding shall be independently single-shielded with a full-width copper electrostatic shield.

d. Neutral rating shall be 200 percent of rated secondary phase current.

2.4 BUCK-BOOST TRANSFORMERS

a. Shall be as specified in Paragraph 2.1, with additional features as specified below.

(1) Suitable for connection as auto transformers to provide the percentages of voltage buck or boost as shown on the drawings.

(2) Insulation system to be 220 degrees C.

(3) Sound levels shall not exceed 36 dB.

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation of transformers shall be in accordance with the NEC, as recommended by the equipment manufacturer and as shown on the drawings.

b. Anchor transformers with rustproof bolts, nuts, and washers, in accordance with manufacturer’s instructions, and as shown on drawings.

e. Install transformers with manufacturer's recommended clearance from wall and adjacent equipment for air circulation. Minimum clearance shall be 6 inches.

f. Install transformers on vibration pads designed to suppress transformer noise and vibrations.

3.2 ACCEPTANCE CHECKS AND TESTS

a. Perform tests in accordance with the manufacturer's recommendations. In addition, include the following:

(1) Visual Inspection and Tests:

(a) Compare equipment nameplate data with specifications and approved shop drawings.

(b) Inspect physical and mechanical condition.

(c) Inspect all field-installed bolted electrical connections, using the calibrated torque-wrench method to verify tightness of accessible bolted electrical connections.

(d) Perform specific inspections and mechanical tests as recommended by manufacturer.

(e) Verify correct equipment grounding.

(f) Verify proper secondary phase-to-phase and phase-to-neutral voltage after energization and prior to connection to loads.

3.3 FOLLOW-UP VERIFICATION

Upon completion of acceptance checks, settings, and tests, the contractor shall demonstrate that the transformers are in good operating condition, and properly performing the intended function.

-- End of Section --

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SECTION 26 23 00
11/22

LOW-VOLTAGE SWITCHGEAR

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, connection, and testing of low-voltage switchgear, indicated as switchgear in this section.

1.2 RELATED WORK

a. Section 03 30 00, CAST-IN-PLACE CONCRETE: Requirements for concrete equipment pads.

b. Section 25 10 10, ADVANCED UTILITY METERING: Electric meters installed in switchgear.

c. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

d. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Low-voltage conductors.

e. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible fault currents.

f. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduits.

g. Section 26 05 73, OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY: Short circuit and coordination study, and requirements for a coordinated electrical system.

h. Section 26 11 16, SECONDARY UNIT SUBSTATIONS: Secondary unit substations.

i. Section 26 43 13, SURGE PROTECTIVE DEVICES: For surge protective devices integral to the switchgear.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 FACTORY TESTS

a. Factory Tests shall be required.

b. Factory Tests shall be in accordance with Paragraph, MANUFACTURED PRODUCTS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

c. Switchgear shall be tested, with the circuit breakers in the connected

position in their compartments. Tests shall be in accordance with IEEE C37.20.1 and NEMA C37.51. Factory tests shall be certified, and shall include the following tests:

(1) Design tests.

(2) Production tests.

(3) Conformance tests.

d. The following additional tests shall be performed:

(1) Verify that circuit breaker sizes and types correspond to drawings, and the Overcurrent Protective Device Coordination Study.

(2) Verify tightness of bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer’s published data.

(3) Confirm correct operation and sequencing of key-type mechanical interlock systems for multiple circuit breakers by attempting closure on locked-open devices, and attempting to open locked-closed devices, and making key exchange with devices operated in off-normal positions.

(4) Verify correct barrier and shutter installation and operation.

(5) Exercise all active components.

(6) Inspect indicating devices for correct operation.

(7) Perform an insulation-resistance test, phase to ground, on each bus section, with phases not under test grounded, in accordance with manufacturer’s published data.

(8) Perform insulation-resistance tests on control wiring with respect to ground. Applied potential shall be 500 V DC for 300-volt rated cable and 1000 V DC for 600-volt rated cable, or as required if solid-state components or control devices cannot tolerate the applied voltage.

(9) If applicable, verify correct function of control transfer relays located in the switchgear with multiple control power sources.

(10) Perform phasing checks on double-ended or dual-source switchgear to insure correct bus phasing from each source.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Housing; G,

Buses; G,

Low-Voltage Power Circuit Breakers; G,

Electric Utility Company Equipment; G,

Battery System; G,

Surge Protective Device; G,

Metering; G,

Control Wiring; G,

Nameplates And Mimic Bus; G,

SD-06 Test Reports

Factory Tests; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.6 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

(1) Shop Drawings:

(a) Switchgear shop drawings shall be submitted simultaneously with or after the Overcurrent Protective Device Coordination Study.

(b) Submit sufficient information to demonstrate compliance with drawings and specifications.

(c) Prior to fabrication of switchgear, submit the following data for approval:

1. Complete electrical ratings.

2. Circuit breaker sizes.

3. Interrupting ratings.

4. Safety features.

5. Accessories and nameplate data.

6. Switchgear one line diagram, showing ampere rating, number

of bars per phase and neutral in each bus run (horizontal and vertical), bus spacing, equipment ground bus, and bus material.

7. Elementary and interconnection wiring diagrams.

8. Technical data for each component.

9. Dimensioned exterior views of the switchgear.

10. Dimensioned section views of the switchgear.

11. Floor plan of the switchgear.

12. Foundation plan for the switchgear.

13. Provisions and required locations for external conduit and wiring entrances.

14. Approximate design weights.

(2) Manuals:

(a) Submit, simultaneously with the shop drawings, companion copies of complete maintenance and operating manuals, including technical data sheets, wiring diagrams, and information for ordering replacement parts.

1. Schematic signal and control diagrams, with all terminals identified, matching terminal identification in the switchgear.

2. Include information for testing, repair, trouble shooting, assembly, disassembly, and factory recommended/required periodic maintenance procedures and frequency.

3. Provide a replacement and spare parts list. Include a list of tools and instruments for testing and maintenance purposes.

(b) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

(3) Certifications: Two weeks prior to final inspection, submit the following.

(a) Certification by the manufacturer that switchgear conforms to the requirements of the drawings and specifications.

(b) Certification by the Contractor that switchgear has been properly installed, adjusted, and tested.

1.7 REFERENCES

a. Publications listed below (including amendments, addenda, revisions, supplements, and errata), form a part of this specification to the extent referenced. Publications are referenced in the text by basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C37.20.1	(2020) Metal-Enclosed Low-Voltage (1000 Vac and below, 3200 Vdc and below) Power Circuit-Breaker Switchgear

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA C37.51	(2018) Switchgear--Metal Enclosed Low-Voltage AC Power, Circuit-Breaker Switchgear Assemblies-Conformance Test Procedures

PART 2 PRODUCTS

2.1 GENERAL

a. Shall be in accordance with IEEE, NEMA, NFPA, UL, as shown on the drawings, and have the following features:

(1) Switchgear shall be a complete, grounded, continuous-duty, integral assembly, metal clad, dead-front, dead-rear, self-supporting, indoor type switchgear assembly, . Incorporate devices shown on the drawings and all related components required to fulfill operational and functional requirements.

(2) Switchgear shall be Type 2 front, side, and rear accessible.

(3) Ratings shall not be less than shown on the drawings. Short circuit ratings shall not be less than 100 kA.

(4) Switchgear shall conform to the arrangements and details shown on the drawings.

(5) Provide advanced utility metering devices, instrument transformers (C.T. and P.T.), communication interfaces, wiring and raceway, and accessories as specified in Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS). Meters shall be multifunction electronic type with local display, remote communications, data logging, power quality monitoring, and alarming capabilities. All new meters shall be fully compatible with, and integrated into, the existing facility SCADA system for real-time monitoring, trending, alarming, and data archiving. Coordinate communication protocol (e.g., Modbus, TCP/IP, BACnet/IP, or facility-standard), point mapping/registers, and programming with the existing SCADA system integrator/vendor. Provide complete point lists, Modbus/BACnet object mappings, and support for HMI screen updates, alarm configuration, and trend setup. Dedicated communication cabling (Ethernet CAT6 or fiber as required by distance and facility standard) shall connect new meters to the existing SCADA network/data aggregation devices; do not share with IT or BAS networks. Meters and integration shall comply with VA Electrical Design Manual (PG-18-10) Section 5.4 POWER MONITORING AND METERING. Contractor shall perform integration testing, verify data accuracy against calibrated reference meters, and demonstrate full SCADA functionality (display, alarms, trends) prior to acceptance, refer to Section 25 08 10 UTILITY MONITORING AND CONTROL SYSTEM TESTING.

(6) Key-type mechanical interlocks for multiple circuit breakers shall be provided as shown on the drawings.

(7) Switchgear shall be assembled, connected, and wired at the factory so that only external circuit connections are required at the construction site. Split the structure only as required for shipping and installation. Packaging shall provide adequate protection against rough handling during shipment.

(8) All non-current-carrying parts shall be grounded per Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS for additional requirements.

2.2 HOUSING

a. Shall have the following features:

(1) Frames and enclosures:

(a) The assembly shall be braced with reinforcing gussets using bolted connections to assure rectangular rigidity.

(b) The enclosure shall be steel, leveled, and not less than the gauge required by applicable publications.

(c) Die-pierce the holes for connecting adjacent structures to insure proper alignment, and to allow for future additions.

(d) All bolts, nuts, and washers shall be zinc-plated steel.

(2) Circuit breaker compartments:

(a) An individual compartment shall be supplied for each circuit breaker and each future circuit breaker as shown on the drawings. Compartments shall be provided with isolated wireways for control wiring between devices.

1. Separate each compartment so that the circuit breaker, buses, and cable terminations are in separate compartments with steel partitions or barriers of approved and properly installed insulation.

2. Each compartment furnished with a circuit breaker (active or spare) shall be fully equipped as noted on drawings and specified below.

3. Each compartment noted as space for future circuit breaker, as shown on drawings, shall be fully equipped for positioning and connecting the breaker. Provide all equipment required to implement the future breaker installation.

(3) Auxiliary compartments:

(a) Compartments shall be provided for auxiliaries, metering, and transition or termination sections as required by the manufacturer, and as shown on drawings. Compartments shall be provided with isolated wireways for control wiring between devices.

(4) Compartment doors:

(a) The doors shall permit convenient removal and interchanging of circuit breakers between compartments. The doors shall be capable of a swing approaching 180 degrees.

(b) Concealed or semi-concealed hinges shall be provided to attach the doors. Weld the hinges to the equipment structure and to the compartment doors.

b. Walk-in Protected Enclosure:

(1) Where indicated on the drawings, provide an outdoor, weatherproof, protected walk-in aisle enclosure, fabricated and coordinated with the switchgear to form an integral enclosure.

(2) Adequate space shall be provided for convenient installation, operation and maintenance of the circuit breakers, batteries, battery charger, circuit breaker test equipment, and the auxiliaries. The aisle area shall be not less than shown on the drawings.

(3) The entire space within the enclosure shall be provided with a steel floor adequately reinforced to allow the circuit breakers to be interchanged and serviced without causing the floor to deflect. The entire floor shall be at the same level.

(4) The roof of the enclosure shall slope to allow for adequate run-off of moisture.

(5) The entire area between the floor and foundation, including feeder conduits, shall be enclosed by structural steel or steel sheets.

(6) The enclosure shall include proper air conditioning, heating, and/or ventilation equipment as shown on the drawings or as recommended by the manufacturer. All ventilation openings shall be provided with suitable filters and rodent screens. The air conditioning and ventilation equipment shall limit the temperature rise to 10 degrees F above ambient, but no higher than 104 degrees F.

(7) Enclosure doors:

(a) Locate a door at each end of the protected aisle.

(b) The doors shall be safety type, steel with concealed or semi-concealed hinges for attachment. Weld the hinges to the equipment structure.

(c) Provide the doors with panic hardware on the inside and grab handle on the exterior. A latch bolt controlled by a key cylinder shall lock the door from the outside. Key the cylinder as directed by the Contracting Officer and as coordinated with the electric utility company, if applicable.

(8) Equipment rear doors:

(a) Provide suitable weatherproof type doors on the rear of the switchgear enclosure for each compartment. Attach the doors by concealed or semi-concealed hinges. Weld the hinges to the

enclosure and to the compartment doors. Provide each door with a three-point latching and locking assembly and provisions for padlocking.

(b) The doors shall be capable of a swing approaching 180 degrees and shall be provided with intermediate doorstops.

(9) Compartment heaters:

(a) Install a thermostatically controlled electric strip heater within each circuit breaker compartment and cable termination compartment to limit excessive humidity during adverse weather conditions. Thermostat shall be set and marked with manufacturer’s recommended setting.

(b) Heater and associated control wiring shall be pre-wired at the factory. Properly fuse the wiring and protect to prevent terminal overheating.

(10) Lighting:

(a) Provide 4 feet, two-fluorescent lamps or LED equivalent, ceiling mounted, 8 feet on centers over the front aisle, with fixtures parallel to the switchgear. Lamps, ballasts or LED driver shall comply with Section 26 51 00, INTERIOR LIGHTING. Connect unswitched circuit to emergency battery packs of the fixtures, and switched circuit to fixture ballast or LED driver.

(b) Fixtures shall be securely mounted (chains or wires are not allowed) and include wire guards to protect lamps in each fixture.

(c) Install a 3-way switch at each enclosure entrance to control the lighting.

(11) Receptacles: Provide one 2P, 3W, 20-amp heavy-duty duplex ground fault current interrupter (GFI) receptacle for each three compartments or fraction thereof. Space receptacles equidistant along the interior wall of the aisle space. Install a separate 20-amp circuit for every three receptacles.

(12) All branch circuit wiring shall be installed in conduit and shall be not less than No. 12 AWG.

c. Finish:

(1) All metal surfaces shall be thoroughly cleaned, phosphatized and factory primed prior to applying baked enamel or lacquer finish.

(2) Provide a light gray finish for indoor switchgear.

(3) Outdoor switchgear:

(a) Interior finish shall be light gray.

(b) The underside of the switchgear and enclosure shall be treated with corrosion resistant compounds, epoxy resin or rubberized sealing compound.

2.3 BUSES

a. Bus Bars and Interconnections:

(1) Provide copper phase and neutral buses, fully rated for the amperage as shown on the drawings for the entire length of the switchgear. Bus laminations shall have a minimum of 1/4 inch spacing.

(2) Mount the buses on appropriately spaced insulators and brace to withstand the available short circuit currents.

(3) The bus and bus compartment shall be designed so that the acceptable NEMA standard temperature rises are not exceeded.

(4) Install a copper ground bus the full length of the switchgear assembly.

(5) Main Bonding Jumper: An un-insulated copper bus, size as shown on drawings, shall interconnect the neutral and ground buses, when the switchgear is used to establish the system common ground point.

(6) All bolts, nuts, and washers shall be zinc-plated steel. Bolts shall be torqued to the values recommended by the manufacturer.

(7) Make provisions for future bus extensions by means of bolt holes or other approved method.

2.4 LOW-VOLTAGE POWER CIRCUIT BREAKERS

a. General: Circuit breakers shall be dead front, drawout, stored energy type with solid state trip devices. Arcing contacts shall be renewable.

b. Rating: Circuit breakers shall be 3 pole, 600 volts AC and below, 60 cycle with frame size, trip rating and functions, and system voltage as shown on drawings. Breakers shall have 30 cycle short time current ratings.

c. Drawout Mounting: Provide a racking mechanism to position and hold the breaker in the connected, test, or disconnected position. Provide an interlock to prevent movement of the breaker into or out of the connected position unless the breaker is tripped open.

d. Trip Devices: Breakers shall be electrically and mechanically trip free and shall have trip devices in each pole. Unless otherwise indicated on drawings, each breaker shall have overcurrent and short-circuit , and integral ground fault trip devices. Trip devices shall be of the solid state type with adjustable pick-up settings, with both long time and short time elements, and integral trip unit testing provisions. Devices shall have time-delay band adjustment. Long-time delay element shall have inverse time characteristics. Main circuit breakers shall not have instantaneous trip function.

e. Position Indicator: Provide a mechanical indicator visible from the front of the unit to indicate whether the breaker is open or closed.

f. Trip Button: Equip each breaker with a mechanical trip button accessible from the front of the door.

g. Padlocking: Provisions shall be included for padlocking the breaker in the open position.

h. Operation: Unless otherwise indicated herein or on the drawings, breakers 1600 ampere frame size and less shall be manually operated. Breakers larger than 1600 ampere frame size shall be electrically operated.

2.5 ELECTRIC UTILITY COMPANY EQUIPMENT

a. Provide separate compartment for electric utility company metering equipment as shown on drawings.

b. Provide suitable arrangements within the electric utility company metering compartment for mounting metering equipment. Obtain the electric utility company's approval of the compartment arrangements prior to fabrication of the switchgear.

c. Allow access to electric utility company personnel as required for installation of utility metering equipment.

2.6 BATTERY SYSTEM

a. Batteries:

(1) Provide high discharge rate type maintenance-free nickel-cadmium batteries. voltage shall be 125 volts nominal. Calculate the battery capacity based on the lowest ambient temperature in the room where it is to be installed. Include a safety margin of 50 percent for reserve capacity.

(a) Provide sufficient battery capacity to carry all continuous loads (lamps, relays, etc.) for 8 hours and then perform the greater of the following duties, with the charger de-energized.

1. Trip all circuit breakers simultaneously or,

2. Close the largest breaker in a line-up of four or less breakers, or close the two largest breakers simultaneously in a line-up of more than four breakers. Breaker closing current shall include both the spring release coil current and the starting current of the spring charging motor.

(2) Provide battery connector covers for protection against external short circuits.

(3) Provide corrosion-resistant steel battery racks.

b. Charger:

(1) Provide a charger of the full wave rectifier type utilizing silicon controlled rectifiers as the power-control elements. Construction shall be modular with plug-in control units for easy replacement.

(2) The charger shall maintain 1/2 of one percent voltage regulation from no load to full load for line voltage variation of 10 percent, and frequency variation of 3 Hz from 60 Hz.

(3) The charger shall maintain a nominal float voltage of 1.4 vpc, and a nominal equalizing voltage of 1.5 vpc.

(4) The charger shall be capable of continuous operation in an ambient temperature of 104 degrees F without derating. The charger shall be installed in a convection cooled NEMA Type 1 ventilated enclosure. The housing is to have a hinged front door with all equipment accessible from the front.

(5) Provide both AC and DC transient protection. Charger shall be able to recharge a fully discharged battery without tripping AC protective devices. AC circuit breaker shall not trip under any DC load condition, including short circuit on output terminals.

(6) The charger shall be capable of supplying the following demand simultaneously:

(a) Recharging a fully discharged battery in 12 hours.

(b) Supervisory panel and control panel.

(c) Steady loads (indicating lamps, relays, etc.).

(7) The charger shall have fused AC input and DC output protection.

(8) The charger shall not discharge the batteries when AC power fails.

(9) The charger shall have the following accessories:

(a) On-off control switch with pilot light.

(b) AC power failure alarm light.

(c) High DC voltage alarm light.

(d) Low DC voltage alarm light.

(e) Ground detection switch and alarm light.

(f) DC ammeter - 2 percent accuracy.

(g) DC voltmeter - 2 percent accuracy: Float/equalize voltage marked in red on voltmeter.

(h) Provisions for activation of remote annunciation of trouble for the above conditions.

2.7 SURGE PROTECTIVE DEVICE

Refer to Section 26 43 13, SURGE PROTECTIVE DEVICES.

2.8 METERING

a. Refer to Section 25 10 10, ADVANCED UTILITY METERING. Refer to drawings for meter locations.

b. As necessary, provide compartment with a front hinged door to provide safe isolated access to meters and all associated terminal and fuse

blocks for maintenance, calibration, or testing.

c. Provide current transformers for each meter. Current transformers shall be wired to shorting-type terminal blocks.

d. Provide voltage transformers including primary fuses and secondary protective devices for metering as shown on the drawings.

2.9 OTHER EQUIPMENT

a. Furnish tools and accessories required for circuit breaker and switchgear test, inspection, maintenance, and proper operation.

b. Circuit breaker removal equipment: Furnish a permanent circuit breaker removal device mounted on top of enclosure for installation and removal of circuit breakers.

2.10 CONTROL WIRING

Switchgear control wires shall not be less than No. 14 AWG copper 600 V rated. Install wiring complete at the factory, adequately bundled and protected. Provide separate control circuit fuses in each breaker compartment and locate for ease of access and maintenance.

2.11 NAMEPLATES AND MIMIC BUS

a. Nameplates: For Normal Power system, provide laminated black phenolic resin with white core with 1/2 inch engraved lettered nameplates next to each circuit breaker. For Essential Electrical System, provide laminated red phenolic resin with white core with 1/2 inch engraved lettered nameplates next to each circuit breaker. Nameplates shall indicate equipment served, spaces, or spares in accordance with one line diagram shown on drawings. Nameplates shall be mounted with plated screws on front of breakers or on equipment enclosure next to breakers. Mounting nameplates only with adhesive is not acceptable.

b. Mimic Bus: Provide an approved mimic bus on front of each switchgear assembly. Color shall be black for the Normal Power system and red for the Essential Electrical System, either factory-painted plastic or metal strips. Plastic tape shall not be used. Use symbols similar to one line diagram shown on drawings. Plastic or metal strips shall be mounted with plated screws.

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation shall be in accordance with the NEC, as shown on the drawings, and manufacturer’s instructions.

b. Anchor switchgear with rustproof bolts, nuts, and washers not less than 1/2 inch diameter, in accordance with manufacturer’s instructions, and as shown on drawings.

c. Exterior Location. Mount switchgear on concrete slab. Unless otherwise indicated, the slab shall be at least 8 inches thick, reinforced with a 6 by 6 inches No. 6 mesh placed uniformly 4 inches from the top of the slab. Slab shall be placed on a 6 inches thick, well-compacted gravel base. The top of the concrete slab shall be

approximately 4 inches above the finished grade. Edges above grade shall have 1/2 inch chamfer. The slab shall be of adequate size to project at least 8 inches beyond the equipment. Provide conduit turnups and cable entrance space required by the equipment to be mounted. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant. Cut off and bush conduits 3 inches above slab surface. Concrete work shall be as specified in Section 03 30 00, CAST-IN-PLACE CONCRETE.

d. Interior Location. Mount switchgear on concrete slab. Unless otherwise indicated, the slab shall be at least 4 inches thick. The top of the concrete slab shall be approximately 4 inches above finished floor. Edges above floor shall have 1/2 inch chamfer. The slab shall be of adequate size to project at least 8 inches beyond the equipment. Provide conduit turnups and cable entrance space required by the equipment to be mounted. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant. Cut off and bush conduits 3 inches above slab surface. Concrete work shall be as specified in Section 03 30 00, CAST-IN-PLACE CONCRETE.

3.2 ACCEPTANCE CHECKS AND TESTS

a. Perform in accordance with the manufacturer's recommendations. In addition, include the following:

(1) Visual Inspection and Tests:

(a) Compare equipment nameplate data with specifications and approved shop drawings.

(b) Inspect physical, electrical, and mechanical condition.

(c) Confirm correct application of manufacturer's recommended lubricants.

(d) Verify appropriate anchorage, required area clearances, and correct alignment.

(e) Verify that circuit breaker sizes and types correspond to approved shop drawings.

(f) Verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey after energization.

(g) Confirm correct operation and sequencing of key-type mechanical interlock systems.

(h) Vacuum-clean switchgear enclosure interior. Clean switchgear enclosure exterior.

(i) Inspect insulators for evidence of physical damage or contaminated surfaces.

(j) Verify correct shutter installation and operation.

(k) Exercise all active components.

(l) Verify the correct operation of all sensing devices, alarms,

and indicating devices.

(m) Verify that vents are clear.

(2) Electrical tests:

(a) Perform insulation-resistance tests on each bus section.

(b) Perform insulation-resistance test on control wiring; do not perform this test on wiring connected to solid-state components.

(c) Perform phasing check on double-ended switchgear to ensure correct bus phasing from each source.

3.3 FOLLOW-UP VERIFICATION

Upon completion of acceptance checks, settings, and tests, the Contractor shall show by demonstration in service that the switchgear is in good operating condition and properly performing the intended function.

3.4 TEMPORARY HEATING

Apply temporary heat to switchgear, according to manufacturer's written instructions, throughout periods when switchgear environment is not controlled for temperature and humidity within manufacturer's stipulated service conditions.

3.5 WARNING SIGN

Mount on each entrance door of the switchgear room, approximately 5 feet above grade or floor, a clearly lettered warning sign for warning personnel. The sign shall be attached with rustproof metal screws.

3.6 ONE LINE DIAGRAM AND SEQUENCE OF OPERATION

a. At final inspection, an as-built one line diagram shall be laminated or mounted under acrylic glass, and installed in a frame mounted in the switchgear room or in the outdoor switchgear enclosure.

b. Furnish a written sequence of operation for the switchgear and connected line side/load side electrical distribution equipment. The sequence of operation shall be laminated or mounted under acrylic glass, and installed in a frame mounted in the switchgear room or in the outdoor switchgear enclosure.

c. Deliver an additional four copies of the as-built one line diagram and sequence of operation to the Contracting Officer.

3.7 AS-LEFT TRIP UNIT SETTINGS

a. The trip unit settings shall be set in the field by an authorized representative of the switchgear manufacturer per the approved Overcurrent Protective Device Coordination Study in accordance with Section 26 05 73, OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY.

b. The trip unit settings of the main breaker(s) shall be reviewed by the electric utility company to assure coordination with the electric utility company primary fusing. Prior to switchgear activation, provide written verification of this review to the Contracting Officer.

c. Post a durable copy of the "as-left" trip unit settings in a convenient location in the switchgear room . Deliver four additional copies of the settings to the Contracting Officer prior to the activation of the switchgear.

3.8 INSTRUCTION

Furnish the services of a factory-trained technician for two, 4-hour training periods for instructing personnel in the maintenance and operation of the switchgear, on the dates requested by the Contracting Officer.

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SECTION 26 24 13
11/22

DISTRIBUTION SWITCHBOARDS

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies the furnishing, installation, connection, and testing of the low-voltage circuit-breaker distribution switchboards, indicated as switchboard(s) in this section.

1.2 RELATED WORK

a. Section 03 30 00, CAST-IN-PLACE CONCRETE: Requirements for concrete equipment pads.

b. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

c. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Low-voltage conductors.

d. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible fault currents.

e. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduit.

f. Section 26 05 73, OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY: Short circuit and coordination study, and requirements for a coordinated electrical system.

g. Section 26 11 16, SECONDARY UNIT SUBSTATIONS: Switchboards as part of secondary unit substations.

h. Section 26 43 13, SURGE PROTECTIVE DEVICES: Surge protective devices for switchboards.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 FACTORY TESTS

a. Factory Tests shall be required.

b. Factory Tests shall be in accordance with Paragraph, MANUFACTURED PRODUCTS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirement:

(1) Tests shall be conducted per NEMA PB 2.

(2) Verify that circuit breaker sizes and types correspond to

drawings, and the Overcurrent Protective Device Coordination Study.

(3) Verify tightness of bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer’s published data.

(4) Confirm correct operation and sequencing of key-type mechanical interlock systems for multiple circuit breakers by attempting closure on locked-open devices, and attempting to open locked-closed devices, and making key exchange with devices operated in off-normal positions.

(5) Exercise all active components.

(6) Perform an insulation-resistance test, phase to ground, on each bus section, with phases not under test grounded, in accordance with manufacturer’s published data.

(7) Perform insulation-resistance tests on control wiring with respect to ground. Applied potential shall be 500 V DC for 300-volt rated cable and 1000 V DC for 600-volt rated cable, or as required if solid-state components or control devices cannot tolerate the applied voltage.

(8) If applicable, verify correct function of control transfer relays located in the switchboard with multiple control power sources.

(9) Perform phasing checks on double-ended or dual-source switchboards to insure correct bus phasing from each source.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Housing; G,

Buses; G,

Main Circuit Breakers; G,

Feeder Circuit Breakers; G,

Electric Utility Company Equipment; G,

Surge Protective Devices; G,

Metering; G,

Control Wiring; G,

Nameplates And Mimic Bus; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.6 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Switchboard shop drawings shall be submitted simultaneously with or after the Overcurrent Protective Device Coordination Study.

(2) Submit sufficient information to demonstrate compliance with drawings and specifications.

(3) Prior to fabrication of switchboards, submit the following data for approval:

(a) Complete electrical ratings.

(b) Circuit breaker sizes.

(c) Interrupting ratings.

(d) Safety features.

(e) Accessories and nameplate data.

(f) Switchboard one line diagram, showing ampere rating, number of bars per phase and neutral in each bus run (horizontal and vertical), bus spacing, equipment ground bus, and bus material.

(g) Elementary and interconnection wiring diagrams.

(h) Technical data for each component.

(i) Dimensioned exterior views of the switchboard.

(j) Dimensioned section views of the switchboard.

(k) Floor plan of the switchboard.

(l) Foundation plan for the switchboard.

(m) Provisions and required locations for external conduit and wiring entrances.

(n) Approximate design weights.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, companion copies of complete maintenance and operating manuals, including technical data sheets, wiring diagrams, and information for ordering replacement parts.

(a) Schematic signal and control diagrams, with all terminals identified, matching terminal identification in the switchboard.

(b) Include information for testing, repair, trouble shooting, assembly, disassembly, and factory recommended/required periodic maintenance procedures and frequency.

(c) Provide a replacement and spare parts list. Include a list of tools and instruments for testing and maintenance purposes.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

c. Certifications: Two weeks prior to final inspection, submit the following.

(1) Certification by the manufacturer that the switchboards conform to the requirements of the drawings and specifications.

(2) Certification by the Contractor that the switchboards have been properly installed, adjusted, and tested.

1.7 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by basic designation only.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA PB 2 (2011) Deadfront Distribution Switchboards

PART 2 PRODUCTS

2.1 GENERAL

a. Shall be in accordance with IEEE, NEMA, NFPA, UL, as shown on the drawings, and have the following features:

(1) Switchboard shall be a complete, grounded, continuous-duty, integral assembly, dead-front, dead-rear, self-supporting, indoor type switchboard assembly, . Incorporate devices shown on the drawings and all related components required to fulfill operational and functional requirements.

(2) Ratings shall not be less than shown on the drawings. Short circuit ratings shall not be less than the available fault current shown in the Overcurrent Protective Device Coordination Study.

(3) Switchboard shall conform to the arrangements and details shown on the drawings.

(4) Provide advanced utility metering devices, instrument transformers (C.T. and P.T.), communication interfaces, wiring and raceway, and accessories as specified in Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS). Meters shall be multifunction electronic type with local display, remote communications, data logging, power quality monitoring, and alarming capabilities. All new meters shall be fully compatible with, and integrated into, the existing facility SCADA system for real-time monitoring, trending, alarming, and data archiving. Coordinate communication protocol (e.g., Modbus, TCP/IP, BACnet/IP, or facility-standard), point mapping/registers, and programming with the existing SCADA system integrator/vendor. Provide complete point lists, Modbus/BACnet object mappings, and support for HMI screen updates, alarm configuration, and trend setup. Dedicated communication cabling (Ethernet CAT6 or fiber as required by distance and facility standard) shall connect new meters to the existing SCADA network/data aggregation devices; do not share with IT or BAS networks. Meters and integration shall comply with VA Electrical Design Manual (PG-18-10) Section 5.4 POWER MONITORING AND METERING. Contractor shall perform integration testing, verify data accuracy against calibrated reference meters, and demonstrate full SCADA functionality (display, alarms, trends) prior to acceptance, refer to Section 25 08 10 UTILITY MONITORING AND CONTROL SYSTEM TESTING.

(5) Switchboards shall be assembled, connected, and wired at the factory so that only external circuit connections are required at the construction site. Split the structure only as required for shipping and installation. Packaging shall provide adequate protection against rough handling during shipment.

(6) All non-current-carrying parts shall be grounded per Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS for additional requirements.

(7) Series rated switchboards are not allowed.

2.2 BASIC ARRANGEMENT

a. Type 1: Switchboard shall be front accessible with the following features:

(1) Device mounting:

(a) Main breaker: Individually mounted and compartmented or group mounted with feeder breakers.

(b) Feeder breakers: Group mounted.

(2) Section alignment: As shown on the drawings.

(3) Accessibility:

(a) Main section line and load terminals: Front and side.

(b) Distribution section line and load terminals: Front.

(c) Through bus connections: Front and end.

(4) Bolted line and load connections.

(5) Full height wiring gutter covers for access to wiring terminals.

b. Type 2: Switchboard shall be rear accessible with the following features:

(1) Device mounting:

(a) Main breaker: Individually mounted and compartmented.

(b) Feeder breakers: Individually mounted and compartmented.

(2) Section alignment: As shown on drawings.

(3) Accessibility:

(a) Main section line and load terminals: Rear.

(b) Distribution section line and load terminals: Rear.

(c) Bus connections: Rear.

(4) Bolted line and load connections.

(5) Full height wiring gutter covers for access to wiring terminals.

2.3 HOUSING

a. Shall have the following features:

(1) Frames and enclosures:

(a) The assembly shall be braced with reinforcing gussets using bolted connections to assure rectangular rigidity.

(b) The enclosure shall be steel, leveled, and not less than the gauge required by applicable publications.

(c) Die-pierce the holes for connecting adjacent structures to insure proper alignment, and to allow for future additions.

(d) All bolts, nuts, and washers shall be zinc-plated steel.

b. Finish:

(1) All metal surfaces shall be thoroughly cleaned, phosphatized and factory primed prior to applying baked enamel or lacquer finish.

(2) Provide a light gray finish for indoor switchboard.

(3) Outdoor switchboard:

(a) Interior finish shall be light gray.

(b) The underside of the switchboard shall be treated with corrosion resistant compounds, epoxy resin or rubberized sealing compound.

2.4 BUSES

Bus Bars and Interconnections:

a. Provide copper phase and neutral buses, fully rated for the amperage as shown on the drawings for the entire length of the switchboard. Bus laminations shall have a minimum of 1/4 inch spacing.

b. Mount the buses on appropriately spaced insulators and brace to withstand the available short circuit currents.

c. The bus and bus compartment shall be designed so that the acceptable NEMA standard temperature rises are not exceeded.

d. Install a copper ground bus the full length of the switchboard assembly.

e. Main Bonding Jumper: An un-insulated copper bus, size as shown on drawings, shall interconnect the neutral and ground buses, when the switchboard is used to establish the system common ground point.

f. All bolts, nuts, and washers shall be zinc-plated steel. Bolts shall be torqued to the values recommended by the manufacturer.

g. Make provisions for future bus extensions by means of bolt holes or other approved method.

2.5 MAIN CIRCUIT BREAKERS

a. Type I or Type II Switchboards: Provide molded case main circuit breakers as shown on the drawings. Circuit breakers shall be the solid state adjustable trip type.

(1) Trip units shall have field adjustable tripping characteristics as follows:

(a) Long time pickup.

(b) Long time delay.

(c) Short time pickup.

(d) Short time delay.

(e) Instantaneous.

(f) Ground fault pickup.

(g) Ground fault delay.

(2) Breakers with same frame size shall be interchangeable with each other.

(3) Breakers shall be fully rated.

b. Type II Switchboards: Provide main power circuit breakers as shown on the drawings. Circuit breakers shall be the solid state adjustable trip type.

(1) General: Circuit breakers shall be dead front, drawout, stored energy type with solid state trip devices. Arcing contacts shall be renewable.

(2) Rating: Circuit breakers shall be 3 pole, 600 V AC and below, 60 cycle with frame size, trip rating and functions, and system voltage as shown on drawings. Breakers shall have 30 cycle short time current ratings.

(3) Drawout Mounting: Provide a racking mechanism to position and hold the breaker in the connected, test, or disconnected position. Provide an interlock to prevent movement of the breaker into or out of the connected position unless the breaker is tripped open.

(4) Trip Devices: Breakers shall be electrically and mechanically trip free and shall have trip devices in each pole. Unless otherwise indicated on drawings, each breaker shall have overcurrent and short-circuit , and integral ground fault trip devices. Trip devices shall be of the solid state type with adjustable pick-up settings, with both long time and short time elements, and integral trip unit testing provisions. Devices shall have time-delay band adjustment. Long-time delay element shall have inverse time characteristics. Main circuit breakers shall not have instantaneous trip function.

(5) Position Indicator: Provide a mechanical indicator visible from the front of the unit to indicate whether the breaker is open or closed.

(6) Trip Button: Equip each breaker with a mechanical trip button accessible from the front of the door.

(7) Padlocking: Provisions shall be included for padlocking the breaker in the open position.

(8) Operation: Unless otherwise indicated herein or on the drawings, breakers shall be manually operated. Breakers larger than 1600 ampere frame size shall be electrically operated.

2.6 FEEDER CIRCUIT BREAKERS

a. Provide molded case circuit breakers as shown on the drawings.

b. Non-adjustable Trip Molded Case Circuit Breakers:

(1) Molded case circuit breakers shall have automatic, trip free, non-adjustable, inverse time characteristics, and instantaneous magnetic trip.

(2) Breaker features shall be as follows:

(a) A rugged, integral housing of molded insulating material.

(b) Silver alloy contacts.

(c) Arc quenchers and phase barriers for each pole.

(d) Quick-make, quick-break, operating mechanisms.

(e) A trip element for each pole, thermal magnetic type with long time delay and instantaneous characteristics, a common trip bar for all poles and a single operator.

(f) Electrically and mechanically trip free.

(g) An operating handle which indicates ON, TRIPPED, and OFF positions.

(h) Line and load connections shall be bolted.

(i) An overload on one pole of a multipole breaker shall automatically cause all the poles of the breaker to open.

c. Adjustable Trip Molded Case Circuit Breakers:

(1) Provide molded case, solid state adjustable trip type circuit breakers.

(2) Trip units shall have field adjustable tripping characteristics as follows:

(a) Long time pickup.

(b) Long time delay.

(c) Short time pickup.

(d) Short time delay.

(e) Instantaneous.

(f) Ground fault pickup.

(g) Ground fault delay.

(3) Breakers with same frame size shall be interchangeable with each other.

2.7 ELECTRIC UTILITY COMPANY EQUIPMENT

a. Provide separate compartment for electric utility company metering equipment as shown on drawings.

b. Provide suitable arrangements within the electric utility company metering compartment for mounting metering equipment. Obtain the electric utility company's approval of the compartment arrangements prior to fabrication of the switchboard.

c. Allow access to electric utility company personnel as required for installation of utility metering equipment.

2.8 SURGE PROTECTIVE DEVICES

Refer to Section 26 43 13, SURGE PROTECTIVE DEVICES.

2.9 METERING

a. Refer to Section 25 10 10, ADVANCED UTILITY METERING. Refer to drawings for meter locations.

b. Provide current transformers for each meter. Current transformers shall be wired to shorting-type terminal blocks.

c. Provide voltage transformers including primary fuses and secondary protective devices for metering as shown on the drawings.

2.10 OTHER EQUIPMENT

a. Furnish tools and accessories required for circuit breaker and switchboard test, inspection, maintenance, and proper operation.

b. Panelboards: Requirements for panelboards shown to be installed in the switchboard shall be as shown on the drawings and in Section 26 24 16, PANELBOARDS.

2.11 CONTROL WIRING

Switchboard control wires shall not be less than No. 14 AWG copper 600 volt rated. Install wiring complete at the factory, adequately bundled and protected. Provide separate control circuit fuses in each breaker compartment and locate for ease of access and maintenance.

2.12 NAMEPLATES AND MIMIC BUS

a. Nameplates: For Normal Power system, provide laminated black phenolic resin with white core with 1/2 inch engraved lettered nameplates next to each circuit breaker. For Essential Electrical System, provide laminated red phenolic resin with white core with 1/2 inch engraved lettered nameplates next to each circuit breaker. Nameplates shall indicate equipment served, spaces, or spares in accordance with one line diagram shown on drawings. Nameplates shall be mounted with plated screws on front of breakers or on equipment enclosure next to breakers. Mounting nameplates only with adhesive is not acceptable.

b. Mimic Bus: Provide an approved mimic bus on front of each switchboard assembly. Color shall be black for the Normal Power system and red for the Essential Electrical System, either factory-painted plastic or metal strips. Plastic tape shall not be used. Use symbols similar to one line diagram shown on drawings. Plastic or metal strips shall be mounted with plated screws.

PART 3 EXECUTION

3.1 INSTALLATION

a. Install switchboards in accordance with the NEC, as shown on the drawings, and as recommended by the manufacturer.

b. Anchor switchboards with rustproof bolts, nuts, and washers not less than 1/2 inch diameter, in accordance with manufacturer’s instructions, and as shown on drawings.

c. Exterior Location. Mount switchboard on concrete slab. Unless otherwise indicated, the slab shall be at least 8 inches thick,

reinforced with a 6 by 6 inches No. 6 mesh placed uniformly 4 inches from the top of the slab. Slab shall be placed on a 6 inches thick, well-compacted gravel base. The top of the concrete slab shall be approximately 4 inches above the finished grade. Edges above grade shall have 1/2 inch chamfer. The slab shall be of adequate size to project at least 8 inches beyond the equipment. Provide conduit turnups and cable entrance space required by the equipment to be mounted. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant. Cut off and bush conduits 3 inches above slab surface. Concrete work shall be as specified in Section 03 30 00, CAST-IN-PLACE CONCRETE.

d. Interior Location. Mount switchboard on concrete slab. Unless otherwise indicated, the slab shall be at least 4 inches thick. The top of the concrete slab shall be approximately 4 inches above finished floor. Edges above floor shall have 1/2 inch chamfer. The slab shall be of adequate size to project at least 8 inches beyond the equipment. Provide conduit turnups and cable entrance space required by the equipment to be mounted. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant. Cut off and bush conduits 3 inches above slab surface. Concrete work shall be as specified in Section 03 30 00, CAST-IN-PLACE CONCRETE.

3.2 ACCEPTANCE CHECKS AND TESTS

a. Perform in accordance with the manufacturer's recommendations. In addition, include the following:

(1) Visual Inspection and Tests:

(a) Compare equipment nameplate data with specifications and approved shop drawings.

(b) Inspect physical, electrical, and mechanical condition.

(c) Verify appropriate anchorage, required area clearances, and correct alignment.

(d) Verify that circuit breaker sizes and types correspond to approved shop drawings.

(e) Verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey after energization.

(f) Vacuum-clean switchboard enclosure interior. Clean switchboard enclosure exterior.

(g) Inspect insulators for evidence of physical damage or contaminated surfaces.

(h) Verify correct shutter installation and operation.

(i) Exercise all active components.

(j) Verify the correct operation of all sensing devices, alarms, and indicating devices.

(k) Verify that vents are clear.

(2) Electrical tests:

(a) Perform insulation-resistance tests on each bus section.

(b) Perform insulation-resistance test on control wiring; do not perform this test on wiring connected to solid-state components.

(c) Perform phasing check on double-ended switchboards to ensure correct bus phasing from each source.

b. Prior to the final inspection for acceptance, a technical representative from the electric utility company shall witness the testing of the equipment to assure the proper operation of the individual components, and to confirm proper operation/coordination with electric utility company’s equipment.

3.3 FOLLOW-UP VERIFICATION

Upon completion of acceptance checks, settings, and tests, the Contractor shall show by demonstration in service that the switchboard is in good operating condition and properly performing the intended function.

3.4 WARNING SIGN

Mount on each entrance door of the switchboard room, approximately 5 feet above grade or floor, a clearly lettered warning sign for warning personnel. The sign shall be attached with rustproof metal screws.

3.5 ONE LINE DIAGRAM AND SEQUENCE OF OPERATION

a. At final inspection, an as-built one line diagram shall be laminated or mounted under acrylic glass, and installed in a frame mounted in the switchboard room or in the outdoor switchboard enclosure.

b. Furnish a written sequence of operation for the switchboard and connected line side/load side electrical distribution equipment. The sequence of operation shall be laminated or mounted under acrylic glass, and installed in a frame mounted in the switchboard room or in the outdoor switchboard enclosure.

c. Deliver an additional four copies of the as-built one line diagram and sequence of operation to the Contracting Officer.

3.6 AS-LEFT TRIP UNIT SETTINGS

a. The trip unit settings shall be set in the field by an authorized representative of the switchboard manufacturer per the approved Electrical System Protective Device Study in accordance with Section 26 05 73, OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY.

b. The trip unit settings of the main breaker(s) shall be reviewed by the electric utility company to assure coordination with the electric utility company primary fusing. Prior to switchboard activation, provide written verification of this review to the Contracting Officer.

c. Post a durable copy of the "as-left" trip unit settings in a convenient location in the switchboard room Contracting Officer. Furnish this information prior to the activation of the switchboard.

3.7 INSTRUCTION

Furnish the services of a factory-trained technician for one, 4-hour training period for instructing personnel in the maintenance and operation of the switchboards, on the dates requested by the Contracting Officer.

-- End of Section --

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SECTION 26 24 16
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PANELBOARDS

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, and connection of panelboards.

1.2 RELATED WORK

a. 09 91 00, PAINTING: Painting of panelboards.

b. Section 25 10 10, ADVANCED UTILITY METERING: Requirements for electrical metering.

c. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

d. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Low-voltage conductors.

e. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

f. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduits.

g. Section 26 05 73, OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY: Short circuit and coordination study, and requirements for a coordinated electrical system.

h. Section 26 09 23, LIGHTING CONTROLS: Lighting controls integral to panelboards.

i. Section 26 43 13, SURGE PROTECTIVE DEVICES: Surge protective devices integral to panelboards.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Enclosures And Trims; G,

Molded Case Circuit Breakers; G,

Surge Protective Devices; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Include electrical ratings, dimensions, mounting details, materials, required clearances, terminations, weight, circuit breakers, wiring and connection diagrams, accessories, and nameplate data.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, complete maintenance and operating manuals including technical data sheets, wiring diagrams, and information for ordering circuit breakers and replacement parts.

(a) Include schematic diagrams, with all terminals identified, matching terminal identification in the panelboards.

(b) Include information for testing, repair, troubleshooting, assembly, and disassembly.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

c. Certifications: Two weeks prior to final inspection, submit the following.

(1) Certification by the manufacturer that the panelboards conform to the requirements of the drawings and specifications.

(2) Certification by the Contractor that the panelboards have been properly installed, adjusted, and tested.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

a. Panelboards shall be in accordance with NFPA 70, NEMA, UL, as specified, and as shown on the drawings.

b. Panelboards shall have main breaker or main lugs, bus size, voltage, phases, number of circuit breaker mounting spaces, top or bottom feed, flush or surface mounting, branch circuit breakers, and accessories as shown on the drawings.

c. Panelboards shall be completely factory-assembled with molded case circuit breakers and integral accessories as shown on the drawings or specified herein.

d. Non-reduced size copper bus bars, rigidly supported on molded insulators, and fabricated for bolt-on type circuit breakers.

e. Bus bar connections to the branch circuit breakers shall be the “distributed phase” or “phase sequence” type.

f. Mechanical lugs furnished with panelboards shall be cast, stamped, or machined metal alloys listed for use with the conductors to which they will be connected.

g. Neutral bus shall be 100 percent rated, mounted on insulated supports.

h. Grounding bus bar shall be equipped with screws or lugs for the connection of equipment grounding conductors.

i. Bus bars shall be braced for the available short-circuit current as shown on the drawings, but not be less than 10,000 A symmetrical for 120/208 V and 120/240 V panelboards, and 14,000 A symmetrical for 277/480 V panelboards.

j. In two-section panelboards, the main bus in each section shall be full size. The first section shall be furnished with subfeed lugs on the line side of main lugs only, or through-feed lugs for main breaker type panelboards, and have field-installed cable connections to the second section as shown on the drawings. Panelboard sections with tapped bus or crossover bus are not acceptable.

k. Series-rated panelboards are not permitted.

l. Provide advanced utility metering devices, instrument transformers (C.T. and P.T.), communication interfaces, wiring and raceway, and accessories as specified in Section 25 10 10 UTILITY MONITORING AND

CONTROL SYSTEM (UMCS). Meters shall be multifunction electronic type with local display, remote communications, data logging, power quality monitoring, and alarming capabilities. All new meters shall be fully compatible with, and integrated into, the existing facility SCADA system for real-time monitoring, trending, alarming, and data archiving. Coordinate communication protocol (e.g., Modbus, TCP/IP, BACnet/IP, or facility-standard), point mapping/registers, and programming with the existing SCADA system integrator/vendor. Provide complete point lists, Modbus/BACnet object mappings, and support for HMI screen updates, alarm configuration, and trend setup. Dedicated communication cabling (Ethernet CAT6 or fiber as required by distance and facility standard) shall connect new meters to the existing SCADA network/data aggregation devices; do not share with IT or BAS networks. Meters and integration shall comply with VA Electrical Design Manual (PG-18-10) Section 5.4 POWER MONITORING AND METERING. Contractor shall perform integration testing, verify data accuracy against calibrated reference meters, and demonstrate full SCADA functionality (display, alarms, trends) prior to acceptance, refer to Section 25 08 10 UTILITY MONITORING AND CONTROL SYSTEM TESTING.

2.2 ENCLOSURES AND TRIMS

a. Enclosures:

(1) Provide galvanized steel enclosures, with NEMA rating as shown on the drawings or as required for the environmental conditions in which installed.

(2) Enclosures shall not have ventilating openings.

(3) Enclosures may be of one-piece formed steel or of formed sheet steel with end and side panels welded, riveted, or bolted as required.

(4) Provide manufacturer’s standard option for prepunched knockouts on top and bottom endwalls.

(5) Include removable inner dead front cover, independent of the panelboard cover.

b. Trims:

(1) Hinged “door-in-door” type.

(2) Interior hinged door with hand-operated latch or latches, as required to provide access only to circuit breaker operating handles, not to energized parts.

(3) Outer hinged door shall be securely mounted to the panelboard enclosure with factory bolts, screws, clips, or other fasteners, requiring a key or tool for entry. Hand-operated latches are not acceptable.

(4) Inner and outer doors shall open left to right.

(5) Trims shall be flush or surface type as shown on the drawings.

2.3 MOLDED CASE CIRCUIT BREAKERS

a. Circuit breakers shall be per UL, NFPA 70, as shown on the drawings, and as specified.

b. Circuit breakers shall be bolt-on type.

c. Circuit breakers shall have minimum interrupting rating as required to withstand the available fault current, but not less than:

(1) 120/208 V Panelboard: 10,000 A symmetrical.

(2) 120/240 V Panelboard: 10,000 A symmetrical.

(3) 277/480 V Panelboard: 14,000 A symmetrical.

d. Circuit breakers shall have automatic, trip free, non-adjustable, inverse time, and instantaneous magnetic trips for less than 400 A frame. Circuit breakers with 400 A frames and above shall have magnetic trip, adjustable from 5x to 10x. Breaker trip setting shall be set in the field, based on the approved protective device study as specified in Section26 05 73 OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY .

e. Circuit breaker features shall be as follows:

(1) A rugged, integral housing of molded insulating material.

(2) Silver alloy contacts.

(3) Arc quenchers and phase barriers for each pole.

(4) Quick-make, quick-break, operating mechanisms.

(5) A trip element for each pole, thermal magnetic type with long time delay and instantaneous characteristics, a common trip bar for all poles and a single operator.

(6) Electrically and mechanically trip free.

(7) An operating handle which indicates closed, tripped, and open positions.

(8) An overload on one pole of a multi-pole breaker shall automatically cause all the poles of the breaker to open.

(9) Ground fault current interrupting breakers, shunt trip breakers, lighting control breakers (including accessories to switch line currents), or other accessory devices or functions shall be provided where shown on the drawings.

(10) For circuit breakers being added to existing panelboards, coordinate the breaker type with existing panelboards. Modify the panel directory accordingly.

2.4 SURGE PROTECTIVE DEVICES

Where shown on the drawings, furnish panelboards with integral surge protective devices. Refer to Section 26 43 13, SURGE PROTECTIVE DEVICES.

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation shall be in accordance with the manufacturer’s instructions, NFPA 70, as shown on the drawings, and as specified.

b. Locate panelboards so that the present and future conduits can be conveniently connected.

c. Install a printed schedule of circuits in each panelboard after approval by the Contracting Officer. Schedules shall reflect final load descriptions, room numbers, and room names connected to each circuit breaker. Schedules shall be printed on the panelboard directory cards and be installed in the appropriate panelboards

d. Mount panelboards such that the maximum height of the top circuit breaker above the finished floor shall not exceed (78 inches).

e. Provide blank cover for each unused circuit breaker mounting space.

f. Panelboard enclosures shall not be used for conductors feeding through, spliced, or tapping off to other enclosures or devices.

3.2 ACCEPTANCE CHECKS AND TESTS

Perform in accordance with the manufacturer's recommendations. In addition, include the following: Visual Inspection and Tests:

a. Compare equipment nameplate data with specifications and approved shop drawings.

b. Inspect physical, electrical, and mechanical condition.

c. Verify appropriate anchorage and required area clearances.

d. Verify that circuit breaker sizes and types correspond to approved shop drawings.

e. To verify tightness of accessible bolted electrical connections, use the calibrated torque-wrench method or perform thermographic survey after energization.

f. Vacuum-clean enclosure interior. Clean enclosure exterior.

3.3 FOLLOW-UP VERIFICATION

Upon completion of acceptance checks, settings, and tests, the Contractor shall demonstrate that the panelboards are in good operating condition and properly performing the intended function.

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WIRING DEVICES

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies the furnishing, installation, connection, and testing of wiring devices.

1.2 RELATED WORK

a. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: General electrical requirements that are common to more than one section of Division 26.

b. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Cables and wiring.

c. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path to ground for possible ground fault currents.

d. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduit and boxes.

e. Section 26 51 00, INTERIOR LIGHTING: Fluorescent ballasts and LED drivers for use with manual dimming controls.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Receptacles; G,

Toggle Switches; G,

Manual Dimming Control; G,

Wall Plates; G,

Surface Multiple-Outlet Assemblies; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Include electrical ratings, dimensions, mounting details, construction materials, grade, and termination information.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, companion copies of complete maintenance and operating manuals, including technical data sheets and information for ordering replacement parts.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

c. Certifications: Two weeks prior to final inspection, submit the following.

(1) Certification by the manufacturer that the wiring devices conform to the requirements of the drawings and specifications.

(2) Certification by the Contractor that the wiring devices have been properly installed and adjusted.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 99	(2024) Health Care Facilities Code

UL SOLUTIONS (UL)

UL 943 (2016; Reprint Sep 2023) UL Standard for Safety Ground-Fault Circuit-Interrupters

PART 2 PRODUCTS

2.1 RECEPTACLES

a. General: All receptacles shall comply with NEMA, NFPA, UL, and as shown on the drawings.

(1) Mounting straps shall be nickel plated brass, brass, nickel plated steel or galvanize steel with break-off plaster ears, and shall include a self-grounding feature. Terminal screws shall be brass, brass plated or a copper alloy metal.

(2) Receptacles shall have provisions for back wiring with separate metal clamp type terminals (four minimum) and side wiring from four captively held binding screws.

b. Duplex Receptacles - Hospital-grade: shall be listed for hospital grade, single phase, 20 ampere, 120 volts, 2-pole, 3-wire, NEMA 5-20R, with break-off feature for two-circuit operation.

(1) Bodies shall be ivory in color.

(2) Switched duplex receptacles shall be wired so that only the top receptacle is switched. The lower receptacle shall be unswitched.

(3) Duplex Receptacles on Emergency Circuit:

(a) In rooms without emergency powered general lighting, the emergency receptacles shall be of the self-illuminated type.

(4) Ground Fault Current Interrupter (GFCI) Duplex Receptacles: Shall be an integral unit, hospital-grade, suitable for mounting in a standard outlet box, with end-of-life indication and provisions to isolate the face due to improper wiring. GFCI receptacles shall be self-test receptacles in accordance with UL 943.

(a) Ground fault interrupter shall consist of a differential current transformer, self-test, solid state sensing circuitry and a circuit interrupter switch. Device shall have nominal sensitivity to ground leakage current of 4-6 milliamperes and shall function to interrupt the current supply for any value of ground leakage current above five milliamperes (plus or minus 1 milliampere) on the load side of the device. Device shall have a minimum nominal tripping time of 0.025 second.

(b) Self-test function shall be automatically initiated within five seconds after power is activated to the receptacles. Self-test function shall be periodically and automatically performed every three hours or less.

(c) End-of-life indicator light shall be a persistent flashing or blinking light to indicate that the GFCI receptacle is no longer in service.

(5) Tamper-Resistant Duplex Receptacles: Bodies shall be gray in color.

(a) Shall permit current to flow only while a standard plug is in the proper position in the receptacle.

(b) Screws exposed while the wall plates are in place shall be the tamperproof type.

c. Duplex Receptacles – Non-hospital Grade: shall be the same as duplex receptacles – hospital grade in accordance with sections 2.1A and 2.1B of this specification, except for the hospital grade listing. Bodies shall be ivory nylon.

d. Receptacles - 20, 30, and 50 ampere, 250 Volts: Shall be complete with appropriate cord grip plug.

e. Weatherproof Receptacles: Shall consist of a duplex receptacle, mounted in box with a gasketed, weatherproof, cast metal cover plate and cap over each receptacle opening. The cap shall be permanently attached to the cover plate by a spring-hinged flap. The weatherproof integrity shall not be affected when heavy duty specification or hospital grade attachment plug caps are inserted. Cover plates on outlet boxes mounted flush in the wall shall be gasketed to the wall in a watertight manner.

2.2 TOGGLE SWITCHES

Toggle switches shall be totally enclosed tumbler type with nylon bodies. Handles shall be ivory in color unless otherwise specified or shown on the drawings.

a. Switches installed in hazardous areas shall be explosion-proof type in accordance with NFPA 70 and as shown on the drawings.

b. Shall be single unit toggle, butt contact, quiet AC type, heavy-duty general-purpose use with an integral self grounding mounting strap with break-off plasters ears and provisions for back wiring with separate metal wiring clamps and side wiring with captively held binding screws.

c. Switches shall be rated 20 amperes at 120-277 Volts AC.

2.3 MANUAL DIMMING CONTROL

a. Electronic full-wave manual slide dimmer with on/off switch and audible frequency and EMI/RFI suppression filters.

b. Manual dimming controls shall be fully compatible with LED dimming driver and be approved by the driver manufacturer, shall operate over full specified dimming range, and shall not degrade the performance or rated life of the electronic dimming ballast and lamp.

c. Provide single-pole, three-way or four-way, as shown on the drawings.

d. Manual dimming control and faceplates shall be ivory in color unless otherwise specified.

2.4 WALL PLATES

a. Wall plates for switches and receptacles shall be type 302 stainless steel or . Oversize plates are not acceptable.

b. Color shall be ivory unless otherwise specified.

c. For receptacles or switches mounted adjacent to each other, wall plates shall be common for each group of receptacles or switches.

d. In areas requiring tamperproof wiring devices, wall plates shall be type 302 stainless steel, and shall have tamperproof screws and beveled edges.

e. Duplex Receptacles on Emergency Circuit: Wall plates shall be type 302 stainless steel, with the word “EMERGENCY” engraved in 1/4 inch red letters.

2.5 SURFACE MULTIPLE-OUTLET ASSEMBLIES

Shall have the following features:

a. Enclosures: Thickness of steel shall be not less than 0.040 inch for base and cover. Nominal dimensions shall be 1-1/2 inches by 2-3/4 inches with inside cross sectional area not less than 3-1/2 square inches. The enclosures shall be thoroughly cleaned, phosphatized, and painted at the factory with primer and the manufacturer's standard baked enamel finish.

b. Receptacles shall be duplex, hospital grade . See paragraph 'RECEPTACLES' in this Section. Device cover plates shall be the manufacturer's standard corrosion resistant finish and shall not exceed the dimensions of the enclosure.

c. Unless otherwise shown on drawings, receptacle spacing shall be 24 inches on centers.

d. Conductors shall be as specified in Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLE.

e. Installation fittings shall be the manufacturer’s standard bends, offsets, device brackets, inside couplings, wire clips, elbows, and other components as required for a complete system.

f. Bond the assemblies to the branch circuit conduit system.

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation shall be in accordance with NFPA 70 and as shown as on the drawings.

b. Install wiring devices after wall construction and painting is complete.

c. The ground terminal of each wiring device shall be bonded to the outlet box with an approved green bonding jumper, and also connected to the branch circuit equipment grounding conductor.

d. Outlet boxes for toggle switches and manual dimming controls shall be mounted on the strike side of doors.

e. Provide barriers in multi-gang outlet boxes to comply with NFPA 70.

f. Coordinate the electrical work with the work of other trades to ensure that wiring device flush outlets are positioned with box openings aligned with the face of the surrounding finish material. Pay special attention to installations in cabinet work, and in connection with laboratory equipment.

g. Exact field locations of floors, walls, partitions, doors, windows, and equipment may vary from locations shown on the drawings. Prior to locating sleeves, boxes and chases for roughing-in of conduit and equipment, the Contractor shall coordinate exact field location of the above items with other trades.

h. Install wall switches 48 inches above floor, with the toggle OFF position down.

i. Install wall dimmers 48 inches above floor.

j. Install receptacles 18 inches above floor, and 6 inches above counter backsplash or workbenches. Install specific-use receptacles at heights shown on the drawings.

k. Install horizontally mounted receptacles with the ground pin to the right.

l. When required or recommended by the manufacturer, use a torque screwdriver. Tighten unused terminal screws.

m. Label device plates with a permanent adhesive label listing panel and circuit feeding the wiring device.

3.2 ACCEPTANCE CHECKS AND TESTS

Perform manufacturer’s required field checks in accordance with the manufacturer's recommendations, and the latest NFPA 99. In addition, include the following:

a. Visual Inspection and Tests:

(1) Inspect physical and electrical conditions.

(2) Vacuum-clean surface metal raceway interior. Clean metal raceway exterior.

(3) Test wiring devices for damaged conductors, high circuit resistance, poor connections, inadequate fault current path, defective devices, or similar problems using a portable receptacle tester. Correct circuit conditions, remove malfunctioning units and replace with new, and retest as specified above.

(4) Test GFCI receptacles.

b. Receptacle testing in the Patient Care Spaces, such as retention force of the grounding blade of each receptacle, shall comply with the

latest NFPA 99.

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SECTION 26 29 11
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MOTOR CONTROLLERS

PART 1 GENERAL

1.1 DESCRIPTION

a. This Section specifies the furnishing, installation, connection, and testing of motor controllers, including all low- and medium-voltage motor controllers and manual motor controllers, indicated as motor controllers in this Section, and low-voltage variable speed motor controllers.

b. Motor controllers, whether furnished with the equipment specified in other sections or otherwise (with the exception of elevator motor controllers specified in Division 14 and fire pump controllers specified in Division 21), shall meet this specification and all related specifications.

1.2 RELATED WORK

a. Section 25 10 10, ADVANCED UTILITY METERING: For electricity metering installed in motor controllers.

b. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

c. Section 26 05 13, MEDIUM-VOLTAGE CABLES: Medium-voltage cables and terminations.

d. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Low-voltage conductors.

e. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

f. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduits.

g. Section 26 13 13, MEDIUM-VOLTAGE CIRCUIT BREAKER SWITCHGEAR: Requirements for medium-voltage circuit breaker switchgear used for motor control.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Motor Controllers; G,

Manual Motor Controllers; G,

Magnetic Motor Controllers; G,

Reduced Voltage Motor Controllers; G,

Medium-Voltage Motor Controllers; G,

Low-Voltage Variable Speed Motor Controllers (Vsmc); G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Include electrical ratings, dimensions, weights, mounting details, materials, overcurrent protection devices, overload relays, sizes of enclosures, wiring diagrams, starting characteristics, interlocking, and accessories.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, companion copies of complete maintenance and operating manuals, including technical data sheets, wiring diagrams, and information for ordering replacement parts.

(a) Wiring diagrams shall have their terminals identified to facilitate installation, maintenance, and operation.

(b) Wiring diagrams shall indicate internal wiring for each item of equipment and interconnections between the items of equipment.

(c) Elementary schematic diagrams shall be provided for clarity of operation.

(d) Include the catalog numbers for the correct sizes of overload

relays for the motor controllers.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

c. Certifications: Two weeks prior to final inspection, submit the following.

(1) Certification by the manufacturer that the motor controllers conform to the requirements of the drawings and specifications.

(2) Certification by the Contractor that the motor controllers have been properly installed, adjusted, and tested.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 519	(2022) Standard for Harmonic Control in Electrical Power Systems

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

UL SOLUTIONS (UL)

UL 1449	(2021; Reprint Dec 2022) UL Standard for Safety Surge Protective Devices

PART 2 PRODUCTS

2.1 MOTOR CONTROLLERS

a. Motor controllers shall comply with IEEE, NEMA, NFPA, UL, and as shown on the drawings.

b. Motor controllers shall be separately enclosed, unless part of another assembly. For installation in motor control centers, provide plug-in, draw-out type motor controllers up through NEMA size 4. NEMA size 5 and above require bolted connections.

c. Motor controllers shall be combination type, with magnetic controller per Paragraph 2.3 below and with circuit breaker disconnecting means, with external operating handle with lock-open padlocking positions and ON-OFF position indicator.

(1) Circuit Breakers:

(a) Bolt-on thermal-magnetic type with a minimum interrupting rating as indicated on the drawings.

(b) Equipped with automatic, trip free, non-adjustable, inverse-time, and instantaneous magnetic trips for less than 400A. The magnetic trip shall be adjustable from 5x to 10x for breakers 400A and greater.

(c) Additional features shall be as follows:

1. A rugged, integral housing of molded insulating material.

2. Silver alloy contacts.

3. Arc quenchers and phase barriers for each pole.

4. Quick-make, quick-break, operating mechanisms.

5. A trip element for each pole, a common trip bar for all poles, and one operator for all poles.

(2) Fused Switches:

(a) Quick-make, quick-break type.

(b) Minimum duty rating shall be NEMA classification General Duty (GD) for 240 Volts and NEMA classification Heavy Duty (HD) for 480 Volts.

(c) Horsepower rated, and shall have the following features:

1. Copper blades, visible in the OFF position.

2. An arc chute for each pole.

3. Fuse holders for the sizes and types of fuses specified or as shown on the drawings.

(3) Motor Circuit Protectors:

(a) Magnetic trip only.

(b) Bolt-on type with a minimum interrupting rating as indicated on the drawings.

(c) Equipped with automatic, adjustable magnetic trip. Magnetic trip shall be adjustable up to 1300 percent of the motor full load amperes.

d. Enclosures:

(1) Enclosures shall be NEMA-type rated 1, 3R, or 12 as indicated on the drawings or as required per the installed environment.

(2) Enclosure doors shall be interlocked to prevent opening unless the disconnecting means is open. A "defeater" mechanism shall allow for inspection by qualified personnel with the disconnect means closed. Provide padlocking provisions.

(3) All metal surfaces shall be thoroughly cleaned, phosphatized, and factory primed prior to applying light gray baked enamel finish.

e. Motor control circuits:

(1) Shall operate at not more than 120 Volts.

(2) Shall be grounded, except where the equipment manufacturer recommends that the control circuits be isolated.

(3) For each motor operating over 120 Volts, incorporate a separate, heavy duty, control transformer within each motor controller enclosure.

(4) Incorporate primary and secondary overcurrent protection for the control power transformers.

f. Overload relays:

(1) Thermal Temperature Probe Thermal Relay Electronic type. Devices shall be NEMA type.

(2) One for each pole.

(3) External overload relay reset pushbutton on the door of each motor controller enclosure.

(4) Overload relays shall be matched to nameplate full-load current of actual protected motor and with appropriate adjustment for duty cycle.

(5) Thermal overload relays shall be tamperproof, not affected by vibration, manual reset, sensitive to single-phasing, and shall have selectable trip classes of 10, 20 and 30.

(6) Electronic overload relays shall utilize internal current transformers and electro-mechanical components. The relays shall have ambient temperature compensation, single-phase protection, manual or automatic reset, and trip classes of 10, 15, 20 and 30. The relay shall provide fault cause indication, including jam/stall, ground fault, phase loss, and overload.

g. Hand-Off-Automatic (H-O-A) switch is required unless specifically stated on the drawings as not required for a particular controller. H-O-A switch shall be operable without opening enclosure door. H-O-A switch is not required for manual motor controllers.

h. Incorporate into each control circuit a 120 Volt, electronic time-delay relay (ON delay), minimum adjustable range from 0.3 to 10 minutes, with transient protection. Time-delay relay is not required where H-O-A switch is not required.

i. Unless noted otherwise, equip each motor controller with not less than two normally open (N.O.) and two normally closed (N.C.) auxiliary contacts.

j. Provide green (RUN) and red (STOP) pilot lights.

k. Motor controllers incorporated within equipment assemblies shall also be designed for the specific requirements of the assemblies.

l. Additional requirements for specific motor controllers, as indicated

in other specification sections, shall also apply.

2.2 MANUAL MOTOR CONTROLLERS

a. Shall be in accordance with applicable requirements of 2.1 above.

b. Manual motor controllers shall have the following features:

(1) Controllers shall be general-purpose Class A, manually operated type with full voltage controller for induction motors, rated in horsepower.

(2) Units shall include thermal overload relays, on-off operator, red pilot light, normally open auxiliary contacts.

c. Fractional horsepower manual motor controllers shall have the following features:

(1) Controllers shall be general-purpose Class A, manually operated type with full voltage controller for fractional horsepower induction motors.

(2) Units shall include thermal overload relays, red pilot light, and toggle operator.

2.3 MAGNETIC MOTOR CONTROLLERS

a. Shall be in accordance with applicable requirements of 2.1 above.

b. Controllers shall be general-purpose, Class A magnetic controllers for induction motors rated in horsepower. Minimum NEMA size 0.

c. Where combination motor controllers are used, combine controller with protective or disconnect device in a common enclosure.

d. Provide phase loss protection for each controller, with contacts to de-energize the controller upon loss of any phase.

e. Unless otherwise indicated, provide full voltage non-reversing across-the-line mechanisms for motors less than 75 HP, closed by coil action and opened by gravity. For motors 75 HP and larger, provide reduced-voltage or variable speed controllers as shown on the drawings. Equip controllers with 120 VAC coils and individual control transformer unless otherwise noted.

2.4 REDUCED VOLTAGE MOTOR CONTROLLERS

a. Shall be in accordance with applicable portions of 2.1 above.

b. Shall have closed circuit transition.

c. Shall limit inrush currents to not more than 70 percent of the locked rotor current.

d. Provide phase loss protection for each motor controller, with contacts to de-energize the motor controller upon loss of any phase.

2.5 MEDIUM-VOLTAGE MOTOR CONTROLLERS

a. Shall be in accordance with applicable portions of 2.1 above, and in accordance with applicable provisions of Section 26 13 13, MEDIUM-VOLTAGE CIRCUIT BREAKER SWITCHGEAR.

b. Interrupting ratings shall be not less than the maximum short circuit currents available as shown on the drawings.

c. Shall have the following additional features:

(1) Metal enclosed, free-standing, vacuum break, reduced-voltage, primary reactor, drawout type combined with non-load break fused disconnect switch.

(2) Shall include the following components:

(a) Three pole, magnetically held, drawout type, with start/run contactor(s).

(b) Equipped for the number of motor speeds as shown on the drawings.

(c) Primary reactor with taps for 50, 65 and 80 percent of line voltage.

(d) Definite time transfer relay.

(e) Three current limiting, type “R” power type fuses with 50,000 amperes interrupting capability or as indicated on drawings.

(f) Control power transformer (CPT), protected with current limiting fuses. The CPT shall be rated 300VA, , , , and shall be rated 60kV BIL.

(g) Three current transformers and overcurrent protective devices.

(h) Zero-sequence current transformers and associated devices for ground fault protection.

(i) Under-voltage protection.

(j) Protection against single phasing.

(k) Stator thermal protection.

(l) Indicating-type ammeter and selector switch.

(m) Red and green indicating lights.

(3) A separate enclosure for each motor controller.

(4) Shall be isolated by an externally operated mechanism. The secondary of the control power transformer shall also be opened by this device.

(5) Suitable and adequate compartments and barriers for medium-voltage components. Isolate the power bus from the normally accessible compartments.

(6) Medium-voltage line connections shall be automatically shuttered closed when the motor controller is in the racked-out position. The disconnection shall be clearly indicated.

(7) Interlocks shall prevent:

(a) Inadvertent operation of the isolating mechanism under load.

(b) Opening the medium-voltage compartment before the controller is isolated.

(c) Closing of the line contactor while the enclosure door is open.

(8) Current and potential transformers for operating remote recording watt-hour and demand meters and the indicating meters at the motor controller.

(9) Provide lock-open padlocking provisions.

(10) Furnish accessories as recommended by the manufacturer of the motor controllers to facilitate convenient operation and maintenance of the controllers.

2.6 LOW-VOLTAGE VARIABLE SPEED MOTOR CONTROLLERS (VSMC)

a. VSMC shall be in accordance with applicable portions of 2.1 above.

b. VSMC shall be electronic, with adjustable frequency and voltage, three phase output, capable of driving standard NEMA B three-phase induction motors at full rated speed. The control technique shall be pulse width modulation (PWM), where the VSMC utilizes a full wave bridge design incorporating diode rectifier circuitry. Silicon controlled rectifiers or other control techniques are not acceptable.

c. VSMC shall be suitable for variable torque loads, and shall be capable of providing sufficient torque to allow the motor to break away from rest upon first application of power.

d. VSMC shall be capable of operating within voltage parameters of plus 10 to minus 15 percent of line voltage, and be suitably rated for the full load amps of the maximum watts (HP) within its class.

e. Minimum efficiency shall be 95 percent at 100 percent speed and 85 percent at 50 percent speed.

f. The displacement power factor of the VSMC shall not be less than 95 percent under any speed or load condition.

g. VSMC current and voltage harmonic distortion shall not exceed the values allowed by IEEE 519.

h. Operating and Design Conditions:

(1) Elevation: 10000feet Above Mean Sea Level (AMSL)

(2) Temperatures: Maximum +90 degrees F Minimum -10 degrees F

(3) Relative Humidity: 95 percent

(4) VSMC Location: Air conditioned space

i. VSMC shall have the following features:

(1) Isolated power for control circuits.

(2) Manually resettable overload protection for each phase.

(3) Adjustable current limiting circuitry to provide soft motor starting. Maximum starting current shall not exceed 200 percent of motor full load current.

(4) Independent acceleration and deceleration time adjustment, manually adjustable from to 2000 seconds. Set timers to the equipment manufacturer's recommended time in the above range.

(5) Control input circuitry that will accept 4 to 20 mA current or 0-10 VDC voltage control signals from an external source.

(6) Automatic frequency adjustment from 1 Hz to 300 Hz.

(7) Circuitry to initiate an orderly shutdown when any of the conditions listed below occur. The VSMC shall not be damaged by any of these electrical disturbances and shall automatically restart when the conditions are corrected. The VSMC shall be able to restart into a rotating motor operating in either the forward or reverse direction and matching that frequency.

(a) Incorrect phase sequence.

(b) Single phasing.

(c) Overvoltage in excess of 10 percent.

(d) Undervoltage in excess of 15 percent.

(e) Running overcurrent above 110 percent (VSMC shall not automatically reset for this condition.)

(f) Instantaneous overcurrent above 150 percent (VSMC shall not automatically reset for this condition).

(g) Short duration power outages of 12 cycles or less (i.e., distribution line switching, generator testing, and automatic transfer switch operations.)

(8) Provide automatic shutdown upon receiving a power transfer warning signal from an automatic transfer switch. VSMC shall automatically restart motor after the power transfer.

(9) Automatic Reset/Restart: Attempt three restarts after VSMC fault or on return of power after an interruption and before shutting down for manual reset or fault correction, with adjustable delay time between restart attempts.

(10) Power-Interruption Protection: To prevent motor from re-energizing after a power interruption until motor has stopped, unless "Bidirectional Autospeed Search" feature is available and

engaged.

(11) Bidirectional Autospeed Search: Capable of starting VSMC into rotating loads spinning in either direction and returning motor to set speed in proper direction, without causing damage to VSMC, motor, or load.

j. VSMC shall include an input circuit breaker which will disconnect all input power, interlocked with the door so that the door cannot be opened with the circuit breaker in the closed position.

k. VSMC shall include a 5 percent line reactor and a RFI/EMI filter.

l. Surge Suppression: Provide three-phase protection against damage from supply voltage surges in accordance with UL 1449.

m. VSMC shall include front-accessible operator station, with sealed keypad and digital display, which allows complete programming, operating, monitoring, and diagnostic capabilities.

(1) Typical control functions shall include but not be limited to:

(a) HAND-OFF-AUTOMATIC-RESET, with manual speed control in HAND mode.

(b) NORMAL-BYPASS.

(c) NORMAL-TEST, which allows testing and adjusting of the VSMC while in bypass mode.

(2) Typical monitoring functions shall include but not be limited to:

(a) Output frequency (Hz).

(b) Motor speed and status (run, stop, fault).

(c) Output voltage and current.

(3) Typical fault and alarm functions shall include but not be limited to:

(a) Loss of input signal, under- and over-voltage, inverter overcurrent, motor overload, critical frequency rejection with selectable and adjustable deadbands, instantaneous line-to-line and line-to-ground overcurrent, loss-of-phase, reverse-phase, and short circuit.

(b) System protection indicators indicating that the system has shutdown and will not automatically restart.

n. VSMC shall include two N.O. and two N.C. dry contacts rated 120 Volts, 10 amperes, 60 Hz.

o. Hardware, software, network interfaces, gateways, and programming to control and monitor the VSMC by control systems specified in other specification sections, including but not limited to Divisions 22 and 23.

p. Network communications ports: As required for connectivity to control

systems specified in other specification sections, including but not limited to Divisions 22 and 23.

q. Communications protocols: As required for communications with control systems specified in other specification sections, including but not limited to Divisions 22 and 23.

r. Bypass controller: Provide contactor-style bypass, arranged to bypass the inverter.

(1) Inverter Output Contactor and Bypass Contactor: Load-break NEMA-rated contactor.

(2) Motor overload relays.

(3) HAND-OFF-AUTOMATIC bypass control.

s. Bypass operation: Transfers motor between inverter output and bypass circuit, manually, automatically, or both. VSMC shall be capable of stable operation (starting, stopping, and running), and control by fire alarm and detection systems, with motor completely disconnected from the inverter output. Transfer between inverter and bypass contactor and retransfer shall only be allowed with the motor at zero speed.

t. Inverter Isolating Switch: Provide non-load-break switch arranged to isolate inverter and permit safe troubleshooting and testing of the inverter, both energized and de-energized, while motor is operating in bypass mode. Include padlockable, door-mounted handle mechanism.

PART 3 EXECUTION

3.1 INSTALLATION

a. Install motor controllers in accordance with NFPA 70, as shown on the drawings, and as recommended by the manufacturer.

b. Install manual motor controllers in flush enclosures in finished areas.

c. Set field-adjustable switches, auxiliary relays, time-delay relays, timers, and electronic overload relay pickup and trip ranges.

d. Program variable speed motor controllers per the manufacturer’s instructions and in coordination with other trades so that a complete and functional system is delivered.

e. Adjust trip settings of circuit breakers and motor circuit protectors with adjustable instantaneous trip elements. Initially adjust at six times the motor nameplate full-load ampere ratings and attempt to start motors several times, allowing for motor cooldown between starts. If tripping occurs on motor inrush, adjust settings in increments until motors start without tripping. Do not exceed eight times the motor full-load amperes (or 11 times for NEMA Premium Efficiency motors if required). Where these maximum settings do not allow starting of a motor, notify Contracting Officer before increasing settings.

f. Set the taps on reduced-voltage autotransformer controllers at 65 percent of line voltage.

3.2 ACCEPTANCE CHECKS AND TESTS

Perform manufacturer’s required field tests in accordance with the manufacturer's recommendations. In addition, include the following: Visual Inspection and Tests:

a. Compare equipment nameplate data with specifications and approved shop drawings.

b. Inspect physical, electrical, and mechanical condition.

c. Verify appropriate anchorage, required area clearances, and correct alignment.

d. Verify that circuit breaker, motor circuit protector, and fuse sizes and types correspond to approved shop drawings.

e. Verify overload relay ratings are correct.

f. Vacuum-clean enclosure interior. Clean enclosure exterior.

g. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer’s published data.

h. Test all control and safety features of the motor controllers.

i. For low-voltage variable speed motor controllers, final programming and connections shall be by a factory-trained technician. Set all programmable functions of the variable speed motor controllers to meet the requirements and conditions of use.

3.3 FOLLOW-UP VERIFICATION

Upon completion of acceptance checks, settings, and tests, the Contractor shall show by demonstration in service that the motor controllers are in good operating condition and properly performing the intended functions.

3.4 SPARE PARTS

Two weeks prior to the final inspection, provide one complete set of spare fuses for each motor controller.

3.5 INSTRUCTION

Furnish the services of a factory-trained technician for two 4-hour training periods for instructing personnel in the maintenance and operation of the motor controllers, on the dates requested by the Contracting Officer.

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SECTION 26 29 21
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ENCLOSED SWITCHES AND CIRCUIT BREAKERS

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, and connection of fused and unfused disconnect switches (indicated as switches in this section), and separately-enclosed circuit breakers for use in electrical systems rated 600 V and below.

1.2 RELATED WORK

a. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

b. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Low-voltage conductors.

c. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground faults.

d. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduits.

e. Section 26 24 16, PANELBOARDS: Molded-case circuit breakers.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Fused Switches Rated 600 Amperes And Less; G,

Unfused Switches Rated 600 Amperes And Less; G,

Fused Switches Rated Over 600 Amperes To 1200 Amperes; G,

Motor Rated Toggle Switches; G,

Cartridge Fuses; G,

Separately-Enclosed Circuit Breakers; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Submit the following data for approval: Electrical ratings, dimensions, mounting details, materials, required clearances, terminations, weight, fuses, circuit breakers, wiring and connection diagrams, accessories, and device nameplate data.

b. Manuals:

(1) Submit complete maintenance and operating manuals including technical data sheets, wiring diagrams, and information for ordering fuses, circuit breakers, and replacement parts.

(a) Include schematic diagrams, with all terminals identified, matching terminal identification in the enclosed switches and circuit breakers.

(b) Include information for testing, repair, troubleshooting, assembly, and disassembly.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

c. Certifications: Two weeks prior to final inspection, submit the following.

(1) Certification by the manufacturer that the enclosed switches and circuit breakers conform to the requirements of the drawings and specifications.

(2) Certification by the Contractor that the enclosed switches and circuit breakers have been properly installed, adjusted, and tested.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent

referenced. Publications are referenced in the text by designation only.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA FU 1 (2012) Low Voltage Cartridge Fuses

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

PART 2 PRODUCTS

2.1 FUSED SWITCHES RATED 600 AMPERES AND LESS

a. Switches shall be in accordance with NEMA, NEC, UL, as specified, and as shown on the drawings.

b. Shall be NEMA classified General Duty (GD) for 240 V switches, and NEMA classified Heavy Duty (HD) for 480 V switches.

c. Shall be horsepower (HP) rated.

d. Shall have the following features:

(1) Switch mechanism shall be the quick-make, quick-break type.

(2) Copper blades, visible in the open position.

(3) An arc chute for each pole.

(4) External operating handle shall indicate open and closed positions, and have lock-open padlocking provisions.

(5) Mechanical interlock shall permit opening of the door only when the switch is in the open position, defeatable to permit inspection.

(6) Fuse holders for the sizes and types of fuses specified.

(7) Solid neutral for each switch being installed in a circuit which includes a neutral conductor.

(8) Ground lugs for each ground conductor.

(9) Enclosures:

(a) Shall be the NEMA types shown on the drawings.

(b) Where the types of switch enclosures are not shown, they shall be the NEMA types most suitable for the ambient environmental conditions.

(c) Shall be finished with manufacturer’s standard gray baked enamel paint over pretreated steel.

(10) Electrically operated switches shall only be installed where shown on the drawings.

2.2 UNFUSED SWITCHES RATED 600 AMPERES AND LESS

Shall be the same as fused switches, but without provisions for fuses.

2.3 FUSED SWITCHES RATED OVER 600 AMPERES TO 1200 AMPERES

Shall be the same as fused switches, and shall be NEMA classified Heavy Duty (HD).

2.4 MOTOR RATED TOGGLE SWITCHES

a. Type 1, general purpose for single-phase motors rated up to 1 horsepower.

b. Quick-make, quick-break toggle switch with external reset button and thermal overload protection matched to nameplate full-load current of actual protected motor.

2.5 CARTRIDGE FUSES

a. Shall be in accordance with NEMA FU 1.

b. Motor Branch Circuits: Class RK5, time delay.

c. Other Branch Circuits: Class J, fast acting .

d. Control Circuits: Class CC, fast acting .

2.6 SEPARATELY-ENCLOSED CIRCUIT BREAKERS

a. Provide circuit breakers in accordance with the applicable requirements in Section 26 24 16, PANELBOARDS.

b. Enclosures shall be the NEMA types shown on the drawings. Where the types are not shown, they shall be the NEMA type most suitable for the ambient environmental conditions.

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation shall be in accordance with NFPA 70, as shown on the drawings, and manufacturer’s instructions.

b. Fused switches shall be furnished complete with fuses. Arrange fuses such that rating information is readable without removing the fuses.

3.2 ACCEPTANCE CHECKS AND TESTS

Perform in accordance with the manufacturer's recommendations. In addition, include the following: Visual Inspection and Tests:

a. Compare equipment nameplate data with specifications and approved shop drawings.

b. Inspect physical, electrical, and mechanical condition.

c. Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method.

d. Vacuum-clean enclosure interior. Clean enclosure exterior.

3.3 SPARE PARTS

Two weeks prior to the final inspection, furnish one complete set of spare fuses for each fused disconnect switch installed on the project. Deliver the spare fuses to the Contracting Officer.

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SECTION 26 33 53

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SECTION 26 33 53
11/22

STATIC UNINTERRUPTIBLE POWER SUPPLY

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, connection, and testing of the static uninterruptible power supply, indicated in this section as UPS.

1.2 RELATED WORK

a. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

b. Section 26 05 19, -VOLTAGE ELECTRICAL POWER CONDUCTORS CABLES: Low-voltage conductors.

c. Section 26 05 26, GROUNDING BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible fault currents.

d. Section 26 05 33, RACEWAY BOXES FOR ELECTRICAL SYSTEMS: Conduits.

e. Section 26 05 73, OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY: Short circuit and coordination study, and requirements for a coordinated electrical system.

f. Section 26 24 13, DISTRIBUTION SWITCHBOARDS: For low-voltage switchboard required for maintenance bypass of multiple module UPS.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 FACTORY TESTS

a. UPS shall be thoroughly tested at the factory to assure that there are no electrical or mechanical defects.

b. Factory Tests shall be in accordance with Paragraph, MANUFACTURED PRODUCTS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements: UPS shall be factory full-load tested to meet the requirements specified using a test battery (not the battery to be supplied with the system) with AC input power and with battery power for a minimum of 8 hours, with meter readings taken every 30 minutes. Should a malfunction occur, the problem shall be corrected and the test shall be repeated. The tests shall encompass all aspects of operation, such as module failure, static bypass operation, battery failure, input power failure and overload ratings.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-06 Test Reports

Factory Tests; G,

Test Reports; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.6 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Include electrical ratings, dimensions, mounting details, materials, required clearances, terminations, weight, plan, front, side, and rear elevations, accessories, and device nameplate data.

(3) Provide detailed and project-specific system diagram, showing maintenance bypass, UPS module(s), battery cabinet(s) and batteries, major circuit protective devices, interconnecting power and control wiring, key-type mechanical interlocks, and connections to power sources and loads, as applicable. Indicate whether interconnections are factory-provided/factory-installed, factory-provided/field-installed, or field-provided/field installed.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, companion copies of complete maintenance and operating manuals including technical data sheets, wiring diagrams, and information for ordering replacement parts.

(a) Wiring diagrams shall have their terminals identified to facilitate installation, maintenance, and operation.

(b) Wiring diagrams shall indicate internal wiring for each item of equipment and the interconnection between the items of equipment.

(c) Provide a clear and concise description of operation, which gives, in detail, the information required to properly operate the UPS, including but not limited to bypass switchboard, UPS, key-type mechanical interlocks, remote devices, emergency power off buttons, fire alarm interface, and other components as applicable.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

(a) Include complete "As Installed" diagrams that indicate all pieces of equipment and their interconnecting wiring.

(b) Include complete diagrams of the internal wiring for each piece of equipment, including "As Installed" revisions of the diagrams.

(c) The wiring diagrams shall identify the terminals to facilitate installation, maintenance, operation, and testing.

c. Test Reports:

(1) Submit certified factory design and production test reports for approval.

(2) Two weeks prior to the final inspection, submit certified field test reports and data sheets to the Contracting Officer.

d. Certifications: Two weeks prior to final inspection, submit the following.

(1) Certification by the manufacturer that the UPS conforms to the requirements of the drawings and specifications.

(2) Certification by the Contractor that the UPS has been properly installed, adjusted, and tested.

1.7 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata), form a part of this specification to the extent referenced. Publications are referenced in the text by basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41.1	(2002; R 2008) Guide on the Surges Environment in Low-Voltage (1000 V and Less) AC Power Circuits
IEEE C62.41.2	(2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

a. System Capacity: Unless stated otherwise, the parameters listed are under full rated output load at a minimum of 0.9 power factor, with batteries fully charged and floating on the DC bus and with nominal input voltage. Overall kVA, kW, redundant, at 40 C.

b. Battery Capacity: Discharge time to end voltage: 15 minutes, at 77 F. Battery shall be capable of delivering 125 percent of full rated output load at initial start-up.

c. System Bus Bracing: Braced for amperes symmetrical interrupting capacity as shown on drawings.

d. AC Input:

(1) Voltage 208 volts line-to-line.

(2) Number of phases: 3-phase, 3-wire, plus ground.

(3) Voltage Range: Plus 10 percent, minus 15 percent, without affecting battery float voltage or output voltage.

(4) Frequency: 60 Hz, plus or minus 5 percent.

(5) Total harmonic current distortion () reflected into the primary line: 5 percent maximum.

e. AC Output

(1) Voltage 208 volts line-to-line, 120 volts line-to-neutral.

(2) Number of phases: 3-phase, 4-wire, plus ground.

(3) Voltage regulation:

(a) Balanced load: Plus or minus 1.0 percent.

(b) 100 percent load imbalance, phase-to-phase: Plus or minus 3 percent.

(4) Frequency: 60 Hz.

(5) Frequency regulation: Plus or minus 0.05 percent.

(6) Harmonic content (RMS voltage): 5 percent maximum total harmonic distortion with 100 percent nonlinear load.

(7) Load power factor operating range: 1.0 to 0.8 lagging.

(8) Phase displacement:

(a) Balanced load: Plus or minus 1 degree of bypass input.

(9) Overload capability (at full voltage) (excluding battery):

(a) 125 percent load for 10 minutes.

(b) 150 percent load for 1 minute.

f. Provide advanced utility metering devices, instrument transformers (C.T. and P.T.), communication interfaces, wiring and raceway, and accessories as specified in Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS). Meters shall be multifunction electronic type with local display, remote communications, data logging, power quality monitoring, and alarming capabilities. All new meters shall be fully compatible with, and integrated into, the existing facility SCADA system for real-time monitoring, trending, alarming, and data archiving. Coordinate communication protocol (e.g., Modbus, TCP/IP, BACnet/IP, or facility-standard), point mapping/registers, and programming with the existing SCADA system integrator/vendor. Provide complete point lists, Modbus/BACnet object mappings, and support for HMI screen updates, alarm configuration, and trend setup. Dedicated communication cabling (Ethernet CAT6 or fiber as required by distance and facility standard) shall connect new meters to the existing SCADA network/data aggregation devices; do not share with IT or BAS networks. Meters and integration shall comply with VA Electrical Design Manual (PG-18-10) Section 5.4 POWER MONITORING AND METERING. Contractor shall perform integration testing, verify data accuracy against calibrated reference meters, and demonstrate full SCADA functionality (display, alarms, trends) prior to acceptance, refer to Section 25 08 10 UTILITY MONITORING AND CONTROL SYSTEM TESTING.

g. Voltage Transient Response:

(1) 100 percent load step: Plus or minus 5 percent.

2.2 UPS

a. General Description: UPS module shall consist of a rectifier/charger unit and a 3-phase inverter module unit with their associated transformers, synchronizing equipment, input and output circuit breakers, and accessories as required for operation.

b. Rectifier/Charger Unit: Rectifier/charger unit shall be solid state and shall provide direct current to the DC bus.

(1) Input Circuit Breaker: Rectifier/charger unit shall be provided with an input circuit breaker. The circuit breaker shall be sized to accept simultaneously the full-rated load and the battery recharge current.

(2) Sizing: Rectifier/charger unit shall be sized for the following two simultaneous operating conditions:

(a) Supplying the full rated load current to the inverter.

(b) Recharging a fully-discharged battery to 95 percent of rated ampere-hour capacity within ten times the discharge time after normal AC power is restored, with the input protective device closed.

c. Inverter Unit: Inverter unit shall be a solid-state device capable of accepting power from the DC bus and providing AC power within specified limits.

(1) Output Overload: The inverter shall be able to sustain an overload as specified across its output terminals.

(2) Synchronism: The inverter shall normally operate in phase-lock and synchronism with the bypass source.

(3) Modular Construction: Each control logic printed circuit board shall be electrically and physically packaged on an individual plug-in module with separate indication and adjustments.

(4) Output Circuit Breaker: The output circuit breaker shall be capable of shunt tripping and shall have interrupting capacity as specified. Circuit breaker shall have provision for locking in the "off" position.

d. External Protection: UPS module shall have built-in self-protection against undervoltage, overvoltage, overcurrent and surges introduced on the AC input source and/or the bypass source. The UPS system shall sustain input surges without damage in accordance with IEEE C62.41.1 and IEEE C62.41.2. The UPS shall also have built-in self-protection against overvoltage and voltage surges introduced at the output terminals by paralleled sources, load switching, or circuit breaker operation in the critical load distribution system.

e. Internal Protection: UPS module shall be self-protected against overcurrent, sudden changes in output load and short circuits at the output terminals. UPS module shall be provided with output reverse power detection which shall cause that module to be disconnected from the critical load bus when output reverse power is present. UPS module shall have built-in protection against permanent damage to itself and the connected load for predictable types of failure within itself and the connected load. At the end of battery discharge limit, the module shall shut down without damage to internal components.

2.3 MAINTENANCE BYPASS - SINGLE-MODULE UPS

a. A maintenance bypass switch shall be provided as an integral part of the UPS and located within the UPS module. The maintenance bypass switch shall provide the capability to continuously support the load from the bypass AC power source while the UPS is isolated for maintenance. The maintenance bypass switch shall be housed in an isolated compartment inside the UPS cabinet. Switch shall contain a maintenance bypass protective device and a module isolation protective device.

b. The maintenance bypass switch shall provide the capability of transferring the load from the UPS static bypass transfer switch to maintenance bypass and then back to the UPS static bypass transfer switch with no interruption to the load.

2.4 MODULE CONTROL PANEL – SINGLE-MODULE UPS

a. The UPS module shall be provided with a LCD control/indicator panel. Meters, controls, alarms, system diagnostics, mimic screen showing one-line diagram of the system, and indicators for operation of the UPS shall be on this panel.

(1) Module Meters:

(a) Meters shall have 1 percent accuracy.

(b) The following functions shall be monitored and displayed:

1. Input voltage, phase-to-phase (all three phases).

2. Input current, all three phases.

3. Input frequency.

4. Battery voltage.

5. Battery current (charge/discharge).

6. Output voltage, phase-to-phase and phase-to-neutral (all three phases).

7. Output current, all three phases.

8. Output frequency.

9. Output kilowatts.

10. Elapsed time meter to indicate hours of operation, 6 digits.

11. Bypass voltage, phase-to-phase and phase-to-neutral (all three phases).

12. Output kilovars.

13. Output kilowatt hours, with 15-minute interval.

(2) Module Controls: Module shall have the following controls:

(a) Alarm test/reset function.

(b) Module input protective device trip function.

(c) Module output protective device trip function.

(d) Battery protective device trip function.

(e) Emergency Power Off (EPO) pushbutton, with guard.

(f) Control power off switch.

(g) Static bypass transfer switch enable/disable selector switch.

(3) Module Alarm Indicators: Module shall have indicators for the following alarm items. Any one of these conditions shall turn on an audible alarm and the appropriate summary indicator. Each new alarm shall register without affecting any previous alarm.

(a) Input AC power source failure.

(b) Input protective device open.

(c) Output protective device open.

(d) Overload.

(e) Overload shutdown.

(f) DC overvoltage.

(g) DC ground fault.

(h) Low battery.

(i) Battery discharged.

(j) Battery protective device open.

(k) Cooling fan failure.

(l) Equipment overtemperature.

(m) Control power failure.

(n) Charger off.

(o) Inverter off.

(p) Emergency off.

(q) UPS on battery.

(r) Load on static bypass.

(s) Static bypass transfer switch disabled.

(t) Inverter output overvoltage, undervoltage, overfrequency, and underfrequency.

(u) Bypass source overvoltage, undervoltage, overfrequency, and underfrequency.

(v) Bypass source to inverter out of synchronization.

2.5 SYSTEM CONTROL CABINET – MULTI-MODULE UPS

The UPS shall be provided with a LCD control/indicator panel. Meters, controls, alarms, system diagnostics, mimic screen showing one-line diagram of the system, and indicators for operation of the UPS shall be on this panel. The system control cabinet shall communicate with each UPS module, the static bypass transfer switch and its bypass disconnect, the UPS input and output circuit breakers, and the UPS maintenance bypass switchboard.

a. Module Meters:

(1) Meters shall have 1 percent accuracy.

(2) The following functions shall be monitored and displayed:

(a) Input voltage, phase-to-phase (all three phases).

(b) Input current, all three phases.

(c) Input frequency.

(d) Battery voltage.

(e) Battery current (charge/discharge).

(f) Output voltage, phase-to-phase and phase-to-neutral (all three phases).

(g) Output current, all three phases.

(h) Output frequency.

(i) Output kilowatts.

(j) Elapsed time meter to indicate hours of operation, 6 digits.

(k) Bypass voltage, phase-to-phase and phase-to-neutral (all three phases).

(l) Output kilovars.

(m) Output kilowatt hours, with 15-minute interval.

b. Module Controls: Module shall have the following controls:

(1) Alarm test/reset function.

(2) Module input protective device trip function.

(3) Module output protective device trip function.

(4) Battery protective device trip function.

(5) Emergency Power Off (EPO) pushbutton, with guard.

(6) Control power off switch.

(7) Static bypass transfer switch enable/disable selector switch.

c. Module Alarm Indicators: Module shall have indicators for the following alarm items. Any one of these conditions shall turn on an audible alarm and the appropriate summary indicator. Each new alarm shall register without affecting any previous alarm.

(1) Input AC power source failure.

(2) Input protective device open.

(3) Output protective device open.

(4) Overload.

(5) Overload shutdown.

(6) DC overvoltage.

(7) DC ground fault.

(8) Low battery.

(9) Battery discharged.

(10) Battery protective device open.

(11) Cooling fan failure.

(12) Equipment overtemperature.

(13) Control power failure.

(14) Charger off.

(15) Inverter off.

(16) Emergency off.

(17) UPS on battery.

(18) Load on static bypass.

(19) Static bypass transfer switch disabled.

(20) Inverter output overvoltage, undervoltage, overfrequency, and underfrequency.

(21) Bypass source overvoltage, undervoltage, overfrequency, and underfrequency.

(22) Bypass source to inverter out of synchronization.

2.6 REMOTE MONITORING PANEL

a. A wall-mounted remote monitoring panel shall be provided to monitor system status.

b. Indicators: Minimum display shall include the following indicators:

(1) Load on UPS.

(2) Load on battery.

(3) Load on bypass.

(4) Low battery.

(5) Summary alarm.

(6) New alarm (to alert the operator that a second summary alarm condition has occurred).

c. Audible Alarm.

d. Any single indicator shall also turn on the audible alarm. An audible alarm test/reset button and lamp test/reset button shall be included.

This reset button shall not affect nor reset the alarm on the system control cabinet.

2.7 BATTERY SYSTEM

a. General: A storage battery with sufficient ampere-hour rating to maintain UPS output at full capacity for the specified duration shall be provided for each UPS module.

b. Battery Type: Lead calcium.

c. Battery Construction: The battery shall be of the valve-regulated, sealed, non-gassing, recombinant type.

d. Battery Cabinet: The batteries shall be furnished in a battery cabinet matching the UPS. The battery cabinet shall be provided with smoke and high temperature alarms.

e. Battery Cables: Battery-to-battery connections shall be stranded cable with proper cable supports.

f. Battery Disconnect: Each battery cabinet or rack shall have a fused disconnect switch or circuit breaker, lockable in the “off” position, provided in a NEMA 1 enclosure.

g. Electrolyte Spill Containment and Neutralization: Provide spill control and neutralization means integral to the battery cabinet or battery rack assembly. Provide spare neutralization materials in a manufacturer’s cabinet.

PART 3 EXECUTION

3.1 INSTALLATION

a. The UPS shall be set in place, wired, and connected in accordance with the approved shop drawings and manufacturer's instructions.

3.2 ACCEPTANCE CHECKS AND TESTS

a. An authorized representative of the UPS manufacturer shall technically supervise and participate during all of the field adjustments and tests. Major adjustments and field tests shall be witnessed by the Contracting Officer. The manufacturer’s representative shall certify in writing that the equipment has been installed, adjusted, and tested in accordance with the manufacturer’s recommendations.

b. Perform manufacturer’s required field tests in accordance with the manufacturer's recommendations. In addition, include the following:

(1) Visual Inspection and Tests:

(a) Compare equipment nameplate data with specifications and approved shop drawings.

(b) Inspect physical, electrical, and mechanical condition.

(c) Verify appropriate anchorage, required area clearances, and correct alignment.

(d) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey after energization.

(e) Verify grounding connections.

(f) Vacuum-clean enclosure interior. Clean enclosure exterior.

(g) Verify the correct operation of all alarms and indicating devices.

(h) Attach a phase rotation meter to the UPS input, output, and bypass buses, and observe proper phase sequences.

(i) Check and test controls for proper operation.

(j) Check doors for proper alignment and operation.

(k) Check and test each protective device for proper mechanical and electrical operation.

(l) Verify protective device overcurrent trip settings against approved coordination study.

(2) Load Test: The UPS shall be load tested for a continuous 24 hour period by means of resistive load banks, sized for full rated output load and provided by the UPS manufacturer. The UPS shall be continuously tested at 1/2 load for 8 hours, 3/4 load for 8 hours and full load for 8 hours. If a failure occurs during the burn-in period, the tests shall be repeated. Instrument readings shall be recorded every half hour for the following:

(a) Input voltage and current (all three phases, for each module).

(b) Input and output frequency.

(c) Battery voltage for each module.

(d) Output voltage and current (all three phases, for each module).

(e) Output kilowatts for each module.

(f) Output voltage and current (all three phases).

(g) Output kilowatts.

(3) Full Load Burn In Test: The UPS shall undergo an additional full load burn-in period of 24 continuous hours by means of resistive load banks, sized for full rated output load and provided by the UPS manufacturer. If a failure occurs during the burn-in period, the tests shall be repeated. Instrument readings shall be recorded every half hour as above. The following tests shall be performed:

(a) With the UPS carrying full rated output load and supplied from the normal source, switch 50 percent of load bank capacity on and off a minimum of five times within the burn-in period.

(4) Full Load Battery Burn In Test: The UPS shall undergo a full load

battery test by means of resistive load banks, sized for full rated output load and provided by the UPS manufacturer. If a failure occurs during the battery discharge time, the tests shall be repeated. Instrument readings shall be recorded every half hour as above.

(a) With the UPS carrying full rated output load and operating on battery power, switch 50 percent of load bank capacity on and off a minimum of five times within the battery discharge time.

(5) Battery Discharge and Recharge Test: With the battery fully charged, the UPS shall undergo a complete battery discharge test to full depletion followed by a full recharge. Instrument readings shall be recorded every minute during discharge for the following:

(a) Battery voltage and current.

(b) Output voltage and current (all three phases).

(c) Output kilowatts.

(d) Output voltage and current (all three phases).

(e) Output kilowatts (system).

(f) Output frequency.

3.3 FOLLOW-UP VERIFICATION

After completion of acceptance checks and tests, the Contractor shall show by demonstration in service that the UPS is in good operating condition and properly performing the intended function.

3.4 ONE LINE DIAGRAM AND SEQUENCE OF OPERATION

a. At final inspection, an as-built one line diagram shall be laminated or mounted under acrylic glass, and installed in a frame mounted near the UPS.

b. Furnish a written sequence of operation for the UPS and connected line side/load side electrical distribution equipment. The sequence of operation shall be laminated or mounted under acrylic glass, and installed in a frame mounted near the UPS.

c. Deliver an additional four copies of the as-built one line diagram and sequence of operation to the Contracting Officer.

3.5 INSTRUCTION

Furnish the services of a factory-trained technician for one 4-hour training period for instructing personnel in the maintenance and operation of the UPS, on the dates requested by the Contracting Officer.

-- End of Section --

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SECTION 26 36 23
11/22

AUTOMATIC TRANSFER SWITCHES

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, connection, and testing of open-transition automatic transfer switches with bypass isolation, indicated as automatic transfer switches or ATS in this Section.

1.2 RELATED WORK

a. Section 03 30 00, CAST-IN-PLACE CONCRETE: Requirements for concrete equipment pads.

b. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

c. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Low-voltage conductors.

d. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personal safety and to provide a low impedance path for possible ground fault currents.

e. Section 26 05 33, RACEWAYS AND BOXES FOR ELECTRICAL SYSTEMS: Conduits.

f. Section 26 05 73, OVERCURRENT PROTECTIVE DEVICE COORDINATION STUDY: Short circuit and coordination study, and requirements for a coordinated electrical system.

g. Section 27 05 11, REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS: General communications requirements that are common to more than one section in Division 27.

h. Section 27 05 33, RACEWAYS AND BOXES FOR COMMUNICATION SYSTEMS: Raceways for communications cabling.

i. Section 27 15 00, COMMUNICATIONS HORIZONTAL CABLING: Communications media for interconnecting automatic transfer switches and remote control and annunciation components.

1.3 QUALITY ASSURANCE

a. Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

b. A factory-authorized representative shall be capable of providing emergency maintenance and repairs at the project site within four hours maximum of notification.

c. Automatic transfer switch, bypass/isolation switch, and annunciation control panels shall be products of the same manufacturer.

1.4 FACTORY TESTS

a. ATS shall be thoroughly tested at the factory to assure that there are no electrical or mechanical defects.

b. Factory Tests shall be in accordance with Paragraph, MANUFACTURED PRODUCTS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

(1) Perform visual inspection to verify that each ATS is as specified.

(2) Perform mechanical test to verify that ATS sections are free of mechanical defects.

(3) Perform insulation resistance test to ensure electrical integrity and continuity of entire system.

(4) Perform main switch contact resistance test.

(5) Perform electrical tests to verify complete system electrical operation.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Bypass-Isolation Switch; G,

Remote Annunciator System; G,

Remote Annunciator And Control System; G,

SD-05 Design Data

Sequence Of Operation; G,

SD-06 Test Reports

Factory Tests; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.6 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Include voltage rating, continuous current rating, number of phases, withstand and closing rating, dimensions, weights, mounting details, conduit entry provisions, front view, side view, equipment and device arrangement, elementary and interconnection wiring diagrams, factory relay settings, and accessories.

(3) For automatic transfer switches that are networked together to a common means of annunciation and/or control, submit interconnection diagrams as well as site and building plans, showing connections for normal and emergency sources of power, load, control and annunciation components, and interconnecting communications paths. Equipment locations on the diagrams and plans shall match the site, building, and room designations on the drawings.

(4) Complete nameplate data, including manufacturer's name and catalog number.

(5) A copy of the markings that are to appear on the automatic transfer switches when installed.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, companion copies of complete maintenance and operating manuals, including technical data sheets, wiring diagrams, and information for ordering replacement parts.

(a) Schematic signal and control diagrams, with all terminals identified, matching terminal identification in the automatic transfer switches.

(b) Include information for testing, repair, troubleshooting, assembly, disassembly, and factory recommended/required periodic maintenance procedures and frequency.

(c) Provide a replacement and spare parts list. Include a list of tools and instruments for testing and maintenance purposes.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

(a) Include complete "As Installed" diagrams that indicate all pieces of equipment and their interconnecting wiring.

(b) Include complete diagrams of the internal wiring for each piece of equipment, including "As Installed" revisions of the diagrams.

(c) The wiring diagrams shall identify the terminals to facilitate installation, maintenance, operation, and testing.

c. Certifications:

(1) When submitting the shop drawings, submit a certified test report from a recognized independent testing laboratory that a representative sample has passed UL 1008 prototype testing.

(2) Two weeks prior to final inspection, submit the following.

(a) Certification by the manufacturer that the ATS conform to the requirements of the drawings and specifications.

(b) Certification by the Contractor that transfer switches have been properly installed, adjusted, and tested.

1.7 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C37.90.1	(2023; ERTA) Standard for Surge Withstand Capability (SWC) Tests for Relays and Relay Systems Associated with Electric Power Apparatus

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

UL SOLUTIONS (UL)

UL 508	(2018; Reprint Jul 2021) UL Standard for Safety Industrial Control Equipment
UL 891	(2019; Reprint Mar 2025) UL Standard for Safety Switchboards
UL 1008	(2022) UL Standard for Safety Transfer Switch Equipment

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

Automatic transfer switches shall comply with IEEE, NEMA, NFPA, UL, and have the following features:

a. Automatic transfer switches shall be open transition switches, 4-pole, draw-out construction, electrically operated, mechanically held open contact type, without integral overcurrent protection. Automatic transfer switches utilizing automatic or non-automatic molded case circuit breakers, insulated case circuit breakers, or power circuit

breakers as switching mechanisms are not acceptable.

b. Automatic transfer switches shall be completely factory-assembled and wired such that only external circuit connections are required in the field.

c. Each automatic transfer switch shall be equipped with an integral bypass/isolation switch.

d. Ratings:

(1) Phases, voltage, continuous current, poles, and withstand and closing ratings shall be as shown on the drawings.

(2) Transfer switches are to be rated for continuous duty at specified continuous current rating on 60Hz systems.

(3) Maximum automatic transfer switch rating: 800 A.

e. Markings: Markings shall be in accordance with UL 1008.

f. Tests: Automatic transfer switches shall be tested in accordance with UL 1008. The contacts of the transfer switch shall not weld during the performance of withstand and closing tests when used with the upstream overcurrent device and available fault current specified.

g. Surge Withstand Test: Automatic transfer switches utilizing solid-state devices in sensing, relaying, operating, or communication equipment or circuits shall comply with IEEE C37.90.1.

h. Housing:

(1) Enclose automatic transfer switches in wall- or floor-mounted steel cabinets, with metal gauge not less than No. 14, in accordance with UL 508, or in a switchboard assembly in accordance with UL 891, as shown on the drawings.

(2) Enclosure shall be constructed so that personnel are protected from energized bypass-isolation components during automatic transfer switch maintenance.

(3) Automatic transfer switch components shall be removable without disconnecting external source or load power conductors.

(4) Finish: Cabinets shall be given a phosphate treatment, painted with rust-inhibiting primer, and finish-painted with the manufacturer's standard enamel or lacquer finish.

(5) Viewing Ports: Provide viewing ports so that contacts may be inspected without disassembly.

i. Operating Mechanism:

(1) Actuated by an electrical operator.

(2) Electrically and mechanically interlocked so that the main contact cannot be closed simultaneously in either normal and emergency position.

(3) Normal and emergency main contacts shall be mechanically locked in position by the operating linkage upon completion of transfer. Release of the locking mechanism shall be possible only by normal operating action.

(4) Contact transfer time shall not exceed six cycles.

(5) Operating mechanism components and mechanical interlocks shall be insulated or grounded.

j. Contacts:

(1) Main contacts: Silver alloy.

(2) Neutral contacts: Silver alloy, with same current rating as phase contacts .

(3) Current carrying capacity of arcing contacts shall not be used in the determination of the automatic transfer switch rating, and shall be separate from the main contacts.

(4) Main and arcing contacts shall be visible for inspection with cabinet door open and barrier covers removed.

k. Manual Operator: Capable of operation by one person in either direction under no load.

l. Replaceable Parts:

(1) Include the main and arcing contacts individually or as units, as well as relays, and control devices.

(2) Automatic transfer switch contacts and accessories shall be replaceable from the front without removing the switch from the cabinet and without removing main conductors.

m. Sensing Features:

(1) Undervoltage Sensing for Each Phase of Normal Source: Sense low phase-to-ground voltage on each phase. Pickup voltage shall be adjustable from 85 to 100 percent of nominal, and dropout voltage is adjustable from 75 to 98 percent of pickup value. Factory set for pickup at 90 percent and dropout at 85 percent.

(2) Adjustable Time Delay: For override of normal-source voltage sensing to delay transfer and engine start signals. Adjustable from zero to six seconds, and factory set for one second.

(3) Voltage/Frequency Lockout Relay: Prevent premature transfer to the engine-generator. Pickup voltage shall be adjustable from 85 to 100 percent of nominal. Factory set for pickup at 90 percent. Pickup frequency shall be adjustable from 90 to 100 percent of nominal. Factory set for pickup at 95 percent.

(4) Time Delay for Retransfer to Normal Source: Adjustable from 0 to 30 minutes, and factory set for 10 minutes to automatically defeat delay on loss of voltage or sustained undervoltage of emergency source, provided normal supply has been restored.

(5) Test Switch: Simulate normal-source failure.

(6) Switch-Position Indication: Indicate source to which load is connected.

(7) Source-Available Indication: Supervise sources via transfer switch normal- and emergency-source sensing circuits.

(8) Normal Power Indication: Indicate "Normal Source Available."

(9) Emergency Power Indication: Indicate "Emergency Source Available."

(10) Transfer Override Control: Overrides automatic retransfer control so that automatic transfer switch shall remain connected to emergency power source regardless of condition of normal source. Control panel shall indicate override status.

(11) Engine Starting Contacts: One isolated and normally closed and one isolated and normally open; rated 5 A at 30 V DC minimum.

(12) Engine Shutdown Contacts: Time delay adjustable from zero to 15 minutes, and factory set for five minutes. Contacts shall initiate shutdown at remote engine-generator controls after retransfer of load to normal source.

(13) Engine-Generator Exerciser: Programmable exerciser starts engine-generator(s) and transfers load to them from normal source for a preset time, then retransfers and shuts down engine-generator(s) after a preset cool-down period. Initiates exercise cycle at preset intervals adjustable from 7 to 30 days. Running periods are adjustable from 10 to 30 minutes. Factory settings shall be for 7-day exercise cycle, 20-minute running period, and five-minute cool-down period.

n. Controls:

(1) Controls shall provide indication of switch status and be equipped with alarm diagnostics.

(2) Controls shall control operation of the automatic transfer switches.

o. Factory Wiring: Train and bundle factory wiring and label either by color-code or by numbered/lettered wire markers. Labels shall match those on the shop drawings.

p. Annunciation, Control, and Programming Interface Components: Devices for communicating with remote programming devices, annunciators, or control panelsand paralleling switchgear shall have open-protocol communication capability matched with remote device.

q. Provide contacts for connection to elevator controllers, one closed when automatic transfer switch is connected to the normal source, and one closed when automatic transfer switch is connected to the emergency source.

r. Elevator Pre-Transfer Signal Relay: Provide a pre-signal relay on all automatic transfer switches that will indicate to an elevator controller or controllers that a transfer or re-transfer is about to

occur.

s. Motor Disconnect and Timing Relay: Controls designate starters so they disconnect motors before transfer and reconnect them selectively at an adjustable time interval after transfer. Control connection to motor starters is through wiring external to the automatic transfer switch. Time delay for reconnecting individual motor loads is adjustable between 1 and 60 seconds, and settings are as indicated. Relay contacts handling motor-control circuit in-rush and seal currents are rated for actual currents to be encountered.

2.2 SEQUENCE OF OPERATION

a. The specified voltage decrease in one or more phases of the normal power source shall initiate the transfer sequence. The automatic transfer switch shall start the engine-generator(s) after a specified time delay to permit override of momentary dips in the normal power source.

b. The automatic transfer switch shall transfer the load from normal to emergency source when the frequency and voltage of the engine-generator(s) have attained the specified percent of rated value.

c. Engine Start: A voltage decrease, at any automatic transfer switch, in one or more phases of the normal power source to less than the specified value of normal shall start the engine-generator(s) after a specified time delay.

d. Transfer to Emergency System Loads: Automatic transfer switches for Emergency System loads shall transfer their loads from normal to emergency source when frequency and voltage of the engine-generator(s) have attained the specified percent of rated value. Only those switches with deficient normal source voltage shall transfer.

e. Transfer to Equipment Branch Loads: Automatic transfer switches for Equipment Branch loads shall transfer their loads to the engine-generator on a time-delayed, staggered basis, after the Emergency System switches have transferred. Only those switches with deficient normal source voltage shall transfer.

f. Retransfer to Normal (All Loads): Automatic transfer switches shall retransfer the load from emergency to normal source upon restoration of normal supply in all phases to the specified percent or more of normal voltage, and after a specified time delay. Should the emergency source fail during this time, the automatic transfer switches shall immediately transfer to the normal source whenever it becomes available. After restoring to normal source, the engine-generator(s) shall continue to run unloaded for a specified interval before shut-down.

2.3 BYPASS-ISOLATION SWITCH

a. Provide each automatic transfer switch with two-way bypass-isolation manual type switch. The bypass-isolation switch shall permit load by-pass to either normal or emergency power source and complete isolation of the automatic transfer switch, independent of transfer switch position. Bypass and isolation shall be possible under all conditions including when the automatic transfer switch is removed from service.

b. Operation: The bypass-isolation switch shall have provisions for operation by one person through the movement of a maximum of two handles at a common dead front panel in no more than 15 seconds. Provide a lock, which must energize to unlock the bypass switch, to prevent bypassing to a dead source. Provide means to prevent simultaneous connection between normal and emergency sources.

(1) Bypass to normal (or emergency): Operation of bypass handle shall allow direct connection of the load to the normal (or emergency) source, without load interruption or by using a break-before-make design, or provide separate load interrupter contacts to momentarily interrupt the load.

(a) Ensure continuity of auxiliary circuits necessary for proper operation of the system.

(b) A red indicating lamp shall light when the automatic transfer switch is bypassed.

(c) Bypassing source to source: If the power source is lost while in the bypass position, bypass to the alternate source shall be achievable without re-energization of the automatic transfer switch service and load connections.

(2) Isolation: Operation of the isolating handle shall isolate all live power conductors to the automatic transfer switch without interruption of the load.

(a) Interlocking: Provide interlocking as part of the bypass-isolation switch to eliminate personnel-controlled sequence of operation, and to prevent operation to the isolation position until the bypass function has been completed.

(b) Padlocking: Include provisions to padlock the isolating handle in the isolated position.

(c) Visual verification: The isolation blades shall be visible in the isolated position.

(3) Testing: It shall be possible to test (normal electrical operation) the automatic transfer switch and engine–generator(s) with the isolation contacts closed and the load bypassed without interruption of power to the load.

c. Ratings: The electrical capabilities and ratings of the bypass-isolation switch shall be compatible with those of the associated automatic transfer switch, including any required additional withstand tests.

2.4 REMOTE ANNUNCIATOR SYSTEM

a. Remote annunciator panel shall annunciate conditions for indicated automatic transfer switches. Annunciation shall include the following:

(1) Sources available, as defined by actual pickup and dropout settings of automatic transfer switch controls.

(2) Switch position.

(3) Switch in test mode.

(4) Failure of communication link.

b. Remote annunciator panel shall be visual and audible type with LED display panel, audible signal, and silencing switch.

(1) Panel shall indicate each automatic transfer switch monitored, the location of automatic transfer switch, and the identity of load it serves.

(2) Mounting: Steel cabinet, flush or surface mounted, as shown on the drawings.

2.5 REMOTE ANNUNCIATOR AND CONTROL SYSTEM

a. Include the following functions for indicated automatic transfer switches:

(1) Indication of sources available, as defined by actual pickup and dropout settings of automatic transfer switch controls.

(2) Indication of automatic transfer switch position.

(3) Indication of automatic transfer switch in test mode.

(4) Indication of failure of communication link.

(5) Key-switch or user-code access to control functions of panel.

(6) Control of automatic transfer switch test initiation.

(7) Control of automatic transfer switch operation in either direction.

(8) Control of time-delay bypass for transfer to normal source.

b. Malfunction of remote annunciator and control system or communication link shall not affect functions of automatic transfer switches. Automatic transfer switch sensing, controlling, or operating functions shall not depend on remote annunciator and control system for proper operation.

c. Remote annunciation and control system shall include the following features:

(1) Touchscreen type operator interface.

(2) Control and indication means grouped together for each automatic transfer switch.

(3) Label each indication and control group. Indicate the automatic transfer switch it controls, the location of the automatic transfer switch, and the identity of the load that it serves.

(4) Digital Communication Capability: Matched to that of automatic transfer switches supervised.

(5) Mounting: Steel cabinet, flush or surface mounted, as shown on

the drawings.

PART 3 EXECUTION

3.1 INSTALLATION

a. Install automatic transfer switches and associated remote components in accordance with NFPA 70, as shown on the drawings, and manufacturer’s instructions.

b. Anchor automatic transfer switches with rustproof bolts, nuts, and washers not less than 1/2 inch diameter, in accordance with manufacturer’s instructions, and as shown on drawings.

c. Mount automatic transfer switches on concrete slab. Unless otherwise indicated, the slab shall be at least 4 inches thick. The top of the concrete slab shall be approximately 4 inches above finished floor. Edges above floor shall have 1/2 inch chamfer. The slab shall be of adequate size to project at least 8 inches beyond the equipment. Provide conduit turnups and cable entrance space required by the equipment to be mounted. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant. Cut off and bush conduits 3 inches above slab surface. Concrete work shall be as specified in Section 03 30 00, CAST-IN-PLACE CONCRETE.

3.2 ACCEPTANCE CHECKS AND TESTS

a. An authorized representative of the automatic transfer switch manufacturer shall technically supervise and participate during all of the field adjustments and tests. Major adjustments and field tests shall be witnessed by the Contracting Officer. The manufacturer’s representative shall certify in writing that the equipment has been installed, adjusted and tested in accordance with the manufacturer’s recommendations.

b. Perform manufacturer’s required field tests in accordance with the manufacturer's recommendations. In addition, include the following:

(1) Visual Inspection and Tests:

(a) Compare equipment nameplate data with specifications and approved shop drawings.

(b) Inspect physical, electrical, and mechanical condition.

(c) Confirm correct application of manufacturer's recommended lubricants.

(d) Verify appropriate anchorage, required area clearances, and correct alignment.

(e) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey after energization.

(f) Verify grounding connections.

(g) Verify ratings of sensors.

(h) Vacuum-clean enclosure interior. Clean enclosure exterior.

(i) Exercise all active components.

(j) Verify that manual transfer warning signs are properly placed.

(k) Verify the correct operation of all sensing devices, alarms, and indicating devices.

(2) Electrical tests:

(a) Perform insulation-resistance tests.

(b) After energizing circuits, demonstrate the interlocking sequence and operational function for each automatic transfer switch at least three times.

1. Test bypass-isolation unit functional modes and related automatic transfer switch operations.

2. Power failure of normal source shall be simulated by opening upstream protective device. This test shall be performed a minimum of five times.

3. Power failure of emergency source with normal source available shall be simulated by opening upstream protective device for emergency source. This test shall be performed a minimum of five times.

4. Low phase-to-ground voltage shall be simulated for each phase of normal source.

5. Operation and settings shall be verified for specified automatic transfer switch operational feature, such as override time delay, transfer time delay, return time delay, engine shutdown time delay, exerciser, auxiliary contacts, and supplemental features.

6. Verify pickup and dropout voltages by data readout or inspection of control settings.

7. Verify that bypass and isolation functions perform correctly, including the physical removal of the automatic transfer switch while in bypass mode.

(c) Ground-fault tests: Verify that operation of automatic transfer switches shall not cause nuisance tripping or alarms of ground fault protection on either source.

(d) When any defects are detected, correct the defects and repeat the tests as requested by the Contracting Officer at no additional cost to the Government.

3.3 FIELD SETTINGS VERIFICATION

The automatic transfer switch settings shall be verified in the field by an authorized representative of the manufacturer.

3.4 FOLLOW-UP VERIFICATION

Upon completion of acceptance checks and tests, the Contractor shall show by demonstration in service that the automatic transfer switches are in good operating condition and properly performing the intended function.

3.5 INSTRUCTION

Furnish the services of a factory-trained technician for one 4-hour training period for instructing personnel in the maintenance and operation of the automatic transfer switches, on the dates requested by the Contracting Officer.

-- End of Section --

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SECTION 26 41 00

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SECTION 26 41 00
11/22

FACILITY LIGHTNING PROTECTION

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing and installation of a complete UL master labeled lightning protection system.

1.2 RELATED WORK

a. Section 07 60 00, FLASHING AND SHEET METAL: Penetrations through the roof.

b. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

c. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path to ground for possible ground faults.

d. Section 26 43 13, SURGE PROTECTIVE DEVICES: Surge protective device installed at the electrical service entrance.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Ul Representative; G,

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Lightning Protection Components; G,

SD-07 Certificates

Certifications; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Show locations of air terminals, connections to required metal surfaces, down conductors, and grounding means.

(3) Show the mounting hardware and materials used to attach air terminals and conductors to the structure.

b. Certifications: Two weeks prior to final inspection, submit the following.

(1) Certification by the manufacturer that the lightning protection system conforms to the requirements of the drawings and specifications.

(2) Certification by the Contractor that the lightning protection system has been properly installed and inspected.

(3) Certification that the lightning protection system has been inspected by a UL representative and has been approved by UL without variation.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 780	(2023) Standard for the Installation of Lightning Protection Systems

UL SOLUTIONS (UL)

UL 96	(2016; May 2023) UL Standard for Safety Lightning Protection Components
UL 96A	(2023) UL Standard for Safety Installation Requirements for Lightning Protection Systems

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

a. Lightning protection components shall conform to NFPA 780 and UL 96, for use on Class IIstructures. Aluminum materials are not allowed.

(1) Class IIconductors: Copper.

(2) Class I air terminals: Solid copper, 18 inches long, not less than 3/8 inch diameter, with sharp nickel-plated points.

(3) Class II air terminals: Solid copper, 18 inches long, not less than 1/2 inch diameter, with sharp nickel-plated points.

(4) Ground rods: Copper-clad steel , 0.75 in diameter by 10 feet long.

(5) Ground plates: Solid copper, not less than 20 gauge.

(6) Bonding plates: Bronze, 8 square inches.

(7) Through roof connectors: Solid copper riser bar, length and type as required to accommodate roof structure and flashing requirements.

(8) Down conductor guards: Stiff copper or brass.

(9) Anchors and fasteners: Bronze bolt and clamp type shall be used for all applications except for membrane roof. Adhesive type are allowed only for attachment to membrane roof materials, using adhesive that is compatible with the membrane material.

(10) Connectors: Bronze clamp-type connectors shall be used for roof conductor splices, and the connection of the roof conductor to air terminals and bonding plates. Crimp-type connectors are not allowed.

(11) Exothermic welds: Exothermic welds shall be used for splicing the roof conductor to the down conductors, splices of the down conductors, and for connection of the down conductors to ground rods, ground plates, and the ground ring.

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation shall be in accordance with NFPA 70, as shown on the drawings, and manufacturer’s instructions.

b. Coordinate installation with the roofing manufacturer and roofing installer.

c. Install the conductors as inconspicuously as practical.

d. Install the down conductors within the concealed cavity of exterior walls where practical. Run the down conductors to the exterior at elevations below the finished grade.

e. Where down conductors are subject to damage or are accessible near grade, protect with down conductor guards to 8 feet above grade. Bond down conductors guards to down conductor at both ends.

f. Make connections of dissimilar metal with bimetallic type fittings to prevent electrolytic action.

g. Install ground rods and ground plates not less than 2 feet deep and a distance not less than 3 feet nor more than 8 feet from the nearest point of the structure. Exothermically weld the down conductors to ground rods and ground plates in the presence of the Contracting Officer.

h. Bond down conductors to metal main water piping where applicable.

i. Bond down conductors to building structural steel.

j. Connect roof conductors to all metallic projections and equipment above the roof as indicated on the drawings.

k. Connect exterior metal surfaces, located within 3 feet of the conductors, to the conductors to prevent flashovers.

l. Maintain horizontal or downward coursing of main conductor and insure that all bends have at least an 8 inches radius and do not exceed 90 degrees.

m. Conductors shall be rigidly fastened every 3 feet along the roof and down to the building to ground.

n. Air terminals shall be secured against overturning either by attachment to the object to be protected or by means of a substantial tripod or other braces permanently and rigidly attached to the building or structure.

o. Install air terminal bases, cable holders and other roof-system supporting means without piercing membrane or metal roofs.

p. Use through-roof connectors for penetration of the roof system. Flashing shall be provided by roofing contractor in accordance with Section 07 60 00, FLASHING AND SHEET METAL.

q. Down conductors coursed on or in reinforced concrete columns or on structural steel columns shall be connected to the reinforcing steel or the structural steel member at its upper and lower extremities. In the case of long vertical members an additional connection shall be made at intervals not exceeding 100 feet.

r. A counterpoise or ground ring, where shown, shall be of No. 1/0 copper cable having suitable resistance to corrosion and shall be laid around the perimeter of the structure in a trench not less than 2 feet deep at a distance not less than 3 feet nor more than 8 feet from the nearest point of the structure.

s. On construction utilizing post tensioning systems to secure precast concrete sections, the post tension rods shall not be used as a path for lightning to ground.

t. Where shown, use the structural steel framework or reinforcing steel as the down conductor.

(1) Weld or bond the non-electrically-continuous sections together and make them electrically continuous.

(2) Verify the electrical continuity by measuring the ground

resistances to earth at the ground level, at the top of the building or stack, and at intermediate points with a sensitive ohmmeter. Compare the resistance readings.

(3) Connect the air terminals together with an exterior conductor connected to the structural steel framework at not more than 60 feet intervals.

(4) Install ground connections to earth at not more than 60 feet intervals around the perimeter of the building.

(5) Weld or braze bonding plates to cleaned sections of the steel and connect the conductors to the plates.

(6) Do not pierce the structural steel in any manner. Connections to the structural steel shall conform to UL 96A.

u. For obstruction lights, the following additional requirements shall apply:

(1) Extend air terminals 1 foot above the top of the light fixtures and securely clamp to the light fixture supports.

(2) Install 600 volt class lightning arresters. Connect the arresters to the lightning circuit conductors at suitable locations, and ground and bond them to the lightning protection system.

v. Where the drawings show the new lightning protection system connected to an existing lightning protection system with or without a UL master label, the new portion of the lightning protection system requires UL inspection and a Letter of Findings.

3.2 ACCEPTANCE CHECKS AND TESTS

a. Test the ground resistance to earth by standard methods, and conform to the ground resistance requirements specified in Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS.

b. A UL representative shall inspect the lightning protection system. Obtain and install a UL numbered master label for each of the lightning protection systems at the location directed by the UL representative and the Contracting Officer.

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SECTION 26 43 13

SURGE PROTECTIVE DEVICES

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SECTION 26 43 13
11/22

SURGE PROTECTIVE DEVICES

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies the furnishing, installation, and connection of Type 2 Surge Protective Devices, as defined in NFPA 70, and indicated as SPD in this section.

1.2 RELATED WORK

a. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

b. Section 26 23 00, LOW-VOLTAGE SWITCHGEAR: For factory-installed or external SPD.

c. Section 26 24 13, DISTRIBUTION SWITCHBOARDS: For factory-installed or external SPD.

d. Section 26 24 16, PANELBOARDS: For factory-installed or external SPD.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G,

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Switchgear/Switchboard Spd; G,

Panelboard Spd; G,

Enclosures; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit sufficient information to demonstrate compliance with drawings and specifications.

(2) Include electrical ratings and device nameplate data.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, companion copies of complete maintenance and operating manuals including technical data sheets, wiring diagrams, and information for ordering replacement parts.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

c. Certifications: Two weeks prior to final inspection, submit the following.

(1) Certification by the manufacturer that the SPD conforms to the requirements of the drawings and specifications.

(2) Certification by the Contractor that the SPD has been properly installed.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplement and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41.2	(2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

UL SOLUTIONS (UL)

UL 1449 (2021; Reprint Dec 2022) UL Standard for Safety Surge Protective Devices

PART 2 PRODUCTS

2.1 SWITCHGEAR/SWITCHBOARD SPD

a. General Requirements:

(1) Comply with IEEE and UL.

(2) Modular design with field-replaceable modules, or non-modular design.

(3) Fuses, rated at 200 kA interrupting capacity.

(4) Bolted compression lugs for internal wiring.

(5) Integral disconnect switch.

(6) Redundant suppression circuits.

(7) LED indicator lights for power and protection status.

(8) Audible alarm, with silencing switch, to indicate when protection has failed.

(9) Form-C contacts rated at 5 A and 250-V ac, one normally open and one normally closed, for remote monitoring of protection status. Contacts shall reverse on failure of any surge diversion module or on opening of any current-limiting device.

(10) Four-digit transient-event counter.

b. Surge Current per Phase: Minimum 240kA per phase.

2.2 PANELBOARD SPD

a. General Requirements:

(1) Comply with UL 1449 and IEEE C62.41.2.

(2) Modular design with field-replaceable modules, or non-modular design.

(3) Fuses, rated at 200 kA interrupting capacity.

(4) Bolted compression lugs for internal wiring.

(5) Integral disconnect switch.

(6) Redundant suppression circuits.

(7) LED indicator lights for power and protection status.

(8) Audible alarm, with silencing switch, to indicate when protection has failed.

(9) Form-C contacts rated at 5 A and 250-V ac, one normally open and one normally closed, for remote monitoring of protection status. Contacts shall reverse on failure of any surge diversion module or on opening of any current-limiting device.

(10) Four-digit transient-event counter.

b. Surge Current per Phase: Minimum 120kA per phase.

2.3 ENCLOSURES

NEMA 1 .

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation shall be in accordance with NFPA 70, as shown on the drawings, and manufacturer’s instructions.

b. Factory-installed SPD: Switchgear, switchboard, or panelboard manufacturer shall install SPD at the factory.

c. Field-installed SPD: Contractor shall install SPD with conductors or buses between SPD and points of attachment as short and straight as possible. Do not exceed manufacturer's recommended lead length. Do not bond neutral and ground. Provide a circuit breaker as a dedicated disconnecting means for TVSS as shown on drawings.

d. Do not perform insulation resistance tests on switchgear, switchboards, panelboards, or feeders with the SPD connected. Disconnect SPD before conducting insulation resistance tests, and reconnect SPD immediately after insulation resistance tests are complete.

3.2 ACCEPTANCE CHECKS AND TESTS

Perform in accordance with the manufacturer's recommendations. In addition, include the following: Visual Inspection and Tests:

a. Compare equipment nameplate data with specifications and approved shop drawings.

b. Inspect physical, electrical, and mechanical condition.

c. Verify that disconnecting means and feeder size and maximum length to SPD corresponds to approved shop drawings.

d. Verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method.

e. Vacuum-clean enclosure interior. Clean enclosure exterior.

f. Verify the correct operation of all sensing devices, alarms, and indicating devices.

3.3 FOLLOW-UP VERIFICATION

After completion of acceptance checks and tests, the Contractor shall show by demonstration in service that SPD are in good operating condition and properly performing the intended function.

3.4 INSTRUCTION

Provide the services of a factory-trained technician for one 2-hour training period for instructing personnel in the maintenance and operation of the SPD, on the date requested by the Contracting Officer.

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SECTION 26 51 00

INTERIOR LIGHTING

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SECTION 26 51 00
3/24

INTERIOR LIGHTING

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, and connection of the interior lighting systems. The terms “lighting fixture,” “fixture,” and “luminaire” are used interchangeably.

1.2 RELATED WORK

a. Section 01 74 19, CONSTRUCTION WASTE MANAGEMENT: Disposal of lamps.

b. Section 02 41 00, DEMOLITION: Removal and disposal of lamps and ballasts.

e. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: Requirements that apply to all sections of Division 26.

f. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES: Low-voltage conductors.

g. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path to ground for possible ground fault currents.

h. Section 26 27 26, WIRING DEVICES: Wiring devices used for control of the lighting systems.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G,

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Lighting Fixtures; G,

Emergency Lighting Unit; G,

Led Exit Light Fixtures; G,

Led Light Fixtures; G,

SD-07 Certificates

Certifications; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit the following information for each type of lighting fixture designated on the LIGHTING FIXTURE SCHEDULE, arranged in order of lighting fixture designation.

(2) Material and construction details, include information on housing and optics system.

(3) Physical dimensions and description.

(4) Wiring schematic and connection diagram.

(5) Installation details.

(6) Energy efficiency data.

(7) Photometric data based on laboratory tests complying with IES Lighting Measurements testing and calculation guides.

(8) Lamp data including lumen output (initial and mean), color rendition index (CRI), rated life (hours), and color temperature (degrees Kelvin).

(9) For LED lighting fixtures, submit US DOE LED Lighting Facts label, and IES L70 rated life.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, complete maintenance and operating manuals, including technical data sheets, wiring diagrams, and information for ordering replacement parts.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

c. Certifications: Two weeks prior to final inspection, submit the following. Certification by the Contractor that the interior lighting systems have been properly installed and tested.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

FCC Part 15	Radio Frequency Devices (47 CFR 15)

ILLUMINATING ENGINEERING SOCIETY (IES)

ANSI/IES LM-79	(2019) Approved Method: Electrical and Photometric Measurements of Solid State Lighting Products
ANSI/IES LM-80	(2020) Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays and Modules

PART 2 PRODUCTS

2.1 LIGHTING FIXTURES

a. Shall be in accordance with NFPA, UL, as shown on drawings, and as specified.

b. Sheet Metal:

(1) Shall be formed to prevent warping and sagging. Housing, trim and lens frame shall be true, straight (unless intentionally curved), and parallel to each other as designed.

(2) Wireways and fittings shall be free of burrs and sharp edges, and shall accommodate internal and branch circuit wiring without damage to the wiring.

(3) When installed, any exposed fixture housing surface, trim frame, door frame, and lens frame shall be free of light leaks.

(4) Hinged door frames shall operate smoothly without binding. Latches shall function easily by finger action without the use of tools.

c. Ballasts and lamps shall be serviceable while the fixture is in its normally installed position. Ballasts shall not be mounted to removable reflectors or wireway covers unless so specified.

d. Lamp Sockets:

(1) Fluorescent: Single slot entry type, requiring a one-quarter turn of the lamp after insertion. Lampholder contacts shall be the biting edge type.

(2) Compact Fluorescent: 4-pin.

(3) High Intensity Discharge (HID): Porcelain.

e. Recessed fixtures mounted in an insulated ceiling shall be listed for use in insulated ceilings.

f. Mechanical Safety: Lighting fixture closures (lens doors, trim frame, hinged housings, etc.) shall be retained in a secure manner by captive screws, chains, aircraft cable, captive hinges, or fasteners such that they cannot be accidentally dislodged during normal operation or routine maintenance.

g. Metal Finishes:

(1) The manufacturer shall apply standard finish (unless otherwise specified) over a corrosion-resistant primer, after cleaning to free the metal surfaces of rust, grease, dirt and other deposits. Edges of pre-finished sheet metal exposed during forming, stamping or shearing processes shall be finished in a similar corrosion resistant manner to match the adjacent surface(s). Fixture finish shall be free of stains or evidence of rusting, blistering, or flaking, and shall be applied after fabrication.

(2) Interior light reflecting finishes shall be white with not less than 85 percent reflectances, except where otherwise shown on the drawing.

(3) Exterior finishes shall be as shown on the drawings.

h. Lighting fixtures shall have a specific means for grounding metallic wireways and housings to an equipment grounding conductor.

2.2 EMERGENCY LIGHTING UNIT

Complete, self-contained unit with batteries, battery charger, one or more local or remote lamp heads with lamps, under-voltage relay, and test switch.

a. Enclosure: Shall be impact-resistant thermoplastic . Enclosure shall be suitable for the environmental conditions in which installed.

b. Lamp Heads: Horizontally and vertically adjustable, mounted on the face of the unit, except where otherwise indicated.

c. Lamps: Shall be sealed-beam MR-16 halogen, rated not less than 12 watts at the specified DC voltage.

d. Battery: Shall be maintenance-free nickel-cadmium. Minimum normal life shall be minimum of 10 years.

e. Battery Charger: Dry-type full-wave rectifier with charging rates to maintain the battery in fully-charged condition during normal operation, and to automatically recharge the battery within 12 hours following a 1-1/2 hour continuous discharge.

f. Integral Self-Test: Automatically initiates test of unit emergency operation at required intervals. Test failure is annunciated by an integral audible alarm and a flashing LED.

2.3 LED LIGHT FIXTURES

a. General:

(1) LED light fixtures shall be in accordance with IES, NFPA, UL, as shown on the drawings, and as specified.

(2) LED light fixtures shall be Reduction of Hazardous Substances (RoHS)-compliant.

(3) LED drivers shall include the following features unless otherwise indicated:

(a) Minimum efficiency: 85 percent at full load.

(b) Minimum Operating Ambient Temperature: -4 degrees F.

(c) Input Voltage: 120 - 277V (plus or minus 10 percent) at 60 Hz.

(d) Integral short circuit, open circuit, and overload protection.

(e) Power Factor: greater than or equal to 0.95.

(f) Total Harmonic Distortion: less than or equal to 20 percent.

(g) Comply with FCC Part 15.

(4) LED modules shall include the following features unless otherwise indicated:

(a) Comply with ANSI/IES LM-79 and ANSI/IES LM-80 requirements.

(b) Minimum CRI 80 and color temperature 3000˚ K unless otherwise specified in LIGHTING FIXTURE SCHEDULE.

(c) Minimum Rated Life: 50,000 hours per IES L70.

(d) Light output lumens as indicated in the LIGHTING FIXTURE SCHEDULE.

b. LED Downlights: Housing, LED driver, and LED module shall be products of the same manufacturer.

c. LED Troffers:

(1) LED drivers, modules, and reflector shall be accessible, serviceable, and replaceable from below the ceiling.

(2) Housing, LED driver, and LED module shall be products of the same manufacturer.

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation shall be in accordance with the NEC, manufacturer's instructions, and as shown on the drawings or specified.

b. Align, mount, and level the lighting fixtures uniformly.

c. Wall-mounted fixtures shall be attached to the studs in the walls, or to a 20 gauge metal backing plate that is attached to the studs in the walls. Lighting fixtures shall not be attached directly to gypsum board.

d. Lighting Fixture Supports:

(1) Shall provide support for all of the fixtures. Supports may be anchored to channels of the ceiling construction, to the structural slab or to structural members within a partition, or above a suspended ceiling.

(2) Shall maintain the fixture positions after cleaning and relamping.

(3) Shall support the lighting fixtures without causing the ceiling or partition to deflect.

(4) Hardware for recessed fluorescent fixtures:

(a) Where the suspended ceiling system is supported at the four corners of the fixture opening, hardware devices shall clamp the fixture to the ceiling system structural members, or plaster frame at not less than four points in such a manner as to resist spreading of the support members and safely lock the fixture into the ceiling system.

(b) Where the suspended ceiling system is not supported at the four corners of the fixture opening, hardware devices shall independently support the fixture from the building structure at four points.

(5) Hardware for surface mounting fluorescent fixtures to suspended ceilings:

(a) In addition to being secured to any required outlet box, fixtures shall be bolted to a grid ceiling system at four points spaced near the corners of each fixture. The bolts shall be not less than 1/4 inch secured to channel members attached to and spanning the tops of the ceiling structural grid members. Non-turning studs may be attached to the ceiling structural grid members or spanning channels by special clips designed for the purpose, provided they lock into place and require simple tools for removal.

(b) In addition to being secured to any required outlet box, fixtures shall be bolted to ceiling structural members at four points spaced near the corners of each fixture. Pre-positioned 1/4 inch studs or threaded plaster inserts secured to ceiling structural members shall be used to bolt the fixtures to the ceiling. In lieu of the above, 1/4 inch toggle bolts may be used on new or existing ceiling provided the plaster and lath can safely support the fixtures without sagging or cracking.

(6) Hardware for recessed lighting fixtures:

(a) All fixture mounting devices connecting fixtures to the ceiling system or building structure shall have a capacity for a

horizontal force of 100 percent of the fixture weight and a vertical force of 400 percent of the fixture weight.

(b) Mounting devices shall clamp the fixture to the ceiling system structure (main grid runners or fixture framing cross runners) at four points in such a manner as to resist spreading of these supporting members. Each support point device shall utilize a screw or approved hardware to "lock" the fixture housing to the ceiling system, restraining the fixture from movement in any direction relative to the ceiling. The screw (size No. 10 minimum) or approved hardware shall pass through the ceiling member (T-bar, channel or spline), or it may extend over the inside of the flange of the channel (or spline) that faces away from the fixture, in a manner that prevents any fixture movement.

(c) In addition to the above, the following is required for fixtures exceeding 20 pounds in weight.

1. Where fixtures mounted in ASTM Standard C635 "Intermediate Duty" and "Heavy Duty" ceilings and weigh between 20 pounds and 56 pounds, provide two 12 gauge safety hangers hung slack between diagonal corners of the fixture and the building structure.

2. Where fixtures weigh over 56 pounds, they shall be independently supported from the building structure by approved hangers. Two-way angular bracing of hangers shall be provided to prevent lateral motion.

(d) Where ceiling cross runners are installed for support of lighting fixtures, they must have a carrying capacity equal to that of the main ceiling runners and be rigidly secured to the main runners.

(7) Surface mounted lighting fixtures:

(a) Fixtures shall be bolted against the ceiling independent of the outlet box at four points spaced near the corners of each unit. The bolts (or stud-clips) shall be minimum 1/4 inch bolt, secured to main ceiling runners and/or secured to cross runners. Non-turning studs may be attached to the main ceiling runners and cross runners with special non-friction clip devices designed for the purpose, provided they bolt through the runner, or are also secured to the building structure by 12 gauge safety hangers. Studs or bolts securing fixtures weighing in excess of 56 pounds shall be supported directly from the building structure.

(b) Where ceiling cross runners are installed for support of lighting fixtures, they must have a carrying capacity equal to that of the main ceiling runners and be rigidly secured to the main runners.

(c) Fixtures less than 15 pounds in weight and occupying less than two square feet of ceiling area may, when designed for the purpose, be supported directly from the outlet box when all the following conditions are met.

1. Screws attaching the fixture to the outlet box pass through round holes (not key-hole slots) in the fixture body.

2. The outlet box is attached to a main ceiling runner (or cross runner) with approved hardware.

3. The outlet box is supported vertically from the building structure.

(d) Fixtures mounted in open construction shall be secured directly to the building structure with approved bolting and clamping devices.

(8) Single or double pendant-mounted lighting fixtures:

(a) Each stem shall be supported by an approved outlet box mounted swivel joint and canopy which holds the stem captive and provides spring load (or approved equivalent) dampening of fixture oscillations. Outlet box shall be supported vertically from the building structure.

(9) Outlet boxes for support of lighting fixtures (where permitted) shall be secured directly to the building structure with approved devices or supported vertically in a hung ceiling from the building structure with a nine gauge wire hanger, and be secured by an approved device to a main ceiling runner or cross runner to prevent any horizontal movement relative to the ceiling.

e. Furnish and install the new lamps as specified for all lighting fixtures installed under this project, and for all existing lighting fixtures reused under this project.

f. The electrical and ceiling trades shall coordinate to ascertain that approved lighting fixtures are furnished in the proper sizes and installed with the proper devices (hangers, clips, trim frames, flanges, etc.), to match the ceiling system being installed.

g. Bond lighting fixtures to the grounding system as specified in Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS.

h. At completion of project, replace all defective components of the lighting fixtures at no cost to the Government.

i. Dispose of lamps per requirements of Section 01 74 19, CONSTRUCTION WASTE MANAGEMENT, and Section 02 41 00, DEMOLITION.

3.2 ACCEPTANCE CHECKS AND TESTS

Perform the following:

a. Visual Inspection:

(1) Verify proper operation by operating the lighting controls.

(2) Visually inspect for damage to fixtures, lenses, reflectors, diffusers, and louvers. Clean fixtures, lenses, reflectors, diffusers, and louvers that have accumulated dust, dirt, or fingerprints during construction.

b. Electrical tests:

(1) Exercise dimming components of the lighting fixtures over full

range of dimming capability by operating the control devices(s) in the presence of the Contracting Officer. Observe for visually detectable flicker over full dimming range, and replace defective components at no cost to the Government.

(2) Burn-in all lamps that require specific aging period to operate properly, prior to occupancy by Government. Burn-in period to be 40 hours minimum, unless specifically recommended otherwise by the lamp manufacturer. Burn-in dimmed fluorescent and compact fluorescent lamps for at least 100 hours at full voltage, unless specifically recommended otherwise by the lamp manufacturer. Replace any lamps and ballasts which fail during burn-in.

3.3 FOLLOW-UP VERIFICATION

Upon completion of acceptance checks and tests, the Contractor shall show by demonstration in service that the lighting systems are in good operating condition and properly performing the intended function.

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SECTION 26 56 00

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SECTION 26 56 00
11/22

EXTERIOR LIGHTING

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the furnishing, installation, and connection of exterior fixtures, poles, and supports. The terms “lighting fixtures”, “fixture” and “luminaire” are used interchangeably.

1.2 RELATED WORK

a. Section 03 30 00, CAST-IN-PLACE CONCRETE.

b. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS: General electrical requirements and items that are common to more than one section of Division 26.

c. Section 26 05 19, LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES (600 VOLTS AND BELOW): Low voltage power and lighting wiring.

d. Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS: Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

e. Section 26 05 33, RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS: Conduits, fittings, and boxes for raceway systems.

f. Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION: Underground handholes and conduits.

g. Section 26 09 23, LIGHTING CONTROLS: Controls for exterior lighting.

1.3 QUALITY ASSURANCE

Quality Assurance shall be in accordance with Paragraph, QUALIFICATIONS (PRODUCTS AND SERVICES) in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G,

SD-02 Shop Drawings

Shop Drawings; G,

SD-03 Product Data

Poles; G,

Foundations For Poles; G,

Luminaires; G,

Led Drivers; G,

Existing Lighting Systems; G,

SD-10 Operation and Maintenance Data

Manuals; G,

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Paragraph, SUBMITTALS in Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, and the following requirements:

a. Shop Drawings:

(1) Submit the following information for each type of lighting fixture designated on the LIGHTING FIXTURE SCHEDULE, arranged in order of lighting fixture designation.

(2) Material and construction details, include information on housing and optics system.

(3) Physical dimensions and description.

(4) Wiring schematic and connection diagram.

(5) Installation details.

(6) Energy efficiency data.

(7) Photometric data based on laboratory tests complying with IES Lighting Measurements testing and calculation guides.

(8) Lamp data including lumen output (initial and mean), color rendition index (CRI), rated life (hours), and color temperature (degrees Kelvin).

(9) For LED lighting fixtures, submit US DOE LED Lighting Facts label, and IES L70 rated life.

(10) Submit site plan showing all exterior lighting fixtures with fixture tags consistent with Lighting Fixture Schedule as shown on drawings. Site plan shall show computer generated point–by-point illumination calculations. Include lamp lumen and light loss factors used in calculations.

b. Manuals:

(1) Submit, simultaneously with the shop drawings, complete maintenance and operating manuals, including technical data

sheets, wiring diagrams, and information for ordering replacement parts.

(2) If changes have been made to the maintenance and operating manuals originally submitted, submit updated maintenance and operating manuals two weeks prior to the final inspection.

c. Certifications: Two weeks prior to final inspection, submit the following. Certification by the Contractor that the exterior lighting systems have been properly installed and tested.

1.6 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C1089	(2013) Standard Specification for Spun Cast Prestressed Concrete Poles

U.S. FEDERAL AVIATION ADMINISTRATION (FAA)

FAA AC 70/7460-1	(2016; Rev L; Change 2) Obstruction Marking and Lighting
FAA AC 150/5345-53	(2012; Rev D) Airport Lighting Equipment Certification Program

ILLUMINATING ENGINEERING SOCIETY (IES)

ANSI/IES LM-79	(2019) Approved Method: Electrical and Photometric Measurements of Solid State Lighting Products
ANSI/IES LM-80	(2020) Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays and Modules

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41.2	(2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

47 CFR 18	Industrial, Scientific, and Medical Equipment

1.7 DELIVERY, STORAGE, AND HANDLING

Provide manufacturer’s standard provisions for protecting pole finishes

during transport, storage, and installation. Do not store poles on ground. Store poles so they are at least 12 inches above ground level and growing vegetation. Do not remove factory-applied pole wrappings until just before installing pole.

PART 2 PRODUCTS

2.1 GENERAL REQUIREMENTS

Luminaires, materials and equipment shall be in accordance with NEC, UL, ANSI, and as shown on the drawings and specified.

2.2 POLES

a. General:

(1) Poles shall be as shown on the drawings, and as specified. Finish shall be as specified on the drawings.

(2) The pole and arm assembly shall be designed for wind loading of 100 mph minimum, as required by wind loading conditions at project site, with an additional 30 percent gust factor and supporting luminaire(s) and accessories such as shields, banner arms, and banners that have the effective projected areas indicated. The effective projected area of the pole shall be applied at the height of the pole base, as shown on the drawings.

(3) Poles shall be anchor-bolt type designed for use with underground supply conductors. Poles shall have handhole having a minimum clear opening of 2.5 x 5 inches. Handhole covers shall be secured by stainless steel captive screws.

(4) Provide a steel-grounding stud opposite handhole openings, designed to prevent electrolysis when used with copper wire.

(5) Provide a base cover that matches the pole in material and color to conceal the mounting hardware pole-base welds and anchor bolts.

(6) Hardware and Accessories: All necessary hardware and specified accessories shall be the product of the pole manufacturer.

(7) Provide manufacturer's standard finish, as scheduled on the drawings.

b. Types:

(2) Steel: Provide round steel poles having minimum 11-gauge steel with minimum yield/strength of 48,000 psi andhot-dipped galvanized factory finish.

(3) Concrete: Provide round concrete poles conforming to ASTM C1089 with integral cast bases. Poles shall have hollow core suitable as a raceway.

2.3 FOUNDATIONS FOR POLES

a. Foundations shall be cast-in-place concrete, having 3000 psi minimum 28-day compressive strength.

b. Foundations shall support the effective projected area of the specified pole, arm(s), luminaire(s), and accessories, such as shields, banner arms, and banners, under wind conditions previously specified in this section.

c. Place concrete in spirally-wrapped treated paper forms for round foundations, and construct forms for square foundations.

d. Rub-finish and round all above-grade concrete edges to approximately 0.25-inch radius.

e. Anchor bolt assemblies and reinforcing of concrete foundations shall be as shown on the drawings. Anchor bolts shall be in a welded cage or properly positioned by the tiewire to stirrups.

f. Prior to concrete pour, install electrode per Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS.

2.4 LUMINAIRES

a. Luminaires shall be weatherproof, heavy duty, outdoor types designed for efficient light utilization, adequate dissipation of lamp and ballast heat, and safe cleaning and relamping.

b. Illumination distribution patterns, BUG ratings and cutoff types as defined by the IESNA shall be as shown on the drawings.

c. Incorporate ballasts in the luminaire housing, except where otherwise shown on the drawings.

d. Lenses shall be frame-mounted, heat-resistant, borosilicate glass, with prismatic refractors, unless otherwise shown on the drawings. Attach the frame to the luminaire housing by hinges or chain. Use heat and aging-resistant, resilient gaskets to seal and cushion lenses and refractors in luminaire doors.

e. Pre-wire internal components to terminal strips at the factory.

f. Bracket-mounted luminaires shall have leveling provisions and clamp-type adjustable slip-fitters with locking screws.

g. Materials shall be rustproof. Latches and fittings shall be non-ferrous metal.

h. Provide manufacturer's standard finish, as scheduled on the drawings. Where indicated on drawings, match finish process and color of pole or support materials.

i. Luminaires shall carry factory labels, showing complete, specific lamp and ballast information.

2.5 LAMPS

a. Install the proper lamps in every luminaire installed and every existing luminaire relocated or reinstalledas shown on the drawings.

b. Lamps shall be general-service, outdoor lighting types.

c. LED sources shall meet the following requirements:

(1) Operating temperature rating shall be between -40 degrees F and 120 degrees F.

(2) Correlated Color Temperature (CCT): 3000K .

(3) Color Rendering Index (CRI): 85.

(4) The manufacturer shall have performed reliability tests on the LEDs luminaires complying with Illuminating Engineering Society ANSI/IES LM-79 for photometric performance and ANSI/IES LM-80 for lumen maintenance and L70 life.

2.6 LED DRIVERS

LED drivers shall meet the following requirements:

a. Drivers shall have a minimum efficiency of 85 percent.

b. Starting Temperature: -40 degrees F.

c. Input Voltage: 120 to 480 (plus or minus 10 percent) volt.

d. Power Supplies: Class I or II output.

e. Surge Protection: The system must survive 250 repetitive strikes of “C Low” (C Low: 6kV/1.2 x 50 microsecond, 10kA/8 x 20 microsecond) waveforms at 1-minute intervals with less than 10 percent degradation in clamping voltage. “C Low” waveforms are as defined in IEEE C62.41.2, Scenario 1 Location Category C.

f. Power Factor (PF): greater than or equal to 0.90.

g. Total Harmonic Distortion (THD): less than or equal to 20 percent.

h. Comply with 47 CFR 18 Non-consumer RFI/EMI Standards.

i. Drivers shall be reduction of hazardous substances (ROHS)-compliant.

2.7 EXISTING LIGHTING SYSTEMS

a. For modifications or additions to existing lighting systems, the new components shall be compatible with the existing systems.

b. New poles and luminaires shall have approximately the same configurations, dimensions, lamping and reflector type as the existing poles and luminaires, except where otherwise shown on the drawings.

2.8 OBSTRUCTION LIGHTING

a. Refer to Section 26 09 23, LIGHTING CONTROLS for control devices.

b. For Buildings:

(1) LED type luminaires shall comply with FAA AC 70/7460-1K, and FAA AC 150/5345-53, and be Type L-810 duplex units with red steady burning light from and LED light source with minimum 50,000 hour lamp life and employing Night Vision Goggles (NVG) friendly technology.

(2) Mount the luminaires on galvanized rigid steel pipe masts attached to the roof of the buildings so the luminaires extend 12 inches above the level of the highest item on the building, including items attached to the roof.

(3) Locate luminaires in accordance with the applicable FAA Standards.

c. For Water Tanks and Cooling Towers: Luminaires shall be FAA AC 70/7460-1K, and FAA AC 150/5345-53, Type L-810 duplex units with incandescent or LED light source.

PART 3 EXECUTION

3.1 INSTALLATION

a. Install lighting in accordance with NFPA 70, as shown on the drawings, and in accordance with manufacturer’s recommendations.

b. Pole Foundations:

(1) Excavate only as necessary to provide sufficient working clearance for installation of forms and proper use of tamper to the full depth of the excavation. Prevent surface water from flowing into the excavation. Thoroughly compact backfill with compacting arranged to prevent pressure between conductor, jacket, or sheath, and the end of conduit.

(2) Set anchor bolts according to anchor-bolt templates furnished by the pole manufacturer.

(3) Install poles as necessary to provide a permanent vertical position with the bracket arm in proper position for luminaire location.

(4) After the poles have been installed, shimmed, and plumbed, grout the spaces between the pole bases and the concrete base with non-shrink concrete grout material. Provide a plastic or copper tube, of not less than 0.375-inch inside diameter through the grout, tight to the top of the concrete base to prevent moisture weeping from the interior of the pole.

c. Install lamps in each luminaire.

d. Adjust luminaires that require field adjustment or aiming.

3.2 GROUNDING

Ground noncurrent-carrying parts of equipment, including metal poles, luminaires, mounting arms, brackets, and metallic enclosures, as specified in Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS. Where copper grounding conductor is connected to a metal other than copper, provide specially-treated or lined connectors suitable and listed for this purpose.

3.3 ACCEPTANCE CHECKS AND TESTS

Verify operation after installing luminaires and energizing circuits.

3.4 WATER TANKS AND COOLING TOWERS

Mount the luminaires at the extreme top of tank and tower as shown on drawings.

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DIVISION 27 - COMMUNICATIONS

SECTION 27 05 11

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SECTION 27 05 11
9/19

REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS

PART 1 GENERAL

1.1 DESCRIPTION

a. This section includes common requirements to communications installations and applies to all sections of Division 27 and Division 28.

b. Provide completely functioning communications systems.

c. Comply with VAAR 852.236.91 and FAR clause 52.236-21 in circumstance of a need for additional detail or conflict between drawings, specifications, reference standards or code.

1.2 ABBREVIATIONS

a. Abbreviations and Acronyms

(1) Refer to http://www.cfm.va.gov/til/sdetail.asp for Division 00, ARCHITECTURAL ABBREVIATIONS.

(2) Additional Abbreviations and Acronyms:

A	Ampere
AC	Alternating Current
AE	Architect and Engineer
AFF	Above Finished Floor
AHJ	Authority Having Jurisdiction
ANSI	American National Standards Institute
AWG	American Wire Gauge (refer to STP and UTP)
AWS	Advanced Wireless Services
BCT	Bonding Conductor for Telecommunications (also Telecommunications Bonding Conductor (TBC))
BDA	Bi-Directional Amplifier
BICSI	Building Industry Consulting Service International
BIM	Building Information Modeling
BOM	Bill of Materials
BTU	British Thermal Units
BUCR	Back-up Computer Room
BTS	Base Transceiver Station
CAD	AutoCAD
CBOPC	Community Based Out Patient Clinic
CBC	Coupled Bonding Conductor
CBOC	Community Based Out Patient Clinic (refer to CBOPC, OPC, VAMC)
CCS	TIP's Cross Connection System (refer to VCCS and HCCS)
CFE	Contractor Furnished Equipment
CFM	US Department of Veterans Affairs Office of Construction and Facilities Management
CFR	Consolidated Federal Regulations
CIO	Communication Information Officer (Facility, VISN or Region)
cm	Centimeters
CO	Central Office

CO	Contracting Officer
CPU	Central Processing Unit
CSU	Customer Service Unit
CUP	Conditional Use Permit(s) – Federal/GSA for VA
dB	Decibel
dBm	Decibel Measured
dBmV	Decibel per milli-Volt
DC	Direct Current
DEA	United States Drug Enforcement Administration
DSU	Data Service Unit
EBC	Equipment Bonding Conductor
ECC	Engineering Control Center (refer to DCR, EMCR)
EDGE	Enhanced Data (Rates) for GSM Evolution
EDM	Electrical Design Manual
EMCR	Emergency Management Control Room (refer to DCR, ECC)
EMI	Electromagnetic Interference (refer to RFI)
EMS	Emergency Medical Service
EMT	Electrical Metallic Tubing or thin wall conduit
ENTR	Utilities Entrance Location (refer to DEMARC, POTS, LEC)
EPBX	Electronic Digital Private Branch Exchange
ESR	Vendor’s Engineering Service Report
FA	Fire Alarm
FAR	Federal Acquisition Regulations in Chapter 1 of Title 48 of Code of Federal Regulations
FMS	VA’s Headquarters or Medical Center Facility’s Management Service
FR	Frequency (refer to RF)
FTS	Federal Telephone Service
GFE	Government Furnished Equipment
GPS	Global Positioning System
GRC	Galvanized Rigid Metal Conduit
GSM	Global System (Station) for Mobile
HCCS	TIP’s Horizontal Cross Connection System (refer to CCS & VCCS)
HDPE	High Density Polyethylene Conduit
HDTV	Advanced Television Standards Committee High-Definition Digital Television
HEC	Head End Cabinets(refer to HEIC, PA)
HEIC	Head End Interface Cabinets(refer to HEC, PA)
HF	High Frequency (Radio Band; Re FR, RF, VHF & UHF)
HSPA	High Speed Packet Access
HZ	Hertz
IBT	Intersystem Bonding Termination (NEC 250.94)
IC	Intercom
ICRA	Infectious Control Risk Assessment
IDEN	Integrated Digital Enhanced Network
IDC	Insulation Displacement Contact
IDF	Intermediate Distribution Frame
ILSM	Interim Life Safety Measures
IMC	Rigid Intermediate Steel Conduit
IRM	Department of Veterans Affairs Office of Information Resources Management
ISDN	Integrated Services Digital Network
ISM	Industrial, Scientific, Medical
IWS	Intra-Building Wireless System
LAN	Local Area Network
LBS	Location Based Services, Leased Based Systems
LEC	Local Exchange Carrier (refer to DEMARC, PBX & POTS)
LED	Light Emitting Diode
LMR	Land Mobile Radio
LTE	Long Term Evolution, or 4G Standard for Wireless Data Communications Technology
M	Meter
MAS	Medical Administration Service

MATV	Master Antenna Television
MCR	Main Computer Room
MCOR	Main Computer Operators Room
MDF	Main Distribution Frame
MH	Manholes or Maintenance Holes
MHz	Megaherts (106 Hz)
mm	Millimeter
MOU	Memorandum of Understanding
MW	Microwave (RF Band, Equipment or Services)
NID	Network Interface Device (refer to DEMARC)
NEC	National Electric Code
NOR	Network Operations Room
NRTL	OSHA Nationally Recognized Testing Laboratory
NS	Nurse Stations
NTIA	U.S. Department of Commerce National Telecommunications and Information Administration
OEM	Original Equipment Manufacturer
OI&T	Office of Information and Technology
OPC	VA’s Outpatient Clinic (refer to CBOC, VAMC)
OSH	Department of Veterans Affairs Office of Occupational Safety and Health
OSHA	United States Department of Labor Occupational Safety and Health Administration
OTDR	Optical Time-Domain Reflectometer
PA	Public Address System (refer to HE, HEIC, RPEC)
PBX	Private Branch Exchange (refer to DEMARC, LEC, POTS)
PCR	Police Control Room (refer to SPCC, could be designated SCC)
PCS	Personal Communications Service (refer to UPCS)
PE	Professional Engineer
PM	Project Manager
PoE	Power over Ethernet
POTS	Plain Old Telephone Service (refer to DEMARC, LEC, PBX)
PSTN	Public Switched Telephone Network
PSRAS	Public Safety Radio Amplification Systems
PTS	Pay Telephone Station
PVC	Poly-Vinyl Chloride
PWR	Power (in Watts)
RAN	Radio Access Network
RBB	Rack Bonding Busbar
RE	Resident Engineer or Senior Resident Engineer
RF	Radio Frequency (refer to FR)
RFI	Radio Frequency Interference (refer to EMI)
RFID	RF Identification (Equipment, System or Personnel)
RMC	Rigid Metal Conduit
RMU	Rack Mounting Unit
RPEC	Radio Paging Equipment Cabinets(refer to HEC, HEIC, PA)
RTLS	Real Time Location Service or System
RUS	Rural Utilities Service
SCC	Security Control Console (refer to PCR, SPCC)
SMCS	Spectrum Management and Communications Security (COMSEC)
SFO	Solicitation for Offers
SME	Subject Matter Experts (refer to AHJ)
SMR	Specialized Mobile Radio
SMS	Security Management System
SNMP	Simple Network Management Protocol
SPCC	Security Police Control Center (refer to PCR, SMS)
STP	Shielded Balanced Twisted Pair (refer to UTP)
STR	Stacked Telecommunications Room
TAC	VA’s Technology Acquisition Center, Austin, Texas
TCO	Telecommunications Outlet
TER	Telephone Equipment Room
TGB	Telecommunications Grounding Busbar (also Secondary Bonding Busbar (SBB))
TIP	Telecommunications Infrastructure Plant

TMGB	Telecommunications Main Grounding Busbar (also Primary Bonding Busbar (PBB))
TMS	Traffic Management System
TOR	Telephone Operators Room
TP	Balanced Twisted Pair (refer to STP and UTP)
TR	Telecommunications Room (refer to STR)
TWP	Twisted Pair
UHF	Ultra High Frequency (Radio)
UMTS	Universal Mobile Telecommunications System
UPCS	Unlicensed Personal Communications Service (refer to PCS)
UPS	Uninterruptible Power Supply
USC	United States Code
UTP	Unshielded Balanced Twisted Pair (refer to TP and STP)
UV	Ultraviolet
V	Volts
VAAR	Veterans Affairs Acquisition Regulation
VACO	Veterans Affairs Central Office
VAMC	VA Medical Center (refer to CBOC, OPC, VACO)
VCCS	TIP’s Vertical Cross Connection System (refer to CCS and HCCS)
VHF	Very High Frequency (Radio)
VISN	Veterans Integrated Services Network (refers to geographical region)
VSWR	Voltage Standing Wave Radio
W	Watts
WEB	World Electronic Broadcast
WiMAX	Worldwide Interoperability (for MW Access)
WI-FI	Wireless Fidelity
WMTS	Wireless Medical Telemetry Service

WSP Wireless Service Providers

b. Definitions:

(1) Access Floor: Pathway system of removable floor panels supported on adjustable pedestals to allow cable placement in area below.

(2) BNC Connector (BNC): United States Military Standard FCC MIL-C-39012/21 bayonet-type coaxial connector with quick twist mating/unmating, and two lugs preventing accidental disconnection from pulling forces on cable.

(3) Bond: Permanent joining of metallic parts to form an electrically conductive path to ensure electrical continuity and capacity to safely conduct any currents likely to be imposed to earth ground.

(4) Bundled Microducts: All forms of jacketed microducts.

(5) Conduit: Includes all raceway types specified.

(6) Conveniently Accessible: Capable of being reached without use of ladders, or without climbing or crawling under or over obstacles such as, motors, pumps, belt guards, transformers, piping, ductwork, conduit and raceways.

(7) Distributed (in house) Antenna System (DAS): An Emergency Radio Communications System installed for Emergency Responder (or first responders and Government personnel) use while inside facility to maintain contact with each respective control point; refer to Section 27 53 19, DISTRIBUTED RADIO ANTENNA (WITHIN BUILDING) EQUIPMENT AND SYSTEMS.

(8) DEMARC, Extended DMARC or ENTR: Service provider's main point of demarcation owned by LEC or service provider and establishes a physical point where service provider's responsibilities for service and maintenance end. This point is called NID, in data networks.

(9) Effectively Grounded: Intentionally bonded to earth through connections of low impedance having current carrying capacity to prevent buildup of currents and voltages resulting in hazard to equipment or persons.

(10) Electrical Supervision: Analyzing a system’s function and components (i.e. cable breaks / shorts, inoperative stations, lights, LEDs and states of change, from primary to backup) on a 24/7/365 basis; provide aural and visual emergency notification signals to minimum two remote designated or accepted monitoring stations.

(11) Electrostatic Interference (ESI) or Electrostatic Discharge Interference: Refer to EMI and RFI.

(12) Emergency Call Systems: Wall units (in parking garages and stairwells) and pedestal mounts (in parking lots) typically provided with a strobe, camera and two-way audio communication functions.

(13) Project 25 (2014) (P25 (TIA-102 Series)): Set of standards for local, state and Federal public safety organizations and agencies digital LMR services. P25 is applicable to LMR equipment authorized or licensed under the US Department of Commerce National Telecommunications and Information Administration or FCC rules and regulations, and is a required standard capability for all LMR equipment and systems.

(14) Grounding Electrode Conductor: (GEC) Conductor connected to earth grounding electrode.

(15) Grounding Electrode System: Electrodes through which an effective connection to earth is established, including supplementary, communications system grounding electrodes and GEC.

(16) Grounding Equalizer or Backbone Bonding Conductor (BBC): Conductor that interconnects elements of telecommunications grounding infrastructure.

(17) Head End (HE): Equipment, hardware and software, or a master facility at originating point in a communications system designed for centralized communications control, signal processing, and distribution that acts as a common point of connection between equipment and devices connected to a network of interconnected equipment, possessing greatest authority for allowing information to be exchanged, with whom other equipment is subordinate.

(18) Microducts: All forms of air blown fiber pathways.

(19) Ohm: A unit of restive measurement.

(20) Received Signal Strength Indication (RSSI): A measurement of power present in a received RF signal.

(21) Service Provider Demarcation Point (SPDP): Not owned by LEC or service provider, but designated by Government as point within facility considered the DEMARC.

(22) Sound (SND): Changing air pressure to audible signals over given time span.

(23) System: Specific hardware, firmware, and software, functioning together as a unit, performing task for which it was designed.

(24) Telecommunications Bonding Backbone (TBB): Conductors of appropriate size (minimum 1/0 AWG) stranded copper wire, that connect to Grounding Electrode System and route to telecommunications main grounding busbar (TMGB) and circulate to interconnect various TGBs and other locations shown on drawings.

(25) Voice over Internet Protocol (VoIP): A telephone system in which voice signals are converted to packets and transmitted over LAN network using Transmission Control Protocol (TCP)/Internet Protocol (IP). VA'S VoIP is not listed or coded for life and public safety, critical, emergency or other protection functions. When VoIP system or equipment is provided instead of PBX system or equipment, each TR (STR) and DEMARC requires increased AC power provided to compensate for loss of PBX's telephone instrument line power; and, to compensate for absence of PBX's UPS capability.

(26) Wide Area Network (WAN): A digital network that transcends localized LANs within a given geographic location. VA'S WAN/LAN is not nationally listed or coded for life and public safety, critical, emergency or other safety functions.

1.3 REFERENCES

UNLESS DOCUMENTS INCLUDE MORE STRINGENT REQUIREMENTS, APPLICABLE CONSTRUCTION INDUSTRY STANDARDS HAVE SAME FORCE AND EFFECT AS IF BOUND OR COPIED DIRECTLY INTO THE DOCUMENTS TO EXTENT REFERENCED. SUCH STANDARDS ARE MADE A PART OF THESE DOCUMENTS BY REFERENCE.

1. Each entity engaged in construction must be familiar with industry standards applicable to its construction activity.

2. Obtain standards directly from publication source, where copies of standards are needed to perform a required construction activity.

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-C-39012/21	Connectors, Receptacle, Electrical, Coaxial, Radio Frequency, (Series BNC (Uncabled), Socket Contact, Jam Nut Mounted, Class 2)

INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)

ICEA S-98-688	(2012) Broadband Twisted Pair Telecommunication Cable, Aircore, Polyolefin Insulated, Copper Conductors Technical Requirements
ICEA S-99-689	(2012) Broadband Twisted Pair Telecommunication Cable Filled, Polyolefin Insulated, Copper Conductors Technical Requirements

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

U.S. DEPARTMENT OF AGRICULTURE (USDA)

RUS Bull 345-72	(1985) Filled Splice Closures (PE-74)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

7 CFR 1755	Telecommunications policies on specifications, acceptable materials, and standard contract forms

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-569	(2019e; Add 1 2022) Telecommunications Pathways and Spaces
TIA-1152	(2016; R 2021) Requirements for Field Test Instruments and Measurements for Balanced Twisted-Pair Cabling

UL SOLUTIONS (UL)

UL 486A-486B	(2025) UL Standard for Safety Wire Connectors

PROJECT 25

P25	(2014) (P25 (TIA-102 Series)): Set of standards for local, state and Federal public safety organizations and agencies digital LMR services

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

1.4 SINGULAR NUMBER

Where any device or part of equipment is referred in singular number (such as " rack"), reference applies to as many such devices as are required to complete installation.

1.5 RELATED WORK

Specification Order of Precedence: FAR Clause 52.236-21, VAAR Clause 852.236-71.

(1) Field Cutting and Patching: Section 09 91 00, PAINTING.

(2) Additional submittal requirements: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA AND SAMPLES.

(3) Availability and source of references and standards specified in applicable publications: Section 01 42 19, REFERENCE STANDARDS.

(4) Control of environmental pollution and damage for air, water, and land resources: Section 01 57 19, TEMPORARY ENVIRONMENTAL CONTROLS.

(5) Requirements for non-hazardous building construction and demolition waste: Section 01 74 19, CONSTRUCTION WASTE MANAGEMENT.

(6) General requirements and procedures to comply with various federal mandates and U.S. Department of Veterans Affairs (VA) policies for sustainable design: Section 01 81 13, SUSTAINABLE DESIGN REQUIREMENTS.

(7) Closures of openings in walls, floors, and roof decks against penetration of flame, heat, and smoke or gases in fire resistant rated construction: Section 07 84 00, FIRESTOPPING.

(8) Sealant and caulking materials and their application: Section 07 92 00, JOINT SEALANTS.

(9) General electrical requirements that are common to more than one section of Division 26: Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

(10) Electrical conductors and cables in electrical systems rated 600 V and below: Section 26 05 21, LOW VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES (600 VOLTS AND BELOW).

(11) Requirements for personnel safety and to provide a low impedance path to ground for possible ground fault currents: Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS.

(12) Conduit and boxes: Section 26 05 33, RACEWAYS AND BOXES FOR ELECTRICAL SYSTEMS.

(13) Wiring devices: Section 26 27 26, WIRING DEVICES.

(14) Underground ducts, raceways, precast manholes and pull boxes: Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION.

(15) Lightning protection: Section 26 41 00, FACILITY LIGHTNING PROTECTION.

(16) General requirements common to more than one section in Division 28: Section 28 05 00, COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY.

(17) Conductors and cables for electronic safety and security systems: Section 28 05 13, CONDUCTORS AND CABLES FOR ELECTRONIC SAFETY AND SECURITY.

(18) Low impedance path to ground for electronic safety and security system ground fault currents: Section 28 05 26, GROUNDING AND BONDING FOR SECURITY SYSTEMS.

(19) Conduits and partitioned telecommunications raceways for Electronic Safety and Security systems: Section 28 05 28.33, CONDUITS AND BACK BOXES FOR ELECTRONIC SAFETY AND SECURITY.

(20) Physical Access Control System field-installed controllers connected by data transmission network: Section 28 13 00, PHYSICAL ACCESS DETECTION.

(21) Detection and screening systems: Section 28 13 53, SECURITY ACCESS DETECTION.

(22) Intrusion sensors and detection devices, and communication links to perform monitoring, alarm, and control functions: Section 28 16 11, INTRUSION DETECTION EQUIPMENT AND SYSTEMS.

(23) Video surveillance system cameras, data transmission wiring, and control stations with associated equipment: Section 28 23 00, VIDEO SURVEILLANCE EQUIPMENT AND SYSTEMS.

(24) Duress-panic alarms, emergency phones or call boxes, intercom systems, data transmission wiring and associated equipment: Section 28 26 00, ELECTRONIC PERSONAL PROTECTION EQUIPMENT AND SYSTEMS.

(25) Alarm initiating devices, alarm notification appliances, control units, fire safety control devices, annunciators, power supplies, and wiring: Section 28 31 00, FIRE DETECTION AND ALARM.

(26) Emergency Call telephones, intercom systems, with blue strobe light and equipment: Section 28 52 31, SECURITY EMERGENCY CALL/DURESS ALARM/COMMUNICATIONS SYSTEM AND EQUIPMENT.

1.6 ADMINISTRATIVE REQUIREMENTS

a. Assign a single communications project manager to serve as point of contact for Government, contractor, and design professional.

b. Be proactive in scheduling work.

(1) Use of premises is restricted at times directed by Contracting Officer.

(2) Movement of materials: Unload materials and equipment delivered to site.

(3) Coordinate installation of required supporting devices and sleeves to be set in poured-in-place concrete and other structural components, as they are constructed.

(4) Sequence, coordinate, and integrate installations of materials and equipment for efficient flow of Work.

(5) Coordinate connection of materials, equipment, and systems with exterior underground and overhead utilities and services. Comply with requirements of governing regulations, franchised service companies, and controlling agencies; provide required connection for each service.

(6) Initiate and maintain discussion regarding schedule for ceiling construction and install cables to meet that schedule.

c. Contact the Office of Telecommunications, Special Communications Team (005OP2H3) (202) 461-5310 to have a Government-accepted Telecommunications Contracting Officer assigned to project for telecommunications review, equipment and system approval and coordination with other VA personnel.

d. Communications Project Manager Responsibilities:

(1) Assume responsibility for overall telecommunications system integration and coordination of work among trades, subcontractors, and authorized system installers.

(2) Coordinate with related work indicated on drawings or specified.

(3) Manage work related to telecommunications system installation in a manner approved by manufacturer.

1.7 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval.Submittals with an "S" are for inclusion in the Sustainability Notebook, in conformance with Section 01 33 29 SUSTAINABILITY REPORTING. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Installer Qualifications; G

Source Quality Control Submittal; G

Test Equipment List; G

Manufacturer’s Qualifications; G

Product And System Qualification; G

Trade Contractor Qualifications; G

System Supplier Qualifications; G

Oem Certifications For Installers; G

SD-02 Shop Drawings

Submittal Drawings; G

SD-03 Product Data

Manufacturer Product Information; G

SD-04 Samples

Material Samples; G

SD-05 Design Data

Delegated Design Submittals; G

Equipment Anchors And Supports; G

Sustainable Design Submittals; S

SD-06 Test Reports

Electronic Certified Test Reports; G

SD-07 Certificates

Warranty Certificate; G

Low Voltage Certificate Of Inspection; G

SD-10 Operation and Maintenance Data

Parts List; G

Maintenance And Operation Manuals; G

SD-11 Closeout Submittals

Project Record Documents

Instruction Manuals And Software; G

Record Wiring Diagrams; G

Service Qualifications; G

Training Schedule; G

1.8 SUBMITTAL INSTRUCTIONS

Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

a. Installer Qualifications: Submit three installations of similar size and complexity furnished and installed by installer; include:

(1) Installation location and name.

(2) Owner’s name and contact information including, address, telephone and email.

(3) Date of project start and date of final acceptance.

(4) System project number.

(5) Three paragraph description of each system related to this project; include function, operation, and installation.

b. Provide Source Quality Control Submittal:

(1) Submit written certification from OEM indicating that proposed supervisor of installation and proposed provider of warranty maintenance are authorized representatives of OEM. Include individual's legal name, contact information and OEM credentials in certification.

(2) Submit written certification from OEM that wiring and connection diagrams meet Government Life Safety Guidelines, NFPA, NEC, NRTL, these specifications, and Joint Commission requirements and instructions, requirements, recommendations, and guidance set forth by OEM for the proper performance of system.

(3) Pre-acceptance Certification: Certification in accordance with procedure outlined in Section 01 00 00, GENERAL REQUIREMENTS and specific Division 27 qualification documentation.

c. Provide delegated design submittals (e.g. seismic support design).

d. Submittal Drawings:

(1) Telecommunications Space Plans/Elevations: Provide enlarged floor plans of telecommunication spaces indicating layout of equipment and devices, including receptacles and grounding provisions. Submit detailed plan views and elevations of telecommunication spaces showing racks, termination blocks, and cable paths. Include following rooms:

(a) Telecommunications rooms.

(b) Building Entrance Facility/Demarcation rooms.

(c) Server rooms/Data Center.

(d) Equipment rooms.

(e) Antenna Head End rooms.

(2) Logical Drawings: Provide logical riser or schematic drawings for all systems.

(a) Provide riser diagrams systems and interconnection drawings for equipment assemblies; show termination points and identify wiring connections.

(3) Access Panel Schedule on Submittal Drawings: Coordinate and prepare a location, size, and function schedule of access panels required to fully service equipment.

e. Provide parts list including quantity of spare parts.

f. Provide manufacturer product information. Government reserves the right to require a list of installations where products have been in operation.

g. Submittals are required for all equipment anchors and supports. Include weights, dimensions, center of gravity, standard connections, manufacturer's recommendations and behavior problems (e.g., vibration, thermal expansion,) associated with equipment or conduit. Anchors and supports to resist seismic load based on seismic design categories per section 4.0 of VA seismic design requirements H-18-8 dated August, 2013.

h. Test Equipment List:

(1) Supply test equipment of accuracy better than parameters to be tested.

(2) Submit test equipment list including make and model number:

(a) ATSC TIA-1152 Level IIIe twisted pair cabling test instrument.

(b) Fiber optic insertion loss power meter with light source.

(c) Optical time domain reflectometer (OTDR).

(d) Volt-Ohm meter.

(e) Digital camera.

(3) Supply only test equipment with a calibration tag from Government-accepted calibration service dated not more than 12 months prior to test.

(4) Provide sample test and evaluation reports.

i. Furnish electronic certified test reports to Contracting Officer prior to final inspection and not more than 90 days after completion of tests.

j. Provide sustainable design submittals.

1.9 CLOSEOUT SUBMITTAL INSTRUCTIONS

a. Provide following closeout submittals prior to project closeout date:

(1) Warranty certificate.

(2) Evidence of compliance with requirements such as low voltage certificate of inspection.

(3) Project record documents.

(4) Instruction manuals and software that are a part of system.

b. Maintenance and Operation Manuals: Submit in accordance with Section 01 00 00, GENERAL REQUIREMENTS.

(1) Prepare a manual for each system and equipment specified.

(2) Furnish on portable storage drive in PDF format or equivalent accepted by Contracting Officer.

(3) Furnish complete manual as specified in specification section, fifteen days prior to performance of systems or equipment test.

(4) Furnish remaining manuals prior to final completion.

(5) Identify storage drive "MAINTENANCE AND OPERATION MANUAL" and system name.

(6) Include name, contact information and emergency service numbers of each subcontractor installing system or equipment and local representatives for system or equipment.

(7) Provide a Table of Contents and assemble files to conform to Table of Contents.

(8) Operation and Maintenance Data includes:

(a) Approved shop drawing for each item of equipment.

(b) Internal and interconnecting wiring and control diagrams with data to explain detailed operation and control of equipment.

(c) A control sequence describing start-up, operation, and shutdown.

(d) Description of function of each principal item of equipment.

(e) Installation and maintenance instructions.

(f) Safety precautions.

(g) Diagrams and illustrations.

(h) Test Results and testing methods.

(i) Performance data.

(j) Pictorial "exploded" parts list with part numbers. Emphasis to be placed on use of special tools and instruments. Indicate sources of supply, recommended spare parts, and name of servicing organization.

(k) Warranty documentation indicating end date and equipment protected under warranty.

(l) Appendix; list qualified permanent servicing organizations for support of equipment, including addresses and certified personnel qualifications.

c. Record Wiring Diagrams:

(1) Red Line Drawings: Keep one E size 36 inches x 48 inches set of floor plans, on site during work hours, showing installation progress marked and backbone cable labels noted. Make these drawings available for examination during construction meetings or field inspections.

(2) General Drawing Specifications: Detail and elevation drawings to be D size 24 inches x 36 inches with a minimum scale of 1/4 inch

= 12 inches). ER, TR and other enlarged detail floor plan drawings to be D size (24″ x 36″) with a minimum scale of 1/4 inch = 12 inches). Building composite floor plan drawings to be D size 24 inches x 36 inches with a minimum scale of 1/8 inch = 1’ 0 inch).

(3) Building Composite Floor Plans: Provide building floor plans showing work area outlet locations and configuration, types of jacks, distance for each cable, and cable routing locations.

(4) Floor plans to include:

(a) Final room numbers and actual backbone cabling and pathway locations and labeling.

(b) Inputs and outputs of equipment identified according to labels installed on cables and equipment

(c) Device locations with labels.

(d) Conduit.

(e) Head-end equipment.

(f) Wiring diagram.

(g) Labeling and administration documentation.

(5) Submit Record Wiring Diagrams within five business days after final cable testing.

(6) Deliver Record Wiring Diagrams as CAD files in.dwg formats as determined by Contracting Officer.

(7) Deliver four complete sets of electronic record wiring diagrams to Contracting Officer on portable storage drive.

d. Service Qualifications: Submit name and contact information of service organizations providing service to this installation within four hours of receipt of notification service is needed.

1.10 MAINTENANCE MATERIAL SUBMITTAL INSTRUCTIONS

After approval and prior to installation, furnish Contracting Officer with the following Material Samples:

a. A 12 inch length of each type and size of wire and cable along with tag from coils of reels from which samples were taken.

b. One coupling, bushing and termination fitting for each type of conduit.

c. Samples of each hanger, clamp and supports for conduit and pathways.

d. Duct sealing compound.

1.11 QUALITY ASSURANCE

a. Manufacturer’s Qualifications: Manufacturer must produce, as a principal product, the equipment and material specified for this

project, and have manufactured item for at least three years.

b. Product and System Qualification:

(1) OEM must have three installations of equipment submitted presently in operation of similar size and type as this project, that have continuously operated for a minimum of three years.

(2) Government reserves the right to require a list of installations where products have been in operation before approval.

(3) Authorized representative of OEM must be responsible for design, satisfactory operation of installed system, and certification.

c. Trade Contractor Qualifications: Trade contractor must have completed three or more installations of similar systems of comparable size and complexity with regards to coordinating, engineering, testing, certifying, supervising, training, and documentation. Identify these installations as a part of submittal.

d. System Supplier Qualifications: System supplier must be authorized by OEM to warranty installed equipment.

e. Telecommunications technicians assigned to system must be trained, and certified by OEM on installation and testing of system; provide written evidence of current OEM certifications for installers.

f. Manufactured Products:

(1) Comply with FAR clause 52.236-5 for material and workmanship.

(2) When more than one unit of same class of equipment is required, units must be product of a single manufacturer.

(3) Equipment Assemblies and Components:

(a) Components of an assembled unit need not be products of same manufacturer.

(b) Manufacturers of equipment assemblies, which include components made by others, to assume complete responsibility for final assembled unit.

(c) Provide compatible components for assembly and intended service.

(d) Constituent parts which are similar must be product of a single manufacturer.

(4) Identify factory wiring on equipment being furnished and on wiring diagrams.

g. Testing Agencies: Government reserves the option of witnessing factory tests. Notify Contracting Officer minimum 15 working days prior to manufacturer performing the factory tests.

(1) When equipment fails to meet factory test and re-inspection is required, contractor is liable for additional expenses, including expenses of Government.

1.12 DELIVERY, STORAGE, AND HANDLING

a. Delivery and Acceptance Requirements:

(1) Government's approval of submittals must be obtained for equipment and material before delivery to job site.

(2) Deliver and store materials to job site in OEM's original unopened containers, clearly labeled with OEM's name and equipment catalog numbers, model and serial identification numbers for Contracting Officer to inventory cable, patch panels, and related equipment.

b. Storage and Handling Requirements:

(1) Equipment and materials must be protected during shipment and storage against physical damage, dirt, moisture, cold and rain:

(a) Store and protect equipment in a manner that precludes damage or loss, including theft.

(b) Protect painted surfaces with factory installed removable heavy kraft paper, sheet vinyl or equivalent.

(c) Protect enclosures, equipment, controls, controllers, circuit protective devices, and other like items, against entry of foreign matter during installation; vacuum clean both inside and outside before testing and operating.

c. Coordinate storage.

1.13 FIELD CONDITIONS

a. Where variations from documents are requested in accordance with GENERAL CONDITIONS and Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES, connecting work and related components must include additions or changes to branch circuits, circuit protective devices, conduits, wire, feeders, controls, panels and installation methods.

b. A contract adjustment or additional time will not be granted because of field conditions pursuant to FAR 52.236-2 and FAR 52.236-3; a contract adjustment or additional time will not be granted for additional work required for complete and usable construction and systems pursuant to FAR 52.246-12.

1.14 WARRANTY

Comply with FAR clause 52.246-21.

PART 2 PRODUCTS

2.1 PERFORMANCE AND DESIGN CRITERIA

a. Provide communications spaces and pathways conforming to TIA-569, at a minimum.

2.2 EQUIPMENT IDENTIFICATION

a. Provide laminated black phenolic resin with a white core nameplates

with minimum 1/4 inch high engraved lettering.

b. Nameplates furnished by manufacturer as standard catalog items, unless other method of identification is indicated.

2.3 UNDERGROUND WARNING TAPE

Standard 4-Mil polyethylene 3 inch wide tape detectable type; red with black letters imprinted with “CAUTION BURIED ELECTRIC LINE BELOW”, orange with black letters imprinted with “CAUTION BURIED TELEPHONE LINE BELOW” or orange with black letters imprinted with “CAUTION BURIED FIBER OPTIC LINE BELOW”, as applicable.

2.4 WIRE LUBRICATING COMPOUND

Provide non-hardening or forming adhesive coating cable lubricants suitable for cable jacket material and raceway.

2.5 FIREPROOFING TAPE

a. Provide flexible, conformable fabric tape of organic composition and coated one side with flame-retardant elastomer.

b. Tape must be self-extinguishing and cannot support combustion; arc-proof and fireproof.

c. Tape cannot deteriorate when subjected to water, gases, salt water, sewage, or fungus; and tape must be resistant to sunlight and ultraviolet light.

d. Application must withstand a 200-ampere arc for minimum 30 seconds.

e. Securing Tape: Glass cloth electrical tape minimum (7 mils) thick and 3/4 inch wide.

2.6 UNDERGROUND CABLES

a. Provide buried closure suitable for enclosing a straight, butt, and branch splice in a container into which can be poured an encapsulating compound.

b. Provide closure of adequate strength to protect splice and maintain cable shield electrical continuity in buried environment.

c. Provide re-enterable encapsulating compound maintaining chemical stability of closure.

d. Provide filled splice cases in accordance with FCC RUS Bull 345-72.

e. Provide gel filled cable meeting requirements of ATSC ICEA S-99-689.

f. In Vault or Manhole:

(1) Provide underground closure suitable to house a straight, butt, and branch splice in a protective housing into which can be poured an encapsulating compound

(2) Closure must be suitable thermoplastic, thermo-set, or stainless steel material supplying structural strength to pass mechanical

and electrical requirements in a vault or maintenance hole (manhole) environment.

g. Re-Enterable Encapsulating Compound: Product maintaining chemical stability of closure.

h. Provide gel-filled splice cases in accordance with RUS Bull 345-72.

2.7 AERIAL (ABOVEGROUND) ENCLOSURES

a. Provide aboveground enclosures constructed of minimum (14 gauge) steel ultraviolet resistant PVC pole mounting in accordance with RUS 7 CFR 1755.

b. Size enclosures and install marker.

c. Secure covers to prevent unauthorized entry.

d. Provide gel filled cable meeting requirements of ICEA S-99-689, ICEA S-98-688, and RUS 7 CFR 1755; except, Figure 8 distribution wire suitable for aerial installation with:

(1) 6,000 pound; or

(2) 6,000 pound Class A galvanized steel; or

(3) 6,000 pound aluminum-clad steel strand.

2.8 TEMPORARY TIP PATHS (OVERHEAD TRACKS, ROAD/PATH BRIDGES, ETC.)

a. Provide for copper, fiber optic, RF, coaxial and designated electronic system cables to maintain facility communications service during construction and install so as to not present a pedestrian and traffic (including construction) safety hazard.

b. TIP temporary cable installations are not required to meet industry standards; but each must be reviewed and accepted, in writing, by Contracting Officer with concurrences from SMCS 005OP2H3, OI&T and facility safety officer, prior to installation.

(1) Be responsible for work associated with each temporary TIP path installation, required by system design and its removal when determined no longer necessary.

(2) Survey outside TIP locations usually encountered, including roads, driveways, marked paths, high traffic passage ways or personnel walkways, and provide Contracting Officer with a plan for temporary paths.

2.9 ACCESS PANELS

a. Panels: 12 inches by 12 inches, or size allowed by location to provide optimum access to equipment for maintenance and service.

b. Provide access panels and doors as required to allow service of materials and equipment that require inspection, replacement, repair or service.

c. Provide access panels where items installed require access and are

concealed in floor, wall, furred space or above ceiling; ceilings consisting of lay-in or removable splined tiles do not require access panels.

d. Provide access panels with same fire rating classification as surface penetrated.

PART 3 EXECUTION

3.1 PREPARATION

a. Penetrations and Sleeves:

(1) Lay out penetration and sleeve openings in advance, to permit provision in work.

(2) Set sleeves in forms before concrete is poured.

(3) Set sleeves prior to installation of structure for passage of pipes, conduit, ducts, etc.

(4) Provide sleeves and packing materials at penetrations of foundations, walls, slabs, partitions, and floors.

(5) Make sleeves that penetrate outside walls, basement slabs, footings, and beams waterproof.

(6) Fill slots, sleeves and other openings in floors or walls if not used.

(a) Fill spaces in openings after installation of conduit or cable.

(b) Provide fill for floor penetrations to prevent passage of water, smoke, fire, and fumes.

(c) Provide fire resistant fill in rated floors and walls, to prevent passage of air, smoke and fumes.

(7) Install sleeves through floors watertight and extend minimum 2 inches above floor surface.

(8) Match and set sleeves flush with adjoining floor, ceiling, and wall finishes where raceways passing through openings are exposed in finished rooms.

(9) Annular space between conduit and sleeve must be minimum (1/4 inch).

(10) Do not provide sleeves for slabs-on-grade, unless specified or indicated otherwise.

(11) Comply with requirements for firestopping, for sleeves through rated fire walls and smoke partitions.

(12) Do not support piping risers or conduit on sleeves.

(13) Identify unused sleeves and slots for future installation.

(14) Provide core drilling if walls are poured or otherwise constructed without sleeves and wall penetration is required; do

not penetrate structural members.

b. Core Drilling:

(1) Avoid core drilling whenever possible.

(2) Coordinate openings with other trades and utilities, and prevent damage to structural reinforcement.

(3) Investigate existing conditions in vicinity of required opening prior to coring, including an x-ray of floor if determined necessary by competent person or Contracting Officer.

(4) Protect areas from damage.

c. Verification of In-Place Conditions:

(1) Verify location, use and status of all material, equipment, and utilities that are specified, indicated, or determined necessary for removal.

(a) Verify materials, equipment, and utilities to be removed are inactive, not required, or in use after completion of project.

(b) Replace with equivalent any material, equipment and utilities that were removed by contractor that are required to be left in place.

(2) Existing Utilities: Do not interrupt utilities serving facilities occupied by Government or others unless permitted under following conditions and then only after arranging to provide temporary utility services, according to requirements indicated:

(a) Notify Contracting Officer in writing at least 14 days in advance of proposed utility interruptions.

(b) Do not proceed with utility interruptions without Government’s written permission.

d. Provide suspended platforms, strap hangers, brackets, shelves, stands or legs for floor, wall and ceiling mounting of equipment as required.

e. Provide steel supports and hardware for installation of hangers, anchors, guides, and other support hardware.

f. Obtain and analyze catalog data, weights, and other pertinent data required for coordination of equipment support provisions and installation.

g. Verify site conditions and dimensions of equipment to ensure access for proper installation of equipment without disassembly that would void warranty.

3.2 INSTALLATION - GENERAL

a. Coordinate systems, equipment, and materials installation with other building components.

b. Install systems, materials, and equipment to conform with approved

submittal data, including coordination drawings.

c. Conform to VAAR 852.236.91 arrangements indicated, recognizing that work may be shown in diagrammatic form or have been impracticable to detail all items because of variances in manufacturers' methods of achieving specified results.

d. Install systems, materials, and equipment level and plumb, parallel and perpendicular to other building systems and components, where installed in both exposed and un-exposed spaces.

e. Install equipment according to manufacturers' written instructions.

f. Install wiring and cabling between equipment and related devices.

g. Install cabling, wiring, and equipment to facilitate servicing, maintenance, and repair or replacement of equipment components. Connect equipment for ease of disconnecting, with minimum interference of adjacent other installations.

h. Provide access panel or doors where units are concealed behind finished surfaces.

i. Arrange for chases, slots, and openings in other building components during progress of construction, to allow for wiring, cabling, and equipment installations.

j. Where mounting heights are not detailed or dimensioned, install systems, materials, and equipment to provide maximum headroom and access for service and maintenance as possible.

k. Install systems, materials, and equipment giving priority to systems required to be installed at a specified slope.

l. Avoid interference with structure and with work or other trades, preserving adequate headroom and clearing doors and passageways to satisfaction of Contracting Officer and code requirements.

m. Install equipment and cabling to distribute equipment loads on building structural members provided for equipment support under other sections; install and support roof-mounted equipment on structural steel or roof curbs as appropriate.

n. Provide supplementary or miscellaneous items, appurtenances, devices and materials for a complete installation.

3.3 EQUIPMENT INSTALLATION

a. Locate equipment as close as practical to locations shown on drawings.

b. Note locations of equipment requiring access on record drawings.

c. Access and Access Panels: Verify access panel locations and construction with Contracting Officer.

d. Inaccessible Equipment:

(1) Where Government determines that contractor has installed equipment not conveniently accessible for operation and

maintenance, equipment must be removed and reinstalled as directed and without additional cost to Government.

(2) Refer to Section 27 11 00, TELECOMMUNICATIONS ROOM FITTINGS for communication equipment cabinet assembly.

(3) Refer to Section 27 11 00, TELECOMMUNICATIONS ROOM FITTINGS for equipment labeling.

3.4 EQUIPMENT IDENTIFICATION

a. Install an identification sign which clearly indicates information required for use and maintenance of equipment.

b. Secure identification signs with screws.

3.5 CUTTING AND PATCHING

a. Perform cutting and patching according to contract general requirements and as follows:

(1) Remove samples of installed work as specified for testing.

(2) Perform cutting, fitting, and patching of equipment and materials required to uncover existing infrastructure in order to provide access for correction of improperly installed existing or new work.

(3) Remove and replace defective work.

(4) Remove and replace non-conforming work.

b. Cut, remove, and legally dispose of selected equipment, components, and materials, including removal of material, equipment, devices, and other items indicated to be removed and items made obsolete by new work.

c. Provide and maintain temporary partitions or dust barriers adequate to prevent spread of dust and dirt to adjacent areas.

d. Protect adjacent installations during cutting and patching operations.

e. Protect structure, furnishings, finishes, and adjacent materials not indicated or scheduled to be removed.

f. Patch finished surfaces and building components using new materials specified for original installation and experienced installers.

3.6 FIELD QUALITY CONTROL

a. Provide work according to VAAR 852.236.91 and FAR clause 52.236-5.

b. Provide minimum clearances and work required for compliance with NFPA 70, National Electrical Code (NEC), and manufacturers' instructions; comply with additional requirements indicated for access and clearances.

c. Verify all field conditions and dimensions that affect selection and provision of materials and equipment, and provide any disassembly, reassembly, relocation, demolition, cutting and patching required to

provide work specified or indicated, including relocation and reinstallation of existing wiring and equipment.

(1) Protect facility, equipment, and wiring from damage.

d. Submit written notice that:

(1) Project has been inspected for compliance with documents.

(2) Work has been completed in accordance with documents.

e. Non-Conforming Work: Conduct project acceptance inspections, final completion inspections, substantial completion inspections, and acceptance testing and demonstrations after verification of system operation and completeness by Contractor.

f. For project acceptance inspections, final completion inspections, substantial completion inspections, and testing/demonstrations that require more than one site visit by Contracting Officer or design professional to verify project compliance for same material or equipment, Government reserves right to obtain compensation from contractor to defray cost of additional site visits that result from project construction or testing deficiencies and incompleteness, incorrect information, or non-compliance with project provisions.

(1) Contracting Officer will notify contractor, of hourly rates and travel expenses for additional site visits, and will issue an invoice to Contractor for additional site visits.

(2) Contractor is not be eligible for extensions of project schedule or additional charges resulting from additional site visits that result from project construction or testing deficiencies/incompleteness, incorrect information, or non-compliance with Project provisions.

g. Tests:

(1) Interim inspection is required at approximately 50 percent of installation.

(2) Request inspection ten working days prior to interim inspection start date by notifying Contracting Officer in writing; this inspection must verify equipment and system being provided adheres to installation, mechanical and technical requirements of construction documents.

(3) Inspection to be conducted by OEM and factory-certified contractor representative, and witnessed by Contracting Officer, facility and SMCS 0050P2H3 representatives.

(4) Check each item of installed equipment to ensure appropriate NRTL listing labels and markings are fixed in place.

(5) Verify cabling terminations in DEMARC, MCR, TER, SCC, ECC, TRs and head end rooms, workstation locations and TCO adhere to color code for T568B pin assignments and cabling connections are in compliance with TIA standards.

(6) Visually confirm minimum Category 6A cable marking at TCOs, CCSs

locations, patch cords and origination locations.

(7) Review entire communications circulating ground system, each TGB and grounding connection, grounding electrode and outside lightning protection system.

(8) Review cable tray, conduit and path/wire way installation practice.

(9) OEM and contractor to perform:

(a) Fiber optical cable field inspection tests via attenuation measurements on factory reels; provide results along with OEM certification for factory reel tests.

(b) Coaxial cable field inspection tests via attenuation measurements on factory reels; provide results along with OEM certification for factory reel tests.

(c) Baseband cable field inspection tests via attenuation measurements on factory reels and provide results along with OEM certification for factory reel tests.

(10) Relocate failed cable reels to a secured location for inventory, as directed by Contracting Officer, and then remove from project site within two working days; provide Contracting Officer with written confirmation of defective cable reels removal from project site.

(11) Provide results of interim inspections to Contracting Officer.

(12) If major or multiple deficiencies are discovered, additional interim inspections could be required until deficiencies are corrected, before permitting further system installation.

(a) Additional inspections are scheduled at direction of Contracting Officer.

(b) Re-inspection of deficiencies noted during interim inspections, must be part of system’s Final Acceptance Proof of Performance Test.

(c) The interim inspection cannot affect the system’s completion date unless directed by Contracting Officer.

(13) Facility Contracting Officer will ensure test documents become a part of system’s official documentation package.

h. Pretesting: Re-align, re-balance, sweep, re-adjust and clean entire system and leave system working for a “break-in” period, upon completing installation of system and prior to Final Acceptance Proof of Performance Test. System RF transmitting equipment must not be connected to keying or control lines during “break-in” period.

(1) Pretesting Procedure:

(a) Verify systems are fully operational and meet performance requirements, utilizing accepted test equipment and spectrum analyzer.

(b) Pretest and verify system functions and performance requirements conform to construction documents and, that no unwanted physical, aural and electronic effects, such as signal distortion, noise pulses, glitches, audio hum, poling noise are present.

(2) Measure and record signal, aural and control carrier levels of each DAS RF, voice and data channel, at each of the following minimum points in system:

(a) Utility provider entrance.

(b) Buried conduit duct locations.

(c) Maintenance Holes (Manholes) and hand holes.

(d) ENTR or DEMARC.

(e) PBX interconnections.

(f) MCR interconnections.

(g) MCOR interconnections.

(h) TER interconnections.

(i) TOR interconnections.

(j) Control room interconnections.

(k) TR interconnections.

(l) System interfaces in locations listed herein.

(m) HE interconnections.

(n) Antenna (outside and inside) interconnections.

(o) System and lightning ground interconnections.

(p) Communications circulating ground system.

(q) UPS areas.

(r) Emergency generator interconnections.

(s) Each general floor areas.

(t) Others as required by AHJ (SMCS 005OP2H3).

(3) Provide recorded system pretest measurements and certification that the system is ready for formal acceptance test to Contracting Officer.

i. Acceptance Test:

(1) Schedule an acceptance test date after system has been pretested, and pretest results and certification submitted to Contracting Officer.

(2) Give Contracting Officer fifteen working days written notice prior to date test is expected to begin; include expected duration of time for test in notification.

(3) Test in the presence of the following:

(a) Contracting Officer.

(b) OEM representatives.

(c) VACO:

1. CFM representative.

2. AHJ–SMCS 005OP2H3, (202) 461-5310.

(d) VISN–CIO, Network Officer and VISN representatives.

(e) Facility:

1. FMS Service Chief, Bio-Medical Engineering and facility representatives.

2. OI&T Service Chief and OI&T representatives.

3. Safety Officer, Police Chief and facility safety representatives.

(f) Local Community Safety Personnel:

1. Fire Marshal representative.

2. Disaster Coordinator representative.

3. EMS Representatives: Police, Sherriff, City, County or State representatives.

(4) Test system utilizing accepted test equipment to certify proof of performance and Life and Public Safety compliance, FCC, NRTL, NFPA and OSHA compliance.

(a) Rate system as acceptable or unacceptable at conclusion of test; make only minor adjustments and connections required to show proof of performance.

1. Demonstrate and verify that system complies with performance requirements under operating conditions.

2. Failure of any part of system that precludes completion of system testing, and which cannot be repaired within four hours, terminates acceptance test of that portion of system.

3. Repeated failures that result in a cumulative time of eight hours to affect repairs is cause for entire system to be declared unacceptable.

4. If system is declared unacceptable, retesting must be rescheduled at convenience of Government and costs borne by the

contractor.

j. Acceptance Test Procedure:

(1) Physical and Mechanical Inspection: The test team representatives must tour major areas to determine system and sub-systems are completely and properly installed and are ready for acceptance testing.

(2) A system inventory including available spare parts must be taken at this time.

(3) Each item of installed equipment must be re-checked to ensure appropriate NRTL (i.e. UL) certification listing labels are affixed.

(4) Confirm that deficiencies reported during Interim Inspections and Pretesting are corrected prior to start of Acceptance Test.

(5) Inventory system diagrams, record drawings, equipment manuals, pretest results.

(6) Failure of system to meet installation requirements of specifications is grounds for terminating testing and to schedule re-testing.

k. Operational Test:

(1) Individual Item Test: VACO AHJ representative (SMCS 005OP2H3) may select individual items of equipment for detailed proof of performance testing until 100 percent of system has been tested and found to meet requirements of the construction documents.

(2) Government’s Condition of Acceptance of System Language:

(a) Without Acceptance: Until system fully meets conditions of construction documents, system’s ownership, use, operation and warranty commences at Government’s final acceptance date.

(b) With Conditional Acceptance: Stating conditions that need to be addressed by contractor or OEM and stating system’s use and operation to commence immediately while its warranty commences only at Government’s agreed final extended acceptance date.

(c) With Full Acceptance: Stating system’s ownership, use, operation and warranty to immediately commence at Government’s agreed to date of final acceptance.

l. Acceptance Test Conclusion: Reschedule testing on deficiencies and shortages with Contracting Officer, after Contracting Officer and SMCS AHJ jointly agree to results of the test, using the generated punch list or discrepancy list. Perform retesting to comply with these specifications at contractor's expense.

m. Proof of Performance Certification:

(1) If system is declared acceptable, AHJ (SMCS 005OP2H3) provides Contracting Officer notice stating system processes to required operating standards and functions and is Government accepted for

use by facility.

(2) Validate items with Contracting Officer needing to be provided to complete project contract (i.e. charts & diagrams, manuals, spare parts, system warranty documents executed, etc.). Once items have been provided, Contracting Officer contacts FMS service chief to turn over system from CFM oversight for beneficial use by facility.

(3) If system is declared unacceptable without conditions, rescheduled testing expenses are to be borne by contractor.

3.7 CLEANING

a. Remove debris, rubbish, waste material, tools, construction equipment, machinery and surplus materials from project site and clean work area, prior to final inspection and acceptance of work.

b. Put building and premises in neat and clean condition.

c. Remove debris on a daily basis.

d. Remove unused material, during progress of work.

e. Perform cleaning and washing required to provide acceptable appearance and operation of equipment to satisfaction of Contracting Officer.

f. Clean exterior surface of all equipment, including concrete residue, dirt, and paint residue, after completion of project.

g. Perform final cleaning prior to project acceptance by Contracting Officer.

h. Remove paint splatters and other spots, dirt, and debris; touch up scratches and mars of finish to match original finish.

i. Clean devices internally using methods and materials recommended by manufacturer.

j. Tighten wiring connectors, terminals, bus joints, and mountings, to include lugs, screws and bolts according to equipment manufacturer’s published torque tightening values for equipment connectors. In absence of published connection or terminal torque values, comply with torque values specified in UL 486A-486B.

3.8 TRAINING

a. Provide training in accordance with subsection, INSTRUCTIONS, of Section 01 00 00, GENERAL REQUIREMENTS.

b. Provide training for equipment or system as required in each associated specification.

c. Develop and submit training schedule for approval by Contracting Officer, at least 30 days prior to planned training.

3.9 PROTECTION

a. Protection of Fireproofing:

(1) Install clips, hangers, clamps, supports and other attachments to surfaces to be fireproofed, if possible, prior to start of spray fireproofing work.

(2) Install conduits and other items that would interfere with proper application of fireproofing after completion of spray fire proofing work.

(3) Patch and repair fireproofing damaged due to cutting or course of work must be performed by installer of fireproofing and paid for by trade responsible for damage.

b. Maintain equipment and systems until final acceptance.

c. Ensure adequate protection of equipment and material during installation and shutdown and during delays pending final test of systems and equipment because of seasonal conditions.

-- End of Section --

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SECTION 27 05 26

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SECTION 27 05 26
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GROUNDING AND BONDING FOR COMMUNICATIONS SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies grounding and bonding requirements of communications installations based on the requirements of TIA-607, Telecommunications Bonding and Grounding (Earthing) for Customer Premises. Work covered by this Section shall consist of furnishing supplies, labor, materials, equipment, labeling, and tools. Testing is required unless otherwise specified. An operable grounding and bonding infrastructure is required as described on the Drawings and/or required by these specifications. All materials shall be listed by a nationally recognized testing laboratory (NRTL).

1.2 SUMMARY

Section Includes:

a. Required Bonding Busbars.

b. Supplemental Bonding Networks.

c. Telecommunications Bonding Conductors.

d. Joining Requirements.

e. Overvoltage Surge Protectors.

1.3 REFERENCES

a. VA Infrastructure Standard for Telecommunications Spaces.

d. International Annealed Copper Standard (IACS).

e. BICSI Information Technology Systems Installation Methods Manual (

ITSIMM), Recommended Testing Procedures and Criteria.

BICSI International Standards Program (BICSI)

ITSIMM	Information Technology Systems Installation Methods Manual

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-607	(2019d) Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

UL SOLUTIONS (UL)

UL 497	(2001; Reprint Jul 2022) UL Standard for Safety Protectors for Paired Conductor Communication Circuits
UL 497A	(2001; Bul. 2019) UL Standard for Safety Secondary Protectors for Communications Circuits

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

EIA-310-D	Standard 19-in. four-post EIA cabinet or rack, with mounting posts

1.4 RELATED WORK

a. Facility grounding and bonding requirements: Section 26 05 26, GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS.

b. Information Technology equipment enclosures: Section 27 11 16, COMMUNICATION CABINETS, RACKS, FRAMES, AND ENCLOSURES.

c. Cable trays: Section 27 05 36, CABLE TRAYS FOR COMMUNICATIONS SYSTEMS.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Plan Indicating Location Of System Grounding; G

SD-03 Product Data

Bonding Busbars; G

Supplemental Bonding Networks; G

Telecommunications Bonding Backbone (Tbb); G

Bonding Conductors; G

Overvoltage Surge Protectors; G

SD-06 Test Reports

Certified Test Reports Of Ground Resistance; G

SD-07 Certificates

Certifications; G

1.6 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 27 05 00, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS.

b. Provide plan indicating location of system grounding electrode connections and routing of aboveground and underground grounding electrode conductors.

c. Closeout Submittals: In addition to Section 27 05 11, REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS provide the following:

(1) Certified test reports of ground resistance.

(2) Certifications: Two weeks prior to final inspection, submit following to Contracting Officer:

(a) Certification materials and installation is in accordance with construction documents.

(b) Certification complete installation has been installed and tested.

PART 2 PRODUCTS

2.1 BONDING BUSBARS.

Bonding busbars are located in each telecommunications space and in each network rack/cabinet. Bonding busbars provide common locations in the telecommunications space for the connection of the telecommunications bonding conductors.

a. Common Requirements.

(1) Provided with holes for use with correctly-matched UL Listed two-hole lugs and hardware.

(2) Constructed of copper or copper alloy with a minimum of 95% conductivity when annealed.

(3) Annealed as specified by the International Annealed Copper Standard (IACS).

(4) Installation resulting in a maximum 4.0Ω to ground resistance measured from any point in the system.

(5) Installation is insulated from the mounting surface through the use of appropriate insulators.

(6) UL Listed.

b. Primary Bonding Busbar (PBB). The PBB is a dedicated extension of the building grounding electrode system for the telecommunications infrastructure. The PBB also serves as the central attachment point for Secondary Bonding Busbars (SBB) via Telecommunications Bonding Backbone (TBB).

(1) Pre-drilled copper with holes to accommodate lug mounting holes.

(2) Sized for current applications and future growth.

(3) Insulated from its supports.

(4) Electro-tin plated is acceptable.

(5) Maintain a 2″ min clearance between busbar and finished wall.

(6) Size must be 0.25″ thick by 4″ wide by 16″ length.

c. Secondary Bonding Busbar (SBB). The SBB is a common point of connection for telecommunications system and equipment bonding to a ground located in each Telecommunications Room.

(1) Pre-drilled holes to accommodate dual-lug mounting holes.

(2) 0.25″ thick x 2″ wide with varying length to be sized for current applications and future growth.

(3) Maintain a 2″ min clearance between the finished wall and busbar.

(4) Copper or tin annealed copper.

d. Rack Bonding Busbar (RBB). The RBB is a busbar located in a cabinet, rack, or frame.

(1) Mounting. Horizontal mounting to an EIA-310-D 19″ equipment rack is specified.

(2) Capacity: 10 Double-hole lugs.

(3) Size & material 0.75″ wide x 19″ length x 0.25″ thick – Copper or tin annealed copper.

2.2 SUPPLEMENTAL BONDING NETWORKS.

Also known as Signal Reference Grids (SRGs), Mesh Bonding Networks (Mesh-BN), or Isolated Bonding Networks (IBNs), the requirements for supplemental bonding networks are not specified in this document. Supplemental bonding networks may not be used to replace a standards-compliant bonding infrastructure.

2.3 TELECOMMUNICATIONS BONDING BACKBONE (TBB).

The TBB bonds the facility Primary Bonding Busbar (PBB) to telecommunications space Secondary Bonding Busbars (SBBs).

a. Material. Stranded copper wire with a green jacket (or per NEC depending on size), run as a continuous conductor.

b. Size. The TBB shall be sized to meet the requirements of TIA-607 and will be as straight as practicable avoiding bends.

c. Bonding the TBB to the PBB and each SBB will require a UL Listed irreversible compression (crimp) dual-lug connector.

2.4 BONDING CONDUCTORS.

a. Telecommunications Equipment Bonding Conductor (TEBC). The TEBC

connects the cabinets and racks in the telecommunications space to the space's bonding busbar (PBB or SBB).

(1) Material. Stranded copper wire with a green jacket (or per NEC depending on size), run as a continuous conductor.

(2) Size. The TBB shall be sized to meet the requirements of TIA-607. The minimum size for the TEBC shall be AWG 6.

(3) Bonding. Bonded to the telecommunications space bonding busbar (PBB or SBB) via a UL Listed two-hole compression lug. Rack Bonding Conductors (RBCs) are connected to the TEBC using UL Listed irreversible compression (crimp) connectors.

b. Rack Bonding Conductor (RBC). The RBC is a bonding conductor from the cabinet or Rack Bonding Busbar (RBB) to the Telecommunications Equipment Bonding Conductor (TEBC).

(1) Material. Stranded copper wire with a green jacket run as a continuous conductor.

(2) Size. The minimum size for the RBC shall be AWG 6.

(3) Bonding. Bonded to the Telecommunications Equipment Bonding Conductor (TEBC) using UL Listed irreversible compression (crimp) connectors.

(a) Where connected to a server cabinet, the RBC extends to the bottom of the server cabinet allowing Equipment Bonding Conductors to be attached at any point in the cabinet.

(b) Where connected to a network rack/cabinet, the Rack Bonding Conductor (RBC) is bonded to the Rack Bonding Busbar (RBB) via a UL Listed two-hole compression lug.

c. Equipment/Unit Bonding Conductor (EBC/UBC). The Equipment/Unit Bonding Conductor connects individual equipment in a cabinet or rack to the Rack Bonding Conductor (RBC) or Rack Bonding Busbar (RBB). Equipment Bonding conductors are also used as cable tray bonding conductors to bond cable tray sections to the Telecommunications Equipment Bonding Conductor (TEBC).

(1) Material. Stranded copper wire with a green jacket run as a continuous conductor.

(2) Size. The minimum size for the RBC shall be AWG 6.

(3) Bonding.

(a) Where used in a server cabinet, bonded to the Rack Bonding Connector (RBC) using UL Listed irreversible compression (crimp) connectors and to IT equipment via a UL Listed two-hole compression lug. (Some IT equipment may require one-hole lugs.)

(b) Where used in a network cabinet/rack, bonded to the Rack Bonding Busbar (RBB) via a UL Listed two-hole compression lug and to IT equipment via a UL Listed two-hole compression lug. (Some IT equipment may require one-hole lugs.)

(c) Where used as a cable tray bonding conductor connecting cable tray sections, bonded to each adjoining section of the cable tray using UL Listed two-hole compression lugs.

(d) Where used as a cable tray bonding conductor connecting cable tray sections to the Telecommunications Equipment Bonding Conductor (TEBC), bonded to the TEBC using UL Listed irreversible compression (crimp) connectors and to the cable tray via a UL Listed two-hole compression lug.

(4) Contractor shall furnish a minimum of ten (10) EBCs for each RBB. Five (5) EBCs shall be outfitted as described here and shall be six (6) feet in length. Five (5) EBCs shall be outfitted as decried here and shall be nine (9) feet in length. Contractor shall use field measurements to determine EBC cable length when directed by VA project managers.

2.5 OVERVOLTAGE SURGE PROTECTORS.

Protectors are voltage-limiting devices intended to protect equipment, wiring, and personnel against the effects of excessive potentials and currents in communications lines caused by lightning, contacts with power conductors, power induction, and rises in ground potential.

(1) Compliance. Protectors shall be compliant with UL 497 and/or UL 497A, as applicable.

(2) Performance. Protectors shall be rated to support copper UTP performance Category 5e at 100MHz.

(3) Protection. Voltage suppression shall be via low capacitance solid state protectors only, rated at 18V for copper backbone applications or 65V for VOIP or PoE applications.

(4) Construction. Provide protection for each pair.

(5) Termination shall be insulation-displacement contact.

(6) Mounting. Protectors shall be wall-mounted.

PART 3 EXECUTION

3.1 IMPLEMENTATION

a. Components of the telecommunications bonding system will be installed and connected using materials and techniques required by TIA-607. The use of antioxidant joint compound is required for all connections excluding protector 110 block connections.

b. Testing.

(1) Perform tests per BICSI Information Technology Systems Installation Methods Manual (ITSIMM), Recommended Testing Procedures and Criteria.

(2) Perform two-point bond test using trained installers qualified to use test equipment.

(3) Conduct continuity tests to verify that metallic pathways in

telecommunications spaces are bonded to PBB or SBB.

(4) Conduct electrical continuity test to verify that PBB is effectively bonded to the facility grounding electrode conductor.

(5) Perform resistance tests to ensure rack and cabinet bonding connection resistance measures less than 4Ω to PBB or SBB.

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SECTION 27 05 33

CONDUITS AND BACKBOXES FOR COMMUNICATIONS SYSTEMS

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SECTION 27 05 33
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CONDUITS AND BACKBOXES FOR COMMUNICATIONS SYSTEMS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C80.1	(2020) American National Standard for Electrical Rigid Steel Conduit (ERSC)
ANSI C80.3	(2020) American National Standard for Electrical Metallic Tubing (EMT)
ANSI C80.5	(2020) American National Standard for Electrical Rigid Aluminum Conduit
ANSI C80.6	(2018) Electrical Intermediate Metal Conduit (ELMC)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 262	(2023) Standard Method of Test for Flame Travel and Smoke of Wires and Cables for Use in Air-Handling Spaces

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-569	(2019e; Add 1 2022) Telecommunications Pathways and Spaces

UL SOLUTIONS (UL)

UL 1	(2005; Reprint Jan 2022) UL Standard for Safety Flexible Metal Conduit
UL 5	(2016; Reprint Jul 2022) UL Standard for Safety Surface Metal Raceways and Fittings
UL 6	(2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel
UL 6A	(2008; Reprint Nov 2024) UL Standard for Safety Electrical Rigid Metal Conduit - Aluminum, Red Brass, and Stainless Steel
UL 50	(2024) UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations

UL 94	(2023; Reprint Jan 2024) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances
UL 360	(2013; Reprint Jan 2024) UL Standard for Safety Liquid-Tight Flexible Metal Conduit
UL 467	(2022) UL Standard for Safety Grounding and Bonding Equipment
UL 514A	(2024) UL Standard for Safety Metallic Outlet Boxes
UL 514B	(2012; Reprint Mar 2024) UL Standard for Safety Conduit, Tubing and Cable Fittings
UL 514C	(2014; Reprint Apr 2024) UL Standard for Safety Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers
UL 651	(2011; Reprint May 2022) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
UL 651A	(2023; Reprint Jan 2024) UL Standard for Safety Schedule 40 and 80 High Density Polyethylene (HDPE) Conduit
UL 797	(2007; Reprint Apr 2023) UL Standard for Safety Electrical Metallic Tubing -- Steel
UL 1242	(2006; Reprint Apr 2022) UL Standard for Safety Electrical Intermediate Metal Conduit -- Steel

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA FB 1	(2014) Standard for Fittings, Cast Metal Boxes, and Conduit Bodies for Conduit, Electrical Metallic Tubing, and Cable
NEMA TC 3	(2021) Polyvinyl Chloride (PVC) Fittings for Use With Rigid PVC Conduit and Tubing

U.S. DEPARTMENT OF VETERANS AFFAIRS (VA)

VA H-18-8	(2023) Seismic Design Requirements

1.2 DESCRIPTION

This section specifies conduit, fittings, and boxes to form raceway systems for communications cabling.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00

SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Layout Of Required Conduit Penetrations; G

Size And Location Of Cabinets, Splice Boxes, And Pull Boxes; G

SD-03 Product Data

Catalog Cuts; G

SD-07 Certificates

Certification; G

1.4 RELATED WORK

a. Bedding of conduits: Section 31 20 00, EARTH MOVING.

b. Sealing around penetrations to maintain integrity of fire rated construction: Section 07 84 00, FIRESTOPPING.

c. Fabrications for deflection of water away from building envelope at penetrations: Section 07 60 00, FLASHING AND SHEET METAL.

d. Sealing around conduit penetrations through building envelope to prevent moisture migration into building: Section 07 92 00, JOINT SEALANTS.

e. Identification and painting of conduit and other devices: Section 09 91 00, PAINTING.

f. Requirements for personnel safety and to provide a low impedance path for possible ground fault currents: Section 27 05 26, GROUNDING AND BONDING FOR COMMUNICATIONS SYSTEMS.

1.5 SUBMITTAL INSTRUCTIONS

a. In accordance with Section 27 05 00, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS, submit the following:

(1) Size and location of cabinets, splice boxes, and pull boxes.

(2) Layout of required conduit penetrations through structural elements.

(3) Catalog cuts marked with specific item proposed and area of application identified.

b. Certification: Provide letter prior to final inspection, certifying material is in accordance with construction documents and properly installed.

PART 2 PRODUCTS

2.1 MATERIAL

a. Minimum Conduit Size: (3/4 inch).

b. Conduit:

(1) Rigid Galvanized Steel: Conform to UL 6, ANSI C80.1.

(2) Rigid Aluminum: Conform to UL 6A, ANSI C80.5.

(3) Rigid Intermediate Steel Conduit (IMC): Conform to UL 1242, ANSI C80.6.

(4) Electrical Metallic Tubing (EMT):

(a) Maximum Size: (4 inches).

(b) Install only for cable rated 600 volts or less.

(c) Conform to UL 797, ANSI C80.3.

(5) Flexible Galvanized Steel Conduit: Conform to UL 1.

(6) Liquid-tight Flexible Metal Conduit: Conform to UL 360.

(7) Direct Burial Plastic Conduit: Conform to UL 651 and UL 651A, heavy wall PVC, or high density polyethylene (HDPE).

(8) Surface Metal Raceway: Conform to UL 5.

c. Conduit Fittings:

(1) Rigid Galvanized Steel and Rigid Intermediate Steel Conduit Fittings:

(a) Provide fittings meeting requirements of UL 514B and NEMA FB 1.

(b) Sealing: Provide threaded cast iron type. Use continuous drain type sealing fittings to prevent passage of water and vapor. In concealed work, install sealing fittings in flush steel boxes with blank cover plates having same finishes as other electrical plates in room.

(c) Standard Threaded Couplings, Locknuts, Bushings, and Elbows: Only steel or malleable iron materials are acceptable. Integral retractable type IMC couplings are also acceptable.

(d) Locknuts: Bonding type with sharp edges for digging into metal wall of an enclosure.

(e) Bushings: Metallic insulating type, consisting of an insulating insert molded or locked into metallic body of fitting. Bushings made entirely of metal or nonmetallic material are not permitted.

(f) Erickson (union-type) and Set Screw Type Couplings:

1. Couplings listed for use in concrete are permitted for use to complete a conduit run where conduit is installed in concrete.

2. Use set screws of case hardened steel with hex head and cup point to seat in conduit wall for positive ground.

(g) Provide OEM approved fittings.

(2) Rigid Aluminum Conduit Fittings:

(a) Standard Threaded Couplings, Locknuts, Bushings, and Elbows: Malleable iron, steel or aluminum alloy materials; Zinc or cadmium plate iron or steel fittings. Aluminum fittings containing more than 0.4 percent copper are not permitted.

(b) Locknuts and Bushings: As specified for rigid steel and IMC conduit.

(c) Set Screw Fittings: Not permitted for use with aluminum conduit.

(3) Electrical Metallic Tubing Fittings:

(a) Conform to UL 514B and NEMA FB 1; only steel or malleable iron materials are acceptable.

(b) Couplings and Connectors: Concrete tight and rain tight, with connectors having insulated throats.

1. Use gland and ring compression type couplings and connectors for conduit sizes 2 inches and smaller.

2. Use set screw type couplings with four set screws each for conduit sizes over (2 inches).

3. Use set screws of case-hardened steel with hex head and cup point to seat in wall of conduit for positive grounding.

(c) Indent type connectors or couplings are not permitted.

(d) Die-cast or pressure-cast zinc-alloy fittings or fittings made of "pot metal" are not permitted.

(e) Provide OEM approved fittings.

(4) Flexible Steel Conduit Fittings:

(a) Conform to UL 514B; only steel or malleable iron materials are acceptable.

(b) Provide clamp type, with insulated throat.

(c) Provide OEM approved fittings.

(5) Liquid-tight Flexible Metal Conduit Fittings:

(a) Conform to UL 514B and NEMA FB 1; only steel or malleable iron materials are acceptable.

(b) Fittings must incorporate a threaded grounding cone, a steel or plastic compression ring, and a gland for tightening.

(c) Provide connectors with insulated throats to prevent damage to cable jacket.

(d) Provide OEM approved fittings.

(6) Direct Burial Plastic Conduit Fittings: Provide fittings meeting requirements of UL 514C and NEMA TC 3, and as recommended by conduit manufacturer.

(7) Expansion and Deflection Couplings:

(a) Conform to UL 467 and UL 514B.

(b) Accommodate 3/4 inch deflection, expansion, or contraction in any direction, and allow 30-degree angular deflections.

(c) Include internal flexible metal braid sized to ensure conduit ground continuity and fault currents in accordance with UL 467, and NEC code tables for ground conductors.

(d) Jacket: Flexible, corrosion-resistant, watertight, moisture and heat resistant molded rubber material with stainless steel jacket clamps.

(8) Rigid Aluminum Fittings:

(a) Provide malleable iron, steel or aluminum alloy materials; zinc or cadmium plate iron or steel fittings. Aluminum fittings containing more than 0.4 percent copper are prohibited.

(b) Locknuts and Bushings: As specified for rigid steel and IMC conduit.

(c) Set Screw Fittings: Not permitted for use with aluminum conduit.

(d) Indent type connectors or couplings are prohibited.

(e) Die-cast or pressure-cast zinc-alloy fittings or fittings made of “pot metal” are not permitted.

(f) Provide OEM approved fittings.

d. Conduit Supports:

(1) Parts and Hardware: Provide zinc-coat or equivalent corrosion protection.

(2) Individual Conduit Hangers: Designed for the purpose, having a pre-assembled closure bolt and nut, and provisions for receiving a hanger rod.

(3) Multiple Conduit (Trapeze) Hangers: Minimum (1-1/2 by 1-1/2 inch), (12 gage) steel, cold formed, lipped channels; with minimum 3/8 inch diameter steel hanger rods.

(4) Solid Masonry and Concrete Anchors: Self-drilling expansion shields, or machine bolt expansion.

e. Outlet, Splice, and Pull Boxes:

(1) Conform to UL 50 and UL 514A.

(2) Cast metal where required by NEC or shown, and equipped with rustproof boxes.

(3) Sheet Metal Boxes: Galvanized steel, except where otherwise shown.

(4) Install flush mounted wall or ceiling boxes with raised covers so that front face of raised cover is flush with wall.

(5) Install surface mounted wall or ceiling boxes with surface style flat or raised covers.

f. Warning Tape: Standard, 4-Mil polyethylene 3 inch wide tape detectable type, red with black letters, and imprinted with “CAUTION BURIED COMMUNICATIONS CABLE BELOW”.

g. Flexible Nonmetallic Communications Raceway (Innerduct) and Fittings:

(1) General: Provide NFPA 262 listed plenum, riser, and general purpose corrugated pliable communications raceway.

(2) Provide Communications Raceway with a factory installed (1250 lb.) tensile pre-lubricated pull tape.

(3) Use only metallic straps, hangers and fittings to support raceway from building structure. Cable ties are not permitted for securing raceway to building structure.

(4) Provide fittings to be installed in spaces used for environmental air made of materials that do not exceed flammability, smoke generation, ignitibility, and toxicity requirements of environmental air space.

(5) Size: Metric Designator 53 (trade size 2) or smaller.

(6) Outside Plant: Plenum-rated where each innerduct is 3 inches and larger.

(7) Inside Plant: Listed and marked for installation in plenum airspaces and minimum 1 inch inside diameter.

(8) Plenum: Non-metallic communications raceway.

(a) Constructed of low smoke emission, flame retardant PVC with corrugated construction.

(b) UL 94 V-O rating for flame spreading limitation.

(9) Provide innerduct reel lengths as necessary to ensure ducts are continuous.

(10) Provide pulling accessories used for innerduct including but not limited to, inner duct lubricants, spreaders, applicators, grips, swivels, harnesses, and line missiles (blown air) compatible with materials being pulled.

h. Outlet Boxes:

(1) Flush wall mounted minimum (4-11/16 inches) square, (3-5/8 inches) deep pressed galvanized steel.

(2) Flush wall mounted 5 inches square x (2-7/8 inches); deep pressed galvanized steel.

(3) 2-Gang Tile Box:

(a) Flush backbox type for installation in block walls.

(b) Minimum (3-5/8 inches) deep.

i. Weatherproof Outlet Boxes: Surface mount two gang, (2-5/8 inches) deep weatherproof cast aluminum with powder coated finish internal threads on hubs 3/4 inch minimum.

PART 3 EXECUTION

3.1 EQUIPMENT INSTALLATION AND REQUIREMENTS

a. Raceways typically required for cabling systems unless otherwise indicated:

System	Specification Section	Installed Method
Grounding	27 05 26	Conduit Not Required
Control, Communication and Signal Wiring	27 15 01	Complete Conduit Allowed in Non-Partitioned Cable Tray or Cable Ladders
Public Address and Mass Notification Systems	27 51 16	Complete conduit
Grounding and Bonding for Electronic Safety and Security	28 05 26	Conduit Not Required Unless Required by Code
Physical Access Control System	28 13 00	Conduit to Cable Tray Partitioned Cable Tray
Video Surveillance	28 23 00	Complete Conduit
Fire Detection and Alarm	28 31 00	Complete Conduit

b. Penetrations:

(1) Cutting or Holes:

(a) Locate holes in advance of installation. Where proposed in structural sections, obtain approval of structural engineer and Contracting Officer prior to drilling through structural sections.

(b) Make holes through concrete and masonry in new structures with a diamond core drill or concrete saw. Pneumatic hammer, impact electric, hand or manual hammer type drills are not permitted; Contracting Officer may grant limited permission by request, in condition of limited working space.

(c) Fire Stop: Where conduits, wireways, and other communications raceways pass through fire partitions, fire walls, smoke partitions, or floors, install a fire stop that provides an effective barrier against spread of fire, smoke and gases as

specified in Section 07 84 00, FIRESTOPPING.

1. Fill and seal clearances between raceways and openings with fire stop material.

2. Install only retrofittable, non-hardening, and reusable firestop material that can be removed and reinstalled to seal around cables inside conduits.

(d) Waterproofing at Floor, Exterior Wall, and Roof Conduit Penetrations: Seal clearances around conduit and make watertight as specified in Section 07 92 00, JOINT SEALANTS or directed by waterproofing manufacturer.

c. Conduit Installation:

(1) Minimum conduit size of (3/4 inch), but not less than size required for 40 percent fill.

(2) Install insulated bushings on all conduit ends.

(3) Install pull boxes after every 180 degrees of bends (two 90-degree bends) or every 100ft. Size boxes per TIA-569.

(4) Extend vertical conduits/sleeves through floors minimum 3 inches above floor and minimum 3 inches below ceiling of floor below.

(5) Terminate conduit runs to and from a backboard in a TR or interstitial space at top or bottom of backboard. Install conduits to enter telecommunication rooms next to wall and flush with backboard.

(6) Where drilling is necessary for vertical conduits, locate holes so as not to affect structural sections.

(7) Seal empty conduits located in telecommunications rooms or on backboards to prevent entrance of moisture and gases and to meet fire resistance requirements.

(8) Provide pull wire in all empty conduits; sleeves through floor are exceptions.

(9) Complete each entire conduit run installation before pulling in cables.

(10) Flattened, dented, or deformed conduit is not permitted.

(11) Ensure conduit installation does not encroach into ceiling height head room, walkways, or doorways.

(12) Cut conduit square with a hacksaw, ream, remove burrs, and draw tight.

(13) Install conduit mechanically continuous.

(14) Independently support conduit at 8 feet on center; do not use other supports (i.e., suspended ceilings, suspended ceiling supporting members, luminaires, conduits, mechanical piping, or mechanical ducts).

(15) Support conduit within 1 foot of changes of direction, and within 1 foot of each enclosure to which connected.

(16) Close ends of empty conduit with plugs or caps to prevent entry of debris, until cables are pulled in.

(17) Conduit installations under fume and vent hoods are prohibited.

(18) Attach conduits to cabinets, splice cases, pull boxes and outlet boxes with bonding type locknuts. For rigid and IMC conduit installations, provide a locknut on inside of enclosure, made up wrench tight. Do not make conduit connections to box covers.

(19) Do not use aluminum conduits in wet locations.

(20) Unless otherwise indicated on drawings or specified herein, conceal conduits within finished walls, floors and ceilings.

(21) Conduit Bends:

(a) Make bends with standard conduit bending machines; observe minimum bend radius for cable type and outside diameter.

(b) Conduit hickey is permitted only for slight offsets, and for straightening stubbed conduits.

(c) Bending of conduits with a pipe tee or vise is not permitted.

(d) Minimum radius of communication conduit bends:

Sizes of Conduit	
3/4	(6)
1	(9)
1-1/4	(14)
1-1/2	(17)
2	(21)
2-1/2	(25)
3	(31)
3-1/2	(36)
4	(45)

(22) Layout and Homeruns - Deviations: Make only where necessary to avoid interferences and only after drawings showing proposed deviations have been submitted and approved by Contracting Officer.

d. Concealed Work Installation:

(1) In Concrete:

(a) Conduit: Rigid steel or IMC.

(b) Align and run conduit in direct lines.

(c) Install conduit through concrete beams only when the following occurs:

1. Where shown on structural drawings.

2. As accepted by Contracting Officer prior to construction, and after submittal of drawing showing location, size, and position of each penetration.

(d) Installation of conduit in concrete that is less than 3 inches thick is prohibited.

1. Conduit outside diameter larger than 1/3 of slab thickness is prohibited.

2. Space between Conduits in Slabs: Approximately six conduit diameters apart, except one conduit diameter at conduit crossings.

3. Install conduits approximately in center of slab to ensure a minimum of 3/4 inch of concrete around conduits.

(e) Make couplings and connections watertight. Use thread compounds that are NRTL listed conductive type to ensure low resistance ground continuity through conduits. Tightening set screws with pliers is not permitted.

e. Furred or Suspended Ceilings and in Walls:

(1) Rigid steel, or rigid aluminum. Different type conduits mixed indiscriminately in same system is not permitted.

(2) Align and run conduit parallel or perpendicular to building lines.

(3) Tightening set screws with pliers is not permitted.

f. Exposed Work Installation:

(1) Unless otherwise indicated on drawings, exposed conduit is only permitted in telecommunications rooms.

(a) Provide rigid steel, IMC or rigid aluminum.

(b) Different type of conduits mixed indiscriminately in system is not permitted.

(2) Align and run conduit parallel or perpendicular to building lines.

(3) Install horizontal runs close to ceiling or beams and secure with conduit straps.

(4) Support horizontal or vertical runs at not over 96 inches intervals.

(5) Painting:

(a) Paint exposed conduit as specified in Section 09 91 00, PAINTING.

(b) Refer to Section 09 91 00, PAINTING for preparation, paint type, and exact color.

(c) Provide labels where conduits pass through walls and floors

and at maximum 20 foot intervals in between.

g. Expansion Joints:

(1) Conduits 3 inches and larger, that are secured to building structure on opposite sides of a building expansion joint, require expansion and deflection couplings. Install couplings in accordance with manufacturer's recommendations.

(2) Provide conduits smaller than 3 inches with pull boxes on both sides of expansion joint. Connect conduits to expansion and deflection couplings as specified.

(3) Install expansion and deflection couplings where shown.

h. Seismic Areas:

(1) In seismic areas, follow VA H-18-8 Seismic Design Requirements.

(2) Rigidly secure conduit to building structure on opposite sides of a building expansion joint with pull boxes on both sides of joint.

(3) Connect conduits to pull boxes with 15 inches of slack flexible conduit.

(4) Install green copper wire minimum #6 AWG in flexible conduit for bonding jumper.

i. Conduit Supports, Installation:

(1) Select AC193 code listed mechanical anchors or fastening devices with safe working load not to exceed 1/4 of proof test load.

(2) Use pipe straps or individual conduit hangers for supporting individual conduits. Maximum distance between supports is 8 foot on center.

(3) Support multiple conduit runs with trapeze hangers. Use trapeze hangers designed to support a load equal or greater than sum of the weights of the conduits, wires, hanger itself, and (200 pounds). Attach each conduit with U-bolts or other accepted fasteners.

(4) Support conduit independent of pull boxes, luminaires, suspended ceiling components, angle supports, duct work, and similar items.

(5) Fastenings and Supports in Solid Masonry and Concrete:

(a) New Construction: Use steel or malleable iron concrete inserts set in place prior to placing concrete.

(b) Existing Construction:

1. Code AC193 listed wedge type steel expansion anchors minimum 1/4 inch bolt size and minimum (1-1/8 inch) embedment.

2. Power set fasteners minimum 1/4 inch diameter with depth of penetration minimum (3 inches).

3. Use vibration and shock resistant anchors and fasteners

for attaching to concrete ceilings.

(6) Fastening to Hollow Masonry: Toggle bolts are permitted.

(7) Fastening to Metal Structures: Use machine screw fasteners or other devices designed and accepted for application.

(8) Bolts supported only by plaster or gypsum wallboard are not acceptable.

(9) Attachment by wood plugs, rawl plug, plastic, lead or soft metal anchors, or wood blocking and bolts supported only by plaster is prohibited.

(10) Do not support conduit from chain, wire, or perforated strap.

(11) Spring steel type supports or fasteners are not permitted except horizontal and vertical supports/fasteners within walls.

(12) Vertical Supports:

(a) Install riser clamps and supports for vertical conduit runs in accordance with NEC.

(b) Provide supports for cable and wire with fittings that include internal wedges and retaining collars.

j. Box Installation:

(1) Boxes for Concealed Conduits:

(a) Flush mounted.

(b) Provide raised covers for boxes to suit wall or ceiling, construction and finish. In spaces not controlled by VA (i.e., common hallways) covers must be lockable.

(2) In addition to boxes shown, install additional boxes where needed to prevent damage to cables during pulling.

(3) Remove only knockouts as required and plug unused openings. Use threaded plugs for cast metal boxes and snap-in metal covers for sheet metal boxes.

(4) Stencil or install phenolic nameplates on covers of boxes identified on riser diagrams; for example "SIG-FA JB No. 1".

(5) Outlet boxes mounted back-to-back in same wall are not permitted. A minimum 24 inches center-to-center lateral spacing must be maintained between boxes.

k. Flexible Nonmetallic Communications Raceway (Innerduct), Installation:

(1) Install Innerduct in cable tray. Innerduct may not be free-hung.

(2) Install only in accessible spaces not subject to physical damage or corrosive influences.

(3) Make bends manually to assure internal diameter of tubing is not

effectively reduced.

(4) Extend each segment of innerduct minimum 12 inches beyond end of service conduit tie or cable tray. Restrain innerduct ends with wall mount clamps and seal when cable is installed.

3.2 TESTING

a. Examine fittings and locknuts for secureness.

b. Test RMC, IMC and EMT systems for electrical continuity and resistance to ground.

-- End of Section --

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DIVISION 27 - COMMUNICATIONS

SECTION 27 05 36

CABLE TRAYS FOR COMMUNICATIONS SYSTEMS

10/18

PART 1 GENERAL

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PART 3 EXECUTION

3.1 IMPLEMENTION

-- End of Section Table of Contents --

SECTION 27 05 36
10/18

CABLE TRAYS FOR COMMUNICATIONS SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies requirements for telecommunications cabling cable tray and raceway equipment.

1.2 SUMMARY

Section Includes:

a. Cable Tray.

b. Fiber Optic Cable Raceway.

c. Ladder Rack.

1.3 RELATED WORK

Section 27 05 26, GROUNDING AND BONDING FOR COMMUNICATIONS SYSTEMS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Cable Tray; G

Fiber Optic Cable Raceway; G

Ladder Rack; G

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Section 27 05 11, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS.

PART 2 PRODUCTS

2.1 CABLE TRAY.

a. Design. Wire basket with tray insert. Maximum 2″ x 4″ wire mesh construction. Constructed of round, smooth wire with continuous top side to minimize cable sheath damage.

b. Finish. Coated to prevent rust.

c. Size. Cable tray shall be sized to hold the maximum expected cable

load at not more than 40% fill and not more than 6″ maximum cable depth.

d. Accessories. Provide and install the following components and accessories as necessary to achieve the design:

(1) Overhead mounting equipment (thread-rod, unistrut, nuts, washers, etc.) suitable for the supporting structural elements.

(2) Tray section splicing hardware.

(3) Tray insert.

(4) Cable waterfalls.

(5) Ground strap kits.

(6) Bend radius limiters.

2.2 FIBER OPTIC CABLE RACEWAY.

a. Design. Raceway system shall be a modular system of channels, fittings, and brackets, able to be assembled with couplers. Fittings for cable spillout shall maintain a minimum 2″ bend radius.

b. Materials. Raceway shall be constructed of rigid PVC and ABS plastics.

c. Size. Raceway shall be nominally sized 4″x4″.

d. Accessories. Provide and install the following components and accessories as necessary to achieve the design:

(1) Overhead mounting equipment (thread-rod, unistrut, nuts, washers, etc.) suitable for the supporting structural elements.

(2) Channel covers.

(3) Raceway section couplers.

(4) Channel corners and intersections.

(5) Spillway fittings.

2.3 LADDER RACK.

a. Design. Ladder rack system shall be a modular system of ladder, turns, splices, supports, and accessories able to be assembled with couplers.

b. Finish. Coated to prevent rust.

c. Materials. Ladder rack shall be manufactured from steel. Stringers will be 3/8″ by 1-1/2″ steel. Cross members will be 1″ x 1/2". Steel elements shall have a minimum 0.065″ wall thickness.

d. Size. Ladder rack shall be nominally sized 12″ width.

e. Accessories. Provide and install the following components and accessories as necessary to achieve the design:

(1) Overhead mounting equipment (thread-rod, unistrut, nuts, washers, etc.) suitable for the supporting structural elements.

(2) Ladder rack section couplers (splices).

(3) Ladder rack radius bends.

(4) Ground strap kits.

PART 3 EXECUTION

3.1 IMPLEMENTION

a. Cable tray shall be used in horizontal applications. Ladder rack is used in vertical applications.

b. Cable tray and ladder rack sections shall be bonded together with bonding conductors and the system bonded to the bonding busbar in the space.

c. Support horizontal cable tray not less than every 5’ (shorter spans if recommended by the equipment manufacturer).

-- End of Section --

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DIVISION 27 - COMMUNICATIONS

SECTION 27 05 53

IDENTIFICATION FOR COMMUNICATIONS SYSTEMS

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1.4 RELATED WORK
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS

PART 2 PRODUCTS

2.1 EQUIPMENT AND COMPONENT LABELS.

PART 3 EXECUTION

3.1 IMPLEMENTION

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SECTION 27 05 53
7/21

IDENTIFICATION FOR COMMUNICATIONS SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies labeling and identification requirements for information technology (IT) equipment cabinets, racks, and cabling.

1.2 SUMMARY

Section Includes:

a. Equipment and component labels.

b. Labeling implementation requirements.

1.3 REFERENCES

VA Infrastructure Standard for Telecommunications Spaces.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-606	(2021d) Administration Standard for Telecommunications Infrastructure

1.4 RELATED WORK

a. Server cabinets and network racks: Section 27 11 16, COMMUNICATIONS CABINETS, RACKS, FRAMES, AND ENCLOSURES.

b. Copper UTP cabling: Section 27 13 13, COMMUNICATIONS COPPER BACKBONE CABLING.

c. Fiber optic cabling: Section 27 13 23, COMMUNICATIONS OPTICAL FIBER BACKBONE CABLING.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Equipment And Component Labels; G

1.6 SUBMITTAL INSTRUCTIONS

Submit in accordance with Section 27 05 11, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS.

PART 2 PRODUCTS

2.1 EQUIPMENT AND COMPONENT LABELS.

Materials. Materials used for labeling equipment, components, and telecommunications cabling shall be appropriate for the installation environment, durable and permanent, and heat-resistant if necessary for labeling in high-temperature environments.

PART 3 EXECUTION

3.1 IMPLEMENTION

a. Governing Standard. Labeling and identification of equipment, components, and telecommunications cabling shall be based on TIA-606, Administration Standard for Telecommunications Infrastructure, and NFPA 70, National Electrical Code, as modified by guidance in the VA Infrastructure Standard for Telecommunications Spaces.

b. All passive infrastructure equipment, components, and telecommunications cabling installed shall be labeled at the time of installation.

c. Telecommunications cable labeling shall include delivery of complete and accurate connection documentation (where each cable terminates at each end) matching the installed labels.

d. Label locations shall be as specified in the VA Infrastructure Standard for Telecommunications Spaces.

-- End of Section --

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DIVISION 27 - COMMUNICATIONS

SECTION 27 08 00.01

DVA/USACE PROJECTS COMMISSIONING OF COMMUNICATIONS SYSTEMS

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PART 3 EXECUTION

3.1 FIELD QUALITY CONTROL
3.2 TRAINING

-- End of Section Table of Contents --

SECTION 27 08 00.01
8/19

DVA/USACE PROJECTS COMMISSIONING OF COMMUNICATIONS SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

a. This section includes requirements for commissioning facility communications systems, related subsystems and related equipment. This Section supplements general requirements specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

b. Complete list of equipment and systems to be commissioned is specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and Specification 27 05 11, REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS.

c. Commissioned Systems:

(1) Commissioning of systems specified in Division 27 and 28 is part of project’s construction process including documentation and proof of performance testing of these systems, as well as training of VA’s Operation and Maintenance personnel in accordance with requirements of Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and Division 27, in cooperation with Government, CxM and CxA.

(2) The facility exterior closure systems commissioning includes communications systems listed in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS and 27 05 11, REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS.

1.2 RELATED WORK

a. System tests: Also, coordinate with the USACE PM for applicable UFGS GENERAL REQUIREMENTS sections..

b. Commissioning process requires review of selected submittals that pertain to systems to be commissioned: UFGS Section 01 33 00 SUBMITTAL PROCEDURES.

c. Construction phase commissioning process and procedures including roles and responsibilities of commissioning team members and user training: Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

1.3 COORDINATION

a. CxM will provide a list of submittals that must be reviewed by CxM and CxA simultaneously with engineering review; do not proceed with work of sections identified without review completion by CxA, CxM and engineering.

b. Commissioning of communications systems require inspection of individual elements of communications system construction throughout

construction period. Coordinate with CxM in accordance with Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and commissioning plan to schedule communications systems inspections as required to support the commissioning process.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Pre-Functional Checklists; G, DO

Contractor's Tests; G, DO

Functional Performance Testing; G, DO

SD-08 Manufacturer's Instructions

Training Agendas And Trainer Resumes; G, DO

1.5 CLOSEOUT SUBMITTAL INSTRUCTIONS

a. Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for submittal requirements for pre-functional checklists, equipment startup reports, and other commissioning documents.

b. Pre-Functional Checklists:

(1) Complete pre-functional checklists provided by CxM to verify systems, subsystems, and equipment installation is complete and systems are ready for Systems Functional Performance Testing.

(2) Submit completed checklists to Contracting Officer, CxM and CxA. CxM and CxA can spot check a sample of completed checklists. If CxM and CxA determine that information provided on the checklist is not accurate, CxM then returns the marked-up checklist to Contractor for correction and resubmission.

(3) If CxM and CxA determine that a significant number of completed checklists for similar equipment are not accurate, CxM in coordination with CxA will select a broader sample of checklists for review.

(4) If CxM and CxA determine that a significant number of broader sample of checklists is also inaccurate, all checklists for the type of equipment will be returned to Contractor for correction and resubmission.

c. Submit training agendas and trainer resumes in accordance with requirements of Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

PART 2 PRODUCTS NOT USED

PART 3 EXECUTION

3.1 FIELD QUALITY CONTROL

a. Contractor's Tests:

(1) Scheduled tests required by other sections of Division 27 must be documented in accordance with Section 01 00 00.01 DVA/USACE Projects GENERAL REQUIREMENTS.

(2) Incorporate all testing into project schedule. Provide minimum seven calendar days' notice of testing for CxM to witness selected Contractor tests at sole discretion of CxM and CxA. The CxA and other Government representative(s) may choose to participate in some or all the testing procedures as observers.

(3) Complete tests prior to scheduling Systems Functional Performance Testing.

b. Systems Functional Performance Testing:

(1) Commissioning process includes Systems Functional Performance Testing that is intended to test systems functional performance under steady state conditions, to test system reaction to changes in operating conditions, and system performance under emergency conditions.

(2) CxM prepares final and detailed Systems Functional Performance Test procedures for review and acceptance by CxA and Contracting Officer.

(3) Provide required labor, materials, and test equipment identified in test procedure to perform tests.

(4) CxM must witness and document the testing.

(a) Provide test reports to CxM and CxA. CxM will sign test reports to verify tests were performed.

3.2 TRAINING

a. Training of Government's operation and maintenance personnel is required in cooperation with Contracting Officer, VA Resident Engineer, and CxM.

b. Provide competent, factory authorized personnel to provide instruction to operation and maintenance personnel concerning location, operation, and troubleshooting of installed systems.

c. Schedule instruction in coordination with Contracting Officer and CxM after submission and approval of formal training plans.

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SECTION 27 10 00

STRUCTURED CABLING

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2.1 [Enter Appropriate Subpart Title Here]
2.2 FACILITY BACKBONE STRUCTURED CABLING.
2.3 FACILITY HORIZONTAL STRUCTURED CABLING.

PART 3 EXECUTION

3.1 IMPLEMENTION

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SECTION 27 10 00
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STRUCTURED CABLING

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies requirements for telecommunications structured cabling systems.

1.2 SUMMARY

Section Includes:

a. Computer Room Backbone Structured Cabling.

b. Computer Room Horizontal Structured Cabling.

c. Facility Backbone Structured Cabling.

d. Facility Horizontal Structured Cabling.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Computer Room Backbone Structured Cabling; G

Computer Room Horizontal Structured Cabling; G

Facility Backbone Structured Cabling; G

Facility Horizontal Structured Cabling; G

1.4 RELATED WORK

a. Cabling labeling and identification: Section 27 05 53, IDENTIFICATION FOR COMMUNICATIONS SYSTEMS.

b. Cable termination locations and equipment: Section 27 11 19, COMMUNICATIONS TERMINATION BLOCKS AND PATCH PANELS.

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Section 27 05 11, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS.

PART 2 PRODUCTS

2.1 [Enter Appropriate Subpart Title Here]Not Used

2.2 FACILITY BACKBONE STRUCTURED CABLING.

Backbone structured cabling (inter-building and intra-building first level backbone) connects the Entrance Rooms to the Main Distribution Areas (MDAs) in the computer room, and connects the Main Distribution Areas to each Telecommunications Room (TR). This specification describes facility backbone structured cabling when the Entrance Rooms and computer room are in the same building, connected entirely by environmentally conditioned pathways (no outside plant pathways are transited).

a. Interior to each telecommunications space, install facility backbone structured cabling in overhead cable tray and fiber raceway systems.

b. All facility backbone structured cabling shall be redundant and follow diverse path routing.

c. Facility backbone cabling shall be field-terminated (fusion spliced).

d. In the Entrance Room, MDA, and TR network cabinets designated for facility backbone distribution, install fiber optic distribution cabinets as required for the quantity of fibers installed.

e. Between each Entrance Room and the computer room:

(1) Provide 2 diversely routed paths of backbone cabling from each Entrance Room. One path will terminate on the A-side MDA and one path will terminate on the B-side MDA.

(2) Install a minimum of 24 strands of multimode fiber optic cabling and 12 strands of single-mode fiber optic cabling per path.

(3) Populate the fiber optic distribution cabinets on each end with a sufficient quantity of multimode OM4 and single-mode OS1 fiber optic splice cassettes per path.

f. Between each MDA and each TR:

(1) No zone/intermediate distribution areas (ZDAs/IDAs) intermediate cross-connects (ICCs) shall be used. All facility backbone cabling between the MDAs and TRs shall be run directly and continuously.

(2) Provide 2 diversely routed paths of backbone cabling, one from the A-side MDA and one from the B-side MDA.

(3) Where the TR is located in the same building as the computer room MDAs (does not transit outside plant pathways) and the path distance is less than 400 meters, install a minimum of 24 strands of multimode fiber optic cabling and 12 strands of single-mode fiber optic cabling per path.

(a) Populate the fiber optic distribution cabinets on each end with a sufficient quantity of multimode OM4 and single-mode OS1 fiber optic splice cassettes per path.

(4) Where the TR is located in a different building as the computer room (transits outside plant pathways) or the path distance is greater than 400m, install a minimum of 24 strands of single-mode fiber optic cabling per path.

(a) Populate the fiber optic distribution cabinets on each end with a sufficient quantity of single-mode OS2 fiber optic splice cassettes per path.

2.3 FACILITY HORIZONTAL STRUCTURED CABLING.

Facility horizontal structured cabling (horizontal distribution) connects the Telecommunications Room (TR) to each end-user Work Area Outlet (WAO).

a. Interior to each TR, install horizontal distribution structured cabling in overhead cable tray systems.

b. All horizontal distribution structured cabling shall have a minimum of 2 cables per WAO.

c. Horizontal distribution structured cabling may be field-terminated.

d. In each TR install sufficient UTP patch panels to support the quantity of WAOs in the TR’s serving zone.

(1) Plan network racks such that patch panels and horizontal cable managers are located in the top 1/3 (RU31-45), switching equipment in the middle 1/3 (RU16-30), and power distribution and other services in the bottom 1/3 (RU1-15).

e. Install a minimum of 2 UTP cables between the TR UTP patch panels and each work area faceplate.

f. Each typical end-user receptacle shall be constructed of a single-gang workbox with one connector chassis, a minimum of two non-proprietary Category 6A 8P8C media interface connectors (RJ-45), and a four-position keystone faceplate.

(1) Modular plug terminated links (MPTL) are permitted to service wireless access points (WAPs), cameras, and other Power over Ethernet (PoE) devices where mating will be very infrequent.

PART 3 EXECUTION

3.1 IMPLEMENTION

a. 100% construction drawing sets shall include drawings detailing the computer room and facility/campus structured cabling systems, including:

(1) Cable media performance categories for each type of run.

(2) Cabling and equipment quantities.

(3) Elevation drawings showing equipment placement in individual enclosures.

(4) Full interconnection diagram for all structured cabling.

(5) Port map and cable label matrices.

b. All cabling and equipment shall be labeled per the requirements of the VA Infrastructure Standard for Telecommunications Spaces.

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SECTION 27 11 00

TELECOMMUNICATIONS ROOM FITTINGS

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SECTION 27 11 00
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TELECOMMUNICATIONS ROOM FITTINGS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

EIA-310-D	Standard 19-in. four-post EIA cabinet or rack, with mounting posts

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-607	(2019d) Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
TIA-606	(2021d) Administration Standard for Telecommunications Infrastructure

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 4	Enclosures for outdoor use where extra protection against moisture and wind driven rain is required

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

1.2 DESCRIPTION

a. This section specifies equipment cabinets, interface enclosures, relay racks, and associated hardware in service provider DEMARC, computer and telecommunications rooms.

b. Telephone system is defined as an Emergency Critical Care Communication System by the National Fire Protection Association (NFPA). Adhere to Seismic reference standards for systems connecting to or extending telephone system and cabling.

1.3 RELATED WORK

a. Wiring devices: Section 26 27 26, WIRING DEVICES.

b. General electrical requirements that are common to more than one section in Division 27: Section 27 05 11, REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS.

c. Requirements for personnel safety and to provide a low impedance path for possible ground fault currents: Section 27 05 26, GROUNDING AND BONDING FOR COMMUNICATIONS SYSTEMS.

d. Lightning protection system: Section 26 41 00, FACILITY LIGHTNING PROTECTION.

e. Conduits for cables and wiring: Section 27 05 33, RACEWAYS AND BOXES FOR COMMUNICATIONS SYSTEMS.

f. Low voltage cabling system infrastructure: Section 27 10 00, CONTROL, COMMUNICATION AND SIGNAL WIRING.

g. Voice communication switching and routing equipment: Section 27 31 00, VOICE COMMUNICATIONS SWITCHING AND ROUTING EQUIPMENT.

h. Extension of a voice communication switching and routing system: Section 27 31 31, VOICE COMMUNICATIONS SWITCHING AND ROUTING EQUIPMENT EXTENSION.

i. Emergency 2-way radio equipment: Section 27 32 41, TWO-WAY RADIO EQUIPMENT AND SYSTEMS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Equipment Technical Literature; G

Equipment Cabinets (Enclosures); G

Environmental Cabinet; G

Wall Mounted Distribution Or System Interface Cabinet; G

Stand Alone Open Equipment Rack; G

Wire Management Equipment; G

SD-05 Design Data

Environmental Requirements; G

Pictorial Layouts; G

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Section 27 05 11, REQUIREMENTS FOR COMMUNICATIONS INSTALLATION.

a. Environmental Requirements: Identify environmental specifications for housing system as initial and expanded system configurations.

(1) Floor loading for batteries and cabinets.

(2) Minimum floor space and ceiling height.

(3) Minimum door size for equipment passage.

b. Separate submittal into sections for each subsystem containing the following:

(1) Pictorial layouts of each Telecommunications Room and Cross Connection Space (VCCS, and HCCS termination cabinets), each distribution cabinet layout, and TCO as each is expected to be installed and configured.

(2) Equipment technical literature detailing electrical and technical characteristics of each item of equipment to be furnished.

PART 2 PRODUCTS

2.1 EQUIPMENT AND MATERIALS

a. Provide components of cabinet system (cabinet, thermal, cable and power management accessories) from a single manufacturer.

b. Equipment Standards and Testing:

(1) Equipment must be listed by a NRTL where a UL standard is in existence; active and passive equipment must conform with each UL standard in effect for equipment, on the submittal date.

(2) Each item of electronic equipment must be labeled by a NRTL that warrants equipment has been tested in accordance with, and conforms to specified standards.

c. Equipment Cabinets (Enclosures):

(1) Fully enclose and physically secure internally mounted and connected, active and passive equipment.

(2) Types of Equipment Enclosures accepted for specific VA Spectrum Management, FMS and OI&T applications in CFM and Facility Projects:

CABINET	FUNCTION
Communications	FMS Special Communications Equipment
Server / Router	OI&T Data/LAN/WAN Equipment
Seismic	Either FMS or OI&T use, specify need
Environmental	Either FMS or OI&T use, specify need

(3) Each cabinet to be:

(a) Provided in each TR at a minimum.

(b) Fabricated with minimum (16 gauge) steel.

(c) Provided with manufacturer's standard painted finish in a color accepted by Contracting Officer with concurrence from FMS Service Chief.

(d) Mounted on floor.

(e) Lockable; tubular locks keyed alike. Provide six keys to Contracting Officer for each cabinet.

(4) Provide equipment mounting shelves; attach to front and rear mounting rails and allowing equipment to be secured to respective mounting rails.

(5) Each enclosure to include:

(a) Floor mounting.

(b) Knock out holes for conduit connections or cable entrance.

(c) Front and rear locking doors; wall mounted cabinets require only front locking door.

(d) Power outlet strips.

(6) Provide quiet ventilation fan with non-disposable locally cleanable air filter.

(7) Size each cabinet in order to contain and maintain internal mounted equipment items.

(8) Provide OEM’s fully assembled unit.

(9) Provide OEM assembled side-by-side enclosures in a single unit, at locations requiring more than two enclosures.

(10) Provide minimum one cabinet with blank rack space, for additional system expansion equipment.

(11) Bond to communications circulating grounding system.

(12) Technical Characteristics:

(a) External:

1. Overall Height:

a) Communications/Server: Maximum (86 inches).

b) Seismic: Maximum (75 inches).

2. Overall Depth:

a) Communications/Server: Maximum (36 inches).

b) Seismic: Maximum (30 inches).

3. Overall Width - All: Maximum (34 inches).

(b) Front Panel Openings:

1. Width:

a) Communications: (19 inches), per EIA.

b) Server: (19 inches), per EIA-310-D.

c) Seismic: (19 inches), per EIA-310-D.

2. Height:

a) Communications/Server: Maximum (78-3/4 inches or 45 Rack Units RU), per EIA-310-D.

b) Seismic: Maximum (66-1/2 inches or 38 RUs), per EIA-310-D.

(c) Heavy Duty Cycle: Maximum (1,200 pounds) capacity.

(d) Certification:

1. NRTL (i.e. UL): For communications and server cabinets.

2. Telcordia Technologies: #63-GR-CORE, (2012) for seismic cabinets.

3. Seismic: Provide cabinet OEM constructed to seismic design category.

(13) Cabinet Internal Components:

(a) AC Power:

1. Standard “Quad AC Box”:

a) Power capacity: 20 Ampere, single phase, 120 VAC continuous duty.

b) Wire gauge: #12 AWG, solid copper, connected to room’s internal AC Power Panel, or as directed by Contracting Officer.

c) Number of AC power outlets: Minimum 4 receptacles.

d) Enclosure: Fully self-contained, metal 4 inch x 4 inches x (2-1/2 inches) with cover

e) Connection: Minimum 1 inch conduit connected to room’s AC Power Panel, or as directed by Contracting Officer

f) Number of boxes: One.

g) Compliance: NRTL (i.e. UL); NFPA 70 (NEC).

(b) AC Outlet Strips:

1. Power Capacity: 15 Ampere, single phase, 120 VAC continuous duty.

2. Wire Gauge: Minimum #12 AWG, solid copper.

3. Number of AC Power Outlets: Minimum 10 “U” grounded.

4. Enclosure: Fully self-contained; typically metal.

5. Connecting Wire: Minimum 6 feet long, with three prong self-grounding AC plug connected to cabinet's internal AC "Quad" box.

6. Number of Strips: 2.

7. Certification: NRTL (i.e. UL).

(c) AC Power Line Surge Protector and Filter Construction:

1. Input Voltage Range: 120 VAC + 15 percent at 50/60 Hz, single phase.

2. Power Service Capacity: 20 AMP, 120 VAC.

3. Voltage Output Regulation: +5.0 percent, instantaneous of input.

4. Circuit Breaker: 15 AMP; may be self-contained.

5. AC Outlets: Minimum four duplex grounded NEMA 5-20R.

6. Response Time: 5.0 nanosecond.

7. Suppression: Isolate and filter any noise, surge spikes

a) Surge: Minimum 20,000 AMP.

b) Noise:

1. Common: -40 dB.

2. Differential: -45 dB.

8. Clamping Voltage: Minimum 300 V.

9. Enclosure: One; self-contained.

10. Mounting: Internal to cabinet floor or on internal mounting rail shelf, allowing two plugs from two plug strips.

11. AC Power Cord: Required; minimum (6 feet), three wire (green ground); minimum #14 AWG stranded.

12. Compliance: NRTL (i.e. UL60950-1).

(d) Uninterruptible Power Supply (UPS): Provide each cabinet with an internal UPS which may be combined with surge protector and filter if system's 50 percent expansion requirement is met. Provide at least one hour continuous full load reserve capacity, in the event of facility primary or emergency AC power failure.

1. UPS to include:

a) On-Off Switch: This function is required to be a part of system's electronic supervision requirements.

b) First/Fast Charge Unit: Must provide clean predicable charge voltage/current. Function is required to be a part of system's electronic supervision requirements.

c) Over Voltage/Current Protect: Cannot short circuit AC power line at any time. This function is required to be a part of system's electronic supervision requirements.

d) Trickle Charge Unit: Must be capable of maintaining a suitable internal battery charge without damaging batteries.

e) Mounting: Provide per OEM's direction.

f) Proper Ventilation: Do not override cabinets' ventilation system.

g) Power Change from AC Input: Accomplish change without interruption to communications link or subsystem being protected. Generate visual and aural alarms in electrical supervision system, local and remote, to annunciating panels via direct connection for trouble indication.

2. Specific requirements for current and surge protection to include:

a) Voltage Protection: Threshold, line to neutral, starts at maximum 200 Volts peak. Transient voltage cannot exceed 330 Volts peak. Furnish documentation on peak clamping Voltage as a function of transient waveform.

b) Peak Power Dissipation: Minimum 35 Joules per phase, as measured for l.0 millisecond at sub branch panels, l00 Joules per phase at branch panels and 300 Joules per phase at service entrance panels. Typically, power dissipation is 12,000 Watts (W) for l.0 mS . Provide explanation of how ratings were measured or empirically derived.

c) Surge Protector (may be combined with On-Off switch of UPS): Must not short circuit AC power line at any time.

1. Components must be minimum silicon semi-conductors.

2. Secondary stages, if used, may include other types of rugged devices.

3. Indicators: Provide visual device indicating surge suppression component is functioning.

4. Electrical Supervision: Required; must be audile and visual, local and remote to annunciating panels via direct connection for trouble indication.

d) Provide current and surge protection on ancillary equipment.

e) Equip each cabinet with the following:

1. Equipment Mounting Rails (Front & Rear): Fully adjustable internal equipment mounting rails allowing front or rear equipment

mounting with pre-drilled EIA-310-D Standard tapped holes. Support entire equipment by supplementary support in addition to face mounting screws on rails.

2. Cabinet Ground: Stainless steel adjustable, lug connected to cabinet's main structure providing an internal cabinet ground for all installed equipment properly bolted to rail and with ground wire connected.

3. Grounding Terminals: A separate mounting hole on equipment mounting rail, with stainless steel connecting bolt bonded by minimum #10 AWG copper wire to cabinet's internal grounding lug.

(14) Ground Interconnection: Bond cabinet's common grounding lug to room's communications circulating ground busbar with a minimum #4 AWG stranded copper wire.

(15) Blank Panels: Provide at every unused rack space.

(a) Match cabinet color.

(b) Provide panels of 1/8 inch thick aluminum with vertical dimensions in increments of one rack unit (RMU) or (1-3/4 inch) with mounting holes spaced to correspond to EIA-310-D Standard 19 inch rack dimensions.

(c) Fill large unused openings with single standard large panel instead of numerous types.

(d) Leave one blank rack space (RMU), covered with a blank panel, between each item of equipment, for minimum internal air flow.

(e) Leave (14 inches)(8.0 RMU) open space, covered with blank cover panel, for additional expansion equipment.

(f) Wire Management: System that connects each item of installed equipment to room wire management system.

(g) Knock-out Holes: Provide for cable entrance/exits via conduits, cable duct/trays.

g. Wire Management Equipment:

(1) Provide an orderly horizontal and vertical interface between outside and inside wires and cables, distribution and interface wires and cables, interconnection wires and cables and associated equipment, jumper cables, and provide an uniform connection media for system fire-retardant wires and cables and other subsystems.

(2) Interface to each cable tray, duct, wireway, or conduit used in the system.

(3) Interconnection or distribution wires and cables must enter system at top (or from a wireway in the floor) via overhead protection system and be uniformly routed down either or both sides at same time, of the frames side protection system, then laterally for termination on rear of each respective terminating assembly.

h. Vertical Cable Managers:

(1) Use same make, style and size of vertical cable manager on rack/frame or in between racks/frames when more than one cable manager is used on a rack/frame or group of racks/frames.

(2) Match color and cover style of racks/frames and cable managers.

i. Horizontal Cable Managers:

(1) Use same make and style of cable manager on rack/frame or racks/frames, when more than one horizontal cable manager is used on a rack/frame or group of racks/frames.

(2) Match color of racks/frames and cable managers.

j. Telecommunication Room (TR): In hostile TR locations, where it has been determined (by Contracting Officer or Facility Chief Engineer) that proper TR climate or external signal radiation cannot be maintained or controlled, provide a minimum of two individual and properly sized self-contained climate controlled equipment cabinet enclosures; one designated for voice, and one designated for data service.

k. Provide gas protection devices on all circuits and cable pairs serving building distribution frames, or in any area served by an unprotected distribution system (maintenance hole, manhole, aerial, etc.).

l. Provide installation hardware when enclosures or racks are attached to structural floor.

m. Provide noise filters and surge protectors for each equipment interface cabinet, switch equipment cabinet, control console, and local and remote active equipment locations to ensure protection from input primary AC power surges so as a consequence noise glitches are not induced into low voltage data circuits.

PART 3 EXECUTION

3.1 PREPARATION

a. Coordinate cabinet installation such that doors fully close and lock, with active and passive equipment installed and connected.

b. Verify equipment dimensions and brackets allow mounting with cabinet doors closed. Front door or rear door of any cabinet that does not close and lock may result in immediate cancellation of inspections or tests.

3.2 INSTALLATION

a. Equipment Cabinets:

(1) Install cabinets in a manner that complies with OEM instructions, requirements of this specification, and in a manner which does not constitute a safety hazard.

(2) Provide weatherproof equipment installed outdoors or install in NEMA 3S rated enclosures with hinged doors and locks with two keys.

(3) Install equipment indoors in NEMA 4 rated metal cabinets with hinged doors and locks with two keys.

b. Grounding:

(1) Bond equipment, including identified Government furnished equipment, to ground so total ground resistance measures maximum 0.1 Ohm.

(a) Install lightning arrestors and grounding in accordance with NFPA.

(b) Install gas protection devices at nearest point of entrance in buildings where protection is required and on same circuits as MDF in telephone switch room.

(c) Do not use AC neutral, including in power panel or receptacle outlet, for system control, subcarrier or audio reference ground.

(d) Use of conduit, signal duct or cable trays as system or electrical ground is not permitted.

(2) Connect each equipment grounding terminal to a separate mounting hole on equipment mounting rail, to right as one looks at it from rear, with a minimum #12 AWG stranded copper wire with protective green jacket.

(3) Extend common ground bus of minimum #10 AWG solid copper wire throughout each equipment cabinet and bond to TGB. Provide a separate isolated ground connection from each equipment cabinet ground bus to system ground. Do not tie equipment ground buses together.

(4) Bond equipment to cabinet bus with copper braid equivalent to #12 AWG. Self-grounding equipment enclosures, racks or cabinets, that provide OEM certified functional ground connections through physical contact with installed equipment, are acceptable alternatives.

(5) Bond cable shields to cabinet ground bus with minimum #12 AWG stranded copper wire at only one end of cable run. Insulate cable shields from each other, faceplates, equipment racks, consoles, enclosures or cabinets, except at system common ground point. Bond coaxial and audio cables only at source; in all cases, keep cable shield ground connections to a minimum.

c. Equipment Assembly:

(1) Cabinets:

(a) Install and adjust cabinet/frame accessories to position, including thermal management accessories, vertical cable managers, vertical power managers and equipment-mounting rails, using manufacturer’s installation instructions prior to baying or placing cabinet for attachment to building and before installing any rack-mount equipment into cabinet. Shelves, horizontal cable managers and filler panels (rack-mount accessories), if used, may be installed after cabinet is placed.

(b) When used in a multi-cabinet bay, attach cabinets side-by-side using baying kits according to manufacturer’s instructions.

(c) Attach overhead ladder rack or cable tray to ceiling or top of cabinet. Maintain minimum 3 inches clearance between top of cabinet and bottom of ladder rack/cable tray. Position ladder rack/cable tray so that it does not interfere with hot air exhaust through cabinet’s top panel. Use radius drops where cable enters or exits ladder rack/cable tray.

(d) In seismic areas, install additional bracing as required by building codes and recommendations of a licensed structural engineer.

(e) Install ladder rack with side stringers facing rack or cabinet so that ladder forms an inverted U-shape and so that welds between stringers (sides) and cross members (middle) face away from cables.

(f) Secure ladder rack to tops of equipment racks or cabinets using manufacturer’s recommended supports and appropriate hardware.

(g) Attach bonding conductor sized per TIA-607 between telecommunications grounding busbar and cabinet. Attach bonding conductor to cabinet using a ground terminal block according to manufacturer’s installation instructions.

(h) Provide bonding conductor and other hardware required to make connections between cabinet and telecommunications grounding busbar.

(i) Install rack mounted equipment normally requiring adjustment or observation so operational adjustments can be conveniently made.

(j) Mount heavy equipment with rack slides or rails to allow servicing from front of enclosure. Provide support in addition to front panel mounting screws for heavy equipment.

(k) Provide with cable slack to permit servicing by removal of installed equipment from front of enclosure.

(l) Install color-matched blank panel spacer 1.75 inches high between each piece of active and passive equipment to ensure adequate air circulation for efficient equipment cooling and air ventilation.

(m) Provide quiet fans and non-disposable air filters at each console or cabinet.

(n) Install enclosures and racks plumb and square, permanently attached to building structure and held in place.

(o) Provide 15 inches of front vertical space opening for additional equipment.

(p) Install equipment located indoors in metal racks or enclosures with hinged doors to allow access for maintenance without causing interference to other nearby equipment.

(q) Cables must enter equipment racks or enclosures in such a manner to allow doors or access panels to open and close without disturbing or damaging cables.

(r) Mount distribution hardware in a manner that allows access to connections for testing and provides room for doors or access panels to open and close without disturbing the cables.

(2) Racks:

(a) Assemble racks according to manufacturer’s instructions.

(b) Verify that equipment mounting rails are sized properly for rack-mount equipment before attaching rack to floor.

(c) Attach assembled racks to floor in four places using appropriate floor mounting anchors. When placed over a raised floor, threaded rods should pass through raised floor tile and be secured in structural floor below.

(d) Bond racks to telecommunications grounding busbar using appropriate hardware provided by contractor.

(e) In seismic areas, install additional bracing as required by building codes and recommendations of a licensed structural engineer.

(f) Ladder rack may be attached to top of rack to deliver cables to rack. Do not drill rack to attach; use appropriate hardware from rack manufacturer.

(g) Provide radius drops to guide cable where cable exits or enters side of overhead ladder rack to access a rack, frame, cabinet or wall-mounted rack, cabinet or termination field.

(h) Evenly distribute equipment load on rack. Place large and heavy equipment towards bottom of rack. Secure equipment to rack with equipment mounting screws. In seismic areas, secure equipment to shelves with additional bracing.

(3) Vertical Cable Managers:

(a) Provide vertical managers so number of cables in each manager does not exceed OEM fill capacity.

(b) Attach vertical cable managers to side of rack/frame using manufacturer’s installation instructions and hardware.

(c) Attach vertical cable manager to both racks/frames when a single vertical cable manager is used between two racks/frames.

(d) Dress cables through openings in between T-shaped guides on manager so that cables make gradual bends as they exit or enter cable manager into rack-mount space (RMU). Do not twist, coil or make sharp bends in cables.

(e) Attach doors to cable manager in closed position after cabling is complete.

(4) Horizontal Cable Managers:

(a) Attach horizontal cable managers to rack/frame with minimum four screws according to manufacturer’s installation instructions. Center each cable manager within allocated rack-mount space (RMU).

(b) Provide horizontal managers located so number of cables each manager supports is less than cable manager’s cable fill capacity.

(c) Dress cables through openings in between T-shaped guides on cable manager so that cables make gradual bends as they exit or enter cable manager into rack-mount space (RMU). Do not twist, coil or make sharp bends in cables.

(d) Attach covers to cable manager in closed position after cabling is complete.

d. Labeling: Permanently label each enclosure in accordance with TIA-606 using thermal ink transfer process; handwritten labels are not acceptable.

(1) Equipment: Label system equipment with contrasting plastic laminate or bakelite material on face of unit corresponding to its source.

(2) Conduit, Cable Duct, and/or Cable Tray: Label conduit, duct and tray, including utilized GFE, with permanent marking devices or spray painted stenciling a minimum of (10 feet), identifying system.

-- End of Section --

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DIVISION 27 - COMMUNICATIONS

SECTION 27 11 16

COMMUNICATIONS CABINETS, RACKS, FRAMES, AND ENCLOSURES

PART 1 GENERAL

1.1 DESCRIPTION
1.2 SUMMARY
1.3 REFERENCES
1.4 SUBMITTALS
1.5 RELATED WORK
1.6 SUBMITTAL INSTRUCTIONS

PART 2 PRODUCTS

2.1 [Enter Appropriate Subpart Title Here]
2.2 NETWORK CABINETS.
2.3 [Enter Appropriate Subpart Title Here]

PART 3 EXECUTION

3.1 INSTALLATION

-- End of Section Table of Contents --

SECTION 27 11 16

COMMUNICATIONS CABINETS, RACKS, FRAMES, AND ENCLOSURES

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies Information Technology (IT) equipment enclosures for use in VA telecommunications spaces.

1.2 SUMMARY

Section Includes:

c. Network Cabinets.

d. Telecommunications Enclosures.

e. Pathway Racks.

1.3 REFERENCES

VA Infrastructure Standard for Telecommunications Spaces.

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

EIA-310-D	Standard 19-in. four-post EIA cabinet or rack, with mounting posts

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Network Cabinets; G

Telecommunications Enclosures (Te); G

Pathway Racks; G

1.5 RELATED WORK

a. Cabling termination equipment, rack-mounted: Section 27 11 19, COMMUNICATIONS TERMINATION BLOCKS AND PATCH PANELS.

b. Grounding and bonding equipment: Section 27 05 26, GROUNDING AND BONDING FOR COMMUNICATIONS SYSTEMS.

c. Labeling and identification requirements: Section 27 05 53, IDENTIFICATION FOR COMMUNICATIONS SYSTEMS.

1.6 SUBMITTAL INSTRUCTIONS

Submit in accordance with Section 27 05 00, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS.

PART 2 PRODUCTS

2.1 [Enter Appropriate Subpart Title Here]Not Used
2.2 NETWORK CABINETS.

a. Height. Cabinets shall provide 45 standard rack units (RU) of space (45U) for mounting equipment.

b. Dimensions. Nominal dimensions for server cabinets are 84″ tall, 31.5″ wide, and 48″ deep, including side panels and doors. Minimum depth is 46″ . Maximum width is 24″ .

c. Doors and Panels.

(1) Front Door. Provide a single locking, latched, hinged metal front door for the server cabinet with minimum 75% open perforated design for airflow.

(2) Rear Door. Provide a single locking, latched, hinged metal rear door for the server cabinet.

(a) If a vertical exhaust duct (VED, “chimney”) is specified for the server cabinet, the rear door shall be solid.

(b) If no VED is specified, the rear door shall be perforated with a minimum of 75% open design for airflow.

(3) Side Panels. Provide removable, locking solid metal side panels. Side panels shall have cable pass-through cut-outs protected with brushed grommets on each side, aligned on each side to allow Rack PDU and communications cables to pass between enclosures to Zone PDUs.

d. Locks. All locks on a server cabinet shall be keyed the same. Provide two keys per cabinet.

e. Color. The cabinet and all components shall be white in color.

f. Equipment Mounting Rack. The cabinet shall provide an EIA-310-D compliant 19″ racking system with front and rear rails using standard 3/8″ x 3/8″ square hole mounting.

g. Components. The cabinet shall be supplied with the following standard equipment:

(a) Air dams for left and right of the front rails.

(b) Bottom panel with brushed grommets.

(c) Top panel with brushed grommets.

(d) Toolless mounting brackets for vertical Rack PDUs.

(e) 45U of 1RU blanking panels.

(f) Cable management accessories as necessary to maintain cable bend radii, uniformity, and neatness (e.g. d-rings, spools, bend radius limiters, finger cable managers, waterfalls, etc.).

2.3 [Enter Appropriate Subpart Title Here]Not Used

PART 3 EXECUTION

3.1 INSTALLATION

a. Where to be installed in structures assigned to Seismic Design Category C, D, E, or F, seismic bracing equivalent to that for other storage racks, SEISMIC RESTRAINT REQUIREMENTS FOR NON-STRUCTURAL COMPONENTS section 3.7 (“Install… to withstand earthquake forces and anchored to the floor or laterally braced from the top to the structural elements”) is required.

b. Casters on floor-mounted enclosures shall be retracted and the enclosures leveled and stabilized.

-- End of Section --

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DIVISION 27 - COMMUNICATIONS

SECTION 27 11 19

COMMUNICATIONS TERMINATION BLOCKS AND PATCH PANELS

PART 1 GENERAL

1.1 DESCRIPTION
1.2 SUMMARY
1.3 RELATED WORK
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS

PART 2 PRODUCTS

2.1 COPPER (UTP) PATCH PANELS.
2.2 FIBER OPTIC DISTRIBUTION CABINETS.
2.3 FIBER OPTIC DISTRIBUTION CASSETTES.
2.4 FIBER OPTIC SPLICE CASSETTES.
2.5 HORIZONTAL CABLE MANAGERS.
2.6 BLANKING PANELS.

PART 3 EXECUTION

3.1 INSTALLATION

-- End of Section Table of Contents --

SECTION 27 11 19

COMMUNICATIONS TERMINATION BLOCKS AND PATCH PANELS

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies cable termination and management equipment to be mounted in information technology (IT) equipment cabinets and racks.

1.2 SUMMARY

Section Includes:

a. Copper UTP Patch Panels.

b. Fiber Optic Distribution Cabinets.

c. Fiber Optic Distribution Cassettes.

d. Fiber Optic Splice Cassettes.

e. Horizontal Cable Managers.

f. Blanking Panels.

1.3 RELATED WORK

a. Server cabinets and network racks: Section 27 11 16, COMMUNICATIONS CABINETS, RACKS, FRAMES, AND ENCLOSURES.

b. Labeling and identification requirements: Section 27 05 53, IDENTIFICATION FOR COMMUNICATIONS SYSTEMS.

c. Copper UTP cabling: Section 27 13 13, COMMUNICATIONS COPPER BACKBONE CABLING.

d. Copper UTP cabling termination: Section 27 13 13.13, COMMUNICATIONS COPPER CABLE SPLICING AND TERMINATIONS.

e. Fiber optic cabling: Section 27 13 23, COMMUNICATIONS OPTICAL FIBER BACKBONE CABLING.

f. Fiber optic cabling termination: Section 27 13 23.13, COMMUNICATIONS OPTICAL FIBER SPLICING AND TERMINATIONS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Copper (Utp) Patch Panels; G

Fiber Optic Distribution Cabinets; G

Fiber Optic Distribution Cassettes; G

Fiber Optic Splice Cassettes; G

Horizontal Cable Managers; G

Blanking Panels; G

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Section 27 05 11, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS.

PART 2 PRODUCTS

2.1 COPPER (UTP) PATCH PANELS.

a. Form Factor. 1U, 19″ rack-mountable. Default form factor for patch panels shall be angled to extend not less than 4″ forward from the mounting rail.

b. Capacity. 48 copper 8P8C ports in groupings of 6.

c. Interface. Copper (UTP) patch panels shall accept pre-terminated single 8P8C keystone jacks and 6-jack cassettes.

d. Accessories. Provide each patch panel installation with the following accessories.

(1) Angled panel cover. Specify angled panel covers at the top and bottom of each stack of angled patch panels. Covers shall mount to the same RU rail positions as the top and bottom patch panels.

(2) Rear cable manager tray. Specify one rear cable manager tray per patch panel. Trays shall mount to the same RU rail positions as the patch panel.

2.2 FIBER OPTIC DISTRIBUTION CABINETS.

a. Form Factor. 1U, 19″ rack-mountable, angled to extend not less than 4″ forward from the mounting rail.

b. Capacity. 144 fibers in a variety of distribution and/or splice cassettes.

c. Door. Cabinets shall be supplied with a locking front cover.

d. Accessories. Provide each distribution cabinet with the following accessories.

(1) Rear cable manager. Specify one rear cable manager with fiber storage rings per distribution cabinet. Trays shall mount to the same RU rail positions as the distribution cabinet.

(2) 0RU patch cord managers. Specify one pair of patch cord managers per distribution cabinet. Managers shall mount to the same RU

rail positions as the distribution cabinet and rear cable manager.

2.3 FIBER OPTIC DISTRIBUTION CASSETTES.

Fiber optic distribution cassettes are used in structured cabling in the computer room to connect Main Distribution Areas (MDAs), Horizontal Distribution Areas (HDAs), and Equipment Distributors (EDs) where pre-terminated cable is used.

a. Form Factor. Designed to install in rack-mounted fiber optic distribution cabinet.

b. Connections. Adapters on the front of the cassette shall accept 6 duplex Lucent connector (LC) fiber pairs.

2.4 FIBER OPTIC SPLICE CASSETTES.

Fiber optic splice cassettes are used in backbone structured cabling connecting the Entrance Rooms to the Main Distribution Areas (MDAs) in the computer room, and the MDAs to Telecommunications Rooms (TRs) where field-terminated (fusion spliced) cable is used.

a. Form Factor. Designed to install in rack-mounted fiber optic distribution cabinet.

b. Connections. Adapters on the front of the cassette shall accept 6 duplex Lucent connector (LC) fiber pairs.

c. Design. Splice cassette modules shall be self-contained to integrate fiber adapter bulkhead and splice holders. No additional splice trays shall be necessary. Each module shall provide 24 fiber splice connections.

d. Fiber Type. Provide multimode OM4 and single-mode OS2 splice cassettes as required by the quantity of each type of fiber installed.

2.5 HORIZONTAL CABLE MANAGERS.

a. Form Factor. 1U, 19” rack-mountable.

b. Door. Horizontal cable managers shall have a double-hinged front cover to protect installed cables.

c. Fingers. 6-port spacing (five fingers).

2.6 BLANKING PANELS.

a. Form Factor. 1U, 19” rack-mountable, flat.

c. Quantity. Provide blanking panels to fill 75% of all RUs in all server cabinets and network cabinets.

PART 3 EXECUTION

3.1 INSTALLATION

Not used.

-- End of Section --

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DIVISION 27 - COMMUNICATIONS

SECTION 27 13 13.13

COMMUNICATIONS COPPER CABLE SPLICING AND TERMINATIONS

PART 1 GENERAL

1.1 DESCRIPTION
1.2 SUMMARY
1.3 REFERENCES
1.4 RELATED WORK
1.5 SUBMITTALS

PART 2 PRODUCTS

PART 3 EXECUTION

3.1 TERMINATION REQUIREMENTS.
3.2 SPLICING.

-- End of Section Table of Contents --

SECTION 27 13 13.13

COMMUNICATIONS COPPER CABLE SPLICING AND TERMINATIONS

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies requirements for terminations of copper (UTP) cabling.

1.2 SUMMARY

Section Includes:

a. Termination Requirements.

b. Splicing.

1.3 REFERENCES

VA Infrastructure Standard for Telecommunications Spaces.

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-568.1	(2020e) Commercial Building Telecommunications Infrastructure Standard

1.4 RELATED WORK

a. Termination equipment: Section 27 11 19, COMMUNICATIONS TERMINATION BLOCKS AND PATCH PANELS.

b. Copper (UTP) cabling requirements: Section 27 13 13, COMMUNICATIONS COPPER BACKBONE CABLING.

c. Copper (UTP) cabling requirements: Section 27 15 13, COMMUNICATIONS COPPER HORIZONTAL CABLING.

1.5 SUBMITTALS

Submit in accordance with Section 27 05 11, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 TERMINATION REQUIREMENTS.

a. Terminate copper (UTP) cabling used for data communications per TIA-568.1.

b. Connections shall be made to equipment certified to performance category 6a.

c. Pre-terminated connections are preferred for computer room structured cabling applications. Field termination for non-computer room applications is acceptable.

3.2 SPLICING.

Splicing of copper (UTP) cabling is prohibited.

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DIVISION 27 - COMMUNICATIONS

SECTION 27 13 13

COMMUNICATIONS COPPER BACKBONE CABLING

PART 1 GENERAL

1.1 DESCRIPTION
1.2 SUMMARY
1.3 REFERENCES
1.4 RELATED WORK
1.5 SUBMITTALS
1.6 SUBMITTAL INSTRUCTIONS

PART 2 PRODUCTS

2.1 COPPER (UTP) BACKBONE CABLES.

PART 3 EXECUTION

3.1 IMPLEMENTATION.

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SECTION 27 13 13

COMMUNICATIONS COPPER BACKBONE CABLING

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies requirements for telecommunications copper (UTP) backbone and horizontal cabling in structured cabling systems in computer rooms.

1.2 SUMMARY

Section Includes:

Copper (UTP) Backbone Cables.

1.3 REFERENCES

VA Infrastructure Standard for Telecommunications Spaces.

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-568.2	(2018d) Balanced Twisted-Pair Telecommunications Cabling and Components Standards
TIA-568.1	(2020e) Commercial Building Telecommunications Infrastructure Standard
TIA/TSB 155	(2010) Guidelines for the Assessment and Mitigation of Installed Category 6 Cabling to Support 10GBASE-T

1.4 RELATED WORK

a. Computer room structured cabling: Section 27 10 00, STRUCTURED CABLING.

b. Termination equipment: Section 27 11 19, COMMUNICATIONS TERMINATION BLOCKS AND PATCH PANELS.

b. Cable labeling requirements: Section 27 05 53, IDENTIFICATION FOR COMMUNICATIONS SYSTEMS.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Copper (Utp) Backbone Cables; G

1.6 SUBMITTAL INSTRUCTIONS

Submit in accordance with Section 27 05 11, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS.

PART 2 PRODUCTS

2.1 COPPER (UTP) BACKBONE CABLES.

Copper (UTP) backbone and horizontal cables are used in computer room structured cabling to connect Main Distribution Areas (MDAs), Horizontal Distribution Areas (HDAs), and server cabinet Equipment Distributors (EDs).

a. Performance Category. Copper (UTP) backbone cables shall meet the Category 6A performance requirements. Category 6A connectors are required.

b. Performance Specifications. Meets or exceeds TIA-568.2 and TIA/TSB 155.

c. Limited Power (LP) Certification. UL Listed as x-LP (0.5A).

d. Termination. Factory pre-terminated TIA-568.1.

e. Testing. Factory certified to meet performance category requirements per TIA-568.2.

f. Conductor Size. 22-24AWG.

g. Jacket Rating. Communications Multipurpose Cable, Plenum (CMP) shall be specified if any portion of the cable passes through an NEC-defined plenum. Communications Multipurpose Cable, Riser (CMR) shall be specified for all other applications. CMP may be used as a substitute for CMR.

h. Bundling and Construction. Copper (UTP) backbone cables shall be provided in 6-cable bundles with individual 8P8C media interface connectors (RJ-45 ‘jacks’) or 6-jack cassettes designed to fit in copper (UTP) patch panels.

i. Color. A-side 8P8C media interface connectors (RJ-45) shall be blue. B-side 8P8C connectors shall be yellow.

j. Length. Backbone cabling used shall be procured to the specific length required by the design (horizontal and vertical paths) with no more than 2 feet of excess length.

PART 3 EXECUTION

3.1 IMPLEMENTATION.

a. A. All cabling and equipment shall be labeled per the requirements of the VA Infrastructure Standard for Telecommunications Spaces.

b. Install first level backbone and horizontal structured cabling in the computer room in cable tray and fiber raceway systems following diverse path routing.

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DIVISION 27 - COMMUNICATIONS

SECTION 27 13 23.13

COMMUNICATIONS OPTICAL FIBER SPLICING AND TERMINATIONS

PART 1 GENERAL

1.1 DESCRIPTION
1.2 SUMMARY
1.3 RELATED WORK
1.4 SUBMITTALS

PART 2 PRODUCTS

PART 3 EXECUTION

3.1 TERMINATION REQUIREMENTS.

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SECTION 27 13 23.13

COMMUNICATIONS OPTICAL FIBER SPLICING AND TERMINATIONS

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies requirements for terminations of fiber optic cabling.

1.2 SUMMARY

Section Includes:

Termination Requirements.

1.3 RELATED WORK

a. Termination equipment: Section 27 11 19, COMMUNICATIONS TERMINATION BLOCKS AND PATCH PANELS.

b. Fiber optic cabling requirements: Section 27 13 23, COMMUNICATIONS OPTICAL FIBER BACKBONE CABLING.

1.4 SUBMITTALS

Submit in accordance with Section 27 05 11, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 TERMINATION REQUIREMENTS.

a. Terminate fiber optic cabling with duplex Lucent connector (LC) adapters as required for the application.

b. Pre-terminated connections are preferred for computer room structured cabling applications. Field termination (fusion splicing) for non-computer room applications where cable lengths cannot be determined in advance is acceptable.

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DIVISION 27 - COMMUNICATIONS

SECTION 27 13 23

COMMUNICATIONS OPTICAL FIBER BACKBONE CABLING

PART 1 GENERAL

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1.2 SUMMARY
1.3 RELATED WORK
1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS

PART 2 PRODUCTS

2.1 MULTIMODE OPTICAL FIBER BACKBONE CABLES.
2.2 SINGLE MODE OPTICAL FIBER BACKBONE CABLES.
2.3 HYBRID OPTICAL FIBER BACKBONE CABLES.

PART 3 EXECUTION

3.1 IMPLEMENTATION.

-- End of Section Table of Contents --

SECTION 27 13 23

COMMUNICATIONS OPTICAL FIBER BACKBONE CABLING

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies requirements for telecommunications optical fiber backbone cabling.

1.2 SUMMARY

Section Includes:

a. Multimode Optical Fiber Backbone Cables.

b. Single Mode Optical Fiber Backbone Cables.

c. Hybrid Optical Fiber Backbone Cables.

1.3 RELATED WORK

a. Computer room structured cabling: Section 27 10 00, STRUCTURED CABLING.

b. Termination equipment: Section 27 11 19, COMMUNICATIONS TERMINATION BLOCKS AND PATCH PANELS.

b. Cable labeling requirements: Section 27 05 53, IDENTIFICATION FOR COMMUNICATIONS SYSTEMS.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Multimode Optical Fiber Backbone Cables; G

Single Mode Optical Fiber Backbone Cables; G

Hybrid Optical Fiber Backbone Cables; G

1.5 SUBMITTAL INSTRUCTIONS

Submit in accordance with Section 27 05 11, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS.

PART 2 PRODUCTS

2.1 MULTIMODE OPTICAL FIBER BACKBONE CABLES.

Multimode fiber optic backbone cables are specified for inside plant use,

for data center backbone use as well as for intrabuilding campus backbone.

a. Performance Category. Laser-optimized OM4 or OM5 cables are specified.

b. Testing. All fiber optic backbone media shall pass all Tier 1 requirements (attenuation with an Optical Loss Test Set or OLTS, verification of cable length and polarity) and all Tier 2 requirements (characterization by an Optical Time Domain Reflectometer or OTDR resulting in indication of the uniformity of cable attenuation and connector insertion loss).

c. Termination. For computer room first level backbone usage, factory pre-terminated cables with MPO connectors are required. Fiber will be field terminated (fusion spliced) when used to connect the computer room to Telecommunications Rooms (TRs) and other telecommunications spaces.

(1) Polarity. For pre-terminated MPO-MPO cables, the polarity of the cables shall be matched to the type of fiber optic distribution cassettes specified.

d. Jacket Color.

(1) OM4. Aqua.

(2) OM5. Lime.

e. Jacket Rating. Communications Multipurpose Cable, Plenum (CMP) shall be specified if any portion of the cable passes through an NEC-defined plenum. Communications Multipurpose Cable, Riser (CMR) shall be specified for all other applications. CMP may be used as a substitute for CMR.

f. Bundling and Construction. Bundles consisting of multiples of sub-bundles (subunits) of 24 strands of fiber are specified. Cables shall be dielectric and tight-buffered. Construction shall match the rear entry requirements of the fiber optic distribution or splice cassette such that the sub-bundle (subunit) remains jacketed into the cassette.

g. Length. Backbone cabling for computer room first level backbone use (pre-terminated) shall be procured to the specific length required by the design (horizontal and vertical paths) with no more than 1 meter of excess length on each end.

2.2 SINGLE MODE OPTICAL FIBER BACKBONE CABLES.

Single mode fiber optic backbone cables are specified for inside and outside plant use, for campus backbone (interbuilding and intrabuilding) use.

a. Performance Category. OS1 is specified for inside plant applications. Inside/outside rated OS2 is specified for applications where the pathway transits an outside plant path.

b. Testing. All fiber optic backbone media shall pass all Tier 1 requirements (attenuation with an Optical Loss Test Set or OLTS, verification of cable length and polarity) and all Tier 2 requirements (characterization by an Optical Time Domain Reflectometer or OTDR

resulting in indication of the uniformity of cable attenuation and connector insertion loss).

c. Termination. Fiber will be field terminated (fusion spliced) when used to connect the computer room to Telecommunications Rooms (TRs) and other telecommunications spaces.

d. Jacket Color. Yellow.

e. Jacket Rating. Communications Multipurpose Cable, Plenum (CMP) shall be specified if any portion of the cable passes through an NEC-defined plenum. Communications Multipurpose Cable, Riser (CMR) shall be specified for all other applications. CMP may be used as a substitute for CMR.

f. Bundling and Construction. Bundles consisting of multiples of sub-bundles (subunits) of 24 strands of fiber are specified. Cables shall be dielectric and tight-buffered for OS1, and loose-tube gel-filled for OS2.

2.3 HYBRID OPTICAL FIBER BACKBONE CABLES.

Hybrid cables containing multiple 24-strand bundles (subunits) of multimode and/or single mode fiber are acceptable.

PART 3 EXECUTION

3.1 IMPLEMENTATION.

a. A. All cabling and equipment shall be labeled per the requirements of the VA Infrastructure Standard for Telecommunications Spaces.

b. Interior to telecommunications spaces, install fiber optic backbone cabling in overhead cable tray and fiber raceway systems.

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DIVISION 27 - COMMUNICATIONS

SECTION 27 15 00

COMMUNICATIONS STRUCTURED CABLING

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PART 1 GENERAL

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1.4 SUBMITTALS
1.5 SUBMITTAL INSTRUCTIONS
1.6 WARRANTY

PART 2 PRODUCTS

2.1 PERFORMANCE AND DESIGN CRITERIA
2.2 EQUIPMENT AND MATERIALS
2.3 DISTRIBUTION EQUIPMENT AND SYSTEMS

PART 3 EXECUTION

3.1 INSTALLATION
3.2 FIELD QUALITY CONTROL
3.3 MAINTENANCE

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SECTION 27 15 00
1/16

COMMUNICATIONS STRUCTURED CABLING

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 99	(2024) Health Care Facilities Code
NFPA 101	(2024) Life Safety Code

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-526-7	(2015a; R 2022) Measurement of Optical Power Loss of Installed Single-Mode Fiber Cable Plant, Adoption of IEC 61280-4-2 edition 2: Fibre-Optic Communications Subsystem Test Procedures - Part 4-2: Installed Cable Plant - Single-Mode Attenuation and Optical Return Loss Measurement
TIA-526-14	(2023d) OFSTP-14A Optical Power Loss Measurements of Installed Multimode Fiber Cable Plant
TIA-568.0	(2020e) Generic Telecommunications Cabling for Customer Premises
TIA-568.1	(2020e) Commercial Building Telecommunications Infrastructure Standard
TIA-568.2	(2018d) Balanced Twisted-Pair Telecommunications Cabling and Components Standards
TIA-568.3	(2022e) Optical Fiber Cabling Components Standard
TIA-606	(2021d) Administration Standard for Telecommunications Infrastructure
TIA/EIA-455	(1998b) Standard Test Procedure for Fiber Optic Fibers, Cables, Transducers, Sensors, Connecting and Terminating Devices, and Other Fiber Optic Components

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

EIA/ECA 310-E	Cabinets, Racks, Panels, and Associated Equipment

1.2 DESCRIPTION

This section specifies a complete and operating voice and digital structured cabling distribution system and associated equipment and hardware to be installed in VA Medical Center New Chiller Plant, here-in-after referred to as the “facility”.

1.3 RELATED WORK

a. Wiring devices: Section 26 27 26, WIRING DEVICES.

b. Lightning protection system: Section 26 41 00, FACILITY LIGHTNING PROTECTION.

c. General electrical requirements that are common to more than one section in Division 27: Section 27 05 11, REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS.

d. Requirements for personnel safety and to provide a low impedance path for possible ground fault currents: Section 27 05 26, GROUNDING AND BONDING FOR COMMUNICATIONS SYSTEMS.

e. Conduits for cables and wiring: Section 27 05 33, RACEWAYS AND BOXES FOR COMMUNICATIONS SYSTEMS.

f. Low voltage cabling system infrastructure: Section 27 10 00, CONTROL, COMMUNICATION AND SIGNAL WIRING.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

List Of Test Equipment; G

SD-02 Shop Drawings

Pictorial Layout Drawing; G

SD-07 Certificates

Certification; G

SD-11 Closeout Submittals

Closeout Submittal; G

1.5 SUBMITTAL INSTRUCTIONS

a. In addition to requirements of Section 27 05 11, REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS provide:

(1) Pictorial layout drawing of each telecommunications room, showing termination cabinets, each distribution cabinet and rack, as each is expected to be installed and configured.

(2) List of test equipment as per 27 05 11, REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS.

b. Certification:

(1) Submit written certification from OEM indicating that proposed supervisor of installation and proposed provider of contract maintenance are authorized representatives of OEM. Include individual's legal name and address and OEM warranty credentials in the certification.

(2) Pre-acceptance Certification: Submit in accordance with test procedures.

(3) Test system cables and certify to Contracting Officer before proof of performance testing can be conducted. Identify each cable as labeled on as-installed drawings.

(4) Provide current and qualified test equipment OEM training certificates and product OEM installation certification for contractor installation, maintenance, and supervisory personnel.

c. Closeout Submittal: Provide document from OEM certifying that each item of equipment installed conforms to OEM published specifications.

1.6 WARRANTY

Work subject to terms of Article "Warranty of Construction," FAR clause 52.246-21.

PART 2 PRODUCTS

2.1 PERFORMANCE AND DESIGN CRITERIA

a. Provide complete system including “punch down” and cross-connector blocks voice and data distribution sub-systems, and associated hardware including telecommunications outlets (TCO); copper and fiber optic distribution cables, connectors, “patch” cables, “break out” devices and equipment cabinets, interface cabinets, and radio relay equipment rack.

b. Industry Standards:

(1) Cable distribution systems provided under this section are connected to systems identified as critical care performing life support functions.

(2) Conform to National and Local Life Safety Codes (whichever are more stringent), NFPA, NEC, this section, Joint Commission Life Safety Accreditation requirements, and OEM recommendations,

instructions, and guidelines.

(3) Provide supplies and materials listed by a nationally recognized testing laboratory where such standards are established for supplies, materials or equipment.

(4) Refer to industry standards and minimum requirements of Section 27 05 11, REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS and guidelines listed.

(5) Active and passive equipment required by system design and approved technical submittal; must conform to each UL standard in effect for equipment, when technical submittal was reviewed and approved by Government or date when Contracting Officer accepted system equipment to be replaced. Where a UL standard is in existence for equipment to be used in completion of this contract, equipment must bear approved NRTL label.

c. System Performance: Provide complete system to meet or exceed TIA-568.2 Category 6A requirements.

d. Provide continuous inter- and/or intra-facility voice, data, and analog service.

(1) Provide voice and data cable distribution system based on a physical “Star" topology.

(2) Provide separate cable distribution system for emergency, safety and protection systems (i.e. emergency bypass phones; police emergency voice communications from parking lots and stairwells personal protection, duress alarms and annunciation systems; etc.)

(3) Contact SMCS 005OP2H3 (202-462-5310) for specific technical assistance and approvals.

e. Specific Subsystem Requirements: Provide products necessary for a complete and functional voice, data, analog and videotele communications cabling system, including backbone cabling system, patch panels and cross-connections, horizontal cabling systems, jacks, faceplates, and patch cords.

f. Coordinate size and type of conduit, pathways and firestopping for maximum 40 percent cable fill with subcontractors.

g. Terminate all interconnecting twisted pair, fiber-optic or coaxial cables on patch panels or punch blocks. Terminate unused or spare conductors and fiber strands. Do not leave unused or spare twisted pair wire, fiber-optic or coaxial cable unterminated, unconnected, loose or unsecured.

h. Color code distribution wiring to conform to TIA-606 and construction documents, whichever is more stringent. Label all equipment, conduit, enclosures, jacks, and cables on record drawings, to facilitate installation and maintenance.

i. In addition to requirements in Section 27 05 11, REQUIREMENTS FOR COMMUNICATION INSTALLATIONS, provide stainless steel faceplates with plastic covers over labels.

2.2 EQUIPMENT AND MATERIALS

a. Cable Systems - Twisted Pair, Fiber optic, Coaxial and Analog:

(1) General:

(a) Provide cable (i.e. backbone, outside plant, and horizontal cabling) conforming to accepted industry standards with regards to size, color code, and insulation.

(b) Some areas can be considered “plenum”. Comply with all codes pertaining to plenum environments. It is contractor’s responsibility to review the VA’s cable requirements with Contracting Officer and OI&T Service prior to installation to confirm type of environment present at each location.

(c) Provide proper test equipment to confirm that cable pairs meet each OEM’s standard transmission requirements, and ensure cable carries data transmissions at required speeds, frequencies, and fully loaded bandwidth.

(2) Telecommunications Rooms (TR):

(a) In TR’s served with UTP fiber optic, coaxial and analog backbone cables, terminate UTP cable on RJ-45, 8-pin connectors of separate 48-port modular patch panels.

(b) Provide 24 port fiber optic modular patch panels with “LC” couplers dedicated for voice, data and FMS applications.

(c) Provide connecting cables required to extend backbone cables (i.e. patch cords, twenty-five pair, etc.), to ensure complete and operational distribution systems.

(d) In TR’s, which are only served by a UTP backbone cable, terminate cable on separate modular connecting devices, Type 110A punch down blocks (or equivalent), dedicated to data applications.

(3) Backbone Copper Cables:

(a) Riser Cable:

1. Provide communication riser cables listed in NEC Table 800, 154(a) for the purpose and suited for electrical connection to a communication network.

2. Provide STP or Unshielded Twisted Pair (UTP), minimum 24 American Wire Gauge (AWG) solid, thermoplastic insulated conductors for communication (analog RF coaxial cable is not to be provided in riser systems) riser cables with a thermoplastic outer jacket.

3. Label and test complete riser cabling system.

(4) Horizontal Cable: Installed from TCO jack to the TR patch panel.

(a) Tested to TIA-568.2 Category 6A requirements including NEXT, ELFEXT (Pair-to-Pair and Power Sum), Insertion Loss (attenuation), Return Loss, and Delay Skew.

(b) Minimum Transmission Parameters: 500 MHz.

(c) Provide four pair (22 AWG) cable

(d) Terminate all four pairs on same port at patch panel in TR.

(e) Terminate all four pairs on same jack, at work area Telecommunication Outlets (TCO):

1. Jacks: Minimum three eight-pin RJ-45 TIA-568.2 Category 6A Type jacks at TCO.

a) Top Port: RJ-45 jack compatible with RJ-11 plug for voice.

b) Bottom Two Ports: Unkeyed RJ-45 jacks for data.

(5) Fiber Optics Backbone Cable:

(a) Provide single mode fiber optic cable 8.3 mm containing at minimum 12 strands of fiber, unless otherwise specified. Terminate single mode fibers at both ends with LC type female connectors installed in an appropriate patch or breakout panel and secured with a cable management system. Provide minimum 2 feet cable loop at each end to allow for future movement.

(b) Install fiber optic cables in TR’s, Voice (Telephone) Switch Room, and Main Computer Room, in rack mounted fiber optic patch panels. Provide female LC couplers in appropriate panel for termination of each strand.

(c) Test all fiber optic strands' cable transmission performance in accordance with TIA standards. Measure attenuation in accordance with fiber optic test procedures TIA/EIA-455 ('-61', or -53). Provide written results to Contracting Officer for review and approval.

b. Cross-Connect Systems (CCS):

(1) Copper Cables: Provide copper CCS sized to connect cables at TR and allow for a minimum of 50 percent anticipated growth.

(2) Maximum DC Resistance per Cable Pair: 28.6 Ohms per (1,000 feet).

(3) Fiber Optic Cables:

(a) Provide fiber CCS sized to connect cables at TR and allow for a minimum of 50 percent anticipated growth.

(b) Install fiber optic cable slack in protective enclosures.

c. Telecommunication Room (TR):

(1) Terminate backbone and horizontal, copper, fiber optic, coaxial and analog cables on appropriate cross-connection systems (CCS) containing patch panels, punch blocks, and breakout devices provided in enclosures and tested, regardless of installation method, mounting, termination, or cross-connecting used. Provide

cable management system as a part of each CCS.

(2) Coordinate location in TR with FMS equipment (i.e. fire alarm, nurse call, code blue, video, public address, radio entertainment, intercom, and radio paging equipment).

d. Coaxial and Analog Cables: Bond equipment to ground per TIA standards, such that all grounding systems comply with all applicable National, Regional, and Local Building and Electrical codes.

(1) Provide current arrester for each copper or coaxial cable that enters from outside of a building regardless if cable is installed underground or aerial.

(2) Provide a gas surge protector/module and bond to earth ground.

e. Main Cross-connection Subsystem (MCCS): MCCS is common point of distribution for inter- and intra-building copper and fiber optic backbone system cables, and connections to the voice (telephone) and data cable systems.

f. Voice (or Telephone) Cable Cross-Connection Subsystem:

(1) Provide Insulation Displacement Connection (IDC) hardware.

(2) Provide the following for each Category 6A for specialized powered systems technically accepted by SMCS 005OP2H3, (202) 461-5310, OI&T and FMS Services and Contracting Officer) Cabling System termination; cross-connection wires, RJ-45 patch cord connector to RJ-45 patch cord connector.

(a) Provide terminations to be accessible without need for disassembly of IDC wafer. Provide IDC wafers removable from their mounts to facilitate testing on either side of connector.

(b) Provide removable designation strips or labels to allow for inspection of terminations.

(c) Provide cable management system as a part of IDC.

(3) Provide IDC connectors capable of re-terminations, without damage, a minimum of 200 IDC insertions or withdrawals on either side of connector panel.

(4) Install using only non-impact terminating tool having both a tactile and an audible feedback to indicate proper termination.

(5) Provide inputs from FTS, Local Voice (Telephone) System, or diverse routed voice distribution systems on left side of IDC (110A blocks with RJ45 connections are acceptable alternates to IDC) of MCCS.

(6) Provide system outputs from MCCS to voice backbone cable distribution system on the right side of same IDC (or 110A blocks) of MCCS.

(7) Do not split pairs within cables between different jacks or connections.

(8) Provide UTP cross connect wire to connect each pair of terminals plus an additional 50 percent spare.

g. Data Cross-Connection Subsystems:

(1) Provide patch panels with modular RJ45 female to 110 connectors for cross-connection of copper data cable terminations with cable management system.

(2) Provide patch panels conforming to EIA/ECA 310-E dimensions and suitable for mounting in standard equipment racks, with 48 RJ45 jacks aligned in two horizontal rows per panel. Provide RJ45 jacks of modular design and capable of accepting and functioning with other modular (i.e. RJ11) plugs without damaging jack.

(a) Provide system inputs from servers, data LAN, bridge, or interface distribution systems on top row of jacks of appropriate patch panel.

(b) Provide backbone cable connections on bottom row of jacks of same patch panel.

(c) Provide patch cords for each system pair of connection jacks with modular RJ45 connectors provided on each end to match panel’s modular RJ45 female jack’s being provided.

h. Fiber-Optic Cross-Connection Subsystems: Provide rack mounted patch or distribution panels installed inside a lockable cabinet or “breakout enclosure” that accommodate minimum 12 strands multimode fiber and 12 strand single mode fiber - these counts do not include 50 percent spare requirement. Provide cable management system for each panel.

(1) Provide panels for minimum 24 female LC connectors, able to accommodate splices and field mountable connectors and have capacity for additional connectors to be added up to OEM’s maximum standard panel size for this type of use. Protect patch panel sides, including front and back, by a cabinet or enclosure.

(2) Provide panels that conform to EIA/ECA 310-E dimensions suitable for installation in standard racks, cabinets, and enclosures. Provide panels for system grounding where armored cables are installed.

(3) Provide patch panels with highest OEM approved density of fiber LC termination’s (maximum of 72 each), while maintaining a high level of manageability. Provide proper LC couplers installed for each pair of fiber optic cable LC connectors.

(a) Provide system inputs from interface equipment or distribution systems on top row of connectors of appropriate patch panel.

(b) Provide backbone cable connections on bottom row of connectors of same patch panel.

(c) Provide patch cords for each pair of fiber optic strands with connector to match couplers.

(4) Provide field installable connectors that are pre-polished.

(a) Terminate every fiber cable with appropriate connector, and test to ensure compliance to specifications and industry standards for fiber optic LC female connector terminated with a fiber optic cable.

(b) Install a terminating cap for each unused LC connector.

i. Copper Outside Plant Cable: Minimum of UTP, 22 AWG solid conductors, solid PVC insulation, and filled core (flex gel - waterproof Rural Electric Association (REA) listed PE 39 code) between outer armor or jacket and inner conductors protective lining.

(1) Provide copper cable system as a Star Topology.

j. Horizontal Cabling (HC):

(1) Horizontal cable length to farthest system outlet to be maximum of (295 ft).

(2) Splitting of pairs within a cable between different jacks is not permitted.

2.3 DISTRIBUTION EQUIPMENT AND SYSTEMS

a. Telecommunication Outlet:

(1) TCO consists of minimum two data RJ45 jacks mounted in a separate steel outlet box 4 inches x 4 inches x (2-1/2 inches) minimum with a labeled stainless steel faceplate. Where shown on drawings, provide a second steel outlet box minimum 4 inches x 4 inches x (2-1/2 inches), with a labeled faceplate, adjacent to first box to ensure system connections and expandability requirements are met.

(2) Provide RJ-45/11 compatible female type voice (telephone) multi-pin connections. Provide RJ-45 female type data multi-pin connections.

(3) Provide wall outlet with a stainless steel face plate and sufficient ports to fit voice (telephone) multi-pin jack, data multi- pin jacks and plastic covers for labels when mounted on outlet box provided (minimum 4 inches x 4 inches for single and 4 inches x 8 inches for dual outlet box applications. Install stainless steel face plate, for prefabricated bedside patient unit installations.

b. Backbone Distribution Cables:

(1) Meet TIA-568.0 transmission performance requirements of Voice Grade Category 6A.

(2) Provide cable listed for environments where it is installed.

(3) Technical Characteristics:

(a) Length: As required, in minimum (3,000 ft.) reels.

(b) Size:

1. Minimum (22 AWG) outside plant installation.

(c) Color Coding: American Telephone and Telegraph Company Standard; Bell System Practices Outside Plant Construction and Maintenance Section G50.607.3, Issue 2 February, 1959.

(d) Minimum Bend Radius: 10X cable outside diameter.

(e) Impedance: 120 Ohms + 15 percent.

(f) DC Resistance: Maximum 8.00 ohms/100 m

(g) Maximum attenuation for 100m at 20° C:

Frequency (MHz)	Category 6A (dB)
.772	-
1	2.1
4	3.8
8	5.3
10	5.9
16	7.5
20	8.4
25	9.4
31.25	10.5
62.5	15.0
100	19.1
200	27.6
250	31.1
300	34.3
400	40.1
500	45.3

(4) Data Multi-Conductor:

(a) Unshielded UTP cable with solid conductors.

(b) Able to handle the power and voltage used over the distance required.

(c) Meets TIA transmission performance requirements of Category 6A.

(d) Technical Characteristics:

1. 22 AWG cable

2. Bend Radius: 10 times cable outside diameter.

3. Impedance: 100 Ohms + 15%, BAL.

4. Bandwidth: 500 MHz.

5. DC Resistance: Maximum 9.38 Ohms/ (328 ft.) at 20 degrees C.

6. Maximum Mutual Capacitance: 5.6 nF per (328 ft.).

7. Shield Coverage:

a) Overall Outside (if OEM specified): 100 percent.

b) Individual Pairs (if OEM specified): 100 percent.

8. Maximum attenuation for 100m (328 ft.) at 20° C:

Frequency (MHz)	Category 6A (dB)
1	2.1
4	3.8
8	5.3
10	5.9
16	7.5
20	8.4
25	9.4
31.25	10.5
62.5	15.0
100	19.1
200	27.6
250	31.1
300	34.3
400	40.1
500	45.3

(5) Fiber Optic:

(a) Single mode Fiber:

1. Provide OS1 Type general purpose single mode fiber optic cable installed in conduit for all system locations with load-bearing support braid surrounding inner tube for strength during cable installation.

2. Technical Characteristics:

a) Bend Radius: Minimum (4 inches).

b) Outer Jacket: PVC.

c) Fiber Diameter: 8.7 microns.

d) Cladding: 125 microns.

e) Attenuation at 850 nanometer: 1.0 dBm per kilometer.

f) Connectors: Ceramic.

c. Outlet Connection Cables:

(1) Voice (Telephone):

(a) Provide a connection cable for each TCO voice (telephone) jack

in system with 10 percent spares able to connect voice (telephone) connection cable from voice (telephone) instrument to TCO voice (telephone) jack. Do not provide voice (telephone) instruments or equipment.

(b) Technical Characteristics:

1. Length: Minimum (6 feet).

2. Cable: Voice Grade.

3. Connector: RJ-11/45 compatible male on each end.

4. Size: Minimum 24 AWG.

5. Color Coding: Required, telephone industry standard.

(2) Data:

(a) Provide a connection cable for each TCO data jack in system with 10 percent spares to connect a data instrument to TCO data jack. Do not provide data terminals/equipment.

(b) Technical Characteristics:

1. Length: Minimum (6 feet).

2. Cable: Data grade Category 5E or on a case-by-case basis Category 6A for specialized powered systems accepted by SMCS 005OP2H3 (202) 461-5310, IT and FMS Services and Contracting Officer.

3. Connector: RJ-45 male on each end.

4. Color Coding: Required, data industry standard.

5. Size: Minimum 24 AWG.

(3) Fiber Optic:

(a) Provide a connection cable for each TCO fiber optic connector in system with 10 percent spares. Provide data connection cable to connect a fiber optic instrument to TCO fiber optic jack. Do not provide fiber optic instruments/equipment.

(b) Technical Characteristics:

1. Length: Minimum (6 feet).

2. Cable: Flexible single conductor with jacket.

3. Connector: LC male on each end.

4. Size: To fit OM1 single mode.

d. System Connectors:

(1) Modular (RJ-45/11 and RJ-45): Provide voice and high speed data transmission applications type modular plugs compatible with voice

(telephone) instruments, computer terminals, and other type devices requiring linking through modular telecommunications outlet to the system compatible with UTP cables.

(a) Technical Characteristics:

1. Number of Pins:

a) RJ-45: Eight.

b) RJ-11/45: Compatible with RJ-45.

2. Dielectric: Surge.

3. Voltage: Minimum 1,000V RMS, 60 Hz at one minute.

4. Current: 2.2A RMS at 30 minutes or 7.0A RMS at 5.0 seconds.

5. Leakage: Maximum 100 µA.

6. Connections:

a) Initial contact resistance: Maximum 20 milli-Ohms.

b) Insulation displacement: Maximum 10 milli-Ohms.

c) Interface: Must interface with modular jacks from a variety of OEMs. RJ-11/45 plugs provide connection when used in RJ-45 jacks.

d) Durability: Minimum 200 insertions/withdrawals.

e. Fiber Optic Terminators:

(1) Pre-polished crimp on type that has proper ferrule to terminate fiber optic cable.

(2) Technical Characteristics:

(a) Frequency: Light wave.

(b) Power Blocking: As required.

(c) Return Loss: 25 dB.

(d) Connectors: LC.

(e) Construction: Ceramic.

f. Conduit and Signal Ducts:

(1) Conduit:

(a) Provide conduit or sleeves for cables penetrating walls, ceilings, floors, interstitial space, fire barriers, etc.

(b) Minimum Conduit Size: (1 inch).

(c) Provide separate conduit and signal ducts for each cable type installation.

(d) When metal (plastic covered, flexible cable protective armor, etc.) systems are authorized to be provided for use in system, follow installation guidelines and standard specified in Section 27 05 33, RACEWAYS AND BOXES FOR COMMUNICATIONS SYSTEMS and NEC.

(e) Maximum 40 percent conduit fill for cable installation.

(2) Signal Duct, Cable Duct, or Cable Tray: Use existing signal duct, cable duct, and cable tray, when identified and accepted by Contracting Officer.

PART 3 EXECUTION

3.1 INSTALLATION

a. Install for ease of operation, maintenance, and testing.

b. Install system to comply with NFPA 70 National Electrical Code, NFPA 99 Health Care Facilities, NFPA 101 Life Safety Code, Joint Commission Manual for Health Care Facilities, and original equipment manufacturers' (OEM) installation instructions.

c. Cable Systems Installation:

(1) Install system cables in cable duct, cable tray, cable runway, conduit or when specifically approved, flexible NEC Article 800 communications raceway. Confirm drawings show sufficient quantity and size of cable pathways. If flexible communications raceway is used, install in same manner as conduit.

(2) Coordinate outside plant and backbone cables to furnish number of cable pairs for system requirements and obtain approval of Contracting OFfcier and IT Service prior to installation.

(3) Bond to ground metallic cable sheaths, etc. (i.e. risers, underground, horizontal, etc.).

(4) Install temporary cable to not present a pedestrian safety hazard and be responsible for all work associated with removal. Temporary cable installations are not required to meet Industry Standards; but, must be reviewed and accepted by Contracting Officer, IT Service, FMS and SMCS 005OP2H3 (202-461-5310) prior to installation.

d. Patient Bedside Prefabricated Units (PBPU) Installation:

(1) Under no circumstances, proceed with installing PBPU without written approval of PBPU OEM and specific instructions regarding attachment to or modifying of PBPU.

(2) Maintain UL integrity of each PBPU. If installation violates UL integrity, obtain on site UL re-certification of violated PBPU at the direction of Contracting Officer.

e. Labeling:

(1) Industry Standard: Provide labeling in accordance with TIA-606-C.

(2) Print lettering of labels with thermal ink transfer process; handwritten labels are not acceptable.

(3) Label both ends of all cables in accordance with industry standard. Provide permanent Labels in contrasting colors and identify according to system “Record Wiring Diagrams”.

(4) Termination Hardware: Label workstation outlets and patch panel connections using color coded labels with identifiers in accordance with industry standard and record on“Record Wiring Diagrams”.

3.2 FIELD QUALITY CONTROL

a. Interim Inspection:

(1) Verify that equipment provided adheres to installation requirements of this section. Interim inspection must be conducted by a factory-certified representative and witnessed by Contracting Officer.

(2) Check each item of installed equipment to ensure appropriate NRTL label.

(3) Verify cabling terminations in telecommunications rooms and at workstations adhere to color code for T568B pin assignments and cabling connections comply with TIA standards.

(4) Visually confirm marking of cables, faceplates, patch panel connectors and patch cords.

(5) Perform fiber optical field inspection tests via attenuation measurements on factory reels and provide results along with manufacturer certification for factory reel tests. Remove failed cable reels from project site upon attenuation test failure.

(6) Notify Contracting Officer of the estimated date the contractor expects to be ready for interim inspection, at least 20 working days before requested inspection date, so interim inspection does not affect systems’ completion date.

(7) Provide results of interim inspection to Contracting Officer. If major or multiple deficiencies are discovered, Contracting Officer can require a second interim inspection before permitting contractor to continue with system installation.

(8) Do not proceed with installation until Contracting Officer determines if an additional inspection is required. In either case, re-inspection of deficiencies noted during interim inspections must be part of the proof of performance test.

b. Pretesting:

(1) Pretest entire system upon completion of system installation.

(2) Verify during system pretest, utilizing the accepted equipment, that system is fully operational and meets system performance

requirements of this section.

(3) Provide Contracting Officer four copies of recorded system pretest measurements and the written certification that system is ready for formal acceptance test.

c. Acceptance Test:

(1) After system has been pretested and the contractor has submitted pretest results and certification to Contracting Officer, then schedule an acceptance test date and give Contracting Officer 30 days' written notice prior to date acceptance test is expected to begin.

(2) Test only in presence of a Contracting Officer.

(3) Test utilizing approved test equipment to certify proof of performance.

(4) Verify that total system meets the requirements of this section.

(5) Include expected duration oftest time, with notification of the acceptance test.

d. Verification Tests:

(1) Test UTP copper cabling for DC loop resistance, shorts, opens, intermittent faults, and polarity between conductors, and between conductors and shield, if cable has an overall shield. Test cables after termination and prior to cross-connection.

(2) Multi-mode Fiber Optic Cable: Perform end-to-end attenuation tests in accordance with TIA-568.3 and TIA-526-14 using Method A, Optical Power Meter and Light Source and Method B, OTDR. Perform verification acceptance test.

(3) Single mode Fiber Optic Cable: Perform end-to-end attenuation tests in accordance with TIA-568.3 and TIA-526-7 using Method A, Optical Power Meter and Light Source and Method B, OTDR. Perform verification acceptance test.

f. Performance Testing:

(1) Perform Category 5E (or on a case by case basis Category 6A for specialized powered systems accepted by SMCS 005OP2H3, (202) 461-5310, IT and FMS Services and Contracting Officer) tests in accordance with TIA-568.1 and TIA-568.2. Include the following tests - wire map, length, insertion loss, return loss, NEXT, PSNEXT, ELFEXT, PSELFEXT, propagation delay and delay skew.

(2) Fiber Optic Links: Perform end-to-end fiber optic cable link tests in accordance with TIA-568.3.

g. Total System Acceptance Test: Perform verification tests for UTP copper cabling systems and fiber optic cabling systems after complete telecommunication distribution system and workstation outlet are installed.

3.3 MAINTENANCE

Accomplish the following minimum requirements during one year warranty period:

(1) Respond and correct on-site trouble calls, during standard work week:

(a) A routine trouble call within one working day of its report. A routine trouble is considered a trouble which causes a system outlet, station, or patch cord to be inoperable.

(b) Standard work week is considered 8:00 A.M. to 5:00 P.M., Monday through Friday exclusive of Federal holidays.

(2) Respond to an emergency trouble call within six hours of its report. An emergency trouble is considered a trouble which causes a subsystem or distribution point to be inoperable at any time.

(3) Respond on-site to a catastrophic trouble call within four hours of its report. A catastrophic trouble call is considered total system failure.

(a) If a system failure cannot be corrected within four hours (exclusive of standard work time limits), provide alternate equipment, or cables within four hours after four hour trouble shooting time.

(b) Routine or emergency trouble calls in critical emergency health care facilities (i.e., cardiac arrest, intensive care units, etc.) are also be deemed as a catastrophic trouble.

(4) Provide Contracting Officer written report itemizing each deficiency found and the corective action performed during each official reported trouble call. Provide Contracting Officer with sample copies of reports for review and approval at beginning of total system acceptance test.

-- End of Section --

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DIVISION 27 - COMMUNICATIONS

SECTION 27 15 13

COMMUNICATIONS COPPER HORIZONTAL CABLING

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SECTION 27 15 13

COMMUNICATIONS COPPER HORIZONTAL CABLING

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies requirements for telecommunications copper (UTP) horizontal distribution cabling outside of the computer room. For copper (UTP) horizontal cabling in structured cabling systems in the computer room, see Section 27 13 13 Communications Copper Backbone Cabling.

1.2 SUMMARY

Section Includes:

a. Copper (UTP) Horizontal Cables.

1.3 REFERENCES

VA Infrastructure Standard for Telecommunications Spaces.

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-568.1	(2020e) Commercial Building Telecommunications Infrastructure Standard
TIA-568.2	(2018d) Balanced Twisted-Pair Telecommunications Cabling and Components Standards
TIA TSB-155	Guidelines for the Assessment and Mitigation of Installed Category 6 Cabling

1.4 RELATED WORK

a. Computer room structured cabling: Section 27 10 00, STRUCTURED CABLING.

b. Termination equipment: Section 27 11 19, COMMUNICATIONS TERMINATION BLOCKS AND PATCH PANELS.

b. Cable labeling requirements: Section 27 05 53, IDENTIFICATION FOR COMMUNICATIONS SYSTEMS.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Copper (Utp) Horizontal Cables; G

1.6 SUBMITTAL INSTRUCTIONS

Submit in accordance with Section 27 05 11, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS.

PART 2 PRODUCTS

2.1 COPPER (UTP) HORIZONTAL CABLES.

Copper (UTP) horizontal cables connect the Telecommunications Room (TR) to each end-user Work Area Outlet (WAO) (horizontal distribution).

a. Performance Category. Copper (UTP) horizontal cables shall meet the Category 6A performance requirements. Category 6A connectors are required.

b. Performance Specifications. Meets or exceeds TIA-568.2 and TIA TSB-155.

c. Limited Power (LP) Certification. UL Listed as x-LP (0.5A).

d. Termination. Field terminated TIA-568.1.

e. Testing. Following installation, each cable shall be field certified to meet performance category requirements per TIA-568.2.

f. Conductor Size. 22-24AWG.

g. Jacket Rating. Communications Multipurpose Cable, Plenum (CMP) shall be specified if any portion of the cable passes through an NEC-defined plenum. Communications Multipurpose Cable, Riser (CMR) shall be specified for all other applications. CMP may be used as a substitute for CMR.

h. Construction. Copper Clad Aluminum (CCA) cable is not permitted.

j. Length. The maximum distance for UTP horizontal cables is limited to 295 feet.

PART 3 EXECUTION

3.1 IMPLEMENTATION.

a. All cabling and equipment shall be labeled per the requirements of the VA Infrastructure Standard for Telecommunications Spaces.

b. Interior to the TR, install horizontal copper (UTP) structured cabling in overhead cable tray.

-- End of Section --

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DIVISION 27 - COMMUNICATIONS

SECTION 27 16 19

COMMUNICATIONS PATCH CORDS, STATION CORDS, AND CROSS CONNECT WIRE

PART 1 GENERAL

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2.1 COPPER (UTP) PATCH CABLES.
2.2 FIBER OPTIC PATCH CABLES.

PART 3 EXECUTION

3.1 IMPLEMENTATION.

-- End of Section Table of Contents --

SECTION 27 16 19

COMMUNICATIONS PATCH CORDS, STATION CORDS, AND CROSS CONNECT WIRE

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies requirements for telecommunications patch cords.

1.2 SUMMARY

Section Includes:

a. Copper (UTP) Patch Cables.

b. Fiber Optic Patch Cables.

1.3 REFERENCES

VA Infrastructure Standard for Telecommunications Spaces.

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-568.1	(2020e) Commercial Building Telecommunications Infrastructure Standard

1.4 RELATED WORK

a. Termination equipment: Section 27 11 19, COMMUNICATIONS TERMINATION BLOCKS AND PATCH PANELS.

b. Cable labeling requirements: Section 27 05 53, IDENTIFICATION FOR COMMUNICATIONS SYSTEMS.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Copper (Utp) Patch Cables; G

Fiber Optic Patch Cables; G

1.6 SUBMITTAL INSTRUCTIONS

Submit in accordance with Section 27 05 11, COMMON WORK RESULTS FOR COMMUNICATIONS SYSTEMS.

PART 2 PRODUCTS

2.1 COPPER (UTP) PATCH CABLES.

a. Performance Category. Patch cables shall meet the Category 6A performance requirements. Category 6A connectors are required.

b. Termination Method. Factory pre-terminated TIA-568.1.

c. Conductor Size. 22-28AWG. Where 28AWG patch cords are specified, the overall horizontal channel distance is reduced.

d. Length. Patch cords of the shortest appropriate length shall be used.

2.2 FIBER OPTIC PATCH CABLES.

a. Construction. Fiber patch cords shall be constructed of pairs (2-strands) single mode laser-optimized 9/125 Micrometers fiber.

b. Performance Category. Multimode patch cords shall meet the OM4 performance requirements.

c. Connectors. Duplex Lucent connector (LC) connectors are required. Where polarity requirements are not known, reversible polarity duplex LC connectors are specified on one end.

d. Termination Method. Factory pre-terminated.

e. Length. Patch cords of the shortest appropriate length shall be used.

PART 3 EXECUTION

3.1 IMPLEMENTATION.

a. Patch cords shall not be used to connect equipment in different server cabinets.

b. Patch cords shall be neatly installed using cable management equipment and following cable management principles.

c. Upon installation patch cords shall be labeled in accordance with the requirements of the VA Infrastructure Standard for Telecommunications Spaces.

-- End of Section --

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PUBLIC ADDRESS

10/18

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3.12 TRAINING

-- End of Section Table of Contents --

SECTION 27 51 16
10/18

PUBLIC ADDRESS

PART 1 GENERAL

1.1 SECTION SUMMARY

a. Work covered by this document includes design, engineering, labor, material and products, equipment warranty and system warranty, training and services for, and incidental to, the extension of the existing campus Public Address System (PAS) and associated equipment (here-in-after referred to as the System) in approved locations indicated on the contract drawings. These items shall be tested and certified capable of receiving, distributing, interconnecting and supporting PAS communications signals generated local and remotely as detailed herein.

b. Work shall be complete, Occupational Safety and Health Administration (OSHA), National Recognized Testing Laboratory (NRTL – i.e. Underwriters Laboratory UL) Listed and Labeled; and VA Central Office (VACO), Telecommunications Voice Engineering (TVE 005OP3B) tested, certified and ready for operation.

c. The System shall be delivered free of engineering, manufacturing, installation, and functional defects. It shall be designed, engineered and installed for ease of operation, maintenance, and testing.

d. The term “provide”, as used herein, shall be defined as: designed, engineered, furnished, installed, certified, and tested, by the Contractor.

e. Specification Order of Precedence: In the event of a conflict between the text of this document and the Project’s Contract Drawings outlined and/or cited herein; THE TEXT OF THIS DOCUMENT TAKES PRECEDENCE. *HOWEVER, NOTHING IN THIS DOCUMENT WILL SUPERSEDE APPLICABLE EMERGENCY LAWS AND REGULATIONS,* SPECIFICALLY NATIONAL AND/OR LOCAL LIFE AND PUBLIC SAFETY CODES. The Local Fire Marshall and/or VA Public Safety Officer are the only authorities that may modify this document’s EMERGENCY CODE COMPLIANCE REQUIREMENTS, on a case by case basis, in writing and confirmed by VA’s PM, RE and TVE-005OP3B. *The VA PM is the only approving authority* for other amendments to this document that may be granted, on a case by case basis, in writhing with technical concurrencies by VA’s RE, TVE-005OP3B and identified Facility Project Personnel.

f. The Original Equipment Manufacturer (OEM) and Contractor shall ensure *that all* management, sales, engineering and installation personnel have read and understand the requirements of this specification *before* the system is designed, engineered, delivered and provided. The Contractor shall furnish a written statement attesting this requirement as a part of the technical submittal that includes each name and certification, including the OEMs.

1.2 RELATED SECTIONS

a. 07 84 00 – Firestopping.

b. 26 41 00 – Facility Lightning Protection.

c. 27 05 11 – Requirements for Communications Installations.

d. 27 05 26 – Grounding and Bonding for Communications Systems.

e. 27 05 33 – Raceways and Boxes for Communications Systems.

f. 27 10 00 – Control, Communication and Signal Wiring.

Section 27 11 00 – Communications Cabling Interface and Equipment Rooms Fittings.

h. 27 15 00 – Horizontal and Vertical Communications Cabling Equipment and Systems.

1.3 DEFINITIONS

a. Provide: Design, engineer, furnish, install, connect complete, test, certify and warranty.

b. Work: Materials furnished and completely installed.

c. Review of contract drawings: A service by the engineer to reduce the possibility of materials being ordered which do not comply with contract documents. The engineer's review shall not relieve the Contractor of responsibility for dimensions or compliance with the contract documents. The reviewer's failure to detect an error does not constitute permission for the Contractor to proceed in error.

d. Headquarters Technical Review, for National and VA communications and security, codes, frequency licensing, standards, guidelines compliance:

Office of Telecommunications
Special Communications Team (005OP2B)
– 3rd Floor

(O) 301-734-0350, (F) 301-734-0360

e. Contractor: Radio Contractor; you; successful bidder

1.4 REFERENCES

The installation shall comply fully with all governing authorities, laws and ordinances, regulations, codes and standards, including, but not limited to:

(1) Federal Law:

(a) Departments of:

1. Commerce, Consolidated Federal Regulations (CFR), Title 15 – Under the Information Technology Management Reform Act (Public Law 104-106), the Secretary of Commerce approves standards and guidelines that are developed by the:

a) Chapter II, National Institute of Standards Technology (NIST – formerly the National Bureau of Standards). Under Section 5131 of the Information Technology Management Reform Act of 1996 and the Federal Information Security Management Act of 2002 (Public Law 107-347), NIST develops – Federal Information Processing Standards Publication (FIPS)

NIST FIPS 140-2 –Security Requirements for Cryptographic Modules.

b) Chapter XXIII, National Telecommunications and Information Administration (NTIA – aka 'Red Book') Chapter 7.8 / 9; CFR, Title 47 Federal communications Commission (FCC) Part 15

FCC Part 15, Radio Frequency Restriction of Use and Compliance in "Safety of Life" Functions & Locations

2. FCC - Communications Act of 1934, as amended, CFR, Title 47 – Telecommunications, in addition to

47 CFR 15 – Restrictions of use for Part 15 listed Radio Equipment in Safety of Life / Emergency Functions / Equipment/ Locations

FCC Part 15 (also see CFR, Title 15 – Department of Commerce, Chapter XXIII – NTIA):

a) Part 15 – Restrictions of use for Part 15 listed Radio Equipment in Safety of Life / Emergency Functions / Equipment/Locations.

b) Part 58 – Television Broadcast Service.

c) Part 90 – Rules and Regulations, Appendix C.

d) Form 854 – Antenna Structure Registration.

3. Health, (Public Law 96-88), CFR, Title 42, Chapter IV Health & Human Services, CFR, Title 46, Subpart 1395(a)(b) JCAHO "a hospital that meets JCAHO accreditation is deemed to meet the Medicare conditions of Participation by meeting Federal Directives: "

a) All guidelines for Life, Personal and Public Safety; and, Essential and Emergency Communications.

4. Labor, CFR, Title 29, Part 1910, Chapter XVII - Occupational Safety and Health Administration (OSHA), Occupational Safety and Health Standard:

a) Subpart 7 - Definition and requirements (for a NRTL – 15 c's, for complete list, contact (http://www.osha.gov/dts/otpca/nrtl/faq_nrtl.html):

1. UL:

a)

UL 44 – Standard for Thermoset-Insulated Wires and Cables.

b)

UL 65 – Standard for Wired Cabinets.

c)

UL 83 – Standard for Thermoplastic-Insulated Wires and Cables.

d)

UL 467 – Standard for Electrical Grounding and Bonding Equipment

e)

UL 468 – Standard for Grounding and Bonding Equipment.

f)

UL 486A-486B – Standard for Wire Connectors and Soldering Lugs for Use with Copper Conductors

g)

UL 486C – Standard for Splicing Wire Connectors.

h)

UL 486D – Standard for Insulated Wire Connector Systems for Underground Use or in Damp or Wet Locations.

i)

UL 486E – Standard for Equipment Wiring Terminals for Use with Aluminum and/or Copper Conductors.

j)

UL 493 – Standard for Thermoplastic-Insulated Underground Feeder and Branch Circuit Cable.

k)

UL 514B – Standard for Fittings for Cable and Conduit.

l)

UL 1069 – Hospital Signaling and Nurse Call Equipment.

m)

UL 1333 – Vertical (Riser) Fire Rating.

n)

UL 1449 – Standard for Transient Voltage Surge Suppressors.

o)

UL 1479 – Standard for Fire Tests of Through-Penetration Fire

Stops.

p)

UL 1863 – Standard for Safety, Communications Circuits Accessories.

q)

UL 2024 – Standard for Optical Fiber Raceways.

r)

UL 60950 – Information Technology Equipment – Safety.

2. Canadian Standards Association (CSA): same tests as for UL.

3. Communications Certifications Laboratory (CCL): same tests as for UL.

4. Intertek Testing Services NA, Inc. (ITSNA formerly Edison Testing Laboratory ETL): same tests as for UL.

b) Subpart 35 – Compliance with

NFPA 101 – Life Safety Code.

c) Subpart 36 - Design and construction requirements for exit routes.

d) Subpart 268 - Telecommunications.

e) Subpart 305 - Wiring methods, components, and equipment for general use.

5. Department of Transportation, CFR, Title 49 (Public Law 89-670), Part 1, Subpart C – Federal Aviation Administration (FAA):

a) Standards AC 110/460-ID & AC 707 / 460-2E – Advisory Circulars for Construction of Antenna Towers.

b) Forms 7450 and 7460-2 – Antenna Construction Registration.

6. Veterans Affairs (Public Law No. 100-527), CFR, Title 38, Volumes I & II:

a) Office of Telecommunications:

1. Handbook 6100 – Telecommunications.

a) Spectrum Management FCC & NTIA Radio Frequency Compliance and Licensing Program.

b) Special Communications Proof of Performance Testing, VACO Compliance and Life Safety Certification(s).

b) Office of Cyber and Information Security (OCIS):

1. Handbook 6500 - Information Security Program.

2. Wireless and Handheld Device Security Guideline Version 3.2, August 15, 2005.

c) VA's National Center for Patient Safety – Veterans Health Administration Warning System, Failure of Medical Alarm Systems using Paging Technology to Notify Clinical Staff, July 2004.

d) VA's Center for Engineering Occupational Safety and Health, concurrence with warning identified in VA Directive 7700.

e) Office of Construction and Facilities Management (CFM):

1. Master Construction Specifications (

VA PG-18-1).

2. Standard Detail and CAD Standards (

VA PG-18-4).

3. Equipment Guide List (

VA PG-18-5).

4. Electrical Design Manual for VA Facilities (

VA PG-18-10), Articles 7 & 8.

5. Minimum Requirements of A/E Submissions (

VA PG-18-15):

a) Volume B, Major New Facilities, Major Additions; and Major Renovations, Article VI, Paragraph B.

b) Volume C - Minor and NRM Projects, Article III, Paragraph S.

c) Volume E - Request for Proposals Design/Build Projects, Article II, Paragraph F.

6. Mission Critical Facilities Design Manual (Final Draft – 2007).

7. Life Safety Protected Design Manual (Final Draft – 2007).

8. Solicitation for Offerors (SFO) for Lease Based Clinics – (05-2009).

(b) Federal Specifications (Fed. Specs.):

1.

A-A-59544-00 - Cable and Wire, Electrical (Power, Fixed Installation).

(2) National Codes:

(a) American Institute of Architects (AIA): Guidelines for Healthcare Facilities.

(b) American National Standards Institute/Electronic Industries Association/Telecommunications Industry Association (ANSI/EIA/TIA):

1.

TIA 568-B - Telecommunications Wiring Standards:

a)

TIA-568.1 – General Requirements.

b)

TIA-568.2 – Balanced twisted-pair cable systems.

c)

TIA-568.3 - Fiber optic cable systems.

2.

TIA-569 - Standard for Telecommunications Pathways and Spaces.

3.

TIA-606 – Administration Standard for the Telecommunications Infrastructure of Communications Buildings.

4.

TIA-607 – Grounding and Bonding Requirements for Telecommunications.

5. REC 127-49 – Power Supplies.

6.

EIA-RS-160 – Sound systems.

7.

EIA/RS-270 – Tools, Crimping, Solderless Wiring Devices, Recommended Procedures for User Certification.

8.

EIA-SE-101-A49 – Amplifier for Sound Equipment

9.

EIA-SE-103-49 – Speakers for Sound Equipment

(c) American Society of Mechanical Engineers (ASME):

1.

ASME A17.4 – Guide for Emergency Personnel.

2.

ASME A17.5/CSA B44.1 – Elevator & Escalator Equipment (prohibition of installing non-elevator equipment in Elevator Equipment Room / Mechanical Penthouse).

(d) American Society of Testing Material (ASTM):

1.

ASTM D2301 - Standard Specification for Vinyl Chloride Plastic Pressure Sensitive Electrical Insulating Tape.

(e) Building Industries Communications Services Installation (BICSI):

1. All standards for smart building wiring, connections and devices for commercial and medical facilities.

2. Structured Building Cable Topologies.

3. In consort with ANSI/EIA/TIA.

(f) Institute of Electrical and Electronics Engineers (IEEE):

1.

ISO/TR 21730: 2007 - Use of mobile wireless communication and computing technology in healthcare facilities - Recommendations for electromagnetic compatibility (management of unintentional electromagnetic interference) with medical devices.

2. 0739-5175/08/©2008 IEEE – Medical Grade – Mission Critical – Wireless Networks.

3.

IEEE C62.41 – Surge Voltages in Low-Voltage AC Power Circuits.

(g) NFPA:

1.

NFPA 70 - National Electrical Code (current date of issue) – Articles 517, 645 & 800.

2.

NFPA 75 - Standard for Protection of Electronic Computer Data-Processing Equipment.

3.

NFPA 77 – Recommended Practice on Static Electricity.

4.

NFPA 99 - Healthcare Facilities.

5.

NFPA 101 - Life Safety Code.

6.

NFPA 1600 – Disaster Management, Chapter 5.9 – Communications and Warning

(3) Code(s).

(4), City and/or.

(5) Accreditation Organization(s):

(a) Joint Commission on Accreditation of Hospitals Organization (JCAHO) – Section VI, Part 3a – Operating Features.

U.S. DEPARTMENT OF VETERANS AFFAIRS (VA)

VA PG-18-10	Design Manuals

U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

FCC Part 15	Radio Frequency Devices (47 CFR 15)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 101	(2024) Life Safety Code

UL SOLUTIONS (UL)

UL 1333	Riser Low Smoke Vertical Rating

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

EIA-310-D	Standard 19-in. four-post EIA cabinet or rack, with mounting posts
EIA/TIA-606-A	(2012) Cable Labeling Standards

1.5 QUALIFICATIONS

a. The OEM shall have had experience with three (3) or more installations of systems of comparable size and complexity with regards to type and design as specified herein. Each of these installations shall have performed satisfactorily for at least one (1) year after final acceptance by the user. Include the names, locations and point of contact for these installations as a part of the submittal.

b. The Contractor shall submit certified documentation that they have

been an authorized distributor and service organization for the OEM for a minimum of three (3) years. The Contractor shall be authorized by the OEM to pass thru the OEM's warranty of the installed equipment to VA. In addition, the OEM and Contractor shall accept complete responsibility for the design, installation, certification, operation, and physical support for the System. This documentation, along with the System Contractor and OEM certifications must be provided in writing as part of the Contractor's Technical submittal.

c. The Contractor's Communications Technicians assigned to the System shall be fully trained, qualified, and certified by the OEM on the engineering, installation, operation, and testing of the System. The Contractor shall provide formal written evidence of current OEM certification(s) for the installer(s) as a part of the submittal or to the RE before being allowed to commence work on the System.

d. The Contractor shall display all applicable national, state and local licenses.

e. The Contractor shall submit copy (s) of Certificate of successful completion of OEM's installation/training school for installing technicians of the System's PA equipment being proposed.

1.6 CODES AND PERMITS

a. Provide all necessary permits and schedule all inspections as identified in the contract's milestone chart, so that the system is proof of performance tested and ready for operation on a date directed by the Owner.

b. The contractor is responsible to adhere to all codes described herein and associated contractual, state and local codes.

c. The Contractor shall display all applicable national, state and local licenses and permits.

1.7 SCHEDULING

a. After the award of contract, the Contractor shall prepare a detailed schedule (aka milestone chart) using "Microsoft Project" software or equivalent. The Contractor Project Schedule (CPS) shall indicate detailed activities for the projected life of the project. The CPS shall consist of detailed activities and their restraining relationships. It will also detail manpower usage throughout the project.

b. It is the responsibility of the Contractor to coordinate all work with the other trades for scheduling, rough-in, and finishing all work specified. The owner will not be liable for any additional costs due to missed dates or poor coordination of the supplying contractor with other trades.

1.8 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Interconnection Methods; G

SD-02 Shop Drawings

Headend And Each Interface Distribution Cabinet Layout Drawing; G

SD-03 Product Data

Equipment Oem Technical Literature; G

SD-05 Design Data

Surveys; G

Engineering Drawings; G

SD-11 Closeout Submittals

Project Record Documents; G

1.9 SUBMITTAL INSTRUCTIONS

(Note: The Contractor is encouraged, but not required, to submit separate technical submittal(s) outlining alternate technical approach(s) to the system requirements stated here-in as long as each alternate technical document(s) is complete, separate, and submitted in precisely the same manner as outlined herein. VA will review and rate each received alternate submittal, which follows this requirement, in exactly the same procedure as outlined herein. Partial, add-on, or addenda type alternates will not be accepted or reviewed.)

a. Submit at one time within 10 days of contract awarding, drawings and product data on all proposed equipment and system. Check for compliance with contract documents and certify compliance with Contractor's "APPROVED" stamp and signature.

b. Support all submittals with descriptive materials, i.e., catalog sheets, product data sheets, diagrams, and charts published by the manufacturer. These materials shall show conformance to specification and drawing requirements.

c. Where multiple products are listed on a single cut-sheet, circle or highlight the one that you propose to use. Provide a complete and through equipment list of equipment expected to be installed in the system, with spares, as a part of the submittal. Special Communications (TVE-005OP3B) will not review any submittal that does not have this list.

d. Provide four (4) copies to the PM for technical review. The PM will provide a copy to the offices identified in Paragraph 1.3.C & D, at a minimum for compliance review as described herein where each responsible individual(s) shall respond to the PM within 10 days of receipt of their acceptance or rejection of the submittal(s).

e. Provide interconnection methods, conduit (where not already installed), junction boxes (J-Boxes), cable, interface fixtures and equipment lists for the: ENR(s) (aka DMARC), TER, TCR, MCR, MCOR,

PCR, ECR, Stacked Telecommunications Rooms (STR), Nurses Stations (NS), Head End Room (HER), Head End Cabinet (HEC), Head End Interface Cabinet (HEIC) and approved TCO locations Telecommunications Infrastructure Plant (TIP) interface distribution layout drawing, as they are to be installed and interconnected to teach other (REFER TO APPENDIX B – SUGGESTED TELECOMMUNI-CAITONS ONE LINE TOPOLOGY pull-out drawing).

f. Headend and each interface distribution cabinet layout drawing, as they are expected to be installed.

g. Equipment OEM technical literature detailing the electrical and technical characteristics of each item of equipment to be furnished.

h. Engineering drawings of the System, showing calculated of expected signal levels at the headend input and output, each input and output distribution point, and signal level at each telecommunications outlet.

i. Surveys Required as a Part of The Technical Submittal:

(1) The Contractor shall provide the following System survey(s) that depict various system features and capacities required *in addition to* the on-site survey requirements described herein. Each survey shall be in writing and contain the following information (the formats are suggestions and may be used for the initial Technical Submittal Survey requirements), as a minimum:

(a) PA Cable System Design Plan:

1. An OEM and contractor designed functioning PA System cable plan to populate the entire TIP empty conduit/pathway distribution systems provided as a part of Specification 27 11 00 shall be provided as a part of the technical proposal. A specific functioning PA: cable, interfaces, J-boxes and back boxes shall coincide with the total growth items as described herein. It is the Contractor's responsibility to provide the Systems' entire PA cable and accessory requirements and engineer a functioning PA distribution system and equipment requirement plan of the following paragraph(s), at a minimum:

2. The required PA Equipment Locations:

EQUIPPED ITEM	**CAPACITY**	**GROWTH**
Master Control Stations		
Telephone Operators Room		
Police Control Room		
Other		
Zone Amplifiers		
All Call (complete Zone 1)		
Admissions (Zone 2)		
Entrance (Zone 2a)		
Pharmacy Dispensing (Zone 2a)		
Agent Cashier (Zone 2a)		
Other (Zone 2a)		
Labs (Zone 3)		
Blood (Zone 3a)		
Dissecting (Zone 3a)		
Other (Zone 3a)		

EQUIPPED ITEM	**CAPACITY**	**GROWTH**
Clinics (Zone 4)		
Dental (Zone 4a)		
Radiology (Zone 4a)		
Oncology (Zone 4a)		
Other (Zone 4a)		
Supervisory Panel(s)		
Trouble Panel(s)		
Locations		
Speakers		
Overhead		
Locations		
Other		
Other		
Outside		
Locations		
Other		
Horn		
Locations		
Other		
Power Supply(s)		
Location		
Other		
UPS(s)		
Location		
Other		
Radio Paging Access **(when pre-approved by TVE-005OP3B)**		
Wireless Access **(when pre-approved by TVE-005OP3B)**		
Maintenance/Programming Console		
Location(s)		
Other		

3. The required PA Cable Plant/Connections:

The Contractor shall clearly and fully indicate this category for each item identified herein as a part of the technical submittal. For this purpose, the following definitions and s mple connections are provided to detail the system’s capability:

EQUIPPED ITEM	**CAPACITY**	**GROWTH**
Central Control Cabinet/Equipment		
Location		

EQUIPPED ITEM	**CAPACITY**	**GROWTH**
Power Supply(s)		
UPS(s)		
Essential Electrical Power Panel(s)		
Other		
Cable Plant		
Supply to Locations Identified herein		
Speaker Locations		
Remote Locations		
Telephone Operator Room		
Police Control Room		
Other		
Maintenance/Program Console		
Location(s)		
Other		
LAN (Local Facility) Access/Equipment/Location **(when pre-approved by TVE-005OP3B)**		
Wireless Access/Equipment/Location **(when pre-approved by TVE-005OP3B)**		
Other		

1.10 PROJECT RECORD DOCUMENTS (AS BUILTS)

a. Throughout progress of the Work, maintain an accurate record of changes in Contract Documents. Upon completion of Work, transfer recorded changes to a set of Project Record Documents.

b. The floor plans shall be marked in pen to include the following:

(1) All device locations with UL labels affixed.

(2) Conduit locations.

(3) Head-end equipment and specific location.

(4) Each interface and equipment specific location.

(5) Facility Entrance (aka DEMARC) Room(s) interface equipment and location(s).

(6) Telephone Equipment Room (TER) interface equipment and specific location.

(7) Main Computer Room (MCR) interface equipment and specific location.

(8) Police Control Room (PCR) interface equipment and specific location.

(9) Engineering Control Room (ECR) interface equipment and specific location

(10) Telecommunication Outlet (s –TCO) equipment and specific location

(11) TIP Wiring diagram(s).

(12) Warranty certificate.

(13) System test results.

(14) System Completion Document(s) or MOU.

1.11 WARRANTIES / GUARANTY

a. The Contractor shall warrant the installation to be free from defect in material and workmanship for a period of two (2) years from the date of acceptance of the project by the owner. The Contractor shall agree to remedy covered defects within four (4) hours of notification of major failures or within twenty-four (24) hours of notification for individual station related problems.

b. The Contractor shall agree to grantee the system according to the guidelines outlined in Article 4 herein.

1.12 USE OF THE SITE

a. Use of the site shall be at the GC’s direction.

b. Coordinate with the GC for lay-down areas for product storage and administration areas.

c. Coordinate work with the GC and their sub-contractors.

d. Access to buildings wherein the work is performed shall be directed by the GC.

1.13 DELIVERY, STORAGE, AND HANDLING

a. Deliver, store, and handle products using means and methods that will prevent damage, deterioration, and loss, including theft.

b. Store products in original containers.

c. Coordinate with the GC for product storage. There may be little or no storage space available on site. Plan to potentially store materials off site.

d. Do not install damaged products. Remove damaged products from the site and replaced with new product at no cost to the Owner.

1.14 PROJECT CLOSE-OUT

a. Prior to final inspection and acceptance of the work, remove all debris, rubbish, waste material, tools, construction equipment, machinery and surplus materials from the project site and thoroughly clean your work area.

b. Before the project closeout date, the Contractor shall submit:

(1) Warranty certificate.

(2) Evidence of compliance with requirements of governing authorities

such as the Low Voltage Certificate of Inspection.

(3) Project record documents.

(4) Instruction manuals and software that is a part of the system.

c. Contractor shall submit written notice that:

(1) Contract Documents have been reviewed.

(2) Project has been inspected for compliance with contract.

(3) Work has been completed in accordance with the contract.

PART 2 – PRODUCTS

2.1 GENERAL REQUIREMENTS FOR EQUIPMENT AND MATERIALS

a. Extend the existing campus public address system as shown on the contract drawings and TCOs WHOSE EMPTY CONDUIT SYSTEM WAS PROVIDED AS A PART OF SPECIFICATION 27 11 00.

b. The specific location for each PA: Central Control Cabinet is in room G245 of building 1, Speakers are located in chiller plant, Zone is Chiller Plant.

c. Coordinate features and select interface components to form an integrated PA system. Match components and interconnections between the systems for optimum performance of specified functions.

d. Expansion Capability: The PA equipment interfaces and cables shall be able to increase number of enunciation points in the future by a minimum of 50 percent (%) above those indicated without adding any internal or external components or main trunk cable conductors.

e. Equipment: Active electronic type shall use solid-state components, fully rated for continuous duty unless otherwise indicated. Select equipment for normal operation on input power usually supplied between 110 to 130 VAC, 60 Hz.

f. Meet all FCC requirements regarding low radiation and/or interference of RF signal(s). The system shall be designed to prevent direct pickup of signals from within and outside the building structure.

g. Weather/Water Proof Equipment: Listed and labeled by an OSHA certified National Recognized Testing Laboratory (NRTL – i.e. UL) for duty outdoors or in damp locations.

h. Deliver a fully functioning and operable PA in the specific locations shown on the drawings.

2.2 SYSTEM DESCRIPTION

a. Extend the existing campus public address system as shown on drawings and described herein. Provide additional require conduit(s) according to Specification 27 11 00.

b. The Contractor is responsible for interfacing the public address system with the telephone System. The interface unit(s) shall be provided by

the Contractor.

c. The Contractor shall continually employ interfacing methods that are approved by the OEM and VA. At a minimum, an acceptable interfacing method requires not only a physical and mechanical connection, but also a matching of signal, voltage, and processing levels with regard to signal quality and impedance. The total PA system shall be configured and installed so that the combination of equipment actually employed does not produce any undesirable visual or aural effects such as signal distortions, noise pulses, glitches, hum, transients, images, etc. The interface points must adhere to all standards described herein for the full separation of Critical Care and Life Safety systems.

d. It is not acceptable to utilize the telephone cable system for the control of Paging system and equipment. The System Contractor shall connect the Telephone System Remote Control System to the Paging Control Unit ensuring that all NFPA and UL Critical Care and Life Safety Circuit and System separation guidelines are satisfied. The System Contractor is not allowed to make any connections to the Telephone System. The Owner shall arrange for the interconnection between the PA and Telephone Systems with the appropriate responsible parties.

e. System hardware shall consist of a standalone (separate) PA communications network comprised of amplifiers, mixers, speakers, volume controls, test sets, telephone private branch exchange (PBX) interface equipment, equipment cabinets/racks, wiring and other options such as, sub zoning in addition to “all call” functions, computer interfaces, printer interfaces and wireless network interfaces, (when specifically approved by 005OP3B and VA Headquarters Spectrum Management 005OP2B – herein after referred to as 005OP2B) as shown on drawings. All necessary equipment required to meet the intent of these specifications, whether or not enumerated within these specifications, shall be supplied and installed to provide a complete and operating nurse/patient communications network.

f. Systems firmware shall be the product of a reputable firmware OEM of record with a proven history of product reliability and sole control over all source code. Manufacturer shall provide, free of charge, product firmware/software upgrades for a period of two (2) years from date of acceptance by VA for any product feature enhancements. System configuration programming changes shall not require any exchange of parts and shall be capable of being executed remotely via a modem connection (when specifically approved first by 005OP3B).

g. The PA Head End Equipment is located in Room G245 of building 1. The PA shall cover the new chiller plant. The PA shall provide zoned, one-way voice paging through distributed loudspeakers. Voice input into the PA shall be by zone using the telephone system.

h. The System shall utilize microprocessor components for all signaling and programming circuits and functions. Self contained or on board system program memory shall be non-volatile and protected from erasure from power outages for a minimum of 24 hours.

i. Provide a backup battery or a UPS for the System (including each distribution cabinet/point, CRT, LCD and Monitor) to allow normal operation and function (as if there was no AC power failure) in the

event of an AC power failure or during input power fluctuations for a minimum of two (2) Hours.

j. The System shall interface with the Facility's existing PAS so that a global page (aka "all call" page) is communicated to the existing PAS and the new System of this project. Arrangements for interconnection of the System and the telephone system(s) shall be coordinated with the owner and the PBX provider.

k. The system shall be designed to provide continuous electrical supervision of the complete and entire system (i.e. light bulbs, wires, contact switch connections, master control stations, wall stations, circuit boards, data, audio, and communication busses, main and UPS power, etc.). All alarm initiating and signaling circuits shall be supervised for open circuits, short circuits, and system grounds. and UPS power circuits shall be supervised for a change in state (i.e. primary to backup, low battery, UPS on line, etc.). When an open, short or ground occurs in any system circuit, an audible and visual fault alarm signal shall be initiated at the main supervisory panel, nurse control station and all remote amplifier locations.

l. All passive distribution equipment shall meet or exceed -80 dB radiation shielding (aka RFI) shielding specifications and be provided with screw type audio connectors.

m. All equipment face plates utilized in the system shall be stainless steel, anodized aluminum or UL approved cycolac plastic for the areas where provided.

n. All trunk, branch, and interconnecting cables and unused equipment ports or taps shall be terminated with proper terminating resistors designed for RF, audio and digital cable systems without adapters.

o. Noise filters and surge protectors shall be provided for each equipment interface cabinet, headend cabinet, control console and local and remote amplifier locations to insure protection from input primary AC power surges and to insure noise glitches are not induced into low voltage data circuits.

p. Plug-in connectors shall be provided to connect all equipment, except coaxial cables and RF transmission line interface points. Coaxial cable distribution points and RF transmission lines shall use coaxial cable connections recommended by the cable OEM and approved by the system OEM. Base band cable systems shall utilize barrier terminal screw type connectors, at a minimum. As an alternate, crimp type connectors installed with a ratchet type installation tool are acceptable provided the cable dress, pairs, shielding, grounding, connections and labeling are the same as the barrier terminal strip connectors. Tape of any type, wire nuts or solder type connections are unacceptable and will not be approved.

q. Audio Level Processing: The control equipment shall consist of audio mixer(s), volume limiter(s) and/or compressor(s), and power amplifier(s) to process, adjust, equalize, isolate, filter, and amplify each audio channel for each sub-zone in the system and distribute them into the System's RF interfacing distribution trunks and amplification circuits. It is acceptable to use identified Telephone System cable pairs designated for Two-Way Radio interface and control use or identified as spare telephone cable pairs by the

Facility's Telephone System Contractor. The use of telephone cable to distribute RF signals, carrying system or sub-system AC or DC voltage is not acceptable and will not be approved. Additionally, each control location shall be provided with the equipment required to insure the system can produce its designed audio channel capacity at each speaker identified on the contract drawings. The Contractor shall provide: a spare set of telephone paging modules as recommended by the OEM (as a minimum provide one spare module for each installed module); one spare audio power amplifier, one spare audio mixer, one spare audio volume limiter and/or compressor, and one spare audio automatic gain adjusting device, and minimum RF equipment recommended by the OEM.

r. Contractor is responsible for pricing all accessories and miscellaneous equipment required to form a complete and operating system. Unless otherwise noted in this Part, equipment quantities shall be as indicated on the drawings.

2.3 SYSTEM PRFORMANCE

a. At a minimum, each distribution, interconnection, interface, terminating point and TCO shall be capable of supporting the Facility's PA system voice and data service as follows:

(1) Shall be compliant with and not degrade the operating parameters of the Public Switched Telephone Network (PSTN) and the Federal Telecommunications System (FTS) at each PSTN and FTS interface, interconnection and terminating locations in the TERs.

(2) Audio Input: The signal level of each audio input channel at each input point shall be a MINIMUM of zero decibels measured (dBm), +0.10 dBm across 150 Ohms, balanced.

(3) Audio Output: The audio signal level at each speaker shall be a MINIMUM of +0.25 Watt (W) and a maximum of +20 W, 600 Ohms balanced impedance, on a 70.7 V audio distribution line Contractor to determine and set each speaker's proper audio signal level (top) based on speaker location and the ambient noise level in speaker coverage area.

(4) The system shall meet the following MINIMUM parameters at each speaker:

(a) Cross Modulation: -46 dB

(b) Hum Modulation: -55 dB

(c) Isolation (outlet-outlet): 24 dB

(d) Impedance:

1. Distribution: 600 Ohm balanced @ 70.7 V audio line level.

2. Speaker: Selectable, as required.

(e) Audio Gain: 10 dB minimum @ mid-range measured with a sound pressure level meter (SPL)

(f) Signal to noise (S/N) ratio: 35 dB, minimum

b. Audio Level Processing: The head-end equipment shall consist of audio mixer(s), volume limiter(s) and/or compressor(s), and power amplifier(s) to process, adjust, equalize, isolate, filter, and amplify each audio channel for each zone or sub-zone in the system and distribute them into the system's distribution trunks. It is acceptable to use identified telephone system cable pairs designated for PA use or identified as spare telephone cable pairs by the Facility's Telephone System Contractor.

(1) THE USE OF TELEPHONE CABLE TO DISTRIBUTE PA SIGNALS CARRYING AC OR DC VOLTAGE IS NOT ACCEPTABLE AND WILL NOT BE APPROVED.

(2) Additionally, each remote location shall be provided with the equipment required to ensure the system supervision and designed audio channel capacity at each speaker identified on the contract drawings.

2.4 MANUFACTURERS

a. The products specified shall be new, FCC and UL Listed, labeled and produced by OEM of record. An OEM of record shall be defined as a company whose main occupation is the manufacture for sale of the items of equipment supplied and which:

(1) Maintains a stock of replacement parts for the item submitted,

(2) Maintains engineering drawings, specifications, and operating manuals for the items submitted, and

(3) Has published and distributed descriptive literature and equipment specifications on the items of equipment submitted at least 30 days prior to the Invitation for Bid (IFB).

b. Specifications contained herein as set forth in this document detail the salient operating and performance characteristics of equipment in order for VA to distinguish acceptable items of equipment from unacceptable items of equipment. When an item of equipment is offered or furnished for which there is a specification contained herein, the item of equipment offered or furnished shall meet or exceed the specification for that item of equipment.

c. Equipment Standards and Testing:

(1) The System has been defined herein as connected to systems identified as an Emergency performing Public Safety Support Functions. Therefore, at a minimum, the system shall conform to all aforementioned National and/or Local Public and Life Safety Codes (which ever are the more stringent), NFPA 101, NFPA 70, this specification, JCAHCO Life Safety Accreditation requirements, and the OEM recommendations, instructions, and guidelines.

(2) All supplies and materials shall be listed, labeled or certified by UL or a nationally recognized testing laboratory (NRTL) where such standards have been established for the supplies, materials or equipment.

(3) The provided equipment required by the System design and approved technical submittal must conform with each UL standard in effect

for the equipment, as of the date of the technical submittal (or the date when the RE approved system equipment necessary to be replaced) was technically reviewed and approved by VA. Where a UL standard is in existence for equipment to be used in completion of this contract, the equipment must bear the approved UL seal.

(4) Each item of electronic equipment to be provided under this contract must bear the approved UL seal or the seal of the testing laboratory that warrants the equipment has been tested in accordance with, and conforms to the specified standards. The placement of the UL Seal shall be a permanent part of the electronic equipment that is not capable of being transportable from one equipment item to another.

2.5 PRODUCTS

a. General.

(1) Contractor is responsible for pricing all accessories and miscellaneous equipment required to extend the existing system. The equipment quantities shall be as indicated on the drawings with the exception of the indicated spare equipment.

(2) Each cabinet shall be provided with internal and external items to maintain a neat and orderly system of equipment, wire, cable and conduit connections and routing.

(3) Contractor Furnished Equipment List (CFEs):

(a) The Contractor is required to provide a list of the CFE equipment to be furnished. The quantity, make and model number of each item is required. Select the required equipment items quantities that will satisfy the needs of the system as described herein and with the OEM’s concurrence applied to the list(s), in writing.

(b) The following equipment items shall be outlined by the contractor to extend the existing system described herein:

Item		Quantity	Unit
(1)	As required Interface Panel(s)		
1.a	As required Trouble Enunciator	Electrical Supervision	
1.a.1.	As required	Equipment Back Box(s)	
1.a.2.	As required	Telephone Access Equipment	
1.a.3.	As required	Radio Paging Access Equipment	
1.a.3.a.	As required	Radio Pager Equipment	
1.a.4.	As required	Wireless Access Equipment	
1.a.5.	As required Equipment	Personal Communicator	
(2)	As required Lightning Arrestor		
(3)	As required Head End Equipment Locations		
3.a	As required	Cabinet(s)	
3.a.1.	As required	AC Power Conditioner & Filter	

3.a.2.	As required	AC Power Strip
3.a.3.	As required	UPS
3.a.3.a	As required	Main Power Amplifiers
3.a.3.b	As required	Remote Power Amplifiers
3.a.3.c	As required	Distributed Amplifiers (When Approved)
3.a.4.	As required	Interconnecting wire Cable(s)
3.a.4.a	As required	Wire Cable Connector(s)
3.a.4.b	As required	Wire Cable Terminator(s)
3.a.4.c	As required	Wire Management System
3.b.	As required	Head End Function(s)

(4) As required Distribution System(s)

4.a	As required	Equipment Back Box(s)
4.a.1.	As required	Speakers
4.a.1.a	As required	Overhead
4.a.1.b	As required	Horn
4.a.1.c	As required	Outside
4.a.1.d	As required	Speaker w/ Microphone

(5) 2 (MIN) Remote Station(s)

5.a.	As required	Spare Items

b. ENT (aka DEMARC) Room(s):

Refer to CFM Physical Security Manual (07-2007) for VA Facilities, Chapters 9.3 & 1) and VA PG-18-10, EDM, Chapters 7- Table 7-1, 8 & Appendix B, Telecommunications One Line Topology for specific Room and TIP Connection Requirements.

c. TER, TCR, TR, SCC, PCR, STR, HER Rooms and Equipment:

Refer to CFM Physical Security Manual (07-2007) for VA Facilities, Chapters 9.3 & 1) and VA PG-18-10, EDM, Chapters 7- Table 7-1, 8 & Appendix B, Telecommunications One Line Topology for specific Room and TIP Connection Requirements.

(1) Interface Equipment:

(a) TER:

1. Paging adaptor:

a) The Contractor shall coordinate the installation of the paging adapter(s) designed for use with the Facility’s telephone system with the Facility Telephone Contractor or local telephone company.

b) The Contractor shall provide and install a paging adapter(s) for each zone and sub zone. The paging adapter(s) shall be accessible by dialing a telephone number provided by the Facility’s Telephone Contractor. The Paging Adapter shall:

1. Monitor each audio input and output on the unit.

2. Be provided with an electrical supervision panel to provide both audio and visual trouble alarms.

3. Be provided as part of the head end equipment and shall be located in the Telephone Switch Room

4. Be provided with Executive (aka emergency) Paging Override of all routine paging calls in progress or being accessed to allow system “all call” (aka global) and radio paging calls designated as (Code One Blue) functions.

5. Be capable of internal time out capability.

6. Function completely with the interface module.

7. Provide one spare adapter.

c) Time Out Device: A time out device/capability shall be provided to prevent system “hang-up” due to an off-hook telephone. The device shall be able to be preset from 30 seconds to two (2) minutes. Its function shall not interfere with or override the required “all call” (aka global) operational capability.

1. Central Processor Module:

2. Controls system operations and holds all programmed parameters.

3. Data link connection to additional CPU modules.

d) Power Module: Provides 12V DC @ 800mA to Central Processor Module.

e) Minimum three (3) Zone Module:

1. Provides a minimum of three (3) paging zone outputs at 70V audio sound level.

2. Background Music inhibit switch for each zone.

2. Audio Monitor Panel:

a) The panel shall be EIA/TIA standard for (19”) cabinet mounting.

b) It shall be provided in the upper portion of the head-end equipment cabinet.

c) Provide one (1) spare panel.

3. Trouble Annunciator Panel:

a) A trouble annunciator panel shall be provided in the head-end cabinet, and at locations as designated on the contract drawings. The panel(s) shall be compatible with or generate electrical and/or electronic supervising signals to continuously monitor the operating condition for the System head-end audio power amplifier(s), remote power amplifier(s), microphone consoles and interconnecting trunks. The panels shall generate an audible and visual signal when the System’s supervising system detects an

amplifier or trunk-line is malfunctioning.

b) Provide one (1) spare panel.

4. Head-End Equipment

a) Provide all required power supplies, communications hubs, network switches, intelligent controllers and other devices necessary to form a complete system listed herein. Head-end components may be rack mounted or wall mounted in a metal enclosure.

b) Provide the head end equipment in the closed telecommunications closet where the PA system is installed to include the minimum equipment listed herein.

c) Provide minimum of 30 minute battery back-up to system components.

5. Equipment Cabinet: Comply with EIA-310-D. Lockable, ventilated metal cabinet houses terminal strips, power supplies, amplifiers, system volume control, and other switching and control devices required for conversation channels and control functions

a) Vertical Equipment Rack, Wall Mounted (to be included inside of the Equipment Cabinet):

b) 74″ (48RU) rack space, Welded Steel construction, Minimum 20″ usable depth, Adjustable front mounting rails.

1. Install the following products in rack provided by same manufacturer or as specified:

2. Security screws w/ nylon isolation bushings.

3. Textured blank panels.

4. Custom mounts for components without rack mount kits.

5. Security covers.

6. Copper Bus Bar.

7. Power Sequencer rack mounted power conditioner and (provide as needed) delayed sequencer(s) with two (2) inswitched outlets each and contact closure control inputs.

8. Rack mounting: Provide rack mount kit.

6. Amplifier Equipment:

a) Paging (aka zone):

1. Inputs for 600-ohm balanced telephone line, LO-Z balanced microphone, and background music.

2. Input Sensitivity: Compatible with master stations and central equipment so amplifier delivers full rated output with sound-pressure level of less than 10 dynes/sq. cm impinging on

master stations speaker microphones, or handset transmitters

3. Automatic Level Control (ALC) for pages, adjustable background music muting level during page, wall or rack mountable.

4. 16-ohm, 25V, 25V center tapped (CT), and 70V outputs. Amplifier quantity and size (output power) as needed. Continuous amplifier power rating shall exceed loudspeaker load on amplifier by at least 25%.

5. Output Power: 70-V balanced line. 80 percent of the sum of wattage settings of connected for each station and speaker connected in all-call mode of operation, plus an allowance for future stations.

6. Total Harmonic Distortion: Less than 5 percent at rated output power with load equivalent to quantity of stations connected in all-call mode of operation.

7. Minimum Signal-to-Noise Ratio: 45 dB, at rated output.

8. Frequency Response: Within plus or minus 3 dB from 70 to 12,000 Hz.

b) Output Regulation: Maintains output level within 2 dB from full to no load.

c) Amplifier Protection: Prevents damage from shorted or open output.

d) Be provided with electronic supervision function(s).

e) Provide one spare amplifier.

7. Wireless (when specifically approved by TVE 005OP3B):

a) Radio Paging Equipment / Systems:

1. The PA system shall have the ability to interface only with VA certified and licensed radio paging system (FCC Part 15 listed pagers and transmitters are not allowed for “Safety of Life” functions or installed in those specific areas – VA Headquarters TVE – 0050PB2 and SM – 0050PB2 are the only approving authorities for this function) and must have the following minimum system features:

a) Ability to pass-through location information (such as a room number) and call-type as well as other text messages simultaneously to shift supervisor identified staff members.

b) System shall allow the operator to select staff members by name and pager number and to select a message consisting of a room number and a condition code (aka priority level). Operator may also choose to type in a unique alpha-numeric text message (the text message shall meet or exceed all HIPA and VAOCIP Communications Security Guidelines for the transmission of Patient or Staff Specific information aka PII – VA Headquarters TVE – 0050P2B is the approving authority for this function) into the system to be read by the holder of the pager unit.

c) While a patient station is connected to the nurse's master station, the system shall allow the operator to automatically page the staff member assigned to the room. An alternate staff member maybe selected for paging purposes in place of the primary staff member. The system must allow an alternate staff member to be paged when the primary staff member is unable to respond to patient's needs within a specified period of time. The System must have the ability to assign any bed to any pager or pager group, and to assign an unlimited amount of pagers to any patient bed.

d) System shall have the ability to send all code blue calls to staff members by predetermined group (as required) automatically by simply pressing one "Code Blue" button. Pager shall indicate room number of code call, and state "Code Blue" in plain English format on pagers (FCC Part 15 listed pagers are not allowed to be used as "Safety of Life" functions or those specific locations – VA Headquarters TVE –0050P2B is the approving authority for this requirement)

(b) TCR:

1. Microphone Paging Console:

a) A console shall be provided in the TCR and PCR's and _____as shown on the drawings.

b) The console shall contain visual enunciators for each connection to the telephone system's Public Address Paging Adapter. The visual enunciators shall display all the System connections to the telephone system being used.

c) The console shall be fully independent of the Facility's telephone system so if the telephone system has a catastrophic failure (aka partial, multiple or total system failure) the microphone console will function normally as if the Facility's telephone system was operating normally. The restoration of the Facility's telephone system shall not affect the System.

d) Each microphone console shall:

2. Be Mounted: Flush unless otherwise indicated, and

suitable for mounting conditions indicated.

3. Have a Faceplate: Stainless steel or anodized aluminum with tamperproof mounting screws.

4. Have a system interface Back Box: Minimum Two-gang galvanized steel with 2-1/2 inch minimum depth.

5. Have an Internal Speaker: 3 inches, 2.3 oz. minimum; permanent magnet.

6. Have a Call Switch: Mount on faceplate. Permits calls to The system.

7. When approved - in lieu of a standalone microphone, provide a Handset with Hook Switch: Have a Handset with Hook Switch: Telephone type with 24-inch-long, permanently coiled cord. Arrange to disconnect speaker when handset is lifted.

8. Be provided with an electrical supervision panel to provide both audio and visual trouble alarms to the Nurse Call /Code Blue electrical supervision system.

9. Be capable of internal time out capability.

10. Be completely compatible with the Telephone Interface unit(s)

11. Electrical Supervision Trouble Annunciator Panel:

a) The Electrical Supervision Trouble Annunciation Panel shall be located in the TCR and PCR’s SCC.

b) The panel(s) shall be compatible with the generated electrical and/or electronic supervising signals to continuously monitor the operating condition for the PA system head-end processing equipment, local/remote control consoles, audio power amplifier(s), UPS, power supplies, dome lights and interconnecting trunks. The panels shall generate an audible and visual signal when the System’s supervising system detects a system trouble or trunk-line is malfunctioning.

c) TRs: Locate the PA floor distribution equipment within each TR as required by system design and OEM direction. Provide secured and lockable cabinet/rack(s) as required.

12. General Equipment: Provide all required power supplies, communications hubs, network switches, intelligent controllers and other devices necessary to form a complete system listed herein. Equipment components may be rack mounted or wall mounted in a metal enclosure.

13. Amplifiers:

a) Panging Amplifier Equipment:

b) Refer to the Amplifier characteristics described herein Paragraph 2.4.G.f.

c) Provide one (1) spare amplifier in addition to the spare Head End Amplifier.

14. Distributed Amplifier:

a) Provide the type and number of the amplifier(S) required to meet the system design. Provide this unit as complete and separate technical submittal during the IFB review portion of the project.

b) Provide one spare amplifier for each 20% (or portion thereof) of amplifiers used in the system.

15. Provide the equipment in the nearest TER where the System

is installed to include the minimum equipment listed herein.

16. Provide minimum of 30 minute battery (UPS) back-up to system components.

17. Equipment Cabinet: Comply with cabinet requirements as aforementioned.

18. Trouble Annunciator Panel: Comply with the panel characteristics identified herein.

(d) SCC, PCR, STR, HER: Refer to VA PG-18-10, Article 7 for specific required equipment and use minimum aforementioned specifications for population.

d. TIP DISTRIBUTION SYSTEM:

(1) System Speakers:

(a) Ceiling Cone-Type:

1. Minimum Axial Sensitivity: 91 dB at one meter, with 1-W input.

2. Frequency Response: Within plus or minus 3 dB from 70 to 15,000 Hz.

3. Minimum Dispersion Angle: 100 degrees.

4. Line Transformer: Maximum insertion loss of 0.5 dB, power rating equal to speaker's, and at least four level taps.

5. Enclosures: Steel housings or back boxes, acoustically dampened, with front face of at least 0.0478-inch steel and whole assembly rust proofed and factory primed; complete with mounting assembly and suitable for surface ceiling, flush ceiling, pendant or wall mounting; with relief of back pressure.

6. Baffle: For flush speakers, minimum thickness of 0.032-inch aluminum with textured white finish. Completely fill the baffle with fiberglass.

7. Vandal-Proof, High-Strength Baffle: For flush-mounted speakers, self-aging cast aluminum with tensile strength of 44,000 psi, 0.025-inch minimum thickness; countersunk heat-treated alloy mounting screws; and textured white epoxy finish.

8. Size: 8 inches with 1-inch voice coil and minimum 5-oz. ceramic magnet.

9. Have a minimum of two (2) safety wires installed to a solid surface or use a flexible conduit from ceiling / wall back box to the speaker back box.

10. The speakers and mounting shall be self contained and wall mounted with flush back box at a minimum of 10 meter intervals and shall match (or contrast with, at the direction of the RE) the color of the adjacent surfaces.

11. Provide one spare speaker, mount, and back box for each 50 speakers or portion thereof.

(b) Wall Mounted Horne-Type:

1. Each horn speaker shall be provided with a means of adjusting the output level over the rated horn speaker range to an appropriate audio level in the area installed.

2. Provide horn speakers in equipment rooms, mechanical room, supply warehouse areas, loading dock, entrance and exit areas, and at other areas as indicated on the drawings.

3. Speakers shall be all-metal, weatherproof construction; complete with universal mounting brackets.

4. Frequency Response: Within plus or minus 3 dB from 275 to 14,000 Hz.

5. Minimum Power Rating of Driver: 15 W, continuous.

6. Minimum Dispersion Angle: 110 degrees.

7. Line Transformer: Maximum insertion loss of 0.5 dB, power rating equal to speaker's, and at least four level taps.

8. Provide one spare speaker, mount, and back box for each 20 speakers or portion thereof.

(c) System Cables: In addition to the TIP provided under Specification Section 27 15 00 – TIP Horizontal and Vertical Communications Cabling, provide the following additional TIP installation and testing requirements, provide the following minimum System TIP cables & interconnections:

1. Line Level Audio and Microphone Cable:

a) Line level audio and microphone cable for inside racks and conduit.

b) Shielded, twisted pair Minimum 22 American Wire Gauge (AWG), stranded conductors and 24 AWG drain wire with overall jacket.

2. Speaker Level (Audio 70.7Volt) Cable, Riser Rated:

a) For use with 70.7 V audio speaker circuits.

b) 18 AWG stranded pair, minimum.

c) UL 1333 listed.

3. Speaker Level Audio Cable, Plenum Rated (70.7V):

a) For use with 70.7 V audio speaker circuits.

b) 18 AWG stranded pair, minimum.

4. All cabling shall be riser rated.

5. Provide one (1) spare 1,000 foot roll of approved System (not microphone) cable only.

(2) Raceways, Back Boxes and conduit:

(a) Raceways:

1. In addition to the Raceways, Equipment Room Fittings provided under Specification Sections 27 15 00 TIP Communication Room Fittings and 27 15 00 – TIP Communications Horizontal and Vertical Cabling, provide the following additional TIP raceway and fittings:

2. Each raceway that is open top, shall be: UL certified for telecommunications systems, partitioned with metal partitions in order to comply with NFPA 70 Parts 517 & 800 to “mechanically separate telecommunications systems of different service, protect the installed cables from falling out when vertically mounted and allow junction boxes to be attached to the side to interface “drop” type conduit cable feeds.

3. Intercommunication System cable infrastructure: EMT above accessible ceilings, 24 inches on center.

4. Junction boxes shall be not less than 2-1/2 inches deep and 6 inches wide by 6 inches long.

5. Flexible metal conduit is prohibited unless specifically approved by 005OP3B.

(b) System Conduit:

1. The PA system is NFPA listed as Emergency / Public Safety Communication System which requires the entire system to be installed in a separate conduit system.

2. The use of centralized mechanically partitioned wireways may be used to augment main distribution conduit on a case by case basis when specifically approved by VA Headquarters (005OP3B).

3. Conduit Sleeves:

a) The AE has made a good effort to identify where conduit sleeves through full-height and fire rated walls on the drawings, and has instructed the electrician to provide the sleeves as shown on the drawings.

b) The contractor is responsible for installing conduit sleeves and fire-proofing where necessary. It is often the case, that due to field conditions, the nurse-call cable may have to be installed through an alternate route. Any conduit sleeves required due to field conditions or those omitted by the engineer shall be provided by the cabling contractor.

(3) Device Back Boxes:

(a) Furnish to the electrical contractor all back boxes required for the PA system devices.

(b) The electrical contractor shall install the back boxes as well as the system conduit. Coordinate the delivery of the back boxes with the construction schedule.

(4) Telecommunication Outlets (TCO): Populate each TCO that is required to perform system operations in the locations that were provided and cabled as a part of Specifications Sections 27 11 00 and 27 15 00. Provide additional TCO equipment, interfaces and connections as required by System design. Provide secured pathway(s) and TCOs as required.

(5) UPS:

(a) Provide a backup battery or a UPS for the System to allow normal operation and function (as if there was no AC power failure) in the event of an AC power failure or during input power fluctuations for a minimum of four (4) hours.

(b) As an alternate solution, the telephone system UPS may be utilized to meet this requirement at the headend location, as long as this function is specifically approved by the Telephone Contractor and the RE.

(c) The PA Contractor shall not make any attachments or connection to the telephone system until specifically directed to do so, in writing, by the RE.

(d) Provide UPS for all active system components including but not limited to:

1. System Amplifiers.

2. Microphone Consoles.

3. Telephone Interface Units.

4. TER, TR & Headend Equipment Rack(s).

e. Installation Kit:

(1) General: The kit shall be provided that, at a minimum, includes all connectors and terminals, labeling systems, audio spade lugs, barrier strips, punch blocks or wire wrap terminals, heat shrink tubing, cable ties, solder, hangers, clamps, bolts, conduit, cable duct, and/or cable tray, etc., required to accomplish a neat and secure installation. All wires shall terminate in a spade lug and barrier strip, wire wrap terminal or punch block. Unfinished or unlabeled wire connections shall not be allowed. Turn over to the RE all unused and partially opened installation kit boxes, coaxial, fiberoptic, and twisted pair cable reels, conduit, cable tray, and/or cable duct bundles, wire rolls, physical installation hardware. The following are the minimum required installation sub-kits:

(2) System Grounding:

(a) The grounding kit shall include all cable and installation hardware required. All radio equipment shall be connected to

earth ground via internal building wiring, according to the NEC.

(b) This includes, but is not limited to:

1. Coaxial Cable Shields.
2. Control Cable Shields.
3. Data Cable Shields.
4. Equipment Racks.
5. Equipment Cabinets.
6. Conduits.
7. Duct.
8. Cable Trays.
9. Power Panels.
10. Connector Panels.
11. Grounding Blocks.

(3) Coaxial Cable: The coaxial cable kit shall include all coaxial connectors, cable tying straps, heat shrink tabbing, hangers, clamps, etc., required to accomplish a neat and secure installation.

(4) Wire and Cable: The wire and cable kit shall include all connectors and terminals, audio spade lugs, barrier straps, punch blocks, wire wrap strips, heat shrink tubing, tie wraps, solder, hangers, clamps, labels etc., required to accomplish a neat and orderly installation.

(5) Conduit, Cable Duct, and Cable Tray: The kit shall include all conduit, duct, trays, junction boxes, back boxes, cover plates, feed through nipples, hangers, clamps, other hardware required to accomplish a neat and secure conduit, cable duct, and/or cable tray installation in accordance with the NEC and this document.

(6) Equipment Interface: The equipment kit shall include any item or quantity of equipment, cable, mounting hardware and materials needed to interface the systems with the identified sub-system(s) according to the OEM requirements and this document.

(7) Labels: The labeling kit shall include any item or quantity of labels, tools, stencils, and materials needed to completely and correctly label each subsystem according to the OEM requirements, as-installed drawings, and this document.

(8) Documentation: The documentation kit shall include any item or quantity of items, computer discs, as installed drawings, equipment, maintenance, and operation manuals, and OEM materials needed to completely and correctly provide the system documentation as required by this document and explained herein.

PART 3 EXECUTION

3.1 PROJECT MANAGEMENT

a. Assign a single project manager to this project who will serve as the point of contact for the Owner, the General Contractor, and the Engineer.

b. The Contractor shall be proactive in scheduling work at the hospital, specifically the Contractor will initiate and maintain discussion with the general contractor regarding the schedule for ceiling cover up and install cables to meet that schedule.

c. Contact the Office of Telecommunications, Special Communications Team (005OP3B) at (301) 734-0350 to have a VA Certified Telecommunications COTR assigned to the project for telecommunications review, equipment and system approval and co-ordination with VA’s Spectrum Management and OCIS Teams.

3.2 COORDINATION WITH OTHER TRADES

a. Coordinate with the cabling contractor the location of the PA system faceplate and the faceplate opening for the PA system back boxes.

b. Coordinate with the cabling contractor the location of TIP equipment in the **TER,, PCR, SCC, ECR, STRs, NSs, HER and TCOs in order to connect to the TIP cable network that was installed as a part of Section Specification 27 11 00. Contact the RE immediately, in writing, if additional location(s) are discovered to be activated that was not previously provided.**

c. Before beginning work, verify the location, quantity, size and access for the following:

(1) Isolated ground AC power circuits provided for systems.

(2) Junction boxes, wall boxes, wire troughs, conduit stubs and other related infrastructure for the systems.

(3) System components installed by others.

(4) Overhead supports and rigging hardware installed by others.

d. Immediately notify the Owner, GC and Consultant(s) in writing of any discrepancies

3.3 NEEDS ASSESSMENT

Provide a one-on-one meeting with the particular manager of each unit affected by the installation of the new PA system. Review the floor plan drawing, educate the nursing manager with the functions of the equipment that is being provided and gather details specific to the individual units; coverage and priorities of calls; staffing patterns; and other pertinent details that will affect system programming and training.

3.4 INSTALLATION

a. General

(1) Execute work in accordance with National, State and local codes, regulations and ordinances.

(2) Install work neatly, plumb and square and in a manner consistent with standard industry practice. Carefully protect work from dust, paint and moisture as dictated by site conditions. The Contractor will be fully responsible for protection of his work during the construction phase up until final acceptance by the Owner.

(3) Install equipment according to OEM's recommendations. Provide any hardware, adaptors, brackets, rack mount kits or other accessories recommended by OEM for correct assembly and installation.

(4) Secure equipment firmly in place, including receptacles, speakers, equipment racks, system cables, etc.

(a) All supports, mounts, fasteners, attachments and attachment points shall support their loads with a safety factor of at least 5: 1.

(b) Do not impose the weight of equipment or fixtures on supports provided for other trades or systems.

(c) Any suspended equipment or associated hardware must be certified by the OEM for overhead suspension.

(d) The Contractor is responsible for means and methods in the design, fabrication, installation and certification of any supports, mounts, fasteners and attachments.

(5) Locate overhead ceiling-mounted loudspeakers as shown on drawings, with minor changes not to exceed 12" in any direction.

(a) Mount transformers securely to speaker brackets or enclosures using screws. Adjust torsion springs as needed to securely support speaker assembly.

(b) Speaker back boxes shall be completely filled with fiberglass insulation.

(c) Seal cone speakers to their enclosures to prevent air passing from one side of the speaker to the other.

(6) Finishes for any exposed work such as plates, racks, panels, speakers, etc. shall be approved by the Architect, Owner and 005OP3B.

(7) Coordinate cover plates with field conditions. Size and install cover plates as necessary to hide joints between back boxes and surrounding wall. Where cover plates are not fitted with connectors, provide grommeted holes in size and quantity required. Do not allow cable to leave or enter boxes without cover plates installed.

(8) Active electronic component equipment shall consist of solid state components, be rated for continuous duty service, comply with the requirements of FCC standards for telephone and data equipment, systems, and service.

(9) Color code all distribution wiring to conform to the PA Industry Standard, EIA/TIA, and this document, whichever is the more stringent. At a minimum, all equipment, cable duct and/or conduit, enclosures, wiring, terminals, and cables shall be clearly and permanently labeled according to and using the provided record drawings, to facilitate installation and maintenance.

(10) Connect the System's primary input AC power to the Facility' Critical Branch of the Emergency AC power distributionsystem as shown on the plans or if not shown on the plans consult with RE regarding a suitable circuit location prior to bidding.

(11) Product Delivery, Storage and Handling:

(a) Delivery: Deliver materials to the job site in OEM's original unopened containers, clearly labeled with the OEM's name and equipment catalog numbers, model and serial identification numbers. The RE may inventory the cable, patch panels, and related equipment.

(b) Storage and Handling: Store and protect equipment in a manner, which will preclude damage as directed by the RE.

(12) Where TCOs are installed adjacent to each other, install one outlet for each instrument.

(13) Equipment installed outdoors shall be weatherproof or installed in weatherproof enclosures with hinged doors and locks with twokeys.

b. Equipment Racks:

(1) Fill unused equipment mounting spaces with blank panels or vent panels. Match color to equipment racks.

(2) Provide security covers for all devices not requiring routine operator control.

(3) Provide vent panels and cooling fans as required for the operation of equipment within the OEM' specified temperature limits. Provide adequate ventilation space between equipment for cooling. Follow manufacturer's recommendations regarding ventilation space between amplifiers.

(4) Provide insulated connections of the electrical raceway to equipment racks.

(5) Provide continuous raceway/conduit with no more than 40% fill between wire troughs and equipment racks for all non-plenum-rated cable. Ensure each system is mechanically separated from each other in the wireway.

(6) Ensure a minimum of 36 inches around each cabinet and/or rack to comply with OSHA Safety Standards. Cabinets and/or Racks installed side by side – the 36" rule applies to around the entire assembly

c. Distribution Frames.

(1) A new stand-alone (i.e., self supporting, free standing) PA rack/frame may be provided in each TR to interconnect the PA, TER, TCR, PCR, SCC, STRs & ECRs. Rack/frames shall be wired in accordance with industry standards and shall employ "latest state-of-the-art" modular cross-connect devices. The PA riser cable shall be sized to satisfy all voice/digital requirements plus not less than 50% spare (growth) capacity in each TR which includes a fiber optic backbone.

(2) The frames/racks shall be connected to the TER/MCR system ground.

d. Wiring Practice - in addition to the MANDATORY infrastructure requirements outlined in VA Construction Specifications 27 10 00 – TIP Structured Communications Cabling, 27 11 00 – TIP Communications Rooms Fittings and 27 15 00 – TIP Horizontal and Vertical Communicators Cabling, the following additional practices shall be adhered too:

(1) Comply with requirements for raceways and boxes specified in Division 26 Section "Raceway and Boxes for Electrical Systems."

(2) Execute all wiring in strict adherence to the NFPA 70, applicable local building codes and standard industry practices.

(3) Wiring shall be classified according to the following low voltage signal types:

(a) Balanced microphone level audio (below -20dBm) or Balanced line level audio (-20dBm to +30dBm)

(b) 70V audio speaker level audio.

(c) Low voltage DC control or power (less than 48VDC)

(4) Where raceway is to be EMT (conduit), wiring of differing classifications shall be run in separate conduit. Where raceway is to be an enclosure (rack, tray, wire trough, utility box) wiring of differing classifications which share the same enclosure shall be mechanically partitioned and separated by at least four (4) inches. Where Wiring of differing classifications must cross, they shall cross perpendicular to one another.

(5) Do not splice wiring anywhere along the entire length of the run. Make sure cables are fully insulated and shielded from each other and from the raceway for the entire length of the run.

(6) Do not pull wire through any enclosure where a change of raceway alignment or direction occurs. Do not bend wires to less than radius recommended by manufacturer.

(7) Replace the entire length of the run of any wire or cable that is damaged or abraided during installation. There are no acceptable methods of repairing damaged or abraided wiring.

(8) Use wire pulling lubricants and pulling tensions as recommended by the OEM.

(9) Use grommets around cut-outs and knock-outs where conduit or chase

nipples are not installed.

(10) Do not use tape-based or glue-based cable anchors.

(11) Ground shields and drain wires to the Facility's signal ground system as indicated by the drawings.

(12) Field wiring entering equipment racks shall be terminated as follows:

(a) Provide ample service loops at harness break-outs and at plates, panels and equipment. Loops should be sufficient to allow plates, panels and equipment to be removed for service and inspection.

(b) Line level and speaker level wiring may be terminated inside the equipment rack using specified terminal blocks (see "Products.") Provide 15% spare terminals inside each rack. Microphone level wiring may only be terminated at the equipment served.

(c) If specified terminal blocks are not designed for rack mounting, utilize 3/4"" plywood or 1/8" thick aluminum plates/blank panels as a mounting surface. Do not mount on the bottom of the rack.

(d) Employ permanent strain relief for any cable with an outside diameter of 1" or greater.

(13) Use only balanced audio circuits unless noted otherwise

(14) Make all connections as follows:

(a) Make all connections using rosin-core solder or mechanical connectors appropriate to the application.

(b) For crimp-type connections, use only tools that are specified by the manufacturer for the application.

(c) Use only insulated spade lugs on screw terminals. Spade lugs shall be sized to fit the wire gauge. Do not exceed two lugs per terminal.

(d) Wire nuts, electrical tape or "Scotch Lock" connections are not acceptable for any application.

(15) Make all connections as follows:

(a) Make all connections using rosin-core solder or mechanical connectors appropriate to the application.

(b) For crimp-type connections, use only tools that are specified by the manufacturer for the application.

(c) Use only insulated spade lugs on screw terminals. Spade lugs shall be sized to fit the wire gauge. Do not exceed two lugs per terminal.

(d) Wire nuts, electrical tape or "Scotch Lock" connections are

not acceptable for any application.

(16) Noise filters and surge protectors shall be provided for each equipment interface cabinet, switch equipment cabinet, control console, local, and remote active equipment locations to ensure protection from input primary AC power surges and noise glitches are not induced into low Voltage data circuits.

(17) Wires or cables **previously approved** to be installed outside of conduit, cable trays, wireways, cable duct, etc:

(a) Only when specifically authorized as described herein, will wires or cables be identified and approved to be installed outside of conduit. The wire or cable runs shall be UL rated plenum and OEM certified for use in air plenums.

(b) Wires and cables shall be hidden, protected, fastened and tied at (24 in.) intervals, maximum, as described herein to building structure.

(c) Closer wire or cable fastening intervals may be required to prevents sagging, maintain clearance above suspended ceilings, remove unsightly wiring and cabling from view and discourage tampering and vandalism. Wire or cable runs, not provided in conduit, that penetrate outside building walls, supporting walls, and two hour fire barriers shall be sleeved and sealed with an approved fire retardant sealant.

(d) Wire or cable runs to system components installed in walls (i.e.: volume attenuators, circuit controllers, signal, or data outlets, etc.) may, when specifically authorized by the RE, be fished through hollow spaces in walls and shall be certified for use in air plenum areas.

(e) Completely test all of the cables after installation and replace any defective cables.

(f) Wires or cables that are installed outside of buildings shall be in conduit, secured to solid building structures. If specifically approved, on a case by case basis, to be run outside of conduit, the wires or cables shall be installed, as described herein. The bundled wires or cables must: Be tied at not less than (18 in.) intervals to a solid building structure; have ultra violet protection and be totally waterproof (including all connections). The laying of wires or cables directly on roof tops, ladders, drooping down walls, walkways, floors, etc. is not allowed and will not be approved.

e. Cable Installation - In addition to the ***MANDATORY*** infrastructure requirements outlined in VA Construction Specifications 27 10 00 – Structured TIP Communications Cabling, 27 11 00 – TIP Communications Rooms and Fittings and 27 15 00 – TIP Communications Horizontal and Vertical Cabling and the following additional practices shall be adhered too:

(1) Support cable on maximum 2’-0” centers. Acceptable means of cable support are cable trays. Velcro wrap cable bundles loosely to the means of support with plenum rated Velcro straps. Plastic tie wraps are not acceptable as a means to bundle cables.

(2) Run cables parallel to walls.

(3) Install maximum of 10 cables in a single row. Provide necessary rows as required by the number of cables.

(4) Do not lay cables on top of light fixtures, ceiling tiles, mechanical equipment, or ductwork. Maintain at least 2′-0″ clearance from all shielded electrical apparatus.

(5) All cables shall be tested after the total installation is fully complete. All test results are to be documented. All cables shall pass acceptable test requirements and levels. Contractor shall remedy any cabling problems or defects in order to pass or comply with testing. This includes the re-pull of new cable as required at no additional cost to the Owner.

(6) Ends of cables shall be properly terminated on both ends per industry and OEM’s recommendations.

(7) Provide proper temporary protection of cable after pulling is complete before final dressing and terminations are complete. Do not leave cable lying on floor. Bundle and tie wrap up off of the floor until you are ready to terminate.

(8) Terminate all conductors; no cable shall contain unterminated elements. Make terminations only at outlets and terminals.

(9) Splices, Taps, and Terminations: Arrange on numbered terminal strips in junction, pull, and outlet boxes; terminal cabinets; and equipment enclosures. Cables may not be spliced.

(10) Bundle, lace, and train conductors to terminal points without exceeding OEM's limitations on bending radii. Install lacing bars and distribution spools.

(11) Cold-Weather Installation: Bring cable to room temperature before dereeling. Heat lamps shall not be used.

(12) Cable shall not be run through structural members or be in contact with pipes, ducts, or other potentially damaging items.

(13) Separation of Wires: (REFER TO RACEWAY INSTALLATION) Separate speaker-microphone, line-level, speaker-level, and power wiring runs. Install in separate raceways or, where exposed or in same enlosure, separate conductors at least 12 inches apart for speaker microphones and adjacent parallel power and telephone wiring. Separate other intercommunication equipment conductors as recommended by equipment manufacturer.

(14) Serve all cables as follows:

(a) Cover the end of the overall jacket with a 1″ (minimum) length of transparent heat-shrink tubing. Cut unused insulated conductors 2″ (minimum) past the heat-shrink, fold back over jacket and secure with cable-tie. Cut unused shield/drain wires 2″ (minimum) past the Heatshrink and serve as indicated below.

(b) Cover shield/drain wires with heat-shrink tubing extending

back to the overall jacket. Extend tubing ¼" past the end of unused wires, fold back over jacket and secure with cable tie.

(c) For each solder-type connection, cover the bare wire and solder connection with heat-shrink tubing.

f. Labeling: Provide labeling in accordance with EIA/TIA-606-A. All lettering for PA circuits shall be stenciled using thermal ink transfer process.

(1) Cable and Wires (Hereinafter referred to as "Cable"): Cables shall be labeled at both ends in accordance with EIA/TIA-606-A. Labels shall be permanent in contrasting colors. Cables shall be identified according to the System "Record Wiring Diagrams."

(2) Equipment: System equipment shall be permanently labeled with contrasting plastic laminate or Bakelite material. System equipment shall be labeled on the face of the unit corresponding to its source.

(a) Clearly, consistently, logically and permanently mark switches, connectors, jacks, relays, receptacles and electronic and other equipment.

(b) Engrave and paint fill all receptacle panels using 1/8" (minimum) high lettering and contrasting paint.

(c) For rack-mounted equipment, use engraved Lamacoid labels with white 1/8" (minimum) high lettering on black background. Label the front and back of all rack-mounted equipment.

(3) Conduit, Cable Duct, and/or Cable Tray: The Contractor shall label all conduit, duct and tray, including utilized GFE, with permanent marking devices or spray painted stenciling a minimum of (10 ft.) identifying it as the System. In addition, each enclosure shall be labeled according to this standard.

(4) Termination Hardware: The Contractor shall label TCOs and patch panel connections using color coded labels with identifiers in accordance with EIA/TIA-606-A and the "Record Wiring Diagrams."

(5) Where multiple pieces of equipment reside in the same rack group, clearly and logically label each indicating to which room, channel, receptacle location, etc. they correspond.

(6) Permanently label cables at each end, including intra-rack connections. Labels shall be covered by the same, transparent heat-shrink tubing covering the end of the overall jacket. Alternatively, computer generated labels of the type which include a clear protective wrap may be used.

(7) Contractor's name shall appear no more than once on each continuous set of racks. The Contractor's name shall not appear on wall plates or portable equipment.

(8) Ensure each OEM supplied item of equipment has appropriate UL Labels / Marks for the service the equipment is performed permanently attached / marked. SYSTEM EQUIPMENT INSTALLED NOT BEARING THESE UL MARKS WILL NOT BE ALLOWED TO BE A PART OF THE

SYSTEM. THE CONTRACTOR SHALL BEAR ALL COSTS REQUIRED TO PROVIDE REPLACEMENT EQUIPMENT WITH APPROVED UL MARKS.

g. Conduit and Signal Ducts: When the Contractor and/or OEM determines additional system conduits and/or signal ducts are required in order to meet the system minimum performance standards outlined herein, the contractor shall provide these items as follows:

(1) Conduit:

(a) The Contractor shall employ the latest installation practices and materials. The Contractor shall provide conduit, junction boxes, connectors, sleeves, weather heads, pitch pockets, and associated sealing materials not specifically identified in this document as GFE. Conduit penetrations of walls, ceilings, floors, interstitial space, fire barriers, etc., shall be sleeved and sealed.

(b) All cables shall be installed in separate conduit and/or signal ducts (exception from the separate conduit requirement to allow PA cables to be installed in partitioned cable tray with voice cables may be granted in writing by the RE if requested). Conduits shall be provided in accordance with Section 27 05 33, RACEWAYS AND BOXES FOR COMMUNICATIONS SYSTEMS, and NFPA 70 Articles 517 for Critical Care and 800 for Communications systems, at a minimum.

(c) When metal, plastic covered, etc., flexible cable protective armor or systems are specifically authorized to be provided for use in the System, their installation guidelines and standards shall be as specified herein, Section 27 05 33, RACEWAYS AND BOXES FOR COMMUNICATIONS SYSTEMS, and the NEC.

(d) When ”interduct” flexible cable protective systems is specifically authorized to be provided for use in the System, it’s installation guidelines and standards shall be as the specified herein, Section 27 05 33, RACEWAYS AND BOXES FOR COMMUNICATIONS SYSTEMS, and NFPA 70.

(e) Conduit fill (including GFE approved to be used in the system) shall not exceed 40%. Each conduit end shall be equipped with a protective insulator or sleeve to cover the conduit end, connection nut or clamp, to protect the wire or cable during installation and remaining in the conduit. Electrical power conduit shall be installed in accordance with NFPA 70. AC power conduit shall be run separate from signal conduit.

(2) Signal Duct, Cable Duct, or Cable Tray:

(a) The Contractor shall use GFE signal duct, cable duct, and/or cable tray, when identified and approved by the RE.

(b) Approved signal and/or cable duct shall be a minimum size of x (4 in. X 4 in.) inside diameter with removable tops or sides, as appropriate. Protective sleeves, guides or barriers are required on all sharp corners, openings, anchors, bolts or screw ends, junction, interface and connection points.

(c) Approved cable tray shall be fully covered, mechanically and

physically partitioned for multiple electronic circuit use, and be UL certified and labeled for use with telecommunication circuits and/or systems. The RE shall approve width and height dimensions.

(d) All cable junctions and taps shall be accessible. Provide an 8″ X 8″ X 4″ (minimum) junction box attached to the cable duct or raceway for installation of distribution system passive equipment. Ensure all equipment and tap junctions are accessible

3.5 PROTECTION OF NETWORK DEVICES

Contractor shall protect network devices during unpacking and installation by wearing manufacturer approved electrostatic discharge (ESD) wrist straps tied to chassis ground. The wrist strap shall meet OSHA requirements for prevention of electrical shock, should technician come in contact with high voltage.

3.6 CUTTING, CLEANING AND PATCHING

a. It shall be the responsibility of the contractor to keep their work area clear of debris and clean area daily at completion of work.

b. It shall be the responsibility of the contractor to patch and paint any wall or surface that has been disturbed by the execution of this work.

c. The Contractor shall be responsible for providing any additional cutting, drilling, fitting or patching required that is not indicated as provided by others to complete the Work or to make its parts fit together properly.

d. The Contractor shall not damage or endanger a portion of the Work or fully or partially completed construction of the Owner or separate contractors by cutting, patching or otherwise altering such construction, or by excavation. The Contractor shall not cut or otherwise alter such construction by the Owner or a separate contractor except with written consent of the Owner and of such separate contractor; such consent shall not be unreasonably withheld. The Contractor shall not unreasonably withhold from the Owner or a separate Contractor the Contractor’s consent to cutting or otherwise altering the Work.

e. Where coring of existing (previously installed) concrete is specified or required, including coring indicated under unit prices, the location of such coring shall be clearly identified in the field and the location shall be approved by the Project Manager prior to commencement of coring work.

3.7 FIREPROOFING

a. Where PA wires, cables and conduit penetrate fire rated walls, floors and ceilings, fireproof the opening.

b. Provide conduit sleeves (if not already provided by electrical contractor) for cables that penetrate fire rated walls and Telecommunications Rooms floors and ceilings. After the cabling installation is complete, install fire proofing material in and around all conduit sleeves and openings. Install fire proofing material thoroughly and neatly. Seal all floor and ceiling penetrations.

c. Use only materials and methods that preserve the integrity of the fire stopping system and its rating.

d. Install fireproofing where low voltage cables are installed in the same manholes with high voltage cables; also cover the low voltage cables with arc proof and fireproof tape.

e. Use approved fireproofing tape of the same type as used for the high voltage cables, and apply the tape in a single layer, one-half lapped or as recommended by the manufacturer. Install the tape with the coated side towards the cable and extend it not less than (one inch) into each duct.

f. Secure the tape in place by a random wrap of glass cloth tape.

3.8 GROUNDING

a. Ground PA cable shields and equipment to eliminate shock hazard and to minimize ground loops, commonmode returns, noise pickup, cross talk, and other impairments as specified in CFM Division 27, Section 27 05 26 – Grounding and Bonding for Communications Systems.

b. Facility Signal Ground Terminal: Locate at main room or area signal ground within the room (i.e. head end and telecommunications rooms) or area(s) and indicate each signal ground location on the drawings.

c. Extend the signal ground to inside each equipment cabinet and/or rack. Ensure each cabinet and/or rack installed item of equipment is connected to the extended signal ground. Isolate the signal ground from power and major equipment grounding systems.

d. When required, install grounding electrodes as specified in CFM Division 26, Section 26 05 26 –Grounding and Bonding for Electrical Systems.

e. Do not use “3rd or 4th” wire internal electrical system conductors for communications signal ground.

f. Do not connect the signal ground to the building’s external lightning protection system.

g. Do Not “mix grounds” of different systems.

h. Insure grounds of different systems are installed as to not violate OSHA Safety and NFPA 70 installation requirements for protection of personnel.

3.9 SYSTEM LISTING

The PA System is NFPA listed as an “Emergency / Public Safety” Communications system. Where Code Blue signals are transmitted, that listing is elevated to “Life Support/Safety.” Therefore, the following testing and guaranty provisions are the minimum to be performed and provided by the contractor and OEM.

3.10 PROOF OF PERFORMANCE TESTING

a. Intermediate Testing:

(1) After completion of 25 – 30% the installation of a head end cabinet(s) and equipment, one microphone console, local and remote enunciation stations, two (2) zones, two (2) sub zones prior to any further work, this portion of the system must be pretested, inspected, and certified. Each item of installed equipment shall be checked to ensure appropriate UL Listing and Certification Labels are affixed as required by NFPA 101 -Life Safety Code 101-3.2 (a) & (b) and JCHCO evaluation guidelines, and proper installation practices are followed. The intermediate test shall include a full operational test.

(2) All inspections and tests shall be conducted by an OEM-certified contractor representative and witnessed by TVE-005OP3B if there is no local Government Representative that processes OEM and VA approved Credentials to inspect and certify the system. The results of the inspection will be officially recorded by the Government Representative and maintained on file by the RE, until completion of the entire project. The results will be compared to the Acceptance Test results. An identical inspection may be conducted between the 65 - 75% of the system construction phase, at the direction of the RE.

b. Pretesting:

(1) Upon completing installation of the PA System, the Contractor shall align, balance, and completely pretest the entire system under full operating conditions.

(2) Pretesting Procedure:

(a) During the System Pretest the Contractor shall verify (utilizing approved test equipment) that the System is fully operational and meets all the System performance requirements of this standard.

(b) The Contractor shall pretest and verify that all PA System functions and specification requirements are met and operational, no unwanted aural effects, such as signal distortion, noise pulses, glitches, audio hum, poling noise, etc. are present. At a minimum, each of the following locations shall be fully pretested:

1. Central Control Cabinets.

2. Local Control Stations.

3. Zone Equipment/Systems.

4. Sub-Zone Equipment/Systems.

5. Remote Control Panels.

a.)TCR.
b.)PCR/SCC.

6. All Networked locations.

7. System interface locations (i.e. TELCO, two way radio,

etc.).

8. System trouble reporting.

9. System Electrical Supervision.

10. UPS operation.

11. STRs.

12. NSs

13. TCOs.

(3) The Contractor shall provide four (4) copies of the recorded system pretest measurements and the written certification that the System is ready for the formal acceptance test shall be submitted to the RE.

c. Acceptance Test:

(1) After the PA System has been pretested and the Contractor has submitted the pretest results and certification to the RE, then the Contractor shall schedule an acceptance test date and give the RE 30 day's written notice prior to the date the acceptance test is expected to begin. The System shall be tested in the presence of TVE 005OP3B and an OEM certified representatives. The System shall be tested utilizing the approved test equipment to certify proof of performance and Emergency / Public Safety compliance. The tests shall verify that the total System meets all the requirements of this specification. The notification of the acceptance test shall include the expected length (in time) of the test.

(2) The acceptance test shall be performed on a "go-no-go" basis. Only those operator adjustments required to show proof of performance shall be allowed. The test shall demonstrate and verify that the installed System does comply with all requirements of this specification under operating conditions. The System shall be rated as either acceptable or unacceptable at the conclusion of the test. Failure of any part of the System that precludes completion of system testing, and which cannot be repaired in four (4) hours, shall be cause for terminating the acceptance test of the System. Repeated failures that result in a cumulative time of eight (8) hours to affect repairs shall cause the entire System to be declared unacceptable. Retesting of the entire System shall be rescheduled at the convenience of the Government.

(3) Retesting of the entire System shall be rescheduled at the convenience of the Government and costs borne by the Contractor at the direction of the SRE.

d. Acceptance Test Procedure:

(1) Physical and Mechanical Inspection:

(a) The TVE 005OP3B Representative will tour all areas where the PA system and all sub-systems are completely and properly

installed to insure they are operationally ready for proof of performance testing. A system inventory including available spare parts will be taken at this time. Each item of installed equipment shall be checked to ensure appropriate UL certification labels are affixed.

(b) The System diagrams, record drawings, equipment manuals, TIP Auto CAD Disks, intermediate, and pretest results shall be formally inventoried and reviewed.

(c) Failure of the System to meet the installation requirements of this specification shall be grounds for terminating all testing.

(2) Operational Test:

(a) After the Physical and Mechanical Inspection, the system head end equipment shall be checked to verify that it meets all performance requirements outlined herein. A spectrum analyzer and sound level meter may be utilized to accomplish this requirement.

(b) Following the head end equipment test, each speaker (or on board speaker) shall be inspected to ensure there are no signal distortions such as intermodulation, data noise, popping sounds, erratic system functions, on any function.

(c) The distribution system shall be checked at each interface, junction, and distribution point, first, middle, and last speaker in each leg to verify the PA distribution system meets all system performance standards.

(d) If the RED system is a part of the system, each volume stepper switches shall be checked to insure proper operation of the pillow speaker, the volume stepper and the RED system (if installed).

(e) Additionally, each installed head end equipment, microphone console; amplifier, mixer, distributed speaker/amplifier, monitor speaker, telephone interface, power supply and remote amplifiers shall be checked insuring they meet the requirements of this specification.

(f) Once these tests have been completed, each installed sub-system function shall be tested as a unified, functioning and fully operating system. The typical functions are: “all call,” three sub-zoned, minimum of 15 minutes of UPS operation, electrical supervision, trouble panel, corridor speakers and audio paging.

(h) Individual Item Test: The TVE 005OP3B Representative will select individual items of equipment for detailed proof of performance testing until 100% of the System has been tested and found to meet the contents of this specification. Each item shall meet or exceed the minimum requirements of this document.

(3) Test Conclusion:

(a) At the conclusion of the Acceptance Test, using the generated punch list (or discrepancy list) the VA and the Contractor shall jointly agree to the results of the test, and reschedule testing on deficiencies and shortages with the RE. Any retesting to

comply with these specifications will be done at the Contractor's expense.

(b) If the System is declared unacceptable without conditions, all rescheduled testing expenses will be borne by the Contractor.

e. Acceptable Test Equipment: The test equipment shall furnished by the Contractor shall have a calibration tag of an acceptable calibration service dated not more than 12 months prior to the test. As part of the submittal, a test equipment list shall be furnished that includes the make and model number of the following type of equipment as a minimum:

(1) Spectrum Analyzer.

(2) Signal Level Meter.

(3) Volt-Ohm Meter.

(4) Sound Pressure Level (SPL) Meter.

(5) Oscilloscope.

(6) Random Noise Generator.

(7) Audio Amplifier with External Speaker.

3.11 WARRANTY

a. Comply with FAR 52.246-21, except that warranty shall be as follows:

b. Contractor’s Responsibility:

(1) The Contractor shall warranty that all provided material and equipment will be free from defects, workmanship and will remain so for a period of two (2) years from date of final acceptance of the System by the VA. The Contractor shall provide OEM’s equipment warranty documents, to the RE (or Facility Contracting Officer if the Facility has taken procession of the building), that certifies each item of equipment installed conforms to OEM published specifications.

(2) The Contractor's maintenance personnel shall have the ability to contact the Contractor and OEM for emergency maintenance and logistic assistance, remote diagnostic testing, and assistance in resolving technical problems at any time. This contact capability shall be provided by the Contractor and OEM at no additional cost to the VA.

(3) All Contractor maintenance and supervisor personnel shall be fully qualified by the OEM and must provide two (2) copies of current and qualified OEM training certificates and OEM certification upon request.

(4) Additionally, the Contractor shall accomplish the following minimum requirements during the two year guaranty period:

(a) Response Time During the ***Two Year*** Guaranty Period:

1. The RE (or Facility Contracting Officer if the system has been turned over to the Facility) is the Contractor's ONLY OFFICIAL reporting and contact official for nurse call system trouble calls, during the guaranty period.

2. A standard work week is considered 8:00 A.M. to 5:00 P.M. or as designated by the RE (or Facility Contracting Officer), Monday through Friday exclusive of Federal Holidays.

3. The Contractor shall respond and correct on-site trouble calls, during the standard work week to:

a) A routine trouble call within one (1) working day of its report. A routine trouble is considered a trouble which causes a power supply; one (1) master System control station, microphone console or amplifier to be inoperable.

b) Routine trouble calls in critical emergency health care facilities (i.e., cardiac arrest, intensive care units, etc.) shall also be deemed as an emergency trouble call. The RE (or Facility Contracting Officer) shall notify the Contractor of this type of trouble call.

c) An emergency trouble call within four (4) hours of its report. An emergency trouble is considered a trouble which causes a sub-zone, zone, distribution point, terminal cabinet, or all call system to be inoperable at anytime.

4. If a PA System component failure cannot be corrected within four (4) hours (exclusive of the standard work time limits), the Contractor shall be responsible for providing alternate System equipment. The alternate equipment/system shall be operational within a maximum of 12 hours after the four (4) hour trouble shooting time and restore the effected location operation to meet the System performance standards. If any sub-system or major system trouble cannot be corrected within one working day, the Contractor shall furnish and install compatible substitute equipment returning the System or sub-system to full operational capability, as described herein, until repairs are complete.

(b) Required On-Site Visits During the ***Two Year*** Guaranty Period

1. The Contractor shall visit, on-site, for a minimum of eight (8) hours, once every 12 weeks, during the guaranty period, to perform system preventive maintenance, equipment cleaning, and operational adjustments to maintain the System according the descriptions identified in this document.

2. The Contractor shall arrange all Facility visits with the RE (or Facility Contracting Officer) prior to performing the required maintenance visits.

3. Preventive maintenance procedure(s) shall be performed by the Contractor in accordance with the OEM's recommended practice and service intervals during non-busy time agreed to by the RE (or Facility Contracting Officer) and Contractor.

4. The preventive maintenance schedule, functions and reports

shall be provided to and approved by the RE (or Facility Contracting Officer).

5. The Contractor shall provide the RE (or Facility Contracting Officer) a type written report itemizing each deficiency found and the corrective action performed during each required visit or official reported trouble call. The Contractor shall provide the RE with sample copies of these reports for review and approval at the beginning of the Acceptance Test. The following reports are the minimum required:

a) The Contractor shall provide a monthly summary all equipment and sub-systems serviced during this warranty period to RE (or Facility Contracting Officer) by the fifth (5th) working day after the end of each month. The report shall clearly and concisely describe the services rendered, parts replaced and repairs performed. The report shall prescribe anticipated future needs of the equipment and systems for preventive and predictive maintenance.

b) The Contractor shall maintain a separate log entry for each item of equipment and each sub-system of the System. The log shall list dates and times of all scheduled, routine, and emergency calls. Each emergency call shall be described with details of the nature and causes of emergency steps taken to rectify the situation and specific recommendations to avoid such conditions in the future.

6. The RE (or Facility Contracting Officer) shall convey to the Facility Engineering Officer, two (2) copies of actual reports for evaluation.

a) The RE (or Facility Contracting Officer) shall ensure a copy of these reports is entered into the System’s official acquisition documents.

b) The Facility Chief Engineer shall ensure a copy of these reports is entered into the System’s official technical record documents.

c. Work Not Included: Maintenance and repair service shall not include the performance of any work due to improper use; accidents; other vendor, contractor, or owner tampering or negligence, for which the Contractor is not directly responsible and does not control. The Contractor shall immediately notify the RE or Facility Contracting Officer in writing upon the discovery of these incidents. The RE or Facility Contracting Officer will investigate all reported incidents and render an official opinion in writing concerning the supplied information.

3.12 TRAINING

a. Provide thorough training of all biomed engineering and electronic technical staff assigned to those nursing units receiving new networked nurse/patient communications equipment. This training shall be developed and implemented to address two different types of staff. Floor nurses/staff shall receive training from their perspective, and likewise, unit secretaries (or any person whose specific responsibilities include answering patient calls and dispatching

staff) shall receive operational training from their perspective. A separate training room will be set up that allows this type of individualized training utilizing in-service training unit, prior to cut over of the new system.

b. Provide the following minimum training times and durations:

(1) **8** hours prior to opening for BME / Electronic Staff Coordinate schedule with Owner.

(2) 4 hours for Telephone Staff – both day and night shifts.

(3) **4** hours for supervisors and system administrators.

-- End of Section --

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DIVISION 27 - COMMUNICATIONS

SECTION 27 78 00

CLOSEOUT SUBMITTALS FOR COMMUNICATIONS

5/24

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SECTION 27 78 00
5/24

CLOSEOUT SUBMITTALS FOR COMMUNICATIONS

PART 1 GENERAL

1.1 DESCRIPTION

This section includes submittal requirements for communications installations and applies to all sections of Division 27. This section is intended to supplement Section 27 05 00 COMMON WORK RESULTS FOR COMMUNICATIONS until such time as that section can be coordinated and republished.

1.2 REFERENCES

VA Infrastructure Standard for Telecommunications Spaces.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

UL SOLUTIONS (UL)

UL 486A-486B	(2025) UL Standard for Safety Wire Connectors

1.3 RELATED WORK

The requirements of this specification shall apply to all sections of Division 27.

a. General requirements: Section 01 00 00, GENERAL REQUIREMENTS.

b. Submittal requirements: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

1.4 ADMINISTRATIVE REQUIREMENTS

a. Assign a single communications project manager to serve as point of contact for Government, contractor, and design professional.

b. Contact the project Contracting Officer to have reviewers from the VA Office of Information Technology (OIT) Data Center & Infrastructure Engineering team assigned to review project submittals and designs.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

System Supplier Qualifications; G

SD-02 Shop Drawings

Submittal Drawings; G

Equipment Anchors And Supports; G

SD-03 Product Data

Submittal Components; G

SD-05 Design Data

Seismic Support Design; G

SD-07 Certificates

Low Voltage Certificate Of Inspection; G

SD-08 Manufacturer's Instructions

Manufacturer Product Information; G

SD-10 Operation and Maintenance Data

Maintenance And Operation Manuals; G

SD-11 Closeout Submittals

Parts List; G

Warranty Certificate; G

Project Record Documents; G

Instruction Manuals And Software; G

Training Schedule; G

1.6 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

b. Government preapproval is required for submittals with a “G” designation; submittals not having a “G” designation are for information only. When used, a designation following the “G” designation identifies the office(s) that will review the submittal for the Government.

c. Submittal Drawings (G): All drawings shall utilize current equipment and telecommunications space nomenclature. G: Data Center & Infrastructure Operations (DCIE), local site Facility Management Service (FMS).

(1) Telecommunications Space Plans/Elevations: Provide enlarged floor plans of telecommunication spaces indicating layout of equipment and devices, including electrical receptacles and grounding

provisions. Submit detailed plan views and elevations of telecommunication spaces showing enclosures, termination equipment, supporting electrical and mechanical infrastructure equipment, and cable paths. Include following rooms:

(a) Telecommunications Rooms (TRs) and Antenna Entrance Rooms.

(b) Telecommunication Entrance Rooms.

(c) Computer Room.

(2) Cable Plant Distribution Plans: Provide facility and campus plan drawings showing inside plant and outside plant diverse path distribution pathways, distances, media types, and equipment.

(3) Computer Room Structured Cabling Plans: Provide logical, schematic, and plan drawings of the structured cabling system in the facility computer room showing diverse path distribution pathways, distances, media types, and equipment.

(4) Logical Drawings: Provide logical riser or schematic drawings for cable plant and grounding/bonding systems. Show termination points and identify wiring connections.

(5) IT Equipment Enclosure Elevation Drawings: Provide elevation drawings of racks and cabinets in all telecommunications spaces showing passive equipment layouts, grounding and bonding details, and power distribution equipment details.

d. Submittal Components(G): Provide manufacturer’s cut sheets or similar for each element below showing conformance with Division 27 and VA Infrastructure Standard for Telecommunications Spaces salient characteristics and requirements. G: Data Center & Infrastructure Operations (DCIE), local site Facility Management Service (FMS).

1. Bonding Busbars (Primary, Secondary, Rack)
2. Overvoltage Surge Protectors
3. Wire Basket Cable Tray and Accessories
4. Server Cabinets and Accessories
5. Network Channel Racks or Network Cabinets and Accessories
6. Telecommunications Enclosures and Accessories
7. Copper UTP Patch Panels and Accessories
8. Fiber Distribution Panels and Accessories
9. Horizontal Cable Managers
10. Blanking Panels
11. Zone Power Distribution Units (zPDUs)
12. Rack Power Distribution Units (rPDUs)
13. Rack-Mounted Uninterruptible Power Supply (UPS) Units

14. Temperature/Humidity Sensors
15. Copper UTP Backbone (Trunk) Cables
16. Multimode Optical Fiber Backbone Cables (data center backbone)
17. Single Mode Optical Fiber Backbone Cable Media (campus backbone)
18. Hybrid MM/SM Optical Fiver Backbone Cable Media (campus backbone)
19. Copper UTP Horizontal Cable Media
20. Copper UTP Patch Cords
21. Fiber Optic Patch Cords

e. Provide parts list including quantity of spare parts.

f. Provide manufacturer product information. Government reserves the right to require a list of installations where products have been in operation.

g. Submittals are required for all equipment anchors and supports, including seismic support design.

h. Furnish electronic certified test reports to Contracting Officer prior to final inspection and not more than 90 days after completion of tests.

1.7 CLOSEOUT SUBMITTALS INSTRUCTIONS

a. Provide following closeout submittals prior to project closeout date:

(1) Warranty certificate.

(2) Evidence of compliance with requirements such as low voltage certificate of inspection.

(3) Project record documents.

(4) Instruction manuals and software that are a part of system.

b. Maintenance and Operation Manuals: Submit in accordance with Section 01 00 00, GENERAL REQUIREMENTS.

(1) Prepare a manual for each system and equipment specified.

(2) Furnish on portable storage drive in PDF format or equivalent accepted by Contracting Officer.

(3) Furnish complete manual as specified in specification section, fifteen days prior to performance of systems or equipment test.

(4) Furnish remaining manuals prior to final completion.

(5) Identify storage drive "MAINTENANCE AND OPERATION MANUAL" and system name.

(6) Include name, contact information and emergency service numbers of each subcontractor installing system or equipment and local representatives for system or equipment.

(7) Provide a Table of Contents and assemble files to conform to Table of Contents.

(8) Operation and Maintenance Data includes (as appropriate for each item):

(a) Approved shop drawing for each item of equipment.

(b) Internal and interconnecting wiring and control diagrams with data to explain detailed operation and control of equipment.

(c) A control sequence describing start-up, operation, and shutdown.

(d) Description of function of each principal item of equipment.

(e) Installation and maintenance instructions.

(f) Safety precautions.

(g) Diagrams and illustrations.

(h) Test Results and testing methods.

(i) Performance data.

(j) Pictorial "exploded" parts list with part numbers. Emphasis to be placed on use of special tools and instruments. Indicate sources of supply, recommended spare parts, and name of servicing organization.

(k) Warranty documentation indicating end date and equipment protected under warranty.

(l) Appendix: List qualified permanent servicing organizations for support of equipment, including addresses and certified personnel qualifications.

1.8 QUALITY ASSURANCE

a. System Supplier Qualifications: System supplier must be authorized by OEM to warranty installed equipment.

b. Manufactured Products:

(1) Comply with FAR clause 52.236-5 for material and workmanship.

(2) When more than one unit of same class of equipment is required, units must be product of a single manufacturer.

(3) Equipment Assemblies and Components:

(a) Components of an assembled unit need not be products of same manufacturer.

(b) Manufacturers of equipment assemblies, which include components made by others, to assume complete responsibility for final assembled unit.

(c) Provide compatible components for assembly and intended service.

(d) Constituent parts which are similar must be product of a single manufacturer.

(4) Identify factory wiring on equipment being furnished and on wiring diagrams.

c. Testing Agencies: Government reserves the option of witnessing factory tests. Notify Contracting Officer minimum 15 working days prior to manufacturer performing the factory tests.

When equipment fails to meet factory test and re-inspection is required, contractor is liable for additional expenses, including expenses of Government.

1.9 DELIVERY, STORAGE, AND HANDLING

a. Delivery and Acceptance Requirements:

(1) Government's approval of submittals must be obtained for equipment and material before delivery to job site.

(2) Deliver and store materials to job site in OEM's original unopened containers, clearly labeled with OEM's name and equipment catalog numbers, model and serial identification numbers for Contracting Officer to inventory equipment.

b. Storage and Handling Requirements:

Equipment and materials must be protected during shipment and storage against physical damage, dirt, moisture, cold and rain:

(1) Store and protect equipment in a manner that precludes damage or loss, including theft.

(2) Protect painted surfaces with factory installed removable heavy kraft paper, sheet vinyl or equivalent.

(3) Protect enclosures, equipment, controls, controllers, circuit protective devices, and other like items, against entry of foreign matter during installation; vacuum clean both inside and outside before testing and operating.

c. Coordinate storage.

1.10 FIELD CONDITIONS

Where variations from documents are requested in accordance with Section 01 00 00, GENERAL CONDITIONS and Section 01 33 23, SHOP DRAWINGS, PRODUCT

DATA, AND SAMPLES, connecting work and related components must include additions or changes to equipment, components, and installation methods.

1.11 WARRANTY

Comply with FAR clause 52.246-21

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 INSTALLATION - GENERAL

a. Coordinate systems, equipment, and materials installation with other building components.

b. Install systems, materials, and equipment to conform with approved submittal data, including coordination drawings.

c. Install systems, materials, and equipment level and plumb, parallel and perpendicular to other building systems and components, where installed in both exposed and un-exposed spaces.

d. Install equipment according to manufacturers' written instructions.

e. Install cabling, wiring, and equipment to facilitate servicing, maintenance, and repair or replacement of equipment components. Connect equipment for ease of disconnecting, with minimum interference of adjacent other installations.

f. Provide supplementary or miscellaneous items, appurtenances, devices and materials for a complete installation.

3.2 FIELD QUALITY CONTROL

a. Provide minimum clearances and work required for compliance with NFPA 70, National Electrical Code (NEC), and manufacturers' instructions; comply with additional requirements indicated for access and clearances.

b. Conduct project acceptance inspections, final completion inspections, substantial completion inspections, and acceptance testing and demonstrations after verification of system operation and completeness by Contractor.

c. Interim Inspection:

(1) Interim inspection is required at approximately 50 percent of installation.

(2) Request inspection ten working days prior to interim inspection start date by notifying Contracting Officer in writing; this inspection must verify equipment and system being provided adheres to installation and technical requirements of construction documents.

(3) Where applicable, check each item of installed (special

telecommunications system) equipment to ensure appropriate NRTL listing labels and markings are fixed in place.

(4) Verify cabling terminations in telecommunications spaces and end-user Work Area Outlets (WAOs) adhere to T568B pin assignments and cabling connections are in compliance with TIA standards.

(5) Visually confirm minimum performance category cable marking of backbone, horizontal, and patch cabling for both copper UTP and fiber optic cabling.

(6) Review communications bonding system in each telecommunications space.

(7) Review cable tray installation practices.

(8) Provide results of interim inspections to Contracting Officer.

(9) If major or multiple deficiencies are discovered, additional interim inspections may be required until deficiencies are corrected, before permitting further system installation.

(a) Additional inspections are scheduled at direction of Contracting Officer.

(b) Re-inspection of deficiencies noted during interim inspections, must be part of system’s Final Acceptance Proof of Performance Test.

(c) The interim inspection cannot affect the system’s completion date unless directed by Contracting Officer.

(10) Facility Contracting Officer will ensure test documents become a part of system’s official documentation package.

3.3 CLEANING

a. Remove debris, rubbish, waste material, tools, construction equipment, machinery and surplus materials from project site and clean work area, prior to final inspection and acceptance of work.

b. Put building and premises in neat and clean condition.

c. Remove debris on a daily basis.

d. Remove unused material, during progress of work.

e. Perform cleaning and washing required to provide acceptable appearance and operation of equipment to satisfaction of Contracting Officer.

f. Clean exterior surface of all equipment, including concrete residue, dirt, and paint residue, after completion of project.

g. Perform final cleaning prior to project acceptance by Contracting Officer.

h. Remove paint splatters and other spots, dirt, and debris; touch up scratches and mars of finish to match original finish.

i. Clean devices internally using methods and materials recommended by manufacturer.

j. Tighten wiring connectors, terminals, bus joints, and mountings, to include lugs, screws and bolts according to equipment manufacturer’s published torque tightening values for equipment connectors. In absence of published connection or terminal torque values, comply with torque values specified in UL 486A-486B.

3.4 TRAINING

a. Provide training in accordance with subsection, INSTRUCTIONS, of Section 01 00 00, GENERAL REQUIREMENTS.

b. Provide training for equipment or system as required in each associated specification.

c. Develop and submit training schedule for approval by Contracting Officer, at least 30 days prior to planned training.

-- End of Section --

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DIVISION 28 - ELECTRONIC SAFETY AND SECURITY

SECTION 28 05 00

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SECTION 28 05 00
4/18

COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY

PART 1 GENERAL

1.1 DESCRIPTION

a. This Section, Common Work Results for Electronic Safety and Security (ESS), applies to all Sections of Division 28.

b. Furnish and install fully functional electronic safety and security cabling system(s), equipment and approved accessories in accordance with the specification section(s), drawing(s), and referenced publications. Capacities and ratings of cable and other items and arrangements for the specified items are shown on each system's required Bill of Materials (BOM) and verified on the approved system drawing(s). If there is a conflict between contract's specification(s) and drawings(s), the contract's specification requirements shall prevail.

c. The Contractor shall provide a fully functional and operating ESS, programmed, configured, documented, and tested as required herein and the respective Safety and Security System Specification(s). The Contractor shall provide calculations and analysis to support design and engineering decisions as specified in submittals. The Contractor shall provide and pay all labor, materials, and equipment, sales and gross receipts and other taxes. The Contractor shall secure and pay for plan check fees, permits, other fees, and licenses necessary for the execution of work as applicable for the project. Give required notices; the Contractor will comply with codes, ordinances, regulations, and other legal requirements of public authorities, which bear on the performance of work.

d. The Contractor shall provide an ESS, installed, programmed, configured, documented, and tested. The security system shall include but not limited to: physical access control, intrusion detection, duress alarms, elevator control interface, video assessment and surveillance, video recording and storage, delayed egress, personal protection system, intercommunication system, fire alarm interface, equipment cabinetry, dedicated photo badging system and associated live camera, report printer, photo badge printer, and uninterruptible power supplies (UPS) interface. Operator training shall not be required as part of the Security Contractors scope and shall be provided by the Owner. The Security Contractor shall still be required to provide necessary maintenance and troubleshooting manuals as well as submittals as identified herein. The work shall include the procurement and installation of electrical wire and cables, the installation and testing of all system components. Inspection, testing, demonstration, and acceptance of equipment, software, materials, installation, documentation, and workmanship, shall be as specified herein. The Contractor shall provide all associated installation support, including the provision of primary electrical input power circuits.

e. Repair Service Replacement Parts On-site service during the warranty

period shall be provided as specified under “Emergency Service”. The Contractor shall guarantee all parts and labor for a term of one (1) year, unless dictated otherwise in this specification from the acceptance date of the system as described in Part 5 of this Specification. The Contractor shall be responsible for all equipment, software, shipping, transportation charges, and expenses associated with the service of the system for one (1) year. The Contractor shall provide 24-hour telephone support for the software program at no additional charge to the owner. Software support shall include all software updates that occur during the warranty period.

f. Section Includes:

(1) Description of Work for Electronic Security Systems,

(2) Electronic security equipment coordination with relating Divisions,

(3) Submittal Requirements for Electronic Security,

(4) Miscellaneous Supporting equipment and materials for Electronic Security,

(5) Electronic security installation requirements.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS. For General Requirements.

b. Section 07 84 00, FIRESTOPPING. Requirements for firestopping application and use.

c. Section 08 11 73, SLIDING METAL FIRE DOORS. Requirements for door installation.

d. Section 08 35 13.13, ACCORDIAN FOLDING DOORS. Requirements for door installation.

e. Section 08 34 59, VAULT DOORS AND DAY GATES. Requirements for door and gate installation.

f. Section 08 51 13, ALUMINUM WINDOWS. Requirements for window installation.

g. Section 08 71 00, DOOR HARDWARE. Requirements for door installation.

h. Section 10 14 00, SIGNAGE. Requirements for labeling and signs.

i. Section 14 21 00, ELECTRIC TRACTION ELEVATORS. Requirements for elevators.

j. Section 14 24 00, HYDRAULIC ELEVATORS. Requirements for elevators.

k. Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS. Requirements for connection of high voltage.

l. Section 26 05 21, LOW VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES (600 VOLTS AND BELOW). Requirements for power cables.

m. Section 26 05 33 – RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS.

Requirements for infrastructure.

n. Section 26 05 41, UNDERGROUND ELECTRICAL CONSTRUCTION. Requirements for underground installation of wiring.

o. Section 26 56 00, EXTERIOR LIGHTING. Requirements for perimeter lighting.

p. Section 28 05 13, CONDUCTORS AND CABLES FOR ELECTRONIC SAFETY AND SECURITY. Requirements for conductors and cables.

q. Section 28 05 26, GROUNDING AND BONDING FOR ELECTRONIC SAFETY AND SECURITY. Requirements for grounding of equipment.

r. Section 28 05 28.33, CONDUITS AND BOXES FOR ELECTRONIC SAFETY AND SECURITY. Requirements for infrastructure.

s. Section 28 08 00, COMMISIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS. Requirements for Commissioning.

t. Section 28 13 00, PHYSICAL ACCESS CONTROL SYSTEMS (PACS). For physical access control integration.

u. Section 28 23 00, VIDEO SURVEILLANCE. Requirements for security camera systems.

1.3 DEFINITIONS

a. AGC: Automatic Gain Control.

b. Basket Cable Tray: A fabricated structure consisting of wire mesh bottom and side rails.

c. BICSI: Building Industry Consulting Service International.

d. CCD: Charge-coupled device.

e. Central Station: A PC with software designated as the main controlling PC of the security access system. Where this term is presented with initial capital letters, this definition applies.

f. Channel Cable Tray: A fabricated structure consisting of a one-piece, ventilated-bottom or solid-bottom channel section.

g. Controller: An intelligent peripheral control unit that uses a computer for controlling its operation. Where this term is presented with an initial capital letter, this definition applies.

h. CPU: Central processing unit.

i. Credential: Data assigned to an entity and used to identify that entity.

j. DGP: Data Gathering Panel – component of the Physical Access Control System capable to communicate, store and process information received from readers, reader modules, input modules, output modules, and Security Management System.

k. DTS: Digital Termination Service: A microwave-based, line-of-sight

communications provided directly to the end user.

l. EMI: Electromagnetic interference.

m. EMT: Electric Metallic Tubing.

n. ESS: Electronic Security System.

o. File Server: A PC in a network that stores the programs and data files shared by users.

p. GFI: Ground fault interrupter.

q. IDC: Insulation displacement connector.

r. Identifier: A credential card, keypad personal identification number or code, biometric characteristic, or other unique identification entered as data into the entry-control database for the purpose of identifying an individual. Where this term is presented with an initial capital letter, this definition applies.

s. I/O: Input/Output.

t. Intrusion Zone: A space or area for which an intrusion must be detected and uniquely identified, the sensor or group of sensors assigned to perform the detection, and any interface equipment between sensors and communication link to central-station control unit.

u. Ladder Cable Tray: A fabricated structure consisting of two longitudinal side rails connected by individual transverse members (rungs).

v. LAN: Local area network.

w. LCD: Liquid-crystal display.

x. LED: Light-emitting diode.

y. Location: A Location on the network having a PC-to-Controller communications link, with additional Controllers at the Location connected to the PC-to-Controller link with RS-485 communications loop. Where this term is presented with an initial capital letter, this definition applies.

z. Low Voltage: As defined in NFPA 70 for circuits and equipment operating at less than 50 V or for remote-control and signaling power-limited circuits.

aa. M-JPEG: Motion – Joint Photographic Experts Group.

bb. MPEG: Moving picture experts group.

cc. NEC: National Electric Code

dd. NEMA: National Electrical Manufacturers Association

ee. NFPA: National Fire Protection Association

ff. NTSC: National Television System Committee.

gg. NRTL: Nationally Recognized Testing Laboratory.

hh. Open Cabling: Passing telecommunications cabling through open space (e.g., between the studs of a wall cavity).

ii. PACS: Physical Access Control System; A system comprised of cards, readers, door controllers, servers and software to control the physical ingress and egress of people within a given space

jj. PC: Personal computer. This acronym applies to the Central Station, workstations, and file servers.

kk. PCI Bus: Peripheral component interconnect; a peripheral bus providing a high-speed data path between the CPU and peripheral devices (such as monitor, disk drive, or network).

ll. PDF: (Portable Document Format.) The file format used by the Acrobat document exchange system software from Adobe.

mm. RCDD: Registered Communications Distribution Designer.

nn. RFI: Radio-frequency interference.

oo. RIGID: Rigid conduit is galvanized steel tubing, with a tubing wall that is thick enough to allow it to be threaded.

pp. RS-232: An TIA/EIA standard for asynchronous serial data communications between terminal devices. This standard defines a 25-pin connector and certain signal characteristics for interfacing computer equipment.

qq. RS-485: An TIA/EIA standard for multipoint communications.

rr. Solid-Bottom or Non-ventilated Cable Tray: A fabricated structure consisting of integral or separate longitudinal side rails, and a bottom without ventilation openings.

ss. SMS: Security Management System – A SMS is software that incorporates multiple security subsystems (e.g., physical access control, intrusion detection, closed circuit television, intercom) into a single platform and graphical user interface.

tt. TCP/IP: Transport control protocol/Internet protocol incorporated into Microsoft Windows.

uu. Trough or Ventilated Cable Tray: A fabricated structure consisting of integral or separate longitudinal rails and a bottom having openings sufficient for the passage of air and using 75 percent or less of the plan area of the surface to support cables.

vv. UPS: Uninterruptible Power Supply

ww. UTP: Unshielded Twisted Pair

xx. Workstation: A PC with software that is configured for specific limited security system functions.

1.4 QUALITY ASSURANCE

a. Manufacturers Qualifications: The manufacturer shall regularly and presently produce, as one of the manufacturer's principal products, the equipment and material specified for this project, and shall have manufactured the item for at least three years.

b. Product Qualification:

(1) Manufacturer's product shall have been in satisfactory operation, on three installations of similar size and type as this project, for approximately three years.

(2) The Government reserves the right to require the Contractor to submit a list of installations where the products have been in operation before approval.

c. Contractor Qualification:

(1) The Contractor or security sub-contractor shall be a licensed security Contractor with a minimum of five years experience installing and servicing systems of similar scope and complexity. The Contractor shall be an authorized regional representative of the Security Management System’s (PACS) manufacturer. The Contractor shall provide four current references from clients with systems of similar scope and complexity which became operational in the past three years. At least three of the references shall be utilizing the same system components, in a similar configuration as the proposed system. The references must include a current point of contact, company or agency name, address, telephone number, complete system description, date of completion, and approximate cost of the project. The owner reserves the option to visit the reference sites, with the site owner’s permission and representative, to verify the quality of installation and the references’ level of satisfaction with the system. The Contractor shall provide copies of system manufacturer certification for all technicians. The Contractor shall only utilize factory-trained technicians to install, program, and service the PACS. The Contractor shall only utilize factory-trained technicians to install, terminate and service controller/field panels and reader modules. The technicians shall have a minimum of five continuous years of technical experience in electronic security systems. The Contractor shall have a local service facility. The facility shall be located within 60 miles of the project site. The local facility shall include sufficient spare parts inventory to support the service requirements associated with this contract. The facility shall also include appropriate diagnostic equipment to perform diagnostic procedures. The Resident Engineer reserves the option of surveying the company’s facility to verify the service inventory and presence of a local service organization.

(2) The Contractor shall provide proof project superintendent with BICSI Certified Commercial Installer Level 1, Level 2, or Technician to provide oversight of the project.

(3) Cable installer must have on staff a Registered Communication Distribution Designer (RCDD) certified by Building Industry Consulting Service International. The staff member shall provide

consistent oversight of the project cabling throughout design, layout, installation, termination and testing.

d. Service Qualifications: There shall be a permanent service organization maintained or trained by the manufacturer which will render satisfactory service to this installation within four hours of receipt of notification that service is needed. Submit name and address of service organizations.

1.5 GENERAL ARANGEMENT OF CONTRACT DOCUMENTS

a. The Contract Documents supplement to this specification indicates approximate locations of equipment. The installation and/or locations of the equipment and devices shall be governed by the intent of the design; specification and Contract Documents, with due regard to actual site conditions, recommendations, ambient factors affecting the equipment and operations in the vicinity. The Contract Documents are diagrammatic and do not reveal all offsets, bends, elbows, components, materials, and other specific elements that may be required for proper installation. If any departure from the contract documents is deemed necessary, or in the event of conflicts, the Contractor shall submit details of such departures or conflicts in writing to the owner or owner’s representative for his or her comment and/or approval before initiating work.

b. Anything called for by one of the Contract Documents and not called for by the others shall be of like effect as if required or called by all, except if a provision clearly designed to negate or alter a provision contained in one or more of the other Contract Documents shall have the intended effect. In the event of conflicts among the Contract Documents, the Contract Documents shall take precedence in the following order: the Form of Agreement; the Supplemental General Conditions; the Special Conditions; the Specifications with attachments; and the drawings.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Manufacturers Qualifications; G

Product Qualification; G

Contractor Qualification; G

Service Qualifications; G

SD-04 Samples

Samples; G

SD-05 Design Data

Group 1 Technical Data Package; G

Group II Technical Data Package; G

Group III Technical Data Package; G

Group IV Technical Data Package; G

Group V Technical Data Package; G

SD-06 Test Reports

Completed System Readiness Checklists; G

SD-07 Certificates

Fips 201 Compliance Certificates; G

SD-11 Closeout Submittals

Warranty; G

1.7 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

b. The Government's approval shall be obtained for all equipment and material before delivery to the job site. Delivery, storage or installation of equipment or material which has not had prior approval will not be permitted at the job site.

c. Submittals for individual systems and equipment assemblies which consist of more than one item or component shall be made for the system or assembly as a whole. Partial submittals will not be considered for approval.

(1) Mark the submittals, "SUBMITTED UNDER SECTION".

(2) Submittals shall be marked to show specification reference including the section and paragraph numbers.

(3) Submit each section separately.

d. The submittals shall include the following:

(1) Information that confirms compliance with contract requirements. Include the manufacturer's name, model or catalog numbers, catalog information, technical data sheets, shop drawings, pictures, nameplate data and test reports as required.

(2) Parts list which shall include those replacement parts recommended by the equipment manufacturer, quantity of parts, current price and availability of each part.

e. Submittals shall be in full compliance of the Contract Documents. All submittals shall be provided in accordance with this section. Submittals lacking the breath or depth these requirements will be considered incomplete and rejected. Submissions are considered multidisciplinary and shall require coordination with applicable

divisions to provide a complete and comprehensive submission package. All submittals shall include adequate descriptive literature, catalog cuts, shop drawings and other data necessary for the Government to ascertain that the proposed equipment and materials comply with specification requirements. Catalog cuts submitted for approval shall be legible and clearly identify equipment being submitted. Additional general provisions are as follows:

(1) The Contractor shall schedule submittals in order to maintain the project schedule. For coordination drawings refer to Specification Section 01 33 10, DESIGN SUBMITTAL PROCEDURES, which outline basic submittal requirements and coordination. Section 01 33 10, DESIGN SUBMITTAL PROCEDURES shall be used in conjunction with this section.

(2) The Contractor shall identify variations from requirements of Contract Documents and state product and system limitations, which may be detrimental to successful performance of the completed work or system.

(3) Each package shall be submitted at one time for each review and include components from applicable disciplines (e.g., electrical work, architectural finishes, door hardware, etc.) which are required to produce an accurate and detailed depiction of the project.

(4) Manufacturer’s information used for submittal shall have pages with items for approval tagged, items on pages shall be identified, and capacities and performance parameters for review shall be clearly marked through use of an arrow or highlighting. Provide space for Resident Engineer and Contractor review stamps.

(5) Technical Data Drawings shall be in the latest version of AutoCAD®, drawn accurately, and in accordance with VA CAD Std, and VA BIM Std. FREEHAND SKETCHES OR COPIED VERSIONS OF THE CONSTRUCTION DOCUMENTS WILL NOT BE ACCEPTED. The Contractor shall not reproduce Contract Documents or copy standard information as the basis of the Technical Data Drawings. If departures from the technical data drawings are subsequently deemed necessary by the Contractor, details of such departures and the reasons thereof shall be submitted in writing to the Resident Engineer for approval before the initiation of work.

(6) Packaging: The Contractor shall organize the submissions according to the following packaging requirements.

(a) Binders: For each manual, provide heavy duty, commercial quality, durable three ring vinyl covered loose leaf binders, sized to receive 8.5 x 11 in paper, and appropriate capacity to accommodate the contents. Provide a clear plastic sleeve on the spine to hold labels describing the contents. Provide pockets in the covers to receive folded sheets.

1. Where two or more binders are necessary to accommodate data; correlate data in each binder into related groupings according to the Project Manual table of contents. Cross-referencing other binders where necessary to provide essential information for communication of proper operation and/or maintenance of the component or system.

2. Identify each binder on the front and spine with printed binder title, Project title or name, and subject matter covered. Indicate the volume number if applicable.

(b) Dividers: Provide heavy paper dividers with celluloid tabs for each Section. Mark each tab to indicate contents.

(c) Protective Plastic Jackets: Provide protective transparent plastic jackets designed to enclose diagnostic software for computerized electronic equipment.

(d) Text Material: Where written material is required as part of the manual use the manufacturer's standard printed material, or if not available, specially prepared data, neatly typewritten on 8.5 inches by 11 inches 20 pound white bond paper.

(e) Drawings: Where drawings and/or diagrams are required as part of the manual, provide reinforced punched binder tabs on the drawings and bind them with the text.

1. Where oversized drawings are necessary, fold the drawings to the same size as the text pages and use as a foldout.

2. If drawings are too large to be used practically as a foldout, place the drawing, neatly folded, in the front or rear pocket of the binder. Insert a type written page indicating the drawing title, description of contents and drawing location at the appropriate location of the manual.

3. Drawings shall be sized to ensure details and text is of legible size. Text shall be no less than 1/16” tall.

(f) Manual Content: Submit in accordance with Section 01 00 00, GENERAL REQUIREMENTS.

1. Maintenance and Operation Manuals: Submit as required for systems and equipment specified in the technical sections. Furnish four copies, bound in hardback binders, (manufacturer's standard binders) or an approved equivalent. Furnish one complete manual as specified in the technical section but in no case later than prior to performance of systems or equipment test, and furnish the remaining manuals prior to contract completion.

2. Inscribe the following identification on the cover: the words "MAINTENANCE AND OPERATION MANUAL," the name and location of the system, equipment, building, name of Contractor, and contract number. Include in the manual the names, addresses, and telephone numbers of each subcontractor installing the system or equipment and the local representatives for the system or equipment.

3. The manuals shall include:

a) Internal and interconnecting wiring and control diagrams with data to explain detailed operation and control of the equipment.

b) A control sequence describing start-up, operation, and shutdown.

c) Description of the function of each principal item of equipment.

d) Installation and maintenance instructions.

e) Safety precautions.

f) Diagrams and illustrations.

g) Testing methods.

h) Performance data.

i) Pictorial "exploded" parts list with part numbers. Emphasis shall be placed on the use of special tools and instruments. The list shall indicate sources of supply, recommended spare parts, and name of servicing organization.

j) Appendix; list qualified permanent servicing organizations for support of the equipment, including addresses and certified qualifications.

(g) Binder Organization: Organize each manual into separate sections for each piece of related equipment. At a minimum, each manual shall contain a title page, table of contents, copies of Product Data supplemented by drawings and written text, and copies of each warranty, bond, certifications, and service Contract issued. Refer to Group I through V Technical Data Package Submittal requirements for required section content.

(h) Title Page: Provide a title page as the first sheet of each manual to include the following information; project name and address, subject matter covered by the manual, name and address of the Project, date of the submittal, name, address, and telephone number of the Contractor, and cross references to related systems in other operating and/or maintenance manuals.

(i) Table of Contents: After the title page, include a type written table of contents for each volume, arranged systematically according to the Project Manual format. Provide a list of each product included, identified by product name or other appropriate identifying symbols and indexed to the content of the volume. Where more than one volume is required to hold data for a particular system, provide a comprehensive table of contents for all volumes in each volume of the set.

(j) General Information Section: Provide a general information section immediately following the table of contents, listing each product included in the manual, identified by product name. Under each product, list the name, address, and telephone number of the installer and maintenance Contractor. In addition, list a local source for replacement parts and equipment.

(k) Drawings: Provide specially prepared drawings where necessary to supplement the manufacturers printed data to illustrate the relationship between components of equipment or systems, or provide control or flow diagrams. Coordinate these drawings with information contained in Project Record Drawings to assure correct

illustration of the completed installation.

(l) Manufacturer’s Data: Where manufacturer's standard printed data is included in the manuals, include only those sheets that are pertinent to the part or product installed. Mark each sheet to identify each part or product included in the installation. Where more than one item in tabular format is included, identify each item, using appropriate references from the Contract Documents. Identify data that is applicable to the installation and delete references to information which is not applicable.

(m) Where manufacturer's standard printed data is not available and the information is necessary for proper operation and maintenance of equipment or systems, or it is necessary to provide additional information to supplement the data included in the manual, prepare written text to provide the necessary information. Organize the text in a consistent format under a separate heading for different procedures. Where necessary, provide a logical sequence of instruction for each operating or maintenance procedure. Where similar or more than one product is listed on the submittal the Contractor shall differentiate by highlighting the specific product to be utilized.

(n) Calculations: Provide a section for circuit and panel calculations.

(o) Loading Sheets: Provide a section for DGP Loading Sheets.

(p) Certifications: Provide section for Contractor’s manufacturer certifications.

(7) Contractor Review: Review submittals prior to transmittal. Determine and verify field measurements and field construction criteria. Verify manufacturer’s catalog numbers and conformance of submittal with requirements of contract documents. Return non-conforming or incomplete submittals with requirements of the work and contract documents. Apply Contractor’s stamp with signature certifying the review and verification of products occurred, and the field dimensions, adjacent construction, and coordination of information is in accordance with the requirements of the contract documents.

(8) Resubmission: Revise and resubmit submittals as required within 15 calendar days of return of submittal. Make resubmissions under procedures specified for initial submittals. Identify all changes made since previous submittal.

(9) Product Data: Within 15 calendar days after execution of the contract, the Contractor shall submit for approval a complete list of all of major products proposed for use. The data shall include name of manufacturer, trade name, model number, the associated contract document section number, paragraph number, and the referenced standards for each listed product.

f. Group 1 Technical Data Package: Group I Technical Data Package shall be one submittal consisting of the following content and organization. Refer to VA Special Conditions Document for drawing format and content requirements. The data package shall include the following:

(1) Section I - Drawings:

(a) General – Drawings shall conform to VA CAD Standards Guide. All text associated with security details shall be 1/8″ tall and meet VA text standard for AutoCAD™ drawings.

(b) Cover Sheet – Cover sheet shall consist of Project Title and Address, Project Number, Area and Vicinity Maps.

(c) General Information Sheets – General Information Sheets shall consist of General Notes, Abbreviations, Symbols, Wire and Cable Schedule, Project Phasing, and Sheet Index.

(d) Floor Plans – Floor plans shall be produced from the Architectural backgrounds issued in the Construction Documents. The contractor shall receive floor plans from the prime A/E to develop these drawing sets. Security devices shall be placed on drawings in scale. All text associated with security details shall be 1/8″ tall and meet VA text standard for AutoCAD™ drawings. Floor plans shall identify the following:

1. Security devices by symbol,

2. The associated device point number (derived from the loading sheets),

3. Wire and cable types and counts

4. Conduit sizing and routing

5. Conduit riser systems

6. Device and area detail call outs

(e) Architectural details – Architectural details shall be produced for each device mounting type (door details for EECS and IDS, Intrusion Detection system (motion sensor, vibration, microwave Motion Sensor and Camera mounting,

(f) Riser Diagrams – Contractor shall provide a riser diagram indicating riser architecture and distribution of the SMS throughout the facility (or area in scope).

(g) Block Diagrams – Contractor shall provide a block diagram for the entire system architecture and interconnections with SMS subsystems. Block diagram shall identify SMS subsystem (e.g., electronic entry control, intrusion detection, closed circuit television, intercom, and other associated subsystems) integration; and data transmission and media conversion methodologies.

(h) Interconnection Diagrams – Contractor shall provide interconnection diagram for each sensor, and device component. Interconnection diagram shall identify termination locations, standard wire detail to include termination schedule. Diagram shall also identify interfaces to other systems such as elevator control, fire alarm systems, and security management systems.

(i) Security Details:

1. Panel Assembly Detail – For each panel assembly, a panel assembly details shall be provided identifying individual panel component size and content.

2. Panel Details – Provide security panel details identify general arrangement of the security system components, backboard size, wire through size and location, and power circuit requirements.

3. Device Mounting Details – Provide mounting detailed drawing for each security device (physical access control system, intrusion detection, video surveillance and assessment, and intercom systems) for each type of wall and ceiling configuration in project. Device details shall include device, mounting detail, wiring and conduit routing.

4. Details of connections to power supplies and grounding

5. Details of surge protection device installation

6. Sensor detection patterns – Each system sensor shall have associated detection patterns.

7. Equipment Rack Detail – For each equipment rack, provide a scaled detail of the equipment rack location and rack space utilization. Use of BISCI wire management standards shall be employed to identify wire management methodology. Transitions between equipment racks shall be shown to include use vertical and horizontal latter rack system.

8. Security Control Room – The contractor shall provide a layout plan for the Security Control Room. The layout plan shall identify all equipment and details associated with the installation.

9. Operator Console – The contractor shall provide a layout plan for the Operator Console. The layout plan shall identify all equipment and details associated with the installation. Equipment room - the contractor shall provide a layout plan for the equipment room. The layout plan shall identify all equipment and details associated with the installation.

10. Equipment Room – Equipment room details shall provide architectural, electrical, mechanical, plumbing, IT/Data and associated equipment and device placements both vertical and horizontally.

(j) Electrical Panel Schedule – Electrical Panel Details shall be provided for all SMS systems electrical power circuits. Panel details shall be provided identifying panel type (Standard, Emergency Power, Emergency/Uninterrupted Power Source, and Uninterrupted Power Source Only), panel location, circuit number, and circuit amperage rating.

(k) Door Schedule – A door schedule shall be developed for each door equipped with electronic security components. At a minimum, the door schedule shall be coordinated with Division 08 work and

include the following information:

1. Item Number

2. Door Number (Derived from A/E Drawings)

3. Floor Plan Sheet Number

4. Standard Detail Number

5. Door Description (Derived from Loading Sheets)

6. Data Gathering Panel Input Number

7. Door Position or Monitoring Device Type and Model Number

8. Lock Type, Model Number and Power Input/Draw (standby/active)

9. Card Reader Type and Model Number

10. Shunting Device Type and Model Number

11. Sounder Type and Model Number

12. Manufacturer

13. Misc. devices as required

a) Delayed Egress Type and Model Number

b) Intercom

c) Camera

d) Electric Transfer Hinge

e) Electric Pass-through device

14. Remarks column indicating special notes or door configurations

(2) Camera Schedule - A camera schedule shall be developed for each camera. Contractors shall coordinate with the Resident Engineer to determine camera starting numbers and naming conventions. All drawings shall identify wire and cable standardization methodology. Color coding of all wiring conductors and jackets is required and shall be communicated consistently throughout the drawings package submittal. At a minimum, the camera schedule shall include the following information:

(a) Item Number

(b) Camera Number

(c) Naming Conventions

(d) Description of Camera Coverage

(e) Camera Location

(f) Floor Plan Sheet Number

(g) Camera Type

(h) Mounting Type

(i) Standard Detail Reference

(j) Power Input and Draw

(k) Power Panel Location

(l) Remarks Column for Camera

(3) Section II – Data Gathering Panel Documentation Package

(a) Contractor shall provide Data Gathering Panel (DGP) input and output documentation packages for review at the Shop Drawing submittal stage and also with the as-built documentation package. The documentation packages shall be provided in both printed and magnetic form at both review stages.

(b) The Contractor shall provide loading sheet documentation package for the associated DGP, including input and output boards for all field panels associated with the project. Documentation shall be provided in current version Microsoft Excel spreadsheets following the format currently utilized by VA. A separate spreadsheet file shall be generated for each DGP and associated field panels.

(c) The spreadsheet names shall follow a sequence that shall display the spreadsheets in numerical order according to the DGP system number. The spreadsheet shall include the prefix in the file name that uniquely identifies the project site. The spreadsheet shall detail all connected items such as card readers, alarm inputs, and relay output connections. The spreadsheet shall include an individual section (row) for each panel input, output and card reader. The spreadsheet shall automatically calculate the system numbers for card readers, inputs, and outputs based upon data entered in initialization fields.

(d) All entries must be verified against the field devices. Copies of the floor plans shall be forwarded under separate cover.

(e) The DGP spreadsheet shall include an entry section for the following information:

1. DGP number

2. First Reader Number

3. First Monitor Point Number

4. First Relay Number

5. DGP, input or output Location

6. DGP Chain Number

7. DGP Cabinet Tamper Input Number

8. DGP Power Fail Input Number

9. Number of Monitor Points Reserved For Expansion Boards

10. Number of Control Points (Relays) Reserved For Expansion Boards

(f) The DGP, input module and output module spreadsheets shall automatically calculate the following information based upon the associated entries in the above fields:

1. System Numbers for Card Readers

2. System Numbers for Monitor Point Inputs

3. System Numbers for Control Points (Relays)

4. Next DGP or input module First Monitor Point Number

5. Next DGP or output module First Control Point Number

(g) The DGP spreadsheet shall provide the following information for each card reader:

1. DGP Reader Number

2. System Reader Number

3. Cable ID Number

4. Description Field (Room Number)

5. Description Field (Device Type i.e.: In Reader, Out Reader, etc.)

6. Description Field

7. DGP Input Location

8. Date Test

9. Date Passed

10. Cable Type

11. Camera Numbers (of cameras viewing the reader location)

(h) The DGP and input module spreadsheet shall provide the following information for each monitor point (alarm input).

1. DGP Monitor Point Input Number

2. System Monitor Point Number

3. Cable ID Number

4. Description Field (Room Number)

5. Description Field (Device Type i.e.: Door Contact, Motion Detector, etc.)

7. DGP or input module Input Location

8. Date Test

9. Date Passed

10. Cable Type

11. Camera Numbers (of associated alarm event preset call-ups)

(i) The DGP and output module spreadsheet shall provide the following information for each control point (output relay).

1. DGP Control Point (Relay) Number

2. System (Control Point) Number

3. Cable ID Number

4. Description Field (Room Number)

5. Description Field (Device: Lock Control, Local Sounder, etc.)

6. Description Field

7. DGP or OUTPUT MODULE Output Location

8. Date Test

9. Date Passed Cable Type

10. Camera Number (of associated alarm event preset call-ups)

(j) The DGP, input module and output module spreadsheet shall include the following information or directions in the header and footer:

1. Header

a) DGP Input and Output Worksheet

b) Enter Beginning Reader, Input, and Output Starting Numbers and Sheet Will Automatically Calculate the Remaining System Numbers.

2. Footer

a) File Name

b) Date Printed

c) Page Number

(4) Section III - Construction Mock-up: In areas with exposed EMT/Conduit Raceways, contractor shall conceal raceway as much as practical and unobtrusively. In addition, historic significance must be considered to determine installation means and methods for approval by the owner.

(5) Section IV - Manufacturers' Data: The data package shall include manufacturers' data for all materials and equipment, including sensors, local processors and console equipment provided under this specification.

(6) Section V - System Description and Analysis: The data package shall include system descriptions, analysis, and calculations used in sizing equipment required by these specifications. Descriptions and calculations shall show how the equipment will operate as a system to meet the performance requirements of this specification. The data package shall include the following:

(a) Central processor memory size; communication speed and protocol description; rigid disk system size and configuration; flexible disk system size and configuration; back-up media size and configuration; alarm response time calculations; command response time calculations; start-up operations; expansion capability and method of implementation; sample copy of each report specified; and color photographs representative of typical graphics.

(b) Software Data: The data package shall consist of descriptions of the operation and capability of the system, and application software as specified.

(c) Overall System Reliability Calculations: The data package shall include all manufacturers' reliability data and calculations required to show compliance with the specified reliability.

(7) Section VI – Certifications and References: All specified manufacturer's certifications shall be included with the data package. Contractor shall provide Project references as outlined in Paragraph 1.4 "Quality Assurance".

g. Group II Technical Data Package

(1) The Contractor shall prepare a report of "Current Site Conditions" and submit a report to the Resident Engineer documenting changes to the site, particularly those conditions that affect performance of the system to be installed. The Contractor shall provide specification sheets, or written functional requirements to support the findings, and a cost estimate to correct those site changes or conditions which affect the installation of the system or its performance. The Contractor shall not correct any deficiency without written permission from the COTR.

(2) System Configuration and Functionality: The contractor shall provide the results of the meeting with VA to develop system requirements and functionality including but not limited to:

(a) Baseline configuration

(b) Access levels

(c) Schedules (intrusion detection, physical access control, holidays, etc.)

(d) Badge database

(e) System monitoring and reporting (unit level and central control)

(f) Naming conventions and descriptors

h. Group III Technical Data Package: Development of Test Procedures: The Contractor will prepare performance test procedures for the system testing. The test procedures shall follow the format of the VA Testing procedures and be customized to the contract requirements. The Contractor will deliver the test procedures to the Resident Engineer for approval at least 60 calendar days prior to the requested test date.

i. Group IV Technical Data Package

(1) Performance Verification Test: Based on the successful completion of the pre-delivery test, the Contractor shall finalize the test procedures and report forms for the performance verification test (PVT) and the endurance test. The PVT shall follow the format, layout and content of the pre-delivery test. The Contractor shall deliver the PVT and endurance test procedures to the Resident Engineer for approval. The Contractor may schedule the PVT after receiving written approval of the test procedures. The Contractor shall deliver the final PVT and endurance test reports within 14 calendar days from completion of the tests. Refer to Part 3 of this section for System Testing and Acceptance requirements.

(2) Training Documentation

(a) New Facilities and Major Renovations: Familiarization training shall be provided for new equipment or systems. Training can include site familiarization training for VA technicians and administrative personnel. Training shall include general information on new system layout including closet locations, turnover of the completed system including all documentation, including manuals, software, key systems, and full system administration rights. Lesson plans and training manuals training shall be oriented to type of training to be provided.

(b) New Unit Control Room:

1. Provide the security personnel with training in the use, operation, and maintenance of the entire control room system (Unit Control and Equipment Rooms). The training documentation must include the operation and maintenance. The first of the training sessions shall take place prior to system turnover and the second immediately after turnover. Coordinate the training sessions with the Owner. Completed classroom sessions will be witnessed and documented by the Architect/Engineer, and approved by the Resident Engineer. Instruction is not to begin until the system is operational as designed.

2. The training documents will cover the operation and the maintenance manuals and the control console operators' manuals and service manuals in detail, stressing all important operational and service diagnostic information necessary for the maintenance and operations personnel to efficiently use and maintain all systems.

3. Provide an illustrated control console operator's manual and service manual. The operator's manual shall be written in laymen's language and printed so as to become a permanent reference document for the operators, describing all control panel switch operations, graphic symbol definitions and all indicating functions and a complete explanation of all software.

4. The service manual shall be written in laymen's language and printed so as to become a permanent reference document for maintenance personnel, describing how to run internal self diagnostic software programs, troubleshoot head end hardware and field devices with a complete scenario simulation of all possible system malfunctions and the appropriate corrective measures.

5. Provide a professional color DVD instructional recording of all the operational procedures described in the operator's manual. All charts used in the training session shall be clearly presented on the video. Any DVD found to be inferior in recording or material content shall be reproduced at no cost until an acceptable DVD is submitted. Provide four copies of the training DVD, one to the architect/engineer and three to the owner.

(3) System Configuration and Data Entry:

(a) The contractor is responsible for providing all system configuration and data entry for the SMS and subsystems (e.g., video matrix switch, intercom, digital video recorders, network video recorders). All data entry shall be performed per VA standards and guidelines. The Contractor is responsible for participating in all meetings with the client to compile the information needed for data entry. These meetings shall be established at the beginning of the project and incorporated in to the project schedule as a milestone task. The contractor shall be responsible for all data collection, data entry, and system configuration. The contractor shall collect, enter, and program and/or configure the following components:

1. Physical Access control system components,

2. All intrusion detection system components,

3. Video surveillance, control and recording systems,

4. Intercom systems components,

5. All other security subsystems shown in the contract documents.

(b) The Contractor is responsible for compiling the card access database for the VA employees, including programming reader configurations, access shifts, schedules, exceptions, card classes and card enrollment databases.

(c) Refer to Part 3 for system programming requirements and planning guidelines.

(4) Graphics: Based on CAD as-built drawings developed for the construction project, create all map sets showing locations of all alarms and field devices. Graphical maps of all alarm points installed under this contract including perimeter and exterior alarm points shall be delivered with the system. The Contractor shall create and install all graphics needed to make the system operational. The Contractor shall utilize data from the contract documents, Contractor's field surveys, and all other pertinent information in the Contractor's possession to complete the graphics. The Contractor shall identify and request from the COTR, any additional data needed to provide a complete graphics package. Graphics shall have sufficient level of detail for the system operator to assess the alarm. The Contractor shall supply hard copy, color examples at least (8 x 10 in) of each type of graphic to be used for the completed Security system. The graphics examples shall be delivered to the Resident Engineer for review and approval at least 90 calendar days prior to the scheduled date the Contractor requires them.

j. Group V Technical Data Package: Final copies of the manuals shall be delivered to the Resident Engineer as part of the acceptance test. The draft copy used during site testing shall be updated with any changes required prior to final delivery of the manuals. Each manual's contents shall be identified on the cover. The manual shall include names, addresses, and telephone numbers of each sub-contractor installing equipment or systems, as well as the nearest service representatives for each item of equipment for each system. The manuals shall include a table of contents and tab sheets. Tab sheets shall be placed at the beginning of each chapter or section and at the beginning of each appendix. The final copies delivered after completion of the endurance test shall include all modifications made during installation, checkout, and acceptance. Six (6) hard-copies and one (1) soft copy of each item listed below shall be delivered as a part of final systems acceptance.

(1) Functional Design Manual: The functional design manual shall identify the operational requirements for the entire system and explain the theory of operation, design philosophy, and specific functions. A description of hardware and software functions, interfaces, and requirements shall be included for all system operating modes. Manufacturer developed literature may be used; however, shall be produced to match the project requirements.

(2) Equipment Manual: A manual describing all equipment furnished including:

(a) General description and specifications; installation and checkout procedures; equipment electrical schematics and layout drawings; system schematics and layout drawings; alignment and calibration procedures; manufacturer's repair list indicating sources of supply; and interface definition.

(3) Software Manual: The software manual shall describe the functions of all software and include all other information necessary to enable proper loading, testing, and operation. The manual shall include:

(a) Definition of terms and functions; use of system and applications software; procedures for system initialization, start-up, and shutdown; alarm reports; reports generation, database format and data entry requirements; directory of all disk files; and description of all communications protocols including data formats, command characters, and a sample of each type of data transfer.

(4) Operator’s Manual: The operator’s manual shall fully explain all procedures and instructions for the operation of the system, including: Computers and peripherals; system start-up and shutdown procedures; use of system, command, and applications software; recovery and restart procedures; graphic alarm presentation; use of report generator and generation of reports; data entry; operator commands’ alarm messages, and printing formats; and system access requirements.

(5) Maintenance Manual: The maintenance manual shall include descriptions of maintenance for all equipment including inspection, recommend schedules, periodic preventive maintenance, fault diagnosis, and repair or replacement of defective components.

(6) Spare Parts and Components Data: At the conclusion of the Contractor's work, the Contractor shall submit to the Resident Engineer a complete list of the manufacturer's recommended spare parts and components required to satisfactorily maintain and service the systems, as well as unit pricing for those parts and components.

(7) Operation, Maintenance and Service Manuals: The Contractor shall provide two complete sets of operating and maintenance manuals in the form of an instructional manual for use by the VA Security Guard Force personnel. The manuals shall be organized into suitable sets of manageable size. Where possible, assemble instructions for similar equipment into a single binder. If multiple volumes are required, each volume shall be fully indexed and coordinated.

(8) Equipment and Systems Maintenance Manual: The Contractor shall provide the following descriptive information for each piece of equipment, operating system, and electronic system:

(a) Equipment and/or system function.

(b) Operating characteristics.

(c) Limiting conditions.

(d) Performance curves.

(e) Engineering data and test.

(f) Complete nomenclature and number of replacement parts.

(g) Provide operating and maintenance instructions including assembly drawings and diagrams required for maintenance and a list of items recommended to stock as spare parts.

(h) Provide information detailing essential maintenance procedures including the following: routine operations, trouble shooting guide, disassembly, repair and re-assembly, alignment, adjusting, and checking.

(i) Provide information on equipment and system operating procedures, including the following; start-up procedures, routine and normal operating instructions, regulation and control procedures, instructions on stopping, shut-down and emergency instructions, required sequences for electric and electronic systems, and special operating instructions.

(j) Manufacturer equipment and systems maintenance manuals are permissible.

(9) Project Redlines: During construction, the Contractor shall maintain an up-to-date set of construction redlines detailing current location and configuration of the project components. The redline documents shall be marked with the words 'Master Redlines' on the cover sheet and be maintained by the Contractor in the project office. The Contractor will provide access to redline documents anytime during the project for review and inspection by the Resident Engineer or authorized Office of Protection Services representative. Master redlines shall be neatly maintained throughout the project and secured under lock and key in the contractor's onsite project office. Any project component or assembly that is not installed in strict accordance with the drawings shall be so noted on the drawings. Prior to producing Record Construction Documents, the contractor will submit the Master Redline document to the Resident Engineer for review and approval of all changes or modifications to the documents. Each sheet shall have Resident Engineer initials indicating authorization to produce "As Built" documents. Field drawings shall be used for data gathering and field changes. These changes shall be made to the master redline documents daily. Field drawings shall not be considered "master redlines".

(10) Record Specifications: The Contractor shall maintain one copy of the Project Specifications, including addenda and modifications issued, for Project Record Documents. The Contractor shall mark the Specifications to indicate the actual installation where the installation varies substantially from that indicated in the Contract Specifications and modifications issued. (Note related Project Record Drawing information where applicable). The Contractor shall pay particular attention to substitutions, selection of product options, and information on concealed installations that would be difficult to identify or measure and record later. Upon completion of the mark ups, the Contractor shall submit record Specifications to the COTR. As with master relines, Contractor shall maintain record specifications for Resident Engineer review and inspection at anytime.

(11) Record Product Data: The Contractor shall maintain one copy of each Product Data submittal for Project Record Document purposes. The Data shall be marked to indicate the actual product installed where the installation varies substantially from that indicated in the Product Data submitted. Significant changes in the product delivered to the site and changes in manufacturer's instructions and recommendations for installation shall be included.

Particular attention will be given to information on concealed products and installations that cannot be readily identified or recorded later. Note related Change Orders and mark up of Record Construction Documents, where applicable. Upon completion of mark up, submit a complete set of Record Product Data to the COTR.

(12) Miscellaneous Records: The Contractor shall maintain one copy of miscellaneous records for Project Record Document purposes. Refer to other Specifications for miscellaneous record-keeping requirements and submittals concerning various construction activities. Before substantial completion, complete miscellaneous records and place in good order, properly identified and bound or filed, ready for use and reference. Categories of requirements resulting in miscellaneous records include a minimum of the following:

(a) Certificates received instead of labels on bulk products.

(b) Testing and qualification of tradesmen. (“Contractor’s Qualifications”)

(c) Documented qualification of installation firms.

(d) Load and performance testing.

(e) Inspections and certifications.

(f) Final inspection and correction procedures.

(g) Project schedule

(13) Record Construction Documents (Record As-Built)

(a) Upon project completion, the contractor shall submit the project master redlines to the Resident Engineer prior to development of Record construction documents. The Resident Engineer shall be given a minimum of a thirty (30) day review period to determine the adequacy of the master redlines. If the master redlines are found suitable by the Resident Engineer, the Resident Engineer will initial and date each sheet and turn redlines over to the contractor for as built development.

(b) The Contractor shall provide the Resident Engineer a complete set of "as-built" drawings and original master redlined marked "as-built" blue-line in the latest version of AutoCAD drawings unlocked on CD or DVD. The as-built drawing shall include security device number, security closet connection location, data gathering panel number, and input or output number as applicable. All corrective notations made by the Contractor shall be legible when submitted to the COTR. If, in the opinion of the COTR, any redlined notation is not legible, it shall be returned to the Contractor for re-submission at no extra cost to the Owner. The Contractor shall organize the Record Drawing sheets into manageable sets bound with durable paper cover sheets with suitable titles, dates, and other identifications printed on the cover. The submitted as built shall be in editable formats and the ownership of the drawings shall be fully relinquished to the owner.

(c) Where feasible, the individual or entity that obtained record data, whether the individual or entity is the installer, sub-contractor, or similar entity, is required to prepare the mark up on Record Drawings. Accurately record the information in a comprehensive drawing technique. Record the data when possible after it has been obtained. For concealed installations, record and check the mark up before concealment. At the time of substantial completion, submit the Record Construction Documents to the COTR. The Contractor shall organize into bound and labeled sets for the COTR's continued usage. Provide device, conduit, and cable lengths on the conduit drawings. Exact in-field conduit placement/routings shall be shown. All conduits shall be illustrated in their entire length from termination in security closets; no arrowed conduit runs shall be shown. Pull box and junction box sizes are to be shown if larger than 4 inch.

k. FIPS 201 Compliance Certificates: Provide Certificates for all software components and device types utilizing credential verification. Provide certificates for:

(1) Fingerprint Capture Station

(2) Card Readers

(3) Facial Image Capturing Camera

(4) PIV Middelware

(5) Template Matcher

(6) Electromagnetically Opaque Sleeve

(7) Certificate Management

(a) CAK Authentication System

(b) PIV Authentication System

(c) Certificate Validator

(d) Cryptographic Module

(8) <list devices and software>

l. Approvals will be based on complete submission of manuals together with shop drawings.

m. After approval and prior to installation, furnish the Resident Engineer with samples of each of the following:

(1) A 12 inch length of each type and size of wire and cable along with the tag from the coils of reels from which the samples were taken.

(2) Each type of conduit and pathway coupling, bushing and termination fitting.

(3) Conduit hangers, clamps and supports.

(4) Duct sealing compound.

n. Completed System Readiness Checklists provided by the Commissioning Agent and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 28 08 00, COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS.

1.8 REFERENCES

The publications listed below (including amendments, addenda, revisions, supplement, and errata) form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

EIA-310-D	Standard 19-in. four-post EIA cabinet or rack, with mounting posts

U.S. DEPARTMENT OF VETERANS AFFAIRS (VA)

VA CAD Std	(2006) VHA National CAD Standard Application Guide
VA BIM Std	(2017) Veterans Affairs BIM Standard - BIM Manual V2.2

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 142	(2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)

NECA NEIS 1	(2015) Standard for Good Workmanship in Electrical Construction

UL SOLUTIONS (UL)

UL 497A	(2001; Bul. 2019) UL Standard for Safety Secondary Protectors for Communications Circuits
UL 497B	(2004; Reprint Feb 2022) UL Standard for Safety Protectors for Data Communications and Fire Alarm Circuits

UL 611	(1996) Standard for Safety Central-Station Burglar-Alarm Systems
UL 752	(2023) UL Standard for Safety Bullet-Resisting Equipment
UL 796	(2020; Reprint Oct 2023) UL Standard for Safety Printed Wiring Boards
UL 827	(2022; Reprint Oct 2023)UL Standard for Safety Central-Station Alarm Services
UL 1449	(2021; Reprint Dec 2022) UL Standard for Safety Surge Protective Devices
UL 1778	(2014; Reprint Apr 2023) UL Standard for Safety Uninterruptible Power Systems
UL 1981	(2014; Reprint Mar 2023) Standard for Central Station Automation Systems
UL 60950	(2000; Reprint Oct 2007) Safety of Information Technology Equipment

1.9 COORDINATION

a. Coordinate arrangement, mounting, and support of electronic safety and security equipment:

(1) To allow maximum possible headroom unless specific mounting heights that reduce headroom are indicated.

(2) To provide for ease of disconnecting the equipment with minimum interference to other installations.

(3) To allow right of way for piping and conduit installed at required slope.

(4) So connecting raceways, cables, wireways, cable trays, and busways will be clear of obstructions and of the working and access space of other equipment.

b. Coordinate installation of required supporting devices and set sleeves in cast-in-place concrete, masonry walls, and other structural components as they are constructed.

c. Coordinate location of access panels and doors for electronic safety and security items that are behind finished surfaces or otherwise concealed.

1.10 MAINTENANCE and SERVICE

a. General Requirements: The Contractor shall provide all services required and equipment necessary to maintain the entire integrated electronic security system in an operational state as specified for a period of one year after formal written acceptance of the system. The Contractor shall provide all necessary material required for performing scheduled adjustments or other non-scheduled work. Impacts on facility operations shall be minimized when performing scheduled

adjustments or other non-scheduled work. See also General Project Requirements.

b. Description of Work: The adjustment and repair of the security system includes all software updates, panel firmware, and the following new items computers equipment, communications transmission equipment and data transmission media (DTM), local processors, security system sensors, physical access control equipment, facility interface, signal transmission equipment, and video equipment.

c. Personnel: Service personnel shall be certified in the maintenance and repair of the selected type of equipment and qualified to accomplish all work promptly and satisfactorily. The Resident Engineer shall be advised in writing of the name of the designated service representative, and of any change in personnel. The Resident Engineer shall be provided copies of system manufacturer certification for the designated service representative.

d. Schedule of Work: The work shall be performed during regular working hours, Monday through Friday, excluding federal holidays.

e. System Inspections: These inspections shall include: The Contractor shall perform two minor inspections at six month intervals or more if required by the manufacturer, and two major inspections offset equally between the minor inspections to effect quarterly inspection of alternating magnitude.

(1) Minor Inspections shall include visual checks and operational tests of all console equipment, peripheral equipment, local processors, sensors, electrical and mechanical controls, and adjustments on printers.

(2) Major Inspections shall include all work described for Minor Inspections and the following: clean all system equipment and local processors including interior and exterior surfaces; perform diagnostics on all equipment; operational tests of the CPU, switcher, peripheral equipment, recording devices, monitors, picture quality from each camera; check, walk test, and calibrate each sensor; run all system software diagnostics and correct all problems; and resolve any previous outstanding problems.

f. Emergency Service: The owner shall initiate service calls whenever the system is not functioning properly. The Contractor shall provide the Owner with an emergency service center telephone number. The emergency service center shall be staffed 24 hours a day 365 days a year. The Owner shall have sole authority for determining catastrophic and non-catastrophic system failures within parameters stated in General Project Requirements.

(1) For catastrophic system failures, the Contractor shall provide same day four hour service response with a defect correction time not to exceed eight hours from notification. Catastrophic system failures are defined as any system failure that the Owner determines will place the facility(s) at increased risk.

(2) For non-catastrophic failures, the Contractor within eight hours with a defect correction time not to exceed 24 hours from notification.

g. Operation: Performance of scheduled adjustments and repair shall verify operation of the system as demonstrated by the applicable portions of the performance verification test.

h. Records and Logs: The Contractor shall maintain records and logs of each task and organize cumulative records for each component and for the complete system chronologically. A continuous log shall be submitted for all devices. The log shall contain all initial settings, calibration, repair, and programming data. Complete logs shall be maintained and available for inspection on site, demonstrating planned and systematic adjustments and repairs have been accomplished for the system.

i. Work Request: The Contractor shall separately record each service call request, as received. The record shall include the serial number identifying the component involved, its location, date and time the call was received, specific nature of trouble, names of service personnel assigned to the task, instructions describing the action taken, the amount and nature of the materials used, and the date and time of commencement and completion. The Contractor shall deliver a record of the work performed within five working days after the work was completed.

j. System Modifications: The Contractor shall make any recommendations for system modification in writing to the Resident Engineer. No system modifications, including operating parameters and control settings, shall be made without prior written approval from the Resident Engineer. Any modifications made to the system shall be incorporated into the operation and maintenance manuals and other documentation affected.

k. Software: The Contractor shall provide all software updates when approved by the Owner from the manufacturer during the installation and 12-month warranty period and verify operation of the system. These updates shall be accomplished in a timely manner, fully coordinated with the system operators, and incorporated into the operations and maintenance manuals and software documentation. There shall be at least one (1) scheduled update near the end of the first year’s warranty period, at which time the Contractor shall install and validate the latest released version of the Manufacturer’s software. All software changes shall be recorded in a log maintained in the unit control room. An electronic copy of the software update shall be maintained within the log. At a minimum, the contractor shall provide a description of the modification, when the modification occurred, and name and contact information of the individual performing the modification. The log shall be maintained in a white 3 ring binder and the cover marked “SOFTWARE CHANGE LOG”.

1.11 MINIMUM REQUIREMENTS

a. References to industry and trade association standards and codes are minimum installation requirement standards.

b. Drawings and other specification sections shall govern in those instances where requirements are greater than those specified in the above standards.

1.12 DELIVERY, STORAGE, and HANDLING

a. Equipment and materials shall be protected during shipment and storage against physical damage, dirt, moisture, cold and rain:

(1) During installation, enclosures, equipment, controls, controllers, circuit protective devices, and other like items, shall be protected against entry of foreign matter; and be vacuum cleaned both inside and outside before testing and operating and repainting if required.

(2) Damaged equipment shall be, as determined by the Resident Engineer, placed in first class operating condition or be returned to the source of supply for repair or replacement.

(3) Painted surfaces shall be protected with factory installed removable heavy craft paper, sheet vinyl or equal.

(4) Damaged paint on equipment and materials shall be refinished with the same quality of paint and workmanship as used by the manufacturer so repaired areas are not obvious.

b. Central Station, Workstations, and Controllers:

(1) Store in temperature and humidity controlled environment in original manufacturer's sealed containers. Maintain ambient temperature between 50 to 85 deg F, and not more than 80 percent relative humidity, non-condensing.

(2) Open each container; verify contents against packing list, and file copy of packing list, complete with container identification for inclusion in operation and maintenance data.

(3) Mark packing list with designations which have been assigned to materials and equipment for recording in the system labeling schedules generated by cable and asset management system.

(4) Save original manufacturer's containers and packing materials and deliver as directed under provisions covering extra materials.

1.13 PROJECT CONDITIONS

a. Environmental Conditions: System shall be capable of withstanding the following environmental conditions without mechanical or electrical damage or degradation of operating capability:

(1) Interior, Controlled Environment: System components, except central-station control unit, installed in temperature-controlled interior environments shall be rated for continuous operation in ambient conditions of 36 to 122 deg F dry bulb and 20 to 90 percent relative humidity, non-condensing. NEMA 250, Type 1 enclosure.

(2) Interior, Uncontrolled Environment: System components installed in non-temperature-controlled interior environments shall be rated for continuous operation in ambient conditions of 0 to 122 deg F dry bulb and 20 to 90 percent relative humidity, non-condensing. NEMA 250, Type 4X enclosures.

(3) Exterior Environment: System components installed in locations exposed to weather shall be rated for continuous operation in ambient conditions of -30 to 122 deg F dry bulb and 20 to 90 percent relative humidity, condensing. Rate for continuous operation where exposed to rain as specified in NEMA 250, winds up to 85 mph and snow cover up to 24 in thick. NEMA 250, Type 4X enclosures.

(4) Hazardous Environment: System components located in areas where fire or explosion hazards may exist because of flammable gases or vapors, flammable liquids, combustible dust, or ignitable fibers shall be rated, listed, and installed according to NFPA 70.

(5) Corrosive Environment: For system components subjected to corrosive fumes, vapors, and wind-driven salt spray in coastal zones, provide NEMA 250, Type 4X enclosures.

b. Security Environment: Use vandal resistant enclosures in high-risk areas where equipment may be subject to damage.

c. Console: All console equipment shall, unless noted otherwise, be rated for continuous operation under ambient environmental conditions of 60 to 85 deg F and a relative humidity of 20 to 80 percent.

1.14 EQUIPMENT AND MATERIALS

a. Materials and equipment furnished shall be of current production by manufacturers regularly engaged in the manufacture of such items, for which replacement parts shall be available.

b. When more than one unit of the same class of equipment is required, such units shall be the product of a single manufacturer.

c. Equipment Assemblies and Components:

(1) Components of an assembled unit need not be products of the same manufacturer.

(2) Manufacturers of equipment assemblies, which include components made by others, shall assume complete responsibility for the final assembled unit.

(3) Components shall be compatible with each other and with the total assembly for the intended service.

(4) Constituent parts which are similar shall be the product of a single manufacturer.

d. Factory wiring shall be identified on the equipment being furnished and on all wiring diagrams.

e. When Factory Testing Is Specified:

(1) The Government shall have the option of witnessing factory tests. The contractor shall notify the VA through the Resident Engineer a minimum of 15 working days prior to the manufacturers making the factory tests.

(2) Four copies of certified test reports containing all test data

shall be furnished to the Resident Engineer prior to final inspection and not more than 90 days after completion of the tests.

(3) When equipment fails to meet factory test and re-inspection is required, the contractor shall be liable for all additional expenses, including expenses of the Government.

1.15 ELECTRICAL POWER

a. Electrical power of 120 Volts Alternating Current (VAC) shall be indicated on the Division 26 drawings. Additional locations requiring primary power required by the security system shall be shown as part of these contract documents. Primary power for the security system shall be configured to switch to emergency backup sources automatically if interrupted without degradation of any critical system function. Alarms shall not be generated as a result of power switching, however, an indication of power switching on (on-line source) shall be provided to the alarm monitor. The Security Contractor shall provide an interface (dry contact closure) between the PACS and the Uninterruptible Power Supply (UPS) system so the UPS trouble signals and main power fail appear on the PACS operator terminal as alarms.

b. Failure of any on-line battery shall be detected and reported as a fault condition. Battery backed-up power supplies shall be provided sized for eight hours of operation at actual connected load. Requirements for additional power or locations shall be included with the Contract to support equipment and systems offered. The following minimum requirements shall be provided for power sources and equipment.

(1) Emergency Generator

(a) Report Printers: Unit Control Room

(b) Video Monitors: Unit Control Room

(c) Intercom Stations

(d) Radio System

(e) Lights: Unit Control Room, Equipment Rooms, and Security Offices

(f) Outlets: Security Outlets dedicated to security equipment racks or security enclosure assemblies.

(g) Security Device Power Supplies (DGP, VASS, Card Access, Lock Power, etc.) powered from the security closets or remotely: various locations

(h) Telephone/Radio Recording Equipment: Unit Control Room.

(i) VASS Camera Power Supplies: Security Closets

(j) VASS Pan/Tilt Units: Various Locations

(k) VASS Outdoor Housing Heaters and Blowers: Various Sites

(l) Intercom Master Control System

(m) Fiber Optic Receivers/Transmitters

(n) Security office Weapons Storage

(o) Outlets that charge handheld radios

(2) Uninterruptible Power Supply (UPS) on Emergency Power

(a) The following 120VAC circuits shall be provided. The Security Contractor shall coordinate exact locations with the Electrical Contractor:

1. Security System Monitors and Keyboards: Control Room

2. CPU: Control Equipment Room

3. Communications equipment: Control Equipment Room and various sites.

4. VASS Matrix Switcher: Control Equipment Room

5. VASS: Control Equipment Room

6. Digital Video Recorders, encoders and decoders: Control Room

7. All equipment Room racked equipment.

8. Network switches

1.16 TRANSIENT VOLTAGE SUPPRESSION, POWER SURGE SUPPLESION, and GROUNDING

a. Transient Voltage Surge Suppression: All cables and conductors extending beyond building façade, except fiber optic cables, which serve as communication, control, or signal lines shall be protected against Transient Voltage surges and have Transient Voltage Surge Suppression (TVSS) protection. The TVSS device shall be UL listed in accordance with Standard TIA 497B (UL 497B) installed at each end. Lighting and surge suppression shall be a multi-strike variety and include a fault indicator. Protection shall be furnished at the equipment and additional triple solid state surge protectors rated for the application on each wire line circuit shall be installed within 3 ft of the building cable entrance. Fuses shall not be used for surge protection. The inputs and outputs shall be tested in both normal mode and common mode to verify there is no interference.

(1) A 10-microsecond rise time by 1000 microsecond pulse width waveform with a peak voltage of 1500 volts and a peak current of 60 amperes.

(2) An 8-microsecond rise time by 20-microsecond pulse width waveform with a peak voltage of 1000 volts and a peak current of 500 amperes.

(3) Maximum series current: 2 AMPS. Provide units manufactured by Advanced Protection Technologies, model # TE/FA 10B or TE/FA 20B.

(4) Operating Temperature and Humidity: -40 to 185 deg F, 0 to 95

percent relative humidity.

b. Grounding and Surge Suppression

(1) The Security Contractor shall provide grounding and surge suppression to stabilize the voltage under normal operating conditions. To ensure the operation of over current devices, such as fuses, circuit breakers, and relays, under ground-fault conditions.

(2) Security Contractor shall engineer and provide proper grounding and surge suppression as required by local jurisdiction and prevailing codes and standards referenced in this document.

(3) Principal grounding components and features. Include main grounding buses and grounding and bonding connections to service equipment.

(4) Details of interconnection with other grounding systems. The lightning protection system shall be provided by the Security Contractor.

(5) Locations and sizes of grounding conductors and grounding buses in electrical, data, and communication equipment rooms and closets.

(6) AC power receptacles are not to be used as a ground reference point.

(7) Any cable that is shielded shall require a ground in accordance with the best practices of the trade and manufactures installation instructions.

(8) Protection should be provided at both ends of cabling.

1.17 COMPONENT ENCLOSURES

a. Construction of Enclosures

(1) Consoles, power supply enclosures, detector control and terminal cabinets, control units, wiring gutters, and other component housings, collectively referred to as enclosures, shall be so formed and assembled as to be sturdy and rigid.

(2) Thickness of metal in-cast and sheet metal enclosures of all types shall not be less than those in Tables I and II, UL 611. Sheet steel used in fabrication of enclosures shall be not less than 14 gauge. Consoles shall be 16-gauge.

(3) Doors and covers shall be flanged. Enclosures shall not have pre-punched knockouts. Where doors are mounted on hinges with exposed pins, the hinges shall be of the tight pin type or the ends of hinge pins shall be tack welded to prevent removal. Doors having a latch edge length of less than 24 in shall be provided with a single construction core. Where the latch edge of a hinged door is more than 24 in or more in length, the door shall be provided with a three-point latching device with construction core; or alternatively with two, one located near each end.

(4) Any ventilator openings in enclosures and cabinets shall conform

to the requirements of UL 611. Unless otherwise indicated, sheet metal enclosures shall be designed for wall mounting with tip holes slotted. Mounting holes shall be in positions that remain accessible when all major operating components are in place and the door is open, but shall be in accessible when the door is closed.

(5) Covers of pull and junction boxes provided to facilitate initial installation of the system shall be held in place by tamper proof post security screws. Stenciled or painted labels shall be affixed to such boxes indicating they contain no connections. These labels shall not indicate the box is part of the Electronic Security System (ESS).

b. Consoles and Equipment Racks: All consoles and vertical equipment racks shall include a forced air-cooling system.

(1) Vertical Equipment Racks:

(a) The forced air blowers shall be installed in the vented top of each cabinet and shall not reduce usable rack space.

(b) The forced air fan shall consist of one fan rated at 105 CFM per rack bay and noise level shall not exceed 55 decibels.

(c) d. Vertical equipment racks are to be provided with full sized clear plastic locking doors and vented top panels as shown on contract drawings.

(2) Console racks:

(a) Forced air fans shall be installed in the top rear of each console bay. The forced air fan shall consist of one fan rated at 105 CFM mounted to a 133 mm vented blank panel the noise level of each fan shall not exceed 55 decibels. The fans shall be installed so air is pulled from the bottom of the rack or cabinet and exhausted out the top.

(b) Console racks are to be provided with flush mounted hinged rear doors with recessed locking latch on the bottom and middle sections of the consoles. Provide code access to support wiring for devices located on the work surfaces.

c. Tamper Provisions and Tamper Switches:

(1) Enclosures, cabinets, housings, boxes and fittings or every product description having hinged doors or removable covers and which contain circuits, or the integrated security system and its power supplies shall be provided with cover operated, corrosion-resistant tamper switches.

(2) Tamper switches shall be arranged to initiate an alarm signal that will report to the monitoring station when the door or cover is moved. Tamper switches shall be mechanically mounted to maximize the defeat time when enclosure covers are opened or removed. It shall take longer than 1 second to depress or defeat the tamper switch after opening or removing the cover. The enclosure and tamper switch shall function together in such a manner as to prohibit direct line of sign to any internal component before the

switch activates.

(3) Tamper switches shall be inaccessible until the switch is activated. Have mounting hardware concealed so the location of the switch cannot be observed from the exterior of the enclosure. Be connected to circuits which are under electrical supervision at all times, irrespective of the protection mode in which the circuit is operating. Be spring-loaded and held in the closed position by the door or cover and be wired so they break the circuit when the door cover is disturbed. Tamper circuits shall be adjustable type screw sets and shall be adjusted by the contractor to eliminate nuisance alarms associated with incorrectly mounted tamper device shall annunciate prior to the enclosure door opening (within 1/4 " tolerance. The tamper device or its components shall not be visible or accessing with common tools to bypass when the enclosure is in the secured mode.

(4) The single gang junction boxes for the portrait alarming and pull boxes with less than 102 square mm will not require tamper switches.

(5) All enclosures over 305 square mm shall be hinged with an enclosure lock.

(6) Control Enclosures: Maintenance/Safety switches on control enclosures, which must be opened to make routing maintenance adjustments to the system and to service the power supplies, shall be push/pull-set automatic reset type.

(7) Provide one enclosure tamper switch for each 609 linear mm of enclosure lock side opening evenly spaced.

(8) All security screws shall be Torx-Post Security Screws.

(9) The contractor shall provide the owner with two (torx-post screwdrivers.

1.18 ELECTRONIC COMPONENTS

All electronic components of the system shall be of the solid-state type, mounted on printed circuit boards conforming to UL 796. Boards shall be plug-in, quick-disconnect type. Circuitry shall not be so densely placed as to impede maintenance. All power-dissipating components shall incorporate safety margins of not less than 25 percent with respect to dissipation ratings, maximum voltages, and current-carrying capacity.

1.19 SUBSTITUTE MATERIALS and EQUIPMENT

a. Where variations from the contract requirements are requested in accordance with the GENERAL CONDITIONS and Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES, the connecting work and related components shall include, but not be limited to additions or changes to branch circuits, circuit protective devices, conduits, wire, feeders, controls, panels and installation methods.

b. In addition to this Section the Security Contractor shall also reference Section II, Products and associated divisions. The Resident Engineer shall have final authority on the authorization or refusal of substitutions. If there are no proposed substitutions, a statement in

writing from the Contractor shall be submitted to the Resident Engineer stating same. In the preparation of a list of substitutions, the following information shall be included, as a minimum:

(1) Identity of the material or devices specified for which there is a proposed substitution.

(2) Description of the segment of the specification where the material or devices are referenced.

(3) Identity of the proposed substitute by manufacturer, brand name, catalog or model number and the manufacturer's product name.

(4) A technical statement of all operational characteristic expressing equivalence to items to be substituted and comparison, feature-by-feature, between specification requirements and the material or devices called for in the specification; and Price differential.

c. Materials Not Listed: Furnish all necessary hardware, software, programming materials, and supporting equipment required to place the specified major subsystems in full operation. Note that some supporting equipment, materials, and hardware may not be described herein. Depending on the manufacturers selected by the COTR, some equipment, materials and hardware may not be contained in either the Contract Documents or these written specifications, but are required by the manufacturer for complete operation according to the intent of the design and these specifications. In such cases, the Resident Engineer shall be given the opportunity to approve the additional equipment, hardware and materials that shall be fully identified in the bid and in the equipment list submittal. The Resident Engineer shall be consulted in the event there is any question about which supporting equipment, materials, or hardware is intended to be included.

d. Response to Specification: The Contractor shall submit a point-by-point statement of compliance with each paragraph of the security specification. The statement of compliance shall list each paragraph by number and indicate "COMPLY" opposite the number for each paragraph where the Contractor fully complies with the specification. Where the proposed system cannot meet the requirements of the paragraph, and does not offer an equivalent solution, the offers shall indicate "DOES NOT COMPLY" opposite the paragraph number. Where the proposed system does not comply with the paragraph as written, but the bidder feels it will accomplish the intent of the paragraph in a manner different from that described, the offers shall indicate "COMPARABLE". The offers shall include a statement fully describing the "comparable" method of satisfying the requirement. Where a full and concise description is not provided, the offered system shall be considered as not complying with the specification. Any submission that does not include a point-by-point statement of compliance, as described above, shall be disqualified. Submittals for products shall be in precise order with the product section of the specification. Submittals not in proper sequence will be rejected.

1.20 LIKE ITEMS

Where two or more items of equipment performing the same function are required, they shall be exact duplicates produced by one manufacturer.

All equipment provided shall be complete, new, and free of any defects.

1.21 WARRANTY

The Contractor shall, as a condition precedent to the final payment, execute a written guarantee (warranty) to the COTR certifying all contract requirements have been completed according to the final specifications. Contract Drawings and the warranty of all materials and equipment furnished under this contract are to remain in satisfactory operating condition (ordinary wear and tear, abuse and causes beyond his control for this work accepted) for one year from the date the Contactor received written notification of final acceptance from the COTR. Demonstration and training shall be performed prior to system acceptance. All defects or damages due to faulty materials or workmanship shall be repaired or replaced without delay, to the COTR's satisfaction, and at the Contractor's expense. The Contractor shall provide quarterly inspections during the warranty period. The contractor shall provide written documentation to the COTR on conditions and findings of the system and device(s). In addition, the contractor shall provide written documentation of test results and stating what was done to correct any deficiencies. The first inspection shall occur 90 calendar days after the acceptance date. The last inspection shall occur 30 calendar days prior to the end of the warranty. The warranty period shall be extended until the last inspection and associated corrective actions are complete. When equipment and labor covered by the Contractor's warranty, or by a manufacturer's warranty, have been replaced or restored because of it’s failure during the warranty period, the warranty period for the replaced or repaired equipment or restored work shall be reinstated for a period equal to the original warranty period, and commencing with the date of completion of the replacement or restoration work. In the event any manufacturer customarily provides a warranty period greater than one year, the Contractor's warranty shall be for the same duration for that component.

1.22 SINGULAR NUMBER

Where any device or part of equipment is referred to in these specifications in the singular number (e.g., "the switch"), this reference shall be deemed to apply to as many such devices as are required to complete the installation as shown on the drawings.

PART 2 – PRODUCTS

2.1 EQUIPMENT AND MATERIALS

a. All equipment associated within the Security Control Room, Security Console and Security Equipment Room shall be UL 827, UL 1981, and UL 60950 compliant and rated for continuous operation. Environmental conditions (i.e. temperature, humidity, wind, and seismic activity) shall be taken under consideration at each facility and site location prior to installation of the equipment.

b. All equipment shall operate on a 120 or 240 volts alternating current (VAC); 50 Hz or 60 Hz AC power system unless documented otherwise in subsequent sections listed within this specification. All equipment shall have a back-up source of power that will provide a minimum of eight hours of run time in the event of a loss of primary power to the facility.

c. The system shall be designed, installed, and programmed in a manner that will allow for ease of operation, programming, servicing, maintenance, testing, and upgrading of the system.

d. All equipment and materials for the system will be compatible to ensure correct operation.

2.2 EQUIPMENT ITEMS

a. The Security Management System shall provide full interface with all components of the security subsystem as follows:

(1) Shall allow for communication between the Physical Access Control System and Database Management and all subordinate work and monitoring stations, enrollment centers for badging and biometric devices as part of the PACS, local annunciation centers, the electronic Security Management System (SMS), and all other VA redundant or backup command center or other workstations locations.

(2) Shall provide automatic continuous communication with all systems that are monitored by the SMS, and shall automatically annunciate any communication failures or system alarms to the SMS operator providing identification of the system, nature of the alarm, and location of the alarm.

(3) Controlling devices shall be utilized to interface the SMS with all field devices.

(4) The Security control room and security console will be supported by an uninterrupted power supply (UPS) or dedicated backup generator power circuit.

(5) The Security Equipment room, Security Control Room, and Security Operator Console shall house the following equipment i.e. refer to individual master specifications for each security subsystem’s specific requirements:

(a) Equipment Racks

(b) Security Network Server and Workstation

(c) CCTV Monitoring, Controlling, and Recording Equipment

(d) PACS Monitoring and Controlling Equipment

(e) IDS Monitoring and Controlling Equipment

(f) Security Access Detection Monitoring Equipment

(g) EPPS Monitoring and Controlling Equipment

(h) Main Panels for all Security Systems

(i) Power Supply Units (PSU) for all field devices

(j) Life safety and power monitoring equipment

(k) All other building systems deemed necessary by the VA to include, but not limited to, heating, ventilation and air

conditioning (HVAC), elevator control, portable radio, fire alarm monitoring, and other potential systems. Police two-way radio control consoles/units.

b. Security Console Bays - shall be EIA-310-D compliant and:

(1) Utilize stand-up, sit-down, and vertical equipment racks in any combination to monitor and control the security subsystems.

(2) Shall be wide enough for equipment that requires a minimum 19 inch mounting area.

(3) Shall be made of metal, furnished with wire ways, a power strip, a thermostatic controlled bottom or top mounted fan units, a hinge mounted rear door, a hinge mounted front door made of Plexiglas, and a louvered top. When possible, pre-fabricated (standard off-the-shelf) security console equipment shall be used in place of customized designed consoles.

(4) A wire management system shall be designed and installed so that all cables are mounted in a manner that they do not interfere with day-to-day operations, are labeled for quick identification, and so that high voltage power cables do not cause signal interference with low voltage and data carrying cables.

(5) Shall be mounted on lockable casters.

(6) Shall be ergonomically designed so that all devices requiring repetitive interaction with by the operator can be easily accessed, observed, and accomplished.

(7) Controls and displays shall be located so that they are not obscured during normal operation. Control and display units installed with a work bench shall be a minimum of 3 in. from all edges of the work bench area.

(8) All security subsystem controls shall be installed within the same operating console bay of their associated equipment.

(9) Video monitors shall be mounted above all controls within a console bay and positioned in a manner that minimum strain is placed on the operator viewing them at the console.

(10) At least one workbench for every three console bays shall be provided free of control equipment to allow for appropriate operator workspace.

(11) All console devices shall be labeled and marked with a minimum of quarter inch bold print.

(12) All non-security related equipment that is required to be monitored shall be installed in a console bay separate from the security subsystem equipment and clearing be identified as such.

(13) Console bays and related equipment shall be arranged in priority order and sequenced based upon their pre-defined security subsystem operations criticality established by the Contracting Officer.

(14) The following minimum console technical characteristics shall be taken into consideration when designing for and installing the security console and equipment racks:

	Stand-Up	Sit-Down	Vertical Equipment Rack
Workstation Height	No Greater than 84 in.	No greater than 72 in.	No greater than 96 in.
Bench board Slope	21 in.	25 in.	N/A
Bench board Angle	15 degrees	15 degrees	N/A
Depth of Console	24 in.	24 in.	N/A
Leg and Feet Clearance	6 sq. ft. from center of Console Slope front	6 sq. ft. from center of Console Slope front	6 sq. ft. from center of Console Slope front
Distance Between Console Rows	96 in.	96 in.	96 in.
Distance Between Console and Wall	36 in. from the rear and/or side of console or rack	36 in. from the rear and/or side of console or rack	36 in. from the rear and/or side of console or rack

c. Security Console Configuration:

(1) The size shall be defined by the number of console bays required to house and operate the security subsystems, as well as any other factors that may influence the overall design of the space. A small Access Control System and Database Management shall contain no more than four security console bays. A large Access Control System and Database Management shall contain no less than five and no more than eight security console bays.

(2) Shall meet the following minimum spacing requirements to ensure that a Access Control System and Database Management is provided to house existing and future security subsystems and other equipment listed in paragraph 2.3.C:

(a) 500 square feet for a large Access Control System and Database Management.

(b) 300 square feet for a small Access Control System and Database Management.

(c) If office, training room and conference space, is a processing

area as well as holding cell space is to be located adjacent to the Access Control System and Database Management, these space requirements also need to be considered.

(3) Shall be located in an area within, at a minimum, the first level/line of security defense defined by the VA. If the Access Control System and Database Management is to be located outside the first level of security, then the area shall be constructed or retrofit to meet or exceed those requirements outlined in associated VA Master Specifications.

(4) Shall not be located within or near an area with little to no blast mitigation standoff space protection, adjacent to an outside wall exposed to vehicle parking and traffic, within a basement or potential flood zone area, in close approximately to major utility areas, or near an exposed air intake(s).

(5) Access shall meet UFAS and accessibility requirements.

(6) Construction shall be slab to slab and free of windows, with the exception of a service window. All penetrations into the room shall be sealed with fire stopping materials. This material shall apply in accordance with Section 07 84 00, FIRESTOPPING.

(7) A service window shall be installed in the wall next to the main entrance of the Access Control System and Database Management or where it best can be monitored and accessed by the security console operator. The window shall meet all requirements set forth in UL 752, to include at a minimum, Class III ballistic level protection. The windows shall be set in a minimum or 4 inches solid concrete units to ceiling height with either masonry or gypsum wall board to the underside of the slab above. It shall also contain a service tray constructed in a manner that only objects no larger than 3 inches in width may pass through it.

(8) The walls making up or surrounding the Access Control System and Database Management shall be made of materials that at a minimum offer Class III ballistic level protection for the security console operator(s).

(9) There will be a main power cut-off button/switch located inside the Access Control System and Database Management in the event of an electrical fire or related event occurs.

(10) Shall have a fire alarm detection unit that is tied into the main building fire alarm system and have at least two fire extinguishers located within it.

(11) Shall utilize a fire suppression system similar to that used by the VA’s computer and telecommunications room operating areas.

(12) The floor shall be raised a minimum of 4 inches (10 cm) from the concrete floor base. Wire ways shall be utilized under the raised floor for separation of signal and power wires and cables.

(13) Access shall be monitored and controlled by the PACS via card reader and fixed camera that utilizes a wide angle lens. A 1 in. (2.5 cm) deadbolt shall be utilized as a mechanical override for the door in the event of electrical failure of the PACS, card

reader, or locking mechanism.

(14) There shall only be one point of ingress and egress to and from the Security Control Room. The door shall be made of solid core wood or better. If a window is required for the door, then the window shall be ballistic resistant with a Millar covering.

(15) A two-way intercom shall be placed at the point of entry into the Security Control Room for access-communication control purposes.

(16) A remote push-button door unlocking device shall not be installed for the electronic PACS locking mechanism providing access control into the Security Control Room.

(17) All controlling equipment and power supplies that must be wall mounted shall be mounted in a manner that maximizes usability of the Security Control Room wall space. All equipment shall be mounted to three quarter inch fire retardant plywood. The plywood shall be fastened to the wall from slab to slab and fixed to the existing walls supports.

d. Security Control Room Ventilation

(1) Shall meet or exceed all requirements laid out in VA Master Specification listed in Division 23, HEATING, VENTILATION, AND AIR CONDITIONING.

(2) Controls shall be via a separate air handling system that provides an isolated supply and return system. The Security Control Room shall have a dedicated thermostat control unit and cut-off switch to be able to shut off ventilation to the control room in the event of a chemical, biological, or radiological (CBR) event or other related emergency.

(3) There shall be a louver installed in the control room door to assist with ventilation of the room. The louver shall be exactly 12 x 12 inches and closeable.

e. Security Control Room and Security Console Lighting:

(1) The following factors shall be taken into consideration for lighting of the Security Control Room and console area:

(a) Shadows: To reduce eye strain and fatigue, shadows shall be avoided.

(b) Glare: The readability of all display panels, labels, and equipment shall not be interfered with or create visibility problems.

(2) The following table shall provide guidance on the amount of footcandles required per work area and type of task performed:

Work Area/Type of Task	Footcandles

Main Operating Panels		50
Secondary Display Panels		50
Seated Workstations		100
Reading	Handwriting	100
	Typed Documents	50
	Visual Display Units	10
Logbook Recording		100
Maintenance Area		50
Emergency/Back-up Lighting		10

f. Remote security console access: For facilities that have a remote, secondary back-up control console or workstation shall apply the following requirements:

(1) The secondary stations shall the requirements outlined in Sections 2.2.A-G.

(2) Installation of an intercom station or telephone line shall be installed and provide direct one touch call-up for communications between the primary Security Control Console and secondary Security Control Console.

(3) Secondary stations shall not have priority over a primary Security Control Console.

(4) The primary Access Control System and Database Management shall have the ability to shut off power and a signal to a secondary control station in the event the area has been compromised.

g. Wires and Cables:

(1) Shall meet or exceed the manufactures recommendation for power and signals.

(2) Shall be carried in an enclosed conduit system, utilizing electromagnetic tubing (EMT) to include the equivalent in flexible metal, rigid galvanized steel (RGS) to include the equivalent of liquid tight, polyvinylchloride (PVC) schedule 40 or 80.

(3) All conduits will be sized and installed per NFPA 70. All security system signal and power cables that traverse or originate in a high security office space will contained in either EMT or RGS conduit.

(4) All conduit, pull boxes, and junction boxes shall be marked with colored permanent tape or paint that will allow it to be distinguished from all other infrastructure conduit.

(5) Conduit fills shall not exceed 50 percent unless otherwise documented.

(6) A pull string shall be pulled along and provided with signal and power cables to assist in future installations.

(7) At all locations where there is a wall penetration or core drilling is conducted to allow for conduit to be installed, fire stopping materials shall be applied to that area.

(8) High voltage and signal cables shall not share the same conduit and shall be kept separate up to the point of connection. High voltage for the security subsystems shall be any cable or sets of cables carrying 30 VDC/VAC or higher.

(9) For all equipment that is carrying digital data between the Security Control Room, Security Equipment Room, Security Console, or at a remote monitoring station, it shall not be less that 20 AWG and stranded copper wire for each conductor. The cable or each individual conductor within the cable shall have a shield that provides 100 percent coverage. Cables with a single overall shield shall have a tinned copper shield drain wire.

2.3 FIBER OPTIC EQUIPMENT

a. 8 Channel Fiber Optic Transcievers (VideoandPTZ Control)

(1) The field-located and central-located fiber optic transceivers shall utilize wave division multiplexing to transmit and receive video and data pan-tilt-zoom control signals over two standard 62.5/125 multimode fibers.

(2) The units shall be capable of operating over a range of 2 km.

(3) The units shall be NTSC color compatible.

(4) The units shall support data rates up to 64 Kbps.

(5) The units shall be surface or rack mountable.

(6) The units shall be UL listed.

(7) The units shall meet or exceed the following specifications:

(a) Video

1. Input/Output: 1 volt pk-pk (75 ohms)

2. Input/Output Channels: 8

3. Bandwidth: 10 Hz - 6.5 MHZ per channel

4. Differential Gain: <2 percent

5. Differential Phase: <0.7 degrees

6. Tilt: <1 percent

7. Signal to Noise Ratio: 60 dB

(b) Data (Control)

1. Data Channels: 2

2. Data Format: RS-232, RS-422, 2 wire or 4 wire RS-485 with Tri-State Manchester Bi-Phase and Sensornet

3. Data Rate: DC - 100 kbps (NRZ)

4. Bit Error Rate: < 1 in 10-9 @ Maximum Optical Loss Budget

5. Operating Mode: Simplex or Full-Duplex

6. Wavelength: 1310/1550 nm, Multimode or Singlemode

7. Optical Emitter: Laser Diode

8. Number of Fibers: 1

(c) Connectors

1. Optical: ST

2. Power and Data: Terminal Block with Screw Clamps

3. Video: BNC (Gold Plated Center-Pin)

(d) Electrical and Mechanical

1. Power: 12 VDC @ 500 mA (stand-alone)

2. Current Protection: Automatic Resettable Solid-State Current Limiters

(e) Environmental

1. MTBF: > 100,000 hours

2. Operating Temp: -40 to 165 deg F

3. Storage Temp: -40 to 185 deg F

4. Relative Humidity: 0 percent to 95 percent (non-condensing)

b. Fiber Optic Transmitters: The central-located fiber optic transmitters shall utilize wave division multiplexing to transmit video and signals over standard 62.5/125 multimode fibers.

(1) The units shall be capable of operating over a range of 4.8 km.

(2) The units shall be NTSC color compatible.

(3) The units shall support data rates up to 64 Kbps.

(4) The units shall be surface or rack mountable.

(5) The units shall be UL listed.

(6) The units shall meet or exceed the following specifications: Video

(a) Input: 1 volt pk-pk (75 ohms)

(b) Bandwidth: 5H2 - 10 MHZ

(c) Differential Gain: <5 percent

(d) Tilt: <1 percent

(e) Signal-Noise: 60db

(f) Wavelength: 850nm

(g) Number of Fibers: 1

(h) Operating Temp: -4 to 158 deg F

(i) Connectors:

1. Power: Female plug with screw clamps

2. Video: BNC

3. Optical: ST

(j) Power: 12 VDC

c. Fiber Optic Receivers: The field-located fiber optic receivers shall utilize wave division multiplexing to receive video signals over standard 62.5/125 multimode fiber.

(1) The units shall be capable of operating over a range of 4.8 km.

(2) The units shall be NTSC color compatible.

(3) The units shall support data rates up to 64 Kbps.

(4) The units shall be surface or rack mountable.

(5) The units shall be UL listed.

(6) The units shall meet or exceed the following specifications:

(a) Video

1. Output: 1 volt pk-pk (75 ohms)

2. Bandwidth: 5H2 - 10 MHZ

3. Differential Gain: <5 percent

4. Tilt: <1 percent

5. Signal-Noise: 60dB

6. Wavelength: 850nm

7. Number of Fibers: 1

8. Surface Mount: 4.2 x 3.5 x 1 in

9. Operating Temp: -4 to 158 deg F)

10. Connectors:

11. Power: Female plug block with screw clamps

12. Video: BNC

13. Optical: ST

14. Power: 12 VAC8 Channel Fiber Optic Transcievers (VideoandPTZ Control)

d. Fiber Optic Sub Rack with Power Supply

(1) The Card Cage Rack shall provide high-density racking for fiber-optic modules. The unit shall be designed to mount in standard 19 in instrument racks and to accommodate the equivalent of 15 1-inch modules.

(a) Specifications

1. Card Orientation: Vertical

2. Construction: Aluminum

3. Current Consumption: 0.99 A

4. Humidity: 95.0 percent RH

5. Input Power: 100-240 VAC, 60/50 Hz

6. Mounting: Mounts in standard 19 in rack using four screws (optional wall brackets purchased separately)

7. Number of Outputs: 1.0

8. Number of Slots 15.0

9. Operating Temperature: -40.0 to 167.0 deg F)

10. Ouput Voltage: 13.5 V

11. Output Current 6.0 A

12. Power Dissipation: 28.0 W

13. Power Factor: 48.0

14. Power Supply: (built-in)

15. Rack Units: 3RU

16. Redundant Capability: Yes

17. Weight: 5.35 lb

18. Width: 19.0 in

2.4 TRANSIENT VOLTAGE SURGE SUPPRESSION DEVICES (TVSS) AND SURGE SUPPRESION

a. Transient Voltage Surge Suppression

(1) All cables and conductors extending beyond building perimeter, except fiber optic cables, which serve as communication, control, or signal lines shall be protected against Transient Voltage surges and have Transient Voltage surge suppression protection (TVSS) UL listed in accordance with UL 497B installed at each end. Lighting and surge suppression shall be a multi-strike variety and include a fault indicator. Protection shall be furnished at the equipment and additional triple solid state surge protectors rated for the application on each wire line circuit shall be installed within 36 in of the building cable entrance. Fuses shall not be used for surge protection. The inputs and outputs shall be tested in both normal mode and common mode using the following waveforms:

(a) A 10-microsecond rise time by 1000 microsecond pulse width waveform with a peak voltage of 1500 volts and a peak current of 60 amperes.

(b) An 8-microsecond rise time by 20-microsecond pulse width waveform with a peak voltage of 1000 volts and a peak current of 500 amperes.

(c) Maximum series current: 2 AMPS. Provide units manufactured by Advanced Protection Technologies, model # TE/FA 10B or TE/FA 20B or approved equivalent.

(d) Operating Temperature and Humidity: -40 to 185 deg F, and 0 to 95 percent relative humidity, non-condensing.

b. Physical Access Control Systems

(1) Suppressors shall be installed on AC power at the point of service and shall meet the following criteria:

(a) UL 1449, listed

(b) UL 1449 S.V.R. of 400 Volts or lower

(c) Status Indicator Light(s)

(d) Minimum Surge Current Capacity: 40,000 Amps (8 x 20 µsec)

(e) Maximum Continuous Current: 15 Amps

(f) MCOV: 125 VAC

(g) Service Voltage: 110-120 VAC

(2) Suppressors shall be installed on the Low Voltage circuit at both the point of entrance and exit of the building. Suppressors shall

meet the following criteria:

(a) UL 497B

(b) Minimum Surge Current Capacity: 2,000 Amps per pair

(c) Maximum Continuous Current: 5 Amps

(d) MCOV: 33 Volts

(e) Service Voltage: 24Volts

(3) Suppressors shall be installed on the communication circuit between the access controller and card reader at both the entrance and exit of the building. Suppressors shall meet the following criteria:

(a) Conforms with UL 497B standards (where applicable)

(b) Clamp level for 12 and 24V power: 18VDC / 38VDC

(c) Clamp level for Data/LED: 6.8VDC

(d) Service Voltage for Power: 12VDC/24VDC

(e) Service Voltage for Data/LED: <5VDC

(f) Clamp level – PoE Access Power: 72V

(g) Clamp level – PoE Access Data: 7.9V

(h) Service Voltage – PoE Access: 48VAC – 54VAC

(i) Service Voltage – PoE Data: <5VDC

c. Intercom Systems

(1) Suppressors shall be installed on the AC power at the point of service and shall meet the following criteria:

(a) UL 1449 Listed

(b) UL 1449 S.V.R. of 400 Volts or lower

(c) Diagnostic Indicator Light(s)

(d) Integrated ground terminating post (where case/chassis ground exists)

(e) Minimum Surge Current Capacity of 13,000 Amps (8 x 20 µSec)

(2) Suppressors shall be installed on incoming central office lines and shall meet the following criteria:

(a) UL 497A Listed

(b) Multi Stage protection design

(c) Auto-reset current protection not to exceed 2 Amps per pair

(d) Minimum Surge Current of 500 Amps per pair (8 x 20 µSec)

(3) Suppressors shall be installed on all telephone/intercom circuits that enter or leave separate buildings and shall meet the following criteria:

(a) UL 497A Listed (where applicable)

(b) UL 497B Listed (horns, strobes, speakers or communication circuits over 300 feet)

(c) Multi Stage protection design

(d) Auto-reset over-current protection not to exceed 5 Amps per pair

(e) Minimum Surge Current of 1000 Amps per pair (8 x 20 µSec)

d. Intrusion Detection Systems

(1) Suppressors shall be installed on AC at the point of service and shall meet the following criteria:

(a) UL 1449, 2nd Edition 2007, listed

(b) UL 1449 S.V.R. of 400 Volts or lower

(c) Status Indicator Lights

(d) Center screw for terminating Class II transformers

(e) Minimum Surge Current Capacity of 32,000 Amps (8 x 20 µSec)

(2) Suppressors shall be installed on all Telephone Communication Interface circuits and shall meet the following criteria:

(a) UL 497A Listed

(b) Multi Stage protection design

(c) Surge Current Capacity: 9,000 Amps (8x20 µSec)

(d) Clamp Voltage: 130Vrms

(e) Auto reset current protection not to exceed 150 milliAmps

(3) Suppressors shall be installed on all burglar alarm initiating and signaling loops and addressable circuits which enter or leave separate buildings. The following criteria shall be met:

(a) UL 497B for data communications or annunciation (powered loops)

(b) Fail-short/fail-safe mode.

(c) Surge Current Capacity: 9,000 Amps (8x20 µSec)

(d) Clamp Voltage: 15 Vrms

(e) Joule Rating: 76 Joules per pair (10x1000 µSec)

(f) Auto-reset current protection not to exceed 150 milliAmps for UL 497A devices.

e. Video Surveillance System

(1) Protectors shall be installed on coaxial cable systems on points of entry and exit from separate buildings. Suppressors shall be installed at each exterior camera location and include protection for 12 and/or 24 volt power, data signal and motor controls (for Pan, Tilt and Zoom systems). SPDs shall protect all modes herein mentioned and contain all modes in a single unit system. Protection for all systems mentioned above shall be incorporated at the head end equipment. Additionally a minimum 450VA battery back up shall be used to protect the DVR or VCR and monitor. Protectors shall meet the following criteria:

(a) Head-End Power

1. UL 1778, cUL (Battery Back Up)

2. Minimum Surge Current Capacity: 65,000 Amps (8x20µsec)

3. Minimum of two NEMA 5-15R Receptacles (one AC power only, one with UPS)

4. All modes protected (L-N, L-G, N-G)

5. EMI/RFI Filtering

6. Maximum Continuous Current: 12 Amps

(b) Camera Power

1. Minimum Surge Current Capacity: 1,000 Amps (8X20µsec); 240 Amps for IP Video/PoE cameras

2. Screw Terminal Connection

3. All protection modes L-G (all Lines)

4. MCOV <40VAC

(c) Video And Data

1. Surge Current Capacity 1,000 Amps per conductor

2. “BNC” Connection (Coax)

3. Protection modes: L-G (Data), Center Pin-G, Shield-G (Coax)

4. Band Pass 0-2GHz

5. Insertion Loss <0.3dB

f. Grounding and Surge Suppression

(1) The Security Contractor shall provide grounding and surge suppression to stabilize the voltage under normal operating conditions. This is to ensure the operation of over current devices, such as fuses, circuit breakers, and relays, underground-fault conditions.

(2) The Contractor shall engineer, provide, ad install proper grounding and surge suppression as required by local jurisdiction and prevailing codes and standards, referenced in this document.

(3) Principal grounding components and features shall include: main grounding buses, grounding, and bonding connections to service equipment.

(4) The Contractor shall provide detail drawings of interconnection with other grounding systems including lightning protection systems.

(5) The Contractor shall provide details of locations and sizes of grounding conductors and grounding buses in electrical, data, and communication equipment rooms and closets.

(6) AC power receptacles are not to be used as a ground reference point.

(7) Any cable that is shielded shall require a ground in accordance with applicable codes, the best practices of the trade, and all manufactures' installation instructions.

g. 120 VAC Surge Suppression

(1) Continuous Current: Unlimited (parallel connection)

(2) Max Surge Current: 13,500 Amps

(3) Protection Modes: L - N, L - G, N - G

(4) Warranty: Ten Year Limited Warranty

(5) Dimension: 2.90 x 1.62 x 2.05 in

(6) Weight: 0.18 lbs

(7) Housing: ABS

2.5 INSTALLATION KIT

a. General:

(1) The kit shall be provided that, at a minimum, includes all connectors and terminals, labeling systems, audio spade lugs, barrier strips, punch blocks or wire wrap terminals, heat shrink tubing, cable ties, solder, hangers, clamps, bolts, conduit, cable duct, and/or cable tray, etc., required to accomplish a neat and secure installation. All wires shall terminate in a spade lug and barrier strip, wire wrap terminal or punch block. Unfinished or unlabeled wire connections shall not be allowed. All unused and partially opened installation kit boxes, coaxial, fiber-optic, and

twisted pair cable reels, conduit, cable tray, and/or cable duct bundles, wire rolls, physical installation hardware shall be turned over to the Contracting Officer. The following sections outline the minimum required installation sub-kits to be used:

(2) System Grounding:

(a) The grounding kit shall include all cable and installation hardware required. All head end equipment and power supplies shall be connected to earth ground via internal building wiring, according to NFPA 70.

(b) This includes, but is not limited to:

1. Coaxial Cable Shields
2. Control Cable Shields
3. Data Cable Shields
4. Equipment Racks
5. Equipment Cabinets
6. Conduits
7. Cable Duct blocks
8. Cable Trays
9. Power Panels
10. Grounding
11. Connector Panels

(3) Coaxial Cable: The coaxial cable kit shall include all coaxial connectors, cable tying straps, heat shrink tabbing, hangers, clamps, etc., required to accomplish a neat and secure installation.

(4) Wire and Cable: The wire and cable kit shall include all connectors and terminals, audio spade lugs, barrier straps, punch blocks, wire wrap strips, heat shrink tubing, tie wraps, solder, hangers, clamps, labels etc., required to accomplish a neat and orderly installation.

(5) Conduit, Cable Duct, and Cable Tray: The kit shall include all conduit, duct, trays, junction boxes, back boxes, cover plates, feed through nipples, hangers, clamps, other hardware required to accomplish a neat and secure conduit, cable duct, and/or cable tray installation in accordance with NFPA 70 and this document.

(6) Equipment Interface: The equipment kit shall include any item or quantity of equipment, cable, mounting hardware and materials needed to interface the systems with the identified sub-system(s) according to the OEM requirements and this document.

(7) Labels: The labeling kit shall include any item or quantity of

labels, tools, stencils, and materials needed to label each subsystem according to the OEM requirements, as-installed drawings, and this document.

(8) Documentation: The documentation kit shall include any item or quantity of items, computer discs, as installed drawings, equipment, maintenance, and operation manuals, and OEM materials needed to provide the system documentation as required by this document and explained herein.

PART 3 – EXECUTION

3.1 COMMON REQUIREMENTS FOR ELECTRONIC SAFETY AND SECURITY INSTALLATION

a. Comply with NECA NEIS 1.

b. Measure indicated mounting heights to bottom of unit for suspended items and to center of unit for wall-mounting items.

c. Headroom Maintenance: If mounting heights or other location criteria are not indicated, arrange and install components and equipment to provide maximum possible headroom consistent with these requirements.

d. Equipment: Install to facilitate service, maintenance, and repair or replacement of components of both electronic safety and security equipment and other nearby installations. Connect in such a way as to facilitate future disconnecting with minimum interference with other items in the vicinity.

e. Right of Way: Give to piping systems installed at a required slope.

f. Equipment location shall be as close as practical to locations shown on the drawings.

g. Inaccessible Equipment:

(1) Where the Government determines that the Contractor has installed equipment not conveniently accessible for operation and maintenance, the equipment shall be removed and reinstalled as directed at no additional cost to the Government.

(2) "Conveniently accessible" is defined as being capable of being reached without the use of ladders, or without climbing or crawling under or over obstacles such as, but not limited to, motors, pumps, belt guards, transformers, piping, ductwork, conduit and raceways.

3.2 FIRESTOPPING

Apply firestopping to penetrations of fire-rated floor and wall assemblies for electronic safety and security installations to restore original fire-resistance rating of assembly. Firestopping materials and installation requirements are specified in Section 07 84 00, FIRESTOPPING.

3.3 COMMISIONING

a. Provide commissioning documentation in accordance with the requirements of Section 28 08 00, COMMISIONIN OF ELECTRONIC SAFETY AND SECURITY SYSTEMS for all inspection, start up, and contractor testing

required above and required by the System Readiness Checklist provided by the Commissioning Agent.

b. Components provided under this section of the specification will be tested as part of a larger system. Refer to Section 28 08 00, COMMISIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS and related sections for contractor responsibilities for system commissioning.

3.4 DEMONSTRATION AND TRAINING

a. Training shall be provided in accordance with Article, INSTRUCTIONS, of Section 01 00 00, GENERAL REQUIREMENTS.

b. Training shall be provided for the particular equipment or system as required in each associated specification.

c. A training schedule shall be developed and submitted by the contractor and approved by the Resident Engineer at least 30 days prior to the planned training.

d. Provide services of manufacturer’s technical representative for <insert hours> hours to instruct VA personnel in operation and maintenance of units.

e. Submit training plans and instructor qualifications in accordance with the requirements of Section 28 08 00, COMMISIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS.

3.5 WORK PERFORMANCE

a. Job site safety and worker safety is the responsibility of the contractor.

b. For work on existing stations, arrange, phase and perform work to assure electronic safety and security service for other buildings at all times. Refer to Article OPERATIONS AND STORAGE AREAS under Section 01 00 00, GENERAL REQUIREMENTS.

c. New work shall be installed and connected to existing work neatly and carefully. Disturbed or damaged work shall be replaced or repaired to its prior conditions, as required by Section 01 00 00, GENERAL REQUIREMENTS.

d. Coordinate location of equipment and conduit with other trades to minimize interferences. See the GENERAL CONDITIONS.

3.6 SYSTEM PROGRAMMING

a. General Programming Requirements

(1) This following section shall be used by the contractor to identify the anticipated level of effort (LOE) required setup, program, and configure the Electronic Security System (ESS). The contractor shall be responsible for providing all setup, configuration, and programming to include data entry for the Security Management System (SMS) and subsystems (e.g., video matrix switch, intercoms, digital video recorders, intrusion devices, including integration of subsystems to the SMS (e.g., camera call up, time synchronization, intercoms). System programming for existing or

new SMS servers shall not be conducted at the project site.

b. Level of Effort for Programming

(1) The Contractor shall perform and complete system programming (including all data entry) at an offsite location using the Contractor's own copy of the SMS software. The Contractor's copy of the SMS software shall be of the Owners current version. Once system programming has been completed, the Contractor shall deliver the data to the Resident Engineer on data entry forms and an approved electronic medium, utilizing data from the contract documents. The completed forms shall be delivered to the Resident Engineer for review and approval at least 90 calendar days prior to the scheduled date the Contractor requires it. The Contractor shall not upload system programming until the Resident Engineer has provided written approval. The Contractor is responsible for backing up the system prior to uploading new programming data. Additional programming requirements are provided as follows:

(a) Programming for New SMS Server: The contractor shall provide all other system related programming. The contractor will be responsible for uploading personnel information (e.g., ID Cards backgrounds, names, access privileges, personnel photos, access schedules, personnel groupings) along with coordinating with Resident Engineer for device configurations, standards, and groupings. VA shall provide database to support Contractor's data entry tasks. The contractor shall anticipate a weekly coordination meeting and working with Resident Engineer to ensure data uploading is performed without incident of loss of function or data loss.

(b) Programming for Existing SMS Servers: The contractor shall perform all related system programming except for personnel data as noted. The contractor will not be responsible for uploading personnel information (e.g., ID Cards backgrounds, names, access privileges, access schedules, personnel groupings). The contractor shall anticipate a weekly coordination meeting and working alongside of Resident Engineer to ensure data uploading is performed without incident of loss of function or data loss. System programming for SMS servers shall be performed by using the Contractor's own server and software. These servers shall not be connected to existing devices or systems at any time.

(2) The Contractor shall identify and request from the Resident Engineer, any additional data needed to provide a complete and operational system as described in the contract documents.

(3) Contractor and Resident Engineer coordination on programming requires a high level of coordination to ensure programming is performed in accordance with VA requirements and programming uploads do not disrupt existing systems functionality. The contractor shall anticipate a minimum a weekly coordination meeting. Contractor shall ensure data uploading is performed without incident of loss of function or data loss. The following Level of Effort Chart is provided to communicate the expected level of effort required by contractors on VA ESS projects. Calculations to determine actual levels of effort shall be confirmed by the contractor before project award.

	Description of Tasks						
Description of Systems	Develop System Loading Sheets	Coordinatio	Initial Set-up onfiguratic	Graphic Maps	System Programming	Final Checks	Level of Effort (Typical Tasks)
SMS Setup and Configurat	e.g., program monitoring stations, programming networks, erconnecti between CCTV,	e.g., retrieve IP addresses, naming conventions standard event escriptions	e.g., Load system Operating System and Application software, general	e.g., develop naming onventions develop file folders, confirming accuracy of AutoCAD Floor Plans, convert file into jpeg file	e.g., program monitoring stations, programming networks, erconnecti between CCTV,	e.g., check all system diagnostics (e.g., clients, panels)	Load and set-up 4-6 CDs and configure servers (to configure Loading
Electronic Entry Control Systems	e.g., setup of device, door groups and schedules, REX, Locks,	e.g., confirming device configurat naming convention event descriptio and	e.g., enter data from loading sheets; configure components link events,		e.g., setup of device, door groups and schedules, REX, Locks,	e.g., performing entry testing to confirm correct set-up and	e.g., creating a door, door configurati adding request to exit, door
Intrusion Detection Systems	e.g., enter door groups and schedules, link devices - REX,	e.g., confirming device configurat naming convention event descriptio and	e.g., enter data from loading sheets; configure components link events,		e.g., enter door groups and schedules, link devices - REX,	e.g., walk test, device position, and masking	e.g., setting up monitoring and control points (e.g., motion
CCTV Systems	e.g., programmin call-ups recording	e.g., confirming device configurat naming convention	e.g., enter data from loading sheets; camera naming convention sequences,		e.g., programmin call-ups recording	e.g., confirm area of coverage, call-up per event generated and recording	e.g., setting up cameras points, recording ratios (e.g., normal,
Intercoms Systems	e.g., programmin events and call-ups	e.g., confirming device configurat naming convention event descriptio and	e.g., enter data from loading sheets; configure components link events,		e.g., programmin events and call-ups	e.g., confirm operation, SMS event generation and camera call-up	e.g., setup linkages, events for activations device troubles, land

Console Monitoring Components	N/A	per monitor	per monitor	per graphic map	N/A	per monitor	N/A
Note: Programming tasks are supported through the contractor's development of the Technical Data Package Submittals.							

Table 1 Contractor Level of Effort

3.7 TESTING AND ACCEPTANCE

a. Performance Requirements

(1) General:

(a) The Contractor shall perform contract field, performance verification, and endurance testing and make adjustments of the completed security system when permitted. The Contractor shall provide all personnel, equipment, instrumentation, and supplies necessary to perform all testing. Written notification of planned testing shall be given to the Resident Engineer at least 60 calendar days prior to the test and after the Contractor has received written approval of the specific test procedures.

(b) The COTR shall witness all testing and system adjustments during testing. Written permission shall be obtained from the Resident Engineer before proceeding with the next phase of testing. Original copies of all data produced during performance verification and endurance testing shall be turned over to the Resident Engineer at the conclusion of each phase of testing and prior to Resident Engineer approval of the test.

(2) Test Procedures and Reports: The test procedures, compliant with VA standard test procedures, shall explain in detail, step-by-step actions and expected results demonstrating compliance with the requirements of the specification. The test reports shall be used to document results of the tests. The reports shall be delivered to the Resident Engineer within seven calendar days after completion of each test.

b. Pre-Delivery Testing

(1) The purpose of the pre-delivery test is to establish that a system is suitable for installation. As such, pre-delivery test shall be a mock-up of the system as planned in the contract documents. The Contractor shall assemble the Security Test System at the Contractors local project within 50-miles of the project site, and perform tests to demonstrate the performance of the system complies with the contract requirements in accordance with the

approved pre-delivery test procedures. The tests shall take place during regular daytime working hours on weekdays. Model numbers of equipment tested shall be identical to those to be delivered to the site. Original copies of all data produced during pre-delivery testing, including results of each test procedure, shall be documented and delivered to the Resident Engineer at the conclusion of pre-delivery testing and prior to Resident Engineer's approval of the test. The test report shall be arranged so all commands, stimuli, and responses are correlated to allow logical interpretation. For Existing System modifications, the contractor shall provide their own server with loaded applicable software to support PDT.

(2) Test Setup: The pre-delivery test setup shall include the following:

(a) All console equipment.

1. At least one of each type of data transmission media (DTM) and associated equipment to provide a fully integrated PACS.

2. The number of local processors shall equal the amount required by the site design.

3. Enough sensor simulators to provide alarm signal inputs to the system equal to the number of sensors required by the design. The alarm signals shall be manually or software generated.

4. Contractor to prove to owner all systems are appropriately sized and configured as sized.

5. Integration of VASS, intercom systems, other subsystems.

(3) During the bidding process the contractor shall submit a request for information to the Owner to determine if a pre-delivery test will be required. If a pre-delivery test is not required, the contractor shall provide a written notification that the Pre-delivery Test is not required in their shop drawings submission.

c. The inspection and test will be conducted by a factory-certified contractor representative and witnessed by a Government Representative. The results of the inspection will be officially recorded by a designated Government Representative and maintained on file by the Resident Engineer (RE), until completion of the entire project. The results will be compared to the Acceptance Test results.

d. Contractor's Field Testing (CFT)

(1) The Contractor shall calibrate and test all equipment, verify DTM operation, place the integrated system in service, and test the integrated system. Ground rods installed by this Contractor within the base of camera poles shall be tested as specified in IEEE 142. The Contractor shall test all security systems and equipment, and provide written proof of a 100 percent operational system before a date is established for the system acceptance test. Documentation package for CFT shall include completed (fully annotated details of test details) for each device and system tested, and annotated loading sheets documenting complete

testing to Resident Engineer approval. CFT test documentation package shall conform to submittal requirements outlined in this Section. The Contractor’s field testing procedures shall be identical to the Resident Engineer’s acceptance testing procedures. The Contractor shall provide the Resident Engineer with a written listing of all equipment and software indicating all equipment and components have been tested and passed. The Contractor shall deliver a written report to theResident Engineer stating the installed complete system has been calibrated, tested, and is ready to begin performance verification testing; describing the results of the functional tests, diagnostics, and calibrations; and the report shall also include a copy of the approved acceptance test procedure. Performance verification testing shall not take place until written notice by contractor is received certifying that a contractors field test was successful.

e. Performance Verification Test (PVT)

(1) Test team:

(a) After the system has been pretested and the Contractor has submitted the pretest results and certification to the Resident Engineer, then the Contractor shall schedule an acceptance test to date and give the Resident Engineer written, notice as described herein, prior to the date the acceptance test is expected to begin. The system shall be tested in the presence of a Government Representative, an OEM certified representative, representative of the Contractor and other approved by the Resident Engineer. The system shall be tested utilizing the approved test equipment to certify proof of performance, FCC, UL and Emergency Service compliance. The test shall verify that the total system meets all the requirements of this specification. The notification of the acceptance test shall include the expected length (in time) of the test.

(2) The Contractor shall demonstrate the completed Physical Access Control System PACS complies with the contract requirements. In addition, the Contractor shall provide written certification that the system is 100 percent operational prior to establishing a date for starting PVT. Using approved test procedures, all physical and functional requirements of the project shall be demonstrated and shown. The PVT will be stopped and aborted as soon as 10 technical deficiencies are found requiring correction. The Contractor shall be responsible for all travel and lodging expenses incurred for out-of-town personnel required to be present for resumption of the PVT. If the acceptance test is aborted, the re-test will commence from the beginning with a retest of components previously tested and accepted.

(3) The PVT, as specified, shall not begin until receipt of written certification that the Contractors Field Testing was successful. This shall include certification of successful completion of testing as specified in paragraph “Contractor’s Field Testing”, and upon successful completion of testing at any time when the system fails to perform as specified. Upon termination of testing by the Resident Engineer or Contractor, the Contractor shall commence an assessment period as described for Endurance Testing Phase II.

(4) Upon successful completion of the acceptance test, the Contractor shall deliver test reports and other documentation, as specified, to the Resident Engineer prior to commencing the endurance test.

(5) Additional Components of the PVT shall include:

(a) System Inventory

1. All Device equipment

2. All Software

3. All Logon and Passwords

4. All Cabling System Matrices

5. All Cable Testing Documents

6. All System and Cabinet Keys

(b) Inspection

1. Contractor shall record an inspection punch list noting all system deficiencies. The contractor shall prepare an inspection punch list format for Resident Engineers approval.

2. As a minimum the punch list shall include a listing of punch list items, punch list item location, description of item problem, date noted, date corrected, and details of how item was corrected.

(6) Partial PVT - At the discretion of Resident engineer, the Performance Verification Test may be performed in part should a 100 percent compliant CFT be performed. In the event that a partial PVT will be performed instead of a complete PVT; the partial PVT shall be performed by testing 10 percent of the system. The contractor shall perform a test of each procedure on select devices or equipment.

f. Endurance Test

(1) The Contractor shall demonstrate the specified probability of detection and false alarm rate requirements of the completed system. The endurance test shall be conducted in phases as specified below. The endurance test shall not be started until the Resident Engineer notifies the Contractor, in writing, that the performance verification test is satisfactorily completed, training as specified has been completed, and correction of all outstanding deficiencies has been satisfactorily completed. VA shall operate the system 24 hours per day, including weekends and holidays, during Phase I and Phase III endurance testing. VA will maintain a log of all system deficiencies. The Resident Engineer may terminate testing at any time the system fails to perform as specified. Upon termination of testing, the Contractor shall commence an assessment period as described for Phase II. During the last day of the test, the Contractor shall verify the appropriate operation of the system. Upon successful completion of the endurance test, the Contractor shall deliver test reports and other documentation as specified to the Resident Engineer

prior to acceptance of the system.

(2) Phase I (Testing): The test shall be conducted 24 hours per day for 15 consecutive calendar days, including holidays, and the system shall operate as specified. The Contractor shall make no repairs during this phase of testing unless authorized in writing by the Resident Engineer. If the system experiences no failures, the Contractor may proceed directly to Phase III testing after receiving written permission from the Resident Engineer.

(3) Phase II (Assessment):

(a) After the conclusion of Phase I, the Contractor shall identify all failures, determine causes of all failures, repair all failures, and deliver a written report to the Resident Engineer. The report shall explain in detail the nature of each failure, corrective action taken, results of tests performed, and recommend the point at which testing should be resumed.

(b) After delivering the written report, the Contractor shall convene a test review meeting at the job site to present the results and recommendations to the Resident Engineer. The meeting shall not be scheduled earlier than five business days after the Resident Engineer receives the report. As part of this test review meeting, the Contractor shall demonstrate all failures have been corrected by performing appropriate portions of the performance verification test. Based on the Contractor’s report and the test review meeting, the Resident Engineer will provide a written determine of either the restart date or require Phase I be repeated.

(4) Phase III (Testing): The test shall be conducted 24 hours per day for 15 consecutive calendar days, including holidays, and the system shall operate as specified. The Contractor shall make no repairs during this phase of testing unless authorized in writing by the COTR.

(5) Phase IV (Assessment):

(1) After the conclusion of Phase III, the Contractor shall identify all failures, determine causes of all failures, repair all failures, and deliver a written report to the COTR. The report shall explain in detail the nature of each failure, corrective action taken, results of tests performed, and recommend the point at which testing should be resumed.

(2) After delivering the written report, the Contractor shall convene a test review meeting at the job site to present the results and recommendations to the COTR. The meeting shall not be scheduled earlier than five business days after receipt of the report by the COTR. As a part of this test review meeting, the Contractor shall demonstrate that all failures have been corrected by repeating appropriate portions for the performance verification test. Based on the review meeting the test should not be scheduled earlier than five business days after the Resident Engineer receives the report. As a part of this test review meeting, the Contractor shall demonstrate all failures have been corrected by repeating appropriate portions of the performance verification test. Based on the Contractor’s report and the test review meeting, the

Resident Engineer will provide a written determine of either the restart date or require Phase III be repeated. After the conclusion of any re-testing which the Resident Engineer may require, the Phase IV assessment shall be repeated as if Phase III had just been completed.

g. Exclusions: The Contractor will not be held responsible for failures in system performance resulting from the following:

(1) An outage of the main power in excess of the capability of any backup power source provided the automatic initiation of all backup sources was accomplished and that automatic shutdown and restart of the PACS performed as specified.

(2) Failure of an Owner furnished equipment or communications link, provided the failure was not due to Contractor furnished equipment, installation, or software.

(3) Failure of existing Owner owned equipment, provided the failure was not due to Contractor furnished equipment, installation, or software.

-- End of Section --

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SECTION 28 05 13
10/18

CONDUCTORS AND CABLES FOR ELECTRONIC SAFETY AND SECURITY

PART 1 GENERAL

1.1 DESCRIPTION

This Section specifies the finishing, installation, connection, testing and certification the conductors and cables required for a fully functional for electronic safety and security (ESS) system.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS. For General Requirements.

b. Section 07 84 00, FIRESTOPPING. Requirements for firestopping application and use.

c. Section 28 05 00, COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY. Requirements for general requirements that are common to more than one section in Division 28.

d. Section 28 05 26, GROUNDING AND BONDING FOR ELECTRONIC SAFETY AND SECURITY. Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

e. Section 28 05 28.33, CONDUITS AND BOXES FOR ELECTRONIC SECURITY AND SAFETY. Requirements for infrastructure.

f. Section 28 08 00, COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS. Requirements for commissioning.

g. Section 31 20 00, EARTH MOVING. For excavation and backfill for cables that are installed in conduit.

1.3 DEFINITIONS

a. BICSI: Building Industry Consulting Service International.

b. EMI: Electromagnetic interference.

c. IDC: Insulation displacement connector.

d. Ladder Cable Tray: A fabricated structure consisting of two longitudinal side rails connected by individual transverse members (rungs).

e. Low Voltage: As defined in NFPA 70 for circuits and equipment operating at less than 50 V or for remote-control and signaling power-limited circuits.

f. Open Cabling: Passing telecommunications cabling through open space (e.g., between the studs of a wall cavity).

g. RCDD: Registered Communications Distribution Designer.

h. Solid-Bottom or Nonventilated Cable Tray: A fabricated structure consisting of integral or separate longitudinal side rails, and a bottom without ventilation openings.

i. Trough or Ventilated Cable Tray: A fabricated structure consisting of integral or separate longitudinal rails and a bottom having openings sufficient for the passage of air and using 75 percent or less of the plan area of the surface to support cables.

j. UTP: Unshielded twisted pair.

1.4 QUALITY ASSURANCE

See Section 28 05 00, COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY Paragraph 1.4.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G

Wiring Diagrams; G

Cable Administration Drawings; G

SD-03 Product Data

Manufacturer's Literature And Data; G

SD-05 Design Data

Project Planning Documents ; G

SD-07 Certificates

Certificates; G

SD-10 Operation and Maintenance Data

Maintenance Data; G

1.6 SUBMITTAL INSTRUCTIONS

a. In accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES, furnish the following:

(1) Manufacturer's Literature and Data: Showing each cable type and rating.

(2) Certificates: Two weeks prior to final inspection, deliver to the Resident Engineer/Contracting Officer four copies of the certification that the material is in accordance with the drawings

and specifications and diagrams for cable management system.

(3) Shop Drawings: Cable tray layout, showing cable tray route to scale, with relationship between the tray and adjacent structural, electrical, and mechanical elements. Include the following:

(a) Vertical and horizontal offsets and transitions.

(b) Clearances for access above and to side of cable trays.

(c) Vertical elevation of cable trays above the floor or bottom of ceiling structure.

(d) Load calculations to show dead and live loads as not exceeding manufacturer's rating for tray and its support elements.

(e) System labeling schedules, including electronic copy of labeling schedules that are part of the cable and asset identification system of the software specified in Parts 2 and 3.

(4) Wiring Diagrams. Show typical wiring schematics including the following:

(a) Workstation outlets, jacks, and jack assemblies.

(b) Patch cords.

(c) Patch panels.

(5) Cable Administration Drawings: As specified in Part 3 "Identification" Article.

(6) Project planning documents as specified in Part 3.

(7) Maintenance Data: For wire and cable to include in maintenance manuals.

1.7 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements and errata) form a part of this specification to the extent referenced. Publications are reference in the text by the basic designation only.

BICSI International Standards Program (BICSI)

BICSI TDMM	(2024; 15th Edition) Telecommunications Distribution Methods Manual

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

ASTM INTERNATIONAL (ASTM)

ASTM B633	(2023) Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)

ICEA S-83-596	(2021) Indoor Optical Cable
ICEA S-90-661	(2021) Category 3 and 5E Individually Unshielded Twisted Pairs, Indoor Cables (With or Without an Overall Shield) for Use in General Purpose and LAN Communications Wiring Systems

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)

NECA NEIS 1	(2015) Standard for Good Workmanship in Electrical Construction
NECA/BICSI 568	(2006) Standard for Installing Building Telecommunications Cabling

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 72	(2022; ERTA 22-1) National Fire Alarm and Signaling Code
NFPA 262	(2023) Standard Method of Test for Flame Travel and Smoke of Wires and Cables for Use in Air-Handling Spaces

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-492AAAA	(2009b) 62.5-um Core Diameter/125-um Cladding Diameter Class 1a Graded-Index Multimode Optical Fibers
TIA-526-14	(2023d) OFSTP-14A Optical Power Loss Measurements of Installed Multimode Fiber Cable Plant
TIA-568.1	(2020e) Commercial Building Telecommunications Infrastructure Standard
TIA-568.2	(2018d) Balanced Twisted-Pair Telecommunications Cabling and Components Standards
TIA-568.3	(2022e) Optical Fiber Cabling Components Standard
TIA-569	(2019e; Add 1 2022) Telecommunications Pathways and Spaces
TIA-598-B	Optical Fiber Cable Color Coding
TIA-604-2	(2004) FOCIS-2 Fiber Optic Connector Intermateability Standard, Type ST

TIA-604-3	(2004; Reaffirmed 2014) FOCIS 3 Fiber Optic Connector Intermateability Standard - Typse SC and SC-APC
TIA-604-12	(2000) FOCIS 12 Fiber Optic Connector Intermateability Standard Type MT-RJ
TIA-606	(2021d) Administration Standard for Telecommunications Infrastructure
TIA-607	(2019d) Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises

UL SOLUTIONS (UL)

UL 83	(2017; Reprint Mar 2020) UL Standard for Safety Thermoplastic-Insulated Wires and Cables
UL 444	(2023) UL Standard for Safety Communications Cables
UL 969	(2017; Reprint May 2023) UL Standard for Safety Marking and Labeling Systems
UL 1424	(2015) Cables for Power-Limited Fire-Alarm Circuits
UL 1581	(2019; Reprint Nov 2023) UL Standard for Safety Reference Standard for Electrical Wires, Cables, and Flexible Cords
UL 1651	(2015) Optical Fiber Cable
UL 1655	(2009) Standard for Community-Antenna Television Cables
UL 1666	(2007; Reprint Sep 2021) UL Standard for Safety Test for Flame Propagation Height of Electrical and Optical-Fiber Cables Installed Vertically in Shafts
UL 2196	(2017) Fire Test for Circuit Integrity of Fire-Resistive Power, Instrumentation, Control and Data Cables

1.8 DELIVERY, STORAGE, AND HANDLING

a. Test cables upon receipt at Project site.

(1) Test optical fiber cable to determine the continuity of the strand end to end. Use optical loss test set.

(2) Test optical fiber cable on reels. Use an optical time domain reflectometer to verify the cable length and locate cable defects, splices, and connector; include the loss value of each. Retain test data and include the record in maintenance data.

(3) Test each pair of UTP cable for open and short circuits.

1.9 PROJECT CONDITIONS

Do not deliver or install UTP, optical fiber, and coaxial cables and connecting materials until wet work in spaces is complete and dry, and temporary HVAC system is operating and maintaining ambient temperature and humidity conditions at occupancy levels during the remainder of the construction period.

PART 2 PRODUCTS

2.1 GENERAL

a. Support of Open Cabling: NRTL labeled for support of Category 6A cabling, designed to prevent degradation of cable performance and pinch points that could damage cable.

(1) Support brackets with cable tie slots for fastening cable ties to brackets.

(2) Lacing bars and spools.

(3) Straps and other devices.

b. Cable Trays:

(1) Cable Tray Materials: Metal, suitable for indoors, and protected against corrosion by electroplated zinc galvanizing, complying with ASTM B633, Type 1, not less than 0.000472 inch thick.

(2) Basket Cable Trays: 12 inches wide and 4 inches deep. Wire mesh spacing shall not exceed 2 by 4 inches.

c. Conduit and Boxes: Comply with requirements in Section 28 05 28.33, CONDUITS AND BACKBOXES FOR ELECTRICAL SYSTEMS. Flexible metal conduit shall not be used. Outlet boxes shall be no smaller than 2 inches wide, 3 inches high, and 2-1/2 inches deep.

2.2 BACKBOARDS

Plywood, fire-retardant treated, 3/4 by 48 by 96 inches . Comply with requirements for plywood backing panels in Section 06 10 00, ROUGH CARPENTRY.

2.3 UTP CABLE

a. Description: 100-ohm, 4-pair UTP, formed into 25-pair binder groups covered with a blue thermoplastic jacket.

(1) Comply with ICEA S-90-661 for mechanical properties.

(2) Comply with TIA-568.1 for performance specifications.

(3) Comply with TIA-568.2, Category 6A.

(4) Listed and labeled by an NRTL acceptable to authorities having jurisdiction as complying with UL 444 and NFPA 70 for the following types:

(a) Communications, General Purpose: Type CM or CMG.

(b) Communications, Plenum Rated: Type CMP, complying with NFPA 262.

(c) Communications, Riser Rated: Type CMR, complying with UL 1666.

(d) Communications, Limited Purpose: Type CMX.

(e) Multipurpose: Type MP or MPG.

(f) Multipurpose, Plenum Rated: Type MPP, complying with NFPA 262.

(g) Multipurpose, Riser Rated: Type MPR, complying with UL 1666.

2.4 UTP CABLE HARDWARE

a. UTP Cable Connecting Hardware: IDC type, using modules designed for punch-down caps or tools. Cables shall be terminated with connecting hardware of the same category or higher.

b. Connecting Blocks: 110-style for Category 6A. Provide blocks for the number of cables terminated on the block, plus 25 percent spare. Integral with connector bodies, including plugs and jacks where indicated.

2.5 OPTICAL FIBER CABLE

a. Description: Multimode, 50/125-micrometer, 24-fiber, nonconductive, tight buffer, optical fiber cable.

(1) Comply with ICEA S-83-596 for mechanical properties.

(2) Comply with TIA-568.3 for performance specifications.

(3) Comply with TIA-492AAAA for detailed specifications.

(4) Listed and labeled by an NRTL acceptable to authorities having jurisdiction as complying with UL 444, UL 1651, and NFPA 70 for the following types:

(a) General Purpose, Nonconductive: Type OFN or OFNG.

(b) Plenum Rated, Nonconductive: Type OFNP, complying with NFPA 262.

(c) Riser Rated, Nonconductive: Type OFNR, complying with UL 1666.

(d) General Purpose, Conductive: Type OFC or OFCG.

(e) Plenum Rated, Conductive: Type OFCP, complying with NFPA 262.

(f) Riser Rated, Conductive: Type OFCR, complying with UL 1666.

(5) Conductive cable shall be steel armored type.

(6) Maximum Attenuation: 3.50 dB/km at 850 nm; 1.5 dB/km at 1300 nm.

(7) Minimum Modal Bandwidth: 160 MHz-km at 850 nm; 500 MHz-km at 1300 nm.

b. Jacket:

(1) Jacket Color: Aqua for 50/125-micrometer cable.

(2) Cable cordage jacket, fiber, unit, and group color shall be according to TIA-598-B.

(3) Imprinted with fiber count, fiber type, and aggregate length at regular intervals not to exceed 40 inches .

2.6 OPTICAL FIBER CABLE HARDWARE

Cable Connecting Hardware: Meet the Optical Fiber Connector Intermateability Standards (FOCIS) specifications of TIA-604-2, TIA-604-3, and TIA-604-12. Comply with TIA-568.3.

(1) Quick-connect, simplex and duplex, Type LC connectors. Insertion loss shall be not more than 0.75 dB.

(2) Type SFF connectors may be used in termination racks, panels, and equipment packages.

2.7 COAXIAL CABLE

a. General Coaxial Cable Requirements: Broadband type, recommended by cable manufacturer specifically for broadband data transmission applications. Coaxial cable and accessories shall have 75-ohm nominal impedance with a return loss of 20 dB maximum from 7 to 806 MHz.

b. RG-11/U: NFPA 70, Type CATV.

(1) No. 14 AWG, solid, copper-covered steel conductor.

(2) Gas-injected, foam-PE insulation.

(3) Double shielded with 100 percent aluminum polyester tape and 60 percent aluminum braid.

(4) Jacketed with sunlight-resistant, black PVC or PE.

(5) Suitable for outdoor installations in ambient temperatures ranging from minus 40 to plus 85 deg C.

c. RG59/U: NFPA 70, Type CATVR.

(1) No. 20 AWG, solid, silver-plated, copper-covered steel conductor.

(2) Gas-injected, foam-PE insulation.

(3) Triple shielded with 100 percent aluminum polyester tape and 95 percent aluminum braid; covered by aluminum foil with grounding strip.

(4) Color-coded PVC jacket.

d. RG-6/U: NFPA 70, Type CATV or CM.

(1) No. 16 AWG, solid, copper-covered steel conductor; gas-injected, foam-PE insulation.

(2) Double shielded with 100 percent aluminum-foil shield and 60 percent aluminum braid.

(3) Jacketed with black PVC or PE.

(4) Suitable for indoor installations.

e. RG59/U: NFPA 70, Type CATV.

(1) No. 20 AWG, solid, copper-covered steel conductor; gas-injected, foam-PE insulation.

(2) Double shielded with 100 percent aluminum polyester tape and 40 percent aluminum braid.

(3) PVC jacket.

f. RG59/U (Plenum Rated): NFPA 70, Type CMP.

(1) No. 20 AWG, solid, copper-covered steel conductor; foam fluorinated ethylene propylene insulation.

(2) Double shielded with 100 percent aluminum-foil shield and 65 percent aluminum braid.

(3) Copolymer jacket.

g. NFPA and UL compliance, listed and labeled by an NRTL acceptable to authorities having jurisdiction as complying with UL 1655, and with NFPA 70 "Radio and Television Equipment" and "Community Antenna Television and Radio Distribution" Articles. Types are as follows:

(1) CATV Cable: Type CATV.

(2) CATV Plenum Rated: Type CATVP, complying with NFPA 262.

(3) CATV Riser Rated: Type CATVR, complying with UL 1666.

(4) CATV Limited Rating: Type CATVX.

2.8 COAXIAL CABLE HARDWARE

Type BNC, 75 ohms.

2.9 RS-232 CABLE

a. Standard Cable: NFPA 70, Type CM.

(1) Paired, 2 pairs, No. 22 AWG, stranded (7x30) tinned copper conductors.

(2) Polypropylene insulation.

(3) Individual aluminum foil-polyester tape shielded pairs with 100 percent shield coverage.

(4) PVC jacket.

(5) Pairs are cabled on common axis with No. 24 AWG, stranded (7x32) tinned copper drain wire.

(6) Flame Resistance: Comply with UL 1581.

b. Plenum-Rated Cable: NFPA 70, Type CMP.

(1) Paired, 2 pairs, No. 22 AWG, stranded (7x30) tinned copper conductors.

(2) Plastic insulation.

(3) Individual aluminum foil-polyester tape shielded pairs with 100 percent shield coverage.

(4) Plastic jacket.

(5) Pairs are cabled on common axis with No. 24 AWG, stranded (7x32) tinned copper drain wire.

(6) Flame Resistance: Comply with NFPA 262.

2.10 RS-485 CABLE

a. Standard Cable: NFPA 70, Type CM.

(1) Paired, 2 pairs, twisted, No. 22 AWG, stranded (7x30) tinned copper conductors.

(2) PVC insulation.

(3) Unshielded.

(4) PVC jacket.

(5) Flame Resistance: Comply with UL 1581.

b. Plenum-Rated Cable: NFPA 70, Type CMP.

(1) Paired, 2 pairs, No. 22 AWG, stranded (7x30) tinned copper conductors.

(2) Fluorinated ethylene propylene insulation.

(3) Unshielded.

(4) Fluorinated ethylene propylene jacket.

(5) Flame Resistance: NFPA 262, Flame Test.

2.11 LOW-VOLTAGE CONTROL CABLE

a. Paired Lock Cable: NFPA 70, Type CMG.

(1) 1 pair, twisted, No. 16 AWG, stranded (19x29) tinned copper conductors.

(2) PVC insulation.

(3) Unshielded.

(4) PVC jacket.

(5) Flame Resistance: Comply with UL 1581.

b. Plenum-Rated, Paired Lock Cable: NFPA 70, Type CMP.

(1) 1 pair, twisted, No. 16 AWG, stranded (19x29) tinned copper conductors.

(2) PVC insulation.

(3) Unshielded.

(4) PVC jacket.

(5) Flame Resistance: Comply with NFPA 262.

c. Paired Lock Cable: NFPA 70, Type CMG.

(1) 1 pair, twisted, No. 18 AWG, stranded (19x30) tinned copper conductors.

(2) PVC insulation.

(3) Unshielded.

(4) PVC jacket.

(5) Flame Resistance: Comply with UL 1581.

d. Plenum-Rated, Paired Lock Cable: NFPA 70, Type CMP.

(1) 1 pair, twisted, No. 18 AWG, stranded (19x30) tinned copper conductors.

(2) Fluorinated ethylene propylene insulation.

(3) Unshielded.

(4) Plastic jacket.

(5) Flame Resistance: NFPA 262, Flame Test.

2.12 CONTROL-CIRCUIT CONDUCTORS

a. Class 1 Control Circuits: Stranded copper, Type THHN-THWN, in raceway complying with UL 83.

b. Class 2 Control Circuits: Stranded copper, Type THHN-THWN, in raceway complying with UL 83.

c. Class 3 Remote-Control and Signal Circuits: Stranded copper, Type TW or TF, complying with UL 83.

2.13 FIRE ALARM WIRE AND CABLE

a. General Wire and Cable Requirements: NRTL listed and labeled as complying with NFPA 70, Article 760.

b. Signaling Line Circuits: Twisted, shielded pair,No. 18 AWG.

Circuit Integrity Cable: Twisted shielded pair, NFPA 70, Article 760, Classification CI, for power-limited fire alarm signal service Type FPL. NRTL listed and labeled as complying with UL 1424 and UL 2196 for a 2-hour rating.

c. Non-Power-Limited Circuits: Solid-copper conductors with 600-V rated, 75 deg C, color-coded insulation.

(1) Low-Voltage Circuits: No. 16 AWG, minimum.

(2) Line-Voltage Circuits: No. 12 AWG, minimum.

(3) Multiconductor Armored Cable: NFPA 70, Type MC, copper conductors, Type TFN/THHN conductor insulation, copper drain wire, copper armor with outer jacket with red identifier stripe, NTRL listed for fire alarm and cable tray installation, plenum rated, and complying with requirements in UL 2196 for a 2-hour rating.

2.14 IDENTIFICATION PRODUCTS

Comply with UL 969 for a system of labeling materials, including label stocks, laminating adhesives, and inks used by label printers.

2.15 SOURCE QUALITY CONTROL

a. Testing Agency: Engage a qualified testing agency to evaluate cables.

b. Factory test UTP and optical fiber cables on reels according to TIA-568.1.

c. Factory test UTP cables according to TIA-568.2.

d. Factory test multimode optical fiber cables according to TIA-526-14 and TIA-568.3.

e. Factory sweep test coaxial cables at frequencies from 5 MHz to 1 GHz. Sweep test shall test the frequency response, or attenuation over frequency, of a cable by generating a voltage whose frequency is varied through the specified frequency range and graphing the results.

f. Cable will be considered defective if it does not pass tests and inspections.

g. Prepare test and inspection reports.

2.16 WIRE LUBRICATING COMPOUND

a. Suitable for the wire insulation and conduit it is used with, and shall not harden or become adhesive.

b. Shall not be used on wire for isolated type electrical power systems.

2.17 FIREPROOFING TAPE

a. The tape shall consist of a flexible, conformable fabric of organic composition coated one side with flame-retardant elastomer.

b. The tape shall be self-extinguishing and shall not support combustion. It shall be arc-proof and fireproof.

c. The tape shall not deteriorate when subjected to water, gases, salt water, sewage, or fungus and be resistant to sunlight and ultraviolet light.

d. The finished application shall withstand a 200-ampere arc for not less than 30 seconds.

e. Securing tape: Glass cloth electrical tape not less than 7 mils thick, and 3/4 inch wide.

PART 3 EXECUTION

3.1 INSTALLATION OF CONDUCTORS AND CABLES

a. Comply with NECA NEIS 1.

b. General Requirements for Cabling:

(1) Comply with TIA-568.1.

(2) Comply with NECA/BICSI 568, 6, "Cable Termination Practices."

(3) Install 110-style IDC termination hardware unless otherwise indicated.

(4) Terminate all conductors; no cable shall contain un-terminated elements. Make terminations only at indicated outlets, terminals, and cross-connect and patch panels.

(5) Cables may not be spliced. Secure and support cables at intervals not exceeding 30 inches and not more than 6 inches from cabinets, boxes, fittings, outlets, racks, frames, and terminals.

(6) Bundle, lace, and train conductors to terminal points without exceeding manufacturer's limitations on bending radii, but not less than radii specified in NECA/BICSI 568, "Cabling Termination Practices" Chapter. Install lacing bars and distribution spools.

(7) Do not install bruised, kinked, scored, deformed, or abraded cable. Do not splice cable between termination, tap, or junction points. Remove and discard cable if damaged during installation and replace it with new cable.

(8) Cold-Weather Installation: Bring cable to room temperature before dereeling. Heat lamps shall not be used for heating.

(9) Pulling Cable:

(a) Comply with NECA/BICSI 568, Ch. 4, "Pulling Cable." Monitor cable pull tensions.

(b) Provide installation equipment that will prevent the cutting or abrasion of insulation during pulling of cables.

(c) Use ropes made of nonmetallic material for pulling feeders.

(d) Attach pulling lines for feeders by means of either woven basket grips or pulling eyes attached directly to the conductors, as approved by the Resident Engineer/Contracting Officer.

(e) Pull in multiple cables together in a single conduit.

c. Splice cables and wires where necessary only in outlet boxes, junction boxes, or pull boxes.

(1) Splices and terminations shall be mechanically and electrically secure.

(2) Where the Government determines that unsatisfactory splices or terminations have been installed, remove the devices and install approved devices at no additional cost to the Government.

d. Seal cable and wire entering a building from underground, between the wire and conduit where the cable exits the conduit, with a non-hardening approved compound.

e. Unless otherwise specified in other sections install wiring and connect to equipment/devices to perform the required functions as shown and specified.

f. Except where otherwise required, install a separate power supply circuit for each system so that malfunctions in any system will not affect other systems.

g. Where separate power supply circuits are not shown, connect the systems to the nearest panel boards of suitable voltages, which are intended to supply such systems and have suitable spare circuit breakers or space for installation.

h. Install a red warning indicator on the handle of the branch circuit breaker for the power supply circuit for each system to prevent accidental de-energizing of the systems.

i. System voltages shall be 120 volts or lower where shown on the drawings or as required by the NFPA 70.

j. UTP Cable Installation:

(1) Comply with TIA-568.2.

(2) Do not untwist UTP cables more than 1/2 inch from the point of termination to maintain cable geometry.

k. Optical Fiber Cable Installation:

(1) Comply with TIA-568.3.

(2) Cable shall be terminated on connecting hardware that is rack or cabinet mounted.

l. Open-Cable Installation:

(1) Install cabling with horizontal and vertical cable guides in telecommunications spaces with terminating hardware and interconnection equipment.

(2) Suspend copper cable not in a wireway or pathway a minimum of 8 inches above ceilings by cable supports not more than 60 inches apart.

(3) Cable shall not be run through structural members or in contact with pipes, ducts, or other potentially damaging items.

m. Installation of Cable Routed Exposed under Raised Floors:

(1) Install plenum-rated cable only.

(2) Install cabling after the flooring system has been installed in raised floor areas.

(3) Coil cable 72 inches long shall be neatly coiled not less than 12 inches in diameter below each feed point.

n. Outdoor Coaxial Cable Installation:

(1) Install outdoor connections in enclosures complying with NEMA 250, Type 4X. Install corrosion-resistant connectors to keep out moisture.

(2) Attach antenna lead-in cable to support structure at intervals not exceeding 36 inches.

o. Separation from EMI Sources:

(1) Comply with BICSI TDMM and TIA-569 recommendations for separating unshielded copper voice and data communication cable from potential EMI sources, including electrical power lines and equipment.

(2) Separation between open communications cables or cables in nonmetallic raceways and unshielded power conductors and electrical equipment shall be as follows:

(a) Electrical Equipment Rating Less Than 2 kVA: A minimum of 5 inches .

(b) Electrical Equipment Rating between 2 and 5 kVA: A minimum of 12 inches .

(c) Electrical Equipment Rating More Than 5 kVA: A minimum of 24 inches .

(3) Separation between communications cables in grounded metallic raceways and unshielded power lines or electrical equipment shall be as follows:

(a) Electrical Equipment Rating Less Than 2 kVA: A minimum of 2-1/2 inches .

(b) Electrical Equipment Rating between 2 and 5 kVA: A minimum of 6 inches .

(c) Electrical Equipment Rating More Than 5 kVA: A minimum of 12 inches .

(4) Separation between communications cables in grounded metallic raceways and power lines and electrical equipment located in grounded metallic conduits or enclosures shall be as follows:

(a) Electrical Equipment Rating Less Than 2 kVA: No requirement.

(b) Electrical Equipment Rating between 2 and 5 kVA: A minimum of 3 inches .

(c) Electrical Equipment Rating More Than 5 kVA: A minimum of 6 inches .

(5) Separation between Cables and Electrical Motors and Transformers, 5 kVA or HP and Larger: A minimum of 48 inches.

(6) Separation between Cables and Fluorescent Fixtures: A minimum of 5 inches .

3.2 FIRE ALARM WIRING INSTALLATION

a. Comply with NECA NEIS 1 and NFPA 72.

b. Wiring Method: Install wiring in metal raceway according to Section 28 05 28.33, CONDUITS AND BACKBOXES FOR ELECTRICAL SYSTEMS."

(1) Install plenum cable in environmental air spaces, including plenum ceilings.

(2) Fire alarm circuits and equipment control wiring associated with the fire alarm system shall be installed in a dedicated raceway system. This system shall not be used for any other wire or cable.

c. Wiring Method:

(1) Cables and raceways used for fire alarm circuits, and equipment control wiring associated with the fire alarm system, may not contain any other wire or cable.

(2) Fire-Rated Cables: Use of 2-hour, fire-rated fire alarm cables, NFPA 70, Types MI and CI, is permitted.

(3) Signaling Line Circuits: Power-limited fire alarm cables shall not be installed in the same cable or raceway as signaling line circuits.

d. Wiring within Enclosures: Separate power-limited and non-power-limited conductors as recommended by manufacturer. Install conductors parallel with or at right angles to sides and back of the enclosure. Bundle, lace, and train conductors to terminal points with no excess. Connect conductors that are terminated, spliced, or interrupted in any enclosure associated with the fire alarm system to terminal blocks. Mark each terminal according to the system's wiring diagrams. Make all connections with approved crimp-on terminal spade lugs,

pressure-type terminal blocks, or plug connectors.

e. Cable Taps: Use numbered terminal strips in junction, pull, and outlet boxes, cabinets, or equipment enclosures where circuit connections are made.

f. Color-Coding: Color-code fire alarm conductors differently from the normal building power wiring. Use one color-code for alarm circuit wiring and another for supervisory circuits. Color-code audible alarm-indicating circuits differently from alarm-initiating circuits. Use different colors for visible alarm-indicating devices. Paint fire alarm system junction boxes and covers red.

g. Risers: Install at least two vertical cable risers to serve the fire alarm system. Separate risers in close proximity to each other with a minimum one-hour-rated wall, so the loss of one riser does not prevent the receipt or transmission of signals from other floors or zones.

h. Wiring to Remote Alarm Transmitting Device: 1-inch (25-mm) conduit between the fire alarm control panel and the transmitter. Install number of conductors and electrical supervision for connecting wiring as needed to suit monitoring function.

3.3 CONTROL CIRCUIT CONDUCTORS

Minimum Conductor Sizes:

(1) Class 1 remote-control and signal circuits, No. 14 AWG.

(2) Class 2 low-energy, remote-control and signal circuits, No. 16 AWG.

(3) Class 3 low-energy, remote-control, alarm and signal circuits, No. 12 AWG.

3.4 CONNECTIONS

a. Comply with requirements in Section 28 13 00, PHYSICAL ACCESS CONTROL for connecting, terminating, and identifying wires and cables.

b. Comply with requirements in Section 28 23 00, VIDEO SURVEILLANCE for connecting, terminating, and identifying wires and cables.

c. Comply with requirements in Section 28 31 00, FIRE DETECTION AND ALARM for connecting, terminating, and identifying wires and cables.

3.5 FIRESTOPPING

a. Comply with requirements in Division 07 Section "PENETRATION FIRESTOPPING."

b. Comply with TIA-569, "Firestopping" Annex A.

c. Comply with BICSI TDMM, "Firestopping Systems" Article.

3.6 GROUNDING

a. For communications wiring, comply with TIA-607-A and with BICSI TDMM, "Grounding, Bonding, and Electrical Protection" Chapter.

b. For low-voltage wiring and cabling, comply with requirements in Section 28 05 28.33, "GROUNDING AND BONDING FOR ELECTRONIC SAFETY AND SECURITY."

3.7 IDENTIFICATION

a. Identify system components, wiring, and cabling complying with TIA-606.

b. Install a permanent wire marker on each wire at each termination.

c. Identifying numbers and letters on the wire markers shall correspond to those on the wiring diagrams used for installing the systems.

d. Wire markers shall retain their markings after cleaning.

e. In each handhole, install embossed brass tags to identify the system served and function.

3.8 FIELD QUALITY CONTROL

a. Testing Agency: Engage a qualified testing agency to perform tests and inspections.

b. Perform tests and inspections.

c. Tests and Inspections:

(1) Visually inspect UTP and optical fiber cable jacket materials for UL or third-party certification markings. Inspect cabling terminations to confirm color-coding for pin assignments, and inspect cabling connections to confirm compliance with TIA-568.1.

(2) Visually inspect cable placement, cable termination, grounding and bonding, equipment and patch cords, and labeling of all components.

(3) Test UTP cabling for DC loop resistance, shorts, opens, intermittent faults, and polarity between conductors. Test operation of shorting bars in connection blocks. Test cables after termination but not cross connection.

Test instruments shall meet or exceed applicable requirements in TIA-568.2. Perform tests with a tester that complies with performance requirements in "Test Instruments (Normative)" Annex, complying with measurement accuracy specified in "Measurement Accuracy (Informative)" Annex. Use only test cords and adapters that are qualified by test equipment manufacturer for channel or link test configuration.

(4) Optical Fiber Cable Tests:

(a) Test instruments shall meet or exceed applicable requirements in TIA-568.1. Use only test cords and adapters that are qualified by test equipment manufacturer for channel or link test configuration.

(b) Link End-to-End Attenuation Tests:

1. Multimode Link Measurements: Test at 850 or 1300 nm in 1 direction according to TIA-526-14, Method B, One Reference Jumper.

2. Attenuation test results for links shall be less than 2.0 dB. Attenuation test results shall be less than that calculated according to equation in TIA-568.1.

d. Document data for each measurement. Print data for submittals in a summary report that is formatted using Table 10.1 in BICSI TDMM as a guide, or transfer the data from the instrument to the computer, save as text files, print, and submit.

e. End-to-end cabling will be considered defective if it does not pass tests and inspections.

f. Prepare test and inspection reports.

3.9 EXISITNG WIRING

Unless specifically indicated on the plans, existing wiring shall not be reused for the new installation. Only wiring that conforms to the specifications and applicable codes may be reused. If existing wiring does not meet these requirements, existing wiring may not be reused and new wires shall be installed.

-- End of Section --

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DIVISION 28 - ELECTRONIC SAFETY AND SECURITY

SECTION 28 05 26

GROUNDING AND BONDING FOR ELECTRONIC SAFETY AND SECURITY

9/11

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SECTION 28 05 26
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GROUNDING AND BONDING FOR ELECTRONIC SAFETY AND SECURITY

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies the finishing, installation, connection, testing and certification of the grounding and bonding required for a fully functional Electronic Safety and Security (ESS) system.

b. “Grounding electrode system” refers to all electrodes required by NEC, as well as including made, supplementary, grounding electrodes.

c. The terms “connect” and “bond” are used interchangeably in this specification and have the same meaning

1.2 RELATED WORK

a. Section 01 00 00 - GENERAL REQUIREMENTS. For General Requirements.

b. Section 26 41 00 - FACILITY LIGHTNING PROTECTION. Requirements for a lightning protection system.

c. Section 28 05 00 - REQUIREMENTS FOR ELECTRONIC SAFETY AND SECURITY INSTALLATIONS. For general electrical requirements, quality assurance, coordination, and project conditions that are common to more than one section in Division 28.

d. Section 28 05 13 - CONDUCTORS AND CABLES FOR ELECTRONIC SAFETY AND SECURITY. Requirements for low voltage power and lighting wiring.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G

SD-03 Product Data

Grounding And Bonding Conductors; G

Ground Rods; G

Splices And Termination Components; G

Equipment Rack And Cabinet Ground Bars; G

Ground Terminal Blocks; G

Splice Case Ground Accessories; G

Computer Room Ground; G

Security Control Room Ground; G

Corrosion Inhibitors; G

SD-06 Test Reports

Test Reports; G

SD-07 Certificates

Certifications; G

1.4 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 28 05 00, COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY.

b. Shop Drawings:

(1) Clearly present enough information to determine compliance with drawings and specifications.

(2) Include the location of system grounding electrode connections and the routing of aboveground and underground grounding electrode conductors.

c. Test Reports: Provide certified test reports of ground resistance.

d. Certifications: Two weeks prior to final inspection, submit four copies of the following to the COTR:

(1) Certification that the materials and installation are in accordance with the drawings and specifications.

(2) Certification by the contractor that the complete installation has been properly installed and tested.

1.5 REFERENCES

Publications listed below (including amendments, addenda, revisions, supplements, and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by designation only.

ASTM INTERNATIONAL (ASTM)

ASTM B1	(2013) Standard Specification for Hard-Drawn Copper Wire
ASTM B8	(2023) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 81	(2012) Guide for Measuring Earth

Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

UL SOLUTIONS (UL)

UL 83	(2017; Reprint Mar 2020) UL Standard for Safety Thermoplastic-Insulated Wires and Cables
UL 467	(2022) UL Standard for Safety Grounding and Bonding Equipment

PART 2 PRODUCTS

2.1 GROUNDING AND BONDING CONDUCTORS

a. Equipment grounding conductors shall be UL 83 insulated stranded copper, except that sizes (10 AWG) and smaller shall be solid copper. Insulation color shall be continuous green for all equipment grounding conductors, except that wire sizes (4 AWG) and larger shall be permitted to be identified per NFPA 70.

b. Bonding conductors shall be ASTM B8 bare stranded copper, except that sizes (10 AWG) and smaller shall be ASTM B1 solid bare copper wire.

2.2 GROUND RODS

a. Copper clad steel, (3/4-inch) diameter by 10 feet long, conforming to UL 467.

b. Quantity of rods shall be as required to obtain the specified ground resistance.

2.3 SPLICES AND TERMINATION COMPONENTS

a. Components shall meet or exceed UL 467 and be clearly marked with the manufacturer, catalog number, and permitted conductor size(s).2.4 ground connections

b. Listed and labeled by an NRTL acceptable to authorities having jurisdiction for applications in which used and for specific types, sizes, and combinations of conductors and other items connected.

c. Below Grade: Exothermic-welded type connectors.

d. Above Grade:

(1) Bonding Jumpers: Compression-type connectors, using zinc-plated fasteners and external tooth lockwashers.

(2) Connection to Building Steel: Exothermic-welded type connectors.

(3) Ground Busbars: Two-hole compression type lugs, using tin-plated copper or copper alloy bolts and nuts.

(4) Rack and Cabinet Ground Bars: One-hole compression-type lugs, using zinc-plated or copper alloy fasteners.

(5) Bolted Connectors for Conductors and Pipes: Copper or copper alloy, pressure type with at least two bolts.

Pipe Connectors: Clamp type, sized for pipe.

(6) Welded Connectors: Exothermic-welding kits of types recommended by kit manufacturer for materials being joined and installation conditions.

2.4 EQUIPMENT RACK AND CABINET GROUND BARS

Provide solid copper ground bars designed for mounting on the framework of open or cabinet-enclosed equipment racks with minimum dimensions of thick by wide 3/8 inch x ¾ inch).

2.5 GROUND TERMINAL BLOCKS

At any equipment mounting location (e.g., backboards and hinged cover enclosures) where rack-type ground bars cannot be mounted, provide screw lug-type terminal blocks.

2.6 SPLICE CASE GROUND ACCESSORIES

Splice case grounding and bonding accessories shall be supplied by the splice case manufacturer when available. Otherwise, use (6 AWG) insulated ground wire with shield bonding connectors.

2.7 COMPUTER ROOM GROUND

Provide (1/0 AWG) bare copper grounding conductors bolted at mesh intersections to form an equipotential grounding grid. The equipotential grounding grid shall form a 24 inch mesh pattern. The grid shall be bonded to each of the access floor pedestals.

2.8 SECURITY CONTROL ROOM GROUND

a. Provide (1/0 AWG) stranded copper grounding conductor(s) color coded with a green jacket, bolted at the Room’s Communications System Grounding Electrode Cooper Plate and circulate to each equipment rack ground buss bar through the wire management system. Connect each equipment rack, wire management system’s cable tray, ladder, etc. to the circulating ground wire with a minimum (4AWG) stranded Cooper Wire, color coded with a green jacket.

(1) Connect each equipment rack ground buss bar to the circulating ground wire a indicated in 2.9.A, and

(2) Connect each additional room item to the circulating ground wire as indicated in 2.9.A.

PART 3 EXECUTION

3.1 GENERAL

a. Ground in accordance with the NEC, as shown on drawings, and as

specified herein.

b. System Grounding:

(1) Secondary service neutrals: Ground at the supply side of the secondary disconnecting means and at the related transformers.

(2) Separately derived systems (transformers downstream from the service entrance): Ground the secondary neutral.

c. Equipment Grounding: Metallic structures, including ductwork and building steel, enclosures, raceways, junction boxes, outlet boxes, cabinets, machine frames, and other conductive items in close proximity with electrical circuits, shall be bonded and grounded.

3.2 INACCESSIBLE GROUNDING CONNECTIONS

Make grounding connections, which are buried or otherwise normally inaccessible (except connections for which periodic testing access is required) by exothermic weld.

3.3 CORROSION INHIBITORS

When making ground and ground bonding connections, apply a corrosion inhibitor to all contact surfaces. Use corrosion inhibitor appropriate for protecting a connection between the metals used.

3.4 CONDUCTIVE PIPING

Bond all conductive piping systems, interior and exterior, to the building to the grounding electrode system. Bonding connections shall be made as close as practical to the equipment ground bus.

3.5 COMPUTER ROOM/SECURITY EQUIPMENT ROOM GROUNDING

Conduit: Ground and bond metallic conduit systems as follows:

(1) Ground metallic service conduit and any pipes entering or being routed within the computer room at each end using (6AWG) bonding jumpers.

(2) Bond at all intermediate metallic enclosures and across all joints using (6 AWG) bonding jumpers.

3.6 WIREWAY GROUNDING

Ground and Bond Metallic Wireway Systems as follows:

(1) Bond the metallic structures of wireway to provide 100 percent electrical continuity throughout the wireway system by connecting a (6 AWG) bonding jumper at all intermediate metallic enclosures and across all section junctions.

(2) Install insulated (6 AWG) bonding jumpers between the wireway system bonded as required in paragraph 1 above, and the closest building ground at each end and approximately every (50 feet).

(3) Use insulated (6 AWG) bonding jumpers to ground or bond metallic wireway at each end at all intermediate metallic enclosures and

cross all section junctions.

(4) Use insulated (6 AWG) bonding jumpers to ground cable tray to column-mounted building ground plates (pads) at each end and approximately every 15 meters.

3.7 LIGHTNING PROTECTION SYSTEM

Bond the lightning protection system to earth ground externally to the building. Under no condition shall the electrical system's third of fourth ground electrode system, or the telecommunications system circulating ground system be connected to the lightning protection system. The Facility's structural steel may be used to connected the lightning protection system at the direction of the Resident Engineer certified by an independent certified grounding contractor.

3.8 EXTERIOR LIGHT/CAMERA POLES

Provide 20 ft of No. 4 bare copper coiled at bottom of pole base excavation prior to pour, plus additional unspliced length in and above foundation as required to reach pole ground stud.

3.9 GROUND RESISTANCE

a. Grounding system resistance to ground shall not exceed 5 ohms. Make any modifications or additions to the grounding electrode system necessary for compliance without additional cost to the Government. Final tests shall ensure that this requirement is met.

b. Resistance of the grounding electrode system shall be measured using a four-terminal fall-of-potential method as defined in IEEE 81. Ground resistance measurements shall be made before the electrical distribution system is energized and shall be made in normally dry conditions not fewer than 48 hours after the last rainfall. Resistance measurements of separate grounding electrode systems shall be made before the systems are bonded together below grade. The combined resistance of separate systems may be used to meet the required resistance, but the specified number of electrodes must still be provided.

c. Services at power company interface points shall comply with the power company ground resistance requirements.

d. Below-grade connections shall be visually inspected by the Resident Engineer prior to backfilling. The contractor shall notify the Resident Engineer 24 hours before the connections are ready for inspection.

3.10 GROUND ROD INSTALLATION

a. Drive each rod vertically in the earth, not less than 10 feet in depth.

b. Where permanently concealed ground connections are required, make the connections by the exothermic process to form solid metal joints. Make accessible ground connections with mechanical pressure type ground connectors.

c. Where rock prevents the driving of vertical ground rods, install angled ground rods or grounding electrodes in horizontal trenches to achieve

the specified resistance.

3.11 GROUNDING FOR RF/EMI CONTROL

a. Install bonding jumpers to bond all conduit, cable trays, sleeves and equipment for low voltage signaling and data communications circuits. Bonding jumpers shall consist of 4 inches wide copper strip or two (10 AWG) copper conductors spaced minimum 4 inches apart. Use (6 AWG) copper where exposed and subject to damage.

b. Comply with the following when shielded cable is used for data circuits.

(1) Shields shall be continuous throughout each circuit.

(2) Connect shield drain wires together at each circuit connection point and insulate from ground. Do not ground the shield.

(3) Do not connect shields from different circuits together.

(4) Shield shall be connected at one end only. Connect shield to signal reference at the origin of the circuit. Consult with equipment manufacturer to determine signal reference.

3.12 LABELING

a. Comply with requirements in Division 26 Section "ELECTRICAL IDENTIFICATION" Article for instruction signs. The label or its text shall be green.

b. Install labels at the telecommunications bonding conductor and grounding equalizer.

(1) Label Text: "If this connector or cable is loose or if it must be removed for any reason, notify the facility manager."

3.13 FIELD QUALITY CONTROL

a. Perform tests and inspections.

b. Tests and Inspections:

(1) After installing grounding system but before permanent electrical circuits have been energized, test for compliance with requirements.

(2) Inspect physical and mechanical condition. Verify tightness of accessible, bolted, electrical connections with a calibrated torque wrench according to manufacturer's written instructions.

(3) Test completed grounding system at each location where a maximum ground-resistance level is specified, at service disconnect enclosure grounding terminal at individual ground rods. Make tests at ground rods before any conductors are connected.

(a) Measure ground resistance no fewer than two full days after last trace of precipitation and without soil being moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural ground resistance.

(b) Perform tests by fall-of-potential method according to IEEE 81.

c. Grounding system will be considered defective if it does not pass tests and inspections.

d. Prepare test and inspection reports.

e. Report measured ground resistances that exceed the following values:

(1) Power Distribution Units or Panel boards Serving Electronic Equipment: 3 ohm(s).

(2) Manhole Grounds: 10 ohms.

f. Excessive Ground Resistance: If resistance to ground exceeds specified values, notify Architect promptly and include recommendations to reduce ground resistance.

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DIVISION 28 - ELECTRONIC SAFETY AND SECURITY

SECTION 28 05 28.33

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SECTION 28 05 28.33
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CONDUITS AND BACKBOXES FOR ELECTRONIC SAFETY AND SECURITY

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies the finishing, installation, connection, testing certification of the conduit, fittings, and boxes to form a complete, coordinated, raceway system(s). Conduits and when approved separate UL Certified and Listed partitioned telecommunications raceways are required for a fully functional Electronic Safety and Security (ESS) system. Raceways are required for all electronic safety and security cabling unless shown or specified otherwise.

b. Definitions: The term conduit, as used in this specification, shall mean any or all of the raceway types specified.

1.2 RELATED WORK

a. Section 01 00 00 - GENERAL REQUIREMENTS. For General Requirements.

b. Section 06 10 00 - ROUGH CARPENTRY. Requirements for mounting board for communication closets.

c. Section 07 84 00 - FIRESTOPPING. Requirements for sealing around penetrations to maintain the integrity of fire rated construction.

d. Section 07 60 00 - FLASHING AND SHEET METAL. Requirements for fabrications for the deflection of water away from the building envelope at penetrations.

e. Section 07 92 00 - JOINT SEALANTS. Requirements for sealing around conduit penetrations through the building envelope to prevent moisture migration into the building.

f. Section 09 91 00 - PAINTING. Requirements for identification and painting of conduit and other devices.

g. Section 28 05 00 - COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY. For general electrical requirements, general arrangement of the contract documents, coordination, quality assurance, project conditions, equipment and materials, and items that is common to more than one section of Division 28.

h. Section 28 05 26 - GROUNDING AND BONDING FOR ELECTRONIC SAFETY AND SECURITY. Requirements for personnel safety and to provide a low impedance path for possible ground fault currents.

i. Section 28 08 00 - COMMISIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS. Requirements for commissioning - systems readiness checklists, and training.

j. Section 31 20 00 - EARTH MOVING. For bedding of conduits.

1.3 DEFINITIONS

a. EMT: Electrical metallic tubing.

b. ENT: Electrical nonmetallic tubing.

c. EPDM: Ethylene-propylene-diene terpolymer rubber.

d. FMC: Flexible metal conduit.

e. IMC: Intermediate metal conduit.

f. LFMC: Liquidtight flexible metal conduit.

g. LFNC: Liquidtight flexible nonmetallic conduit.

h. NBR: Acrylonitrile-butadiene rubber.

i. RNC: Rigid nonmetallic conduit.

1.4 QUALITY ASSURANCE

Refer to Paragraph 1.4 Quality Assurance, in Section 28 05 00, COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Shop Drawings; G

Coordination Drawings; G

SD-03 Product Data

Product Data; G

SD-06 Test Reports

Completed System Readiness Checklists; G

Source Quality-Control Test Reports; G

SD-07 Certificates

Certification; G

Manufacturer Seismic Qualification Certification; G

1.6 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 28 05 00, COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY and Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES. Furnish the following:

b. Shop Drawings:

(1) Size and location of main feeders;

(2) Size and location of panels and pull boxes

(3) Layout of required conduit penetrations through structural elements.

(4) The specific item proposed and its area of application shall be identified on the catalog cuts.

c. Certification: Prior to final inspection, deliver to the Resident Engineer/Contracting Officer four copies of the certification that the material is in accordance with the drawings and specifications and has been properly installed.

d. Completed System Readiness Checklists provided by the Commissioning Agent and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 28 08 00 COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS.

e. Product Data: For surface raceways, wireways and fittings, floor boxes, hinged-cover enclosures, and cabinets.

f. Shop Drawings: For the following raceway components. Include plans, elevations, sections, details, and attachments to other work.

(1) Custom enclosures and cabinets.

(2) Handholes and boxes for underground wiring, including the following:

(a) Duct entry provisions, including locations and duct sizes.

(b) Frame and cover design.

(c) Grounding details.

(d) Dimensioned locations of cable rack inserts, and pulling-in and lifting irons.

(e) Joint details.

g. Coordination Drawings: Conduit routing plans, drawn to scale, on which the following items are shown and coordinated with each other, based on input from installers of the items involved:

(1) Structural members in the paths of conduit groups with common supports.

(2) HVAC and plumbing items and architectural features in the paths of conduit groups with common supports.

h. Manufacturer Seismic Qualification Certification: Submit certification that enclosures and cabinets and their mounting provisions, including those for internal components, will withstand

seismic forces defined in Division 16 Section "Electrical Supports and Seismic Restraints." Include the following:

(1) Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.

(a) The term "withstand" means "the cabinet or enclosure will remain in place without separation of any parts when subjected to the seismic forces specified and the unit will retain its enclosure characteristics, including its interior accessibility, after the seismic event."

(2) Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.

(3) Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.

i. Source quality-control test reports.

1.7 REFERENCES

a. Publications listed below (including amendments, addenda, revisions, supplements and errata) form a part of this specification to the extent referenced. Publications are referenced in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM C1107/C1107M	(2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C80.1	(2020) American National Standard for Electrical Rigid Steel Conduit (ERSC)
ANSI C80.3	(2020) American National Standard for Electrical Metallic Tubing (EMT)
ANSI C80.5	(2020) American National Standard for Electrical Rigid Aluminum Conduit
ANSI C80.6	(2018) Electrical Intermediate Metal Conduit (ELMC)
ANSI/NEMA OS 2	(2013; R 2020) Nonmetallic Outlet Boxes, Device Boxes, Covers, and Box Supports
NEMA TC-3-04	PVC Fittings for Use with Rigid PVC Conduit and Tubing
NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NEMA FB 1	(2014) Standard for Fittings, Cast Metal Boxes, and Conduit Bodies for Conduit, Electrical Metallic Tubing, and Cable

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code

UL SOLUTIONS (UL)

UL 1	(2005; Reprint Jan 2022) UL Standard for Safety Flexible Metal Conduit
UL 5	(2016; Reprint Jul 2022) UL Standard for Safety Surface Metal Raceways and Fittings
UL 6	(2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel
UL 6A	(2008; Reprint Nov 2024) UL Standard for Safety Electrical Rigid Metal Conduit - Aluminum, Red Brass, and Stainless Steel
UL 50	(2024) UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations
UL 360	(2013; Reprint Jan 2024) UL Standard for Safety Liquid-Tight Flexible Metal Conduit
UL 467	(2022) UL Standard for Safety Grounding and Bonding Equipment
UL 514A	(2024) UL Standard for Safety Metallic Outlet Boxes
UL 514B	(2012; Reprint Mar 2024) UL Standard for Safety Conduit, Tubing and Cable Fittings
UL 514C	(2014; Reprint Apr 2024) UL Standard for Safety Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers
UL 651	(2011; Reprint May 2022) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
UL 651A	(2023; Reprint Jan 2024) UL Standard for Safety Schedule 40 and 80 High Density Polyethylene (HDPE) Conduit
UL 797	(2007; Reprint Apr 2023) UL Standard for Safety Electrical Metallic Tubing -- Steel
UL 1242	(2006; Reprint Apr 2022) UL Standard for Safety Electrical Intermediate Metal Conduit -- Steel

SOCIETY OF CABLE TELECOMMUNICATIONS ENGINEERS (SCTE)

ANSI/SCTE 77 (2013) Specification for Underground Enclosure Integrity

PART 2 PRODUCTS

2.1 GENERAL

a. Conduit Size: In accordance with the NEC, but not less than 3/4 inch unless otherwise shown.

b. Rigid galvanized steel: Shall Conform to UL 6, ANSI C80.1.

c. Rigid aluminum: Shall Conform to UL 6A, ANSI C80.5.

d. Rigid intermediate steel conduit (IMC): Shall Conform to UL 1242, ANSI C80.6.

e. Electrical metallic tubing (EMT): Shall Conform to UL 797, ANSI C80.3. Maximum size not to exceed 4 inches and shall be permitted only with cable rated 600 volts or less.

f. Flexible galvanized steel conduit: Shall Conform to UL 1.

g. Liquid-tight flexible metal conduit: Shall Conform to UL 360.

h. Direct burial plastic conduit: Shall conform to UL 651 and UL 651A, heavy wall PVC or high density polyethylene (PE).

i. Surface metal raceway: Shall Conform to UL 5.

j. Rigid steel and IMC conduit fittings:

(1) Fittings shall meet the requirements of UL 514B and NEMA FB 1.

(2) Standard threaded couplings, locknuts, bushings, and elbows: Only steel or malleable iron materials are acceptable. Integral retractable type IMC couplings are also acceptable.

(3) Locknuts: Bonding type with sharp edges for digging into the metal wall of an enclosure.

(4) Bushings: Metallic insulating type, consisting of an insulating insert molded or locked into the metallic body of the fitting. Bushings made entirely of metal or nonmetallic material are not permitted.

(5) Erickson (union-type) and set screw type couplings: Approved for use in concrete are permitted for use to complete a conduit run where conduit is installed in concrete. Use set screws of case hardened steel with hex head and cup point to firmly seat in conduit wall for positive ground. Tightening of set screws with pliers is prohibited.

(6) Sealing fittings: Threaded cast iron type. Use continuous drain type sealing fittings to prevent passage of water vapor. In concealed work, install fittings in flush steel boxes with blank

cover plates having the same finishes as that of other electrical plates in the room.

k. Rigid aluminum conduit fittings:

(1) Standard threaded couplings, locknuts, bushings, and elbows: Malleable iron, steel or aluminum alloy materials; Zinc or cadmium plate iron or steel fittings. Aluminum fittings containing more than 0.4 percent copper are prohibited.

(2) Locknuts and bushings: As specified for rigid steel and IMC conduit.

(3) Set screw fittings: Not permitted for use with aluminum conduit.

l. Electrical metallic tubing fittings:

(1) Fittings shall meet the requirements of UL 514B and NEMA FB 1.

(2) Only steel or malleable iron materials are acceptable.

(3) Couplings and connectors: Concrete tight and rain tight, with connectors having insulated throats. Use gland and ring compression type couplings and connectors for conduit sizes 2 inches and smaller. Use set screw type couplings with four set screws each for conduit sizes over (2 inches). Use set screws of case-hardened steel with hex head and cup point to firmly seat in wall of conduit for positive grounding.

(4) Indent type connectors or couplings are prohibited.

(5) Die-cast or pressure-cast zinc-alloy fittings or fittings made of "pot metal" are prohibited.

m. Flexible steel conduit fittings:

(1) Conform to UL 514B. Only steel or malleable iron materials are acceptable.

(2) Clamp type, with insulated throat.

n. Liquid-tight flexible metal conduit fittings:

(1) Fittings shall meet the requirements of UL 514B and NEMA FB 1.

(2) Only steel or malleable iron materials are acceptable.

(3) Fittings must incorporate a threaded grounding cone, a steel or plastic compression ring, and a gland for tightening. Connectors shall have insulated throats.

o. Direct burial plastic conduit fittings:

(1) Fittings shall meet the requirements of UL 514C and NEMA TC-3-04.

(2) As recommended by the conduit manufacturer.

p. Surface metal raceway fittings: As recommended by the raceway manufacturer.

q. Expansion and deflection couplings:

(1) Conform to UL 467 and UL 514B.

(2) Accommodate, 0.75 inch deflection, expansion, or contraction in any direction, and allow 30 degree angular deflections.

(3) Include internal flexible metal braid sized to guarantee conduit ground continuity and fault currents in accordance with UL 467, and the NEC code tables for ground conductors.

(4) Jacket: Flexible, corrosion-resistant, watertight, moisture and heat resistant molded rubber material with stainless steel jacket clamps.

2.2 CONDUIT SUPPORTS

a. Parts and hardware: Zinc-coat or provide equivalent corrosion protection.

b. Individual Conduit Hangers: Designed for the purpose, having a pre-assembled closure bolt and nut, and provisions for receiving a hanger rod.

c. Multiple conduit (trapeze) hangers: Not less than (1-1/2 by 1-1/2 inch), 12 gage steel, cold formed, lipped channels; with not less than 3/8 inch diameter steel hanger rods.

d. Solid Masonry and Concrete Anchors: Self-drilling expansion shields, or machine bolt expansion.

2.3 OUTLET, JUNCTION, AND PULL BOXES

a. UL 50 and UL 514A.

b. Cast metal where required by the NEC or shown, and equipped with rustproof boxes.

c. Nonmetallic Outlet and Device Boxes: ANSI/NEMA OS 2.

d. Metal Floor Boxes: Cast or sheet metal, semi-adjustable, rectangular.

e. Sheet metal boxes: Galvanized steel, except where otherwise shown.

f. Flush mounted wall or ceiling boxes shall be installed with raised covers so that front face of raised cover is flush with the wall. Surface mounted wall or ceiling boxes shall be installed with surface style flat or raised covers.

2.4 CABINETS

a. NEMA 250, Type 1, galvanized-steel box with removable interior panel and removable front, finished inside and out with manufacturer's standard enamel.

b. Hinged door in front cover with flush latch and concealed hinge.

c. Key latch to match panelboards.

d. Metal barriers to separate wiring of different systems and voltage.

e. Accessory feet where required for freestanding equipment.

2.5 WIREWAYS

Equip with hinged covers, except where removable covers are shown.

2.6 WARNING TAPE

Standard, 4-Mil polyethylene 3 inches wide tape non-detectable type, red with black letters, and imprinted with “CAUTION BURIED ELECTRONIC SAFETY AND SECURITY CABLE BELOW”.

2.7 HANDHOLES AND BOXES FOR EXTERIOR UNDERGROUND WIRING

a. Description: Comply with ANSI/SCTE 77.

(1) Color of Frame and Cover: Gray.

(2) Configuration: Units shall be designed for flush burial and have closed bottom, unless otherwise indicated.

(3) Cover: Weatherproof, secured by tamper-resistant locking devices and having structural load rating consistent with enclosure.

(4) Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.

(5) Cover Legend: Molded lettering, as indicated for each service. <Insert legend.>

(6) Conduit Entrance Provisions: Conduit-terminating fittings shall mate with entering ducts for secure, fixed installation in enclosure wall.

(7) Handholes (2 inches wide by 24 inches long) <Insert dimensions> and larger shall have inserts for cable racks and pulling-in irons installed before concrete is poured.

b. Polymer-Concrete Handholes and Boxes with Polymer-Concrete Cover: Molded of sand and aggregate, bound together with polymer resin, and reinforced with steel or fiberglass or a combination of the two.

c. Fiberglass Handholes and Boxes with Polymer-Concrete Frame and Cover: Sheet-molded, fiberglass-reinforced, polyester-resin enclosure joined to polymer-concrete top ring or frame.

d. Fiberglass Handholes and Boxes: Molded of fiberglass-reinforced polyester resin, with covers of polymer concrete.

2.8 SLEEVES FOR RACEWAYS

a. Steel Pipe Sleeves: ASTM A53/A53M, Type E, Grade B, Schedule 40, galvanized steel, plain ends.

b. Cast-Iron Pipe Sleeves: Cast or fabricated "wall pipe," equivalent to ductile-iron pressure pipe, with plain ends and integral waterstop,

unless otherwise indicated.

c. Sleeves for Rectangular Openings: Galvanized sheet steel with minimum 0.052- or 0.138-inch thickness as indicated and of length to suit application.

d. Coordinate sleeve selection and application with selection and application of firestopping specified in Division 07 84 00 "FIRESTOPPING."

2.9 SLEEVE SEALS

a. Description: Modular sealing device, designed for field assembly, to fill annular space between sleeve and cable.

(1) Sealing Elements: EPDM interlocking links shaped to fit surface of cable or conduit. Include type and number required for material and size of raceway or cable.

(2) Pressure Plates: Plastic. Include two for each sealing element.

(3) Connecting Bolts and Nuts: Carbon steel with corrosion-resistant coating of length required to secure pressure plates to sealing elements. Include one for each sealing element.

2.10 GROUT

ASTM C1107/C1107M, factory-packaged, nonmetallic aggregate grout, noncorrosive, nonstaining, mixed with water to consistency suitable for application and a 30-minute working time. WIRELINE DATA TRANSMISSION MEDIA FOR SECURITY SYSTEMS

PART 3 EXECUTION

3.1 PENETRATIONS

a. Cutting or Holes:

(1) Locate holes in advance where they are proposed in the structural sections such as ribs or beams. Obtain the approval of the Resident Engineer/Contracting Officer prior to drilling through structural sections.

(2) Cut holes through concrete and masonry in new and existing structures with a diamond core drill or concrete saw. Pneumatic hammer, impact electric, hand or manual hammer type drills are not allowed, except where permitted by the Resident Engineer/Contracting Officer as required by limited working space.

b. Fire Stop: Where conduits, wireways, and other electronic safety and security raceways pass through fire partitions, fire walls, smoke partitions, or floors, install a fire stop that provides an effective barrier against the spread of fire, smoke and gases as specified in Section 07 84 00, FIRESTOPPING, with rock wool fiber or silicone foam sealant only. Completely fill and seal clearances between raceways and openings with the fire stop material.

c. Waterproofing: At floor, exterior wall, and roof conduit penetrations, completely seal clearances around the conduit and make

watertight as specified in Section 07 92 00, “JOINT SEALANTS”.

3.2 INSTALLATION, GENERAL

a. Install conduit as follows:

(1) In complete runs before pulling in cables or wires.

(2) Flattened, dented, or deformed conduit is not permitted. Remove and replace the damaged conduits with new undamaged material.

(3) Assure conduit installation does not encroach into the ceiling height head room, walkways, or doorways.

(4) Cut square with a hacksaw, ream, remove burrs, and draw up tight.

(5) Mechanically continuous.

(6) Independently support conduit at 8 foot on center. Do not use other supports i.e., (suspended ceilings, suspended ceiling supporting members, lighting fixtures, conduits, mechanical piping, or mechanical ducts).

(7) Support within 12 inches of changes of direction, and within 12 inches of each enclosure to which connected.

(8) Close ends of empty conduit with plugs or caps at the rough-in stage to prevent entry of debris, until wires are pulled in.

(9) Conduit installations under fume and vent hoods are prohibited.

(10) Secure conduits to cabinets, junction boxes, pull boxes and outlet boxes with bonding type locknuts. For rigid and IMC conduit installations, provide a locknut on the inside of the enclosure, made up wrench tight. Do not make conduit connections to junction box covers.

(11) Flashing of penetrations of the roof membrane is specified in Section 07 60 00, “FLASHING AND SHEET METAL”.

(12) Do not use aluminum conduits in wet locations.

(13) Unless otherwise indicated on the drawings or specified herein, all conduits shall be installed concealed within finished walls, floors and ceilings.

b. Conduit Bends:

(1) Make bends with standard conduit bending machines.

(2) Conduit hickey may be used for slight offsets, and for straightening stubbed out conduits.

(3) Bending of conduits with a pipe tee or vise is prohibited.

c. Layout and Homeruns:

(1) Install conduit with wiring, including homeruns, as shown.

(2) Deviations: Make only where necessary to avoid interferences and only after drawings showing the proposed deviations have been submitted approved by the Resident Engineer/Contracting Officer.

d. Fire Alarm:

(1) Fire alarm conduit shall be painted red (a red “top-coated” conduit from the conduit manufacturer may be used in lieu of painted conduit) in accordance with the requirements of Section 28 31 00, “FIRE DETECTION AND ALARM”.

3.3 CONCEALED WORK INSTALLATION

a. In Concrete:

(1) Conduit: Rigid steel, IMC or EMT. Do not install EMT in concrete slabs that are in contact with soil, gravel or vapor barriers.

(2) Align and run conduit in direct lines.

(3) Install conduit through concrete beams only when the following occurs:

(a) Where shown on the structural drawings.

(b) As approved by the Resident Engineer/Contracting Officer prior to construction, and after submittal of drawing showing location, size, and position of each penetration.

(4) Installation of conduit in concrete that is less than 3 inch thick is prohibited.

(a) Conduit outside diameter larger than 1/3 of the slab thickness is prohibited.

(b) Space between conduits in slabs: Approximately six conduit diameters apart, except one conduit diameter at conduit crossings.

(c) Install conduits approximately in the center of the slab so that there will be a minimum of 3/4 inch of concrete around the conduits.

(5) Make couplings and connections watertight. Use thread compounds that are UL approved conductive type to insure low resistance ground continuity through the conduits. Tightening set screws with pliers is prohibited.

b. Furred or Suspended Ceilings and in Walls:

(1) Conduit for conductors above 600 volts:

(a) Rigid steel or rigid aluminum.

(b) Aluminum conduit mixed indiscriminately with other types in the same system is prohibited.

(2) Conduit for conductors 600 volts and below:

(a) Rigid steel, IMC, rigid aluminum, or EMT. Different type

conduits mixed indiscriminately in the same system is prohibited.

(3) Align and run conduit parallel or perpendicular to the building lines.

(4) Connect recessed lighting fixtures to conduit runs with maximum 6 feet of flexible metal conduit extending from a junction box to the fixture.

(5) Tightening set screws with pliers is prohibited.

3.4 EXPOSED WORK INSTALLATION

a. Unless otherwise indicated on the drawings, exposed conduit is only permitted in mechanical and electrical rooms.

b. Conduit for Conductors 600 volts and below:

(1) Rigid steel, IMC, rigid aluminum, or EMT. Different type of conduits mixed indiscriminately in the system is prohibited.

c. Align and run conduit parallel or perpendicular to the building lines.

d. Install horizontal runs close to the ceiling or beams and secure with conduit straps.

e. Support horizontal or vertical runs at not over (eight foot) intervals.

f. Surface metal raceways: Use only where shown.

g. Painting:

(1) Paint exposed conduit as specified in Section 09 91 00, “PAINTING”.

(2) Paint all conduits containing cables rated over 600 volts safety orange. Refer to Section 09 91 00, “PAINTING” for preparation, paint type, and exact color. In addition, paint legends, using (two inch) high black numerals and letters, showing the cable voltage rating. Provide legends where conduits pass through walls and floors and at maximum 20 foot intervals in between.

3.5 EXPANSION JOINTS

a. Conduits 3 inches and larger, that are secured to the building structure on opposite sides of a building expansion joint, require expansion and deflection couplings. Install the couplings in accordance with the manufacturer's recommendations.

b. Provide conduits smaller than 3 inches with junction boxes on both sides of the expansion joint. Connect conduits to junction boxes with sufficient slack of flexible conduit to produce 5 inch vertical drop midway between the ends. Flexible conduit shall have a copper green ground bonding jumper installed. In lieu of this flexible conduit, expansion and deflection couplings as specified above for 15 inches and larger conduits are acceptable.

c. Install expansion and deflection couplings where shown.

d. Seismic Areas: In seismic areas, provide conduits rigidly secured to

the building structure on opposite sides of a building expansion joint with junction boxes on both sides of the joint. Connect conduits to junction boxes with 15 inches of slack flexible conduit. Flexible conduit shall have a copper green ground bonding jumper installed.

3.6 CONDUIT SUPPORTS, INSTALLATION

a. Safe working load shall not exceed 1/4 of proof test load of fastening devices.

b. Use pipe straps or individual conduit hangers for supporting individual conduits. Maximum distance between supports is 8 foot on center.

c. Support multiple conduit runs with trapeze hangers. Use trapeze hangers that are designed to support a load equal to or greater than the sum of the weights of the conduits, wires, hanger itself, and (200 pounds). Attach each conduit with U-bolts or other approved fasteners.

d. Support conduit independently of junction boxes, pull boxes, fixtures, suspended ceiling T-bars, angle supports, and similar items.

e. Fasteners and Supports in Solid Masonry and Concrete:

(1) New Construction: Use steel or malleable iron concrete inserts set in place prior to placing the concrete.

(2) Existing Construction:

(a) Steel expansion anchors not less than 1/4 inch bolt size and not less than (1-1/8 inch) embedment.

(b) Power set fasteners not less than 1/4 inch diameter with depth of penetration not less than (3 inches).

(c) Use vibration and shock resistant anchors and fasteners for attaching to concrete ceilings.

f. Hollow Masonry: Toggle bolts are permitted.

g. Bolts supported only by plaster or gypsum wallboard are not acceptable.

h. Metal Structures: Use machine screw fasteners or other devices specifically designed and approved for the application.

i. Attachment by wood plugs, rawl plug, plastic, lead or soft metal anchors, or wood blocking and bolts supported only by plaster is prohibited.

j. Chain, wire, or perforated strap shall not be used to support or fasten conduit.

k. Spring steel type supports or fasteners are prohibited for all uses except: Horizontal and vertical supports/fasteners within walls.

l. Vertical Supports: Vertical conduit runs shall have riser clamps and supports in accordance with the NEC and as shown. Provide supports for cable and wire with fittings that include internal wedges and retaining collars.

3.7 BOX INSTALLATION

a. Boxes for Concealed Conduits:

(1) Flush mounted.

(2) Provide raised covers for boxes to suit the wall or ceiling, construction and finish.

b. In addition to boxes shown, install additional boxes where needed to prevent damage to cables and wires during pulling in operations.

c. Remove only knockouts as required and plug unused openings. Use threaded plugs for cast metal boxes and snap-in metal covers for sheet metal boxes.

d. Outlet boxes in the same wall mounted back-to-back are prohibited. A minimum (24 inch), center-to-center lateral spacing shall be maintained between boxes).

e. Minimum size of outlet boxes for ground fault interrupter (GFI) receptacles is 4 inches square by (2-1/8 inches) deep, with device covers for the wall material and thickness involved.

f. Stencil or install phenolic nameplates on covers of the boxes identified on riser diagrams; for example "SIG-FA JB No. 1".

g. On all Branch Circuit junction box covers, identify the circuits with black marker.

3.8 ELECTRONIC SAFETY AND SECURITY CONDUIT

a. Install the electronic safety and security raceway system as shown on drawings.

b. Minimum conduit size of (3/4 inch), but not less than the size shown on the drawings.

c. All conduit ends shall be equipped with insulated bushings.

d. All (four inch) conduits within buildings shall include pull boxes after every two 90 degree bends. Size boxes per the NFPA 70.

e. Vertical conduits/sleeves through closets floors shall terminate not less than 3 inches below the floor and not less than 3 inches below the ceiling of the floor below.

f. Terminate conduit runs to/from a backboard in a closet or interstitial space at the top or bottom of the backboard. Conduits shall enter communication closets next to the wall and be flush with the backboard.

g. Where drilling is necessary for vertical conduits, locate holes so as not to affect structural sections such as ribs or beams.

h. All empty conduits located in communications closets or on backboards shall be sealed with a standard non-hardening duct seal compound to prevent the entrance of moisture and gases and to meet fire resistance requirements.

i. Conduit runs shall contain no more than four quarter turns (90 degree bends) between pull boxes/backboards. Minimum radius of communication conduit bends shall be as follows (special long radius):

Sizes of Conduit	
¾	(6)
1	(9)
1-1/4	(14)
1-1/2	(17)
2	(21)
2-1/2	(25)
3	(31)
3-1/2	(36)
4	(45)

j. Furnish and install 3/4 inch thick fire retardant plywood specified in on the wall of communication closets where shown on drawings. Mount the plywood with the bottom edge (one foot) above the finished floor.

k. Furnish and pull wire in all empty conduits. (Sleeves through floor are exceptions).

3.9 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 28 08 00 – “COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS” for all inspection, start up, and contractor testing required above and required by the System Readiness Checklist provided by the Commissioning Agent.

b. Components provided under this section of the specification will be tested as part of a larger system. Refer to Section 28 08 00, “COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS” and related sections for contractor responsibilities for system commissioning.

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SECTION 28 08 00.01
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DVA/USACE PROJECTS COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

a. The requirements of this Section apply to all sections of Division 28.

b. This project will have selected building systems commissioned. The complete list of equipment and systems to be commissioned is specified in Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS. The commissioning process, which the Contractor is responsible to execute, is defined in Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS. A Commissioning Agent (CxA) appointed by the Government and Commissioning Manager (CxM) as indicated in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS will manage the commissioning process.

1.2 RELATED WORK

a. UFGS GENERAL REQUIREMENTS sections

b. Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS

c. Section UFGS 01 33 00, SUBMITTAL PROCEDURES

1.3 SUMMARY

a. This Section includes requirements for commissioning the Facility electronic safety and security systems, related subsystems and related equipment. This Section supplements the general requirements specified in Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

b. Refer to Section 01 91 00.01, DVA/USACE GENERAL COMMISSIONING REQUIREMENTS for more details regarding processes and procedures as well as roles and responsibilities for all Commissioning Team members.

1.4 DEFINITIONS

Refer to Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for definitions.

1.5 COMMISSIONED SYSTEMS

a. Commissioning of a system or systems specified in Division 28 is part of the construction process. Documentation and testing of these systems, as well as training of the VA’s Operation and Maintenance personnel in accordance with the requirements of Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and of Division 28, is required in cooperation with the DVA/USACE and the CxA and CxM.

b. The Facility exterior closure systems commissioning will include the

systems listed in Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS:

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Construction Inspections; G, DO

Pre-Functional Checklists; G, DO

Contractors Tests; G, DO

Systems Functional Performance Testing; G, DO

SD-08 Manufacturer's Instructions

Training Agendas And Trainer Resumes; G, DO

1.7 SUBMITTAL INSTRUCTIONS

a. The commissioning process requires review of selected Submittals that pertain to the systems to be commissioned. The CxA will provide a list of submittals that will be reviewed by the CxA and CxM. This list will be reviewed and approved by the VA prior to forwarding to the Contractor. Refer to UFGS Section 01 33 00, SUBMITTAL PROCEDURES for further details.

b. The commissioning process requires Submittal review simultaneously with engineering review. Specific submittal requirements related to the commissioning process are specified in Section 01 91 00.01, DVA/USACE GENERAL COMMISSIONING REQUIREMENTS.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 CONSTRUCTION INSPECTIONS

Commissioning of Electronic Safety and Security systems will require inspection of individual elements of the electronic safety and security systems throughout the construction period. The Contractor shall coordinate with the CxA and CxM in accordance with Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and the Commissioning plan to schedule electronic safety and security systems inspections as required to support the Commissioning Process.

3.2 PRE-FUNCTIONAL CHECKLISTS

The Contractor shall complete Pre-Functional Checklists to verify systems, subsystems, and equipment installation is complete and systems are ready for Systems Functional Performance Testing. The CxM will prepare Pre-Functional Checklists to be used to document equipment installation. The Contractor shall complete the checklists. Completed checklists shall

be submitted to the DVA/USACE, CxA, and CxM for review. The CxM and CxA may spot check a sample of completed checklists. If the CxM and CxA determine that the information provided on the checklist is not accurate, the CxM will return the marked-up checklist to the Contractor for correction and resubmission. If the CxM and CxA determine that a significant number of completed checklists for similar equipment are not accurate, the CxM in coordination with CxA will select a broader sample of checklists for review. If the CxM and CxA determine that a significant number of the broader sample of checklists is also inaccurate, all the checklists for the type of equipment will be returned to the Contractor for correction and resubmission. Refer to Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for submittal requirements for Pre-Functional Checklists, Equipment Startup Reports, and other commissioning documents.

3.3 CONTRACTORS TESTS

Contractor tests as required by other Sections of Division 28 shall be scheduled and documented in accordance with UFGS GENERAL REQUIREMENTS sections. All testing shall be incorporated into the project schedule. Contractor shall provide no less than seven calendar days' notice of testing. The CxM will witness selected Contractor tests at the sole discretion of the CxM and CxA. The CxA and other Government representative(s) may choose to participate in some or all the testing procedures as observers. Contractor tests shall be completed prior to scheduling Systems Functional Performance Testing.

3.4 SYSTEMS FUNCTIONAL PERFORMANCE TESTING

The Commissioning Process includes Systems Functional Performance Testing that is intended to test systems functional performance under steady state conditions, to test system reaction to changes in operating conditions, and system performance under emergency conditions. The CxM will prepare final and detailed Systems Functional Performance Test procedures for review and approval by the CxA and Contracting Officer. The Contractor shall review and comment on the tests prior to approval. The Contractor shall provide the required labor, materials, and test equipment identified in the test procedure to perform the tests. The CxM will witness and document the testing. The Contractor shall sign the test reports to verify tests were performed. See Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS, for additional details.

3.5 TRAINING OF VA PERSONNEL

Training of the VA operation and maintenance personnel is required in cooperation with the Contracting Officer, VA Resident Engineer, and CxM. Provide competent, factory authorized personnel to provide instruction to operation and maintenance personnel concerning the location, operation, and troubleshooting of the installed systems. Contractor shall submit training agendas and trainer resumes in accordance with the requirements of Section 01 91 00.01, DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS. The instruction shall be scheduled in coordination with the Contracting Officer and VA Resident Engineer after submission and approval of formal training plans. Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and Division 28 Sections for additional Contractor training requirements.

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SECTION 28 13 00
10/11

PHYSICAL ACCESS CONTROL SYSTEM

PART 1 – GENERAL

1.1 DESCRIPTION

a. This Section specifies the finishing, installation, connection, testing and certification of a complete and fully operating extenstion to the existing Physical Access Control System, hereinafter referred to as the PACS.

b. This Section includes a Physical Access Control System consisting of a system server, operating system and application software, and field-installed Controllers connected by a high-speed electronic data transmission network. The PACS shall have the following:

(1) Physical Access Control:

(a) Regulating access through doors

(b) Anti-passback

(c) Visitor assignment

(d) Surge and tamper protection

(e) Secondary alarm annunciator

(f) Credential cards and readers

(g) Biometric identity verification equipment

(h) Push-button switches

(i) RS-232 ASCII interface

(j) Credential creation and credential holder database and management

(k) Monitoring of field-installed devices

(l) Reporting

c. System Architecture:

(1) Criticality, operational requirements, and/or limiting points of failure may dictate the development of an enterprise and regional server architecture as opposed to system capacity. Provide server and workstation configurations with all necessary connectors, interfaces and accessories as shown.

d. PACS shall provide secure and reliable identification of Federal employees and contractors by utilizing credential authentication per FIPS-201.

e. Physical Access Control System (PACS) shall consist of:

(1) Head-End equipment server,

(2) One or more networked PC-based workstations,

(3) Physical Access Control System and Database Management Software,

(4) Credential validation software/hardware,

(5) Field installed controllers,

(6) PIV Middelware,

(7) Card readers,

(8) Biometric identification devices,

(9) PIV <PIV-I>, <Legacy CAC>, <CAC NG>, <CAC EP>, <TWIC>, <FRAC> cards.

(10) Supportive information system,

(11) Door locks and sensors,

(12) Power supplies,

(13) Interfaces with:

(a) Video Surveillance and Assessment System,

(b) Gate, turnstile, and traffic arm controls,

(c) Automatic door operators,

(d) Intrusion Detection System,

(e) Intercommunication System

(f) Fire Protection System,

(g) HVAC,

(h) Building Management System,

(i) Elevator Controls,

(j) <list interfaced systems>.

f. Head-End equipment server, workstations and controllers shall be connected by a high-speed electronic data transmission network.

g. Information system supporting PACS, Head-End equipment server, workstations, network switches, routers and controllers shall comply with NIST FIPS 200 requirements (Minimum Security Requirements for Federal Information and Information Systems) and NIST SP 800-53 (Recommended Security Controls for Federal Information Systems).

h. PACS system shall support:

(1) Multiple credential authentication modes,

(2) Bidirectional communication with the reader,

(3) Incident response policy implementation capability; system shall have capability to automatically change access privileges for certain user groups to high security areas in case of incident/emergency.

(4) Visitor management,

i. All security relevant decisions shall be made on “secure side of the door”. Secure side processing shall include;

(1) Challenge/response management,

(2) PKI path discovery and validation,

(3) Credential identifier processing,

(4) Authorization decisions.

j. For locations where secure side processing is not applicable the tamper switches and certified cryptographic processing shall be provided per NIST FIPS 140-2.

k. System Software: Based on central-station, workstation operating system, server operating system, and application software.

l. Software and controllers shall be capable of matching full 56 bit FASC-N plus minimum of 32 bits of public key certificate data.

m. Software shall have the following capabilities:

(1) Multiuser multitasking to allow for independent activities and monitoring to occur simultaneously at different workstations.

(2) Support authentication and enrolment;

(a) PIV verification,

(b) Expiration date check,

(c) Biometric check,

(d) Digital photo display/check,

(e) Validate digital signatures of data objects (Objects are signed by the Trusted Authority

(f) Private key challenge (CAK & PAK to verify private key public key pairs exist and card is not a clone)

(3) Support CRL validation via OCSP or SCVP on a scheduled basis and automatically deny access to any revoked credential in the system.

(4) Graphical user interface to show pull-down menus and a menu tree

format that complies with interface guidelines of Microsoft Windows operating system.

(5) System license shall be for the entire system and shall include capability for future additions that are within the indicated system size limits specified in this Section.

(6) System shall have open architecture that allows importing and exporting of data and interfacing with other systems that are compatible with <insert operating system> operating system.

(7) Operator login and access shall be utilized via integrated smart card reader and password protection.

n. Systems Networks:

(1) A standalone system network shall interconnect all components of the system. This network shall include communications between a central station and any peer or subordinate workstations, enrollment stations, local annunciation stations, portal control stations or redundant central stations.

o. Number of points:

(1) PACS shall support multiple autonomous regional servers that can connect to a master command and controller server.

(2) Unlimited number of access control readers, unlimited number of inputs or outputs, unlimited number of client workstations, unlimited number of cardholders.

(3) Total system solution to enable enterprise-wide, networked, multi-user access to all system resources via a wide range of options for connectivity with the customer’s existing LAN and WAN.

p. Console Network:

Console network, if required, shall provide communication between a central station and any subordinate or separate stations of the system. Where redundant central or parallel stations are required, the console network shall allow the configuration of stations as master and slave. The console network may be a part of the field device network or may be separate depending upon the manufacturer's system configuration.

q. Network(s) connecting PCs and Controllers shall comply with NIST SP 800-53 (Recommended Security Controls for Federal Information Systems) and consist of one or more of the following:

(1) Local area,IEEE 802.3 Fast Ethernet 100 BASE-TX, star topology network based on TCP/IP.

(2) Direct-connected, RS-232 cable from the COM port of the Central Station to the first Controller, then RS-485 to interconnect the remainder of the Controllers at that Location.

1.2 RELATED WORK

a. Section 01 00 00, - GENERAL REQUIREMENTS. For General Requirements.

b. Section 07 84 00, - FIRESTOPPING. Requirements for firestopping application and use.

c. Section 08 11 73, - SLIDING METAL FIRE DOORS. Requirements for door installation.

d. Section 08 34 59, - VAULT DOORS AND DAY GATES. Requirements for door and gate installation.

e. Section 08 35 13.13, - ACCORDIAN FOLDING DOORS. Requirements for door installation.

f. Section 08 71 00, - DOOR HARDWARE. Requirements for door installation.

g. Section 10 14 00, - SIGNAGE. Requirements for labeling and signs.

h. Section 14 21 00, ELECTRIC TRACTION ELEVATORS. Requirements for elevators.

i. Section 14 24 00, - HYDRAULIC ELEVATORS. Requirements for elevators.

j. Section 26 05 11, - REQUIREMENTS FOR ELECTRICAL INSTALLATIONS. Requirements for connection of high voltage.

k. Section 26 05 21, - LOW VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES (600 VOLTS AND BELOW). Requirements for power cables.

l. Section 26 05 33, – RACEWAYS AND BOXES FOR ELECTRICAL SYSTEMS. Requirements for infrastructure.

m. Section 26 05 41, - UNDERGROUND ELECTRICAL CONSTRUCTION. Requirements for underground installation of wiring.

n. Section 26 56 00, - EXTERIOR LIGHTING. Requirements for perimeter lighting.

o. Section 28 05 00, - COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY. For general requirements that are common to more than one section in Division 28.

p. Section 28 05 13, - CONDUCTORS AND CABLES FOR ELECTRONIC SAFETY AND SECURITY. Requirements for conductors and cables.

q. Section 28 05 26, - GROUNDING AND BONDING FOR ELECTRONIC SAFETY AND SECURITY. Requirements for grounding of equipment.

r. Section 28 05 28.33, - CONDUITS AND BOXES FOR ELECTRONIC SAFETY AND SECURITY. Requirements for infrastructure.

s. Section 28 23 00, - VIDEO SURVEILLANCE. Requirements for security camera systems.

t. Section 28 31 00, - FIRE DETECTION AND ALARM. Requirements for integration with fire detection and alarm system.

1.3 QUALITY ASSURANCE

a. The Contractor shall be responsible for providing, installing, and the

operation of the PACS as shown. The Contractor shall also provide certification as required.

b. The security system will be installed and tested to ensure all components are fully compatible as a system and can be integrated with all associated security subsystems, whether the security system is stand-alone or a part of a complete Information Technology (IT) computer network.

c. Manufacturers Qualifications: The manufacturer shall regularly and presently produce, as one of the manufacturer's principal products, the equipment and material specified for this project, and shall have manufactured the item for at least three years.

d. Product Qualifications:

(1) Manufacturer's product shall have been in satisfactory operation, on three installations of similar size and type as this project, for approximately three years.

(2) The Government reserves the right to require the Contractor to submit a list of installations where the products have been in operation before approval.

e. Contractor Qualifications:

(1) The Contractor or security sub-contractor shall be a licensed security Contractor with a minimum of five (5) years experience installing and servicing systems of similar scope and complexity. The Contractor shall be an authorized regional representative of the Security Management System’s (PACS) manufacturer. The Contractor shall provide four (4) current references from clients with systems of similar scope and complexity which became operational in the past three (3) years. At least three (3) of the references shall be utilizing the same system components, in a similar configuration as the proposed system. The references must include a current point of contact, company or agency name, address, telephone number, complete system description, date of completion, and approximate cost of the project. The owner reserves the option to visit the reference sites, with the site owner’s permission and representative, to verify the quality of installation and the references’ level of satisfaction with the system. The Contractor shall provide copies of system manufacturer certification for all technicians. The Contractor shall only utilize factory-trained technicians to install, program, and service the PACS. The Contractor shall only utilize factory-trained technicians to install, terminate and service controller/field panels and reader modules. The technicians shall have a minimum of five (5) continuous years of technical experience in electronic security systems. The Contractor shall have a local service facility. The facility shall be located within 60 miles of the project site. The local facility shall include sufficient spare parts inventory to support the service requirements associated with this contract. The facility shall also include appropriate diagnostic equipment to perform diagnostic procedures. The Resident Engineer reserves the option of surveying the company’s facility to verify the service inventory and presence of a local service organization.

(a) The Contractor shall provide proof project superintendent with BICSI Certified Commercial Installer Level 1, Level 2, or Technician to provide oversight of the project.

(b) Cable installer must have on staff a Registered Communication Distribution Designer (RCDD) certified by Building Industry Consulting Service International. The staff member shall provide consistent oversight of the project cabling throughout design, layout, installation, termination and testing.

f. Service Qualifications: There shall be a permanent service organization maintained or trained by the manufacturer which will render satisfactory service to this installation within four hours of receipt of notification that service is needed. Submit name and address of service organizations.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Manufacturers Qualifications; G

Product Qualifications; G

Contractor Qualifications; G

SD-02 Shop Drawings

Shop Drawing And As-Built Packages; G

SD-03 Product Data

Manufacturer Security System Product Cut-Sheets; G

SD-05 Design Data

Pre-Installation And As-Built Design Package; G

Pre-Installation Design Packages; G

Group 1 Technical Data Package; G

Group II Technical Data Package; G

Group III Technical Data Package; G

Group IV Technical Data Package; G

Group V Technical Data Package; G

SD-06 Test Reports

Completed System Readiness Checklists; G

SD-07 Certificates

Certificates Of Compliance; G

Manufacture’s Certification; G

Nist Fips; 200 Compliance Certificates; G

1.5 SUBMITTAL INSTRUCTIONS

a. Submit below items in conjunction with Master Specification Sections 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES, Section 02 41 00, DEMOLITION, and Section 28 05 00, COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY.

b. Provide certificates of compliance with Section 1.3, Quality Assurance.

c. Provide a complete and thorough pre-installation and as-built design package in both electronic format and on paper, minimum size 48 x 48 inches ; drawing submittals shall be per the established project schedule.

d. Shop drawing and as-built packages shall include, but not be limited to:

(1) Index Sheet that shall:

(a) Define each page of the design package to include facility name, building name, floor, and sheet number.

(b) Provide a complete list of all security abbreviations and symbols.

(c) Reference all general notes that are utilized within the design package.

(d) Specification and scope of work pages for all individual security systems that are applicable to the design package that will:

1. Outline all general and job specific work required within the design package.

2. Provide a detailed device identification table outlining device Identification (ID) and use for all security systems equipment utilized in the design package.

(2) Drawing sheets that will be plotted on the individual floor plans or site plans shall:

(a) Include a title block as defined above.

(b) Clearly define the drawings scale in both standard and metric measurements.

(c) Provide device identification and location.

(d) Address all signal and power conduit runs and sizes that are associated with the design of the electronic security system and other security elements (e.g., barriers, etc.).

(e) Identify all pull box and conduit locations, sizes, and fill capacities.

(f) Address all general and drawing specific notes for a particular drawing sheet.

(3) A detailed riser drawing for each applicable security subsystem shall:

(a) Indicate the sequence of operation.

(b) Relationship of integrated components on one diagram.

(c) Include the number, size, identification, and maximum lengths of interconnecting wires.

(d) Wire/cable types shall be defined by a wire and cable schedule. The schedule shall utilize a lettering system that will correspond to the wire/cable it represents (example: A = 18 AWG/1 Pair Twisted, Unshielded). This schedule shall also provide the manufacturer’s name and part number for the wire/cable being installed.

(4) A detailed system drawing for each applicable security system shall:

(a) Clearly identify how all equipment within the system, from main panel to device, shall be laid out and connected.

(b) Provide full detail of all system components wiring from point-to-point.

(c) Identify wire types utilized for connection, interconnection with associate security subsystems.

(d) Show device locations that correspond to the floor plans.

(e) All general and drawing specific notes shall be included with the system drawings.

(5) A detailed schedule for all of the applicable security subsystems shall be included. All schedules shall provide the following information:

(a) Device ID.

(b) Device Location (e.g. site, building, floor, room number, location, and description).

(c) Mounting type (e.g. flush, wall, surface, etc.).

(d) Power supply or circuit breaker and power panel number.

(e) In addition, for the PACS, provide the door ID, door type (e.g. wood or metal), locking mechanism (e.g. strike or electromagnetic lock) and control device (e.g. card reader or biometrics).

(6) Detail and elevation drawings for all devices that define how they were installed and mounted.

e. Pre-installation design packages shall go through a full review process conducted by the Contractor along with a VA representative to ensure all work has been clearly defined and completed. All reviews shall be conducted in accordance with the project schedule. There shall be four (4) stages to the review process:

(1) 35 percent

(2) 65 percent

(3) 90 percent

(4) 100 percent

f. Provide manufacturer security system product cut-sheets. Submit for approval at least 30 days prior to commencement of formal testing, a Security System Operational Test Plan. Include procedures for operational testing of each component and security subsystem, to include performance of an integrated system test.

g. Submit manufacture’s certification of Underwriters Laboratories, Inc. (UL) listing as specified. Provide all maintenance and operating manuals per Section 01 00 00, GENERAL REQUIREMENTS, and Section 28 05 00COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY.

h. Completed System Readiness Checklists provided by the Commissioning Agent and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 28 08 00 COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS.

i. General: Submittals shall be in full compliance of the Contract Documents. All submittals shall be provided in accordance with this section. Submittals lacking the breath or depth these requirements will be considered incomplete and rejected. Submissions are considered multidisciplinary and shall require coordination with applicable divisions to provide a complete and comprehensive submission package. Additional general provisions are as follows:

(1) The Contractor shall schedule submittals in order to maintain the project schedule. For coordination drawings refer to Specification Section 01 33 10, DESIGN SUBMITTAL PROCEDURES, which outline basic submittal requirements and coordination. Section 01 33 10 shall be used in conjunction with this section.

(2) The Contractor shall identify variations from requirements of Contract Documents and state product and system limitations, which may be detrimental to successful performance of the completed work or system.

(3) Each package shall be submitted at one (1) time for each review and include components from applicable disciplines (e.g., electrical work, architectural finishes, door hardware, etc.) which are required to produce an accurate and detailed depiction of the project.

(4) Manufacturer’s information used for submittal shall have pages with items for approval tagged, items on pages shall be identified, and capacities and performance parameters for review shall be clearly marked through use of an arrow or highlighting. Provide space for Resident Engineer and Contractor review stamps.

(5) Technical Data Drawings shall be in the latest version of AutoCAD®, drawn accurately, and in accordance with VA CAD Standards. FREEHAND SKETCHES OR COPIED VERSIONS OF THE CONSTRUCTION DOCUMENTS WILL NOT BE ACCEPTED. The Contractor shall not reproduce Contract Documents or copy standard information as the basis of the Technical Data Drawings. If departures from the technical data drawings are subsequently deemed necessary by the Contractor, details of such departures and the reasons thereof shall be submitted in writing to the Resident Engineer for approval before the initiation of work.

(6) Packaging: The Contractor shall organize the submissions according to the following packaging requirements.

(a) Binders: For each manual, provide heavy duty, commercial quality, durable three (3) ring vinyl covered loose leaf binders, sized to receive 8.5 x 11 in paper, and appropriate capacity to accommodate the contents. Provide a clear plastic sleeve on the spine to hold labels describing the contents. Provide pockets in the covers to receive folded sheets.

1. Where two (2) or more binders are necessary to accommodate data, correlate data in each binder into related groupings according to the Project Manual table of contents. Cross-referencing other binders where necessary to provide essential information for communication of proper operation and or maintenance of the component or system.

2. Identify each binder on the front and spine with printed binder title, Project title or name, and subject matter covered. Indicate the volume number if applicable.

(b) Dividers: Provide heavy paper dividers with celluloid tabs for each Section. Mark each tab to indicate contents.

(c) Protective Plastic Jackets: Provide protective transparent plastic jackets designed to enclose diagnostic software for computerized electronic equipment.

(d) Text Material: Where written material is required as part of the manual use the manufacturer's standard printed material, or if not available, specially prepared data, neatly typewritten on 8.5 inches by 11 inches 20 pound white bond paper.

(e) Drawings: Where drawings and/or diagrams are required as part of the manual, provide reinforced punched binder tabs on the drawings and bind them with the text.

1. Where oversized drawings are necessary, fold the drawings to the same size as the text pages and use as a foldout.

2. If drawings are too large to be used practically as a foldout, place the drawing, neatly folded, in the front or rear

pocket of the binder. Insert a type written page indicating the drawing title, description of contents and drawing location at the appropriate location of the manual.

3. Drawings shall be sized to ensure details and text is of legible size. Text shall be no less than 1/16" tall.

(f) Manual Content: In each manual include information specified in the individual Specification section, and the following information for each major component of building equipment and controls:

1. General system or equipment description.

2. Design factors and assumptions.

3. Copies of applicable Shop Drawings and Product Data.

4. System or equipment identification including: manufacturer, model and serial numbers of each component, operating instructions, emergency instructions, wiring diagrams, inspection and test procedures, maintenance procedures and schedules, precautions against improper use and maintenance, repair instructions, sources of required maintenance materials and related services, and a manual index.

(g) Binder Organization: Organize each manual into separate sections for each piece of related equipment. At a minimum, each manual shall contain a title page, table of contents, copies of Product Data supplemented by drawings and written text, and copies of each warranty, bond, certifications, and service Contract issued. Refer to Group I through V Technical Data Package Submittal requirements for required section content.

(h) Title Page: Provide a title page as the first sheet of each manual to include the following information; project name and address, subject matter covered by the manual, name and address of the Project, date of the submittal, name, address, and telephone number of the Contractor, and cross references to related systems in other operating and/or maintenance manuals.

(i) Table of Contents: After the title page, include a type written table of contents for each volume, arranged systematically according to the Project Manual format. Provide a list of each product included, identified by product name or other appropriate identifying symbols and indexed to the content of the volume. Where more than one (1) volume is required to hold data for a particular system, provide a comprehensive table of contents for all volumes in each volume of the set.

(j) General Information Section: Provide a general information section immediately following the table of contents, listing each product included in the manual, identified by product name. Under each product, list the name, address, and telephone number of the installer and maintenance Contractor. In addition, list a local source for replacement parts and equipment.

(k) Drawings: Provide specially prepared drawings where necessary to supplement the manufacturers printed data to illustrate the

relationship between components of equipment or systems, or provide control or flow diagrams. Coordinate these drawings with information contained in Project Record Drawings to assure correct illustration of the completed installation.

(l) Manufacturer’s Data: Where manufacturer's standard printed data is included in the manuals, include only those sheets that are pertinent to the part or product installed. Mark each sheet to identify each part or product included in the installation. Where more than one (1) item in tabular format is included, identify each item, using appropriate references from the Contract Documents. Identify data that is applicable to the installation and delete references to information which is not applicable.

(m) Where manufacturer's standard printed data is not available and the information is necessary for proper operation and maintenance of equipment or systems, or it is necessary to provide additional information to supplement the data included in the manual, prepare written text to provide the necessary information. Organize the text in a consistent format under a separate heading for different procedures. Where necessary, provide a logical sequence of instruction for each operating or maintenance procedure. Where similar or more than one product is listed on the submittal the Contractor shall differentiate by highlighting the specific product to be utilized.

(n) Calculations: Provide a section for circuit and panel calculations.

(o) Loading Sheets: Provide a section for DGP Loading Sheets.

(p) Certifications: Provide section for Contractor’s manufacturer certifications.

(7) Contractor Review: Review submittals prior to transmittal. Determine and verify field measurements and field construction criteria. Verify manufacturer’s catalog numbers and conformance of submittal with requirements of contract documents. Return non-conforming or incomplete submittals with requirements of the work and contract documents. Apply Contractor’s stamp with signature certifying the review and verification of products occurred, and the field dimensions, adjacent construction, and coordination of information is in accordance with the requirements of the contract documents.

(8) Resubmission: Revise and resubmit submittals as required within 15 calendar days of return of submittal. Make resubmissions under procedures specified for initial submittals. Identify all changes made since previous submittal.

(9) Product Data: Within 15 calendar days after execution of the contract, the Contractor shall submit for approval a complete list of all of major products proposed for use. The data shall include name of manufacturer, trade name, model number, the associated contract document section number, paragraph number, and the referenced standards for each listed product.

j. Group 1 Technical Data Package: Group I Technical Data Package shall be one submittal consisting of the following content and

organization. Refer to VA Special Conditions Document for drawing format and content requirements. The data package shall include the following:

Section I - Drawings:

(a) General – Drawings shall conform to VA Special Conditions and CAD Standards Documents. All text associated with security details shall be 1/8″ tall and meet VA text standard for AutoCAD™ drawings.

(b) Cover Sheet – Cover sheet shall consist of Project Title and Address, Project Number, Area and Vicinity Maps.

(c) General Information Sheets – General Information Sheets shall consist of General Notes, Abbreviations, Symbols, Wire and Cable Schedule, Project Phasing, and Sheet Index.

(d) Floor Plans – Floor plans shall be produced from the Architectural backgrounds issued in the Construction Documents. The contractor shall receive floor plans from the prime A/E to develop these drawing sets. Security devices shall be placed on drawings in scale. All text associated with security details shall be 1/8″ tall and meet VA text standard for AutoCAD™ drawings. Floor plans shall identify the following:

1. security devices by symbol,

2. the associated device point number (derived from the loading sheets),

3. wire & cable types and counts

4. conduit sizing and routing

5. conduit riser systems

6. device and area detail call outs

(e) Architectural details – Architectural details shall be produced for each device mounting type (door details for doors with physical access control, reader pedestals and mounts, security panel and power supply details).

(f) Riser Diagrams – Contractor shall provide a riser diagram indicating riser architecture and distribution of the physical access control system throughout the facility (or area in scope).

(g) Block Diagrams – Contractor shall provide a block diagram for the entire system architecture and interconnections with SMS subsystems. Block diagram shall identify SMS subsystem (e.g., physical access control, intrusion detection, closed circuit television, intercom, and other associated subsystems) integration; and data transmission and media conversion methodologies.

(h) Interconnection Diagrams – Contractor shall provide interconnection diagram for each sensor, and device component. Interconnection diagram shall identify termination locations,

standard wire detail to include termination schedule. Diagram shall also identify interfaces to other systems such as elevator control, fire alarm systems, and security management systems.

(i) Security Details:

1. Panel Assembly Detail – For each panel assembly, a panel assembly details shall be provided identifying individual panel component size and content.

2. Panel Details – Provide security panel details identify general arrangement of the security system components, backboard size, wire through size and location, and power circuit requirements.

3. Device Mounting Details – Provide mounting detailed drawing for each security device (physical access control system, intrusion detection, video surveillance and assessment, and intercom systems) for each type of wall and ceiling configuration in project. Device details shall include device, mounting detail, wiring and conduit routing.

4. Details of connections to power supplies and grounding

5. Details of surge protection device installation

6. Sensor detection patterns – Each system sensor shall have associated detection patterns.

7. Equipment Rack Detail – For each equipment rack, provide a scaled detail of the equipment rack location and rack space utilization. Use of BISCI wire management standards shall be employed to identify wire management methodology. Transitions between equipment racks shall be shown to include use vertical and horizontal latter rack system.

8. Security Control Room – The contractor shall provide a layout plan for the Security Control Room. The layout plan shall identify all equipment and details associated with the installation.

9. Operator Console – The contractor shall provide a layout plan for the Operator Console. The layout plan shall identify all equipment and details associated with the installation. Equipment room - the contractor shall provide a layout plan for the equipment room. The layout plan shall identify all equipment and details associated with the installation.

10. Equipment Room – Equipment room details shall provide architectural, electrical, mechanical, plumbing, IT/Data and associated equipment and device placements both vertical and horizontally.

(j) Electrical Panel Schedule – Electrical Panel Details shall be provided for all SMS systems electrical power circuits. Panel details shall be provided identifying panel type (Standard, Emergency Power, Emergency/Uninterrupted Power Source, and Uninterrupted Power Source Only), panel location, circuit number, and circuit amperage rating.

(k) Door Schedule – A door schedule shall be developed for each door equipped with electronic security components. At a minimum, the door schedule shall be coordinated with Division 08 work and include the following information:

1. Item Number

2. Door Number (Derived from A/E Drawings)

3. Floor Plan Sheet Number

4. Standard Detail Number

5. Door Description (Derived from Loading Sheets)

6. Data Gathering Panel Input Number

7. Door Position or Monitoring Device Type & Model Number

8. Lock Type, Model Number & Power Input/Draw (standby/active)

9. Card Reader Type & Model Number

10. Shunting Device Type & Model Number

11. Sounder Type & Model Number

12. Manufacturer

13. Misc. devices as required

(a) Delayed Egress Type & Model Number

(b) Intercom

(c) Camera

(d) Electric Transfer Hinge

(e) Electric Pass-through device

14. Remarks column indicating special notes or door configurations

(2) Camera Schedule - A camera schedule shall be developed for each camera. Contractors shall coordinate with the Resident Engineer to determine camera starting numbers and naming conventions. All drawings shall identify wire and cable standardization methodology. Color coding of all wiring conductors and jackets is required and shall be communicated consistently throughout the drawings package submittal. At a minimum, the camera schedule shall include the following information:

(a) Item Number

(b) Camera Number

(c) Naming Conventions

(d) Description of Camera Coverage

(e) Camera Location

(f) Floor Plan Sheet Number

(g) Camera Type

(h) Mounting Type

(i) Standard Detail Reference

(j) Power Input & Draw

(k) Power Panel Location

(l) Remarks Column for Camera

(3) Section II – Data Gathering Panel Documentation Package

(a) Contractor shall provide Data Gathering Panel (DGP) input and output documentation packages for review at the Shop Drawing submittal stage and also with the as-built documentation package. The documentation packages shall be provided in both printed and magnetic form at both review stages.

(b) The Contractor shall provide loading sheet documentation package for the associated DGP, including input and output boards for all field panels associated with the project. Documentation shall be provided in current version Microsoft Excel spreadsheets following the format currently utilized by VA. A separate spreadsheet file shall be generated for each DGP and associated field panels.

(c) The spreadsheet names shall follow a sequence that shall display the spreadsheets in numerical order according to the DGP system number. The spreadsheet shall include the prefix in the file name that uniquely identifies the project site. The spreadsheet shall detail all connected items such as card readers, alarm inputs, and relay output connections. The spreadsheet shall include an individual section (row) for each panel input, output and card reader. The spreadsheet shall automatically calculate the system numbers for card readers, inputs, and outputs based upon data entered in initialization fields.

(d) All entries must be verified against the field devices. Copies of the floor plans shall be forwarded under separate cover.

(e) The DGP spreadsheet shall include an entry section for the following information:

1. DGP number

2. First Reader Number

3. First Monitor Point Number

4. First Relay Number

5. DGP, input or output Location

6. DGP Chain Number

7. DGP Cabinet Tamper Input Number

8. DGP Power Fail Input Number

9. Number of Monitor Points Reserved For Expansion Boards

10. Number of Control Points (Relays) Reserved For Expansion Boards

(f) The DGP, input module and output module spreadsheets shall automatically calculate the following information based upon the associated entries in the above fields:

1. System Numbers for Card Readers

2. System Numbers for Monitor Point Inputs

3. System Numbers for Control Points (Relays)

4. Next DGP or input module First Monitor Point Number

5. Next DGP or output module First Control Point Number

(g) The DGP spreadsheet shall provide the following information for each card reader:

1. DGP Reader Number

2. System Reader Number

3. Cable ID Number

4. Description Field (Room Number)

5. Description Field (Device Type i.e.: In Reader, Out Reader, etc.)

6. Description Field

7. DGP Input Location

8. Date Test

9. Date Passed

10. Cable Type

11. Camera Numbers (of cameras viewing the reader location)

(h) The DGP and input module spreadsheet shall provide the following information for each monitor point (alarm input).

1. DGP Monitor Point Input Number

2. System Monitor Point Number

3. Cable ID Number

4. Description Field (Room Number)

5. Description Field (Device Type i.e.: Door Contact, Motion Detector, etc.)

6. DGP or input module Input Location

7. Date Test

8. Date Passed

9. Cable Type

10. Camera Numbers (of associated alarm event preset call-ups)

(i) The DGP and output module spreadsheet shall provide the following information for each control point (output relay).

1. DGP Control Point (Relay) Number

2. System (Control Point) Number

3. Cable ID Number

4. Description Field (Room Number)

5. Description Field (Device: Lock Control, Local Sounder, etc.)

6. Description Field

7. DGP or OUTPUT MODULE Output Location

8. Date Test

9. Date Passed Cable Type

10. Camera Number (of associated alarm event preset call-ups)

(j) The DGP, input module and output module spreadsheet shall include the following information or directions in the header and footer:

1. Header

(a) DGP Input and Output Worksheet

(b) Enter Beginning Reader, Input, and Output Starting Numbers and Sheet Will Automatically Calculate the Remaining System Numbers.

2. Footer

(a) File Name

(b) Date Printed

(c) Page Number

(4) Section III - Construction Mock-up: In areas with exposed EMT/Conduit Raceways, contractor shall conceal raceway as much as practical and unobtrusively. In addition, historic significance must be considered to determine installation means and methods for approval by the owner.

(5) Section IV - Manufacturers' Data: The data package shall include manufacturers' data for all materials and equipment, including sensors, local processors and console equipment provided under this specification.

(6) Section V - System Description and Analysis: The data package shall include system descriptions, analysis, and calculations used in sizing equipment required by these specifications. Descriptions and calculations shall show how the equipment will operate as a system to meet the performance requirements of this specification. The data package shall include the following:

(a) Central processor memory size; communication speed and protocol description; rigid disk system size and configuration; flexible disk system size and configuration; back-up media size and configuration; alarm response time calculations; command response time calculations; start-up operations; expansion capability and method of implementation; sample copy of each report specified; and color photographs representative of typical graphics.

(b) Software Data: The data package shall consist of descriptions of the operation and capability of the system, and application software as specified.

(c) Overall System Reliability Calculations: The data package shall include all manufacturers' reliability data and calculations required to show compliance with the specified reliability.

(7) Section VI – Certifications & References: All specified manufacturer's certifications shall be included with the data package. Contractor shall provide Project references as outlined in Paragraph 1.4 "Quality Assurance".

k. Group II Technical Data Package

(1) The Contractor shall prepare a report of "Current Site Conditions" and submit a report to the Resident Engineer documenting changes to the site, particularly those conditions that affect performance of the system to be installed. The Contractor shall provide specification sheets, or written functional requirements to support the findings, and a cost estimate to correct those site changes or conditions which affect the installation of the system or its performance. The Contractor shall not correct any deficiency without written permission from the Contracting Officer.

(2) System Configuration and Functionality: The contractor shall provide the results of the meeting with VA to develop system requirements and functionality including but not limited to:

(a) Baseline configuration

(b) Access levels

(c) Schedules (intrusion detection, physical access control, holidays, etc.)

(d) Badge database

(e) System monitoring and reporting (unit level and central control)

(f) Naming conventions and descriptors

l. Group III Technical Data Package

(1) Development of Test Procedures: The Contractor will prepare performance test procedures for the system testing. The test procedures shall follow the format of the VA Testing procedures and be customized to the contract requirements. The Contractor will deliver the test procedures to the Resident Engineer for approval at least 60 calendar days prior to the requested test date.

m. Group IV Technical Data Package

(1) Performance Verification Test

(a) Based on the successful completion of the pre-delivery test, the Contractor shall finalize the test procedures and report forms for the performance verification test (PVT) and the endurance test. The PVT shall follow the format, layout and content of the pre-delivery test. The Contractor shall deliver the PVT and endurance test procedures to the Resident Engineer for approval. The Contractor may schedule the PVT after receiving written approval of the test procedures. The Contractor shall deliver the final PVT and endurance test reports within 14 calendar days from completion of the tests. Refer to Part 3 of this section for System Testing and Acceptance requirements.

(2) Training Documentation

(a) New Facilities and Major Renovations: Familiarization training shall be provided for new equipment or systems. Training can include site familiarization training for VA technicians and administrative personnel. Training shall include general information on new system layout including closet locations, turnover of the completed system including all documentation, including manuals, software, key systems, and full system administration rights. Lesson plans and training manuals training shall be oriented to type of training to be provided.

(b) New Unit Control Room:

1. Provide the security personnel with training in the use, operation, and maintenance of the entire control room system (Unit Control and Equipment Rooms). The training documentation must include the operation and maintenance. The first of the training

sessions shall take place prior to system turnover and the second immediately after turnover. Coordinate the training sessions with the Owner. Completed classroom sessions will be witnessed and documented by the Architect/Engineer, and approved by the Resident Engineer. Instruction is not to begin until the system is operational as designed.

2. The training documents will cover the operation and the maintenance manuals and the control console operators' manuals and service manuals in detail, stressing all important operational and service diagnostic information necessary for the maintenance and operations personnel to efficiently use and maintain all systems.

3. Provide an illustrated control console operator's manual and service manual. The operator's manual shall be written in laymen's language and printed so as to become a permanent reference document for the operators, describing all control panel switch operations, graphic symbol definitions and all indicating functions and a complete explanation of all software.

4. The service manual shall be written in laymen's language and printed so as to become a permanent reference document for maintenance personnel, describing how to run internal self diagnostic software programs, troubleshoot head end hardware and field devices with a complete scenario simulation of all possible system malfunctions and the appropriate corrective measures.

5. Provide a professional color DVD instructional recording of all the operational procedures described in the operator's manual. All charts used in the training session shall be clearly presented on the video. Any DVD found to be inferior in recording or material content shall be reproduced at no cost until an acceptable DVD is submitted. Provide four copies of the training DVD, one to the architect/engineer and three to the owner.

(3) System Configuration and Data Entry:

(a) The contractor is responsible for providing all system configuration and data entry for the SMS and subsystems (e.g., video matrix switch, intercom, digital video recorders, network video recorders). All data entry shall be performed per VA standards & guidelines. The Contractor is responsible for participating in all meetings with the client to compile the information needed for data entry. These meetings shall be established at the beginning of the project and incorporated in to the project schedule as a milestone task. The contractor shall be responsible for all data collection, data entry, and system configuration. The contractor shall collect, enter, & program and/or configure the following components:

1. Physical Access control system components,

2. All intrusion detection system components,

3. Video surveillance, control and recording systems,

4. Intercom systems components,

5. All other security subsystems shown in the contract

documents.

(b) The Contractor is responsible for compiling the card access database for the VA employees, including programming reader configurations, access shifts, schedules, exceptions, card classes and card enrollment databases.

(c) Refer to Part 3 for system programming requirements and planning guidelines.

(4) Graphics: Based on CAD as-built drawings developed for the construction project, create all map sets showing locations of all alarms and field devices. Graphical maps of all alarm points installed under this contract including perimeter and exterior alarm points shall be delivered with the system. The Contractor shall create and install all graphics needed to make the system operational. The Contractor shall utilize data from the contract documents, Contractor's field surveys, and all other pertinent information in the Contractor's possession to complete the graphics. The Contractor shall identify and request from the Contracting Officer, any additional data needed to provide a complete graphics package. Graphics shall have sufficient level of detail for the system operator to assess the alarm. The Contractor shall supply hard copy, color examples at least (8 x 10 in) of each type of graphic to be used for the completed Security system. The graphics examples shall be delivered to the Resident Engineer for review and approval at least 90 calendar days prior to the scheduled date the Contractor requires them.

n. Group V Technical Data Package: Final copies of the manuals shall be delivered to the Resident Engineer as part of the acceptance test. The draft copy used during site testing shall be updated with any changes required prior to final delivery of the manuals. Each manual's contents shall be identified on the cover. The manual shall include names, addresses, and telephone numbers of each sub-contractor installing equipment or systems, as well as the nearest service representatives for each item of equipment for each system. The manuals shall include a table of contents and tab sheets. Tab sheets shall be placed at the beginning of each chapter or section and at the beginning of each appendix. The final copies delivered after completion of the endurance test shall include all modifications made during installation, checkout, and acceptance. Six (6) hard-copies and one (1) soft copy of each item listed below shall be delivered as a part of final systems acceptance.

(1) Functional Design Manual: The functional design manual shall identify the operational requirements for the entire system and explain the theory of operation, design philosophy, and specific functions. A description of hardware and software functions, interfaces, and requirements shall be included for all system operating modes. Manufacturer developed literature may be used; however, shall be produced to match the project requirements.

(2) Equipment Manual: A manual describing all equipment furnished including:

General description and specifications; installation and checkout procedures; equipment electrical schematics and layout drawings; system schematics and layout drawings; alignment and calibration

procedures; manufacturer’s repair list indicating sources of supply; and interface definition.

(3) Software Manual: The software manual shall describe the functions of all software and include all other information necessary to enable proper loading, testing, and operation. The manual shall include:

Definition of terms and functions; use of system and applications software; procedures for system initialization, start-up, and shutdown; alarm reports; reports generation, database format and data entry requirements; directory of all disk files; and description of all communications protocols including data formats, command characters, and a sample of each type of data transfer.

(4) Operator’s Manual: The operator’s manual shall fully explain all procedures and instructions for the operation of the system, including:

Computers and peripherals; system start-up and shutdown procedures; use of system, command, and applications software; recovery and restart procedures; graphic alarm presentation; use of report generator and generation of reports; data entry; operator commands’ alarm messages, and printing formats; and system access requirements.

(5) Maintenance Manual: The maintenance manual shall include descriptions of maintenance for all equipment including inspection, recommend schedules, periodic preventive maintenance, fault diagnosis, and repair or replacement of defective components.

(6) Spare Parts & Components Data: At the conclusion of the Contractor's work, the Contractor shall submit to the Resident Engineer a complete list of the manufacturer's recommended spare parts and components required to satisfactorily maintain and service the systems, as well as unit pricing for those parts and components.

(7) Operation, Maintenance & Service Manuals: The Contractor shall provide two (2) complete sets of operating and maintenance manuals in the form of an instructional manual for use by the VA Security Guard Force personnel. The manuals shall be organized into suitable sets of manageable size. Where possible, assemble instructions for similar equipment into a single binder. If multiple volumes are required, each volume shall be fully indexed and coordinated.

(8) Equipment and Systems Maintenance Manual: The Contractor shall provide the following descriptive information for each piece of equipment, operating system, and electronic system:

(a) Equipment and/or system function.

(b) Operating characteristics.

(c) Limiting conditions.

(d) Performance curves.

(e) Engineering data and test.

(f) Complete nomenclature and number of replacement parts.

(g) Provide operating and maintenance instructions including assembly drawings and diagrams required for maintenance and a list of items recommended to stock as spare parts.

(h) Provide information detailing essential maintenance procedures including the following: routine operations, trouble shooting guide, disassembly, repair and re-assembly, alignment, adjusting, and checking.

(i) Provide information on equipment and system operating procedures, including the following; start-up procedures, routine and normal operating instructions, regulation and control procedures, instructions on stopping, shut-down and emergency instructions, required sequences for electric and electronic systems, and special operating instructions.

(j) Manufacturer equipment and systems maintenance manuals are permissible.

(9) Project Redlines: During construction, the Contractor shall maintain an up-to-date set of construction redlines detailing current location and configuration of the project components. The redline documents shall be marked with the words ‘Master Redlines’ on the cover sheet and be maintained by the Contractor in the project office. The Contractor will provide access to redline documents anytime during the project for review and inspection by the Resident Engineer or authorized Office of Protection Services representative. Master redlines shall be neatly maintained throughout the project and secured under lock and key in the contractor’s onsite project office. Any project component or assembly that is not installed in strict accordance with the drawings shall be so noted on the drawings. Prior to producing Record Construction Documents, the contractor will submit the Master Redline document to the Resident Engineer for review and approval of all changes or modifications to the documents. Each sheet shall have Resident Engineer initials indicating authorization to produce “As Built” documents. Field drawings shall be used for data gathering & field changes. These changes shall be made to the master redline documents daily. Field drawings shall not be considered “master redlines”.

(10) Record Specifications: The Contractor shall maintain one (1) copy of the Project Specifications, including addenda and modifications issued, for Project Record Documents. The Contractor shall mark the Specifications to indicate the actual installation where the installation varies substantially from that indicated in the Contract Specifications and modifications issued. (Note related Project Record Drawing information where applicable). The Contractor shall pay particular attention to substitutions, selection of product options, and information on concealed installations that would be difficult to identify or measure and record later. Upon completion of the mark ups, the Contractor shall submit record Specifications to the Contracting Officer. As with master relines, Contractor shall maintain record

specifications for Resident Engineer review and inspection at anytime.

(11) Record Product Data: The Contractor shall maintain one (1) copy of each Product Data submittal for Project Record Document purposes. The Data shall be marked to indicate the actual product installed where the installation varies substantially from that indicated in the Product Data submitted. Significant changes in the product delivered to the site and changes in manufacturer's instructions and recommendations for installation shall be included. Particular attention will be given to information on concealed products and installations that cannot be readily identified or recorded later. Note related Change Orders and mark up of Record Construction Documents, where applicable. Upon completion of mark up, submit a complete set of Record Product Data to the Contracting Officer.

(12) Miscellaneous Records: The Contractor shall maintain one (1) copy of miscellaneous records for Project Record Document purposes. Refer to other Specifications for miscellaneous record-keeping requirements and submittals concerning various construction activities. Before substantial completion, complete miscellaneous records and place in good order, properly identified and bound or filed, ready for use and reference. Categories of requirements resulting in miscellaneous records include, a minimum of the following:

(a) Certificates received instead of labels on bulk products.

(b) Testing and qualification of tradesmen. (“Contractor’s Qualifications”)

(c) Documented qualification of installation firms.

(d) Load and performance testing.

(e) Inspections and certifications.

(f) Final inspection and correction procedures.

(g) Project schedule

(13) Record Construction Documents (Record As-Built)

(a) Upon project completion, the contractor shall submit the project master redlines to the Resident Engineer prior to development of Record construction documents. The Resident Engineer shall be given a minimum of a thirty (30) day review period to determine the adequacy of the master redlines. If the master redlines are found suitable by the Resident Engineer, the Resident Engineer will initial and date each sheet and turn redlines over to the contractor for as built development.

(b) The Contractor shall provide the Resident Engineer a complete set of "as-built" drawings and original master redlined marked "as-built" blue-line in the latest version of AutoCAD drawings unlocked on CD or DVD. The as-built drawing shall include security device number, security closet connection location, data gathering panel number, and input or output number as applicable.

All corrective notations made by the Contractor shall be legible when submitted to the Contracting Officer. If, in the opinion of the Contracting Officer, any redlined notation is not legible, it shall be returned to the Contractor for re-submission at no extra cost to the Owner. The Contractor shall organize the Record Drawing sheets into manageable sets bound with durable paper cover sheets with suitable titles, dates, and other identifications printed on the cover. The submitted as built shall be in editable formats and the ownership of the drawings shall be fully relinquished to the owner.

(c) Where feasible, the individual or entity that obtained record data, whether the individual or entity is the installer, sub-contractor, or similar entity, is required to prepare the mark up on Record Drawings. Accurately record the information in a comprehensive drawing technique. Record the data when possible after it has been obtained. For concealed installations, record and check the mark up before concealment. At the time of substantial completion, submit the Record Construction Documents to the Contracting Officer. The Contractor shall organize into bound and labeled sets for the Contracting Officer's continued usage. Provide device, conduit, and cable lengths on the conduit drawings. Exact in-field conduit placement/routings shall be shown. All conduits shall be illustrated in their entire length from termination in security closets; no arrowed conduit runs shall be shown. Pull box and junction box sizes are to be shown if larger than 100mm (4 inch).

o. NIST FIPS 200 Compliance Certificates

(1) Provide Certificates for all software components and device types utilizing credential verification. Provide certificates for:

(a) Fingerprint Capture Station

(b) Card Readers

(c) Facial Image Capturing Camera

(d) PIV Middelware

(e) Template Matcher

(f) Electromagnetically Opaque Sleeve

(g) Certificate Management

1. CAK Authentication System

2. PIV Authentication System

3. Certificate Validator

4. Cryptographic Module

(h) <list devices and software>

p. Approvals will be based on complete submission of manuals together with shop drawings.

q. Completed System Readiness Checklists provided by the Commissioning Agent and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 28 08 00 COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS.

1.6 REFERENCES

a. The publications listed below (including amendments, addenda, revisions, supplement, and errata) form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

SECURITY INDUSTRY ASSOCIATION (SIA)

ANSI/SIA CP-01	(2019) Security System Standard - Features for False Alarm Reduction

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-568.1	(2020e) Commercial Building Telecommunications Infrastructure Standard
TIA-569	(2019e; Add 1 2022) Telecommunications Pathways and Spaces
TIA-606	(2021d) Administration Standard for Telecommunications Infrastructure

INTERNATIONAL CODE COUNCIL (ICC)

ICC A117.1	(2017) Standard And Commentary Accessible and Usable Buildings and Facilities

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 101	(2024) Life Safety Code

OFFICE OF MANAGEMENT AND BUDGET (OMB)

M-04-04	E-Authentication Guidance for Federal Agencies

UL SOLUTIONS (UL)

UL 294	(2023) UL Standard for Safety Access Control System Units
UL 1610	(2016; Reprint Apr 2021) UL Standard for Safety Central-Station Burglar-Alarm Units

U.S. DEPARTMENT OF HOMELAND SECURITY (DHS)

HSPD-12	(2004) Homeland Security Presidential Directive 12: Policy for a Common

	Identification Standard for Federal Employees and Contractors

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST FIPS 140-2	(2001) Security Requirements for Cryptographic Modules
NIST FIPS 200	Minimum Security Requirements for Federal Information and Information Systems
NIST FIPS 201-2	(2013) Personal Identity Verification (PIV) of Federal Employees and Contractors
NIST SP 800-53	Security and Privacy Controls for Information Systems and Organizations
NIST SP 800-63	(2023) Electronic Authentication Guideline
NIST SP 800-73-3	Interfaces for Personal Identity Verification (4 Parts)
NIST SP 800-73-4	(2016) Interfaces for Personal Identity Verification (4 Parts)
NIST SP 800-104A	Scheme for PIV Visual Card Topography
NIST SP 800-116	(2018; Rev 1) Guidelines for the Use of PIV Credentials in Facility Access

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.3	(2022; CORR 2024) Ethernet
IEEE 1100	(2005) Emerald Book IEEE Recommended Practice for Powering and Grounding Electronic Equipment
IEEE C2	(2023) National Electrical Safety Code

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ANSI ISO/IEC 7816	(R 2009) Identification Cards - Integrated Circuit Cards
ISO 14443	Identification Cards - Contactless Integrated Circuit Cards - Proximity Cards
ISO 15693	Identification/Contactless/Vicinity/RFID Cards Standard

U.S. DEPARTMENT OF JUSTICE (DOJ)

UFAS	(1984) Uniform Federal Accessibility Standards
ADA	ADA Standards for Accessible Design

1.7 DEFINITIONS

a. ABA Track: Magnetic stripe that is encoded on track 2, at 75-bpi density in binary-coded decimal format; for example, 5-bit, 16-character set.

b. Access Control List: A list of (identifier, permissions) pairs associated with a resource or an asset. As an expression of security policy, a person may perform an operation on a resource or asset if and only if the person’s identifier is present in the access control list (explicitly or implicitly), and the permissions in the (identifier, permissions) pair include the permission to perform the requested operation.

c Access Control: A function or a system that restricts access to authorized persons only.

d. API Application Programming Interface

e. Assurance Level (or E-Authentication Assurance Level): A measure of trust or confidence in an authentication mechanism defined in OMB Memorandum M-04-04 and NIST SP 800-63, in terms of four levels: M-04-04

(1) Level 1: LITTLE OR NO confidence

(2) Level 2: SOME confidence

(3) Level 3: HIGH confidence

(4) Level 4: VERY HIGH confidence

f. Authentication: A process that establishes the origin of information, or determines an entity’s identity. In this publication, authentication often means the performance of a PIV authentication mechanism.

g. Authenticator: A memory, possession, or quality of a person that can serve as proof of identity, when presented to a verifier of the appropriate kind. For example, passwords, cryptographic keys, and fingerprints are authenticators.

h. Authorization: A process that associates permission to access a resource or asset with a person and the person’s identifier(s).

i. BIO or BIO-A: A NIST FIPS 201-2 authentication mechanism that is implemented by using a Fingerprint data object sent from the PIV Card to the PACS. Note that the short-hand “BIO (-A)” is used throughout the document to represent both BIO and BIO-A authentication mechanisms.

j. Biometric: An authenticator produced from measurable qualities of a living person.

k. CAC EP – CAC End Point with end point PIV applet

l. CAC NG – CAC Next Generation with transitional PIV applet

m. Card Authentication Key (CAK): A PIV authentication mechanism (or the PIV Card key of the same name) that is implemented by an asymmetric or

symmetric key challenge/response protocol. The CAK is an optional mechanism defined in NIST SP 800-73-3. NIST SP 800-73-4 NIST strongly recommends that every PIV Card contain an asymmetric CAK and corresponding certificate, and that agencies use the asymmetric CAK protocol, rather than a symmetric CAK protocol, whenever the CAK authentication mechanism is used with PACS.

n. CCTV: Closed-circuit television.

o. Central Station: A PC with software designated as the main controlling PC of the PACS. Where this term is presented with initial capital letters, this definition applies.

p. Controller: An intelligent peripheral control unit that uses a computer for controlling its operation. Where this term is presented with an initial capital letter, this definition applies.

q. CPU: Central processing unit.

r. Credential: Data assigned to an entity and used to identify that entity.

s. File Server: A PC in a network that stores the programs and data files shared by users.

t. FIPS Federal Information Processing Standards

u. FRAC – First Responder Authentication Credential

v. HSPD Homeland Security Presidential Directive

w. I/O: Input/Output.

x. Identifier: A credential card, keypad personal identification number or code, biometric characteristic, or other unique identification entered as data into the entry-control database for the purpose of identifying an individual. Where this term is presented with an initial capital letter, this definition applies.

y. IEC International Electrotechnical Commission

z. ISO International Organization for Standardization

aa. KB Kilobyte

ab. kbit/s Kilobits / second

ac. LAN: Local area network.

ad. LED: Light-emitting diode.

ae. Legacy CAC – Contact only Common Access Card with v1 and v2 applets

af. Location: A Location on the network having a PC-to-Controller communications link, with additional Controllers at the Location connected to the PC-to-Controller link with RS-485 communications loop. Where this term is presented with an initial capital letter, this definition applies.

ag. NIST: National Institute of Standards and Technology

ah. PACS: Physical Access Control System

ai. PC/SC: Personal Computer / Smart Card

aj. PC: Personal computer. This acronym applies to the Central Station, workstations, and file servers.

ak. PCI Bus: Peripheral component interconnect; a peripheral bus providing a high-speed data path between the CPU and peripheral devices (such as monitor, disk drive, or network).

al. PDF: (Portable Document Format.) The file format used by the Acrobat document exchange system software from Adobe.

am. PIV: Personal Identification Verification

an. PIV-I – PIV Interoperable credential

ao. PPS: Protocol and Parameters Selection

ap. RF: Radio frequency.

aq. ROM: Read-only memory. ROM data are maintained through losses of power.

ar. RS-232: An TIA/EIA standard for asynchronous serial data communications between terminal devices. This standard defines a 25-pin connector and certain signal characteristics for interfacing computer equipment.

as. RS-485: An TIA/EIA standard for multipoint communications.

at. TCP/IP: Transport control protocol/Internet protocol incorporated into Microsoft Windows.

au. TPDU: Transport Protocol Data Unit

av. TWIC – Transportation Worker Identification Credential

aw. UPS: Uninterruptible power supply.

ax. Vcc: Voltage at the Common Collector

ay. WAN: Wide area network.

az. WAV: The digital audio format used in Microsoft Windows.

ba. Wiegand: Patented magnetic principle that uses specially treated wires embedded in the credential card.

bb. Windows: Operating system by Microsoft Corporation.

bc. Workstation: A PC with software that is configured for speciflimited security system functions.

1.8 COORDINATION

a. Coordinate arrangement, mounting, and support of electronic safety and security equipment:

(1) To allow maximum possible headroom unless specific mounting heights that reduce headroom are indicated.

(2) To provide for ease of disconnecting the equipment with minimum interference to other installations.

(3) To allow right of way for piping and conduit installed at required slope.

(4) So connecting raceways, cables, wireways, cable trays, and busways will be clear of obstructions and of the working and access space of other equipment.

b. Coordinate installation of required supporting devices and set sleeves in cast-in-place concrete, masonry walls, and other structural components as they are constructed.

c. Coordinate location of access panels and doors for electronic safety and security items that are behind finished surfaces or otherwise concealed.

1.9 MAINTENANCE & SERVICE

a. General Requirements

(1) The Contractor shall provide all services required and equipment necessary to maintain the entire integrated electronic security system in an operational state as specified for a period of one (1) year after formal written acceptance of the system. The Contractor shall provide all necessary material required for performing scheduled adjustments or other non-scheduled work. Impacts on facility operations shall be minimized when performing scheduled adjustments or other non-scheduled work. See also General Project Requirements.

b. Description of Work

The adjustment and repair of the security system includes all software updates, panel firmware, and the following new items computers equipment, communications transmission equipment and data transmission media (DTM), local processors, security system sensors, physical access control equipment, facility interface, signal transmission equipment, and video equipment.

c. Personnel

Service personnel shall be certified in the maintenance and repair of the selected type of equipment and qualified to accomplish all work promptly and satisfactorily. The Resident Engineer shall be advised in writing of the name of the designated service representative, and of any change in personnel. The Resident Engineer shall be provided copies of system manufacturer certification for the designated service representative.

d. Schedule of Work

The work shall be performed during regular working ours, Monday through Friday, excluding federal holidays. These inspections shall include:

(a) The Contractor shall perform two (2) minor inspections at six (6) month intervals or more if required by the manufacturer, and two (2) major inspections offset equally between the minor inspections to effect quarterly inspection of alternating magnitude.

1. Minor Inspections shall include visual checks and operational tests of all console equipment, peripheral equipment, local processors, sensors, electrical and mechanical controls, and adjustments on printers.

2. Major Inspections shall include all work described for Minor Inspections and the following: clean all system equipment and local processors including interior and exterior surfaces; perform diagnostics on all equipment; operational tests of the CPU, switcher, peripheral equipment, recording devices, monitors, picture quality from each camera; check, walk test, and calibrate each sensor; run all system software diagnostics and correct all problems; and resolve any previous outstanding problems.

e. Emergency Service

(1) The owner shall initiate service calls whenever the system is not functioning properly. The Contractor shall provide the Owner with an emergency service center telephone number. The emergency service center shall be staffed 24 hours a day 365 days a year. The Owner shall have sole authority for determining catastrophic and non-catastrophic system failures within parameters stated in General Project Requirements.

(a) For catastrophic system failures, the Contractor shall provide same day four (4) hour service response with a defect correction time not to exceed eight (8) hours from notification. Catastrophic system failures are defined as any system failure that the Owner determines will place the facility(s) at increased risk.

(b) For non-catastrophic failures, the Contractor within eight (8) hours with a defect correction time not to exceed 24 hours from notification.

f. Operation

Performance of scheduled adjustments and repair shall verify operation of the system as demonstrated by the applicable portions of the performance verification test.

g. Records & Logs

The Contractor shall maintain records and logs of each task and organize cumulative records for each component and for the complete system chronologically. A continuous log shall be submitted for all devices. The log shall contain all initial

settings, calibration, repair, and programming data. Complete logs shall be maintained and available for inspection on site, demonstrating planned and systematic adjustments and repairs have been accomplished for the system.

h. Work Request

The Contractor shall separately record each service call request, as received. The record shall include the serial number identifying the component involved, its location, date and time the call was received, specific nature of trouble, names of service personnel assigned to the task, instructions describing the action taken, the amount and nature of the materials used, and the date and time of commencement and completion. The Contractor shall deliver a record of the work performed within five (5) working days after the work was completed.

i. System Modifications

The Contractor shall make any recommendations for system modification in writing to the Contracting Officer. No system modifications, including operating parameters and control settings, shall be made without prior written approval from the Contracting Officer. Any modifications made to the system shall be incorporated into the operation and maintenance manuals and other documentation affected.

j. Software

The Contractor shall provide all software updates when approved by the Owner from the manufacturer during the installation and 12-month warranty period and verify operation of the system. These updates shall be accomplished in a timely manner, fully coordinated with the system operators, and incorporated into the operations and maintenance manuals and software documentation. There shall be at least one (1) scheduled update near the end of the first year’s warranty period, at which time the Contractor shall install and validate the latest released version of the Manufacturer’s software. All software changes shall be recorded in a log maintained in the unit control room. An electronic copy of the software update shall be maintained within the log. At a minimum, the contractor shall provide a description of the modification, when the modification occurred, and name and contact information of the individual performing the modification. The log shall be maintained in a white 3 ring binder and the cover marked “SOFTWARE CHANGE LOG”.

1.10 PERFORMANCE REQUIREMENTS

a. PACS shall provide support for multiple authentication modes and bidirectional communication with the reader. PACS shall provide implementation capability for enterprise security policy and incident response.

b. All processing of authentication information must occur on the “safe side” of a door

c. Physical Access Control System shall provide access to following Security Areas:

(1) Controlled

(2) Limited

(3) Exclusion

d. PACS shall provide:

(1) One authentication factor for access to Controlled security areas

(2) Two authentication factors for access to Limited security areas

(3) Three authentication factors for access to Exclusion security areas

e. PACS shall provide Credential Validation and Path Validation per NIST SP 800-116.

f. The PACS System shall have an Enterprise Path Validation Module (PVM) component that processes X.509 certification paths composed of X.509 v3 certificates and X.509 v2 CRLs. The PVM component MUST support the following features:

(1) Name chaining;

(2) Signature chaining;

(3) Certificate validity;

(4) Key usage, basic constraints, and certificate policies certificate extensions;

(5) Full CRLs; and

(6) CRLs segmented on names.

g. Distributed Processing: System shall be a fully distributed processing system so that information, including time, date, valid codes, access levels, and similar data, is downloaded to Controllers so that each Controller makes access-control decisions for that Location. Do not use intermediate Controllers for physical access control. If communications to Central Station are lost, all Controllers shall automatically buffer event transactions until communications are restored, at which time buffered events shall be uploaded to the Central Station.

h. Number of Locations: Support unlimited number of separate Locations using a single PC with combinations of direct-connect, dial-up, or TCP/IP LAN connections to each Location.

Each Location shall have its own database and history in the Central Station. Locations may be combined to share a common database.

i. System Network Requirements:

(1) Interconnect system components and provide automatic communication of status changes, commands, field-initiated interrupts, and other communications required for proper system operation.

(2) Communication shall not require operator initiation or response,

and shall return to normal after partial or total network interruption such as power loss or transient upset.

(3) System shall automatically annunciate communication failures to the operator and identify the communication link that has experienced a partial or total failure.

(4) Communications Controller may be used as an interface between the Central Station display systems and the field device network. Communications Controller shall provide functions required to attain the specified network communications performance.

j. Central Station shall provide operator interface, interaction, display, control, and dynamic and real-time monitoring. Central Station shall control system networks to interconnect all system components, including workstations and field-installed Controllers.

k. Field equipment shall include Controllers, sensors, and controls. Controllers shall serve as an interface between the Central Station and sensors and controls. Data exchange between the Central Station and the Controllers shall include down-line transmission of commands, software, and databases to Controllers. The up-line data exchange from the Controller to the Central Station shall include status data such as intrusion alarms, status reports, and entry-control records. Controllers are classified as alarm-annunciation or entry-control type.

l. System Response to Alarms: Field device network shall provide a system end-to-end response time of 1 second(s) or less for every device connected to the system. Alarms shall be annunciated at the Central Station within 1 second of the alarm occurring at a Controller or device controlled by a local Controller, and within 100 ms if the alarm occurs at the Central Station. Alarm and status changes shall be displayed within 100 ms after receipt of data by the Central Station. All graphics shall be displayed, including graphics-generated map displays, on the console monitor within 5 seconds of alarm receipt at the security console.

m. False Alarm Reduction: The design of Central Station and Controllers shall contain features to reduce false alarms. Equipment and software shall comply with ANSI/SIA CP-01.

n. Error Detection: A cyclic code error detection method shall be used between Controllers and the Central Station, which shall detect single- and double-bit errors, burst errors of eight bits or less, and at least 99 percent of all other multibit and burst error conditions. Interactive or product error detection codes alone will not be acceptable. A message shall be in error if one bit is received incorrectly. System shall retransmit messages with detected errors. A two-digit decimal number shall be operator assignable to each communication link representing the number of retransmission attempts. When the number of consecutive retransmission attempts equals the assigned quantity, the Central Station shall print a communication failure alarm message. System shall monitor the frequency of data transmission failure for display and logging.

o. Data Line Supervision: System shall initiate an alarm in response to opening, closing, shorting, or grounding of data transmission lines.

p. Door Hardware Interface: Coordinate with Division 08 Sections that

specify door hardware required to be monitored or controlled by the PACS. The Controllers in this Section shall have electrical characteristics that match the signal and power requirements of door hardware. Integrate door hardware specified in Division 08 Sections to function with the controls and PC-based software and hardware in this Section.

q. References to industry and trade association standards and codes are minimum installation requirement standards.

r. Drawings and other specification sections shall govern in those instances where requirements are greater than those specified in the above standards.

1.11 EQUIPMENT AND MATERIALS

a. Refer to Secion 28 05 00 COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY, Part 1

1.12 WARRANTY OF CONSTRUCTION.

a. Warrant PACS work subject to the Article “Warranty of Construction” of FAR clause 52.246-21.

b. Demonstration and training shall be performed prior to system acceptance.

1.13 GENERAL REQUIREMENTS

a. For general requirements that are common to more than one section in Division 28 refer to Section 28 05 00, REQUIREMENTS FOR ELECTRONIC SAFETY AND SECURITY INSTALLATIONS.

b. General requirements applicable to this section include:

(1) General Arrangement Of Contract Documents,

(2) Delivery, Handling and Storage,

(3) Project Conditions,

(4) Electrical Power,

(5) Lightning, Power Surge Suppression, and Grounding,

(6) Electronic Components,

(7) Substitute Materials and Equipment, and

(8) Like Items.

PART 2 – PRODUCTS

2.1 GENERAL

a. All equipment and materials for the system will be compatible to ensure correct operation as outlined in NIST FIPS 201-2, March 2006 and HSPD-12.

Extend the existing campus access control system into the chiller plant and outlined herein and shown on the drawings. The existing system is Johnson Controls P2000 system, C-Cure 9000 as installed by JCI.

2.2 SECURITY MANAGEMENT SYSTEM (SMS)

a. Extend the existing SMS system (C-Cure 9000). Provide additional licences as required for a complete and operating system.

2.3 [Enter Appropriate Subpart Title Here]

2.4 PACS SERVER HARDWARE

a. The PACS server hardware is existing. Expand existing as required to provide a complete and operating system.

2.5 STANDARD WORKSTATION HARDWARE

a. Standard workstation hardware is existing. Expand existing as required to provide a complete and operating system.

2.6 COMMUNICATIONS WORKSTATION

a. Communication workstation is existing. Expand existing as requried to provide a complete and operating system.

2.7 CONTROLLERS

a. Controllers: Intelligent peripheral control unit, complying with UL 294, that stores time, date, valid codes, access levels, and similar data downloaded from the Central Station or workstation for controlling its operation.

b. Subject to compliance with requirements in this Article, manufacturers may use multipurpose Controllers.

c. Battery Backup: Sealed, lead acid; sized to provide run time during a power outage of 90 minutes, complying with UL 924.

d. Alarm Annunciation Controller:

(1) The Controller shall automatically restore communication within 10 seconds after an interruption with the field device network with dc line supervision on each of its alarm inputs.

(a) Inputs: Monitor dry contacts for changes of state that reflect alarm conditions. Provides at least eight alarm inputs, which are suitable for wiring as normally open or normally closed contacts for alarm conditions.

(b) Alarm-Line Supervision:

1. Supervise the alarm lines by monitoring each circuit for changes or disturbances in the signal by monitoring for abnormal open, grounded, or shorted conditions using dc change measurements. System shall initiate an alarm in response to an abnormal current, which is a dc change of 5 percent or more for longer than 500 ms.

2. Transmit alarm-line-supervision alarm to the Central

Station during the next interrogation cycle after the abnormal current condition.

(c) Outputs: Managed by Central Station software.

(2) Auxiliary Equipment Power: A GFI service outlet inside the Controller enclosure.

e. Entry-Control Controller:

(1) Function: Provide local entry-control functions including one- and two-way communications with access-control devices such as card readers, keypads, biometric personal identity verification devices, door strikes, magnetic latches, gate and door operators, and exit push-buttons.

(a) Operate as a stand-alone portal Controller using the downloaded database during periods of communication loss between the Controller and the field-device network.

(b) Accept information generated by the entry-control devices; automatically process this information to determine valid identification of the individual present at the portal:

1. On authentication of the credentials or information presented, check privileges of the identified individual, allowing only those actions granted as privileges.

2. Privileges shall include, but not be limited to, time of day control, day of week control, group control, and visitor escort control.

(c) Maintain a date-, time-, and Location-stamped record of each transaction. A transaction is defined as any successful or unsuccessful attempt to gain access through a controlled portal by the presentation of credentials or other identifying information.

(2) Inputs:

(a) Data from entry-control devices; use this input to change modes between access and secure.

(b) Database downloads and updates from the Central Station that include enrollment and privilege information.

(3) Outputs:

(a) Indicate success or failure of attempts to use entry-control devices and make comparisons of presented information with stored identification information.

(b) Grant or deny entry by sending control signals to portal-control devices and mask intrusion alarm annunciation from sensors stimulated by authorized entries.

(c) Maintain a date-, time-, and Location-stamped record of each transaction and transmit transaction records to the Central Station.

(d) Door Prop Alarm: If a portal is held open for longer than 20 seconds, alarm sounds.

(4) With power supplies sufficient to power at voltage and frequency required for field devices and portal-control devices.

(5) Data Line Problems: For periods of loss of communications with Central Station, or when data transmission is degraded and generating continuous checksum errors, the Controller shall continue to control entry by accepting identifying information, making authentication decisions, checking privileges, and controlling portal-control devices.

Store up to 1000 transactions during periods of communication loss between the Controller and access-control devices for subsequent upload to the Central Station on restoration of communication.

(6) Controller Power: NFPA 70, Class II power supply transformer, with 12- or 24-V ac secondary, backup battery and charger.

(a) Backup: Premium, valve-regulated, recombinant-sealed, lead-calcium battery; spill proof; with a full 1-year warranty and a pro rata 19-year warranty. With single-stage, constant-voltage-current, limited battery charger, comply with battery manufacturer's written instructions for battery terminal voltage and charging current recommendations for maximum battery life.

(b) Backup: Valve-regulated, recombinant-sealed, lead-acid battery; spill proof. With single-stage, constant-voltage-current, limited battery charger, comply with battery manufacturer's written instructions for battery terminal voltage and charging current recommendations for maximum battery life.

(c) Backup Power Supply Capacity: 5 minutes of battery supply. Submit battery and charger calculations.

(d) Power Monitoring: Provide manual dynamic battery load test, initiated and monitored at the control center; with automatic disconnection of the Controller when battery voltage drops below Controller limits. Report by using local Controller-mounted LEDs and by communicating status to Central Station. Indicate normal power on and battery charger on trickle charge. Indicate and report the following:

1. Trouble Alarm: power off load assumed by battery.

2. Trouble Alarm: Low battery.

3. Alarm: Power off.

2.8 PIV MIDDLEWARE

a. PIV Middleware shall provide three-factor authentication, including biometric matching using a fingerprint capture device capable of single fingerprint capture. Unit shall enable digital certificates can to be verified by security personnel using the issuer's certificate authority, SCVP, OCSP responder/repeater, orthe TSA hot

list for TWIC cardholders. All cards shall be validated using NIST FIPS 201-2 challenge-response protocol in order to identify forged or cloned cards. PIV Middleware solution shall validate all PIV, TWIC, NG CAC, and FRAC cards. TWIC card FASC-Ns shall also be verified against a live or cached TSA hot list.

b. PIV Middleware shall have ability to:

(1) Verify cardholder identity and validates NIST FIPS 201-2-compliant PIV-II, next-generation (NG) CAC, TWIC, or FRAC credentials in real-time

(2) Perform three-factor authentication of cardholder using PIN, biometrics, and certificate (or serial numbers) detecting forged or cloned cards

(3) Enroll FASC-N, photo, and pertinent cardholder information into PACS software

(4) Automatically suspend a cardholder’s badge if his or her PIV, TWIC, or CAC card certificate serial number is on the Certificate Revocation List (CRL)

(5) Upload a cardholder transaction audit trail to central database or exports it to a.csv file for centralized transaction management

(6) Be compatible with biometric mobile terminal for off-site verification and enrollment

(7) Re-validate imported cardholder certificates on a periodic basis via the Internet

(8) Operate with commercial, off-the-shelf (COTS) NIST FIPS 201-2 PIV-II and ANSI INCITS 378-compliant fingerprint capture devices

(9) Revalidate imported cardholder certificates at regular intervals, ensuring that the credentials used in PACS system are backed by a valid set of digital certificates. Digital certificates are verified against local OCSP repeater/validation authority using the issuer's validation authority, or Microsoft Crypto Application Programming Interface (API) on Windows XP SP3 or Vista.

(10) Certificate Manager shall fully support SCVP and OCSP for fast, online validation.

(11) Provide verification of TWIC credentials against a live TSA hot list.

(12) Support uploading local transactions to a central database for consolidated activity reporting. This application shall support a variety of ODBC- or ADO-compliant databases, including Oracle, SQL Server 2005, Informix, DB2, and Firebird.

(13) Provide user with ability to produce canned transaction log queries as well as creating queries directly from the SQL database.

c. PIV Middleware PC requirements:

(1) PIV Middleware software shall operate on Intel-based PC with

minimum 1.8 GHz CPU, 1 GB RAM, 40 GB hard disk, and Microsoft Windows XP SP2 with Microsoft.NET Framework 2.0

(2) Unit shall fingerprint capture devices and smart card reader.

d. PIV Middleware shall be NIST FIPS 201-2 approved product.

2.9 CARD READERS

a. Power: Card reader shall be powered from its associated Controller, including its standby power source.

b. Response Time: Card reader shall respond to passage requests by generating a signal that is sent to the Controller. Response time shall be 800 ms or less, from the time the card reader finishes reading the credential card until a response signal is generated.

c. Enclosure: Suitable for surface, semiflush, or pedestal mounting. Mounting types shall additionally be suitable for installation in the following locations:

(1) Indoors, controlled environment.

(2) Indoors, uncontrolled environment.

(3) Outdoors, with built-in heaters or other cold-weather equipment to extend the operating temperature range as needed for operation at the site.

d. Display: LED or other type of visual indicator display shall provide visual status indications and user prompts. Indicate power on/off, whether user passage requests have been accepted or rejected, and whether the door is locked or unlocked.

e. Shall be utilized for controlling the locking hardware on a door and allows for reporting back to the main control panel with the time/date the door was accessed, the name of the person accessing the point of entry, and its location.

f. Will be fully programmable and addressable, locally and remotely, and hardwired to the system.

g. Shall be individually home run to the main panel.

h. Shall be installed in a manner that they comply with:

(1) The Uniform Federal Accessibility Standards (UFAS)

(2) The Americans with Disabilities Act ()

(3) The Standards for Accessible Design

i. Shall support a variety of card readers that must encompass a wide functional range. The PACS may combine any of the card readers described below for installations requiring multiple types of card reader capability (i.e., card only, card and/or PIN, card and/or biometrics, card and/or pin and/or biometrics, supervised inputs, etc.). These card readers shall be available in the approved technology to meet NIST FIPS 201-2, and is ISO 14443 A or B,

ANSI ISO/IEC 7816 compliant. The reader output can be Wiegand, RS-22, 485 or TCP/IP.

j. Shall be housed in an aluminum bezel with a wide lead-in for easy card entry.

k. Shall contain read head electronics, and a sender to encode digital door control signals.

l. LED’s shall be utilized to indicate card reader status and access status.

m. Shall be able to support a user defined downloadable off-line mode of operation (e.g. locked, unlocked), which will go in effect during loss of communication with the main control panel.

n. Shall provide audible feedback to indicate access granted/denied decisions. Upon a card swipe, two audible tones or beeps shall indicate access granted and three tones or beeps shall indicate access denied. All keypad buttons shall provide tactile audible feedback.

o. Shall have a minimum of two programmable inputs and two programmable outputs.

p. All card readers that utilize keypad controls along with a reader and shall meet the following specifications:

(1) Entry control keypads shall use a unique combination of alphanumeric and other symbols as an identifier. Keypads shall contain an integral alphanumeric/special symbols keyboard with symbols arranged in ascending ASCII code ordinal sequence. Communications protocol shall be compatible with the local processor.

q. Shall include a Light Emitting Diode (LED) or other type of visual indicator display and provide visual or visual and audible status indications and user prompts. The display shall indicate power on/off, and whether user passage requests have been accepted or rejected. The design of the keypad display or keypad enclosure shall limit the maximum horizontal and vertical viewing angles of the keypad. The maximum horizontal viewing angle shall be plus and minus five (5) degrees or less off a vertical plane perpendicular to the plane of the face of the keypad display. The maximum vertical viewing angle shall be plus and minus 15 degrees or less off a horizontal plane perpendicular to the plane of the face of the keypad display.

(1) Shall respond to passage requests by generating a signal to the local processor. The response time shall be 800 milliseconds or less from the time the last alphanumeric symbol is entered until a response signal is generated.

(2) Shall be powered from the source as designed and shall not dissipate more than 150 Watts.

(3) Shall be suitable for surface, semi-flush, pedestal, or weatherproof mounting as required.

(4) Shall provide a means for users to indicate a duress situation by entering a special code.

r. PIV Contact Card Reader

(1) Application Protocol Data Unit (APDU) Support: At a minimum, the contact interface shall support all card commands for contact based access specified in Section 7, End-point PIV Card Application Card Command Interface of NIST SP 800-73-4, Interfaces for Personal Identity Verification.

(2) Buffer Size: The reader must contain a buffer large enough to receive the maximum size frame permitted by International Organization for Standardization International Electrotechnical Commission ANSI ISO/IEC 7816, Section 9.4.

(3) Programming Voltage: PIV Readers shall not generate a Programming Voltage.

(4) Support for Operating Class: PIV Readers shall support cards with Class A Vccs as defined in ANSI ISO/IEC 7816 and ANSI ISO/IEC 7816 /Amd 1:2002.

(5) Retrieval Time: Retrieval time for 12.5 kilobytes (KB) of data through the contact interface of the reader shall not exceed 2.0 seconds.

(6) Transmission Protocol: The PIV Reader shall support both the character-based T=0 protocol and block-based T=1 protocol as defined in ANSI ISO/IEC 7816.

(7) Support for PPS Procedure: The reader shall support Protocol and Parameters Selection (PPS) procedure by having the ability to read character TA1 of the Answer to Reset (ATR) sent by the card as defined in ANSI ISO/IEC 7816.

s. Contactless Smart Cards and Readers

(1) Smart card readers shall read credential cards whose characteristics of size and technology meet those defined by ANSI ISO/IEC 7816, ISO 14443, ISO 15693.

(2) The readers shall have "flash" download capability to accommodate card format changes.

(3) The card reader shall have the capability of reading the card data and transmitting the data to the main monitoring panel.

(4) The card reader shall be contactless and meet or exceed the following technical characteristics:

(a) Data Output Formats: NIST FIPS 201-2 low outputs the FASC-N in an assortment of Wiegand bit formats from 40 – 200 bits. NIST FIPS 201-2 medium outputs a combination FASC-N and HMAC in an assortment of Wiegand bit formats from 32 – 232 bits. All Wiegand formats or the upgradeability from Low to Medium Levels can be field configured with the use of a command card.

(b) NIST FIPS 201-2 readers shall be able to read, but not be limited to, DESfire and iCLASS cards.

(c) Reader range shall comply with ANSI ISO/IEC 7816, ISO 14443, and ISO 15693, and also take into consideration conditions, are at a minimum 1″ to 2″ .

(d) APDU Support: At a minimum, the contactless interface shall support all card commands for contactless based access specified in Section 7, End-point PIV Card Application Card Command Interface of NIST SP 800-73-4, Interfaces for Personal Identity Verification.

(e) Buffer Size: The reader shall contain a buffer large enough to receive the maximum size frame permitted by ANSI ISO/IEC 7816, Section 9.4.

(f) ISO 14443 Support: The PIV Reader shall support parts (1 through 4) of ISO 14443 as amended in the References of this publication.

(g) Type A and B Communication Signal Interfaces: The contactless interface of the reader shall support both the Type A and Type B communication signal interfaces as defined in ISO 14443.

(h) Type A and B Initialization and Anti-Collision The contactless interface of the reader shall support both Type A and Type B initialization and anti-collision methods as defined in ISO 14443.

(i) Type A and B Transmission Protocols: The contactless interface of the reader shall support both Type A and Type B transmission protocols as defined in ISO 14443.

(j) Retrieval Time: Retrieval time for 4 KB of data through the contactless interface of the reader shall not exceed 2.0 seconds.

(k) Transmission Speeds: The contactless interface of the reader shall support bit rates of fc/128 (~106 kbits/s), fc/64(~212 kbits/s), and configurable to allow activation/deactivation.

(l) Readibility Range: The reader shall not be able to read PIV card more than (4inch) from the readers

2.10 [Enter Appropriate Subpart Title Here]Not Used
2.11 KEYPADS

a. Designed for use with unique combinations of alphanumeric and other symbols as an Identifier. Keys of keypads shall contain an integral alphanumeric/special symbol keyboard with symbols arranged in ascending ASCII-code ordinal sequence. Communications protocol shall be compatible with Controller.

(1) Keypad display or enclosure shall limit viewing angles of the keypad as follows:

(a) Maximum Horizontal Viewing Angle: 5 degrees or less off in either direction of a vertical plane perpendicular to the plane of the face of the keypad display.

(b) Maximum Vertical Viewing Angle: 15 degrees or less off in either direction of a horizontal plane perpendicular to the plane of the face of the keypad display.

(2) Duress Codes: Provide duress situation indication by entering a special code.

2.12 CREDENTIAL CARDS

a. Personal Identity Verification (PIV) credential cards shall comply to Federal Information Processing Standards Publication NIST FIPS 201-2.

b. Visual Card Topography shall be compliant with NIST SP 800-104A.

c. PIV logical credentials shall contain multiple data elements for the purpose of verifying the cardholder's identity at graduated assurance levels. These mandatory data elements shall collectively comprise the data model for PIV logical credentials, and include the following:

(1) CHUID

(2) PIN

d. The credential card (PIV) shall be an ISO 14443 type smart card with contactless interface that operates at 13.56 MHZ.

2.13 SYSTEM SENSORS AND RELATED EQUIPMENT

a. The PACS (Physical Access Control System) and related Equipment provided by the Contractor shall meet or exceed the following performer specifications:

b. Request to Exit Detectors:

(1) Passive Infrared Request to Exit Motion Detector (REX PIR) (1) The Contractor shall provide a surface mounted motion detector to signal the physical access control system request to exit input. The motion detector shall be a passive infrared sensor designed for wall or ceiling mounting (7 to 15 ft) height. The detector shall provide two (2) form “C” (SPDT) relays rated one (1) Amp. @ 30 VDC for DC resistive loads. The detectors relays shall be user adjustable with a latch time from 1-60 seconds. The detector shall also include a selectable relay reset mode to follow the timer or absence of motion. The detection pattern shall be adjustable plus or minus fourteen (± 14) degrees. The detector shall operate on 12 VDC with approximately 26 mA continuous current draw. The detector shall have an externally visible activation LED. The motion detector shall measure approximately (1.5 x 6.25 x 1.5 in). The detector shall be immune to radio frequency interference. The detector shall not activate or set-up on critical frequencies in the range 26 to 950 Megahertz using a 50 watt transmitter located (1 ft) from the unit or attached wiring. The detector shall be available on gray or black enclosures. The color of the housing shall be coordinated with the surrounding surface.

d. Delayed Egress (DE)

(1) General:

(a) The delay egress locking hardware shall provide a method to secure emergency exits and provide an approved delayed emergency

exit method. The package shall be Underwriters Laboratories listed as a delay egress-locking device. The delay egress device shall be available to support configurations with both rated and non-rated fire doors. The delay egress device shall comply with Life Safety Codes (NFPA 101, BOCA) as it applies to special locking arrangements for delay egress locks. Unless specifically identified as a non-fire rated opening, all doors shall be equipped with fire rated door hardware. The Contractor shall be responsible for providing all equipment and installation to provide a fully functioning system. Need to amend to use crashbars type mechanical release switches.

(2) The delay-locking device shall include all of the following features:

(a) Delay Egress Mode

1. The delayed egress device shall be a SDC 101V Series Exit Check with wall mounted control module. Upon activation of an approved panic bar the delay locking device shall begin a delay sequence of 30 seconds; a flush mounted wall LED panel adjacent to the door will indicate initiation of the countdown time. During the 30 second delay period, a local sounding device shall annunciate a tone activation of the delay cycle and verbal exit instructions. At the end of the delay cycle the locking device shall unlock and allow free egress. The reset of the local sounding device shall be user definable and include options to select either local sound until silenced by reset or local sounder silenced upon opening of the door. Unless otherwise indicated the local delay sounder shall be silenced upon opening of the door. The SDC’s device trigger output shall be connected to the SMS DGP alarm panel for pre-activation warning. The contractor shall specify the bond sensor option when ordering the delayed egress hardware; this output shall be wired to the SMS DGP to activate an alarm if the door does not lock. Use of reset panel not top mounted device.

2. Delayed egress doors will have bond sensors.

3. Delayed egress activation shall also trigger CCTV call –up.

(b) Fire Alarm Mode

1. Upon activation of the facility’s fire evacuation and water flow alarm signal the delay locking devices shall immediately unlock and provide free egress. The Contractor shall provide any required fire alarm relays or interface devices.

(c) Reset Mode

1. The delay egress device shall be manually reset by the Delayed Egress controller located at the door via key switch.

2. The delay egress device shall automatically reset upon fire alarm system reset.

3. The delayed egress shall be resettable through the SMS.

(d) The Contractor shall provide a Master Open Switch for all the

facility's delayed egress hardware, with protective cover and permanent labeling in the Unit Control Room. The switch shall be wired into the fire alarm system to activate the evacuation alarms. When the switch is pressed all delayed egress or evacuation doors shall unlock and generate an alarm at the security console monitor showing and recording time and date of when the switch was pressed. The contractor is responsible for coordinating the wiring and connection with the fire alarm contactor. The Master Open Switch shall be linked to the fire alarm panel for the release of doors locks.

(e) Each individual delayed egress door shall have the ability to unlock through a manual action on the SMS.

(f) Unless otherwise indicated the Contractor shall provide all of the above reset methods for each door. All signs will meet the latest ADA requirements.

(g) Signs

1. The delay egress package shall be provided with a warning sign complying with local code requirements. The warning sign shall be attached to the interior side of the controlled door. The sign shall be located on the interior side of the door above and within (12 in) of the panic bar. The sign shall read:

EMERGENCY EXIT.
PUSH UNTIL
ALARM SOUNDS
DOOR CAN BE OPENED,
IN 30 SECONDS.

2. Signs shall be coordinated and comply with the building's existing sign specifications. Signs shall include grade 2 Braille.

3. Signs shall meet the current ADA requirements.

4. In instances of code and specification conflicts, the life safety code requirement shall prevail.

5. The Division 10 Contractor shall provide samples for approval with their submittal package.

(3) Physical Access Control Interface

(a) The delay egress device shall be capable of interface with card access control systems.

(b) The system shall include a bypass feature that is activated via a dry contact relay output from the physical access control system. This bypass shall allow authorized personnel to pass through the controlled portal without creating an alarm condition or activating the delay egress cycle. The bypass shall include internal electronic shunts or door switches to prevent activation (re-arming) until the door returns to the closed position. An unused access event shall not cause a false alarm and shall automatically rearm the delay egress lock upon expiration of the programmed shunt time. The delay egress physical access control interface shall support extended periods of automated and/or

manual lock and unlock cycles.

e. Crash Bar:

(1) Emergency Exit with Alarm (Panic):

(a) Entry control portals shall include panic bar emergency exit hardware as designed.

(b) Panic bar emergency exit hardware shall provide an alarm shunt signal to the PACS and SMS.

(c) The panic bar shall include a conspicuous warning sign with one (1) inch high, red lettering notifying personnel that an alarm will be annunciated if the panic bar is operated.

(d) Operation of the panic bar hardware shall generate an intrusion alarm that reports to both the SMS and Intrusion Detection System. The use of a micro switch installed within the panic bar shall be utilized for this.

(e) The panic bar shall utilize a fully mechanical connection only and shall not depend upon electric power for operation.

(f) The panic bar shall be compatible with mortise or rim mount door hardware and shall operate by retracting the bolt manually by either pressing the panic bar or with a key by-pass. Refer to Section 2.2.I.9 for key-bypass specifications.

(g) Normal Exit:

1. Entry control portals shall include panic bar non-emergency exit hardware as designed.

2. Panic bar non-emergency exit hardware shall be monitored by and report to the SMS.

3. Operation of the panic bar hardware shall not generate a locally audible or an intrusion alarm within the IDS.

4. When exiting, the panic bar shall depend upon a mechanical connection only. The exterior, non-secure side of the door shall be provided with an electrified thumb latch or lever to provide access after the credential I.D. authentication by the SMS.

5. The panic bar shall be compatible with mortise or rim mount door hardware and shall operate by retracting the bolt manually by either pressing the panic bar or with a key by-pass. Refer to Section 2.2.I.9 for key-bypass specifications. The strikes/bolts shall include a micro switch to indicate to the system when the bolt is not engaged or the strike mechanism is unlocked. The signal switches shall report a forced entry to the system in the event the door is left open or accessed without the identification credentials.

f. Key Bypass:

(1) Shall be utilized for all doors that have a mortise or rim mounted door hardware.

(2) Each door shall be individually keyed with one master key per secured area.

(3) Cylinders shall be six (6)-pin and made of brass or equivalent. Keys for the cylinders shall be constructed of solid material and produced and cut by the same distributor. Keys shall not be purchased, cut, and supplied by multiple dealers.

(4) All keys shall have a serial number cut into the key. No two serial numbers shall be the same.

(5) All keys and cylinders shall be stored in a secure area that is monitored by the Intrusion Detection System.

g. Automatic Door Opener and Closer:

(1) Shall be low energy operators.

(2) Door closing force shall be adjustable to ensure adequate closing control.

(3) Shall have an adjustable back-check feature to cushion the door opening speed if opened violently.

(4) Motor assist shall be adjustable from 0 to 30 seconds in five (5) second increments. Motor assist shall restart the time cycle with each new activation of the initiating device.

(5) Unit shall have a three-position selector mode switch that shall permit unit to be switched “ON” to monitor for function activation, switched to “H/O” for indefinite hold open function or switched to “OFF,” which shall deactivate all control functions but will allow standard door operation by means of the internal mechanical closer.

(6) Door control shall be adjustable to provide compliance with the requirements of the Americans with Disabilities Act (ADA) and ICC A117.1.

(7) All automatic door openers and closers shall:

(a) Meet UL standards.

(b) Be fire rated.

(c) Have push and go function to activate power operator or power assist function.

(d) Have push button controls for setting door close and door open positions.

(e) Have open obstruction detection and close obstruction detection built into the unit.

(f) Have door closer assembly with adjustable spring size, back-check valve, sweep valve, latch valve, speed control valve and pressure adjustment valve to control door closing.

(g) Have motor start-up delay, vestibule interface delay; electric lock delay and door hold open delay up to 30 seconds. All operators shall close door under full spring power when power is removed.

(h) Are to be hard wired with power input of 120 VAC, 60Hz and connected to a dedicated circuit breaker located on a power panel reserved for security equipment.

h. Door Status Indicators:

(1) Shall monitor and report door status to the SMS.

(2) Door Position Sensor:

(a) Shall provide an open or closed indication for all doors operated on the PACS and report directly to the SMS.

(b) Shall also provide alarm input to the Intrusion Detection System for all doors operated by the PACS and all other doors that require monitoring by the intrusion detection system.

(c) Switches for doors operated by the PACS shall be double pole double throw (DPDT). One side of the switch shall monitor door position and the other side if the switch shall report to the intrusion detection system. For doors with electromagnetic locks a magnetic bonding sensor (MBS) can be used in place of one side of a DPDT switch, in turn allowing for the use of a single pole double throw (SPDT) switch in it place of a DPDT switch.

(d) Switches for doors not operated by the PACS shall be SPDT and report directly to the IDS.

(e) Shall be surface or flush mounted and wide gap with the ability to operate at a maximum distance of up to 2″ .

2.14 PUSH BUTTON SWITCHES

a. Push-Button Switches: Momentary-contact back-lighted push buttons, with stainless-steel switch enclosures.

(1) Electrical Ratings:

(a) Minimum continuous current rating of 10 A at 120 V ac or 5 A at 240-V ac.

(b) Contacts that will make 720 VA at 60 A and that will break at 720 VA at 10 A.

(2) Enclosures: Flush or surface mounting. Push buttons shall be suitable for flush mounting in the switch enclosures.

(3) Enclosures shall additionally be suitable for installation in the following locations:

(a) Indoors, controlled environment.

(b) Indoors, uncontrolled environment.

(c) Outdoors.

(4) Power: Push-button switches shall be powered from their associated Controller, using dc control.

2.15 PORTAL CONTROL DEVICES

a. Shall be used to assist the PACS.

b. Such devices shall:

(1) Provide a means of monitoring the doors status.

(2) Allow for exiting a space via either a push button, request to exit, or panic/crash bar.

(3) Provide a means of override to the PACS via a keypad or key bypass.

(4) Assist door operations utilizing automatic openers and closures.

(5) Provide a secondary means of access to a space via a keypad.

c. Shall be connected to and monitored by the main PACS panel.

d. Shall be installed in a manner that they comply with:

(1) The Uniform Federal Accessibility Standards (UFAS)

(2) The Americans with Disabilities Act (ADA)

(3) The ADA Standards for Accessible Design

e. Shall provide a secondary means of physical access control within a secure area.

f. Push-Button Switches:

(1) Shall be momentary contact, back lighted push buttons, and stainless steel switch enclosures for each push button as shown. Buttons are to be utilized for secondary means of releasing a locking mechanism.

(a) In an area where a push button is being utilized for remote access of the locking device then no more than two (2) buttons shall operate one door from within one secure space. Buttons will not be wired in series with one other.

(b) In an area where locally stationed guards control entry to multiple secure points via remote switches. An interface board shall be designed and constructed for only the amount of buttons it shall house. These buttons shall be flush mounted and clearly labeled for ease of use. All buttons shall be connected to the PACS and SMS system for monitoring purposes.

(c) Shall have double-break silver contacts that will make 720 VA at 60 amperes and break 720 VA at 10 amperes.

g. Entry Control Devices:

(1) Shall be hardwired to the PACS main control panel and operated by either a card reader or a biometric device via a relay on the main control panel.

(2) Shall be fail-safe in the event of power failure to the PACS system.

(3) Shall operate at 24 VCD, with the exception of turnstiles and be powered by a separate power supply dedicated to the door control system. Each power supply shall be rated to operate a minimum of two doors simultaneously without error to the system or overload the power supply unit.

(4) Shall have a diode or metal-oxide veristor (MOV) to protect the controller and power supply from reverse current surges or back-check.

(5) Electric Strikes/Bolts: Shall be:

(a) Made of heavy-duty construction and tamper resistant design.

(b) Tested to over one million cycles.

(c) Rated for a minimum of 1000 lbs. holding strength.

(d) Utilize an actuating solenoid for the strike/bolt. The solenoid shall move from fully open to fully closed position and back in not more than 500 milliseconds and be rated for continuous duty.

(e) Utilize a signal switch that will indicate to the system if the strike/bolt is not engaged or is unlocked when it should be secured.

(f) Flush mounted within the door frame.

(6) Electric Mortise Locks: Shall be installed within the door and an electric transfer hinge shall be utilized to allow the wires to be transferred from the door frame to the lock. If utilized with a double door then the lock shall be installed inside the active leaf. Electric Mortise Locks shall:

(a) These locks shall be provided and installed by the Division 8 “DOOR HARDWARE” Contractor.

(b) Have integrated Request to Exit switch for doors receiving physical access control devices.

(b) Provide integration of the Electric Mortise Locks with the PACS for:

1. Lock Power

2. Request to Exit switch.

(7) Electromagnetic Locks:

(a) These locks shall be without mechanical linkage utilizing no moving parts, and securing the door to its frame solely on

electromagnetic force.

(b) Shall be comprised of two pieces, the mag-lock and the door plate. The electromagnetic locks shall be surface mounted to the door frame and the door plate shall be surface mounted to the door.

(c) Ensure a diode is installed in line with the DC voltage supplying power to the unit in order to prevent back-check on the system when the electromagnetic lock is powered.

(d) Shall utilize a magnetic bonding sensor (MBS) to monitor the door status and report that status to the SMS.

(e) Electromagnetic locks shall meet the following minimum technical characteristics:

Operating Voltage		24 VDC
Current Draw		.5A
Holding Force	Swing Doors	(1500 lbs)
	Sliding Doors	(500 lbs)

(8) Turnstiles:

(a) Shall operate at 110 VAC, 60 Hz or 220 VAC, 50 Hz supplied from a dedicated circuit breaker on a security power panel. This device does not require a back-up power source.

(b) Shall be utilized as a means of monitoring and controlling access in a lobby.

(c) Shall meet the following minimum requirements:

1. Be UFAS compliant.

2. Provide either an audible or visual confirmation that access has been granted to a cleared individual.

3. Provide an audible alarm in the event a non-cleared individual is attempting to gain access.

4. Interface with the SMS and utilize a card reader for accessing and exiting a facility, and provide a recorded event of personnel accessing these points.

5. Have a built-in step-down transformer to provide power to a card reader unit.

6. Have built-in signal wiring chassis to allow for plug and play capabilities with the PACS.

7. Have the ability to detect tailgating within one quarter on an inch to prevent unauthorized access to a facility.

(9) Vehicle Gate Operator: Interface electrical operation of gate with controls of this Section. Vehicle gate operators shall be connected, monitored, and controlled, by the security access Controllers. Vehicle gate and accessories are specified in Division 32 Section "Chain Link Fences and Gates."

2.16 SECONDARY ALARM ANNUNCIATOR

A workstation with limited I/O capacity, consisting of a secondary alarm annunciation workstation to allow the operator to duplicate functions of the main operator interface, and to show system status changes.

2.17 INTERFACES

a. CCTV System Interface

(1) An RS232 interface associated driver, and controller shall be provided for connection of the SMS Central Computer to the CCTV Alarm interface and switcher. The interface shall provide alarm data to the CCTV Alarm interface for automatic camera call-up. If required the Security Contractor shall be responsible for programming the command strings into the SMS Server.

b. Intercom System Interface

(1) The CCTV call-up from intercom stations shall be through the intercom unit via RS232 communications interface to the SMS system, then through the matrix switcher.

(a) Application Software

1. Provides the interface between the Alarm Annunciation System and Operator; all sensors, local processors and data links, drive displays, report alarms, and report generation.

2. Software is categorized as System Software and Application Software. System Software must consist of software to support set-up, operation, hard drive back-ups and maintenance processor. Application Software must consist of software to provide the completion of Physical Access Control System.

c. Power Supplies:

(1) Shall be UL rated and able to adequately power (enter number) entry control devices on a continuous base without failure.

(2) Shall meet the following minimum technical characteristics:

INPUT POWER	110 VAC 60 HZ (enter amperage)A
OUTPUT VOLTAGE	12 VDC Nominal (13.8 VDC) 24 VDC Nominal (27.6 VDC) Filtered and Regulated
	Dependant on Output Voltage shall provide up to 5 Ah
OUTPUT CURRENT	10 amp max. @ 13.8 VDC 5 amp max. @ 27.6 VDC
PRIMARY FUSE SIZE	6.3 amp (non-removable)
FUSE SIZE	12 amp, 3AG

CHARGING CIRCUIT	Built-in standard

2.18 FLOOR SELECT ELEVATOR CONTROL

a. Elevator access control shall be integral to security access.

(1) System shall be capable of providing full elevator security and control through dedicated Controllers without relying on the control-station host PC for elevator control decisions.

(2) Access-control system shall enable and disable car calls on each floor and floor select buttons in each elevator car, restricting passengers' access to the floors where they have been given access.

(3) System setup shall, through programming, automatically secure and unsecure each floor select button of a car individually by time and day. Each floor select button within a car shall be separately controlled so that some floors may be secure while others remain unsecure.

(4) When a floor select button is secure, it shall require the passenger to use his/her access code and have access to that floor before the floor select button will operate. The passenger's credential shall determine which car call and floor select buttons are to be enabled, restricting access to floors unless authorized by system's access code database. Floor select button shall be enabled only in the car where the credential holder is the passenger.

b. PACS shall record which call button is pressed, along with credential and time information.

(1) System Controller shall record elevator access data.

(2) The Controller shall reset all additional call buttons that may have been enabled by the user's credential.

(3) The floor select elevator control shall allow for manual override either individually by floor or by cab as a group from a workstation PC.

2.19 AFTER-HOURS HVAC CONTROL

a. After-Hours HVAC Control: Provide for any credential read to activate or control individual HVAC zones based on access level. This control module shall control and record the after-hours use of the heating and cooling system in zones or tenant space.

(1) This control shall give the administrator the ability to determine how much extra energy consumption each tenant is responsible for. This information can be used in billing tenants for the extra after-hour usage.

(2) At the specified time every day, the HVAC shall automatically go into its after-hours mode. It shall then revert into its normal business hours mode by a tenant using an access code or card at a designated keypad or reader.

(3) Once enabled, the tenant's HVAC zone shall be under thermostat control for a preset amount of time. When the preset amount of time elapses, the HVAC for that zone shall revert back to after-hours mode unless a tenant uses his/her code or card again. This shall continue until the unit automatically returns to its normal business hours operation.

b. Control module activates the HVAC system after a valid access by any of three methods; however, the HVAC control shall always allow for manual override from the PC.

(1) By time expiration after access of an adjustable period from 1 second to 546 minutes (9.1 hours).

(2) By use of the card or code again at the same or different reader or keypad.

(3) By system returning to its normal business hours operation.

c. After-hours HVAC control shall operate with all other features running simultaneously and use the central-station PC that controls access for the building but shall not rely on the host PC for any HVAC control decisions.

2.20 REAL TIME GUARD TOUR

a. Guard tour module shall provide the ability to plan, track, and route tours. Module shall input an alarm during tour if guard fails to make a station. can be programmed for sequential or random tour-station order.

(1) Guard tour setup shall define specific routes or tours for the guard to take, with time restrictions in which to reach every predefined tour station.

(2) Guard tour activity shall be automatically logged to the central-station PC's hard drive.

(3) If the guard is early or late to a tour station, a unique alarm per station shall appear at the Central Station to indicate the time and station.

(4) Guard tour setup shall allow the tours to be executed sequentially or in a random order with an overall time limit set for the entire tour instead of individual times for each tour station.

(5) Setup shall allow recording of predefined responses that will display for the operator at the control station should a "Failed to Check-in" alarm occur.

b. A tour station is a physical location a guard shall reach and perform an action indicating that the guard has arrived. This action, performed at the tour station, shall be 1 of 13 different events with any combination of station types within the same tour. A tour station shall be one of the following event types:

(1) Access Granted.

(2) Access Denied Code.

(3) Access Denied Card plus PIN.

(4) Access Denied Time Zone.

(5) Access Denied Level.

(6) Access Denied Facility.

(7) Access Denied Code Timer.

(8) Access Denied Anti-Passback.

(9) Access Granted Passback Violation.

(10) Alarm.

(11) Restored.

(12) Input Normal.

(13) Input Abnormal.

c. Guard tour and other system features shall operate simultaneously with no interference.

d. Guard Tour Module Capacity: 999 possible guard tour definitions with each tour having up to 99 tour stations. System shall allow all 999 tours to be running at same time.

2.21 VIDEO AND CAMERA CONTROL

a. Control station or designated workstation displays live video from a CCTV source.

(1) Control Buttons: On the display window, with separate control buttons to represent Left, Right, Up, Down, Zoom In, Zoom Out, Scan, and a minimum of two custom command auxiliary controls.

(2) Provide at least seven icons to represent different types of cameras, with ability to import custom icons. Provide option for display of icons on graphic maps to represent their physical location.

(3) Provide the alarm-handling window with a command button that will display the camera associated with the alarm point.

b. Display mouse-selectable icons representing each camera source, to select source to be displayed. For CCTV sources that are connected to a video switcher, control station shall automatically send control commands through a COM port to display the requested camera when the camera icon is selected.

c. Allow cameras with preset positioning to be defined by displaying a different icon for each of the presets. Provide control with Next and Previous buttons to allow operator to cycle quickly through the preset positions.

2.22 WIRES AND CABLES

a. Refer to section 28 05 13 "CONDUCTORS AND CABLES FOR ELECTRONIC SAFETY AND SECURITY".

PART 3 EXECUTION

3.1 GENERAL

a. The Contractor shall install all system components and appurtenances in accordance with the manufacturers' instructions, IEEE C2, and shall furnish all necessary interconnections, services, and adjustments required for a complete and operable system as specified. Control signals, communications, and data transmission lines grounding shall be installed as necessary to preclude ground loops, noise, and surges from affecting system operation. Equipment, materials, installation, workmanship, inspection, and testing shall be in accordance with manufacturers' recommendations and as modified herein.

b. Consult the manufacturers' installation manuals for all wiring diagrams, schematics, physical equipment sizes, etc., before beginning system installation. Refer to the Riser/Connection diagram for all schematic system installation/termination/wiring data.

c. All equipment shall be attached to walls and ceiling/floor assemblies and shall be held firmly in place (e.g., sensors shall not be supported solely by suspended ceilings). Fasteners and supports shall be adequate to support the required load.

3.2 CURRENT SITE CONDITIONS

The Contractor shall visit the site and verify that site conditions are in agreement with the design package. The Contractor shall report all changes to the site or conditions which will affect performance of the system to the Owner in a report as defined in paragraph Group II Technical Data Package. The Contractor shall not take any corrective action without written permission from the Owner.

3.3 EXAMINATION

a. Examine pathway elements intended for cables. Check raceways, cable trays, and other elements for compliance with space allocations, installation tolerances, hazards to cable installation, and other conditions affecting installation.

b. Examine roughing-in for LAN and control cable conduit systems to PCs, Controllers, card readers, and other cable-connected devices to verify actual locations of conduit and back boxes before device installation.

c. Proceed with installation only after unsatisfactory conditions have been corrected.

3.4 PREPARATION

a. Comply with recommendations in ANSI/SIA CP-01.

b. Comply with TIA-606, "Administration Standard for the Telecommunications Infrastructure of Commercial Buildings."

c. Obtain detailed Project planning forms from manufacturer of access-control system; develop custom forms to suit Project. Fill in all data available from Project plans and specifications and publish as Project planning documents for review and approval.

(1) Record setup data for control station and workstations.

(2) For each Location, record setup of Controller features and access requirements.

(3) Propose start and stop times for time zones and holidays, and match up access levels for doors.

(4) Set up groups, linking, and list inputs and outputs for each Controller.

(5) Assign action message names and compose messages.

(6) Set up alarms. Establish interlocks between alarms, intruder detection, and video surveillance features.

(7) Prepare and install alarm graphic maps.

(8) Develop user-defined fields.

(9) Develop screen layout formats.

(10) Propose setups for guard tours and key control.

(11) Discuss badge layout options; design badges.

(12) Complete system diagnostics and operation verification.

(13) Prepare a specific plan for system testing, startup, and demonstration.

(14) Develop acceptance test concept and, on approval, develop specifics of the test.

(15) Develop cable and asset management system details; input data from construction documents. Include system schematics and Technical Drawings.

d. In meetings with Architect and Owner, present Project planning documents and review, adjust, and prepare final setup documents. Use final documents to set up system software.

3.5 CABLING

a. Comply with NECA 1, "Good Workmanship in Electrical Contracting."

b. Install cables and wiring according to requirements in Division 28 Section "Conductors and Cables for Electronic Safety and Security."

c. Wiring Method: Install wiring in raceway and cable tray except within consoles, cabinets, desks, and counters. Conceal raceway and wiring except in unfinished spaces.

d. Wiring Method: Install wiring in raceway and cable tray except within

consoles, cabinets, desks, and counters and except in accessible ceiling spaces and in gypsum board partitions where unenclosed wiring method may be used. Use NRTL-listed plenum cable in environmental air spaces, including plenum ceilings. Conceal raceway and cables except in unfinished spaces.

e. Install LAN cables using techniques, practices, and methods that are consistent with Category 5E rating of components and that ensure Category 5E performance of completed and linked signal paths, end to end.

f. Install cables without damaging conductors, shield, or jacket.

g. Boxes and enclosures containing security system components or cabling, and which are easily accessible to employees or to the public, shall be provided with a lock. Boxes above ceiling level in occupied areas of the building shall not be considered to be accessible. Junction boxes and small device enclosures below ceiling level and easily accessible to employees or the public shall be covered with a suitable cover plate and secured with tamperproof screws.

h. Install end-of-line resistors at the field device location and not at the Controller or panel location.

3.6 CABLE APPLICATION

a. Comply with TIA-569, " Standard for Telecommunications Pathways and Spaces."

b. Cable application requirements are minimum requirements and shall be exceeded if recommended or required by manufacturer of system hardware.

c. RS-232 Cabling: Install at a maximum distance of 50 feet .

d. RS-485 Cabling: Install at a maximum distance of 4000 feet .

e. Card Readers and Keypads:

(1) Install number of conductor pairs recommended by manufacturer for the functions specified.

(2) Unless manufacturer recommends larger conductors, install No. 22 AWG wire if maximum distance from Controller to the reader is 250 feet , and install No. 20 AWG wire if maximum distance is 500 feet .

(3) For greater distances, install "extender" or "repeater" modules recommended by manufacturer of the Controller.

(4) Install minimum No. 18 AWG shielded cable to readers and keypads that draw 50 mA or more.

f. Install minimum No. 16 AWG cable from Controller to electrically powered locks. Do not exceed 250 feet.

g. Install minimum No. 18 AWG ac power wire from transformer to Controller, with a maximum distance of 25 feet.

3.7 GROUNDING

a. Comply with Section 26 05 26,Grounding and Bonding for Electrical Systems.

b. Comply with IEEE 1100, "Power and Grounding Sensitive Electronic Equipment."

c. Ground cable shields, drain conductors, and equipment to eliminate shock hazard and to minimize ground loops, common-mode returns, noise pickup, cross talk, and other impairments.

d. Signal Ground:

(1) Terminal: Locate in each equipment room and wiring closet; isolate from power system and equipment grounding.

(2) Bus: Mount on wall of main equipment room with standoff insulators.

(3) Backbone Cable: Extend from signal ground bus to signal ground terminal in each equipment room and wiring closet.

3.8 INSTALLATION

a. System installation shall be in accordance with UL 294, manufacturer and related documents and references, for each type of security subsystem designed, engineered and installed.

b. Components shall be configured with appropriate “service points” to pinpoint system trouble in less than 30 minutes.

c. The Contractor shall install all system components including Government furnished equipment, and appurtenances in accordance with the manufacturer's instructions, documentation listed in Sections 1.4 and 1.5 of this document, and shall furnish all necessary connectors, terminators, interconnections, services, and adjustments required for a operable system.

d. The PACS will be designed, engineered, installed, and tested to ensure all components are fully compatible as a system and can be integrated with all associated security subsystems, whether the system is a stand alone or a network.

e. For integration purposes, the PACS shall be integrated where appropriate with the following associated security subsystems:

(1) CCTV:

(a) Provide 24 hour coverage of all entry points to the perimeter and agency buildings. As well as all emergency exits utilizing a fixed color camera.

(b) Be able to monitor, control and record cameras on a 24 hours basis.

(c) Be programmed automatically call up a camera when an access point is but into an alarm state.

(d) For additional PACS system requirements as they relate to the CCTV, refer to Section 28 23 00, VIDEO SURVEILLANCE.

(2) IDS:

(a) Be able monitor door control sensors.

(b) Be able to monitor and control the IDS on a 24 hours basis.

(c) Be programmed to go into an alarm state when an IDS device is put into an alarm state, and notify the operator via an audible alarm.

(d) For additional PACS system requirements as they relate to the IDS, refer to Section 28 16 11, INTRUSION DETECTION SYSTEM.

(3) Security Access Detection:

(a) Be able to monitor all objects that have been screened with an x-ray machine and be able to monitor all data acquired by the bomb detection unit.

(b) For additional PACS system requirements as they relate to the Security Access Detection, refer to Section 28 13 53, SECURITY ACCESS DETECTION.

(4) EPPS:

(a) Be programmed to go into an alarm state when an emergency call box or duress alarm/panic device is activated, and notify the Physical Access Control System and Database Management of an alarm event.

(b) For additional PACS requirements as they relate to the EPPS, refer to Section 28 26 00, ELECTRONIC PERSONAL PROTECTION SYSTEM.

f. Integration with these security subsystems shall be achieved by computer programming or the direct hardwiring of the systems.

g. For programming purposes refer to the manufacturers requirements for correct system operations. Ensure computers being utilized for system integration meet or exceed the minimum system requirements outlined on the systems software packages.

h. The Contractor shall visit the site and verify that site conditions are in agreement with the design package. The Contractor shall report all changes to the site or conditions that will affect performance of the system. The Contractor shall not take any corrective action without written permission from the Government.

i. The Contractor shall visit the site and verify that site conditions are in agreement/compliance with the design package. The Contractor shall report all changes to the site or conditions that will affect performance of the system to the Contracting Officer in the form of a report. The Contractor shall not take any corrective action without written permission received from the Contracting Officer.

j. Existing Equipment:

(1) The Contractor shall connect to and utilize existing door equipment, control signal transmission lines, and devices as outlined in the design package. Door equipment and signal lines that are usable in their original configuration without modification may be reused with Contracting Officer approval.

(2) The Contractor shall perform a field survey, including testing and inspection of all existing door equipment and signal lines intended to be incorporated into the PACS, and furnish a report to the Contracting Officer as part of the site survey report. For those items considered nonfunctioning, provide (with the report) specification sheets, or written functional requirements to support the findings and the estimated cost to correct the deficiency. As part of the report, the Contractor shall include a schedule for connection to all existing equipment.

(3) The Contractor shall make written requests and obtain approval prior to disconnecting any signal lines and equipment, and creating equipment downtime. Such work shall proceed only after receiving Contracting Officer approval of these requests. If any device fails after the Contractor has commenced work on that device, signal or control line, the Contractor shall diagnose the failure and perform any necessary corrections to the equipment.

(4) The Contractor shall be held responsible for repair costs due to Contractor negligence, abuse, or improper installation of equipment.

(5) The Contracting Officer shall be provided a full list of all equipment that is to be removed or replaced by the Contractor, to include description and serial/manufacturer numbers where possible. The Contractor shall dispose of all equipment that has been removed or replaced based upon approval of the Contracting Officer after reviewing the equipment removal list. In all areas where equipment is removed or replaced the Contractor shall repair those areas to match the current existing conditions.

k. Enclosure Penetrations: All enclosure penetrations shall be from the bottom of the enclosure unless the system design requires penetrations from other directions. Penetrations of interior enclosures involving transitions of conduit from interior to exterior, and all penetrations on exterior enclosures shall be sealed with rubber silicone sealant to preclude the entry of water and will comply with VA Master Specification 07 84 00, Firestopping. The conduit riser shall terminate in a hot-dipped galvanized metal cable terminator. The terminator shall be filled with an approved sealant as recommended by the cable manufacturer and in such a manner that the cable is not damaged.

l. Cold Galvanizing: All field welds and brazing on factory galvanized boxes, enclosures, and conduits shall be coated with a cold galvanized paint containing at least 95 percent zinc by weight.

m. Control Panels:

(1) Connect power and signal lines to the controller.

(2) Program the panel as outlined by the design and per the manufacturer’s programming guidelines.

n. SMS:

(1) Coordinate with the VA agency's IT personnel to place the computer on the local LAN or Intranet and provide the security system protection levels required to insure only authorized VA personnel have access to the system.

(2) Program and set-up the SMS to ensure it is in fully operation.

o. Card Readers:

(1) Connect all signal inputs and outputs as shown and specified.

(2) Terminate input signals as required.

(3) Program and address the reader as per the design package.

(4) Readers shall be surface or flushed mounted and all appropriate hardware shall be provided to ensure the unit is installed in an enclosed conduit system.

p. Biometrics:

(1) Connect all signal input and output cables along with all power cables.

(2) Program and ensure the device is in operating order.

q. Portal Control Devices:

(1) Install all signal input and output cables as well as all power cables.

(2) Devices shall be surface or flush mounted as per the design package.

(3) Program all devices and ensure they are working.

r. Door Status Indicators:

(1) Install all signal input and output cables as well as all power cables.

(2) RTE's shall be surface mounted and angled in a manner that they cannot be compromised from the non-secure side of a windowed door, or allow for easy release of the locking device from a distance no greater than 6 feet from the base of the door.

(3) Door position sensors shall be surface or flush mounted and wide gap with the ability to operate at a maximum distance of up to 2″ .

s. Entry Control Devices:

(1) Install all signal input and power cables.

(2) Strikes and bolts shall be mounted within the door frame.

(3) Mortise locks shall be mounted within the door and an electric

transfer hinge shall be utilized to transfer the wire from within the door frame to the mortise lock inside the door.

(4) Electromagnetic locks shall be installed with the mag-lock mounted to the door frame and the metal plate mounted to the door.

t. System Start-Up:

(1) The Contractor shall not apply power to the PACS until the following items have been completed:

(a) PACS equipment items and have been set up in accordance with manufacturer's instructions.

(b) A visual inspection of the PACS has been conducted to ensure that defective equipment items have not been installed and that there are no loose connections.

(c) System wiring has been tested and verified as correctly connected as indicated.

(d) All system grounding and transient protection systems have been verified as installed and connected as indicated.

(e) Power supplies to be connected to the PACS have been verified as the correct voltage, phasing, and frequency as indicated.

(2) Satisfaction of the above requirements shall not relieve the Contractor of responsibility for incorrect installation, defective equipment items, or collateral damage as a result of Contractor work efforts.

(3) The Commissioning Agent will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Resident Engineer and Commissioning Agent. Provide a minimum of 7 days prior notice.

u. Supplemental Contractor Quality Control:

(1) The Contractor shall provide the services of technical representatives who are familiar with all components and installation procedures of the installed PACS; and are approved by the Contracting Officer.

(2) The Contractor will be present on the job site during the preparatory and initial phases of quality control to provide technical assistance.

(3) The Contractor shall also be available on an as needed basis to provide assistance with follow-up phases of quality control.

(4) The Contractor shall participate in the testing and validation of the system and shall provide certification that the system installed is fully operational as all construction document requirements have been fulfilled.

3.9 SYSTEM SOFTWARE

Install, configure, and test software and databases for the complete and

proper operation of systems involved. Assign software license to Owner.

3.10 FIELD QUALITY CONTROL

a. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust field-assembled components and equipment installation, including connections, and to assist in field testing. Report results in writing.

b. Testing Agency: Engage a qualified testing and inspecting agency to perform field tests and inspections and prepare test reports:

c. Perform the following field tests and inspections and prepare test reports:

(1) LAN Cable Procedures: Inspect for physical damage and test each conductor signal path for continuity and shorts. Use Class 2, bidirectional, Category 5 tester. Test for faulty connectors, splices, and terminations. Test according to TIA-568.1, "Commercial Building Telecommunications Cabling Standards - Part 1 General Requirements." Link performance for UTP cables must comply with minimum criteria in TIA-568.1.

(2) Test each circuit and component of each system. Tests shall include, but are not limited to, measurements of power supply output under maximum load, signal loop resistance, and leakage to ground where applicable. System components with battery backup shall be operated on battery power for a period of not less than 10 percent of the calculated battery operating time. Provide special equipment and software if testing requires special or dedicated equipment.

(3) Operational Test: After installation of cables and connectors, demonstrate product capability and compliance with requirements. Test each signal path for end-to-end performance from each end of all pairs installed. Remove temporary connections when tests have been satisfactorily completed.

3.11 PROTECTION

Maintain strict security during the installation of equipment and software. Rooms housing the control station, and workstations that have been powered up shall be locked and secured, with an activated burglar alarm and access-control system reporting to a Central Station complying with UL 1610, "Central-Station Burglar-Alarm Units," during periods when a qualified operator in the employ of Contractor is not present.

3.12 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 28 08 00,COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS for all inspection, start up, and contractor testing required above and required by the System Readiness Checklist provided by the Commissioning Agent.

b. Components provided under this section of the specification will be tested as part of a larger system. Refer to Section 28 08 00,COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS and related sections for contractor responsibilities for system commissioning.

3.13 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for four hours to instruct VA personnel in operation and maintenance of units.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 28 08 00, COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS.

c. Develop separate training modules for the following:

(1) Computer system administration personnel to manage and repair the LAN and databases and to update and maintain software.

(2) Operators who prepare and input credentials to man the control station and workstations and to enroll personnel.

(3) Security personnel.

(4) Hardware maintenance personnel.

(5) Corporate management.

d. All testing and training shall be compliant with the VA General Requirements,Section 01 00 00, GENERAL REQUIREMENTS.

-- End of Section --

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SECTION 28 23 00
9/11

VIDEO SURVEILLANCE

PART 1 – GENERAL

1.1 DESCRIPTION

a. Provide and install a complete extension of the existing campus Video Surveillance System, which is identified as the Video Assessment and Surveillance System hereinafter referred to as the VASS System as specified in this section.

b. This Section includes video surveillance system consisting of cameras, data transmission wiring, and a control station with its associated equipment.

c. Video surveillance system Video assessment & surveillance system shall be integrated with monitoring and control system specified in Division 28 Section PHYSICAL ACCESS CONTROL that specifies systems integration.

1.2 RELATED WORK

a. Section 01 00 00 - GENERAL REQUIREMENTS. For General Requirements.

b. Section 07 84 00 - FIRESTOPPING. Requirements for firestopping application and use.

c. Section 10 14 00 - SIGNAGE. Requirements for labeling and signs.

d. Section 14 21 00 - ELECTRIC TRACTION ELEVATORS. Requirements for elevators.

e. Section 14 24 00 - HYDRAULIC ELEVATORS. Requirements for elevators.

f. Section 26 05 11 - REQUIREMENTS FOR ELECTRICAL INSTALLATIONS. Requirements for connection of high voltage.

g. Section 26 05 21 - LOW VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES (600 VOLTS AND BELOW). Requirements for power cables.

h. Section 26 05 41 - UNDERGROUND ELECTRICAL CONSTRUCTION. Requirements for underground installation of wiring.

i. Section 26 56 00 - EXTERIOR LIGHTING. Requirements for perimeter lighting.

j. Section 28 05 00 – COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY. Requirements for general requirements that are common to more than one section in Division 28.

k. Section 28 05 13 - CONDUCTORS AND CABLES FOR ELECTRONIC SAFETY AND SECURITY. Requirements for conductors and cables.

l. Section 28 05 26 - GROUNDING AND BONDING FOR ELECTRONIC SAFETY AND SECURITY. Requirements for grounding of equipment.

m. Section 28 05 28.33 - CONDUITS AND BACKBOXES FOR ELECTRONIC SAFETY AND SECURITY. Requirements for infrastructure.

n. Section 28 08 00 - COMMISIONING OF ELECTRONIC SAFETY AND SECURITY. Requirements for commissioning, systems readiness checklists, and training.

o. Section 28 13 00 - PHYSICAL ACCESS CONTROL SYSTEM. Requirements for physical access control system integration.

1.3 DEFINITIONS

a. AGC: Automatic gain control.

b. B/W: Black and white.

c. CCD: Charge-coupled device.

d. CIF: Common Intermediate Format CIF images are 352 pixels wide and 88/240 (PAL/NTSC) pixels tall (352 x 288/240).

e. 4CIF: resolution is 704 pixels wide and 576/480 (PAL/NTSC) pixels tall (704 x 576/480).

f. H.264 (also known as MPEG4 Part 10): a encoding format that compresses video much more effectively than older (MPEG4) standards.

g. ips: Images per second.

h. MPEG: Moving picture experts group.

i. MPEG4: a video encoding and compression standard that uses inter-frame encoding to significantly reduce the size of the video stream being transmitted.

j. NTSC: National Television System Committee.

k. UPS: Uninterruptible power supply.

l. PTZ: refers to a movable camera that has the ability to pan left and right, tilt up and down, and zoom or magnify a scene.

1.4 QUALITY ASSURANCE

a. The Contractor shall be responsible for providing, installing, and the operation of the VASS System as shown. The Contractor shall also provide certification as required.

b. The security system shall be installed and tested to ensure all components are fully compatible as a system and can be integrated with all associated security subsystems, whether the security system is stand-alone or a part of a complete Information Technology (IT) computer network.

c. The Contractor or security sub-contractor shall be a licensed security Contractor as required within the state or jurisdiction of where the installation work is being conducted.

d. Manufacturers Qualifications: The manufacturer shall regularly and presently produce, as one of the manufacturer's principal products, the equipment and material specified for this project, and shall have manufactured the item for at least three years.

e. Product Qualification:

(1) Manufacturer's product shall have been in satisfactory operation, on three installations of similar size and type as this project, for approximately three years.

(2) The Government reserves the right to require the Contractor to submit a list of installations where the products have been in operation before approval.

f. Contractor Qualification:

(1) The Contractor or security sub-contractor shall be a licensed security Contractor with a minimum of five (5) years experience installing and servicing systems of similar scope and complexity. The Contractor shall be an authorized regional representative of the Video Assessment and Surveillance System’s (VASS) manufacturer. The Contractor shall provide four (4) current references from clients with systems of similar scope and complexity which became operational in the past three (3) years. At least three (3) of the references shall be utilizing the same system components, in a similar configuration as the proposed system. The references must include a current point of contact, company or agency name, address, telephone number, complete system description, date of completion, and approximate cost of the project. The owner reserves the option to visit the reference sites, with the site owner’s permission and representative, to verify the quality of installation and the references’ level of satisfaction with the system. The Contractor shall provide copies of system manufacturer certification for all technicians. The Contractor shall only utilize factory-trained technicians to install, program, and service the VASS. The Contractor shall only utilize factory-trained technicians to install, terminate and service cameras, control, and recording equipment. The technicians shall have a minimum of five (5) continuous years of technical experience in electronic security systems. The Contractor shall have a local service facility. The facility shall be located within 60 miles of the project site. The local facility shall include sufficient spare parts inventory to support the service requirements associated with this contract. The facility shall also include appropriate diagnostic equipment to perform diagnostic procedures. The COTR reserves the option of surveying the company’s facility to verify the service inventory and presence of a local service organization.

(2) The Contractor shall provide proof project superintendent with BICSI Certified Commercial Installer Level 1, Level 2, or Technician to provide oversight of the project.

(3) Cable installer must have on staff a Registered Communication Distribution Designer (RCDD) certified by Building Industry Consulting Service International. The staff member shall provide consistent oversight of the project cabling throughout design,

layout, installation, termination and testing.

g. Service Qualifications: There shall be a permanent service organization maintained or trained by the manufacturer which will render satisfactory service to this installation within [four] [eight] hours of receipt of notification that service is needed. Submit name and address of service organizations.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Security System Operational Test Plan; G

Service Qualifications; G

Manufacturers Qualifications; G

Product Qualification; G

Contractor Qualification; G

SD-03 Product Data

Manufacturer Security System Product Cut-Sheets; G

SD-05 Design Data

Pre-Installation Design Packages; G

SD-06 Test Reports

Completed System Readiness Checklists; G

SD-07 Certificates

Certificates Of Compliance; G

Manufacture’s Certification Of Underwriters Laboratories, Inc. (Ul) Listing; G

SD-11 Closeout Submittals

As-Built Design Package; G

1.6 SUBMITTAL INSTRUCTIONS

a. Submit below items in conjunction with Master Specification Sections 01 33 23, Shop Drawings, Product Data, and Samples, and Section 02 41 00, Demolition Drawings.

b. Provide certificates of compliance with Section 1.4, Quality Assurance.

c. Provide a pre-installation and as-built design package in both

electronic format and on paper, minimum size (48 x 48 inches); drawing submittals shall be per the established project schedule.

d. Pre-installation design and as-built packages shall include, but not be limited to:

(1) Index Sheet that shall:

(a) Define each page of the design package to include facility name, building name, floor, and sheet number.

(b) Provide a list of all security abbreviations and symbols.

(c) Reference all general notes that are utilized within the design package.

(d) Specification and scope of work pages for all security systems that are applicable to the design package that will:

1. Outline all general and job specific work required within the design package.

2. Provide a device identification table outlining device Identification (ID) and use for all security systems equipment utilized in the design package.

(2) Floor plans, site plans, and enlarged plans shall:

(a) Include a title block as defined above.

(b) Define the drawings scale in both standard and metric measurements.

(c) Provide device identification and location.

(d) Address all signal and power conduit runs and sizes that are associated with the design of the electronic security system and other security elements (e.g., barriers, etc.).

(e) Identify all pull box and conduit locations, sizes, and fill capacities.

(f) Address all general and drawing specific notes for a particular drawing sheet.

(3) A riser drawing for each applicable security subsystem shall:

(a) Indicate the sequence of operation.

(b) Relationship of integrated components on one diagram.

(c) Include the number, size, identification, and maximum lengths of interconnecting wires.

(d) Wire/cable types shall be defined by a wire and cable schedule. The schedule shall utilize a lettering system that will correspond to the wire/cable it represents (example: A = 18 AWG/1 Pair Twisted, Unshielded). This schedule shall also provide the manufacturer’s name and part number for the wire/cable being

installed.

(4) A system drawing for each applicable security system shall:

(a) Identify how all equipment within the system, from main panel to device, shall be laid out and connected.

(b) Provide full detail of all system components wiring from point-to-point.

(c) Identify wire types utilized for connection, interconnection with associate security subsystems.

(d) Show device locations that correspond to the floor plans.

(e) All general and drawing specific notes shall be included with the system drawings.

(5) A schedule for all of the applicable security subsystems shall be included. All schedules shall provide the following information:

(a) Device ID.

(b) Device Location (e.g. site, building, floor, room number, location, and description).

(c) Mounting type (e.g. flush, wall, surface, etc.).

(d) Power supply or circuit breaker and power panel number.

(e) In addition, for the VASS Systems, provide the camera ID, camera type (e.g. fixed or pan/tilt/zoom (P/T/Z), lens type (e.g. for fixed cameras only) and housing model number.

(6) Detail and elevation drawings for all devices that define how they were installed and mounted.

e. Pre-installation design packages shall be reviewed by the Contractor along with a VA representative to ensure all work has been clearly defined and completed. All reviews shall be conducted in accordance with the project schedule. There shall be four (4) stages to the review process:

(1) 35 percent

(2) 65 percent

(3) 90 percent

(4) 100 percent

f. Provide manufacturer security system product cut-sheets. Submit for approval at least 30 days prior to commencement of formal testing, a Security System Operational Test Plan. Include procedures for operational testing of each component and security subsystem, to include performance of an integrated system test.

g. Submit manufacture’s certification of Underwriters Laboratories, Inc. (UL) listing as specified. Provide all maintenance and operating

manuals per the VA General Requirements, Section 01 00 00, GENERAL REQUIREMENTS.

h. Submit completed System Readiness Checklists provided by the Commissioning Agent and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 28 08 00 COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS.

1.7 REFERENCES

a. The publications listed below (including amendments, addenda, revisions, supplement, and errata) form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C2	(2023) National Electrical Safety Code
IEEE C62.41	(1991; R 1995) Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits
IEEE 802.3	(2022; CORR 2024) Ethernet

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)

NECA 303	(2005) Installing Closed Circuit Television (CCTV) Systems

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST FIPS 140-2	(2001) Security Requirements for Cryptographic Modules

UL SOLUTIONS (UL)

UL 1	(2005; Reprint Jan 2022) UL Standard for Safety Flexible Metal Conduit

U.S. DEPARTMENT OF VETERANS AFFAIRS (VA)

VA PG-18-10	Design Manuals

1.8 COORDINATION

a. Coordinate arrangement, mounting, and support of video surveillance equipment:

(1) To allow maximum possible headroom unless specific mounting heights that reduce headroom are indicated.

(2) To provide for ease of disconnecting the equipment with minimum

interference to other installations.

(3) To allow right of way for piping and conduit installed at required slope.

(4) So connecting raceways, cables, wireways, cable trays, and busways will be clear of obstructions and of the working and access space of other equipment.

b. Coordinate installation of required supporting devices and set sleeves in cast-in-place concrete, masonry walls, and other structural components as they are constructed.

c. Coordinate location of access panels and doors for video surveillance items that are behind finished surfaces or otherwise concealed.

1.9 WARRANTY OF CONSTRUCTION

a. Warrant VASS System work subject to the Article “Warranty of Construction” of FAR clause 52.246-21.

b. Demonstration and training shall be performed prior to system acceptance.

PART 2 – PRODUCTS

2.1 GENERAL

a. Video signal format shall comply with the NTSC standard composite video, interlaced. Composite video signal termination shall be 75 ohms.

b. Surge Protection: Protect components from voltage surges originating external to equipment housing and entering through power, communication, signal, control, or sensing leads. Include surge protection for external wiring of each conductor entry connection to components.

c. Power Connections: Comply with requirements in Section 28 05 00 COMMON WORK REQUIREMENTS FOR ELECTRONIC SAFETY AND SECURITY, Part 2, as recommended by manufacturer for type of line being protected.

d. Tamper Protection: Tamper switches on enclosures, control units, pull boxes, junction boxes, cabinets, and other system components shall initiate a tamper-alarm signal when unit is opened or partially disassembled. Control-station, control-unit alarm display shall identify tamper alarms and indicate locations.

2.2 CAMERAS

a. All Cameras will be EIA 330 and UL 1. Minimum Protection for Power Connections 120 V and more: Auxiliary panel suppressors shall comply with requirements in Section 28 05 00 COMMON WORK REQUIREMENTS FOR ELECTRONIC SAFETY AND SECURITY, Part 2.

b. Minimum Protection for Communication, Signal, Control, and Low-Voltage 983 compliant as well as:

(1) Will be charge coupled device (CCD cameras and shall conform to

National Television System Committee (NTSC) formatting.

(2) Fixed cameras shall be color and the primary choice for monitoring following the activities described below. Pan/Tilt/Zoom (P/T/Z) cameras shall be color and are to be utilized to complement the fixed cameras.

(3) Shall be powered by either 12 volts direct current (VDC) or 24 volts alternate current (VAC). Power supplies shall be Class 2 and UL compliant and have a back-up power source to ensure cameras are still operational in the event of loss of primary power to the VASS System.

(4) Shall be powered over Ethernet. Network switches supporting PoE cameras shall have a back-up power source to ensure cameras are still operational in the event of loss of primary power to the VASS System.

(5) Shall be rated for continuous operation under the environmental conditions listed in Part 1, Project Conditions.

(6) Will be home run to a monitoring and recording device via a controlling device such as a matrix switcher or network server and monitored on a 24 hour basis at a designated Security Management System location.

(7) Each function and activity shall be addressed within the system by a unique user defined name, with minimum of twenty (20) characters. The use of codes or mnemonics identifying the VASS action shall not be accepted.

(8) Shall come with built-in video motion detection that shall automatically monitor and process information from each camera. The camera motion detection shall detect motion within the camera's field of view and provide automatic visual, remote alarms as a result of detected motion.

(9) Shall be programmed to digitally flip from color to black and white at dusk and vice versa at low light conditions.

(10) Will be fitted with AI/DC lenses to ensure the image quality under different light conditions.

(11) P/T/Z cameras shall be utilized in a manner that they complement fixed cameras and shall not be used as a primary means of monitoring activity.

(12) Dummy or fake cameras will not be utilized at any time.

(13) Appropriate signage shall be designed, provided, and posted that notifies people that an area is under camera surveillance.

2.3 VIDEO MANAGEMENT SYSTEM

a. Extend the existing Johnson Controls VidSys Video Management System (VMS) shall be extended to provide all existing features and functions as currently exist on the system.:

2.4 DIGITAL BASED VIDEO MANAGEMENT SYSTEM

a. Extend exisitng system. Provide new licenses as required to extend existing system with all exisitng features.

2.5 VIDEO DISPLAY EQUIPMENT

a. Extend existing video display equipment to include all new cameras.

2.6 CONTROLLING EQUIPMENT

a. Extend existing controlling equipment as required to operate all new cameras.

2.7 Not Used
2.8 Network Cameras

a. Megapixel High Definition Integrated Digital Network Camera

(1) The network camera shall offer dual video streams with up to 3.1 megapixel resolution (2048 x 1536) in progressive scan format.

(2) An alarm input and relay output shall be built in for integration with hard wired external sensors.

(3) The network camera shall be capable of firmware upgrades through a network using a software-based device utility.

(4) The network camera shall offer auto back focus (ABF) functionality through a push button on the camera. ABF parameters shall also be configurable through a standard Web browser interface.

(5) The network camera shall offer a video output port providing an NTSC/PAL analog video output signal for adjusting field of view and focus at the camera.

(6) The network camera shall provide advanced low-light capabilities for color and day/night models with sensitivity down to 0.12 lux in color and 0.03 lux in black-white (B-W).

(7) The network camera shall have removable IR cut filter mechanism for increased sensitivity in low-light installations. The sensitivity of IR cut filter removal shall be configurable through a Web browser.

(8) The network camera shall support two simultaneous, configurable video streams. H.264 and MJPEG compression formats shall be available for primary and secondary streams with selectable unicast and multicast protocols. The streams shall be configurable in a variety of frame rates and bit rates.

(9) The network camera shall support industry standard Power over Ethernet (PoE)

(10) IEEE 802.3 to supply power to the camera over the network. The network camera shall also offer a 24 VAC power input for optional use.

(11) The network camera shall use a standard Web browser interface for remote administration and configuration of camera parameters.

(12) The network camera shall have a window blanking feature to conceal user-defined privacy areas that cannot be viewed by an operator. The network camera shall support up to four blanked windows. A blanked area shall appear on the screen as a solid gray window.

(13) The network camera shall support standard IT protocols.

(14) The network camera shall support open architecture best practices with a published API available to third-party network video recording and management systems.

(15) Megapixel High Definition Integrated Digital Network Camera Technical Specifications:

Imaging Device	1/3-inch, effective
Imager Type	CMOS, Progressive scan
Maximum Resolution	2048 x 1536
Signal-to-Noise Ratio	50 dB
Auto Iris Lens Type	DC drive
Electronic Shutter Range	1~1/100,000 sec
Wide Dynamic Range	60 dB
White Balance Range	2,000° to 10,000°K
Sensitivity	f/1.2; 2,850K; SNR >24dB Color (1x/33ms) 0.50 lux Color SENS (15x/500 ms) 0.12 lux Mono SENS (15x/500 ms) Mono
Dome Attenuation	Clear Zero light loss Smoke f/1.0 light loss
Compression	H.264 in base profile and MJPEG

Video Streams	Up to 2 simultaneous streams, the second Stream variable based on the setup of the primary stream
Frame Rate	Up to 30, 25, 24, 15, 12.5, 12, 10, 8, 7.5, 6.5, 4, 3, 2, and 1 (depending upon coding, resolution, and stream configuration
Available Resolutions	3.1 MPx2048 x 1536; 4:3 aspect ratio; 2.0 ips max., 10.0 Mbps bit rate for MJPEG; 3.0 ips max., 2.6 Mbps bit rate H.264
Supported Protocols	TCP/IP, UDP/IP (Unicast, Multicast IGMP), UPnP, DNS, DHCP, RTP, RTSP, NTP,IPv4, SNMP, QoS, HTTP, HTTPS, LDAP(client), SSH, SSL, STMP, FTP,
Security Access	Password protected
Software Interface	Web browser view and setup, up to 16 cameras
Connectors	RJ-45 for 100Base-TX, Auto MDI/MDI-X
Cable	Cat6A cable for 100Base-TX
Input Voltage	24 VAC or PoE (IEEE 802.3 class 3)
Power Consumption	6 to 10 W
Current Consumption	PoE <200 mA maximum 24 VAC <295 mA nominal; <390 mA maximum
Alarm Input	10 VDC maximum, 5 mA maximum
Alarm Output	0 to 15 VDC maximum, 75 mA maximum
Lens Mount	CS mount, adjustable
Pan/Tilt Adjustment	Pan 368° Tilt 160° (10° to 170°) Rotate 355°

(16) Accessories

(a) Pendant mount

(b) Wall mount for pendant

(c) Corner adapter for wall mount

(d) Pole adapter for wall mount

(17) Recommended Lenses

(a) Megapixel lens, varifocal, 2.2~6.0 mm, f/1.3~2.0

(b) Megapixel lens, varifocal, 2.8~8.0 mm, f/1.1~1.9

(c) Megapixel lens, varifocal, 2.8~12.0 mm, f/1.4~2.7

(d) Megapixel lens, varifocal, 15.0~50.0 mm, f/1.5~2.1

b. LENSES

(1) Camera Field of View shall be set by the Contractor to produce full view of door or window opening and anyone entering or leaving through it. Follow the project construction drawings for design intent.

(2) Camera Lenses shall be of the type supplied with the camera from the manufacture. All cameras which are not supplied with lenses from the factory are specified in this specification. The lens shall be equipped with an auto-iris mechanism unless otherwise specified. Lenses having auto-iris, DC iris, or motor zoom functions shall be supplied with connectors, wiring, receiver/drivers, and controls as needed to operate the lens functions. Lenses shall have sufficient circle of illumination to cover the image sensor evenly. Lenses shall not be used on a camera with an image format larger than the lens is designed to cover. Lenses shall be provided with pre-set capability.

(3) Lenses shall have optical-quality coated optics, designed specifically for video surveillance applications, and matched to specified camera. Provide color-corrected lenses with color cameras, megapixel lenses for megapixel cameras, and lenses with day/night for color/b&w cameras.

(4) Auto-Iris Lens: Electrically controlled iris with circuit set to maintain a constant video level in varying lighting conditions.

(5) Zoom Lenses: Motorized, remote-controlled units, rated as "quiet operating." Features include the following:

(a) Electrical Leads: Filtered to minimize video signal interference.

(b) Motor Speed: Variable.

(c) Lens shall be available with preset positioning capability to recall the position of specific scenes.

(6) Lenses: Shall be utilized in a manner that provides maximum coverage of the area being monitored by the camera. The lenses shall:

(a) Be 1/3" to fit CCD fixed camera.

(b) Be all glass with coated optics.

(c) Have mounts that are compatible with the camera selected.

(d) Be packaged and supplied with the camera.

(e) Have a maximum f-stop of f/1.3 for fixed lenses, and a maximum f-stop of f/1.6 for variable focus lenses.

(f) Be equipped with an auto-iris mechanism.

(g) Have sufficient circle of illumination to cover the image sensor evenly.

(h) Not be used on a camera with an image format larger than the lens is designed to cover.

(i) Be provided with pre-set capability.

(7) Two types of lenses shall be utilized for both interior and exterior fixed cameras:

(a) Manual Variable Focus

(b) Auto Iris Fixed

(8) Manual Variable Focus:

(a) Shall be utilized in large areas that are being monitored by the camera. Examples of this are perimeter fence lines, vehicle entry points, parking areas, etc.

(b) Shall allow for setting virtually any angle of field, which maximizes surveillance effects.

(c) Technical Characteristics:

Image format	1/3 inch
Focal length	5–50mm
Iris range	F1.4 to close
Focus range	(3.3 ft)
Back focus distance	(0.4 in)
Angle view Wide (1/3 in)	53.4 x 40.1
Angle view Tele (1/3 in)	5.3 x 4.1
Iris control	manual
Focus ctrl	manual
Zoom ctrl	manual

c. CAMERA HOUSINGS AND MOUNTS

(1) This section pertains to all interior and exterior housings, domes, and applicable wall, ceiling, corner, pole, and rooftop

mounts associated with the housing. Housings and mounts shall be specified in accordance to the type of cameras used.

(2) All cameras and lenses shall be enclosed in a tamper resistant housing. Any additional mounting hardware required to install the camera housing at its specified location shall be provided along with the housing.

(3) The camera and lens contained inside the housing shall be installed on a camera mount. All additional mounting hardware required to install the camera housing at its specified location shall be provided along with the housing.

(4) Shall be manufactured in a manner that are capable of supporting a maximum of three (3) cameras with housings, and meet environmental requirements for the geographical area the camera support equipment is being installed on or within.

(5) Environmentally Sealed

(a) Shall be designed in manner that it provides a condensation free environment for correct camera operation.

(b) Shall be operated in a 100 percent condensing humidity atmosphere.

(c) Shall be constructed in a manner that:

1. Has a fill valve to allow for the introduction of nitrogen into the housing to eliminate existing atmospheric air and pressurize the housing to create moisture free conditions.

2. Has an overpressure valve to prevent damage to the housing in the event of over pressurization.

3. Is equipped with a humidity indicator that is visible to the eye to ensure correct atmospheric conditions at all times.

4. The leak rate of the housing is not to be greater than or 2 pounds per square inch at sea level within a 90 day period.

5. It shall contain camera mounts or supports as needed to allow for correct positioning of the camera and lens.

6. The housing and sunshield are to be white in color.

(6) All electrical and signal cables required for correct operations shall be supplied in a hardened carrier system from the controller to the camera.

(7) The mounting bracket shall be adjustable to allow for the housing weight of the camera and the housing unit it is placed in.

(8) Accessibility to the camera and mounts shall be taken into consideration for maintenance and service purposes.

d. Indoor Mounts

(1) Ceiling Mounts:

(a) This enclosure and mount shall be installed in a finished or suspended ceiling.

(b) The enclosure and mount shall be fastened to the finished ceiling, and shall not depend on the ceiling tile grid for complete support.

(c) Suspended ceiling mounts shall be low profile, and shall be suitable for replacement of 2 foot by 2 foot ceiling tiles.

(2) Wall Mounts:

(a) The enclosure shall be installed in manner that it matches the existing décor and placed at a height that it will be unobtrusive, unable to cause personal harm, and prevents tampering and vandalism.

(b) The mount shall contain a manual pan/tilt head that will provide 360 degrees of horizontal and vertical positioning from a horizontal position, and has a locking bar or screw to maintain its fixed position once it has been adjusted.

e. Interior Domes

(1) The interior dome shall be a pendant mount, pole mount, ceiling mount, surface mount, or corner mounted equipment.

(2) The lower portion of the dome that provides camera viewing shall be made of black opaque acrylic and shall have a light attenuation factor of no more that 1 f-stop.

(3) The housing shall be equipped with integral pan/tilt capabilities complete with wiring, wiring harness, connectors, receiver/driver, pan/tilt control system, pre-position cards, or any other hardware and equipment as needed to fully provide a fully functional pan/tilt dome.

(4) The pan/tilt mechanism shall be:

(a) Constructed of heavy duty bearings and hardened steel gears.

(b) Permanently lubricated to ensure smooth and consistent movement of all parts throughout the life of the product.

(c) Equipped with motors that are thermally or impedance protected against overload damage.

(d) Pan movements shall be 360 degrees and tilt movement shall no be less than +/- 90 degrees.

(e) Pan speed shall be a minimum of 10 degrees per second.

f. Exterior Domes

(1) The exterior dome shall meet all requirements outlined in the interior dome paragraph above.

(2) The housing shall be constructed to be dust and water tight, and

fully operational in 100 percent condensing humidity.

g. Exterior Wall Mounts

(1) Shall have an adjustable head for mounting the camera.

(2) Shall be constructed of aluminum, stainless steel, or steel with a corrosion-resistant finish.

(3) The head shall be adjustable for not less than plus and minus 90 degrees of pan, and not less than plus and minus 45 degrees of tilt. If the bracket is to be used in conjunction with a pan/tilt, the bracket shall be supplied without the adjustable mounting head, and shall have a bolt-hole pattern to match the pan/tilt base.

(4) Shall be installed at a height that allows for maximum coverage of the area being monitored.

2.9 POWER SUPPLIES

a. Power supplies shall be a low-voltage power supplies matched for voltage and current requirements of cameras and accessories, type as recommended by camera, infrared illuminator, and lens manufacturer.

b. Technical specifications:

(1) Input: 115VAC, 50/60Hz, 2.7 amps

(2) Outputs:

(a) Number of outputs, 16

(b) Fuse/PTC protected, power limited

(c) Output voltage & power:

1. 24VAC @ 12.5 amps (300VA) or 28VAC @ 10 amp (280VA) supply current

(3) Illuminated power disconnect circuit breaker with manual reset

(4) Surge suppression

(5) Camera synchronization

(6) Wall/Rack mount.

(7) Enclosure: NEMA 250, Type 1.

2.10 [Enter Appropriate Subpart Title Here]Not Used

2.11 NETWORK SERVER

a. Camera system is existing and shall be extended into the new Chiller Plant. Provide all required upgrades to the existing server, including licences to accomodate the expansion.

2.12 RECORDING DEVICES

a. All cameras on the VASS System shall be recorded in real time using a Digital Video Recorder (DVR), Network Video Recorder (NVR), or attached storage. The type of recording device utilized should be determined by the size and type of VASS System designed and installed, and to what extent the system is to be utilized.

b. All recording devices shall be 19 inch rack-mountable.

c. All DVR's and NVR's that are viewable over an Intranet or Internet will be routed through an encryptor.

d. Encryptors shall:

(1) Comply with NIST FIPS 140-2.

(2) Support TCP/IP.

(3) Directly interfaces to low-cost commercial routers.

(4) Provide packet-based crypto synchronization.

(5) Encrypt source and destination IP addresses.

(6) Support web browser based management requiring no additional software.

(7) Have a high data sustained throughput – 1.544 Mbps (T1) full duplex data rate.

(8) Provide for both bridging and routing network architecture support.

(9) Support Electronic Key Management System (EKMS) compatible.

(10) Have remote management ability.

(11) Automatically reconfigure when secure network or wide area network changes.

e. Digital Video Recorder (DVR)

(1) Shall record video to a hard drive-based digital storage medium in either NTSC or MPEG format.

(2) Shall meet the following minimum requirements:

(a) Record at minimum rate of 30 images per second (IPS).

(b) Have a minimum of eight (8) to 16 looping inputs.

(c) Have a minimum of eight (8) to 16 alarm inputs and two (2) relay outputs.

(d) Shall provide instantaneous playback of all recorded images.

(e) Be IP addressable, if part of a VASS network.

(f) Have built-in digital motion detection with masking and

sensitivity adjustments.

(g) Provide easy playback and forward/reverse search capabilities.

(h) Complete audit trail database, with minimum of a six-month history that tracks all events related to the alarm; specifically who, what, where and when.

(i) DVR management capability providing automatic video routing to a back-up spare recorder in case of failure.

(j) Accessible locally and remotely via the Internet, Intranet, or a personal digital assistant (PDA).

(k) Records all alarm events in real time, ensuring 60 seconds before and after the event are included in the recording.

(l) Utilize RS-232 or fiber optic connections for integration with the SMS computer station via a remote port on a network hub.

(m) Allow for independently adjustable frame rate settings.

(n) Be compatible with the matrix switcher utilized to operate the cameras. The DVR could be utilized as a matrix switcher only if it meets all of the requirements listed in the matrix switcher section.

(3) Technical Characteristics:

Compression	MPEG-4
Internal Storage Capacities.	1 TB, and 2 TB. Available USB hard drive up to 250 GB. Optional internal DVD available
Digital Recording	Up to 16 video and 8 audio channels.
Full real-time video recording	Up to 400 IPS@352 x 288: PAL
Multiple simultaneous functions	Live viewing, Recording, playback, network transmission, back-up
Search functions	Date/time search, event search, bookmark search, smart (pixel) Search
PTZ Control	Third party PTZ control

User ID security	3 levels
Connectivity to external devices:	Sixteen 16 video input and looping output channels. VGA and dual monitor BNC outputs. Eight (8) audio inputs and one (1) audio output. Ethernet 10/100BaseT network connection. Sixteen (16) alarm inputs and eight (8) relay outputs. Biphase connection to control Bosch PTZ cameras. Third party PTZ control via RS-422/RS-485 connection. Front and back USB connectors to connect to a PC mouse, or archive video to a USB memory stick or similar device.
PC requirements	Windows 2000 or above; DirectX 8.1 or above. Intel Pentium III or above, AMD Athlon with 800 MHz or faster CPU.
Electrical	Power Input: 100 to 240 VAC; 50/60 Hz Power consumption: 120W Max. 1.2A
Video	Video standard: PAL or NTSC selectable. Resolution: 704 x 576 PAL, 704 x 480 NTSC Compression: MPEG-4 Inputs: 8 or 16 composite video 0.5-2 Vpp, 75 Ohm automatic termination. Outputs 8 or 16 composite video 1 Vpp, 75 Ohm.
Audio	Inputs: 4 or 8 line in, 30 kOhm Output: 1 line, 100 kOhm
Monitors	VGA: analog RGB 800x600 MON A: CVBS 1 Vpp0.1 V, 75 Ohm, BNC Monitor A multi-screen (VGA or CVBS) MON B: CVBS 1 Vpp0.1 V, 75 Ohm, BNC Monitor B spot/alarm
Frame Rate and Resolution	16-channels PAL: Up to 400 IPS@352x288, up to 200 IPS@704x288, up to 100 IPS@704x576.

Alarm inputs	16 configurable NO/NC, max. input 5 VDC.
Alarm outputs	8 relay outputs, configurable NO/NC, max. rated 1A, 125 VAC.
Connections	Ethernet: RJ45 modular jack 8 pins shielded, 10/100 Base-T. Biphase: Screw terminal connector (5
Network:	Transmission speed: up to 120 IPS@352x240 Bandwidth control: Automatic Remote users: Maximum 5 simultaneous connected Control Center users.

Processor	Intel Pentium III 750 MHz
Memory	256 MB RAM
Operating System	Windows 98, NT, ME, 2000, and XP
Video Card	4 MB of RAM capable of 24-bit true color
Free Hard Disk Space	160 MB for software installation
Network Card	10Base-T network for LAN operation
Archiving	80 GB, 160 GB, 320 GB and 640 GB Hard Drive;
Video Input	1.0 Vpp (signal 714mV, sync 286mV) 75 ohms
Video Output Level	1.0 Vpp +/-10%,75 ohms(BNC unbalanced)
Impedance	75 ohms/Hi- impedance x 16 switchable
Network Interface	Ethernet (RJ-45, 10/100M)
Network Protocol	TCP/IP, DHCP, HTTP, UDP
Network Capabilities	Live/Playback/P/T/Z control
Recording Rate	30 ips for 720 x 240 (NTSC)
Password Protection	Menu Setup, Remote Access
Recording Capacity	160 (1 or 2 fixed HDD) 1 CD-RW
Power Interrupt	Auto recovered to recording mode

f. Network Video Recorder (NVR)

(1) Shall record video to a hard drive-based digital storage medium in MPEG, MPEG4 or H.264 format.

(2) Shall meet the following minimum requirements:

(a) Record at minimum rate of 30 IPS.

(b) Have a minimum of 16 looping inputs.

(c) Have a minimum of 16 alarm inputs and two (2) relay outputs.

(d) Shall provide instantaneous playback of all recorded images.

(e) Be IP addressable, if part of a VASS network.

(f) Have built-in digital motion detection with masking and sensitivity adjustments.

(g) Easy playback and forward/reverse search capabilities.

(h) Complete audit trail database, with minimum of a six-month history that tracks all events related to the alarm; specifically who, what, where and when.

(i) NVR management capability providing automatic video routing to a back-up spare recorder in case of failure.

(j) Accessible locally and remotely via the internet, intranet, or a personal digital assistant (PDA).

(k) Records all alarm events in real time, ensuring 60 seconds before and after the event are included in the recording.

(l) Utilize RS-232 or fiber optic connections for integration with the SMS computer station via a remote port on a network hub.

(m) Allow for independently adjustable frame rate settings.

(n) Be compatible with the matrix switcher utilized to operate the cameras.

(3) Technical Characteristics:

Hardware/CPU	Pentium III Xeon or IV, 1.8 GHz
HDD Interface	IDE or better; optional: SCSI II, SCSI Ultra, or Fiber Channel
RAM	1024 MB
Operating System	Windows 2000/XP Professional/Server 2003 Standard
Graphic	Card VGA
Ethernet Card	100/1000 MB
Memory	20 MB
Software Setup	Centralized setup from each authorized PC; access via integrated web server
Storage Media	All storage media possible (e.g., HD, RAID), depending on operating system
Storage Mode	Linear mode, ring mode (capacity-based)
Recording Configuration	Camera name assignment, bandwidth limit, frame rate, video quality
Recording Content	Video and/or audio data
Search Parameters	Time, date, event
Playback	Playback via any IP network (LAN/WAN) simultaneous recording, playback, and backup
Network Interface	Ethernet (RJ-45, 10/100M)
Network Protocol	TCP/IP, DHCP, HTTP, UDP
Network Capabilities	Live/Playback/P/T/Z control
Recording Rate	30 ips for 720 x 240 (NTSC)
Password Protection	Menu Setup, Remote Access
Recording Capacity	160 (1 or 2 fixed HDD) 1 CD-RW
Power Interrupt	Auto recovered to recording mode

2.13 WIRES AND CABLES

a. Shall meet or exceed the manufactures recommendation for power and signal.

b. Will be carried in an enclosed conduit system, utilizing electromagnetic tubing (EMT) to include the equivalent in flexible metal, rigid galvanized steel (RGS) to include the equivalent of liquid tight, polyvinylchloride (PVC) schedule 40 or 80.

c. All conduits will be sized and installed per the NEC. All security system signal and power cables that traverse or originate in a high security office space will contained in either EMT or RGS conduit.

d. All conduit, pull boxes, and junction boxes shall be clearly marked with colored permanent tape or paint that will allow it to be distinguished from all other conduit and infrastructure.

e. Conduit fills shall not exceed 50 percent unless otherwise documented.

f. A pull string shall be pulled along and provided with signal and power cables to assist in future installations.

g. At all locations where there is a wall penetration or core drilling is conducted to allow for conduit to be installed, fire stopping materials shall be applied to that area

h. High voltage and signal cables shall not share the same conduit and shall be kept separate up to the point of connection. High voltage for the security system shall be defined as any cable or sets of cables carrying 30 VDC/VAC or higher.

i. For all equipment that is carrying digital data between the Physical Access Control System and Database Management or at a remote monitoring station, shall not be less that 20 AWG and stranded copper wire for each conductor. The cable or each individual conductor within the cable shall have a shield that provides 100% coverage. Cables with a single overall shield shall have a tinned copper shield drain wire.

j. All cables and conductors, except fiber optic cables, that act as a control, communication, or signal lines shall include surge protection. Surge protection shall be furnished at the equipment end and additional triple electrode gas surge protectors rated for the application on each wire line circuit shall be installed within (3 ft.) of the building cable entrance. The inputs and outputs shall be tested in both normal and common mode using the following wave forms:

   (1) A 10 microsecond rise time by 1000 microsecond pulse width waveform with a peak voltage of 1500 watts and peak current of 60 amperes.

   (2) An 8 microsecond rise time by 20 microsecond pulse width wave form with a peak voltage of 1000 volts and peak current of 500 amperes.

k. The surge suppression device shall not attenuate or reduce the video or sync signal under normal conditions. Fuses and relays shall not be used as a means of surge protection.

l. Coaxial Cables

(1) All video signal cables for the VASS System, with exception to the PoE cameras, shall be a coaxial cable and have a characteristic impedance of 75 ohms plus or minus 3 ohms.

(2) For runs up to 750 feet use of an RG-59/U is required. The RG-59/U shall be shielded which provides a minimum of 95 percent coverage, with a stranded copper center conductor of a minimum 23 AWG, polyethylene insulation, and black non-conductive polyvinylchloride (PVC) jacket.

(3) For runs between 750 feet and 1250 feet, RG-6/U is required. RG-6/U shall be shielded which provides a minimum of 95 percent coverage, with a stranded copper center conductor of a minimum 18 AWG, polyethylene insulation, and black non-conductive polyvinylchloride (PVC) jacket.

(4) For runs of 1250 to 2750 feet, RG-11/U is required. RG-11/U shall be shielded which provides a minimum of 95 percent coverage, with a stranded copper center conductor of a minimum 14 AWG, polyethylene insulation, and black non-conductive polyvinylchloride (PVC) jacket.

(5) All runs greater than 2750 feet will be substituted with a fiber optic cable. If using fiber optics as a signal carrier then the following equipment will be utilized:

(a) Multimode fiber optic cable a minimum size of 62 microns

(b) Video transmitter, installed at the camera that utilizes 12 VDC or 24 VAC for power.

(c) Video receiver, installed at the switcher.

(6) RG-59/U Technical Characteristics

AWG	22
Stranding	7x29
Conductor Diameter	.031 in.
Conductor Material	BCC
Insulation Material	Gas-injected FHDPE
Insulation Diameter	.145 in.
Outer Shield Type	Braid/Braid
Outer Jacket Material	PVC
Overall Nominal Diameter	.242 in.
UL Temperature Rating	75°C
Nom. Characteristic Impedance	75 Ohms
Nom. Inductance	0.094 μH/ft
Nom. Capacitance	Conductor to Shield 17.0 pF/ft
Nom. Velocity of Propagation	80 %
Nom. Delay	1.3 ns/ft
Nom. Conductor DC Resistance @	12.2 Ohms/1000 ft
Nom. Outer Shield DC Resistance @	2.4 Ohms/1000 ft

Max. Operating Voltage	UL 300 V RMS

(7) RG-6/U Technical Characteristics:

AWG	18
Stranding	7x27
Conductor Diameter	.040 in.
Conductor Material	BC
Insulation Material	Gas-injected FHDPE
Insulation Diameter	.180 in.
Outer Shield Material	Trade Name Duofoil
Outer Shield Type	Tape/Braid
Outer Shield %Coverage	100 %
Outer Jacket Material	PVC
Overall Nominal Diameter	.274 in.
Nom. Characteristic Impedance	75 Ohms
Nom. Inductance	0.106 μH/ft
Nom. Capacitance	Conductor to Shield 16.2 pF/ft
Nom. Velocity of Propagation	82 %
Nom. Delay	1.24 ns/ft
Nom. Conductor DC Resistance	6.4 Ohms/1000 ft
Nominal Outer Shield DC	2.8 Ohms/1000 ft
Max. Operating Voltage	UL 300 V RMS

(8) RG-11/U Technical Characteristics:

AWG	15
Stranding	19x27
Conductor Diameter	.064 in.
Conductor Material	BC
Insulation Material	Gas-injected FHDPE
Insulation Diameter	.312 in.
Inner Shield Type	Braid
Inner Shield Material	BC - Bare Copper
Inner Shield %Coverage	95 %
Inner Jacket Material	PE – Polyethylene
Inner Jacket Diameter	.391 in.
Outer Shield Type	Braid
Outer Shield Material	BC - Bare Copper
Outer Shield %Coverage	95 %
Outer Jacket Material	Trade Name Belflex
Outer Jacket Material	PVC Blend
Overall Nominal Diameter	.520 in.
Operating Temperature Range	-35°C To +75°C
Non-UL Temperature Rating	75°C
Nom. Characteristic Impedance	75 Ohms
Nom. Inductance	0.097 μH/ft
Nom. Capacitance	Conductor to Shield 17.3 pF/ft
Nom. Velocity of Propagation	78 %
Nom. Delay	1.30 ns/ft
Nom. Conductor DC Resistance	3.1 Ohms/1000 ft

Nom. Inner Shield DC Resistance	1.8 Ohms/1000 ft
Nom. Outer Shield DC Resistance	1.4 Ohms/1000 ft
Max. Operating Voltage Non-UL	300 V RMS

(9) Signal Cables:

(a) Signal wiring for PoE cameras depends on the distance the camera is being installed from either a hub or the server.

(b) If the camera is up to 300 ft from a hub or the server, then use an unshielded UTP category 6A cable a with standard RJ-45 connector at each end. The cable with comply with the Power over Ethernet, IEEE 802.3, Standard.

(c) If the camera is over 300 ft from a hub or server then utilize a multimode fiber optic cable with a minimum size of 62 microns.

(d) Provide a separate cable for power.

(e) CAT-5 Technical Characteristics:

Number of Pairs	4
Total Number of Conductors	8
AWG	24
Stranding	Solid
Conductor Material	BC - Bare Copper
Insulation Material	PO – Polyolefin
Overall Nominal Diameter	.230 in.
IEC Specification	11801 Category 5
TIA/EIA Specification	568-B.2 Category 5e
Max. Capacitance Unbalance	(pF/100 m) 150 pF/100 m
Nom. Velocity of Propagation	70 %
Max. Delay	(ns/100 m) 538 @ 100MHz
Max. Delay Skew	(ns/100m) 45 ns/100 m
Max. Conductor DC Resistance	9.38 Ohms/100
Max. DCR Unbalance@ 20°C	3 %
Max. Operating Voltage	UL 300 V RMS

(10) Fiber Optic Cables Technical Characteristics:

Fiber Type	62.5 Micron
Number of Fibers	4
Core Diameter 6	2.5 +/- 2.5 microns
Core Non-Circularity	5% Maximum
Clad Diameter	125 +/- 2 microns
Clad Non-Circularity	1% Maximum
Core-clad Offset	1.5 Microns Maximum
Primary Coating Material	Acrylate

Primary Coating Diameter	245 +/- 10 microns
Secondary Coating Material	Engineering Thermoplastic
Secondary Coating Diameter	900 +/- 50 microns
Strength Member Material	Aramid Yarn
Outer Jacket Material	PVC
Outer Jacket Color	Orange
Overall Diameter	.200 in.
Numerical Aperture	.275
Maximum Gigabit Ethernet	300 meters
Maximum Gigabit Ethernet	550 meters

(11) Power Cables

(a) Will be sized accordingly and shall comply with the NEC. High voltage power cables will be a minimum of three conductors, 14 AWG, stranded, and coated with a non-conductive polyvinylchloride (PVC) jacket. Low voltage cables will be a minimum of 18 AWG, stranded and non-conductive polyvinylchloride (PVC) jacket.

(b) Will be utilized for all components of the VASS System that require either a 110 VAC 60 Hz or 220 VAC 50 Hz input. Each feed will be connected to a dedicated circuit breaker at a power panel that is primarily for the security system.

(c) All equipment connected to AC power shall be protected from surges. Equipment protection shall withstand surge test waveforms described in IEEE C62.41. Fuses shall not be used as a means of surge protection.

(d) Shall be rated for either 110 or 220 VAC, 50 or 60 Hz, and shall comply with VA Master Spec 26 05 21 Low Voltage Electrical Power Conductors and Cables (600 Volts and Below).

(e) Low Voltage Power Cables

1. Shall be a minimum of 18 AWG, Stranded and have a polyvinylchloride outer jacket.

2. Cable size shall determined using a basic voltage over distance calculation and shall comply with the NEC’s requirements for low voltage cables.

PART 3 EXECUTION

a. Installation: The Contractor shall install all system components including Owner furnished equipment, and appurtenances in accordance with the manufacturer’s instructions, IEEE C2 and as shown, and shall furnish all necessary connectors, terminators, interconnections, services, and adjustments required for a complete and operable data transmission system.

b. Identification and Labeling: The Contractor shall supply permanent identification labels for each cable at each end that will appear on the as-built drawings. The labeling format shall be identified and a complete record shall be provided to the Owner with the final documentation. Each cable shall be identified by type or signal being

carried and termination points. The labels shall be printed on letter size label sheets that are self laminated vinyl that can be printed from a computer data base or spread sheet. The labels shall be E-Z code WES12112 or equivalent.

(1) The Contractor shall provide all personnel, equipment, instrumentation, and supplies necessary to perform all testing.

c. Transient Voltage Surge Suppressors (TVSS): The Contractor shall mount TVSS within (118 in) of equipment to be protected inside terminal cabinets or suitable NEMA 1 enclosures. Terminate off-premise conductors on input side of device. Connect the output side of the device to the equipment to be protected. Connect ground lug to a low impedance earth ground (less than 10 ohms) via Number 12 AWG insulated, stranded copper conductor.

d. Contractor’s Field Test: The Contractor shall verify the complete operation of the data transmission system during the Contractor’s Field Testing. Field test shall include a bit error rate test. The Contractor shall perform the test by sending a minimum of 1,000,000 bits of data on each DTM circuit and measuring the bit error rate. The bit error rate shall not be greater than one (1) bit out of each 100,000 bits sent for each dial-up DTM circuit, and one (1) bit out of 1,000,000 bits sent for each leased or private DTM circuit. The Contractor shall submit a report containing results of the field test.

e. Acceptance Test and Endurance Test: The wire line data transmission system shall be tested as a part of the completed IDS and EECS during the Acceptance test and Endurance Test as specified.

f. Identification and Labeling: The Contractor shall supply identification tags or labels for each cable. Cable shall be labeled at both end points and at intermediate hand holes, manholes, and junction boxes. The labeling format shall be identified and a complete record shall be provided to the Owner with the final documentation. Each cable shall be identified with type of signal being carried and termination points.

3.1 INSTALLATION

a. System installation shall be in accordance with NECA 303, manufacturer and related documents and references, for each type of security subsystem designed, engineered and installed.

b. Components shall be configured with appropriate “service points” to pinpoint system trouble in less than 30 minutes.

c. The Contractor shall install all system components including Government furnished equipment, and appurtenances in accordance with the manufacturer's instructions, documentation listed in Sections 1.5 of this document, and shall furnish all necessary connectors, terminators, interconnections, services, and adjustments required for a complete and operable system.

d. The VASS System will be designed, engineered, installed, and tested to ensure all components are fully compatible as a system and can be integrated with all associated security subsystems, whether the system is a stand alone or a complete network.

e. For integration purposes, the VASS System shall be integrated where appropriate with the following associated security subsystems:

(1) PACS:

(a) Provide 24 hour coverage of all entry points to the perimeter and agency buildings, as well as all emergency exits utilizing a fixed color camera.

(b) Record cameras on a 24 hours basis.

(c) Be programmed go into an alarm state when an emergency exit is opened, and notify the Physical Access Control System and Database Management of an alarm event.

(2) IDS:

(a) Provide a recorded alarm event via a color camera that is connected to the IDS system by either direct hardwire or a security system computer network.

(b) Record cameras on a 24 hours basis.

(c) Be programmed to go into an alarm state when an IDS device is put into an alarm state, and notify the PACS.

(d) For additional VASS System requirements as they relate to the IDS, refer to Section 28 16 00 “INTRUSION DETECTION”.

(3) Security Access Detection:

(a) Provide full coverage of all vehicle and lobby entrance screening areas utilizing a fixed color camera.

(b) Record cameras on a 24 hours basis.

(c) The VASS System should have facial recognition software to assist in identifying individuals for current and future purposes.

(4) EPPS:

(a) Provide a recorded alarm event via a color camera that is connected to the EPPS system by either direct hardwire or a security system computer network.

(b) Record cameras on a 24 hours basis.

(c) Be programmed to go into an alarm state when an emergency call box or duress alarm/panic device is activated, and notify the Physical Access Control System and Database Management of an alarm event.

f. Integration with these security subsystems shall be achieved by computer programming or the direct hardwiring of the systems.

g. For programming purposes refer to the manufacturers requirements for correct system operations. Ensure computers being utilized for system integration meet or exceed the minimum system requirements outlined on the systems software packages.

h. A complete VASS System shall be comprised of, but not limited to, the following components:

(1) Cameras

(2) Lenses

(3) Video Display Equipment

(4) Camera Housings and Mounts

(5) Controlling Equipment

(6) Recording Devices

(7) Wiring and Cables

i. The Contractor shall visit the site and verify that site conditions are in agreement/compliance with the design package. The Contractor shall report all changes to the site or conditions that will affect performance of the system to the Contracting Officer in the form of a report. The Contractor shall not take any corrective action without written permission received from the Contracting Officer.

j. Existing Equipment

(1) The Contractor shall connect to and utilize existing video equipment, video and control signal transmission lines, and devices as outlined in the design package. Video equipment and signal lines that are usable in their original configuration without modification may be reused with Contracting Officer approval.

(2) The Contractor shall perform a field survey, including testing and inspection of all existing video equipment and signal lines intended to be incorporated into the VASS System, and furnish a report to the Contracting Officer as part of the site survey report. For those items considered nonfunctioning, provide (with the report) specification sheets, or written functional requirements to support the findings and the estimated cost to correct the deficiency. As part of the report, the Contractor shall include a schedule for connection to all existing equipment.

(3) The Contractor shall make written requests and obtain approval prior to disconnecting any signal lines and equipment, and creating equipment downtime. Such work shall proceed only after receiving Contracting Officer approval of these requests. If any device fails after the Contractor has commenced work on that device, signal or control line, the Contractor shall diagnose the failure and perform any necessary corrections to the equipment.

(4) The Contractor shall be held responsible for repair costs due to Contractor negligence, abuse, or incorrect installation of equipment.

(5) The Contracting Officer shall be provided a full list of all equipment that is to be removed or replaced by the Contractor, to include description and serial/manufacturer numbers where

possible. The Contractor shall dispose of all equipment that has been removed or replaced based upon approval of the Contracting Officer after reviewing the equipment removal list. In all areas where equipment is removed or replaced the Contractor shall repair those areas to match the current existing conditions.

k. Enclosure Penetrations: All enclosure penetrations shall be from the bottom of the enclosure unless the system design requires penetrations from other directions. Penetrations of interior enclosures involving transitions of conduit from interior to exterior, and all penetrations on exterior enclosures shall be sealed with rubber silicone sealant to preclude the entry of water and will comply with VA Master Specification 07 84 00, Firestopping. The conduit riser shall terminate in a hot-dipped galvanized metal cable terminator. The terminator shall be filled with an approved sealant as recommended by the cable manufacturer and in such a manner that the cable is not damaged.

l. Cold Galvanizing: All field welds and brazing on factory galvanized boxes, enclosures, and conduits shall be coated with a cold galvanized paint containing at least 95 percent zinc by weight.

m. Interconnection of Console Video Equipment: The Contractor shall connect signal paths between video equipment as specified by the OEM. Cables shall be as short as practicable for each signal path without causing strain at the connectors. Rack mounted equipment on slide mounts shall have cables of sufficient length to allow full extension of the slide rails from the rack.

n. Cameras:

(1) Install the cameras with the focal length lens as indicated for each zone.

(2) Connect power and signal lines to the camera.

(3) Aim camera to give field of view as needed to cover the alarm zone.

(4) Aim fixed mounted cameras installed outdoors facing the rising or setting sun sufficiently below the horizon to preclude the camera looking directly at the sun.

(5) Focus the lens to give a sharp picture (to include checking for day and night focus and image quality) over the entire field of view

(6) Synchronize all cameras so the picture does not roll on the monitor when cameras are selected.

(7) PTZ cameras shall have all preset positions and privacy areas defined and programmed.

o. Monitors:

(1) Install the monitors as shown and specified in design and construction documents.

(2) Connect all signal inputs and outputs as shown and specified.

(3) Terminate video input signals as required.

(4) Connect the monitor to AC power.

p. Switcher:

(1) Install the switcher as shown in the design and construction documents, and according to the OEM.

(2) Connect all subassemblies as specified by the manufacturer and as shown.

(3) Connect video signal inputs and outputs as shown and specified; terminate video inputs as required.

(4) Connect alarm signal inputs and outputs as shown and specified; connect control signal inputs and outputs for ancillary equipment or secondary control/monitoring sites as specified by the manufacturer and as shown.

(5) Connect the switcher CPU and switcher subassemblies to AC power.

(6) Load all software as specified and required for an operational VASS System configured for the site and building requirements, including data bases, operational parameters, and system, command, and application programs.

(7) Provide the original and 2 backup copies for all accepted software upon successful completion of the endurance test.

(8) Program the video annotation for each camera.

q. Video Encoder/Decoder

(1) Install the Video Encoder/Decoder per design and construction documents, and as specified by the OEM.

(2) Connect analog camera inputs to video encoder.

(3) Connect network camera to video decoder.

(4) Connect video encoder to VASS network.

(5) Connect video decoder to video matrix, DVR, monitor etc.

(6) Connect unit to AC power (UPS).

(7) Configure the video encoder/decoder per manufacturer’s recommendation and project requirements.

r. Video Server:

(1) Install the video server per design and construction documents, and as specified by the OEM.

(2) Connect video server to AC power (UPS).

(3) Connect to VASS network.

(4) Install operating system and Video Management Software.

(5) Provide Video Management Software programming per VA guidance and the requirements provided by the Owner. Programming shall include:

(a) Camera names

(b) Screen views

(c) Camera recording schedules (continuous and event) driven recording. Events include alarms from other systems (sensors), manual input, and video motion detection.

(d) Video detection zones for each camera requiring video motion detection

(e) Alarm interface

(f) Alarm outputs

(g) GUI maps, views, icons and actions

(h) PTZ controls (presets, time schedules for privacy zones etc.)

(i) Reports

s. Video Workstation:

(1) Install the video workstation per design and construction documents, and as specified by the OEM.

(2) Connect video workstation to AC power (UPS).

(3) Connect to VASS network.

(4) Install operating system and application software.

(5) Provide application software programming per VA guidance and the requirements provided by the Owner. Programming shall include:

(a) Screen views

(b) Graphical User Interface (GUI) maps, views, icons and actions

(c) Alarm outputs

(d) Reports

t. Network Switch:

(1) Install the network switch per design and construction documents, and as specified by the OEM.

(2) Connect network switch to AC power (UPS).

(3) Connect network cameras to network switch.

(4) Configure the network switch per manufacturer’s recommendation and project requirements.

u. Network Recording Equipment

(1) Install the NVR or video storage unit as shown in the design and construction documents, and as specified by the OEM.

(2) Connect recording device to AC power (UPS).

(3) Connect recording device to network switch as shown and specified.

(4) Configure network connections

(5) Provide recording unit programming per VA guidance and the requirements provided by the Owner. Programming shall include:

(a) Camera names

(b) Screen views

(c) Camera recording schedules (continuous and event) driven recording. Events include alarms from other systems (sensors), manual input, and video motion detection.

(d) Video detection zones for each camera requiring video motion detection

(e) Alarm interface

(f) Alarm outputs

(g) GUI maps, views, icons and actions

(h) PTZ controls (presets, time schedules for privacy zones etc.)

(i) Reports

v. Video Recording Equipment:

(1) Install the video recording equipment as shown in the design and construction documents, and as specified by the OEM.

(2) Connect video signal inputs and outputs as shown and specified.

(3) Connect alarm signal inputs and outputs as shown and specified.

(4) Connect video recording equipment to AC power.

(5) Program the video recording equipment;

(a) Recording schedules

(b) Camera caption

w. Video Signal Equipment:

(1) Install the video signal equipment as shown in the design and construction documents, and as specified by the OEM.

(2) Connect video or signal inputs and outputs as shown and specified.

(3) Terminate video inputs as required.

(4) Connect alarm signal inputs and outputs as required.

(5) Connect control signal inputs and outputs as required

(6) Connect electrically powered equipment to AC power.

x. Camera Housings, Mounts, and Poles:

(1) Install the camera housings and mounts as specified by the manufacturer and as shown, provide mounting hardware sized appropriately to secure each camera, housing and mount with maximum wind and ice loading encountered at the site.

(2) Provide a foundation for each camera pole as specified and shown.

(3) Provide a ground rod for each camera pole and connect the camera pole to the ground rod as specified in DIVISION 26 of the VA Master Specification and the VA PG-18-10.

(4) Provide electrical and signal transmission cabling to the mount location via a hardened carrier system from the Physical Access Control System and Database Management to the device.

(5) Connect signal lines and AC power to the housing interfaces.

(6) Connect pole wiring harness to camera.

3.2 SYSTEM START-UP

a. The Contractor shall not apply power to the VASS System until the following items have been completed:

(1) VASS System equipment items and have been set up in accordance with manufacturer's instructions.

(2) A visual inspection of the VASS System has been conducted to ensure that defective equipment items have not been installed and that there are no loose connections.

(3) System wiring has been tested and verified as correctly connected as indicated.

(4) All system grounding and transient protection systems have been verified as installed and connected as indicated.

(5) Power supplies to be connected to the VASS System have been verified as the correct voltage, phasing, and frequency as indicated.

b. The Commissioning Agent will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with the Resident Engineer and Commissioning Agent. Provide a minimum of 7 days prior notice.

c. Satisfaction of the above requirements shall not relieve the Contractor of responsibility for incorrect installation, defective

equipment items, or collateral damage as a result of Contractor work efforts.

3.3 SUPLEMENTAL CONTRACTOR QUIALITY CONTROL

a. The Contractor shall provide the services of technical representatives who are familiar with all components and installation procedures of the installed VASS System; and are approved by the Contracting Officer.

b. The Contractor will be present on the job site during the preparatory and initial phases of quality control to provide technical assistance.

c. The Contractor shall also be available on an as needed basis to provide assistance with follow-up phases of quality control.

d. The Contractor shall participate in the testing and validation of the system and shall provide certification that the system installed is fully operational as all construction document requirements have been fulfilled.

3.4 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 28 08 00 – COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS for all inspection, start up, and contractor testing required above and required by the System Readiness Checklist provided by the Commissioning Agent.

b. Components provided under this section of the specification will be tested as part of a larger system. Refer to Section 28 08 00 – “COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS” and related sections for contractor responsibilities for system commissioning.

3.5 DEMONSTRATION AND TRAINING

a. All testing and training shall be compliant with the VA General Requirements, Section 01 00 00, “GENERAL REQUIREMENTS”.

b. Provide services of manufacturer’s technical representative for four hours to instruct VA personnel in operation and maintenance of units.

c. Submit training plans and instructor qualifications in accordance with the requirements of Section 28 08 00 – “COMMISSIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS”.

-- End of Section --

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SECTION 28 31 00
10/11

FIRE DETECTION AND ALARM

PART 1 GENERAL

1.1 DESCRIPTION
This section addresses requirements for configuration/conditions (1), (2), and (3) as defined above.

a. This section of the specifications includes the furnishing, installation, and connection of the fire alarm equipment to form a complete coordinated system ready for operation. It shall include, but not be limited to, alarm initiating devices, alarm notification appliances, control units, fire safety control devices, annunciators, power supplies, and wiring as shown on the drawings and specified. The fire alarm system shall not be combined with other systems such as building automation, energy management, security, etc.

b. Fire alarm systems shall comply with requirements of the most recent VA FIRE PROTECTION DESIGN MANUAL and NFPA 72 unless variations to NFPA 72 are specifically identified within these contract documents by the following notation: "variation". The design, system layout, document submittal preparation, and supervision of installation and testing shall be provided by a technician that is certified NICET level III or a registered fire protection engineer. The NICET certified technician shall be on site for the supervision and testing of the system. Factory engineers from the equipment manufacturer, thoroughly familiar and knowledgeable with all equipment utilized, shall provide additional technical support at the site as required by the Resident Engineer or his authorized representative. Installers shall have a minimum of 2 years experience installing fire alarm systems.

c. Fire alarm signals:

(1) Building(s) 17 shall have a general evacuation fire alarm signal in accordance with ASA S3.41 to notify all occupants in the respective building to evacuate.

d. Alarm signals (by device), supervisory signals (by device) and system trouble signals (by device not reporting) shall be distinctly transmitted to the main fire alarm system control unit located in the security office .

e. The main fire alarm control unit shall automatically transmit alarm signals to a listed central station using a digital alarm communicator transmitter in accordance with NFPA 72.

1.2 SCOPE

a. A fully addressable fire alarm system shall be designed and installed

in accordance with the specifications and drawings. Device location and wiring runs shown on the drawings are for reference only unless specifically dimensioned. Actual locations shall be in accordance with NFPA 72 and this specification.

b. All existing fire alarm equipment, wiring, devices and sub-systems that are not shown to be reused shall be removed. All existing fire alarm conduit not reused shall be removed.

c. Existing fire alarm bells, chimes, door holders, 120VAC duct smoke detectors, valve tamper switches and waterflow/pressure switches may be reused only as specifically indicated on the drawings and provided the equipment:

(1) Meets this specification section

(2) Is UL listed or FM APP GUIDE

(3) Is compatible with new equipment being installed

(4) Is verified as operable through contractor testing and inspection

(5) Is warranted as new by the contractor.

d. Existing 120 VAC duct smoke detectors, waterflow/pressure switches, and valve tamper switches reused by the Contractor shall be equipped with an addressable interface device compatible with the new equipment being installed.

e. Existing reused equipment shall be covered as new equipment under the Warranty specified herein.

f. Basic Performance:

(1) Alarm and trouble signals from each building fire alarm control panel shall be digitally encoded by UL listed electronic devices onto a multiplexed communication system.

(2) Response time between alarm initiation (contact closure) and recording at the main fire alarm control unit (appearance on alphanumeric read out) shall not exceed 5 seconds.

(3) The signaling line circuits (SLC) between building fire alarm control units shall be wired Style 7 in accordance with NFPA 72. Isolation shall be provided so that no more than one building can be lost due to a short circuit fault.

(4) Initiating device circuits (IDC) shall be wired Style C in accordance with NFPA 72.

(5) Signaling line circuits (SLC) within buildings shall be wired Style 4 in accordance with NFPA 72. Individual signaling line circuits shall be limited to covering 22,500 square feet of floor space or 3 floors whichever is less.

(6) Notification appliance circuits (NAC) shall be wired Style Y in accordance with NFPA 72.

1.3 RELATED WORK

a. Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

Requirements for procedures for submittals.

b. Section 07 84 00 - FIRESTOPPING. Requirements for fire proofing wall penetrations.

c.

d. Section 21 13 13 - WET-PIPE SPRINKLER SYSTEMS. Requirements for sprinkler systems.

e. Section 28 05 00 – COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY. Requirements for general requirements that are common to more than one section in Division 28.

f. Section 28 05 13 - CONDUCTORS AND CABLES FOR ELECTRONIC SAFETY AND SECURITY. Requirements for conductors and cables.

g. Section 28 05 26 - GROUNDING AND BONDING FOR ELECTRONIC SAFETY AND SECURITY. Requirements for grounding of equipment.

h. Section 28 05 28.33 - CONDUITS AND BACKBOXES FOR ELECTRONIC SAFETY AND SECURITY. Requirements for infrastructure.

i. Section 28 05 13 - CONDUCTORS AND CABLES FOR ELECTRONIC SAFETY AND SECURITY. Requirements for conductors and cables.

j. Section 28 08 00, COMMISIONING OF ELECTRONIC SAFETY AND SECURITY SYSTEMS. Requirements for commissioning - systems readiness checklists, and training.

k. Section 28 13 00, PHYSICAL ACCESS CONTROL SYSTEMS (PACS). Requirements for integration with physical access control system.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Drawings;

SD-03 Product Data

Conduit, Boxes, And Wire;

Fire Alarm Control Unit;

Standby Power Supply;

Annunciator;

Voice Communication System (Vcs);

Alarm Notification Appliances;

Alarm Initiating Devices;

Supervisory Devices;

Address Reporting Interface Device;

Smoke Barrier Door Control;

Utility Locks And Keys;

Spare And Replacement Parts;

SD-07 Certificates

Certifications;

SD-10 Operation and Maintenance Data

Manuals;

1.5 SUBMITTAL INSTRUCTIONS

a. General: Submit 5 copies in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES, and Section 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS.

b. Drawings:

(1) Prepare drawings and include all contractors information. Layering shall be by VA criteria as provided by the Contracting Officer (CO). Bid drawing files will be provided to the Contractor at the pre-construction meeting. The contractor shall be responsible for verifying all critical dimensions shown on the drawings provided by VA.

(2) Floor plans: Provide locations of all devices (with device number at each addressable device corresponding to control unit programming), appliances, panels, equipment, junction/terminal cabinets/boxes, risers, electrical power connections, individual circuits and raceway routing, system zoning; number, size, and type of raceways and conductors in each raceway; conduit fill calculations with cross section area percent fill for each type and size of conductor and raceway. Only those devices connected and incorporated into the final system shall be on these floor plans. Do not show any removed devices on the floor plans. Show all interfaces for all fire safety functions.

(3) Riser diagrams: Provide, for the entire system, the number, size and type of riser raceways and conductors in each riser raceway and number of each type device per floor and zone. Show door holder interface, elevator control interface, HVAC shutdown interface, fire extinguishing system interface, and all other fire safety interfaces. Show wiring Styles on the riser diagram for all circuits. Provide diagrams both on a per building and campus wide basis.

(4) Detailed wiring diagrams: Provide for control panels, modules, power supplies, electrical power connections, auxiliary relays and annunciators showing termination identifications, size and type conductors, circuit boards, LED lamps, indicators, adjustable controls, switches, ribbon connectors, wiring harnesses, terminal strips and connectors, spare zones/circuits. Diagrams shall be drawn to a scale sufficient to show spatial relationships between components, enclosures and equipment configuration.

(5) Two weeks prior to final inspection, the Contractor shall deliver to the CO 3 sets of as-built drawings and one set of the as-built drawing computer files. As-built drawings (floor plans) shall show all new and/or existing conduit used for the fire alarm system.

c. Manuals:

(1) Submit simultaneously with the shop drawings, companion copies of complete maintenance and operating manuals including technical data sheets for all items used in the system, power requirements, device wiring diagrams, dimensions, and information for ordering replacement parts.

(a) Wiring diagrams shall have their terminals identified to facilitate installation, operation, expansion and maintenance.

(b) Wiring diagrams shall indicate internal wiring for each item of equipment and the interconnections between the items of equipment.

(c) Include complete listing of all software used and installation and operation instructions including the input/output matrix chart.

(d) Provide a clear and concise description of operation that gives, in detail, the information required to properly operate, inspect, test and maintain the equipment and system. Provide all manufacturer's installation limitations including but not limited to circuit length limitations.

(e) Complete listing of all digitized voice messages.

(f) Provide standby battery calculations under normal operating and alarm modes. Battery calculations shall include the magnets for holding the doors open for one minute.

(g) Include information indicating who will provide emergency service and perform post contract maintenance.

(h) Provide a replacement parts list with current prices. Include a list of recommended spare parts, tools, and instruments for testing and maintenance purposes.

(i) A computerized preventive maintenance schedule for all equipment. The schedule shall be provided on disk in a computer format acceptable to the VAMC and shall describe the protocol for preventive maintenance of all equipment. The schedule shall include the required times for systematic examination, adjustment and cleaning of all equipment. A print out of the schedule shall also be provided in the manual. Provide the disk in a pocket

within the manual.

(j) Furnish manuals in 3 ring loose-leaf binder or manufacturer's standard binder.

(k) A print out for all devices proposed on each signaling line circuit with spare capacity indicated.

(2) Two weeks prior to final inspection, deliver 4 copies of the final updated maintenance and operating manual to the CO.

(a) The manual shall be updated to include any information necessitated by the maintenance and operating manual approval.

(b) Complete "As installed" wiring and schematic diagrams shall be included that shows all items of equipment and their interconnecting wiring. Show all final terminal identifications.

(c) Complete listing of all programming information, including all control events per device including an updated input/output matrix.

(d) Certificate of Installation as required by NFPA 72 for each building. The certificate shall identify any variations from the National Fire Alarm Code.

(e) Certificate from equipment manufacturer assuring compliance with all manufacturers installation requirements and satisfactory system operation.

d. Certifications:

(1) Together with the shop drawing submittal, submit the technician's NICET level III fire alarm certification as well as certification from the control unit manufacturer that the proposed performer of contract maintenance is an authorized representative of the major equipment manufacturer. Include in the certification the names and addresses of the proposed supervisor of installation and the proposed performer of contract maintenance. Also include the name and title of the manufacturer’s representative who makes the certification.

(2) Together with the shop drawing submittal, submit a certification from either the control unit manufacturer or the manufacturer of each component (e.g., smoke detector) that the components being furnished are compatible with the control unit.

(3) Together with the shop drawing submittal, submit a certification from the major equipment manufacturer that the wiring and connection diagrams meet this specification, UL and NFPA 72 requirements.

1.6 WARRANTY

All work performed and all material and equipment furnished under this contract shall be free from defects and shall remain so for a period of one year from the date of acceptance of the entire installation by the Contracting Officer.

1.7 GUARANTY PERIOD SERVICES

a. Complete inspection, testing, maintenance and repair service for the fire alarm system shall be provided by a factory trained authorized representative of the manufacturer of the major equipment for a period of 5 years from the date of acceptance of the entire installation by the Contracting Officer.

b. Contractor shall provide all necessary test equipment, parts and labor to perform required inspection, testing, maintenance and repair.

c. All inspection, testing, maintenance and permanent records required by NFPA 72, and recommended by the equipment manufacturer shall be provided by the contractor. Work shall include operation of sprinkler system alarm and supervisory devices . It shall include all interfaced equipment including but not limited to elevators, HVAC shutdown, and extinguishing systems.

d. Maintenance and testing shall be performed in accordance with NFPA 72. A computerized preventive maintenance schedule shall be provided and shall describe the protocol for preventive maintenance of equipment. The schedule shall include a systematic examination, adjustment and cleaning of all equipment.

e. Non-included Work: Repair service shall not include the performance of any work due to improper use, accidents, or negligence for which the contractor is not responsible.

f. Service and emergency personnel shall report to the Engineering Office or their authorized representative upon arrival at the hospital and again upon the completion of the required work. A copy of the work ticket containing a complete description of the work performed and parts replaced shall be provided to the VA Resident Engineer or his authorized representative.

g. Emergency Service:

(1) Warranty Period Service: Service other than the preventative maintenance, inspection, and testing required by NFPA 72 shall be considered emergency call-back service and covered under the warranty of the installation during the first year of the warranty period, unless the required service is a result of abuse or misuse by the Government. Written notification shall not be required for emergency warranty period service and the contractor shall respond as outlined in the following sections on Normal and Overtime Emergency Call-Back Service. Warranty period service can be required during normal or overtime emergency call-back service time periods at the discretion of the Resident Engineer or his authorized representative.

(2) Normal and overtime emergency call-back service shall consist of an on-site response within 2 hours of notification of a system trouble.

(3) Normal emergency call-back service times are between the hours of 7: 30 a.m. and 4:00 p.m., Monday through Friday, exclusive of federal holidays. Service performed during all other times shall be considered to be overtime emergency call-back service. The cost of all normal emergency call-back service for years 2 through

5 shall be included in the cost of this contract.

(4) Overtime emergency call-back service shall be provided for the system when requested by the Government. The cost of the first 40 manhours per year of overtime call-back service during years 2 through 5 of this contract shall be provided under this contract. Payment for overtime emergency call-back service in excess of the 40 man hours per year requirement will be handled through separate purchase orders. The method of calculating overtime emergency call-back hours is based on actual time spent on site and does not include travel time.

h. The contractor shall maintain a log at each fire alarm control unit. The log shall list the date and time of all examinations and trouble calls, condition of the system, and name of the technician. Each trouble call shall be fully described, including the nature of the trouble, necessary correction performed, and parts replaced.

i. In the event that VA modifies the fire alarm system post-Acceptance but during the 5 year Guaranty Period Service period, Contractor shall be required to verify that the system, as newly modified or added, is consistent with the manufacturer's requirements; any verification performed will be equitably adjusted under the Changes clause. The post-Acceptance modification or addition to the fire alarm system shall not void the continuing requirements under this contract set forth in the Guarantee Period Service provision for the fire alarm system as modified or added. The contract will be equitably adjusted under the Changes clause for such additional performance.

1.8 REFERENCES

a. The publications listed below (including amendments, addenda, revisions, supplements and errata) form a part of this specification to the extent referenced. The publications are referenced in text by the basic designation only and the latest editions of these publications shall be applicable.

FM GLOBAL (FM)

FM APP GUIDE	(updated on-line) Approval Guide https://www.approvalguide.com/

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 20	(2025) Standard for the Installation of Stationary Pumps for Fire Protection
NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 72	(2022; ERTA 22-1) National Fire Alarm and Signaling Code
NFPA 90A	(2024) Standard for the Installation of Air Conditioning and Ventilating Systems
NFPA 101	(2024) Life Safety Code

ACOUSTICAL SOCIETY OF AMERICA (ASA)

ASA S3.41 (2015) Audible Emergency Evacuation Signal (ASA 96)

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2024) International Building Code

PART 2 PRODUCTS

2.1 NEW EQUIPMENT AND MATERIALS, GENERAL

All equipment and components shall be new and the manufacturer's current model. All equipment shall be tested and listed by Underwriters Laboratories, Inc. or Factory Mutual Research Corporation for use as part of a fire alarm system. The authorized representative of the manufacturer of the major equipment shall certify that the installation complies with all manufacturers' requirements and that satisfactory total system operation has been achieved.

2.2 CONDUIT, BOXES, AND WIRE

a. Conduit shall be in accordance with Section 28 05 28.33 CONDUIT AND BACKBOXES FOR ELECTRONIC SAFETY AND SECURITY and as follows:

(1) All new conduits shall be installed in accordance with NFPA 70.

(2) Conduit fill shall not exceed 40 percent of interior cross sectional area.

(3) All new conduits shall be 3/4 inch minimum.

b. Wire:

(1) Wiring shall be in accordance with NFPA 70 NEC article 760, Section 28 05 13, CONDUCTORS AND CABLES FOR ELECTRONIC SAFETY AND SECURITY, and as recommended by the manufacturer of the fire alarm system. All wires shall be color coded. Number and size of conductors shall be as recommended by the fire alarm system manufacturer, but not less than 18 AWG for initiating device circuits and 14 AWG for notification device circuits.

(2) Addressable circuits and wiring used for the multiplex communication loop shall be twisted and shielded unless specifically excepted by the fire alarm equipment manufacturer in writing.

(3) Any fire alarm system wiring that extends outside of a building shall have additional power surge protection to protect equipment from physical damage and false signals due to lightning, voltage and current induced transients. Protection devices shall be shown on the submittal drawings and shall be UL listed or in accordance with written manufacturer's requirements.

(4) All wire or cable used in underground conduits including those in concrete shall be listed for wet locations.

c. Terminal Boxes, Junction Boxes, and Cabinets:

(1) Shall be galvanized steel in accordance with UL requirements.

(2) All boxes shall be sized and installed in accordance with NFPA 70.

(3) covers shall be repainted red in accordance with Section 09 91 00, PAINTING and shall be identified with white markings as "FA" for junction boxes and as "FIRE ALARM SYSTEM" for cabinets and terminal boxes. Lettering shall be a minimum of 3/4 inch high.

(4) Terminal boxes and cabinets shall have a volume 50 percent greater than required by the NFPA 70. Minimum sized wire shall be considered as 14 AWG for calculation purposes.

(5) Terminal boxes and cabinets shall have identified pressure type terminal strips and shall be located at the base of each riser. Terminal strips shall be labeled as specified or as approved by the CO.

2.3 FIRE ALARM CONTROL UNIT

a. General:

(1) Each building shall be provided with a fire alarm control unit and shall operate as a supervised zoned fire alarm system.

(2) Each power source shall be supervised from the other source for loss of power.

(3) All circuits shall be monitored for integrity.

(4) Visually and audibly annunciate any trouble condition including, but not limited to main power failure, grounds and system wiring derangement.

(5) Transmit digital alarm information to the main fire alarm control unit.

b. Enclosure:

(1) The control unit shall be housed in a cabinet suitable for both recessed and surface mounting. Cabinet and front shall be corrosion protected, given a rust-resistant prime coat, and manufacturer's standard finish.

(2) Cabinet shall contain all necessary relays, terminals, lamps, and legend plates to provide control for the system.

c. Operator terminal at main control unit:

(1) Operator terminal shall consist of the central processing unit, display screen, keyboard and printer.

(2) Display screen shall have a minimum 15-inch diagonal non-glare screen capable of displaying 24 lines of 80 characters each.

(3) Keyboard shall consist of 60 alpha numeric and 12 user/functional control keys.

(4) Printer shall be the automatic type, printing the date, time and location for all alarm, supervisory, and trouble conditions.

d. Power Supply:

(1) The control unit shall derive its normal power from a 120 volt, 60 Hz dedicated supply connected to the emergency power system. Standby power shall be provided by a 24 volt DC battery as hereinafter specified. The normal power shall be transformed, rectified, coordinated, and interfaced with the standby battery and charger.

(2) The door holder power shall be arranged so that momentary or sustained loss of main operating power shall not cause the release of any door.

(3) Power supply for smoke detectors shall be taken from the fire alarm control unit.

(4) Provide protectors to protect the fire alarm equipment from damage due to lightning or voltage and current transients.

(5) Provide new separate and direct ground lines to the outside to protect the equipment from unwanted grounds.

e. Circuit Supervision: Each alarm initiating device circuit, signaling line circuit, and notification appliance circuit, shall be supervised against the occurrence of a break or ground fault condition in the field wiring. These conditions shall cause a trouble signal to sound in the control unit until manually silenced by an off switch.

f. Supervisory Devices: All sprinkler system valves, standpipe control valves, post indicator valves (PIV), and main gate valves shall be supervised for off-normal position. Closing a valve shall sound a supervisory signal at the control unit until silenced by an off switch. The specific location of all closed valves shall be identified at the control unit. Valve operation shall not cause an alarm signal. Low air pressure switches and duct detectors shall be monitored as supervisory signals. The power supply to the elevator shunt trip breaker shall be monitored by the fire alarm system as a supervisory signal.

g. Trouble signals:

(1) Arrange the trouble signals for automatic reset (non-latching).

(2) System trouble switch off and on lamps shall be visible through the control unit door.

h. Function Switches: Provide the following switches in addition to any other switches required for the system:

(1) Remote Alarm Transmission By-pass Switch: Shall prevent transmission of all signals to the main fire alarm control unit when in the "off" position. A system trouble signal shall be energized when switch is in the off position.

(2) Alarm Off Switch: Shall disconnect power to alarm notification circuits on the local building alarm system. A system trouble

signal shall be activated when switch is in the off position.

(3) Trouble Silence Switch: Shall silence the trouble signal whenever the trouble silence switch is operated. This switch shall not reset the trouble signal.

(4) Reset Switch: Shall reset the system after an alarm, provided the initiating device has been reset. The system shall lock in alarm until reset.

(5) Lamp Test Switch: A test switch or other approved convenient means shall be provided to test the indicator lamps.

(6) Drill Switch: Shall activate all notification devices without tripping the remote alarm transmitter. This switch is required only for general evacuation systems specified herein.

(7) Door Holder By-Pass Switch: Shall prevent doors from releasing during fire alarm tests. A system trouble alarm shall be energized when switch is in the abnormal position.

(8) Elevator recall By-Pass Switch: Shall prevent the elevators from recalling upon operation of any of the devices installed to perform that function. A system trouble alarm shall be energized when the switch is in the abnormal position.

(9) HVAC/Smoke Damper By-Pass: Provide a means to disable HVAC fans from shutting down and/or smoke dampers from closing upon operation of an initiating device designed to interconnect with these devices.

i. Remote Transmissions:

(1) Provide capability and equipment for transmission of alarm, supervisory and trouble signals to the main fire alarm control unit.

(2) Transmitters shall be compatible with the systems and equipment they are connected to such as timing, operation and other required features.

j. Remote Control Capability: Each building fire alarm control unit shall be installed and programmed so that each must be reset locally after an alarm, before the main fire alarm control unit can be reset. After the local building fire alarm control unit has been reset, then the all system acknowledge, reset, silence or disabling functions can be operated by the main fire alarm control unit

k. System Expansion: Design the control units and enclosures so that the system can be expanded in the future (to include the addition of 20 percent more alarm initiating, alarm notification and door holder circuits) without disruption or replacement of the existing control unit and secondary power supply.

2.4 STANDBY POWER SUPPLY

a. Uninterrupted Power Supply (UPS):

(1) The UPS system shall be comprised of a static inverter, a

precision battery float charger, and sealed maintenance free batteries.

(2) Under normal operating conditions, the load shall be filtered through a ferroresonant transformer.

(3) When normal AC power fails, the inverter shall supply AC power to the transformer from the battery source. There shall be no break in output of the system during transfer of the system from normal to battery supply or back to normal.

(4) Batteries shall be sealed, gel cell type.

(5) UPS system shall be sized to operate the central processor, CRT, printer, and all other directly connected equipment for 5 minutes upon a normal AC power failure.

b. Batteries:

(1) Battery shall be of the sealed, maintenance free type, 24-volt nominal.

(2) Battery shall have sufficient capacity to power the fire alarm system for not less than 24 hours plus 5 minutes of alarm to an end voltage of 1.14 volts per cell, upon a normal AC power failure.

(3) Battery racks shall be steel with an alkali-resistant finish. Batteries shall be secured in seismic areas 2B, 3, or 4 as defined by the ICC IBC.

c. Battery Charger:

(1) Shall be completely automatic, with constant potential charger maintaining the battery fully charged under all service conditions. Charger shall operate from a 120-volt, 60 hertz emergency power source.

(2) Shall be rated for fully charging a completely discharged battery within 48 hours while simultaneously supplying any loads connected to the battery.

(3) Shall have protection to prevent discharge through the charger.

(4) Shall have protection for overloads and short circuits on both AC and DC sides.

(5) A trouble condition shall actuate the fire alarm trouble signal.

(6) Charger shall have automatic AC line voltage regulation, automatic current-limiting features, and adjustable voltage controls.

2.5 ANNUNCIATION

a. Annunciator, Alphanumeric Type (System):

(1) Shall be a supervised, LCD display containing a minimum of 2 lines of 40 characters for alarm annunciation in clear English text.

(2) Message shall identify building number, floor, zone, etc on the

first line and device description and status (pull station, smoke detector, waterflow alarm or trouble condition) on the second line.

(3) The initial alarm received shall be indicated as such.

(4) A selector switch shall be provided for viewing subsequent alarm messages.

(5) The display shall be UL listed for fire alarm application.

(6) Annunciators shall display information for all buildings connected to the system. Local building annunciators, for general evacuation system buildings, shall be permitted when shown on the drawings and approved by the CO.

b. Printers:

(1) System printers shall be high reliability digital input devices, UL approved, for fire alarm applications. The printers shall operate at a minimum speed of 30 characters per second. The printer shall be continually supervised.

(2) Printers shall be programmable to either alarm only or event logging output.

(a) Alarm printers shall provide a permanent (printed) record of all alarm information that occurs within the fire alarm system. Alarm information shall include the date, time, building number, floor, zone, device type, device address, and condition.

(b) Event logging printers shall provide a permanent (printed) record of every change of status that occurs within the fire alarm system. Status information shall include date, time, building number, floor, zone, device type, device address and change of status (alarm, trouble, supervisory, reset/return to normal).

(3) System printers shall provide tractor drive feed pins for conventional fan fold 8-1/2" x 11" paper.

(4) The printers shall provide a printing and non-printing self test feature.

(5) Power supply for printers shall be taken from and coordinated with the building emergency service.

(6) Each printer shall be provided with a stand for the printer and paper.

(7) Spare paper and ribbons for printers shall be stocked and maintained as part of the one year guarantee period services in addition to the one installed after the approval of the final acceptance test.

2.6 ALARM NOTIFICATION APPLIANCES

a. Bells:

(1) Shall be electric, single-stroke or vibrating, heavy-duty, under-dome, solenoid type.

(2) Unless otherwise shown on the drawings, shall be 6 inches diameter and have a minimum nominal rating of 80 dBA at 10 feet .

(3) Mount on removable adapter plates on outlet boxes.

(4) Bells located outdoors shall be weatherproof type with metal housing and protective grille.

(5) Each bell circuit shall have a minimum of 20 percent spare capacity.

b. Speakers:

(1) Shall operate on either 25 VRMS or 70.7 VRMS with field selectable output taps from 0.5 to 2.0W and originally installed at the 1/2 watt tap. Speakers shall provide a minimum sound output of 80 dBA at 10 feet with the 1/2 watt tap.

(2) Frequency response shall be a minimum of 400 HZ to 4,000 HZ.

(3) Four inches or 8 inches cone type speakers ceiling mounted with white colored baffles in areas with suspended ceilings and wall mounted in areas without ceilings.

c. Strobes:

(1) Xenon flash tube type minimum 15 candela in toilet rooms and 75 candela in all other areas with a flash rate of 1 HZ. Strobes shall be synchronized where required by the National Fire Alarm Code (NFPA 72).

(2) Backplate shall be red with 1/2 inch permanent red letters. Lettering to read "Fire", be oriented on the wall or ceiling properly, and be visible from all viewing directions.

(3) Each strobe circuit shall have a minimum of 20 percent spare capacity.

(4) Strobes may be combined with the audible notification appliances specified herein.

d. Fire Alarm Horns:

(1) Shall be electric, utilizing solid state electronic technology operating on a nominal 24 VDC.

(2) Shall be a minimum nominal rating of 80 dBA at 10 feet .

(3) Mount on removable adapter plates on conduit boxes.

(4) Horns located outdoors shall be of weatherproof type with metal housing and protective grille.

(5) Each horn circuit shall have a minimum of 20 percent spare capacity.

2.7 ALARM INITIATING DEVICES

a. Manual Fire Alarm Stations:

(1) Shall be non-breakglass, address reporting type.

(2) Station front shall be constructed of a durable material such as cast or extruded metal or high impact plastic. Stations shall be semi-flush type.

(3) Stations shall be of single action pull down type with suitable operating instructions provided on front in raised or depressed letters, and clearly labeled "FIRE."

(4) Operating handles shall be constructed of a durable material. On operation, the lever shall lock in alarm position and remain so until reset. A key shall be required to gain front access for resetting, or conducting tests and drills.

(5) Unless otherwise specified, all exposed parts shall be red in color and have a smooth, hard, durable finish.

(6)

b. Smoke Detectors:

(1) Smoke detectors shall be photoelectric type and UL listed for use with the fire alarm control unit being furnished.

(2) Smoke detectors shall be addressable type complying with applicable UL Standards for system type detectors. Smoke detectors shall be installed in accordance with the manufacturer's recommendations and NFPA 72.

(3) Detectors shall have an indication lamp to denote an alarm condition. Provide remote indicator lamps and identification plates where detectors are concealed from view. Locate the remote indicator lamps and identification plates flush mounted on walls so they can be observed from a normal standing position.

(4) All spot type and duct type detectors installed shall be of the photoelectric type.

(5) Photoelectric detectors shall be factory calibrated and readily field adjustable. The sensitivity of any photoelectric detector shall be factory set at 3.0 plus or minus 0.25 percent obscuration per foot.

(6) Detectors shall provide a visual trouble indication if they drift out of sensitivity range or fail internal diagnostics. Detectors shall also provide visual indication of sensitivity level upon testing. Detectors, along with the fire alarm control units shall be UL listed for testing the sensitivity of the detectors.

c. Heat Detectors:

(1) Heat detectors shall be of the addressable restorable rate compensated fixed-temperature spot type.

(2) Detectors shall have a minimum smooth ceiling rating of 2,500 square feet .

(3) Ordinary temperature (135 degrees F) heat detectors shall be utilized in elevator shafts and elevator mechanical rooms. Intermediate temperature rated (200 degrees F) heat detectors shall be utilized in all other areas.

(4) Provide a remote indicator lamp, key test station and identification nameplate (e.g. “Heat Detector - Elevator P-_____) for each elevator group. Locate key test station in plain view on elevator machine room wall.

d. Water Flow and Pressure Switches:

(1) Wet pipe water flow switches and dry pipe alarm pressure switches for sprinkler systems shall be connected to the fire alarm system by way of an address reporting interface device.

(2) All new water flow switches shall be of a single manufacturer and series and non-accumulative retard type.

(3) All new switches shall have an alarm transmission delay time that is conveniently adjustable from 0 to 60 seconds. Initial settings shall be 30-45 seconds. Timing shall be recorded and documented during testing.

e. Extinguishing System Connections:

(1) Kitchen Range Hood and Duct Suppression Systems:

(a) Each suppression system shall be equipped with a micro-switch connected to the building fire alarm control unit. Discharge of a suppression system shall automatically send a alarm signal to the building fire detection and alarm system for annunciation.

(b) Operation of this suppression system shall also automatically shut off all sources of fuel and heat to all equipment requiring protection under the same hood.

(2) Each gaseous suppression system shall be monitored for system alarm and system trouble conditions via addressable interface devices.

2.8 SUPERVISORY DEVICES

a. Duct Smoke Detectors:

(1) Duct smoke detectors shall be provided and connected by way of an address reporting interface device. Detectors shall be provided with an approved duct housing mounted exterior to the duct, and shall have perforated sampling tubes extending across the full width of the duct (wall to wall). Detector placement shall be such that there is uniform airflow in the cross section of the duct.

(2) Interlocking with fans shall be provided in accordance with NFPA 90A and as specified hereinafter under Part 3.2, "TYPICAL OPERATION".

(3) Provide remote indicator lamps, key test stations and identification nameplates (e.g. "DUCT SMOKE DETECTOR AHU-X") for all duct detectors. Locate key test stations in plain view on walls or ceilings so that they can be observed and operated from a normal standing position.

b. Sprinkler and Standpipe System Supervisory Switches:

(1) Each sprinkler system water supply control valve, riser valve or zone control valve, and each standpipe system riser control valve shall be equipped with a supervisory switch. Standpipe hose valves, and test and drain valves shall not be equipped with supervisory switches.

(2) PIV (post indicator valve) or main gate valve shall be equipped with a supervisory switch.

(3) Valve supervisory switches shall be connected to the fire alarm system by way of address reporting interface device. See Section 21 13 13, WET-PIPE SPRINKLER SYSTEMS for new switches to be added. Connect tamper switches for all control valves shown on the approved shop drawings.

(4) The mechanism shall be contained in a weatherproof die-cast aluminum housing that shall provide a 3/4 inch tapped conduit entrance and incorporate the necessary facilities for attachment to the valves.

(5) The entire installed assembly shall be tamper-proof and arranged to cause a switch operation if the housing cover is removed or if the unit is removed from its mounting.

(6) Where dry-pipe sprinkler systems are installed, high and low air pressure switches shall be provided and monitored by way of an address reporting interface devices.

(7) Fire supervisory signals required by NFPA 20 and monitored by the pump controller shall be provided and monitored by way of address reporting interface devices for the fire pump

2.9 ADDRESS REPORTING INTERFACE DEVICE

a. Shall have unique addresses that reports directly to the building fire alarm panel.

b. Shall be configurable to monitor normally open or normally closed devices for both alarm and trouble conditions.

c. Shall have terminal designations clearly differentiating between the circuit to which they are reporting from and the device that they are monitoring.

d. Shall be UL listed for fire alarm use and compatibility with the panel to which they are connected.

e. Shall be mounted in weatherproof housings if mounted exterior to a building.

2.10 SMOKE BARRIER DOOR CONTROL

a. Electromagnetic Door Holders:

(1) New Door Holders shall be standard wall mounted electromagnetic type. In locations where doors do not come in contact with the wall when in the full open position, an extension post shall be added to the door bracket.

(2) Operation shall be by 24 volt DC supplied from a battery located at the fire alarm control unit. Door holders shall be coordinated as to voltage, ampere drain, and voltage drop with the battery, battery charger, wiring and fire alarm system for operation as specified.

b. A maximum of twelve door holders shall be provided for each circuit. Door holders shall be wired to allow releasing doors by smoke zone.

c. Door holder control circuits shall be electrically supervised.

d. Smoke detectors shall not be incorporated as an integral part of door holders.

2.11 UTILITY LOCKS AND KEYS

a. All key operated test switches, control units, annunciator panels and lockable cabinets shall be provided with a single standardized utility lock and key.

b. Key-operated manual fire alarm stations shall have a single standardized lock and key separate from the control equipment.

c. All keys shall be delivered to the CO.

2.12 SPARE AND REPLACEMENT PARTS

a. Provide spare and replacement parts as follows:

(1) Manual pull stations - 5

Heat detectors - 2 of each type

(3) Fire alarm strobes - 5

(4) Fire alarm bells - 5

(5) Fire alarm speakers - 5

(6) Smoke detectors - 20

(7) Duct smoke detectors with all appurtenances - 1

(7) Sprinkler system water flow switch - 1 of each size

(8) Sprinkler system water pressure switch - 1 of each type

(9) Sprinkler valve tamper switch - 1 of each type

(10) Control equipment utility locksets - 5

(11) Control equipment keys - 25

(12) 2.5 oz containers aerosol smoke - 12

(13) Printer paper - 3 boxes

(14) Printer replacement ribbons - 3

(15) Monitor modules - 3

(16) Control modules - 3

(17) Fire alarm SLC cable (same as installed) – 500 feet

Spare and replacement parts shall be in original packaging and submitted to the CO.

c. Furnish and install a storage cabinet of sufficient size and suitable for storing spare equipment. Doors shall include a pad locking device. Padlock to be provided by the VA. Location of cabinet to be determined by the CO.

2.13 INSTRUCTION CHART

Provide typewritten instruction card mounted behind a Lexan plastic or glass cover in a stainless steel or aluminum frame with a backplate. Install the frame in a conspicuous location observable from each control unit where operations are performed. The card shall show those steps to be taken by an operator when a signal is received under all conditions, normal, alarm, supervisory, and trouble. Provide an additional copy with the binder for the input output matrix for the sequence of operation. The instructions shall be approved by the CO before being posted.

PART 3 EXECUTION

3.1 INSTALLATION

a. Installation shall be in accordance with NFPA 70,NFPA 72,NFPA 90A, and NFPA 101 as shown on the drawings, and as recommended by the major equipment manufacturer. Fire alarm wiring shall be installed in conduit. All conduit and wire shall be installed in accordance with, Section 28 05 13 CONDUCTORS AND CABLES FOR ELECTRONIC SAFETY AND SECURITY, Section 28 05 26 GROUNDING AND BONDING FOR ELECTRONIC SAFETY AND SECURITY, Section 28 05 28.33 CONDUIT AND BACKBOXES FOR ELECTRONIC SAFETY AND SECURITY, and all penetrations of smoke and fire barriers shall be protected as required by Section 07 84 00, FIRESTOPPING.

b. All conduits, junction boxes, conduit supports and hangers shall be concealed in finished areas and may be exposed in unfinished areas.

c. All new and reused exposed conduits shall be painted in accordance with Section 09 91 00, PAINTING to match surrounding finished areas and red in unfinished areas.

All fire detection and alarm system devices, control units and remote annunciators shall be flush mounted when located in finished areas and may be surface mounted when located in unfinished areas. Exact

locations are to be approved by the CO.

e. Speakers shall be ceiling mounted and fully recessed in areas with suspended ceilings. Speakers shall be wall mounted and recessed in finished areas without suspended ceilings. Speakers may be surface mounted in unfinished areas.

f. Strobes shall be flush wall mounted with the bottom of the unit located 80 inches above the floor or 6 inches below ceiling, whichever is lower. Locate and mount to maintain a minimum 36 inches clearance from side obstructions.

g. Manual pull stations shall be installed not less than 42 inches or more than 48 inches from finished floor to bottom of device and within 60 inches of a stairway or an exit door.

h. Where possible, locate water flow and pressure switches a minimum of 12 inches from a fitting that changes the direction of the flow and a minimum of 36 inches from a valve.

i. Mount valve tamper switches so as not to interfere with the normal operation of the valve and adjust to operate within 2 revolutions toward the closed position of the valve control, or when the stem has moved no more than 1/5 of the distance from its normal position.

j. Connect flow and tamper switches installed under Section 21 13 13, WET-PIPE SPRINKLER SYSTEMS.

3.2 TYPICAL OPERATION

a. Activation of any manual pull station, water flow or pressure switch, heat detector, kitchen hood suppression system, gaseous suppression system, or smoke detector shall cause the following operations to occur:

(1) Operate the alarm system in Buildings 17. For sprinkler protected buildings, flash strobes continuously only in the zone of alarm. For buildings without sprinkler protection throughout, flash strobes continuously only on the floor of alarm.

(2) Continuously sound a temporal pattern general alarm and flash all strobes in the building in alarm until reset at the local fire alarm control unit in Buildings 17.

(4) Transmit a separate alarm signal, via the main fire alarm control unit to the fire department.

(5) Unlock the electrically locked exit doors within the zone of alarm.

b. Heat detectors in elevator machine rooms shall, in addition to the above functions, disconnect all power to all elevators served by that machine room after a time delay. The time delay shall be programmed within the fire alarm system programming and be equal to the time it takes for the car to travel from the highest to the lowest level, plus 10 seconds.

c. Smoke detectors in the primary elevator lobbies of Buildings 17 shall, in addition to the above functions, return all elevators in the bank

to the secondary floor.

d. Smoke detectors in the remaining elevator lobbies, elevator machine room, or top of hoistway shall, in addition to the above functions, return all elevators in the bank to the primary floor.

f. Operation of duct smoke detectors shall cause a system supervisory condition and shut down the ventilation system and close the associated smoke dampers as appropriate.

g. Operation of any sprinkler or standpipe system valve supervisory switch, high/low air pressure switch, or fire pump alarm switch shall cause a system supervisory condition.

h. Alarm verification shall not be used for smoke detectors installed for the purpose of early warning.

3.3 TESTS

a. Provide the service of a NICET level III, competent, factory-trained engineer or technician authorized by the manufacturer of the fire alarm equipment to technically supervise and participate during all of the adjustments and tests for the system. Make all adjustments and tests in the presence of the CO.

b. When the systems have been completed and prior to the scheduling of the final inspection, furnish testing equipment and perform the following tests in the presence of the CO. When any defects are detected, make repairs or install replacement components, and repeat the tests until such time that the complete fire alarm systems meets all contract requirements. After the system has passed the initial test and been approved by the CO, the contractor may request a final inspection.

(1) Before energizing the cables and wires, check for correct connections and test for short circuits, ground faults, continuity, and insulation.

(2) Test the insulation on all installed cable and wiring by standard methods as recommended by the equipment manufacturer.

(3) Run water through all flow switches. Check time delay on water flow switches. Submit a report listing all water flow switch operations and their retard time in seconds.

(4) Open each alarm initiating and notification circuit to see if trouble signal actuates.

(5) Ground each alarm initiation and notification circuit and verify response of trouble signals.

3.4 FINAL INSPECTION AND ACCEPTANCE

a. Prior to final acceptance a minimum 30 day "burn-in" period shall be provided. The purpose shall be to allow equipment to stabilize and potential installation and software problems and equipment malfunctions to be identified and corrected. During this diagnostic period, all system operations and malfunctions shall be recorded.

Final acceptance will be made upon successful completion of the "burn-in" period and where the last 14 days is without a system or equipment malfunction.

b. At the final inspection a factory trained representative of the manufacturer of the major equipment shall repeat the tests in Article 3.3 TESTS and those required by NFPA 72. In addition the representative shall demonstrate that the systems function properly in every respect. The demonstration shall be made in the presence of a VA representative.

3.5 INSTRUCTION

a. The manufacturer's authorized representative shall provide instruction and training to the VA as follows:

(1) Six 1-hour sessions to engineering staff, security police and central attendant personnel for simple operation of the system. Two sessions at the start of installation, 2 sessions at the completion of installation and 2 sessions 3 months after the completion of installation.

(2) Four 2-hour sessions to engineering staff for detailed operation of the system. Two sessions at the completion of installation and 2 sessions 3 months after the completion of installation.

(3) Three 8-hour sessions to electrical technicians for maintaining, programming, modifying, and repairing the system at the completion of installation and one 8-hour refresher session 3 months after the completion of installation.

b. The Contractor and/or the Systems Manufacturer's representative shall provide a typewritten "Sequence of Operation" including a trouble shooting guide of the entire system for submittal to the VA. The sequence of operation will be shown for each input in the system in a matrix format and provided in a loose leaf binder. When reading the sequence of operation, the reader will be able to quickly and easily determine what output will occur upon activation of any input in the system. The INPUT/OUTPUT matrix format shall be as shown in Appendix A to NFPA 72.

c. Furnish the services of a competent instructor for instructing personnel in the programming requirements necessary for system expansion. Such programming shall include addition or deletion of devices, zones, indicating circuits and printer/display text.

-- End of Section --

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO T 180	(2025) Standard Method of Test for Moisture-Density Relations of Soils Using a 4.54-kg (10-lb) Rammer and a 457-mm (18-in.) Drop

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C600	(2023) Installation of Ductile-Iron Mains and Their Appurtenances

ASTM INTERNATIONAL (ASTM)

ASTM C33/C33M	(2024a) Standard Specification for Concrete Aggregates
ASTM C117	(2023) Standard Test Method for Materials Finer than 75-um (No. 200) Sieve in Mineral Aggregates by Washing
ASTM C136/C136M	(2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM D698	(2012; R 2021) Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/cu. ft. (600 kN-m/cu. m.))
ASTM D1140	(2017) Standard Test Methods for Determining the Amount of Material Finer than 75-µm (No. 200) Sieve in Soils by Washing
ASTM D1556/D1556M	(2015; E 2016) Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method
ASTM D1557	(2012; E 2015) Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft3) (2700 kN-m/m3)

ASTM D2216	(2019) Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
ASTM D2321	(2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications
ASTM D2487	(2017; R 2025) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D2974	(2020; E 2020) Moisture, Ash, and Organic Matter of Peat and Other Organic Soils
ASTM D4253	(2016; E 2019) Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table
ASTM D4254	(2016) Standard Test Methods for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density
ASTM D4318	(2017; E 2018) Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
ASTM D4829	(2021) Standard Test Method for Expansion Index of Soils
ASTM D5268	(2019) Topsoil Used for Landscaping Purposes
ASTM D6938	(2017a) Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)
ASTM D8167/D8167M	(2023) Standard Test Method for In-Place Bulk Density of Soil and Soil-Aggregate by a Low-Activity Nuclear Method (Shallow Depth)

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA SW-846.3-3	(1999, Third Edition, Update III-A) Test Methods for Evaluating Solid Waste: Physical/Chemical Methods

1.2 DEFINITIONS

1.2.1 Structural Fill

Soil material placed to support buildings, walls, pads, and other similar facilities.

1.2.2 Porous Fill

Free-draining material placed for subsurface drainage, as a capillary break, or another specific purpose.

1.2.3 Topsoil

Surface layer of primarily organic soil capable of supporting vegetation growth.

1.2.4 Utility Bedding Material

Fill placed to directly support pipes, conduits, cables, and appurtenant structures. Bedding may also be used to provide a cushion between utilities and bedrock, obstacles, obstructions and other unyielding materials.

1.2.5 Flowable Fill

Fill placed in a plastic or liquid form that flows to near its final placement location with limited assistance and subsequently cures or solidifies to provide a stable or impermeable barrier.

1.2.6 Satisfactory Materials

Satisfactory materials for fill, backfill, and/or any in-situ soils to remain in place comprise any materials classified by ASTM D2487 as GW, GP, SW, SP. Maximum particle size to be no greater than one-half of the allowable lift thickness 6 inches in any dimension.

1.2.7 Unsatisfactory Materials

Materials which do not comply with the requirements for satisfactory materials are unsatisfactory. Unsatisfactory materials also include man-made fills; trash; refuse; backfills from previous construction; roots and other organic matter or frozen material. Notify the Contracting Officer when encountering any contaminated materials.

1.2.8 Cohesionless Materials

Cohesionless materials include materials classified in ASTM D2487 as GW, GP, SW, and SP. Materials classified as GM and SM will be identified as cohesionless only when the fines are nonplastic. Perform testing, required for classifying materials, in accordance with ASTM D4318, ASTM C117, ASTM C136/C136M and ASTM D1140.

1.2.9 Cohesive Materials

Cohesive materials include materials classified as GC, SC, ML, CL, MH, and CH. Materials classified as GM and SM will be identified as cohesive only when the fines are plastic. Perform testing, required for classifying materials, in accordance with ASTM D4318, ASTM C117, ASTM C136/C136M and

ASTM D1140.

1.2.10 Hard/Unyielding Materials

Hard/Unyielding materials comprise weathered rock, dense consolidated deposits, or conglomerate materials which are not included in the definition of "rock" with stones greater than 3 inch in any dimension or as defined by the pipe manufacturer, whichever is smaller. These materials usually require the use of heavy excavation equipment, ripper teeth, or jack hammers for removal.

1.2.11 Unstable Material

Unstable materials are too weak to adequately support the utility pipe, conduit, equipment, or appurtenant structure. Satisfactory material may become unstable due to ineffective drainage, dewatering, becoming frozen, excessive loading.

1.2.12 Expansive Soils

Expansive soils are defined as soils that have an expansion index greater than 20 when tested in accordance with ASTM D4829.

1.2.13 Rock

Solid homogeneous interlocking crystalline material with firmly cemented, laminated, or foliated masses or conglomerate deposits, neither of which can be removed without systematic drilling, drilling and the use of expansion jacks or feather wedges, or the use of backhoe-mounted pneumatic hole punchers or rock breakers; also large boulders, buried masonry, or concrete other than pavement exceeding 1/2 cubic yard in volume. Removal of hard material will not be considered rock excavation because of intermittent drilling that is performed merely to increase production.

1.2.14 Capillary Water Barrier

A layer of clean, poorly graded crushed rock, stone, or natural sand or gravel having a high porosity which is placed beneath a building slab with or without a vapor barrier to cut off the capillary flow of pore water to the area immediately below a slab.

1.2.15 Degree of Compaction (Proctor)

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum dry density obtained by the test procedure presented in ASTM D1557 abbreviated as a percent of laboratory maximum dry density. Since ASTM D1557 applies only to soils that have 30 percent or less by weight of their particles retained on the 3/4 inch sieve, express the degree of compaction for material having more than 30 percent by weight of their particles retained on the 3/4 inch sieve as a percentage of the maximum dry density in accordance with AASHTO T 180-21 paragraph 1.5, Note 1.

1.2.16 Degree of Compaction (Relative Density)

Degree of compaction required for soils with less than 5 percent passing the No. 200 sieve, is expressed as a relative percentage of the maximum index density/dry unit weight and minimum index density/dry unit weight, obtained by the test procedures in accordance with ASTM D4253 and

ASTM D4254, respectively, abbreviated as a percent of laboratory relative density.

1.2.17 Overhaul

Overhaul is the authorized transportation of satisfactory excavation or borrow materials in excess of the free-haul limit of West Haven VA. stations. Overhaul is the product of the quantity of materials hauled beyond the free-haul limit, and the distance such materials are hauled beyond the free-haul limit, expressed in mile yards.

1.2.18 Borrow

Soil brought to the project site from an external location for the purposes of project construction.

1.2.19 Subgrade

Earth materials directly below foundations and directly below granular base materials in building slab and pavement areas including shoulders.

1.3 SUBSURFACE DATA

Subsurface soil boring logs are shown in project plans and in an Attachment to these specifications. This data represents available subsurface information; however, variations may exist between boring locations and across the site.

1.4 CRITERIA FOR BIDDING

Base bids on the following criteria:

a. Surface elevations are as indicated; however, variations amy exist throughout the site.

b. Unknown pipes or other obstructions may be encountered. Known locations of pipes and artificial obstructions are approximate.

c. Groundwater elevations indicated in the boring logs are based on readings taken at the time of drilling and at various times after the end of drilling. Readings within the area of the chiller plant were taken from monitoring wells installed in boring locations IB-101 through IB-103. Fluctations in the observed groundwater levels will occur due to variations in precipitation, tempurature, and other factors different from those existing at the time the measuremenst were made.

d. Material characteristics indicated within the boring logs may vary from the soils encountered across the site.

e. Material character is indicated by the boring logs.

f. Hard materials and rock will be encountered. Depth, elevation, and material characteristics may vary across the site.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00

SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Excavation and Trenching Plan; G

Dewatering Work Plan; G

Rock Excavation Plan; G

Disposition of Surplus Materials; G

Preconstruction Meeting; G

Geotechnical Data Validation; G

Construction Stormwater General Permit; G

Pre-Construction Survey; G

Underpinning Plan; G

SD-06 Test Reports

Dewatering Performance Records; G

Material Test Report; G

Borrow Site Testing; G

Pipe Inspection Report; G

Geotechnical Evaluation Report; G

Structural Instrumentation and Monitoring; G

1.6 QUALITY CONTROL

1.6.1 Geotechnical Engineer

Contractor shall retain a Professional Geotechnical Engineer to provide inspection of excavations and soil/groundwater conditions throughout construction. The Geotechnical Engineer is responsible for performing pre-construction and periodic site visits throughout construction to assess site conditions. The Geotechnical Engineer is responsible for preparing and updating the Excavation and Trenching Plan and Dewatering Work Plan as construction progresses to reflect changing conditions and submit an updated plan if necessary. The Geotechnical Engineer shall monitor and document vibration levels generated by excavation, rock removal, compaction, and other earthwork operations. The Contractor shall promptly notify the Contracting Officer of any readings exceeding established thresholds or conditions that may adversely affect nearby structures or utilities. Submit a monthly Geotechnical Evaluation report, informing the Contractor and Contracting Officer of the status of the plan and an accounting of the Contractor's adherence to the plan addressing any present or potential problems. The Contractor is responsible for arranging meetings with the Geotechnical Engineer and Contracting Officer throughout the contract duration.

1.6.2 Qualified Technician

Provide a Qualified Technician to inspect, monitor, sample, and performing field testing. The technician qualifications need to be one of the following: a current National Institute for Certification in Engineering Technologies (NICET) Level II minimum certification in Construction Materials Testing Soils; a Geologist-in-Training with minimum one-year experience; an Engineer-in-Training with minimum one-year experience; a Registered Geologist; or a Professional Engineer. The Qualified Technician must be supervised by a Geotechnical Engineer who meet sthe qualifications of Geotechnical Engineer defined herein.

1.6.3 Lab Validation

Perform testing by a Corps validated commercial testing laboratory or Contractor established testing laboratory meeting the requirements of Section 01 45 00 (or similar number) entitled QUALITY CONTROL and approved by the Contracting Officer. Submit testing laboratory validation for the testing to be performed. Do not permit work requiring testing until testing facilities have been inspected, Corps validated and approved by the Contracting Officer.

1.6.4 Geotechnical Data Validation

Contractors may perform additional subsurface exploration efforts that meet the requirements of Section 01 45 00 (or similar number) entitled QUALITY CONTROL and approved by the Contracting Officer. Submit collected boring information to the Contracting Officer via OpenGround. Do not permit work requiring subsurface exploration until utility clearance has been performed and the work plan has been Contracting Officer validated and approved by the Contracting Officer.

1.6.5 Preconstruction Meeting

Conduct a preconstruction meeting at the jobsite at least five business days prior to the start of earthwork operations on the project. The preconstruction meeting is to be arranged by the Contractor and is to follow the written agenda submitted prior to the meeting. The purpose of this meeting is to review the requirements of this specification and the associated plans. The following individuals must be in attendance at this meeting: Contractor's Project Manager and Project Superintendent, earthwork subcontractor's Project Manager and Site Foreman, Contractor's Geotechnical Engineer and Testing Agency, Contracting Officer, Geotechnical Engineer and Civil Engineer, and Contracting Officer, Construction Manager and Engineering Technician.

The minutes of this meeting are to be recorded by the Contractor and published via email within 48 hours to all attendees. The minutes must be re-published within 48 hours via email pending any subsequent comments from the attendees.

1.6.6 Pre-Construction Survey

Conduct a pre-construction survey as indicated in Section 01 11 00 (or similar number) entitled PRE-CONSTRUCTION PHOTOGRAPHIC SURVEY.

1.6.7 Structural Instrumentation and Monitoring

Install instrumentation on existing buildings and structures to monitor

for vertical, horizontal and angular movements to prepare for when earthwork such as deep excavations, shoring and sheeting installation and removal, and underpinning installations are performed next to the existing buildings. Do not permit work until an Instrumentation and Monitoring plan has been reviewed and approved by the Contracting Officer. Submit a daily Monitoring and Instrumentation Report to the Contracting Officer.

1.6.8 Construction Stormwater General Permit

Prepare a Construction Stormwater General Permit (CSGP) application including supporting documents and submit to Connecticut Department of Energy & Environmental Affairs (CTDEEP). Provide the approved CSGP to the Contracting Officer and have one copy available onsite during construction.

PART 2 PRODUCTS

2.1 SOIL MATERIALS

2.1.1 Structural Fill

Materials classified as GW, GP, GM, GC, SW, SP, or SM in accordance with ASTM D2487. Select material type appropriate for the intended purpose.

2.1.2 Porous Fill

Materials containing less than 5 percent passing the No. 200 sieve. Provide the gradation as appropriate for the intended purpose. Porous Fill material will be in accordance with ASTM D5268.

2.1.3 Topsoil

Material suitable for topsoil obtained from offsite areas is defined as: Natural, friable soil representative of productive, well-drained soils in the area, free of subsoil, stumps, rocks larger than one inch diameter, brush, weeds, toxic substances, and other material detrimental to plant growth. Amend topsoil pH range to obtain a pH of 5.5 to 7. Topsoil material will be in accordance with ASTM D5268.

2.1.4 Capillary Water Barrier

Conform to ASTM C33/C33M for fine aggregate grading with a maximum of 3 percent by weight passing ASTM D1140, No. 200 sieve.

2.1.5 Utility Bedding Material

Except as specified otherwise in the individual piping section, provide bedding for buried piping in accordance with AWWA C600 and ASTM D2321. Install bedding for plastic piping to spring line of pipe. Provide geotextile fabric below bedding layer where indicated. Utility bedding material may include the following:

2.1.5.1 Class I

Angular, 0.25 to 1.5 inch, graded stone, including a number of fill materials that have regional significance such as coral, slag, cinders, crushed stone, and crushed shells.

2.1.5.2 Class II

Coarse sands and gravels with maximum particle size of 1.5 inch, including various graded sands and gravels containing small percentages of fines, generally granular and noncohesive, either wet or dry. Soil Types GW, GP, SW, and SP are included in this class as specified in ASTM D2487.

2.1.5.3 Sand

Clean, coarse-grained sand classified as SW or SP by ASTM D2487.

2.1.5.4 Gravel and Crushed Stone

Clean, coarsely graded natural gravel, crushed stone or a combination thereof identified as GW or GP in accordance with ASTM D2487. Do not exceed maximum particle size of 3 inches.

2.2 BURIED WARNING AND IDENTIFICATION MARKERS

Provide polyethylene plastic and metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic warning tape manufactured specifically for warning and identification of buried utility lines. Provide tape on rolls, 3 inches minimum width, color coded as specified below for the intended utility with warning and identification imprinted in bold black letters continuously over the entire tape length. Warning and identification to read, "CAUTION, BURIED (intended service) LINE BELOW" or similar wording. Provide permanent color and printing, unaffected by moisture or soil.

Warning Tape Color Codes	
Red	Electric
Yellow	Gas, Oil; Dangerous Materials
Orange	Telephone and Other Communications
Blue	Water Systems
Green	Sewer Systems
White	Steam Systems
Gray	Compressed Air

2.2.1 Warning Tape for Metallic Piping

Provide acid and alkali-resistant polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.003 inch and a minimum strength of 1500 psi lengthwise, and 1250 psi crosswise, with a maximum 350 percent elongation.

2.2.2 Detectable Warning Tape for Non-Metallic Piping

Provide polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.004 inch, and a minimum strength of 1500 psi lengthwise and 1250 psi

crosswise. Manufacture tape with integral wires, foil backing, or other means of enabling detection by a metal detector when tape is buried up to 3 feet deep. Encase metallic element of the tape in a protective jacket or provide with other means of corrosion protection.

2.2.3 Detection Wire for Non-Metallic Piping

Insulate a single strand, solid copper detection wire with a minimum of 12 AWG.

2.3 BORROW

Provide borrow materials from sources located outside of Government property meeting the requirements of paragraph 2.1 Soil Meterials.

PART 3 EXECUTION

3.1 PROTECTION

Perform all work specified in accordance with applicable requirements of the Corps of Engineers publication EM 385-1-1 Safety and Health Requirements Manual. Provide a Geotechnical Engineer to monitor construction activities and to prepare necessary work plans and reports; see paragraph QUALITY CONTROL.

Use equipment of type and size appropriate for the site conditions (soil character and moisture content). Maintenance of exposed subgrades and fills is the responsibility of the Contractor. The Contractor is required to prevent damage by ineffective drainage, dewatering, and heavy loads and equipment by implementing precautionary measures. Repair or replace any defects or damage.

3.1.1 Underground Utilities

Location of the existing utilities indicated is approximate. Physically verify the location and elevation of the existing utilities indicated prior to starting construction. The Contractor is responsible for protecting utilities from damage during construction.

3.1.2 Drainage and Dewatering

Provide for the collection and disposal of surface and subsurface water encountered during construction.

3.1.2.1 Drainage

Provide for the collection and disposal of surface and subsurface water encountered during construction. Construct storm drainage features (ponds/basins) at the earliest stages of site development, and throughout construction. Grading of the construction area to provide positive surface water runoff away from the construction activity and/or provide temporary ditches, swales, and other drainage features and equipment as required to keep soils from becoming unstable, prevent erosion, or undermining of foundations. Remove unstable material from working platforms for equipment operation and soil support for subsequent construction features and provide new material as specified herein. It is the responsibility of the Contractor to assess the site conditions to employ necessary measures to permit construction to proceed.

3.1.2.2 Dewatering

Control groundwater flowing toward or into excavations to prevent sloughing of excavation slopes and walls, boils, uplift and heave in the excavation, and to eliminate interference with orderly progress of construction. French drains, sumps, ditches or trenches are not allowed within 3 feet of the foundation of any structure, except with specific written approval, and after specific contractual provisions for restoration of the foundation area have been made. Perform control measures by the time the excavation reaches the water level in order to maintain the integrity of the in-situ material. While the excavation is open, maintain the water level continuously, at least 2 feet below the working level. Submit a Dewatering Work Plan outlining procedures for accomplishing dewatering work. Operate dewatering system continuously until construction work below existing water levels is complete. Measure and record performance of dewatering system at same time each day by use of observation wells or piezometers installed in conjunction with the dewatering system. Submit dewatering performance records weekly.

3.1.3 Shoring and Sheeting

Submit an Excavation and Trenching Plan to stabilize features, prevent undermining or unintended horizontal and vertical movement of adjacent structures, and prevent slippage or movement in banks or slopes adjacent to the excavation. Submit drawings and calculations, certified by a professional engineer registered in the State of Connecticut, describing the methods for shoring and sheeting of excavations. Shoring and sheeting shall be in compliance with OSHA Trenching and Excavation Regulations. Drawings to include material sizes and types, arrangement of members, and the sequence and method of installation and removal. Calculations are to include data and references used.

3.1.4 Underpinning

Submit an Underpinning Plan to stabilize existing foundations and floor slabs where excavations of new foundations undermine or encroach into the footing bearing zone of the existing footings and floor slabs. Underpinning options include methods such as conventional concrete-filled pits, drilled micro-piles or grouting. A specialty contractor experienced in underpinning is required.Submit drawings and calculations, certified by a professional engineer registered in the State of Connecticut, describing the methods for underpinning. Drawings to include material sizes and types, arrangement of members, and the sequence and method of installation. Calculations are to include data and references used.

3.1.5 Protection of Graded Surfaces

Protect newly backfilled, graded, and topsoiled areas from traffic, erosion, and settlements that may occur. Repair or reestablish damaged grades, elevations, or slopes.

3.2 BORROW

Select borrow material to meet the requirements and conditions of the fill or embankment for which it is to be used. Obtain borrow material from selected by the Contractor approved private sources. Unless otherwise provided in the contract, the Contractor is responsible for obtaining the right to procure material, pay royalties and other charges involved, and bear the expense of developing the sources, including rights-of-way for

hauling from the owners. Unless specifically provided, do not obtain borrow within the limits of the project site without prior written approval.

3.2.1 Contractor Furnished Borrow Area(s)

Obtain approved borrow materials from approved offsite sources. If a borrow source is selected that is not a commercial entity from which soil material is directly purchased, submit a Borrow Plan that includes the borrow source location, geotechnical test results showing the fill material meets the Contract requirements, environmental test results in accordance with paragraph ENVIRONMENTAL REQUIREMENTS FOR OFF-SITE SOIL, and any Federal, State, and local permits required for excavation and reclamation of the borrow area.

3.2.2 Environmental Requirements for Off-Site Soil

Test offsite soils brought in for use as backfill for Total Petroleum Hydrocarbons (TPH), Benzene, Toluene, Ethyl Benzene, and Xylene (BTEX) and full Toxicity Characteristic Leaching Procedure (TCLP) including ignitability, corrosivity and reactivity. Backfill may not contain concentrations above appropriate State and EPA criteria, and for hazardous waste characteristics 10 parts per million (ppm) of total petroleum hydrocarbons (TPH) and 10 ppm of the sum of Benzene, Toluene, Ethyl Benzene, and Xylene (BTEX) and pass the TCLP test. Determine TPH concentrations by using EPA Method 8015. Determine BTEX concentrations by using EPA SW-846.3-3 Method 5030/8020/8260B. Perform TCLP in accordance with EPA SW-846.3-3 Method 1311. Perform hazardous waste characteristic tests for ignitability, corrosivity, and reactivity in accordance with accepted standard methods. Perform PCB testing in accordance with accepted standard methods for sampling and analysis of bulk solid samples. Provide borrow site testing for petroleum hydrocarbons and BTEX from a grab sample of material from the area most likely to be contaminated at the borrow site (as indicated by visual or olfactory evidence), with at least one test from each borrow site. Provide borrow site testing for hazardous waste characteristics (TPH, BTEX and TCLP) from a composite sample of material, collected in accordance with standard soil sampling techniques. Do not bring borrow material to project site until Borrow Plan containing environmental test results has been received and approved by the Contracting Officer.

3.3 SURFACE PREPARATION

3.3.1 Clearing and Grubbing

Remove trees, stumps, logs, shrubs, brush and vegetation and other items that would interfere with construction operations. Remove stumps entirely. Grub out matted roots and roots over 3 inches in diameter to at least 18 inches below existing surface.

3.3.2 Stripping

Strip site where indicated on the plans. Strip existing surface materials to a depth of 4 inches below the existing ground surface in areas that have lawn as shown on the drawings. All stripped materials not suitable for reuse as topsoil will be disposed of offsite by the Contractor. Screen all stripped soils to remove roots and organic materials prior disposal.

Strip suitable soil from the site where excavation or grading is indicated and stockpile separately from other excavated material. Protect topsoil and keep in segregated piles until needed.

3.3.3 Stockpiling Operations

Place and grade stockpiles of satisfactory materials as specified. Keep stockpiles in a neat and well drained condition, giving due consideration to drainage at all times. Clear, grub, and seal by rubber-tired equipment, the ground surface at stockpile locations; separately stockpile excavated satisfactory and unsatisfactory materials. Protect stockpiles of satisfactory materials from contamination which may destroy the quality and fitness of the stockpiled material. Do not create stockpiles that could obstruct the flow of any stream, endanger a partly finished structure, impair the efficiency or appearance of any structure, or be detrimental to the completed work in any way. If the Contractor fails to protect the stockpiles, and any material becomes unsatisfactory, remove and replace such material with satisfactory material from approved sources.

3.4 EXCAVATION

Excavate to contours, elevation, and dimensions indicated. Excavate soil disturbed or weakened by Contractor's operations, and soils softened or made unstable for subsequent construction due to exposure to weather. Use material removed from excavations meeting the specified requirements in the construction of fills, subgrades, shoulders, bedding (as backfill), and for similar purposes to minimize surplus material and to minimize additional material to brought on site. Do not excavate below indicated depths except to remove unstable material as determined by theGeotechnical Engineer and confirmed by the Contracting Officer. Remove and replace excavations below the grades shown with appropriate materials as directed by the Contracting Officer.

If at any time during excavation, including excavation from borrow areas, the Contractor encounters material that may be classified as rock or as hard/unyielding material, uncover such material, and notify the Contracting Officer. Do not proceed with the excavation of this material until the Contracting Officer has classified the materials as common excavation or rock excavation. Failure on the part of the Contractor to uncover such material, notify the Contracting Officer, and allow sufficient time for classification and delineation of the undisturbed surface of such material will cause the forfeiture of the Contractor's right of claim to any classification or volume of material to be paid for other than that allowed by the Contracting Officer for the areas of work in which such deposits occur.

3.4.1 Ditches, Gutters, and Channel Changes

Finish excavation of ditches, gutters, and channel changes by cutting accurately to the cross sections, grades, and elevations shown. Do not excavate below grades shown. Backfill excessive excavation as directed by the Contracting Officer, with satisfactory, compacted, material or with suitable stone or cobble to grades shown. Dispose excavated material as shown or as directed. Do not allow material to be deposited within 4 feet from edge of a ditch. Maintain excavations free from detrimental quantities of leaves, brush, sticks, trash, and other debris until final acceptance of the work.

3.4.2 Trench Excavation Requirements

Excavate the trench as recommended by the manufacturer of the pipe to be installed. Slope trench walls below the top of the pipe, or make vertical, and of such width as recommended by the manufacturer. Provide vertical trench walls where no manufacturer installation instructions are available. Do not exceed the trench width of 24 inches below the top pipe plus pipe outside diameter (O.D.) for pipes of less than 24 inches inside diameter, and do not exceed 36 inches plus pipe outside diameter for pipe sizes larger than 24 inches inside diameter. Where recommended trench widths are exceeded, provide redesign, stronger pipe, or special installation procedures. The Contractor is responsible for the cost of redesign, stronger pipe, or special installation procedures without any additional cost to the Government.

3.4.2.1 Bottom Preparation

Grade the bottoms of trenches accurately to provide uniform bearing and support for the bottom quadrant of each section of the pipe. Excavate bell holes to the necessary size at each joint or coupling to eliminate point bearing. Remove stones of 1.5 inch or greater in any dimension, or as recommended by the pipe manufacturer, whichever is smaller, to avoid point bearing.

3.4.2.2 Removal of Unyielding Material

Where unyielding material is encountered in the bottom of the trench, notify the Contracting Officer. Following approval, remove such material 6 inches below the required grade and replaced with suitable materials as provided in paragraph FILLING AND COMPACTION.

3.4.2.3 Removal of Unstable Material

Where unstable material is encountered in the bottom of the trench, remove such material to the depth directed and replace it to the proper grade with suitable material as provided in paragraph FILLING AND COMPACTION. When removal of unstable material is required due to the Contractor's fault or neglect in performing the work, the Contractor is responsible for excavating the resulting material and replacing it without additional cost to the Government.

3.4.2.4 Excavation for Appurtenances

Provide excavation for manholes, catch-basins, inlets, or similar structures sufficient to leave at least 12 inches clear between the outer structure surfaces and the face of the excavation or support members or of sufficient size to permit the placement and removal of forms for the full length and width of structure footings and foundations as shown.

3.4.2.5 Gas Distribution

Excavate trenches to a depth that will provide a minimum 18 inches of cover in rock excavation and a minimum 24 inches of cover in other excavation.

3.4.2.6 Water Lines

Excavate trenches to a depth that provides a minimum cover of 4 feet from the existing ground surface, or from the indicated finished grade,

whichever is lower, to the top of the pipe.

3.4.3 Underground Utilities

Perform work adjacent to utilities as indicated and in accordance with procedures outlined by utility owner. Excavation made with power-driven equipment is not permitted within 2 feet of known utility or subsurface construction. For work immediately adjacent to or for excavations exposing a utility or other buried obstruction, excavate by hand. Start hand excavation on each side of the indicated obstruction and continue until the obstruction is uncovered or until clearance for the new grade is assured. Support uncovered lines or other existing work affected by the contract excavation until approval for backfill is granted by the Contracting Officer. Report damage to utility lines or subsurface construction immediately to the Contracting Officer.

3.4.4 Structural Excavation

Following general excavation and rough grading activities, excavate the entire building footprint area and at least 5 feet beyond. Scarify the exposed surface to a depth of 6 to 8 inches, moisture-condition, and compact to at least 95 percent of laboratory maximum dry density. Do not excavate to final grade until just before concrete is to be placed. Roughen level surfaces. Cut sloped surfaces as indicated into rough steps or benches to provide a satisfactory bond for compacting materials. For new pavement areas including exterior concrete pads, over-excavate to a minimum of 12 inches below bottom of new pavement/pad base course, scarify, moisture-condition, and compact to at least 95 percent.

Make excavations to the lines, grades, and elevations shown, or as directed. Provide trenches and foundation pits of sufficient size to permit the placement and removal of forms for the full length and width of structure footings and foundations as shown. Clean rock or other hard foundation material of loose debris and cut to a firm, level, stepped, or serrated surface. Remove loose disintegrated rock and thin strata.

Concrete placement is not allowed until footing subgrades have been inspected and approved by the Contracting Officer.

3.4.5 Rock Excavation and Blasting

Excavate rock encountered in the cut section to a depth of 6 inches below finished grade and replace with satisfactory material. Submit a Rock Excavation Plan, prepared and sealed by a registered professional engineer. Blasting will not be permitted.

3.5 SUBGRADE PREPARATION

3.5.1 General Requirements

Shape subgrade to line, grade, and cross section as indicated. Remove unsatisfactory and unstable material in surfaces to receive fill or in excavated areas, and replaced with satisfactory materials or structural fill as indicated. Do not place material on surfaces that are muddy, frozen, contain frost, or otherwise containing unstable material. Scarify the surface to a depth of 4 inches prior to placing fill. Step or bench sloped surfaces steeper than 1 vertical to 4 horizontal prior to scarifying. Place 4 inches of loose fill and blend with scarified material. When subgrade is part fill and part excavation or natural

ground, scarify to a depth of 8 inches.

3.5.2 Subgrade for Structures, Spread Footings, and Concrete Slabs

Do not excavate below depth shown for structures, spread footings, and concrete slabs. If over excavation occurs, notify the Contracting Officer and remove, replace, and compact as directed. Compact disturbed material to 95 percent of ASTM D1557. After final rolling, the surface of the subgrade for buildings and pavements must not show deviations greater than 0.05 foot when tested with a 12-foot straightedge applied both parallel and at right angles to the centerline of the area.

3.5.3 Subgrade for Pavements

Compact top 12 inches of subgrade for pavements to at least 95 percent of ASTM D1557. After compaction, the surface of the subgrade for buildings and pavements must not show deviations greater than 0.05 foot when tested with a 12-foot straightedge applied both parallel and at right angles to the centerline of the area.

3.5.4 Subgrade for Shoulders

Compact the upper 6 inches of subgrade for shoulders to at least 95 percent of ASTM D1557 for the full depth of the shoulder.

3.5.5 Subgrade Filter Fabric

Place filter fabric as indicated directly on prepared subgrade free of vegetation, stumps, rocks larger than 2 inch diameter and other debris which may puncture or otherwise damage the fabric. Repair damaged fabric by placing an additional layer of fabric to cover the damaged area a minimum of 3 feet overlap in all directions. Overlap fabric at joints a minimum of 3 feet. Obtain approval of filter fabric installation before placing fill or backfill. Place fill or backfill on fabric in the direction of overlaps and compact as specified herein. Follow manufacturer's recommended installation procedures.

3.6 FILLING AND COMPACTION

Prepare ground surface on which backfill is to be placed and provide compaction requirements for backfill materials in conformance with the applicable portions of paragraphs for SUBGRADE PREPARATION. Do not place material on surfaces that are muddy, frozen, or contain frost. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, or other approved equipment well suited to the soil being compacted. Moisten material as necessary to provide the moisture content that will readily facilitate obtaining the specified compaction with the equipment used. Fill and backfill to contours, elevations, and dimensions indicated. Compact and test each lift before placing overlaying lift.

3.6.1 Trench Backfill

Backfill trenches to the grade shown. Backfill the trench to a depth above the top of pipe to allow for safe testing prior to performing the required pressure tests. Leave the joints and couplings uncovered during the pressure test.

3.6.1.1 Replacement of Unyielding Material

Replace unyielding material removed from the bottom of the trench with satisfactory material or initial backfill material.

3.6.1.2 Replacement of Unstable Material

Replace unstable material removed from the bottom of the trench or excavation with satisfactory material placed in layers not exceeding 6 inches loose thickness.

3.6.1.3 Bedding and Initial Backfill

Provide bedding of the type and thickness shown. Place initial backfill material and compact it with approved tampers to a height of at least one foot above the utility pipe or conduit. Bring up the backfill evenly on both sides of the pipe for the full length of the pipe. Take care to ensure thorough compaction of the fill under the haunches of the pipe. Except where shown or when specified otherwise in the individual piping section, provide bedding for buried piping in accordance with PART 2 paragraph UTILITY BEDDING MATERIAL. Compact backfill to top of pipe to 95 percent of ASTM D698 maximum dry density. Provide plastic piping with bedding to spring line of pipe.

3.6.1.4 Final Backfill

Do not begin backfill until construction below finish grade has been approved, underground utilities systems have been inspected, tested and approved, forms removed, and the excavation cleaned of trash and debris. Bring backfill to indicated finish grade. Where pipe is coated or wrapped for protection against corrosion, the backfill material up to an elevation 2 feet above sewer lines and one foot above other utility lines need to be free from stones larger than one inch in any dimension. Heavy equipment for spreading and compacting backfill are not to be operated closer to foundation or retaining walls than a distance equal to the height of backfill above the top of footing; compact remaining area in layers not more than 4 inches in compacted thickness with power-driven hand tampers suitable for the material being compacted. Place backfill carefully around pipes or tanks to avoid damage to coatings, wrappings, or tanks. Do not place backfill against foundation walls prior to 7 days after completion of the walls. As far as practicable, bring backfill up evenly on each side of the wall and sloped to drain away from the wall.

Fill the remainder of the trench, except for special materials for buildings and pavements with satisfactory material. Place backfill material and compact as follows:

3.6.1.4.1 Buildings and Pavements

Place backfill up to the required elevation as specified. Do not permit water flooding or jetting methods of compaction. Compact as specified for Structural Fill.

3.6.1.4.2 Turfed or Seeded Areas and Miscellaneous Areas

Deposit backfill in layers of a maximum of 12 inches loose thickness, and compact it to 85 percent maximum dry density for cohesive soils and 90 percent maximum dry density for cohesionless soils. Apply this requirement to all other areas not specifically designated above.

3.6.1.5 Heat Distribution System

Free initial backfill material of stones larger than 1/4 inch in any dimension.

3.6.1.6 Electrical Distribution System

Provide a minimum cover of 24 inches from the finished grade to direct burial cable and conduit or duct line, unless otherwise indicated.

3.6.1.7 Displacement of Features

After other required tests have been performed and the trench backfill compacted to the finished grade surface, inspect the pipe to determine whether unexpected or damaging displacement has occurred. Conduct walk-through inspection of pipe sizes larger than 48 inches. Inspect pipes smaller than 48 inches using remote methods using closed circuit television, sonar, or hybrid that can provide a 360-degree inspection of the pipe. Prepare and submit a pipe inspection report consisting of digital video or photos. If, in the judgment of the Contracting Officer, the interior of the pipe shows poor alignment or any other defects that would cause improper functioning of the system, replace or repair the defects as directed at no additional cost to the Government.

3.6.1.8 Buried Tape And Detection Wire

3.6.1.8.1 Buried Warning and Identification Tape

Provide buried utility lines with utility identification tape. Bury tape 12 inches below finished grade; under pavements and slabs, bury tape 6 inches below top of subgrade.

3.6.1.8.2 Buried Detection Wire

Bury detection wire directly above non-metallic piping at a distance not to exceed 12 inches above the top of pipe. Extend the wire continuously and unbroken, from manhole to manhole. Terminate the ends of the wire inside the manholes at each end of the pipe, with a minimum of 3 feet of wire, coiled, remaining accessible in each manhole. Furnish insulated wire over its entire length. Install wires at manholes between the top of the corbel and the frame, and extend up through the chimney seal between the frame and the chimney seal. For force mains, terminate the wire in the valve pit at the pump station end of the pipe.

3.6.2 Structural Fill Placement

Place fill and backfill beneath and adjacent to structures in successive horizontal layers of loose material not more than 8 inches in depth, or in loose layers not more than 4 inches in depth when using hand-operated compaction equipment. Do not place over wet or frozen materials. Compact to at least 90 percent of laboratory maximum dry density for cohesive materials or 95 percent of laboratory maximum dry density for cohesionless materials, except as otherwise specified. Perform compaction in such a manner as to prevent wedging action or eccentric loading upon or other damage to the structure. Moisture condition fill and backfill material to a moisture content that will readily facilitate obtaining the specified compaction.

3.6.3 Backfill for Appurtenances

After the manhole, catchbasin, inlet, or similar structure has been constructed and the concrete has been allowed to cure for 7 days if cast-in-place, place backfill in such a manner that the structure is not be damaged by the shock of falling earth. Deposit the backfill material, compact it as specified for final backfill, and bring up the backfill evenly on all sides of the structure to prevent eccentric loading and excessive stress.

3.6.4 Porous Fill Placement

Provide under floor and area-way slabs on a compacted subgrade. Place in a single lift and compact with a minimum of two passes of a hand-operated plate-type vibratory compactor.

3.6.5 Compaction

3.6.5.1 General Site

Compact underneath areas designated for vegetation and areas outside the 5 foot line of the paved area or structure to 85 percent of ASTM D1557.

3.6.5.2 Adjacent Areas

Compact areas within 5 feet of structures to 95 percent of ASTM D1557.

3.7 FINISHING/FINISH OPERATIONS

During construction, keep excavations shaped and drained. Maintain ditches and drains along subgrade to drain effectively at all times. Do not disturb the finished subgrade by traffic or other operation. Protect and maintain the finished subgrade in a satisfactory condition until ballast, subbase, base, or pavement is placed. Do not permit the storage or stockpiling of materials on finished subgrade. Do not lay subbase, base course, ballast, or pavement until the subgrade has been checked and approved, and in no case place subbase, base, surfacing, pavement, or ballast on a muddy, spongy, frozen or otherwise unstable subgrade.

Finish the surface of excavations, and subgrades to a smooth and compact surface in accordance with the lines, grades, and cross sections or elevations shown. Provide the degree of finish for graded areas within 0.1 foot of the grades and elevations indicated except as indicated for subgrades specified in paragraph SUBGRADE PREPARATION. Finish gutters and ditches in a manner that will result in effective drainage. Finish the surface of areas to be turfed to a smoothness suitable for the application of turfing materials. Repair graded, topsoiled, or backfilled areas prior to acceptance of the work, and re-established grades to the required elevations and slopes.

3.7.1 Capillary Water Barrier

Place a capillary water barrier under concrete floor and area-way slabs grade directly on the subgrade and compact with a minimum of two passes of a hand-operated plate-type vibratory compactor.

3.7.2 Grading Around Structures

Construct areas within 5 feet outside of each building and structure line

true-to-grade, shape to drain, and maintain free of trash and debris until final inspection has been completed and the work has been accepted.

3.7.3 Shoulder Construction

Construct shoulders of satisfactory material. Submit advanced notice on shoulder construction for rigid pavements. Construct shoulders immediately after adjacent paving is complete. In the case of rigid pavements, do not construct shoulders until permission of the Contracting Officer has been obtained. Compact the entire shoulder area to at least the percentage of maximum dry density as specified in paragraph SUBGRADE PREPARATION above, for specific ranges of depth below the surface of the shoulder. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment. Finish shoulder construction in proper sequence in such a manner that adjacent ditches will be drained effectively and that no damage of any kind is done to the adjacent completed pavement. Align the completed shoulders true to grade and shaped to drain in conformity with the cross section shown.

3.7.4 Grading

Finish grades as indicated within one-tenth of one foot. Grade areas to drain water away from structures. Maintain areas free of trash and debris. For existing grades that will remain but which were disturbed by Contractor's operations, grade as directed.

3.7.5 Topsoil and Seed

On areas to receive topsoil, prepare the compacted subgrade soil to a 2 inches depth for bonding of topsoil with subsoil. Spread topsoil evenly to a thickness of 4 inch after compaction and grade to the elevations and slopes shown. Do not spread topsoil when frozen or excessively wet or dry. Keep topsoil separate from other excavated materials, brush, litter, objectionable weeds, roots, stones larger than 2 inches in diameter, and other materials that would interfere with planting and maintenance operations. Stockpile in locations indicated any surplus of topsoil from excavations and gradings. Obtain material required for topsoil in excess of that produced by excavation within the grading limits from offsite areas.

3.8 DISPOSITION OF SURPLUS MATERIAL

Remove from Government property all surplus or other soil material not required or not suitable for filling or backfilling, along with brush, refuse, stumps, roots, and timber. Properly disposed of in accordance with all applicable laws and regulations. Prepare plan for Disposition of Surplus Materials to include permissions document to dispose of nonsalable products.

3.9 TESTING

Perform testing as indicated in Table 1. Submit Material Test Reports within 7 days of tests being completed.In addition, submit test data in accordance with Section 01 33 00 SUBMITTAL PROCEDURES

Material Type list materials to be tested as identified in paragraph DEFINITIONS	Location of Material	Test Method	Test Frequency
Fills and Satisfactory Material		Density - ASTM D1556/D1556M	One test per 2000 square feet, or fraction thereof, of each lift of fill or backfill areas compacted by other than hand-operated machines. Double testing frequency for areas compacted by hand-operated machines. If ASTM D6938 or ASTM D8167/D8167M is used, check in-place densities by ASTM D1556/D1556M as follows: One check test per lift for every 6 tests. Where ASTM D8167/D8167M is used, provide water content verification in accordance with ASTM D2216 for each test.
Fills and Satisfactory Materials		Moisture Content - ASTM D2216	Two tests per day for each type of fill and backfill. Sample taken immediately prior to compaction after moisture conditioning.

Material Type list materials to be tested as identified in paragraph DEFINITIONS	Location of Material	Test Method	Test Frequency
Fills and Satisfactory Materials		Moisture Density Relationship - ASTM D1557	One representative test per 500 cubic yards of fill and backfill, or when any change in material occurs which may affect the optimum moisture content or laboratory maximum dry density. Sample to be taken from stockpile or location of placement.

Material Type list materials to be tested as identified in paragraph DEFINITIONS	Location of Material	Test Method	Test Frequency
Fills and Satisfactory Materials		Gradation - ASTM C136/C136M	One representative test per 500 cubic yards of fill and backfill, or when any change in material occurs which may affect the optimum moisture content or laboratory maximum dry density. Sample to be taken from stockpile or location of placement.

Material Type list materials to be tested as identified in paragraph DEFINITIONS	Location of Material	Test Method	Test Frequency
Topsoil		Organic Content Test - ASTM D2974, Method C	One representative test per 20 yards .

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DEWATERING

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies performance of dewatering required to lower and control ground water table levels and hydrostatic pressures to permit excavation, backfill, and construction to be performed in the dry. Control of surface water shall be considered as part of the work under this specification.

1.2 SUMMARY

a. The work to be completed by the Contractor includes, but is not necessarily limited to the following:

(1) Implementation of the Erosion and Sedimentation Control Plan.

(2) Dewater excavations, including seepage and precipitation.

b. The Contractor shall be responsible for providing all materials, equipment, labor, and services necessary for care of water and erosion control. Excavation work shall not begin before the Erosion and Sedimentation Control Plan is in place.

1.3 REQUIREMENT

a. Dewatering system shall be of sufficient size and capacity necessary to lower and maintain ground water table to an elevation at least 1 foot below lowest foundation subgrade or bottom of pipe trench and to allow material to be excavated and concrete placed in a reasonably dry condition. Materials to be removed shall be sufficiently dry to permit excavation to grades shown and to stabilize excavation slopes where sheeting is not required. Operate dewatering system continuously until backfill work has been completed.

b. Reduce hydrostatic head below any excavation to the extent that water level in the construction area is a minimum of 1 foot below prevailing excavation surface.

c. Prevent loss of fines, seepage, boils, quick conditions or softening of foundation strata.

d. Maintain stability of sides and bottom of excavation.

e. Construction operations are performed in the dry.

f. Control of surface and subsurface water is part of dewatering requirements. Maintain adequate control so that:

(1) The stability of excavated and constructed slopes are not adversely affected by saturated soil, including water entering prepared subbase and subgrades where underlying materials are not

free draining or are subject to swelling or freeze-thaw action.

(2) Erosion is controlled.

(3) Flooding of excavations or damage to structures does not occur.

(4) Surface water drains away from excavations.

(5) Excavations are protected from becoming wet from surface water, or insure excavations are dry before additional work is undertaken.

g. Permitting Requirements: The contractor shall comply with and obtain the required State and County permits where the work is performed.

1.4 RELATED WORK

a. Materials testing and inspection during construction: Section 01 45 29, TESTING LABORATORY SERVICES.

b. Safety Requirements: Section 00 72 00, GENERAL CONDITIONS, Article, ACCIDENT PREVENTION.

c. Submittal requirements as specified in Section 01 33 23 SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

d. Protection of existing utilities, fire protection services, existing equipment, roads, and pavements: Section 01 00 00, GENERAL REQUIREMENTS.

e. Subsurface Investigation: Section 01 00 00, GENERAL REQUIREMENTS, Article 1.11, PHYSICAL DATA.

f. Excavation, backfilling, site grade and utilities: Section 31 20 00, EARTHWORK.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Permits; G,

Drawings; G

Correction Action Plan; G

SD-03 Product Data

Dewatering System; G,

Standby Equipment; G,

SD-05 Design Data

Design Data; G,

SD-06 Test Reports

Inspection Reports; G,

1.6 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, AND SAMPLES.

(1) All required permits.

(2) Drawings and Design Data:

a. Submit drawings and data showing the method to be employed in dewatering excavated areas 30 days before commencement of excavation.

b. Material shall include: location, depth and size of wellpoints, headers, sumps, ditches, size and location of discharge lines, capacities of pumps and standby units, and detailed description of dewatering methods to be employed to convey the water from site to adequate disposal.

c. Include a written report outlining control procedures to be adopted if dewatering problem arises.

d. Capacities of pumps, prime movers, and standby equipment.

e. Design calculations proving adequacy of system and selected equipment. The dewatering system shall be designed using accepted and professional methods of design and engineering consistent with the best modern practice. The dewatering system shall include the deep wells, wellpoints, and other equipment, appurtenances, and related earthwork necessary to perform the function.

f. Detailed description of dewatering procedure and maintenance method.

g. Materials submitted shall be in a format acceptable for inclusion in required permit applications to any and all regulatory agencies for which permits for discharge water from the dewatering system are required due to the discharge reaching regulated bodies of water.

(3) Inspection Reports.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 INSTALLATION

a. Install a dewatering system to lower and control ground surface water in order to permit excavation, construction of structure, and placement of backfill materials to be performed under dry conditions. Make the dewatering system adequate to pre-drain the water-bearing strata above and below the bottom of structure foundations, utilities and other excavations.

b. In addition, reduce hydrostatic pressure head in water-bearing strata below structure foundations, utility lines, and other excavations, to extent that water levels in construction area are a minimum of 1 foot below prevailing excavation surface at all times.

3.2 OPERATION

a. Prior to any excavation below the ground water table, place system into operation to lower water table as required and operate it continuously 24 hours a day, 7 days a week until utilities and structures have been satisfactorily constructed, which includes the placement of backfill materials and dewatering is no longer required.

b. Place an adequate weight of backfill material to prevent buoyancy prior to discontinuing operation of the system.

3.3 WATER DISPOSAL

a. Dispose of water removed from the excavations in such a manner as:

(1) Will not endanger portions of work under construction or completed.

(2) Will cause no inconvenience to Government or to others working near site.

(3) Will comply with the stipulations of required permits for disposal of water.

(4) Will Control Runoff: The Contractor shall be responsible for control of runoff in all work areas including but not limited to: excavations, access roads, parking areas, laydown, and staging areas. The Contractor shall provide, operate, and maintain all ditches, basins, sumps, culverts, site grading, and pumping facilities to divert, collect, and remove all water from the work areas. All water shall be removed from the immediate work areas and shall be disposed of in accordance with applicable permits.

b. Excavation Dewatering:

(1) The Contractor shall be responsible for providing all facilities required to divert, collect, control, and remove water from all construction work areas and excavations.

(2) Drainage features shall have sufficient capacity to avoid flooding of work areas.

(3) Drainage features shall be so arranged and altered as required to avoid degradation of the final excavated surface(s).

(4) The Contractor shall utilize all necessary erosion and sediment control measures as described herein to avoid construction related degradation of the natural water quality.

c. Dewatering equipment shall be provided to remove and dispose of all surface and ground water entering excavations, trenches, or other parts of the work during construction. Each excavation shall be kept dry during subgrade preparation and continually thereafter until the structure to be built, or the pipe to be installed therein, is

completed to the extent that no damage from hydrostatic pressure, flotation, or other cause will result.

3.4 STANDBY EQUIPMENT

Provide complete standby equipment, installed and available for immediate operation, as may be required to adequately maintain de-watering on a continuous basis and in the event that all or any part of the system may become inadequate or fail.

3.5 CORRECTIVE ACTION

If dewatering requirements are not satisfied due to inadequacy or failure of the dewatering system (loosening of the foundation strata, or instability of slopes, or damage to foundations or structures), perform work necessary for reinstatement of foundation soil and damaged structure or damages to work in place resulting from such inadequacy or failure by Contractor, at no additional cost to Government. Provide a Correction Action Plan prior to starting work to the Contracting Officer for review and approval.

3.6 DAMAGES

Immediately repair damages to adjacent facilities caused by dewatering operations.

3.7 REMOVAL

Insure compliance with all conditions of regulating permits and provide such information to the Resident Engineer. Obtain written approval from Resident Engineer before discontinuing operation of dewatering system.

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FLOWABLE FILL

PART 1 GENERAL

1.1 INTRODUCTION

a. Flowable fill refers to a cementitious slurry consisting of a mixture of fine aggregate or filler, water, and cementitious material(s), which is used as a fill or backfill in lieu of compacted earth. This mixture is capable of filling all voids in irregular excavations and hard to reach places (such as under undercuts of existing slabs), is self-leveling, and hardens in a matter of a few hours without the need for compaction in layers. Flowable fill is sometimes referred to as controlled density fill (CDF), controlled low strength material (CLSM), lean concrete slurry, and unshrinkable fill.

b. Flowable fill materials will be used as only as a structural fill replacement on Government projects. to produce a comparable compressive strength to the surrounding soil after hardening, making excavation at a later time possible.

1.2 DESCRIPTION

Furnish and place flowable fill in a fluid condition, that sets within the required time and, after curing, obtains the desired strength properties as evidenced by the laboratory testing of the specific mix design, at locations shown on the plans or as directed by the Resident Engineer, in writing . This section specifies flowable fill for use as structural fill to remain easily excavatable using a backhoe as would be utilized for adjoining earth .

1.3 RELATED WORK

a. Materials testing and inspection during construction: Section 01 45 29, TESTING LABORATORY SERVICES.

b. Earthwork, excavation and backfill and compaction requirements: Section 31 20 00, EARTHWORK.

1.4 DEFINITIONS

a. Flowable fill - Ready-mix Controlled Low Strength Material used as an alternative to compacted soil, and is also known as controlled density fill, and several other names, some of which are trademark names of material suppliers. Flowable fill (Controlled Low Strength Material) differs from portland cement concrete as it contains a low cementitious content to reduce strength development for possible future removal. Unless specifically approved otherwise, by the Resident Engineer, flowable fill shall be designed as a permanent material, not designed for future removal. Design strength for this permanent type flowable fill shall be a compressive strength of (300 psi) minimum at 28 days. Chemical admixtures may also be used in flowable fill to modify performance properties of strength, flow, set

and permeability.

b. Excavatable Flowable fill – flowable fill designed with a compressive strength that will allow excavation as either machine tool excavatable at compressive strength of (200 psi) maximum at 1 year, or hand tool excavatable at compressive strength of (100 psi) maximum at 1 year.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Admixture Supplier; G,

Pre-Approval Procedures; G,

SD-03 Product Data

Flowable Fill; G,

Portland Cement; G,

Mixing Water; G,

Air-Entraining Admixture; G,

Chemical Admixtures; G,

Aggregate; G,

SD-04 Samples

Sampling And Acceptance; G,

SD-05 Design Data

Flowable Fill Mix Design; G,

SD-06 Test Reports

Test And Performance; G,

SD-07 Certificates

Manufacturer's Certificates; G,

1.6 SUBMITTAL INSTRUCTIONS

a. Submit in accordance with Section 01 33 00 SUBMITTAL PROCEDURES

(1) Provide documentation that the admixture supplier has experience of at least one year, with the products being provided and any equipment required to obtain desired performance of the product.

(2) Flowable fill Mix Design: Provide flowable fill mix design

containing cement and water. At the contractor’s option, it may also contain fly ash, aggregate, or chemical admixtures in any proportions such that the final product meets the strength and flow consistency, and shrinkage requirements included in this specifications. The mix design should state the sources and proportions of each of the flowable fill constituents. The coefficient of permeability of flowable fill shall be that of uniform fine sand, (0.16 in/sec) or as indicated to provide a backfill material with permeability equal to or greater than that of the surrounding soil.

(3) Test and Performance - Submit the following data:

a. Flowable fill shall have a minimum strength of (300 psi) according to ASTM C39/C39M at 28 days after placement.

b. Flowable fill shall have minimal subsidence and bleed water shrinkage. Evaporation of bleed water shall not result in shrinkage of more than 1/8 inch per ft.) of flowable fill depth (for mixes containing high fly ash content). Measurement of a Final Bleeding shall be as measured in Section 10 of ASTM C940 “Standard Test Method for Expansion and Bleeding of Freshly Mixed Grouts for Preplaced-Aggregate Concrete in the Laboratory.

c. Flowable fill shall have a unit weight of (115 – 145 lb/feet 3) measured at the point of placement after a 60 minute ready-mix truck ride.

(4) Manufacturer's Certificates: Provide Resident Engineer with a certification that the materials incorporated in the flowable fill, following achievement of the required strength, do not represent a threat to groundwater quality.

1.7 REFERENCES

Publications listed below form a part of this specification to extent referenced. Publications are referenced in text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D4832	(2016; E 2018) Standard Test Method for Preparation and Testing of Controlled Low Strength Material (CLSM) Test Cylinders
ASTM C260/C260M	(2024) Standard Specification for Air-Entraining Admixtures for Concrete
ASTM C150/C150M	(2024) Standard Specification for Portland Cement
ASTM C33/C33M	(2024a) Standard Specification for Concrete Aggregates
ASTM C39/C39M	(2024) Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
ASTM C94/C94M	(2024d) Standard Specification for Ready-Mixed Concrete

ASTM C494/C494M	(2024) Standard Specification for Chemical Admixtures for Concrete
ASTM C685/C685M	(2024) Standard Specification for Concrete Made by Volumetric Batching and Continuous Mixing
ASTM C940	(2022) Standard Test Method for Expansion and Bleeding of Freshly Mixed Grouts for Preplaced-Aggregate Concrete in the Laboratory

AMERICAN CONCRETE INSTITUTE (ACI)

ACI SP-150	(1994) Controlled Low-Strength Materials

1.8 QUALITY ASSURANCE

a. Manufacturer: Flowable fill shall be manufactured by a ready-mix concrete producer with a minimum of 1 year experience in the production of similar products.

b. Materials: For each type of material required for the work of this Section, provide primary materials that are the products of one manufacturer. If not otherwise specified here, materials shall comply with recommendations of ACI SP-150, “Controlled Low Strength Materials.”

c. Pre-Approval Procedures: The use of flowable fill during any part of the project shall be restricted to those incidences where, due to field conditions, the Contractor has made the Resident Engineer aware of the conditions for which he recommends the use of the flowable, and the Resident Engineer has confirmed those conditions and approved the use of the flowable fill, in advance. During the submittal process, the contractor shall prepare and submit various flowable fill mix designs corresponding to required conditions or if the contractor desires to use flowable fill due to economics. Approval for the strength of the flowable fill shall be obtained from the Resident Engineer when the contractor desires, or is required, to use flowable fill at specific location(s) within the project. Prior to commencement of field operations the contractor shall establish procedures to maintain optimum working conditions and to coordinate this work with related and adjacent work.

d. Sampling and Acceptance: Flowable fill shall be samples and testing in the field in conformance with either ASTM C94/C94M or ASTM C685/C685M. Samples for tests shall be taken for every (150 cubic yards) of material, or fraction thereof, for each day’s placement. Tests shall include temperature reading and four compressive strength cylinders. Compressive strength sampling and testing shall conform to ASTM D4832 with one specimen tested at 7 days, two at 28 days, and one held for each batch of four specimens. Sampling and testing shall be performed by a qualified, independent commercial testing laboratory. Test results should be submitted within 48 hours of completion of testing.

1.9 DELIVERY, STORAGE, AND HANDLING

Deliver and handle all products and equipment required, in strict compliance with manufacturer's recommendations. Protect from damage due to weather, excessive temperatures, and construction operations.

1.10 PROJECT CONDITIONS

Perform installation of flowable fill only when approved by the Resident Engineer, and when existing and forecasted weather conditions are within the limits established by the manufacturer of the materials and products used.

PART 2 PRODUCTS

2.1 MATERIALS

a. Provide flowable fill containing, at a minimum, cementitious materials and water. Cementitious materials shall be portland cement, pozzolanic materials, or other self-cementing materials, or combinations thereof, at the contractor's option , and following approval by the Resident Engineer. The flowable fill mix design may also contain, fine aggregate or filler, and/or chemical admixtures in any proportions such that the final product meets the strength, flow consistency and shrinkage requirements included in this specification, as approved by the Resident Engineer.

b. Portland Cement: ASTM C150/C150M, Meeting Connecticut State DOT standards.

c. Mixing Water: Fresh, clean, and potable.

d. Air-Entraining Admixture: ASTM C260/C260M.

e. Chemical Admixtures: ASTM C494/C494M.

f. Aggregate: ASTM C33/C33M.

2.2 FLOWABLE FILL MIXTURE

a. Mix design shall produce a consistency that will result in a flowable product at the time of placement which does not require manual means to move it into place.

b. Flowable fill shall have a minimum strength of (300 psi) according to ASTM C39/C39M at 28 days after placement.

c. Flowable fill shall have minimal subsidence and bleed water shrinkage. Evaporation of bleed water shall not result in shrinkage of more than 1/8 inch per foot) of flowable fill depth (for mixes containing high fly ash content). Measurement of a Final Bleeding shall be as measured in Section 10 of ASTM C940 "Standard Test Method for Expansion and Bleeding of Freshly Mixed Grouts for Preplaced-Aggregate Concrete in the Laboratory.

d. Flowable fill shall have a unit weight of (115 – 145 lbs/feet3) measured at the point of placement after a 60 minute ready-mix truck ride. In the absence of strength data the cementitious content shall be a maximum of (150 lbs/cy).

e. Flowable fill shall have an in-place yield of a maximum of 110% of design yield for removable types at 1 year.

f. Provide equipment as recommended by the Manufacturer and comply with manufacturer’s recommendations for the addition of additives, whether at the production plant or prior to placement at the site.

PART 3 EXECUTION

3.1 EXAMINATION

Examine conditions of substrates and other conditions under which work is to be performed and notify Resident Engineer, in writing, of circumstances detrimental to the proper completion of the work. Do not proceed until unsatisfactory conditions are corrected.

3.2 APPLICATION OF FLOWABLE FILL

Secure tanks, pipes and other members to be encased in flowable fill. Insure that there are no exposed metallic pipes, conduits, or other items that will be in contact with the flowable fill after placement. If so, replace with non-metallic materials or apply manufacturers recommended coating to protect metallic objects before placing the flowable fill. Replacement or protection of metallic objects is subject to the approval of the Resident Engineer.

3.3 PROTECTION AND CURING

Protect exposed surfaces of flowable fill from premature drying, wash by rain or running water, wind, mechanical injury, and excessively hot or cold temperature. Curing method shall be subject to approval by Resident Engineer.

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CEMENT AND CONCRETE FOR EXTERIOR IMPROVEMENTS

PART 1 GENERAL

1.1 SUMMARY

a. Section Includes:

(1) Subbase for concrete pavements.

(2) Curbs, gutters, and combination curbs and gutters.

(3) Pedestrian Pavement: Walks, pedestrian crossings, wheelchair curb ramps, steps.

(4) Vehicular Pavement: Driveways and loading docks.

(5) Equipment Pads: Transformers and generator pads.

1.2 RELATED REQUIREMENTS

a. Field Testing: Section 01 45 29, TESTING LABORATORY SERVICES.

b. Step Nosings and Railings: Section 05 50 00, METAL FABRICATIONS.

c. Subgrade Preparation and Subbase Compaction: Section 31 20 00, EARTHWORK.

1.3 REFERENCES

Comply with references to extent specified in this section.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO M 147	Materials for Aggregate and Soil-Aggregate Subbase, Base and Surface Courses
AASHTO M 233	(1986; R 2023) Standard Specification for Boiled Linseed Oil Mixture for Treatment of Portland Cement Concrete

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 211.1	(1991; R 2009) Standard Practice for Selecting Proportions for Normal, Heavyweight and Mass Concrete
ACI 305R	(2020) Guide to Hot Weather Concreting
ACI 306R	(2016) Guide to Cold Weather Concreting

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B101.3	(2012) Historical Test Method for Wet DOCF of Common Hard Surface Floor Materials (Including Action and Limit Thresholds for the Suitable Assessment of the Measured Values)

ASTM INTERNATIONAL (ASTM)

ASTM A615/A615M	(2024) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A996/A996M	(2016) Standard Specification for Rail-Steel and Axle-Steel Deformed Bars for Concrete Reinforcement
ASTM A1064/A1064M	(2024) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
ASTM C33/C33M	(2024a) Standard Specification for Concrete Aggregates
ASTM C94/C94M	(2025) Standard Specification for Ready-Mixed Concrete
ASTM C143/C143M	(2020) Standard Test Method for Slump of Hydraulic-Cement Concrete
ASTM C150/C150M	(2024) Standard Specification for Portland Cement
ASTM C171	(2020) Standard Specification for Sheet Materials for Curing Concrete
ASTM C260/C260M	(2024) Standard Specification for Air-Entraining Admixtures for Concrete
ASTM C309	(2019) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete
ASTM C494/C494M	(2024) Standard Specification for Chemical Admixtures for Concrete
ASTM C618	(2023; E 2023) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
ASTM C979/C979M	(2024) Standard Specification for Pigments for Integrally Colored Concrete
ASTM C989/C989M	(2024) Standard Specification for Slag Cement for Use in Concrete and Mortars
ASTM C1240	(2020) Standard Specification for Silica

	Fume Used in Cementitious Mixtures
ASTM D1751	(2018) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)
ASTM D5893/D5893M	(2016) Standard Specification for Cold Applied, Single Component, Chemically Curing Silicone Joint Sealant for Portland Cement Concrete Pavements
ASTM D6690	(2021) Standard Specification for Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavements

1.4 PREINSTALLATION MEETINGS

a. Conduct preinstallation meeting at project site minimum 30 days before beginning Work of this section.

(1) Required Participants:

(a) Contracting Officer.

(d) Contractor.

(e) Installer.

(f) Other installers responsible for adjacent and intersecting work, including excavation, plantings, traffic markings, and .

(2) Meeting Agenda: Distribute agenda to participants minimum 3 days before meeting.

(a) Installation schedule.

(b) Installation sequence.

(c) Preparatory work.

(d) Protection before, during, and after installation.

(e) Installation.

(f) Terminations.

(g) Transitions and connections to other work.

(h) Inspecting and testing.

(i) Other items affecting successful completion.

(3) Document and distribute preinstallation meeting minutes to participants to record decisions affecting installation.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G,

Proposed Hot And Cold Weather Concreting Methods; G,

Land Surveyor's Construction Staking Notes, Before Placing Concrete; G,

Installer Qualifications; G,

Land Surveyor; G,

Preconstruction Testing; G,

Preinstallation Meeting Minutes; G,

SD-02 Shop Drawings

Submittal Drawings; G,

SD-03 Product Data

Manufacturer's Literature And Data; G,

POROUS GRASS PAVERS; G

SOLAR REFLECTIVE CONCRETE ADMIXTURE FOR URBAN HEAT ISLAND EFFECT REDUCTION; G

SD-05 Design Data

Concrete Mix Design; G,

Select Subbase Job-Mix Design; G,

SD-06 Test Reports

Test Reports; G,

SD-07 Certificates

Certificates; G,

1.6 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 00 SUBMITTAL PROCEDURES

b. Submittal Drawings:

(1) Show size, configuration, and fabrication and installation details.

(2) Show reinforcing.

(3) Include jointing plan for concrete pavements, curbs and gutters.

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Installation instructions.

e. Test reports: Certify products comply with specifications.

(1) Concrete materials.

(2) Select subbase materials.

(3) Field test reports.

f. Certificates: Certify products comply with specifications.

(1) Expansion joint filler.

(2) Reinforcement.

(3) Curing materials.

(4) Concrete protective coating.

g. Qualifications: Substantiate qualifications comply with specifications.

(1) Installer with project experience list.

(2) Land surveyor.

h. Concrete mix design.

i. Select subbase job-mix design.

j. Proposed hot and cold weather concreting methods.

k. Land surveyor's construction staking notes, before placing concrete.

l. Identify discrepancies between field conditions and Drawings.

1.7 QUALITY ASSURANCE

a. Installer Qualifications:

(1) Regularly installs specified products.

(2) Installed specified products with satisfactory service on five similar installations.

Project Experience List: Provide contact names and addresses for completed projects.

b. Land Surveyor: Professional land surveyor or engineer registered to provide land surveys in jurisdiction where project is located.

c. Preconstruction Testing:

(1) Engage independent testing laboratory to perform tests and submit reports.

Deliver samples to laboratory in number and quantity required for testing.

(2) Concrete mix design.

(3) Select subbase job-mix design. Report the following:

(a) Material sources.

(b) Gradation.

(c) Plasticity index.

(d) Liquid limit.

(e) Laboratory compaction curves indicating maximum density at optimum moisture content.

1.8 DELIVERY

a. Deliver steel reinforcement to prevent damage.

b. Before installation, return or dispose of distorted or damaged steel reinforcement.

c. Bulk Products: Deliver bulk products away from buildings, utilities, pavement, and existing turf and planted areas. Maintain dry bulk product storage away from contaminants.

1.9 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight facility.

b. Protect products from damage during handling and construction operations.

1.10 FIELD CONDITIONS

a. Hot Weather Concreting Procedures: ACI 305R.

b. Cold Weather Concreting Procedures: ACI 306R.

(1) Use non-corrosive, non-chloride accelerator admixture.

(2) Do not use calcium chloride, thiocyanates or admixtures containing more than 0.05 % chloride ions.

1.11 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 CONCRETE MATERIALS

a. Portland Cement: ASTM C150/C150M, Type I or II.

b. Pozzolans:

(1) Fly Ash: ASTM C618, Class C or F including supplementary optional physical requirements.

(2) Slag: ASTM C989/C989M; Grade 120.

(3) Silica Fume: ASTM C1240.

c. Coarse Aggregate: ASTM C33/C33M; size 3/4 inch.

d. Fine Aggregate: ASTM C33/C33M.

e. Mixing Water: Fresh, clean, and potable.

f. Air-Entraining Admixture: ASTM C260/C260M.

g. Chemical Admixtures: ASTM C494/C494M.

h. Reinforcing Steel: ASTM A615/A615M or ASTM A996/A996M, Grade 420 (60) deformed.

i. Welded Wire Fabric: ASTM A1064/A1064M, Grade 450 (65); sized as indicated.

j. Expansion Joint Filler: ASTM D1751.

k. Sheet Materials for Curing Concrete: ASTM C171.

l. Color Pigment: ASTM C979/C979M, colored and white powder pigments.

2.2 SELECT SUBBASE

a. Subbase: AASHTO M 147; Grade C or Grade D.

Select granular material composed of sand, sand-gravel, crushed stone, crushed or granulated slag, with or without soil binder, or combinations of these materials.

SUBBASE GRADING REQUIREMENTS

Sieve Size		Percentage Passing by Mass					
		Grades					
(mm)	(in)	A	B	C	D	E	F
50	2	100	100				
25	1		75-95	100	100	100	100
9.5	3/8	30-65	40-75	50-85	60-100		

SUBBASE GRADING REQUIREMENTS							
Sieve Size		Percentage Passing by Mass					
		Grades					
4.47	No. 4	25-55	30-60	35-65	50-85	55-100	70-100
2.00	No. 10	15-40	20-45	25-50	40-70	40-100	55-100
0.425	No. 40	8-20	15-30	15-30	25-45	20-50	30-70
0.075	No. 200	2-8	5-20	5-15	5-20	6-20	8-25

b. Other Acceptable Gradations: Materials within three to five percent, plus or minus, of specified gradation, or as recommended by the geotechnical engineer and approved by the Contracting Officer.

2.3 FORMS

a. Forms: Wood, plywood, metal, or other materials, approved by Contracting Officer, of grade or type suitable to obtain type of finish specified.

(1) Plywood: Exterior grade, free of defects and patches on contact surface.

(2) Lumber: Sound, grade-marked, S4S stress graded softwood, minimum 2 inches thick, free from warp, twist, loose knots, splits, or other defects.

(3) Form Coating: As recommended by Architect/Engineer.

b. Provide forms suitable in cross-section, depth, and strength to resist springing during depositing and consolidating concrete.

(1) Do not use forms varying from straight line more than 1/8 inch in 10 feet, horizontally and vertically.

c. Provide flexible or curved forms for forming radii.

2.4 CONCRETE CURING MATERIALS

a. Concrete curing materials, conform to one of the following:

(1) Burlap: Minimum 7 ounces/sq. yd. dry.

(2) Sheet Materials for Curing Concrete: ASTM C171.

(3) Curing Compound: ASTM C309, Type 2; liquid membrane forming type, without paraffin or petroleum.

2.5 CONCRETE MIXES

a. Design concrete mixes according to ASTM C94/C94M, Option C.

b. Concrete Type: Air-entrained. See Table I.

TABLE I - CONCRETE TYPES

Concrete Type	Minimum 28 Day Compressive Strength f'c psi	Non-Air-Entrained		Air-Entrained	
		Min. Cement lbs./cu. yd.	Max. Water Cement Ratio	Min. Cement lbs./cu. yd.	Max. Water Cement Ratio
A	5000 1,3	630	0.45	650	0.40
B	4000 1,3	550	0.55	570	0.50
C	3000 1,3	470	0.65	490	0.55
D	3000 1,2	500	*	520	*

Footnotes:
1. If trial mixes are used, achieve compressive strength 1,200 psi in excess of f'c. For concrete strengths greater than 5,000 psi, achieve compressive strength 1,400 psi in excess of f'c.
2. For Concrete Exposed to High Sulfate Content Soils: Maximum water cement ratio is 0.44.
3. Laboratory Determined according to ACI 211.1 for normal weight concrete.

c. Maximum Slump: ASTM C143/C143M. See Table II.

TABLE II - MAXIMUM SLUMP

APPLICATION	MAXIMUM SLUMP
Curb & Gutter	3 inches
Pedestrian Pavement	3 inches
Vehicular Pavement	2 inches Machine Finished 4 inches Hand Finished
Equipment Pad	3 to 4 inches

2.6 ACCESSORIES

a. Equipment and Tools: Obtain Contracting Officer's, approval of equipment and tools needed for handling materials and performing work before work begins.

b. Maintain equipment and tools in satisfactory working condition.

c. Sealants:

(1) Concrete Paving Expansion Joints: ASTM D5893/D5893M, Type SL, single component, self-leveling, silicone joint sealant.

(2) Concrete Paving Joints: ASTM D6690, Type IV, hot-applied, single component joint sealant.

d. Concrete Protective Coating: AASHTO M 233 linseed oil mixture.

e. Solar Reflective Concrete Admixture for Urban Heat Island Effect Reduction

(1) The Solar Reflective Concrete Admixture shall be used in all new exterior concrete that is part of new paving work. The product shall provide solar reflective methods that will stay cool longer than standard concrete mixtures and have lower maximum temperatures to help reduce urban heat island effect and be in compliance with LEED requirements. The selected solar reflectance color shall have a minimum Solar Reflectance Index (SRI) of 63 and a minimum Solar Reflectance (SR) of 0.525. The selected material shall be weather resistant, UV stable, lightfast and alkali resistant. It shall not contain any materials that initiate, accelerate or promote the corrosion of steel, coated metal, plastic, or any concrete reinforcements. All pigments shall conform with ASTM C979.

2.7 POROUS GRASS PAVERS

a. Porous grass pavers shall be made of 100% recycled polyethylene. Runoff coefficient of 0.15 for the grass paver at 2.5 inches per hour of rain. The porous grass pavers shall be rated to withstand H-20 load rates at a minimum. Engineering base material will be made of an 8-inch base of homogenous mixture consisting of open graded crushed aggregate having an AASHTO #5 or similar material blended with pulverized topsoil and a void component containing air and/or water. This aggregate portion shall have a particle range from 0.375 to 1.0 inch (9.5 mm to 25 mm) with a D50 if 0.5 inches (13mm). The percentage void-space of the aggregate portion when compacted shall be at least 30%. The pulverized topsoil portion shall be 33% of the total volume and be added and blended to produce a homogenous mixture prior to placement. Additional details shall be in compliance with manufacturer’s specifications.

2.8 SOLAR REFLECTIVE CONCRETE ADMIXTURE FOR URBAN HEAT ISLAND EFFECT REDUCTION

a. The Solar Reflective Concrete Admixture shall be used for all exterior new paving and provide solar reflective methods that will stay cool longer than standard concrete mixtures and have lower maximum temperatures that will help reduce urban heat island effect and be in compliance with LEED requirements. The selected solar reflectance color shall have a minimum Solar Reflectance Index (SRI) of 63 and a minimum Solar Reflectance (SR) of 0.525. The selected material shall be weather resistant, UV stable, lightfast and alkali resistant. It shall not contain any materials that initiate, accelerate or promote the corrosion of steel, coated metal, plastic, or any concrete reinforcements. All pigments shall conform with ASTM C979.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

b. Protect existing construction and completed work from damage.

c. Prepare, construct, and finish subgrade. See Section 31 20 00,

EARTHWORK.

d. Maintain subgrade in smooth, compacted condition, in conformance with the required section and established grade until the succeeding operation has been accomplished.

3.2 SELECT SUBBASE

a. Placing:

(1) Place subbase material on prepared subgrade in uniform layer to required contour and grades, and to maximum 8 inches loose depth.

(2) When required compacted thickness exceeds 6 inches, place subbase material in equal thickness layers.

(3) When subbase elevation is 1/2 inch or more below required grade, excavate subbase minimum 3 inches deep. Place and compact subbase to required grade.

b. Compaction:

(1) Perform compaction with approved hand or mechanical equipment well suited to the material being compacted.

(2) Maintain subbase at optimum moisture content for compaction.

(3) Compact each subbase layer to minimum 95 percent or 100 percent of maximum density as specified in Section 31 20 00, EARTHWORK.

c. Subbase Tolerances:

(1) Variation from Indicated Grade: Maximum 3/8 inch.

(2) Variation from Indicated Thickness: Maximum 1/2 inch.

d. Protection:

(1) Protect subbase from damage until concrete is placed.

(2) Reconstruct damaged subbase before placing concrete.

3.3 SETTING FORMS

a. Form Substrate:

(1) Compact form substrate to uniformly support forms along entire length.

(2) Correct substrate imperfections and variations by cutting, filling, and compacting.

b. Form Setting:

(1) Set forms to indicated line and grade with tight joints. Rigidly brace forms preventing movement.

(2) Remove forms when removal will not damage concrete and when required for finishing.

(3) Clean and oil forms before each use.

(4) Correct forms, when required, immediately before placing concrete.

c. Land Surveyor: Establish control, alignment, and grade for forms .

(1) Notify Contracting Officer immediately when discrepancies exist between field conditions and drawings.

(2) Correct discrepancies greater than 1 inch before placing concrete.

d. Form Tolerances:

(1) Variation from Indicated Line: Maximum 1/4 inch.

(2) Variation from Indicated Grade: Maximum 1/8 inchin 10 feet.

3.4 PLACING REINFORCEMENT

a. Keep reinforcement clean from contamination preventing concrete bond.

b. Install reinforcement shown on drawings.

c. Support and securely tie reinforcing steel to prevent displacement during concrete placement.

d. Obtain Contracting Officer's reinforcement placement approval before placing concrete.

3.5 JOINTS - GENERAL

a. Place joints, where shown on approved submittal Drawings.

(1) Conform to details shown.

(2) Install joints perpendicular to finished concrete surface.

b. Make joints straight and continuous from edge to edge of pavement.

3.6 CONSTRUCTION JOINTS

a. Locate longitudinal and transverse construction joints between slabs of vehicular pavement as shown on approved submittal Drawings.

b. Place transverse construction joints of type shown, where indicated, and whenever concrete placement is suspended for more than 30 minutes.

c. Provide butt-type joint with dowels in curb and gutter at planned joint locations.

d. Provide keyed joints with tie bars when joint occurs in middle third of planned curb and gutter joint interval.

3.7 CONTRACTION JOINTS

a. Tool or cut joints to width, depth, and radius edge shown on drawings using grooving tool, jointer, or saw.

b. Construct joints in curbs and gutters by inserting 1/8 inch steel plates conforming to curb and gutter cross sections.

Keep plates in place until concrete can hold its shape.

c. Finish joint edges with edging tool.

d. Score pedestrian pavement with grooving tool or jointer.

3.8 EXPANSION JOINTS

a. Form expansion joints with expansion joint filler of thickness shown on drawings.

(1) Locate joints around perimeter of structures and features abutting site work concrete.

(2) Create complete, uniform separation between structure and site work concrete.

b. Extend expansion joint material full depth of concrete with top edge of joint filler below finished concrete surface where sealant is indicated on Drawings.

c. Cut and shape material matching cross section.

d. Anchor with approved devices to prevent displacing during placing and finishing operations.

e. Round joint edges with edging tool.

3.9 PLACING CONCRETE - GENERAL

a. Preparation before Placing Concrete:

(1) Obtain Contracting Officer's approval.

(2) Remove debris and other foreign material.

(3) Uniformly moisten substrate, without standing water.

b. Convey concrete from mixer to final location without segregation or loss of ingredients. Deposit concrete to minimize handling.

c. During placement, consolidate concrete by spading or vibrating to minimize voids, honeycomb, and rock pockets.

(1) Vibrate concrete against forms and along joints.

(2) Avoid excess vibration and handling causing segregation.

d. Place concrete continuously between joints without bulkheads.

e. Install construction joint in concrete placement suspended for more than 30 minutes.

f. Replace concrete with cracks, chips, bird baths, and other defects to nearest joints, approved by Contracting Officer.

3.10 PLACING CONCRETE FOR CURB AND GUTTER, PEDESTRIAN PAVEMENT, AND EQUIPMENT PADS

a. Place concrete in one layer conforming to cross section shown on Drawings after consolidating and finishing.

b. Deposit concrete near joints without disturbing joints. Do not place concrete directly onto joint assemblies.

c. Strike concrete surface to proper section ready for consolidation.

d. Consolidate concrete by tamping and spading or with approved mechanical finishing equipment.

e. Finish concrete surface with wood or metal float.

f. Construct concrete pads and pavements with sufficient slope to drain, preventing standing water.

3.11 PLACING CONCRETE FOR VEHICULAR PAVEMENT

a. Deposit concrete as close as possible to its final position.

b. Place concrete continuously between construction joints without cold joints.

c. Strike and consolidate concrete with finishing machine, vibrating screed, or by hand-finishing.

d. Finish concrete surface to elevation and crown shown on drawings.

e. Deposit concrete near joints without disturbing joints. Do not place concrete directly onto joint assemblies.

f. Obtain Contracting Officer's approval before placing adjacent lanes.

g. Curb-Forming Machines: Curb-forming machines for constructing integral curbs curbs and gutter will be approved based on trial use on the project. When equipment produces unsatisfactory results, discontinue use of the equipment at any time during construction and accomplish work by hand method construction. Remove unsatisfactory work and reconstruct full length between regularly scheduled joints. Dispose of removed portions off the project site.

3.12 FORM REMOVAL

a. Keep forms in place minimum 12 hours after concrete placement. Remove forms without damaging concrete.

b. Do not use bars or heavy tools against concrete to remove forms. Repair damage concrete found after form removal.

3.13 CONCRETE FINISHING - GENERAL

a. Follow operation sequence below, unless otherwise indicated on Drawings:

(1) Consolidating, floating, striking, troweling, texturing, and joint edging.

b. Use edging tool with 1/4 inch radius, unless otherwise shown on Drawings.

c. Keep finishing equipment and tools clean and suitable for use.

3.14 CONCRETE FINISHING - PEDESTRIAN PAVEMENT

a. Walks, Wheelchair Curb Ramps,:

(1) Finish concrete surfaces with metal float, troweled smooth, and finished with a broom moistened with clear water.

(2) Finish slab edges and formed transverse joints with edger.

(3) Broom surfaces transverse to traffic direction.

(a) Use brooming to eliminate flat surface produced by edger.

(b) Produce uniform corrugations, maximum 1/16 inch deep profile.

(4) Provide surface uniform in color and free of surface blemishes, form marks, and tool marks.

(5) Paving Tolerances:

(a) Variation from Indicated Plane: Maximum 3/16 inch in 10 feet.

(b) Variation from Indicated Thickness: Maximum 1/4 inch.

(6) Replace paving within joint boundary when paving exceeds specified tolerances.

b. Step Treads, Risers and Sidewalls: Finish as specified for pedestrian pavement, except as follows:

(1) Remove riser forms sequentially, starting with top riser.

(2) Rub riser face with wood or concrete rubbing block and water. Remove blemishes, form marks, and tool marks. Use outside edger to round nosing; use inside edger to finish bottom of riser.

(3) Apply uniform brush finish to treads, risers, and sidewall.

(a) Apply stiff brush finish to treads to provide slip resistant surface complying with ANSI B101.3.

(4) Step Tolerance:

(a) Variation from Indicated Plane: Maximum 3/16 inch in 10 feet.

3.15 CONCRETE FINISHING - VEHICULAR PAVEMENT

a. Align finish surfaces where new and existing pavements abut.

b. Longitudinally float pavement surface to profile and grade indicated on drawings.

c. Straighten surface removing irregularities and maintaining specified

tolerances while concrete is plastic.

d. Finish pavement edges and joints with edging tool.

e. Broom finish concrete surface after bleed water dissipates and before concrete hardens.

(1) Broom surface transverse to traffic direction.

(a) Use brooming to eliminate flat surface produced by edger.

(b) Produce uniform corrugations, maximum 1/8 inch deep profile.

f. Pavement Tolerances:

(1) Variation from Indicated Plane: Maximum 1/4 inchin 10 feet tested parallel and perpendicular to traffic direction at maximum 5 feet intervals.

(2) Variation from Indicated Thickness: Maximum 1/4 inch.

g. Replace paving within joint boundary when paving exceeds specified tolerances.

3.16 CONCRETE FINISHING - CURBS AND GUTTERS

a. Round edges of gutter and top of curb with edging tool.

b. Gutter and Curb Top:

(1) Float surfaces and finish with smooth wood or metal float until true to grade and section and uniform color.

(2) Finish surfaces, while still plastic, longitudinally with bristle brush.

c. Curb Face:

(1) Remove curb form and immediately rub curb face with wood or concrete rubbing block removing blemishes, form marks, and tool marks and providing uniform color.

(2) Brush curb face, while still plastic, matching gutter and curb top.

d. Curb and Gutter Tolerances: Except at grade changes or curves.

(1) Variation from Indicated Plane and Grade:

(a) Gutter: Maximum 1/8 inch in 10 feet.

(b) Curb Top and Face: Maximum 1/4 inch in 10 feet.

e. Replace curbs and gutters within joint boundary when curbs and gutters exceed specified tolerances.

f. Correct depressions causing standing water.

3.17 CONCRETE FINISHING - EQUIPMENT PADS

a. Strike pad surface to elevation shown on Drawings.

b. Provide smooth, dense float finish, free from depressions or irregularities.

c. Finish pad edges with edger.

d. After removing forms, rub pad edge faces with wood or concrete rubbing block, removing blemishes, form marks, and tool marks and providing uniform color.

e. Pad Tolerances:

(1) Variation from Indicated Plane: Maximum 1/8 inchin 10 feet.

(2) Variation from Indicated Elevation: Maximum 1/4 inch.

(3) Variation from Indicated Thickness: Maximum 1/4 inch.

f. Replace pads when pads exceed specified tolerances.

3.18 CONCRETE CURING

a. Concrete Protection:

(1) Protect unhardened concrete from rain and flowing water.

(2) Provide sufficient curing and protection materials available and ready for use before concrete placement begins.

(3) Protect concrete to prevent pavement cracking from ambient temperature changes during curing period.

(a) Replace pavement damaged by curing method allowing concrete cracking.

(b) Employ another curing method as directed by Contracting Officer.

b. Cure concrete for minimum seven days by one of the following methods appropriate to weather conditions preventing moisture loss and rapid temperature change:

(1) Burlap Mat: Provide minimum two layers kept saturated with water during curing period. Overlap Mats at least 6 inches.

(2) Sheet Materials:

(a) Wet exposed concrete surface with fine water spray and cover with sheet materials.

(b) Overlap sheets minimum 12 inches.

(c) Securely anchor sheet materials preventing displacement.

(3) Curing Compound:

(a) Protect joints indicated to receive sealants preventing

contamination from curing compound.

(b) Insert moistened paper or fiber rope into joint or cover joint with waterproof paper.

(c) Apply curing compound before concrete dries.

(d) Apply curing compound in two coats at right angles to each other.

(e) Application Rate: Maximum 200 sq. ft./gallon, both coats.

(f) Immediately reapply curing compound to surfaces damaged during curing period.

3.19 CONCRETE PROTECTIVE COATING

a. Apply protective coating of linseed oil mixture to exposed-to-view concrete surfaces, drainage structures, and features that project through, into, or against concrete exterior improvements to protect the concrete against deicing materials.

b. Complete backfilling and curing operation before applying protective coating.

c. Dry and thoroughly clean concrete before each application.

d. Apply two coats, with maximum coverage of 50 sq. yds./gal.; first coat, and maximum 70 sq. yds./gal.; second coat, except apply commercially prepared mixture according to manufacturer's instructions.

e. Protect coated surfaces from vehicular and pedestrian traffic until dry.

f. Do not heat protective coating, and do not expose protective coating to open flame, sparks, or fire adjacent to open containers or applicators. Do not apply material at temperatures lower than 50 degrees F.

3.20 POROUS GRASS PAVERS

a. Per manufacturer’s installation guidelines.

3.21 SOLAR REFLECTIVE CONCRETE ADMIXTURE FOR URBAN HEAT ISLAND EFFECT REDUCTION

a. Per manufacturer’s installation guidelines.

3.22 FIELD QUALITY CONTROL

a. Field Tests: Performed by testing laboratory specified in Section 01 45 29, TESTING LABORATORY SERVICES.

(1) Compaction.

(a) Pavement subgrade.

(b) Curb, gutter, and sidewalk.

(2) Concrete:

(a) Delivery samples.

(b) Field samples.

(3) Slip Resistance: Steps and pedestrian paving.

3.23 CLEANING

After completing curing:

a. Remove burlap and sheet curing materials.

b. Sweep concrete clean, removing foreign matter from the joints.

c. Seal joints as specified.

3.24 PROTECTION

a. Protect exterior improvements from traffic and construction operations.

Prohibit traffic on paving for minimum seven days after placement, or longer as directed by the Contracting Officer.

b. Remove protective materials immediately before acceptance.

c. Repair damage.

Replace concrete containing excessive cracking, fractures, spalling, and other defects within joint boundary, when directed by Contracting Officer, and at no additional cost to the Government.

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ASPHALT PAVING

PART 1 GENERAL

1.1 DESCRIPTION

This work shall cover the composition, mixing, construction upon the prepared subgrade, and the protection of hot asphalt concrete pavement. The hot asphalt concrete pavement shall consist of an aggregate or asphalt base course and asphalt surface course constructed in conformity with the lines, grades, thickness, and cross sections as shown. Each course shall be constructed to the depth, section, or elevation required by the drawings and shall be rolled, finished, and approved before the placement of the next course.

1.2 RELATED WORK

a. Laboratory and field testing requirements: Section 01 45 29, TESTING LABORATORY SERVICES.

b. Subgrade Preparation: Paragraph 3.3 and Section 31 20 00, EARTH MOVING.

c. Pavement Markings: Section 32 17 23, PAVEMENT MARKINGS.

1.3 INSPECTION OF PLANT AND EQUIPMENT

The Resident Engineer shall have access at all times to all parts of the material producing plants for checking the mixing operations and materials and the adequacy of the equipment in use.

1.4 ALIGNMENT AND GRADE CONTROL

The Contractor’s Registered Professional Land Surveyor shall establish and control the pavement (aggregate or asphalt base course and asphalt surface course) alignments, grades, elevations, and cross sections as shown on the Drawings.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Job-Mix Formula; G,

Material Safety Data Sheets; G,

SD-05 Design Data

State Highway Department Specifications; G,

SD-06 Test Reports

Aggregate Base Course; G,

Asphalt Base/Surface Course; G,

SD-07 Certificates

Certifications; G,

1.6 SUBMITTAL INSTRUCTIONS

a. In accordance with Section 01 33 00 SUBMITTAL PROCEDURES, furnish the following:

b. Data and Test Reports:

(1) Aggregate Base Course: Sources, gradation, liquid limit, plasticity index, percentage of wear, and other tests required by State Highway Department.

(2) Asphalt Base/Surface Course: Aggregate source, gradation, soundness loss, percentage of wear, and other tests required by State Highway Department.

(3) Job-mix formula.

c. Certifications:

(1) Asphalt prime and tack coat material certificate of conformance to State Highway Department requirements.

(2) Asphalt cement certificate of conformance to State Highway Department requirements.

(3) Job-mix certification - Submit plant mix certification that mix equals or exceeds the State Highway Specification.

d. One copy of State Highway Department Specifications.

e. Provide MSDS (Material Safety Data Sheets) for all chemicals used on ground.

PART 2 PRODUCTS

2.1 GENERAL

Aggregate base, Asphaltic base, and asphalt concrete materials shall conform to the requirements of the following and other appropriate sections of the latest version of the State Highway Material Specifications, including amendments, addenda and errata. Where the term "Engineer" or "Commission" is referenced in the State Highway Specifications, it shall mean the Government Resident Engineer or Government Contracting Officer.

2.2 AGGREGATES

a. Provide aggregates consisting of crushed stone, gravel, sand, or other

sound, durable mineral materials processed and blended, and naturally combined.

b. Subbase aggregate (where required) maximum size: 1-1/2 inches.

c. Base aggregate maximum size:

(1) Base course over 6 inches thick: 1-1/2 inches;

(2) Other base courses: 3/4 inches.

d. Asphaltic base course:

(1) Maximum particle size not to exceed 1 inch.

(2) Where conflicts arise between this specification and the requirements in the latest version of the State Highway Specifications, the State Specifications shall control.

e. Aggregates for asphaltic concrete paving: Provide a mixture of sand, mineral aggregate, and liquid asphalt mixed in such proportions that the percentage by weight will be within:

Sieve Sizes	Percentage Passing
3/4"	100
3/8"	67 to 85
1/4"	50 to 65
No. 8 mesh	37 to 50
No. 30 mesh	15 to 25
No. 200 mesh	3 to 8

plus 50/60 penetration liquid asphalt at 5 percent to 6-1/2 percent of the combined dry aggregates.

2.3 ASPHALTS

Comply with provisions of Asphalt Institute Specification SS2:

a. Asphalt cement: Penetration grade 50/60

b. Prime coat: Cut-back type, grade MC-250

c. Tack coat: Uniformly emulsified, grade SS-1H

2.4 SEALER

a. Provide a sealer consisting of suitable fibrated chemical type asphalt base binders and fillers having a container consistency suitable for troweling after thorough stirring, and containing no clay or other deleterious substance.

b. Where conflicts arise between this specification and the requirements in the latest version of the State Highway Specifications, the State Specifications shall control.

PART 3 EXECUTION

3.1 GENERAL

The Asphalt Concrete Paving equipment, weather limitations, job-mix formula, mixing, construction methods, compaction, finishing, tolerance, and protection shall conform to the requirements of the appropriate sections of the State Highway Specifications for the type of material specified.

3.2 MIXING ASPHALTIC CONCRETE MATERIALS

Provide hot plant-mixed asphaltic concrete paving materials.

a. Temperature leaving the plant: 290 degrees F minimum, 320 degrees F maximum.

b. Temperature at time of placing: 280 degrees F minimum.

3.3 SUBGRADE

a. Shape to line and grade and compact with self-propelled rollers.

b. All depressions that develop under rolling shall be filled with acceptable material and the area re-rolled.

c. Soft areas shall be removed and filled with acceptable materials and the area re-rolled.

d. Should the subgrade become rutted or displaced prior to the placing of the subbase, it shall be reworked to bring to line and grade.

e. Proof-roll the subgrade with maximum 50 ton gross weight dump truck as directed by Government Resident Engineer or Government Contracting Officer. If pumping, pushing, or other movement is observed, rework the area to provide a stable and compacted subgrade.

3.4 BASE COURSES

a. Subbase (when required)

(1) Spread and compact to the thickness shown on the drawings.

(2) Rolling shall begin at the sides and continue toward the center and shall continue until there is no movement ahead of the roller.

(3) After completion of the subbase rolling there shall be no hauling over the subbase other than the delivery of material for the top course.

b. Base

(1) Spread and compact to the thickness shown on the drawings.

(2) Rolling shall begin at the sides and continue toward the center and shall continue until there is no movement ahead of the roller.

(3) After completion of the base rolling there shall be no hauling over the base other than the delivery of material for the top

course.

c. Thickness tolerance: Provide the compacted thicknesses shown on the Drawings within a tolerance of minus 0.0 inches to plus 0.5 inches.

d. Smoothness tolerance: Provide the lines and grades shown on the Drawings within a tolerance of 3/16 inch in 10 feet.

e. Moisture content: Use only the amount of moisture needed to achieve the specified compaction.

3.5 PLACEMENT OF ASPHALTIC CONCRETE PAVING

a. Remove all loose materials from the compacted base.

b. Apply the specified prime coat, and tack coat where required, and allow to dry in accordance with the manufacturer’s recommendations as approved by the Architect or Engineer.

c. Receipt of asphaltic concrete materials:

(1) Do not accept material unless it is covered with a tarpaulin until unloaded, and unless the material has a temperature of not less than 280 degrees F.

(2) Do not commence placement of asphaltic concrete materials when the atmospheric temperature is below 50 degrees F, not during fog, rain, or other unsuitable conditions.

d. Spreading:

(1) Spread material in a manner that requires the least handling.

(2) Where thickness of finished paving will be 3 inches or less, spread in one layer.

e. Rolling:

(1) After the material has been spread to the proper depth, roll until the surface is hard, smooth, unyielding, and true to the thickness and elevations shown own the drawings.

(2) Roll in at least two directions until no roller marks are visible.

(3) Finished paving smoothness tolerance:

(a) No depressions which will retain standing water.

(b) No deviation greater than 1/8 in 6 feet.

3.6 APPLICATION OF SEAL COAT

a. Prepare the surfaces, mix the seal coat material, and apply in accordance with the manufacturer’s recommendations as approved by the Architect or Engineer.

b. Achieve a finished surface seal which, when dry and thoroughly set, is smooth, tough, resilient, of uniform black color, and free from coarse textured areas, lap marks, ridges, and other surface irregularities.

c. When sealing new asphalt paving wait an entire year to allow for the expansion and contraction of a year’s cycle of both warm and cool temperatures. This allows for the asphalt’s oils to properly cure and begin oxidation before applying a seal coat.

d. When seal coating in less than a year apply two coats, spray applied. This application method is preferred for less than a year application when there is still plenty of asphalt cement present for the seal coat to bond to.

e.

f.

3.7 PROTECTION

Protect the asphaltic concrete paved areas from traffic until the sealer is set and cured and does not pick up under foot or wheeled traffic.

3.8 FINAL CLEAN-UP

Remove all debris, rubbish, and excess material from the work area.

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CONCRETE PAVEMENT FOR FUEL STORAGE CONTAINMENT DIKES
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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 305R	(2020) Guide to Hot Weather Concreting
ACI 306R	(2016) Guide to Cold Weather Concreting

ASTM INTERNATIONAL (ASTM)

ASTM A615/A615M	(2024) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A775/A775M	(2022) Standard Specification for Epoxy-Coated Steel Reinforcing Bars
ASTM A996/A996M	(2016) Standard Specification for Rail-Steel and Axle-Steel Deformed Bars for Concrete Reinforcement
ASTM A1064/A1064M	(2024) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
ASTM C31/C31M	(2025a) Standard Practice for Making and Curing Concrete Test Specimens in the Field
ASTM C33/C33M	(2024a) Standard Specification for Concrete Aggregates
ASTM C39/C39M	(2024a) Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
ASTM C94/C94M	(2025) Standard Specification for Ready-Mixed Concrete
ASTM C150/C150M	(2024) Standard Specification for Portland Cement
ASTM C172/C172M	(2017) Standard Practice for Sampling Freshly Mixed Concrete
ASTM C260/C260M	(2024) Standard Specification for

	Air-Entraining Admixtures for Concrete
ASTM C309	(2019) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete
ASTM C494/C494M	(2024) Standard Specification for Chemical Admixtures for Concrete
ASTM C595/C595M	(2024) Standard Specification for Blended Hydraulic Cements
ASTM C618	(2023; E 2023) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
ASTM C989/C989M	(2024) Standard Specification for Slag Cement for Use in Concrete and Mortars
ASTM C1116/C1116M	(2023) Standard Specification for Fiber-Reinforced Concrete
ASTM C1240	(2020) Standard Specification for Silica Fume Used in Cementitious Mixtures
ASTM C1260	(2023) Standard Test Method for Potential Alkali Reactivity of Aggregates (Mortar-Bar Method)
ASTM C1567	(2025) Standard Test Method for Determining the Potential Alkali-Silica Reactivity of Combinations of Cementitious Materials and Aggregate (Accelerated Mortar-Bar Method)
ASTM D1751	(2018) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)
ASTM D1752	(2018) Standard Specification for Preformed Sponge Rubber, Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Aggregate Sources; G

SD-06 Test Reports

Independent Laboratory Test Results; G

Placing, Spreading And Vibrating; G.

SD-07 Certificates

Concrete Sampling and Testing; G.

Certified Test Results; G

1.3 CONCRETE SAMPLING AND TESTING

1.3.1 General

Provide certification of sampling and testing of all concrete and concrete materials, including design of concrete mixes, conforming to the requirements specified in ASTM C94/C94M Option A and submit to the Contracting Officer for approval. During actual concrete operations, no substitutions are allowed in the materials or proportions that were used in the mix design without additional testing unless specifically approved or directed by the Contracting Officer. In lieu of performing new concrete mix design studies, a concrete mix design from a current project at the military base may be used provided the required concrete strength is obtained and the materials proposed for use in this project are identical to those used in the concrete mix design. Perform quality control sampling and testing in accordance with Section 01 45 00 QUALITY CONTROL and as specified herein. The Government may perform verification tests as considered necessary.

1.3.2 Certification for Additional Pavement Materials

Prior to the use of materials not listed in ASTM C94/C94M, but listed in this section, submit Certified Test Results for each lot as directed by the Contracting Officer.

PART 2 PRODUCTS

2.1 SYSTEM REQUIREMENTS

2.1.1 General Requirements

Provide concrete and the equipment, workmanship, materials and quality control conforming to the applicable requirements of ASTM C94/C94M, except as otherwise specified herein. Use concrete composed of cement, supplementary cementitious materials (SCM), fine aggregate, coarse aggregate, water, and an air entraining mixture. State final mix proportions by weight and batch; cementitious materials by weight. Do not exceed a water-cementitious materials ratio (WCR) of 0.45. Maintain the air content of the concrete by volume at 6.0 percent plus or minus 1.0 percent. Do not exceed a concrete slump of 3 inches for fixed form paving. Do not increase the slump of transit-mixed concrete because of the inadequacy of mixing, discharge, or placing equipment.

2.1.2 Strength Requirements

Provide concrete with the following average compressive strength:
Containment Dikes and Basin 4000 psi at 28 days

2.2 MATERIALS

2.2.1 Aggregate

2.2.1.1 Gradation

Provide coarse and fine aggregate conforming to ASTM C33/C33M with a maximum nominal size of 1 inch.

2.2.1.2 Quality

Provide coarse and fine aggregate conforming to ASTM C33/C33M, Class 4.

2.2.1.3 Alkali-Silica Reactivity (ASR)

a. Evaluate and test fine and coarse aggregates to be used for alkali-aggregate reactivity. Evaluate fine and coarse aggregates separately, using ASTM C1260. Test results of the individual aggregates must show a measured expansion equal to or less than 0.08 percent after 28 days of immersion in a 1N NaOH solution. Should the test data indicate an expansion of greater than 0.08 percent, reject the aggregate(s) or perform additional testing as follows: utilize the Contractor's proposed low alkali portland cement, blended cement, and/or SCM in combination with each individual aggregate and test in accordance with ASTM C1567. Determine the quantity that will meet all the requirements of these specifications and that will lower the expansion equal to or less than 0.08 percent after 28 days of immersion in a 1N NaOH solution. Base the mixture proportioning on the highest percentage of SCM required to mitigate ASR.

b. If any of the above options does not lower the expansion to less than 0.08 percent after 28 days of immersion in a 1N NaOH solution, reject the aggregate(s) and submit new aggregate sources for retesting. Submit the results of testing to the Contracting Officer for evaluation and acceptance.

2.2.2 Admixtures

a. Provide air-entraining admixture conforming to ASTM C260/C260M.

b. Use accelerating admixture conforming to ASTM C494/C494M, Type C, only when cold weather protection is required and only when approved in writing. Do not use admixtures containing the chlorine ion.

c. Provide water-reducing or retarding admixtures conforming to ASTM C494/C494M, Type A, B, or D.

d. When using Silica Fume, provide a high-range water-reducing admixture (HRWRA) meeting the requirements of ASTM C494/C494M, Type F or G. Provide the HRWRA that is a synthesized, sulfonated complex polymer type and free from chlorides and alkalies. Add the HRWRA to the concrete as a single component at the batch plant. Add the admixture to the concrete mixture only when its use is approved or directed, and only when it has been used in mixture proportioning studies to arrive at approved mixture proportions. Submit certified copies of the independent laboratory test results required for compliance with ASTM C494/C494M.

2.2.3 Cementitious Materials

Provide cementitious materials of portland cement, blended cement, or only portland cement in combination with supplementary cementitious materials (SCM), and conforming to appropriate specifications listed below. New submittals are required when the cementitious materials sources or types change.

2.2.3.1 Portland Cement

Provide portland cement conforming to ASTM C150/C150M, Type I or II, low alkali . Low alkali cement is required if the proposed aggregates are found to have greater than 0.04 percent expansion when tested in accordance with paragraph: Alkali-Silica Reactivity above.

2.2.3.2 Blended Cement

Provide blended cement conforming to ASTM C595/C595M, Type IP or IS, including the optional requirement for mortar expansion and sulfate soundness. Provide statement in writing from the manufacturer that the amount of pozzolan in the finished cement will not vary more than plus or minus 5 mass percent of the finished cement from lot to lot or within a lot. No change is allowed in the percentage and type of mineral admixture used in the blend from that submitted for the aggregate evaluation and mixture proportioning.

2.2.4 Supplementary Cementitious Materials (SCM)

2.2.4.1 Fly Ash

Provide fly ash conforming to ASTM C618, Class F, including the optional requirements for uniformity and effectiveness in controlling Alkali-Silica reaction and not have a loss on ignition exceeding 3 percent. For use in mitigating Alkali-Silica Reactivity, provide a Calcium Oxide (CaO) content of less than 13 percent and a total equivalent alkali content less than 3 percent.

2.2.4.2 Raw or Calcined Natural Pozzolan

Provide raw or calcined natural pozzolan conforming to ASTM C618, Class N, including the optional requirements for uniformity and effectiveness in controlling Alkali-Silica reaction and not have a loss on ignition exceeding 3 percent. Provide a Calcium Oxide (CaO) content of less than 13 percent and a total equivalent alkali content less than 3 percent for use in mitigating Alkali-Silica reactivity.

2.2.4.3 Ground Granulated Blast Furnace Slag (GGBFS)

Provide Ground Granulated Blast-Furnace Slag conforming to ASTM C989/C989M, Grade 100 or Grade 120.

2.2.4.4 Silica Fume

Provide silica fume conforming to ASTM C1240, including the optional limits on reactivity with cement alkalis. Silica fume may be furnished as a dry, densified material or as a slurry. Provide at the Contractor's expense the services of a manufacturer's technical representative, experienced in mixing, proportioning, placement procedures, and curing of concrete containing silica fume. This representative must be present on

the project prior to and during at least the first 4 days of concrete production and placement using silica fume.

2.2.4.5 Supplementary Cementitious Materials (SCM) Content

The Contractor may elect to use one of the SCMs listed below, unless the SCM is required to mitigate ASR. The use of SCMs is encouraged in accordance with Section 01 81 13 SUSTAINABLE CONSTRUCTION REQUIREMENTS

TABLE 2 SUPPLEMENTARY CEMENTITIOUS MATERIALS CONTENT

Supplementary Cementitious Material	Minimum, percent	Maximum Content, percent
Class N Pozzolan and Class F Fly Ash		
$SiO_2 + Al_2O_3 + Fe_2O_3 > 70$ percent	25	35
$SiO_2 + Al_2O_3 + Fe_2O_3 > 80$ percent	20	35
$SiO_2 + Al_2O_3 + Fe_2O_3 > 90$ percent	15	35
GGBFS	40	50
Silica Fume	7	10

2.2.5 Reinforcement Steel and Dowels

Provide reinforcement bars conforming to ASTM A615/A615M Grade 40 or 60. Provide welded steel wire fabric conforming to ASTM A1064/A1064M. Provide dowels that are plain (non-deformed) steel bars conforming to ASTM A615/A615M, Grade 40 or 60; ASTM A996/A996M, Grade 50 or 60. Provide epoxy coated dowel bars in conformance with ASTM A775/A775M. Use grout retention rings of fully circular metal or plastic devices capable of supporting the dowel until the epoxy hardens. Dowel sleeves or inserts are not permitted.

2.2.6 CURING MATERIALS

Only approved white pigmented membrane-forming curing compound materials conforming to the requirements specified in ASTM C309, Type 2, Class A or B may be used.

2.2.7 Joint Filler

For expansion joints use a preformed joint filler material conforming to ASTM D1751 or ASTM D1752.

2.2.8 SYNTHETIC FIBER REINFORCEMENT

ASTM C1116/C1116M. Use 100 percent virgin nylon or polypropylene fibers, 23 micron diameter, 3/4 inch length with a minimum tensile strength of 70 ksi. Add fibers to the concrete mix at the batch plant at the rate of 1.5 lbs/cubic yard.

PART 3 EXECUTION

3.1 GRADE CONTROL

Using bench-mark elevations furnished by the Contracting Officer, establish and maintain the lines and grades shown for the pavement by means of line and grade stakes placed at the jobsite. Construct pavements to the thicknesses and elevations indicated.

3.2 SUBGRADE, FORMS AND STRINGLINE

3.2.1 Underlying Material

3.2.1.1 General

Test the surface of the subgrade as to elevation and density in advance of setting forms. Keep the prepared surface free of foreign matter, waste concrete and/or cement, and debris at all times and thoroughly wet down sufficiently in advance to insure a firm, moist condition when the concrete is placed. In cold weather, prepare and protect the underlying material so that it will be entirely free from frost when the concrete is placed. The use of chemicals to eliminate frost in the underlying material will not be permitted.

3.2.1.2 Liner System

Where an impermeable liner system is placed directly under the concrete, take precautions to protect the underlying liner system from damage. See Section 33 56 19 FUEL IMPERMEABLE LINER SYSTEM for additional requirements.

3.2.2 Forms for Fixed-Form Paving

Use either steel or wood forms and forms subject to approval. Provide one-piece forms that are equal in depth to the edge thickness of the slab as shown on the drawings. Under no conditions are forms other than the depth of the pavement to be used and adjusted by filling or excavating under the forms to an elevation other than the bottom of the pavement slab. Do not vary the top surface of a form by more than 1/8 inch in 10 feet from a true line and the face by more than 1/4 inch in 10 feet from a true plane.

3.2.2.1 Steel Forms

Furnish steel forms in sections not less than 10 feet in length, except that on curves the sections are required to be flexible or curved to the proper radius. Provide each form section with form braces, pin sockets, and rigid joint locking devices.

3.2.2.2 Wood Forms

Furnish wood forms made of not less than 2 inches nominal thickness, well-seasoned, surfaced plank or plywood, straight, and free from warp or bend. Use wood forms that have the strength and rigidity to resist the impact and vibrations of concrete placing, spreading and finishing without springing, weaving or settling.

3.2.2.3 Form Setting

Set the forms on firm material cut true to grade so that each form section

when placed will be firmly in contact with the underlying layer for its entire length and base width. Setting forms on blocks or on built-up spots of subgrade and then attempting to fill and compact under forms after they are in place will not be permitted under any condition. Lock the form sections tightly together. When tested by a 10-foot straightedge, provide the top of the form conforming to the requirements specified for the finished surface of the concrete, and do not vary the longitudinal axis of the upstanding leg by more than 1/4 inch from the straightedge. Clean and oil the forms each time before concrete is placed. Do not place concrete until setting of forms has been approved. Do not drive form stakes in areas underlain by a geomemebrane. Anchoring of the forms in these areas is required to be approved by the Contracting Officer prior to installation.

3.3 PLACING, SPREADING AND VIBRATING

Submit a document detailing proposed concrete placement procedures. At a minimum, address form setting, protection of geomembrane, conveyance/pumping, construction joints, expansion joints, placement of reinforcement, curing and joint sealing procedures.

3.3.1 General

Place concrete between stationary forms. Deposit concrete between the forms within 90 minutes from the time all ingredients are charged into the mixing drum. Deposit concrete as close as possible to its final position in the pavement cross section. Place concrete at a continuous and uniform rate. Spread and vibrate concrete immediately after placement.

3.3.2 Vibration

Consolidate concrete by properly designed vibrating screeds, internal vibrators, or other approved techniques immediately after spreading. Cease forward motion of the paver as soon as a vibrator becomes inoperable. Maintain additional vibrators at the site at all times.

3.3.2.1 Slabs 8 Inches Thick

Consolidate concrete, greater than 8 inches in thickness, with mechanical vibrating equipment immediately after spreading. Provide internal type mechanical vibrating equipment and the number of units and adequate power of each unit to properly consolidate all of the concrete. Automatically control the vibrators and/or tamping elements so that they will be stopped as forward motion ceases. Do not exceed vibrator unit spacing of 30 inches, and provide a space from the outside unit and the edge of the slab of approximately 1 foot. Insert vibrators into the concrete to a depth that will provide the best consolidation, but not closer to the underlying material than 2 inches. Change the depth and angle of vibrators whenever directed by the Contracting Officer.

3.3.2.2 Slabs Less Than 8 Inches Thick

Consolidate concrete 8 inches or less in thickness with properly designed and operating vibratory screeds immediately after spreading.

3.3.2.3 Hand Placement

Vibrate concrete in odd shaped slabs, or lanes 50 feet or less in length or in locations inaccessible to the above vibrating equipment with a

hand-manipulated vibrator operated from a bridge spanning the concrete placement. Do not have workmen walk in the fresh concrete. Do not use vibrators to transport or spread the concrete in the forms. Do not operate vibrators in the concrete at one location for more than 20 seconds.

3.3.3 Placing Reinforcing Steel

Position reinforcement steel on suitable chairs prior to concrete placement or it may be installed by the strike-off method wherein the concrete is deposited on the underlying material, consolidated and struck to the indicated elevation of the steel reinforcement. When using the strike-off method, lay the reinforcement upon the prestruck surface, and place and finish the remaining concrete in the required manner. Remove and replace any portions of the bottom layer of concrete, that was placed more than 30 minutes without being covered with the top layer, with newly mixed concrete at no additional cost to the Government. Regardless of placement procedure, keep the reinforcing steel free from coatings which could impair bond between the steel and concrete and provide laps in the reinforcement as indicated.

3.3.4 Placing During Cold Weather

Place concrete in cold weather in accordance with ACI 306R. Do not place concrete on base course or subgrade containing frost or frozen material. Include provision to protect the concrete from freezing during the specified curing period. Remove and replace concrete damaged by freezing at no cost to the Government.

3.3.5 Placing During Warm Weather

Place concrete during warm weather in accordance with ACI 305R. During warm weather, produce concrete at the lowest temperature practicable under the existing conditions. Cool the mixing water and/or aggregates, if necessary, to maintain a satisfactory placing temperature. Place concrete continuously and rapidly at a rate of not less than 100 feet of paving lane per hour. Keep the finished surfaces of newly placed pavement damp by applying a waterfog or mist with approved spraying equipment until the pavement is covered by the curing medium.

3.4 FINISHING

Start finishing operations immediately after placing, spreading and vibrating of the concrete. Finish by the machine method except that, as specifically approved, the hand method may be used for lanes 50 feet or less in length, minor amounts of narrow slabs, irregular slab widths or shapes and separate, isolated slabs during removal and replacement type repair operations. Maintain finishing equipment and tools clean and in an approved condition.

3.4.1 Machine Finishing - Fixed Forms

3.4.1.1 Equipment

Conform to applicable requirements specified in subparagraph: PAVER FIXED-FORM METHOD of paragraph: PLACING, SPREADING AND VIBRATING above. Check screed and float adjustments of these machines at the start of each day's paving operations and more often as required. When finishing machines ride the edge of a previously constructed slab, include provision to protect the surface of these slabs.

3.4.1.2 Transverse Finishing

As soon as placed, accurately strike off and screed the concrete to the crown and cross section shown and to such elevation that when consolidated and finished, the surface of the pavement will be free from porous places and will be at the required grade. Excessive manipulation that brings to the surface an excess of mortar and water will not be permitted. Keep the top of the form or pavement edge upon which the finishing machine travels clean.

3.4.1.3 Mechanical Floating

Operate the mechanical float to smooth and finish the pavement to grade and maintain surface contact at all times. Do not use rotating pipe or tube floats or finishers, such as: Clary screeds, rotating "bridge deck finishers" and similar equipment.

3.4.1.4 Other Types of Finishing Equipment

Except for rotating pipe or tube floats or finishers, concrete finishing equipment of types other than specified above may be used on a trial basis, when specifically approved. Replace equipment that fails to produce finished concrete of the required quality with the approved equipment before specified herein.

3.4.2 HAND FINISHING

3.4.2.1 Finishing and Floating

As soon as placed and vibrated, strike off and screed the concrete to the crown and cross section and to such elevation above grade that, when consolidated and finished, the surface of the pavement will be at the required elevation. Tamp the entire surface. Continue the tamping operation until the required compaction and reduction of internal and surface voids are accomplished. Immediately following the final tamping of the surface, float the pavement longitudinally from bridges resting on the side forms and spanning but not touching the concrete. If necessary, place and screed additional concrete, and the float operated until a satisfactory surface has been produced. Advance the floating operation to not more than half the length of the float, and the floating continued over the new and previously floated surfaces.

3.4.3 Surface Correction and Testing

After all other finishing is completed but while the concrete is still plastic, eliminate minor irregularities and score marks in the pavement surface by means of straight-edges. Use straightedges 10 feet in length rigidly constructed to prevent deflection in any direction during use, and operate from the sides of the pavement and from bridges. After straight-edge finishing appears complete, test the entire surface for trueness with a 10 foot straightedge held in successive positions parallel and at right angles to the centerline of the pavement, and the whole area covered as necessary to detect variations. Advance the straightedge along the pavement in successive stages of not more than one-half the length of the straightedge. Continue the straightedge testing and finishing until the entire surface of the concrete is free from observable departure from the straightedge and conforms to the surface requirements specified under subparagraph: SURFACE TESTS AND CORRECTIONS below.

3.4.4 Texturing

Before the surface sheen has disappeared and before the concrete becomes nonplastic, give the surface of the pavement a stiff broom finish.

3.4.5 Edging

After texturing has been completed, the edge of slabs along the forms, and at the joints, where indicated or directed, carefully finish with an edging tool to form a smooth rounded surface of the required radius. Eliminate tool marks, and the provide edges that are smooth and true to line.

3.5 FORM REMOVAL

Keep forms in place at least 12 hours after the concrete has been placed or for a longer period, if directed by the Contracting Officer. Remove forms without injuring the concrete. Repair any concrete found defective after form removal promptly, using approved procedures.

3.6 CURING

3.6.1 General

Protect concrete against loss of moisture and rapid temperature changes for at least 7 days commencing immediately after finishing is complete. Protect unhardened concrete from rain and flowing water. Keep all equipment needed for adequate curing and protection of the concrete on hand and ready to use before actual concrete placement begins. If the curing materials and procedures used do not provide proper curing and protection against concrete cracking caused by temperature changes during the curing period, remove and replace the damaged pavement and employ another method of curing as directed.

3.6.2 Membrane Curing

Apply a uniform coating of white pigmented membrane curing compound to the entire exposed surface of the concrete as soon after finishing as free water has disappeared from the finished surface. Coat formed surfaces immediately after the forms are removed and in no case longer than 1 hour after removal of forms. Do not allow the concrete to dry before the application of the membrane. Apply the curing compound to the finished surfaces by means of an approved automatic spraying machine as soon as the free water has disappeared. Thoroughly and continuously agitate the curing compound in the drum used for the spraying operation mechanically throughout the full depth of the drum during the application. Air agitation may be used only to supplement mechanical agitation. Apply the curing compound with an overlapping coverage that will give a two-coat application at a coverage of not more than 400 square feet/gallon for each coat. The application of curing compound by hand-operated pressure sprayers will be permitted only on odd widths or shapes of slabs where specifically approved, and on concrete surfaces exposed by the removal of forms. When application is made by hand-operated sprayers, apply the second coat in a direction approximately at right angles to the direction of the first coat. Respray curing compound that has pinholes, abrasions, or other discontinuities, that was subjected to heavy rainfall within 3 hours of application, or was damaged by subsequent construction operations by the method and at the coverage specified above. Take necessary

precautions to insure that the concrete is properly cured at sawed joints, but that no curing compound enters the joints. Tightly seal the top of the joint opening and the joint groove at exposed edges by approved procedures using a temporary sealer or filler before the concrete in the region of the joint is resprayed with curing compound. Use the method for sealing the joint groove that prevents loss of moisture from the joint during the entire specified curing period. Provide approved standby facilities for curing concrete pavement at an accessible location at the jobsite for use in the event of mechanical failure of the spraying equipment or other conditions that might prevent correct application of the membrane curing compound at the proper time. Protect concrete surfaces to which membrane curing compounds have been applied adequately during the entire curing period from pedestrian and vehicular traffic, except as required for joint-sawing operations and surface tests, and from any other possible damage to the continuity of the membrane.

3.7 GRADE AND SURFACE-SMOOTHNESS REQUIREMENTS AND TESTS

3.7.1 General

Provide pavements that are smooth and true to grade and cross section. When tested with a 10 foot straightedge on lines 5 feet apart parallel with and at right angles to the centerline of the pavement, do not vary the surface no more than 1/4 inch from the testing edge of the straightedge.

3.7.2 Surface Tests And Corrections

Not later than 24 hours after concrete has been placed, test the surface of the pavement in the presence of the Contracting Officer using an approved straightedge or other approved device that will reveal all surface irregularities varying from the testing edge exceeding tolerances specified above for concrete pavements. Plainly mark high spots indicated by the testing edge in excess of applicable tolerances. Reduce high areas by approved methods or remove and replace the pavement at no cost to the Government.

3.8 TOLERANCES IN PAVEMENT THICKNESS

Provide pavements of the thicknesses indicated on the plans. Treat deficiencies in the thickness as described below.

3.8.1 Thickness Determination

Determine the anticipated thickness of the concrete prior to placement by passing a template through the formed section. When measurements indicate that the completed concrete section is deficient in thickness by more than 1/4 inch, the deficient section will be removed, between regularly scheduled joints, and replaced at no cost to the Government.

3.9 FIELD TEST SPECIMENS

3.9.1 General

Except as modified hereinafter, perform tests to determine the slump, air content, and strength of the concrete in accordance with the requirements of ASTM C94/C94M. Complete tests for slump and air content each time cylinders are fabricated and at such other times as directed by the Contracting Officer.

3.9.2 Specimens for Strength Tests

Take compressive test cylinders not less than once a day nor less than once for each 250 cubic yards of concrete or fraction thereof. Take the samples of strength tests in accordance with ASTM C172/C172M. Mold and cure cylinders for acceptance tests in accordance with ASTM C31/C31M. Test cylinders in accordance with ASTM C39/C39M by an approved testing laboratory at no cost to the Government. Mold sufficient cylinders each time to provide two compressive-strength tests at each test age. Test ages are 7, 14, and 28 days.

3.10 JOINTS

3.10.1 General

Provide joints conforming to the details indicated and that are perpendicular to the finished grade of the pavement. Provide transverse expansion and contraction joints that are straight and continuous from edge to edge of the pavement.

3.10.2 Construction Joints

Install transverse construction joints at the end of each day's placing operations and at any other points within a paving lane when concrete placement is interrupted for 30 minutes or longer. Install transverse construction joints in the location of a planned joint.

3.10.3 Expansion Joints

Form expansion joints by means of a preformed filler material.

3.10.4 Contraction Joints

Provide transverse and longitudinal contraction joints of the weakened-plane, and constructed as indicated hereinafter in subparagraph: SAWED JOINTS. Construct longitudinal contraction joints by sawing a groove in the hardened concrete with a power-driven saw in conformance with subparagraph: SAWED JOINTS below, unless otherwise approved.

3.10.4.1 Sawed Joints

Construct sawed joints by sawing a groove in the concrete with a 1/8 inch blade to full depth as indicated, without chipping, spalling, or tearing the concrete adjacent to the joint. After expiration of the curing period, widen the upper portion of the groove by sawing to the width and depth indicated. Vary the time of sawing depending on existing and anticipated weather conditions, and such as to prevent uncontrolled cracking of the pavement. Saw the joints at the required spacing consecutively in the sequence of the concrete placement. Do not vary the saw cut by more than 1/2 inch from the true joint alignment. Do not saw joints if a crack has occurred near the joint location and discontinue sawing when a crack develops ahead of the saw cut. Immediately after joint is sawed, thoroughly flush the saw cut and adjacent concrete surface with water until all waste from sawing is removed from the joint. Seal the top of the joint opening immediately as specified in subparagraph: MEMBRANE CURING. Provide an ample supply of saw blades and at least one standby sawing unit in good working order at the jobsite at all times during concrete paving operations.

3.10.4.2 Sealing Joints

Seal joints immediately following curing of the concrete, or as soon as weather conditions permit, as directed. Accomplish sawing of the reservoir or space for seals, immediately before sealing of the joints. Perform sawing by a multi-blade concrete saw. Perform sawing to the width specified and to the depth indicated in one pass. Readily adjust the cutting unit for width by the addition and removal of spacers or by other suitable means. Equip the machine with a mechanical guide which will keep the cutting unit aligned so as to cut equal widths from each side of the joint groove.

a. Sealing Dike Berms: Seal the joints in the concrete dike berms with a non-sag type fuel resistant sealant.

b. Sealing Dike Basins: Seal joints in the concrete surface within the diked areas with fuel resistant sealant.

3.11 PAVEMENT PROTECTION

Protect the pavement against all damage, prior to final acceptance of the work by the Government. No vehicular traffic will be allowed on the 4 inch concrete pavement at any time.

3.12 REPAIR OF DAMAGED SLABS

Repair new pavement slabs that are broken, have spalled edges, or contain cracks as specified hereinafter at no cost to the Government.

3.12.1 Cracked Slabs

Rout and seal cracks in reinforced slabs. Do not epoxy inject these cracks.

3.12.2 Spalled Slabs

Where directed, repair spalls along joints of new slabs, along edges of adjacent existing concrete, and along parallel cracks by first making a vertical saw cut at least 1 inch outside the spalled area and to a depth of at least 2 inches. Provide straight line saw cuts forming rectangular areas. Chip out the concrete between the saw cut and the joint, or crack, to remove all unsound concrete and into at least 1/2 inch of visually sound concrete.

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SECTION 32 17 23
8/16

PAVEMENT MARKINGS

PART 1 GENERAL

1.1 SUMMARY

Section Includes:

Paint and reflective glass beads on pavement surfaces, in form of traffic lanes, parking bays, areas restricted to handicapped persons, crosswalks, and other detail pavement markings.

1.2 RELATED REQUIREMENTS

a. Paint VOC Limits: Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

b. Paint Color: As indicated on drawings.

1.3 REFERENCES

a. Comply with references to extent specified in this Section.

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS TT-B-1325	(Rev D; Notice 1; Notice 2 2017) Beads (Glass Spheres) Retro-Reflective (Metric)
TT-P-1952F	(2015) Paint, Traffic, and Airfield Marking, Waterborne

MASTER PAINTERS INSTITUTE (MPI)

MPI 97	(2012) Traffic Marking Paint, Latex

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualifications; G,

Manufacturer's Instructions Deviate From Specifications; G,

SD-02 Shop Drawings

Submittal Drawings; G,

Field Measurements; G,

SD-03 Product Data

Manufacturer's Literature And Data; G,

SD-04 Samples

Samples; G,

SD-06 Test Reports

Product Temperature; G,

SD-07 Certificates

Certificates; G,

1.5 SUBMITTAL INSTRUCTIONS

a. Submittal Procedures: Section 01 33 00 SUBMITTAL PROCEDURES

b. Submittal Drawings:

(1) Show pavement marking configuration and dimensions.

(2) Show international symbol of accessibility at designated parking spaces.

c. Manufacturer's Literature and Data:

(1) Description of each product.

(2) Application instructions.

d. Samples:

Paint: 8 inches square, each type and color.

e. Certificates: Certify products comply with specifications.

f. Qualifications: Substantiate qualifications comply with specifications.

Installer with project experience list.

1.6 QUALITY ASSURANCE

Installer Qualifications:

a. Regularly installs specified products.

b. Installed specified products with satisfactory service on five similar installations for minimum five years.

Project Experience List: Provide contact names and addresses for completed projects.

1.7 DELIVERY

a. Deliver products in manufacturer's original sealed packaging.

b. Mark packaging, legibly. Indicate manufacturer's name or brand, type, color, production run number, and manufacture date.

c. Before installation, return or dispose of products within distorted, damaged, or opened packaging.

1.8 STORAGE AND HANDLING

a. Store products indoors in dry, weathertight conditioned facility.

b. Protect products from damage during handling and construction operations.

1.9 FIELD CONDITIONS

Environment:

a. Product Temperature: Minimum 55 degrees F for minimum 48 hours before installation.

b. Surface to be painted and ambient temperature: Minimum 50 degrees F and maximum 95 degrees F.

c. Field Measurements: Verify field conditions affecting traffic marking installation. Show field measurements on Submittal Drawings.

1.10 WARRANTY

FAR clause 52.246-21, "Warranty of Construction."

PART 2 PRODUCTS

2.1 SYSTEM PERFORMANCE

Design paint complying with specified performance:

Application: TT-P-1952F.

2.2 PRODUCTS - GENERAL

a. Provide each product from one manufacturer and from one production run.

(1) In accordance with Manual on Uniform Traffic Control Devices (MUTCD) 11th Edition the materials used for markings shall provide the specific color throughout their useful life. Consideration shall be given to selecting pavement marking materials that will minimize tripping or loss of traction for road users, including pedestrians, bicyclists, and motorcyclists.

(1) Low Pollutant-Emitting Materials: Comply with VOC limits specified in Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS for the following products:

(a) Paints and coatings.

2.3 SANDBLASTING EQUIPMENT

Air compressor, hoses, and nozzles of proper size and capacity as required for cleaning painted surfaces. Compressor to provide minimum 150 cfm of air at pressure of minimum 90 psi at each nozzle used.

2.4 PAINT APPLICATOR

Apply marking paint with approved mechanical equipment. Provide equipment with constant agitation of paint and travel at controlled speeds. Synchronize one or more paint "guns" to automatically begin and cut off paint flow in case of skip lines. Equipment to have manual control to apply continuous lines of varying length and marking widths as indicated on Drawings. Provide pneumatic spray guns for hand application of paint in areas where mobile paint applicator cannot be used. Use separate piece of equipment when equipment does not have glass bead dispenser. Adjust and synchronize equipment with paint applicator to distribute reflective beads on paint lines uniformly within ten seconds without any waste.

2.5 PAINT

MPI 97. For obliterating existing markings comply with TT-P-1952F. Provide minimum 5 gallons containers.

2.6 REFLECTIVE GLASS BEADS

Comply with FS TT-B-1325, Type I, Gradation A. In regions of high humidity, coat beads with silicone or other suitable waterproofing material to ensure free flow. Provide glass beads in containers suitable for handling and strong enough to prevent loss during shipment.

PART 3 EXECUTION

3.1 PREPARATION

a. Examine and verify substrate suitability for product installation.

Allow new pavement surfaces to cure for period of minimum 14 days before application of marking materials.

b. Protect existing construction and completed work from damage.

c. Clean substrates. Remove contaminants capable of affecting subsequently installed product's performance.

(1) Remove dust, dirt, and other granular surface deposits by sweeping, blowing with compressed air, rinsing with water, or combination of these methods.

(2) Completely remove rubber deposits, existing paint markings, and other coatings adhering to pavement with scrapers, wire brushings, sandblasting, mechanical abrasion, or approved chemicals as directed by Contracting Officer.

(3) As an option, comply with TT-P-1952F for removal of existing paint markings on asphalt pavement. Apply black paint in as many coats as necessary to completely obliterate existing markings.

(4) Scrub affected areas with several applications of trisodium

phosphate solution or other approved detergent or degreaser, and rinse thoroughly after each application, Where oil or grease are present on old pavements to be marked,.

After cleaning, seal oil-soaked areas with cut shellac to prevent bleeding through new paint.

(5) Clean and dry surface before pavement marking.Do not begin any marking until Contracting Officer inspected surface and gives permission to proceed.

3.2 TEMPORARY PAVEMENT MARKING

a. Apply Temporary Pavement Markings of colors, widths and lengths shown on drawings or directed by Contracting Officer. After temporary marking has served its purpose and when so ordered by Contracting Officer, remove temporary marking by carefully controlled sandblasting, approved grinding equipment, or other approved method to prevent damage on applied surface.

b. As an option, provide approved preformed pressure sensitive, reflective, adhesive tape type of temporary pavement marking of required colors, widths and lengths in lieu of temporary painted and reflective marking. Continuous durability and effectiveness of such marking is required during period for which its use is required. Remove any unsatisfactory tape type marking and replace with painted and reflective markings.

3.3 INSTALLATION - GENERAL

Install products according to manufacturer's instructions and approved submittal drawings.

When manufacturer's instructions deviate from specifications, submit proposed resolution for Contracting Officer's consideration.

3.4 PAINT APPLICATION

a. Apply uniformly painted and reflective pavement marking of required colors, length, and width with true, sharp edges and ends on properly cured, prepared, and dried surfaces.

b. Comply with details as indicated on drawings and established control points.

c. Apply paint at wet film thickness of 0.015 inch. Disperse reflective glass beads evenly on wet paint at rate of 6 pounds per gal. of paint. Apply paint in one coat. When directed by Contracting Officer, apply additional coats at markings showing light spots. Comply with paint manufacturer's maximum drying time requirements to prevent undue softening of asphalt, and pick-up, displacement, or discoloration by tires of traffic.

d. When deficiency in marking drying occurs, discontinue paint operations until cause of slow drying is determined and corrected.

e. Remove and replace marking applied less than minimum material rates, deviates from true alignment, exceeds stipulated length and width tolerances, or shows light spots, faulty distribution of beads,

smears, or other deficiencies or irregularities.

f. Remove marking by carefully controlled sandblasting, approved grinding equipment, or other approve method to prevent damage on applied surface.

3.5 DETAIL PAVEMENT MARKING APPLICATION

a. Apply Detail Pavement Markings, exclusive of actual traffic lane marking as follows:

(1) At exit and entrance islands and turnouts.

(2) On curbs.

(3) At crosswalks.

(4) At parking bays.

(5) Other locations as indicated on drawings.

b. Apply International Handicapped Symbol at indicated parking spaces. Color as shown on drawings. Apply paint for symbol using suitable template that will provide pavement marking with true, sharp edges and ends.

c. Install detail pavement markings of colors, widths and lengths, and design pattern at locations indicated on drawings.

3.6 TOLERANCES

a. Length and Width of Lines: Plus or minus 3 inches and plus or minus 1/8 inch, respectively, in case of skip markings.

b. Length of intervals exceeding line length tolerance are not acceptable.

3.7 CLEANING

Remove excess paint before paint sets.

3.8 PROTECTION

a. Protect pavement markings from traffic and construction operations.

(1) Protect newly painted markings from vehicular traffic until paint is dry and track free.

(2) Place warning signs at beginning of wet line, and at points well in advance of marking equipment for alerting approaching traffic from both directions.

(3) Place small flags or other similarly effective small objects near freshly applied markings at frequent intervals to reduce crossing by traffic.

b. Repair damage.

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CHAIN LINK FENCES AND GATES

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SECTION 32 31 13

CHAIN LINK FENCES AND GATES
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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A392	(2011; R 2022a) Standard Specification for Zinc-Coated Steel Chain-Link Fence Fabric
ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM C94/C94M	(2025) Standard Specification for Ready-Mixed Concrete
ASTM F567	(2023) Standard Practice for Installation of Chain Link Fence
ASTM F626	(2014; R 2023) Standard Specification for Fence Fittings
ASTM F668	(2017; R 2022) Standard Specification for Polyvinyl Chloride (PVC) and Other Polymer-Coated Steel Chain Link Fence Fabric
ASTM F883	(2013; R 2022) Standard Performance Specification for Padlocks
ASTM F934	(2016; R 2022) Standard Specification for Standard Colors for Polymer-Coated Chain Link Fence Materials

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS RR-F-191/1	(Rev F) Fencing, Wire and Post, Metal (Chain-Link Fence Fabric)
FS RR-F-191/2	(Rev E) Fencing, Wire and Post, Metal (Chain-Link Fence Gates)
FS RR-F-191/3	(2023; Rev F) Fencing, Wire and Post, Metal (Chain-Link Fence Posts, Top Rails and Braces)
FS RR-F-191/4	(Rev F) Fencing, Wire and Post, Metal (Chain-Link Fence Accessories)

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fence Assembly; G

Location of Gate, Corner, End, and Pull Posts; G

Gate Assembly; G

Gate Hardware and Accessories; G

Erection/Installation Drawings; G

SD-03 Product Data

Fence Assembly; G

Gate Assembly; G

Gate Hardware and Accessories; G

Zinc Coating; G

PVC Coating; G

Fabric; G

Stretcher Bars; G

Concrete; G

Padlocks; G

Fabric; G

Posts; G

Braces; G

Line Posts; G

Sleeves; G

Top Rail; G

Bottom Rail; G

Tension Wire; G

Stretcher Bars; G

Gate Posts; G

Gate Hardware and Accessories; G

Wire Ties; G

SD-07 Certificates

Certificates of Compliance; G

SD-08 Manufacturer's Instructions

Fence Assembly; G

Gate Assembly; G

Hardware Assembly; G

Accessories; G

SD-11 Closeout Submittals

Recycled Material Content; G

1.3 QUALITY CONTROL

1.3.1 Certificates of Compliance

Submit certificates of compliance in accordance with the applicable reference standards and descriptions of this section for the following:

a. Zinc coating

b. PVC coating

c. Aluminum alloy coating

d. Fabric

e. Stretcher bars

f. Gate hardware and accessories

g. Concrete

1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to site in an undamaged condition. Store materials off the ground to provide protection against oxidation caused by ground contact.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Submit reports of listing chain-link fencing and accessories regarding weight in ounces for zinc coating, thickness of PVC coating.

Submit manufacturer's catalog data for complete fence assembly, gate assembly, hardware assembly and accessories.

2.2 FENCES

2.2.1 Fabric

FS RR-F-191/1; Type IV, Class 2b polyvinyl chloride (PVC) coated over zinc- or aluminum-coated steel, 9-gauge core wire size, conforming to ASTM F668. Color to be determined by Government. complying with ASTM F934. . Provide selvage twisted and barbed at both selvages . Height of fabric, as indicated.

Provide fabric consisting of wires woven into a 2 inch diamond mesh. Provide one-piece fabric widths for fence heights up to 12 feet.

2.2.1.1 Top and Bottom Selvages

Provide knuckled selvages at top and bottom for fabric with 2 inch mesh and up to 60 inches high, and if over 60 inches high, provide twisted and barbed on both selvages.

Knuckle top and bottom selvages for 1-3/4 inch and 1 inch mesh fabric.

2.2.2 Posts

2.2.2.1 Metal

2.2.2.1.1 Line Posts

Provide line posts complying with FS RR-F-191/3. Provide Class 1, steel pipe, Grade A, High Strength.

2.2.2.1.2 End, Corner and Pull Posts

Provide end, corner, and pull posts in minimum sizes listed in FS RR-F-191/3 for each class and grade. Provide Class 1, steel pipe, Grade A High Strength.

2.2.2.2 Post Tops

Provide steel, wrought iron, or malleable iron tops and designed as a weathertight closure cap. Post top to have finish and coating matching rails and posts. Provide one cap for each post, unless equal protection is provided by a combination post-cap and wire supporting arm. Provide caps with an opening to permit through passage of the top rail.

2.2.3 Braces and Rails

Class 1, steel pipe, Grade A, High Strength, in minimum sizes listed in FS RR-F-191/3 for each class and grade.

2.2.3.1 Top Rail

Provide top rail conforming to minimum sizes specified in FS RR-F-191/3 for each class and grade unless members are to be oversized. Provide expansion couplings 6 inches long at each joint in top rails.

2.2.3.2 Center Rails Between Line Posts

Provide center rail conforming to minimum sizes specified in FS RR-F-191/3 for each class and grade unless members are to be oversized.

2.2.3.3 Bottom Rail

Provide bottom rail conforming to minimum sizes specified in FS RR-F-191/3 for each class and grade unless members are to be oversized.

2.2.3.4 Post-Brace Assembly

Provide bracing conforming to minimum sizes specified in FS RR-F-191/3 for each class and grade, and 3/8 inch adjustable truss rods and turnbuckles.

2.2.4 Wire Ties

Provide 9-gauge wire for tying fabric to line posts,spaced 12 inches on center. For tying fabric to rails and braces, space wire ties 24 inches on center. For tying fabric to tension wire, space 0.105-inch hog rings 24 inches on center. Manufacturer's standard procedure will be accepted if of equal strength and durability.

FS RR-F-191/4. Provide wire ties constructed of the same material and coating as the fencing fabric.

2.2.5 Sleeves

Provide sleeves for setting into concrete construction of the same material as post sections, sized 1 inch greater than the diameter or dimension of the post. Weld flat plates to each sleeve base to provide anchorage and prevent intrusion of concrete.

2.2.6 Stretcher Bars

Provide bars that have one-piece lengths equal to the full height of the fabric with a minimum cross section of 3/16 by 3/4 inch, in accordance with ASTM F626.

2.2.7 Stretcher Bar Bands

Provide bar bands for securing stretcher bars to posts that are steel, wrought iron, or malleable iron spaced not over 15 inches on center. Bands may also be used in conjunction with special fittings for securing rails to posts. Provide bands with projecting edges chamfered or eased.

2.2.8 Tension Wire

Provide zinc-coated steel wire with zinc coating that weighs not less than 2.0 ounces per square foot.

2.2.9 Miscellaneous Hardware

Provide miscellaneous hot-dip galvanized hardware as required.

2.3 GATES

FS RR-F-191/2; Type I, single swing II, double swing. Shape and size of gate frame, as indicated. Framing and bracing members, square of steel alloy. Steel member finish, zinc-coated. Provide gate frames and braces of minimum sizes listed in FS RR-F-191/3 for each Class and Grade, except that steel pipe frames are a minimum of 1.90 inches o.d., 0.120 inches minimum wall thickness and aluminum pipe frames and intermediate braces are

1.869 inches o.d. minimum, 0.940 lb/ft of length. Provide intermediate members as necessary for gate leaves more than 8 feet wide, to provide rigid construction, free from sag or twist. Provide truss rods or intermediate braces for gate leaves less than 8 feet wide.

2.3.1 Gate Posts

Provide gate posts for supporting each gate leaf in minimum sizes listed in FS RR-F-191/3 for each material class and grade. Gate post material class, grade and finish to match other fence posts.

2.3.2 Gate Fabric

Gate fabric, is as specified for fencing fabric. Attach gate fabric to gate frame in accordance with manufacturer's standards, except that welding is not permitted.

2.3.3 Gate Frame

Provide gate frame assembly that is welded or assembled with special malleable or pressed-steel fittings and rivets to provide rigid connections. Install fabric with stretcher bars at vertical edges; stretcher bars may also be used at top and bottom edges. Attach stretcher bars and fabric to gate frames on all sides at intervals not exceeding 15 inches. Attach hardware with rivets or by other means which provides equal security against breakage or removal.

2.3.4 Gate Bracing

Provide diagonal cross-bracing, consisting of 3/8 inch diameter adjustable-length truss rods on welded gate frames, where necessary to obtain frame rigidity without sag or twist. Provide nonwelded gate frames with diagonal bracing.

2.3.5 Padlocks

Provide padlocks conforming to ASTM F883, with chain.

2.3.6 Gate Hardware and Accessories

Provide gate hardware and accessories that conforms to ASTM A392 and ASTM F626, and as specified. Coating for steel latches, stops, hinges, keepers, and accessories, is galvanized

a. Provide malleable iron or forged steel hinges to suit gate size, non-lift-off type, offset to permit 180-degree opening. Provide hinge with stainless steel pin.

b. Provide latch that permits accessibility and operation from either side of the gate regardless of the latching arrangement, and with a padlock eye provided as an integral part of the latch. Provide plunger bar type gate latches.

c. Provide stops and holders of malleable iron for vehicular gates. Provide stops that automatically engage the gate and hold it in the open position until manually released.

d. Provide double gates with a cane bolt and ground-set keeper, with latch or locking device and padlock eye designed as an integral part.

2.4 MATERIALS

2.4.1 Zinc Coating

Provide zinc-coated ferrous metal components and accessories that are factory coated after fabrication, except as otherwise specified.

For galvanizing field repairs, provide material that is cold-applied zinc-rich coating conforming to ASTM A780/A780M.

2.4.2 Concrete

Provide concrete conforming to ASTM C94/C94M, and obtaining a minimum 28-day compressive strength of 3,000 psi.

2.4.3 Grout

Provide grout of proportions one part portland cement to three parts clean, well-graded sand and a minimum amount of water to produce a workable mix.

PART 3 EXECUTION

Submit manufacturer's erection/installation drawings and instructions that detail proper assembly and materials in the design for fence, gate, hardware and accessories.

Provide complete installation conforming to ASTM F567.

3.1 PREPARATION

Ensure final grading and established elevations are complete prior to commencing fence installation.

3.1.1 Clearing and Grading

Clear fence line of trees, brush, and other obstacles to install fencing for a distance of 3 feet outside the fence. Establish a graded, compacted fence line prior to fencing installation.

3.2 INSTALLATION

3.2.1 Security

Install new chain link fencing, remove existing fencing, and perform related work to provide continuous security for facility. Schedule and fully coordinate work with Contracting Officer and cognizant Security Officer.

3.2.2 Fence Installation

Install fence on prepared surfaces to line and grade indicated. Install fence in accordance with fence manufacturer's written installation instructions except as modified herein.

3.2.2.1 Post Spacing

Provide line posts spaced equidistantly apart, not exceeding 10 feet on

center. Provide gate posts spaced as necessary for size of gate openings. Do not exceed 500 feet on straight runs between braced posts. Provide corner or pull posts, with bracing in both directions, for changes in direction of 15 degrees or more, or for abrupt changes in grade. Submit drawings showing location of gate, corner, end, and pull posts.

3.2.2.2 Top and Bottom Tension Wire

Install top and bottom tension wires before installing chain-link fabric, and pull wires taut. Place top and bottom tension wires within 8 inches of respective fabric line.

3.2.3 Excavation

Provide excavations for post footings which are drilled holes in virgin or compacted soil, of minimum sizes as indicated. Space footings for line posts 10 feet on center maximum and at closer intervals when indicated, with bottoms of the holes approximately 3 inches below the bottoms of the posts. Set bottom of each post not less than 36 inches below finished grade when in firm, undisturbed soil. Set posts deeper, as required, in soft and problem soils and for heavy, lateral loads. Remove excavated soil from Government property.

When solid rock is encountered near the surface, drill into the rock at least 12 inches for line posts and at least 18 inches for end, pull, corner, and gate posts. Drill holes at least 1 inch greater in diameter than the largest dimension of the placed post. If solid rock is below the soil overburden, drill to the full depth required except that penetration into rock need not exceed the minimum depths specified above.

3.2.4 Setting Posts

Remove loose and foreign materials from holes and moisten the soil prior to placing concrete. Provide tops of footings that are trowel finished and sloped or domed to shed water away from posts. Set hold-open devices, sleeves, and other accessories in concrete.

Keep exposed concrete moist for at least 7 calendar days after placement or cured with a membrane curing material, as approved. Grout all posts set into sleeved holes in concrete with an approved grouting material. Maintain vertical alignment of posts in concrete construction until concrete has set.

3.2.4.1 Earth and Bedrock

Provide concrete bases of dimensions indicated on the manufactures installation drawings, except in bedrock. Compact concrete to eliminate voids, and finish to a dome shape. In bedrock, set posts with a minimum of 1 inch of grout around each post. Work grout into hole to eliminate voids, and finish to a dome shape.

3.2.4.2 Concrete Slabs and Walls

Set posts into zinc-coated sleeves, set in concrete slab or wall, to a minimum depth of 12 inches. Fill sleeve joint with lead, nonshrink grout, or other approved material. Set posts for support of removable fence sections into sleeves that provide a tight sliding joint and hold posts aligned and plumb without use of lead or setting material.

3.2.4.3 Bracing

Brace gate, corner, end, and pull posts to nearest post with a horizontal brace used as a compression member, placed at least 12 inches below top of fence, and a diagonal tension rod.

3.2.4.4 Tolerances

Provide posts that are straight and plumb within a vertical tolerance of 1/4 inch after the fabric has been stretched. Provide fencing and gates that are true to line with no more than 1/2 inch deviation from the established centerline between line posts. Repair defects as directed.

3.2.5 Concrete Strength

Provide concrete that has attained at least 75 percent of its minimum 28-day compressive strength, but in no case sooner than 7 calendar days after placement, before rails, tension wire, or fabric are installed. Do not stretch fabric and wires or hang gates until the concrete has attained its full design strength.

Sample and test concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.2.6 Top Rails

Provide top rails that run continuously through post caps or extension arms, bending to radius for curved runs. Provide expansion couplings as recommended by the fencing manufacturer.

3.2.7 Center Rails

Provide single piece center rails between posts set flush with posts on the fabric side, using special offset fittings where necessary.

3.2.8 Brace Assembly

Provide bracing assemblies at end and gate posts and at both sides of corner and pull posts, with the horizontal brace located at midheight of the fabric.

Install brace assemblies so posts are plumb when the diagonal rod is under proper tension. Provide two complete brace assemblies at corner and pull posts where required for stiffness and as indicated.

3.2.9 Tension Wire Installation

Install tension wire by weaving them through the fabric and tying them to each post with not less than 7-gauge galvanized wire or by securing the wire to the fabric with 10-gauge ties or clips spaced 24 inches on center.

3.2.10 Fabric Installation

Provide fabric in single lengths between stretch bars with bottom barbs placed approximately 1-1/2 inches above the ground line. Pull fabric taut and tied to posts, rails, and tension wire with wire ties and bands.

Install fabric on the security side of fence, unless otherwise directed. Ensure fabric remains under tension after the pulling force is released.

3.2.11 Stretcher Bar Installation

Thread stretcher bars through or clamped to fabric 4 inches on center and secured to posts with metal bands spaced 15 inches on center.

3.2.12 Gate Installation

Install gates plumb, level, and secure, with full opening without interference. Install ground set items in concrete for anchorage as recommended by the fence manufacturer. Adjust hardware for smooth operation and lubricated where necessary.

3.2.13 Tie Wires

Provide tie wires that are U-shaped to the pipe diameters to which attached. Twist ends of tie wires not less than two full turns and bent so as not to present a hazard.

3.2.14 Fasteners

Install nuts for tension bands and hardware on the side of the fence opposite the fabric side. Peen ends of bolts to prevent removal of nuts.

3.2.15 Zinc-Coating Repair

Clean and repair galvanized surfaces damaged by welding or abrasion, and cut ends of fabric, or other cut sections with specified galvanizing repair material applied in strict conformance with the manufacturer's printed instructions.

3.2.16 Accessories Installation

3.2.16.1 Post Caps

Design post caps to accommodate top rail. Install post caps as recommended by the manufacturer.

3.2.16.2 Padlocks

Provide padlocks for gate openings and provide chains that are securely attached to gate or gate posts. Provide padlocks keyed alike, and provide two keys for each padlock.

3.2.17 Grounding

Ground fencing as indicated on drawings and specified.

Ground all fences crossed by overhead power lines in excess of 600 volts, and all electrical equipment attached to the fence. Ground fences on each side of all gates, at each corner, at the closest approach to each building located within 50 feet of the fence, and where the fence alignment changes more than 15 degrees. Grounding locations can not exceed 650 feet. Bond each gate panel with a flexible bond strap to its gate post. Ground fences crossed by power lines of 600 volts or more at or near the point of crossing and at distances not exceeding 150 feet on each side of crossing. Provide ground conductor consisting of No. 6 AWG solid copper wire. Provide solid copper rod grounding electrodes3/4 inch by 10 foot long. Drive electrodes into the earth so that the top of the

electrode is at least 6 inches below the grade. Where driving is impracticable, bury electrodes a minimum of 12 inches deep and radially from the fence, with top of the electrode not less than 2 feet or more than 8 feet from the fence. Clamp ground conductor to the fence and electrodes with bronze grounding clamps to create electrical continuity between fence posts, fence fabric, and ground rods. Total resistance of the fence to ground cannot exceed 25 ohms.

3.3 CLOSEOUT ACTIVITIES

Remove waste fencing materials and other debris from the work site.

Submit manufacturer's data indicating percentage of recycled material content in protective fence materials, including chain link fence, fabric, and gates to verify affirmative procurement compliance.

-- End of Section --

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DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 31 40

HIGH-SECURITY FENCES AND GATES

11/21

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SECTION 32 31 40

HIGH-SECURITY FENCES AND GATES
11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM C94/C94M	(2025) Standard Specification for Ready-Mixed Concrete
ASTM F567	(2023) Standard Practice for Installation of Chain Link Fence
ASTM F883	(2013; R 2022) Standard Performance Specification for Padlocks
ASTM F2408	(2016) Standard Specification for Ornamental Fences Employing Galvanized Steel Tubular Pickets
ASTM F2814	(2009; R 2015) Standard Guide for Design and Construction of Ornamental Steel Picket Fence Systems for Security Purposes

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fence Installation Drawings; G

SD-03 Product Data

Padlocks; G

Ornamental Fence Systems; G

Swing Gates; G

Sleeves; G

SD-07 Certificates

Gate Hardware and Accessories; G

Concrete; G

Gate Operator; G

SD-10 Operation and Maintenance Data

Electro-Mechanical Locks; G

Gate Operator; G

Operating and maintenance instructions; G

1.3 DELIVERY, STORAGE, AND HANDLING

Deliver materials to site in an undamaged condition. Store materials elevated off of the ground to protect against oxidation caused by ground contact.

PART 2 PRODUCTS

2.1 COMPONENTS

2.1.1 Ornamental Fence Systems

Submit manufacturer's catalog data. Provide ASTM F2814 structural components consisting of tubular steel ornamental pickets and rails. Provide ASTM F2408 industrial class pickets with a minimum cross-sectional area of 1 sq in and a minimum wall thickness of 14 gauge. Provide pickets with spear-pointed tips extending a minimum of 6 in above the top rail of the fence. Mount pickets to a top and bottom rail spaced a maximum of 80 inches apart. Space pickets along rails with a maximum gap not to exceed 2.25 inches. Secure pickets to rails by inaccessible, tamper-proof fasteners. Provide all items and accessories finished by PVC powder-coating in color selected by Government.

2.1.2 Ornamental Fence Gates

2.1.2.1 Swing Gates

Submit manufacturer's catalog data. Fabricate swing gates by welding 2 sq in tubular steel ends and rails. Use pickets that match the adjacent fence construction. Reinforce gates to ensure assembly sags no more than 1% of the gate leaf width or 2 in, whichever is less. Size gate posts to accommodate the weight and width of each gate leaf. Mount gates to posts with weldable steel plates or blocks, pressed steel, or malleable iron hinges. Hot-dip galvanize all hinges with a minimum zinc weight of 1.20 oz/sq ft. Provide hinge with stainless steel pin. Secure all tamper points by welding or peening the threads. Use swing gate latches and drop bar guides manufactured of pressed steel, hot-dipped galvanized with a minimum zinc weight of 1.20 oz/sq ft. Finish all gate hardware in the same color/coating as the fence system.

2.1.3 Padlocks

Provide padlocks conforming to ASTM F883, Type PO1, Options A, B, and G, Grade 6. Size 1-3/4 inch. Key all padlocks alike.

2.1.4 Gate Operator

Provide electric gate operators for sliding gates as follows: Provide electric gate operators with a right angle gearhead instantly reversing motor with magnetic drum-type brake, friction disc clutch, reversing starter with thermal overload protection, and a chain-driven geared rotary-type automatic limit switch. Use only hardened steel machine cut worm and mating bronze gears that operate in a bath of oil. Gate operators with V-belt pulleys are not allowed. Equip gate operators with an emergency release to allow the gate to be operated manually that is also capable of being locked in the engaged or disengaged position. Provide positive stops on the gate tracks as a backup to the limit switches.

2.1.5 Electro-Mechanical Locks

Provide electro-mechanical locking devices for sliding gates and personnel gates that are solenoid actuated such that the deadbolt retracts when the solenoid is energized and remains electrically retracted until the gate is closed. Provide continuous duty type solenoid, rated for 120V ac, 60Hz operation. Ensure the locking device is unlockable by key and keyed on both sides. Monitor status of the electro-mechanical lock by two limit switches (integral to the locking device) wired in series. Ensure one switch monitors the deadlock lever and the other monitors the locking tongue.

2.2 MATERIALS

2.2.1 Concrete

ASTM C94/C94M, using 3/4 inch maximum size aggregate, and having minimum compressive strength of 3000 psi at 28 days. Use grout consisting of one part portland cement to three parts clean, well-graded sand and the minimum amount of water to produce a workable mix.

PART 3 EXECUTION

3.1 PREPARATION

Perform complete installation conforming to ASTM F567.

3.1.1 Line and Grade

Install fence to the lines and grades indicated. Clear the area on either side of the fence line to the extent indicated. Space line posts equidistant at intervals not exceeding 10 feet. Set terminal (corner, gate, and pull) posts whenever abrupt changes in vertical and horizontal alignment are encountered. Provide continuous fabric between terminal posts; however, ensure runs between terminal posts do not exceed 500 feet. Repair any damage to galvanized surfaces, including welding, with paint containing zinc dust in accordance with ASTM A780/A780M.

3.1.2 Excavation

Excavate holes to depths indicated. Clear all post holes of loose material and spread waste material where directed. Eliminate ground surface irregularities along the fence line to the extent necessary to maintain a 2 inch clearance between the bottom of the fabric and finish

grade.

3.2 INSTALLATION

3.2.1 Installation Drawings

Submit complete Fence Installation Drawings for review and approval by the Contracting Officer prior to shipment. Submit drawing details that include, but are not limited to the following information: Fence Installation Drawings, Location of gate, corner, end, and pull posts, Gate Assembly, Turnstiles, and Gate Hardware and Accessories. Install fence system per approved drawings.

3.2.2 Security Fencing

Install new security fencing, remove existing security fencing, and perform related work to provide continuous security for facility. Schedule and fully coordinate work with Contracting Officer.

3.2.3 Posts

3.2.3.1 Earth and Bedrock

a. Set posts plumb and in alignment. Except where solid rock is encountered, set posts in concrete to the depth indicated on the drawings. Where solid rock is encountered with no overburden, set posts to a minimum depth of 18 inches in rock. Where solid rock is covered with an overburden of soil or loose rock, set posts to the minimum depth indicated on the drawing unless a penetration of 18 inches in solid rock is achieved before reaching the indicated depth, in which case terminate depth of penetration. Grout all portions of posts set in rock.

b. Set portions of posts not set in rock in concrete from the rock to ground level. Set posts in holes not less than the diameter shown on the drawings. Make diameters of holes in solid rock at least 1 inch greater than the largest cross section of the post. Thoroughly consolidate concrete and grout around each post, free of voids and finished to form a dome. Allow concrete and grout to cure for 72 hours prior to attachment of any item to the posts. Group II line posts may be mechanically driven, for temporary fence construction only, if rock is not encountered. Set driven posts to a minimum depth of 3 feet and protect with drive caps when setting.

c. Test fence post rigidity by applying a 50 pound force on the post, perpendicular to the fabric, at 5 feet above ground. Ensure post movement measured at the point where the force is applied is less than or equal to 3/4 inch from the relaxed position. Test every tenth post for rigidity. When a post fails this test, make further tests on the next four posts on either side of the failed post. Remove, replace, and retest all failed parts at the Contractor's expense.

3.2.3.2 Concrete Slabs and Walls

When installed in concrete slabs or walls, set posts in zinc-coated sleeves, to a minimum depth of 12 inches. Fill sleeve joint with lead, nonshrink grout, or other approved material. Set posts for support of removable fence sections in sleeves that provide a tight sliding joint and hold posts aligned and plumb without use of lead or setting material.

3.2.4 Rails

Install in accordance with manufacturers instructions.

3.2.5 Gate Installation

a. Install gates at the locations shown. Mount gates to swing as indicated. Install latches, stops, and keepers as required. Install gates as recommended by the manufacturer.

b. Secure hinge pins, and hardware assembly to prevent removal.

c. Submit 6 copies of operating and maintenance instructions. Outline the step-by-step procedures required for system startup, operation, and shutdown. Include the manufacturer's name, model number, service manual, parts list, and brief description of all equipment and their basic operating features. Include in the maintenance instructions routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guide. Also include the general gate layout, equipment layout and simplified wiring and control diagrams of the system as installed.

3.2.6 Grounding

a. Ground fencing as specified.

b. Ground fences crossed by overhead powerlines in excess of 600 volts and ground all electrical equipment attached to the fence.

c. Ground fences on each side of all gates, at each corner, at the closest approach to each building located within 50 feet of the fence, and where the fence alignment changes more than 15 degrees. Ensure grounding locations are located no more than 650 feet apart. Bond each gate panel with a flexible bond strap to its gate post. Ground fences crossed by powerlines of 600 volts or more at or near the point of crossing and at distances not exceeding 150 feet on each side of crossing.

d. Provide ground conductor consisting of No. 8 AWG solid copper wire. Use grounding electrodes that measures 3/4 inch by 10 foot long and are a solid copper rod. Drive electrodes into the earth so that the top of the electrode is at least 6 inches below the grade. Where driving is impracticable, bury electrodes a minimum of 12 inches deep and radially from the fence. Install the top of the electrode to be less than 2 feet or more than 8 feet from the fence. Clamp ground conductor to the fence and electrodes with bronze grounding clamps to create electrical continuity between fence posts, fence fabric, and ground rods. Measure total resistance of the fence to ground and ensure it is not greater than 25 ohms.

3.3 CLOSEOUT ACTIVITIES

3.3.1 Cleanup

Remove waste fencing materials and other debris from the work site each workday.

-- End of Section --

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DIVISION 33 - UTILITIES

SECTION 33 08 00.01

DVA/USACE PROJECTS COMMISSIONING OF SITE UTILITY SYSTEMS

8/19

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SECTION 33 08 00.01
8/19

DVA/USACE PROJECTS COMMISSIONING OF SITE UTILITY SYSTEMS

PART 1 GENERAL

1.1 DESCRIPTION

a. The requirements of this Section apply to all sections of Division 31.

b. This project will have selected building systems commissioned. The complete list of equipment and systems to be commissioned is specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS. The commissioning process, which the Contractor is responsible to execute, is defined in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS. A Commissioning Agent (CxA) appointed by the Government and Commissioning Manager (CxM) as indicated in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIRMENTS will manage the commissioning process.

1.2 RELATED WORK

a. UFGS GENERAL REQUIREMENTS sections

b. Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS

c. UFGS Section 01 33 00 SUBMITTAL PROCEDURES

1.3 SUMMARY

a. This Section includes requirements for commissioning the Facility site utilities systems, related subsystems and related equipment. This Section supplements the general requirements specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

b. Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for more details regarding processes and procedures as well as roles and responsibilities for all Commissioning Team members.

1.4 DEFINITIONS

Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for definitions.

1.5 COMMISSIONED SYSTEMS

a. Commissioning of a system or systems specified in Division 31 is part of the construction process. Documentation and testing of these systems, as well as training of the VA’s Operation and Maintenance personnel in accordance with the requirements of Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and of Division 31, is required in cooperation with the DVA/USACE and the CxA and CxM.

b. The Facility site utilities systems commissioning will include the systems listed in Section 01 91 00.01 DVA/USACE Projects GENERAL

COMMISSIONING REQUIREMENTS:

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Construction Inspections; G, DO

Pre-Functional Checklists; G, DO

Contractors Tests; G, DO

Systems Functional Performance Testing; G, DO

SD-08 Manufacturer's Instructions

Factory Authorized Personnel; G, DO

Training Agendas And Trainer Resumes; G, DO

Training Plans; G, DO

1.7 SUBMITTAL INSTRUCTIONS

1. The commissioning process requires review of selected Submittals that pertain to the systems to be commissioned. The CxA will provide a list of submittals that will be reviewed by the CxA and CxM. This list will be reviewed and approved by the VA prior to forwarding to the Contractor. Refer to UFGS Section 01 33 00 SUBMITTAL PROCEDURES for further details.

2. The commissioning process requires Submittal review simultaneously with engineering review. Specific submittal requirements related to the commissioning process are specified in Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 CONSTRUCTION INSPECTIONS

Commissioning of Site Utility systems will require inspection of individual elements of the site utility systems construction throughout the construction period. The Contractor shall coordinate with the CxA and CxM in accordance with Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and the Commissioning plan to schedule site utility systems inspections as required to support the Commissioning Process.

3.2 PRE-FUNCTIONAL CHECKLISTS

The Contractor shall complete Pre-Functional Checklists to verify systems, subsystems, and equipment installation is complete and systems are ready for Systems Functional Performance Testing. The CxM will prepare

Pre-Functional Checklists to be used to document equipment installation. The Contractor shall complete the checklists. Completed checklists shall be submitted to the DVA/USACE, CxA, and CxM for review. The Commissioning Agent may spot check a sample of completed checklists. If the CxM and CxA determine that the information provided on the checklist is not accurate, the CxM will return the marked-up checklist to the Contractor for correction and resubmission. If the CxM and CxA determine that a significant number of completed checklists for similar equipment are not accurate, the CxM in coordination with CxA will select a broader sample of checklists for review. If the CxM and CxA determine that a significant number of the broader sample of checklists is also inaccurate, all the checklists for the type of equipment will be returned to the Contractor for correction and resubmission. Refer to SECTION 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS for submittal requirements for Pre-Functional Checklists, Equipment Startup Reports, and other commissioning documents.

3.3 CONTRACTORS TESTS

Contractor tests as required by other sections of Division 31 shall be scheduled and documented in accordance with UFGS GENERAL REQUIREMENTS sections. All testing shall be incorporated into the project schedule. Contractor shall provide no less than 7 calendar days' notice of testing. The CxM will witness selected Contractor tests at the sole discretion of the CxM and CxA. The CxA and other Government representative(s) may choose to participate in some or all the testing procedures as observers. Contractor tests shall be completed prior to scheduling Systems Functional Performance Testing.

3.4 SYSTEMS FUNCTIONAL PERFORMANCE TESTING

The Commissioning Process includes Systems Functional Performance Testing that is intended to test systems functional performance under steady state conditions, to test system reaction to changes in operating conditions, and system performance under emergency conditions. The CxM will prepare final and detailed Systems Functional Performance Test procedures for review and approval by the CxA and Contracting Officer. The Contractor shall review and comment on the tests prior to approval. The Contractor shall provide the required labor, materials, and test equipment identified in the test procedure to perform the tests. The CxM will witness and document the testing. The Contractor shall sign the test reports to verify tests were performed. See Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS, for additional details.

3.5 TRAINING OF VA PERSONNEL

Training of the VA operation and maintenance personnel is required in cooperation with the Contracting Officer, VA Resident Engineer and CxM. Provide competent, factory authorized personnel to provide instruction to operation and maintenance personnel concerning the location, operation, and troubleshooting of the installed systems. Contractor shall submit training agendas and trainer resumes in accordance with the requirements of Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS. The instruction shall be scheduled in coordination with the Contracting Officer and VA Resident Engineer after submission and approval of formal training plans. Refer to Section 01 91 00.01 DVA/USACE Projects GENERAL COMMISSIONING REQUIREMENTS and Division 31 Sections for additional Contractor training requirements.

----- END -----

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SECTION 33 10 00
3/17

WATER UTILITIES

PART 1 GENERAL

1.1 DESCRIPTION

This section specifies materials and procedures for construction of underground water distribution for domestic and/or fire supply systems outside the building that are complete and ready for operation. This includes piping, structures, appurtenances and all other incidentals.

1.2 RELATED WORK

a. Excavation, Trench Widths, Pipe Bedding, Backfill, Shoring, Sheeting, Bracing: Section 31 20 00, EARTH MOVING.

b. Concrete: Section 03 30 00, CAST IN-PLACE CONCRETE.

c. Fire Protection System connection: Section 21 12 00, FIRE-SUPPRESSION STANDPIPES.

d. General plumbing: Section 22 05 11, COMMON WORK RESULTS FOR PLUMBING.

e. Submittals: Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA AND SAMPLES.

f. Metering: Section 25 10 10, ADVANCED UTILITY METERING SYSTEM.

g. Erosion and Sediment Control: Section 01 57 19, TEMPORARY ENVIRONMENTAL CONTROLS.

1.3 DEFINITIONS

a. Water distribution system: Pipelines and appurtenances which are part of the distribution system outside the building for potable water and fire supply.

b. Water service line: Pipeline from main line to 5 feet outside of building.

1.4 ABBREVIATIONS

a. DI: Ductile iron pipe.

1.5 DELIVERY, STORAGE AND HANDLING

a. Ensure that valves are dry and internally protected against rust and corrosion. Protect valves against damage to threaded ends and flange faces.

b. Use a sling to handle valves and fire hydrants if size requires handling by crane or lift. Rig valves to avoid damage to exposed parts. Do not use handwheels or stems as lifting or rigging points.

c. Deliver piping with factory-applied end caps. Maintain end caps

through shipping, storage, and handling to prevent pipe-end damage and to prevent entrance of dirt, debris, and moisture.

d. Protect stored piping from moisture and dirt by elevating above grade. Protect flanges, fittings, and specialties from moisture and dirt.

e. Store plastic piping protected from direct sunlight and support to prevent sagging and bending.

f. Cleanliness of Piping and Equipment Systems:

(1) Care shall be exercised in the storage and handling of equipment and piping material to be incorporated in the work. Debris arising from cutting, threading and welding of piping shall be removed.

(2) Piping systems shall be flushed, blown or pigged as necessary to deliver clean systems.

1.6 COORDINATION

a. Coordinate connection to water main with Public Utility company.

b. Coordinate water service lines with building contractor.

1.7 QUALITY ASSURANCE

a. Products Criteria:

(1) When two or more units of the same type or class of materials or equipment are required, these units shall be products of one manufacturer.

(2) A nameplate bearing manufacturer's name or trademark, including model number, shall be securely affixed in a conspicuous place on equipment. In addition, the model number shall be either cast integrally with equipment, stamped, or otherwise permanently marked on each item of equipment.

b. Materials and equipment shall be the standard products of a manufacturer regularly engaged in the manufacture of the products for at least three years. Digital electronic devices, software and systems such as controls, instruments or computer work stations shall be the current generation of technology and basic design that has a proven satisfactory service record of at least three years.

c. Regulatory requirements:

(1) Comply with the rules and regulations of the public utility company having jurisdiction over the connection to public water lines and the extension and/or modifications to public utility systems.

(2) Comply with the rules and regulations of the Federal Stateand/or Local Health Department having jurisdiction for potable water-service.

(3) Comply with rules and regulations of Federal State and/or Local

authorities having jurisdiction for fire-suppression water-service piping including materials, hose threads, installation and testing.

d. Provide certification of factory hydrostatic testing of not less than 500 psi in accordance with AWWA C151/A21.51. Piping materials shall bear the label, stamp or other markings of the specified testing agency.

e. Before any welding is performed, contractor shall submit a certificate certifying that welders comply with the following requirements:

(1) Qualify welding processes and operators for piping according to ASME "Boiler and Pressure Vessel Code", Section IX, "Welding and Brazing Qualifications".

(2) Comply with provisions of ASME B31 series "Code for Pressure Piping".

(3) Provide Welder Qualifications to certify that each welder has passed American Welding Society (AWS) qualification tests for the welding processes involved, and that certification is current.

(4) All welds shall be stamped according to the provisions of the American Welding Society.

f. Where installation procedures or any part thereof are required to be in accordance with the recommendations of the manufacturer of the material being installed, printed copies of these recommendations shall be furnished to the Resident Engineer prior to installation.

g. Applicable codes:

(1) Plumbing Systems: IPC, International Plumbing Code.

(2) Electrical components, devices and accessories shall be listed and labeled as defined in NFPA 70, Article 100, by a testing agency acceptable to authorities having jurisdiction and marked for intended use.

(3) Fire-service main products shall be listed in the FM Global “Approval Guide” or Underwriters Laboratories (UL) “Fire Protection Equipment Directory”.

h. Provide Material and Test Certificate for Underground Piping in accordance with NFPA 24.

1.8 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced in the text by the basic designation only.

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-60	Connecting Flange Joint Between Tapping Sleeves and Tapping Valves

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A112.6.3	(2022) Floor Drains
ASME B16.1	(2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250
ASME B31	Code for Pressure Piping Standards

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A123/A123M	(2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A674	Polyethylene Encasement for Ductile Iron Pipe for Water or Other Liquids
ASTM C891	(2020) Standard Practice for Installation of Underground Precast Concrete Utility Structures
ASTM C858	(2010; E 2012) Standard Specification for Underground Precast Concrete Utility Structures
ASTM C1802	(2023; E 2023) Standard Specification for Design, Testing, Manufacture, Selection, and Installation of Horizontal Fabricated Metal Access Hatches for Utility, Water, and Wastewater Structures.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA B300	(2024) Hypochlorites
AWWA B301	(2024) Liquid Chlorine
AWWA C104/A21.4	(2022) Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for Water
AWWA C105/A21.5	(2018) Polyethylene Encasement for Ductile-Iron Pipe Systems
AWWA C110/A21.10	(2021) Ductile-Iron and Gray-Iron Fittings
AWWA C111/A21.11	(2023) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings
AWWA C115/A21.15	(2020) Flanged Ductile-Iron Pipe With Ductile-Iron or Gray-Iron Threaded Flanges
AWWA C151/A21.51	(2023) Ductile-Iron Pipe, Centrifugally Cast
AWWA C153/A21.53	(2019) Ductile-Iron Compact Fittings for

	Water Service
AWWA C502	(2024) Dry-Barrel Fire Hydrants
AWWA C509	(2023) Resilient-Seated Gate Valves for Water Supply Service
AWWA C512	(2015) Air-Release, Air/Vacuum, and Combination Air Valves for Water and Wastewater Service
AWWA C550	(2024) Protective Interior Coatings for Valves and Hydrants
AWWA C600	(2023) Installation of Ductile-Iron Mains and Their Appurtenances
AWWA C651	(2023) Standard for Disinfecting Water Mains
AWWA M6	(2012) Water Meters - Selection, Installation, Testing, and Maintenance
AWWA M23	(2020) Manual: PVC Pipe - Design and Installation - Third Edition
AWWA M41	(2009; 3rd Ed) Ductile-Iron Pipe and Fittings
AWWA M44	Distribution Valves: Selection, Installation, Field Testing and Maintenance

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 24	(2025) Standard for the Installation of Private Fire Service Mains and Their Appurtenances
NFPA 70	(2023; ERTA 1 2024; TIA 24-1) National Electrical Code
NFPA 1963	(2019) Standard for Fire Hose Connections

NSF INTERNATIONAL (NSF)

NSF/ANSI 14	(2023) Plastics Piping System Components and Related Materials
NSF 372	(2022) Drinking Water System Components - Lead Content
NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects

UL SOLUTIONS (UL)

UL 789	(2004; Reprint May 2024) UL Standard for Safety Indicator Posts for Fire-Protection Service

1.9 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Welder Qualifications; G,

SD-03 Product Data

Safety Guards; G,

Lifting Attachments; G,

Ductile Iron Pipe And Fittings; G,

Valves; G

Concrete Vaults; G,

Fire Hydrants; G,

Disinfection Chlorine; G,

Warning Tape; G,

SD-04 Samples

Water Samples; G,

SD-07 Certificates

Certification Of Factory Hydrostatic Testing; G,

Material and Test Certificate for Underground Piping in accordance with NFPA 24; G,

SD-11 Closeout Submittals

Warranty; G,

1.10 WARRANTY

The Contractor shall remedy any defect due to faulty material or workmanship and pay for any damage to other work resulting therefrom within a period of one year from final acceptance. Further, the Contractor will furnish all manufacturers’ and supplier's written guarantees and warranties covering materials and equipment furnished under this Contract.

PART 2 PRODUCTS

2.1 MATERIALS

a. Material or equipment containing a weighted average of greater than

0.25 percent lead shall not be used in any potable water system intended for human consumption, and shall be certified in accordance with NSF/ANSI 61 or NSF 372.

b. Plastic pipe, fittings, and solvent cement shall meet NSF/ANSI 14 and shall be NSF listed for the service intended.

2.2 FACTORY-ASSEMBLED PRODUCTS

Standardization of components shall be maximized to reduce spare part requirements. The contractor shall guarantee performance of assemblies of components, and shall repair or replace elements of the assemblies as required to deliver specified performance of the complete assembly.

2.3 SAFETY GUARDS

All equipment shall have moving parts protected to prevent personal injury. Pump shafts and couplings shall be fully guarded by a sheet steel guard, covering coupling and shaft but not bearings. Material shall be minimum 16-gauge sheet steel; ends shall be braked and drilled and attached to pump base with minimum of four 1/4 inch bolts. Reinforce guard as necessary to prevent side play forcing guard onto couplings.

2.4 LIFTING ATTACHMENTS

Equipment shall be provided with suitable lifting attachments to enable equipment to be lifted in its normal position. Lifting attachments shall withstand any handling conditions that might be encountered, without bending or distortion of shape, such as rapid lowering and braking of load.

2.5 DUCTILE IRON PIPE AND FITTINGS

a. Mechanical-Joint, Ductile-Iron Pipe: AWWA C151/A21.51, with mechanical-joint bell and plain spigot end unless grooved or flanged ends are indicated, 350 psi .

(1) Mechanical-Joint, Ductile-Iron Fittings: AWWA C110/A21.10, ductile- or gray-iron standard pattern or AWWA C153/A21.53, ductile-iron compact pattern.

(2) Glands, Gaskets, and Bolts: AWWA C111/A21.11, ductile- or gray-iron glands, rubber gaskets, and steel bolts.

b. Push-on-Joint, Ductile-Iron Pipe: AWWA C151/A21.51, with push-on-joint bell and plain spigot end unless grooved or flanged ends are indicated, 350 psi .

(1) Push-on-Joint, Ductile-Iron Fittings: AWWA C110/A21.10, ductile- or gray-iron standard pattern or AWWA C153/A21.53, ductile-iron compact pattern.

(2) Gaskets: AWWA C111/A21.11, rubber.

c. Flanged Ductile Iron Pipe: AWWA C115/A21.15, with factory applied screwed long hub flanges.

(1) Flanges: ASME B16.1 for or 250 psi pressure ratings, as necessary.

(2) Wall Sleeve Castings, size and types shown on the drawings, shall be hot dipped galvanized per ASTM A123/A123M.

(3) Pipe and fittings exposed to view in the finished work are to be painted in accordance with Section 09 91 00, PAINTING. Pipe shall be shop primed with one coat of rust inhibitive primer. Final paint color shall match the final wall color.

e. Cement Mortar Internal Lining: Cement mortar lining and bituminous seal coat as per AWWA C104/A21.4.

f. Exterior Pipe Coating: The exterior of pipe shall have the standard asphaltic coating.

2.6 VALVES

a. Gate Valves: AWWA C509, Non-rising Stem, Resilient Seat, 200 psi .

(1) Valves 3 inches and larger: Resilient seat valve with gray- or ductile iron body and bonnet; cast iron or bronze double-disc gate; bronze gate rings; non-rising bronze stem and stem nut.

(2) Interior and exterior coating: AWWA C550, thermo-setting or fusion epoxy.

(3) Underground valve nut: Furnish valves with 2 inch nut for socket wrench operation.

(5) End connections shall be mechanical joint be push on .

b. Gate Valve Accessories and Specialties

(1) Tapping-Sleeve Assembly: MSS SP-60; sleeve and valve to be compatible with the drilling matching.

(a) Tapping Sleeve: Stainless-Steel, two-piece bolted sleeve. Sleeve to match the size and type of pipe material being tapped.

(b) Valve shall include one raised face flange mating tapping-sleeve flange.

(2) Valve Boxes: AWWA M44 with top section, adjustable extension of length required for depth of burial of valve, plug with lettering "WATER," and bottom section with base that fits over valve and with a barrel.

(3) Operating Wrenches: Steel, tee-handle with one pointed end, stem of length to operate deepest buried valve, and socket matching valve operating nut. (Provide two wrenches for Project.)

(4) Indicator Posts: UL 789, FMG approved, vertical-type, cast iron body with operating wrench, extension rod, and adjustable cast iron barrel of length required for depth of burial of valve.

h. Post-Indicator: NFPA 24 and be fully compatible with the valve and supervisory switches.

i. Water Meter: SECTION 25 10 10, ADVANCED UTILITY METERING SYSTEM.

(2) Water Meter will be furnished by Water Service Utility and installed by Contractor.

(3) Furnish and install meter approved by the Water Service Utility. Forward approval of meter to Government Contracting Officer.

l. Relief Valves

(1) Air-Release Valve: AWWA C512, Hydromechanical device to automatically release accumulated air.

(2) Pressure Rating: 300 psi

(3) Body material: Cast iron

(4) Trim material: Stainless steel

(5) Water inlet size: 2 to 3 inch

(6) Air outlet size: 2-inches

(7) Orifice size: 1/16 to 3/8 inch

(8) Design air-release capacity: Between 5 and 150 cfm at existing main pressure

2.7 CONCRETE VAULTS

Precast, reinforced-concrete vault: ASTM C858, designed for AASHTO H20-44 load designation.

(1) Ladder: ASTM A36/A36M, steel or polyethylene-encased steel steps.

(2) Drain: ASME A112.6.3, cast iron floor drain with outlet. Include body anchor flange, light-duty cast iron grate, bottom outlet, and integral or field-installed bronze ball or clapper-type backwater valve.

(3) Provide access hatch in accordance with ASTM C1802, size as indicated on drawings.

2.8 FIRE HYDRANTS

a. All hydrants shall have removable interiors capable of replacement without digging up the hydrant and be packable under pressure. Threaded joints or spindles shall be bronze and upper and lower barrels shall be of equal diameter. Upper barrel shall be of sufficient length to permit setting hydrant with barrel flange not more than 4 inches above finished grade. All fire hydrants shall have 6 inch bottom connection. Provide 250 psi. Hydrant valve shall open by turning operating nut to left or counterclockwise. Exterior finish shall be red alkyd-gloss enamel paint, unless otherwise indicated. Outlet threads shall meet NFPA 1963, with external hose thread used by local fire department. Include cast iron caps with steel chains and Pentagon, 1-1/2 inch point to flat operating and cap nuts.

b. Dry-Barrel Fire Hydrants:

(1) AWWA C502, freestanding, one NPS 4-1/2 (DN 115) and two NPS 2-1/2

(DN 65) outlets, 5-1/4 inch main valve, drain valve, and NPS 6 (DN 150) mechanical-joint inlet; interior coating according to AWWA C550; cast iron body, compression-type valve opening against pressure and closing.

2.9 DISINFECTION CHLORINE

a. Liquid chlorine: AWWA B301.

b. Sodium Hypochlorite: AWWA B300 with 5 percent to 15 percent available chlorine.

c. Calcium hypochlorite: AWWA B300 supplied in granular form of 5 g. tablets, and shall contain 65 percent chlorine by weight.

2.10 WARNING TAPE

Warning tape shall be standard, 4 mil. Polyethylene, 3 inch wide tape, non-detectable type, blue with black letters and imprinted with “CAUTION BURIED WATER LINE BELOW”.

PART 3 EXECUTION

3.1 PIPING APPLICATIONS

a. Use pipe, fittings, and joining methods for piping systems according to the following applications.

(1) Transition couplings and special fittings with pressure ratings at least equal to piping pressure rating may be used, unless otherwise indicated.

(2) Do not use flanges or unions for underground piping.

(3) Flanges, unions, grooved-end-pipe couplings, and special fittings may be used, instead of joints indicated, on aboveground piping and piping in vaults.

b. Underground water-service piping shall be the following:

(2) Ductile iron, push-on-joint pipe; ductile iron, push-on-joint fittings; and gasketed mechanical-joint pipe; ductile iron, mechanical-joint fittings; and mechanical joints.

c. Underground Fire-Service-Main Piping shall be the following:

(1) Ductile iron, push-on-joint pipe; ductile iron, push-on-joint fittings; and gasketed mechanical-joint pipe; ductile iron, mechanical-joint fittings; and mechanical joints.

3.2 VALVE APPLICATIONS

a. Use mechanical-joint-end valves for NPS 3 (DN 80) and larger underground installation. Use threaded- or flanged-end valves for installation in vaults. Use UL/FMG, non-rising-stem gate valves for installation with indicator posts. Use corporation valves and curb valves with ends compatible with piping, for NPS 2 (DN 50) and smaller installation.

b. Drawings indicate valve types to be used. Where specific valve types are not indicated, the following requirements apply:

(1) Underground Valves, NPS 3 (DN 80) and Larger: AWWA, cast iron, non-rising-stem, resilient -seated gate valves with valve box.

(2) Underground Valves, NPS 4 (DN 100) and Larger, for Indicator Posts: UL/FMG, cast iron, non-rising-stem gate valves with indicator post.

3.3 DUCTILE IRON PIPE

a. Install Ductile Iron, water-service piping according to AWWA C600 and AWWA M41-3rd Edition.

(1) Install PE corrosion-protection encasement according to ASTM A674 or AWWA C105/A21.5.

b. Pipe shall be sound and clean before laying. When laying is not in progress, the open ends of the pipe shall be closed by watertight plug or other approved means.

c. When cutting pipe is required, the cutting shall be done by machine, leaving a smooth cut at right angles to the axis of the pipe. Bevel cut ends of pipe to be used with push-on bell to conform to the manufactured spigot end. Cement lining shall be undamaged.

d. Push on joints shall be made in strict accordance with the manufacturer’s instruction. Pipe shall be laid with bell ends looking ahead.

3.4 ANCHORAGE INSTALLATION

a. Install water-distribution piping with restrained joints. Anchorages and restrained-joint types that may be used include: concrete thrust blocks, locking mechanical joints, set-screw mechanical retainer glands, bolted flanged joints, pipe clamps and tie rods, .

b. Install anchorages for tees, plugs and caps, bends, crosses, valves, and hydrant branches. Include anchorages for the following piping systems:

(1) Gasketed-Joint, Ductile Iron, Water-Service Piping: According to AWWA C600.

(2) Gasketed-Joint, PVC Water-Service Piping: According to AWWA M23.

(3) Fire-Service-Main Piping: According to NFPA 24.

c. Apply full coat of asphalt or other acceptable corrosion-resistant material to surfaces of installed ferrous anchorage devices.

3.5 VALVE INSTALLATION

a. AWWA Valves: Install each underground valve with stem pointing up and with valve box.

b. UL/FMG, Valves: Install each underground valve and valves in vaults with stem pointing up and with vertical cast iron indicator post.

c. MSS Valves: Install as component of connected piping system.

d. Corporation Valves and Curb Valves: Install each underground curb valve with head pointed up and with service box.

g. Raise or lower existing valve and curb stop boxes and fire hydrants to finish grade in areas being graded.

3.6 WATER METER INSTALLATION

a. Install water meters, piping, and specialties according to utility company's written instructions.

d. Install detector-type water meters in meter vault according to AWWA M6. Include shutoff valves on water meter inlets and outlets and full-size valved bypass around meters. Support meters, valves, and piping on brick or concrete piers.

3.7 ROUGHING-IN FOR WATER METERS

Rough-in piping and specialties for water meter installation according to utility company's written instructions.

3.8 CONCRETE VAULT INSTALLATION

Install precast concrete vaults according to ASTM C891.

3.9 FIRE HYDRANT INSTALLATION

a. Install each fire hydrant with separate gate valve in supply pipe, anchor with restrained joints or thrust blocks, and support in upright position.

b. Fully open and close each hydrant under system water pressure in accordance with NFPA 24.

3.10 CONNECTIONS

Drawings indicate general arrangement of piping, fittings, and specialties. Install water service lines to a point of connection within approximately 5 feet outside of building(s) to which service is to be connected and make connections thereto. If building services have not been installed provide temporary caps and mark for future connection.

3.11 FIELD QUALITY CONTROL

a. Conduct piping tests before joints are covered and after concrete thrust blocks have hardened sufficiently. Fill pipeline 24 hours before testing and apply test pressure to stabilize system. Use only potable water.

b. Prior to final acceptance, provide a video record of all piping from the building to the municipal connection to show the lines are free from obstructions, properly sloped and joined.

c. Perform hydrostatic tests at not less than one-and-one-half times working pressure for two hours.

(1) Increase pressure in 50-psi increments and inspect each joint between increments. Hold at test pressure for 1 hour; decrease to 0 psi . Slowly increase again to test pressure and hold for 1 more hour. Maximum allowable leakage is 2 quarts per hour per 100 joints. Remake leaking joints with new materials and repeat test until leakage is within allowed limits.

d. For private fire service mains, perform hydrostatic tests at 200 psi or 50 psi in excess of the system work pressure, whichever is greater, in accordance with NFPA 24.

e. For private fire serviced mains, forward flow test backflow preventer(s) in accordance with NFPA 24.

f. Prepare reports of testing activities.

3.12 IDENTIFICATION

Install continuous underground warning tape 12 inches directly over piping.

3.13 CLEANING

a. Purge new water-distribution piping systems and parts of existing systems that have been altered, extended, or repaired before use.

b. Use purging and disinfecting procedure prescribed by local utility provider or other authorities having jurisdiction or, if method is not prescribed by authorities having jurisdiction, use procedure described in AWWA C651 or do as follows:

(1) Fill the water system with a water/chlorine solution containing at least 50 ppm of chlorine; isolate and allow to stand for 24 hours.

(2) Drain the system of the previous solution and refill with water/chlorine solution containing at least 200 ppm of chlorine; isolate and allow system to stand for 3 hours.

(3) After standing time, flush system with clean, potable water until no chlorine remains in water coming from system.

(4) Submit water samples in sterile bottles to authorities having jurisdiction. Repeat procedure if biological examination shows evidence of contamination.

c. Prepare reports of purging and disinfecting activities.

-- End of Section --

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DIVISION 33 - UTILITIES

SECTION 33 30 00

SANITARY SEWERAGE

05/18

PART 1 GENERAL

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SECTION 33 30 00

SANITARY SEWERAGE
05/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C111/A21.11	(2023) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings
AWWA C600	(2023) Installation of Ductile-Iron Mains and Their Appurtenances
AWWA C605	(2021) Underground Installation of Polyvinyl Chloride (PVC) and Molecularly Oriented Polyvinyl Chloride (PVCO) Pressure Pipe and Fittings
AWWA C900	(2022) Polyvinyl Chloride (PVC) Pressure Pipe, and Fabricated Fittings, 4 In. Through 60 In. (100 mm Through 1,500 mm)

ASTM INTERNATIONAL (ASTM)

ASTM A48/A48M	(2022) Standard Specification for Gray Iron Castings
ASTM A123/A123M	(2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A536	(2024) Standard Specification for Ductile Iron Castings
ASTM C94/C94M	(2025) Standard Specification for Ready-Mixed Concrete
ASTM C443	(2021) Standard Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets
ASTM C478	(2018) Standard Specification for Circular Precast Reinforced Concrete Manhole Sections
ASTM C478M	(2018) Standard Specification for Precast Reinforced Concrete Manhole Sections (Metric)

ASTM C972	(2024) Standard Test Method for Compression-Recovery of Tape Sealant
ASTM C1244	(2020) Standard Test Method for Concrete Sewer Manholes by the Negative Air Pressure (Vacuum) Test Prior to Backfill
ASTM C1644	(2006; R 2017) Standard Specification for Resilient Connectors Between Reinforced Concrete On-Site Wastewater Tanks and Pipes
ASTM D412	(2016; R 2021) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension
ASTM D624	(2000; R 2020) Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers
ASTM D2321	(2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications
ASTM D3034	(2016) Standard Specification for Type PSM Poly(Vinyl Chloride) (PVC) Sewer Pipe and Fittings
ASTM D3212	(2020) Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals
ASTM D4101	(2017) Standard Classification System and Basis for Specification for Polypropylene Injection and Extrusion Materials
ASTM F477	(2014; R 2021) Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe
ASTM F949	(2025) Standard Specification for Poly(Vinyl Chloride) (PVC) Corrugated Sewer Pipe with a Smooth Interior and Fittings

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.27	(Nov 2016) Scaffolds and Roope Descent Systems

UNI-BELL PVC PIPE ASSOCIATION (UBPPA)

UBPPA UNI-B-6	(1998) Recommended Practice for Low-Pressure Air Testing of Installed Sewer Pipe

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00

SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor's License; G

SD-02 Shop Drawings

Installation Drawings; G

SD-03 Product Data

Precast Concrete Manholes; G

Frames, Covers, and Gratings; G

Gravity Pipe; G

SD-06 Test Reports

Precast Concrete Sewer Manhole Test; G

Hydrostatic Sewer Test; G

Negative Air Pressure Test; G

Low-Pressure Air Tests; G

1.3 QUALITY CONTROL

1.3.1 Installer Qualifications

Install specified materials by a licensed underground utility Contractor licensed for such work in the state where the work is to be performed. Verify installing Contractor's License is current and state certified or state registered.

1.4 DELIVERY, STORAGE, AND HANDLING

1.4.1 Delivery and Storage

Check upon arrival; identify and segregate as to types, functions, and sizes. Store off the ground in a manner affording easy accessibility and not causing excessive rusting or coating with grease or other objectionable materials.

1.4.1.1 Piping

Inspect materials delivered to site for damage; store with minimum of handling. Store materials on site in enclosures or under protective coverings. Store plastic piping and jointing materials and rubber gaskets under cover out of direct sunlight. Do not store materials directly on the ground. Keep inside of pipes and fittings free of dirt and debris.

1.4.2 Handling

Handle pipe, fittings, and other accessories in such manner as to ensure delivery to the trench in sound undamaged condition. Take special care not to damage linings of pipe and fittings; if lining is damaged, make satisfactory repairs. Carry, do not drag, pipe to trench. Store solvents, solvent compounds, lubricants, elastomeric gaskets, and any similar materials required to install the plastic pipe in accordance with the manufacturer's recommendation and discard those materials if the storage period exceeds the recommended shelf life. Discard solvents in use when the recommended pot life is exceeded.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

2.1.1 Sanitary Sewer Gravity Pipeline

Provide building connections of polyvinyl chloride (PVC) plastic pipe. Provide new and modify existing exterior sanitary gravity sewer piping and appurtenances. Provide each system complete and ready for operation. The exterior sanitary gravity sewer system includes equipment, materials, installation, and workmanship as specified herein more than 5 feet outside of building walls.

2.2 MATERIALS

Provide materials conforming to the respective specifications and other requirements specified below. Submit manufacturer's product specification,standard drawings or catalog cuts.

2.2.1 Gravity Pipe

2.2.1.1 PVC Gravity Sewer Piping

2.2.1.1.1 PVC Gravity Pipe and Fittings

ASTM D3034, SDR 35, or ASTM F949 with ends suitable for elastomeric gasket joints.

2.2.1.1.2 PVC Gravity Joints and Jointing Material

Provide joints conforming to ASTM D3212. Gaskets are to conform to ASTM F477.

2.2.2 Portland Cement Concrete

Provide portland cement concrete conforming to ASTM C94/C94M, compressive strength of 4000 psi at 28 days, except for concrete cradle and encasement or concrete blocks for manholes. Concrete used for cradle and encasement is to have a compressive strength of 2500 psi minimum at 28 days. Protect concrete in place from freezing and moisture loss for 7 days.

2.2.3 Precast Concrete Manholes

Provide precast concrete manholes, risers, base sections, and tops conforming to ASTM C478; base and first riser are to be monolithic.

2.2.4 Gaskets and Connectors

Provide gaskets for joints between manhole sections conforming to ASTM C443. Resilient connectors for making joints between manhole and pipes entering manhole are to conform toASTM C1644.

2.2.5 External Preformed Rubber Joint Seals

An external preformed rubber joint seal is an accepted method of sealing cast iron covers to precast concrete sections to prevent ground water infiltration into sewer systems. All finished and sealed manholes constructed in accordance with paragraph entitled "Manhole Construction" are to be tested for leakage in the same manner as pipelines as described in paragraph entitled "Leakage Tests." The seal is to be multi-section with a neoprene rubber top section and all lower sections made of Ethylene Propylene Diene Monomer (EPDM) rubber with a minimum thickness of 60 mils. Each unit is to consist of a top and bottom section and have mastic on the bottom of the bottom section and mastic on the top and bottom of the top section. The mastic is to be a non-hardening butyl rubber sealant and seal to the cone/top slab of the manhole/catch basin and over the lip of the casting. Extension sections are to cover up to two more adjusting rings. Properties and values are listed in the following table:

Properties, Test Methods and Minimum Values for Rubber used in Preformed Joint Seals				
Physical Properties	Test Methods	EPDM	Neoprene	Butyl Mastic
Tensile, psi	ASTM D412	1840	2195	--
Elongation, percent	ASTM D412	553	295	350
Tear Resistance, ppi	ASTM D624 (Die B)	280	160	--
Rebound, percent, 5 minutes	ASTM C972 (mod.)	--	--	11
Rebound, percent, 2 hours	ASTM C972	--	--	12

2.2.6 Frames, Covers, and Gratings for Manholes

Submit certification on the ability of frame and cover to carry the imposed live load. Frame and cover are to be cast gray iron, ASTM A48/A48M, Class 35B, cast ductile iron, ASTM A536, Grade 65-45-12, or reinforced concrete, ASTM C478 ASTM C478M. Frames and covers are to be circular without vent holes. Size are to be as indicated on the plans. Stamp or cast the words "Sanitary Sewer" into covers so that it is plainly visible.

2.2.7 Manhole Steps

Zinc-coated steel conforming to 29 CFR 1910.27 with a plastic or rubber coating pressure-molded to the steel is to be used. Provide plastic coating conforming to ASTM D4101, copolymer polypropylene. Rubber is to conform to ASTM C443, except shore A durometer hardness is to be 70 plus

or minus 5. Aluminum steps or rungs will not be permitted. Steps are not required in manholes less than 4 feet deep.

2.2.8 Manhole Ladders

Provide a steel ladder where the depth of a manhole exceeds 12 feet. The ladder is not to be less than 16 inches in width, with 3/4 inch diameter rungs spaced 12 inches apart. The two stringers are to be a minimum 3/8 inch thick and 2 inches wide. Galvanize ladders and inserts after fabrication in conformance with ASTM A123/A123M.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Installation Drawings

Submit Installation Drawings showing complete detail, both plan and side view details with proper layout and elevations.

3.2 INSTALLATION

Backfill after inspection by the Contracting Officer. Before, during, and after installation, protect plastic pipe and fittings from any environment that would result in damage or deterioration to the material. Keep a copy of the manufacturer's instructions available at the construction site at all times and follow these instructions unless directed otherwise by the Contracting Officer.

3.2.1 Connections to Existing Lines

Obtain approval from the Contracting Officer before making connection to existing line. Conduct work so that there is minimum interruption of service on existing line.

3.2.2 General Requirements for Installation of Pipelines

These general requirements apply except where specific exception is made in the following paragraphs entitled "Special Requirements."

3.2.2.1 Location

Terminate the work covered by this section at a point approximately 5 feet from the building, unless otherwise indicated.

3.2.2.1.1 Sanitary Sewer Manholes

No water piping must pass through or come in contact with any part of a sanitary sewer manhole.

3.2.2.2 Earthwork

Perform earthwork operations in accordance with Section 31 20 00 EARTHWORK.

3.2.2.3 Pipe Laying and Jointing

Inspect each pipe and fitting before and after installation; replace those found defective and remove from site. Provide proper facilities for lowering sections of pipe into trenches. Lay nonpressure pipe with the

bell ends in the upgrade direction. Adjust spigots in bells to give a uniform space all around. Blocking or wedging between bells and spigots will not be permitted. Replace by one of the proper dimensions, pipe or fittings that do not allow sufficient space for installation of joint material. At the end of each work day, close open ends of pipe temporarily with wood blocks or bulkheads. Provide batterboards not more than 25 feet apart in trenches for checking and ensuring that pipe invert elevations are as indicated. Laser beam method may be used in lieu of batterboards for the same purpose. Construct branch connections by use of regular fittings or solvent cemented saddles as approved. Provide saddles for PVC pipe conforming to Table 4 of ASTM D3034.

3.2.3 Special Requirements

3.2.3.1 Installation of PVC Piping

Install pipe and fittings in accordance with paragraph entitled "General Requirements for Installation of Pipelines" of this section and with the requirements of ASTM D2321 for laying and joining pipe and fittings. Make joints with the gaskets specified for joints with this piping and assemble in accordance with the requirements of ASTM D2321 for assembly of joints. Make joints to other pipe materials in accordance with the recommendations of the plastic pipe manufacturer.

3.2.3.2 Pipe 4 inch Diameter And Larger

Make push-on joints with AWWA C900 pipe with intrgral elastomeric gasket. For pipe-to-pipe push-on joint connections, use only pipe with push-on joint ends having factory-made bevel. For push-on joint connections to fittings, use cut spigot end of pipe off square, marked to match the manufacturer's insertion line and beveled to match factory supplied bevel. Use an approved lubricant recommended by the pipe manufacturer for push-on joints. Assemble push-on joints for pipe-to-pipe joint connections in accordance with the requirements of AWWA C605. Assemble push-on joints for connection to fittings in accordance with the requirements of AWWA C605 for joining PVC pipe to fittings and accessories and with the applicable requirements of AWWA C600 for joint assembly. Make mechanical-joints or flanged jointswith the gaskets, glands, bolts, nuts, and internal stiffeners specified for this type joint and assemble in accordance with the requirements of AWWA C605 for joining PVC pipe to fittings and accessories or with the applicable requirements of AWWA C600 for ductile iron joint assembly, and with the recommendations of Appendix A to AWWA C111/A21.11. Cut off spigot end of pipe for mechanical-joint or flanged jointconnections and do not bevel.

3.2.4 Concrete Work

Cast-in-place concrete is included in Section 03 30 00 CAST-IN-PLACE CONCRETE. Support the pipe on a concrete cradle, or encased in concrete where indicated or directed.

3.3 FIELD QUALITY CONTROL

The Contracting Officer will conduct field inspections and witness field tests specified in this section. Be able to produce evidence, when required, that each item of work has been constructed in accordance with the drawings and specifications.

3.3.1 Tests

Perform field tests and provide labor, equipment, and incidentals required for testing.

3.3.1.1 Hydrostatic Sewer Test

When unusual conflicts are encountered between sanitary sewer and waterlines a hydrostatic pressure sewer test will be performed in accordance with the applicable AWWA standard for the piping material or AWWA C600.

3.3.1.2 Leakage Tests for Nonpressure Lines

Test lines for leakage by either negative air pressure tests or by low-pressure air tests. When necessary to prevent pipeline movement during testing, place additional backfill around pipe sufficient to prevent movement, but leaving joints uncovered to permit inspection. When leakage or pressure drop exceeds the allowable amount specified, make satisfactory correction and retest pipeline section in the same manner. Correct visible leaks regardless of leakage test results.

3.3.1.2.1 Negative Air Pressure Test

3.3.1.2.1.1 Precast Concrete Manholes

Test precast concrete sewer manhole test in accordance with ASTM C1244. The allowable vacuum drop is located in ASTM C1244 Make calculations in accordance with the Appendix to ASTM C1244.

3.3.1.2.2 Low-Pressure Air Tests

3.3.1.2.2.1 PVC Pipelines

Test PVC pipe in accordance with UBPPA UNI-B-6. The allowable pressure drop is located in UBPPA UNI-B-6. Make calculations in accordance with the Appendix to UBPPA UNI-B-6.

3.3.2 Field Tests for Cast-In-Place Concrete

Field testing requirements are covered in Section 03 30 00 CAST-IN-PLACE CONCRETE

3.3.3 Inspection

Check each straight run of pipeline for gross deficiencies by holding a light in a manhole; the light must show a practically full circle of light through the pipeline when viewed from the adjoining end of line.

-- End of Section --

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DIVISION 33 - UTILITIES

SECTION 33 40 00

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11/21

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SECTION 33 40 00

STORMWATER UTILITIES
11/21

PART 1 GENERAL

1.1 UNIT PRICES

1.1.1 Storm Drains

The length of pipe installed will be measured along the centerlines of the pipe from end to end of pipe without deductions for diameter of manholes. Pipe will be paid for at the contract unit price for the number of linear feet of storm drains placed in the accepted work.

1.1.2 Storm Drainage Structures

The quantity of manholes and inlets will be measured as the total number of manholes and inlets of the various types of construction, complete with frames and gratings or covers and, where indicated, with fixed side-rail ladders in the accepted work. The depth of manholes and inlets will be measured from the top of grating or cover to invert of outlet pipe. Manholes and inlets constructed to depths greater than the depth specified above will be paid for as units at the contract unit price for manholes and inlets, plus an additional amount per linear foot for the measured depth beyond what is indocated.

1.1.3 Sheeting and Bracing

Payment will be made for that sheeting and bracing ordered to be left in place, based on the number of square feet of sheeting and bracing remaining below the surface of the ground.

1.1.4 Rock Excavation

Payment will be made for the number of cubic yards of material acceptably excavated, as specified and defined as rock excavation in Section 31 20 00 EARTHWORK, measured in the original position, and computed by allowing actual width of rock excavation with the following limitations: maximum rock excavation width, 30 inches for pipe of 12 inch or less nominal diameter; maximum rock excavation width, 16 inches greater than outside diameter of pipe of more than 12 inch nominal diameter. Measurement will include authorized overdepth excavation. Payment will also include all necessary drilling and blasting, and all incidentals necessary for satisfactory excavation and disposal of authorized rock excavation. No separate payment will be made for backfill material required to replace rock excavation; include this cost in the unit price bid per cubic yard for rock excavation. In rock excavation for manholes and other appurtenances, 1 foot will be allowed outside the wall lines of the structures.

1.1.5 Backfill Replacing Unstable Material

Payment will be made for the number of cubic yards of select granular material required to replace unstable material for foundations under pipes or drainage structures, which will constitute full compensation for this

backfill material, including removal and disposal of unstable material and all excavating, hauling, placing, compacting, and all incidentals necessary to complete the construction of the foundation satisfactorily.

1.1.6 Concrete Ditch Lining

Payment will be made for the number of linear feet of concrete ditch lining including any steel reinforcing accepted in the completed work measured along the centerline of the ditch.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO HB-17	(2002; Errata 2003; Errata 2005, 17th Edition) Standard Specifications for Highway Bridges

ASTM INTERNATIONAL (ASTM)

ASTM A48/A48M	(2022) Standard Specification for Gray Iron Castings
ASTM A123/A123M	(2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A536	(2024) Standard Specification for Ductile Iron Castings
ASTM A807/A807M	(2019) Standard Practice for Installing Corrugated Steel Structural Plate Pipe for Sewers and Other Applications
ASTM B26/B26M	(2018; E 2018) Standard Specification for Aluminum-Alloy Sand Castings
ASTM C231/C231M	(2024) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C270	(2025a) Standard Specification for Mortar for Unit Masonry
ASTM C425	(2021) Standard Specification for Compression Joints for Vitrified Clay Pipe and Fittings
ASTM C443	(2021) Standard Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets
ASTM C443M	(2021) Standard Specification for Joints for Concrete Pipe and Manholes, Using

	Rubber Gaskets (Metric)
ASTM C478/C478M	(2022) Standard Specification for Circular Precast Reinforced Concrete Manhole Sections
ASTM C923/C923M	(2020; R 2025) Standard Specification for Resilient Connectors Between Reinforced Concrete Manhole Structures, Pipes and Laterals
ASTM C990	(2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants
ASTM C990M	(2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants (Metric)
ASTM D1751	(2018) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)
ASTM D1752	(2018) Standard Specification for Preformed Sponge Rubber, Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction
ASTM D2321	(2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications
ASTM D3034	(2016) Standard Specification for Type PSM Poly(Vinyl Chloride) (PVC) Sewer Pipe and Fittings
ASTM D3212	(2020) Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals
ASTM F1417	(2025) Standard Practice for Installation Acceptance of Plastic Non-pressure Sewer Lines Using Low-Pressure Air

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Precast Reinforced Concrete Manholes And Outlet Control Structures; G

SUBSURFACE STORMWATER INFILTRATION CHAMBER; G

SD-06 Test Reports

Leakage Test; G

SD-07 Certificates

Hydrostatic Test on Watertight Joints; G

Frame and Cover or Gratings; G

SD-08 Manufacturer's Instructions

Placing Pipe; G

SD-11 Closeout Submittals

Post-Installation Inspection Report; G

1.4 DELIVERY, STORAGE, AND HANDLING

1.4.1 Delivery and Storage

Inspect materials delivered to site for damage and unload and store materials with minimumal handling. Do not store materials directly on the ground. Keep the inside of pipes and fittings free of dirt and debris. Before, during, and after installation, protect plastic pipe and fittings from any environment that would result in damage or deterioration to the material. Keep a copy of the manufacturer's instructions available at the construction site at all times and follow these instructions unless directed otherwise by the Contracting Officer. Store solvents, solvent compounds, lubricants, elastomeric gaskets, and any similar materials required to install plastic pipe in accordance with the manufacturer's recommendations and discard if the storage period exceeds the recommended shelf life. Discard solvents in use when the recommended pot life is exceeded.

1.4.2 Handling

Handle materials in a manner that ensures delivery to the trench in sound, undamaged condition. Carry pipe to the trench.

PART 2 PRODUCTS

2.1 PIPE FOR STORM DRAINS

Pipe sizes for storm drains are indicated on the drawings.

2.1.1 Poly Vinyl Chloride (PVC) Pipe

2.1.1.1 Type PSM PVC Pipe

ASTM D3034, maximum SDR 35.

2.2 PIPE JOINTS

Provide joints that have been tested for and meet the requirements of

paragraph HYDROSTATIC TEST ON WATERTIGHT JOINTS.

2.3 SUBSURFACE STORMWATER INFILTRATION CHAMBER

Stormwater Infiltration Chamber shall meet the requirements of ASTM F2787 and ASTM F3430. The load configuration shall include Instantaneous AASHTO Design Truck Live Load at minimum cover, Maximum Permanent (50-year) cover load and 1-week parked AASHTO Design Truck Load. The selected chamber system shall be structurally designed to withstand AASHTO H-20 loading at a minimum.

The stormwater infiltration chamber shall be open bottomed to allow for infiltration. Storage capacity shall be as indicated in the project drawings and supporting documents. Provide non-woven geotextile around stone of the proposed system. The selected chamber must have inspection ports for maintenance and future service. The Contractor shall submit manufacturer’s analysis for anti-buoyancy to determine if the stormwater chamber requires stormwater anchors. Additional details shall be in compliance with manufacturer’s specifications.

2.4 MISCELLANEOUS MATERIALS

2.4.1 Concrete

Unless otherwise specified, provide concrete and reinforced concrete conforming to the requirements for 3,000 psi concrete under Section 03 30 00 CAST-IN-PLACE CONCRETE. Provide air content by volume of concrete mixture, based on measurements made immediately after discharge from the mixer, of 5 to 7 percent when maximum size of coarse aggregate exceeds 1-1/2 inches. Determine air content in accordance with ASTM C231/C231M. Provide a minimum concrete covering over steel reinforcing of not less than 1 inch thick for covers and not less than 1-1/2 inches thick for walls and flooring. For concrete deposited directly against the ground, provide a covering thickness of at least 3 inches between steel and ground. Provide expansion-joint filler material conforming to ASTM D1751, or ASTM D1752, or provide be resin-impregnated fiberboard conforming to the physical requirements of ASTM D1752.

2.4.2 Mortar

Mortar is not allowed for pipe joints. Provide mortar for pipe connections to drainage structures conforming to ASTM C270, Type M, except that the maximum placement time will be 1 hour. Provide a sufficient quantity of water in the mixture to produce a stiff workable mortar but in no case may the quantity exceed the manufacturers recommendations per sack of cement. Use water that is clean and free of harmful acids, alkalis, and organic impurities. Use the mortar within 30 minutes after the ingredients are mixed with water.

2.4.3 Precast Reinforced Concrete Manholes and Outlet Control Structures

Provide precast reinforced concrete manholes conforming to ASTM C478/C478M. Provide joints between precast concrete risers and tops that are made with flexible watertight, rubber-type gaskets meeting the requirements of paragraph PIPE JOINTS.

Install precast outlet control structures with an internal weir wall with orifice and top elevation as shown in the project drawings. Weir wall shall be structurally reinforced.

2.4.4 Frame and Cover or Gratings

Submit certification on the ability of frame and cover or gratings to carry the imposed live load indicated on the drawings. Provide frame and cover or gratings made of cast gray iron, ASTM A48/A48M, Class 35B; cast ductile iron, ASTM A536, Grade 65-45-12; or cast aluminum, ASTM B26/B26M, Alloy 356.0-T6. Provide curb inlet grates conforming to the weight, shape, size, and waterway openings indicated on the plans. Stamp or cast the word "Storm Sewer" into covers so that it is plainly visible.

2.4.5 Steel Ladder

Provide a steel ladder where the depth of the storm drainage structure exceeds 12 feet. Provide ladders not less than 16 inches in width, with 3/4 inch diameter rungs spaced 12 inches apart. Provide two stringers that are a minimum 3/8 inch thick and 2-1/2 inches wide. Galvanize ladders and inserts after fabrication in conformance with ASTM A123/A123M.

2.4.6 Resilient Connectors

Provide flexible, watertight connectors conforming to ASTM C923/C923M for connecting pipe to manholes and inlets.

2.5 TESTS, INSPECTIONS, AND VERIFICATIONS

2.5.1 Hydrostatic Test on Watertight Joints

Perform a hydrostatic test on the watertight joint types as proposed. This test will be conducted at the plant or by an independent laboratory. Only one sample joint of each type needs testing; however, if the sample joint fails because of faulty design or workmanship, an additional sample joint may be tested.

2.5.1.1 Concrete, Clay, PVC, PE, SRPE and PP Pipe

Provide joints in reinforced and nonreinforced concrete pipe meeting the performance requirements in ASTM C990M ASTM C990 or ASTM C443M ASTM C443. Provide joints in clay pipe meeting the test requirements in ASTM C425. Provide joints in PVC, PE, SRPE, and PP plastic pipe meeting the test requirements in ASTM D3212.

PART 3 EXECUTION

3.1 STORM DRAINS AND DRAINAGE STRUCTURES

Excavate trenches, excavate for appurtenances and backfill for storm drains, in accordance with the applicable portions of Section 31 20 00 EARTHWORK and the requirements specified below.

3.1.1 Trenching

Excavate trenches to the width indicated on the drawings or as specified herein. Trench width should permit satisfactory jointing and thorough tamping of the bedding material under and around the pipe. Place sheeting and bracing, where required, within the trench width as specified, without any overexcavation.

3.1.2 Removal of Rock

Replace rock in either ledge or boulder formation with suitable materials to provide a compacted earth cushion. Provide a compacted earth cushion between unremoved rock and the pipe with a thickness of at least 8 inches or 1/2 inch for each foot of fill over the top of the pipe, whichever is greater, but not more than three-fourths the nominal diameter of the pipe. Maintain the cushion under the bell as well as under the straight portion of the pipe where bell-and-spigot pipe is used. Provide a compacted earth cushion between unremoved rock and the box culvert of at least 8 inches in thickness for concrete box culverts. Excavate rock as specified and defined in Section 31 20 00 EARTHWORK.

3.1.3 Removal of Unstable Material

Where wet or otherwise unstable soil incapable of properly supporting the pipe or box culvert, as determined by the Contracting Officer, is unexpectedly encountered in the bottom of a trench, remove such material to the depth required and replace with select granular material to the proper grade. Compact select granular material as specified in paragraph FINAL BACKFILL. When removal of unstable material is due to the fault or neglect of the Contractor while performing shoring and sheeting, water removal, or other specified requirements, perform such removal and replacement at no additional cost to the Government.

3.2 BEDDING AND INITIAL BACKFILL

Provide a firm bedding foundation of uniform density throughout the entire length of the pipe or box culvert.

3.2.1 Plastic Pipe

Provide bedding for PVC, PE, SRPE and PP pipe meeting the requirements of ASTM D2321. Use Class IB or II material for PVC, PE, SRPE pipe bedding, haunching, and initial backfill. Use Class I, II, or III material for PP pipe bedding, haunching and initial backfill.

3.3 PLACING PIPE

Submit printed copies of the pipe manufacturer's recommended pipe installation procedures prior to installation. Thoroughly examine each section of pipe or box culvert before being laid; do not use defective or damaged pipe. Protect plastic pipe, excluding SRPE pipe, from exposure to direct sunlight prior to laying, if necessary to maintain adequate pipe stiffness and meet installation deflection requirements. Lay pipelines to the grades and alignment indicated. Provide proper facilities for lowering sections of pipe into trenches. Do not lay pipe in water or when trench conditions or weather are unsuitable for such work. Divert drainage or dewater trenches during construction as necessary. Deflection of installed flexible pipe must not exceed the following limits:

TYPE OF PIPE	MAXIMUM ALLOWABLE DEFLECTION (percent)

TYPE OF PIPE	MAXIMUM ALLOWABLE DEFLECTION (percent)
Plastic (PVC, HDPE, SRPE, and PP)	5

3.3.1 Concrete, Clay, PVC, Ribbed PVC, Ductile Iron Pipe

Lay pipe proceeding upgrade with spigot ends of bell-and-spigot pipe and tongue ends of tongue-and-groove pipe pointing in the direction of the flow.

3.3.2 Structural-Plate Steel

Install structural plate in accordance with ASTM A807/A807M. Assemble structural plate in accordance with instructions furnished by the manufacturer. Instructions must show the position of each plate and the order of assembly. Tighten bolts progressively and uniformly, starting at one end of the structure after all plates are in place. Repeat the operation to ensure that all bolts are tightened to meet the torque requirements of 200 foot-pounds plus or minus 50 foot-pounds. Check power wrenches used by the use of hand torque wrenches or long-handled socket or structural wrenches for amount of torque produced. Check and adjust power wrenches frequently as needed, according to type or condition, to ensure proper adjustment to supply the required torque.

3.4 JOINTING

3.4.1 Concrete and Clay Pipe

3.4.1.1 Plastic Sealing Compound Joints for Tongue-and-Grooved Pipe and Box Culverts

Follow the recommendation of the particular manufacturer in regard to sealing compound special installation requirements. When lubricants, primers, or adhesives are used, only apply on surfaces that are dry and clean. Affix sealing compounds to the pipe or box culvert not more than 3 hours prior to installation of the pipe or box culvert. Protect sealing compounds from the sun, blowing dust, and other deleterious agents at all times. Inspect sealing compounds before installation of the pipe or box culvert, and remove and replace any loose or improperly affixed sealing compound. Align the pipe or box culvert with the previously installed pipe or box culvert, and pull the joint together.

3.4.1.2 Flexible Watertight Joints

Use lubricants, cements, adhesives, and other special installation requirements for gaskets and jointing materials as recommended by the manufacturer. When lubricants, cements, or adhesives are used, only apply on surfaces that are clean and dry. Affix gaskets and jointing materials to the pipe not more than 24 hours prior to the installation of the pipe, and protect from the sun, blowing dust, and other deleterious agents at all times. Inspect gaskets and jointing materials before installing the pipe; remove and replace any loose or improperly affixed gaskets and jointing materials. Align the pipe with the previously installed pipe,

and push the joint home. If the gasket becomes visibly dislocated when joining sections of pipe, remove the pipe and remake the joint.

3.5 DRAINAGE STRUCTURES

3.5.1 Manholes, Inlets and Outlet Control Structures

Construct manholes of precast reinforced concrete. Construct inlets of precast reinforced concrete. Provide manholes and inlets complete with frames and covers or gratings; and with fixed galvanized steel ladders as indicated. Make pipe connections to concrete manholes and inlets with flexible, watertight connectors.

3.6 INSTALLATION OF TRACER WIRE AND WARNING TAPE

Install a continous length of tracer wire for the full length of each run of nonmetallic pipe in accordance with Section 31 20 00 EARTHWORK. Attach wire to top of pipe in such a manner that it will not be displaced during construction operations. Install warning tape above all storm drain pipe in accordance with Section 31 20 00 EARTHWORK.

3.7 SUBSURFACE STORMWATER INFILTRATION CHAMBER

Per manufacturer’s installation guidelines.

3.8 FINAL BACKFILL

Backfill trenches with satisfactory material deposited in layers of a maximum of 8 inches loose thickness and compacted to 90 percent of maximum density for cohesive soils and 95 percent of maximum density for cohesionless soils in accordance with Section 31 20 00 EARTHWORK. Testing is the responsibility of the Contractor and will be performed at no additional cost to the Government. Unless otherwise specified, determine field in-place density of final backfill at a frequency of one test per 50 linear feet, or fraction thereof, of each lift of backfill. Submit test results in accordance with Section 31 20 00 EARTHWORK. Do not displace or damage pipe or box when compacting final backfill by rolling or operating heavy equipment parallel with the pipe or box. Movement of construction machinery over a culvert or storm drain at any stage of construction will be at the Contractor's risk. Repair or replace any damaged pipe. Protect concrete pipes with a minimum of 3 feet of cover prior to permitting heavy construction equipment to pass over them during construction. Provide the minimum cover for construction loads over corrugated steel pipes as specified in Section 26, Division II of AASHTO HB-17. Provide minimum cover for construction loads over plastic pipes as specified in ASTM D2321.

3.9 FIELD QUALITY CONTROL

3.9.1 Tests

Testing is the responsibility of the Contractor. Perform all testing and retesting at no additional cost to the Government.

3.9.1.1 Leakage Test

Test pipe lines for leakage prior to completing backfill by performing a low pressure air pipeline test. Submit leakage test results to the

Contracting Officer.

3.9.1.1.1 Low Pressure Air Pipeline Tests

Perform low pressure air testing for plastic pipe in accordance with ASTM F1417.

3.9.1.2 Deflection Testing

Conduct deflection test no sooner than 30 days after completion of final backfill and compaction testing. Clean or flush all lines prior to testing. Perform a deflection test on entire length of installed flexible pipeline upon completion of work adjacent to and over the pipeline, including backfilling, placement of fill, grading, paving, placement of concrete, and any other superimposed loads. Deflection of pipe in the installed pipeline under external loads must not exceed the limits in paragraph PLACING PIPE above as percent of the average inside diameter of pipe. Use a laser profiler to determine if allowable deflection has been exceeded.

3.9.1.2.1 Laser Profiler

Inspect pipe interior with laser profiling equipment. Utilize low barrel distortion video equipment for pipe diameters 48 inches or less.

3.9.1.3 Tracer Wire Continuity

Test tracer wire for continuity after initial and final backfilling of pipes. Verify that tracer wire is locatable with electronic utility location equipement. Repair breaks or separations and re-test for continuity.

3.9.2 Inspection

3.9.2.1 Post-Installation Inspection

Perform a CCTV inspection and video recording of pipes with diameters 48 inches or less. Visually inspect pipes with diameters larger than 48 inches. Inspect each segment of pipe for alignment, settlement, joint separations, soil migration through the joint, cracks, buckling, bulging and deflection. An engineer must evaluate all defects to determine if any remediation or repair is required.

3.9.2.1.1 Flexible Pipe

Check each flexible pipe (PE, PVC, PP, corrugated steel and aluminum) for rips, tears, joint separations, soil migration throught the joint, cracks, localized buckling, bulges, settlement and alignment.

3.9.2.1.2 Post-Installation Inspection Report

The deflection results and final post installation inspection report must include: a copy of all video taken, pipe location identification, equipment used for inspection, inspector name, deviation from design, grade, deviation from line, deflection and deformation of flexible pipe, inspector notes, condition of joints, condition of pipe wall (e.g. distress, cracking, wall damage dents, bulges, creases, tears, holes, etc.).

3.9.3 Repair of Defects

3.9.3.1 Leakage Test

When leakage exceeds the maximum amount specified, correct source of excess leakage by replacing damaged pipe and gaskets and retest.

3.9.3.2 Deflection Testing

When deflection readings are in excess of the allowable deflection of average inside diamater of pipe are obtained, remove pipe which has excessive deflection and replace with new pipe. Retest 30 days after completing backfill, leakage testing and compaction testing.

3.9.3.3 Inspection

Replace pipe or repair defects indicated in the Post-Installation Inspection Report.

3.9.3.3.1 Flexible Pipe

Replace pipes having cracks or splits.

3.10 PROTECTION

Protect storm drainage piping and adjacent areas from superimposed and external loads during construction.

3.11 WARRANTY PERIOD

Pipe segments found to have defects during the warranty period must be replaced with new pipe and retested.

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FOUNDATION DRAINAGE
05/18

PART 1 GENERAL

1.1 UNIT PRICES

1.1.1 Pipe Subdrains

Measure the length of pipe installed from end to end along the centerlines without any deduction for the diameter of the manholes. Pipe will be paid for according to the number of linear feet of subdrains placed in the accepted work. Payment for bedding and drainage layer materials, except geotextiles, will be included in the payment for the pipe subdrain system.

1.1.2 Flushing and Observation Risers

Flushing and observation risers to be paid for will be the number of flushing and observation risers completed with frames and covers (where specified) constructed in the accepted work.

1.1.3 Geotextile

Measure geotextile for payment by the square yard in place. Measure overlapped joints and seams as a single layer of cloth.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO M 252	(2025) Standard Specification for Corrugated Polyethylene Pipe, 75- to 250-mm (3- to 10-in.) Diameter
AASHTO M 288	(2021) Standard Specification for Geosynthetic Specification for Highway Applications

ASTM INTERNATIONAL (ASTM)

ASTM C33/C33M	(2024a) Standard Specification for Concrete Aggregates
ASTM C136/C136M	(2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM D2321	(2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications

ASTM D3034	(2016) Standard Specification for Type PSM Poly(Vinyl Chloride) (PVC) Sewer Pipe and Fittings
ASTM F758	(2014; R 2023) Smooth-Wall Poly(Vinyl Chloride) (PVC) Plastic Underdrain Systems for Highway, Airport, and Similar Drainage
ASTM F949	(2025) Standard Specification for Poly(Vinyl Chloride) (PVC) Corrugated Sewer Pipe with a Smooth Interior and Fittings

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-04 Samples

Geotextile; G

Pipe and Pipe Fittings; G

SD-07 Certificates

Geotextile; G

Pipe and Pipe Fittings; G

1.4 DELIVERY, STORAGE, AND HANDLING

1.4.1 Delivery and Storage

Inspect materials delivered to site for damage; unload, and store with minimum handling. Do not store materials directly on the ground. Keep the inside of pipes and fittings sfree of dirt and debris. Keep, during shipment and storage, geotextile wrapped in burlap or similar heavy duty protective covering. Protect the geotextile from mud, soil, dust, and debris. Do not store geotextile materials in direct sunlight. Install plastic pipe within 6 months from the date of manufacture unless otherwise approved.

1.4.2 Handling

Handle materials in such a manner as to ensure delivery to the trench in sound undamaged condition. Carry pipe to the trench.

PART 2 PRODUCTS

2.1 PIPE FOR SUBDRAINS

Submit samples of pipe and pipe fittings, before starting the work. Provide type and sizes of subdrain pipe indicated. Submit certifications from the manufacturers attesting that materials meet specification requirements. Certificates are required for drain pipe and fittings.

2.1.1 Plastic

Provide plastic pipe containing ultraviolet inhibitor to provide protection from exposure to direct sunlight. Provide pipe with with bell and spigot or solvent cement joints. Provide manufacturer's standard type fittings conforming to the indicated specification.

2.1.1.1 Polyvinyl Chloride (PVC) and Fittings

ASTM D3034, ASTM F949 or ASTM F758, Type PS 46.

2.1.1.2 Corrugated Polyethylene (PE) and Fittings

AASHTO M 252, Type S or SP as indicated.

2.1.1.3 Pipe Perforations

Provide pipe perforations with a minimum water inlet area of 0.5 square inch per linear foot and as specified below.

2.1.1.3.1 Circular Perforations in Plastic Pipe

Cleanly cut circular holes not more than 3/8 inch or less than 3/16 inch in diameter and arrange in rows parallel to the longitudinal axis of the pipe. Provide pipe with perforations spaced uniformly along rows. Unless otherwise recommended by the pipe manufacturer, provide pipe with rows approximately 1-1/2 inches apart and arranged in a staggered pattern so that all perforations lie at the midpoint between perforations in adjacent rows. Space the rows over not more than 155 degrees of circumference. Provide pipe that is not perforated for a length equal to the depth of the socket at the spigot or tongue end and provide perforations that continue at uniform spacing over the entire length of the pipe.

2.1.1.3.2 Slotted Perforations in Plastic Pipe

Cleanly cut circumferential slots so as not to restrict the inflow of water and uniformly spaced along the length and circumference of the pipe. Provide pipe with slots not exceeding 1/8 inch nor less than 1/32 inch in width. Provide pipe with individual slot lengths not exceeding 10 percent of the pipe inside nominal circumference on 6 to 8 inch diameter pipe, and 2-1/2 inches on 10 inch diameter pipe. Symmetrically space rows of slots so that they are fully contained in 2 quadrants of the pipe. Center slots in the valleys of the corrugations of profile wall pipe.

2.2 GEOTEXTILE

Provide geotextile conforming to AASHTO M 288 and meeting the subsurface drainage requirements.

Submit samples of geotextile and certifications from the manufacturers attesting that geotextile meets specification requirements.

2.3 SUBDRAIN FILTER AND BEDDING MATERIAL

Provide subdrain filter and bedding material composed of washed sand, sand and gravel, crushed stone, crushed stone screenings, or slag composed of hard, tough, durable particles free from adherent coatings. Filter material may not contain corrosive agents, organic matter, or soft, friable, thin, or elongated particles. Provide filter material that is

evenly graded between the limits specified in TABLE I. Gradation curves will exhibit no abrupt changes in slope denoting skip or gap grading. Provide filter materials that are clean and free from soil and foreign materials. Remove and replace filter blankets found to be dirty or otherwise contaminated with material meeting the specific requirements, at no additional cost to the Government.

TABLE I

	Type I Gradation E 11 ASTM C33/C33M	Type II Gradation 57 ASTM C33/C33M	Type III Gradation
ASTM C136/C136M Sieve Size	Percent Passing	Percent Passing	Percent Passing
1-1/2 inch	--	100	
1 inch	--	90 - 100	
3/8 inch	100	25 - 60	
No. 4	95 - 100	5 - 40	
No. 8	--	0 - 20	
No. 16	45 - 80	--	
No. 50	10 - 30	--	
No. 100	0 - 10	--	

2.4 DRAINAGE STRUCTURES

Refer to Section 33 40 00 STORMWATER UTILITIES

PART 3 EXECUTION

3.1 EXCAVATION AND BEDDING FOR SUBDRAIN SYSTEMS

Excavate trenches, including the removal of rock and unstable material, in accordance with Section 31 20 00 EARTHWORK. Place bedding material in the trench as indicated or as required as replacement materials used in those areas where unstable materials were removed. Compact bedding material as specified for cohesionless material in Section 31 20 00 EARTHWORK.

3.2 FLUSHING AND OBSERVATION RISERS

3.2.1 Flushing and Observation Risers

Install flushing and observation riser pipes with frames and covers at the locations indicated. Construct risers of non-perforated plastic pipe. Join riser pipes to the subdrain system as indicated.

3.3 INSTALLATION OF GEOTEXTILE AND PIPE FOR SUBDRAINS

3.3.1 Installation of Geotextile

3.3.1.1 Trench Lining and Overlaps

Grade trenches to be lined with geotextile to obtain smooth side and bottom surfaces so that the geotextile will not bridge cavities in the soil or be damaged by projecting rock. Lay the geotextile flat but not stretched on the soil, and secure it with anchor pins in accordance with manufacturer's instructions. Overlap at least 6 to 12 inches, and secure with anchor pins along the overlaps.

3.3.2 Installation of Pipe for Subdrains

3.3.2.1 Pipelaying

Install pipe in accordance with the manufacturer's recommendations. Thoroughly examine each section of pipe before being laid; do not use defective or damaged pipe. Do not lay pipe when the trench conditions or weather is unsuitable for such work. Remove water from trenches by sump pumping or other approved methods. Lay the pipe to the grades and alignment as indicated. Bed the pipe to the established gradeline. Center perforations on the bottom of the pipe. Lay bell-and-spigot type with the bell ends upstream. Approval of all in-place pipes by the Contracting Officer is required prior to backfilling.

3.4 INSTALLATION OF DRAINAGE LAYER FILTER MATERIAL AND BACKFILLING FOR PERFORATED SUBDRAINS

After perforated pipe for subdrains has been laid, inspected, and approved, place drainage layer material around and over the pipe to the depth indicated. Place the drainage layer material in layers not to exceed 8 inches thick. Thoroughly compact each layer using mechanical tampers or rammers.

3.5 INSTALLATION OF BEDDING AND BACKFILL FOR NON-PERFORATED SUBRAIN OUTFALL PIPE

3.5.1 Plastic Pipe

Place and compact pipe embedment for plastic pipe in accordance with ASTM D2321. Use Class IB or II embedment materials.

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FUEL IMPERMEABLE LINER SYSTEM

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SECTION 33 56 19

FUEL IMPERMEABLE LINER SYSTEM
11/22

PART 1 GENERAL

1.1 MEASUREMENT AND PAYMENT

Measure the total surface area in square feet covered by liner system. Final quantities will be based on as-built conditions. Allowance will be made for liner system in anchor and drainage trenches; however, no allowance will be made for waste, overlap, repairs, or materials used for the convenience of the Contractor. Liner system installed and accepted by the Contracting Officer will be paid for at the respective contract unit price in the bidding schedule.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C920	(2018; R 2024) Standard Specification for Elastomeric Joint Sealants
ASTM D1004	(2021) Initial Tear Resistance of Plastic Film and Sheeting
ASTM D1505	(2018) Standard Test Method for Density of Plastics by the Density-Gradient Technique
ASTM D1603	(2020) Carbon Black Content in Olefin Plastics
ASTM D3895	(2014) Oxidative-Induction Time of Polyolefins by Differential Scanning Calorimetry
ASTM D4218	(2020) Determination of Carbon Black Content in Polyethylene Compounds by the Muffle-Furnace Technique
ASTM D4437/D4437M	(2016; R 2018) Standard Practice for Non-destructive Testing (NDT) for Determining the Integrity of Seams Used in Joining Flexible Polymeric Sheet Geomembranes
ASTM D4533	(2011) Trapezoid Tearing Strength of Geotextiles
ASTM D4632/D4632M	(2015a) Grab Breaking Load and Elongation of Geotextiles

ASTM D4833/D4833M	(2007; R 2020) Standard Test Method forIndex Puncture Resistance of Geomembranes and Related Products
ASTM D5199	(2012) Measuring Nominal Thickness of Geosynthetics
ASTM D5261	(2010; R 2018) Standard Test Method for Measuring Mass Per Unit Area of Geotextiles
ASTM D5397	(2019a) Standard Test Method for Evaluation of Stress Crack Resistance of Polyolefin Geomembranes Using Notched Constant Tensile Load Test
ASTM D5596	(2003; R 2016) Standard Test Method For Microscopic Evaluation of the Dispersion of Carbon Black in Polyolefin Geosynthetics
ASTM D5641/D5641M	(2016) Standard Practice for Geomembrane Seam Evaluation by Vacuum Chamber
ASTM D5721	(2008; R 2013) Air-Oven Aging of Polyolefin Geomembranes
ASTM D5820	(1995; R 2018) Standard Practice for Pressurized Air Channel Evaluation of Dual Seamed Geomembranes
ASTM D5885/D5885M	(2017) Standard Test Method for Oxidative Induction Time of Polyolefin Geosynthetics by High-Pressure Differential Scanning Calorimetry
ASTM D5893/D5893M	(2016) Standard Specification for Cold Applied, Single Component, Chemically Curing Silicone Joint Sealant for Portland Cement Concrete Pavements
ASTM D6241	(2014) Standard Test Method for the Static Puncture Strength of Geotextiles and Geotextile-Related Products Using a 50-mm Probe
ASTM D6392	(2012; R 2018) Standard Test Method for Determining the Integrity of Nonreinforced Geomembrane Seams Produced Using Thermo-Fusion Methods
ASTM D6693/D6693M	(2004; R 2015; E 2015) Standard Test Method for Determining Tensile Properties of Nonreinforced Polyethylene and Nonreinforced Flexible Polypropylene Geomembranes
ASTM D7238	(2006; R 2012) Standard Test Method for Effect of Exposure of Unreinforced Polyolefin Geomembrane Using Fluorescent UV Condensation Apparatus

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Installation; G

Liner Projections; G

As-Built Drawings; G

SD-03 Product Data

Non-Reinforced Smooth Geomembrane Liner; G

Nonwoven Geotextile; G

Liner Fittings; G

Mounting Strip System; G

Sealant; G

Manufacturer's Warranty; G

Installer's Warranty; G

SD-04 Samples

Destructive Field Seam Test Sample; G

SD-06 Test Reports

Trial Seam Logs; G

Non-Destructive Field Seam; G

Destructive Field Seam; G

SD-07 Certificates

Field Engineer Qualifications; G

Installer Qualifications; G

Liner Manufacturer's Certification; G

Surface Preparation; G

SD-08 Manufacturer's Instructions

Liner Manufacturer's Installation Instructions; G

SD-10 Operation and Maintenance Data

Geomembrane Liner; G

1.4 QUALITY ASSURANCE

1.4.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of a manufacturer regularly engaged in the manufacturing of such products, that are of a similar material, design and workmanship. Materials and equipment must have been in satisfactory commercial or industrial use for a minimum 2 years prior to bid opening. The 2 year period must include applications of the equipment and materials under similar circumstances and of similar size. Materials and equipment must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period. Products having less than a 2 year field service record will be acceptable if a certified record of satisfactory field operation, for not less than 6000 hours, exclusive of the manufacturer's factory tests, can be shown.

1.4.2 Field Engineer Qualifications

Provide a field engineer who has successfully completed manufacturer's training on handling and installing of the fuel impermeable liner to be installed. Demonstrate that the engineer has at least one million square feet of liner installation experience. Submit a letter providing evidence of the field engineer's experience, training, and licensing. In regard to the field engineer's experience, include in the submittal a point of contact, a phone number, the address, the type of installation, and the current status of each installation mentioned.

1.4.3 Installer Qualifications

The installer is responsible for field handling, deploying, seaming, anchoring, and field Quality Control (QC) testing of the geomembrane. Demonstrate that the installer has installed the proposed geomembrane material for at least 5 completed projects and a total minimum area of 1 million square feet. At least one seamer must have experience seaming a minimum of 500,000 square feet of the proposed geomembrane using the same type of seaming equipment and same geomembrane specified for this project.

1.4.4 Liner Manufacturer's Certification

Following the successful installation and testing of the liner, an authorized representative from the liner manufacturer must submit a letter certifying that the liner installation and testing results are satisfactory and that each meets the company's quality expectations and warranty. The letter must also certify that the liner installed is compatible with and recommend for use with the fuel to be stored. Include in the letter the representative's name, address, phone number, and qualifications for being a manufacturer's representative.

1.5 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer.

1.5.1 Delivery

The QC inspector must be present during delivery and unloading of the geomembrane. Label each geomembrane roll/panel with the manufacturer's name, product identification number, roll/panel number, and roll/panel dimensions.

1.5.2 Storage

Store geomembranes and geotextiles elevated off of the ground and covered to provide protection from precipitation, sunlight/ultraviolet light, puncture/abrasion, undesirable chemicals (as recommended by the manufacturer), and flames/welding sparks. Storage in temperatures between 32 degrees F and 160 degrees F, or as recommended by the manufacturer. Storage must not result in crushing the core of roll goods or flattening of the rolls. Do not store rolls more than two high. Store palleted materials on level surfaces and not stacked on top of one another. Remove damaged geomembrane from the site and replace with geomembrane that meets the specified requirements.

1.5.3 Handling

Do not drag, lift by one end, or drop rolls and panels. Use a pipe or solid bar, of sufficient strength to support the full weight of a roll without significant bending but small enough to be easily inserted through the core of the roll, for all handling activities. Link the ends of the pipe or bar to the ends of a spreader bar with chains. Use a spreader bar wide enough to prevent the chains from rubbing against the ends of the roll. Alternatively, a stinger bar protruding from the end of a forklift or other equipment may be used. Use a stinger bar at least three-fourths the length of the core and capable of supporting the full weight of the roll without significant bending. If recommended by the manufacturer, a sling handling method utilizing appropriate loading straps may be used.

1.6 AMBIENT CONDITIONS

Do not deploy or field-seam geomembrane in the presence of excess moisture (i.e., rain, fog, dew), in areas of ponded water, or in the presence of winds above 12 mph. The relative humidity must be less than 80 percent, and temperature above the dew point. Unless authorized by the Contracting Officer, do not attempt placement or seaming at ambient temperatures below 40 degrees F or above 104 degrees F. Measure ambient temperature at a height no greater than 6 inches above the ground or geomembrane surface. Seaming is only allowed below 40 degrees F if recommended by the geomembrane manufacturer and if destructive tests of trial seams at the proposed temperature meet the seam property requirements listed in Table 2.

1.7 WARRANTY

State in the manufacturer's warranty that the installed geomembrane liner is warranted for 10 years against deterioration as installed for containment of the intended liquid. State in the installer's warranty that the geomembrane liner won't fail due to improper installation within 2 years.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Non-Reinforced Smooth Geomembrane Liner

Provide non-reinforced smooth geomembrane liner of unreinforced high density polyethylene (HDPE), manufactured as wide as possible to minimize factory and field seams. Geomembrane sheets must be uniform in color, thickness, and surface texture. The sheets must be free of and resistant to fungal or bacterial attack and free of cuts, abrasions, holes, blisters, contaminants and other imperfections. Provide liner meeting the physical properties of Table 2.

Make resin used in manufacturing geomembrane sheets of virgin uncontaminated ingredients. Use no more than 10 percent regrind, reworked, or trim material in the form of chips or edge strips to manufacture the geomembrane sheets. All regrind, reworked, or trim materials must be from the same manufacturer and exactly the same formulation as the geomembrane sheet being produced. Do not use post consumer materials or water-soluble ingredients to produce the geomembrane. For geomembranes with plasticizers, use only primary plasticizers that are resistant to migration. Submit a copy of the test reports and QC certificates for materials used in the manufacturing of the geomembrane shipped to the site.

TABLE 2 - SMOOTH HDPE GEOMEMBRANE PROPERTIES (ENGLISH)

PROPERTY	TEST VALUE	TEST METHOD
Thickness (min ave)	60 mils	ASTM D5199
Lowest individual of 10 values	-10 percent	ASTM D5199
Density (min)	0.940 g/cc	ASTM D1505
Tensile Properties(1)(min ave)		ASTM D6693/D6693M Type IV
Yield Strength	126 lb/in	
Break Strength	228 lb/in	
Yield Elongation	12 percent	
Break Elongation	700 percent	
Tear Resistance (min ave)	42 lb	ASTM D1004
Puncture Resistance(min ave)	108 lb	ASTM D4833/D4833M
Stress Crack Resistance (2)	300 hr	ASTM D5397 (Appendix)
Carbon Black Content	2.0-3.0 percent	ASTM D1603 (3)
Carbon Black Dispersion	Note (4)	ASTM D5596

TABLE 2 - SMOOTH HDPE GEOMEMBRANE PROPERTIES (ENGLISH)		
PROPERTY	TEST VALUE	TEST METHOD
Oxidative Induction Time (OIT)(min ave)(5)		
Std OIT	100 min	ASTM D3895
High Pres OIT	400 min	ASTM D5885/D5885M
Oven Aging at 185 deg F (min ave) (5), (6)		ASTM D5721
Std OIT	55 percent at 90 days	ASTM D3895
or High Pres OIT	80 percent at 90 days	ASTM D5885/D5885M
UV Resistance (min ave) (7)		ASTM D7238
High Pres OIT(8)(9)	50 percent at 1600 hours	ASTM D5885/D5885M
Seam Shear Strength (min) Hot Wedge, Hot Air, Ultrasonic, Extrusion Fillet (10)(12)	120 lb/in	ASTM D6392
Seam Peel Strength (min) Hot Wedge, Hot Air, Ultrasonic (10)(11)(12)	91 lb/in	ASTM D6392
Seam Peel Strength (min) Extrusion Fillet (10)(11)(12)	78 lb/in	ASTM D6392

(1) Base minimum average machine direction and minimum average cross machine direction values on 5 test specimens in each direction. For HDPE geomembrane, yield elongation is calculated using a gauge length of 1.3 inches. For HDPE geomembrane, break elongation is calculated using a gauge length of 2.0 inches.

(2) For HDPE geomembrane, use the manufacturer's mean value yield stress used to calculate the applied load for test method ASTM D5397 (Appendix).

(3) Other methods such as ASTM D4218 or microwave methods are acceptable if an appropriate correlation to ASTM D1603 can be established.

(4) Carbon black dispersion for 10 different views:
- minimum 8 of 10 in Categories 1 or 2
- all 10 in Categories 1,2, or 3

(5) The manufacturer has the option to select either one of the OIT methods to evaluate the antioxidant content.

(6) Evaluate samples at 30 and 60 days and compare with the 90 day response.

(7) The condition of the test must be a 20 hour UV cycle at 167 degrees F followed by a 4 hour condensation cycle at 140 degrees F.

TABLE 2 - SMOOTH HDPE GEOMEMBRANE PROPERTIES (ENGLISH)		
PROPERTY	TEST VALUE	TEST METHOD
(8) Do not use the standard OIT test (ASTM D3895) in determining UV resistance.		
(9) UV resistance is based on percent retained value regardless of the original HP-OIT value.		
(10) Seam tests for peel and shear must fail in the Film Tear Bond mode. This is a failure in the ductile mode of one of the bonded sheets by tearing or breaking prior to complete separation of the bonded area.		
(11) Where applicable, test both tracks of a double hot wedge seam for peel adhesion.		
(12) For five samples tested, four must meet or exceed the given value, with the fifth within 80 percent of the given value.		

2.1.2 Nonwoven Geotextile

Provide a nonwoven, polypropylene, needle punched geotextile fabric. Provide geotextile having the following physical properties as a minimum. No substitute methods are allowed for verification of any property.

2.1.2.1 General Properties

Retard the growth of mildew and be compatible with the soil in contact.

2.1.2.2 Nominal Unit Weight

Provide a nominal unit weight of 12 oz/yd2 as measured in accordance with ASTM D5261.

2.1.2.3 Grab Tensile Strength

Provide a minimum grab tensile strength of 300 lbf when tested in accordance with ASTM D4632/D4632M.

2.1.2.4 Grab Tensile Elongation

Provide a minimum grab tensile elongation of 50 percent when tested in accordance with ASTM D4632/D4632M.

2.1.2.5 Puncture Strength

Provide a minimum puncture strength of 140 lbf when tested in accordance with the pin strength methods of ASTM D4833/D4833M, or 800 lbf when tested in accordance with the CBR methods of ASTM D6241.

2.1.2.6 Trapezoid Tear Strength

Provide a minimum trapezoid tear strength of 115 lbf when tested in accordance with ASTM D4533.

2.1.2.7 Ultraviolet (UV) Resistance

Maintain 70 percent of its original strength after 500 hours of testing in accordance with ASTM D4533.

2.2 ACCESSORIES

2.2.1 Liner Fittings

Provide liner fittings (for example, boots and sleeves) that are factory prefabricated components produced from the same manufacturer that produces the fuel impermeable liner and having the same fabrication characteristics as the liner.

2.2.2 Batten / Mounting Strip System

Provide minimum 1/4-inch thick by 2 inches wide stainless steel mounting strips. Provide pre-punched bolt holes in the strip at maximum 12 inches on center to accommodate the concrete anchor bolts.

2.2.2.1 Gasket

Provide a minimum 1/4-inch thick by 2-inches wide nitrile gasket on one or both sides of the geomembrane as recommended by the geomembrane manufacturer at mounting strip locations to seal and protect the geomembrane from the concrete and the mounting strip.

2.2.2.2 Concrete Anchor Bolts

Use minimum 3/8-inch diameter mechanical wedge-type concrete anchors, constructed of type 304 or 316 stainless steel. Provide nuts, washers, and other accessories of the same materials.

2.2.3 Sealant

Use sealant conforming to ASTM C920, Type S, Grade NS, Class 25 or better, or ASTM D5893/D5893M, Type NS, or other sealant recommended by the geomembrane manufacturer. Sealant used must be compatible with the material being stored, as recommended by the sealant manufacturer.

2.2.4 Permanent Ballast

Provide permanent ballast as indicated below.

2.2.4.1 Cast in Place Concrete

Provide concrete cover over the containment geomembrane in accordance with Section 32 13 15.20 CONCRETE PAVEMENTS FOR CONTAINMENT DIKES.

2.3 EQUIPMENT

Utilize equipment in performance of the work in accordance with the geomembrane manufacturer's recommendations and maintain in satisfactory working condition.

PART 3 EXECUTION

3.1 INSTALLATION

Make equipment/parts subject to degradation or requiring adjustment, inspection or repair accessible and capable of convenient removal. Prior to any fabrication or erection, submit detailed installation drawings that show the proposed panel layout of the liner over the entire containment area. As a minimum, indicate the direction and location of factory and field fabricated seams, the termination of the panels at the perimeter of lined areas, details and methods of sealing around penetrations, details and methods for anchoring, and any applicable site specific installation instructions.

3.1.1 Field Engineer

Provide a field engineer to supervise the complete installation of the liner and perform each liner inspection and test.

3.1.2 Surface Preparation

Prepare surfaces in accordance with Section 31 20 00 EARTHWORK. Surfaces to be covered must be free of vegetation, gravel, rocks, debris, etc., graded true, compacted, and be smooth with no abrupt projections. Install the geotextile in direct contact with the existing subgrade to be covered. Install the geotextile in strict accordance with manufacturer's recommendations. Install geotextile to closely fit around projections (for example, pipe penetrations, concrete foundations/pads, conduit penetrations, etc.). Submit a signed letter from the field engineer, prior to placing any geotextile or liner, that states the subgrade and surface preparation were adequate and in accordance with the liner manufacturer's recommendations.

3.1.3 Embedment Strip Installation

Install embedment strips in new cast-in-place concrete structures using methods and materials recommended by the embedment strip/geomembrane manufacturer. Extrusion weld all joints and intersections of embedment material together or otherwise join as recommended by the manufacturer to provided a continuous surface for attachment of the geomembrane. Fill any holes through the face of the embed material resulting from the concrete forming or placement process with material recommended by the manufacturer.

3.1.4 Liner Installation

See paragraph AMBIENT CONDITIONS. Place the liner over the prepared surface in accordance with the liner manufacturer's installation instructions. The procedures and equipment used cannot elongate, wrinkle, scratch, or otherwise damage the geomembrane, geotextile, or the underlying subgrade. Place the liner in such a manner as to assure minimum handling and field seams. Use sand bags (or other manufacturer approved means) to hold the lining down in position during installation. Do not drag or slide materials, equipment or other items across the surface of the liner at anytime. Do not allow personnel walking or working on the lining to damage it. Replace or repair geomembrane damaged during installation. Allow adequate slack in the geomembrane to prevent the creation of tensile stress and avoid "trampolining" or "bridging". Do not exceed a wrinkle height to width ratio of 0.5 for installed

geomembrane. In addition, geomembrane wrinkles must not exceed 3 inches in height. Cut and repair wrinkles that do not meet the above criteria in accordance with paragraph REPAIRS.

3.1.4.1 Liner Projections

Install lining sheets to closely fit around liner projections (for example, pipe penetrations, concrete foundations/pads, conduit penetrations, etc.). Install and center manufacturer supplied sleeves/boots around liner projections in strict accordance with manufacturer's recommendations. Compress the end of pipe sleeves to a pipe with a stainless steel band clamp assembly. For liner anchorage to precast or existing concrete, install a batten / mounting strip around the liner edge and mount with concrete anchor bolts. Apply sealant to the perimeter edge of exposed liner to include the edge of sleeves/boots. For liner anchorage to new cast-in-place concrete, install an embedment strip in the concrete using methods and materials recommended by the embedment strip/geomembrane manufacturer.

3.2 FIELD SEAMING

3.2.1 Trial Seams

Make trial seams under field conditions on strips of excess geomembrane. Make trial seams each day prior to production seaming, whenever there is a change in seaming personnel or seaming equipment settings, whenever environmental conditions change significantly (more than 25 degrees), and at least once every four hours, by each seamer and each piece of seaming equipment used that day. Collect and test trial seam samples in accordance with ASTM D6392 for non-reinforced liner. Obtain one sample from each trial seam. This sample must be at least 18 inches long by 12 inches wide with the seam centered lengthwise. Cut ten random specimens from the sample, 1 inch wide. Field test five seam specimens for seam shear strength and field test 5 seam specimens for seam peel adhesion. To be acceptable, 4 out of 5 replicate test specimens must meet seam strength requirements specified in Table 2. If the field tests fail to meet these requirements, repeat the entire operation. If the additional trial seam fails, do not use the seaming apparatus or seamer until the deficiencies are corrected by the installer and 2 consecutive successful trial seams are achieved. Maintain logs of trial seams, which include date, time, weather, seaming personnel, seaming equipment, equipment settings, and trail seam field test results. Submit trial seam logs following the completion of installation.

3.2.2 Field Seams

Seam panels in accordance with the geomembrane manufacturer's recommendations. Comply with the geomembrane manufacturer's recommnedation for the overlap between panels being seamed together; however, in no overlap less than 2 inches. In sumps, corners and odd-shaped geometric locations, minimize the number of field seams. Extend seaming to the outside edge of panels. Compact soft subgrades prior to seaming. Provide a clean and free of moisture, dust, dirt, debris, markings, and foreign material area at the time of seaming. Completely unroll and layout adjacent liner panels/sheets before performing field seam welds. Repair fish mouths in seams.

3.3 FIELD QUALITY CONTROL

3.3.1 Visual Inspection of Field Seams

Visually inspect each field seam to confirm that the seams are tightly bonded. Perform the inspection of a seam within 30 hours after the manufacturer's suggested application, curing, and cooling time. Repair and re-inspect seams found to be defective in accordance with manufacturer's recommendations.

3.3.2 Non-Destructive Field Seam Testing

Perform non-destructive testing on all field seams over the full seam length and on any other areas showing damage or other distresses. Allowable methods are stated below. Alternate methods approved by the geomembrane manufacturer may be submitted upon approval by the Contracting Officer. Submit non-destructive field seam test reports following the completion of installation and prior to covering the geomembrane liner.

3.3.2.1 Liner Vacuum Box Test

Perform a vacuum box test in accordance with ASTM D5641/D5641M on each field seam, the area around the seams, and each liner surface showing injury due to scuffing, penetration by foreign objects, or distress from rough subgrade. If the vacuum box test indicates a continuous stream of bubbles on repeated testing at the same location, then the area being tested is considered damaged and must be repaired and retested. Perform repairs in accordance with manufacturer's recommendations.

3.3.2.2 Liner Air Lance Test

Perform an air lance test on seams to detect an unbonded area in accordance with ASTM D4437/D4437M. Perform the test using a minimum 50 psig jet of air regulated and directed through a 3/16-inch diameter nozzle. Apply the jet of air to the lip of a seam in a near perpendicular direction to the length of the seam. Hold the nozzle a maximum of 2 inches from the seam and travel at a rate of not to exceed 40 fpm. Inflation of any section of the seam by the impinging air stream is indicative of an unbonded area. Repair unbonded areas in accordance with manufacturer's recommendations and retest.

3.3.2.3 Liner Air Pressure Test

Perform air pressure testing in accordance with ASTM D5820. Upon completion of the test, relieve pressure from the opposite end of the seam being tested to verify continuity of the seam.

3.3.2.4 Liner Point Stress Test

Where other non-destructive test methods are not possible, perform point stress testing in accordance with ASTM D4437/D4437M.

3.3.3 Destructive Field Seam Testing

Obtain a minimum of one destructive test sample per 500 feet of field seam at locations specified by the Contracting Officer. Recommended locations include the ends of seams or anchor trenches, so as to minimize the impact of required repairs to the geomembrane; however, final sample locations will be determined by the Contracting Officer. Sample locations

will not be identified prior to seaming. Collect samples a minimum of 12 inches wide by 36 inches long with the seam centered lengthwise. Cit each sample into two equal pieces, with one piece retained by the installer for field testing and the remaining destructive field seam test sample given to the Contracting Officer for QA testing and/or permanent record. Number each sample and cross reference to a field log which identifies: (1) panel number; (2) seam number; (3) date and time cut; (4) ambient temperature within 6 inches above the geomembrane; (5) seaming unit designation; (6) name of seamer; and (7) seaming apparatus temperature and pressures (where applicable). Field test in accordance with ASTM D6392 for non-reinforced liner. Cut ten 1 inch wide replicate specimens from the installer's sample. Test five specimens for shear strength and 5 for peel adhesion. Jaw separation speed must be in accordance with the approved QC manual. To be acceptable, 4 out of 5 replicate test specimens must meet the seam strength requirements specified in Table 2. If the field tests fail, repair the seam in accordance with paragraph REPAIRS. Repair holes for destructive seam samples the same day they are cut. Submit destructive field seam test reports following completion of installation and prior to covering the geomembrane liner.

3.4 REPAIRS

Patch damaged areas, destructive testing areas, or other areas requiring repair with the geomembrane material. Round the corners of the patch material and extend a minimum of 6 inches in each direction from the damaged area. Small holes or snags less than 1/4 inch in diameter may be repaired by extrusion welding where such process is approved by the geomembrane manufacturer.

3.5 PROTECTION AND BACKFILLING

Prior to installation of permanent ballast material, the deployed geomembrane must be in intimate contact with the underlying surface, with no areas in sufficient tension to form "bridges" or "trampolines". Minimize waves/wrinkles in the geomembrane prior to placement of ballast materials; in no case allow wrinkles to fold over during placement of ballast materials. Install the cover geotextile in direct contact with the geomembrane liner to be covered, in strict accordance with manufacturer's recommendations. Install geotextile to closely fit around projections (for example, pipe penetrations, concrete foundations/pads, conduit penetrations, etc.). Cover the deployed and seamed geomembrane with the specified permanent ballast material within 14 calendar days of acceptance. Do not drop ballast material onto the geomembrane or overlying geotextile from a height greater than 3 feet. Push the material out over the geomembrane in an outward/upward tumbling motion. Place ballast material from the bottom of the slope upward.

3.5.1 Ballast Placement Equipment

Place ballast material utilizing construction equipment that is positioned off of the deployed geomembrane. Do not drive construction equipment on or above the geomembrane at any time.

3.6 PERMANENT BALLAST

3.6.1 Cast in Place Concrete

Place concrete cover over containment geomembrane in accordance with Section 32 13 15.20 CONCRETE PAVEMENTS FOR FUEL STORAGE CONTAINMENT DIKES.

3.7 PROJECT CLOSEOUT

Provide As-Built Drawings which show the as constructed panel layout, factory and field seam locations, penetrations, destructive field seam testing location, and repair locations. Provide operation and maintenance data package for the geomembrane liner to include material cut sheets, liner's routine maintenance requirements as well as procedures for liner repair and troubleshooting.

-- End of Section --

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SECTION 33 63 00
2/24

STEAM ENERGY DISTRIBUTION

PART 1 GENERAL

1.1 DESCRIPTION

a. This section specifies materials and procedures for construction of underground fuel oil, steam distribution, and condensate return piping system, including manholes, outside the buildings. System shall be: precast concrete shallow trenches with removable coversand pre-engineered, direct-buried piping systems.

b. A complete listing of common acronyms and abbreviations are included in Section 23 05 10, COMMON WORK RESULTS FOR BOILER PLANT AND STEAM GENERATION.

c. Definitions:

(1) System: The complete underground steam and condensate distribution system including all components such as carrier piping, pipe supports, insulation, protective enclosures, anchors, corrosion protection, stress analysis, and accessories.

(2) Pre-Engineered Direct-Buried System: A factory-fabricated system.

(3) Walk-through Concrete Tunnels: A system located below grade with sufficient space for carrier pipes, other services, and space to walk upright along the entire length of the system.

(4) Carrier Pipe: Pipe carrying the steam or condensate.

1.2 RELATED WORK

a. Section 01 00 00, GENERAL REQUIREMENTS.

b. Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA AND SAMPLES.

c. Section 01 57 19, TEMPORARY ENVIRONMENTAL CONTROLS: Erosion and Sediment Controls.

d. Section 01 74 19, CONSTRUCTION WASTE MANAGEMENT.

e. Section 01 81 13, SUSTAINABLE CONSTRUCTION REQUIREMENTS.

f. Section 01 91 00, GENERAL COMMISSIONING REQUIREMENTS.

g. SECTION 02 82 11, TRADITIONAL ASBESTOS ABATEMENT.

h. Section 03 30 00, CAST-IN-PLACE CONCRETE: Concrete Work, Reinforcing, Placement and Finishing.

i. Section 05 50 00, METAL FABRICATIONS: Steel for trench and tunnel pipe supports.

j. Section 09 91 00, PAINTING, Painting exposed steel and other surfaces.

k. Section 13 05 41 SEISMIC RESTRAINT REQUIREMENTS FOR NON-STRUCTURAL COMPONENTS: Bracing and concrete anchors.

l. Section 23 05 10, COMMON WORK RESULTS FOR BOILER PLANT AND STEAM GENERATION.

m. Section 25 10 10, ADVANCED UTILITY METERING SYSTEM: Metering.

n. Section 26 42 00, CATHODIC PROTECTION: Cathodic Protection of DDT Pre-Engineered Direct-Buried Systems.

o. Section 31 20 00, EARTHWORK: Excavation, Trench Widths, Pipe Bedding, Backfill, Shoring, Sheeting, Bracing.

p. Section 33 08 00, COMMISSIONING OF SITE UTILITY SYSTEMS.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred in the text by the basic designation only. Where conflicts occur these specifications and the VHA standard will govern.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.20.1	(2013; R 2018) Pipe Threads, General Purpose (Inch)
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2024) Factory-Made Wrought Buttwelding Fittings
ASME B16.11	(2022) Forged Fittings, Socket-Welding and Threaded
ASME B31.1	(2024) Power Piping
ASME B31.9	(2020) Building Services Piping
ASME B40.100	(2022) Pressure Gauges and Gauge Attachments
ASME BPVC SEC VIII D1	(2023) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1
ASME BPVC SEC IX	(2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon

Structural Steel

ASTM A53/A53M	(2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A105/A105M	(2023) Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A126	(2004; R 2023) Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings
ASTM A193/A193M	(2024a) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications
ASTM A194/A194M	(2024) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both
ASTM A234/A234M	(2024) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
ASTM A240/A240M	(2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications
ASTM B61	(2015; R 2021) Standard Specification for Steam or Valve Bronze Castings
ASTM C920	(2018; R 2024) Standard Specification for Elastomeric Joint Sealants

AMERICAN WELDING SOCIETY (AWS)

AWS B2.1/B2.1M	(2021) Specification for Welding Procedure and Performance Qualification
AWS D10.12M/D10.12	(2000) Guideline for Welding Mild Steel Pipe
AWS Z49.1	(2021) Safety in Welding, Cutting and Allied Processes

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

CID A-A-60005	(Basic; Notice 2) Frames, Covers, Gratings, Steps, Sump And Catch Basin, Manhole

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-58	(2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation

NACE INTERNATIONAL (NACE)

NACE SP0169	(2024) Control of External Corrosion on Underground or Submerged Metallic Piping Systems

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 4	Enclosures for outdoor use where extra protection against moisture and wind driven rain is required

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 2	(2024) Hand Tool Cleaning

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contact Persons; G, AE

Welding Qualifications; G, AE

SD-02 Shop Drawings

Shop Drawings; G, AE

Coordination/Shop Drawings; G, AE

SD-03 Product Data

Manufacturers’ Literature And Data; G, AE

SD-05 Design Data

Stress And Anchor Force Calculations; G, AE

SD-06 Test Reports

Completed System Readiness Checklist; G, AE

Test And Inspection Reports; G, AE

SD-07 Certificates

Certification; G, AE

Asme Compliance; G, AE

Manufacturer’s Certification; G, AE

SD-08 Manufacturer's Instructions

Training Plans And Instructor Qualifications; G, AE

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G, AE

SD-11 Closeout Submittals

As-Built Documentation; G, AE

1.5 SUBMITTAL INSRUCTIONS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit all submittals in conformance with Section 01 33 29 SUSTAINABILITY REPORTING. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

a. Submittals, including number of required copies, shall be submitted in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA, and SAMPLES.

b. Information and material submitted under this section shall be marked "SUBMITTED UNDER SECTION 33 63 00, STEAM ENERGY DISTRIBUTION”, with applicable paragraph identification.

c. Prior to submitting shop drawings for approval, contractor shall provide certification in writing that manufacturers of all major items of equipment have each reviewed drawings and specifications, and have jointly coordinated and properly integrated their equipment and controls to provide a complete and efficient installation.

d. Installing Contractor shall provide lists of previous installations for selected items of equipment. Contact persons who will serve as references, with telephone numbers and e-mail addresses shall be submitted with the references. COMMON WORK RESULTS FOR PLUMBING", with applicable paragraph identification.

e. Manufacturers’ Literature and Data including: Full item description and optional features and accessories of the complete system including, but not limited to, dimensions, weights, materials, applications, standard compliance, model numbers, size and capacity. Submit as one package for pipes, fittings and appurtenances, including jointing materials, insulation, hangars, expansion and power set fasteners, and other miscellaneous items.

f. Submittals and shop drawings for interdependent items, containing applicable descriptive information, shall be furnished together and complete in a group. Coordinate and properly integrate materials and equipment in each group to provide a completely compatible and

efficient installation. Final review and approvals will be made only by groups.

g. Coordination/Shop Drawings:

(1) Submit complete consolidated and coordinated shop drawings for all new systems, and for existing systems that are in the same areas.

(2) The coordination/shop drawings shall include plan views, elevations and sections of all systems and shall be on a scale of not less than 1: 32 3/8 inch equal to one foot). Clearly identify and dimension the proposed locations of the principal items of equipment. The drawings shall clearly show the proposed locations and adequate clearance for all equipment, controls, piping, pumps, valves and other items. All equipment requiring service shall be provided with an access door sized for the complete removal of device, component, or servicing of the equipment. Access for service and access for removal of components may be separate as necessary. Provide detailed coordination/shop drawings and loading calculations for all piping systems. The drawings should include all lockout/tagout points for all energy/hazard sources for each piece of equipment. Coordinate lockout/tagout procedures and practices with local VA requirements.

(3) Do not install equipment foundations, equipment or piping until coordination/shop drawings have been approved.

h. Complete operating and maintenance manuals including wiring diagrams, technical data sheets, information for ordering replacement parts, and troubleshooting guide:

(1) Include complete list indicating all components of the systems.

(2) Include complete diagrams of the internal wiring for each item of equipment.

(3) Diagrams shall have their terminals identified to facilitate installation, operation and maintenance.

i. Completed System Readiness Checklist provided by the Commissioning Agent and completed by the contractor, signed by a qualified technician and dated on the date of completion, in accordance with the requirements of Section 33 08 00, COMMISSIONING OF SITE UTILITY SYSTEMS. This shall include pressure testing and cleaning of piping systems.

j. Submit training plans and instructor qualifications in accordance with the requirements of Section 33 08 00, COMMISSIONING OF SITE UTILITY SYSTEMS.

k. Contractor shall make all necessary field measurements and investigations to assure that the equipment and assemblies will meet contract requirements and will fit the space available.

l. If equipment is submitted which differs in arrangement from that shown, provide drawings that show the rearrangement of all associated systems. Approval will be given only if all features of the equipment and associated systems, including accessibility, are equivalent to

that required by the contract.

1.6 QUALITY ASSURANCE

a. Products Criteria:

(1) Standard Products: Material and equipment shall be the standard products of a manufacturer regularly engaged in the manufacture, supply and servicing of the specified products for at least 5 years. However, digital electronics devices, software and systems such as controls and instruments, shall be the current generation of technology and basic design that has a proven satisfactory service record of at least 5 years.

(2) All items furnished shall be free from defects that would adversely affect the performance, maintainability and appearance of individual components and overall assembly.

(3) The products and execution of work specified in Division 33 shall conform to the referenced codes and standards as required by the specifications. Local codes and amendments shall be enforced, along with requirements of local utility companies. The most stringent requirements of these specifications, local codes, or utility company requirements shall always apply. Any conflicts shall be brought to the attention of the Contracting Officer.

(4) When two or more units of the same type or class of materials or equipment are required, these units shall be products of one manufacturer.

(5) Assembled Units: Manufacturers of equipment assemblies, which use components made by others, assume complete responsibility for the final assembled product.

(6) A nameplate bearing manufacturer's name or trademark, including model number, shall be securely affixed in a conspicuous place on equipment. In addition, the model number shall be cast integrally with equipment, stamped, or otherwise permanently marked on each item of equipment.

(7) Asbestos products or equipment or materials containing asbestos shall not be used.

b. Contractor shall restore damaged items to as-new operating condition or replace damaged items as directed by the Contracting Officer, at no additional cost or time to the Government.

c. Welding Qualifications: Before any welding is performed, contractor shall submit a certificate certifying that welders comply with the following requirements:

(1) Qualify welding processes and operators for piping according to ASME BPVC SEC IX, AWS Z49.1 and AWS B2.1/B2.1M.

(2) Comply with provisions in ASME B31.9 or ASME B31.1.

(3) Certify that each welder and welding operator has passed AWS qualification tests for welding processes involved and that certification is current and recent. Submit documentation to the

Contracting Officer.

(4) All welds shall be stamped according to the provisions of the American Welding Society.

d. ASME Compliance: Comply with ASME B31.9 or ASME B31.1 for materials, products, and installation. Safety valves and pressure vessels shall bear appropriate ASME labels.

1.7 DELIVERY, STORAGE AND HANDLING

a. Equipment and material placed on the job site shall remain in the custody of the Contractor until phased acceptance, whether or not the Government has reimbursed the Contractor for the equipment and material. The Contractor is solely responsible for the protection of equipment and material against damage or theft.

b. Protect piping systems against the entry of water, mud or other foreign substances by installing watertight covers on open ends at all times. Both inside and outside shall be cleaned before painting or placing equipment in operation. Protect direct-buried system coatings from ultraviolet light (sunlight). Existing equipment worked on by the Contractor or in the Contractor's working area shall be considered to be in the custody and responsibility of the Contractor and shall be protected as required for new work.

c. Damaged equipment shall be replaced with an identical unit as determined and directed by the Contracting Officer. All insulated piping systems exposed to water must be replaced prior to installation at no additional cost or time to the Government.

1.8 AS-BUILT DOCUMENTATION

a. Submit manufacturer’s literature and data updated to include submittal review comments and any equipment substitutions.

b. Submit operation and maintenance data updated to include submittal review comments, VA approved substitutions and construction revisions shall be four hard copies and one electronic copy of each inserted into a three-ring binder. All aspects of system operation and maintenance procedures, including applicable piping isometrics, wiring diagrams of all circuits, a written description of system design, control logic, and sequence of operation shall be included in the operation and maintenance manual. The operations and maintenance manual shall include troubleshooting techniques and procedures for emergency situations. Notes on all special systems or devices shall be included. A list of recommended spare parts (manufacturer, model number, and quantity) shall be furnished. Information explaining any special knowledge or tools the owner will be required to employ shall be inserted into the As-Built documentation.

c. The installing contractor shall maintain as-built drawings of each completed phase for verification; and, shall provide the complete set at the time of final systems certification testing. Should the installing contractor engage the testing company to provide as-built or any portion thereof, it shall not be deemed a conflict of interest or breach of the ‘third party testing company’ requirement. Provide record drawings as follows:

(1) Red-lined, hand-marked drawings are to be provided, with one paper copy and a scanned PDF version of the hand-marked drawings provided on CD or DVD.

(2)

(3)

d. The as-built drawings shall indicate the location and type of all lockout/tagout points for all energy sources for all equipment and pumps to include breaker location and numbers, valve tag numbers, etc. Coordinate lockout/tagout procedures and practices with local VA requirements.

e. Certification documentation shall be provided to Contracting Officer 21 working days prior to submitting the request for final inspection. The documentation shall include all test results, the names of individuals performing work for the testing agency on this project, detailed procedures followed for all tests, and provide documentation/certification that all results of tests were within limits specified. Test results shall contain written sequence of test procedure with written test results annotated at each step along with the expected outcome or setpoint. The results shall include all readings, including but not limited to data on device (make, model and performance characteristics), normal pressures, switch ranges, trip points, amp readings, and calibration data to include equipment serial numbers or individual identifications, etc.

1.9 COORDINATION

Coordinate exterior steam lines and associated systems and connections to building services up to the actual extent of building wall.

1.10 UTILITY LOCATION SERVICES

Prior to any demolition or excavation, provide for utility location services to mark on the ground with fluorescent paint the location of existing underground utilities, and their identification. The term “utility(ies)” includes both public utilities and VA-owned utilities, for all underground services.

PART 2 PRODUCTS

2.1 PRE-ENGINEERED, FACTORY-FABRICATED, DIRECT-BUIRED SYSTEMS

a. Complete factory-fabricated steam and condensate piping system with carrier pipes,carrier pipe insulation, pipe supports, outer conduit, outer conduit coating, outer conduit insulation and jacket, and diffuser barrier. The system supplier shall have at least ten years' experience fabricating underground high temperature distribution systems. Do not locate condensate pipes in casings that contain steam pipes. All straight sections, fittings, anchors and other accessories shall be prefabricated to job dimensions. Each system layout shall be computer analyzed by the piping system manufacturer to determine stresses and movements of the service pipe. The system design shall be in strict confirmance with ANSI B31.1, latest edition, and stamped by a Registered Progessinal Engineer.

b. All components of system shall be suitable for carrier pipe pressures

and temperatures as follows:

(1) Steam System: (150 psig); (366 degrees F).

(2) Condensate System: (50 psig); (310 degrees F).

c. Steam Carrier Pipes and Condensate Carrier Pipes: No piping joints are allowed in factory-fabricated straight sections of pre-engineered direct-buried systems.

d. Subassemblies: End seals, gland seals and anchors shall be designed and facotry prefabricated to prevent the ingress of moisture into the system. All subassemblies shall be designed to allow for complete draining and drying of the conduit system.

e. Carrier Pipe Insulation:

(1) Conform to minimum insulation thickness of 1-1/2" or as required for service temperature in carrier pipe.

(2) Section A: Steam temperature is (350 degrees F), steam pressure is 110 (psig). Pumped condensate temperature is (200 degrees F). Drip return temperature is (212 degrees F).

f. Carrier Pipe Insulation: Carrier pipe insulation shall be a high temperature insulation blanket formed of silica aerogel and reinforced with a non-woven, glass-fiber batting. The insulation shall be held in place by a stainless steel bands or staples installed not more than 18 inches apart.

g. Pipe Supports: All pipes within the outer conduit shall be suppoarted to allow for continuatous drainage of the conduit in place. Supports shall be the type where the service pipe insulation thermally isolates the service pipe from the outer conduit. No calcium silicate or other type of insulation shall be allowed. The surface of the support insulation shall be protected by a steel sleeve not less than 12 inches long.

h. Outer Conduit: The steel conduit casing shall be smooth wall, welded steel conduit of the thickness specified below. Changes in casing size, as required at oversized casing to allow for service pipe expansion shall be accomplished by accentric and/or concentric fittings and shall provide for continuous drainage.

Conduit Size (inches)	**Conduit Thickness (inches)**
(6 – 26)	(10 Gauge)
(28 - 36)	(6 Gauge)
(38 - 42)	(4 Gauge)

i. Outer Conduit Coating: The exterior steel conduit surface shall be abrasive blast-cleaned to a minium of a near white surface, SSPC-SP10-63T. Profile must be a minimum of 1.5 mil peak to valley range. Any areas of rust bloom or oil shall be wiped and reblasted. After blasting, the steel conduit shall be coated with epoxy. The epoxy coating shall be a two part coating consisting of a base material and curing agent spray applied to a minimum thickness of 8-12 mils. The coated conduit shall be holiday tested at 1,000 volts to

ensure a void free coating. Areas of the conduit not passing the holiday test shall be patch coated and retested.

j. Outer Conduit Insulation and Jacket: Conduit insulation shall be spray applied polyurethane foam having a minimum density of 2lbs/ft3 for the straight lengths and fittings. The insulation thickness shall be 1 inch maximum. The polyurethane foam shall have a maximum initial K value of 0.18, minimum density of 2 lbs/ft3 and a minimum closed cell content of 90%. The outer jacket shall be fiberglass reinforced polymer (FRP) and shall be applied directly onto the urethane foam insulatioin. No PVC or polyethylene jacket shall be allowed. ALl straights and fittings shall be factory jacketed.

k. Diffusion Barrier: An aluminum diffusion barrier shall be applied on the outside of the insulation before application of the outer jacket. The barrier shall prevent the diffusion of the blowing agent out of the foam to prevent the foam from aging. The diffusion barrier shall be of composite construction iwth a minimum 12 micron aluminum layer sandwiched between two layers of polyethylene each a minimum of 50 microns thick. The polyethylene layers shall be corona treated to guarantee bonding between the foam insulation and the outer jacket.

l. Leak Detection and Location System: The pre-engineered direct buried piping system shall be provided with leak detection by means of manufacturering into the system a copper wire through each piece of pre-insulated pipe and fittings. The piping system manufacturer shall install the wire ina manner that has the wire embadded in the outer polyurethane foam insulation and not touching the steel conduit. The condtractor shall connect the wire together at each field joint with a recommended crimping tool. Provide with advanced electronic monitoring system that provides full time surveillance for system integrity. Monitoring system shall continuously monitor for the pressence of moisture in the conduit and will signal the location of any moisture. Location of the leak detection system monitor shall be as indicated on the contract drawings.

l. Carrier pipe guides and supports shall be maximum spacing 10 feet on centers, no greater than 5 feet from pipe ends, minimum of three guides per elbow section. Designed to permit thermal expansion without damage, provide proper pipe guiding and support, and to allow horizontal movement in two directions as necessary at expansion loops and bends. Design of guides and supports must permit continuous drainage of water in bottom of casing. Pipe insulation shall extend through the pipe guides and supports and be protected by steel sleeves. Design of guides and supports shall be such that no metal-to-metal contact exists between the casing and the carrier pipe. Insulation or non-metallic material used to ensure no metal to metal contact shall be designed to not be compressed by the weight of the carrier pipe when full of water.

m. Anchor plates shall be ASTM A36/A36M steel, welded to carrier pipe and casing, 1/2 inch minimum thickness, passages for air flow and water drainage through the annular air space in the system. Coated with same coating material as the casing. Locate (3 to 5 feet from piping entrance to manhole or building wall. Walls of manholes and buildings cannot be utilized as anchor points.

n. Field connection of casing sections shall be steel section conforming to casing specification, welded to casing sections, coated on all

surfaces with system manufacturer’s coating field repair compound, and covered with a 0.05 inch minimum thickness polyethylene shrink sleeve designed for a service temperature exceeding (176 degrees F).

o. Manhole and building wall penetrations shall provide steel leak plates welded to wall sleeves or to casings. Where a wall sleeve is utilized, allow sufficient annular space between the sleeve and the casing and install a watertight seal, rated for (250 degrees F) minimum. Manhole and building walls cannot be used as anchor points.

p. Provide sacrificial anode type cathodic protection system with dielectric isolation devices and test stations for all systems. Design system for 25 years of service, assume two percent bare metal. System shall comply with NACE SP0169.

q. Provide embossed brass or stainless-steel tag hung by a brass or stainless-steel chain at each end of each casing or insulated piping in the manholes and buildings. The tag shall identify system manufacturer’s name, date of installation, government contract, and manufacturer’s project number.

r. All branch piping connections must be located in manholes.

s. Ensure the Pre-Engineered Direct Buried System manufacturer is responsible for the complete design of the direct buried system, including, but not limited to, the product to be supplied, fabrication, installation, supervision, and testing of the system within the design parameters established by the contract documents, and in compliance with the detailed design. The complete design of the system shall be sealed by a Professional Engineer in the employ of the Pre-Engineered Direct Buried System manufacturer.

t. Furnish thermal expansion calculations for the steam and condensate piping using the design characteristics indicated in this section and installation temperature no higher than the ambient temperature at the site: (50 degrees F).

u. Pre-Engineered Direct Buried System manufacturer shall submit a complete description of the design and assembly of the system, materials of construction and field installation instructions. Include sufficient system details to show that the specified minimum insulation thickness has been met. A detailed design layout of the system (plan and elevation views) showing size, type, elevations and location of each component to be used in the system, the design and location of anchors, pipe guides, pipe supports, expansion loops, Z-bends, L-bends, end seals, leak plates, joint locations, pipe and insulation thickness and sizes, types, and movements, connection to manhole and building wall penetrations, and including, if applicable, details of transition point to aboveground or other type systems. Detailed design layout drawings shall be stamped by a registered Professional Engineer.

v. Expansion Loops and Bends: Pipe-stress and system-expansion calculations for each expansion compensation elbow using a finite element computer generated 3-dimensional analysis (FEA). Demonstrate with calculations that pipe stresses from temperature changes are within the allowable requirements in ASME B31.1 and that the anchors and the guides will withstand the resultant forces. Detailed design layout drawings shall include all analysis node points. As a minimum,

computer analysis results shall include node stresses, forces, moments and displacements. Calculations shall be stamped by a registered Professional Engineer in the employ of the DDT manufacturer.

w. Reinforced Concrete Manholes: Not less than 8 inches thick. Pour monolithically where possible. Place waterproof membrane between mud slab and bottom concrete slab, and continue up sides to top of sidewalls. Joints between manhole walls and casings or concrete trench sections shall be watertight. Steel manholes or prefabricated concrete manholes are prohibited.

x. Accessories for Manholes: Cast iron manhole frames and solid covers, not less than 28 inches clear openings. Unless otherwise shown on the drawings, frames and covers shall be as follows:

(1) For non-traffic applications:

(a) CID A-A-60005 NOT1, Frame Type IV, Size 28

(b) CID A-A-60005 NOT1, Cover Type E, Size 28, cast identification “STEAM”.

(2) For traffic applications:

(a) CID A-A-60005 NOT1, Frame Type I, Style A, Size 27A.

(b) CID A-A-60005 NOT1, Cover Type A, Size 27A, cast identification “STEAM”.

(3) Manhole steps shall be standard, cast iron.

y. Manhole Ventilation: As indicated on Drawings. Construct ventilation ducts of galvanized steel sheet metal and in accordance with ASHRAE Handbook recommendations for low pressure ducts. Gravity ventilators shall be factory-fabricated of aluminum or galvanized steel and arranged as indicated on drawings. Ventilating pipes shall be standard weight black steel and installed as shown on drawings.

z. Drainage as shown on drawings. Provide a 24 inches square by 24 inches deep sump pit in each manhole where indicated on drawings. Provide larger sump pit if necessary to accommodate required electric sump pumps.

aa. Electric Sump Pumps with Automatic Controls and High-Water Alarm:

(1) Type: High temperature submersible duplex pumps and automatic controls.

(2) Service: Continuous operation at required flows and pressures while completely submerged at (200 degrees F). All pumps and pump controls shall have demonstrated 200,000 cycles of operation while totally submerged in (200 degrees F) water.

(3) Capacity and Pressure: Pumps shall be capable of passing 3/8 inch spheres.

(4) Pumps: Epoxy-coated cast iron casing, cast iron impeller, stainless steel shaft, carbon/ceramic shaft seal, stainless steel hardware, permanently lubricated bearings, screened inlets.

Schedule 80 discharge pipe protected from corrosion.

(5) Motors: Non-overloading at all points on the pump performance curve. Include overload protection.

(6) Controls: Automatic alternating lead-lag, with damp-proof electrical service. Mount non-submerged control components in a NEMA 4 enclosure on adjacent wall or dedicated galvanized steel support stand.

(7) High Water Alarm Switch: Set at level below lowest steam or condensate pipe in the manhole. Switch shall activate weatherproof red alarm light mounted above grade as shown. Provide contacts and connect to engineering control center.

2.2 CONCRETE SHALLOW TRENCHES

Cast-in-Place Trench: Reinforced concrete with minimum dimensions as called for on drawings.

(1) Trench Covers in Grass or Sidewalk Areas: Precast, U-shaped, steel reinforced concrete sections with removable concrete lid, set to existing grade, flat and true at all points of contact on trench wall. Trench sections and covers shall be rated to resist AASHTO HS-20 design load. Concrete shall have a 28-day compressive strength of 5,000 PSI minimum.

(2) Trench sections shall be fabricated such that sections fit end-to-end with reasonably watertight connections.

(3) Fabricate each trench section to have at least one 12"x12" square opening per section to allow water to pass through to gravel sub-base below.

(4) Gaskets and Sealants: ASTM C920, 1/4 inch thick neoprene pads with a minimum width of 2 inches between covers and tops of walls; elastomeric sealants that are available as a one or two component system. Asphaltic sealants are prohibited. Sealants must resist 50 percent total joint movement. Non-sagging sealant must be used for vertical joints. Self-leveling sealant must be used for trench top butt joints.

(5) Fabricate trench with weldable ASTM A36 steel embedded plates for pipe support. Plates shall be of sizes shown on drawings. Embeds shall be spaced not more than 10'-0" on center in direction of pipe, but each trench section shall have at least one set of pipe support embeds. Embeds shall be located per mechanical contractor's direction to achieve the required pipe slope and clearances.

(6) Fabricate trench with weldable ASTM A36 steel embed plates for tie plates between each trench section. Provide steel tie plates to be installed by the contractor.

(7) Fabricate trench at slope change noted on civil drawings by mitering one trench end or contractor shall trim precast trench in field to provide reasonably watertight miter.

(8) Trenches shall be installed in accordance with manufacturer's

instructions. Use rope-style mastic to seal joints at trench sections meeting concrete walls.2.3 STEAM PIPING

a. Pipe: ASTM A53/A53M, steel, seamless, Grade B, Schedule 40.. Grade F, furnace butt-welded pipe is prohibited. Use Schedule 80 pipe and fittings for threaded joints.

b. Joints:

(1) In trenches: Butt-weld joints. Socket weld is required for pipe sizes 2 inches and less.

(2) In tunnels, manholes, and open areas: Butt weld pipe sizes (2-1/2 inches) and greater; thread or socket weld pipe sized 2 inches and less. Use Schedule 80 pipe and fittings for threaded joints.

c. Fittings:

(1) Butt welded joints: ASTM A234/A234M or ASME B16.9, steel, Grade B, same schedule as adjoining pipe. All elbows shall be long radius unless otherwise indicated. Tees shall be full size or reducing as required, having interior surfaces smoothly contoured.

(2) Threaded joints: ASME B16.11, forged steel fittings, Class 2000. Use Schedule 80 pipe only.

(3) Socket welded joints: ASME B16.11, forged steel, 2000 psig class.

d. Flanges and Bolts: ASTM A105/A105M, pressure class (150 psig). Bolts shall be high strength ASTM A193/A193M, Class 2, Grade B7. Nuts shall be ASTM A194/A194M.

e. Unions: Pipe 2 inches and less shall be threaded, steel, 297 psig class.

2.4 STEAM CONDENSATE PIPING

a. Pipe: ASTM A53/A53M, seamless, Grade B , Schedule 80. Grade F, furnace butt-welded pipe is prohibited. Use Schedule 80 pipe and fittings for threaded joints.

b. Joints:

(1) In trenches: Butt weld joints. Socket weld is required for pipe sizes 2 inches and less.

(2) In tunnels, manholes, and open areas: Butt weld pipe sizes (2-1/2 inches) and greater. For system pressures of (15 psi) or less, thread or socket weld pipe sizes 2 inches and less. For system pressures of (15 psi) and no greater than (100 psig), socket weld pipe sizes 1 inch and greater, and thread pipe sizes less than (1 inch). For higher system pressure, socket weld pipe sizes of 2 inches or less.

c. Fittings:

(1) Welded joints: ASTM A234/A234M, steel, Grade B, or ASME B16.9, same schedule as adjoining pipe.

(2) Threaded joints: ASME B16.11, forged steel fittings, Class 2000. Use Schedule 80 pipe and fittings only.

(3) Socket welded joints: ASME B16.11, forged steel, 2000 psig class.

d. Unions (Except in Trenches): Pipe 2 inches and less, 297 psig steel.

e. Flanges: Weld neck ASME B16.5 or ASTM A105/A105M, forged steel, (150 psig).

2.5 EXPANSION LOOPS AND BENDS

a. Stresses: Less than the maximum allowable stress in accordance with ASME B31.1. Submit shop drawings and stress and anchor force calculations for all loops and bends. Show locations of all anchors, guides and supports. Base calculations on 150 psig and (366 degrees F) for steam line loops and bends and 50 psig and (310 degrees F) for condensate return line loops and bends. Base calculations on actual pressures and temperatures if they are higher than those listed above. Stress analysis shall cover all conditions under which the system can conceivably experience during its lifetime.

b. Steam systems 15 psig and less: ASME B31.9, base calculations for steam and condensate on 15 psig and (250 degrees F).2.6 EXPANSION JOINTS

a. Provide factory-built or field-fabricated guides located along the pipelines to restrain lateral pipe motion and direct the axial pipe movement into the expansion joints.

b. Minimum Service Requirements:

(1) Pressure containment:

(b) Steam service (31 to 123 psig): Rated 150 psig at (366 degrees F).

(d) Condensate service: Rated 100 psig at (309 degrees F).

(2) Number of full reverse cycles without failure: Minimum 1000.

(3) Movement: Allowed as recommended safety factor of the manufacturer.

c. Internally pressurized bellows shall have:

(1) ASTM A240/A240M, multiple corrugations, Type 304 or 321 stainless steel.

(2) Internal stainless-steel sleeve running the entire length of bellows.

(3) External steel equalizing rings for services exceeding (50 psig).

(4) Welded ends, flanged ends for 2 inches and greater pipes.

(5) External tie rods: Design to withstand pressure thrust force upon anchor failure if one or both anchors for the joint are at change in direction of pipeline and integral external cover.

d. Externally pressurized bellows shall have:

(1) ASTM A240/A240M, multiple corrugations, Type 304 stainless steel.

(2) Internal and external guides integral with joint.

(3) Design for external pressurization of bellows to eliminate squirm.

(4) Welded ends, flanged ends for 2 inches and greater pipes.

(5) Include threaded connection at bottom, 1 inch minimum, for drain or drip point and integral external cover and internal sleeve.

2.7 VALVES

a. Gate Valves (ASTM A126):

(1) Type 101:

(a) Type applies to steam valves with sizes (2-1/2 inches) and greater.

(b) Steel body, rated 150 psig at (500 degrees F), 11-1/2 to 13 percent chromium stainless steel flexible wedge and hard faced (stellite) or nickel copper alloy seats, 150 psig flanged ends, OS&Y, rising stem, bolted bonnet.

(c) Factory installed globe valved bypass on all steam valves greater than (3 inches).

(d) Drill and tap bosses for connection of drains where shown.

(2) Type 102 is not used.

(3) Type 103:

(a) Type applies to condensate valves with sizes (2-1/2 inches) and greater.

(b) Forged steel body, Class B, rated for 123 psig saturated steam, 200 psig WOG, bronze or bronze face wedge and seats, 123 psig ASME flanged ends, OS&Y, rising stem, bolted bonnet, renewable seat rings.

(4) Type 104:

(a) Type applies to condensate valves with sizes 2 inches and less.

(b) Forged steel body, rated for 200 psig saturated steam, 400 psig WOG, bronze wedges and Monel or stainless-steel seats, threaded ends, rising stem, union bonnet.

(5) Type 105 is not used.

(6) Type 106:

(a) Type applies to steam valves with sizes 2 inches and less.

(b) Forged steel body, rated for 300 psig at (420 degrees F) minimum Class 600 psig or Class (800 psig), hardened stainless steel or satellite wedge and seats, threaded ends, OS&Y, rising stem, bolted bonnet.

b. Globe Valves (ASTM A126):

(1) Type 201:

(a) Type applies to steam valves with sizes (2-1/2 inches) and greater.

(b) Carbon steel body, rated 150 psig at (500 degrees F), 11-1/2 to 13 percent chromium stainless steel or stellite disc and seat, 150 psig ASME flanged ends, OS&Y, rising stem, bolted bonnet, renewable seat rings. Drill and tap bosses for connection of drains.

(2) Type 202 is not used.

(3) Type 203:

(a) Type applies to condensate valves with sizes (2-1/2 inches) and greater.

(b) Steel body, rated for 123 psig saturated steam, 200 psig WOG, bronze or bronze-faced disc (Teflon or composition facing permitted) and seat, 123 psig ASME flanged ends, OS&Y, rising stem, bolted bonnet, renewable seat rings.

(4) Type 204:

(a) Type applies to steam valves and condensate valves with sizes 2 inches and less.

(b) ASTM B61, Forged steel body, rated for 200 psig saturated steam, 400 psig WOG, hardened stainless steel disc and seat, threaded ends, rising stem, union bonnet, renewable seat rings.

c. Check Valves (ASTM A126):

(1) Type 401:

(a) Type applies to steam valves with sizes (2-1/2 inches) and greater.

(b) Steel body, swing-type, rated for 150 psig at (500 degrees F), stainless steel or stainless steel - faced disc and seat, 150 psig ASME flanged ends, bolted cover, renewable disc.

(2) Type 402 is not used.

(3) Type 403:

(a) Type applies to condensate valves with sizes (2-1/2 inches)

and greater.

(b) Forged Steel body, Class B, swing-type, rated for 123 psig saturated steam, 200 psig WOG, bronze or bronze-faced disc and seat, 123 psig ASME flanged ends, bolted cover, renewable disc and seat.

(4) Type 404:

(a) Type applies to steam valves and condensate valves with sizes 2 inches and less.

(b) Forged Steel body, swing-type, rated for 200 psig saturated steam, 400 psig WOG, bronze disc, threaded ends, regrinding disc.

d. Ball Valves (ASTM A126):

(1) Type 501 is not used.

(2) Type 502:

(a) Type applies to steam valves and condensate valves with sizes 2 inches and less.

(b) Forged steel body, rated for 150 psig at (365 degrees F), 250 psig at (250 degrees F); reinforced TFE seat, stem seal and thrust washer; end entry, threaded ends, one-fourth turn to open.

(3) Type 503 is not used.

(4) Type 504:

(a) Type applies to steam valves and condensate valves with sizes (2-1/2 inches) and greater.

(b) Carbon steel or ductile iron body, saturated steam service, rated for (150 psig), stainless steel ball and stem, Polyfil seat, live-loaded stem seal, 150 psig ASME flanged ends.

2.8 STEAM PRESSURE REDUCING VALVES

a. Valves: Single seated, diaphragm operated, spring loaded, steam pilot controlled, normally closed, packless, adjustable set pressure. Pilot shall sense controlled pressure downstream of main valve.

b. Controlled reduced pressure to steam piping systems: Design for saturated steam at pressures shown on drawings.

c. Pressure Control: Smooth, continuous. Maximum 10 percent deviation from set pressure over a 10 to 1 turndown. Refer to schedules on drawings for flow and pressure requirements. Downstream safety valve to be sized to equal or exceed the maximum total flow capacity of the pressure reducing station.

d. Construction:

(1) Main Valve:

(a) Pipe sizes 2 inches and less: Steel body rated for (250 psig),

threaded ends. Globe body valve and seat shall be replaceable, Type 316 stainless steel and include stainless steel stem.

(b) Pipe sizes greater than (2 inches): Steel body rated for 150 psig ASME flanged ends, or steel body 250 psig ASME flanged ends, globe body valve and seat shall be replaceable, Type 316 stainless steel and include stainless steel stem.

(2) Pilot Valve: Valve plug and seat shall be replaceable, stainless steel.

(3) Select pressure reducing valves to develop less than 85 db(A) at 5 feet elevation above adjacent floor, and 5 feet distance in any direction. Inlet and outlet piping for steam pressure reducing valves shall be Schedule 80 minimum for required distance to achieve required levels or sound attenuators shall be applied.

e. Direct Digital Control PRV Valves: May be furnished in lieu of steam operated valves. All specification requirements for steam operated valves shall apply. In the event of signal failure, failsafe device accessory in the actuator to stroke valve to predetermined position indicated . Install per manufacturer’s recommendation.

2.9 STEAM TRAPS

a. Apply at steam line drip points.

b. Fixed orifice or Venturi type traps are prohibited.

c. Construct inverted bucket type with thermostatic vent in bucket, except closed-float-thermostatic on discharge side of pressure reducing stations. Each type furnished by a single manufacturer. Select the traps for pressures and capacities as shown or required.

d. Traps: Stainless steel bodies. Construction shall permit ease of removal and servicing working parts without disturbing connecting piping. Include stainless steel floats, hardened chrome steel valves, stainless steel mechanisms and bi-metallic air vent on inverted bucket traps.

e. Provide electronic trap performance monitoring devices that are compatible with the existing monitoring system. Trap malfunctions shall be automatically transmitted to and properly interpreted by the existing monitoring system. Provide all necessary power sources, transmitting and retransmitting devices and batteries to achieve a properly operating system. .

f. All traps shall include ports for future installation of monitoring devices. To facilitate future removal of plugs, remove plugs, install Teflon tape on the threads, and reinstall the plugs.

g. Label each trap at the factory with an identification number keyed to the contract drawings. Label shall be a metal tag permanently attached to the trap.

2.10 STRAINERS, Y TYPE

a. Provide as shown on steam and condensate piping systems.

b. Include open end removable cylindrical screen and threaded blow off connection.

c. For steam service up to 150 psig and at drip traps, strainer shall be rated for minimum 150 psig saturated steam; rated for (150 psig), flanged ends, steel, for pipe sizes greater than (2 inches). Use forged steel, rated for 250 psig saturated steam, threaded ends, for pipe sizes 2 inches and less.

d. For condensate service, strainer shall be rated for 123 psig saturated steam, 175 psig WOG. Provide steel, flanged ends, for pipe sizes greater than (2 inches). Provide forged steel, threaded ends, for pipe sizes 2 inches and less.

e. Strainer screen shall be stainless steel, with a free area not less than 2-1/2 times flow area of pipe. Diameter of openings shall be 0.05 inch or less on steam service and 0.06 inch or less on water service.

f. Include gate type valve and quick couple hose connection on all blowoff connections.

2.11 SAFETY VALVES AND VENT CONNECTORS

a. Safety Valves: Conform to the requirements of ASME BPVC SEC VIII D1, Unfired Pressure Vessels) and be approved by the National Board of Boiler and Pressure Vessel Inspectors.

b. Relieving Capacity: Not less than that shown on the drawings with a pressure rise above set pressure not to exceed 10 percent of set pressure.

2.12 PRESSURE GAUGES

a. Provide gauges immediately downstream of each steam line isolation valve, before and after each steam pressure reducing station and where shown on the drawings.

b. Gauges: ASME B40.100.

(1) Solid armored front between measuring element and dial, blowout back, bottom connection, phenol turret type.

(2) Non-corrosive, (4-1/2 inches) diameter face with black markings on white background.

(3) Bourdon tube measuring element designed for service. Provide bellows for pressure ranges less than (15 psig).

(4) Stainless steel, rotary movement.

(5) Micrometer adjustable, black color pointer.

(6) Plastic window.

(7) Provide liquid filled gauges at outlet of all pumps.

(8) Factory calibrated and certified.

c. Accuracy: Grade 2A, 1/2 percent, on all gauges; except Grade A, one percent permitted on diaphragm actuated gauges, liquid filled gauges, and compound gauges.

d. Include:

(1) Red set hands on gauges located at automatic pressure regulator valve outlets.

(2) Needle valve or gauge cock rated for the service.

(3) Syphon on all steam gauges.

(4) Overload stop on all pressure gauges.

e. Pressure gauge ranges shall be selected such that the normal operating pressure for each gauge is displayed near the midpoint of each gauge's range. Gauges with ranges selected such that the normal pressure is displayed at less than 30 percent or more than 70 percent of the gauge's range are prohibited. The units of pressure shall be psig.

2.13 THERMOMETERS, PIPE OR TANK MOUNTED

a. Thermometer locations are shown on the drawings.

b. Thermometers:

(1) Industrial type, separable well and socket.

(2) Red reading mercury combination Celsius/Fahrenheit scale, 9 inches long.

(3) Stainless steel corrosion resistant case with safety glass front.

(4) Adjustable angle for ease of viewing.

(5) Wells sized to suit pipe diameter without restricting flow, or provide oversized pipe at well location. Snug sliding fit between socket and well. Well should be 316 stainless steel.

(6) Accuracy shall be one percent of scale range.

(7) (32 to 300 degrees F).

(8) Factory calibrated and certified.

2.14 PIPE HANGERS AND SUPPORTS

a. Requirements: MSS SP-58 and ASME B31.1. Refer to Section 13 05 41, SEISMIC RESTRAINT REQUIREMENTS FOR NON-STRUCTURAL COMPONENTS.

b. Applies to all piping not in factory-fabricated direct-buried system. All systems shall be completely supported. Arrange supports so that all loads due to weight, thermal expansion, seismic shock (if applicable), and pressure are transferred from the support system to the structure. The design and location of supports shall at all times prevent excessive forces, moments, and stresses from being imposed on the equipment, structure, supported system, and supports. Heated systems generally require resilient or roller/slide supports.

c. Manufacturer Certification: Factory built products of a manufacturer whose principle business is pipe supports for 10 years. All components must have published load ratings. For concrete trenches, non-factory built products that comply with details may be utilized.

d. Components:

(1) Roller supports: MSS SP-58, Type 41, 43, and 46. Provide vertical adjustment for Type 41 with threaded studs and nuts adjacent to the roller.

(2) Variable spring support assembly: MSS SP-58, Type 51 variable spring, Type 3 pipe clamp, Type 1 clevis, and Type 53 variable spring trapeze. Locate Type 51 variable spring within 1 foot above pipe attachment. Attach rod to top of variable spring with Type 14 clevis.

e. Spring Cushion Support Assembly: MSS SP-58.

(1) Double rod assembly: Type 41 and 49.

(2) Single rod assembly: Type 48 spring cushion Type 3 pipe clamp Type 1 clevis. Locate spring cushion within 1 foot above pipe attachment.

f. Clevis Supports: MSS SP-58, Type 1.

g. Wall Brackets: MSS SP-58, Type 33.

h. Pipe Stands: MSS SP-58, Type 38.

i. Riser Clamp: MSS SP-58, Type 42.

j. Alignment Guides: Welded steel as shown to restrain movement perpendicular to the long axis of the piping. If not welded, provide steel spider clamped to pipe, enclosed within steel sleeve that is bolted to structural support. Must provide lateral force equal to minimum of 15 percent of anchor loading.

k. Trapeze Supports: MSS SP-58, may be used where pipes are close together and parallel, structural steel channels or angles. Bolt roller supports to steel to support piping subject to horizontal thermal expansion. Attach other piping with “U” bolts.

l. Pipe Covering Protection Saddles: MSS SP-58, Type 39. Provide at all support points on insulated pipe except where Type 3 pipe clamps are provided.

m. Sliding Supports: MSS SP-58, Type 35. Welded steel attachments to pipe and structure with Teflon or graphite sliding surfaces bonded to the attachments. Provide steel guides, except at expansion bends, to prevent lateral movement of the pipe.

n. Pipe Racks and Miscellaneous Supports: ASTM A36/A36M, structural steel shapes. Manufactured strut systems are acceptable if they have the required load carrying ability.

o. Supports, including all structural steel, in trenches and manholes:

Hot-dip galvanized.

p. Seismic Restraints: Refer to Section 13 05 41, SEISMIC RESTRAINT REQUIREMENTS FOR NON-STRUCTURAL COMPONENTS.

(1) Provide bracing as required. Refer to details on drawings.

(2) Shock Absorbers: MSS SP-58, Type 50. Mechanical or hydraulic type rated for shock loads. Pipe attachments shall be MSS SP-58, Type 3.

2.15 BURIED UTILITY WARNING TAPE

0.004 inch thick, 6 inches wide, yellow polyethylene with a ferrous metallic core, acid and alkali-resistant and shall have a minimum strength of 1740 psig lengthwise and 1500 psig crosswise with an elongation factor of 350 percent. Provide bold black letters on the tape identifying the type of system. Tape color and lettering shall be unaffected by moisture and other substances contained in the backfill material.

PART 3 EXECUTION

3.1 GENERAL

a. If an installation is unsatisfactory to the Contracting Officer, the Contractor shall correct the installation at no additional cost or time to the Government.

b. Connect new work to existing work in a neat and workmanlike manner. Where an existing structure must be cut or existing utilities interfere, such obstruction shall be bypassed, removed, replaced or relocated, patched and repaired. Piping connections shall be made only in manholes, tunnels or buildings.

c. Coordinate the location of all items of equipment and work of all trades. Maintain operability and maintainability of the equipment and systems. The contractor at his cost shall perform any relocation of equipment or systems to comply with the requirement of operability and maintainability.

d. Unless otherwise shown in the contract documents, steam lines shall be graded downward not less than 2 inches in 40 feet in direction of the flow. Provide eccentric reducing fittings on steam mains and branches, (except on vertical piping). Install said fittings to maintain continuity of grade in bottom of pipeline. Provide risers with drip pockets and steam traps on steam lines where space restrictions prevent continuous grading. All steam traps must be located in manholes or tunnels.

3.2 DEMOLITION

a. Perform work in accordance with requirements for phasing and the Drawings. Phasing shown on drawings is an outline from which the contractor will use to provide more details in executing the work in each phase.

b.

c. Completely remove all pipe, valves, fittings, insulation, and all

hangers including the connection to the structure and any fastenings.

d. Seal all openings in manhole or building walls after removal of piping.

e. All material and equipment removed shall become the property of the Contractor and shall be removed from Government property and shall not be stored in operating areas unless designated as being turned over to the owner.

f. All flame cutting shall be performed with facility burn permit in place and adequate fire protection facilities available as required by safety codes and Contracting Officer.

3.3 PIPING INSTALLATION

a. Drawings indicate general location and arrangement of piping systems. For field installed insulation, install piping insulation as specified herein and/or as recommended by the manufacturer.

b. Remove all standing water in the bottom of trench.

c. Pipe Bedding: Minimum 6 inch layer of sand. Provide compaction and leveling of virgin soil prior to installing bedding.

d. Clearance: Minimum (12 inches)clearance between the external insulation/casing of pipes 12 inches or smaller, and if no valves and fittings are present along the pipes. Otherwise provide the required service clearances between valves/fittings on adjacent pipes to allow future replacement. Obtain minimum service clearances from valve/fitting manufacturer.

e. Testing: Do not insulate piping or backfill piping trench until field quality-control and cleaning, followed by testing has been completed and results approved by Contracting Officer.

f. Grade:

(1) Install condensate piping at uniform grade of 0.4 percent downward in direction of flow.

(2) Install steam piping at uniform grade of 0.2 percent downward in direction of flow or as indicated on the Drawings.

g. Drain Valves and Air Vents: For carrier piping, install at low points and air vents at high points.

h. Install components with pressure rating equal to or greater than system design pressure.

i. Install piping free of sags and bends.

j. Install fittings for changes in direction and branch connections.

k. On direct-buried pipes, secure anchors with concrete thrust blocks.

3.4 DRAIN VALVES AND VENT VALVES

a. Provide (1-1/2 inch) minimum pipe size drain valves on condensate return carrier pipes at all low points in manholes. Provide 1 inch

minimum manual air vent valves in manholes at all high points in condensate return carrier piping.

b. Do not install any valves at the outer casing piping vents.

3.5 PIPE SUPPORT INSTALLATION (IN TRENCHES, TUNNELS, MANHOLES)

a. Coordinate support locations prior to erection of piping. Hanger parts must be marked at the factory with a numbering system keyed to hanger layout drawings. Layout drawings must be available at the site during construction.

b. Upper Attachments to Structure:

(1) New reinforced concrete construction shall have concrete inserts.

(2) For existing reinforced concrete construction, upper attachment shall be welded or clamped to steel clip angles or other construction methods/detail shown on the drawings that are expansion bolted to the concrete. Expansion bolting shall be located so that bolts are shear loaded.

(3) For steel deck and structural framing, upper attachments shall be welded or clamped to structural steel members.

c. In existing concrete construction, expansion fasteners may be used for hanger loads up to one third the manufacturer's rated strength of the expansion fastener. Power set fasteners may be used for loads up to one fourth of rated load. When greater hanger loads are encountered, additional fasteners may be used and interconnected with steel members combining to support the hanger.

d. Special Supports:

(1) Secure horizontal pipes where necessary to prevent vibration or excess sway.

(2) Where hangers cannot be adequately secured as specified, make special provisions for hanging and supporting pipe to be submitted to Contracting Officer for A/E review.

(3) Do not attach pipe supports, hangers, clamps or anchors to equipment unless specified for that equipment or unless the Contracting Officer gives written permission.

e. Locate spring hangar units within 1 foot of the pipe attachment, except in locations where spring assemblies interfere with pipe insulation.

f. Seismic Braces and Restraints: Do not insulate piping within 1 foot of device until device has been inspected by Contracting Officer. Refer to Section 13 05 41, SEISMIC RESTRAINT REQUIREMENTS FOR NON-STRUCTURAL COMPONENTS.

g. Minimum Clearances in Tunnels and Trenches:

(1) Floor to bottom of pipe support beam: (6 inches).

(2) Floor to bottom of pipe insulation jacket: (6 inches).

(3) Wall to side of pipe insulation jacket: (6 inches.)

(4) Ceiling to top of pipe insulation jacket: (6 inches).

(5) Greater clearances shall be provided to meet maintenance, repair/replacement, and service requirements of valves, fittings and equipment as recommended by their manufacturers.

3.6 PAINTING EXPOSED STEEL SURFACES IN MANHOLES, TUNNELS AND CONCRETE SHALLOW TRENCHES

a. For manholes, provide surface cleaning and preparation in accordance with SSPC SP 2 and apply prime coat of rust resistant metal primer.

b. For concrete shallow trenches, provide surface cleaning and preparation in accordance with SSPC SP 2 and apply primer and finish coat of zinc-rich paint.

3.7 DIRECT-BURIED SYSTEM INSTALLATION

a. The Contractor shall deliver, store, install and test the system as per manufacturer’s recommendations. All work shall be in strict accordance with the requirements specified by the manufacturer. Printed instructions must be available on site prior to delivery of system components. Any changes required to the design and layout of the system due to site conditions must be approved in writing by the manufacturer and submitted for review by the Contracting Officer. All branch piping connections, valves and drip traps must be located within manholes.

b. Excavation, Trenching, and Backfilling: Perform all excavation, trenching, and backfilling as required by the system manufacturer’s design. Beach sand or any sand with large amounts of chlorides is prohibited. Install per manufacturers recommendations and as shown on contract drawings based on soil conditions. Foundation for system must be firm and stable. Foundation and backfill must be free from rocks. Concrete anchor and thrust blocks must be installed in undisturbed earth. Backfilling must not commence until elevations have been surveyed and accepted and system has been satisfactorily pressure tested and cleaned, including hydrostatic testing of carrier pipes and air testing of casings.

c. Maintain constant slope of carrier pipes as shown or specified. Prior to backfilling over the top of the casing, but after removal of temporary supports, Contractor shall measure and record elevations of top of casing in the trench. Elevations shall be taken at every field joint, 1/3 points along each pipe section, and at tops of elbows. These measurements shall be checked against contract drawings and shall confirm that the casing system has been installed to the elevations shown on the contract drawings unless approved by the Contracting Officer. Slope shall be uniform within 0.1 percent. Measurements shall be recorded by the Contractor, accompanied with photo documentation of each piping with the elevation instrument reading visible, included in the direct-buried system manufacturer representative’s daily report, and given to the Contracting Officer prior to covering the top of the casing with backfill.

d. Provide cathodic protection for all steel casing systems and all

buried exposed metal. Provide dielectric pipe flanges and unions and isolation devices at all points necessary. Provide test stations at grade on each section of the piping system. Isolation flanges and unions shall be rated for the carrier pipe service temperature and pressure.

e. Completely remove all dirt, scale, oil, coatings, and other foreign matter from inside the piping by use of a pipe swab, pipe “pig”, brush, scrapers, or chemical cleaning before connecting pipe sections, valves, or fittings.

f. Sections of system that have been fully or partially submerged in water must be replaced. Moisture content of insulation during installation shall not exceed five percent by weight.

g. At each casing termination (end plate) in buildings and manholes, plug the casing drain openings with brass plugs and extend 1 inch pipe size galvanized vent pipes (ASTM A53/A53M) from the casing vents through the tops of the manholes or 1 foot above the casing in buildings. Terminate the outside vents in 180-degree bends.

h. Provide reports to the Contracting Officer that include:

(1) Daily Written Report: Prepared daily and signed by the Contractor. Submit the original report to the Contracting Officer two days after it is prepared. Provide one set of field pictures of work daily. Format shall be in digital PDF files and hardcopy printout. Digital files can be submitted through email, FTP, USB stick or CD.

(2) Report Contents: State whether or not the condition and quality of the materials used and the delivery, storage, installation and testing of the system are in accordance with the manufacturer’s recommendations, changes to drawings and specifications, any corrective action that was taken of the system, identify any conditions that could result in an unsatisfactory installation.

(3) Report Certification: Daily reports are to be reviewed, signed and sealed by the Professional Engineer responsible for the system installation.

(4) Report Submittals and Stop Order: Daily reports shall be submitted with the payment requests. All work must stop if daily reports are not furnished and requests for payments shall be denied if the daily reports are not furnished.

(5) Certification of Compliance: Upon completion of the work and 30 days prior to final acceptance, deliver to Contracting Officer a notarized Certificate of Compliance signed by principal officers of Contractor, stating that the installation is satisfactory and in accordance with plans, specifications, and manufacturer’s instructions.

(6) The Contractor shall retain copies of all the daily reports and the Certificate of Compliance for 5 years after final acceptance of the system by the Government.

3.8 JOINT CONSTRUCTION

a. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.

b. Remove scale, slag, dirt, and debris from inside and outside of pipe and fittings before assembly.

c. Threaded joints: ASME B1.20.1, tapered pipe threads. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:

(1) Apply appropriate tape or thread compound to external pipe threads unless dry seal threading is specified. Joints made with oil and graphite pipe joint compound shall have compound applied to male threads only.

(2) Do not use pipe or pipe fittings with threads that are corroded or damaged. Do not use pipe sections that have cracked or open welds.

(3) Pipe threads shall be cut to give proper engagement in threaded fittings. Clean pipe and fittings before installation and ream pipe after cutting threads. Threaded pipe shall have clean-cut threads; dull or damaged pipe dies shall not be used.

d. Construct Welded Joints: AWS D10.12M/D10.12, using qualified processes and welding operators according to paragraph, QUALITY ASSURANCE. Branch connections shall be made with either welding tees or welding outlet fittings. Welding outlet fittings shall be forged, integrally reinforced to provide 100 percent pipe strength, beveled for full penetration welding and funneled at inlet for full fluid flow.

e. Flanged Joints: Select gasket material, size, type, and thickness for service application. Install gasket concentrically positioned. Use suitable lubricants on bolt threads. Gaskets and bolting shall be applied in accordance with the recommendations of the gasket manufacturer and bolting standards of ASME B31.1. Strains shall be evenly applied without overstress of bolts. Gaskets shall cover entire area of mating faces of flanges.

f. Location, Spacing and Cold Set of Ball Joints: Conform to layout drawings approved by manufacturer of ball joints. Representative of manufacturer shall visit site and verify that installation is proper. Locate to allow access to all packing injection devices, when provided.

g. Expansion Joints (Bellows and Slip Type):

(1) Type, quantity and spacing of anchors and guides as recommended by manufacturer of expansion joint and as shown. A Professional Engineer shall verify in writing that anchors and guides are properly designed for forces and moments that will be imposed.

(2) Cold setting of joint travel at installation as recommended by the manufacturer for the ambient temperature during the installation.

(3) Prepare for service by cleaning all sliding surfaces, add packing as necessary. Remove all apparatus provided to restrain joint during shipping or installation.

(4) Expansion joints must be located in readily accessible manhole or in walk-through tunnel. Locate joints to permit access without removing piping or other devices. Allow clear space to permit replacement of joints and to permit access to devices for inspection of all surfaces and for adding packing.

h. Piping joints shall be assembled in sections.

i. All pipe intersections and changes in direction shall be made with factory-built-reinforced fittings. Field-fabricated fittings and miters are prohibited.

3.9 INSTALLATION – SAFETY VALVES

a. Valves must be upright and oriented so that lifting levers are accessible from nearest walkway.

b. Provide drip pan elbow as necessary. Support vent line from above. Provide drain line to nearest floor drain from drip pan elbow. Provide separate vent line from each safety valve to atmosphere unless otherwise shown. Piping weight on safety valve outlet shall not exceed that allowed by valve manufacturer.

c. Provide union or flanged connection at safety valve outlet to allow removal of safety valves without disassembling vents.

3.10 INSTALLATION – PRESSURE GAUGES

Locate at inlet and outlet of each pressure reducing station, on each pump discharge and suction, and after main stop valves (gate valves) on steam distribution lines. Orient gauges so that dials are upright and visible from nearest walkway and from the main steam stop valves. Provide gauge cock. Provide siphon on steam service. Provide liquid filled gauges on pump discharge and suction.

3.11 INSTALLATION - THERMOMETERS

Orient thermometers so that scales are upright and visible from nearest walkway. Locate wells in flow stream.

3.12 INSTALLATION - VALVES

a. Do not locate valve stems below the horizontal centerline of the pipe.

b. Locate valves to permit access for operation, maintenance, and replacement.

c. Provide 3/4 inch globe-valved warm-up bypasses at all steam gate valves 3 inch pipe size and greater.

d. Provide 3/4 inch gate or ball-valved drains at each side of steam gate valves where condensate could collect, due to the slope of the pipeline, when the main valve is shut.

3.13 THERMAL INSULATION

a. Steam, condensate and drip return piping, other than in pre-engineered direct-buried systems, shall be insulated as follows:

(1) Piping in concrete trenches and manholes: Insulated with calcium silicate, cellular glass pipe insulation, or aluminum jacket.

(2) Exposed piping in walk through tunnels: Insulated with mineral wool, calcium silicate, fiberglass, or cellular glass pipe insulation, aluminum jacket. Condensate return piping may be insulated with mineral wool, calcium silicate, fiberglass, or cellular glass pipe insulation, aluminum jacket.

(3) Piping in manholes: Insulated with calcium silicate or cellular glass pipe insulation, aluminum jacket.

(4) Minimum insulation thickness: Insulation thicknesses given in Table 1 and 2 are minimum nominal thickness.

b. Parts not to be insulated are:

(1) Threaded valves

(2) Steam traps

(3) Check valves

(4) Unions

(5) Threaded strainers

(6) Strainer basket removal cover and bolting

(7) Dielectric flanges and unions

(8) Expansion joints

(9) Flexible connectors

(10) Ball joints except piping between joints

c. Installation of Insulation:

(1) Pressure Tests: Complete all pressure tests and cleaning before installing.

(2) Insulation Material: New, clean, dry and stored in a clean dry environment; jacketing materials to be clean and unmarred; store adhesives in original containers. Materials shall not have exceeded the predicted shelf life as set by manufacturer.

(3) Identify all materials incorporated in the job on manufacturer’s container by name, type and description.

(4) Apply materials on clean, dry surfaces from which all dirt, loose scale, construction debris has been removed by wire brushing.

(5) The installation shall be neat, thermally and structurally tight without sag, neatly finished at all hanger or other penetrations and shall provide a smooth finished surface primed as required to receive specified painting.

(6) Do not use scrap insulation. Repair any work damaged by welding,

burning, compressing due to concentrated construction loads.

(7) Apply pipe covering protection saddles, MSS SP-58, Type 39, at all hanger points. Fill space between saddle and piping with high density insulation, thoroughly packed. Terminate jacket clear of saddle bearing area.

(8) Insulation and jacket shall terminate hard and tight at all anchor points.

(9) Insulation termination at piping facilities not to be insulated shall stop short, and be finished with 45-degree chamfered section of insulating and finishing cement, and covered with jacket.

(10) Flanged fittings and valves shall be insulated with sections of pipe insulation cut, fitted and arranged neatly, and firmly wired in place. Insulating cement shall fill all cracks, voids and outer surface for covering with glass cloth. Insulation of valve bonnet shall terminate on valve side of bonnet flange to permit valve repair.

(11) On calcium silicate, and cellular glass insulated piping systems, fittings shall be insulated with field or factory-shaped sections of insulation, finished with specified insulating and finishing cements and covered with specified jacket.

(12) On mineral wool insulated piping systems fittings over 2 inches shall be insulated with specified molded pipe fitting insulation or compressed blanket, finished with specified insulating and finishing cements and covered with jacket. On sizes 2 inches and less apply insulating and finishing cements and cover with specified jacket.

(13) Apply glass cloth jacket using an approved adhesive. Glass cloth shall be smooth, tight and neatly finished at all edges; prime cloth to receive paint.

3.14 WELDING

a. The Contractor is entirely responsible for the quality of the welding and shall:

(1) Conduct tests of the welding procedures used on the project, verify the suitability of the procedures used, verify that the welds made will meet the required tests, and also verify that the welding operators have the ability to make sound welds under standard conditions.

(2) Perform all welding operations required for construction and installation of the distribution system.

b. Welder Qualifications: All welders shall be qualified as per ASME B31.1, AWS B2.1/B2.1M, and AWS Z49.1.

c. Field Bevels and Shop Bevels: Done by mechanical means or by flame cutting. Where beveling is done by flame cutting, surfaces shall be thoroughly cleaned of scale and oxidation just prior to welding. Conform to specified standards.

d. Provide approved welding method for field joints on all carrier pipes greater than 2 inches to assure proper alignment, complete weld penetration, and prevention of weld spatter reaching the interior of the pipe. Make field joints 2 inches and smaller with welding sockets.

e. Piping shall not be split, bent, flattened, or otherwise damaged either before, during, or after installation. Where the pipe temperature falls to (32 degrees F) or lower, the pipe shall be heated to approximately (100 degrees F) for a distance of 1 foot on each side of the weld before welding, and the weld shall be finished before the pipe cools to (32 degrees F).

f. Replace and reinspect defective welds. Repairing defective welds by adding weld material over the defect or by peening are prohibited. Welders responsible for defective welds must be requalified.

g. Electrodes shall be stored in a dry heated area, and be kept free of moisture and dampness during fabrication operations. Discard electrodes that have lost part of their coating.

h. An approved independent testing firm regularly engaged in radiographic testing shall perform radiographic examination of field welds in the steam and condensate piping of the systems, in manholes and in walk-through tunnels, in accordance with ASME B31.1. Perform radiographic examination of 50 percent of the first 10 welds made and 10 percent of all additional welds made. The Contracting Officer reserves the right to identify individual welds for which the radiographic examination must be performed. All welds will be visually inspected by the Contracting Officer. The VA reserves the right to require testing on additional welds up to 100 percent if more than 25 percent of the examined welds fail the inspection. Furnish a set of films or pictures showing each weld inspected, a report evaluating the quality of each weld, and a location plan showing the physical location where each weld is to be found in the completed project, prior to installing casing field joints, trench covers, backfilling and hydrostatic testing. All radiographs shall be reviewed and interpreted by an ASNT Certified Level III radiographer, employed by the testing firm, who shall sign the reading report. The Contracting Officer or their representative shall review all inspection records, and if any welds inspected are found unacceptable they shall be removed, rewelded, and radiographically reexamined at no cost to the Government.

3.15 CLEANING OF PIPING

a. Clean pipe and fittings inside and outside before and after assembly. Remove all dirt, scale, and other foreign matter from inside the piping by use of a pipe swab, pipe “pig”, brush, scrapers or chemical cleaners before connecting pipe sections, valves, equipment or fittings. For carbon steel piping, use detergent to remove any oil and caustic solution to remove the mill scale.

b. Cleaning of piping shall be witnessed by the Contracting Officer, their representative, or the Commissioning Agent.

3.16 IDENTIFICATION FOR UTILITY LOCATING

Install continuous plastic underground warning tapes during back filling of trenches for underground steam and condensate distribution piping.

Locate tapes 12 inches below finished grade, directly over piping.

3.17 IDENTIFICATION TAGS

a. Valves: Provide laminated brass tags, with engraved lettering not less than 3/16 inch high, on all isolating valves on steam and condensate return system, identifying building or area served. Attach to the valves with corrosion-resistant chains.

b. Pipes: Label service of all pipes in manholes and walk-through tunnels every 25 feet and at every change in direction. Label shall include flow direction arrows.

3.18 FIELD QUALITY CONTROL

a. Demonstrate leak-tightness of all piping systems by performing hydrostatic tests at 1-1/2 times the design pressure and operational tests. All labor, material and test instruments must be furnished by the Contractor. All instruments must be approved by the Contracting Officer.

b. Pressure test direct-buried systems in conformance with requirements stated in this specification and in printed instructions for the system supplied. Tests must include carrier piping and casing.

c. Holiday testing of direct-buried system steel casings: Test entire surface of casings for faults in coating after installation in trench prior to backfilling. Use test method and voltage recommended by coating manufacturer. Repair any holidays found and retest. System shall not be backfilled until all holidays are eliminated.

d. Before conducting steam system operating test, remove steam trap elements or use bypass connections around traps; then flush lines with water until discharge shows no foreign matter to the satisfaction of Contracting Officer.

e. Steam and condensate piping shall be tested hydrostatically before insulation is applied at field joints and shall be proved tight at a pressure 1-1/2 times the design pressure for a period not less than 2 hours with no pressure decay.

(1) Test piping located in concrete trenches prior to installing trench covers. Test direct-buried systems prior to backfilling.

(2) Remove or isolate any elements of the system such as expansion joints and spring hangers which are not designed for the test pressure.

(3) Prior to acceptance of installation, Contractor shall subject system to operating tests as may be required by Contracting Officer to demonstrate satisfactory functional and operating efficiency. These operating tests shall cover a period of not less than 8 hours for each portion of system tested. Conduct tests at times as the Contracting Officer may direct.

(4) Provide calibrated instruments, equipment, facilities and labor, at no additional cost or time to the Government. Test gauge shall read in increments not exceeding (0.15 psig).

(5) Repeat tests when failures occur.

(6) After completion of satisfactory test, replace all elements that have been removed prior to testing.

f. Pneumatic Testing of Pre-Engineered Direct Buried System Casings:

(1) Perform test on all sections of the system before field-coating the field joints and before back-filling.

(2) Test shall be with compressed air at 15 psig maximum for 2 hours with pressure source disconnected and with decay in pressure not to exceed 5 percent. Corrections to the readings are permissible to compensate for significant ambient temperature changes during the test period.

(3) Pressure shall be measured with a gauge with reading increments of (0.15 psig).

(4) Each casing field joint shall be tested for leaks by means of soap solution or equivalent.

g. NACE-accredited corrosion specialist shall test cathodic protection systems and demonstrate proper operation and protection in accordance with the recommendations and criteria in NACE SP0169.

h. Deficiencies discovered shall be corrected at the Contractor's expense, to satisfaction of Contracting Officer. Major deficiencies or failure to correct deficiencies, to the satisfaction of the Contracting Officer, may be considered cause for rejecting the entire installation.

i. Contractor will engage a qualified testing agency to perform tests and inspections.

j. Engage a factory-authorized service representative to inspect, test, and adjust components, assemblies, and equipment installations for the system.

k. Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.

l. Tests and Inspections:

(1) Steam and condensate piping for testing: ASME B31.1 and ASME B31.9 and as follows:

(a) Leave joints, including welds, uninsulated and exposed for examination during test.

(b) Isolate equipment. Do not subject equipment to test pressure.

(c) Install relief valve set at pressure no more than one-third higher than test pressure and replace safety valves of the appropriate pressure. Reset pressure setpoint of all relief valves to the appropriate pressures and replace safety valves after all tests have been completed. Contractor to provide written report of the reset with date and time stamp for each

relief valve and replacement of the safety valves.

(d) Fill system with normal temperature water between (40 degrees F) and (90 degrees F). Ensure the water filled pipe system is protected against freezing during the test such as providing heat tracing along the pipe.

(e) Use vents installed at high points to release trapped air while filling system. Use drip legs installed at low points for complete removal of liquid.

(2) Test steam and condensate piping as follows:

(a) Subject steam and condensate piping to hydrostatic test pressure that is not less than 1-1/2 times the design pressure.

(b) After hydrostatic test pressure has been applied for 10 minutes, examine joints for leakage. Remake leaking joints using new materials and repeat hydrostatic test until no leaks exist.

(3) Test outer casing as follows: Seal vents and drains and subject casing to 15 psig of air for 4 hours with no loss of pressure. Repair leaks and retest as required.

m. Prepare and submit test and inspection reports to the Contracting Officer within 5 working days of test completion and prior to covering the pipe.

n. All tests shall be witnessed by the Contracting Officer, their representative, or the Commissioning Agent and be documented by each section tested, date tested, and list or personnel present.

3.19 STARTUP AND TESTING

a. Perform tests as recommended by product manufacturer and listed standards and under actual or simulated operating conditions and prove full compliance with design and specified requirements. Tests of the various items of equipment shall be performed simultaneously with the system of which each item is an integral part.

b. When any defects are detected, correct defects and repeat test at no additional cost or time to the Government.

c. The Commissioning Agent will observe startup and contractor testing of selected equipment. Coordinate the startup and contractor testing schedules with Contracting Officer and Commissioning Agent. Provide a minimum notice of 10 working days prior to startup and testing.

3.20 COMMISSIONING

a. Provide commissioning documentation in accordance with the requirements of Section 33 08 00, COMMISSIONING OF SITE UTILITY SYSTEMS.

b. Components provided under this section of the specification will be tested as part of a larger system.

3.21 DEMONSTRATION AND TRAINING

a. Provide services of manufacturer’s technical representative for 4 hour s to instruct each VA personnel responsible in operation and maintenance of the system.

b. Submit training plans and instructor qualifications in accordance with the requirements of Section 33 08 00, COMMISSIONING OF SITE UTILITY SYSTEMS.

3.22 NOT USED

3.23 APPENDIX II – CLASSIFICATIONS FOR DIRECT-BURIED SYSTEMS

a. Groundwater Conditions:

CLASSIFICATIONS FOR DIRECT-BURIED SYSTEMS	
Site Classification	**General Conditions for Such Classifications**
A - Severe	1. The water table is expected to be frequently above the bottom of the system and surface water is expected to accumulate and remain for long periods in the soil surrounding the system, or
	2. The water table is expected to be occasionally above the bottom of the system and surface water is expected to accumulate and remain for long periods in the soil surrounding the system.
B - Bad	1. The water table is expected to be occasionally above the bottom of the system and surface water is expected to accumulate and remain for short periods (or not at all) in the soil surrounding the system, or
	2. The water table is expected never to be above the bottom of the system but surface water is expected to accumulate and remain for long periods in the soil surrounding the system.
C - Moderate	The water table is expected never to be above the bottom of the system but surface water is expected to accumulate and remain for short periods in the soil surrounding the system.
D - Mild	The water table is expected never to be above the bottom of the system and surface water is not expected to accumulate or remain in the soil surrounding the system.

b. System Temperature Classifications: High (261 to 450 degrees F); Medium (201 to 260 degrees F); Low (200 degrees F) or lower.

c. Type of Underground System Allowed: Drainable-Dryable-Testable (DDT) shall be allowed for Site Classifications A, B, C, D.

d. Soil Conditions:

(1) Soil Corrosiveness Classification:

(a) The soil at the site should be classified as corrosive or noncorrosive on the basis of the following criteria:

(b) Corrosive: The soil resistivity is less than 30,000 ohm-cm or stray direct currents can be detected underground.

(c) Noncorrosive: The soil resistivity is 30,000 ohm-cm or greater and no stray direct currents can be detected underground.

(d) The classification should be made by an experienced corrosion engineer based on a field survey of the site carried out in accordance with recognized guidelines for conducting such surveys.

(2) Soil pH: If there is any reason to suspect that the soil pH will be less than 5.0 anywhere along the proposed path of the system, pH measurements should be made at pipeline depth at close intervals along the proposed route, and all locations at which the pH is less than 5.0 should be indicated in the contract documents. An experienced soils engineer, preferably the same engineer responsible for other soil engineering work, should determine soil pH.

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO HB-17	(2002; Errata 2003; Errata 2005, 17th Edition) Standard Specifications for Highway Bridges

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318M	(2019; Errata 2022) Building Code Requirements for Structural Concrete & Commentary
ACI MNL-66	(2020) ACI Detailing Manual

ASSOCIATION OF EDISON ILLUMINATING COMPANIES (AEIC)

AEIC CS8	(2013) Specification for Extruded Dielectric Shielded Power Cables Rated 5 Through 46 kV

ASTM INTERNATIONAL (ASTM)

ASTM B1	(2013) Standard Specification for Hard-Drawn Copper Wire
ASTM B3	(2013; R 2024) Standard Specification for Soft or Annealed Copper Wire
ASTM B8	(2023) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft
ASTM B496	(2016; R 2021) Standard Specification for Compact Round Concentric-Lay-Stranded Copper Conductors
ASTM C32	(2023) Standard Specification for Sewer and Manhole Brick (Made from Clay or Shale)
ASTM C309	(2019) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete

ASTM C478	(2018) Standard Specification for Circular Precast Reinforced Concrete Manhole Sections
ASTM C857	(2016) Standard Practice for Minimum Structural Design Loading for Underground Precast Concrete Utility Structures
ASTM C990	(2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants
ASTM F512	(2019; R 2024) Standard Specification for Smooth-Wall Poly (Vinyl Chloride) (PVC) Conduit and Fittings for Underground Installation
ASTM F2160	(2022a) Standard Specification for Solid Wall High Density Polyethylene (HDPE) Conduit Based on Controlled Outside Diameter (OD)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 48	(2020) Test Procedures and Requirements for Alternating-Current Cable Terminations Used on Shielded Cables Having Laminated Insulation Rated 2.5 kV through 765 kV or Extruded Insulation Rated 2.5 kV through 500 kV
IEEE 81	(2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System
IEEE 386	(2016) Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV
IEEE 400.2	(2024) Guide for Field Testing of Shielded Power Cable Systems Using Very Low Frequency (VLF)
IEEE 404	(2012) Standard for Extruded and Laminated Dielectric Shielded Cable Joints Rated 2500 V to 500,000 V
IEEE C2	(2023) National Electrical Safety Code
IEEE C37.20.3	(2013) Standard for Metal-Enclosed Interrupter Switchgear
IEEE Stds Dictionary	(2009) IEEE Standards Dictionary: Glossary of Terms & Definitions

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS	(2025) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C119.1	(2023) Electric Connectors - Sealed Insulated Underground Connector Systems Rated 600 Volts
ANSI/NEMA WC 71/ICEA S-96-659	(2014; R 2022) Standard for Nonshielded Cables Rated 2001-5000 Volts for use in the Distribution of Electric Energy
NEMA C119.4	(2011) Electric Connectors - Connectors for Use Between Aluminum-to-Aluminum or Aluminum-to-Copper Conductors Designed for Normal Operation at or Below 93 Degrees C and Copper-to-Copper Conductors Designed for Normal Operation at or Below 100 Degrees C
NEMA RN 1	(2018) Polyvinyl-Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit
NEMA TC 2	(2020) Standard for Electrical Polyvinyl Chloride (PVC) Conduit
NEMA TC 6 & 8	(2020) Standard for Polyvinyl Chloride (PVC) Plastic Utilities Duct for Underground Installations
NEMA TC 7	(2021) Smooth-Wall Coilable and Straight Electrical Polyethylene Conduit
NEMA TC 9	(2020) Standard for Fittings for Polyvinyl Chloride (PVC) Plastic Utilities Duct for Underground Installation
NEMA WC 70	(2021) Power Cable Rated 2000 Volts or Less for the Distribution of Electrical Energy
NEMA WC 74/ICEA S-93-639	(2022) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2026) National Electrical Code

SOCIETY OF CABLE TELECOMMUNICATIONS ENGINEERS (SCTE)

ANSI/SCTE 77	(2013) Specification for Underground Enclosure Integrity

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-758	(2012b) Customer-Owned Outside Plant Telecommunications Infrastructure Standard

U.S. DEPARTMENT OF AGRICULTURE (USDA)

RUS Bull 1751F-644	(2002) Underground Plant Construction

UL SOLUTIONS (UL)

UL 6	(2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel
UL 83	(2017; Reprint Mar 2020) UL Standard for Safety Thermoplastic-Insulated Wires and Cables
UL 94	(2023; Reprint Jan 2024) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances
UL 467	(2022) UL Standard for Safety Grounding and Bonding Equipment
UL 486A-486B	(2025) UL Standard for Safety Wire Connectors
UL 510	(2020; Dec 2022) UL Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape
UL 514A	(2024) UL Standard for Safety Metallic Outlet Boxes
UL 514B	(2012; Reprint Mar 2024) UL Standard for Safety Conduit, Tubing and Cable Fittings
UL 651	(2011; Reprint May 2022) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
UL 854	(2020; Reprint Jul 2024) Standard for Service-Entrance Cables
UL 1072	(2006; Reprint Oct 2024) Medium-Voltage Power Cables
UL 1242	(2006; Reprint Apr 2022) UL Standard for Safety Electrical Intermediate Metal Conduit -- Steel

1.2 DEFINITIONS

a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE Stds Dictionary.

b. In the text of this section, the words conduit and duct are used interchangeably and have the same meaning.

c. In the text of this section, "medium voltage cable splices," and "medium voltage cable joints" are used interchangeably and have the same meaning.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Precast Underground Structures; G

SD-03 Product Data

Medium Voltage Cable; G

Medium Voltage Cable Joints; G

Medium Voltage Cable Terminations; GPrecast Concrete Structures; G

Sealing Material

Pulling-In Irons

Manhole Frames and Covers; G

Handhole Frames and Covers; GComposite/Fiberglass Handholes; G

Cable Supports (racks, arms and insulators); GSD-06 Test Reports

Medium Voltage Cable Qualification and Production Tests; G

Field Acceptance Checks and Tests; G

Arc-proofing Test for cable fireproofing tape; GSD-07 Certificates

Cable splicer/terminator; G

Cable Installer Qualifications; G1.4 QUALITY ASSURANCE

1.4.1 Precast Underground Structures

Submittal required for each type used. Provide calculations and drawings for precast manholes and handholes bearing the seal of a registered professional engineer including:

a. Material description (i.e., f'c and Fy)

b. Manufacturer's printed assembly and installation instructions

c. Design calculations

d. Reinforcing shop drawings in accordance with ACI MNL-66

e. Plans and elevations showing opening and pulling-in iron locations and details

1.4.2 Certificate of Competency for Cable Splicer/Terminator

The cable splicer/terminator must have a certification from the National Cable Splicing Certification Board (NCSCB) in the field of splicing and terminating shielded medium voltage (5 kV to 35 kV) power cable using pre-manufactured kits (pre-molded, heat-shrink, cold shrink). Submit "Proof of Certification" for approval, for the individuals that will be performing cable splicer and termination work, 30 days before splices or terminations are to be made.

1.4.3 Cable Installer Qualifications

Provide at least one onsite person in a supervisory position with a documentable level of competency and experience to supervise all cable pulling operations. Provide a resume showing the cable installers' experience in the last three years, including a list of references complete with points of contact, addresses and telephone numbers. Cable installer must demonstrate experience with a minimum of three medium voltage cable installations. The Contracting Officer reserves the right to require additional proof of competency or to reject the individual and call for an alternate qualified cable installer.

1.4.4 Directional Boring Certificate of Conformance

Provide certification of compliance with the registered Professional Engineer's design requirements for each directional bore, including: HDPE conduit size and type, bend radius, elevation changes, vertical and horizontal path deviations, conductor size and type and any conductor derating due to depth of conduit. Record location and depth of all directional-bore installed HDPE conduits using Global Positioning System (GPS) recording means with "resource grade" accuracy.

1.4.5 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of IEEE C2 and NFPA 70 unless more stringent requirements are specified or indicated.

1.4.6 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.4.6.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.4.6.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable, unless specified otherwise.

PART 2 PRODUCTS

2.1 CONDUIT, DUCTS, AND FITTINGS

2.1.1 Rigid Metal Conduit

UL 6.

2.1.1.1 Rigid Metallic Conduit, PVC Coated

NEMA RN 1, Type A40, except that hardness must be nominal 85 Shore A durometer, dielectric strength must be minimum 400 volts per mil at 60 Hz, and tensile strength must be minimum 3500 psi.

2.1.2 Intermediate Metal Conduit

UL 1242.

2.1.2.1 Intermediate Metal Conduit, PVC Coated

NEMA RN 1, Type A40, except that hardness must be nominal 85 Shore A durometer, dielectric strength must be minimum 400 volts per mil at 60 Hz, and tensile strength must be minimum 3500 psi.

2.1.3 Plastic Conduit for Direct Burial and Riser Applications

UL 651 and NEMA TC 2, EPC-40 or EPC-80 as indicated.

2.1.4 Plastic Duct for Concrete Encasement

Provide Type EB-35 per UL 651, ASTM F512, and NEMA TC 6 & 8.

2.1.5 High Density Polyethylene (HDPE) Electrical Conduit for Directional Boring

Smoothwall, approved/listed for directional boring, minimum Schedule 80, ASTM F2160, NEMA TC 7.

2.1.6 Duct Sealant

UL 94, Class HBF. Provide high-expansion urethane foam duct sealant that expands and hardens to form a closed, chemically and water resistant, rigid structure. Sealant must be compatible with common cable and wire jackets and capable of adhering to metals, plastics and concrete. Sealant must be capable of curing in temperature ranges of 35 degrees F to 95 degrees F. Cured sealant must withstand temperature ranges of -20 degrees

F to 200 degrees F without loss of function.

2.1.7 Fittings

2.1.7.1 Metal Fittings

UL 514B.

2.1.7.2 PVC Conduit Fittings

UL 514B, UL 651.

2.1.7.3 PVC Duct Fittings

NEMA TC 9.

2.1.7.4 Outlet Boxes for Steel Conduit

Outlet boxes for use with rigid or flexible steel conduit must be cast-metal cadmium or zinc-coated if of ferrous metal with gasketed closures and must conform to UL 514A.

2.2 LOW VOLTAGE INSULATED CONDUCTORS AND CABLES

Insulated conductors must be rated 600 volts and conform to the requirements of NFPA 70, including listing requirements, or in accordance with NEMA WC 70. Wires and cables manufactured more than 12 months prior to date of delivery to the site are not acceptable. Service entrance conductors must conform to UL 854, type USE.

2.2.1 Conductor Types

Cable and duct sizes indicated are for copper conductors and THHN/THWN unless otherwise noted. Conductors No. 10 AWG and smaller must be solid. Conductors No. 8 AWG and larger must be stranded. All conductors must be copper.

2.2.2 Conductor Material

Unless specified or indicated otherwise or required by NFPA 70, wires in conduit, other than service entrance, must be 600-volt, Type THWN/THHN conforming to UL 83. Copper conductors must be annealed copper complying with ASTM B3 and ASTM B8.

2.2.3 Cable Marking

Insulated conductors must have the date of manufacture and other identification imprinted on the outer surface of each cable at regular intervals throughout the cable length.

Identify each cable by means of a fiber, laminated plastic, or non-ferrous metal tags in each manhole, handhole, junction box, and each terminal. Each tag must contain the following information; cable type, conductor size, circuit number, circuit voltage, cable destination and phase identification.

Color code conductors. Provide conductor identification within each enclosure where a tap, splice, or termination is made. Conductor identification must be by color-coded insulated conductors, plastic-coated

self-sticking printed markers, colored nylon cable ties and plates, heat shrink type sleeves,or colored electrical tape. Properly identify control circuit terminations. Color must be green for grounding conductors and white for neutrals; except where neutrals of more than one system are installed in same raceway or box, other neutrals may be white with a different colored (not green) stripe for each. Color of ungrounded conductors in different voltage systems are as follows:

a. 208/120 volt, three-phase

(1) Phase A - black

(2) Phase B - red

(3) Phase C - blue

b. 480/277 volt, three-phase

(1) Phase A - brown

(2) Phase B - orange

(3) Phase C - yellow

c. 120/240 volt, single phase: Black and red

2.3 LOW VOLTAGE WIRE CONNECTORS AND TERMINALS

Provide a uniform compression over the entire conductor contact surface. Use solderless terminal lugs on stranded conductors.

a. For use with copper conductors: UL 486A-486B.

2.4 LOW VOLTAGE SPLICES

Provide splices in conductors with a compression connector on the conductor and by insulating and waterproofing using one of the following methods which are suitable for continuous submersion in water and comply with ANSI C119.1.

2.4.1 Heat Shrinkable Splice

Provide heat shrinkable splice insulation by means of a thermoplastic adhesive sealant material applied in accordance with the manufacturer's written instructions.

2.4.2 Cold Shrink Rubber Splice

Provide a cold-shrink rubber splice which consists of EPDM rubber tube which has been factory stretched onto a spiraled core which is removed during splice installation. The installation must not require heat or flame, or any additional materials such as covering or adhesive. It must be designed for use with inline compression type connectors, or indoor, outdoor, direct-burial or submerged locations.

2.5 MEDIUM VOLTAGE CABLE

Cable (conductor) sizes are designated by American Wire Gauge (AWG) and Thousand Circular Mils (Kcmil). Conductor and conduit sizes indicated are

for copper conductors unless otherwise noted. Insulated conductors must have the date of manufacture and other identification imprinted on the outer surface of each cable at regular intervals throughout cable length. Wires and cables manufactured more than 12 months prior to date of delivery to the site are not acceptable. Provide single conductor type cables unless otherwise indicated.

2.5.1 Cable Configuration

Provide Type MV cable, conforming to NEMA WC 74/ICEA S-93-639 and UL 1072. Provide cables manufactured for use in duct applications as indicated. Cable must be rated 15 kV with 133 percent insulation level.

2.5.2 Conductor Material

Provide concentric-lay-stranded, Class B compact round conductors. Provide soft drawn copper cables complying with ASTM B3 and ASTM B8 for regular concentric and compressed stranding or ASTM B496 for compact stranding.

2.5.3 Insulation

Provide ethylene-propylene-rubber (EPR) insulation conforming to the requirements of ANSI/NEMA WC 71/ICEA S-96-659 and AEIC CS8.

2.5.4 Shielding

Cables rated for 2 kV and above must have a semiconducting conductor shield, a semiconducting insulation shield, and an overall copper tape shield for each phase.

2.5.5 Neutrals

Neutral conductors must be copper, employing the same insulation and jacket materials as phase conductors, except that a 600-volt insulation rating is acceptable.

2.5.6 Jackets

Provide cables with a PVC jacket. Provide PVC jackets with a separator that prevents contact with underlying semiconducting insulating shield.

2.6 MEDIUM VOLTAGE CABLE TERMINATIONS

IEEE 48 Class 1; of the molded elastomer, prestretched elastomer, or heat-shrinkable elastomer. Acceptable elastomers are track-resistant silicone rubber or track-resistant ethylene propylene compounds, such as ethylene propylene rubber or ethylene propylene diene monomer. Separable insulated connectors may be used for apparatus terminations, when such apparatus is provided with suitable bushings. Provide terminations, where required, with mounting brackets suitable for the intended installation and with grounding provisions for the cable shielding, metallic sheath, or armor. Provide terminations in a kit, including: skirts, stress control terminator, ground clamp, connectors, lugs, and complete instructions for assembly and installation. Terminations must be the product of one manufacturer, suitable for the type, diameter, insulation class and level, and materials of the cable terminated. Do not use separate parts of copper or copper alloy in contact with aluminum alloy parts in the construction or installation of the terminator.

2.6.1 Cold-Shrink Type

Terminator must be a one-piece design, utilizing the manufacturer's latest technology, where high-dielectric constant (capacitive) stress control is integrated within a skirted insulator made of silicone rubber. Termination must not require heat or flame for installation. Termination kit must contain all necessary materials (except for the lugs). Design termination for installation in low or highly contaminated indoor and outdoor locations and must resist ultraviolet rays and oxidative decomposition.

2.6.2 Heat Shrinkable Type

Terminator must consist of a uniform cross section heat shrinkable polymeric construction stress relief tubing and environmentally sealed outer covering that is nontracking, resists heavy atmospheric contaminants, ultra violet rays and oxidative decomposition. Provide heat shrinkable sheds or skirts of the same material. Design termination for installation in low or highly contaminated indoor or outdoor locations.

2.6.3 Separable Insulated Connector Type

IEEE 386. Provide connector with steel reinforced hook-stick eye, grounding eye, test point, and arc-quenching contact material. Provide connectors of the loadbreak or deadbreak type as indicated, of suitable construction for the application and the type of cable connected, and that include cable shield adaptors. Provide external clamping points and test points. Do no use separable connectors in manholes/handholes.

a. 600 Ampere deadbreak connector ratings: Voltage: 15 kV, 95 kV BIL. Short time rating: 25,000 rms symmetrical amperes. Connectors must have 200 ampere bushing interface for surge arresters.

2.7 MEDIUM VOLTAGE CABLE JOINTS

Provide joints (splices) in accordance with IEEE 404 suitable for the rated voltage, insulation level, insulation type, and construction of the cable. Joints must be certified by the manufacturer for waterproof, submersible applications. Upon request, supply manufacturer's design qualification test report in accordance with IEEE 404. Connectors for joint must be tin-plated electrolytic copper, having ends tapered and having center stops to equalize cable insertion.

2.7.1 Heat-Shrinkable Joint

Consists of a uniform cross-section heat-shrinkable polymeric construction with a linear stress relief system, a high dielectric strength insulating material, and an integrally bonded outer conductor layer for shielding. Replace original cable jacket with a heavy-wall heat-shrinkable sleeve with hot-melt adhesive coating.

2.7.2 Cold-Shrink Rubber-Type Joint

Joint must be of a cold shrink design that does not require any heat source for its installation. Splice insulation and jacket must be of a one-piece factory formed cold shrink sleeve made of black EPDM rubber. Splice should be packaged three splices per kit, including complete

installation instructions.

2.8 TELECOMMUNICATIONS CABLING

Refer to Division 27.

2.9 TAPE

2.9.1 Insulating Tape

UL 510, plastic insulating tape, capable of performing in a continuous temperature environment of 80 degrees C.

2.9.2 Buried Warning and Identification Tape

Refer to Division 27.

2.9.3 Fireproofing Tape

Provide tape composed of a flexible, conformable, unsupported intumescent elastomer. Tape must be not less than .030 inch thick, noncorrosive to cable sheath, self-extinguishing, noncombustible, adhesive-free, and must not deteriorate when subjected to oil, water, gases, salt water, sewage, and fungus.

2.10 PULL ROPE

Plastic or flat pull line (bull line) having a minimum tensile strength of 200 pounds.

2.11 GROUNDING AND BONDING

2.11.1 Driven Ground Rods

Provide copper-clad steel ground rods conforming to UL 467 not less than 3/4 inch in diameter by 10 feet in length. Sectional type rods may be used for rods 20 feet or longer.

2.11.2 Grounding Conductors

Stranded-bare copper conductors must conform to ASTM B8, Class B, soft-drawn unless otherwise indicated. Solid-bare copper conductors must conform to ASTM B1 for sizes No. 8 and smaller. Insulated conductors must be of the same material as phase conductors and green color-coded, except that conductors must be rated no more than 600 volts. Aluminum is not acceptable.

2.12 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE. In addition, provide concrete for encasement of underground ducts with 3000 psi minimum 28-day compressive strength. Concrete associated with electrical work for other than encasement of underground ducts must be 4000 psi minimum 28-day compressive strength unless specified otherwise.

2.13 UNDERGROUND STRUCTURES

Provide precast concrete underground structures or standard type

cast-in-place manhole types as indicated, conforming to ASTM C857 and ASTM C478. Top, walls, and bottom must consist of reinforced concrete. Walls and bottom must be of monolithic concrete construction. Locate duct entrances and windows near the corners of structures to facilitate cable racking. Covers must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Install a pulling-in iron in the wall opposite each duct line entrance. Cable racks, including rack arms and insulators, must be adequate to accommodate the cable.

2.13.1 Cast-In-Place Concrete Structures

Concrete must conform to Section 03 30 00 CAST-IN-PLACE CONCRETE. Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers.

2.13.2 Precast Concrete Structures, Risers and Tops

Precast concrete underground structures may be provided in lieu of cast-in-place subject to the requirements specified below. Precast units must be the product of a manufacturer regularly engaged in the manufacture of precast concrete products, including precast manholes.

2.13.2.1 General

Precast concrete structures must have the same accessories and facilities as required for cast-in-place structures. Likewise, precast structures must have plan area and clear heights not less than those of cast-in-place structures. Concrete materials and methods of construction must be the same as for cast-in-place concrete construction, as modified herein. Slope in floor may be omitted provided precast sections are poured in reinforced steel forms. Concrete for precast work must have a 28-day compressive strength of not less than 4000 psi. Structures may be precast to the design and details indicated for cast-in-place construction, precast monolithically and placed as a unit, or structures may be assembled sections, designed and produced by the manufacturer in accordance with the requirements specified. Structures must be identified with the manufacturer's name embedded in or otherwise permanently attached to an interior wall face.

2.13.2.2 Design for Precast Structures

ACI 318M. In the absence of detailed on-site soil information, design for the following soil parameters/site conditions:

a. Angle of Internal Friction (phi) = 30 degrees

b. Unit Weight of Soil (Dry) = 110 pcf, (Saturated) = 130 pcf

c. Coefficient of Lateral Earth Pressure (Ka) = 0.33

d. Ground Water Level = 3 feet below ground elevation

e. Vertical design loads must include full dead, superimposed dead, and live loads including a 30 percent magnification factor for impact.

Live loads must consider all types and magnitudes of vehicular (automotive, industrial, or aircraft) traffic to be encountered. The minimum design vertical load must be for H20 highway loading per AASHTO HB-17.

f. Horizontal design loads must include full geostatic and hydrostatic pressures for the soil parameters, water table, and depth of installation to be encountered. Also, horizontal loads imposed by adjacent structure foundations, and horizontal load components of vertical design loads, including impact, must be considered, along with a pulling-in iron design load of 6000 pounds.

g. Each structural component must be designed for the load combination and positioning resulting in the maximum shear and moment for that particular component.

h. Design must also consider the live loads induced in the handling, installation, and backfilling of the manholes. Provide lifting devices to ensure structural integrity during handling and installation.

2.13.2.3 Construction

Provide a uniform thickness for structure top, bottom, and wall not less than 6 inches. Thin-walled knock-out panels for designed or future duct bank entrances are not permitted. Provide quantity, size, and location of duct bank entrance windows as directed, and cast completely open by the precaster. Size of windows must exceed the nominal duct bank envelope dimensions by at least 12 inches vertically and horizontally to preclude in-field window modifications made necessary by duct bank misalignment. However, the sides of precast windows must be a minimum of 6 inches from the inside surface of adjacent walls, floors, or ceilings. Form the perimeter of precast window openings to have a keyed or inward flared surface to provide a positive interlock with the mating duct bank envelope. Provide welded wire fabric reinforcing through window openings for in-field cutting and flaring into duct bank envelopes. Provide additional reinforcing steel comprised of at least two No. 4 bars around window openings. Provide drain sumps a minimum of 12 inches in diameter and 4 inches deep for precast structures.

2.13.2.4 Joints

Provide tongue-and-groove joints on mating edges of precast components. Shiplap joints are not allowed. Design joints to firmly interlock adjoining components and to provide waterproof junctions and adequate shear transfer. Seal joints watertight using preformed plastic strip conforming to ASTM C990. Install sealing material in strict accordance with the sealant manufacturer's printed instructions. Provide waterproofing at conduit/duct entrances into structures, and where access frame meets the top slab, provide continuous grout seal.

2.13.3 Manhole Frames and Covers

Provide cast iron frames and covers for manholes conforming to AASHTO H-20. Cast the words "ELECTRIC" or "TELECOMMUNICATIONS" in the top face of power and telecommunications manhole covers, respectively.

2.13.4 Handhole Frames and Covers

Frames and covers of steel must be welded by qualified welders in accordance with standard commercial practice. Provide rolled-steel floor plate covers having an approved antislip surface. Hinges must be of stainless steel with bronze hinge pin , 5 by 5 inches by approximately 3/16 inch thick, without screw holes, and must be for full surface application by fillet welding. Hinges must have nonremovable pins and five knuckles. The surfaces of plates under hinges must be true after the removal of raised antislip surface, by grinding or other approved method.

2.13.5 Brick for Manhole Collar

Provide sewer and manhole brick conforming to ASTM C32, Grade MS.

2.13.6 Composite/Fiberglass Handholes and Covers

ANSI/SCTE 77. Provide handholes and covers of polymer concrete, reinforced with heavy weave fiberglass with a design load (Tier rating) appropriate for or greater than the intended use. All covers are required to have the Tier level rating embossed on the surface which must not exceed the design load of the box.

2.14 CABLE SUPPORTS (RACKS, ARMS, AND INSULATORS)

Zinc coat the metal portion of racks and arms after fabrication.

2.14.1 Cable Rack Stanchions

The wall bracket or stanchion must be 4 inches by approximately 1-1/2 inch by 3/16 inch channel steel, or 4 inches by approximately 1 inch glass-reinforced nylon with recessed bolt mounting holes, 48 inches long (minimum) in manholes. Space slots for mounting cable rack arms at 8 inch intervals.

2.14.2 Rack Arms

Cable rack arms must be steel or malleable iron or glass reinforced nylon and must be of the removable type. Rack arm length must be a minimum of 8 inches and a maximum of 12 inches.

2.14.3 Insulators

Insulators for metal rack arms must be dry-process glazed porcelain. Insulators are not required for nylon arms.

2.15 CABLE TAGS IN MANHOLES

Provide polyethylene tags for each power cable located in manholes. Do not provide handwritten letters. The first position on the power cable tag denotes the voltage. The second through sixth positions on the tag identifies the circuit. The next to last position denotes the phase of the circuit and include the Greek "phi" symbol. The last position denotes the cable size. As an example, a tag could have the following designation: "11.5 NAS 1-8(Phase A)500," denoting that the tagged cable is on the 11.5kV system circuit number NAS 1-8, underground, Phase A, sized at 500 kcmil.

2.15.1 Polyethylene Cable Tags

Provide tags of polyethylene having an average tensile strength of 3250 pounds per square inch; and that are 0.08 inch thick (minimum), non-corrosive non-conductive; resistive to acids, alkalis, organic solvents, and salt water; and distortion resistant to 170 degrees F. Provide 0.05 inch (minimum) thick black polyethylene tag holder. Provide a one-piece nylon, self-locking tie at each end of the cable tag, having a minimum loop tensile strength of 175 pounds and black block letters, numbers, and symbols one inch high on a yellow background. Letters, numbers, and symbols must not fall off or change positions regardless of the cable tags' orientation.

2.16 MEDIUM VOLTAGE ABOVE GROUND CABLE TERMINATING CABINETS

Cable terminating cabinets must be hook-stick operable, deadfront construction conforming to the requirements of IEEE C37.20.3, Category A. Provide cabinets with 600 A. dead-break junctions and elbow-type separable dead-break connectors, cable parking stands, and grounding lugs. Provide cable terminating equipment in conformance with IEEE 386.

Ratings at 60 Hz must be:

Nominal voltage (kV)	
Rated maximum voltage (kV)	15
Rated continuous current (A)	600
One-second short-time current-carrying capacity (kA)	25kA
BIL (kV)	95kV

2.17 LOW VOLTAGE ABOVE GROUND TERMINATION PEDESTAL

Provide copolymer polypropylene, low voltage above ground termination pedestal manufactured through an injection molding process. Pedestals must resist fertilizers, salt air environments and ultra-violet radiation. Pedestal top must be imprinted with a "WARNING" and "ELECTRIC" identification. Pedestal must contain three lay-in six port connectors, NEMA C119.4, Class "A", dual rated for aluminum or copper, and capable of terminating conductors ranging from 10 AWG to 500 kcmil. Protect each connector with a clear, hard lexan (plastic) cover. Provide pedestal with rust-free material and stainless steel hardware that is lockable.

2.18 SOURCE QUALITY CONTROL

2.18.1 Arc-Proofing Test for Cable Fireproofing Tape

Manufacturer must test one sample assembly consisting of a straight lead tube 12 inches long with a 2 1/2 inch outside diameter, and a 1/8 inch thick wall, and covered with one-half lap layer of arc and fireproofing tape per manufacturer's instructions. The arc and fireproofing tape must withstand extreme temperature of a high-current fault arc 13,000 degrees K for 70 cycles as determined by using an argon directed plasma jet capable of constantly producing and maintaining an arc temperature of 13,000 degrees K. Temperature (13,000 degrees K) of the ignited arc between the

cathode and anode must be obtained from a dc power source of 305 (plus or minus 5) amperes and 20 (plus or minus 1) volts. Direct the arc toward the sample assembly accurately positioned 5 (plus or minus 1) millimeters downstream in the plasma from the anode orifice by fixed flow rate of argon gas (0.18 g per second). Test each sample assembly at three unrelated points. Start time for tests must be taken from recorded peak current when the specimen is exposed to the full test temperature. Surface heat on the specimen prior to that time must be minimal. The end point is established when the plasma or conductive arc penetrates the protective tape and strikes the lead tube. Submittals for arc-proofing tape must indicate that the test has been performed and passed by the manufacturer.

2.18.2 Medium Voltage Cable Qualification and Production Tests

Results of AEIC CS8 qualification and production tests as applicable for each type of medium voltage cable.

PART 3 EXECUTION

3.1 INSTALLATION

Install equipment and devices in accordance with the manufacturer's published instructions and with the requirements and recommendations of NFPA 70 and IEEE C2 as applicable. In addition to these requirements, install telecommunications in accordance with TIA-758 and RUS Bull 1751F-644.

3.2 CABLE INSPECTION

Inspect each cable reel for correct storage positions, signs of physical damage, and broken end seals prior to installation. If end seal is broken, remove moisture from cable prior to installation in accordance with the cable manufacturer's recommendations.

3.3 CABLE INSTALLATION PLAN AND PROCEDURE

Obtain from the manufacturer an installation manual or set of instructions which addresses such aspects as cable construction, insulation type, cable diameter, bending radius, cable temperature limits for installation, lubricants, coefficient of friction, conduit cleaning, storage procedures, moisture seals, testing for and purging moisture, maximum allowable pulling tension, and maximum allowable sidewall bearing pressure. Perform pulling calculations and prepare a pulling plan and submit along with the manufacturer's instructions in accordance with SUBMITTALS. Install cable strictly in accordance with the cable manufacturer's recommendations and the approved installation plan.

Calculations and pulling plan must include:

a. Site layout drawing with cable pulls identified in numeric order of expected pulling sequence and direction of cable pull.

b. List of cable installation equipment.

c. Lubricant manufacturer's application instructions.

d. Procedure for resealing cable ends to prevent moisture from entering cable.

e. Cable pulling tension calculations of all cable pulls.

f. Cable percentage conduit fill.

g. Cable sidewall bearing pressure.

h. Cable minimum bend radius and minimum diameter of pulling wheels used.

i. Cable jam ratio.

j. Maximum allowable pulling tension on each different type and size of conductor.

k. Maximum allowable pulling tension on pulling device.

3.4 UNDERGROUND FEEDERS SUPPLYING BUILDINGS

Terminate underground feeders supplying building at a point 5 feet outside the building and projections thereof, except that conductors must be continuous to the terminating point indicated. Provide PVC, Type EPC-40 conduit from the supply equipment to a point 5 feet outside the building and projections thereof. Protect ends of underground conduit with plastic plugs until connections are made.

Encase the underground portion of the conduit in a concrete envelope and bury as specified for underground duct with concrete encasement.

3.5 UNDERGROUND STRUCTURE CONSTRUCTION

Provide standard type cast-in-place construction as specified herein and as indicated, or precast construction as specified herein. Horizontal concrete surfaces of floors must have a smooth trowel finish. Cure concrete by applying two coats of white pigmented membrane forming-curing compound in strict accordance with the manufacturer's printed instructions, except that precast concrete may be steam cured. Curing compound must conform to ASTM C309. Locate duct entrances and windows in the center of end walls (shorter) and near the corners of sidewalls (longer) to facilitate cable racking and splicing. Covers for underground structures must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Manhole locations, as indicated, are approximate. Coordinate exact manhole locations with other utilities and finished grading and paving.

3.5.1 Cast-In-Place Concrete Structures

Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers.

3.5.2 Precast Concrete Construction

Set commercial precast structures on 6 inches of level, 90 percent compacted granular fill, 3/4 inch to 1 inch size, extending 12 inches beyond the structure on each side. Compact granular fill by a minimum of four passes with a plate type vibrator. Installation must additionally

conform to the manufacturer's instructions.

3.5.3 Pulling-In Irons

Provide steel bars bent as indicated, and cast in the walls and floors. Alternatively, pipe sleeves may be precast into the walls and floors where required to accept U-bolts or other types of pulling-in devices possessing the strengths and clearances stated herein. The final installation of pulling-in devices must be made permanent. Cover and seal exterior projections of thru-wall type pulling-in devices with an appropriate protective coating. In the floor, locate the irons a minimum of 6 inches from the edge of the sump, and in the walls, locate the irons within 6 inches of the projected center of the duct bank pattern or precast window in the opposite wall. However, the pulling-in iron must not be located within 6 inches of an adjacent interior surface, or duct or precast window located within the same wall as the iron. If a pulling-in iron cannot be located directly opposite the corresponding duct bank or precast window due to this clearance limitation, locate the iron directly above or below the projected center of the duct bank pattern or precast window the minimum distance required to preserve the 6 inch clearance previously stated. In the case of directly opposing precast windows, pulling-in irons consisting of a 3 foot length of No. 5 reinforcing bar, formed into a hairpin, may be cast-in-place within the precast windows simultaneously with the end of the corresponding duct bank envelope. Irons installed in this manner must be positioned directly in line with, or when not possible, directly above or below the projected center of the duct bank pattern entering the opposite wall, while maintaining a minimum clear distance of 3 inches from any edge of the cast-in-place duct bank envelope or any individual duct. Pulling-in irons must have a clear projection into the structure of approximately 4 inches and must be designed to withstand a minimum pulling-in load of 6000 pounds. Hot-dip galvanize irons after fabrication.

3.5.4 Cable Racks, Arms and Insulators

Cable racks, arms and insulators must be sufficient to accommodate the cables. Space racks in power manholes not more than 3 feet apart, and provide each manhole wall with a minimum of two racks. Space racks in signal manholes not more than 16 1/2 inches apart with the end rack being no further than 12 inches from the adjacent wall. Methods of anchoring cable racks are as follows:

a. Provide a 5/8 inch diameter by 5 inch long anchor bolt with 3 inch foot cast in structure wall with 2 inch protrusion of threaded portion of bolt into structure. Provide 5/8 inch steel square head nut on each anchor bolt. Coat threads of anchor bolts with suitable coating immediately prior to installing nuts.

b. Provide concrete channel insert with a minimum load rating of 800 pounds per foot. Insert channel must be steel of the same length as "vertical rack channel;" and cast flush in structure wall. Provide 5/8 inch steel nuts in channel insert to receive 5/8 inch diameter by 3 inch long steel, square head anchor bolts.

c. Provide concrete "spot insert" at each anchor bolt location, cast flush in structure wall. Each insert must have minimum 800 pound load rating. Provide 5/8 inch diameter by 3 inch long steel, square head anchor bolt at each anchor point. Coat threads of anchor bolts with suitable coating immediately prior to installing bolts.

3.5.5 Field Painting

Clean cast-iron frames and covers not buried in concrete or masonry of mortar, rust, grease, dirt and other deleterious materials, and coat with bituminous paint.

3.6 UNDERGROUND CONDUIT AND DUCT SYSTEMS

3.6.1 Requirements

Run conduit in straight lines except where a change of direction is necessary. Provide numbers and sizes of ducts as indicated. Provide a 4/0 AWG bare copper grounding conductor above medium-voltage distribution duct banks. Bond bare copper grounding conductor to ground rings (loops) in all manholes and to ground rings (loops) at all equipment slabs (pads). Route grounding conductor into manholes with the duct bank (sleeving is not required). Ducts must have a continuous slope downward toward underground structures and away from buildings, laid with a minimum slope of 3 inches per 100 feet. Depending on the contour of the finished grade, the high-point may be at a terminal, a manhole, a handhole, or between manholes or handholes. Terminate all PVC conduit end points in utility holes, switching cabinets, transform handholes and buildings with end bells. The bell end of the conduits that enter manholes and handholes must be flush with the wall.

Perform changes in ductbank direction as follows:

a. Short-radius manufactured 90-degree duct bends may be used only for pole or equipment risers, unless specifically indicated as acceptable.

b. The minimum manufactured bend radius must be 18 inches for ducts of less than 3 inch diameter, and 36 inches for ducts 3 inches or greater in diameter.

c. As an exception to the bend radius required above, provide field manufactured longsweep bends having a minimum radius of 25 feet for a change of direction of more than 5 degrees, either horizontally or vertically, using a combination of curved and straight sections. Maximum manufactured curved sections allowed for use in field manufactured longsweep bend: 30 degrees.

3.6.2 Treatment

Keep ducts clean of concrete, dirt, or foreign substances during construction. Make field cuts requiring tapers with proper tools and match factory tapers. Use a coupling recommended by the duct manufacturer whenever an existing duct is connected to a duct of different material or shape. Store ducts to avoid warping and deterioration with ends sufficiently plugged to prevent entry of any water or solid substances. Thoroughly clean ducts before being laid. Store plastic ducts on a flat surface and protected from the direct rays of the sun.

3.6.3 Conduit Cleaning

As each conduit run is completed, for conduit sizes 3 inches and larger, draw a flexible testing mandrel approximately 12 inches long with a diameter less than the inside diameter of the conduit through the conduit. After which, draw a stiff bristle brush through until conduit is

clear of particles of earth, sand and gravel; then immediately install conduit plugs. For conduit sizes less than 3 inches, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs.

3.6.4 Jacking and Drilling Under Roads and Structures

Conduits to be installed under existing paved areas which are not to be disturbed, and under roads and railroad tracks, must be zinc-coated, rigid steel, jacked into place. Where ducts are jacked under existing pavement, install rigid steel conduit because of its strength. To protect the corrosion-resistant conduit coating, predrilling or installing conduit inside a larger iron pipe sleeve (jack-and-sleeve) is required. For crossings of existing railroads and airfield pavements greater than 50 feet in length, the predrilling method or the jack-and-sleeve method will be used. Separators or spacing blocks must be made of steel, concrete, plastic, or a combination of these materials placed not farther apart than 4 feet on centers. Hydraulic jet method must not be used.

3.6.5 Galvanized Conduit Concrete Penetrations

Galvanized conduits which penetrate concrete (slabs, pavement, and walls) in wet locations must be PVC coated and extend from at least 2 inches within the concrete to the first coupling or fitting outside the concrete (minimum of 6 inches from penetration).

3.6.6 Multiple Conduits

Separate multiple conduits by a minimum distance of 3 inches, except that light and power conduits must be separated from control, signal, and telephone conduits by a minimum distance of 12 inches. Stagger the joints of the conduits by rows (horizontally) and layers (vertically) to strengthen the conduit assembly. Provide plastic duct spacers that interlock vertically and horizontally. Spacer assembly must consist of base spacers, intermediate spacers, ties, and locking device on top to provide a completely enclosed and locked-in conduit assembly. Install spacers per manufacturer's instructions, but provide a minimum of two spacer assemblies per 10 feet of conduit assembly.

3.6.7 Conduit Plugs and Pull Rope

Provide new conduit indicated as being unused or empty with plugs on each end. Plugs must contain a weephole or screen to allow water drainage. Provide a plastic pull rope having 3 feet of slack at each end of unused or empty conduits.

3.6.8 Conduit and Duct Without Concrete Encasement

Depths to top of the conduit must be not less than 24 inches below finished grade. Provide not less than 3 inches clearance from the conduit to each side of the trench. Grade bottom of trench smooth; where rock, soft spots, or sharp-edged materials are encountered, excavate the bottom for an additional 3 inches, fill and tamp level with original bottom with sand or earth free from particles, that would be retained on a 1/4 inch sieve. The first 6 inch layer of backfill cover must be sand compacted as previously specified. The rest of the excavation must be backfilled and compacted in 3 to 6 inch layers.

3.6.8.1 Encasement Under Roads and Structures

Under roads, paved areas, and railroad tracks, install conduits in concrete encasement of rectangular cross-section providing a minimum of 3 inch concrete cover around ducts. Extend concrete encasement at least 5 feet beyond the edges of paved areas and roads, and 12 feet beyond the rails on each side of railroad tracks. Depths to top of the concrete envelope must be not less than 24 inches below finished grade.

3.6.8.2 Directional Boring

HDPE conduits must be installed below the frostline and as specified herein.

For distribution voltages greater than 1000 volts and less than 34,500 volts, depths to the top of the conduit must not be less than 48 inches in pavement-covered areas and not less than 120 inches in non-pavement-covered areas.

3.6.9 Duct Encased in Concrete

Construct underground duct lines of individual conduits encased in concrete. Depths to top of the concrete envelope must be not less than 18 inches below finished grade, except under roads and pavement, concrete envelope must be not less than 24 inches below finished grade. Do not mix different kinds of conduit in any one duct bank. Concrete encasement surrounding the bank must be rectangular in cross-section and provide at least 3 inches of concrete cover for ducts. Separate conduits by a minimum concrete thickness of 3 inches. Before pouring concrete, anchor duct bank assemblies, prevent floating during concrete pouring by driving reinforcing rods adjacent to duct spacer assemblies and attaching the rods to the spacer assembly.

3.6.9.1 Connections to Manholes

Duct bank envelopes connecting to underground structures must be flared to have enlarged cross-section at the manhole entrance to provide additional shear strength. Dimensions of the flared cross-section must be larger than the corresponding manhole opening dimensions by no less than 12 inches in each direction. Perimeter of the duct bank opening in the underground structure must be flared toward the inside or keyed to provide a positive interlock between the duct bank and the wall of the structure. Use vibrators when this portion of the encasement is poured to assure a seal between the envelope and the wall of the structure.

3.6.9.2 Connections to Existing Underground Structures

For duct bank connections to existing structures, break the structure wall out to the dimensions required and preserve steel in the structure wall. Cut steel and extend into the duct bank envelope. Chip the perimeter surface of the duct bank opening to form a key or flared surface, providing a positive connection with the duct bank envelope.

3.6.9.3 Connections to Existing Concrete Pads

For duct bank connections to concrete pads, break an opening in the pad out to the dimensions required and preserve steel in pad. Cut the steel and extend into the duct bank envelope. Chip out the opening in the pad to form a key for the duct bank envelope.

3.6.9.4 Connections to Existing Ducts

Where connections to existing duct banks are indicated, excavate the banks to the maximum depth necessary. Cut off the banks and remove loose concrete from the conduits before new concrete-encased ducts are installed. Provide a reinforced concrete collar, poured monolithically with the new duct bank, to take the shear at the joint of the duct banks. Remove existing cables which constitute interference with the work.

3.6.9.5 Partially Completed Duct Banks

During construction wherever a construction joint is necessary in a duct bank, prevent debris such as mud, and, and dirt from entering ducts by providing suitable conduit plugs. Fit concrete envelope of a partially completed duct bank with reinforcing steel extending a minimum of 2 feet back into the envelope and a minimum of 2 feet beyond the end of the envelope. Provide one No. 4 bar in each corner, 3 inches from the edge of the envelope. Secure corner bars with two No. 3 ties, spaced approximately one foot apart. Restrain reinforcing assembly from moving during concrete pouring.

3.6.9.6 Removal of Ducts

Where duct lines are removed from existing underground structures, close the openings to waterproof the structure. Chip out the wall opening to provide a key for the new section of wall.

3.6.10 Duct Sealing

Seal all electrical penetrations for radon mitigation, maintaining integrity of the vapor barrier, and to prevent infiltration of air, insects, and vermin.

3.7 CABLE PULLING

Test existing duct lines with a mandrel and thoroughly swab out to remove foreign material before pulling cables. Pull cables down grade with the feed-in point at the manhole or buildings of the highest elevation. Use flexible cable feeds to convey cables through manhole opening and into duct runs. Do not exceed the specified cable bending radii when installing cable under any conditions, including turnups into switches, transformers, switchgear, switchboards, and other enclosures. Cable with tape shield must have a bending radius not less than 12 times the overall diameter of the completed cable. If basket-grip type cable-pulling devices are used to pull cable in place, cut off the section of cable under the grip before splicing and terminating.

3.7.1 Cable Lubricants

Use lubricants that are specifically recommended by the cable manufacturer for assisting in pulling jacketed cables.

3.8 CABLES IN UNDERGROUND STRUCTURES

Do not install cables utilizing the shortest path between penetrations, but route along those walls providing the longest route and the maximum spare cable lengths. Form cables to closely parallel walls, not to interfere with duct entrances, and support on brackets and cable

insulators. Support cable splices in underground structures by racks on each side of the splice. Locate splices to prevent cyclic bending in the spliced sheath. Install cables at middle and bottom of cable racks, leaving top space open for future cables, except as otherwise indicated for existing installations. Provide one spare three-insulator rack arm for each cable rack in each underground structure.

3.8.1 Cable Tag Installation

Install cable tags in each manhole as specified, including each splice. Tag wire and cable provided by this contract. Install cable tags over the fireproofing, if any, and locate the tags so that they are clearly visible without disturbing any cabling or wiring in the manholes.

3.9 CONDUCTORS INSTALLED IN PARALLEL

Group conductors such that each conduit of a parallel run contains one Phase A conductor, one Phase B conductor, one Phase C conductor, and one neutral conductor.

3.10 LOW VOLTAGE CABLE SPLICING AND TERMINATING

Make terminations and splices with materials and methods as indicated or specified herein and as designated by the written instructions of the manufacturer. Do not allow the cables to be moved until after the splicing material has completely set. Make splices in underground distribution systems only in accessible locations such as manholes, handholes, or aboveground termination pedestals.

3.10.1 Terminating Aluminum Conductors

a. Use particular care in making up joints and terminations. Remove surface oxides by cleaning with a wire brush or emery cloth. Apply joint compound to conductors, and use UL-listed solid aluminum connectors for connecting aluminum conductors. When connecting aluminum to copper conductors, use connectors specifically designed for this purpose.

b. Terminate aluminum conductors to copper bus either by: (1) in line splicing a copper pigtail to the aluminum conductor (copper pigtail must have a ampacity at least that of the aluminum conductor); or (2) using a circumferential compression type, aluminum bodied terminal lug UL listed for AL/CU and steel Belleville spring washers, flat washers, bolts, and nuts. Belleville spring washers must be cadmium-plated hardened steel. Install the Belleville spring washers with the crown up toward the nut or bolt head, with the concave side of the Belleville bearing on a heavy-duty, wide series flat washer of larger diameter than the Belleville. Tighten nuts sufficient to flatten Belleville and leave in that position. Lubricate hardware with joint compound prior to making connection. Wire brush and apply joint compound to conductor prior to inserting in lug.

c. Terminate aluminum conductors to aluminum bus by using all-aluminum nuts, bolts, washers, and lugs. Wire brush and apply inhibiting compound to conductor prior to inserting in lug. Lubricate hardware with joint compound prior to making connection; if bus contact surface is unplated, scratch-brush and coat with joint compound (without grit).

3.11 MEDIUM VOLTAGE CABLE TERMINATIONS

Make terminations in accordance with the written instruction of the termination kit manufacturer.

3.12 MEDIUM VOLTAGE CABLE JOINTS

Provide power cable joints (splices) suitable for continuous immersion in water. Make joints only in accessible locations in manholes or handholes by using materials and methods in accordance with the written instructions of the joint kit manufacturer.

3.12.1 Joints in Shielded Cables

Cover the joined area with metallic tape, or material like the original cable shield and connect it to the cable shield on each side of the splice. Provide a bare copper ground connection brought out in a watertight manner and grounded to the manhole grounding loop as part of the splice installation. Ground conductors, connections, and rods must be as specified elsewhere in this section. Wire must be trained to the sides of the enclosure to prevent interference with the working area.

3.13 CABLE END CAPS

Cable ends must be sealed at all times with coated heat shrinkable end caps. Cables ends must be sealed when the cable is delivered to the job site, while the cable is stored and during installation of the cable. The caps must remain in place until the cable is spliced or terminated. Sealing compounds and tape are not acceptable substitutes for heat shrinkable end caps. Cable which is not sealed in the specified manner at all times will be rejected.

3.14 FIREPROOFING OF CABLES IN UNDERGROUND STRUCTURES

Fireproof (arc proof) wire and cables which will carry current at 2200 volts or more in underground structures.

3.14.1 Fireproofing Tape

Tightly wrap strips of fireproofing tape around each cable spirally in half-lapped wrapping. Install tape in accordance with manufacturer's instructions.

3.14.2 Tape-Wrap

Tape-wrap metallic-sheathed or metallic armored cables without a nonmetallic protective covering over the sheath or armor prior to application of fireproofing. Wrap must be in the form of two tightly applied half-lapped layers of a pressure-sensitive 10 mil thick plastic tape, and must extend not less than one inch into the duct. Even out irregularities of the cable, such as at splices, with insulation putty before applying tape.

3.15 GROUNDING SYSTEMS

NFPA 70 and IEEE C2, except provide grounding systems with a resistance to solid earth ground not exceeding 25 ohms.

3.15.1 Grounding Electrodes

Provide cone pointed driven ground rods driven full depth plus 6 inches, installed to provide an earth ground of the appropriate value for the particular equipment being grounded.
If the specified ground resistance is not met, provide an additional ground rod in accordance with the requirements of NFPA 70 (placed not less than 6 feet from the first rod). Should the resultant (combined) resistance exceed the specified resistance, measured not less than 48 hours after rainfall, notify the Contracting Officer immediately.

3.15.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, by exothermic weld or compression connector.

a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.

b. Make compression connections using a hydraulic compression tool to provide the correct circumferential pressure. Tools and dies must be as recommended by the manufacturer. An embossing die code or other standard method must provide visible indication that a connector has been adequately compressed on the ground wire.

3.15.3 Grounding Conductors

Provide bare grounding conductors, except where installed in conduit with associated phase conductors. Ground cable sheaths, cable shields, conduit, and equipment with No. 6 AWG. Ground other noncurrent-carrying metal parts and equipment frames of metal-enclosed equipment. Ground metallic frames and covers of handholes and pull boxes with a braided, copper ground strap with equivalent ampacity of No. 6 AWG. Provide direct connections to the grounding conductor with 600 v insulated, full-size conductor for each grounded neutral of each feeder circuit, which is spliced within the manhole.

3.15.4 Ground Cable Crossing Expansion Joints

Protect ground cables crossing expansion joints or similar separations in structures and pavements by use of approved devices or methods of installation which provide the necessary slack in the cable across the joint to permit movement. Use stranded or other approved flexible copper cable across such separations.

3.15.5 Manhole Grounding

Loop a 4/0 AWG grounding conductor around the interior perimeter, approximately 12 inches above finished floor. Secure the conductor to the manhole walls at intervals not exceeding 36 inches. Connect the conductor to the manhole grounding electrode with 4/0 AWG conductor. Connect all incoming 4/0 grounding conductors to the ground loop adjacent to the point of entry into the manhole. Bond the ground loop to all cable shields, metal cable racks, and other metal equipment with a minimum 6 AWG conductor.

3.16 EXCAVATING, BACKFILLING, AND COMPACTING

Provide in accordance with NFPA 70 .

3.16.1 Reconditioning of Surfaces

3.16.1.1 Unpaved Surfaces

Restore to their original elevation and condition unpaved surfaces disturbed during installation of duct or direct burial cable. Preserve sod and topsoil removed during excavation and reinstall after backfilling is completed. Replace sod that is damaged by sod of quality equal to that removed. When the surface is disturbed in a newly seeded area, re-seed the restored surface with the same quantity and formula of seed as that used in the original seeding, and provide topsoiling, fertilizing, liming, seeding, sodding, sprigging, or mulching.

3.16.1.2 Paving Repairs

Where trenches, pits, or other excavations are made in existing roadways and other areas of pavement where surface treatment of any kind exists , restore such surface treatment or pavement the same thickness and in the same kind as previously existed, except as otherwise specified, and to match and tie into the adjacent and surrounding existing surfaces.

3.17 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.17.1 Concrete Slabs (Pads) for Equipment

Unless otherwise indicated, the slab must be at least 8 inches thick, reinforced with a 6 by 6 - W2.9 by W2.9 mesh, placed uniformly 4 inches from the top of the slab. Place slab on a 6 inch thick, well-compacted gravel base. Top of concrete slab must be approximately 4 inches above finished grade with gradual slope for drainage. Edges above grade must have 1/2 inch chamfer. Slab must be of adequate size to project at least 8 inches beyond the equipment.

Stub up conduits, with bushings, 2 inches into cable wells in the concrete pad. Coordinate dimensions of cable wells with transformer cable training areas.

3.17.2 Sealing

When the installation is complete, seal all conduit and other entries into the equipment enclosure with an approved sealing compound. Seals must be of sufficient strength and durability to protect all energized live parts of the equipment from rodents, insects, or other foreign matter.

3.18 FIELD QUALITY CONTROL

3.18.1 Performance of Field Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

3.18.1.1 Medium Voltage Cables

Perform tests after installation of cable, splices, and terminators and before terminating to equipment or splicing to existing circuits.

a. Visual and Mechanical Inspection

(1) Inspect exposed cable sections for physical damage.

(2) Verify that cable is supplied and connected in accordance with contract plans and specifications.

(3) Inspect for proper shield grounding, cable support, and cable termination.

(4) Verify that cable bends are not less than ICEA or manufacturer's minimum allowable bending radius.

(5) Inspect for proper fireproofing.

(6) Visually inspect jacket and insulation condition.

(7) Inspect for proper phase identification and arrangement.

b. Electrical Tests

(1) Perform a shield continuity test on each power cable by ohmmeter method. Record ohmic value, resistance values in excess of 10 ohms per 1000 feet of cable must be investigated and justified.

(2) Perform acceptance test on new cables before the new cables are connected to existing cables and placed into service, including terminations and joints. Perform maintenance test on complete cable system after the new cables are connected to existing cables and placed into service, including existing cable, terminations, and joints. Tests must be very low frequency (VLF) alternating voltage withstand tests in accordance with IEEE 400.2. VLF test frequency must be 0.05 Hz minimum for a duration of 60 minutes using a sinusoidal waveform. Test voltages must be as follows:

CABLE RATING AC TEST VOLTAGE for ACCEPTANCE TESTING	
5 kV	10kV rms(peak)
8 kV	13kV rms(peak)
15 kV	20kV rms(peak)
25 kV	31kV rms(peak)
35 kV	44kV rms(peak)

CABLE RATING AC TEST VOLTAGE for MAINTENANCE TESTING	
5 kV	7kV rms(peak)
8 kV	10kV rms(peak)
15 kV	16kV rms(peak)
25 kV	23kV rms(peak)
35 kV	33kV rms(peak)

3.18.1.2 Low Voltage Cables, 600-Volt

Perform tests after installation of cable, splices and terminations and before terminating to equipment or splicing to existing circuits.

a. Visual and Mechanical Inspection

(1) Inspect exposed cable sections for physical damage.

(2) Verify that cable is supplied and connected in accordance with contract plans and specifications.

(3) Verify tightness of accessible bolted electrical connections.

(4) Inspect compression-applied connectors for correct cable match and indentation.

(5) Visually inspect jacket and insulation condition.

(6) Inspect for proper phase identification and arrangement.

b. Electrical Tests

(1) Perform insulation resistance tests on wiring No. 6 AWG and larger diameter using instrument which applies voltage of approximately 1000 volts dc for one minute.

(2) Perform continuity tests to insure correct cable connection.

3.18.1.3 Grounding System

a. Visual and mechanical inspection

Inspect ground system for compliance with contract plans and specifications.

b. Electrical tests

Perform ground-impedance measurements utilizing the fall-of-potential method in accordance with IEEE 81. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground

resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. The instrument must be equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test. Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance.

3.18.2 Follow-Up Verification

Upon completion of acceptance checks and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. As an exception to requirements stated elsewhere in the contract, the Contracting Officer must be given 5 working days advance notice of the dates and times of checking and testing.

.... -- End of Section --